DIAScreen
Software User Manual

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Related Documents

Document Name	Document ID
DIAInstaller User Manual	DIAS-Manual-0005-EN
DIADesigner Manual	DIAS-Manual-0003-EN

Revision History

Version	Issue Date	Revision
1.0	2020/12/08	The first version was published.
1.1.0	2021/09/30	Added Chapter 2 ~ Chapter 30: Added description of DOP series/AX series/IMP series/DXMC series.
1.1.1	2021/12/30	1. Updated Section 2.2.2.1: Update the description of Clipboard. 2. Updated Section 2.2.2.7: Update Select All instructions. 3. Updated Section 24.3: Update the description of macro instructions. 4. Updated Section 28.6: Add descriptions of new functions of the Text Bank. 5. Added Section 29.2.3: Added OPC UA variable setting function.
1.1.2	2022/09/30	1. Updated the overall manual structure in response to software UI adjustments. 2. Update Section 2.1.1: Added description of DIAScreen startup screen. 3. Added Section 2.1.1.1: Added Update Manager.
1.3.0	2022/12/22	Standalone TP series user manual.
1.3.1 1.3.2	2023/07/28	1. Added Custom Confirmation window section. 2. Updated Enhanced Recipe section. 3. Added Font Template section. 4. Added Installment section. 5. Added DIADeigner-AX Tag Automatic Synchronization section. 6. Added User Login Screen section.
1.4.0	2023/12/19	1. Updated Address Setting section. 2. Updated Alarm Setting section. 3. Added Copy Multi-Language Font section. 4. Updated Create Screen Data File section. 5. Updated MODBUS TCP / COM Mapping Table section. 6. Updated MQTT Settings section and added Retain Message and Will Message Function sections. 7. Added Retain Permissions and Password Table Settings after Screen Download section.
1.5.0	2024/10/17	1. Updated Report List section.

Version	Issue Date	Revision
		2. Updated Data Display section. 3. Updated IIoT section. 4. Updated OPC UA section. 5. Added EIP Exchange Table section. 6. Updated MQTT Settings section. 7. Updated Install and Uninstallation section.
1.5.2	2025/02/11	1. Updated Communication section and renamed it to Device Communication. 2. Updated History Buffer section. 3. Updated Sampling section. 4. Updated Setting Up History Buffer Example section. 5. Added File Brower and Picture Viewer sections. 6. Updated OPC UA section. 7. Update MQTT Settings section.
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1.6.1	10/03/2025	1. Added Screen section. 2. Updated Configuration section. 3. Added Environment Settings section. 4. Updated Compile section. 5. Added Transmission section. 6. Added Picture Bank section. 7. Added Text Bank section. 8. Updated Font Management section. 9. Added Multimedia Setting section. 10. Added Scheduler section. 11. Updated Installment section. 12. Updated Button element section. 13. Added Meter element section.

Version	Issue Date	Revision
		14. Added Pipe element section. 15. Updated Pie element section, add the description of Pie Chart element. 16. Added Indicator element section. 17. Updated Data Display element section. 18. Added Graphic element section. 19. Added Input element section. 20. Added Curve element section, 21. Added Keypad element section. 22. Updated List element section. 23. Added Basic Shape elements section. 24. Added Drawing element section. 25. Restructured Alarm element section, add Notification section. 26. Deleted Enhanced Recipe section and added Recipe section. 27. Added Operation Log Setup section. 28. Renamed IIoT section to DIACloud and update the content, add the description of cloud remaining traffic. 29. Added Change Model section. 30. Updated All Tags section. 31. Added Security Settings section. 32. Updated Address Settings section: Delete the description of Variable Sharing. 33. Added Appendix section. 34. Updated DOP-300 Series HMI Special Features section.
1.7.0	2026/03/05	1. Revised Install section: Revised the content of Update Manager. 2. Updated New Project section. 3. Updated Create Screen Data File section and rename it to Create Transfer Data. 4. Updated Screen section. 5. Updated Configuration section: Updated the description of Elements are automatically resized based on text. 6. Updated Environment Setting section: Add the description of high security password setting. 7. Updated Compile section: Added the description of COMMGR Simulator Connection, updated the description of Download All Data.

Version	Issue Date	Revision
		8. Added FileSlot Management section. 9. Revised Installment section. 10. Added Macro section. 11. Updated Button element section: Moved the description of Batch Multistate to common properties, updated the description of Import / Export FileSlot. 12. Updated Meter element section: Added the description of scale setting. 13. Updated Data Display and Graphic elements sections: Added the description of Batch Multistate to common properties. 14. Updated Curve element section: Added the description of Dot Weight setting. 15. Updated Keyboard and List elements sections: Updated the description of element color. 16. Updated DIACloud section: Added the description of creating data Center. 17. Updated Security Settings section: Updated the description of security setting. 18. Updated Address Setting section: Added internal parameter NET1_IPSETTING to Ethernet Setting, modified the description of Non-volatile Internal Register ($M).

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DIAScreen Overview

The DIAScreen application is an intuitive visualization software, which can configure the Delta DOP series, TP series, and AX series controller soft-HMIs.

DIAScreen Features

DIAScreen has the following features:

- Tag sharing with the variables of DIADesigner and DIADesigner-AX
- Convenient User Interface, and rich 3D image library
- Smoother display for Meter and other elements
- Enhanced software download speed
- Clear output results
- Various control elements and functions, such as Recipe and Macros
- Support multi-language, such as English, Traditional Chinese and Simplified Chinese
- Provide passwords and permission levels to enhanced security.

Supported Devices

DIAScreen version 1.7.0 supports the following devices.

Type	Product Series
Controller soft-HMI	AX series
Touch type high color HMI	DOP-100 series
IoT type high color HMI	DOP-300 series
Handheld high color HMI	DOP-H Series (Handheld)
Motion Control development platform	IMP series
Text Panel (TP) HMI	TP series

Note:

- Support AX / IMP series soft-HMI planning functions. Refer to AX series products manual and IMP manual for detail.
- Support DOP-100, DOP-300 and DOP-H series soft-HMI planning functions. Refer to HMI product manual for detail.

- Supports TP series HMI planning functions. Refer to TP series operation manual.
- To avoid network attacks, the HMI must be connected and run within a firewall to slave devices or upper systems.
- If the environmental equipment uses flow control, it will affect the human-machine operating performance.

Installation

This section introduces how to download, install and uninstall DIAScreen in DIAInstaller.

System Requirements

To install and run DIAScreen, your computer should meet the following minimum hardware and software requirements.

Item	System Requirements
Operating System	Windows 10/11 (32-bit and 64-bit)
CPU	Intel Core i5 M520 2.4 GHz (min.)
Memory	Recommend to use 4 GB or above
Hard Disk Drive	15 GB free for DIAScreen
Monitor	Resolution: 1024 x 768 Pixels, 1920 x 1080 Pixels
Keyboard/Mouse	General keyboard mouse or Windows-compatible device
Printer	Printer that can be driven by Windows system (if you need to print project content)
PC Interface	Ethernet, USB, Serial port (depends on product interface)
Software	Need to install Microsoft .Net Framework 4.6.2 Need to install WinPcap V4.1.3* Need to install NDIS V5.0

***Note**: The upload and download function of the DOP-100 Gen2 and DOP-300 series HMI can transmit data through ZeroIP (zero address network) without network configuration and IP addresses. You must install and use the WinPcap package to realize this ZeroIP technology.

WinPcap download link: winpcap.org/install/bin/WinPcap_4_1_3.exe

Install DIAScreen

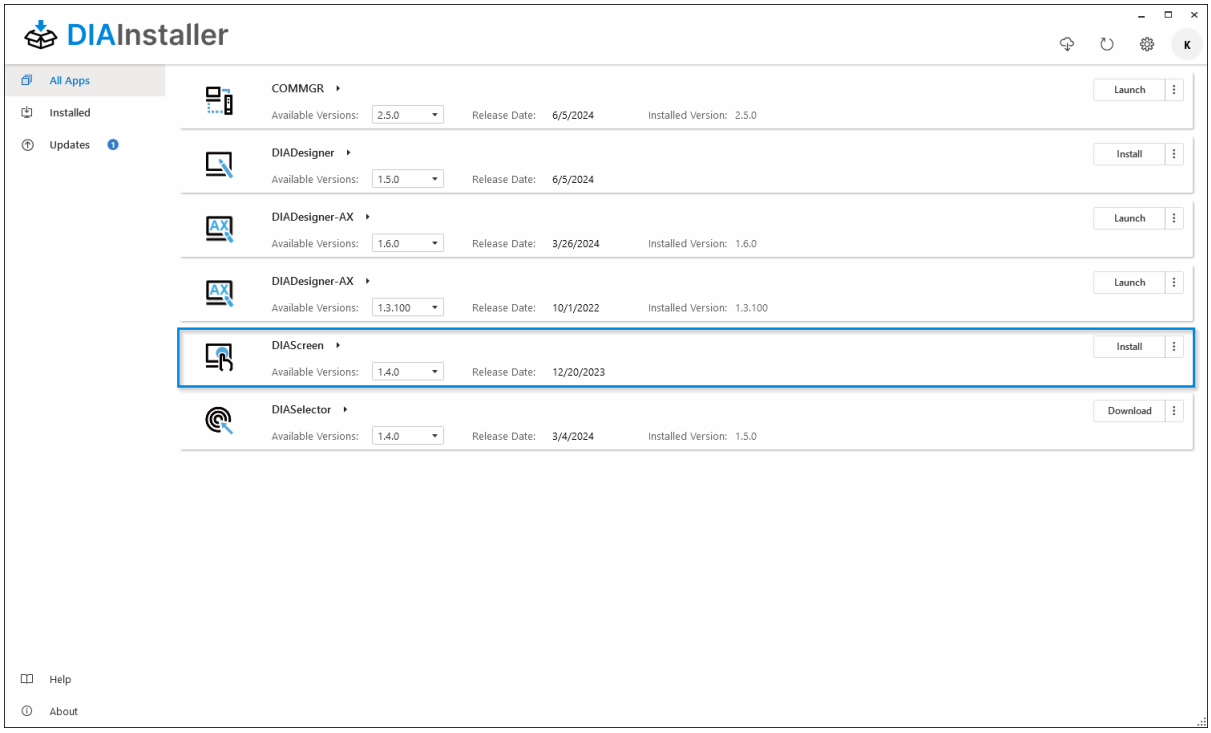
DIAInstaller is used to easily download, install, uninstall, and update all DIAStudio software. You can download DIAInstaller from diastudio.deltaww.com. For more information about DIAInstaller, see *DIAInstaller Download and Installation Manual*.

Prerequisites

- DIAInstaller is installed.

To install DIAScreen

1. Open DIAInstaller.
2. Click at the upper-right corner to sign in.
3. Go to **All Apps** and find DIAScreen.
4. In **Available Versions**, select a version.
5. Click and then select **Install**.
6. Select components to install. You can select the full components or customize the components to install.
7. In the **DIAScreen** dialog, follow the on-screen instructions to complete the installation.
8. Open DIAScreen and update the firmware version. For the firmware update method, see Update Manager.



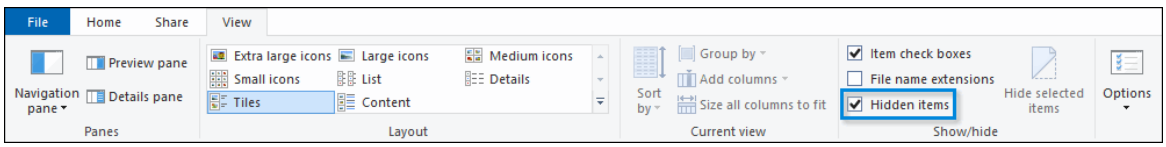
Uninstall DIAScreen

Use the following steps to uninstall DIAScreen in DIAInstaller.

Prerequisites

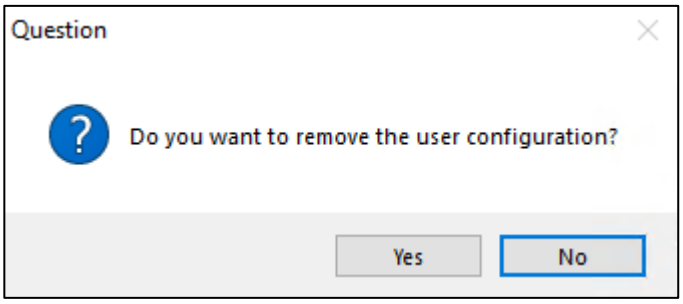
- Back up the picture bank or font templates created by users.
  - Picture bank file path: `C:\ProgramData\Delta Industrial Automation\HMI\Common\Pic`
  - Font templates file path: `C:\ProgramData\Delta Industrial Automation\HMI\Common\FontTemplate.xlsx`

  Note: In the folder, click **View** and select the **Hidden Items** checkbox to view the ProgramData folder.



To uninstall DIAScreen

1. In DIAInstaller, go to **Installed**, and select **Uninstall**.
2. Follow the on-screen instructions to uninstall.
3. A message dialog appears, asking whether to delete the user configuration. Click **No** to keep the user configuration. User configurations include picture bank and font templates.



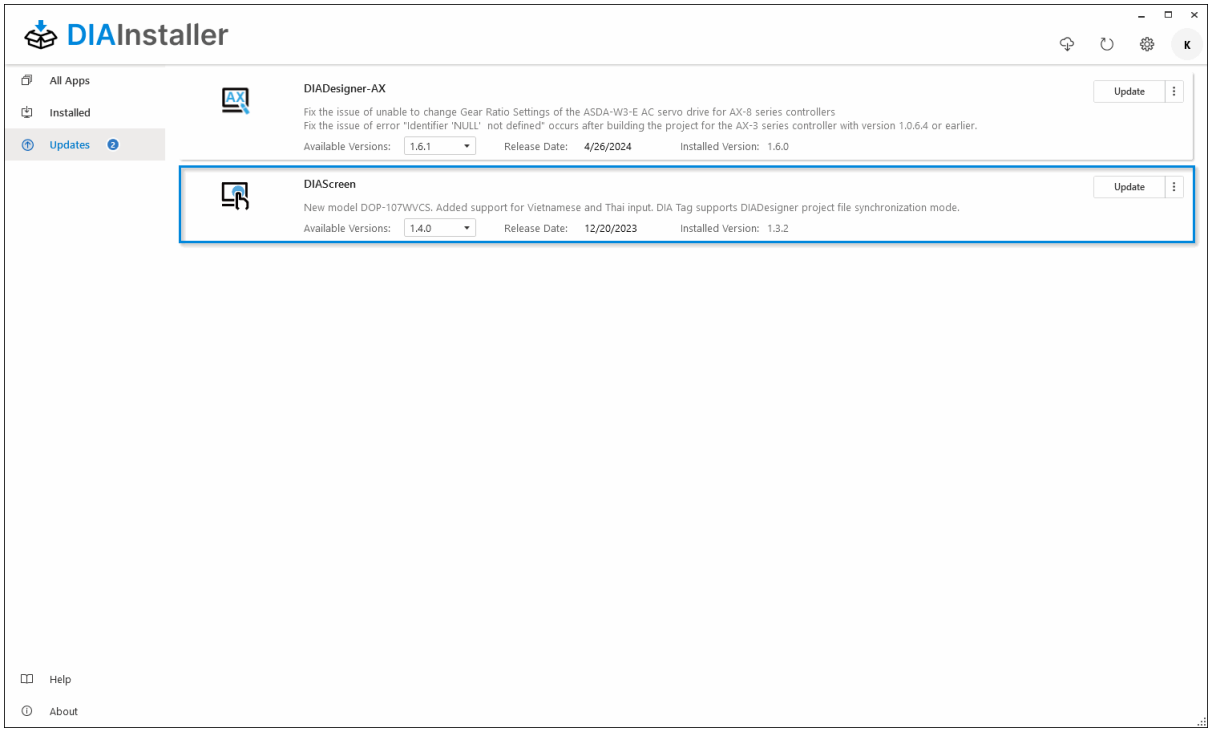
4. If you choose to remove user configuration during uninstallation, after reinstalling DIAScreen, overwrite the picture bank or font templates files to be restored to the above file path.

Update DIAScreen

Use the following steps to update DIAScreen.

To update DIAScreen

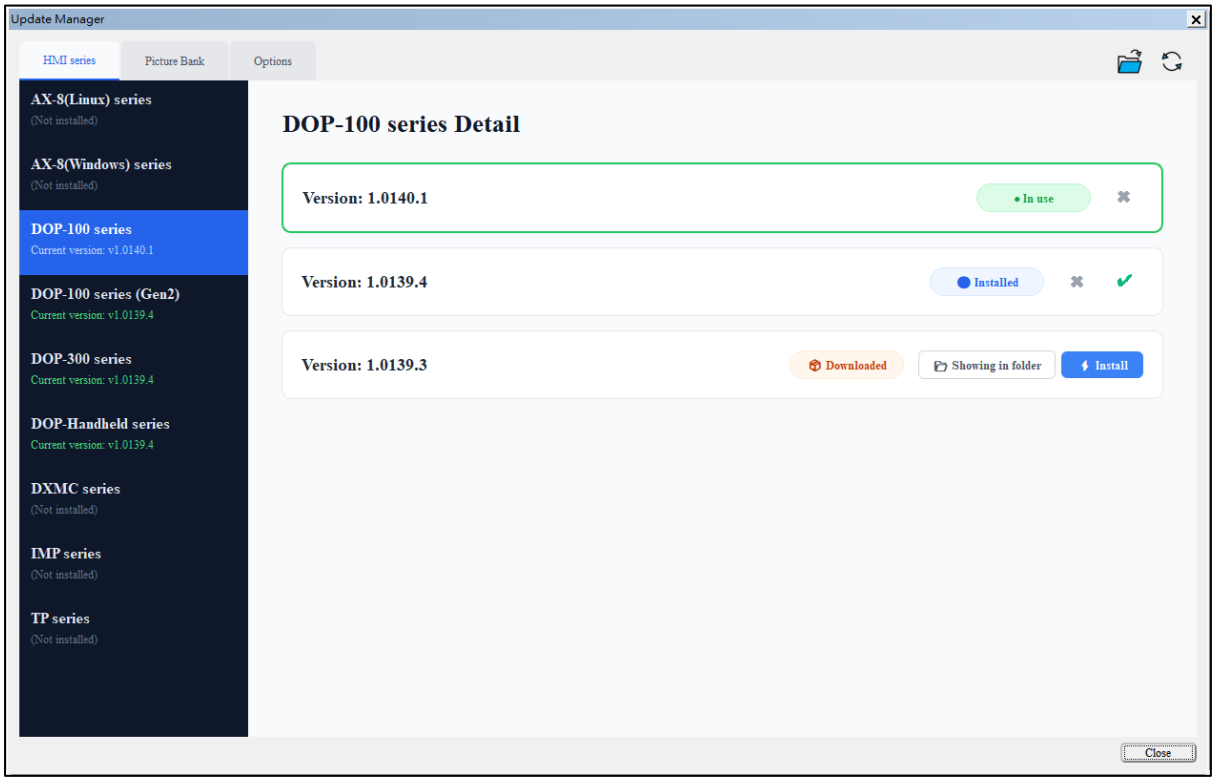
1. In DIAInstaller, go to **Updates**.
2. Find DIAScreen.
3. In **Available Versions**, select a version.
4. Do one of the following:
   - Select **Update** and then follow the on-screen instructions to complete the update.
   - Click ⋮ and select **Download** to download the installation file to the local path, and then select **Show in folder** to find the .exe file to install.



Update Manager

DIAScreen only has built-in firmware files for DOP-100 Series models, and you can download, install, and update the firmware for each series through **Update Manager** to edit the HMI screen of other models**.**

Note: Your computer must have a network connection to use the update manager to download the firmware file.

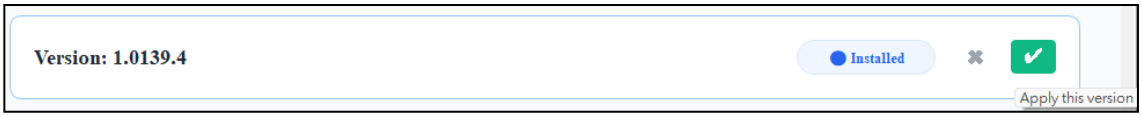


Follow these steps to download and install the HMI firmware.

To download and install the HMI firmware

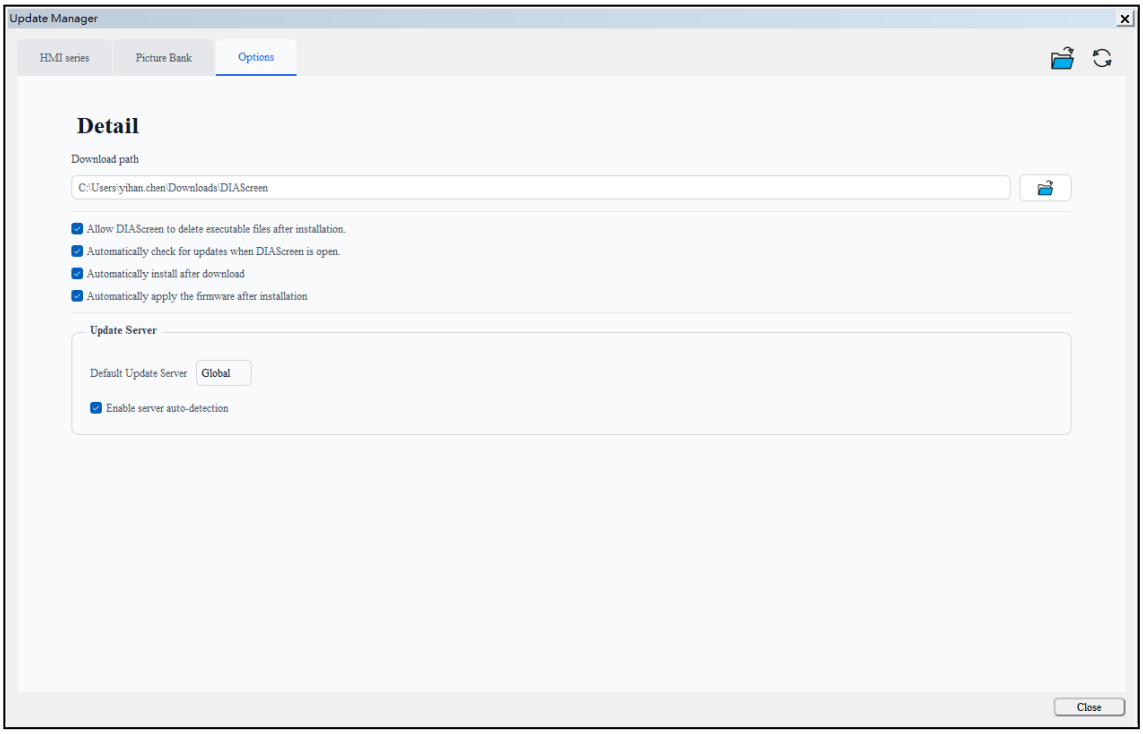
1. On the home page of DIAScreen, click **Update Manager**.
2. Select the HMI series in the list on the left of the HMI series tab in the **Update Manager** dialog, and then **Download** the required firmware version in the firmware list on the right The firmware version will be automatically installed after downloading. After the installation is completed, the firmware will be in **In use** status.

   Note: To download the firmware file without installing it, deselect the **Automatically install after download** checkbox in the **Options** tab.
3. You can also download other firmware versions. Click ✔ in the version list to apply the updated version.



4. After the firmware package is downloaded, you can click to open the download folder to view the firmware installation file.

   Note: The **Download path** can be set in the **Options** tab. To view the firmware files, deselect the **Allow DIAScreen to delete executable files after installation** checkbox.



Get Started

New Project

To create a DIAScreen project, do any of the following:

- Click the **New Project** button on the DIAScreen startup page.

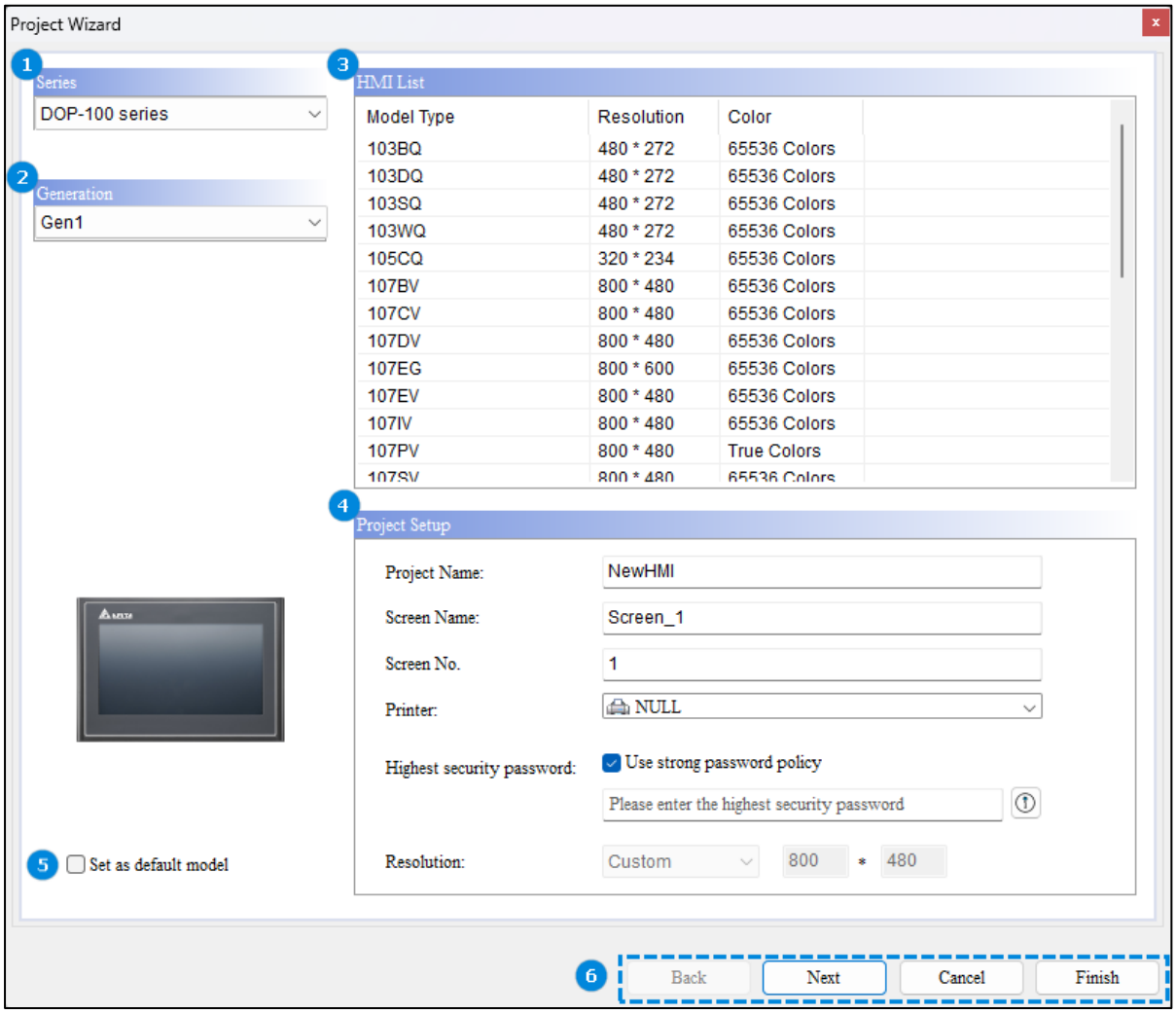
Or

- Click **New Project** in the menu in the upper left corner of the software.

Or

- Click icon on the toolbar in the upper left corner of the software.

The following table lists the settings in the **Project Wizard** dialog with their description.

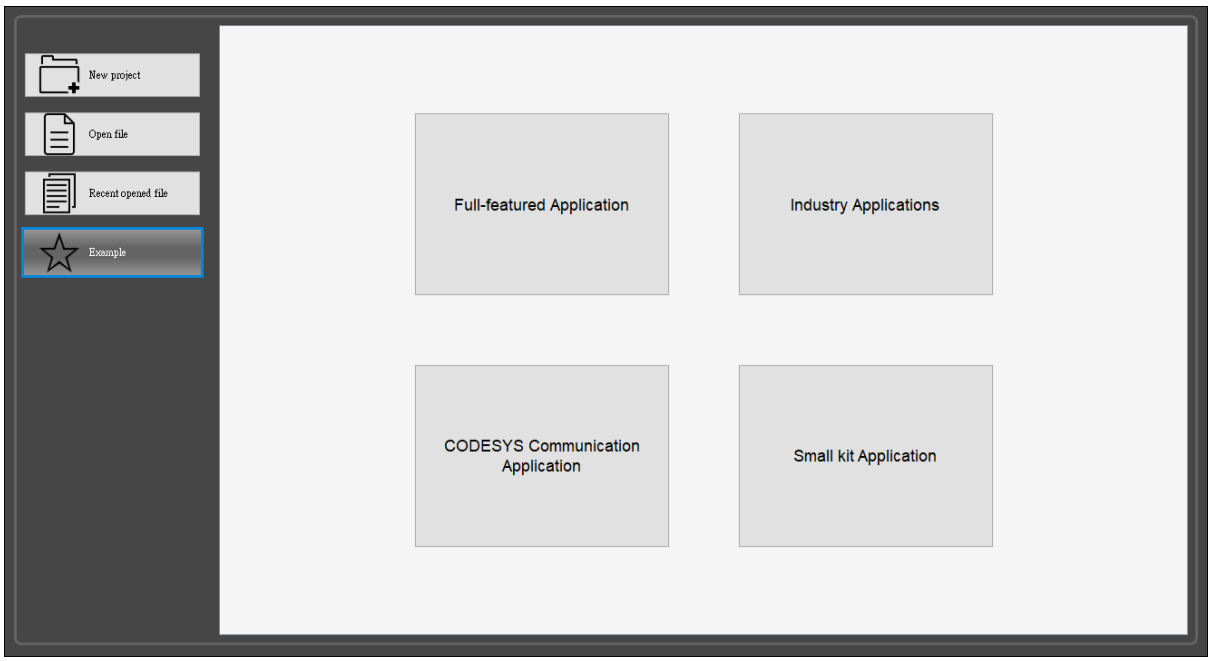


Legend	Setting	Description
❶	Series	Lists available series according to the HMI firmware installed in the **Update Manager**.
❷	Generation	The generation can only be selected when selecting **DOP-100 Series**.
❸	HMI List	The list is displayed according to the selected **HMI series**. After selecting the HMI, the preview of the selected HMI will be displayed in the lower left corner of the dialog.
❹	Project Setup	• Enter the **Project Name** and the **Screen Name** and **Screen No**. of the default screen. • Enter the **Highest security password** of the HMI. The **Use strong password policy** checkbox is selected by default, the password must be at least 8 characters, including one uppercase letter, one lowercase letter, one number, and one symbol (!, $, #, %). **Note**: You can set the default **Highest security password** in the **Environment Settings**. • For AX-8 Series HMI, the **Resolution** can be set.
❺	Set as default model	After selecting an HMI model, select **Set as default model** checkbox, and the default model will be automatically selected the next time a new project is created.
❻	Action	After selecting the HMI and setting the project, click the buttons to perform the actions. • **Next**: Enter the **Device Communication** page to configure the settings related to communication between HMI and device. Refer to Device Communication. • **Finish**: Complete the project creation and enter the screen editing interface.

Sample Project

To help users quickly set up projects, DIAScreen provides the following four sample projects:

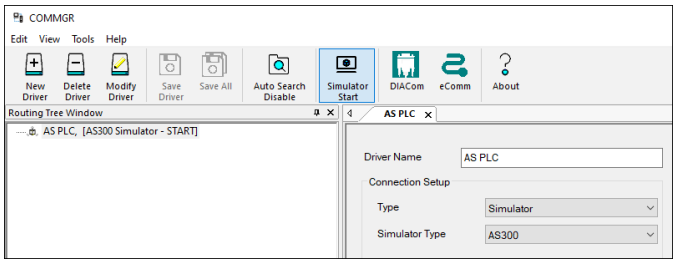
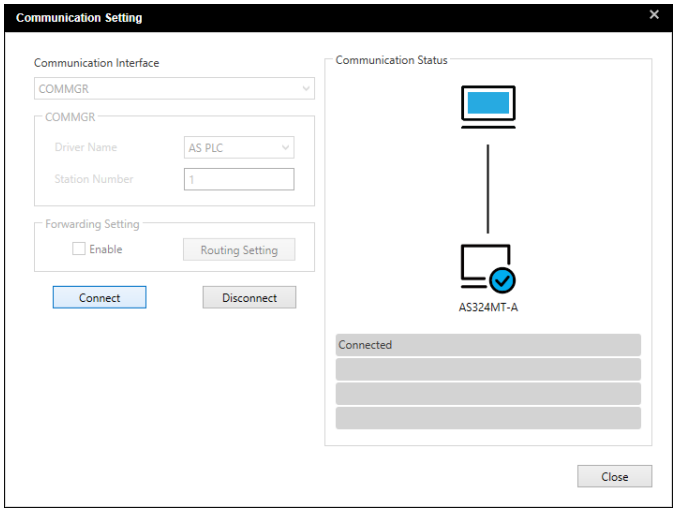
- Full-featured Application
- Industry Applications
- CODESYS Communication Application
- Small kit Application

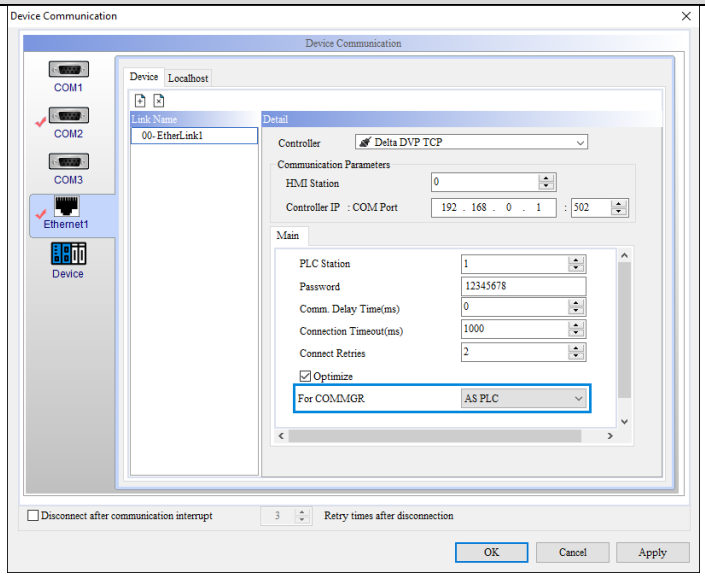


The following table lists the functions demonstrated by each project and how they operate.

- **Full-featured Application**

Project	Description
Alarm Function	Configure **discrete alarms**, trigger alarms through value changes or button clicks on the screen, and display alarm information on the alarm element.
Full_Element_Demo	Configure various elements on the screen and apply different styles. After loading the project into the HMI, switch between different pages to view the operating effects of each element.

Project	Description
Temperature_Control_Demo	DOP series HMI connects AS controller to perform temperature control operations. Follow these steps to connect to the simulator. 1. Open sample project in DIAScreen. 2. Open the controller project in DIADesigner. Project Path: `C:\ProgramData\Delta Industrial Automation\HMI\DIAScreen (version)\ScrEditApp\Example\1\ Example3_Temperature_Control_Demo\Temperature_Control_Demo.diade` 3. Create an AS300 simulator in COMMGR and start it.  4. In DIADesigner, select **AS PLC** Simulator in **Communication Setting**, and click **Connect**.  5. Perform **Project Download** and then perform **Online**. 6. Open the sample project in DIAScreen and go to **General** > **Device Communication**. In **Ethernet1** page of the **Device Communication** dialog, select **AS PLC** in **For COMMGR**.

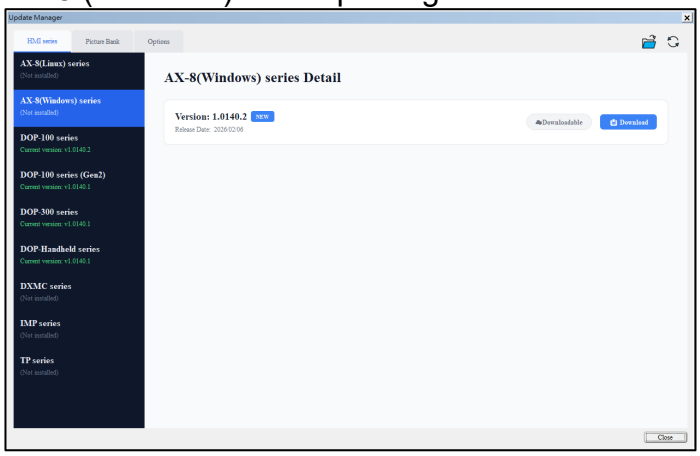
Project	Description
	 7. Perform **Off-line Simulation** in DIAScreen. 8. Perform temperature control operations through HMI simulator.
History Function	Presents historical data of various data types through the **Historical Trend Graph** element, and different styles of historical trend graphs are also configured to facilitate quick understanding of the functions provided by the history buffer.

- **Industry Applications**

Project	Description
Labeling machine	Demonstrates the operating interface used for HMI monitoring equipment status in different industries. You can refer to the page configuration, copy the screen and apply it to their own designed projects.
Extrusion rubber plate machine	
Electrical industry	

- **CODESYS Communication Application**

Project	Description
CODESYS_Data_ Exchange	Demonstrates data exchange between AX-8's Soft-HMI and CODESYS controller. Follow these steps to configure the project.

Project	Description
	1. Use **Update Manager** to download and install AX-8 (Windows) series package in DIAScreen.  2. Open sample project in DIAScreen and then download the project directly to the AX-8 device. 3. Open the controller project in DIADesigner-AX, and then download the controller project to the AX-8 device. Project Path: `C:\ProgramData\Delta Industrial Automation\HMI\DIAScreen (version)\ScrEditApp\Example\3 \Example1_CODESYS_Data_Exchange_Demo \CODESYS_Data_Exchange_Demo.project` 4. Open HMIManager on the AX-8 device to monitor the CODESYS controller through the HMI.

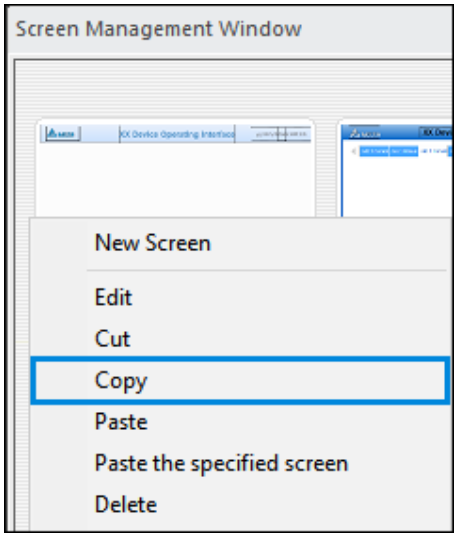
- **Small kit Application**

  This category of projects provides a variety of template styles. You can directly copy the template screens into the development project to achieve rapid project configuration.

  Follow these steps to copy a template.

To copy a template

1. Open the sample project. Select the screen in **Screen Management Window**. Right-click the selected screen and select **Copy** (or use the shortcut key Ctrl + C).



2. Open another DIAScreen project and paste the screen directly in **Screen Management Window** (or use the shortcut key Ctrl + V) to configure the template screen in the development project.

Project	Description
StartUp	This project provides a variety of **Startup interface** template styles.
Template	This project provides a variety of **Operating Interface** template styles.

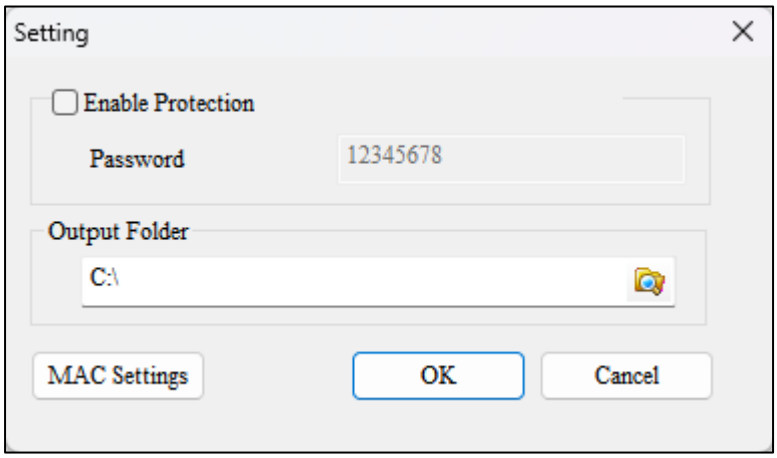
Create Transfer Data

Screen Update File / Automatic Screen Update File

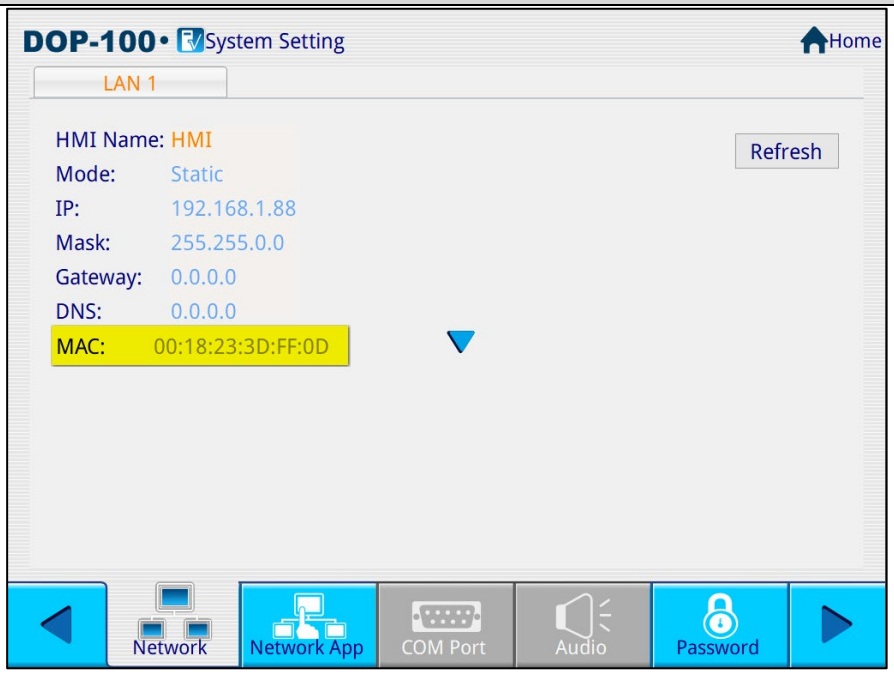
After storing the files compiled by the project in an external storage device (USB or SD card) and inserting it into the HMI, you can update the HMI screen through Screen Update File and Automatic Screen Update File.

- **Screen Update File**: Need to enter the system menu to update the HMI screen and firmware.
- **Automatic Screen Update File:** Automatically update the HMI screen and firmware without entering the system menu.

The following table lists the settings in the **Setting** dialog with their descriptions.



Setting	Description
Enable Protection	Enable password protection. When updating the screen on the HMI, you need to enter this copy password.
Output Folder	Select the folder path to save the download screen executable file.
MAC Settings	Enter the MAC of the HMI to update screen data. In the **Configuration** dialog, click and enter the MAC code of the HMI to be bound, the file can only perform screen updates on the specified HMI; if the MAC code is not set, the HMI model is not restricted. **Note**: You can obtain the MAC code from **System Menu** > **System Setting** > **Network** on the HMI.

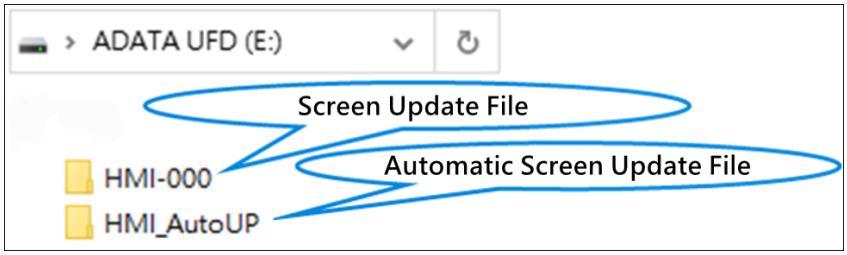
Setting	Description
	

Follow these steps to create the screen update file and automatic screen update file.

- **DIASceen Operation**

1. In the main menu, click **Create Transfer Data** > **Screen Update File** / **Automatic Screen Update File**.
2. In the **Setting** dialog, make the related settings, and then click **OK**.

   The Screen Update File / Automatic Screen Update File are generated.

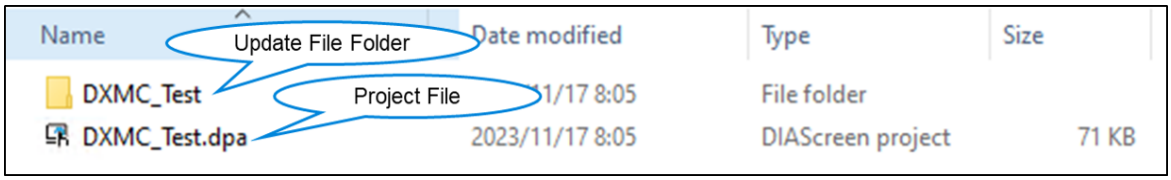


Note:

- If there are multiple data files in the HMI-000 folder, the software will number them sequentially from 000, for example, 000, 001, 002 and so on.

- After executing **Automatic Screen Update File**, the HMI only recognizes the folder name HMI_AutoUP. If you repeatedly execute this function, the contents of this folder will be overwritten.

If the project model is the DXMC series, after creating the project and save project, then execute the compilation, the software will automatically create a folder with the same name as the project under the project path and store the current **Screen Update File** in the folder. You can directly import this update file into **DXMC Model Software-DMARS (DELTA Motion And Robot Software)** to update HMI screen.



HMI Operation

- **Screen Update File**

  After inserting the storage device into the HMI, go to **System Setting** > **File Manager** > **File Copy** on the HMI, and copy the screen data file in the storage device to the HMI.

- **Automatic Screen Update File**

  1. Inserting the external storage device into the HMI.

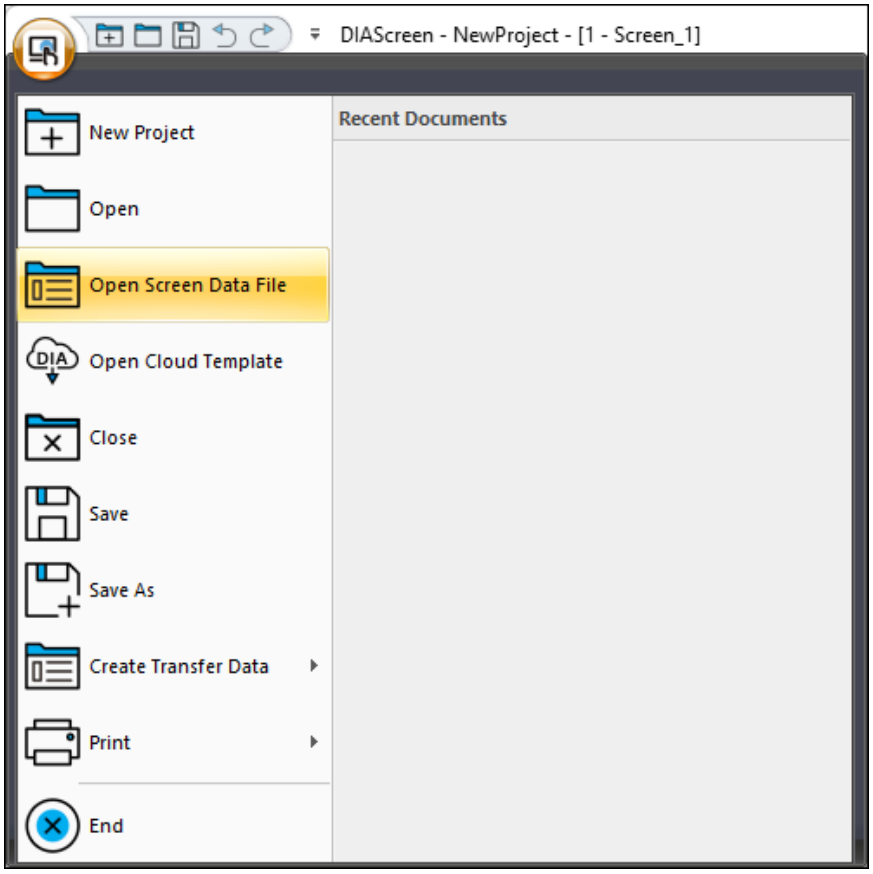
     When the HMI detects an external storage device with an automatic screen update file (HMI_AutoUP), a message dialog displays and ask you whether to execute the auto update.

  2. Click **Yes**, the firmware and screen will be updated automatically.

     Note: If the folder HMI_AutoUP is stored in both USB and SD card, when updating the firmware and screen files, the HMI will first update the USB.

Open Screen Data File

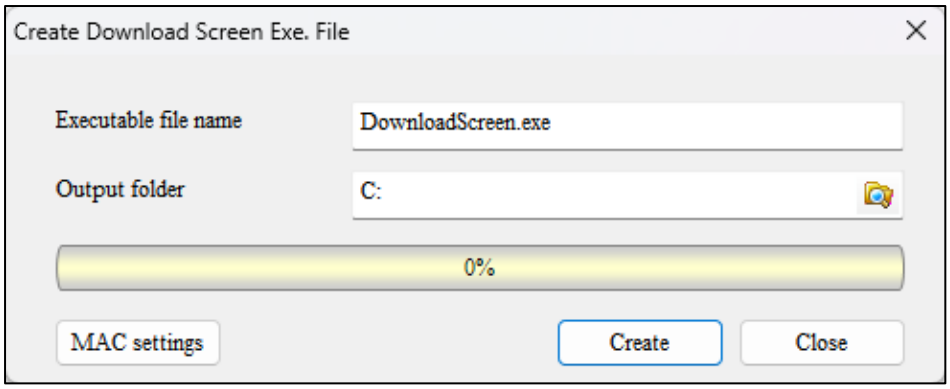
The generated **Screen Update File** / **Automatic Screen Update File** can also be edited in the software by **Open Screen Data File** function.

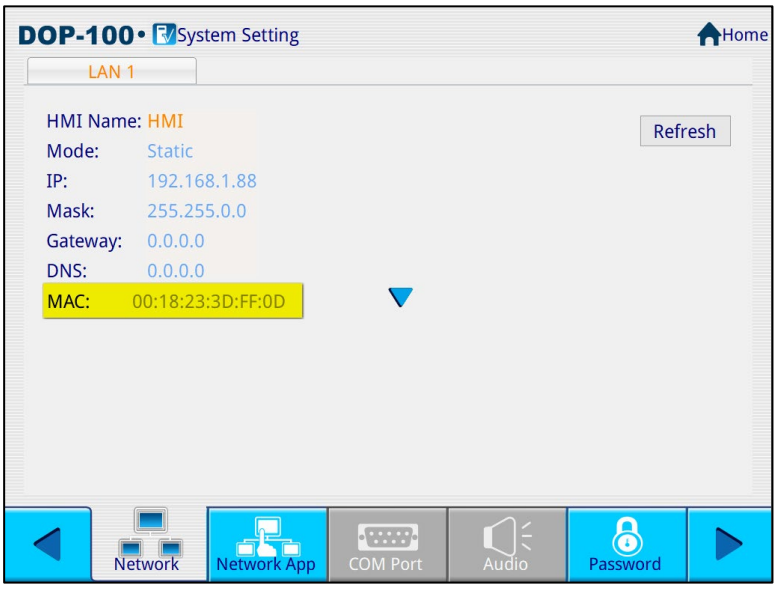


Download Screen Executable File

You can update the HMI screen or firmware to the HMI through **Download Screen Executable File** without installing the DIAScreen software.

The following table lists the settings in the **Create Download Screen Exe. File** dialog and their descriptions.

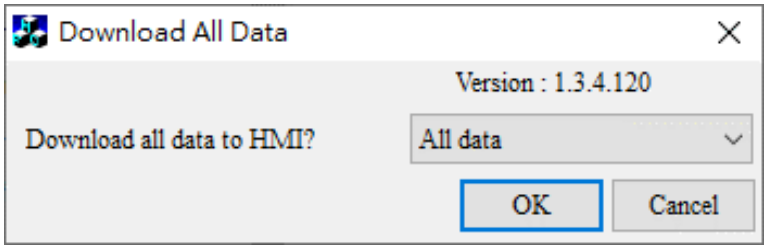


Settings	Description
Executable file name	Set the name of the generated executable file.
Output folder	Select the folder path to save the download screen executable file.
MAC settings	Set the HMI to execute download screen executable file. In the **Configuration** dialog, click and enter the MAC code of the HMI to be bound, the file can only perform screen download on the specified HMI; if the MAC code is not set, the HMI model is not restricted. **Note**: You can obtain the MAC code from **System Menu** > **System Setting** > **Network** on the HMI. 

Follow these steps to create download screen executable file.

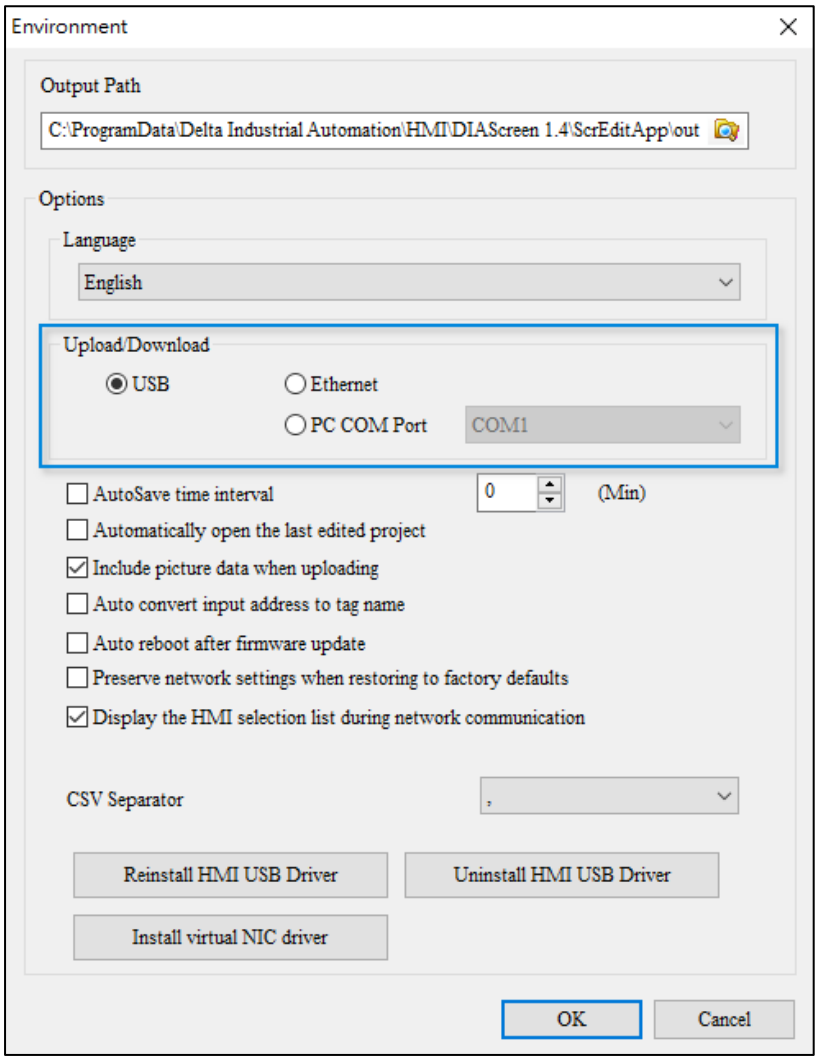
To create download screen executable file

1. In the main menu, click **Create Transfer Data File** > **Download Screen Executable File**.
2. In the **Create Download Screen Exe. File** dialog, make the related settings.
3. In the output folder, execute the executable file (the default file name is DownloadScreen.exe).
4. In the pup-up dialog, do one of the following:



- Selecting **All data** will automatically check the HMI firmware version. If the version matches the version that can be executed by the executable, the firmware update will be executed, and the screen data will be downloaded to the HMI.
- Selecting **Only upgrade firmware** will update the HMI firmware update directly.

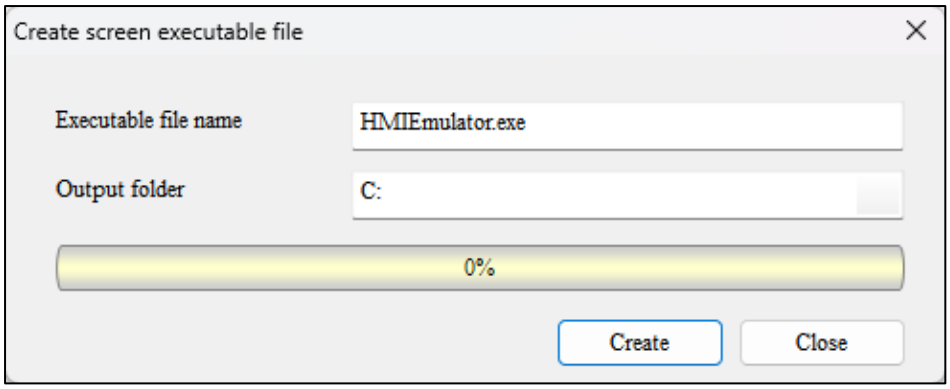
Note: Before executing **Download Screen Executable File**, set the HMI screen upload / download mode first. Go to **General** > **Environment** on the toolbar to set.



Simulated Screen Executable File

You can execute auto operation of the HMI screen on the computer through **Simulated Screen Executable File** without installing the DIAScreen software.

The following table lists the settings in the **Create screen executable file** dialog and their descriptions.

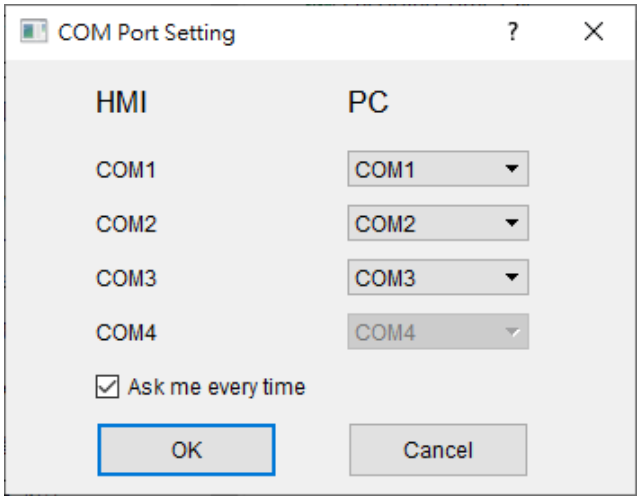


Settings	Description
Executable file name	Set the name of the generated executable file.
Output folder	Select the folder path to save the download screen executable file.

Follow these steps to create simulated screen executable file.

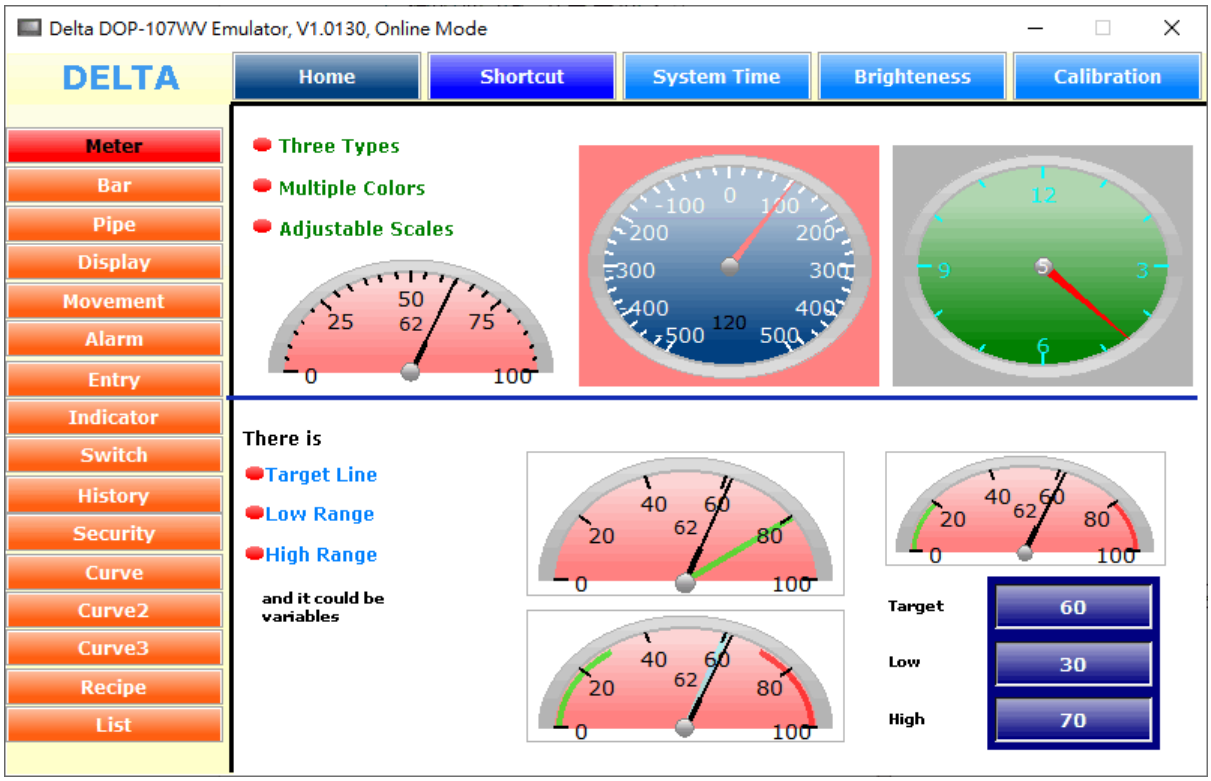
To create simulated screen executable file

1. In the main menu, click **Create Transfer Data** > **Simulated Screen Executable File**.
2. In the **Create screen executable file** dialog, make the related settings.
3. In the output folder, execute the executable file (the default file name is HMIEmulator.exe).
4. In the **COM Port Setting** dialog, set the communication port of the executable file.



Note: If communicating via network, you can ignore this dialog. Click **Cancel** to communicate with the IP set in the project.

The execution screen is as follows.



Duplicate

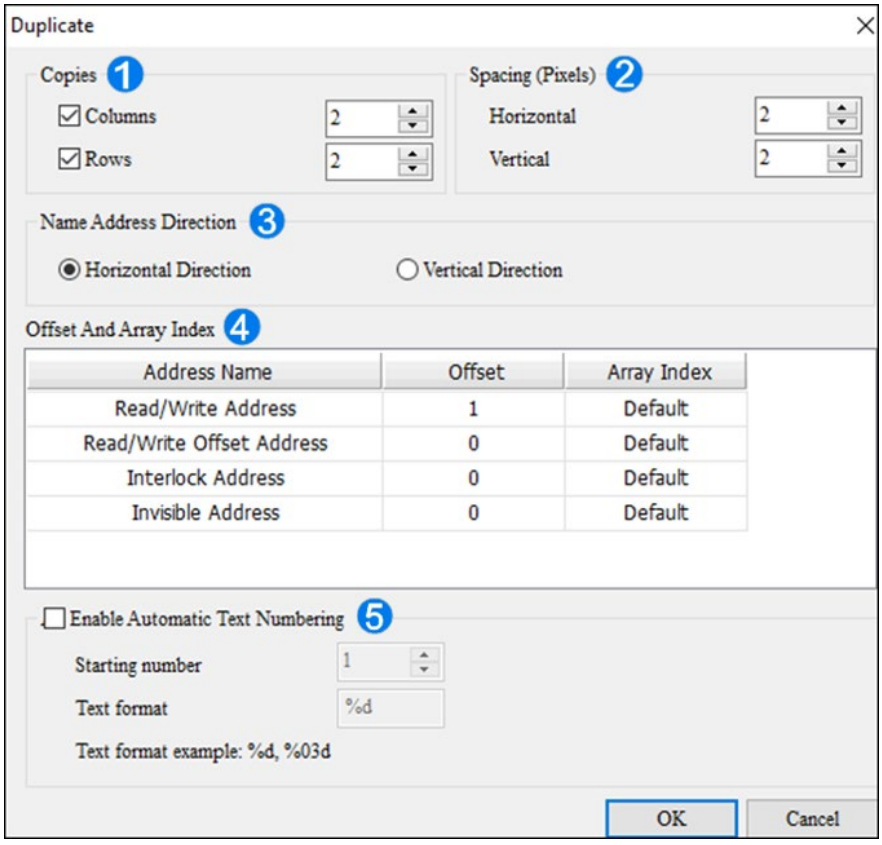
The duplicate feature allows you to perform duplication of an element, and then automatically increments or decrements the address according to your needs.

To duplicate

Do one of the following methods to perform duplication of an element:

- Right-click on the element and select **Duplicate**.
- Select the element and click **General** > **Duplicate** in the menu bar.

The following table lists the settings in the **Duplicate** dialog and their descriptions.



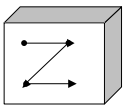
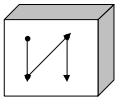
1 Copies

Setting	Description
Columns	Select or enter the number of horizontal (X) and vertical (Y) to get the total number of X*Y elements. You can also select / unselect the checkbox to single-select columns or rows according to your needs.
Rows	

② Spacing (Pixels)

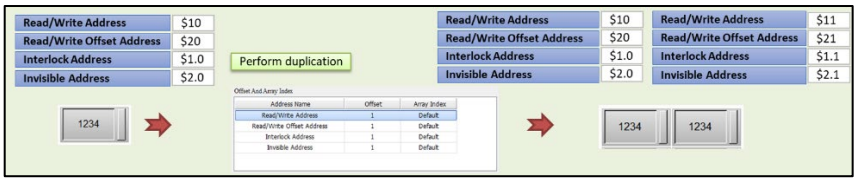
Setting	Description
Horizontal	Select or enter the spacing between each element. Once the duplication is complete, spacing generates automatically according to the settings.
Vertical	

③ Name Address Direction

Setting	Description
Horizontal Direction	Set the duplicate direction of addresses. The duplicated element address increments (positive) or decrements (negative) according to the offset and gets duplicated horizontally or vertically.
Vertical Direction	<table><tr><th>Horizontal</th><th>Vertical</th></tr><tr><td>  </td><td>  </td></tr></table>

④ Offset And Array index

Setting	Description
Offset	Select or enter the amount that addresses increment or decrement automatically. • If elements are set as Word units, the increments or decrements are calculated in Word units. • If elements are set as Bit units, the increments or decrements are calculated in bit units.

Setting	Description
Array Index	Select the array index location. The default value is default. • Select **Default**, and the address is offset in the element unit.  • Select **1–4** for the Tag address. The array indexes are arranged from right to left. **Note**: You can specify the offset of the array index. Currently available for CODESYS Tag only.

❺ Enable Automatic Text Numbering

Setting	Description
Starting number	Select or enter the starting value of the number displayed on the element.
Text format	Enter the text format of the number displayed on the element.

Screen

Type

DIAScreen provides a variety of screens to apply to various operational needs. Do any of the following to add a new screen.

- On the toolbar, click **General** > **New Screen**.

Or

- In **Project** pane, right-click the Screen node, and then select New Screen.

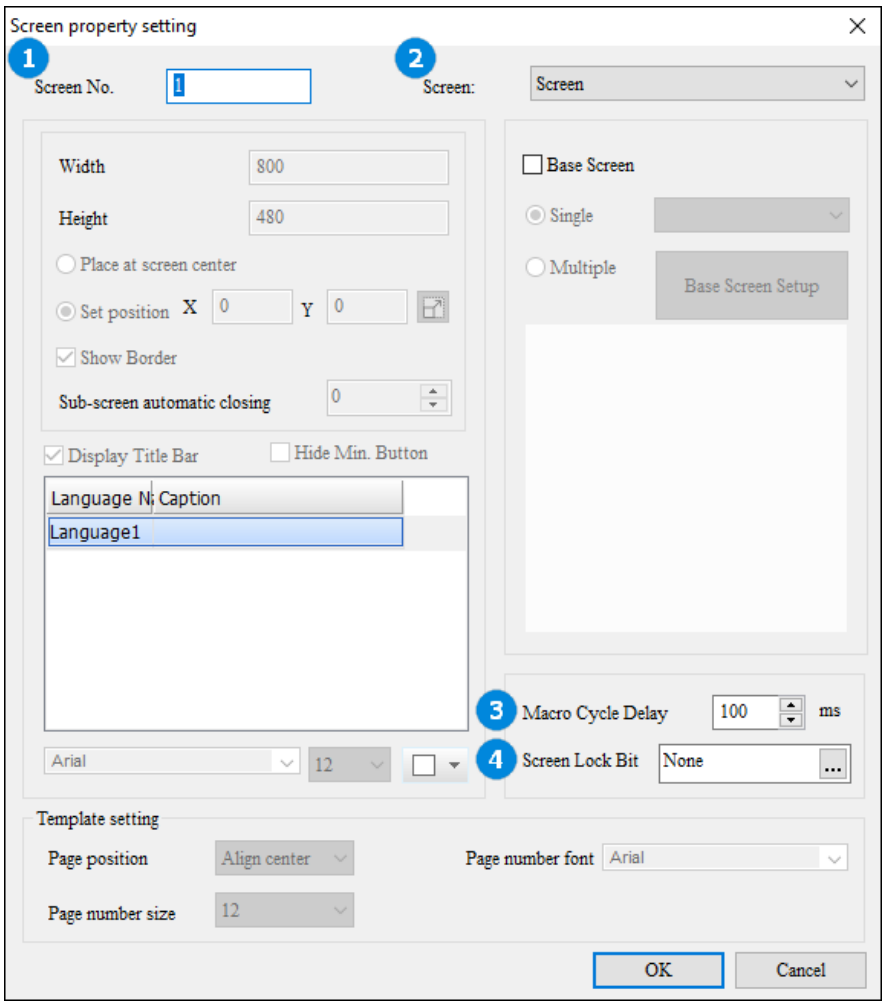
Or

- Right click the blank space on the **Screen Management** pane, and then select **New Screen**.

Screen

The **Screen** is the default display screen of the HMI, and the screen size is fixed according to the model size.

The following table lists the common properties of each type of screen in the **Screen property setting** dialog with their description.



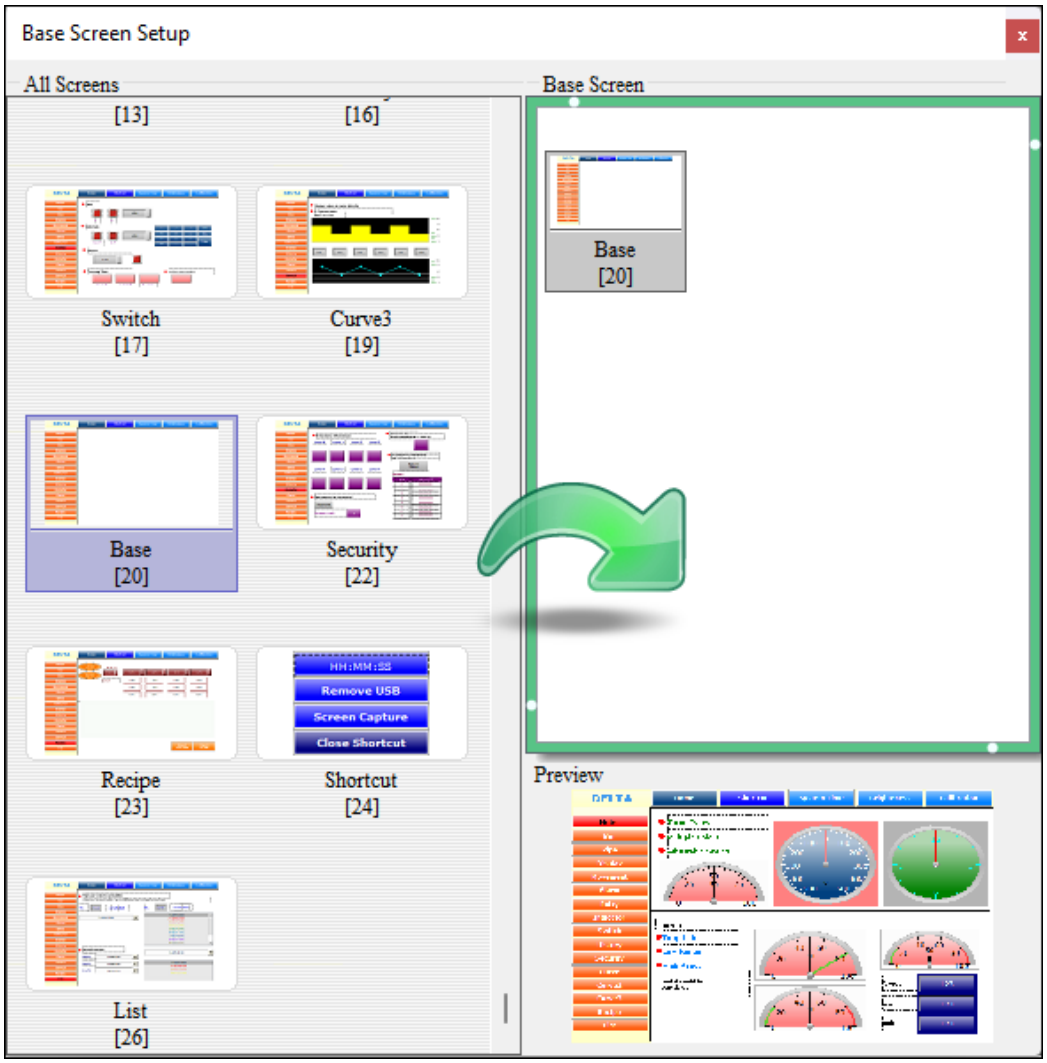
Legend	Setting	Description
1	Screen No.	Enter the screen number, and the Screen Management pane will num **Note**: The screen number cannot be repeated.
2	Screen	Select the screen type. **Note**: The screen type of the **Template Screen** cannot be changed.
3	Macro Cycle Delay	Select or enter the delay time for repeated execution of screen Clock macro.
4	Screen Lock Bit	When set to On, the current screen will be locked and cannot be changed to other screens.

- **Base Screen**

  If the **Screen** is used, you can select the **Basic Screen** checkbox and select **Single** or **Multiple** base screen.

In the **Base Screen Setup** dialog, drag the screen from the **All Screens** area to the **Base Screen** area to configure it on the current screen.

Note: The properties of elements on the base screen cannot be set on the general screen.

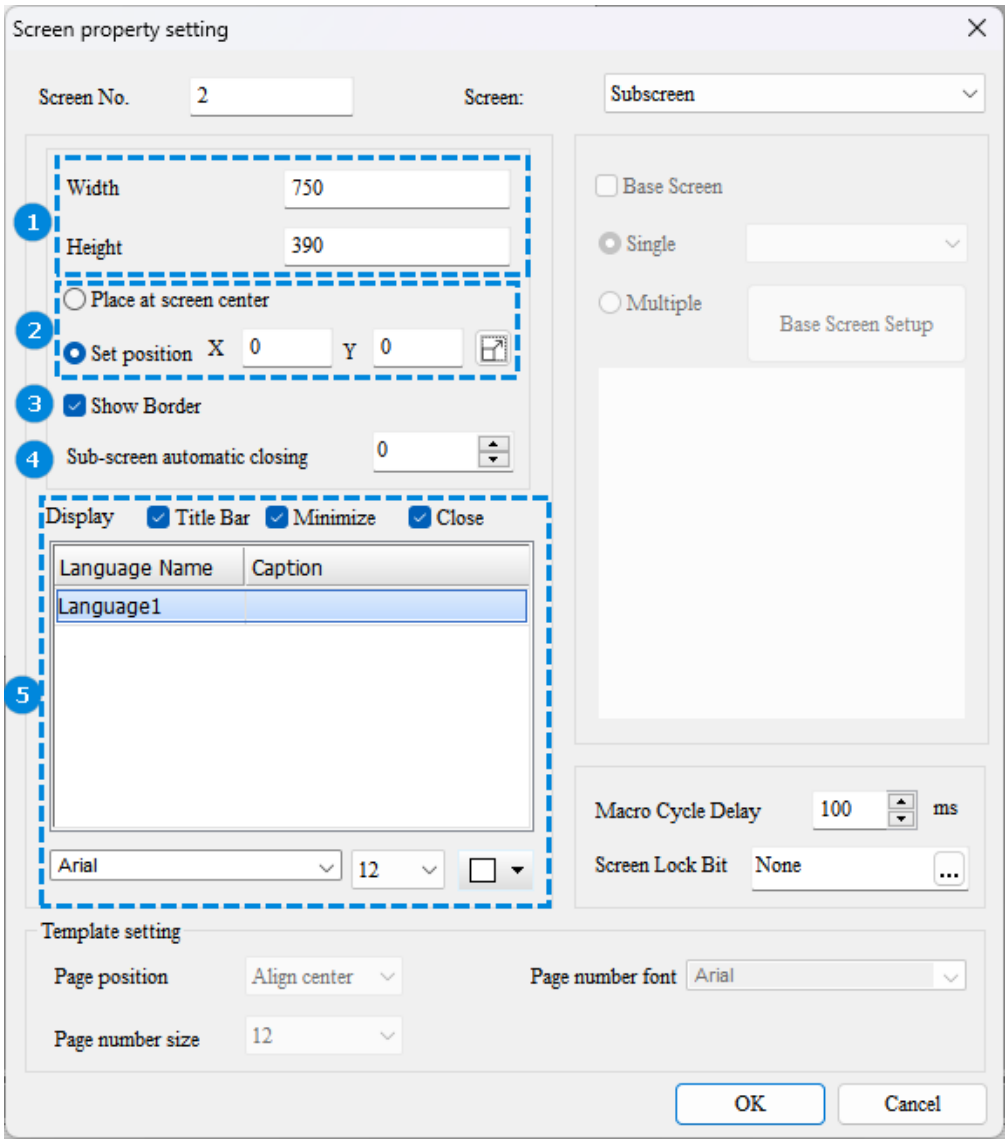


Subscreen

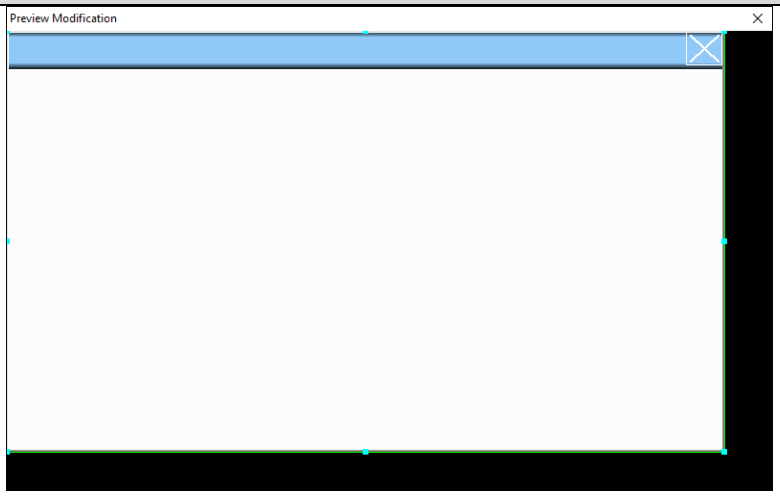
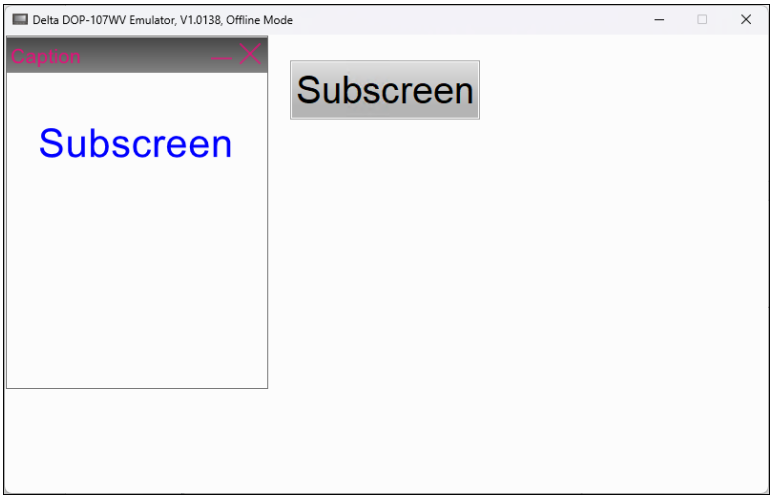
The sub-screen is a floating window displayed on the screen, which can be used with the alarm function or called up by Macro. You can also use an **Embedded Subscreen** element to display the sub-screen on the main screen.

Note: Elements such as **Embedded Subscreen**, **Picture Viewer**, and **Gantt Chart** cannot be configured on the sub-screen.

The following table lists the special properties of the **Subscreen** with their description.



Legend	Setting	Description
1	Width / Height	Enter the width and height of the sub-screen. The maximum value is the width and height of the general screen.
2	Position	Enter the display position of the screen. You can select **Place at screen center** or set the XY position of the upper left corner of the screen. Click in the **Preview Modification** dialog to drag the mouse to adjust the size and display position of the sub-screen.

Legend	Setting	Description
		
❸	Show Border	Select to display the border of the sub-screen.
❹	Sub-screen automatic closing	Select or enter the time to automatically close the sub-screen. Set to 0 not to automatically close the sub-screen.
❺	Title Bar Settings	Select the **Title Bar** checkbox, and the subscreen displays the title bar on the HMI. You can also use the multi-language setting to set the title in different languages, and then set the text font, size, and color of the title in each language. **Note**: You can choose whether to display the **Minimize** and **Close** button on the title bar of the subscreen. 

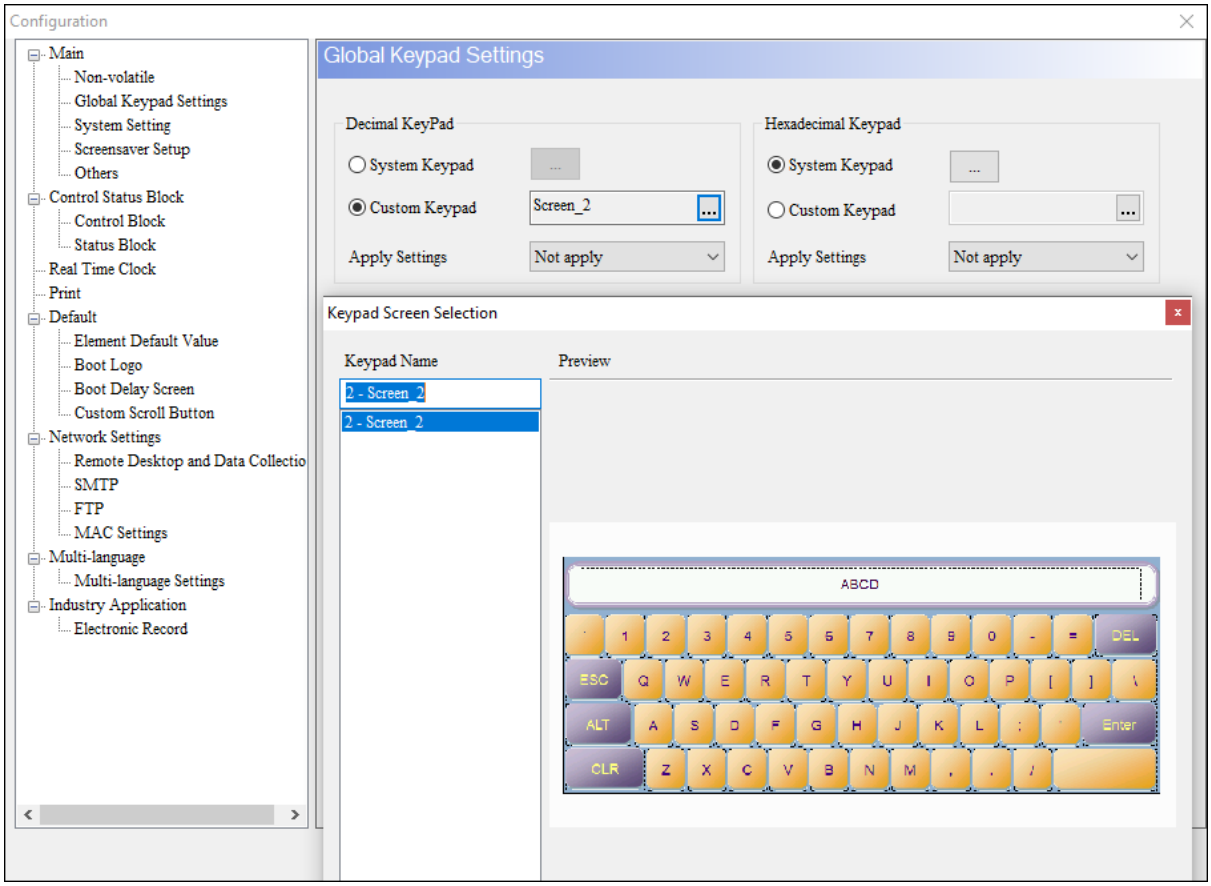
Keypad Screen

When the **Custom Keypad** set in the **Global Keyboard Settings**, or the **Custom Keypad** are selected in the **Numeric Entry**, **Character Entry**, **Barcode Input** and **Set Value** button elements, the keypad screen can be selected through the **Keypad Screen Selection** dialog. You can select a keypad in the **Element Bank** and drag it to the keypad screen or directly create a **Keypad** element. After clicking the Input element on the HMI, the customized keypad screen will be displayed.

Note:

- The Keypad Screen cannot be the default boot screen.
- For Keypad property settings, refer to the property description of Subscreen.

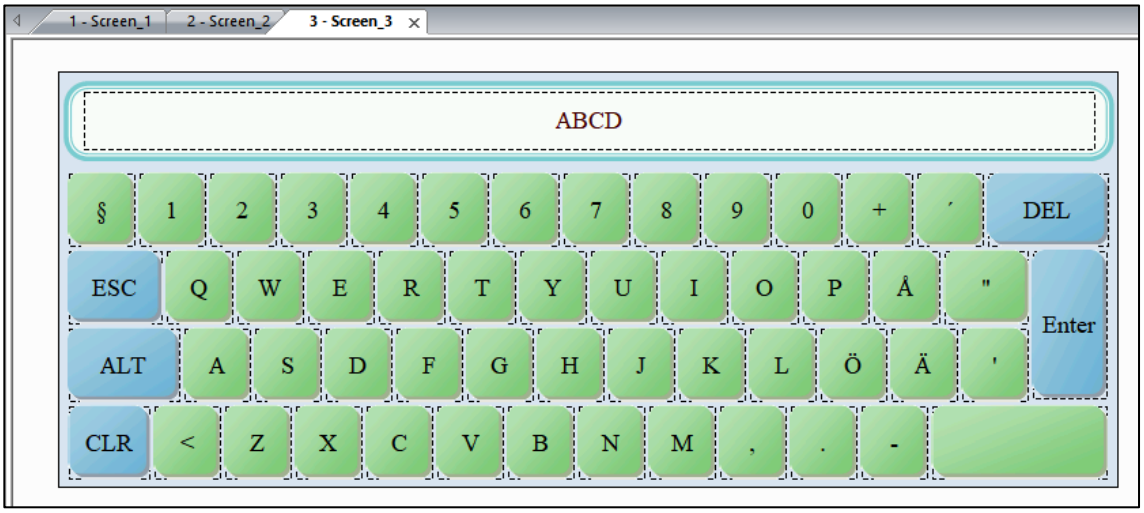
Go to **General** > **Configuration** > **Global Keypad Settings** to set the screen for various types of keypads.



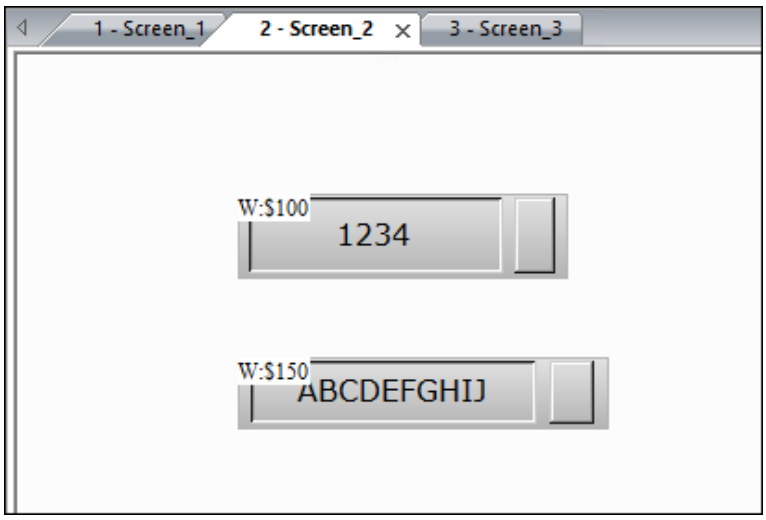
Follow these steps to set the keypad screen.

Set the keypad screen

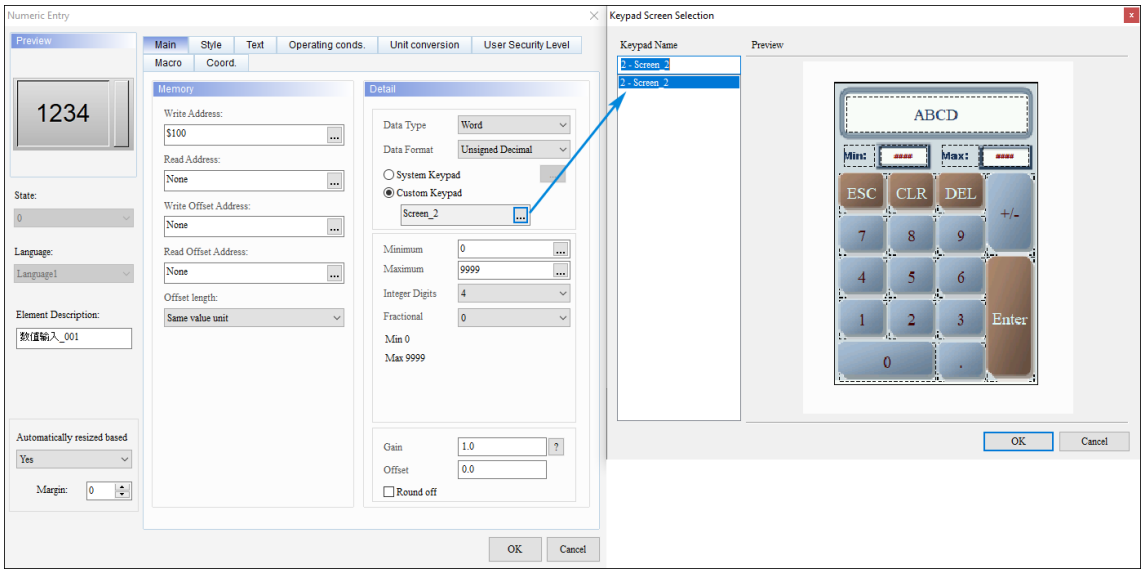
1. Create 2 **Keypad** elements on the **Keypad Screen**.
2. Double-click **Keypad-Template** in the **Element Bank** pane, drag KP(1)_01_Big keypad to the first keypad screen, and then drag KP_Swedish_Big keypad to the second keypad screen.

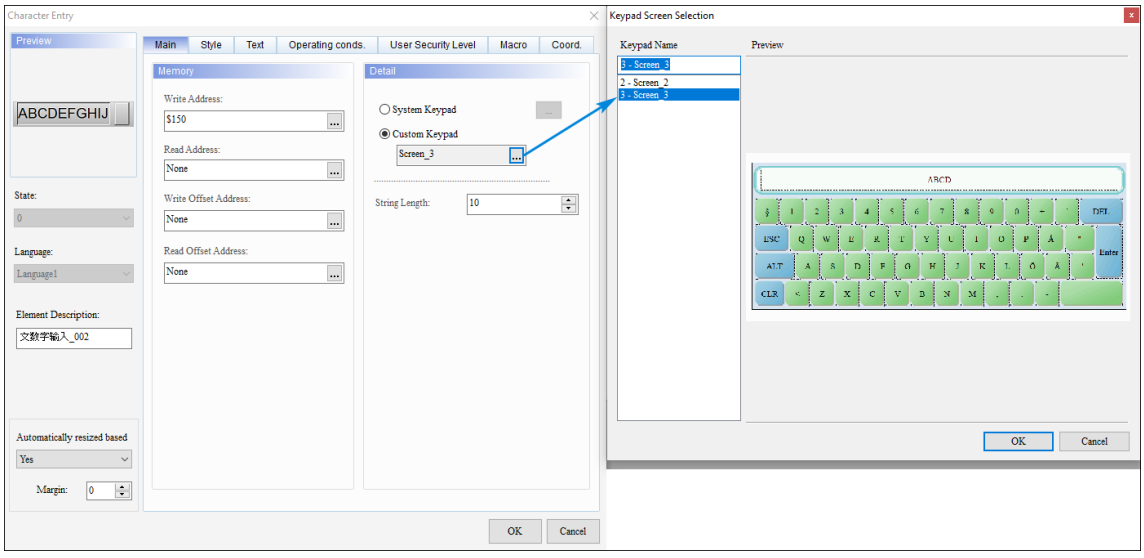


3. Create **Numeric Entry** and **Character Entry** elements on the Screen, and then respectively set the **Write Address** to **$100** and **$150**.



4. In the element properties dialog, set the **Custom Keypad** to the keypad screen of the Screen_2 and Screen_3.



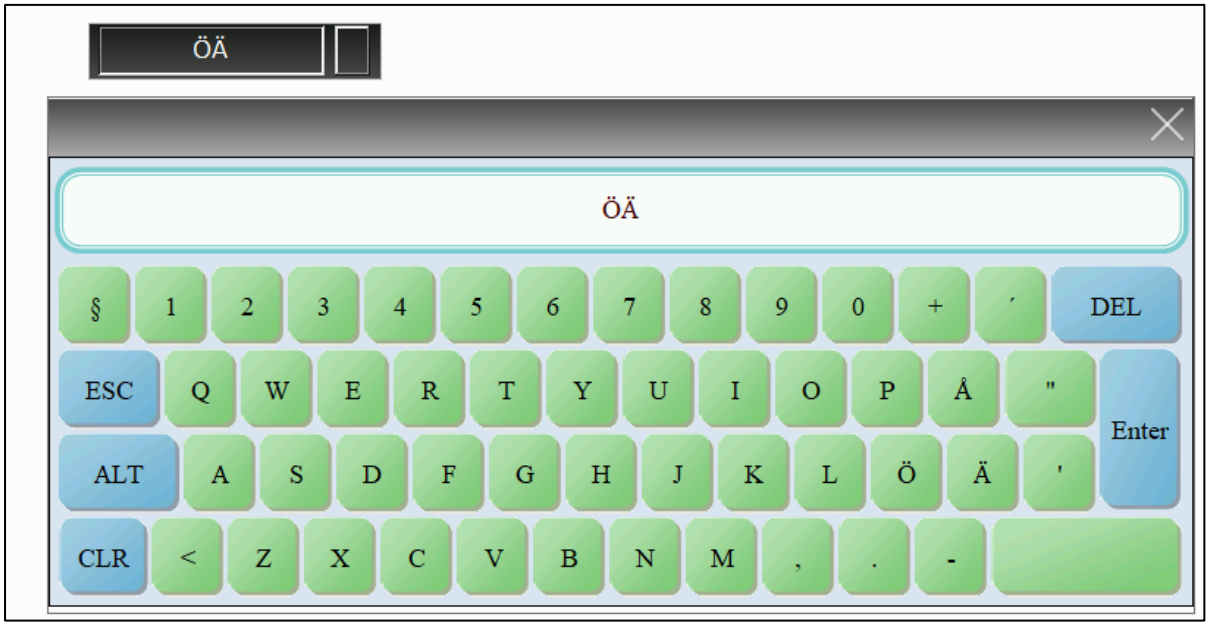


5. Compile and download the screen to HMI.

The execution result is as follows.

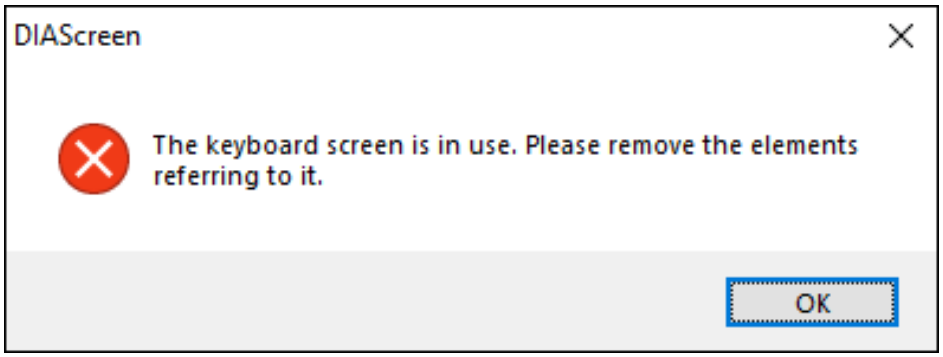
Click the $100 **Numeric Entry** element on the HMI screen, and the keypad style that pops up is customized keypad style.

Click the $150 **Character Entry** element, and you can also enter ASCII characters to display.



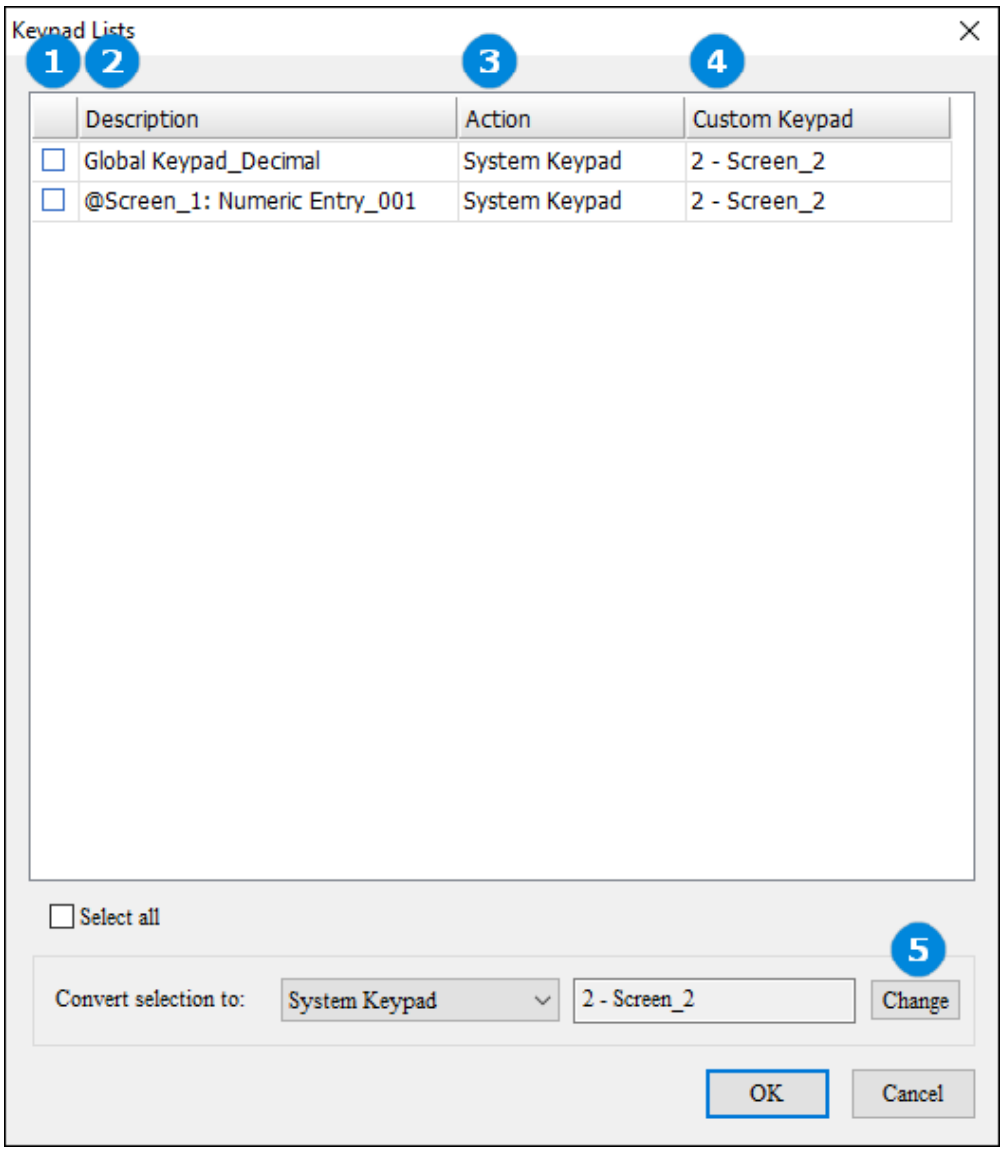
To remove the used keypad screen

If **Global Keypad Settings** or the Input element has applied the created keypad screen, the following message will appear when you delete the screen.

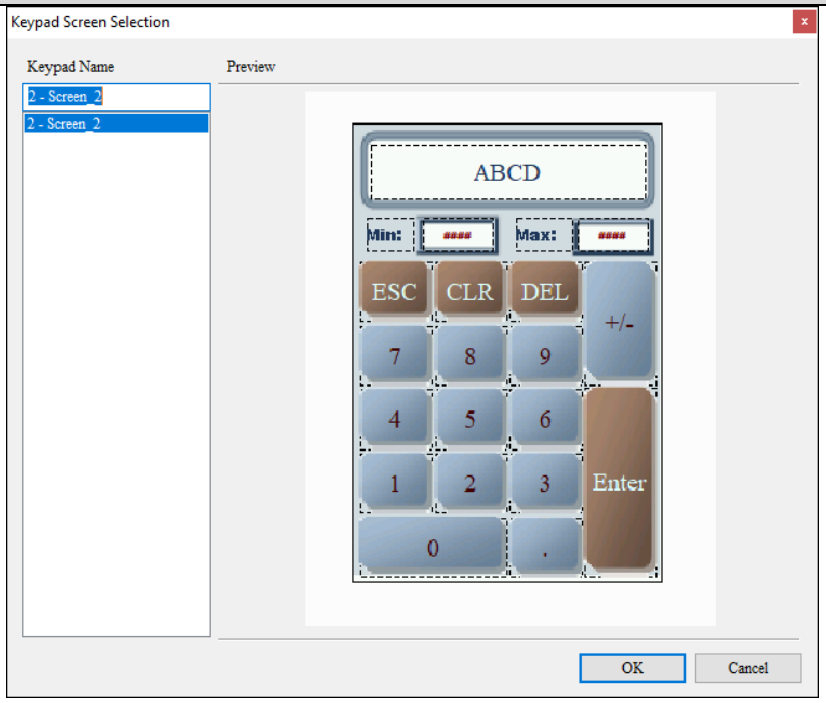


Click **OK** in the dialog, and the **Keypad Lists** dialog appears for you to change the keypad.

The following table lists the settings in the **Keypad Lists** dialog with their description.



Legend	Setting	Description
1	Selection	Select the element to change in batch. After selecting, click **Change** in the **Convert selection** to field.
2	Description	Displays the description of the element or global keypad. When you click the description of the element, a corresponding screen will open for you to select the element.
3	Action	Select the action to change. • **No Action**: The current settings do not change. • **System Keypad**: Changes to the system keypad. • **Custom Keypad**: The **Keypad Screen Selection** dialog pops up for you to select the customized keypad.

Legend	Setting	Description
		
❹	Custom Keypad	Displays the number and name of the currently set keypad screen. The changed screen name is only displayed when the **Custom Keypad** is selected in the **Action** field.
❺	Change	Click to perform the change of action according to the selected item. **Note**: After the change, you must click OK to take effect.

Print Screen

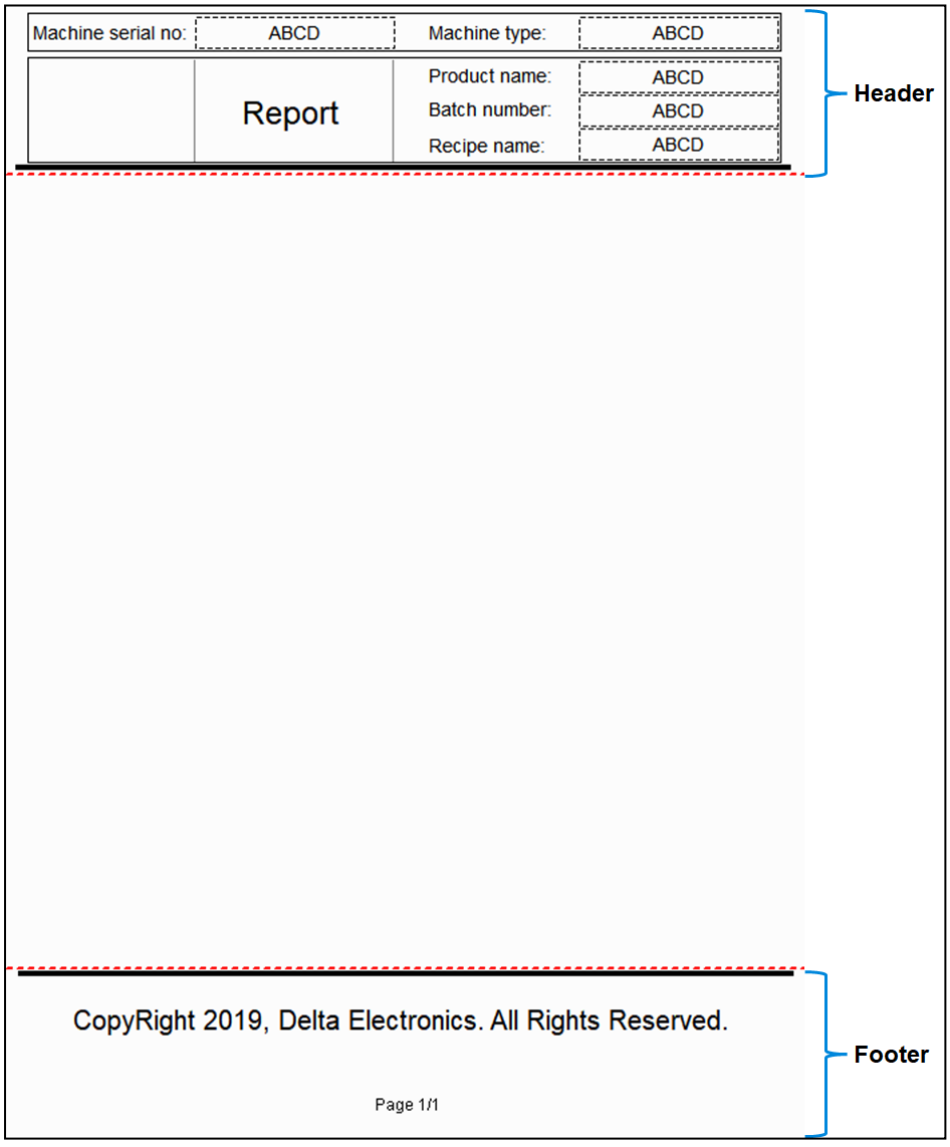
Use the **Print Screen** function to design the screen style printed by the printer and configure multiple elements, as well as set the read address of the element. Then, you can print out the current reading status. For the screen print settings, refer to Print Settings.

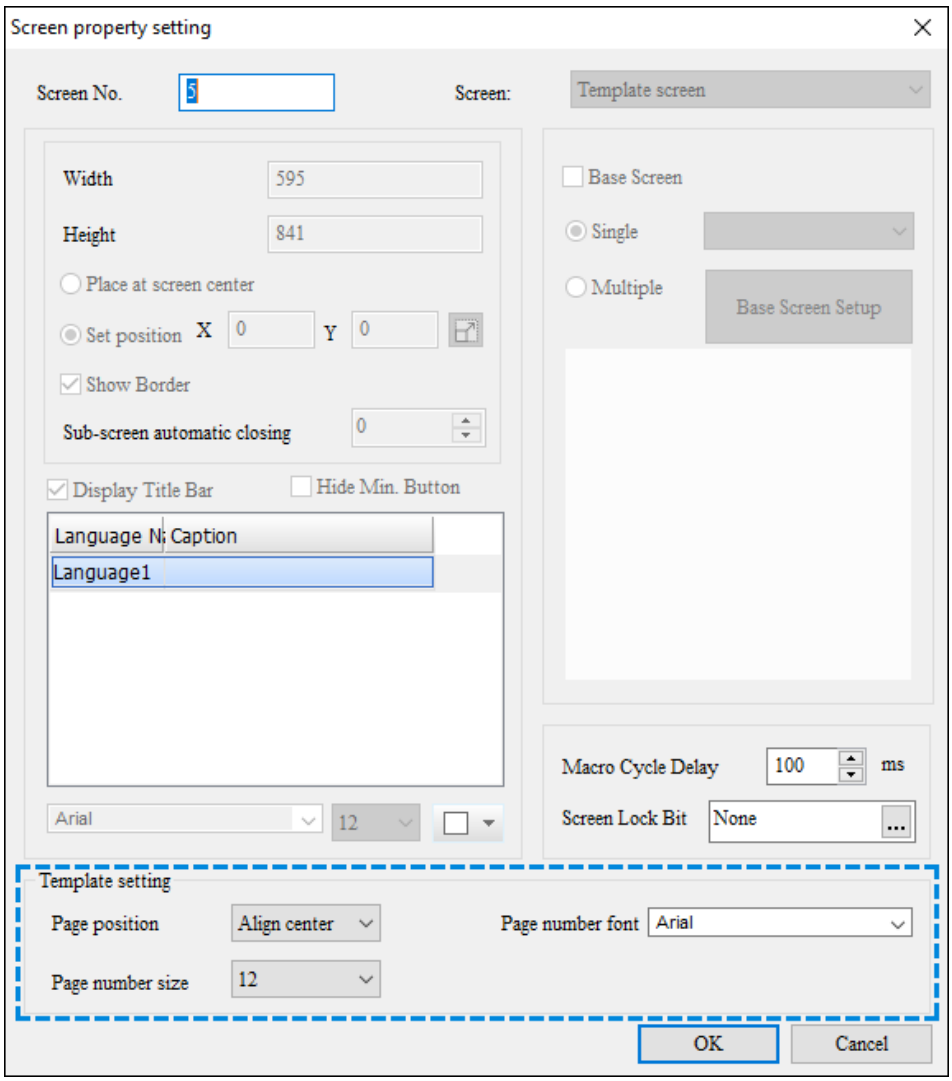
Note: This function only applies to the HMI that can connect to the printer.

Template Screen

The page style of the output PDF file can be designed through the **Template Screen**. The output PDF file can contain **Historical Data**, **Recipe**, **Alarm**, and **Operation Log**.

The Template Screen is divided into two parts by the red line. Above the red line is the header, where you can configure the data display element to record the read address data in the PDF file; below the red line is the footer, which automatically displays the page number. You can set the position, size, and font of the page number in the **Screen property setting** dialog. The area between the red lines is the data recording area, and the content is set in the property dialog of the **Template Output** button element.



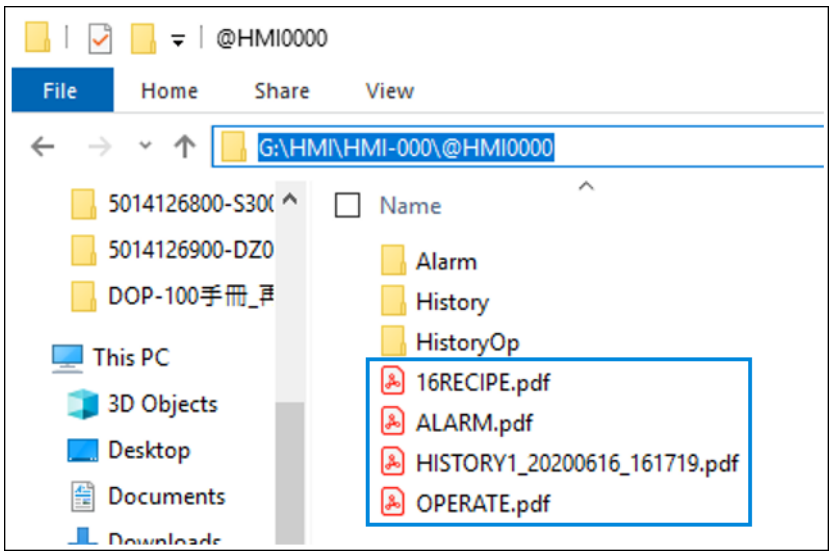


The following describes how to use the **Template Screen**.

1. Add a **Template Screen** and configure the footer, and then set the position size, and font of the page number in **Template setting** area of the **Page property setting** dialog.
2. Create a **Template Output** button element on the Screen, and then configure the template setting in the element properties dialog.

The execution result is as follows.

Click the **Template Output** element on the HMI to output the PDF file. The output PDF file is stored in the report device path `\HMI\HMI-000\@HMI0000`.



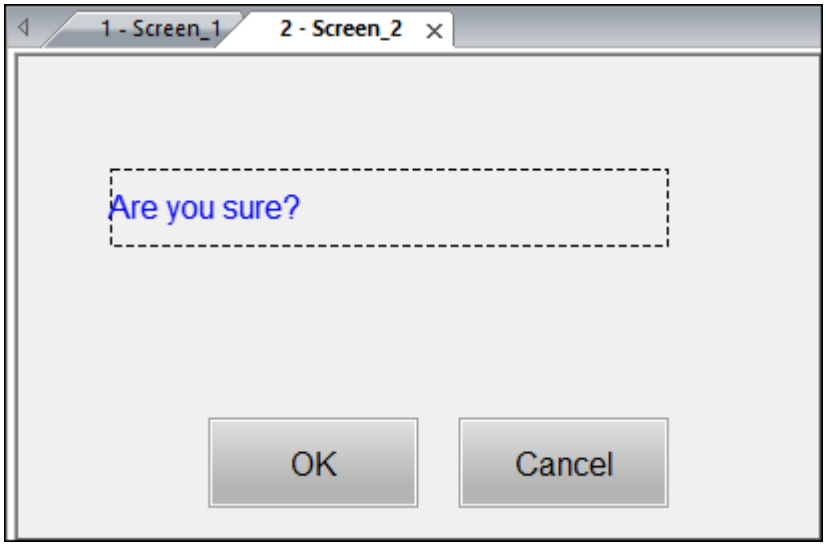
Confirmation Window Screen

The display confirmation screen when the element is operated can be designed through the **Confirmation Window Screen**. Some operating elements can enable a **Confirmation Window** to be displayed after triggering the element on the HMI.

Note: For the screen property of the **Confirmation Window Screen**, refer to the property description of Subscreen.

The following describes how to use the **Confirmation Window Screen**.

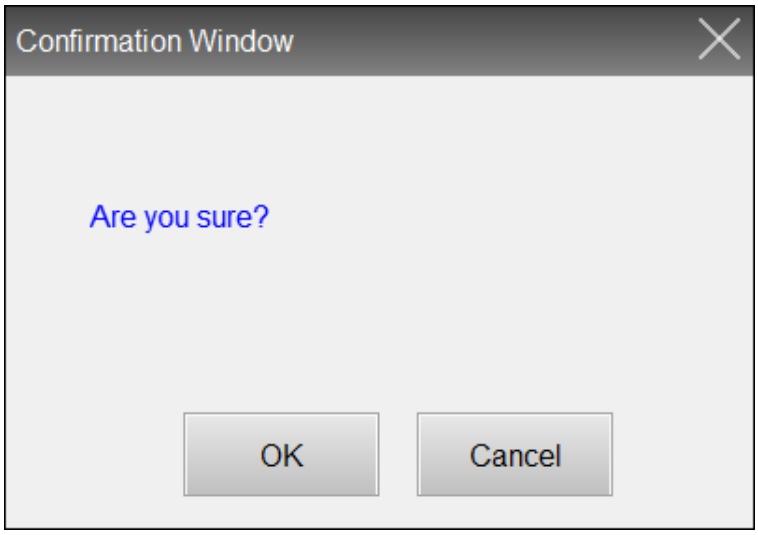
1. Add a **Confirmation Window Screen**.



2. Create a **Maintained** button element on the Screen.
3. In the **Operating conds**. tab of the element properties dialog, select **Custom** for **Confirmation Window** and select the created **Confirmation Window Screen**.
4. Compile and download the screen to HMI.

The execution result is as follows.

Click the **Maintained** button element on the HMI, and the customized **Confirmation Window** displays.



Login / Logout Screen

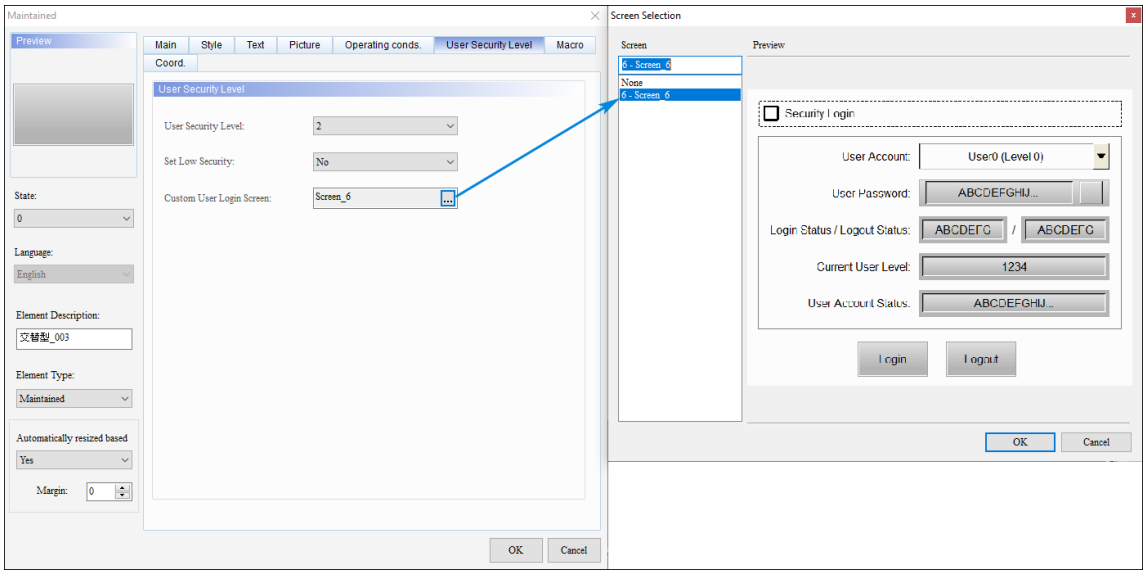
The operation screen for security login can be designed through the **Login / Logout Screen**. When executing an element action and the user's security is insufficient, you can log in to the corresponding security through the login window to perform action, or you can use the **Goto Screen** button element to call the **Login / Logout Screen** to switch security.

Note: For the screen property of the **Login / Logout Screen**, refer to the property description of Subscreen.

The following describes how to use the **Login / Logout Screen**.

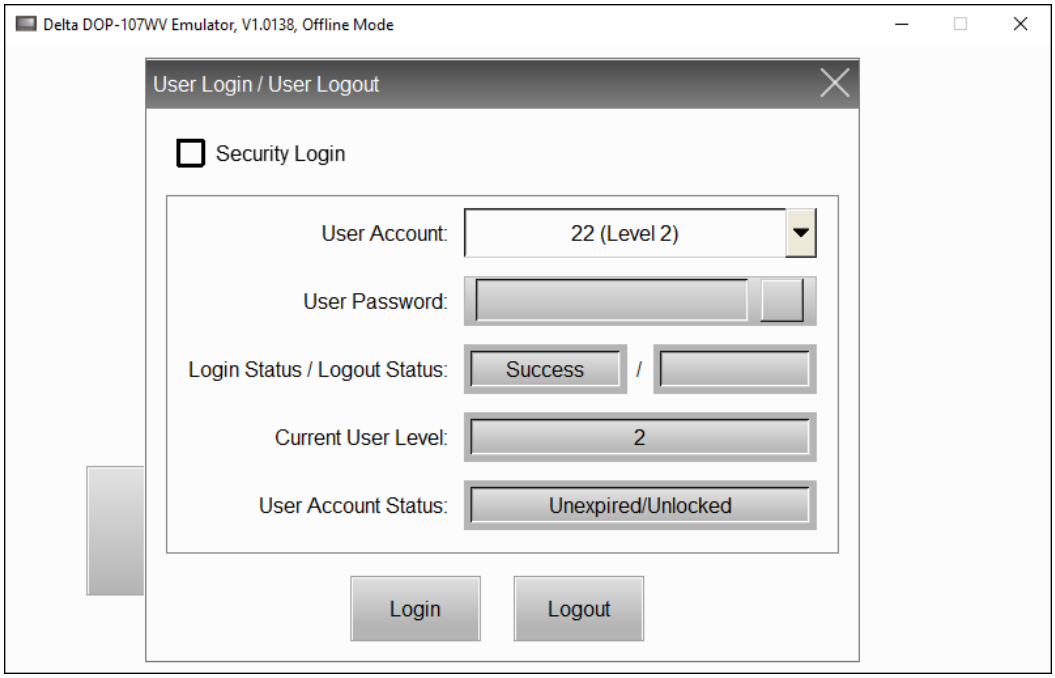
1. Add a **Login / Logout Screen**.
2. Create a **Maintained** button element on the Screen.

3. In the **User Security Level** tab of the element properties dialog, select **2** for **User Security Level**, and then select **Custom User Login Screen**.

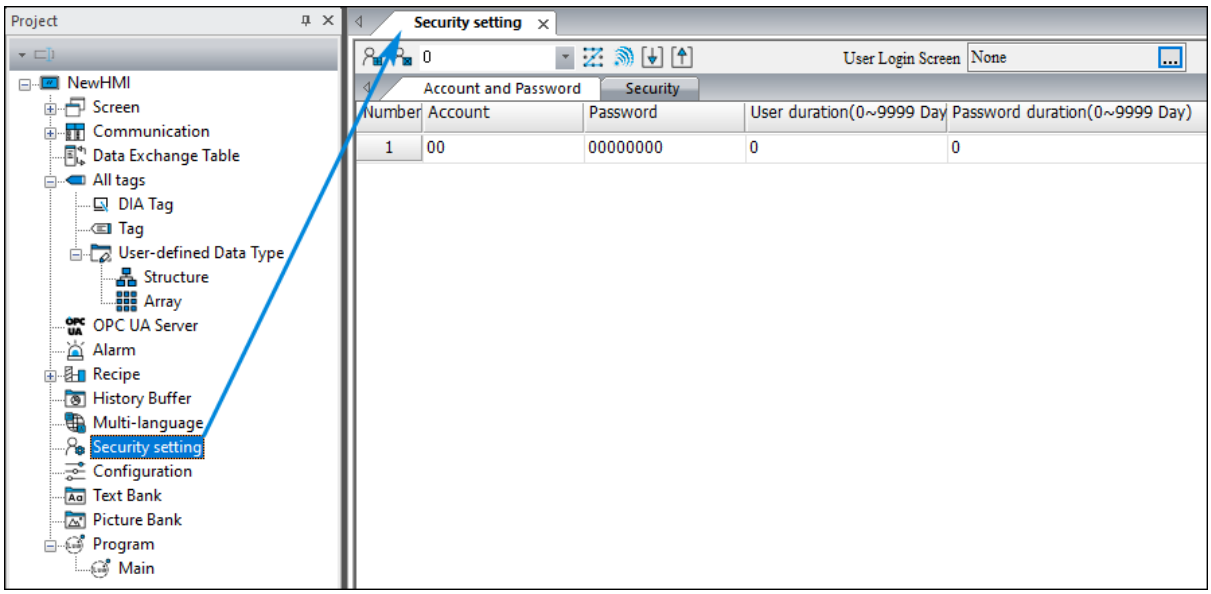


The execution result is as follows.

Click the **Maintained** button element on the HMI, and the customized **Login / Logout Screen** displays. Log in with an account above the set user security level to perform the element action.



You can also set the **User Login Screen** for certain level on the **Security setting** page. When the element is set to this level, the customized screen is displayed.



Note: If the login screen set in the element and the **Security setting** page are different, the login screen set in the element will be displayed.

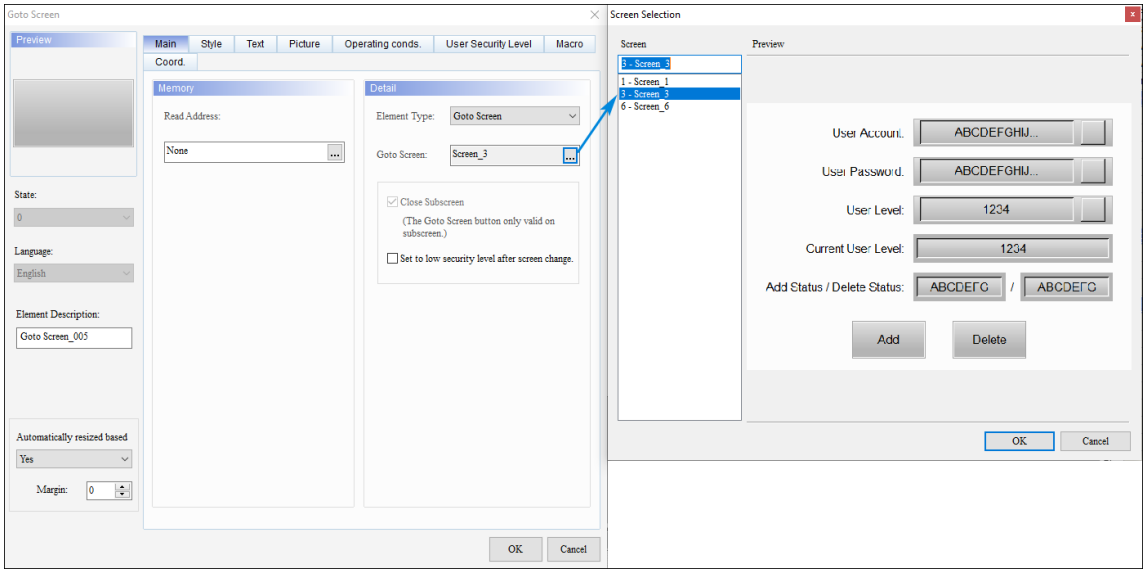
User Account Screen

You can customize the screen display mode through the **Add / Delete User Account Screen** and **Change Password screen** and quickly set the password table on the HMI.

Note: For the screen property of the **Add / Delete User Account Screen** and **Change Password Screen**, refer to the property description of Subscreen.

The following describes how to use the **Add / Delete User Account Screen**. The same steps are used for the **Change Password screen**.

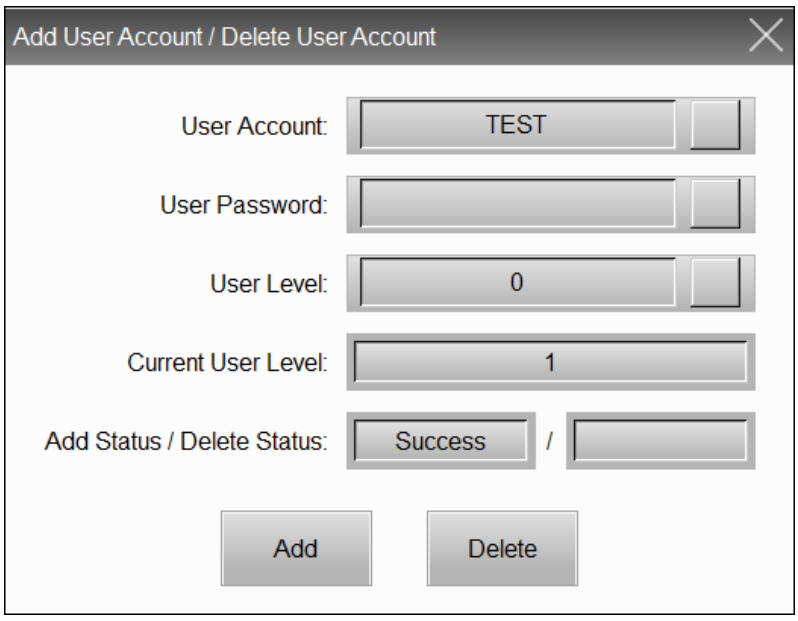
1. Add a **Add / Delete User Account Screen**.
2. Create a **Goto Screen** button element on the Screen.
3. In the Main tab of the element property dialog, set the **Goto Screen** to the set **Add / Delete User Account Screen**.



4. Create a **Password Table button** element to confirm whether the added account is written into the HMI.

The execution result is as follows.

1. Click the **Goto Screen** element on the HMI, and the **Add / Delete User Account Screen** displays.
2. Add / delete accounts below the login level. For example, if the login level is 2, only level 0 and level 1 accounts can be added. If you add a level 2 account, the **Add Status** will display **Fail**.

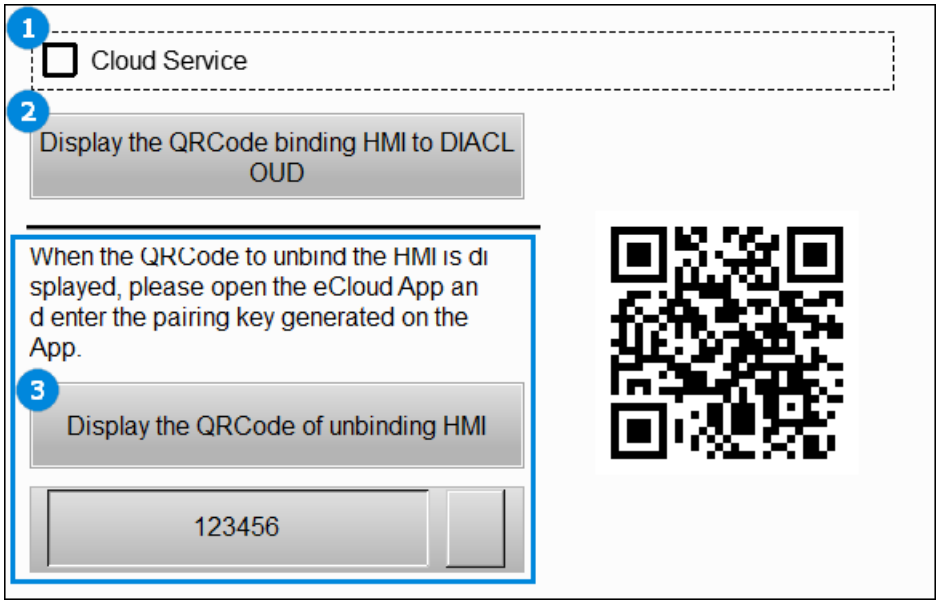


3. Click the **Password Table** button element on the HMI and confirm in the pop-up password table that the added or deleted account has been successfully written into the HMI.

Bind Cloud Service Screen

You can enable the cloud service function on the HMI screen through the **Bind Cloud Service Screen** and bind the HMI to the DIACloud platform.

The following table lists the functions of the **Bind Cloud Service Screen** with their description.

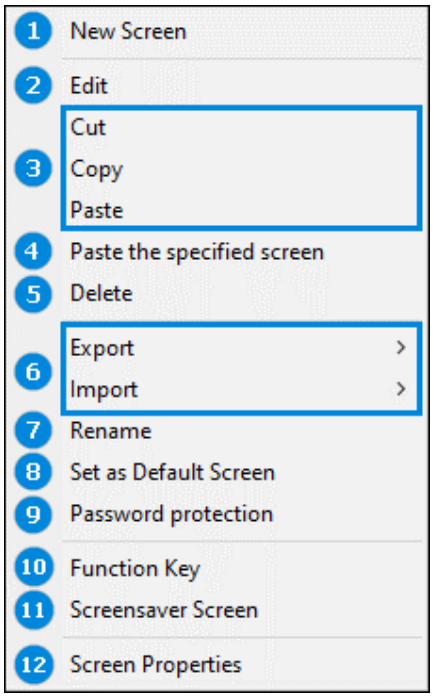


Legend	Setting	Description
❶	Cloud Service	Select to enable the cloud service function on the HMI.
❷	Display the QRCode binding HMI to DIACLOUD	After clicking the button on the HMI to display the binding QR Code, use the eCloud APP to scan and bind the HMI to the DIACloud platform.
❸	Display the QRCode of unbinding HMI	To unbind the HMI, obtain the PAIRING_KEY in the eCloud APP first and enter the 6-digit code on the **Bind Cloud Service Screen** to display the unbinding QR Code.

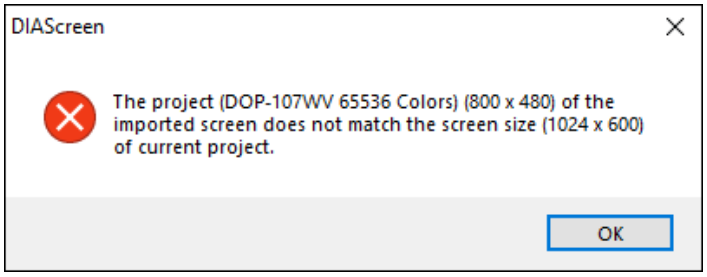
Screen Management

The **Screen Management** pane displays the preview of all screens. You can also edit the screen in this pane.

The following table lists the functions in the context menu of the **Screen Management** pane with their description.



Legend	Setting	Description
❶	New Screen	Enter the screen name, screen number, and screen type in the new screen dialog to create a new screen.
❷	Edit	Open the selected screen in the screen editing area.
❸	Cut / copy / paste	Cut, copy, and paste the selected screen. Pressing Shift can select multiple screens and perform the actions. **Note**: You can use shortcut keys to perform actions.
❹	Paste the specified screen	Paste the cut or copied screen to the specified screen number. If the specified number is occupied, the screen number will be automatically added later. For example, if the specified screen number is 10, and there are screens numbered 10 and 11, it will be automatically pasted to number 12.
❺	Delete	Delete the selected screen, and the screen cannot be restored after the deletion.

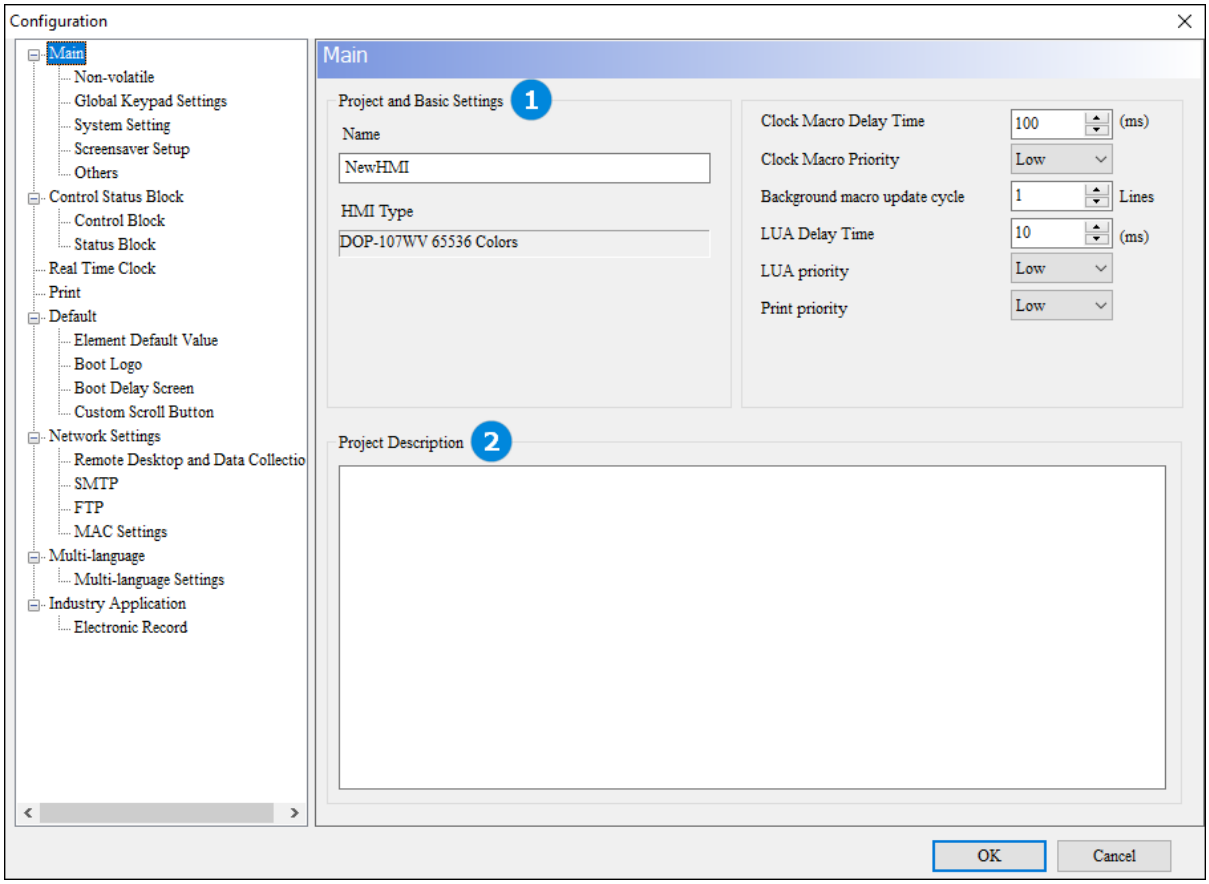
Legend	Setting	Description
⑥	Export / Import	• **Screen Picture** ○ **Export**: Export selected screen to a .bmp file. ○ **Import**: Import the .bmp, .jpg, .gif, .ico, .png, and .svg format files as the screen base picture. **Note**: To clear the screen picture, go to **General** > **Screen Export/Import** > **Clear Import Dada**. • **Screen Data** ○ **Export**: Export selected screen to a .dpi file. ○ **Import**: Import a .dpi file. **Note**: • Only **Screen** and **Subscreen** can export screen data. • If the exported screen contains a base screen, the base screen will be imported together when importing the screen. • If some components in the exported screen refer to other screens, such as the **Goto Screen** button element, the screen map will be disabled after importing, and an error will be reported during compilation. • If the imported .dpi file does match the resolution of the current project, an alarm message appears. 
⑦	Rename	Enter the screen name. Sscreen names support multi-languages. You can go to **General** > **Language** list to swtich languages and set names for different languages.
⑧	Set as Default screen	Set the selected screen as the default screen after starting the HMI.
⑨	Password protection	Enable / disable the screen password protection. If enabled, you need to enter the correct password to edit the screen when opening the screen. The password can be set up to 8 characters (including numbers 0~9 and English letters A~F).

Legend	Setting	Description
⑩	Function Key	Click to open the function key settings dialog. **Note**: This function only applies to the DOP-Handheld series HMI.
⑪	Screensaver Screen	Click to open the Screensaver Setup dialog. For the related settings, refer to Screensaver Setup.
⑫	Screen Properties	Click to open the **Screen property setting** dialog of the selected screen.

Configuration

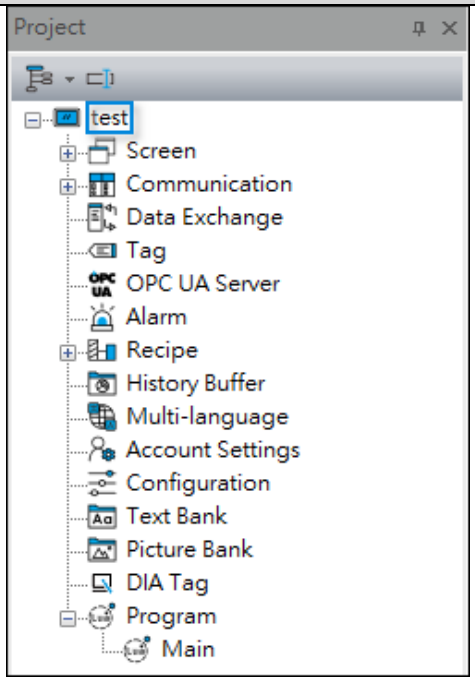
Go to **General** > **Configuration** to configure the operation settings for the HMI screen. The functions of each parameter are described as follows.

Main



1 Project and Basic Settings

Function	Description
Name	Set the name of project. The project name on the **Project tree will be changed synchronously**.

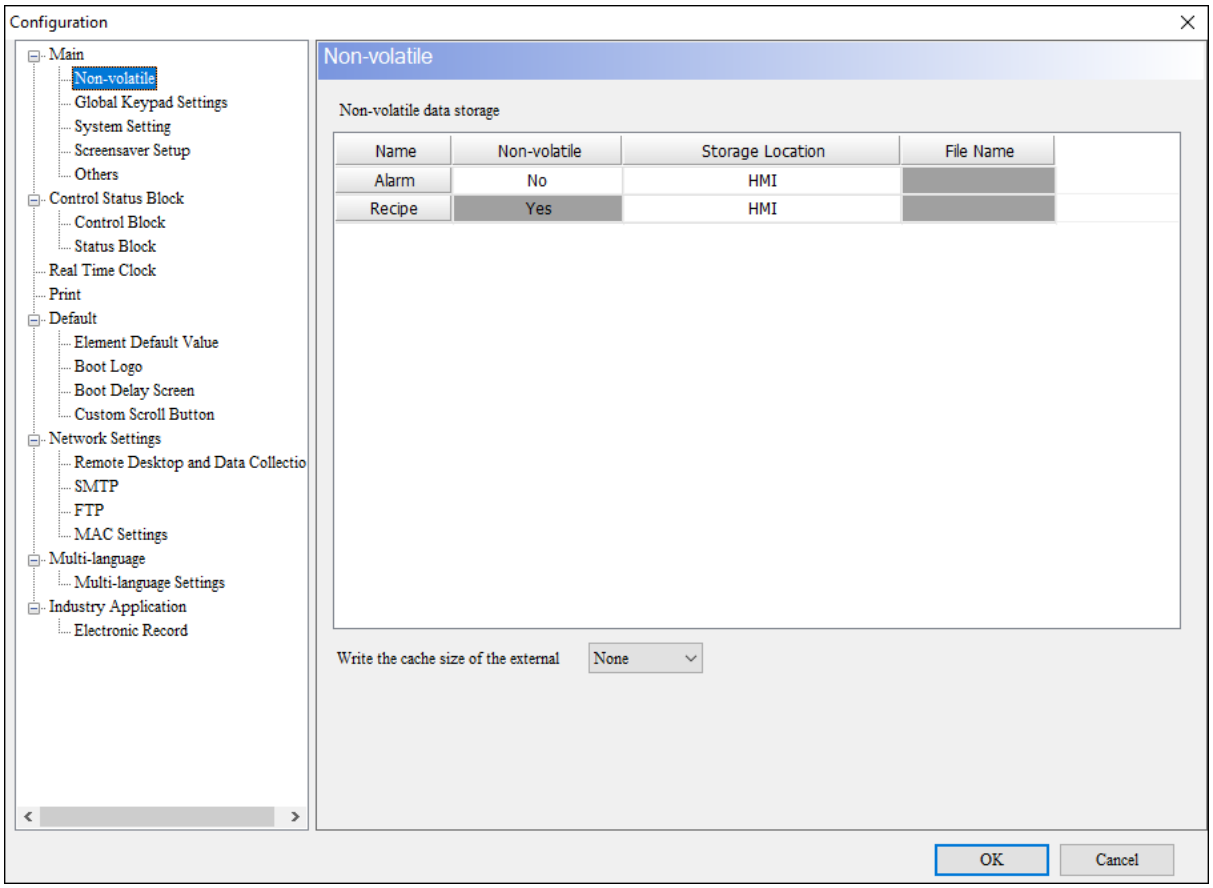
Function	Description
	
HMI Type	Display the currently edited HMI model.
Clock Macro Delay Time	Set the interval time after each execution of Clock and Macro. The range value is 50–65535, and the default value is **100**.
Clock Macro Priority	Set the priority of the Clock macro execution. The higher the priority, the more accurate the **Clock Macro Delay Time**.
Background macro update cycle	Set the number of lines executed per cycle of the background macro. The range value is 1–512, and the default value is **1**.
LUA Delay Time	Set the interval time after each execution of LUA. The range value is 0–10, and the default value is **10**.
LUA priority	Set the priority of LUA execution. The higher the priority, the more accurate the **LUA Delay Time.**
Print priority	Set the priority of print execution.

② Project Description

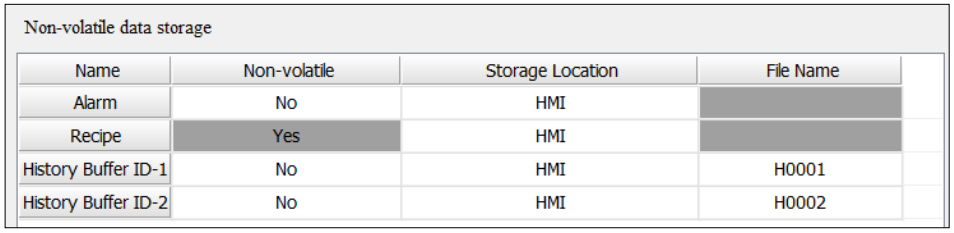
Function	Description
Project Description	Set the purpose of the HMI screen. After running DIAScreen, you can see the description of the project in **Recent opened file**.

Function	Description

Non-volatile



Function	Description
Non-volatile data storage	Non-volatile data can be stored in **Alarm**, **Recipe**, and **History Buffer**. Storage location depends on the devices supported by the model. The available options include HMI, USB Disk, USB Disk 2, and SD Card.

Function	Description
	The part of historical data will be determined if you have established a History Buffer.  Non-volatile data storage Name / Non-volatile / Storage Location / File Name Alarm / No / HMI / Recipe / Yes / HMI / History Buffer ID-1 / No / HMI / H0001 History Buffer ID-2 / No / HMI / H0002
Write the cache size of the external	Select the external storage devices (USB Disk and SD Card). The data written by the HMI to the external storage device will be temporarily placed in the cache area. However, this setting value depends on the amount of data in the cache area. If the amount of data does not reach this setting value, the data will not be written to the external storage device. This method can prevent the external storage device from being damaged due to continuous writing. Assuming that the amount of data to be accessed is less than the capacity of the buffer or there is unexpected power failure, it may result in partial data loss. To avoid this situation, you can forcibly trigger the bit5 of the general control flag of the control area at regular intervals to write data to external storage device to ensure the existence of data.

Global Keypad Settings

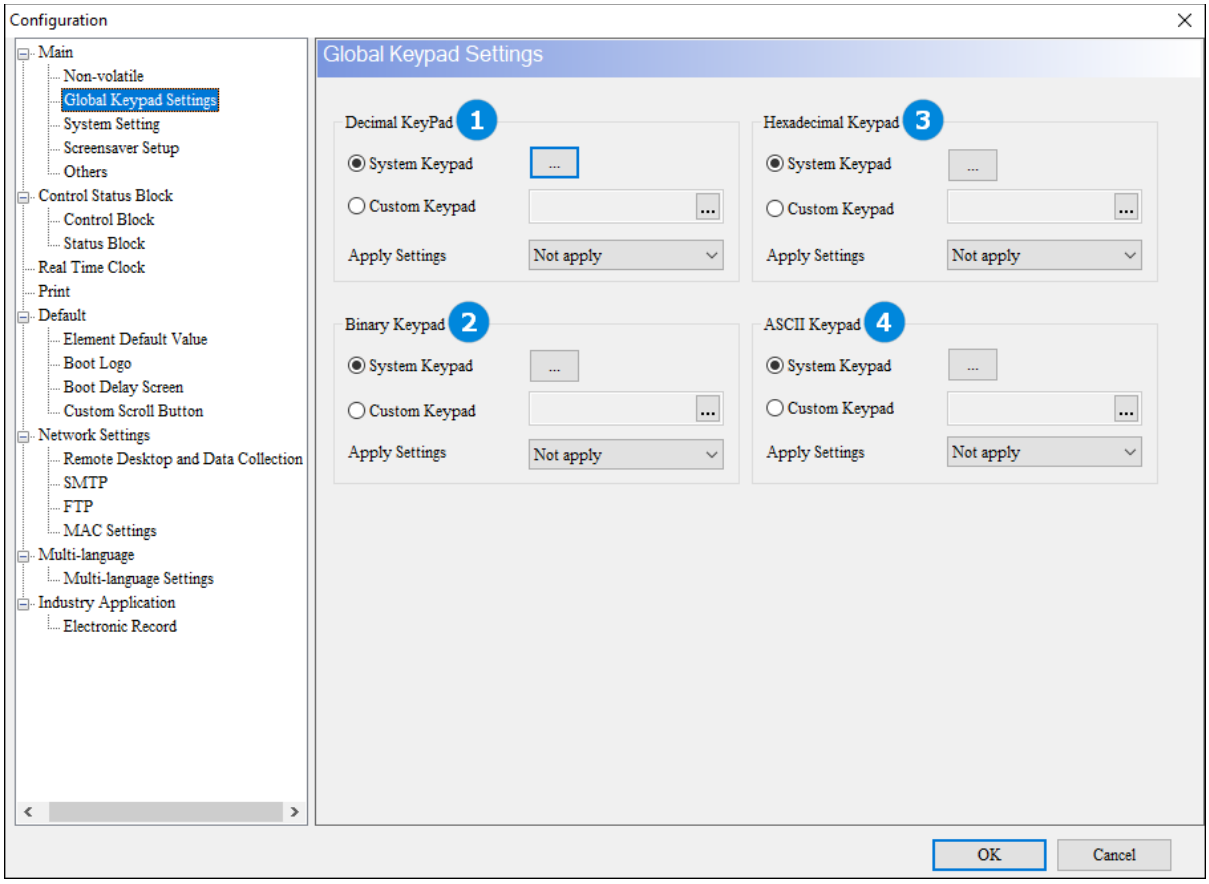
You can set the keypad styles through **Global Keypad Settings** or element property dialog. The keypad types include Decimal, Hexadecimal, Binary and ASCII. **Global Keypad Settings** also provides multiple application methods to update the keypad styles of elements.

Follow the steps to set the global keypad style through element property dialog.

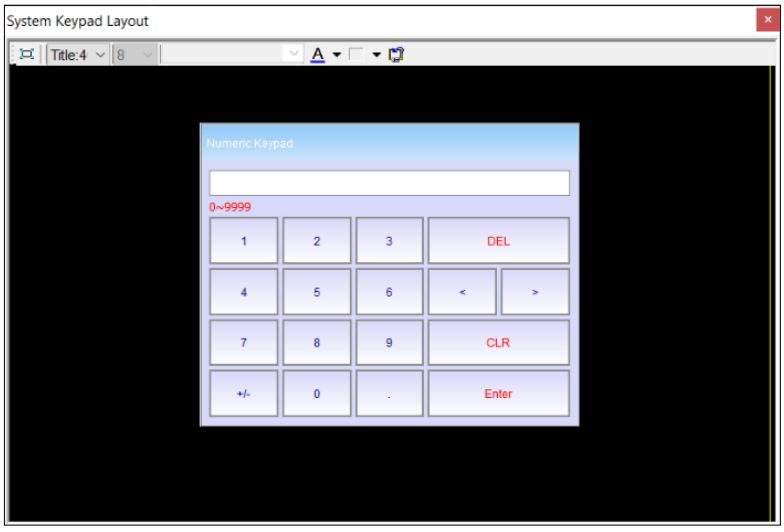
To set the global keypad style through element property dialog

1. Create **Numeric Entry** or **Character Entry** elements on the editing screen, double-click the elements.
2. In the element property dialog, click ... in the **Detail** area in the **Main** tab.
3. In the **System Keypad Layout** dialog, set the keypad style according to your needs. Or you can edit through **General** > **Configuration** > **Global Keypad Settings.**

The following table lists the functions of the **Global Keypad Settings** page with their description.

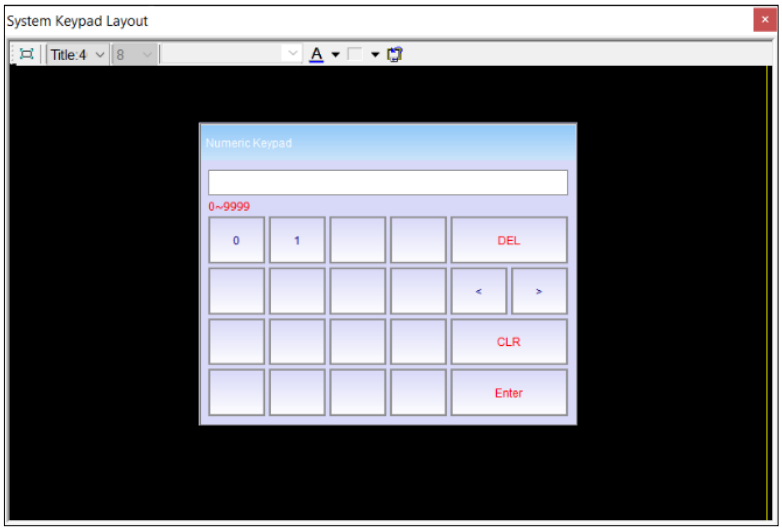


1 Decimal Keypad

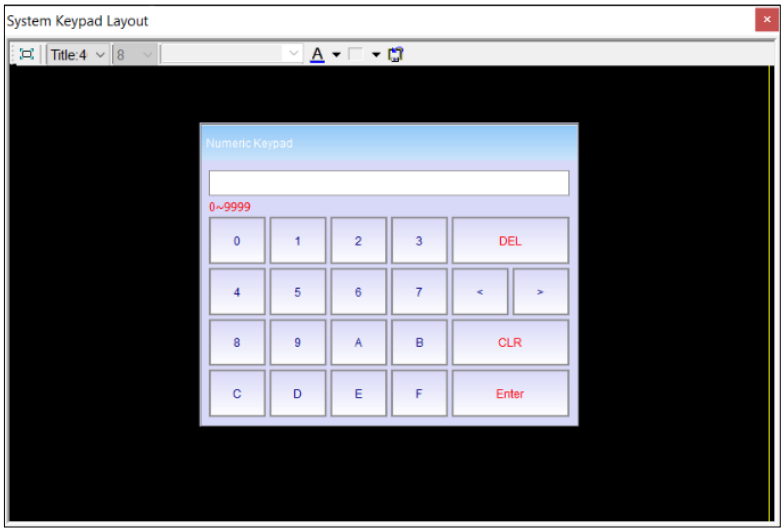
Function	Description
System Keypad	This keypad is applied to the **Numeric Keypad** element when the place value is set to decimal. 

Function	Description
Custom Keypad	Click ... to select the custom screen. Go to **General** > **New Screen** > **Keypad Screen** to customize screen.
Apply Settings	Set the keypad style applied method. • **Not apply**: Maintain the original style of the system keypad. • **Apply to all**: Replace the old keypad style with the new style. • **Apply to new**: Only applies to newly created keypads.

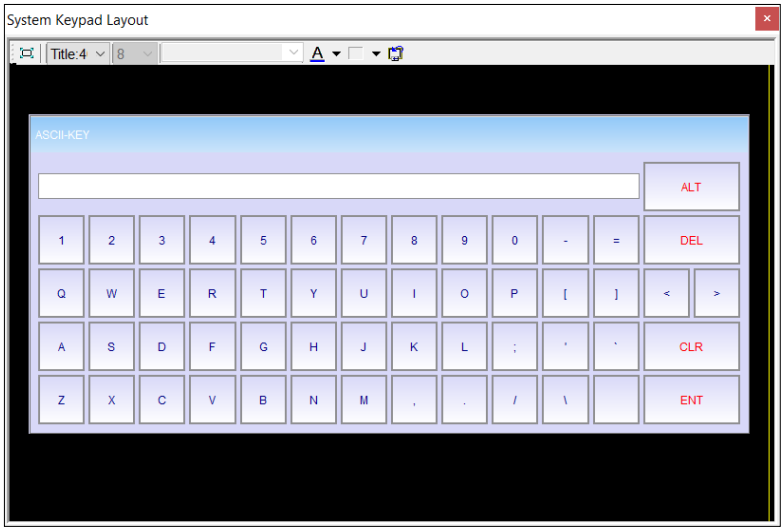
② Binary Keypad

Function	Description
System Keypad	This keypad is applied to the **Numeric Keypad** element when the place value is set to binary. 
Custom Keypad	Click ... to select the custom screen. Go to **General** > **New Screen** > **Keypad Screen** to customize screen.
Apply Settings	Set the keypad style applied method. • **Not apply**: Maintain the original style of the system keypad. • **Apply to all**: Replace the old keypad style with the new style.

③ Hexadecimal Keypad

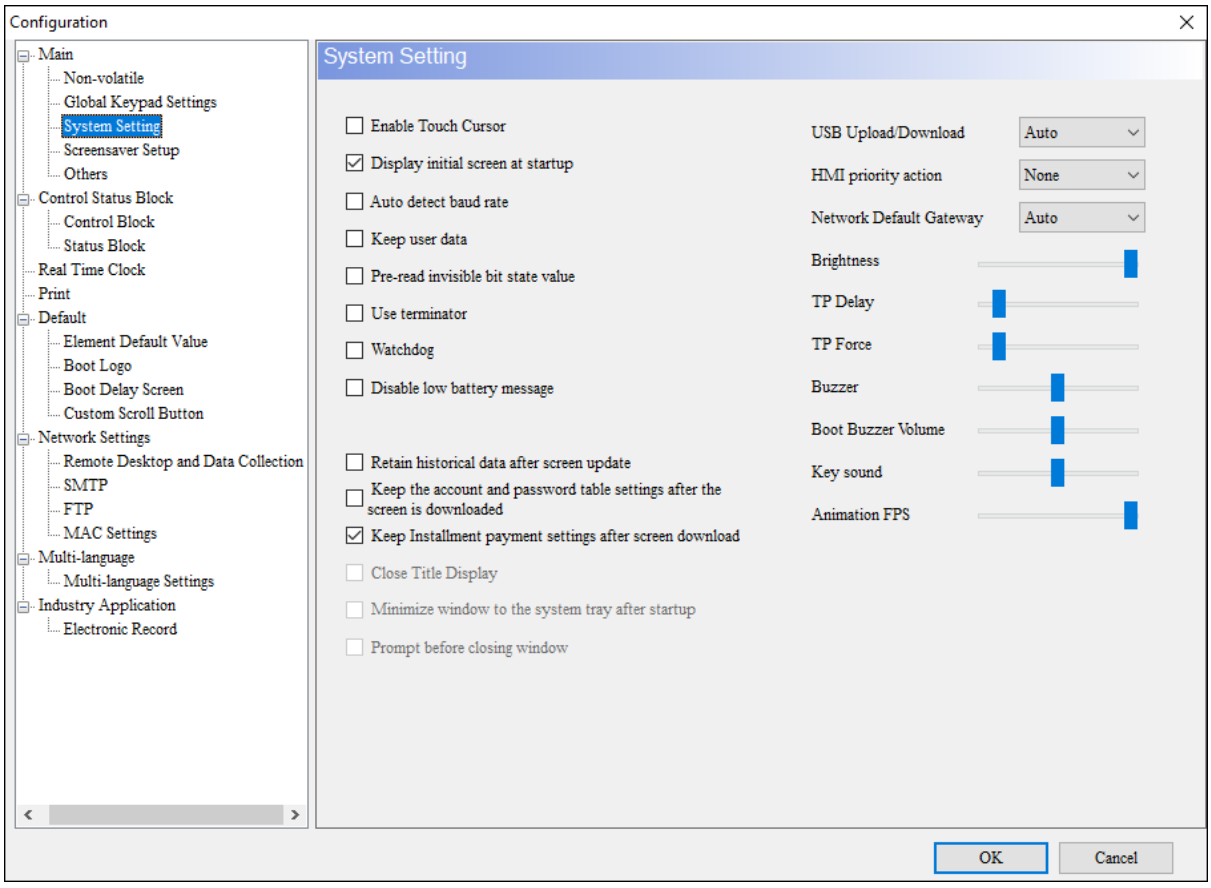
Function	Description
System Keypad	This keypad is applied to the **Numeric Keypad** element when the place value is set to hexadecimal. 
Custom Keypad	Click ... to select the custom screen. Go to **General** > **New Screen** > **Keypad Screen** to customize screen.
Apply Settings	Set the keypad style applied method. • **Not apply**: Maintain the original style of the system keypad. • **Apply to all**: Replace the old keypad style with the new style.

4 ASCII Keypad

Function	Description
System Keypad	This keypad is applied to the **Character Keypad** element when the place value is set to ASCII. 

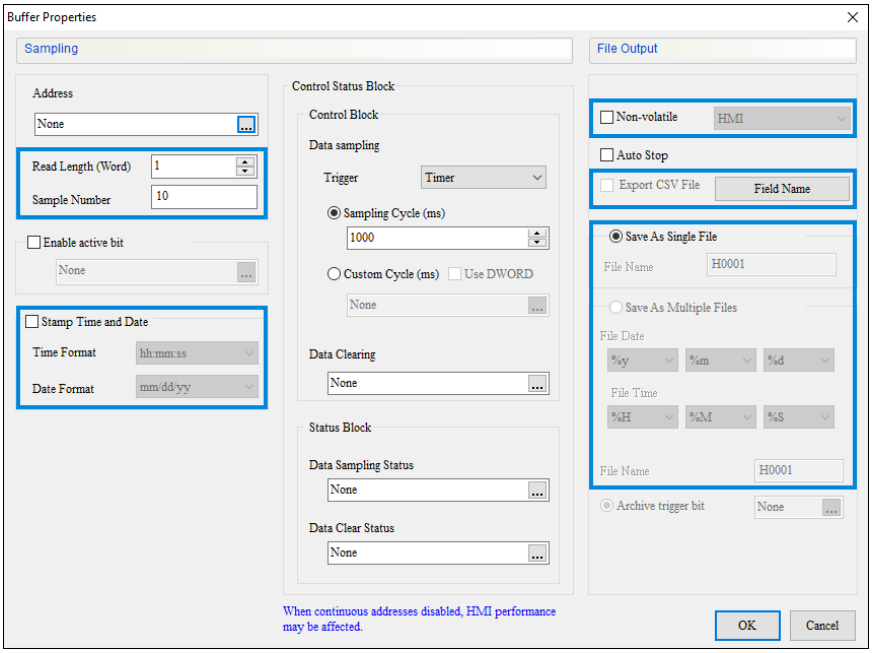
Function	Description
Custom Keypad	Click ... to select the custom screen. Go to **General** > **New Screen** > **Keypad Screen** to customize screen.
Apply Settings	Set the keypad style applied method. • **Not apply**: Maintain the original style of the system keypad. • **Apply to all**: Replace the old keypad style with the new style. • **Apply to new**: Only applies to newly created keypads.

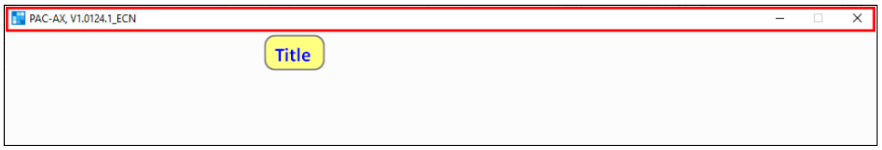
System Setting

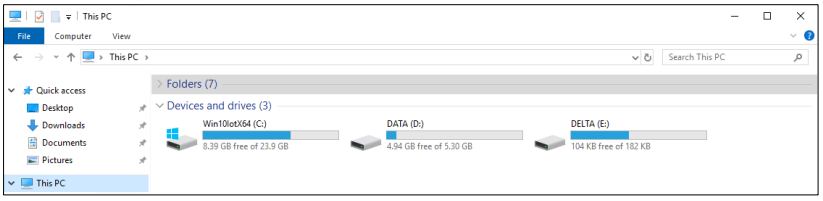
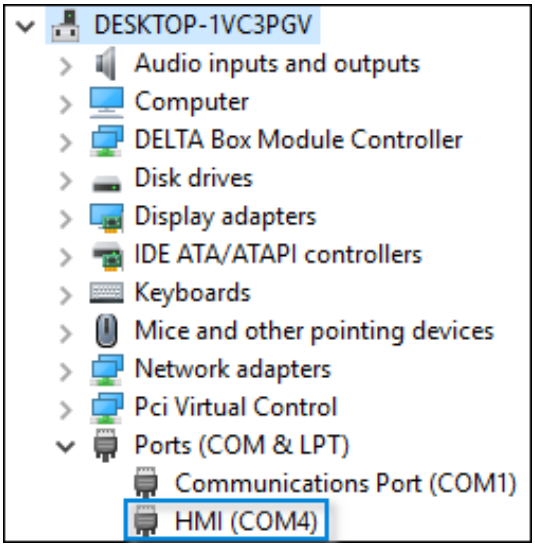


Function	Description
Enable Touch Cursor	If selected, when the HMI screen is pressed, the same mouse icon as the Windows operating system pointer appears. HMI supports wireless mouse with Unifying receiver.

Function	Description
Display initial screen at startup	If selected, the HMI displays the initial screen every time it is powered on.
Auto detect baud rate	If selected, a message displays on the HMI and asks whether to detect procedure. When the baud rate of the HMI is different from that of the controller, it will be automatically adjusted to the same.
Keep user data	If selected, the files generated through LUA do not delete while downloading the screen. It will only be deleted while formatting or restoring to factory settings.
Pre-read invisible bit state value	If selected, when the invisible address is set to the controller address, switching the screen can avoid the transient state of the element display, but it affects the screen display speed.
Use terminator	If selected, character display/input, historical data display, and historical data CSV storage are applied the end character '\0'.
Watchdog	If selected, the HMI automatically restarts when it crashes.
Disable low battery message	If selected, the HMI does not display the low battery message when it is initializing.

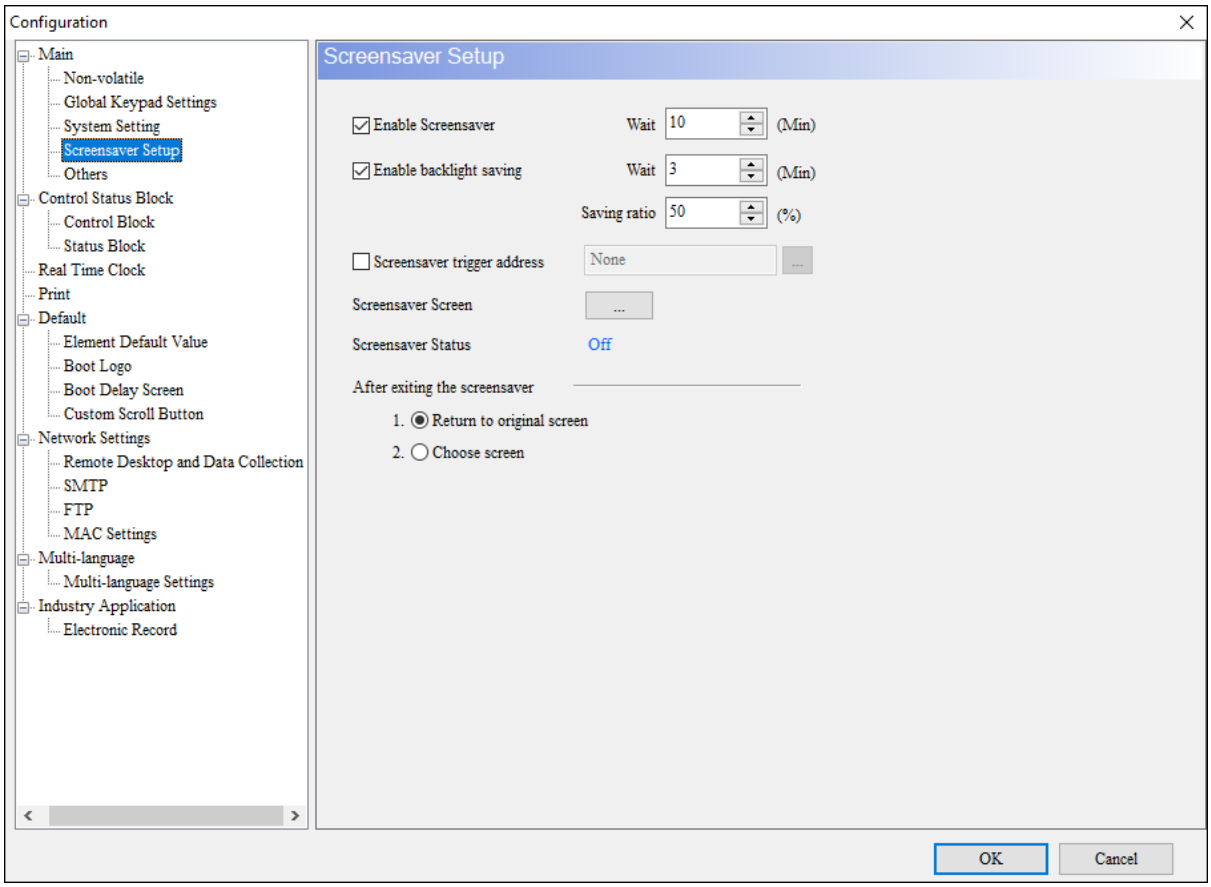
Function	Description
Retain historical data after screen update	If selected, the historical data on the HMI is retained when downloading the screen. To retain and continue to use the historical data, the historical data will be compared with the history buffer settings before updating the HMI screen. The following are the comparison principles. • The process of database saving and the export of CSV files are not affected. • Historical elements may still display or behave normally. After changing Read Length, Sample Number, Stamp Time and Date, Non-volatile, Export CSV File (including Field Name), and Save As Single File / Multiple File, the historical data will not be retained even if this setting is selected. 
Keep the account and password table settings after the screen is downloaded	If selected, the settings of the account and password table do not change while downloading the screen.
Keep Installment payment settings after screen download	If selected, the installment data on the HMI is retained while downloading the screen.

Function	Description
Close Title Display	If selected, the HMI screen does not display the title bar.  **Note**: This setting is only available for AX-8 Windows and IMP Series models.
Minimize window to the system tray after startup	If selected, the HMI screen minimizes to the system tray when it starts. **Note**: This setting is only available for AX-8 Windows and IMP Series models.
Prompt before closing window	If selected, a message displays when the HMI screen is closed, informing that the HMI application will be closed. **Note**: This setting is only available for AX-8 Windows and IMP Series models.

Function	Description
USB Upload/Download	Select the USB upload / download mode. • **Auto** ○ If selected, after downloading the project, the HMI maintains the currently set mode to upload and download. ○ The factory default value of other HMIs is Auto. • **Disk** If selected, after downloading the project, the HMI uses this mode to upload and download (it is USBCommMode = 1.). You can view “DELTA” removable storage device in **This PC**.  • **CDC** If selected, after downloading the project, the HMI uses this mode to upload and download (it is USBCommMode = 2). Right-click the blank space in **The PC** and go to **Content** > **Device Manager.** In the **Device Manager** dialog, check whether there is **HMI device** under **Ports (COM & LPT)**.  • **None** If selected, the USB upload and download cannot be used.
HMI priority action	Select the HMI priority action. • **None**—The HMI does not take any actions first. • **Touch**—The HMI receives the touch action first, and then process the system operation.

Function	Description
Network Default Gateway	Select network default gateway. The options are displayed according to the number of network ports of models. If extra network accessories are attached on the HMI, you can select Wi-Fi or 4G. The default value is **Auto**.
Brightness	Set the brightness of the HMI backlight.
TP Delay	Set the delay time for the HMI to process pressing the move messages. Drag the delay to the right to decrease, the speed becomes faster; drag the delay to the left to increase the speed, and the speed becomes slower.
TP Force	Set the force for pressing the HMI screen. Drag the force to the left to decrease, the HMI screen becomes easier to press; drag the force to the right to increase, and the HMI screen becomes difficult to press.
Buzzer	Set the buzzer volume of the HMI. Drag to the left to increase the volume and drag to the right to decrease the volume.
Boot Buzzer Volume	Set the boot buzzer volume of the HMI. Drag to the left to increase the volume and drag to the right to decrease the volume.
Key sound	Set the sound when the HMI is pressed. Drag to the left to increase the volume and drag to the right to decrease the volume.
Animation FPS	Set the update speed of the animation element setting frequency.

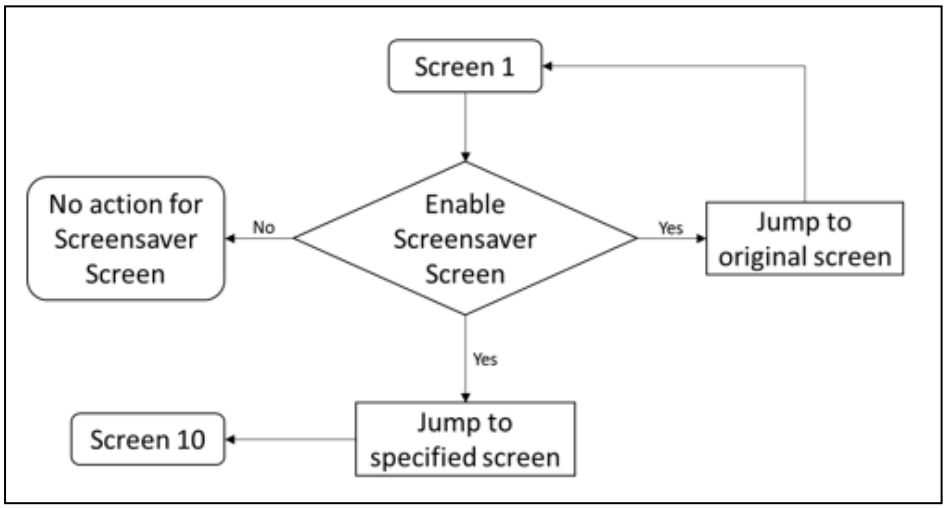
Screensaver Setup



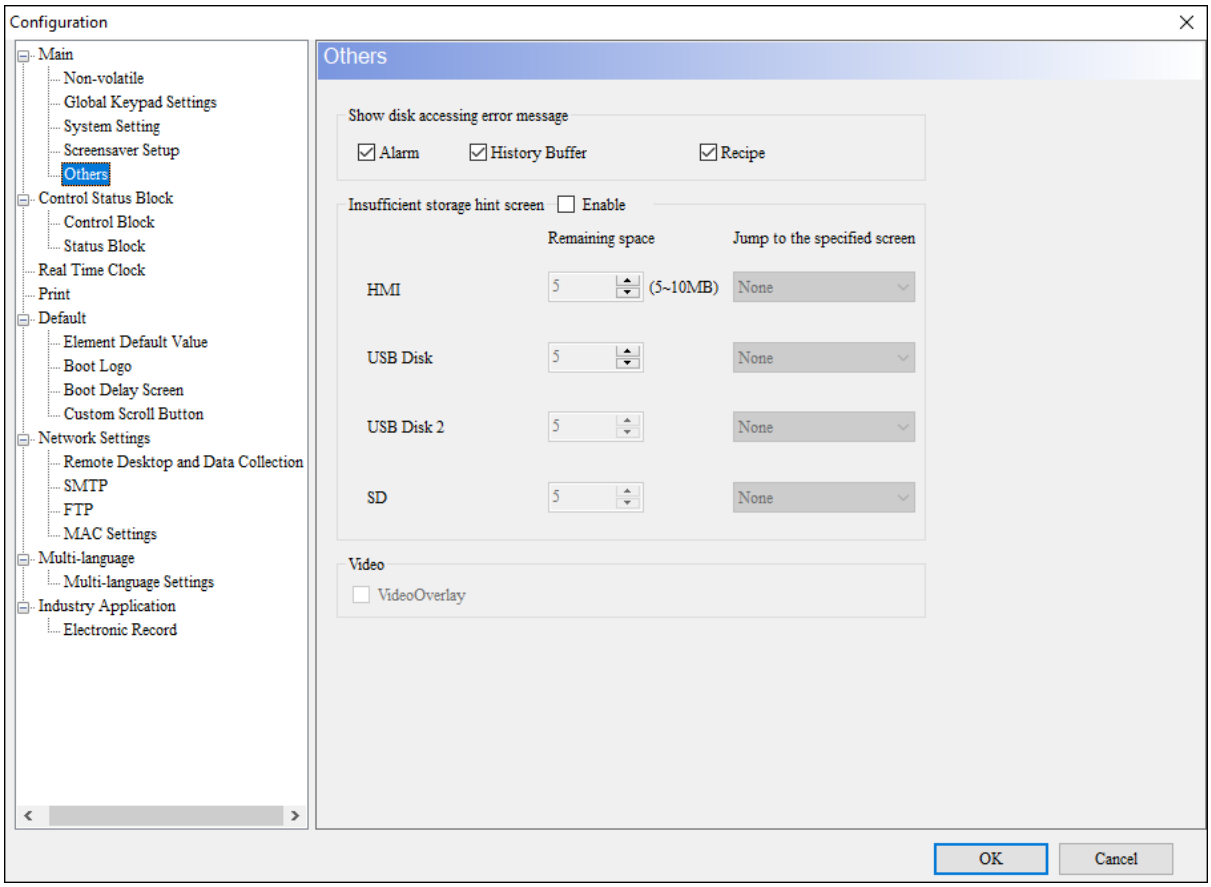
Function	Description
Enable Screensaver	If selected, you can set how long it takes to enter the screen saver screen without pressing the screen. The value range is 1–100, and the default value is **10**. The screen saver can be configured by clicking ... of **Screensaver Screen** or in **General** > **Screen saver Screen**. **Note**: If **Enable Screensaver** is not selected, but the screen saver is configured, the screen saver will not be activated.
Enable backlight saving	If selected, you can set how long it takes to enable the backlight saving function without pressing the screen. • The value range is 1–10, and the default value is **3**. • The saving ratio range is 1–99, and the default value is **50**.
Screensaver trigger address	Set Word trigger address, 0 is to turn off the screen saver, and 1 is to turn on the screen saver.

Function	Description
After exiting the screensaver	• **Return to original screen**: If selected, after the screen saver is over, it jumps to the original screen before starting the screen saver. • **Choose screen**: If selected, you can specify the screen number, so that the designated screen will be displayed after the screen saver ends. **Note**: If you want to use the specified screen, create a screen saver screen first.

The following figure displays the flow chart of the screen saver screen.



Others

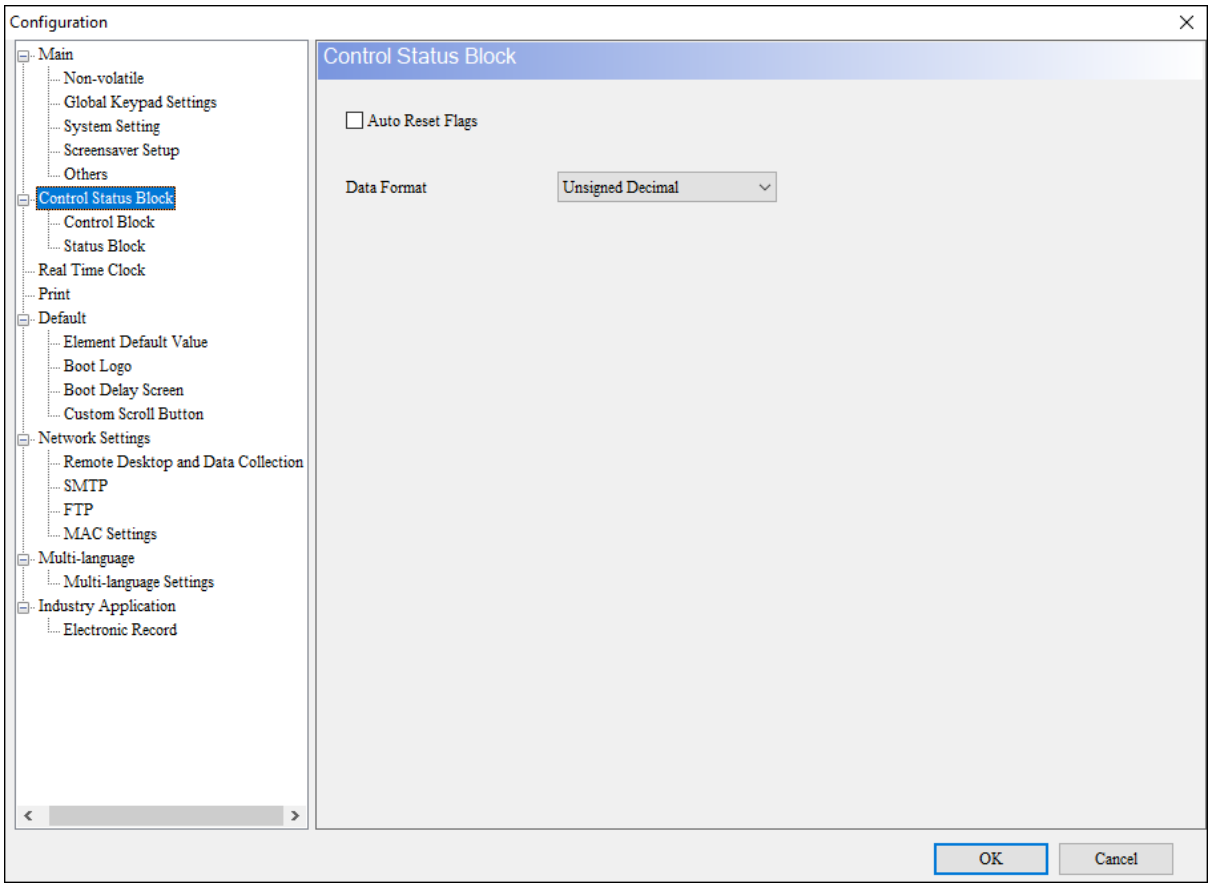


Function	Description
Show disk accessing error message	Since alarms, history buffers and recipes can be set to save data as USB Disk or SD Card, if the external disk access fails, you can decide whether to display the warning message through this setting. The disk access failure means that when the non-volatile data location of history buffer is set to USB Disk, the HMI cannot detect the USB Disk after powering on and cannot write data. If this option is not selected, when the above situation occurs, no error message appears.
Insufficient storage hint screen	Set how many MB of storage space is left to jump to the specified screen. The storage spaces can be specified to HMI, USB Disk, USB Disk 2, and SD.
Video	If **Video Overlay** is selected, the VGA or Camera element superimposes on the element, and the display can be triggered through the invisible address. This setting enables / disables according to different models.

Control Status Block

The **Control Status Block** is used to execute or monitor the system operation and status. Select any of functions according to your needs, for example, in **Control Block**, you only need to select the **Screen No.** and **Recipe Number**, the **control status block** automatically configures the addresses continuously and only enables the **Goto Screen** and **Recipe Group Control** functions.

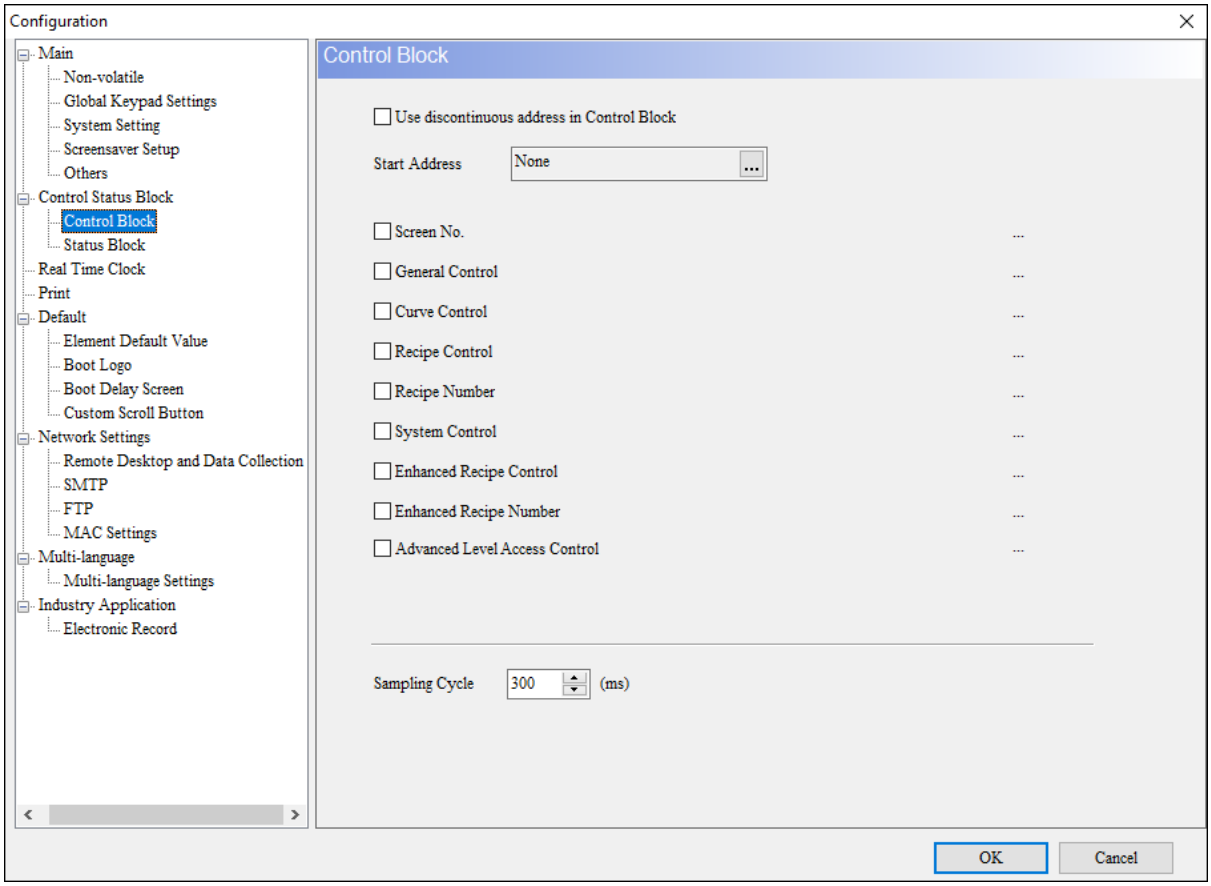
If an additional function is selected, such as **General Control**, the memory addresses are arranged continuously from top to bottom.



Function	Description
Auto Reset Flags	If selected, the HMI automatically resets the flags. To reactivate some functions in the **Control Block**, first turn the corresponding flags OFF, and then turn it ON again.
Data Format	Set the **Data Format** of the control status block. The default value is **Unsigned Decimal**.

Control Block

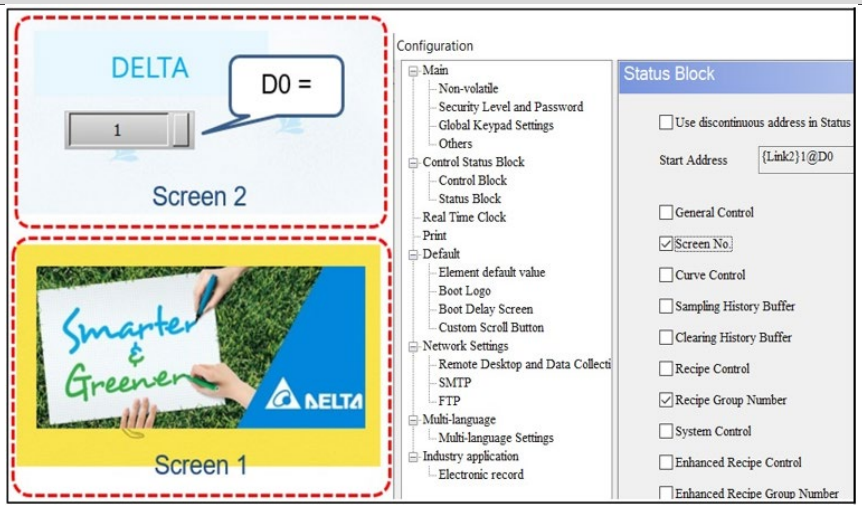
The **Control Block** is used to control the actions of the HMI, such as Goto screen, curve, recipe control, and user level.



Function	Description
Use discontinuous address in Control Block	If selected, the settings in the Control Block are available, and you can define the register addresses in the controller or the HMI internal memory.
Sampling Cycle	Set the interval time between the sampling actions. The value range is 100–1000, and the default value is **300**.

The following table lists the flags of screen no. register with its description.

Screen No. Register	
b0~b15 – Screen No.	• Write the specified screen number into this register, and the HMI switches to the specified screen. • If a **Numeric Entry** element is set to D0 and the input value is 1, the HMI switches to the first screen.

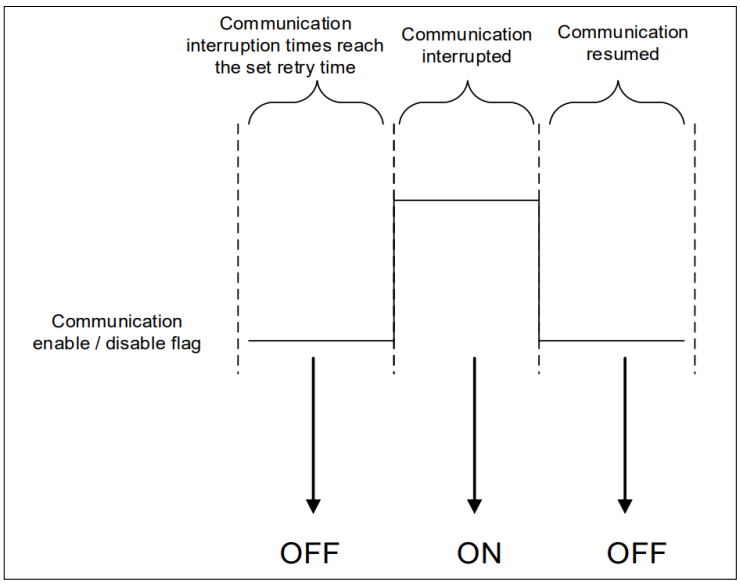
Screen No. Register	
	

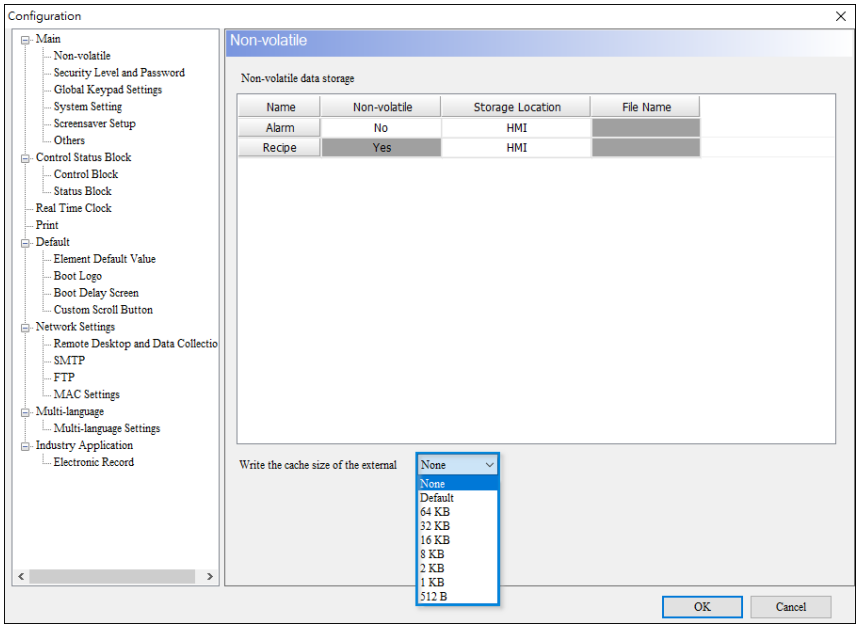
The following table lists the flags of general control register with their description.

General Control Register

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
(10)				(9)				(8)	(7)	(6)	(5)	(4)	(3)	(2)	(1)

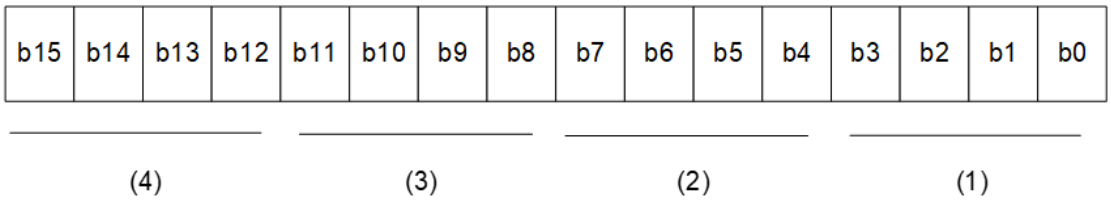
(1) b0 Communication enable / disable flag
(2) b1 Backlight enable / disable flag
(3) b2 Buzzer enable / disable flag
(4) b3 Alarm buffer clear flag
(5) b4 Alarm counter clear flag
(6) b5 External storage device cache write flag
(7) b6 Remote control lock
(8) b7 Reserved
(9) b8~b11 Set User Security Level
(10) b12~b15 Reserved

General Control Register	
b0 – Communication enable / disable flag	Enable / disable the HMI communication. • To use this flag, click **General** > **Device Communication**, and then select **Disconnect after communication interrupt checkbox** and set the **Retry times after disconnection.** • When the HMI communicates with the controller, the HMI automatically stops the communication with the controller and turns the flag ON after the number of interruption times reaches the set retry times without the communication error message (this does not affect the communication between the HMI and other controllers). You can restore the communication by turning the flag OFF.  **Note**: • This flag is only for restoring communication when it is automatically stopped. It is not possible to directly stop communication between the HMI and any controller by turning the flag ON. • The **Auto Reset Flags** function is not applicable to this flag.
b1 – Backlight enable / disable flag	Enable / disable the HMI backlight. • Set to ON, the HMI backlight is enabled. • Set to OFF, the HMI backlight is disabled. **Note**: The **Auto Reset Flags** function is not applicable to this flag.
b2 – Buzzer enable / disable flag	Enable / disable the HMI buzzer. • Set to ON, the HMI buzzer is enabled. • Set to OFF, the HMI buzzer is disabled. **Note**: The **Auto Reset Flags** function is not applicable to this flag.

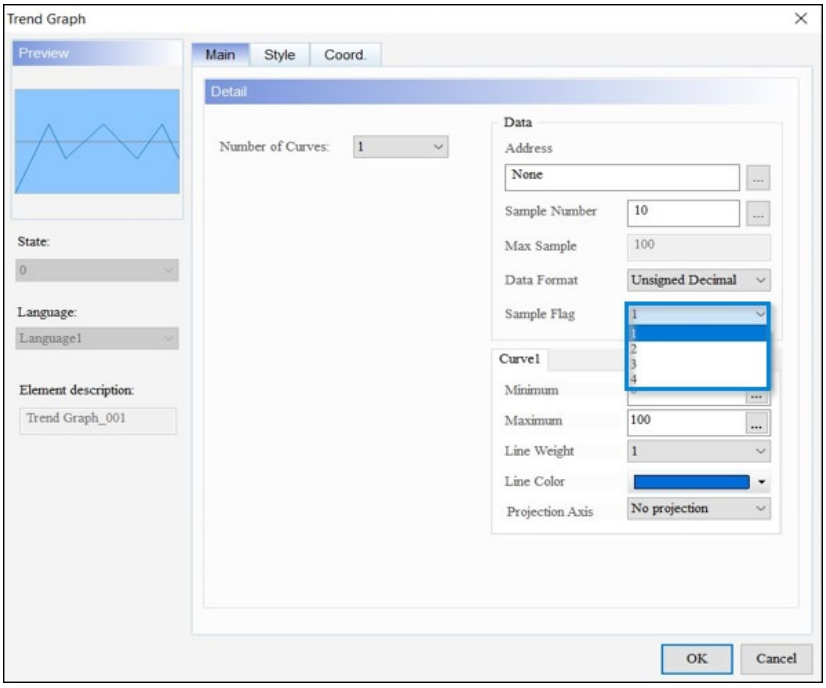
General Control Register	
b3 – Alarm buffer clear flag	Clear the alarm buffer data of the HMI. Set to ON to clear the data in the alarm buffer. To reactivate this function, turn the flag OFF and then ON again.
b4 – Alarm counter clear flag	Clear the alarm frequency table data of the HMI. Set to ON to clear the data in the alarm frequency table. To reactivate this function, turn the flag OFF and then ON again.
b5 – External storage device cache write flag	Update the HMI cache data into an external storage device (USB Disk or SD Card) in real time. • If the alarm buffer, history buffer, or recipe function is activated, and the non-volatile storage location is set to USB Disk or SD, when the flag is set to ON, the HMI updates the data temporarily stored in the cache into an external storage device in real time. To reactivate this function, turn the flag OFF and then ON again. • The data written into USB or SD Card by the HMI is temporarily stored in the cache first. Before the cache data size reaches the set limit (as shown in the following figure), the data is not written into an external storage device. This is to prevent the external storage device from being damaged by frequent overwriting. However, if the data volume accessed is less than the buffer capacity or the power is cut off unexpectedly, part of the data may be lost. To keep the data, turn ON the flag in a cyclic pattern to write the data into the external storage device. 

<table>
<tr><th colspan="2">General Control Register</th></tr>
<tr><td>b6 –
Remote control lock</td><td>Control whether the eRemote can be operated.
Set to ON, eRemote can only be monitored and cannot be operated.
Set to OFF, eRemote can be operated normally.
Note: The Auto Reset Flags function is not applicable to this flag.</td></tr>
<tr><td>b8~b11 –
Set User Level</td><td>You can set the current user level of the HMI. The HMI internal security levels are divided as follows:
Security level 0–7: Controlled by Bit 8, Bit 9, and Bit 10 flags. Level 0 is the lowest security level
Highest security level: Controlled by Bit 11 flag.
The following table lists the setting method or the security level.
<table>
<tr><th rowspan="2">Security Level</th><th colspan="3">Flag Control</th></tr>
<tr><th>Bit 10</th><th>Bit 9</th><th>Bit 8</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>2</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>3</td><td>0</td><td>1</td><td>1</td></tr>
<tr><td>4</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>5</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>6</td><td>1</td><td>1</td><td>0</td></tr>
<tr><td>7</td><td>1</td><td>1</td><td>1</td></tr>
</table>
Note: If enable Advanced Level Access Control, the user level control here will be invalid.</td></tr>
</table>

The following table lists the flags of curve control register with their description.

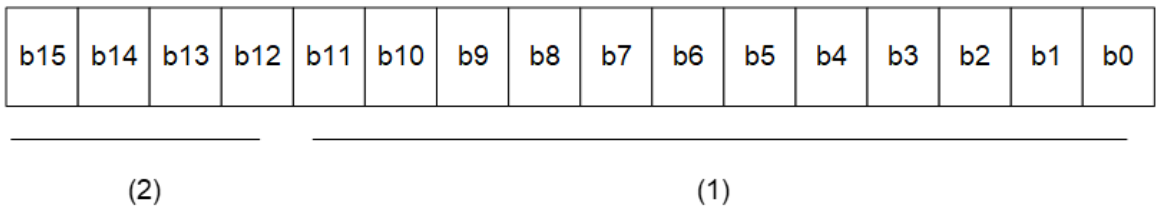
<table>
<tr><th>Curve Control Register</th></tr>
<tr><td>


(1) b0~b3 Curve sampling flags 1~4
(2) b4~b7 Reserved
(3) b8~b11 Curve clear flags 1~4
(4) b12~15 Reserved
</td></tr>
</table>

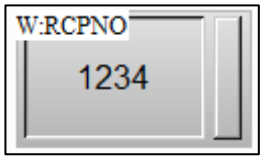
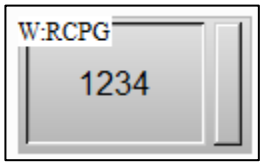
Curve Control Register	
b0~b3 – Curve sampling flags (1~4)	Control the curve drawing action. • The curves include the Trend Graph and X-Y Chart. • If the flag is set to ON, the corresponding curve element samples and draws the curve. To reactivate this function, turn the flag OFF and then ON again. • The sampling flag 1 of the **Trend Graph** element corresponds to the curve clear flag 1; the sampling flag 2 of the **Trend Graph** element corresponds to the curve clear flag 2, and so forth. 
b8~b11 – Curve clear flags (1~4)	Control the curve clear action. If the flag is set to ON, the corresponding curve element clears the curve. To reactivate this function, turn the flag OFF and then ON again.

The following table lists the flags of recipe control register with their description.

Recipe Control Register

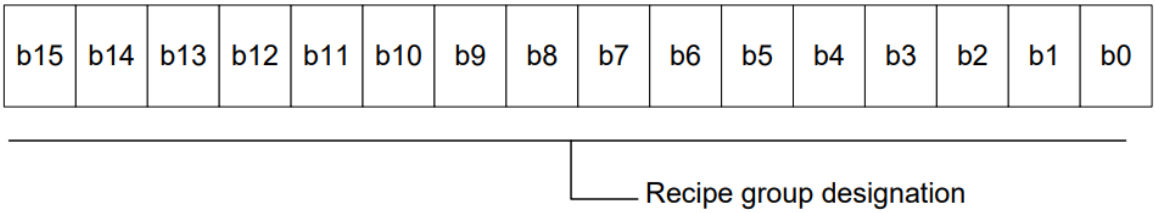


(1) b0 Recipe number change flag
(2) b1 Recipe read flag (PLC → HMI)

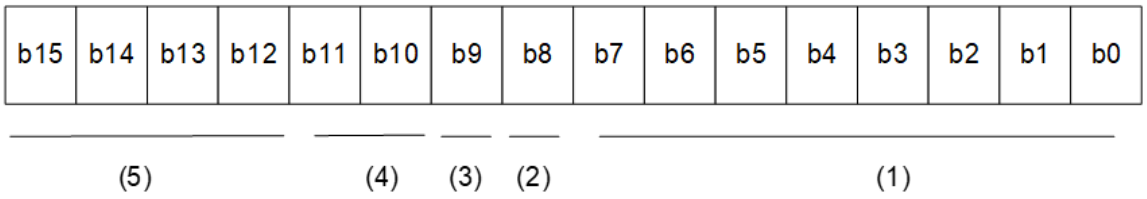
Recipe Control Register	
(3) b2 Recipe write flag (HMI → PLC) (4) b3 Recipe group change flag (5) b4~b7 Reserved (6) b8~b15 Specify Recipe group number	
b0 – Recipe number change flag	Change the HMI recipe number. This is applicable to 16-Bit recipes. You must write the recipe number to be changed into the recipe number control register and then trigger the recipe number change flag. Do one of the following to call or change the recipe number: • Use the HMI internal register **RCPNO.**  • Use this recipe number change flag. When the recipe number change flag is turned ON, the recipe number is changed according to the value defined in the recipe number control register, and the number of RCPNO internal register is changed automatically. To reactivate the function, turn the flag OFF and then ON again.
b1 – Recipe read flag (PLC → HMI)	When the recipe read flag is turned ON, the HMI reads the controller recipe data and writes the data into the designated recipe data register. To reactivate the function, turn the flag OFF and then ON again.
b2 – Recipe write flag (HMI → PLC)	When the recipe write flag is turned ON, the HMI writes the designated recipe data into the controller register. To reactivate the function, turn the flag OFF and then ON again.
b3 – Recipe group change flag	Change the HMI recipe group number. This is applicable to 32-Bit recipes. Do one of the following to call and change the recipe group: • Use the HMI internal register **RCPG.**  • Use the recipe group change flag. When the recipe group change flag is turned ON, the recipe group is changed according to the value defined in the recipe group change bits (b8~b15), and the number of the **RCPG** internal register is changed automatically. To reactivate the function, turn the flag OFF and then ON again.

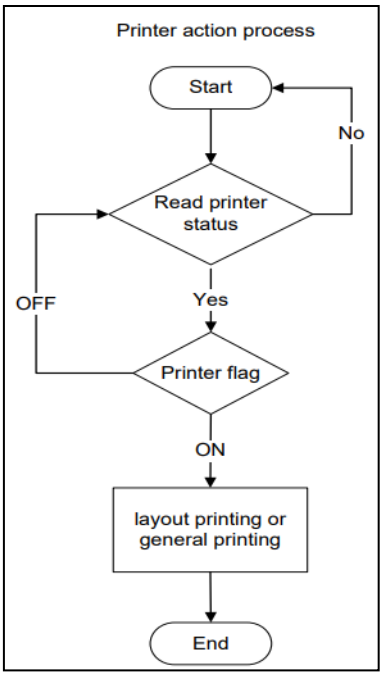
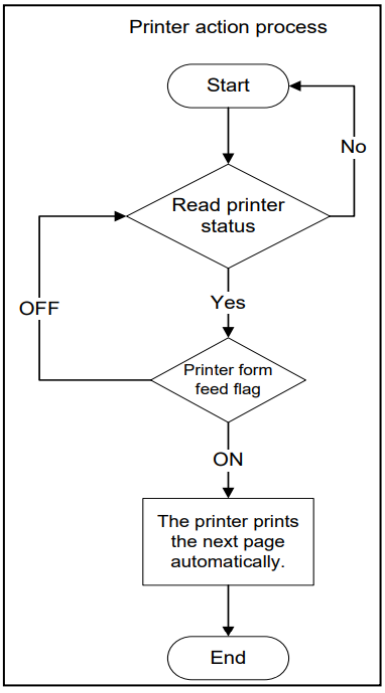
Recipe Control Register	
b8~b15 – Designate recipe group to be changed	Designate the recipe group to be changed with the high byte (Bits 8~15) from the recipe control register. By activating the recipe group change flag, the HMI changes the number of the RCPG internal register, it is changing the recipe group.

The following table lists the flags of recipe number register with their description.

Recipe Number Register	
	
b0~b15 – Recipe designation	Designate the group number of the recipe group to be changed through the recipe number register. By activating the recipe group change flag, the HMI automatically changes the number of the RCPG internal register, it is changing the recipe group.

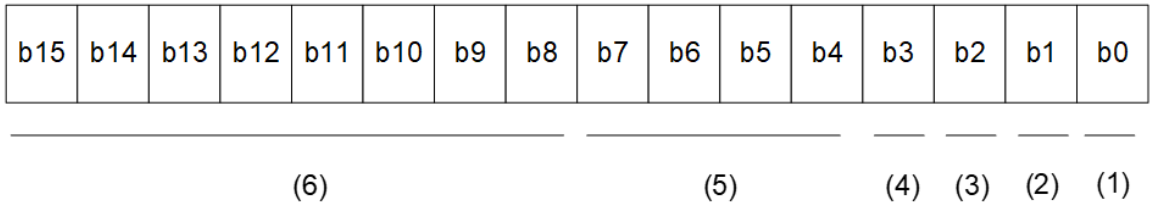
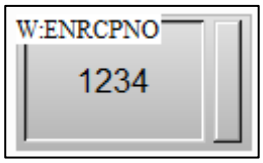
The following table lists the flags of system control register with their description.

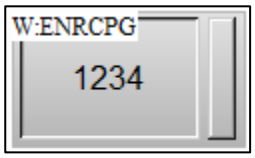
System Control Register	
	(1) b0~b7 Multi-language setting values (2) b8 Printer flag (3) b9 Printer form feed flags (4) b10~b11 Retrieve Ethernet address (5) b12 Wi-Fi retrieve network address (6) b13~b15 Reserved
b0~b7 – Multi-language setting values	Change the **multi-language setting values** to switch languages. You can set the multi-languages in **Multi-language setting page of Configuration** dialog. **Note**: Supports adding up to maximum 32 languages.

System Control Register	
b8 – Printer flag	Set the printing mode, which is divided into general printing and screen layout printing. • Set to ON, the printer runs according to the set mode. • Set to OFF, the printer is idled.  **Note**: Only one mode is enabled at a time. The screen layout printing is prioritized.
b9 – Printer form feed flag	• Set to ON, the printer ejects the paper and flips the paper. • Set to OFF, the printer is idled. 

System Control Register	
b10~b12 – Retrieve network address	When the HMI fails to obtain the IP address assigned by the DHCP server, trigger this bit to retrieve the IP address. • b10 is for obtaining the IP address with LAN1 • b11 is for obtaining the IP address with LAN2. • b12 is for obtaining the IP address with Wi-Fi.

The following table lists the flags of enhanced recipe control register with their description.

Enhanced Recipe Control Register	
 (1) b0 Enhanced Recipe number change flag (2) b1 Enhanced Recipe read flag (PLC → HMI) (3) b2 Enhanced Recipe write flag (HMI → PLC) (4) b3 Enhanced Recipe group change flag (5) b4~b7 Reserved (6) b8~b15 Specify group number of Enhanced Recipe	
b0 – Enhanced Recipe number change flag	Change HMI recipe group. This is applicable to enhanced recipes. Do one of the following to call or change the enhanced recipe number: • Use the HMI internal register **ENRCPNO**.  • Use enhanced recipe group change flag. When the enhanced recipe group change flag is turned ON, the enhanced recipe group is changed according to the number defined in the enhanced recipe group control register, and the number in the internal register ENRCPNO is changed automatically. To reactivate the function, turn the flag OFF and then ON again.

Enhanced Recipe Control Register	
b1 – Enhanced Recipe read flag (PLC → HMI)	When the enhanced recipe read flag is turned ON, the HMI reads the controller enhanced recipe data and writes it into the designated enhanced recipe data register. To reactivate the function, turn the flag OFF and then ON again.
b2 – Enhanced Recipe write flag (HMI → PLC)	When the enhanced recipe write flag is turned ON, the HMI writes the designated enhanced recipe data into the controller register. To reactivate the function, turn the flag OFF and then ON again.
b3 – Enhanced Recipe group number change flag	Change the HMI recipe group number. This is applicable to enhanced recipes. Do one of the following to call or change the enhanced recipe group number: • Use the HMI internal register ENRCPG.  • Use enhanced recipe group number change flag. When the enhanced recipe group number change flag is turned ON, the enhanced recipe group number is changed according to the value defined in the enhanced recipe group number bits (b8~b15), and the number in the internal register ENRCPG is changed automatically. To reactivate the function, turn the flag OFF and then ON again.
b8~b15 – Designate enhanced recipe group number to be changed	Designate the enhanced recipe group number to be changed with the high byte (Bits 8~15) from the enhanced recipe control register. By activating the enhanced recipe group number change flag, the HMI changes the number in the internal register ENRCPG, it is changing the enhanced recipe group.

The following table lists the enhanced recipe control register with their description.

Enhanced Recipe Control Register															
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Enhanced recipe group designation															

Enhanced Recipe Control Register	
b0~b15 – Enhanced Recipe designation	Designate the enhanced recipe group number to be changed from the enhanced recipe group control register. By activating the enhanced recipe group change flag, the HMI changes the number in the internal register ENRCPNO, it is changing the enhanced recipe group.

The following table lists the advanced level access control register with their description.

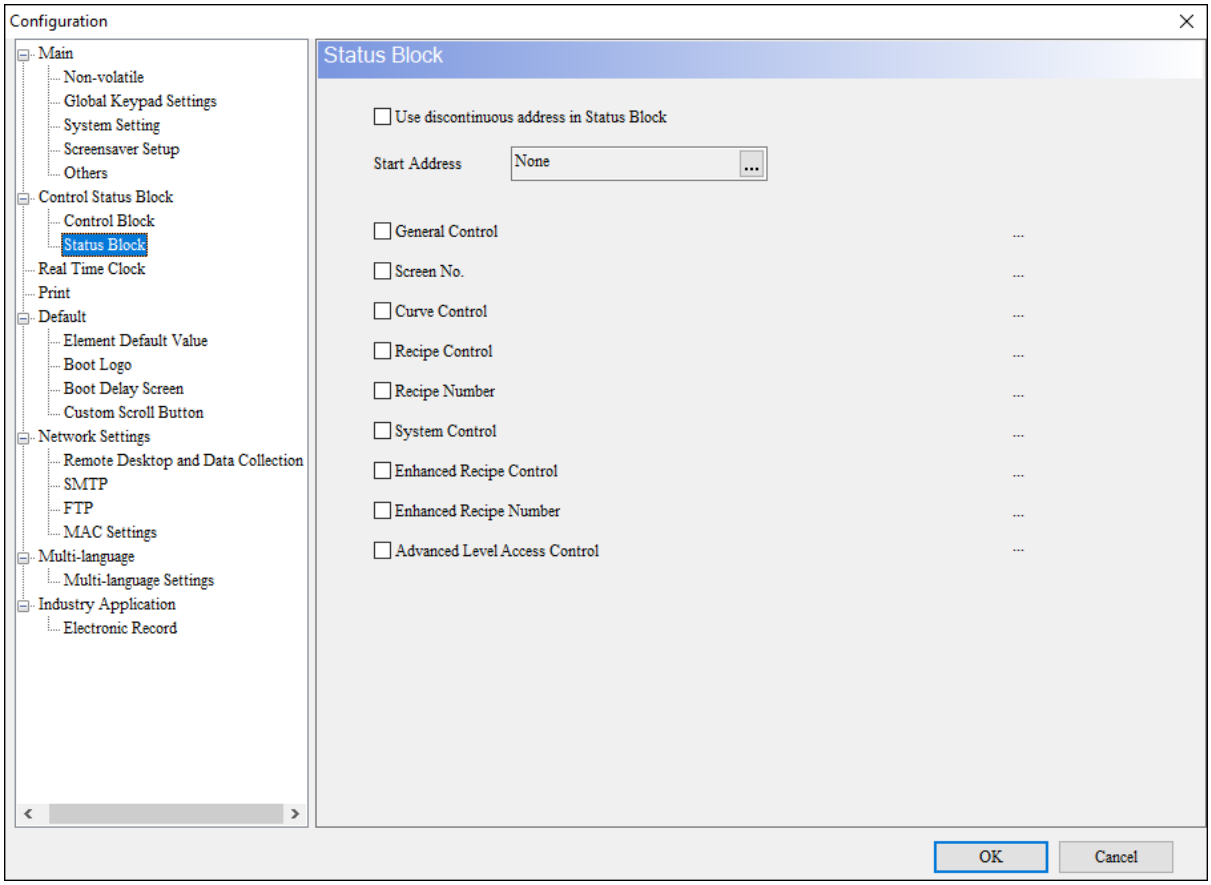
<table>
<tr><th colspan="2">Advanced Level Access Control</th></tr>
<tr><td>b0~b3–
Set User Level</td><td>Change the user level of the current HMI by triggering the Bit 0, Bit 1, Bit 2 and Bit 3 flags.
The following table lists how to set security levels.
<table>
<tr><th rowspan="2">Security Level</th><th colspan="4">Flag Control</th></tr>
<tr><th>Bit 3</th><th>Bit 2</th><th>Bit 1</th><th>Bit 0</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>2</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>1</td><td>1</td></tr>
<tr><td>4</td><td>0</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>5</td><td>0</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>6</td><td>0</td><td>1</td><td>1</td><td>0</td></tr>
<tr><td>7</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>8</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>9</td><td>1</td><td>0</td><td>0</td><td>1</td></tr>
</table></td></tr>
<tr><td>b15
Set User Highest Level</td><td>Highest security level control.</td></tr>
</table>

Note: If **Advanced Level Access Control** is enabled, user level of **General Control** becomes invalid.

Status Block

The **Status Block** is used to monitor the present HMI status, such as current screen number, curves, recipe control, and user level.

Note: If the **Control Block** is not set, the **Status Block** cannot monitor the status. Moreover, the addresses of the **Control Block** and **Status Block** cannot be the same.



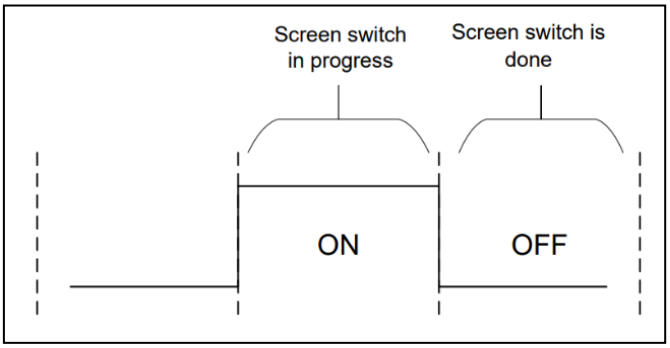
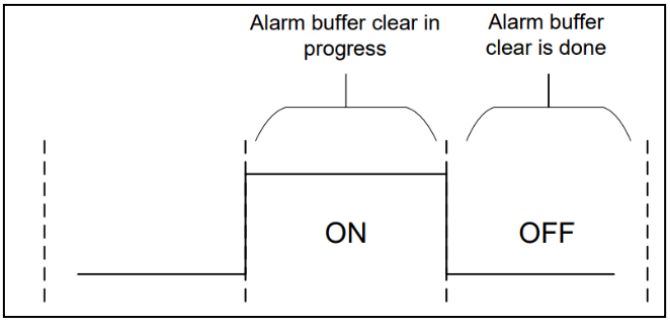
Function	Description
Use discontinuous address in Status Block	If selected, the settings in the Status Block are available, and you can define the register addresses in the controller or the HMI internal memory.

The following table lists the flags of general control status register with their description.

General Control Status Register

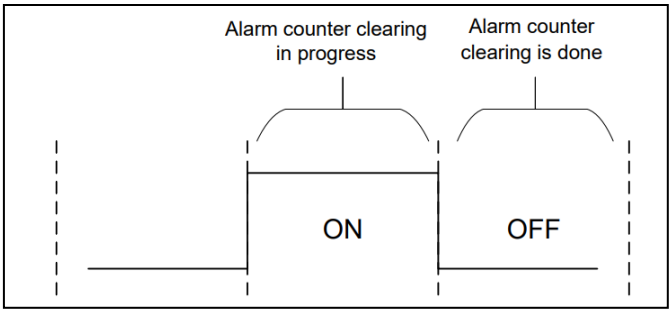
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
(9)				(8)				(7)	(6)	(5)	(4)	(3)	(2)		(1)

(1) b0 Screen switch flag
(2) b1~b2 Reserved
(3) b3 Alarm buffer clear flag
(4) b4 Alarm counter clear flag
(5) b5 External storage device cache write flag
(6) b6 Remote control lock flag
(7) b7 Reserved
(8) b8~b11 Set user level
(9) b12~b15 Reserved

b0 – Screen switch flag	• When the screen is switching, the flag is turned ON, • When the screen switching is completed, the flag is turned OFF. 
b3 – Alarm buffer clear flag	• When the HMI is clearing the alarm buffer, the flag is turned ON. • When the alarm buffer clearing is completed, the flag is turned OFF. 

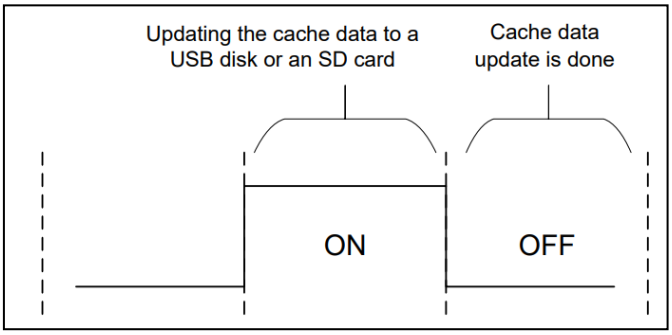
<table>
<tr><th colspan="2">General Control Status Register</th></tr>
<tr><td>b4 – Alarm counter clear flag</td><td>

When the HMI is clearing the alarm counter, the flag is turned ON.
When the alarm counter clearing is completed, the flag is turned OFF.



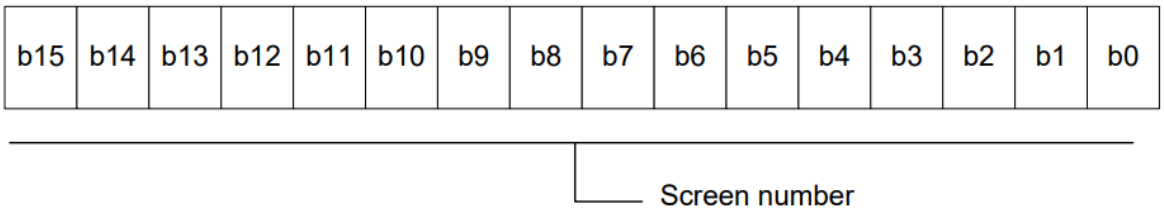
</td></tr>
<tr><td>b5 – External storage device cache write flag</td><td>

When the HMI is updating the cache data into the USB or SD card, the flag is turned ON.
When data updating is completed, the flag is turned OFF.

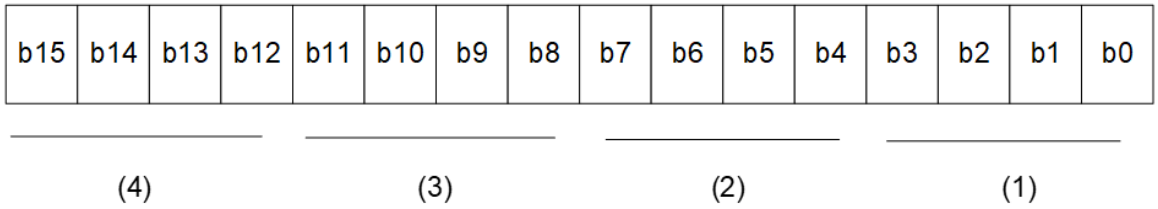


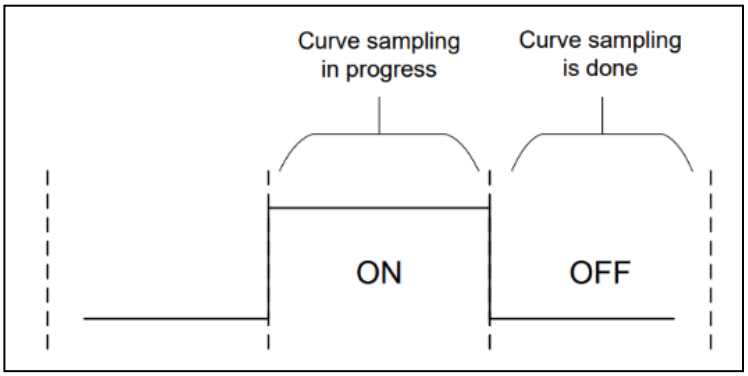
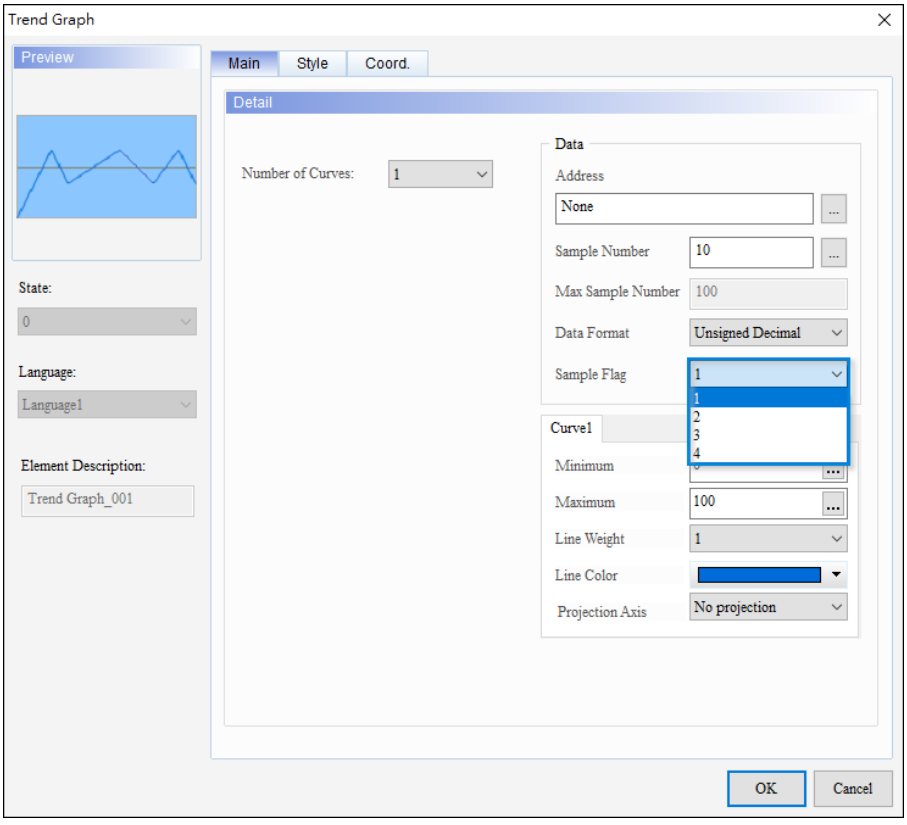
</td></tr>
<tr><td>b8~b11 – Set user level</td><td>
Reads the current user level operated by the HMI. The following table lists the bit state corresponding to the user level operated by the HMI.
<table>
<tr><th rowspan="2">Security Level</th><th colspan="4">Flag Control</th></tr>
<tr><th>Bit 11</th><th>Bit 10</th><th>Bit 9</th><th>Bit 8</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>2</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>1</td><td>1</td></tr>
<tr><td>4</td><td>0</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>5</td><td>0</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>6</td><td>0</td><td>1</td><td>1</td><td>0</td></tr>
<tr><td>7</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>Highest security level</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
</table>
Note: If Advanced Level Access Control is enabled, the user level here will be invalid.
</td></tr>
</table>

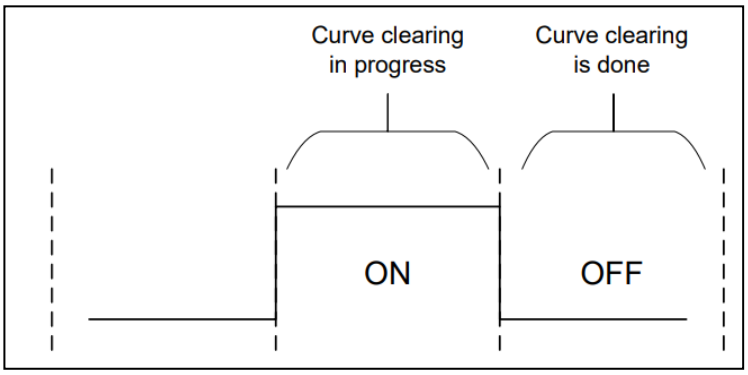
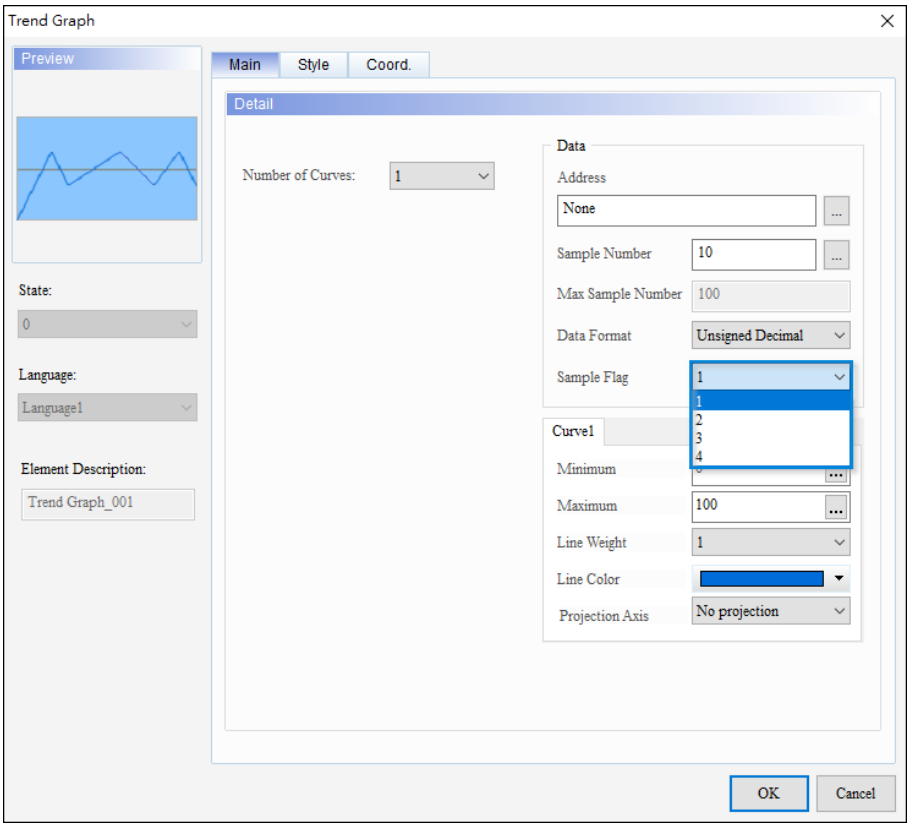
The following table lists the flags of screen No. status register with their description.

Screen No. Status Register	
	
b0~b15 – Screen No. status	Check the last opened screen number with this status register.

The following table lists the flags of the curve control status register with their description.

Curve Control Status Register

(1) b0~b3 Curve sampling status flags 1~4 (b0 refers to flag 1, and so forth) (2) b4~b7 Reserved (3) b8~b11 Curve clear status flags 1~4 (b8 refers to flag 1, and so forth) (4) b12~b15 Reserved

Curve Control Status Register	
b0~b3 – Curve sampling status flags (1~4)	When the **Trend Graph** or **X-Y Chart** elements sample the data, the HMI turns the corresponding curve sampling status flag ON. When the sampling is completed, the curve sampling status flag is turned OFF.  The sampling flag 1 of the **Trend Graph** element corresponds to the curve sampling status flag 1; the sampling flag 2 of the **Trend Graph** element corresponds to the curve sampling status flag 2, and so forth. 

Curve Control Status Register	
b8~b11 – Curve clear status flags (1~4)	When the **Trend Graph** or **X-Y Chart** elements clear the data, the HMI turns the corresponding curve clear status flag ON. When the clearing is completed, the curve clear status flag is turned OFF.  The clear flag 1 of the **Trend Graph** element corresponds to the curve clear status flag 1; the clear flag 2 of the **Trend Graph** element corresponds to the curve clear flag 2, and so forth. 

The following table lists the flags of recipe control status register with their description.

Recipe Control Status Register	

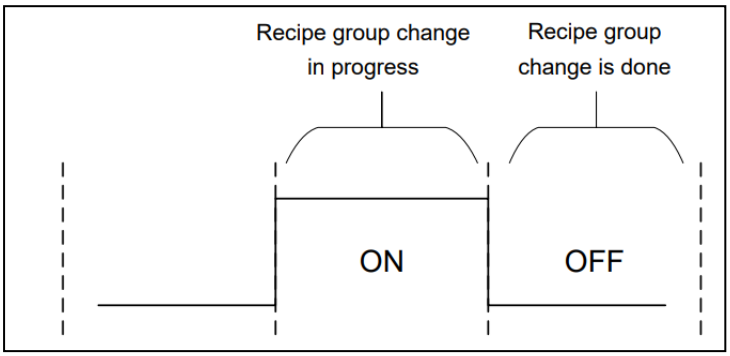
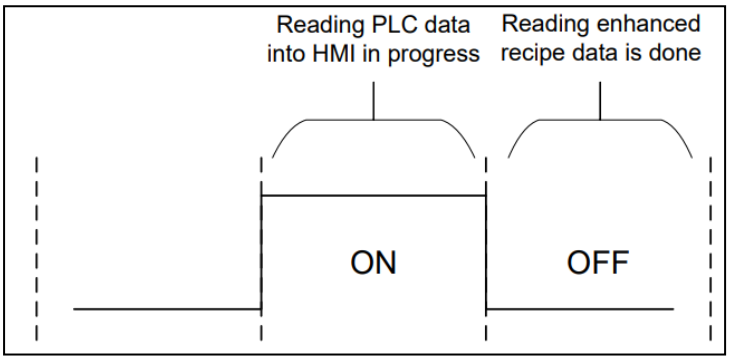
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
(4)													(3)	(2)	(1)

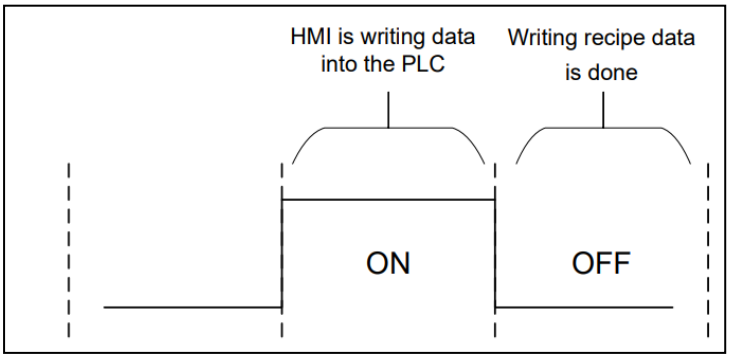
(1) b0 Recipe change status flag

(2) b1 Recipe read status flag (PLC → HMI)

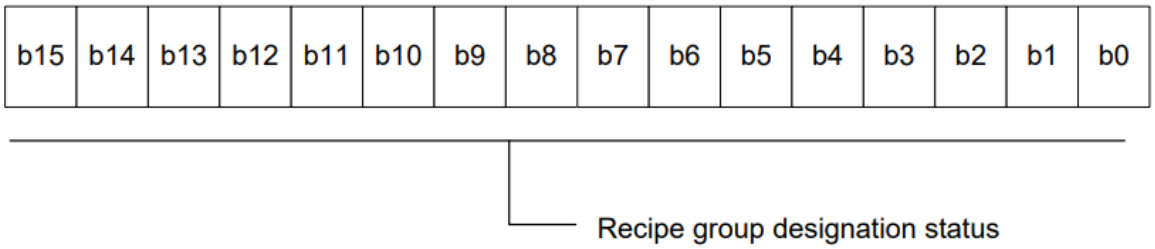
(3) b2 Recipe write status flag (HMI → PLC)

(4) b3~b15 Reserved

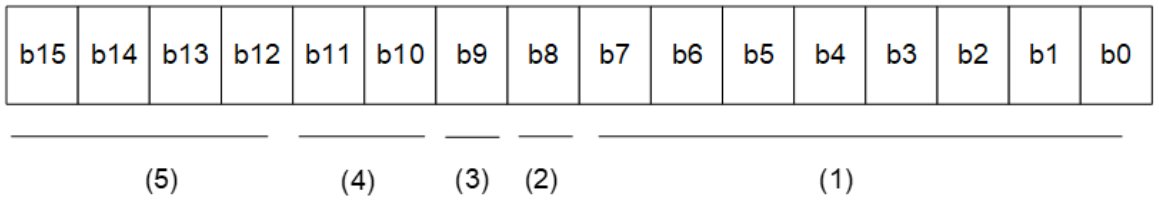
b0 – Recipe change status flag	When the control recipe group in the **control block** is changed, the recipe group change status flag is turned ON; when the HMI changes the recipe group and completes updating the RCPNO number, the recipe group change status flag is turned OFF. 
b1 – Recipe read status flag	When the HMI reads back one set of recipe data from the control block, the recipe read status flag is turned ON; when the HMI completes reading and saving the recipe from the control block, the recipe read status flag is turned OFF. 

Recipe Control Status Register	
b2 – Recipe write status flag	When the HMI sends one set of designated recipe data to the control block, the recipe write status flag is turned ON; when the HMI completes writing the recipe into the control block, the recipe write status flag is turned OFF. 

The following table lists the flags of recipe control status register with their description.

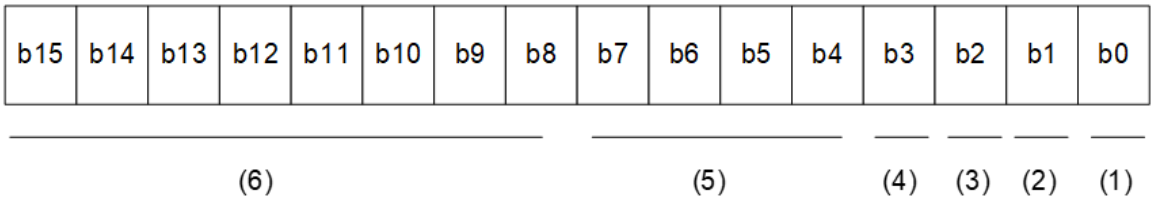
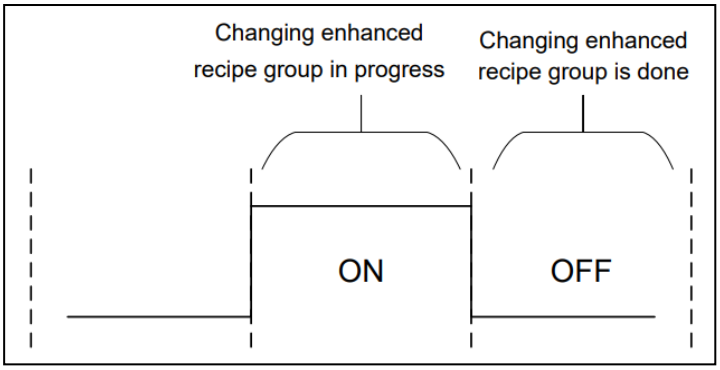
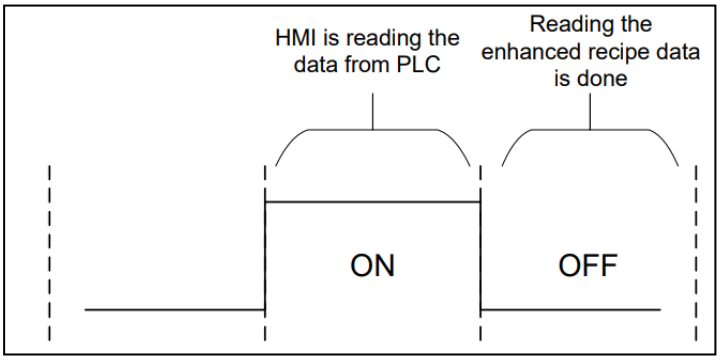
Recipe Control Status Register	
	
b0~b15 – Recipe number designation status	• Whether you change the value of the recipe designation register RCPNO by the control block or HMI, the recipe group control status register updates its value so that the control block finds the current recipe number from this register. • This recipe designation flag must work with the recipe change flag.

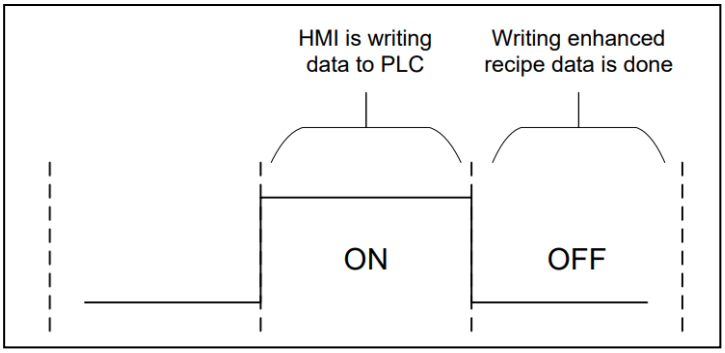
The following table lists the flags of system control status register with their description.

System Control Status Register
 (1) b0~b7 Multi-language status value (2) b8 Printer status flag (3) b9 Printer form feed status flag (4) b10~b11 Retrieve Ethernet address status (5) b12 Wi-Fi retrieve network address status (6) b13~b15 Reserved

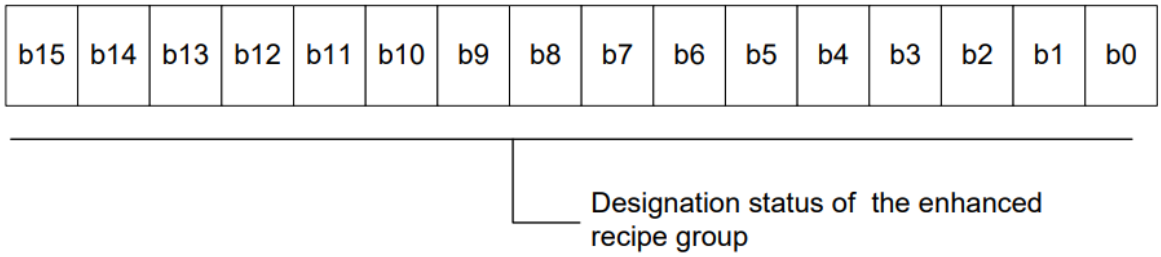
b0~b7 – Multi-language status value	Display the corresponding status value of the language in use.
b8 – Printer status flag	• Set to ON, the printer is printing the display screen or the edited screen of the HMI. • Set to OFF, the printer is idled.
b9 – Printer form feed status flag	• Set to ON, the printer is ejecting the paper and flipping the paper automatically. • Set to OFF, the printer is idled.
b10~b12 – Retrieve network address status value	Displays the status value of IP retrieve. When the Control Block is triggered to retrieve the network address, the status value is 1, and the status value is toggled to 0 after the address is obtained. • b10 is the LAN1 status. • b11 is the LAN2 status. • b12 is the Wi-Fi status.

The following table lists the flags of enhanced recipe control status register with their description.

Enhanced Recipe Control Status Register	
 (1) b0 Enhanced Recipe change status flag (2) b1 Enhanced passive read status flag (PLC → HMI) (3) b2 Enhanced Recipe write status flag (HMI → PLC) (4) b3 Enhanced Recipe group change status flag (5) b4~b7 Reserved (6) b8~b15 Designate the enhanced recipe group number to be changed.	
b0 – Enhanced Recipe change status flag	When the control-enhanced recipe group in the **control block** is changed, the recipe group change status flag is turned ON; when the HMI group is changed and the **RCPNO** number is updated, the recipe group change status flag is turned OFF. 
b1 – Enhanced Recipe read status flag	When the HMI reads one set of recipe data from the **control block**, the enhanced recipe read status flag is turned ON; when the reading and saving of the enhanced recipe via the **control block** is completed, the enhanced recipe read status flag is turned OFF. 

Enhanced Recipe Control Status Register	
b2 – Enhanced Recipe write status flag	When the HMI sends one set of designated enhanced recipe data to the **control block**, the recipe write status flag is turned ON; after the enhanced recipe is written into the **control block**, the enhanced recipe write status flag is turned OFF. 

The following table lists the flags of enhanced recipe control status register with their description.

Enhanced Recipe Control Status Register	
	
b0~b15 – Enhanced Recipe designation status	• Whether you change the value of the enhanced recipe group designation register ENRCPNO by the **control block** or HMI, the enhanced recipe group status register updates its value so that the **control block** finds the recipe group number from this register. • The enhanced recipe designation status flag must work with the enhanced recipe group change status flag.

The following table lists the advanced level access control with their description.

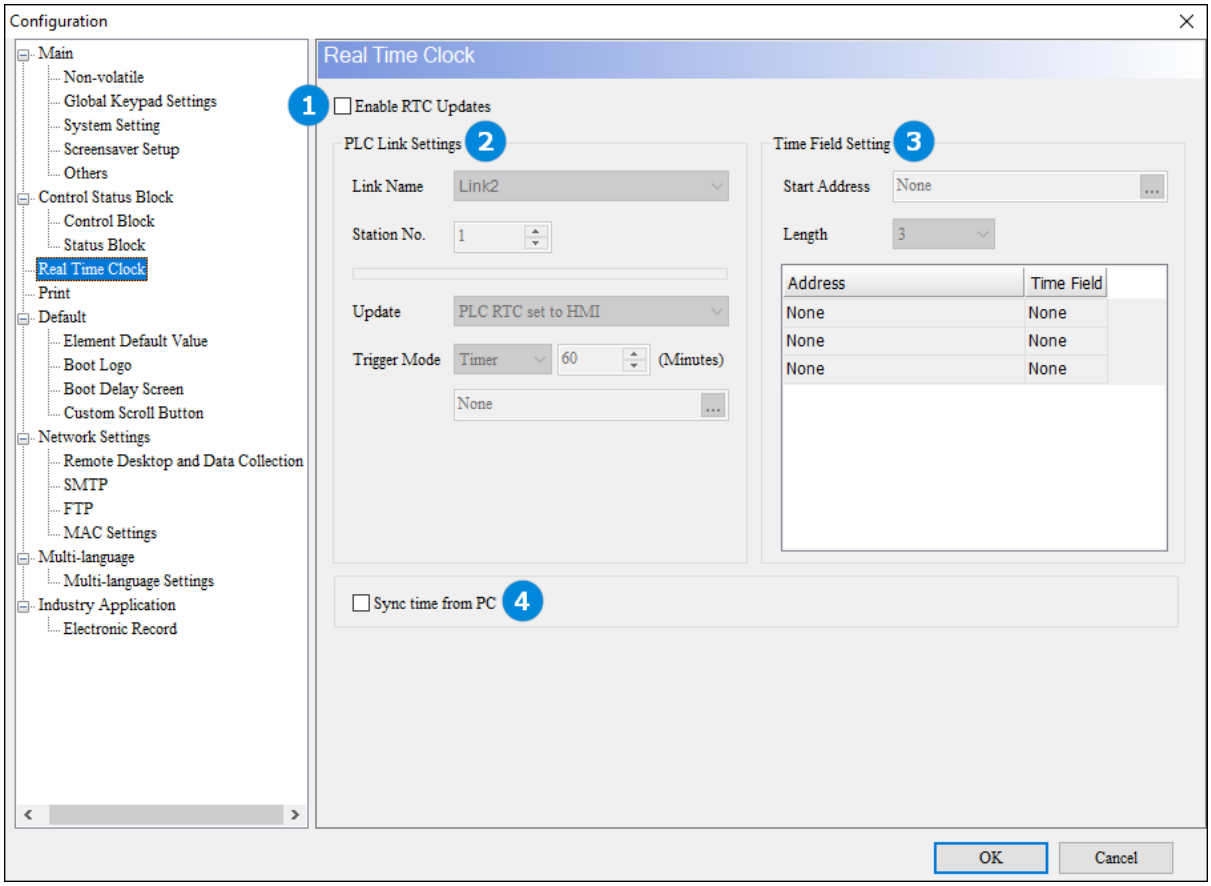
<table>
<tr><th colspan="2">Advanced Level Access Control</th></tr>
<tr><td>b0~b3–
Set User Level</td><td>Use Bit 0, Bit 1, Bit 2 and Bit 3 to read the current user authority level of the HMI.
The following table lists how to set security levels.
<table>
<tr><th rowspan="2">Security Level</th><th colspan="4">Flag Control</th></tr>
<tr><th>Bit 3</th><th>Bit 2</th><th>Bit 1</th><th>Bit 0</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>2</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>1</td><td>1</td></tr>
<tr><td>4</td><td>0</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>5</td><td>0</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>6</td><td>0</td><td>1</td><td>1</td><td>0</td></tr>
<tr><td>7</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>8</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>9</td><td>1</td><td>0</td><td>0</td><td>1</td></tr>
</table></td></tr>
<tr><td>b15
Set User Highest Level</td><td>Highest authority level access.</td></tr>
</table>

Note: If **Advanced Level Access** is enabled, user level of **General Control** becomes invalid.

Real Time Clock

Controller has a built-in real-time clock (RTC), allowing you to synchronize the HMI RTC time to the controller or the controller RTC time to the HMI. Some controllers do not have a built-in real-time clock, so this function cannot be used in operations related to time setting, for example, turn on and off the machine regularly every day, manage the access control, and so on.

Note: Some old Delta controller models (ES / SS / EC) do not support real time clock update.



1 Enable RTC Updates

Function	Description
Enable RTC Updates	Select to enable real-time clock update. **Note**: If the Delta controller is selected, the **Time Field Setting** area is disabled, and the address will be set to the special registers D1319 ~ D1313 of the real time clock of Delta controller.

2 PLC Link Settings

Function	Description
Link Name	Select the **link name, even** if the selected controller is COM1, COM2, COM3 or Ethernet.
Station No.	Set the **station number** of the controller for which you want to update the time. If the selected controller does not support station number setting, this setting is not available.
Update	Select the RTC update mode. The default is **PLC RTC time set to HMI**.

Function	Description
Trigger Mode	Select the RTC trigger mode. The default is **Timer**. • If **Timer** is selected, you can set how often to perform automatic update. The range is 1~1440 minutes, and the default value is **60** minutes. • If **Bit On** is selected, when the Bit is ON, it triggers to update the settings. • **If Bit Off** is selected, when the Bit is OFF, it triggers to update the settings. **Note**: If **Bit On** or **Bit Off** is selected, the trigger address must be set. The trigger address can be the internal memory or the register address of the controller.

③ Time Field Setting

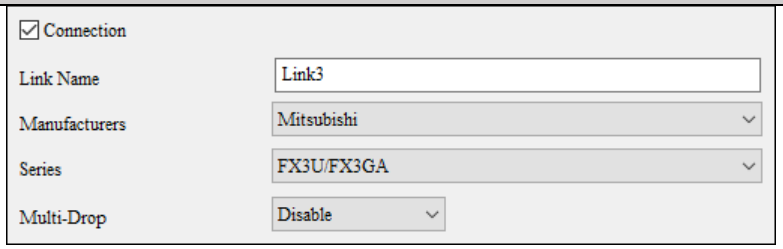
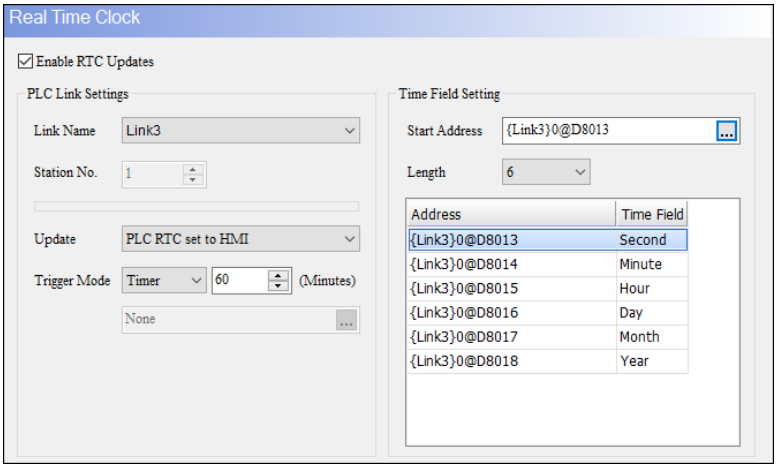
Function	Description
Start Address	Set the register address of the controller for RTC time synchronization. **Note**: If Delta controller is selected, setting the **Start Address** is not required.
Length	Select the **length** according to the number of time field to synchronize. The value range is 1–7, and the default value is **3**. **Note**: If Delta controller is selected, setting **Length** is not required.

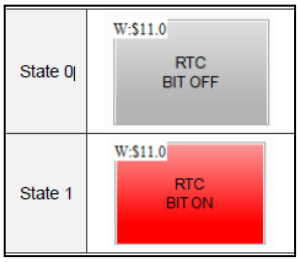
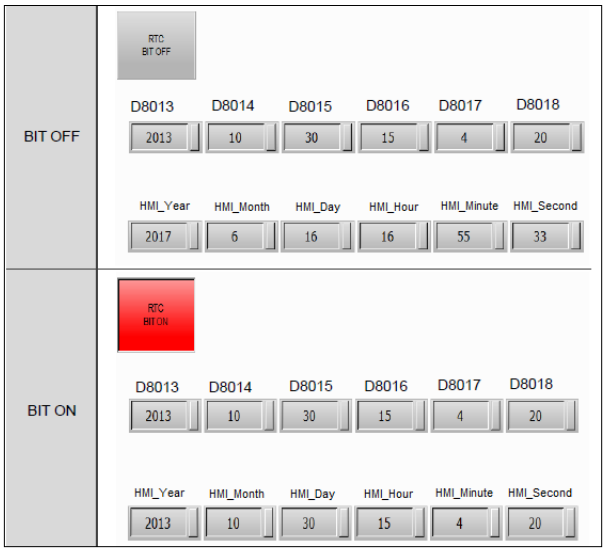
④ Sync time from PC

Function	Description
Sync time from PC	If selected, after downloading the project, the HMI time synchronizes with the PC.

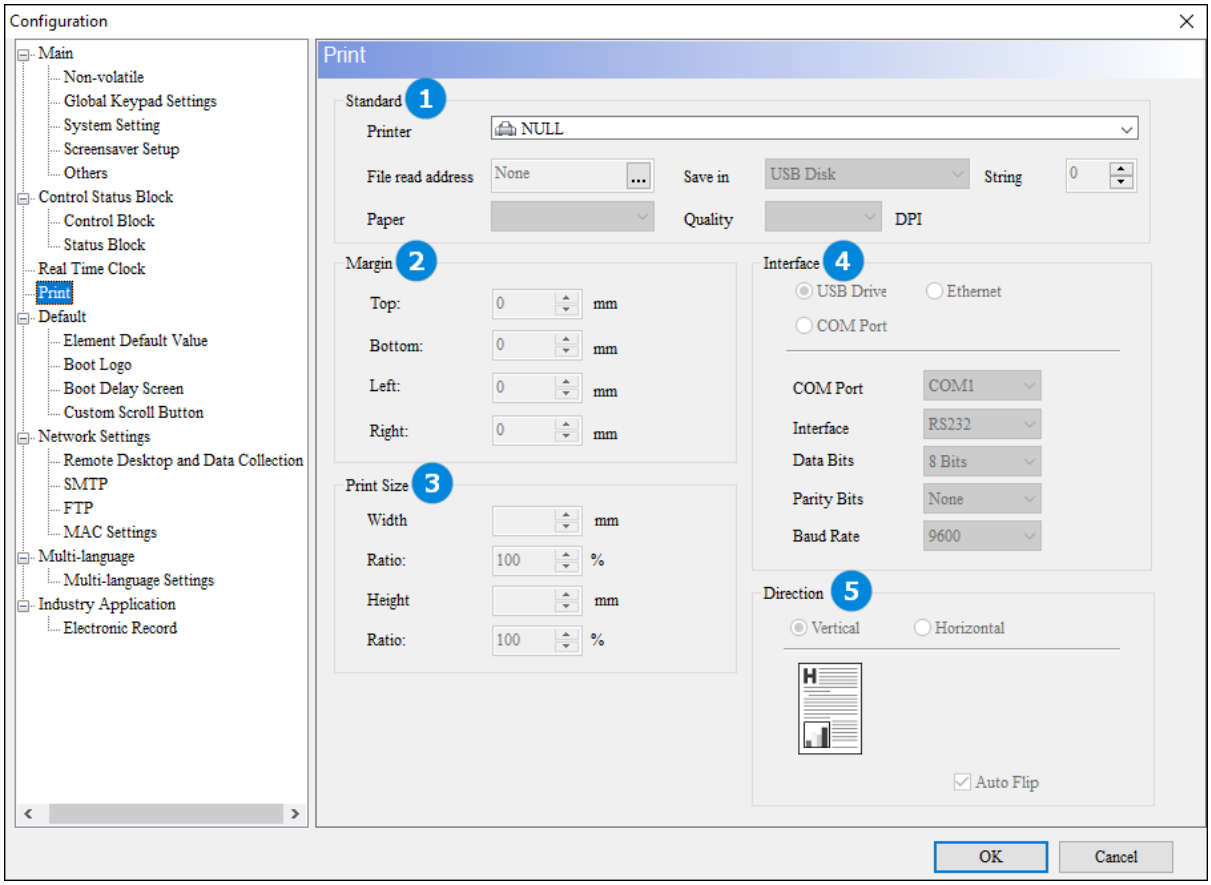
The following table lists the setting steps with examples.

Setting Step	Example
Real Time Clock Update	1. On the toolbar, go to **General** > **Device Communication**, and then configure the communication setting in the **Device Communication** dialog.

Setting Step	Example
	 2. On the toolbar, go to **General** > **Configuration**. 3. In the **Real Time Clock** setting page, select **Enable RTC Updates** checkbox. 4. In the **PLC Link Settings** area, select **Link3** (Mitsubishi – FX3U / FX3GA) for **Link Name**. 5. Select **PLC RTC time set to HMI** for **Update**. 6. Select **Bit On** for **Trigger Mode**. Click ... to set the trigger address to **$11.0**. 7. In the **Time Field Setting** area, set **{Link3}1@D8013** for the start address and **6 for Length**. 8. In the address list, select **Second** for the Time Field corresponding to {Link3}1@D8013. Select **Minute** for the Time Field corresponding to {Link3}1@D8014. Select **Hour** for the Time Field corresponding to {Link3}1@D8015. Select **Day** for the Time Field corresponding to {Link3}1@D8016. Select **Month** for the Time Field corresponding to {Link3}1@D8017. Select **Year** for the Time Field corresponding to {Link3}1@D8018.  9. Click **OK**.

Setting Step	Example
Create button to trigger the real time clock	1. Create a **Maintained Button** element and set its **Write Address** to **$11.0**. 2. Edit the **State 0** text as **RTC BIT OFF**, **State 1** text as **RTC BIT ON.** 3. Set the **Foreground Color** as red, representing the state is triggered ON. 
Create Numeric Entry element	1. Create 6 **Numeric Entry** elements and set their addresses as **{Link3}1@D8013 ~ {Link3}1@D8018** in order. 2. Create 6 more **Numeric Entry** elements, and set their **Internal parameter** as **TIME_YEAR**, **TIME_MONTH**, **TIME_DAY**, **TIME_HOUR**, **TIME_MINUTE**, **TIME_SECOND** in order.
Result	After completing the creation of all elements, compile and download to the HMI. When the RTC **Maintained Button** (BITON) is triggered, the system will set the RTC time of the controller to the HMI. The HMI internal parameter TIME_YEAR, TIME_MONTH, TIME_DAY, TIME_HOUR, TIME_MINUTE, TIME_SECOND will be synchronized with the RTC of the controller. 

Print



1 Standard

Function	Description
Printer	DOP-100 series models support Honeywell, HP, EPSON, ZEBRA, GoDEX, Micro Printer, PDF Writer, and ePrinter printers.
Paper	The paper size varies depending on the selected printer. The available options include A4, LETTER, CUSTOM, and so on.
Quality	Set the resolution of the printer. 72 and 203 DPI can be selected according to the selected printers.

2 Margin

Function	Description
Top / Bottom / Left / Right	Set the margins of the paper as the reserved area during printing, which means that it is displayed as blank and not printed. The value range is 0–550, and the default value is **0**.

❸ Print Size

Function	Description
Width / Height	Set the width and height. This setting is available when **Paper** is set as CUSTOM.
Ratio	Set the ratio of the width and height. The value range is 10–400, and the default value is **100**.

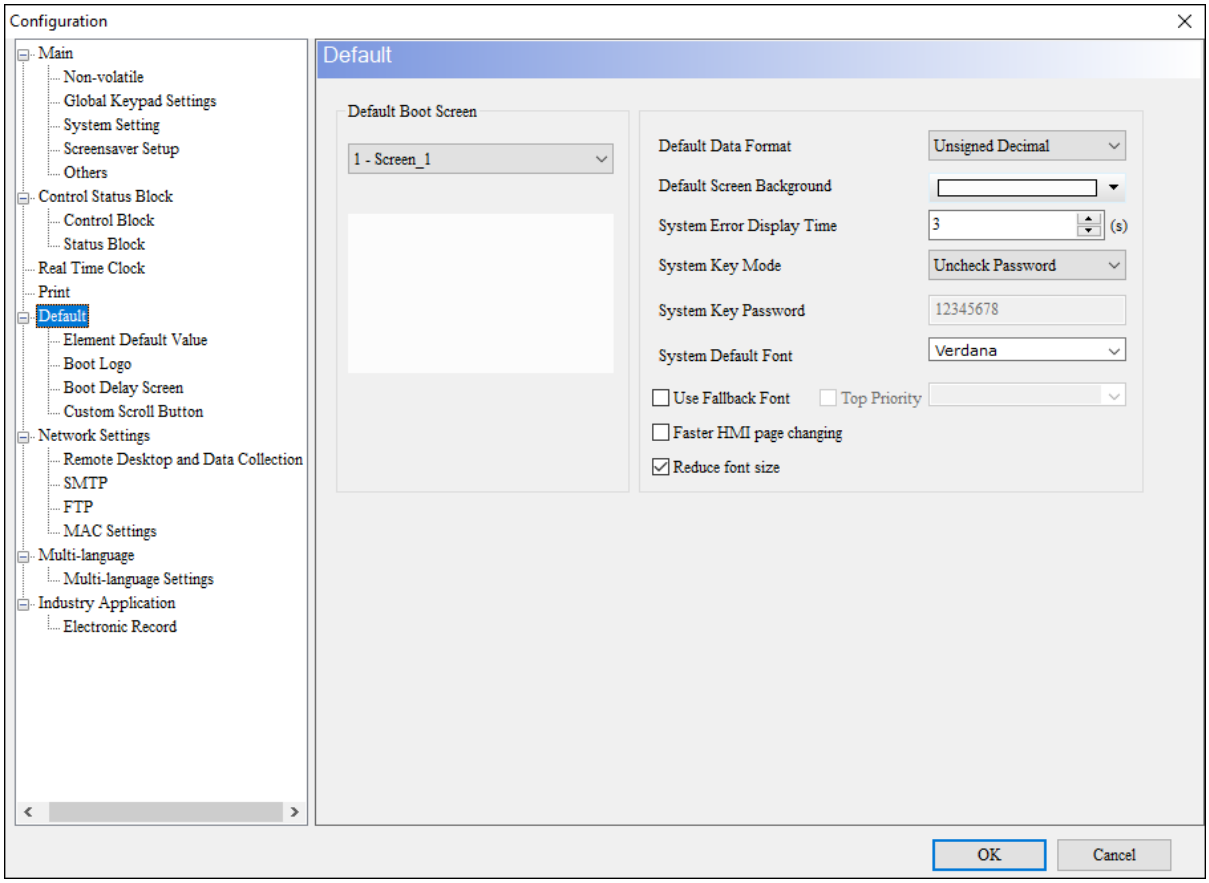
❹ Interface

Function	Description
Interface	USB, Ethernet, and COM Port can be selected according to the selected printer. If you choose ePrinter to print, you need to fill in the IP address and printer port (the default value is **85**) of the PC you are using.

❺ Direction

Function	Description
Direction	Set the printing **direction** as vertical or horizontal.
Auto Flip	The printer provides an automatic paper ejection function, helping you to flip the paper to print. • If selected, the printer will automatically flip the paper and continue printing after the paper printing is completed. • If not selected, the printer will forcibly eject the paper after the paper printing is completed, and you need to flip the paper manually.

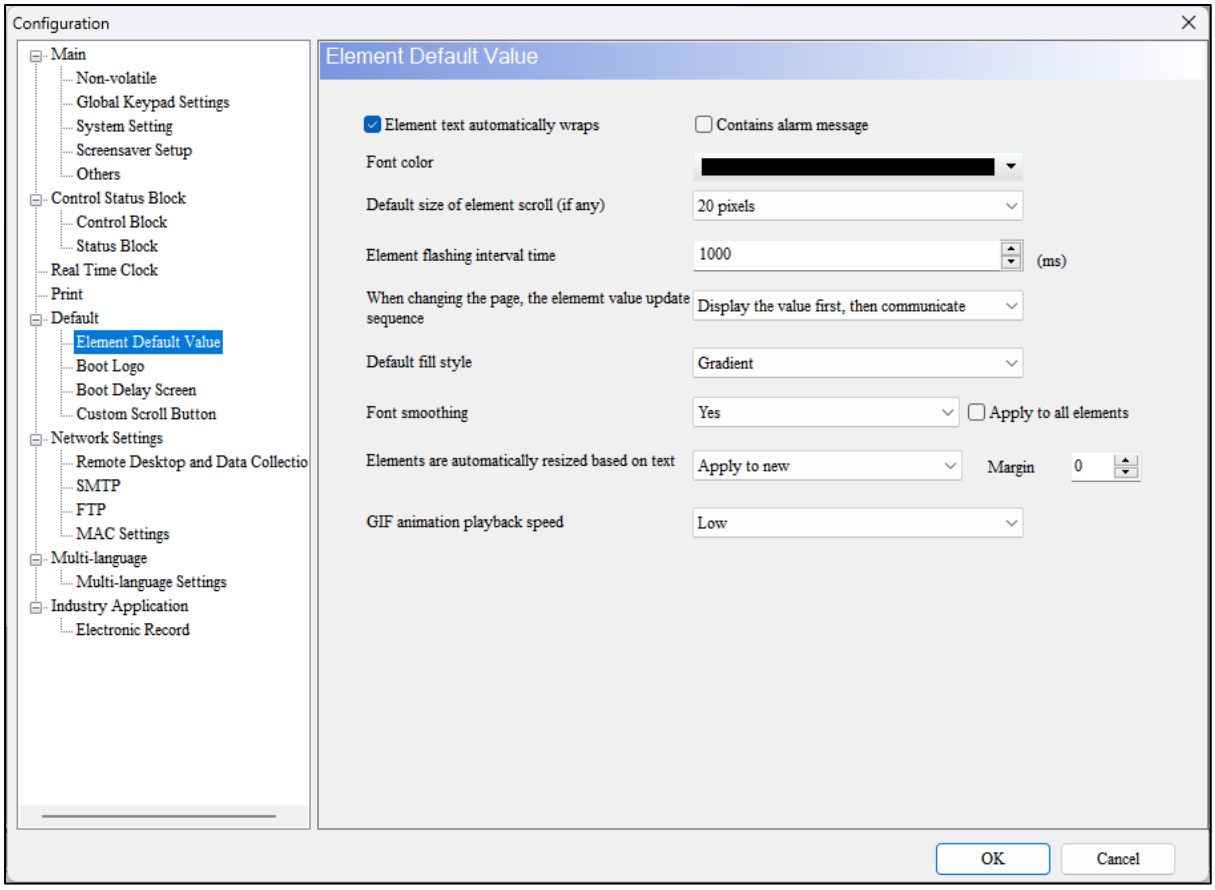
Default



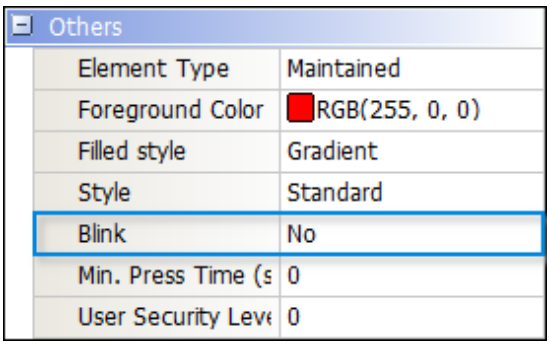
Function	Description
Default Boot Screen	Set the initial screen when the HMI is turned on. You can choose another screen to set as the initial screen. The default is **1-Screen_1**.
Default Data Format	Set the **default data format** when creating an element. The default is **Unsigned Decimal**.
Default Screen Background	Set the default background color when editing the screen on the HMI. The default is .
System Error Display Time	Set the time the message stays when an error message appears. The value range is 0–5, and the default value is **3**. **Note**: When it is set to 0, if there is any error in the system, no message appears on the HMI.

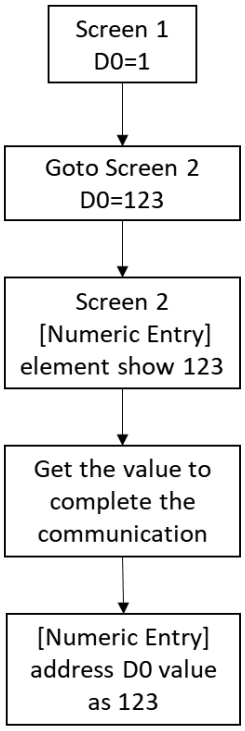
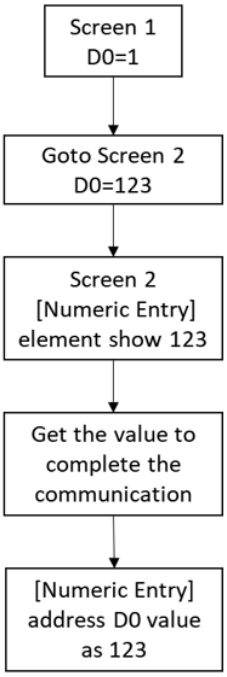
<table>
<tr><th>Function</th><th>Description</th></tr>
<tr><td>System Key Mode</td><td>Set the response the HMI should have when pressing the system key. The available options include Disabled, Enable the password check, and Uncheck Password. The default is Uncheck Password.
<table>
<tr><td>Disabled</td><td>If selected, pressing the system key cannot enter the system screen of the HMI.</td></tr>
<tr><td>Enable password check</td><td>If selected, after pressing the system key, the HMI asks you to enter the system key password.</td></tr>
<tr><td>Uncheck Password</td><td>If selected, after pressing the system key, you can enter the system screen of the HMI without entering the system key password.</td></tr>
</table></td></tr>
<tr><td>System Key Password</td><td>Set the system key password. This is the password that the HMI asks you to enter when the System Key Mode is set to Enable Password Check. The default value is 12345678.</td></tr>
<tr><td>System Default Font</td><td>Set the system default font. The default is Verdana.</td></tr>
<tr><td>Use Fallback Font</td><td>Currently, the DOP-100 series is to download the existing fonts to display on the HMI. When the operating system lacks fonts set in the elements, it makes the HMI unable to display the text normally after downloading, so the firmware loads a set of default fonts.
If Use Fallback Font is selected, when the firmware default fonts fail, the software enables the backup font.
If Top Priority is selected, the priority of the backup fonts of the software will be higher than the default fonts of the firmware.</td></tr>
<tr><td>Faster HMI page changing</td><td>If selected, when there are a lot of graphics and static text on the screen, the display speed can be speeded up when switching pages.</td></tr>
<tr><td>Reduce font size</td><td>If selected, when the project is downloaded to the HMI, only the fonts used in the project will be downloaded.</td></tr>
</table>

Element Default Value

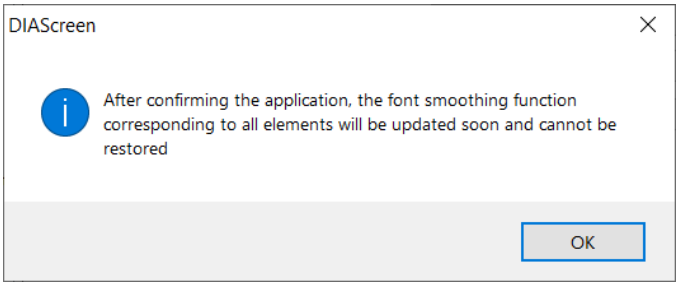


Function	Description
Element text automatically wraps	• If selected, the element text automatically wraps as long as it encounters the boundaries. • If not selected, when the text encounters the boundaries, the text does not wrap, but it gets extended.
Font color	Set the **font color**. The default is .
Default size of element scroll (if any)	Set the default size of the element scroll. The value range is 20–60 pixels, and the default value is **20 pixels**. **Note**: This setting is applied to the elements with scrolls, such as historical data and alarm tables.

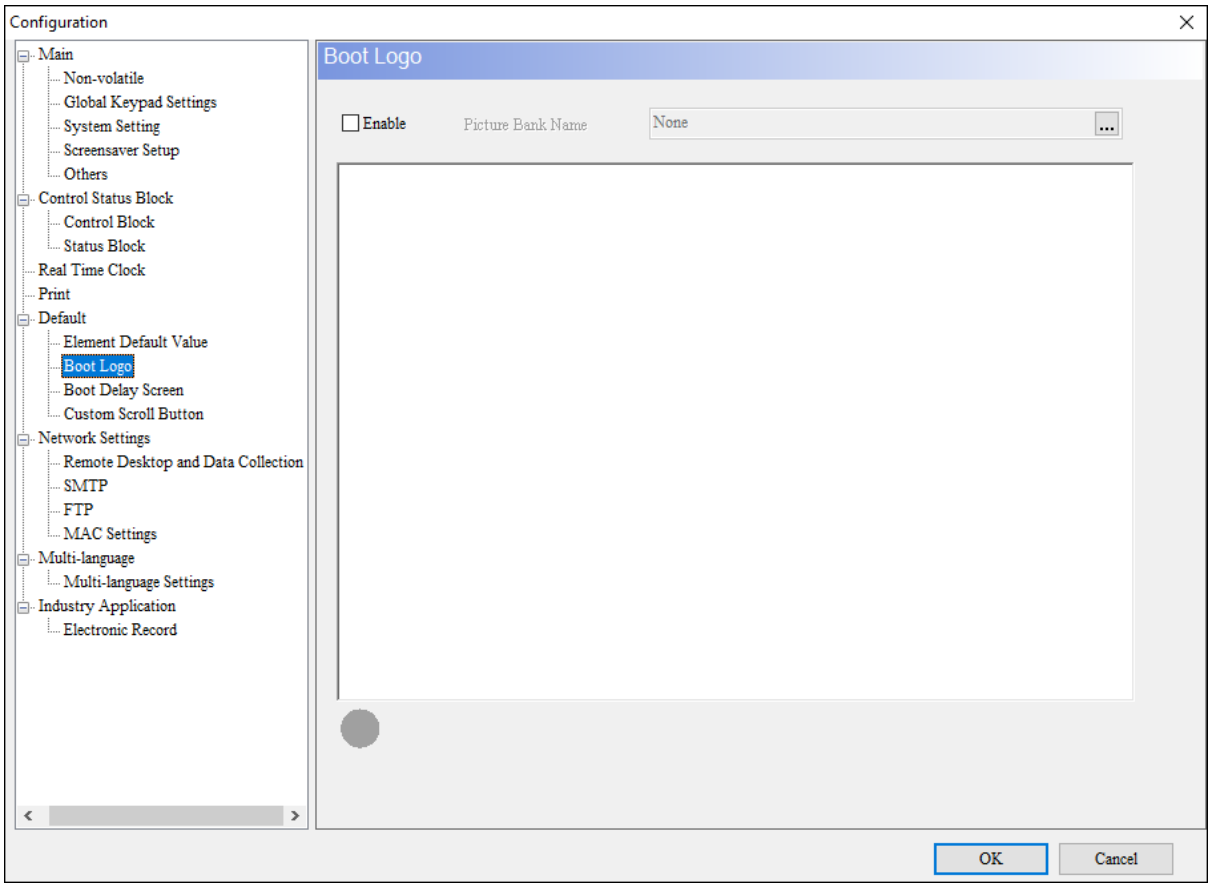
Function	Description
Element flashing interval time	Set the interval time for element flashing. The range value is 500–5000, and the default value is **1000**. **Note**: This setting is available only when the **Blink** property of the element is set to **Yes**. 

When changing the page, the element value update sequence	Set the update order of the element value when the HMI changes the page. The default is **Communicate first, then display the value**. • **Display the value first, then communicate**: The value is read normally after switching the screen and is not affected by the communication.  • **Communication first, then display the value**: The value display of the **Numeric Entry** elements pauses due to the switching of the screen. 

<table>
<tr><th>Function</th><th>Description</th></tr>
<tr><td></td><td>The following table lists the elements that currently support this feature.
<table>
<tr><td rowspan="5">Button element</td><td>Set to ON</td></tr>
<tr><td>Set to Off</td></tr>
<tr><td>Momentary</td></tr>
<tr><td>Maintained</td></tr>
<tr><td>Set value</td></tr>
<tr><td>Meter element</td><td>Meter(1) ~ (4)</td></tr>
<tr><td>Pipe Diagram element</td><td>Pipe Diagram (1), Pipe Diagram (2)</td></tr>
<tr><td rowspan="3">Indicator element</td><td>Multistate Indicator</td></tr>
<tr><td>Range Indicator</td></tr>
<tr><td>Simple Indicator</td></tr>
<tr><td rowspan="4">Data Display element</td><td>Numerical Display</td></tr>
<tr><td>Character Display</td></tr>
<tr><td>General Message display</td></tr>
<tr><td>Moving Sign</td></tr>
<tr><td rowspan="2">Graphic Display element</td><td>State Graphic</td></tr>
<tr><td>Animated Graphic</td></tr>
<tr><td rowspan="3">Input element</td><td>Numeric Entry</td></tr>
<tr><td>Character Entry</td></tr>
<tr><td>Barcode Input</td></tr>
<tr><td>Curve element</td><td>Curve Input</td></tr>
<tr><td>Analog element</td><td>Slider</td></tr>
</table></td></tr>
<tr><td>Default fill style</td><td>Set the default filled style of the HMI. The default is Gradient.
The settings of this page are global. After Filled Style is set, the filled style of the elements created later will be based on the setting here.</td></tr>
</table>

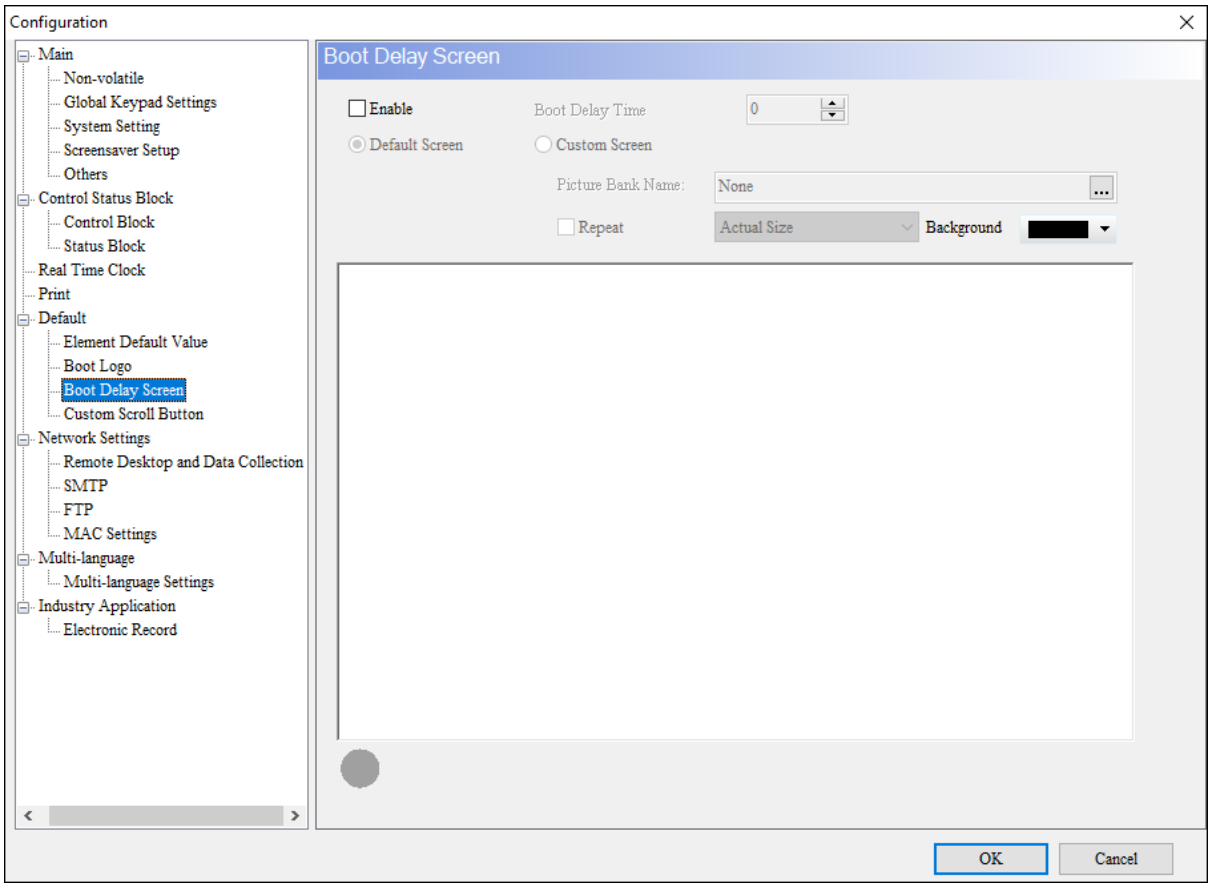
Function	Description
Font smoothing	Set whether the text font of the HMI is smooth. If set to **Yes,** the text is rendered anti-aliased. If **Apply to all elements is** selected, it will be applied to all elements and pops up a message. Otherwise, it will be applied to the **Static Text** elements. 
Elements are automatically resized based on text	• **Don't Apply**: After setting it, the **automatically resized based on text** will be changed to **No** for all elements in the project. • **Apply to all**: the **automatically resized based on text** will be changed to **Yes** for all elements in the project The element is automatically resized according to the **Margin** to show the complete text. • **Apply to new**: The added elements are automatically resized.

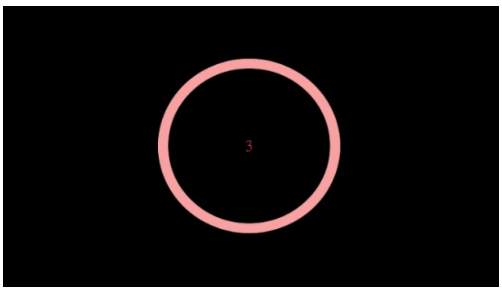
Boot Logo



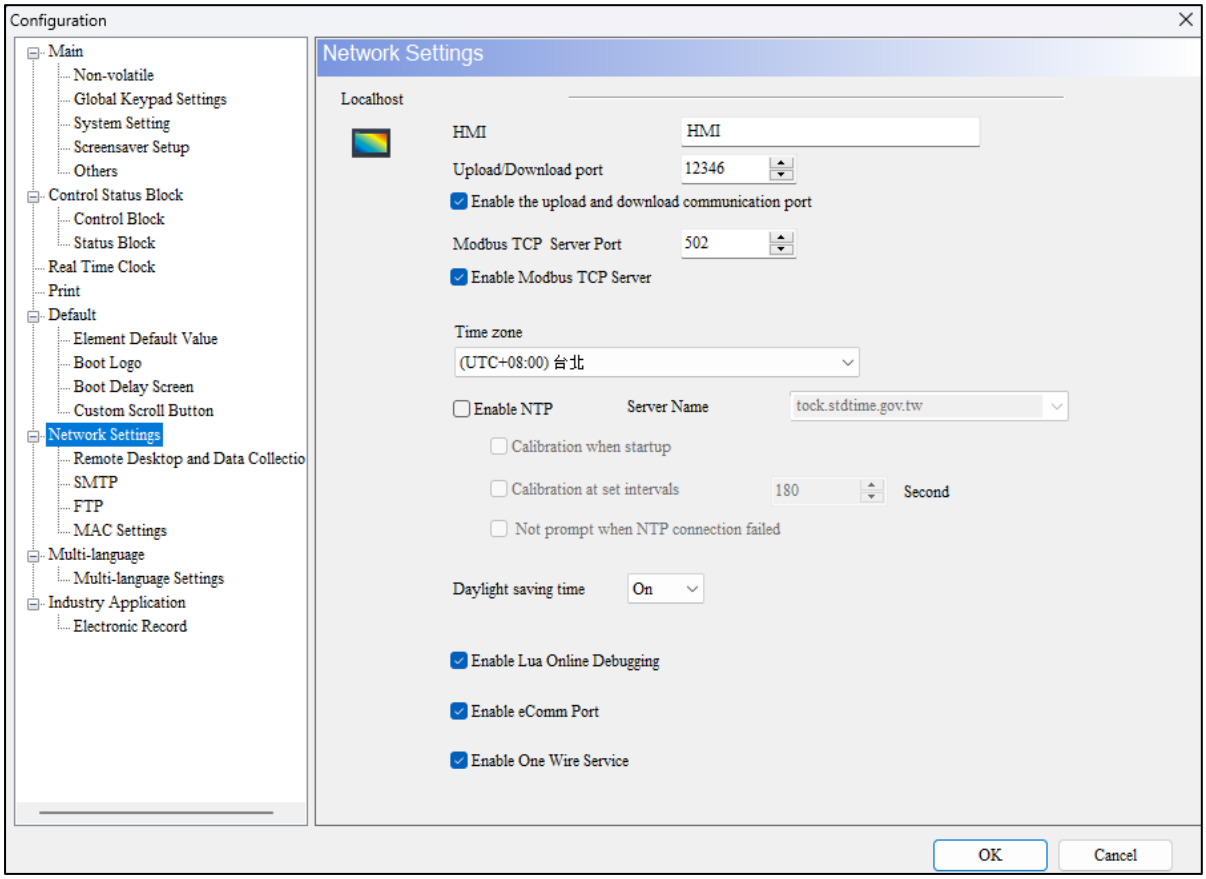
Function	Description
Enable	If selected, you can choose the boot screen from the picture bank. You can also: • Go to **Project** > **Download the Boot Screen** to replace the HMI boot screen. • Go to **Project** > **Download Screen** to download the boot screen. When a GIF file is selected, the gray circle at the bottom left corner displays as , you can preview the GIF action on the software. **Note**: • To use files not in the picture bank, import the picture to the picture bank first. • After downloading the boot screen, power on the HMI again. • Supported image file formats are BMP, JPG, GIF, ICO, and PNG. • Only the boot screen after conversion by the software less than 3 Mbytes can be imported.

Boot Delay Screen



Function	Description
Enable	If selected, the **boot delay time** can be set.
Boot Delay Time	Set the **boot delay time** to wait for the controller to start. The value range is 0–255.
Default screen	If the **Boot Delay Time** is set to 3 seconds, and the system default boot delay screen is used, then after downloading to the HMI, the HMI screen appears after counting down from 3 seconds to 0 seconds. 
Custom Screen	Define the boot delay screen, and set the graphic display mode, background color, and whether to repeat.

Network Settings



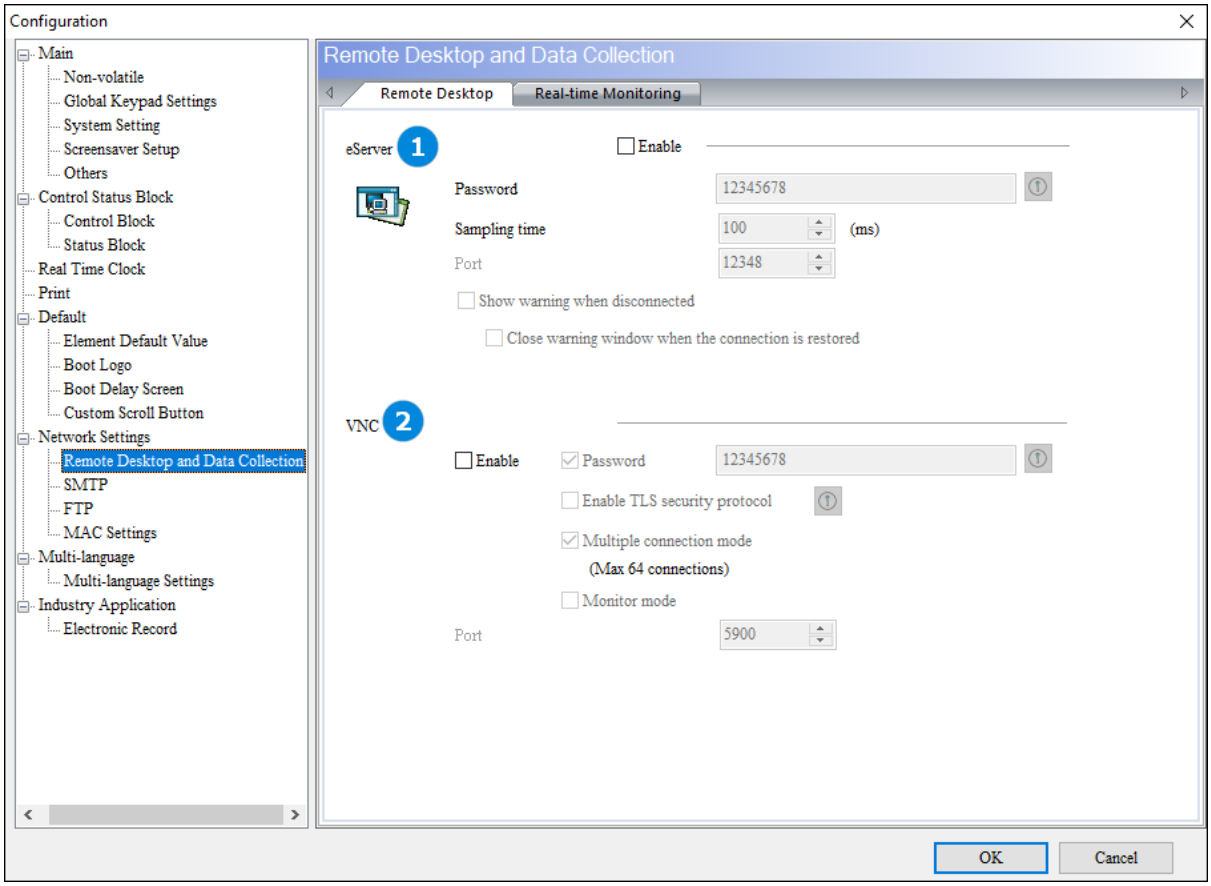
Function	Description
HMI	Set the name of the HMI. When using network remote monitoring or data sampling, use the HMI name to quickly identify which HMI is being monitored or is accessing data.
Upload/Download port	Set the communication port. The default value is **12346 and is in the ON state**. If selected enable the upload and download communication port, you can't use ethernet to upload and download screen. **Note**: There is a total of 65536 communication ports. Some specific communication ports are only used by specific programs.
MODBUS TCP Server Port	Set the communication port number. The default value is **502**. **Note**: • This communication port must be the same as that of the MODBUS TCP/IP controller. • If the HMI communicates with the MODBUS application on the PC side, change the communication port here.
Time	Select the time of the HMI to display according to the local time zone.

Function	Description
Enable NTP	If selected, the time of HMI can be calibrated through the network. **Note**: You must first check whether the network of HMI is smooth.
Server Name	Select the server provided in the software or enter the local NTP server name.
Calibration when startup	If selected, the time of HMI is calibrated when the HMI is turned on.
Calibration at set intervals	If selected, set the number of seconds. The calibration timing is when the HMI is turned on and enters the screen. The value range is 10–99999, and the default value is **180**.
Daylight saving time	Choose whether to enable daylight-saving time. The default is **off**.
Enable Lua Online Debugging	If selected, you can set breakpoints in the **Program** page to perform the **Online debug** function.
Enable eComm Port	If selected, you can enable the HMI forward function to forward data to the terminal device through the HMI, and then establish the forward tunnel through eComm of the COMMGR software.
Enable One Wire Service	If selected, you can open the QIP function and scan the devices connected to the HMI communication port through the COMMGR software.

Remote Desktop and Data Collection

The following table lists the settings in the **Remote Desktop** and **Real-time Monitoring** tabs with their description.

Remote Desktop



1 eServer

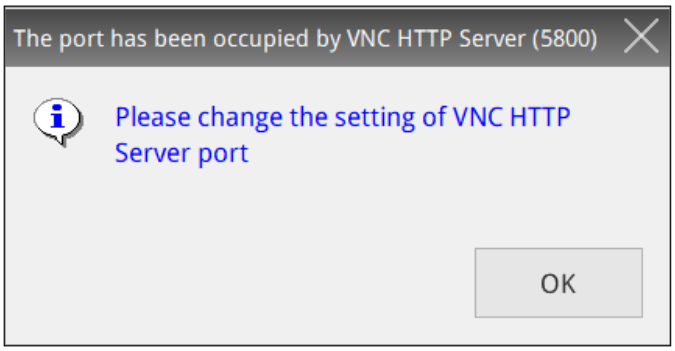
Function	Description
Enable	Select to enable the eServer function.
Password	Set the password to monitor or access data according to your needs. • To comply with CyberSecurity specifications, the password must be set up to 8 characters and contain an uppercase letter, a lowercase letter, a number, and a symbol (!, $, #, %). • This is the password that must be entered to obtain the HMI project data and execute eServer and eRemote.
Sampling time	Set how often eServer and eRemote take to perform sampling. The range value is 100–5000, and the default value is **100**.
Port	Set the connection ports of eServer and eRemote. The default value is **12348**. **Note**: The upload / download communication ports of the HMI are different from this communication port. The communication ports are different according to different programs.

Function	Description
Show warning when disconnected	This option is available only when **Show warning when disconnected** checkbox is selected. When the HMI is disconnected from the eServer or eRemote, the HMI displays a disconnection warning message.
Close warning window when the connection is restored	This option can only be enabled when **Show warning when disconnected** is checked. • If selected, an error message keeps popping up on the HMI until the connection between the HMI and the eServer or eRemote is restored. • If not selected, an error message only pops up once on the HMI.

② VNC

Virtual Network Computing (VNC) function is an application that can remotely monitor and operate the HMI. This application can transmit keyboard and mouse actions and real-time screens through the network. When operating VNC using web pages, the browser must support JAVA installation, otherwise it cannot be opened.

Function	Description
Enable	If selected, you can remotely monitor to operate the HMI through VNC.
Password	Set the connection password according to your needs. To comply with CyberSecurity specifications, the password must be set up to 8 characters and contain an uppercase letter, a lowercase letter, a number, and a symbol (!, $, #, %).
Multiple connection mode	Select to allow multiple HMIs to be connected. Up to 64 HMIs can be connected.
Monitor mode	If selected, you can only monitor the HMI when VNC is connecting remotely.
Port	Set the connection port. The default value is **5900**. VNC Viewer provides web page operation mode. In this mode, you only need to enter the IP Address of the HMI into the browser and set the port as 5800 (even if the communication port is not set as default value **5900**) to connect, such as http://192.168.123.148:5800. **Note**: • The connection port must be changed to 5902 when using VNC Viewer to connect.

Function	Description
	• Do not use 5800 when setting the connection port on the software side. If it is set to 5800, after downloading the screen to the HMI, it asks you to change the port. 
Network Use	Select the port to open the VNC function. It applies to the HMI models with two Ethernet ports.

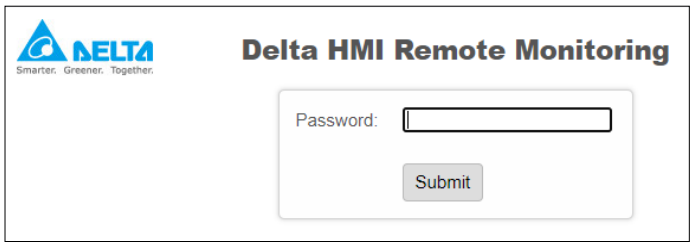
Real-time Monitoring

The real-time monitoring function allows you to write the value to the HMI on the web page or monitor the change of the value by the web page after the HMI writes the value. The data formats supported by the monitoring address include BCD, Signed Decimal, Unsigned Decimal, Hexadecimal, Floating, and Char. You can set the reading length for each data format to decide whether to read Word or Double Word. When the number of reading is 1, the integer number of digits can be set up to 5, which means that the reading is Word; when the number of readings is 2, the integer number of digits can be set up to 10, which means that the reading is Double Word. Address input provides Word and Bit, which can support internal memory address and external controller address.

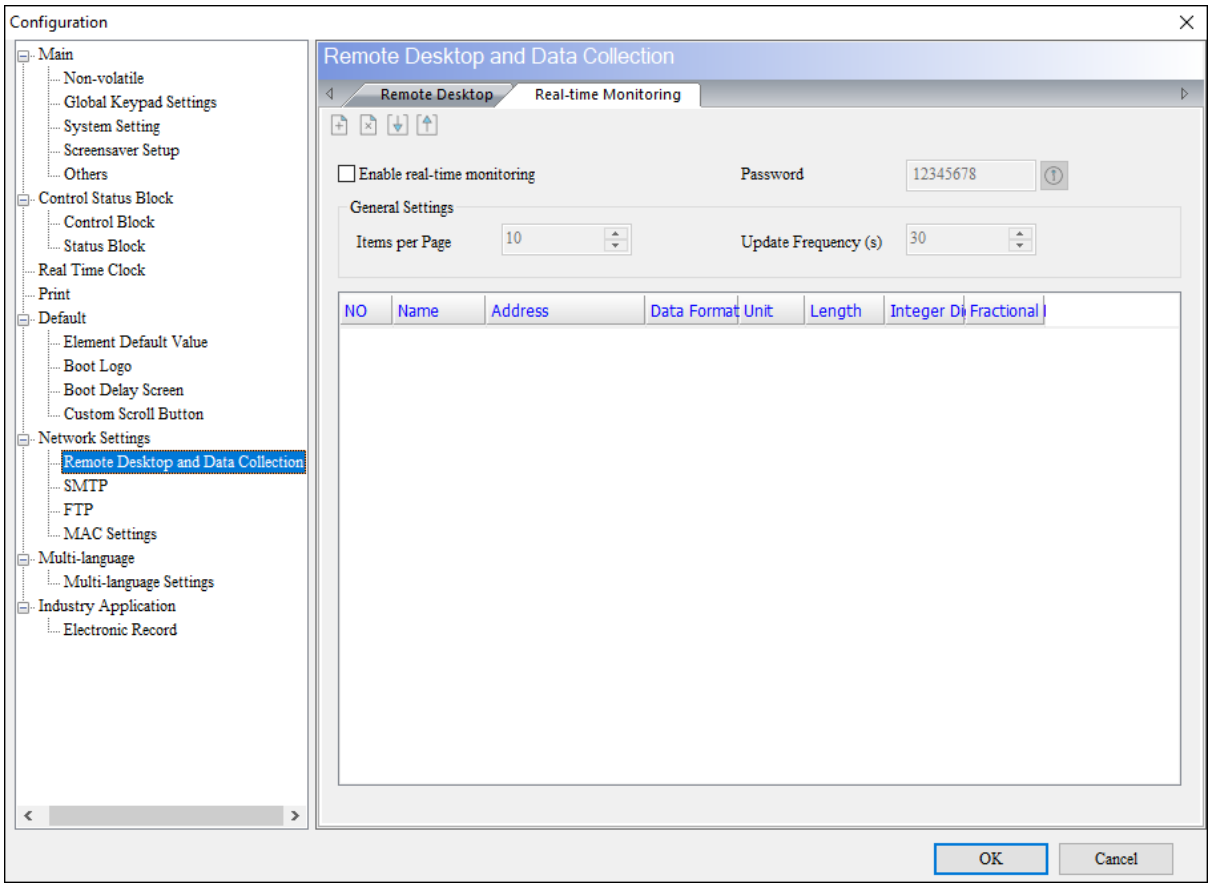
How to use network real-time monitoring?

1. In the **Real-time Monitoring** tab, select **Enable real-time monitoring** checkbox and set the address.
2. Enter **http: //[HMI IP]/RemoteMon/** on the browser.

   Note: The two characters R and M must be capital letters.
3. After the connection is successful, the login screen appears. Enter the password of the web application to log in.



The following table lists the settings in the **Real-time Monitoring** tab with their description.



Function	Description
Enable real-time monitoring	If selected, the real-time monitoring settings are available.
	Click to add monitoring address. You can enter the name and address. The name can be entered up to 30 characters.
	Select the items you want to delete, and then click this icon to delete the monitoring address.
	Click to import a CSV file.
	Click to export the monitoring address content to a CSV file.

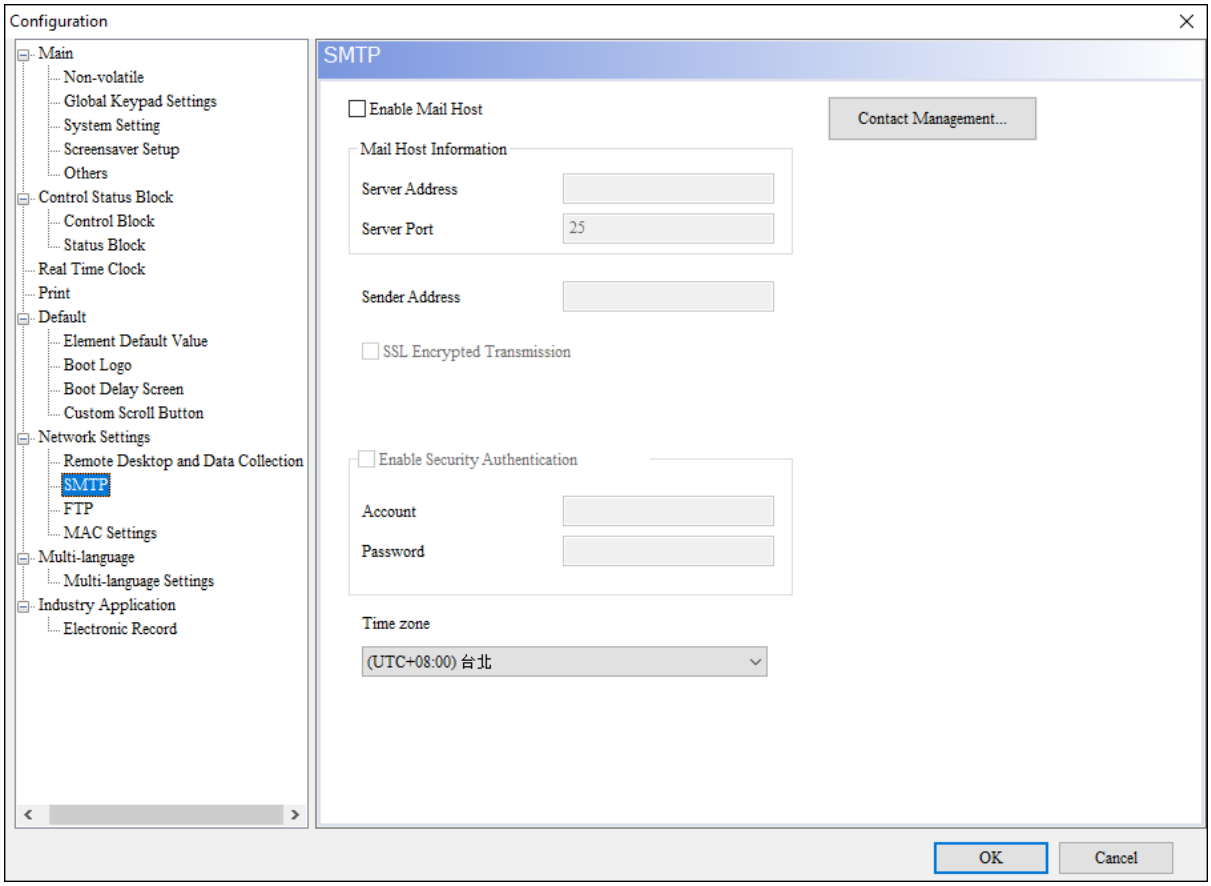
Function	Description
Password	Set the password that must be entered when entering the connection address using the web page. To comply with CyberSecurity specifications, the password must be set up to 8 characters and contain an uppercase letter, a lowercase letter, a number, and a symbol (!, $, #, %).
Items per Page	Set the number of monitoring addresses to be displayed on one page. The value range is 1–20, and the default value is **10**.
Update	Set how often the screen updates after updating the value. The value range is 1–30, and the default value is **30**.

SMTP

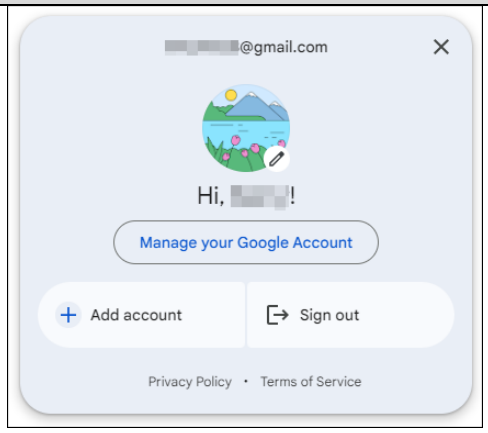
SMTP (Simple Mail Transfer Protocol) is used to transmit email and define the rules for sending email from a source address to a destination address to control how email is relayed. This function allows you to obtain an alarm message when an alarm occurs by receiving email.

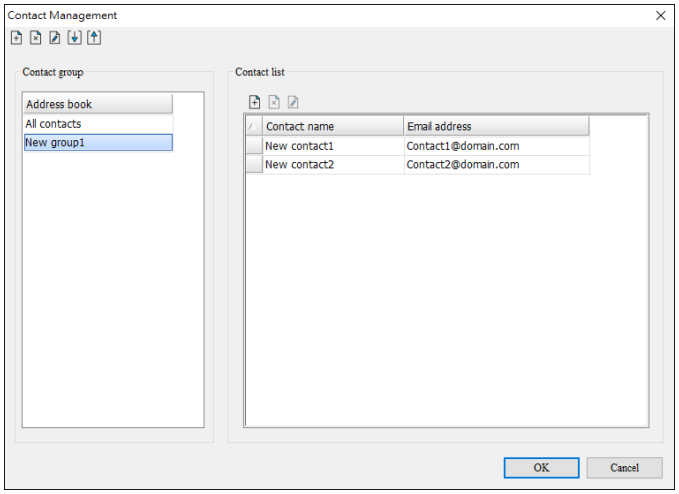
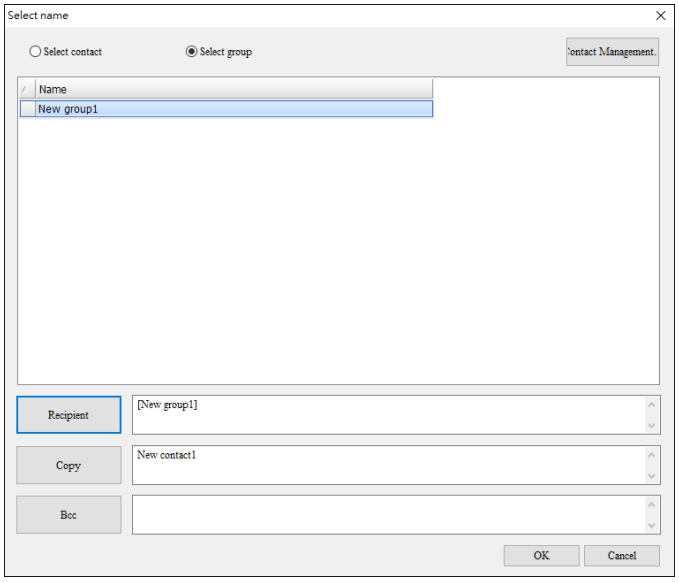
Note: Before using the SMTP function, set up a Mail Server environment or search for a free Mail Server on the Internet.

The following table lists the settings in the **SMTP** setting page with their description.

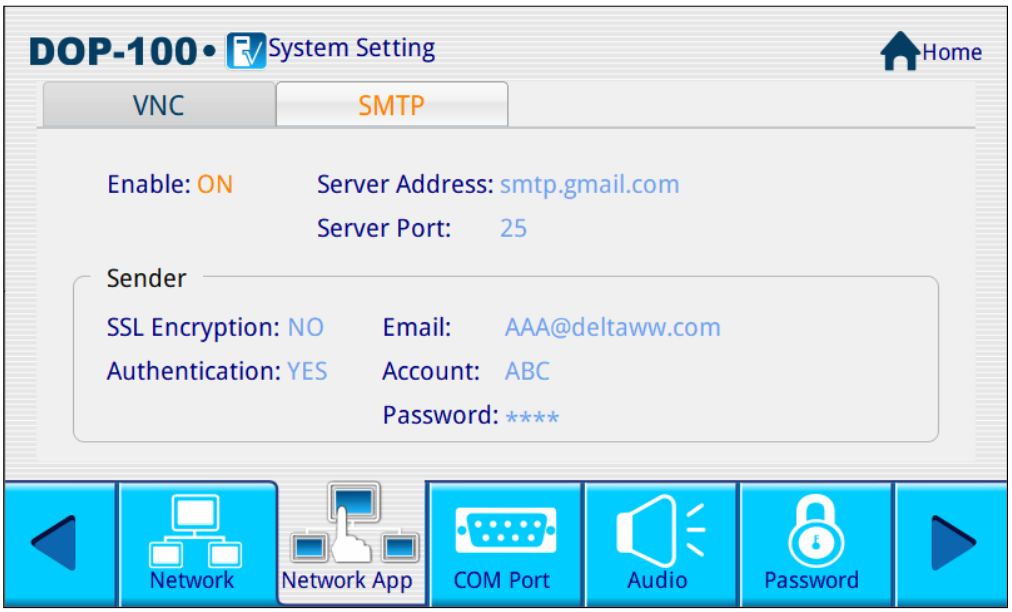


Function	Description
Enable Mail Host	If selected, the mail host settings are available.
Server Address	Set the server address. This IP address is the Mail Server IP created by users. **Note**: Before using the SMTP function, set up a Mail Server environment or search for a free Mail Server on the Internet.
Server Port	Set the server port. This is the general SMTP communication port. The default value is **25**
Sender Address	Set the sender's email address.
SSL Encrypted Transmission	SSL (Secure Socket Layer) is an encryption-based Internet security protocol, which ensures the confidentiality and integrity of communication between two applications, as well as verifying the identity of the server. To use Gmail to send email, do the following: 1. After logging into your Gmail account, click **Manage your Google Account**.

Function	Description
	 2. **On the left navigation panel,** select **Security**. 3. Enable **2-Step Verification** function under **How you sign in to Google**. For the detailed operation method, please see Sign in with app passwords. After the above setting is completed, you can use Gmail to receive alarm emails. **Note**: The e-mail must support SSL function.
Enable Security Authentication	If selected, the account and password can be set. If the SMTP server is set up with an account and password to verify the identity, this option must be selected.
Account	If the security authentication checkbox that requires an account and password is selected when setting up the SMTP Mail Server, first enter the account and password. This account and password are used to check whether the recipient is a legitimate user of the back-end mail system to prevent uncollected emails from occupying the system resources to cause hidden mail security problems. **Note**: • The account and password must be set according to the requirements of the SMTP server. • The account format varies depending on the format of each SMTP Mail Server. You can look up the related rules of MIS.
Password	
Time	Select the local time zone so the time to send the alarm email will be more accurate.
Contact Management	Set the contact address for the mail information of the alarm. Follow the steps to set the contact address and its mail information. 1. In the **Contact list** area, click to add new contact. 2. On the toolbar, click to set the group name in the **Group settings** dialog.

Function	Description
	3. In **All contacts** area, choose the contact and click << to add it to the **Group contact**. You can also click >> to remove the contact from the **Group contact**. 4. In the **Contact Management** dialog, choose the group name in the **Address book**, the contact in the group lists in the **Contact list**.  5. Set the mail recipient in the **Mail** field of the **Alarm Settings** page. The contacts or groups can be selected as **Recipient**. 

After the SMTP settings are completed and downloaded to the HMI, go to **System Menu** > **System Setting** > **Network App** > **SMTP** to access the settings. You can also change the SMTP settings on the HMI and then upload the settings to DIAScreen.



FTP

FTP (File transfer protocol) is a standard communication protocol that is used for downloading alarms, historical data, recipes, and operation logs in a USB or SD card through the network to the PC for viewing. You can also upload files from your PC to a USB or SD card.

The following table lists the rules of FTP with their description.

FTP Rule	Description	
Supported HMI	Network-based HMI	
Network-based HMI	File transfer application	
	Windows File Explorer	
	DOS Command Line	
Connection Limit	Up to 3 FTP Clients can be connected at the same time	
	Automatically disconnect when idle for more than 90 seconds	
Login method	Anonymous login	Adding of directories is not allowed.
		Uploading of files is not allowed.
		Downloading of files is not allowed.
		Deletion of files is not allowed.
		Change of filename is allowed.

FTP Rule	Description	
	Account login	Adding of directories is allowed.
		Uploading of files is allowed.
		Downloading of files is allowed.
		Deletion of files is allowed.
		Change of filename is allowed.
File transfer rules	Unlimited traffic	
	Support resume	
	Unlimited size of the file to be sent.	
	The maximum length of file name is 260 bytes.	
	Change of filename is allowed.	
	Support Chinese filename.	
	Encryption is not supported.	
	Support active / passive mode connection.	
	When FTP is transferring, you can enter and exit the system menu.	

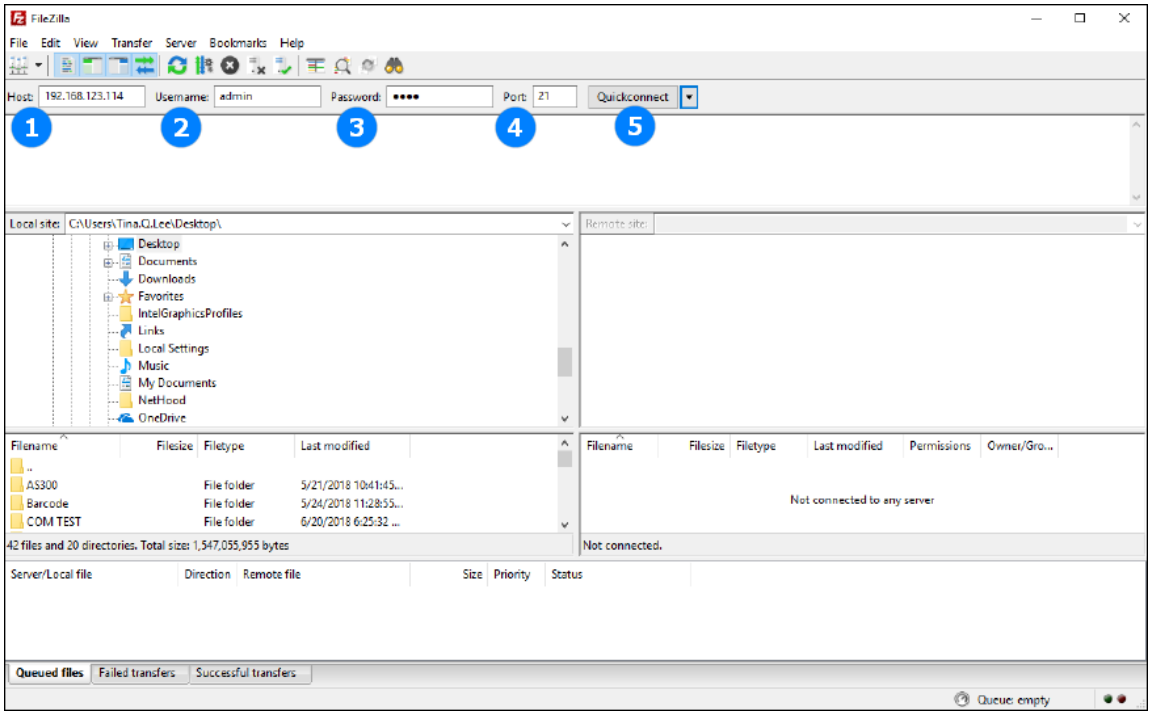
The FTP Server supports connection through file transfer applications, Windows File Explorer, and DOS Command Line.

- **File transfer applications**

  Upload / download files through file transfer applications.

  Note: The FTP Server provided by HMI needs to be used with FTP Client application.

  The following figure is an example using the file transfer application FileZilla, which is free and can be downloaded from filezilla-project.org/download.php.



The following table lists the functions in the FileZilla application with their description.

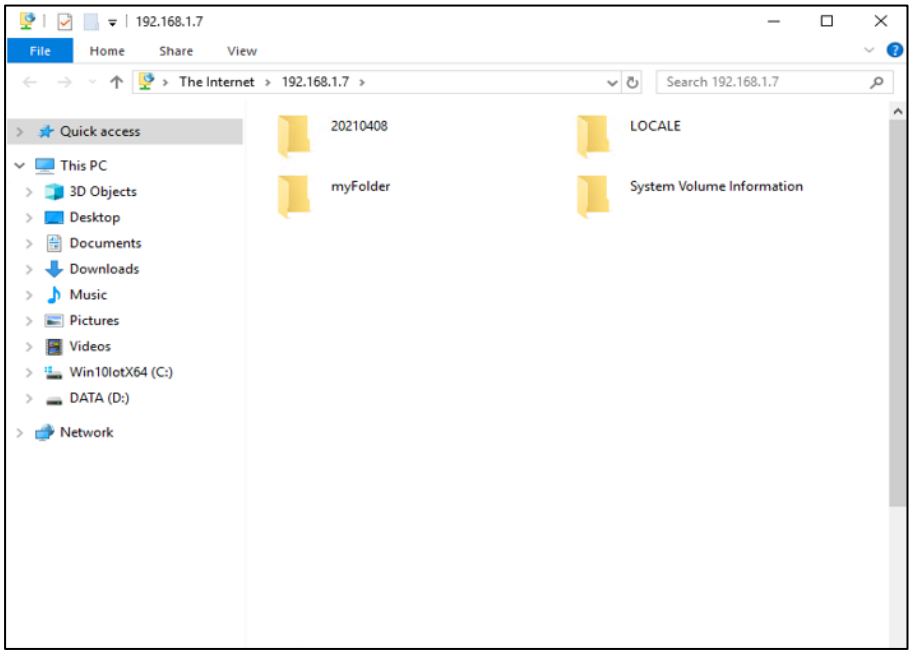
Legend	Function	Description
1	Host	Set the HMI IP address, the example is 192.168.123.114.
2	Username	Set the username as admin.
3	Password	Set the password as 1234.
4	Port	Set the port as 21.
5	Quick connect	Click to quickly connect to the FTP Server. **Note**: Before executing this function, complete the above settings.

- **Windows File Explorer**

  Upload / download files by connecting to the FTP Server through Windows File Explorer.

  1. Open the file manager and enter **ftp://192.168.1.7/** in the input box for the file path.
  2. Enter the account and password to log in to FTP.

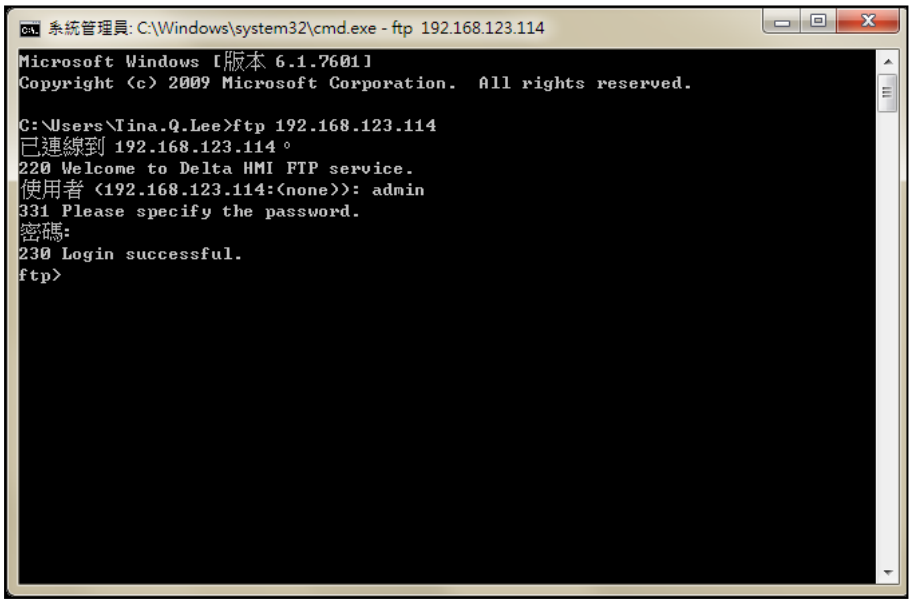
     After logging in, all files in the USB will be displayed.



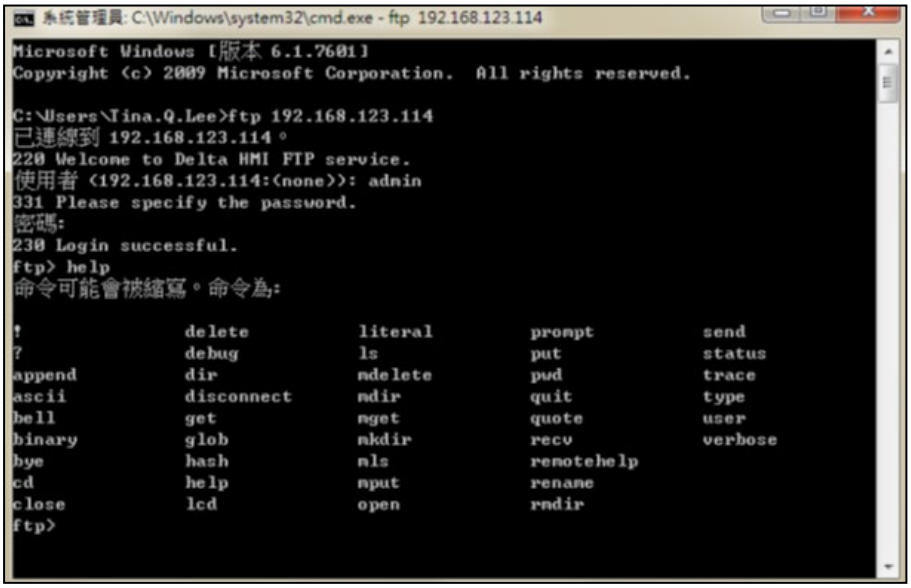
- **DOS Command Line**

  Upload / download the files by connecting to the FTP Server through DOS Command Line.

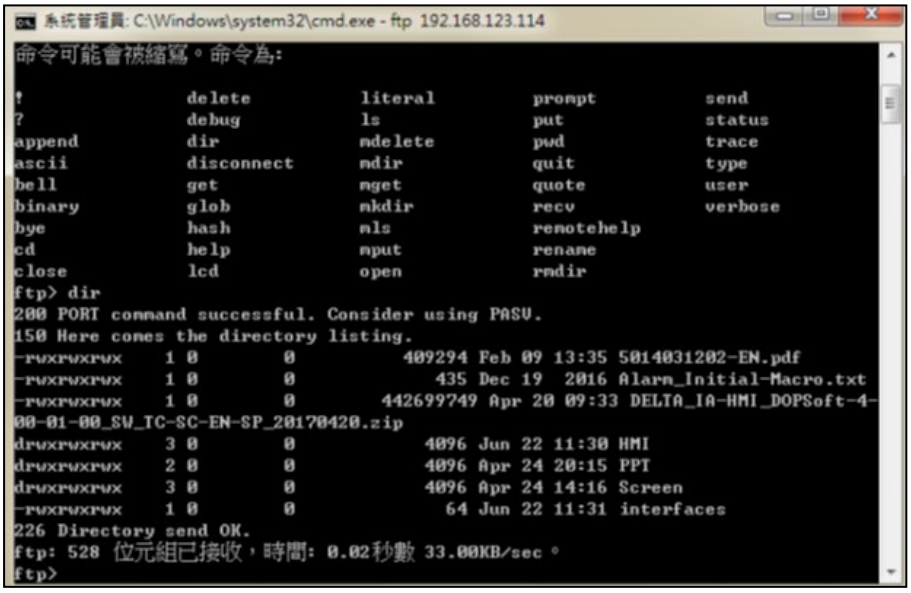
  1. Open the command prompt and enter **ftp 192.168.123.114**.
  2. Enter users account **admin** and password **1234**.



  According to the FTP command, use help to view the supported related commands.



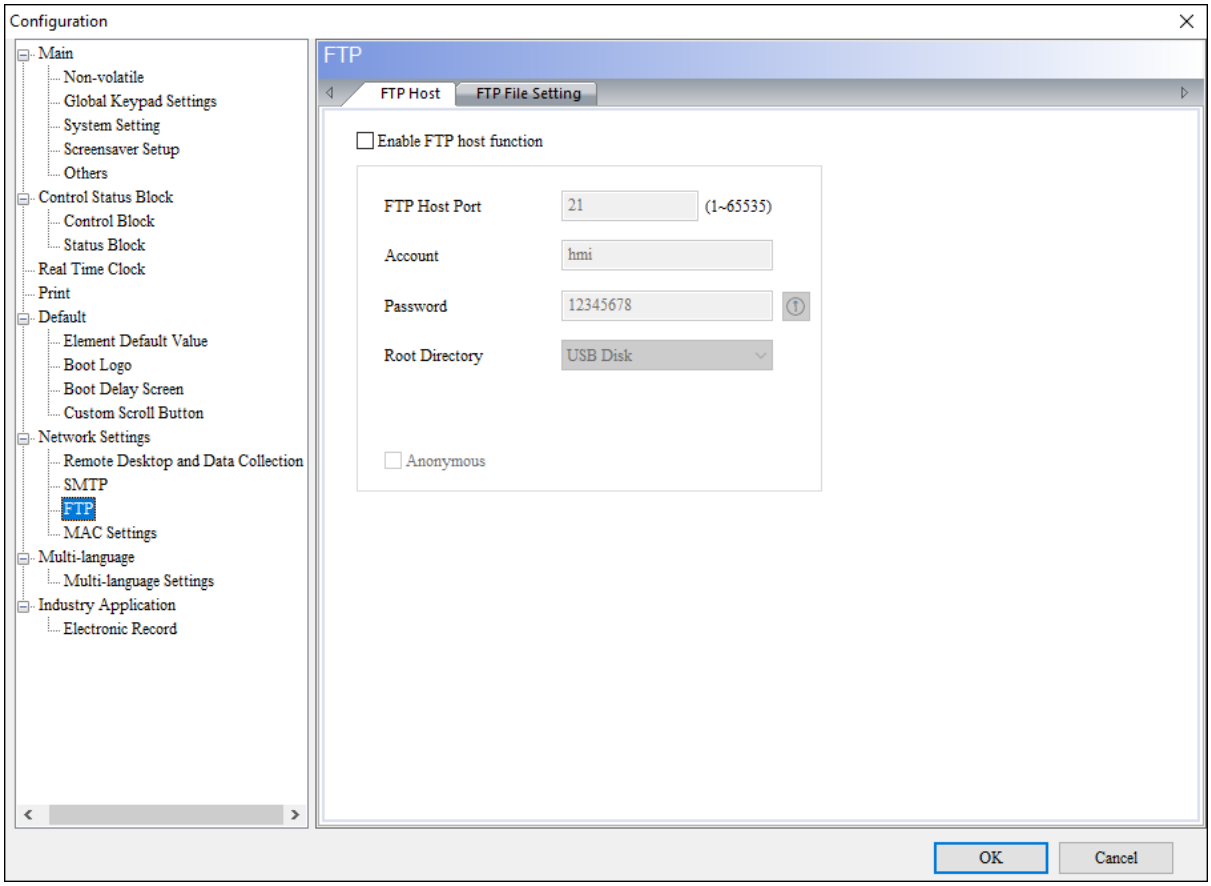
Enter the dir command to list all files currently in the USB.



To download files from a USB or SD Card, use the get command. If you want to upload files from PC to USB or SD Card, use the put command.

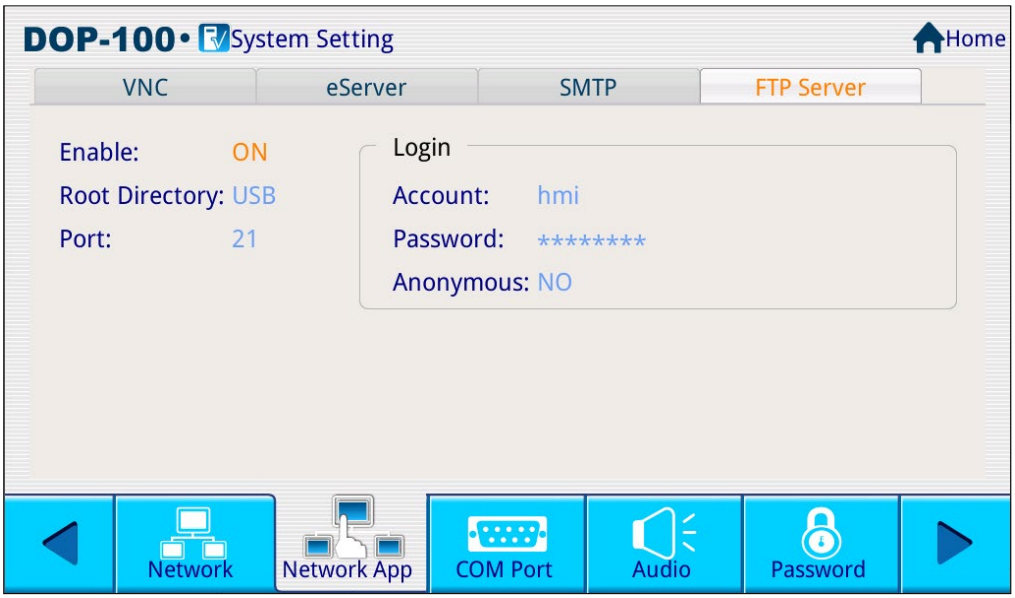
The following table lists the functions of the **FTP Host** tab with their description.

FTP Host



Function	Description
Enable FTP host function	If selected, the FTP settings are available.
FTP Host Port	Set the FTP host port. The default value is **21**.
Account	Set the account name.
Password	Set the password. To comply with CyberSecurity specifications, the password must be set up to 8 characters and contain an uppercase letter, a lowercase letter, a number, and a symbol (!, $, #, %).
Root Directory	Select the storage location of HMI files. The available options include **USB Disk**, USB Disk 2, SD, and HMI. The default is **USB Disk**.
Anonymous	If selected, you can log in to FTP without entering an account and password. **Note**: After using an anonymous connection to log into FTP, you cannot upload / download files, delete files or add new directories.

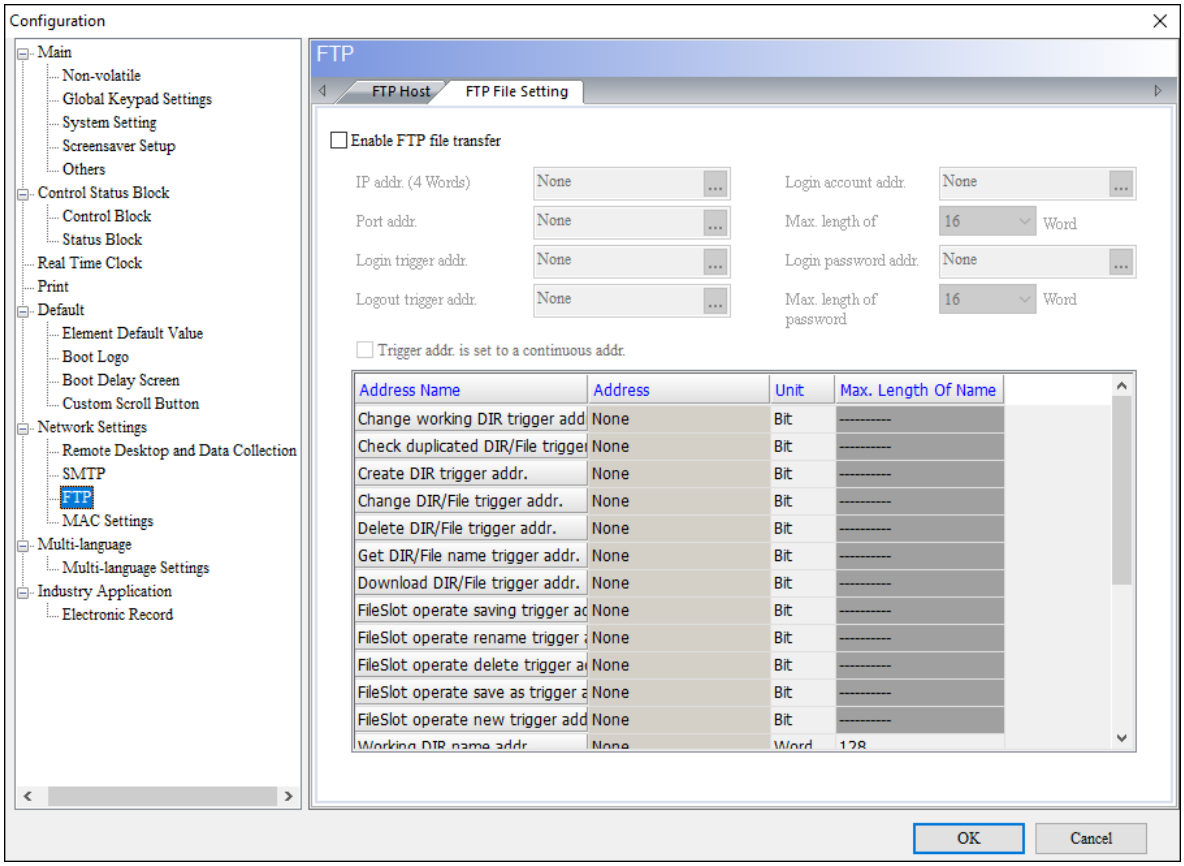
After the FTP settings are completed and loaded into the HMI, go to **System Menu** > **System Setting** > **Network App** > **FTP Server** to access the settings. You can also change the FTP settings on the HMI and then upload the settings to DIAScreen.



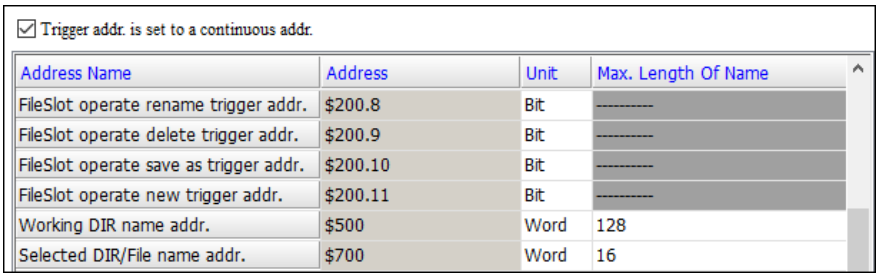
FTP File Setting

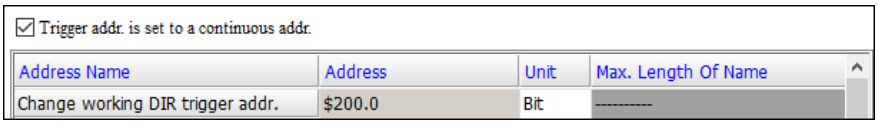
FTP file transfer provides functions such as account login / logout, file upload / download, file deletion, file management / rename, directory (folder) management, etc. File transfer can be carried out through DOP series HMI and FTP Server.

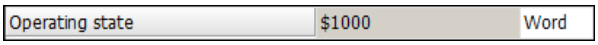
Note: FTP file transfer should be used with the **FTP File List** element.



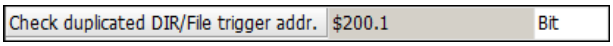
Function	Description	
Enable FTP file transfer function	Check to enable FTP Client function.	
Host Login	IP addr.	Set the FTP Server IP address to which to connect.
	Port addr.	Set the FTP Server connection port to which to connect.
	Login trigger addr.	The input value type is Bit. When this bit is turned On, it will be logged in.
	Logout trigger addr.	The input value type is Bit. When this bit is turned On, the system will log out.
	Login account addr.	Set the starting address of the account. The maximum length supports 16 words.
	Max. length of account	
	Login password addr.	Set the starting address of the password. The maximum length supports 16 words.
	Max. length of password	
Change working DIR / File	Change working DIR trigger addr.	When this bit is On, the working directory is changed and automatically cleared after completion.

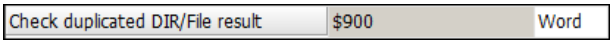
<table>
<tr><th>Function</th><th colspan="2">Description</th></tr>
<tr><td></td><td colspan="2">
Example
1. Set the string content of Working DIR name addr. ($500) to /lua.


2. Set Change working DIR trigger addr. ($200.0) to On.


3. The FTP File List element will be switched to the lua directory. If the directory does not exist, the operating state ($1000) becomes 0x0022.


4. Once Change working DIR trigger addr. ($200.0) operation is completed, it will be automatically cleared to Off.
</td></tr>
<tr><td rowspan="2">Check duplicate DIR/File</td><td>Check duplicated DIR/File trigger addr.</td><td>When this bit is On, the duplicated directory / file is checked and automatically cleared after completion.</td></tr>
<tr><td colspan="2">
Example
1. Set the string content of Change/Create/Check name addr. ($750) to lua.

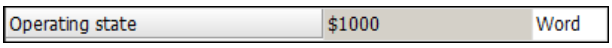
2. Set Check duplicated DIR/File trigger addr. ($200.1) to On.


3. If the Lua folder already exists, the content of Check duplicate DIR/File result ($900) becomes 1; otherwise, it becomes 0.


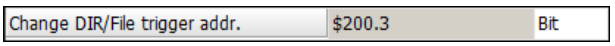
4. Once Check duplicated DIR/File trigger addr. ($200.1) operation is completed, it will be automatically cleared to Off.
</td></tr>
<tr><td rowspan="2">Create DIR / Change DIR / Delete DIR / File</td><td>Create DIR trigger addr.</td><td>When this bit is On, a new directory is created and automatically cleared after completion.</td></tr>
<tr><td colspan="2">
Example
1. Set the string content of Change/Create/Check name addr. ($750) to TestHMI.

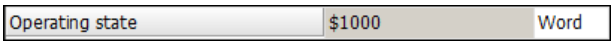
</td></tr>
</table>

<table>
<tr><th>Function</th><th>Description</th></tr>
<tr><td></td><td>
2. Set Create DIR trigger addr. ($200.2) to On.

3. Add the TestHMI folder in the FTP Server and update the FTP File List element. If the TestHMI folder already exists, Operating state ($1000) changes to 0x0042.


4. Once Create DIR trigger addr. ($200.2) operation is completed, it will be automatically cleared to Off.
</td></tr>
<tr><td>Change DIR/File trigger addr.</td><td>When this bit is On, the directory / file name is changed and automatically cleared after completion.</td></tr>
<tr><td></td><td>
Example
1. Use the FTP File List element to select the directory TestHMI to be changed.
2. Set the string content of Change/Create/Check name addr. ($750) to ChangeHMI.

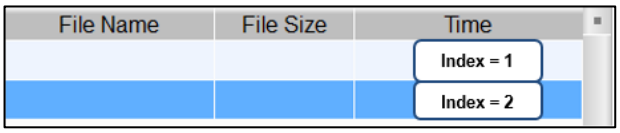
3. Set Change DIR/File trigger addr. ($200.3) to On.


4. Change the name of the TestHMI folder to ChangeHMI, and the HMI updates the FTP File List elements at the same time. If the ChangeHMI folder already exists, the Operating state ($1000) changes to 0x0052.


5. Once Change DIR/File trigger addr. ($200.3) operation is completed, it will be automatically cleared to Off.
Note: This example operates on a directory. You can also use the trigger address to operate on files.
</td></tr>
<tr><td>Delete DIR/File trigger addr.</td><td>When this bit is On, the directory / file name is deleted and automatically cleared after completion.</td></tr>
<tr><td></td><td>
Example
1. Set the string content of Selected DIR/File name addr. ($700) to ChangeHMI.

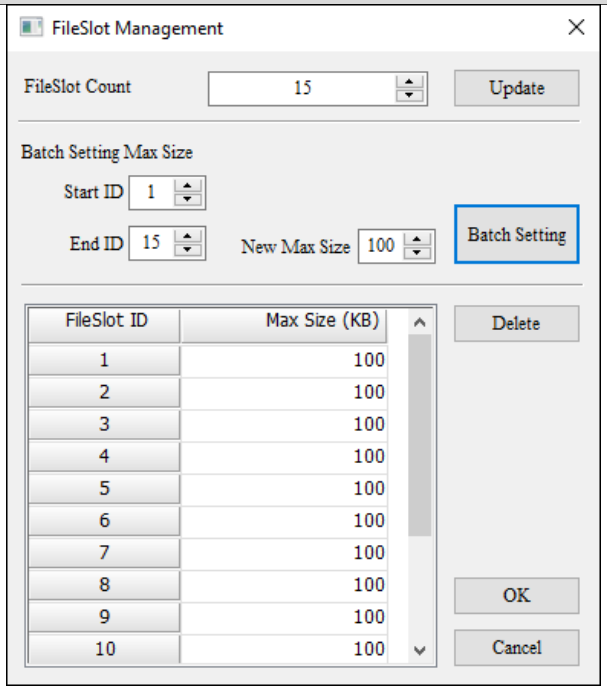
2. Set Delete DIR/File trigger addr. ($200.4) to On.

3. Delete the ChangeHMI folder of the FTP Server, and the HMI updates the FTP File List element at the same time.
4. Once Delete DIR/File trigger addr. ($200.4) operation is completed, it will be automatically cleared to Off.
</td></tr>
</table>

<table>
<tr><th>Function</th><th colspan="2">Description</th></tr>
<tr><td></td><td colspan="2">Note: This example operates on a directory. You can also use the trigger address to operate on files.</td></tr>
<tr><td rowspan="2">Get DIR / Filename</td><td>Get DIR/File name trigger addr.</td><td>When this bit is On, the directory / file name is obtained and automatically cleared after completion.</td></tr>
<tr><td colspan="2">Example
The following figure shows the FTP File List.
 

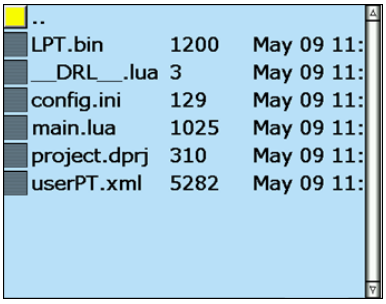
1. Set Index of DIR/File name addr. ($921) to 2.
Index of DIR/File name addr. | $921 | Word
If the index value is set to 0, the Operating state ($1000) becomes 0x0027.
Operating state | $1000 | Word
2. Set Get DIR/File name trigger addr. ($200.5) to On.
Get DIR/File name trigger addr. | $200.5 | Bit
3. Get the folder name with FTP Server index value 2, Get DIR/File name result ($770) content is lua.
Get DIR/File name result | $770 | Word | 16
Note:
• This example operates on directory lines. You can also use this trigger address to operate on files.
• The index value is based on the FTP Server. The sorting displayed in the FTP File List of the HMI may not be the same as the sorting of the FTP Server.</td></tr>
<tr><td rowspan="2">Download DIR / File</td><td>Download directory/file trigger address</td><td>When this bit is On, the HMI starts downloading files from the FTP Server and automatically clear them after completion.</td></tr>
<tr><td colspan="2">To use this function, first set up the FileSlot file function. In the menu bar, click Project > Other Settings > FileSlot Management to make related settings.</td></tr>
</table>

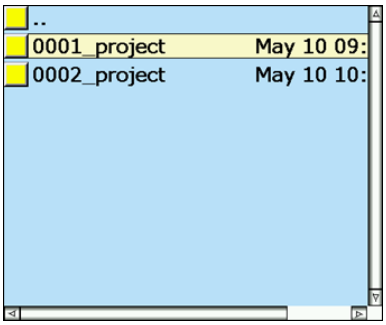
Function	Description



Example

1. If the FTP Server file is stored in /lua/0001_project, switch to the /lua directory and select the 0001_project directory.



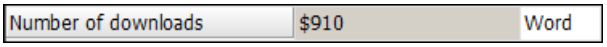


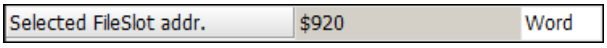
2. Set **Download DIR/File trigger addr.** ($200.6) to On.

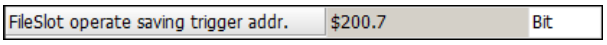
Download DIR/File trigger addr.	$200.6	Bit

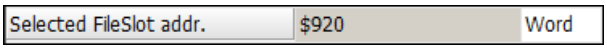
3. The HMI downloads the file from the FTP Server to the **Selected FileSlot addr.** ($920). If the directory contains multiple files, the **Selected FileSlot addr.** ($920) represents the starting FileSlot position.

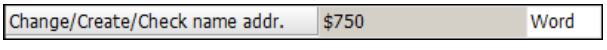
Selected FileSlot addr.	$920	Word

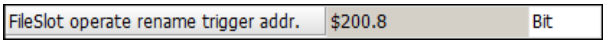
<table>
<tr><th>Function</th><th colspan="2">Description</th></tr>
<tr><td></td><td colspan="2">After the action is completed, the Number of downloads ($910) becomes 6.
 

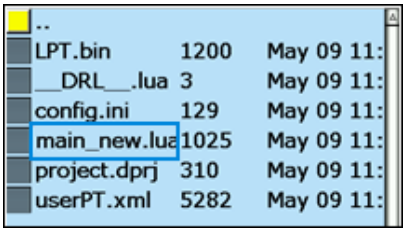
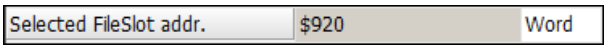
4. After the file download is completed, the file will only be stored in the FileSlot address. To edit the downloaded file, you need to use the Text List element. To read out the file name of a downloaded file, use the FileSlotGetName macro command.
Note:
• This example operates on a directory, and you can also use the trigger address to operate on files.
• If the name of the downloaded file is known, the HMI can obtain the location information (ID) of the file stored in FileSlot through the FileSlotGetID macro command.</td></tr>
<tr><td rowspan="4">FileSlot Operations</td><td>FileSlot operate saving trigger addr.</td><td>When this bit is On, the selected FileSlot file will be uploaded to the FTP Server and automatically cleared after completion.</td></tr>
<tr><td colspan="2">To use this function, you need to set up the FileSlot file function and execute the download directory/file function.
Example
1. Set Selected FileSlot addr. ($920) to 4.
 

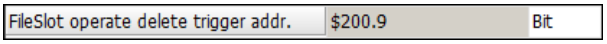
2. Set FileSlot operate saving trigger addr. ($200.7) to On.
 

3. The HMI uploads the file (main.lua) with FileSlotID 4 to the FTP Server.</td></tr>
<tr><td>FileSlot operate rename trigger addr.</td><td>When this bit is On, the selected FileSlot file is renamed and uploaded to the FTP Serverm and automatically cleared after completion.</td></tr>
<tr><td colspan="2">To use this function, first set up the FileSlot file function and execute the download directory/file function.
Example
1. Set Selected FileSlot addr. ($920) to 4.
 

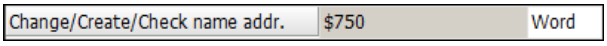
2. Set Change/Create/Check name addr. ($750) to main_new.lua.
 

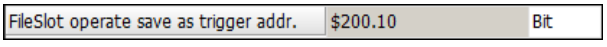
3. Set FileSlot operate rename trigger addr. ($200.8) to On.
 
</td></tr>
</table>

<table>
<tr><th>Function</th><th colspan="2">Description</th></tr>
<tr><td rowspan="3"></td><td colspan="2">4. The HMI changes the name of the main.lua file to main_new.lua, and simultaneously rename the FileSlot and updates the FTP File List.
 
</td></tr>
<tr><td>FileSlot operate delete trigger addr.</td><td>When this bit is On, the selected FileSlot file is deleted from the FTP Server and automatically cleared after completion.</td></tr>
<tr><td colspan="2">To use this function, you need to set up the FileSlot file function and execute the download directory/file function.
Example
1. Set Selected FileSlot addr. ($920) to 1.
 

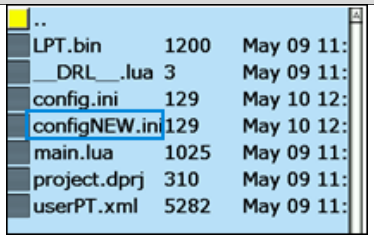
2. Set FileSlot operate delete trigger addr. ($200.9) to On.
 

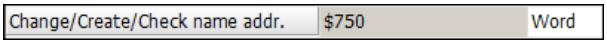
3. The HMI deletes the file LPT.bin on the FTP Server and updates the FTP File List simultaneously.</td></tr>
<tr><td rowspan="3">FileSlot Operations</td><td>FileSlot operate save as trigger addr.</td><td>When this bit is On, the selected FileSlot file is saved in the FTP Server and automatically cleared after completion.</td></tr>
<tr><td colspan="2">To use this function, set up the FileSlot file function and execute the download directory / file function.
Example
1. Set Selected FileSlot addr. ($920) to 3.

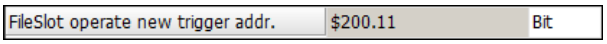
2. Set string content of the Change/Create/Check name addr. ($750) to configNEW.ini.
 

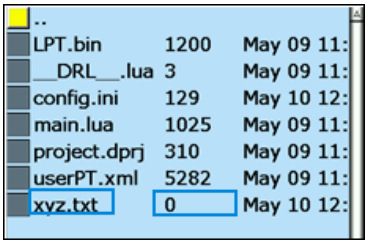
3. Set FileSlot operate save as trigger addr. ($200.10) to On.
 

4. The HMI adds a new file configNEW.ini on the FTP Server and updates the FTP file list simultaneously.</td></tr>
</table>

<table>
<tr><th>Function</th><th colspan="2">Description</th></tr>
<tr><td rowspan="3"></td><td colspan="2"> 
</td></tr>
<tr><td>FileSlot operate new trigger addr.</td><td>When this bit is On, the selected FileSlot file is added to the FTP Server and automatically cleared after completion.</td></tr>
<tr><td colspan="2">To use this function, set up the FileSlot file function first.
Example
1. Set Selected FileSlot addr. ($920) to 10.

2. Set Change/Create/Check name addr. ($750) to xyz.txt.
 

3. Set FileSlot operate new trigger addr. ($200.11) to On.
 

4. The HMI adds a file xyz.txt with a file size of 0 on the FTP Server, and updates the FTP File List simultaneously.
 
</td></tr>
<tr><td rowspan="5">Operation directory/name address setting</td><td>Working DIR name addr.</td><td>Set the working directory name and Max. Length Of Name.</td></tr>
<tr><td>Max. Length Of Name</td><td>Note: It should be used with Change working DIR trigger addr. Refer to the example operation of Change working DIR trigger addr.</td></tr>
<tr><td>Selected DIR/File name addr.</td><td rowspan="2">Set the directory or file to be deleted and the maximum length.
Note: It should be used with Delete DIR/File trigger addr. Refer to the example operation of Delete DIR/File trigger addr.</td></tr>
<tr><td>Max. Length Of Name</td></tr>
<tr><td>Change/Create/Check name addr.</td><td>Set the directory or file to be changed / created / checked and the maximum length.</td></tr>
</table>

Function	Description	
	Max. Length Of Name	**Note**: It should be used with **Check duplicated DIR/File trigger addr.** or **Create DIR trigger addr.** Refer to the example operations for **Check duplicated DIR/File trigger addr.** or **Create DIR trigger addr.**
	Selected FileSlot addr.	Enter data type as Word and set the selected FileSlot. **Note**: It should be used with **Download DIR/File trigger addr.** Refer to the example operation of **Download DIR/File trigger addr.**
	Index of DIR/File name addr.	Enter data type as Word and set **Index of DIR/File name addr.** **Note**: It should be used with **Get DIR/File name trigger addr.** Refer to the example operation of **Get DIR/File name trigger addr.**
Execution result setting	Operating state	When the actions in the FTP transfer settings are completed, the operating state changes accordingly. If the operation is successful, 0 is displayed; otherwise, an error code is displayed.
	Check duplicated DIR/File result	Returns the result of **Check duplicated DIR/File trigger addr.**
	Number of downloads	Returns the number of downloaded files. **Note**: It should be used with **Download DIR/File trigger addr.**
	Get DIR/File name addr.	Returns the result of **Get DIR/File name trigger addr.**

The following table lists the FTP Client's return values with description.

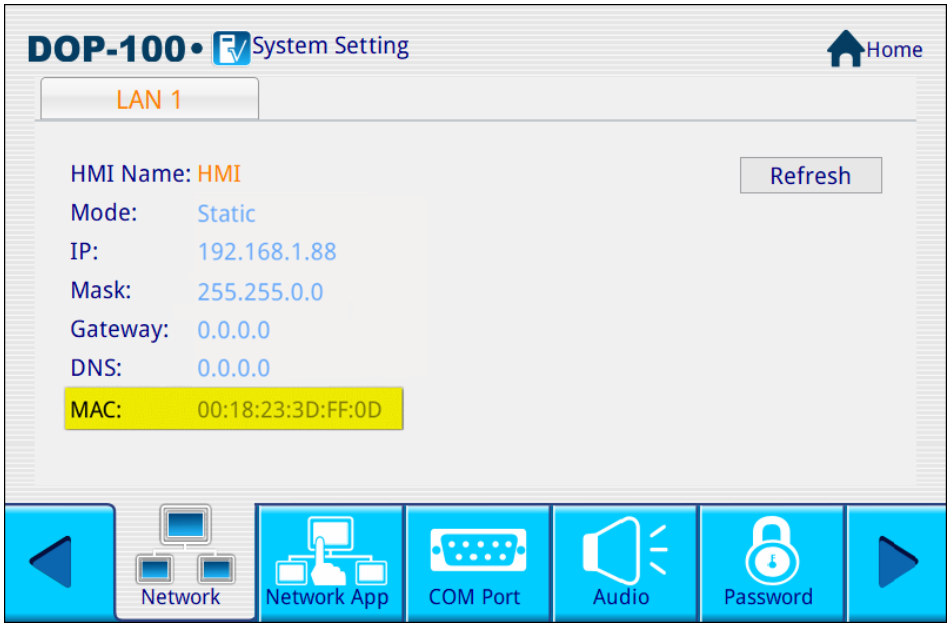
Return Value		Description
Decimal	Hexadecimal	
0	0	Completed successfully
20	32	Login failed
21	33	Logout failed
22	34	Failed to switch working directory
23	35	Check for duplicate directory / file names failed

Return Value		Description
Decimal	Hexadecimal	
24	36	Unable to obtain directory / file information
25	37	Connection abnormality
26	38	Insufficient system memory space
27	39	Failed to query directory / file name
40	64	Failed to create directory / file
41	65	Unable to create file
42	66	Unable to create directory
50	80	Failed to rename directory / file
51	81	File cannot be renamed
52	82	Directory cannot be renamed
60	96	Failed to delete directory / file
61	97	Unable to delete file
62	98	Unable to delete directory
70	112	Fail to download directory / file
71	113	Unable to download file
72	114	Unable to download directory
73	115	FIleSlot Insufficient storage space
80	128	Fail to save FIleSlot
81	129	Unable to save FileSlot
82	130	Unable to save as FileSlot
A1	161	Unable to create FileSlot
A2	162	Unable to rename FileSlot
A3	163	Unable to delete FileSlot
A4	164	Unable to open FileSlot
A5	165	Unable to copy FileSlot
A6	166	Unable to obtain FileSlot ID
A7	167	Unrecognized FileSlot ID
A8	168	Unable to obtain FileSlot name
A9	169	Unable to set FileSlot name
AA	170	Unable to obtain the maximum number of FileSlots allowed

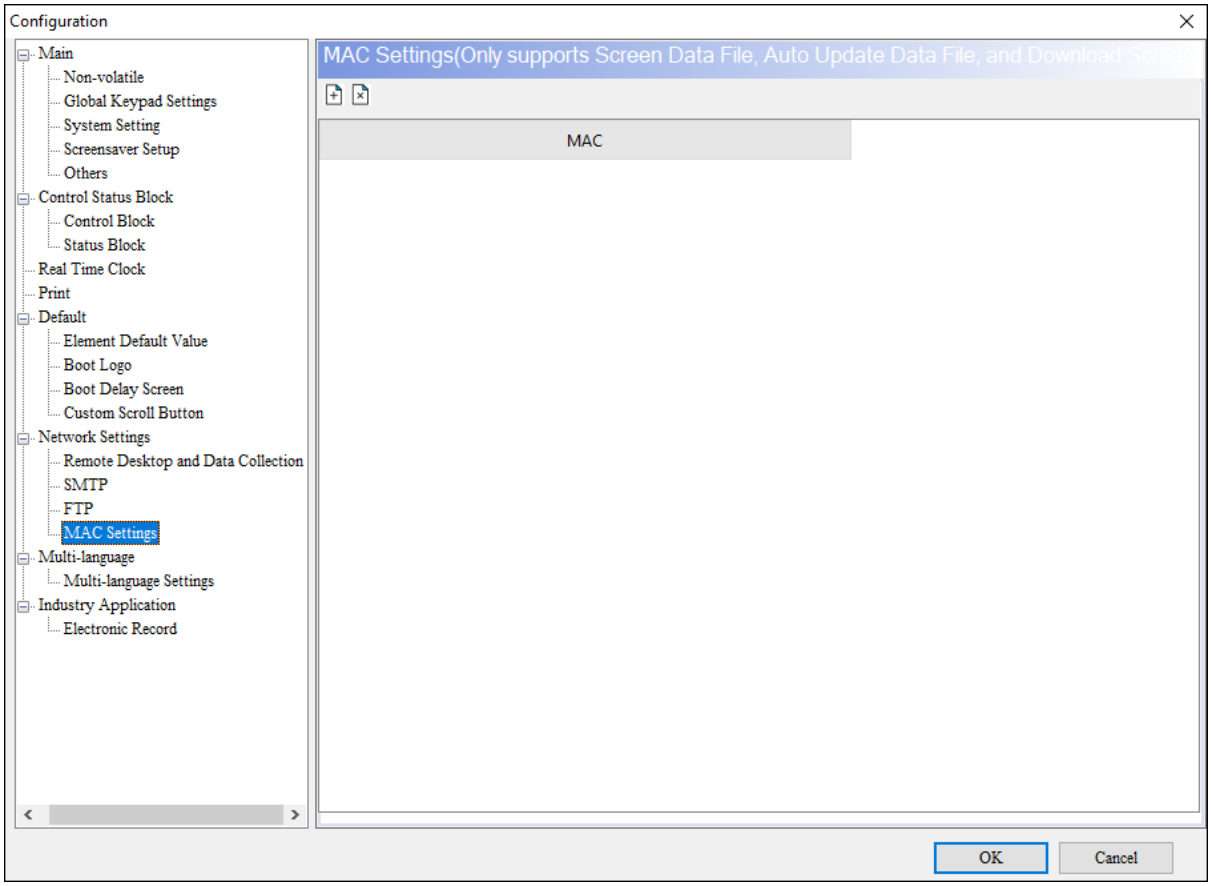
Return Value		Description
Decimal	Hexadecimal	
AB	171	Unable to add FileSlot, the maximum number of FileSlots allowed has been exceeded.
AC	172	Unable to obtain FileSlot space limit ed size
AD	173	FileSlot name duplicate

MAC Settings

In the **MAC Settings** page, set the MAC of the HMI to be loaded when creating screen data file, the file only can update the screen on the specified HMI. If the MAC is not set, the HMI model is not restricted. You can obtain the MAC from **System Menu** > **System Setting** > **Network** on the HMI.



The following table lists the functions in the **MAC Settings** page with their description.

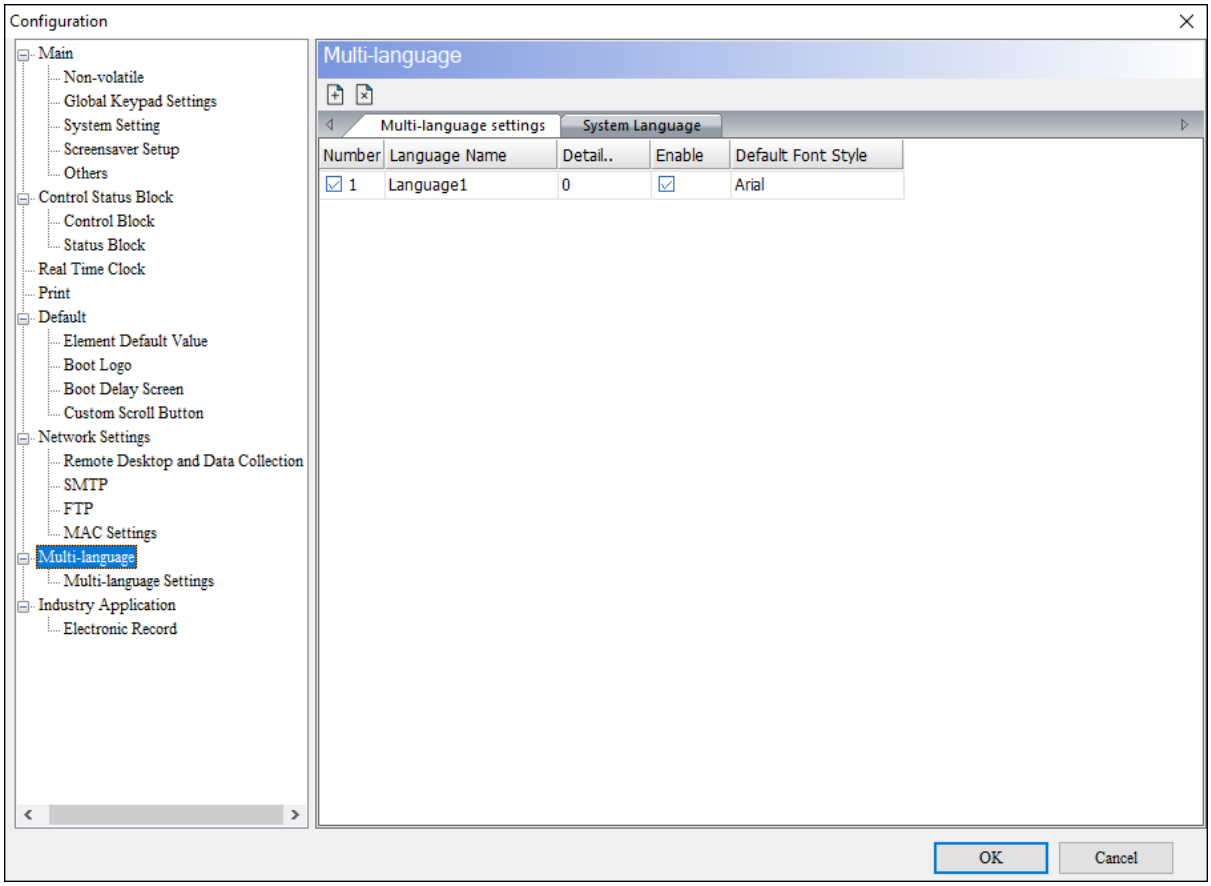


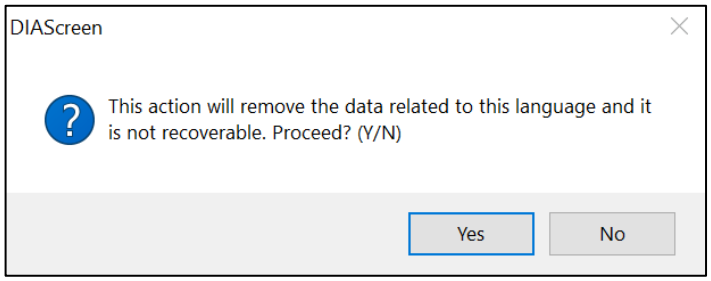
Function	Description
	Click to add the MAC.
	Click to delete the MAC.

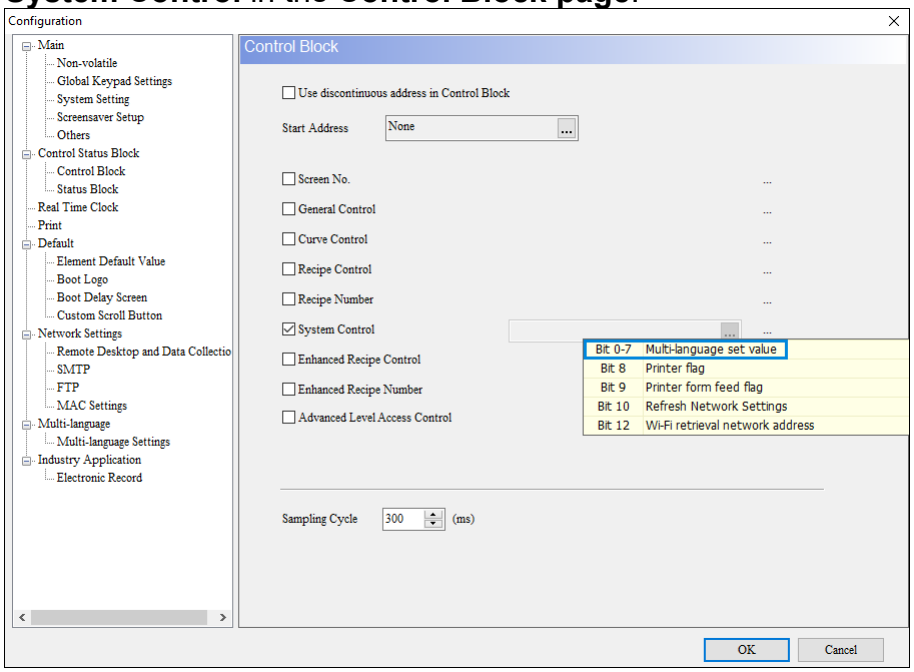
Multi-language

The following table lists the functions in the **Multi-language settings** tab with their description.

Multi-language settings



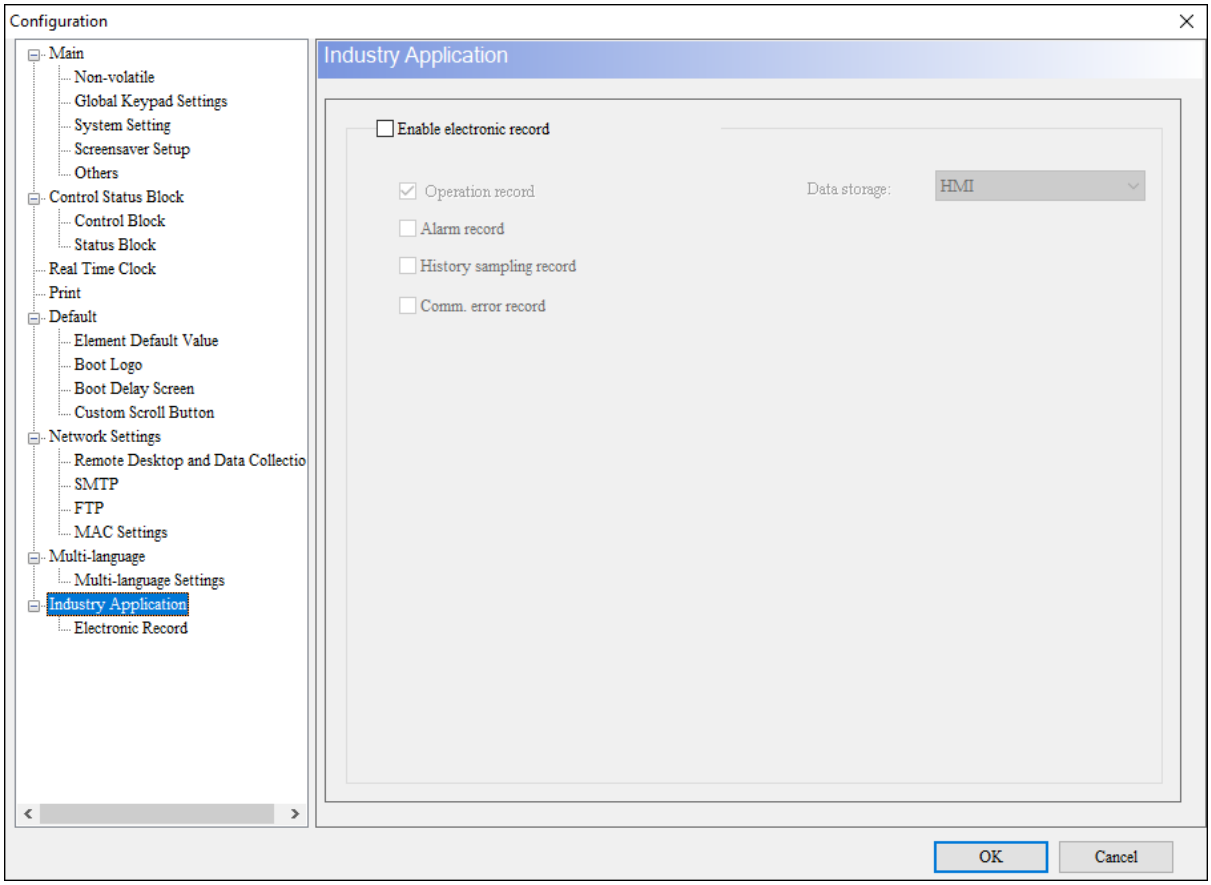
Function	Description
	Click to add a language. It supports adding up to 32 languages.
	Select the item to be deleted and click the icon to delete, a warning message appears and asks whether to proceed. 
Language Name	Set the language name according to your needs.

Function	Description
Detail	Detail column switches the language according to its value. To switch the language, you can: • Switch the language to the corresponding settings through the **System Control** in the **Control Block page**.  • Click **General** on the toolbar and select the language to be switched in the **Language** field of the **State/Language** pane.
Enable	The first row is the default language, the **Enable** checkbox cannot be unselected, or a warning message displays, informing that the language cannot be disabled.
Default Font Style	Apply the **font name** and **font size** according to the selected language. • If you only choose to apply the **Font Name** or **Font Size**, the font setting only applies to the newly created elements. • If you choose both **Font Name** and **Font Size to apply**, the font setting applies to the existing and newly created elements. **Note**: The **font name** and **font size** settings are applied to the elements that you can enter the text.

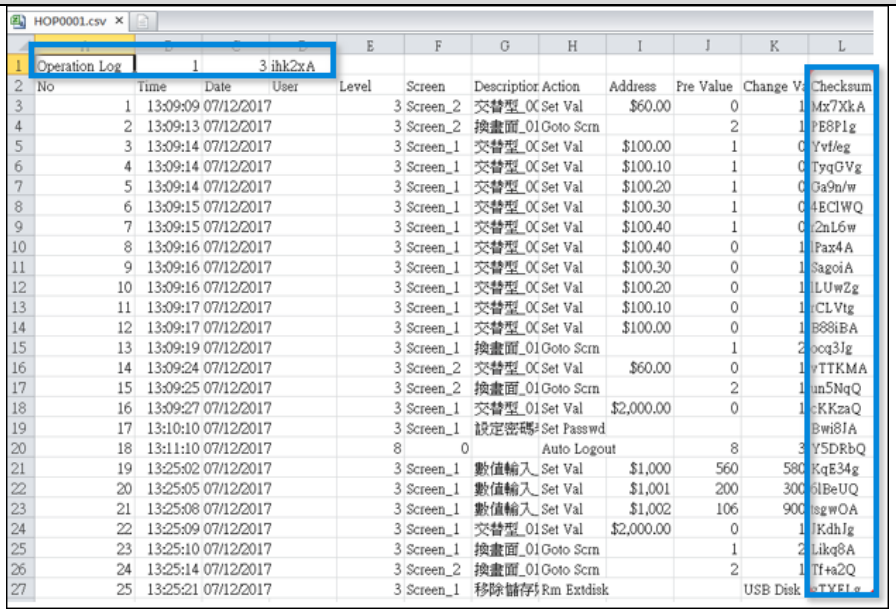
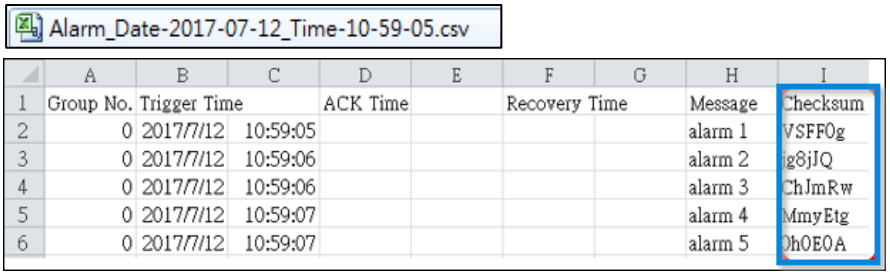
System Language Settings

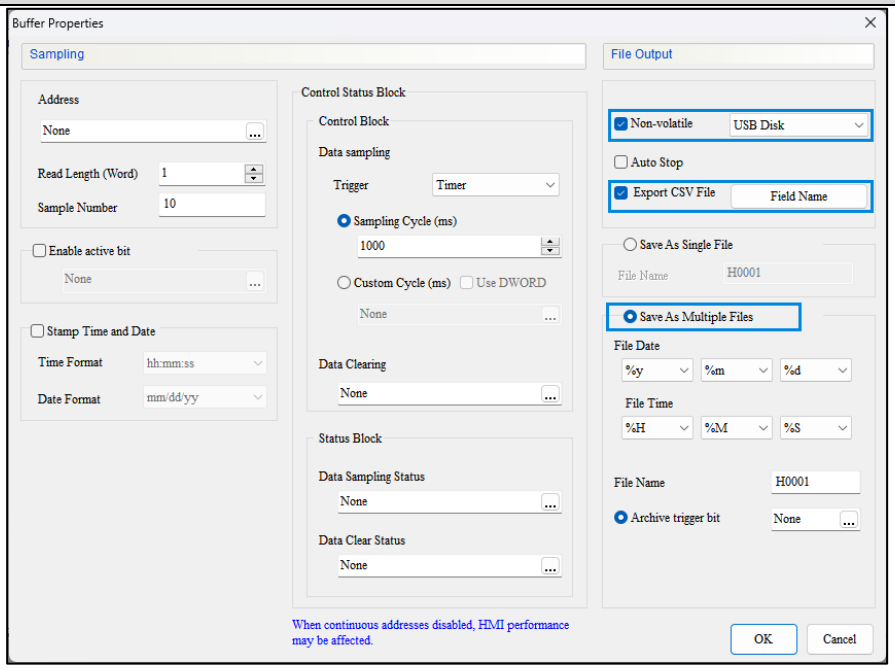
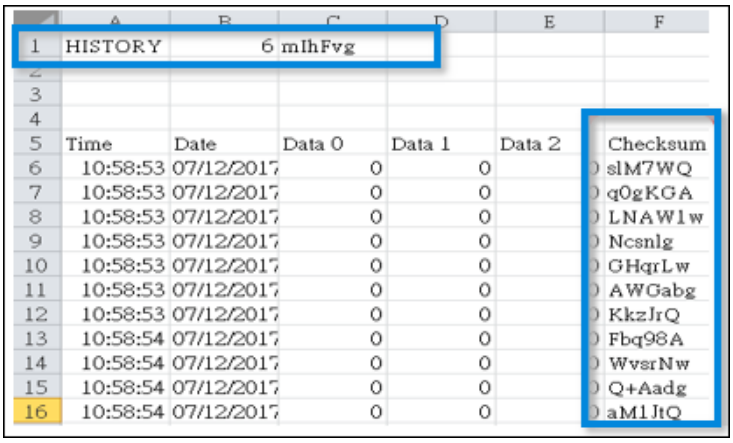
System languages include 6 languages, which can be applied to the system menu of the HMI and the pop-up message when operating the HMI.

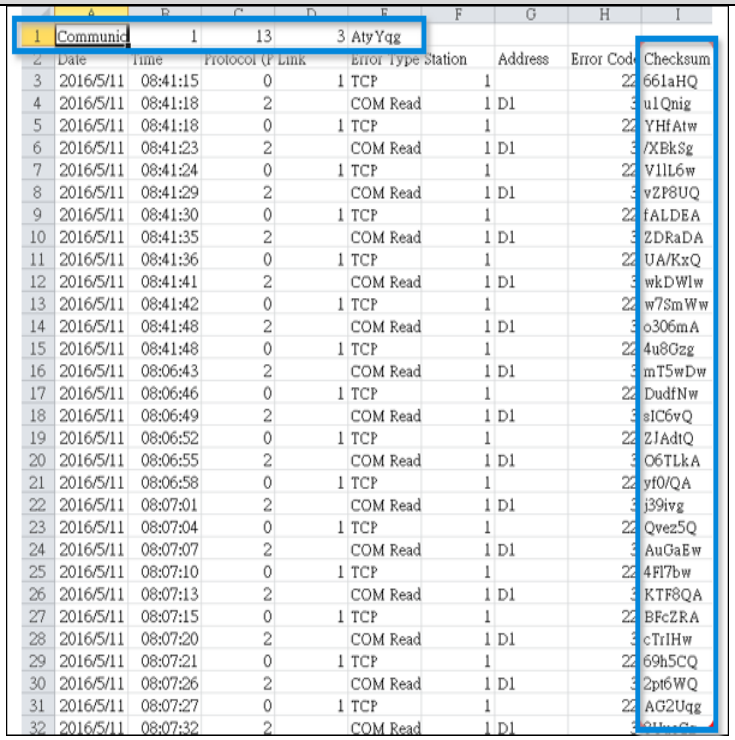
Industry Application



Function	Description
Enable electronic record	If selected, the electronic record settings are available. **Note**: If enabled, **Advanced Level Access Control** of Control and Status Block becomes invalid.
Operation record	The **Operation record** function records all operations of the HMI, including numerical changes and user level. Checksum and Operation Log data will be added to the operation record file.

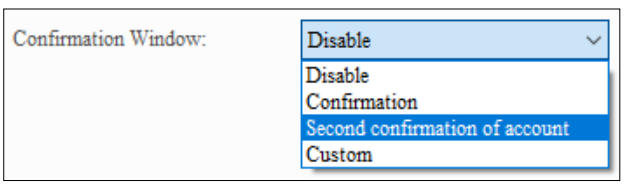
Function	Description
	
Alarm record	If selected, the HMI alarm data will be recorded, and the **Export CSV File** field in the **Alarm Settings** page will be forced to be set to **Yes**. To cope with electronic signatures, the file name of the exported alarm data file will be created according to the year, month, day, hour, minute and second. Checksum data will be added.  **Note**: The timing of generating this file is based on the maximum number of records set by the alarm.
History sampling record	If selected, the HMI historical data will be recorded. In the **Buffer Properties** dialog of the **History Buffer** setting page, • **Non-volatile** will be forced to be set to HMI, SD, USB Disk or USB Disk 2 (based on the setting of **Data storage**), • **Export CSV File** checkbox will be forced to be selected, • The file storage method will also be forced to be set to **Save As Multiple Files**.

Function	Description
	 The historical data of **Save As Multiple Files** will be stored according to year, month, day, hour, minute, and second. Checksum and HISTORY data will be added. H0001_Date-2017-07-12_Time-10-59-22.csv 
Comm. error record	If selected, the HMI will automatically store the communication error data in the HMI, USB, or SD card, and the storage location is HMI\HMI-000\CommEventLog. Checksum and Communication data will be added to the communication error record file. CommunicationEvents.csv

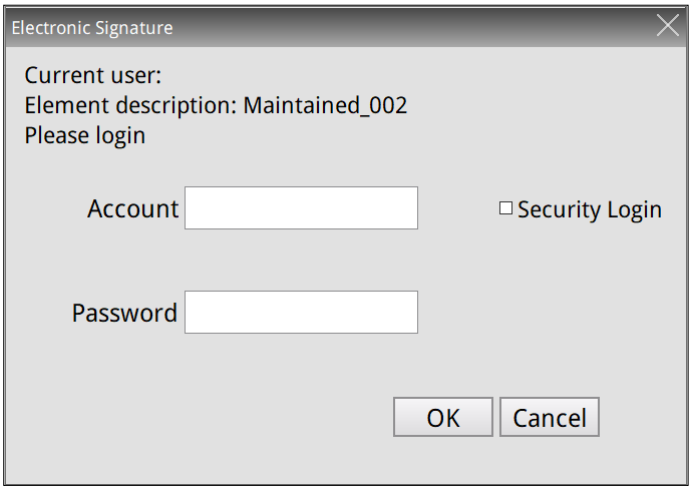
Function	Description
	
Data storage	Select the storage location of all stored data in electronic records. The available options include HMI, USB Disk, USB Disk 2, and SD. The default value is **HMI**.

The electronic record function mainly ensures that the data are consistent and accurately generated and saved, and can be viewed at any time, which is different from the daily records. Therefore, the generated record data will be added with a checksum mechanism and exported to a CSV file.

Enable the Enable electronic record function will also enable the **Second confirmation of account** function for the **Confirmation Window** in the element dialog.

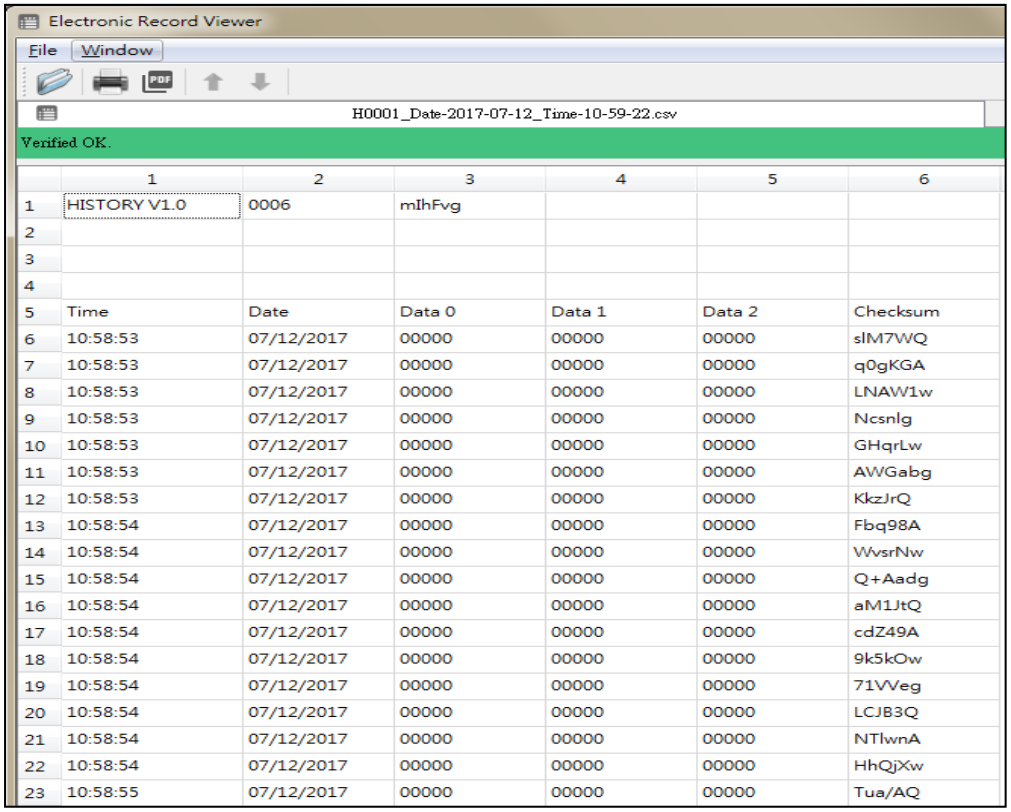


If the account is set to **Second confirmation of account**, you will be asked to enter the password twice.



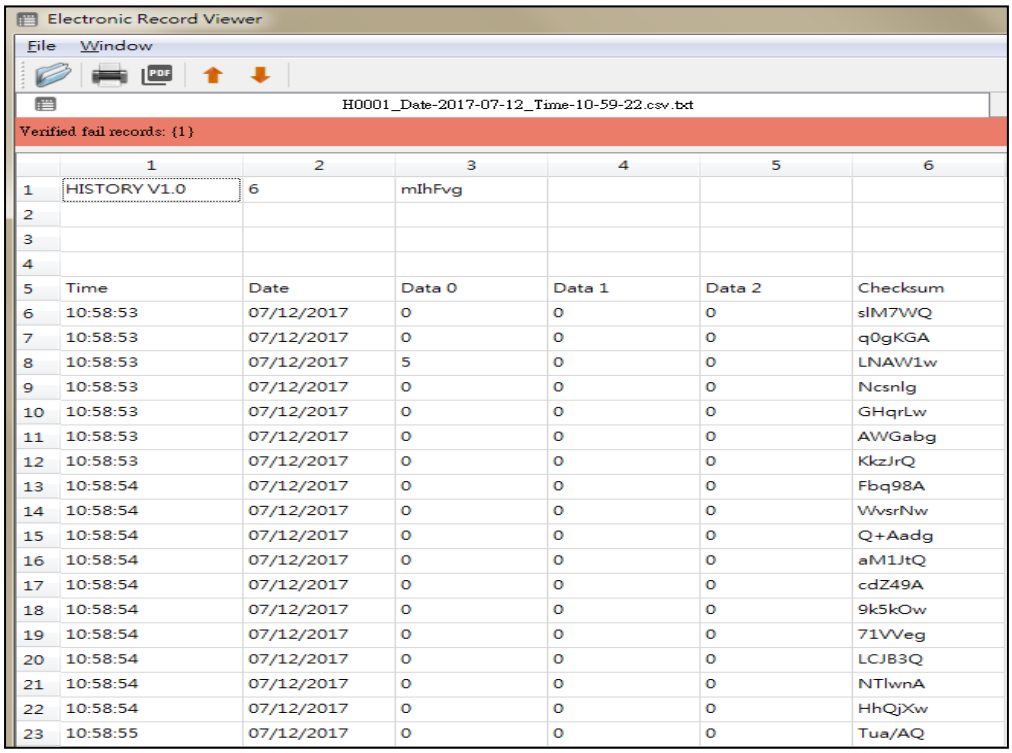
The files generated by the electronic signature can be opened by Excel or the electronic record viewing tool eRecordViewer. The file storage location is `C:\Program Files (x86)\Delta Industrial Automation\DIAStudio\DIAScreen (version)\Utility \eRecordViewer.`

If the loaded data has not been tampered with, eRecordViewer will prompt "**Verified OK"**.



	1	2	3	4	5	6
1	HISTORY V1.0	0006	mIhFvg			
2						
3						
4						
5	Time	Date	Data 0	Data 1	Data 2	Checksum
6	10:58:53	07/12/2017	00000	00000	00000	slM7WQ
7	10:58:53	07/12/2017	00000	00000	00000	q0gKGA
8	10:58:53	07/12/2017	00000	00000	00000	LNAW1w
9	10:58:53	07/12/2017	00000	00000	00000	Ncsnlg
10	10:58:53	07/12/2017	00000	00000	00000	GHqrLw
11	10:58:53	07/12/2017	00000	00000	00000	AWGabg
12	10:58:53	07/12/2017	00000	00000	00000	KkzJrQ
13	10:58:54	07/12/2017	00000	00000	00000	Fbq98A
14	10:58:54	07/12/2017	00000	00000	00000	WvsrNw
15	10:58:54	07/12/2017	00000	00000	00000	Q+Aadg
16	10:58:54	07/12/2017	00000	00000	00000	aM1JtQ
17	10:58:54	07/12/2017	00000	00000	00000	cdZ49A
18	10:58:54	07/12/2017	00000	00000	00000	9k5kOw
19	10:58:54	07/12/2017	00000	00000	00000	71VVeg
20	10:58:54	07/12/2017	00000	00000	00000	LCJB3Q
21	10:58:54	07/12/2017	00000	00000	00000	NTlwnA
22	10:58:54	07/12/2017	00000	00000	00000	HhQjXw
23	10:58:55	07/12/2017	00000	00000	00000	Tua/AQ

If the data has been tampered with, for example, eRecordViewer detects that the Checksum of the data in the 8th column of the file does not match, it will display "**Verified fail records: {1}**". The number in the braces is the total number of error data.



	1	2	3	4	5	6
1	HISTORY V1.0	6	mIhFvg			
2						
3						
4						
5	Time	Date	Data 0	Data 1	Data 2	Checksum
6	10:58:53	07/12/2017	0	0	0	slM7WQ
7	10:58:53	07/12/2017	0	0	0	q0gKGA
8	10:58:53	07/12/2017	5	0	0	LNAW1w
9	10:58:53	07/12/2017	0	0	0	Ncsnlg
10	10:58:53	07/12/2017	0	0	0	GHqrLw
11	10:58:53	07/12/2017	0	0	0	AWGabg
12	10:58:53	07/12/2017	0	0	0	KkzJrQ
13	10:58:54	07/12/2017	0	0	0	Fbq98A
14	10:58:54	07/12/2017	0	0	0	WvsrNw
15	10:58:54	07/12/2017	0	0	0	Q+Aadg
16	10:58:54	07/12/2017	0	0	0	aM1JtQ
17	10:58:54	07/12/2017	0	0	0	cdZ49A
18	10:58:54	07/12/2017	0	0	0	9k5kOw
19	10:58:54	07/12/2017	0	0	0	71VVeg
20	10:58:54	07/12/2017	0	0	0	LCJB3Q
21	10:58:54	07/12/2017	0	0	0	NTlwnA
22	10:58:54	07/12/2017	0	0	0	HhQjXw
23	10:58:55	07/12/2017	0	0	0	Tua/AQ

You can also save the electronic records as PDF files through eRecordViewer and print them.

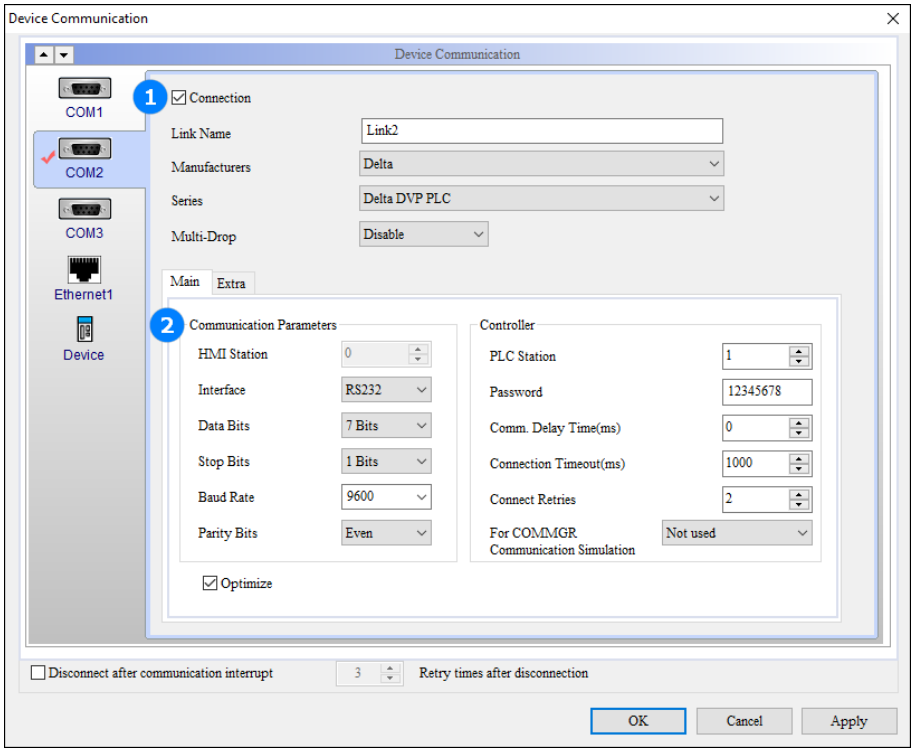
Device Communication

The **Device Communication** function is used for configuring the related setting of communication between the HMI and devices, including COM Port, Ethernet, and other interfaces.

This section describes the parameter settings for each communication interface.

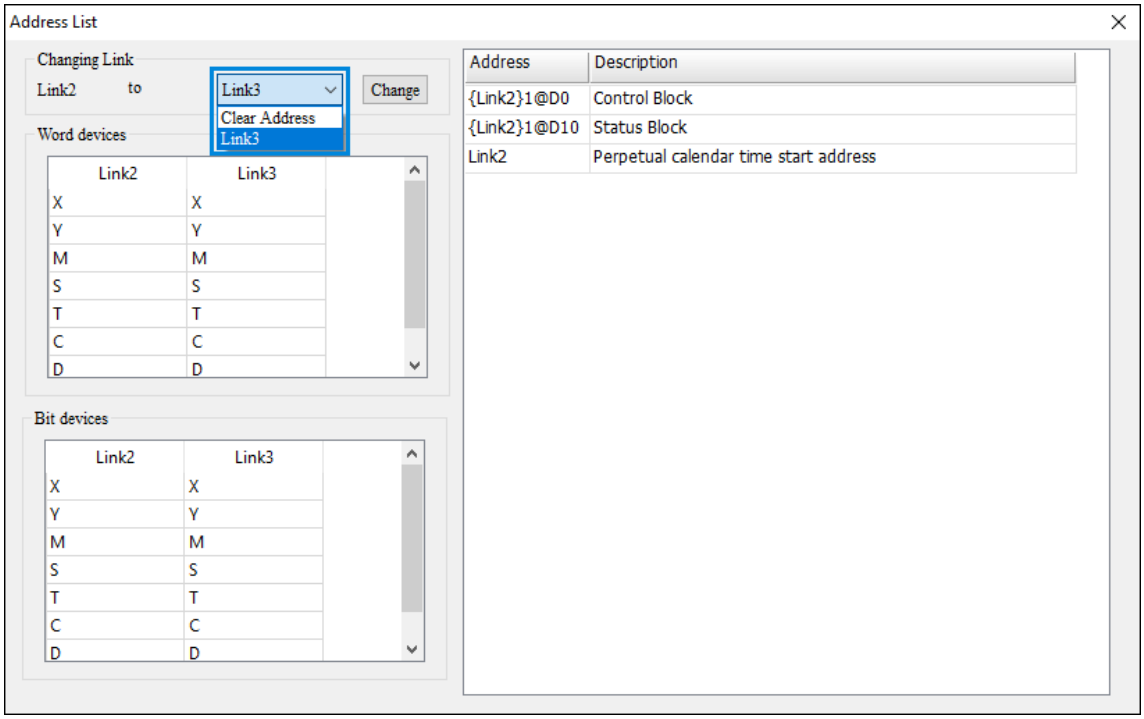
COM Port

The following table lists the functions in the COM Port setting page with the description.



1 Connection

- The settings of COM Port are only available when **Connection** is selected.
- You can customize the **Link Name** and select the device to connect.
- If **Connection** is unselected, DIAScreen detects whether the current connection is being used. If so, a warning message appears, and the **Address List** dialog displays. In the **Address List** dialog, you can choose to change the address to another connection or **Clear Address**.



② Communication Parameters

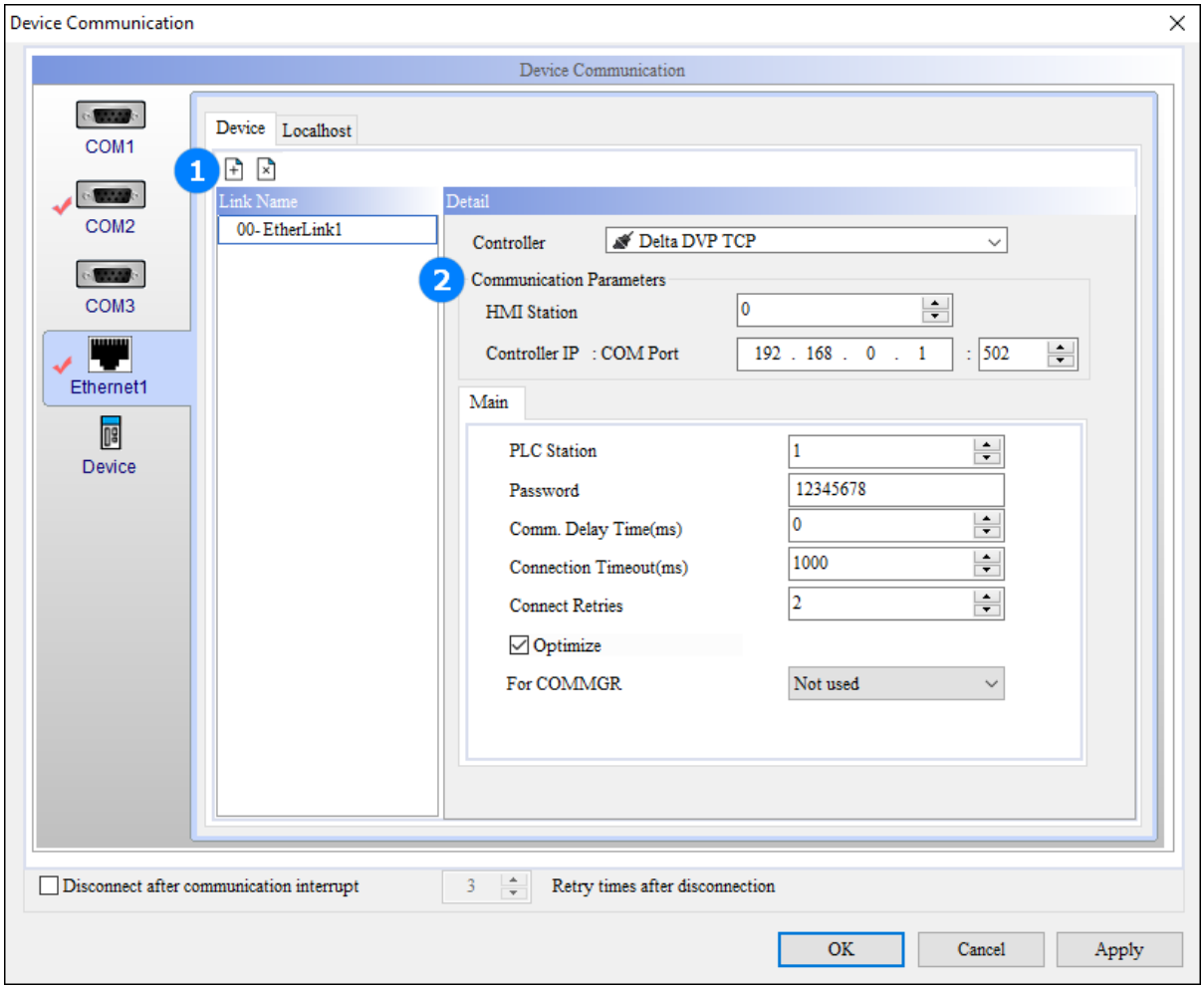
Function	Description
HMI Station	Select or enter the HMI station ID. The range value is 1–255 and the default value is **0**.
Interface	Select the communication interface (transmission method). The available options are RS232, RS422, and RS485. For COM1, only RS232 can be selected; for COM2 and COM3, RS232, RS422, and RS485 can be selected.
Data Bits	Select the length of the received data packet.
Stop Bits	Select the stop bits that inform the receiver of the end of the data signal.
Baud Rate	Select the data transmission speed.
Parity Bits	Select the parity bits. It is used to check for errors in the transmitted data. The available options are None, Odd, Even, Mark, and Space.

	COM1	COM2	COM3
RS232	V	V	V
RS422	-	V	V
RS485	-	V	V

For other parameter settings, see Common Properties.

Ethernet Port

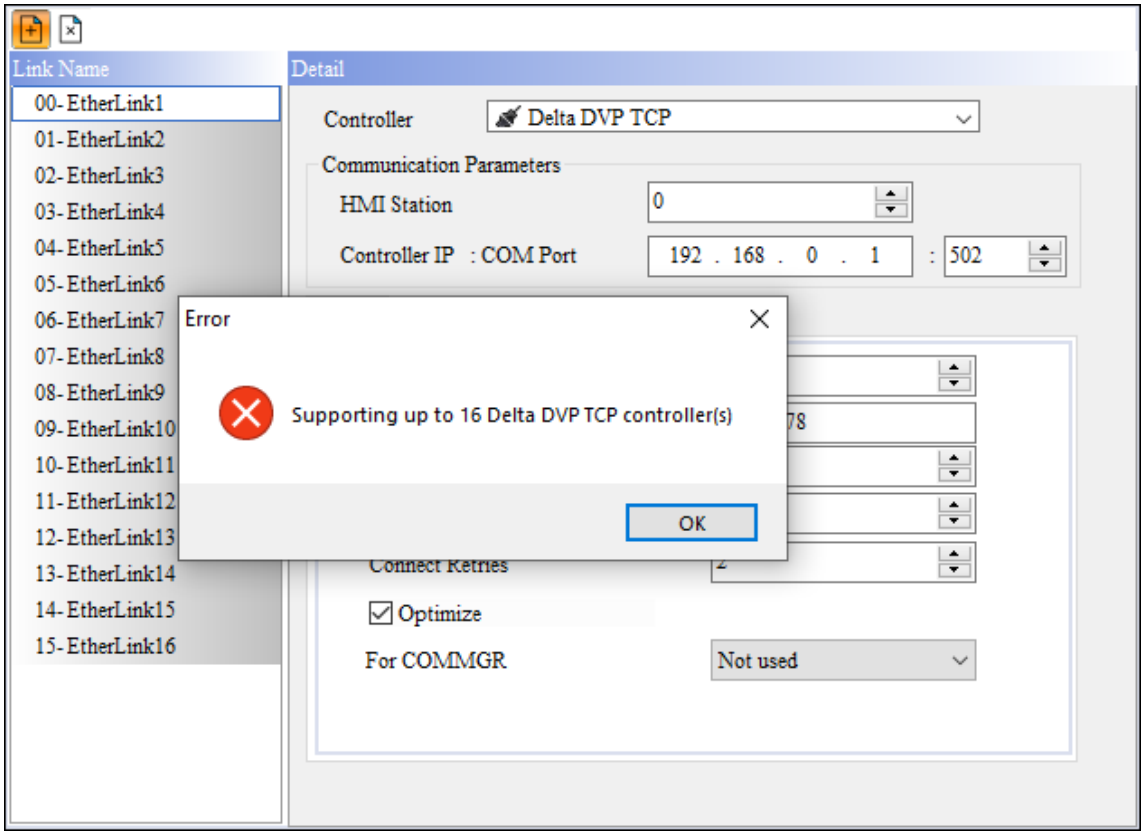
The following table lists the functions in the Ethernet Port setting page with the description.



Device

1 Device

- Click to add a link device named EtherLink1. The **Link Name** EtherLink1 can be changed according to your need.
- Up to 16 links are supported for each controller, and up to 4 types of controllers are supported for one network port.

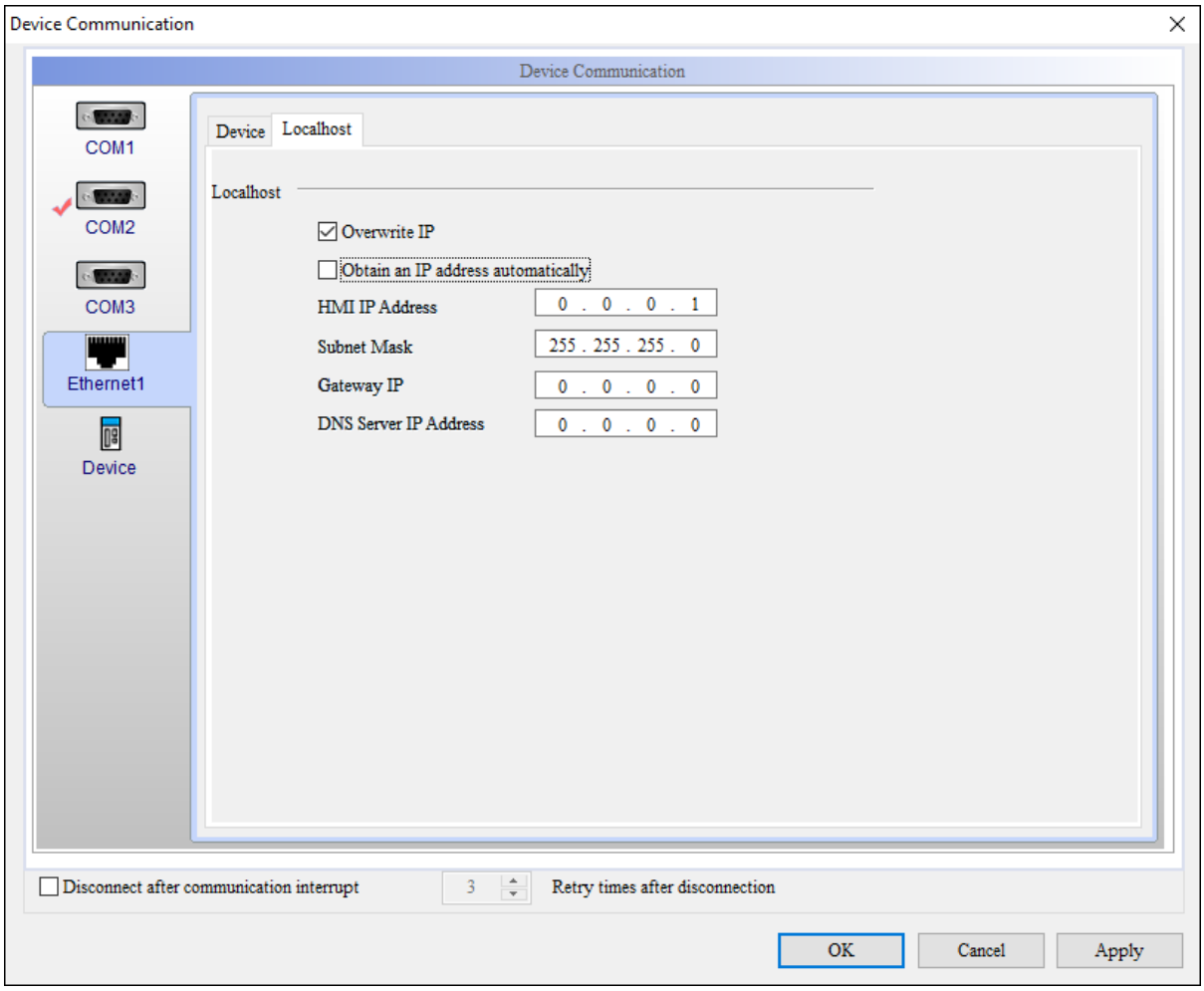


② Communication Parameters

Function	Description
HMI Station	Select or enter the HMI station ID. The range value is 1–255 and the default value is **0**.
Controller IP	Set the IP address of controller. **Note**: The controller IP address must be set in the same network segment as the HMI IP address.
COM Port	Select or enter the communication port. You can also set it corresponding to the controller's port. **Note**: The **COM Port** varies depending on the selected controller.

For the descriptions of the **Main** tab, see Common Properties.

Localhost

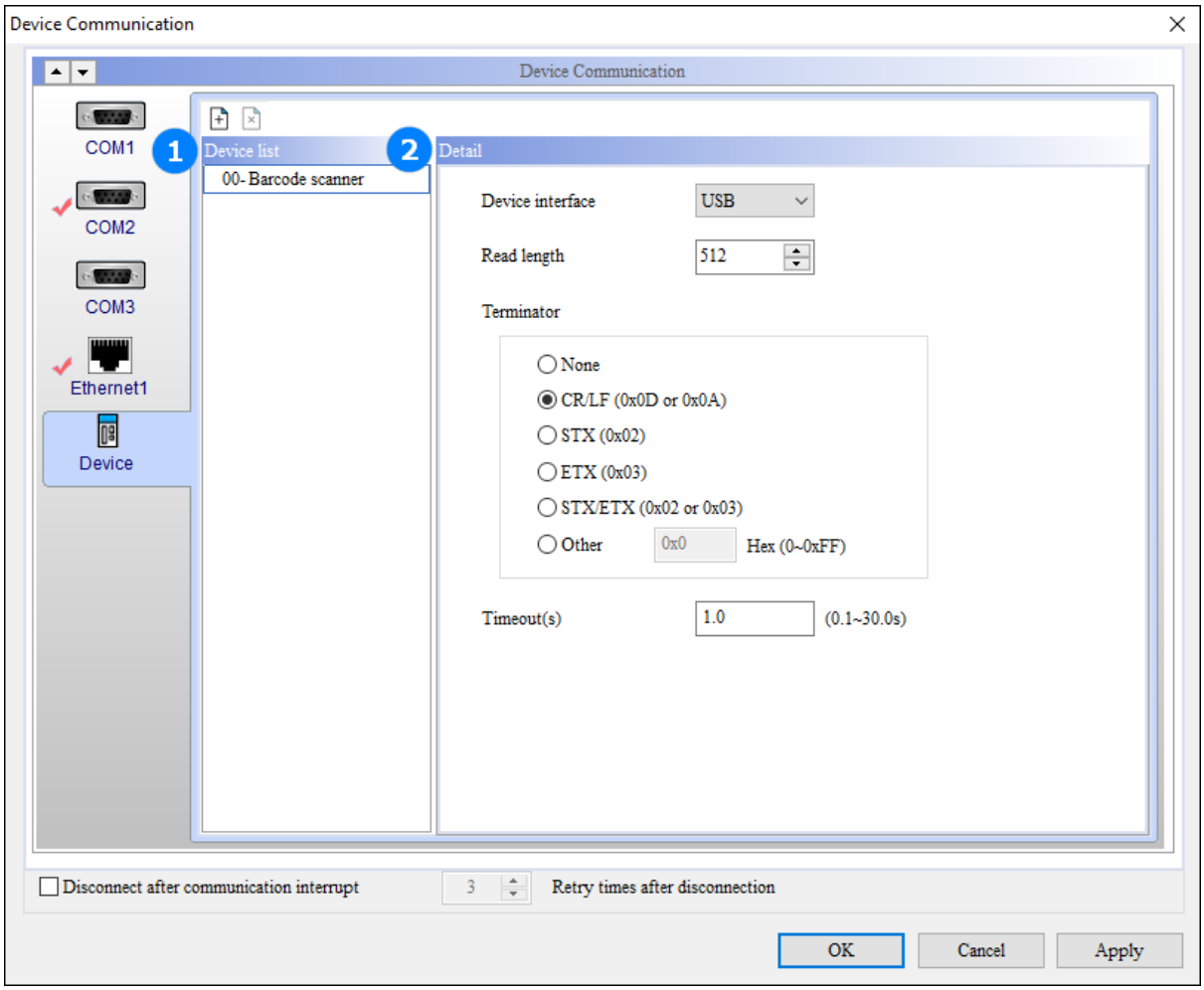


Function	Description
Overwrite IP	• If selected, the network IP address of the HMI is written according to the screen file settings when DIAScreen downloads the project or updates the screen file. • If not selected, the network IP address does not change due to project download and screen file updates. Go to **System Menu** > **System Setting** > **Network** on the HMI to change its IP address.
Obtain an IP address automatically	• This setting is only available when **Overwrite IP** is selected. • If **Obtain an IP address automatically** is selected, the HMI uses the DHCP mode to get the IP address.
HMI IP address	Set the IP address of the HMI. If **Obtain an IP address automatically** is not selected, the fixed IP address can be set. **Note**: The **HMI IP address** must be set in the same network segment as the controller IP address.

<table>
<tr><th>Function</th><th>Description</th></tr>
<tr><td>Subnet Mask</td><td>Set the subnet mask. The subnet mask is used to segment the network and identify the destination address. It has the same format as the IP address, which is represented by four bytes separated by decimal points.
<table>
<tr><th>IP Class</th><th>Network Address</th><th>Subnet Mask</th></tr>
<tr><td>Class A</td><td>1.x.x.x~126.x.x.x</td><td>255.0.0.0</td></tr>
<tr><td>Class B</td><td>128.0.x.x~191.255.x.x</td><td>255.255.0.0</td></tr>
<tr><td>Class C</td><td>192.0.0.x~223.255.255.x</td><td>255.255.255.0</td></tr>
</table>
Subnet Mask must be set when the IP address is set on each computer. For Class C in the table above, the first 3 bytes of the IP address are Network ID. Therefore, the first 3 bytes of the Subnet Mask are all 255. The last byte is the Host ID, which is 0.</td></tr>
</table>

Device

Set the connection parameters between HMI and external devices.



❶ Device list

- A USB device has been added by default and cannot be deleted.
- Click to add a COM Port device. When all COM Ports are occupied, new devices cannot be added.

Note: The device name is defaulted as Barcode scanner and cannot be changed.

❷ Detail

Function	Description
Device interface	• The default interface is **USB**. • After adding a device, select an unoccupied COM Port.
Communication Parameter	• This setting is only available when using a COM Port device. • This setting is the same as that of COM Port.
Read length	Select or enter the string length (not including the terminator) for reading data. The range value is 10–1024.
Terminator	Choose the terminator. The available options are None (No Terminator), CR/LF, STX, ETX, STX/ETX, and Other. **Note**: **Other** is custom hexadecimal encoding, and the range value is 0X00~0XFF.
Timeout	Enter the time interval between scanning and reading the first character and the terminator. The range value is 0.1–30.0.

Note: For the reading setting of the external devices, see the content of **Internal Parameter** > **Input Device** in the Address Settings.

Common Properties

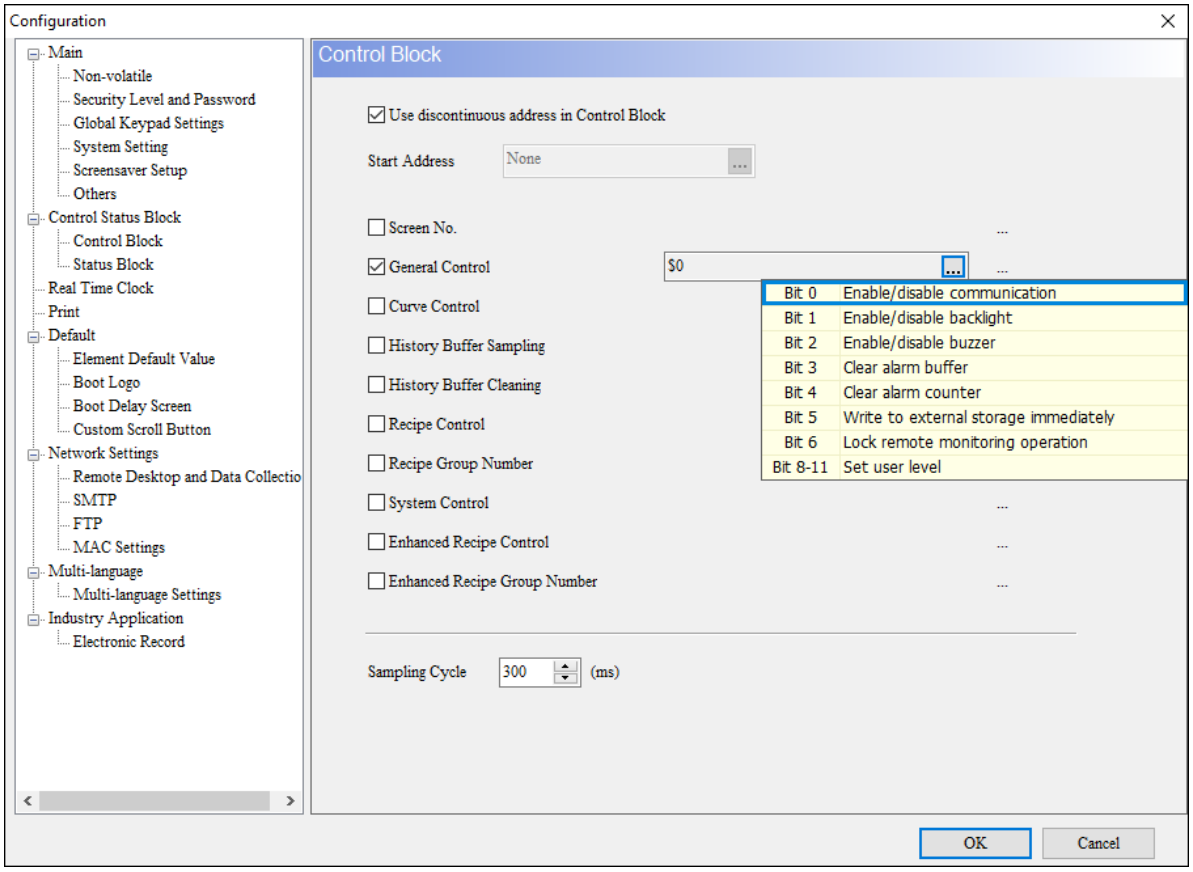
Controller Settings

Function	Description
PLC Station	**PLC Station** is the default station ID generated automatically after selecting the controller and can be changed. The range value is 0–255.
Password	Enter the password. If the set PLC requires a password check, you must set the corresponding password for communication in the software. The default value is **12345678**.

Function	Description
Comm. Delay Time	Select or enter the time interval between each communication. The range value is 0–255 and the default value is **0**.
Connection Timeout	Select or enter the connection timeout that HMI communicates with the device. The range value is 10–2000 and the default value is **1000**.
Connect Retries	Select or enter the retries. If the controller fails to respond to the communication, the HMI sends the communication command again. If reaching the set number of retries, the HMI displays a warning message of abnormal communication. The range value is 0–15 and the default value is **2**.
Optimize	• If selected, the process of reading the address of all elements referring to this link gets optimized to facilitate communication. • If unselected, the speed for reading the elements becomes slow.
For COMMGR Communication Simulation	Select the simulator driver configured in COMMGR to communicate with COMMGR simulator through the **Off-line Simulation** function. **Note**: Download and install COMMGR in DIAInstaller.

Disconnect after communication interrupt

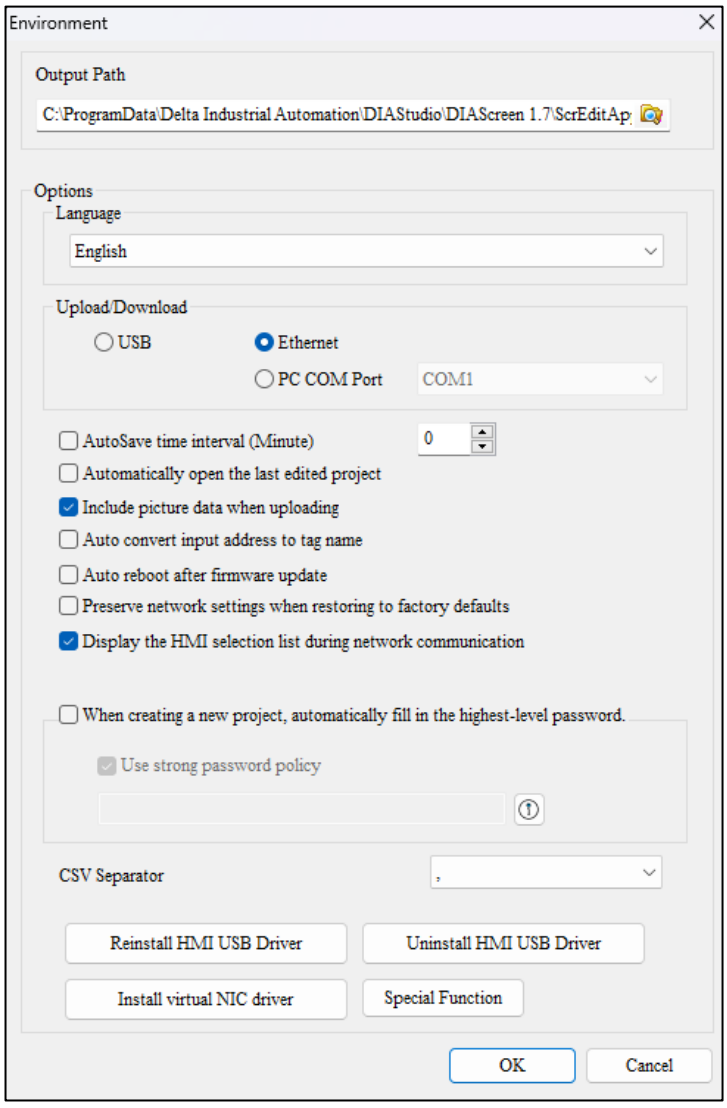
- If **Disconnect after communication interrupt** is selected, **Connect Retries** can be set. When the communication is interrupted and the set **Connect Retries** is reached, the HMI will no longer connect with the controller. The range value of **Connect Retries** is 0–255 and the default value is **3**.
- When the communication between the HMI and controller stops due to reaching the **Connect Retries**, use Bit 0 in the **Control Block** to enable / disable the communication to restore the connection.


Configuration
Main
Non-volatile
Security Level and Password
Global Keypad Settings
System Setting
Screensaver Setup
Others
Control Status Block
Control Block
Status Block
Real Time Clock
Print
Default
Element Default Value
Boot Logo
Boot Delay Screen
Custom Scroll Button
Network Settings
Remote Desktop and Data Collectio
SMTP
FTP
MAC Settings
Multi-language
Multi-language Settings
Industry Application
Electronic Record
Control Block
Use discontinuous address in Control Block
Start Address
None
Screen No.
General Control
$0
Curve Control
History Buffer Sampling
History Buffer Cleaning
Recipe Control
Recipe Group Number
System Control
Enhanced Recipe Control
Enhanced Recipe Group Number
Sampling Cycle
300
(ms)
Bit 0 Enable/disable communication
Bit 1 Enable/disable backlight
Bit 2 Enable/disable buzzer
Bit 3 Clear alarm buffer
Bit 4 Clear alarm counter
Bit 5 Write to external storage immediately
Bit 6 Lock remote monitoring operation
Bit 8-11 Set user level
OK
Cancel

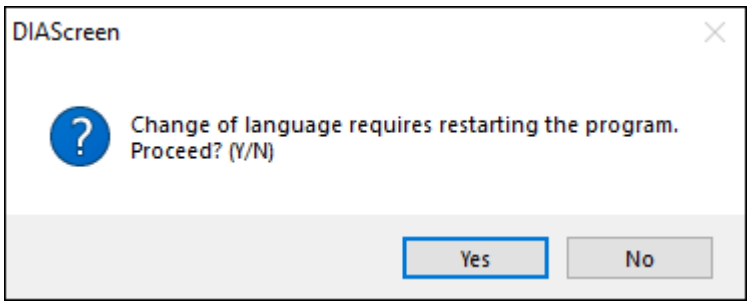
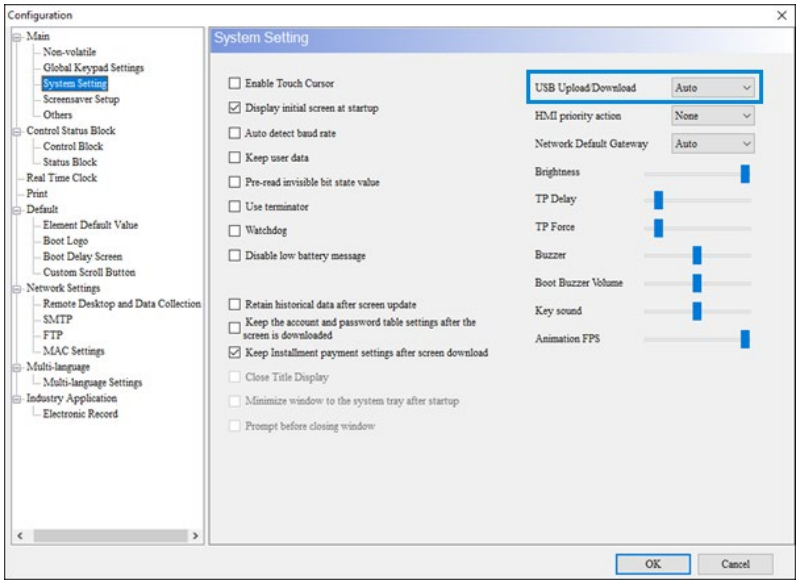
Environment Setting

You can set the relevant environment parameter of the system such as the interface language of the software, download mode, upload / download setting, and USB driver installation through the **Environment** function.

The following table lists the settings in the **Environment** dialog with their description.



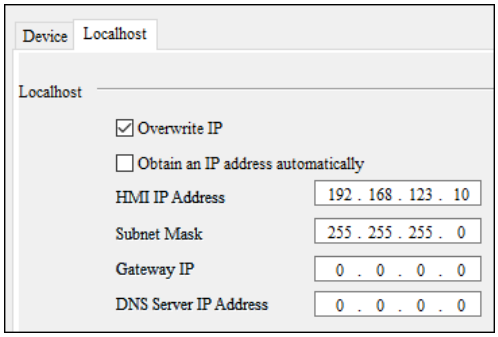
Setting	Description
Output Path	Select the CIN file path to be output after compiling the screen. Some functions of the software, such as on-line / off-line simulation and file upload / download, will refer to the data under this path. If it is not necessary or the version has not changed, do not modify this path to avoid program execution or file not found errors.

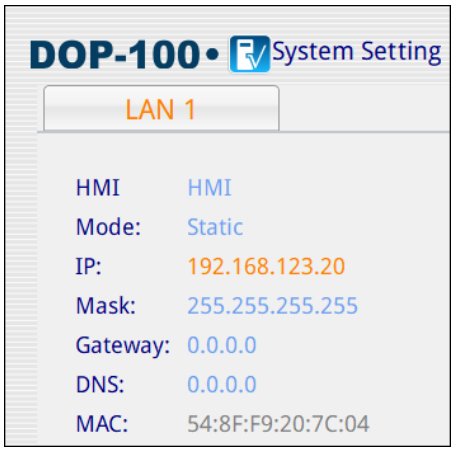
<table>
<tr><th>Setting</th><th colspan="2">Description</th></tr>
<tr><td>Language</td><td colspan="2">Select the interface language of the software.
After selection, a message dialog will pop up asking whether to proceed. Click Yes, and the software will automatically close and open with the language changed.
 
</td></tr>
<tr><td>Upload / Download</td><td>USB</td><td>USB is the default transmission mode for data uploading / downloading between the software and HMI. You can go to Configuration > USB Upload/Download to set transmission mode:
 

AUTO: The software uses the USB upload / download mode currently set in the HMI system menu.
USB Disk (USBCommMode = 1): Upload / download the HMI programs without installing the driver.
USB CDC (USBCommMode = 2): You can click Reinstall HMI USB Driver to install the driver for transmission.
None: You can’t use USB to upload and download.
</td></tr>
</table>

<table>
<tr><th>Setting</th><th colspan="3">Description</th></tr>
<tr><td></td><td>Ether net</td><td colspan="2">Use the Ethernet to upload / download the data to the HMI. If the Ethernet is used, you must set the IP address of both the HMI and the computer on the same network segment. You can customize or automatically obtain the HMI IP address.</td></tr>
<tr><td></td><td></td><td>Customize IP</td><td>

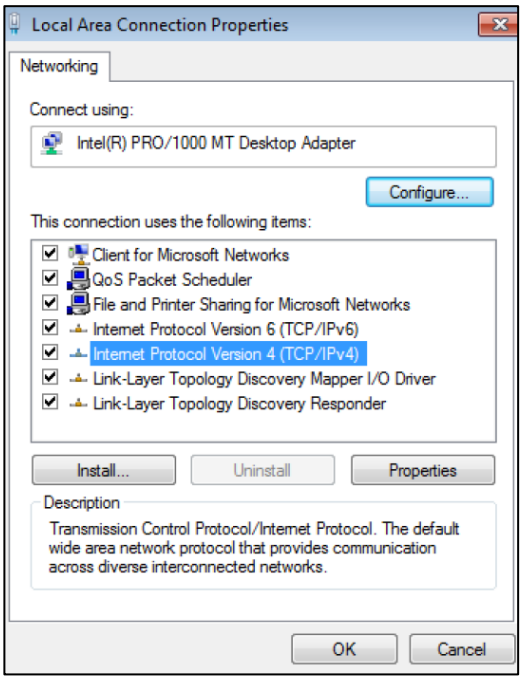
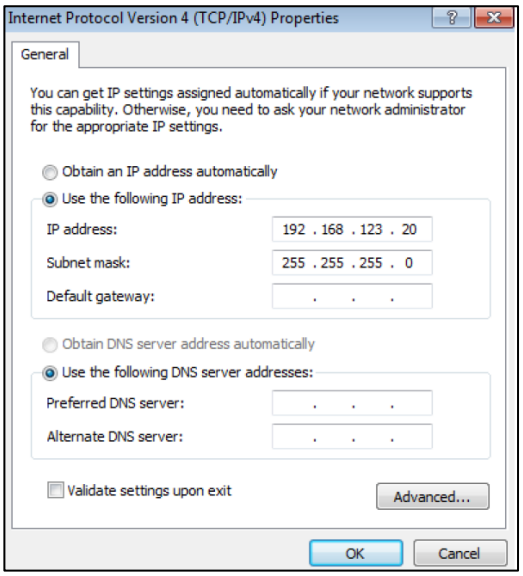
HMI

1. Go to General > Device Communication.
2. Set the IP address. Do any of the following:
○ On the Ethernet1 page of the Device Communication window, customize the IP address in the Local tab.


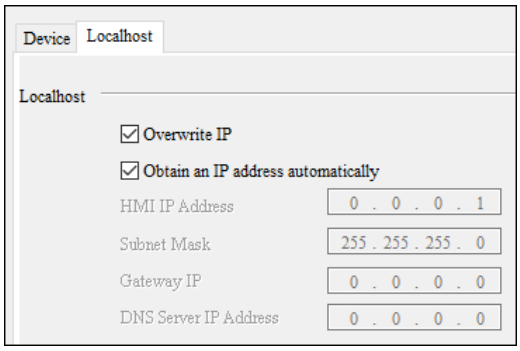
Or
○ On the HMI screen, go to System Setting > Network to set the IP address.


Computer

1. Select Network and Sharing Center in the control panel, and then select Local Area in the Network Connections dialog. Select Internet Protocol Version 4 (TCP/IPv4) in</td></tr>
</table>

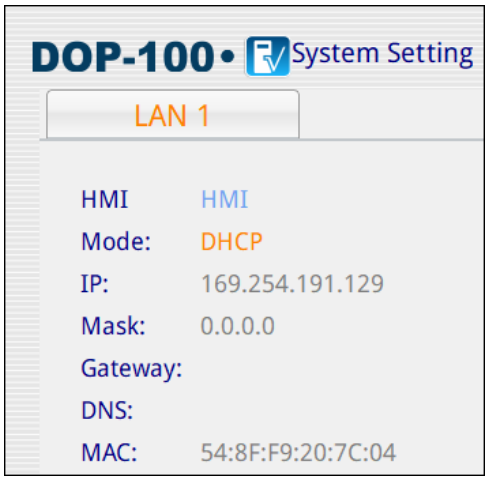
				the **Local Area Connection Properties** dialog.  2. Click **Properties** to set TCP/IP. 
			Obtain IP automati-cally	• **HMI** 1. Go to **General** > **Device Communication**. 2. Set to obtain an IP address automatically, meaning you do not need to set the IP address. DHCP dynamically assigns the IP address to the HMI instead. Do any of the following: ○ On the **Ethernet1** page of the **Device Communication** dialog,

select the **Obtain an IP address automatically** checkbox in the Localhost tab.



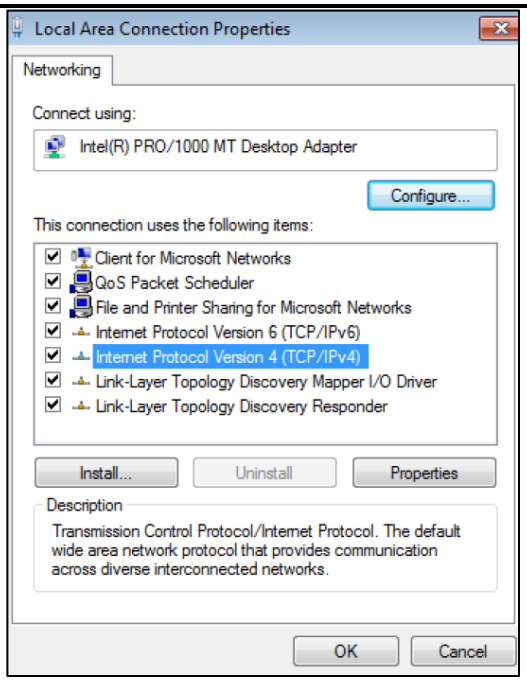
Or

- On the HMI screen, go to **System Setting** > **Network** to set the **Mode** to **DHCP**.

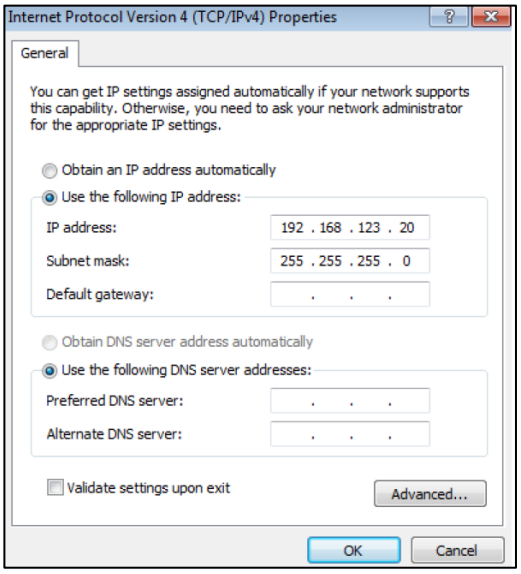


- **Computer**

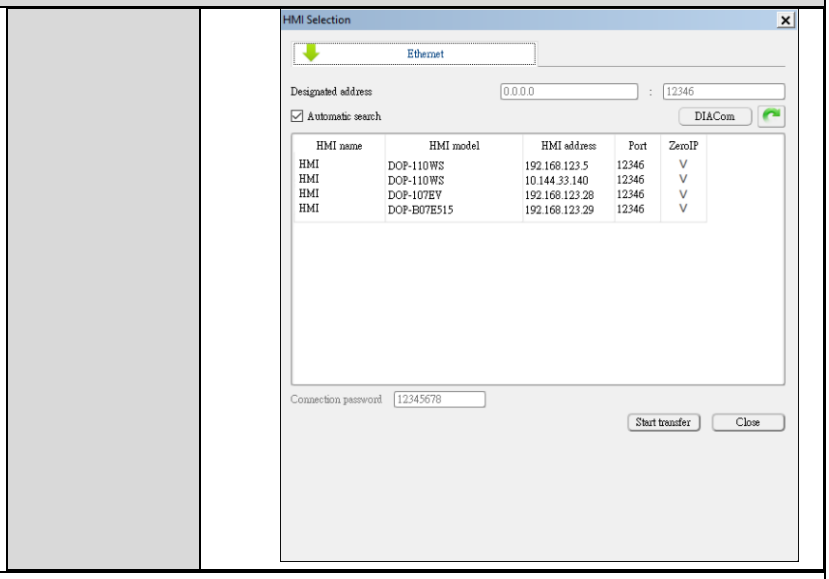
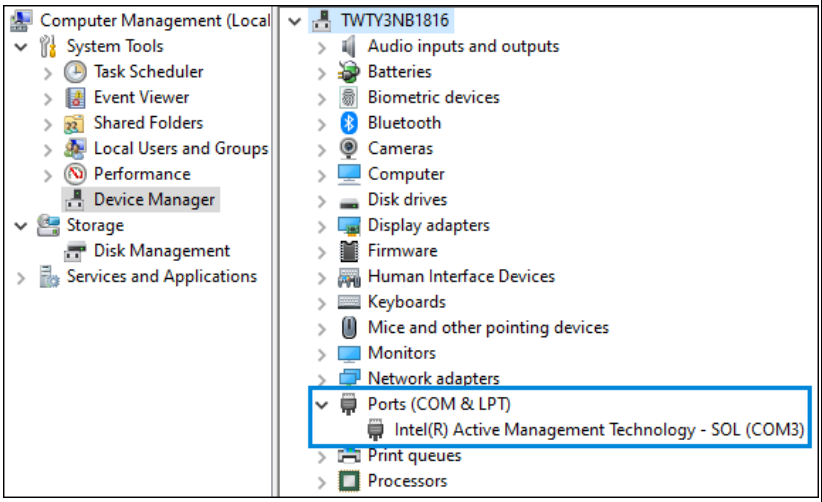
1. Select **Network and Sharing Center** in the control panel, and then select **Local Area** in the **Network Connections** dialog. Select **Internet Protocol Version 4 (TCP/IPv4)** in the **Local Area Connection Properties** dialog.

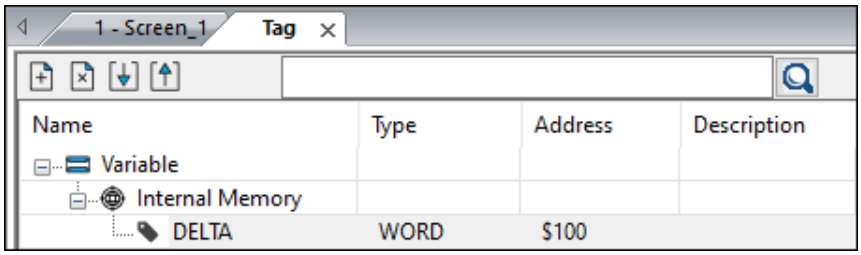
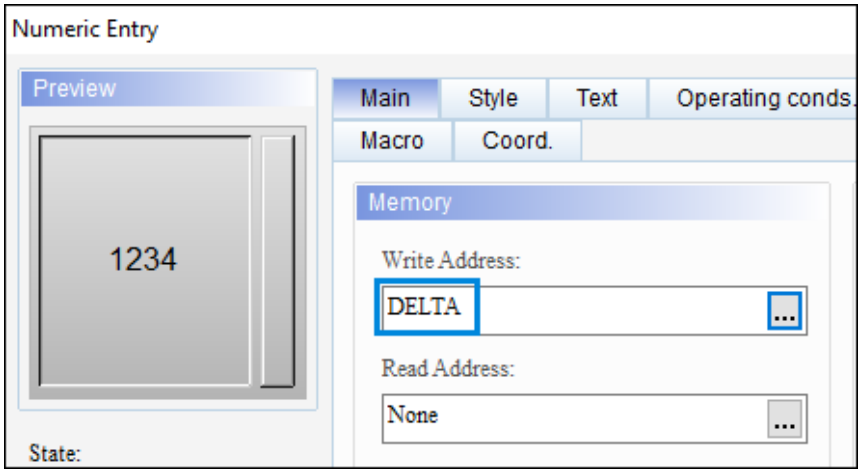


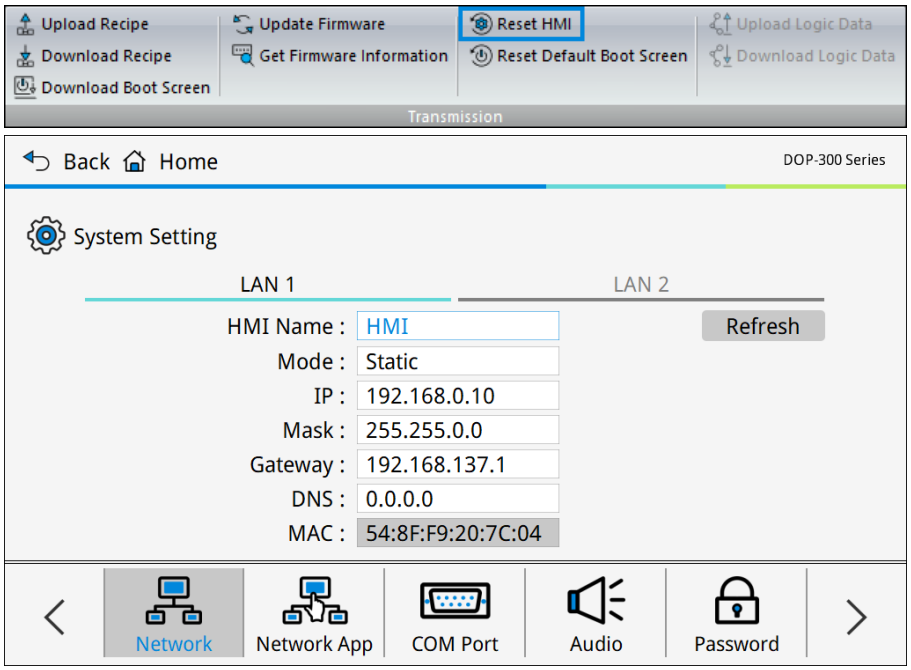
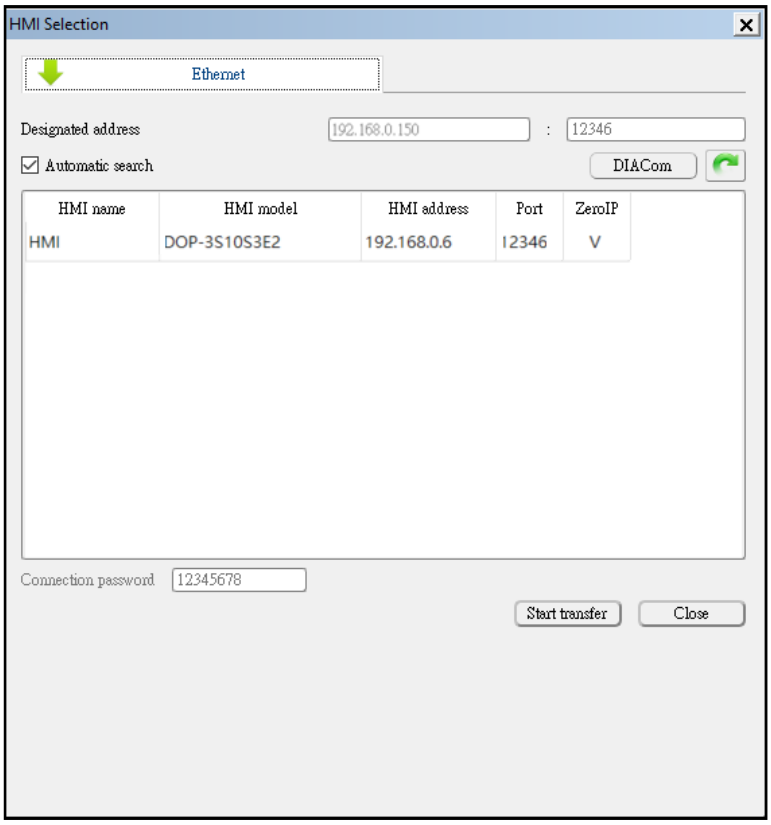
2. Click **Properties** to set TCP/IP.

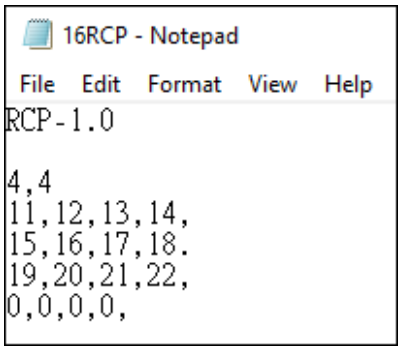
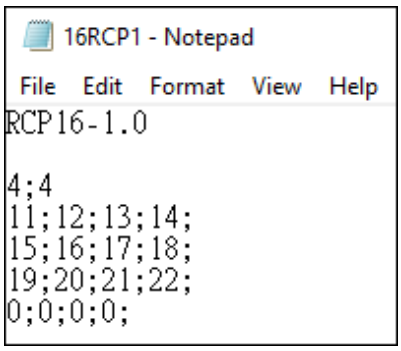
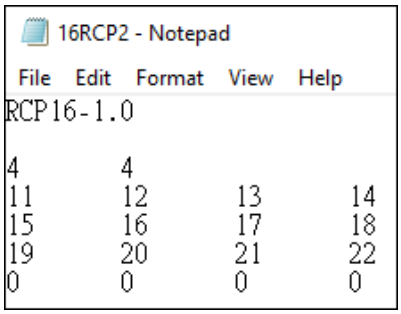
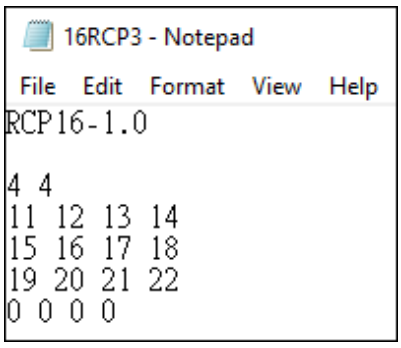


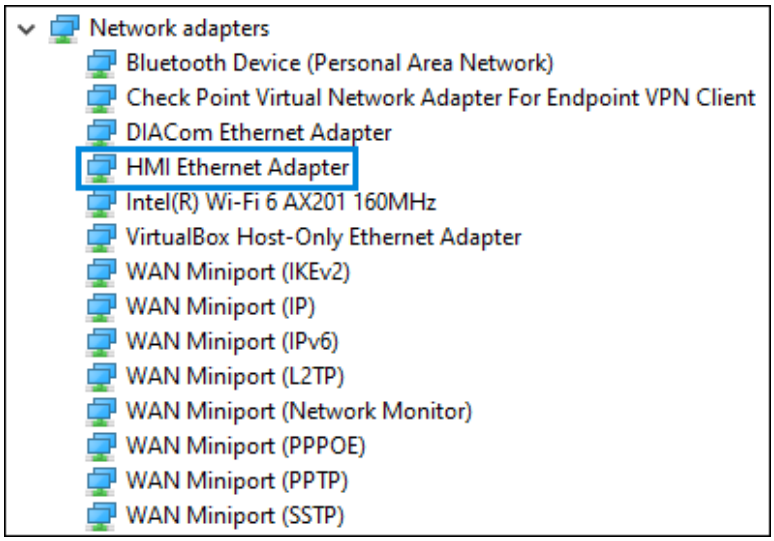
3. After the IP addresses of the HMI and computer are set in the same network segment, the HMI upload / download can be done via Ethernet. When clicking **Download Screen** or **Upload All Data** on DIAScreen, the software will automatically search for the HMI with the same network segment. Click **Start transfer** to download or upload data to HMI.

Setting	Description
	
PC COM Port	PC COM Port is for data upload / download between the PC COM port and HMI. If **PC COM Port** is selected, you must first enter the HMI system screen > **Up/Download** > **Standard Mode** to select COM1 or COM2 and wait for the software to set the port number for the PC COM Port. Right-click **This PC**, and then select **Manage**. You can find the port number in the **Computer Management** dialog. 
AutoSave time interval	Select and set the time interval, and the software will automatically save the project according to the set time interval. • If not selected, the minimum value is 0. • If selected, the minimum value is 3, up to 120 can be set.
Automatically open the last edited project	If selected, the software will automatically open the previously edited project. When closing the project and restarting

Setting	Description
	DIAScreen, the screen will display the previously edited project.
Include picture data when uploading	If not selected, if the edited project contains picture data, the picture data content will be lost when the screen data is uploaded to the HMI.
Auto convert input address to tag name	If selected, the memory address of the element to be created will be automatically displayed as the Tag name Follow these steps to change the input address to Tag. 1. Double-click **Tag** in the **Project** pane and edit the tag data.  2. Go to **General > Environment** and select **Auto convert input address to tag name** checkbox. 3. Create a **Numeric Entry** element, and then open the Input dialog to set **Write Address** to **$100**. After closing the Input dialog, the address will display as the created Tag name. 
Auto reboot after firmware update	If selected, the HMI will restart after the firmware is updated and the message “Update Firmware Succeed!!!” will not show.

Setting	Description
Preserve network settings when restoring to factory defaults	If selected, the **Network** setting in the **System Settings** can be preserved through the **Reset HMI** function. 
Display the HMI selection list during network communication	• If not selected, the **HMI Selection** dialog will display when connecting to the network.  • If selected, the H**MI Selection** dialog will not display when connecting to the correct models.

<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>When creating a new project, automatically fill in the</td><td>• If selected, you can enter the default highest security password, and it will be automatically filled in when creating a project.
• If not selected, the highest security password must be entered to create a project.</td></tr>
<tr><td>CSV Separator</td><td>Select the separator to be displayed in the output CSV file.
<table>
<tr><td>,</td><td> 
</td></tr>
<tr><td>;</td><td> 
</td></tr>
<tr><td>TAB</td><td> 
</td></tr>
<tr><td>SPACE</td><td> 
</td></tr>
</table>
Note: You can use Notepad to open and view the separator displayed in the CSV file.</td></tr>
</table>

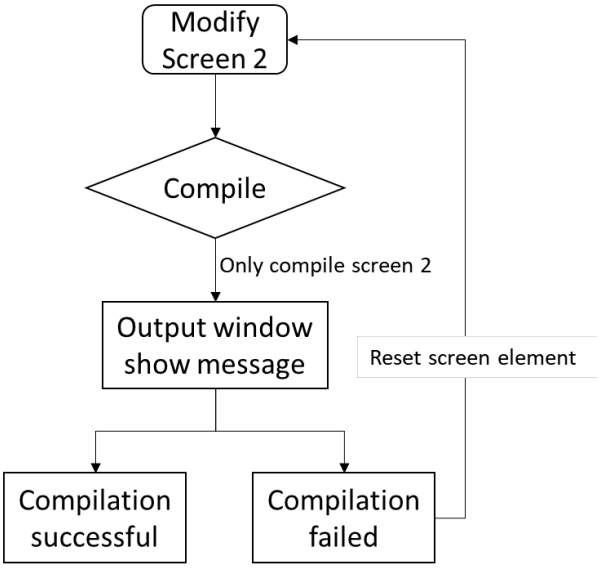
<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Reinstall HMI USB Driver</td><td rowspan="2">After clicking Reinstall HMI USB Driver, the system will reinstall the HMI USB driver.
After clicking Uninstall HMI USB Driver, the system will uninstall the HMI USB driver and reinstall.
Note: You can uninstall and reinstall the HMI USB driver when you are unable to upload / download via USB transmission. Available only when the USB upload / download is in Normal and CDC transmission mode.</td></tr>
<tr><td>Uninstall HMI USB Driver</td></tr>
<tr><td>Install virtual NIC driver</td><td>Click to reinstall the virtual NIC driver.
 
</td></tr>
</table>

Compile

DIAScreen provides the compiling function for convenient operation and use.

Compile

If only one screen among several screens is edited, you can use the **Compile** function to compile only that screen to save compilation time.



How to use the **Compile** function?

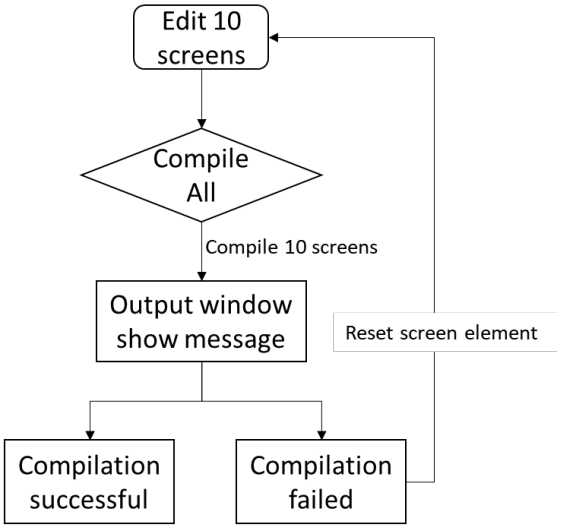
- Click **Project** > **Compile** on the menu bar.

Or

- Press **Ctrl + F7**.

Compile All

Use the **Compile All** function to compile all screens. If an error occurs during compilation, an error messages will be displayed in the **Output** pane to remind users. Click the message to link to the element with the error.

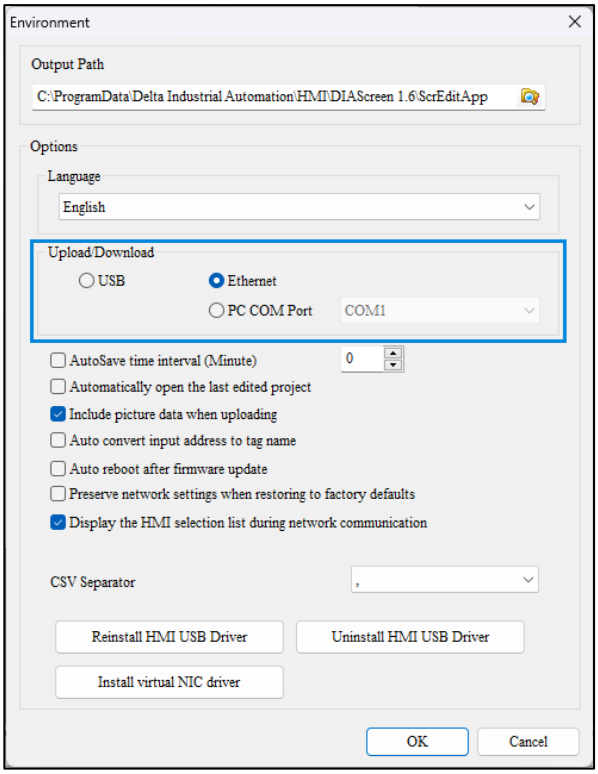


Download All Data

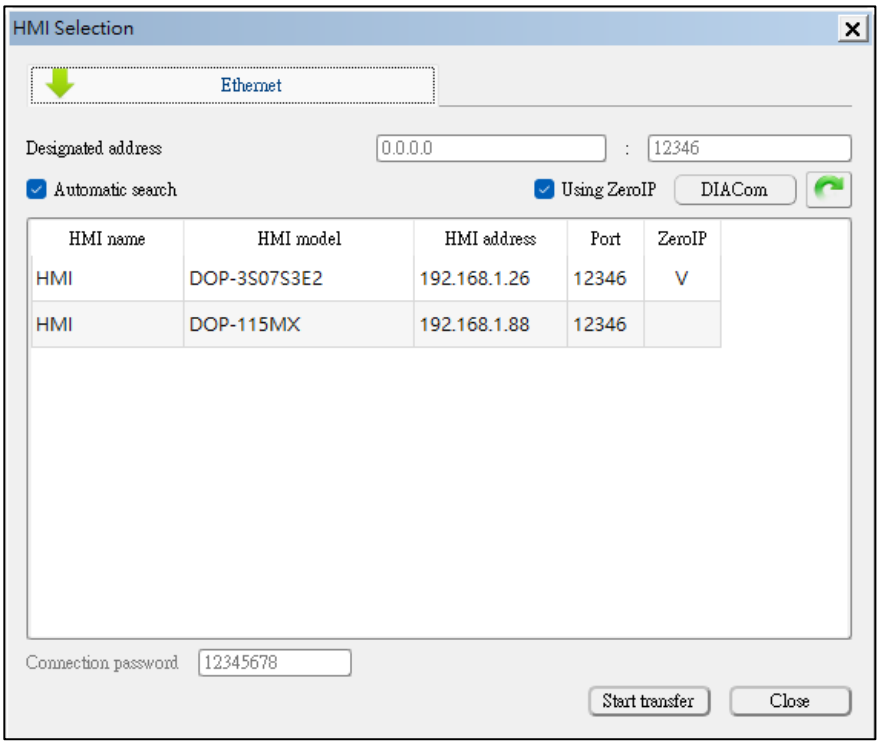
Use the **Download All Data** function to download the screen data and recipe to the HMI.

Use the Download All Data function

1. Click **General** > **Environment** on the menu bar and select the communication interface in the **Upload/Download** area.

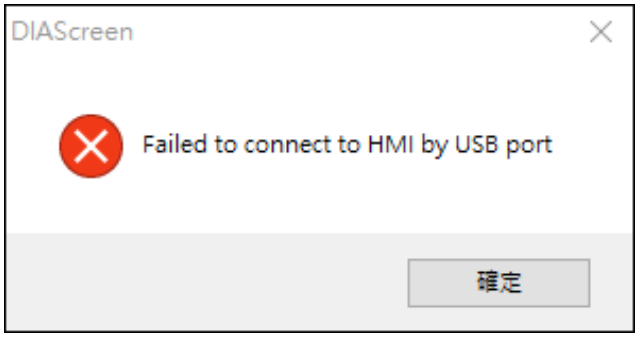


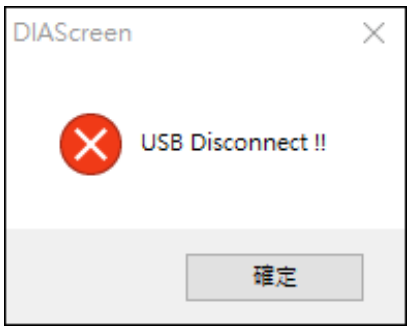
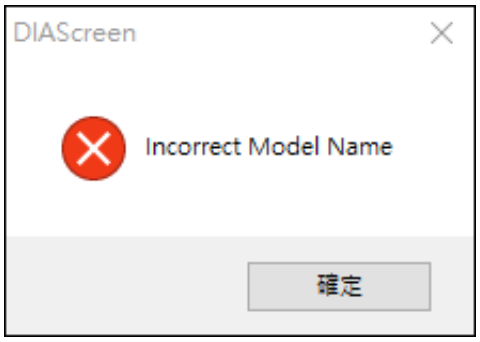
2. Click **Project** > **Download All** on the menu bar or press **Ctrl + F8**.
3. If select **Ethernet** for the **Upload/Download**, you can enter the HMI IP address in the **Designated address** in the **HMI Selection** dialog or select the **Automatic search** checkbox to search for available HMI.



Note: If DOP-100 series Gen2 or DOP-300 series HMI is being used, you can select the **Using ZeroIP** checkbox to download the screen.

When downloading all data, the software will detect the connection between the HMI and the computer. If the transmission interface between the HMI and the computer is not turned on, an error message will appear. The following table lists the check items with their description.

Check Item	Description
USB Tunnel	USB is turned on incorrectly. 

Check Item	Description
	During the download process, the transmission line is unplugged, or the communication is interrupted. 
Model Setting	The HMI model is set incorrectly. 

Note: Perform the DIAScreen uploading and downloading on an authorized HMI.

Download Screen

The **Download Screen** function downloads only screen data. Refer to Download All Data for the download method.

How to use the **Download Screen** function?

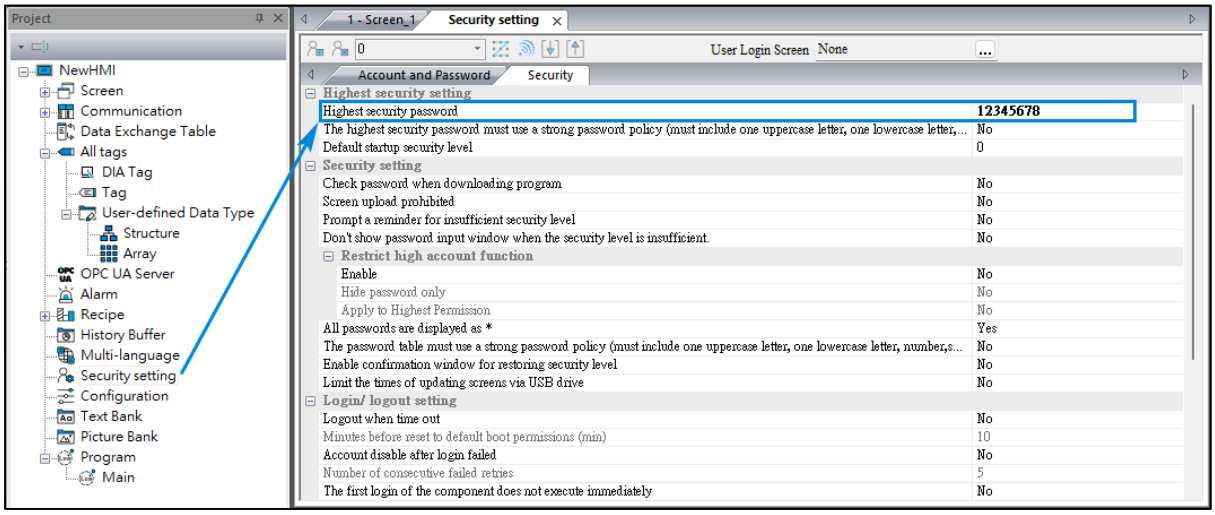
- Click **Project** > **Download All** on the menu bar.

Or

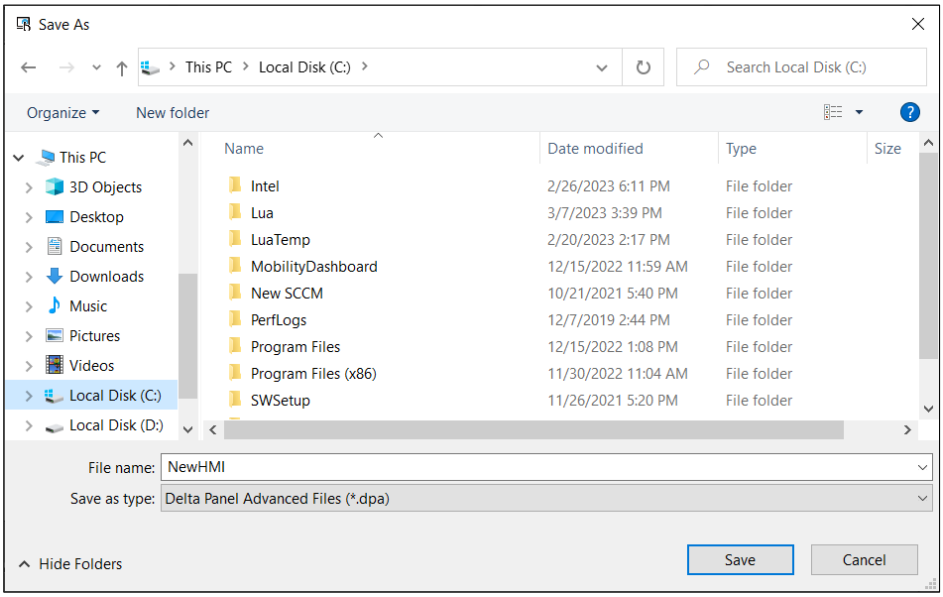
- Press **Ctrl + F9**.

Upload All Data

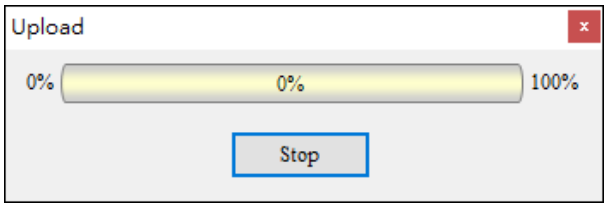
When uploading all data, the software will ask you to enter **highest security password**. To set a password, go to **Project** > **Security Setting** > **Security**.



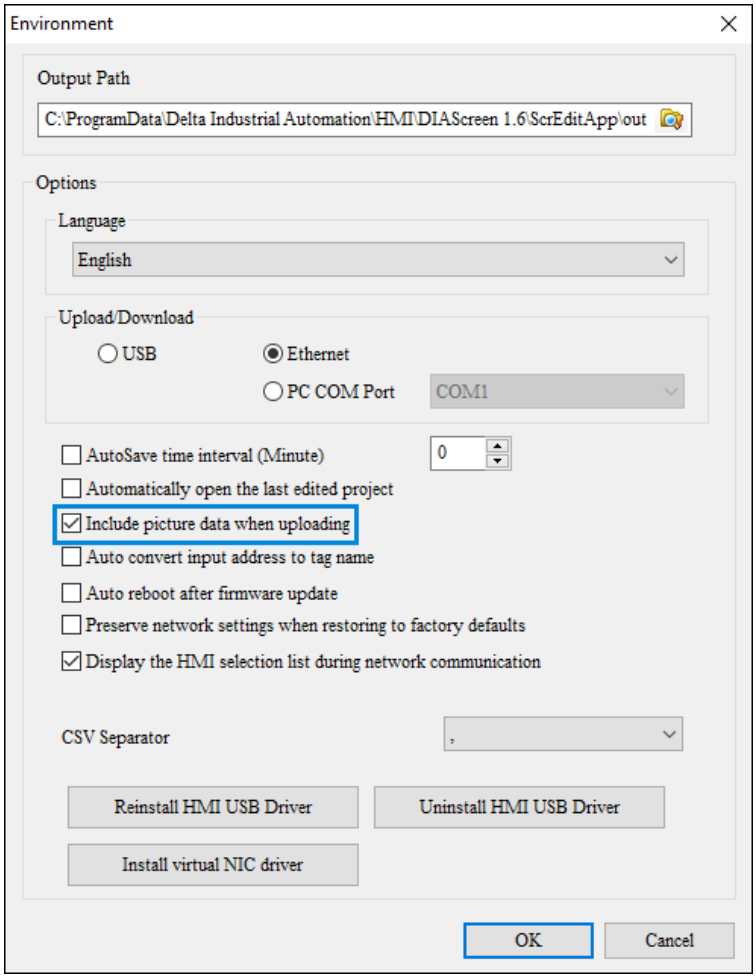
After entering the password, save the screen file to upload.



After saving the file, the screen data starts to upload. You can click **Stop** to stop uploading the screen data.

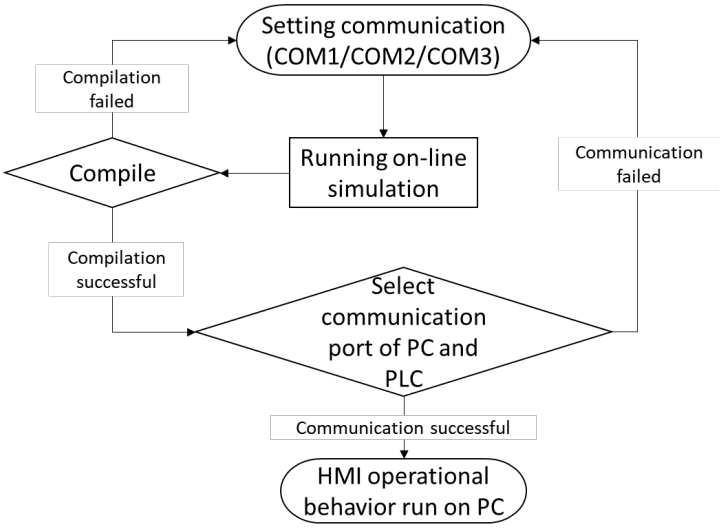


Besides uploading screen data, you can also go to **General** > **Environment** to set whether to include picture data.



On-line Simulation

Use the **On-line Simulation** function to simulate the computer as an HMI and communicate with the controller through the computer's COM port. If the communication is normal, the computer can be used as an HMI for actual operation.



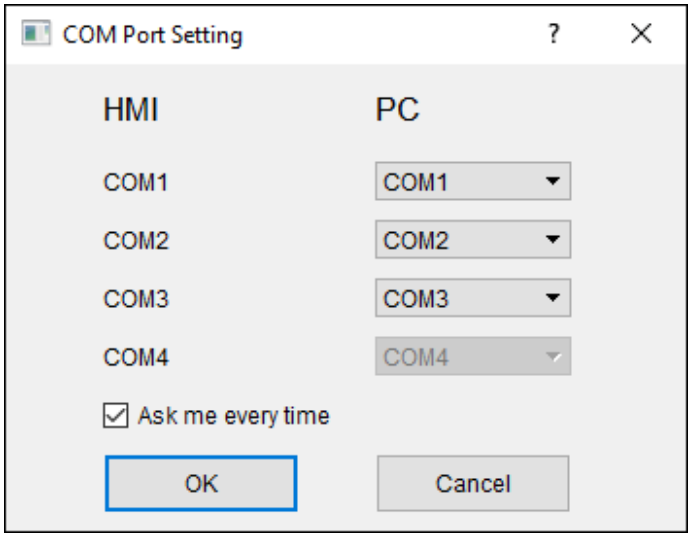
How to use the **On-line Simulation** function?

- Click **Project** > **On-line Simulation**.

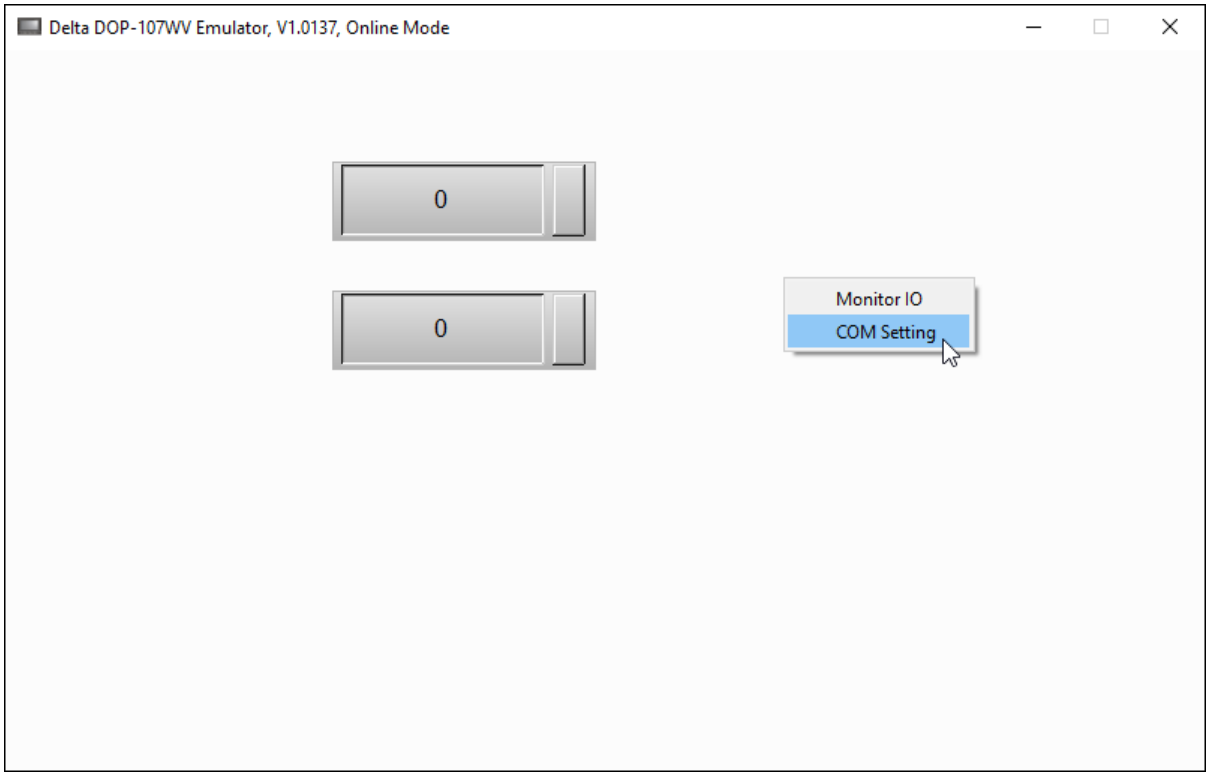
Or

- Press **Ctrl + F4**.

When performing an online simulation, the software will first compile and check whether the screen has errors. Then set the COM port number of the computer. The HMI will communicate with the controller if the setting is correct.

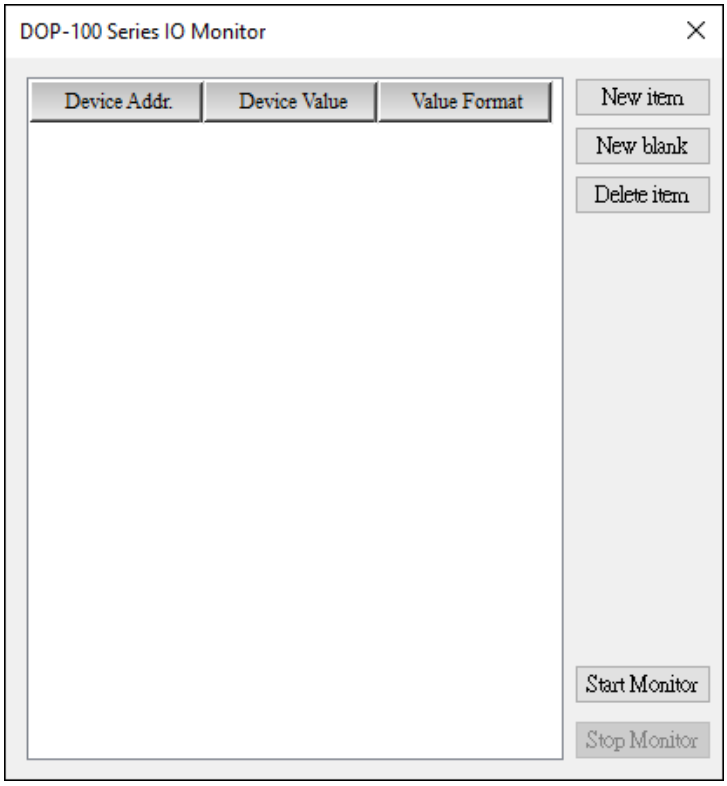


In the **COM Port Setting** dialog, if the **Ask me every time** checkbox is selected, this dialog displays every time you perform an online simulation. If unselected, the dialog will not display. You can right-click the blank space on the online mode screen and select **COM Setting** to open this dialog again.



Online simulation also provides a monitoring function. Right-click the blank space on the screen and select **Monitor IO**. After setting the monitoring address, you can monitor the content value of the IO device.

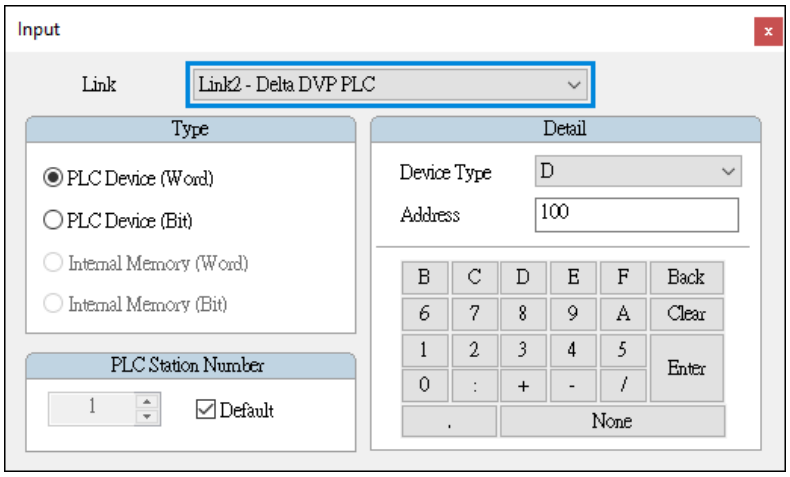
The following table lists the setting in the **IO Monitor** dialog with their description.



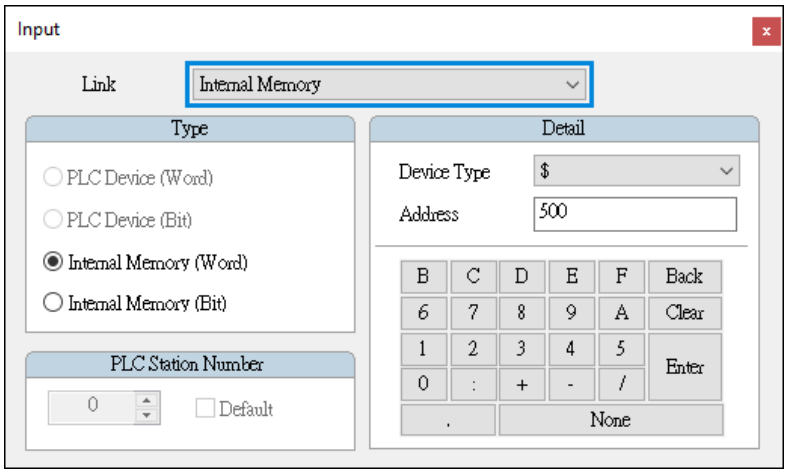
Setting	Description
New item	Click to add the address to monitor in the **Input** dialog.
New blank	Click to add a new item row, and you can directly enter the monitor address in the **Device Addr.** field.
Delete item	Click to delete the selected monitor address.
Start Monitor	Click to start monitoring the device.
Stop Monitor	Click to stop monitoring the device.
Device Addr.	Set the internal memory or controller register address.
Device Value	Enter the content value of the internal address or controller register. You can also change the content value of the device in real-time. **Note**: The Length setting is not required if you are using a Delta controller.
Value Format	Select the value format.

The following describes the operation example of **Monitor IO**.

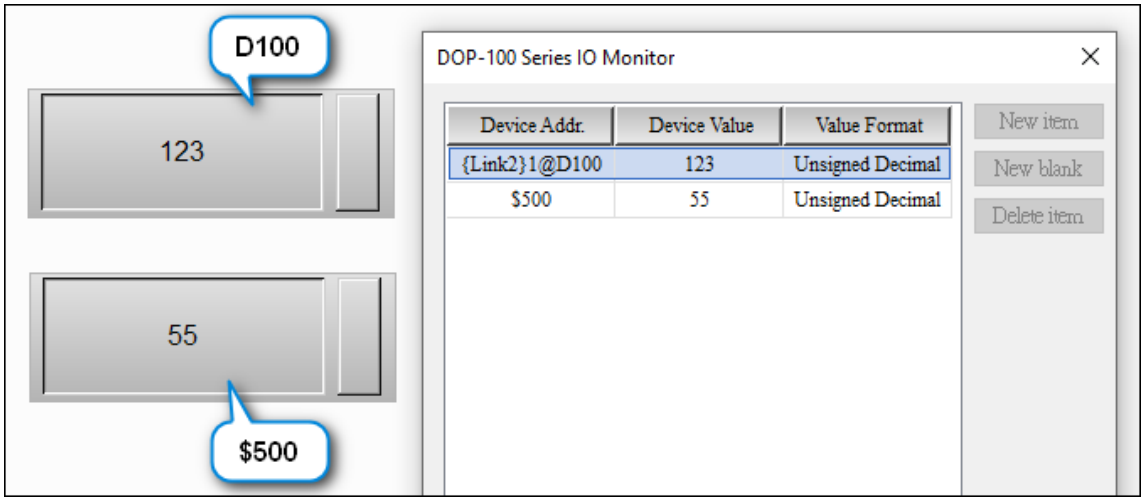
1. Create 2 **Numeric Entry** elements on the screen, and then set the **Write Address** to **$500** and **{Link2}2@D100** respectively.
2. Click **Project** > **On-line Simulation** on the menu bar.
3. On the online simulation screen, right-click the bank space and select **Monitor IO**.
4. Click **New Item** to add a monitor address in the **IO Monitor** dialog.
5. Set the monitor address to **{Link2}2@D100** in the **Input** dialog.



6. Repeat steps 1~2 to set another monitor address to **$500**.



7. Click **Start Monitor** to start monitoring the values of **{Link2}2@D100** and **$500** in the **Device Value** field. You can also manually change device values.



Off-line Simulation

Use the **Off-line Simulation** function to check whether the edited screen, the read and write memory address, and the Macro are correct. No communication with the controller is required.

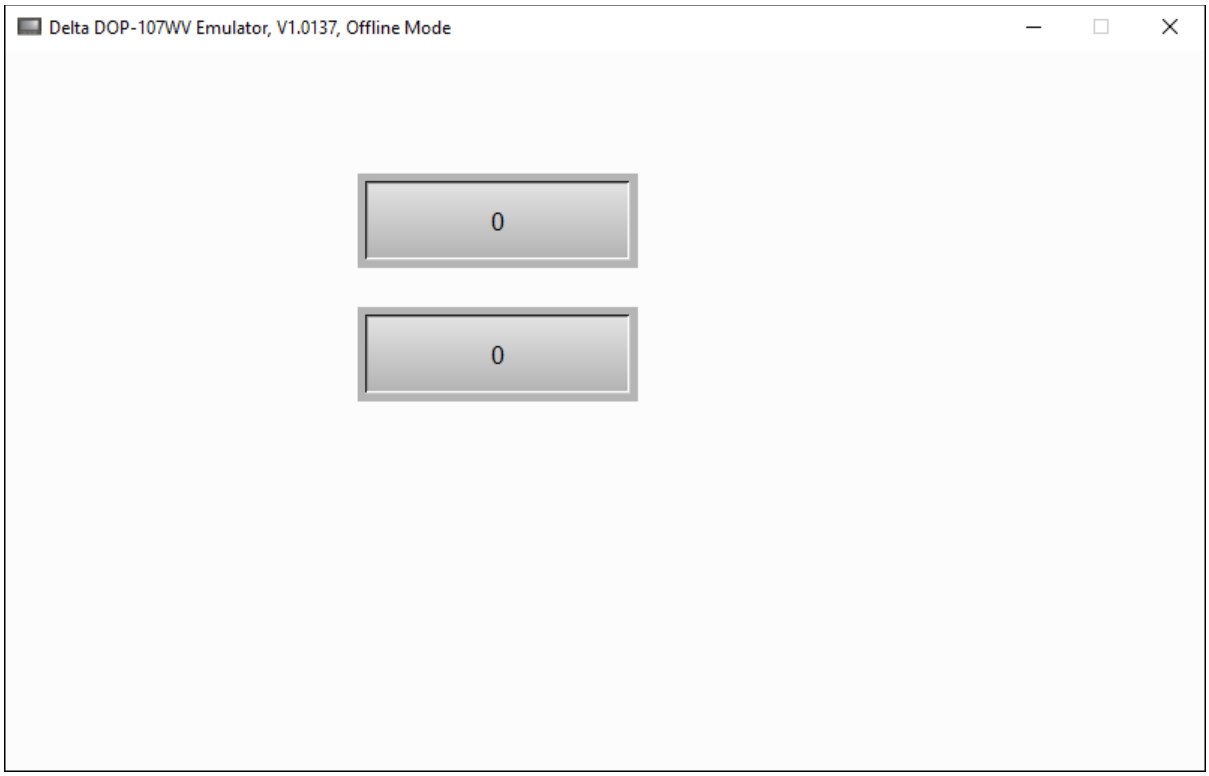
How to use the **Off-line Simulation** function?

- Click **Project** > **On-line Simulation**.

Or

- Press **Ctrl + F5**.

When performing an offline simulation, the software will first compile the screen and then enter the offline mode screen.

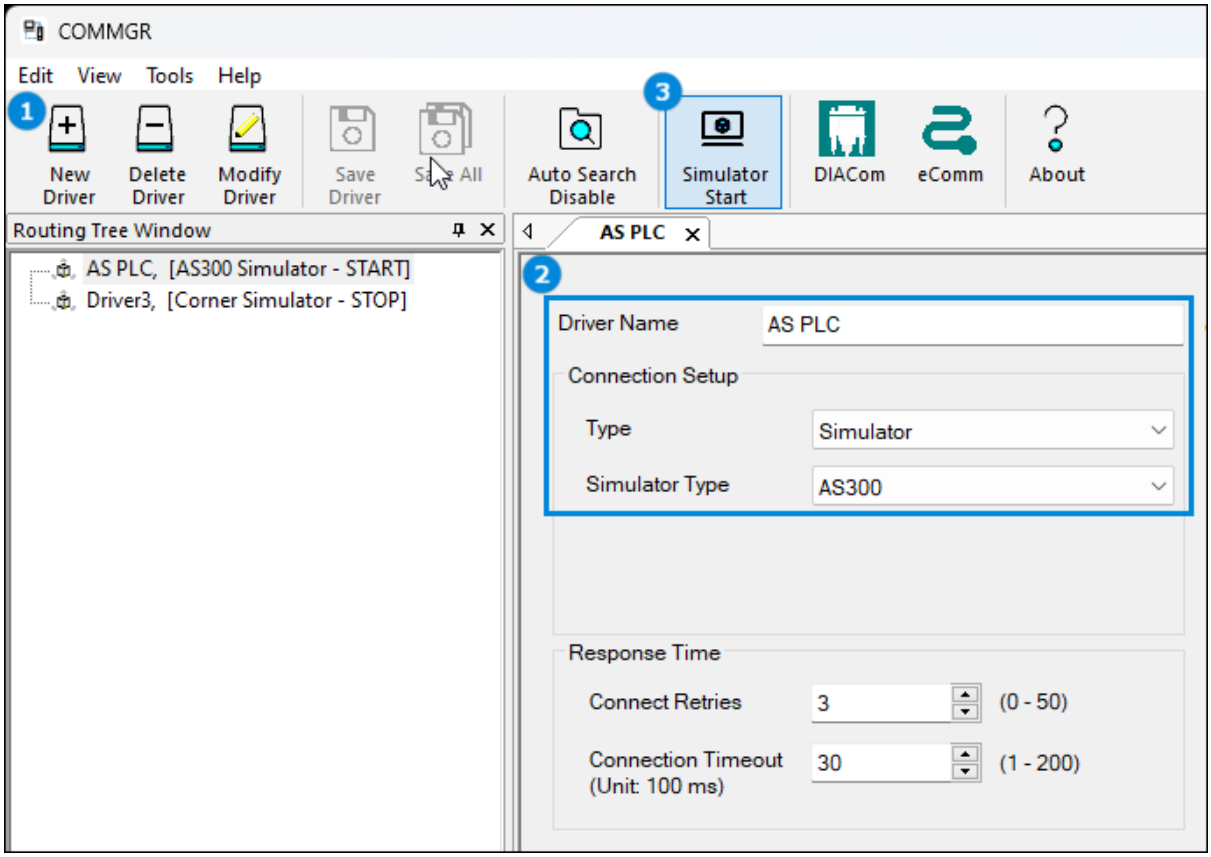


COMMGR Simulator Connection

The **Off-line Simulation** function of DIAScreen can communicate with the COMMGR simulator to realize the simulation application of HMI and PLC communication on the computer. The following describes the operation example of simulator connection.

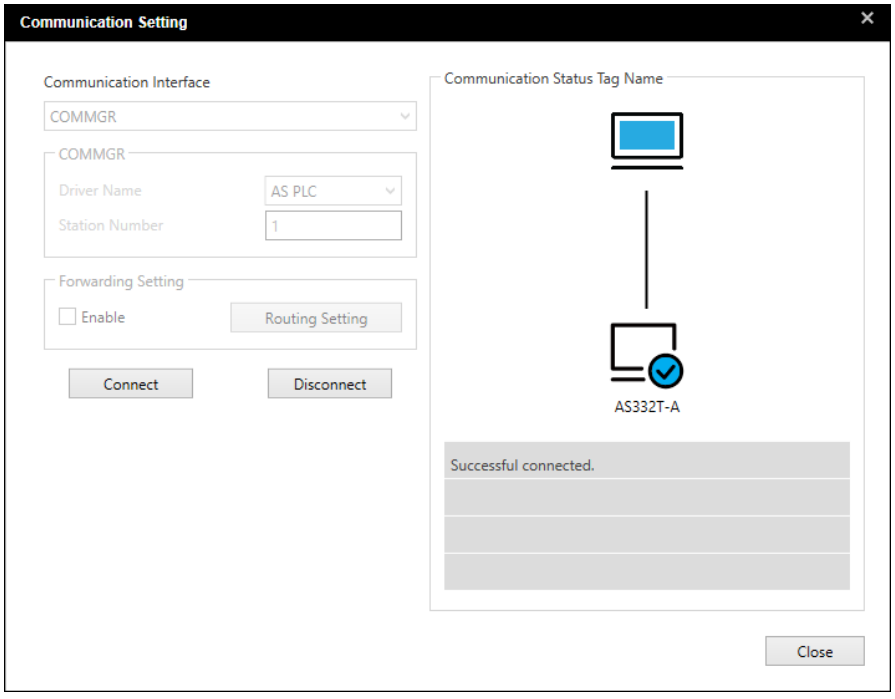
COMMGR Operation

1. In COMMGR, add a drive.
2. Enter the **Drive Name**, select **Simulator** for the connection type, and then select **Simulator Type**.
3. On the toolbar, click **Simulator** to start the simulator.

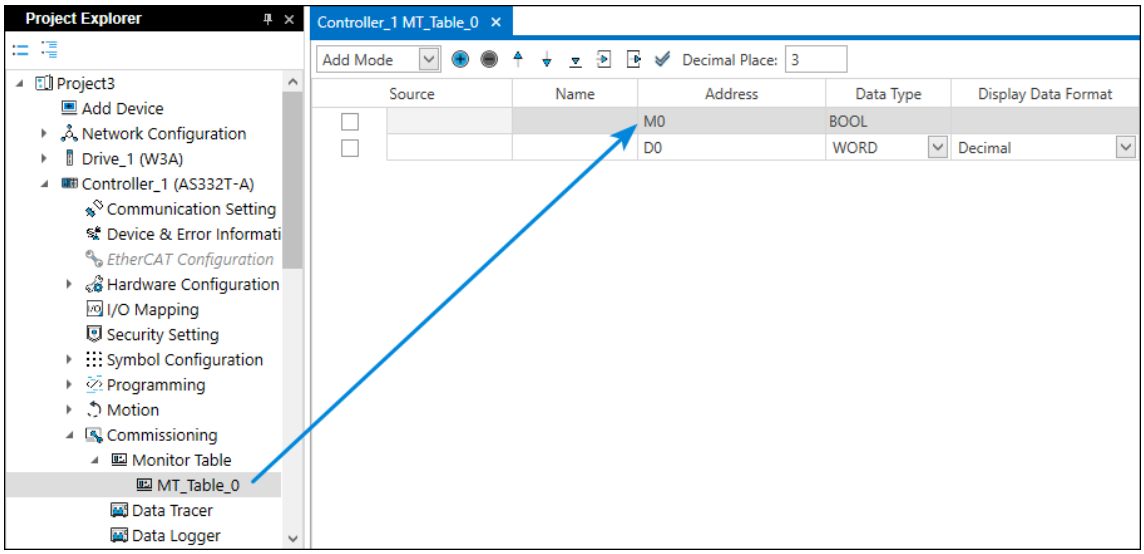


DIADesigner Operation

1. In DIADesigner, create a project and add a controller.
2. In the **Communication Setting** dialog, select the simulator drive created in COMMGR to establish the connection.

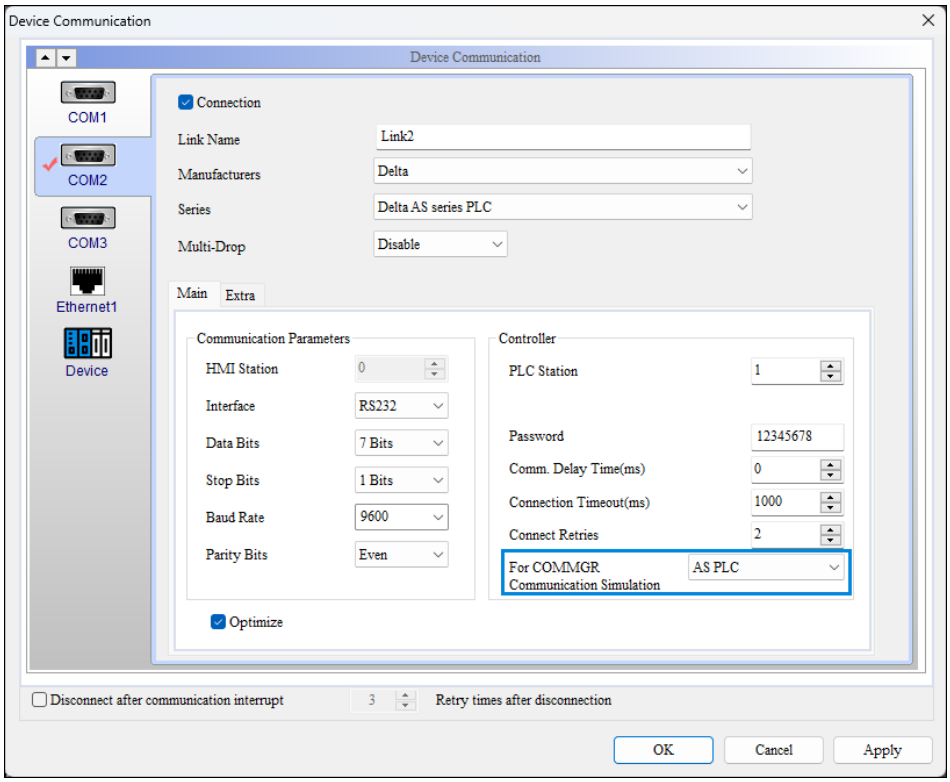


3. In **Project Explorer**, double-click **Controller** > **Commission**.
4. Right-click the **Monitor Table**, and then select **Add Monitor Table**.
5. In the **Monitor Table** page, add registers D0 and M0.

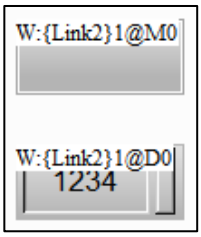


DIAScreen Operation

1. In DIAScreen, create a project.
2. In the **Device Communication** dialog, select a **Delta** controller, and then select simulator drive for the **For COMMGR Communication Simulation**.



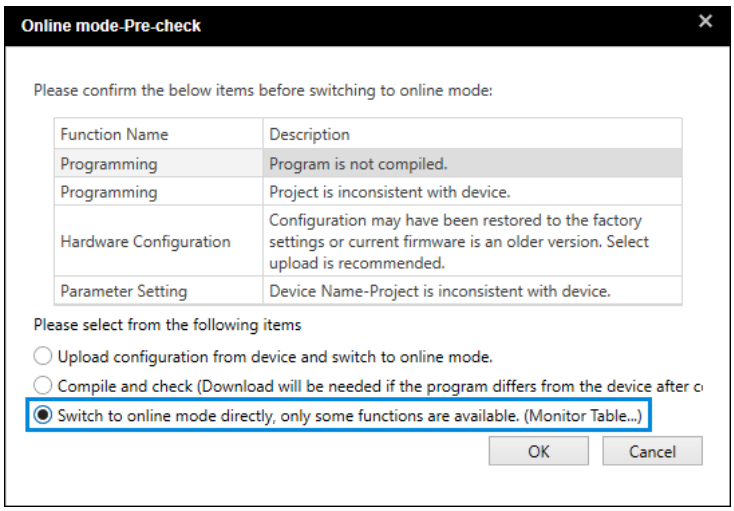
3. Create elements on the screen and set the read and write address to the address of PLC communication respectively.



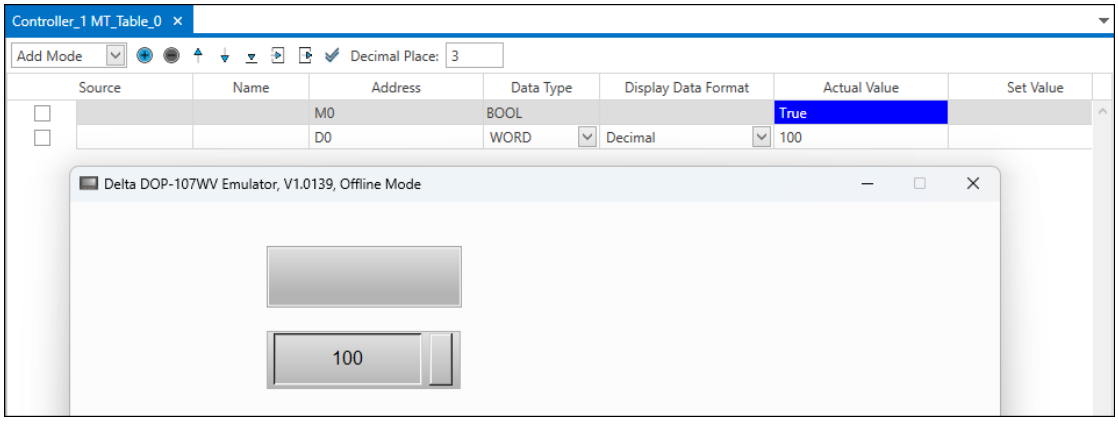
Execution Result

1. Enter the online mode in DIADesigner. Click

on the toolbar, and then select **Switch to online mode directly, only some functions are available** in the **Online mode-Pre-check** dialog.



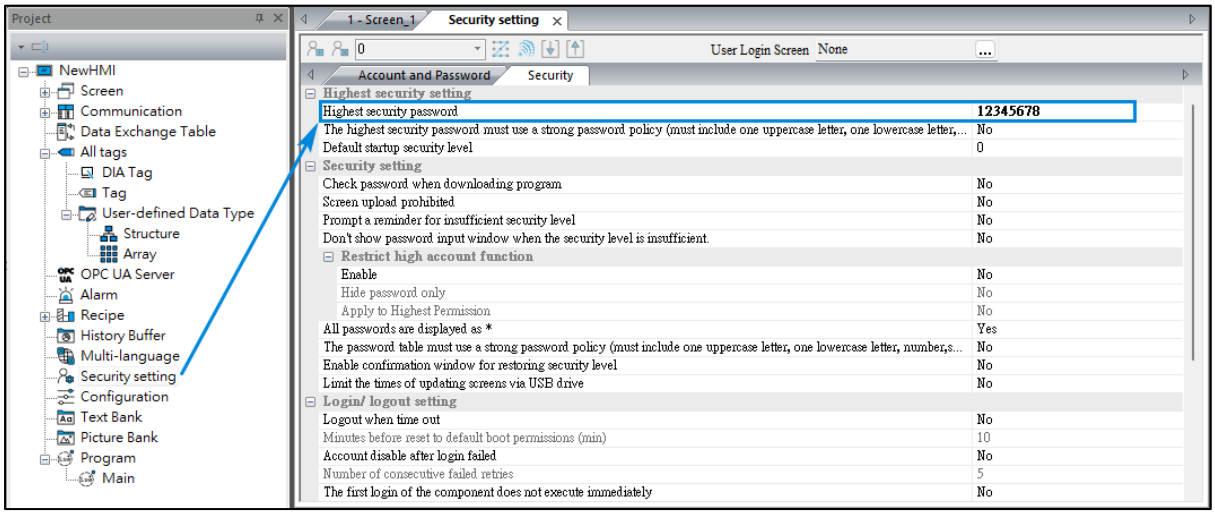
2. Perform offline simulation in DIAScreen. By entering the value into the operating element on the HMI simulator, you can view value changes in the monitor table of DIADesigner.



Transmission

Upload Recipe

When uploading the recipe, the software will ask you to enter **highest security password**. To set a password, go to **Project** > **Security Setting** > **Security**.



Note: The 16-bit recipes are stored as RCP files, and the 32-bit recipes or enhanced recipes are stored as CSV files.

Download Recipe

Use the **Download Recipe** function to only download the recipe.
How to use the **Download Recipe** function?

1. On the toolbar, click **Project** > **Download Recipe**.
2. In the **Open** dialog, select the recipe file to download to the HMI. For 16-bit recipe, select the .rcp file; for 32-bit recipe or enhanced recipe, select the .csv file.

Download Boot Screen

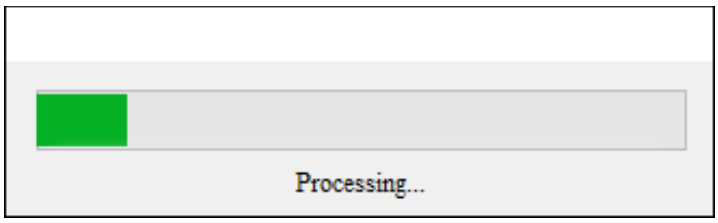
Use the **Download Boot Screen** function to only download the boot screen.

How to use the **Download Boot Screen** function?

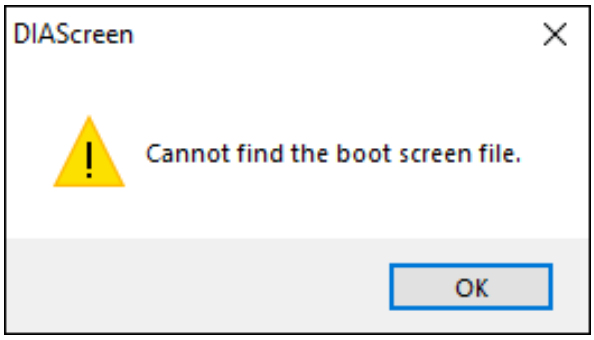
1. Go to **General** > **Configuration** > **Default** > **Boot Logo** and select the **Enable** checkbox.

2. On the toolbar, click **Project** > **Download Boot Screen**.

After performing, the screen displays the download dialog.



If the boot screen is enabled but no boot screen is selected, the following message dialog appears when downloading the boot screen.



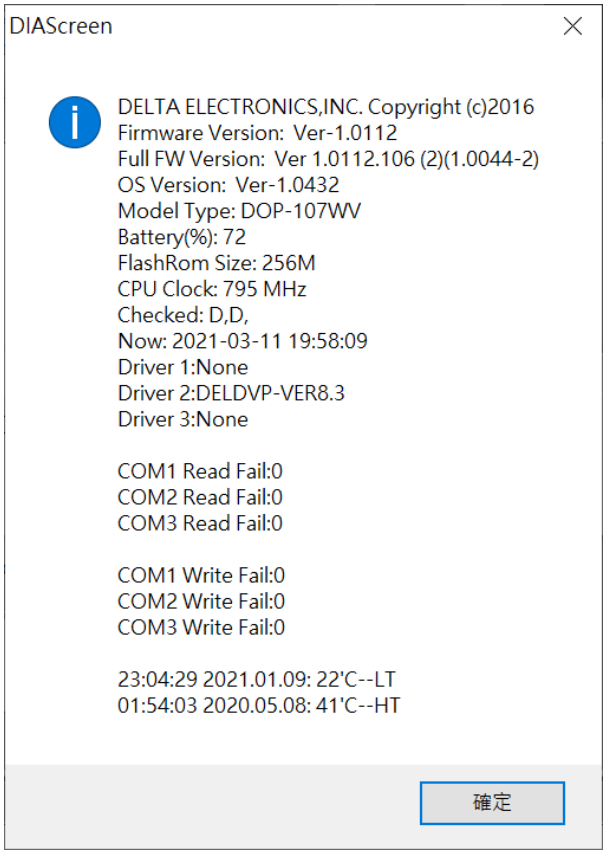
- **Note**:
  - Restart the HMI after downloading the boot screen or resetting the default boot screen.
  - The AX-8 series models do not support the boot screen.
  - The boot screen supports files in BMP, JPG, GIF, ICO, PNG, and SVG format.

Update Firmware

Use the **Update Firmware** function to update the firmware version of the HMI to ensure that the firmware version of the HMI is kept up to date.

Get Firmware Information

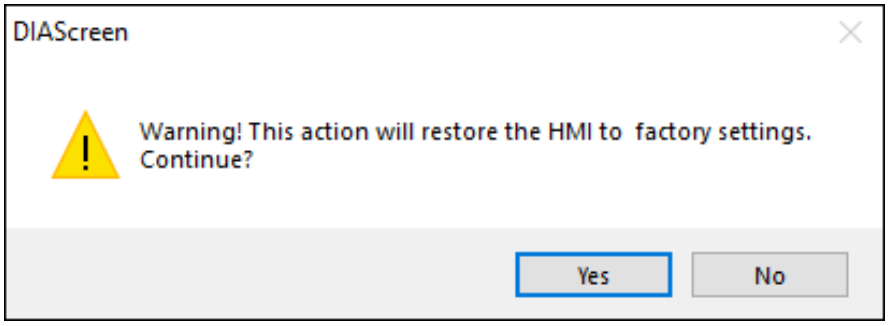
Use the **Get Firmware Information** function to obtain information about the current firmware version.



Reset HMI

Use the **Reset HMI** function to reset the HMI to its original state, including file initialization, clearing all screen data, clearing password, and so on.

When resetting the HMI, a message dialog will appear, asking whether to continue restoring the HMI. Click **Yes** to reset and clear all screen data. If there are files for which the password has been forgotten, they cannot be retrieved.



Reset Default Boot Screen

Use the **Reset Default Boot Screen** function to use the default screen of Delta HMI.

Upload Logic Data

When uploading Logic data, the software will ask you to enter **highest security password**. To set a password, go to **Project** > **Security Setting** > **Security**.

Download Logic Data

Use the **Download Logic Data** function to only download the Logic data.

How to use the **Download Logic Data** function?

1. Go to **Project** > **Other Setting** > **Edit Logic Data** and add the Logic data.
2. On the toolbar, click **Project** > **Download Logic Data**.

Note: The **Upload / Download Logic Data** function is only supported on the IMP series models.

Language Management

Language Management provides **Import / Export Multi-language Text** and **Copy Multi-language Text functions, allowing you to** quickly edit multi-language text in the project with multiple languages and states.

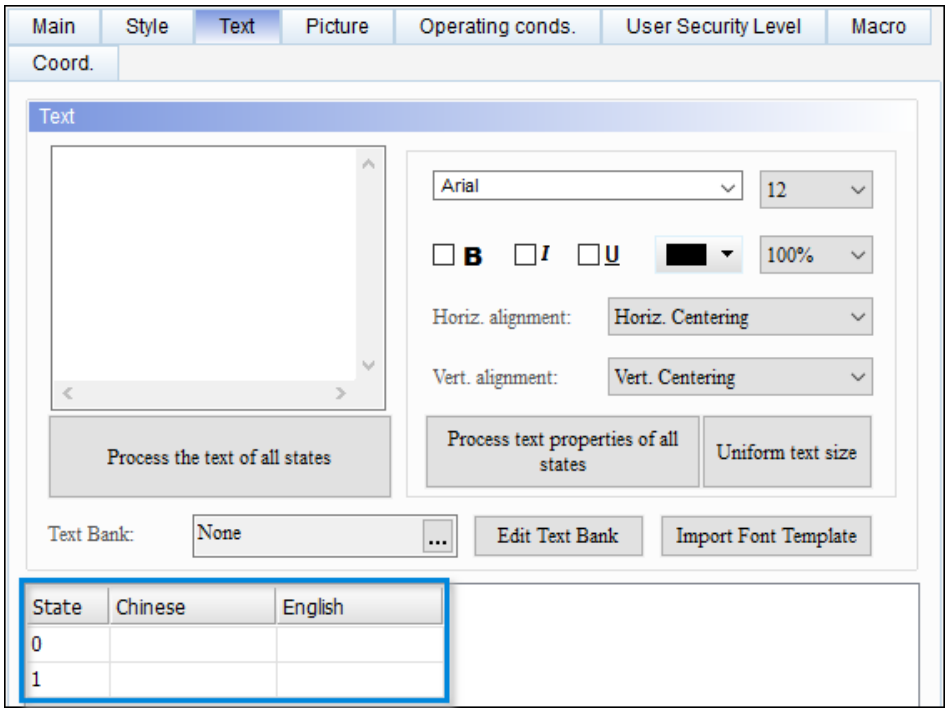
Multi-language Configuration

Follow these steps to configure multi-language.

To configure multi-language

1. In the **Project** pane, double-click **Multi-language**.
2. In the **Multi-language** setting page, click to add a language.
3. Configure languages according to your needs.

   In the **Text** tab in the property dialog of the element, you can view the language configuration of each state.



Export Multi-language Text

With the **Export Multi-language Text** function, you can export the element that multiple-language can be configured in the project as a file and edit.

Follow these steps to export multi-language text.

To export multi-language text

1. Create an element on the screen and configure its multi-language.
2. Click **Project** > **Language Management** > **Export Multi-language Text** on the toolbar.

   The multi-language content of all elements in the current project is exported as a file.

	F	G	H
1	English	繁中	简中
2	Manual	手動	手动
3	28	40	16
4	Arial	Arial	Arial
5	Auto Range	自動量程	自动量程
6	28	16	16
7	Arial	Arial	Arial
8	Time to heat	加熱時間	加热时间
9	28	16	16
10	Arial	Arial	Arial
11	Time to cool	冷卻時間	冷却时间
12	28	16	16
13	Arial	Arial	Arial
14	Auto	自動	自动
15	28	16	16
16	Arial	Arial	Arial
17	Multistate_015	Multistate_015	Multistate_015

Export Multi-language Text (excluding element font settings)

With the **Export Multi-language Text (excluding element font settings)** function, the file size can be reduced when exporting / importing multiple-language text. The exported file only contains element description and multi-language content.

Follow these steps to export multi-language text excluding element font settings.

To export multi-language text excluding element font settings

1. Create an element on the screen and configure its multi-language.
2. Click **Project** > **Language Management** > **Export Multi-language Text (excluding element font settings)** on the toolbar.

   The multi-language content (excluding font settings) for all elements in the current project is exported.

	F	G	H
1	English	繁中	简中
2	Manual	手動	手动
3	Auto Range	自動量程	自动量程
4	Time to heat	加熱時間	加热时间
5	Time to cool	冷卻時間	冷却时间
6	Auto	自動	自动
7	Multistate_015	Multistate_015	Multistate_015

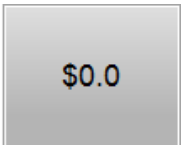
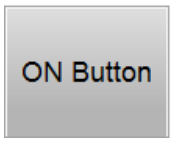
Import Multi-language Text

After importing the multi-language text file, check whether the edited data has been imported completely.

To import multi-language text file

- Click **Project** > **Language Management** > **Import Multi-language Text** on the toolbar.

The following table lists the state before import and after import.

State	Chinese	English
Before Import		
After Import		

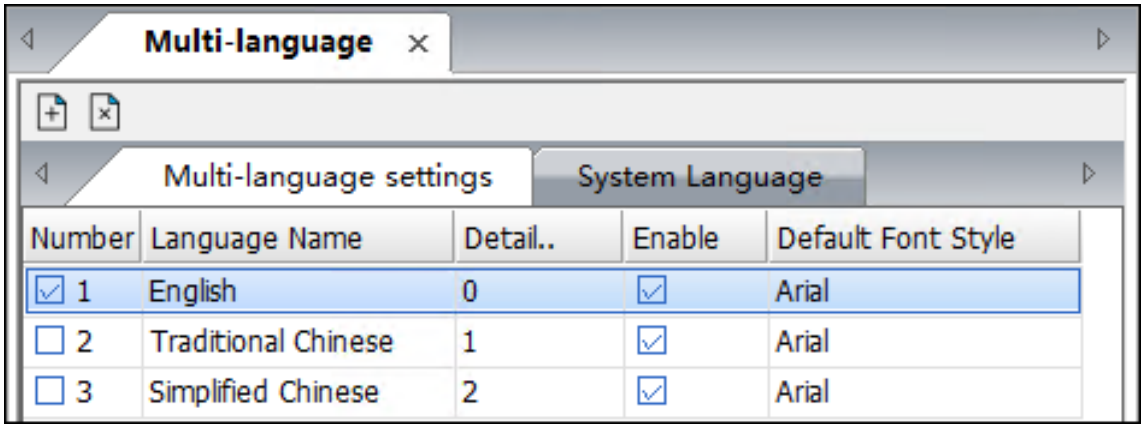
Copy Multi-Language Font

The **Copy Multi-Language Font** function helps you quickly apply the font settings of the existing language to the imported language.

Follow these steps to copy multi-language font.

To copy multi-language font

1. In the **Project** pane, double-click **Multi-language**.
2. In the **Multi-language** setting page, click to add a language.



3. Edit the screen in the first language and click **Project** > **Language Management** > **Export Multi-language Text** on the toolbar to export the language settings as an Excel file.
4. Edit the text content of each language in the Excel file and save,
5. Click **Project** > **Language Management** > **Import Multi-language Text** on the toolbar to update the edited content to the project.

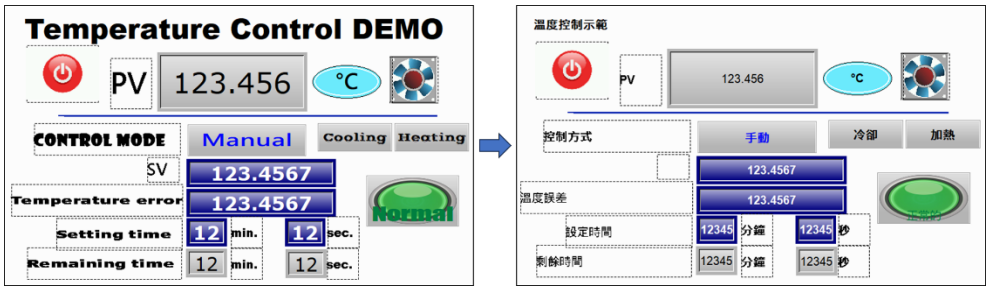
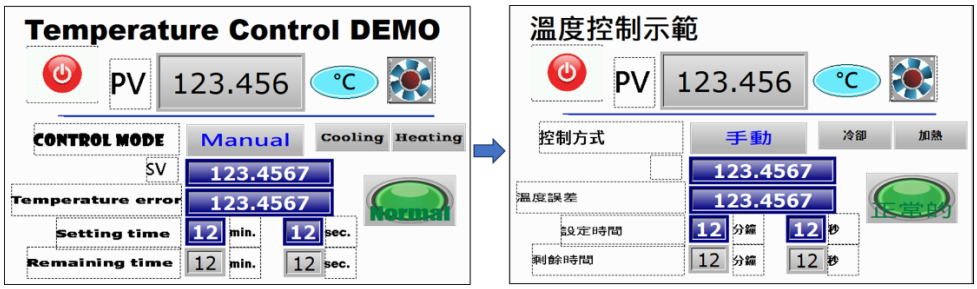
6. Click **Project** > **Language Management** > **Copy Multi-Language Font** on the toolbar.
7. In the **Copy Language Font Setting** dialog, select the **Source Language** and **Target Language**, and then click **OK**.

   The font settings of the source language are applied to the target language.

Note:

- If there is only one language in the project, the **Copy Multi-Language Font** function is not available.
- If a text bank is applied in the element, the font settings will not be applied to the element after executing the **Copy Multi-Language Font** function.

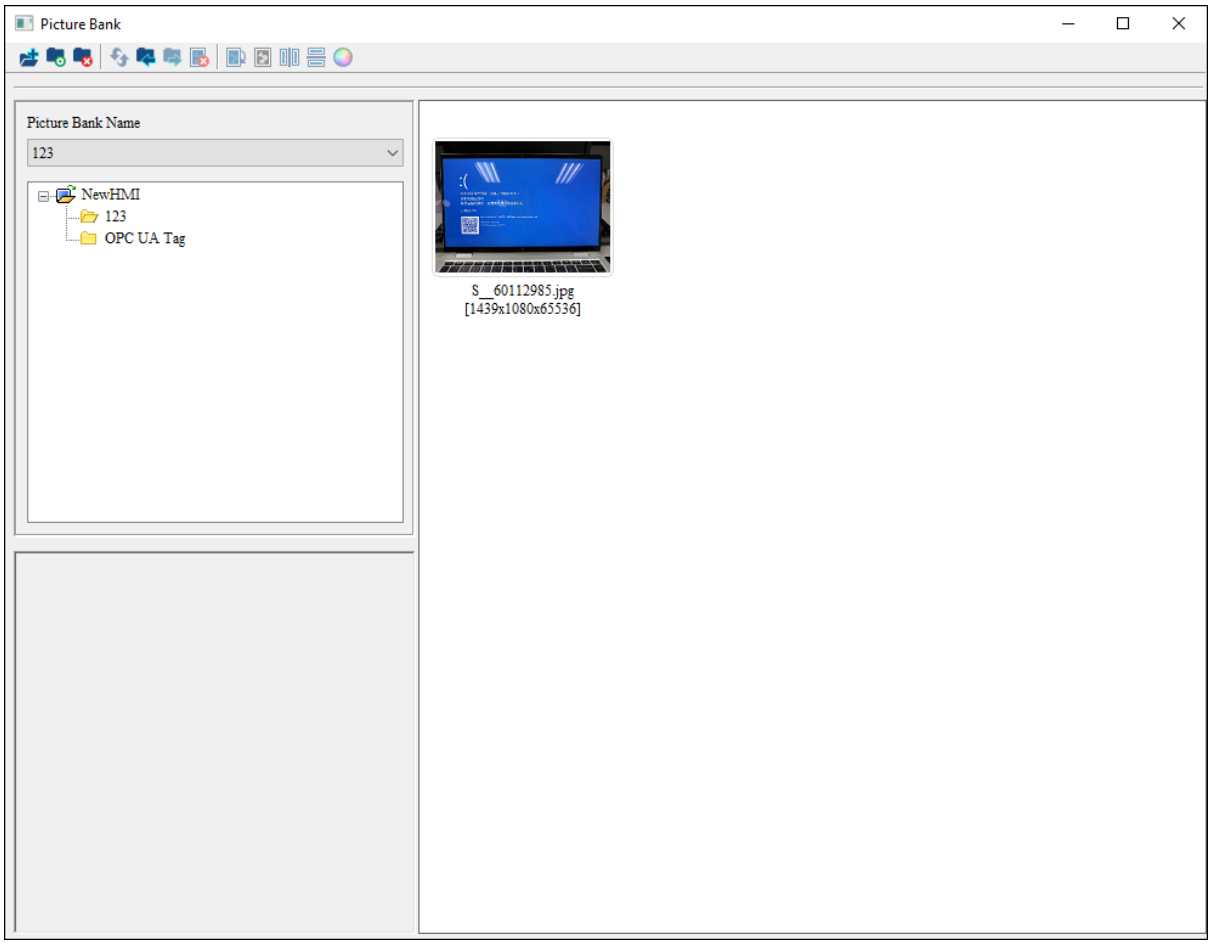
The following table lists the execution result.

State	Result
Before Execution	
After Execution	

Picture Bank

The **Picture Bank** allows you to quickly apply pictures to the element. You can also import non-built-in pictures and perform simple editing on the picture, such as color inversion, converting to 256 colors grayscale, horizontal mirroring, and vertical mirroring.

The following table lists the icon in the **Picture Bank** dialog with their description.

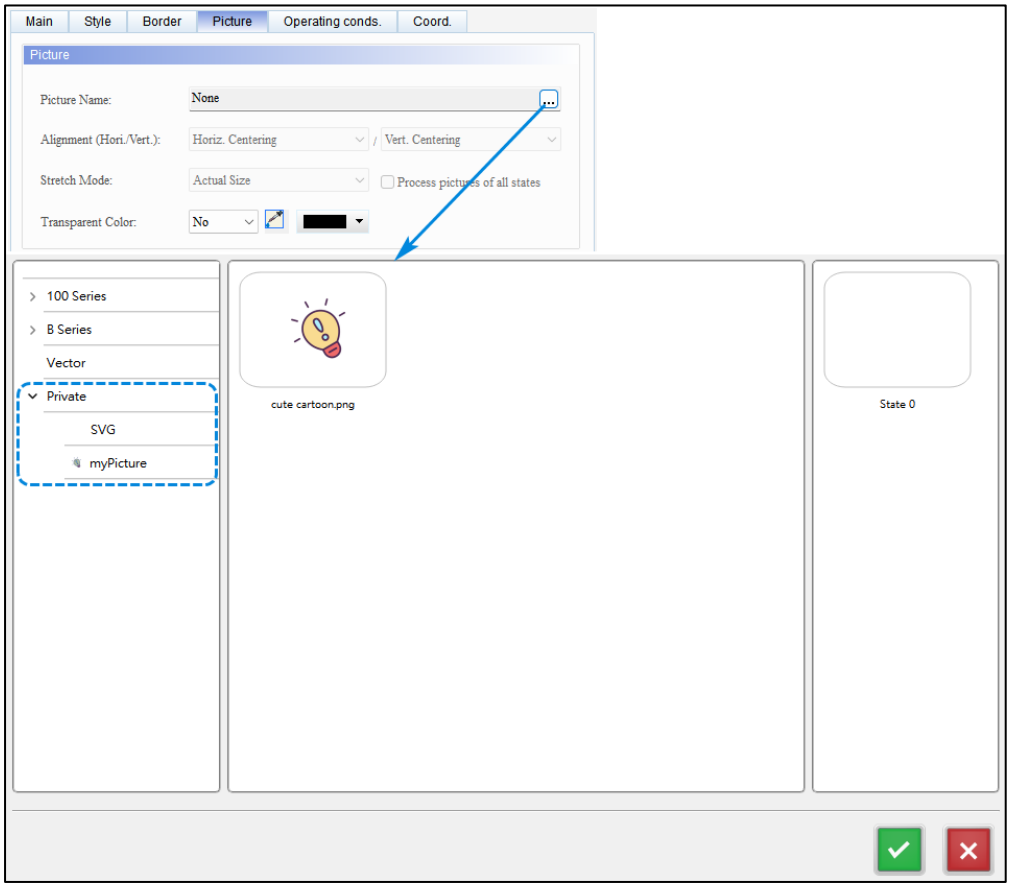


Icon	Description
	Click to add new picture bank. After adding, you can click to import pictures into the picture bank.
	Click to open the picture bank created by other projects. **Note**: You must select the path whether existing picture bank is stored.
	Click to remove the picture bank.
	Click to import pictures data into the picture bank. Supported formats: bmp, jpg, gif, ico, png, emf, svg.

<table>
<tr><th>Icon</th><th>Description</th></tr>
<tr><td> </td><td>After importing the picture data, click this icon to export the content of the picture bank in .bmp format.</td></tr>
<tr><td> </td><td>Click to delete the imported picture data and pictures.</td></tr>
<tr><td> </td><td>Click to invert the original color of the picture to present a Negative effect.
<table>
<tr><th>Before Color Inverse</th><th>After Color Inverse</th></tr>
<tr><td> </td><td> </td></tr>
</table></td></tr>
<tr><td> </td><td>Click to convert the original color of the picture to present a grayscale (256 levels)
<table>
<tr><th>Before Converting to 256 Colors Grayscale</th><th>After Converting to 256 Colors Grayscale</th></tr>
<tr><td> </td><td> </td></tr>
</table></td></tr>
<tr><td> </td><td>Click to mirror the original picture horizontally.
<table>
<tr><th>Before Horizontal Mirroring</th><th>After Horizontal Mirroring</th></tr>
<tr><td> </td><td> </td></tr>
</table></td></tr>
<tr><td> </td><td>Click to mirror the original picture vertically.
<table>
<tr><th>Before Vertical Mirroring</th><th>After Vertical Mirroring</th></tr>
<tr><td> </td><td> </td></tr>
</table></td></tr>
</table>

<table>
<tr><th>Icon</th><th>Description</th></tr>
<tr><td> </td><td>Click to adjust the saturation of the original picture. Saturation refers to the chroma of the colors or the amount of a single color within a specific color. The higher the amount, the higher the color saturation is for this color; whereas, the lower the amount, the lower the color saturation is for this color.
The following table demonstrates the effect of adjusting the picture saturation to 100.
<table>
<tr><th>Before Adjusting Saturation</th><th>After Adjusting Saturation</th></tr>
<tr><td> </td><td> </td></tr>
</table>
</td></tr>
</table>

After the picture bank is set, you can select picture of the picture bank to apply to the element on the **picture** page of the element.



Text Bank

Use the **Text Bank** to store commonly used texts so that the strings can be directly referred from the text bank to elements. Once the **Text Bank** is created, right-click the element in the screen editing area and select **Import from Text Bank**, or click **General** > **Font** > on the toolbar to import text bank content into element text.

Note:

- If multi-languages are created, you can also edit their text data in the text bank.
- When using the text replace function and an element uses a text bank, replacement will not be performed.

How to open the **Text Bank** settings page?

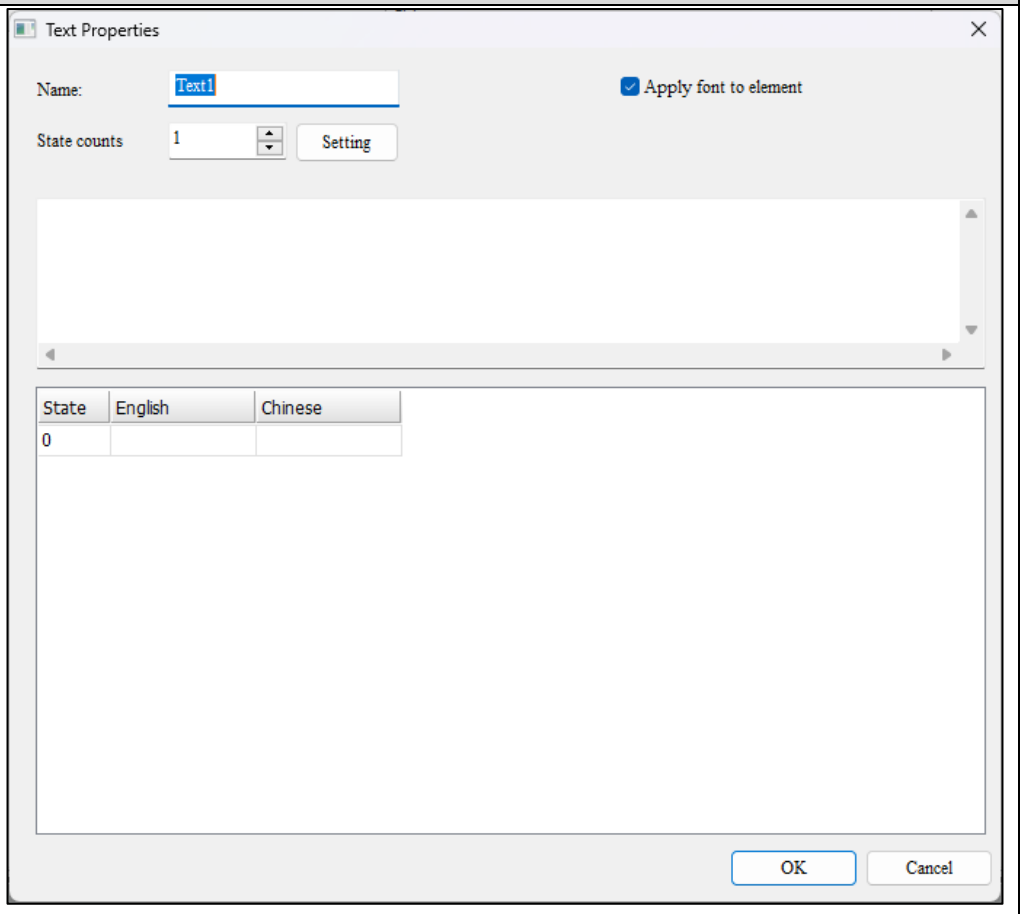
- On the toolbar, click **Project** > **Text Bank**.

Or

- In the **Project** pane, double-click **Text Bank**.

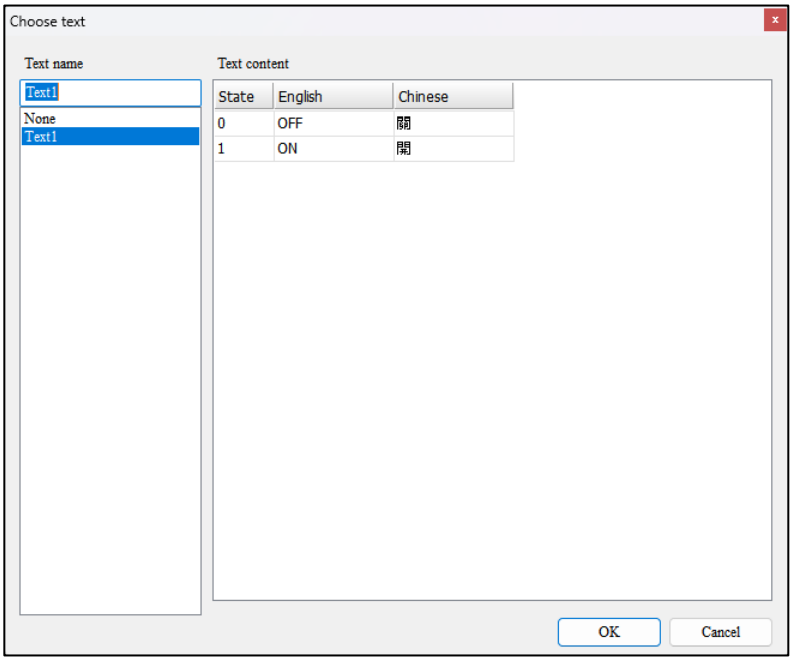
The following table lists the special properties of the **Text Bank** with their description.

Setting	Description
	Click to add text bank item. 1. In the **Text Properties** dialog, enter a name for the text. 2. Select or enter **states counts**, and then click **Setting**. The content box will display the field corresponding to the state number. 3. In the text editing field, you can enter text in multi-languages as well as select the **Apply font to element** checkbox to apply a multi-languages font to the element. **Note:** The Text content supports multiple lines of text.

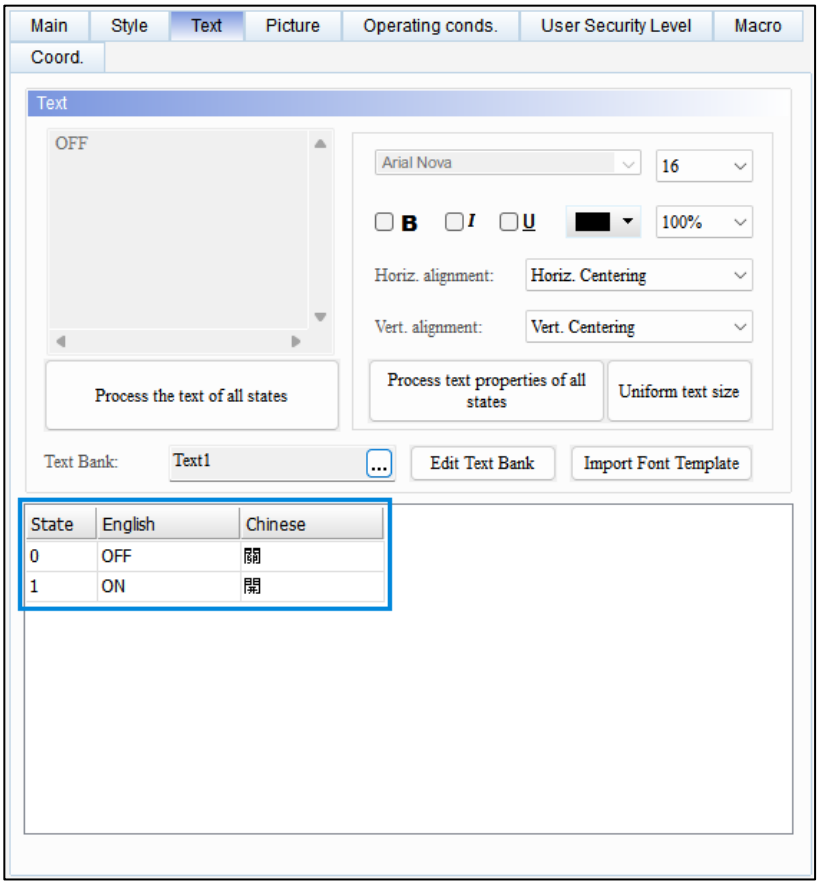
Setting	Description
	
	Click to delete the selected text items.
	Click to import a text bank in .tbk, .txt or .xlsx format.
	Click to export text bank as a .tbk or .xlsx file.
	After selecting a text item, click this icon to open the **Text attributes** dialog to modify the text content.
Font	Select the text font in the language field.
State	Switch to display the text content of each state.

The following describes how to use the **Text Bank** in elements.

1. Create an element on the screen that supports text settings.
2. Double-click the element and then click **Text Bank** on the **Text** tab of the element properties dialog to open the **Choose text** dialog.



3. Select the text name to use and click **OK** to apply the text bank to the element text.



4. Click **Edit Text Bank** to open the **Text Bank** editing dialog. Edit the text content and apply it to the element.

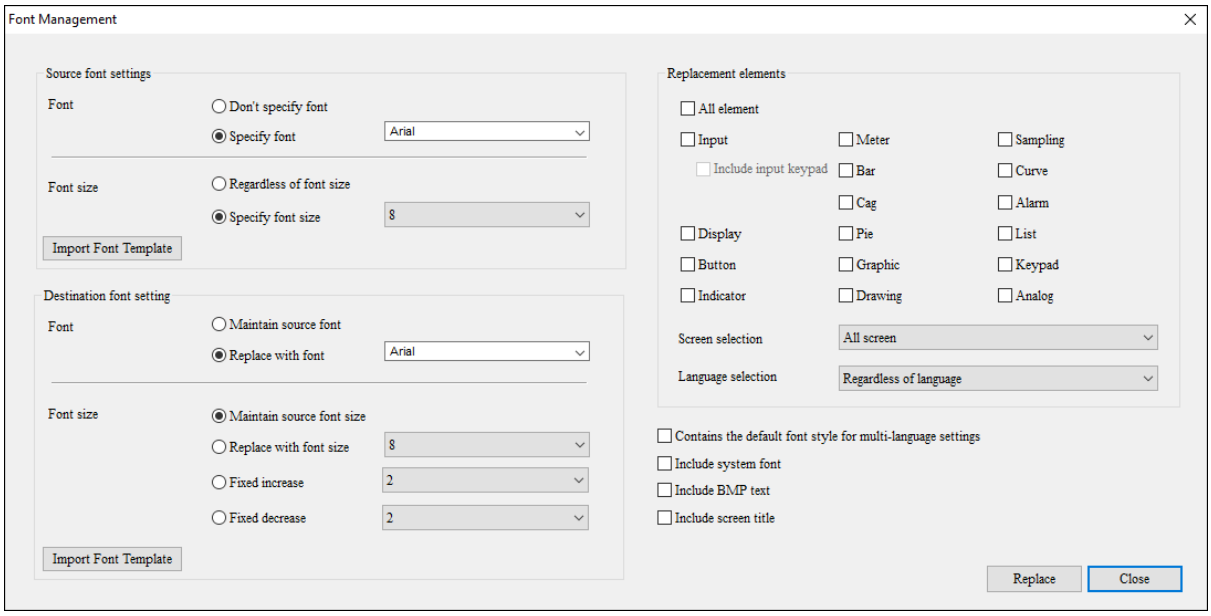
Font Management

The specific source fonts of all text can be converted into target fonts through the font management function.

How to open the **Font Management** function?

- On the menu bar, click **Project** > **Font Management**.

In the **Replacement elements** area of the **Font Management** dialog, you can select the checkbox of part of the elements or the **All element** checkbox to replace their fonts with the target fonts, and you can choose whether to include the default font style for multi-language settings, system font, BMP text, and screen title.



Font Template

DIAScreen provides a wide range of fonts. To save users' time in selecting fonts, DIAScreen provides font templates to help users to customize frequently used font settings that improve the efficiency of screen design.

Follow the steps to set the font templates.

To set the font templates

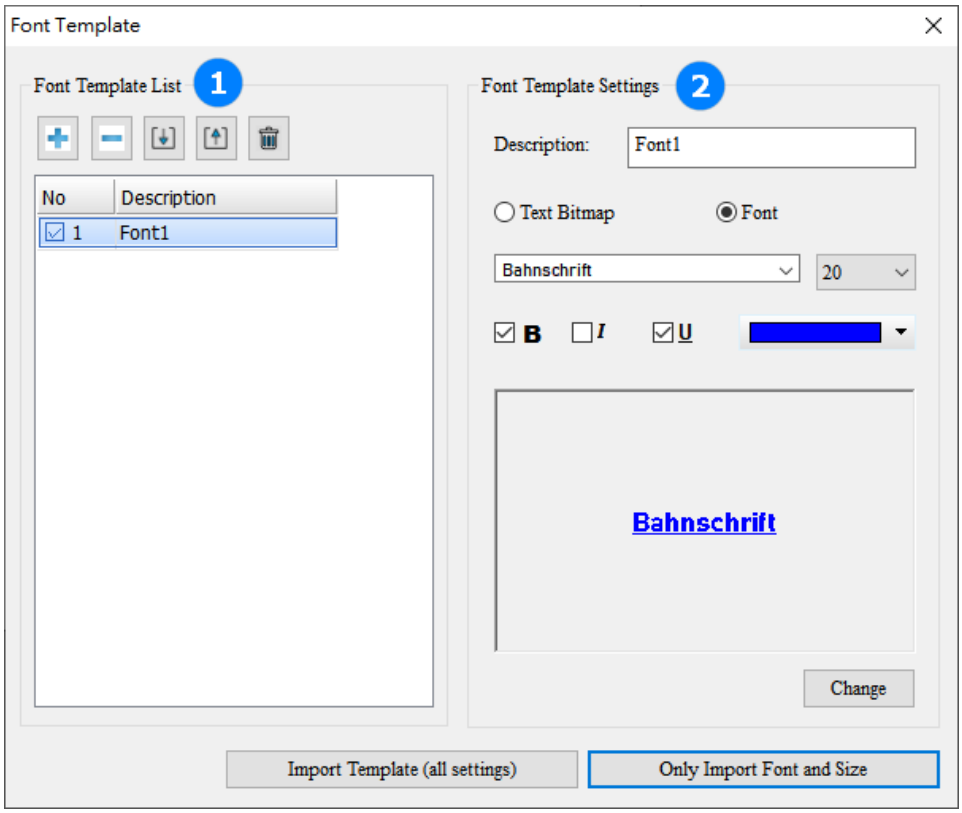
1. On the toolbar, click **General** > **Font** > **Import font template** .

2. In the **Font Template** dialog, click to add a new font template in **Font Template List**, and set the font, size, style, and color in **Font Template Settings**.
3. Click **Change** to save the font template.
4. Create an element and double-click it.
5. In the property dialog of the element, click **Import Font Template** in the **Text** tab.
6. In the **Font Template** dialog, select the desired font template, and then click **Import Template (all settings)** or **Only Import Font and Size** according to your needs.

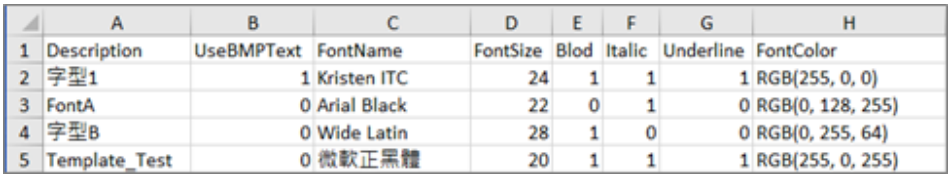
   The font settings are imported to the text of the selected status.

Note: Font template settings are stored in the computer with DIAScreen installed. The same set of font template settings can be used when opening different projects on the same computer.

The following table lists the settings in the **Font Template** dialog and their descriptions.



❶ Font Template List

Setting	Description
	Click to add the font template. You can configure up to 100 templates.
	Select the template and click this icon to delete.
	Click to import the font template.
	Click to export the template as a .xlsx file. The following displays the format content of the font template. 
	Click to clear all font templates.

	A	B	C	D	E	F	G	H
1	Description	UseBMPText	FontName	FontSize	Blod	Italic	Underline	FontColor
2	字型1	1	Kristen ITC	24	1	1	1	RGB(255, 0, 0)
3	FontA	0	Arial Black	22	0	1	0	RGB(0, 128, 255)
4	字型B	0	Wide Latin	28	1	0	0	RGB(0, 255, 64)
5	Template_Test	0	微軟正黑體	20	1	1	1	RGB(255, 0, 255)

❷ Font Template Settings

Setting	Description
Description	Set the description of the font template.
Text Bitmap / Font	Choose to display a text in bitmap or font.
Font setting	Set the font, size, style, and color of a text.
Preview	Preview the effect of font template settings.

Multimedia Setting

Camera Device

The DOP-100 series HMI models provide 12-inch and 15-inch screens, including DOP-112MX, DOP-112WX, DOP-115MX, and DOP-115WX models. M indicates a multimedia model, W indicates a narrow-framed model, and X indicates the model has an XGA TFT display. The multimedia type models support functions such as Analog Camera, IP Camera, VGA display, Video Play, and Event trigger. For DOP-112WX and DOP-115WX models, the functions supported by the multimedia type models are not available.

The following takes the 112MX or 115MX model with the Multimedia function as an example.

The **Camera Device** function must be used with the **Camera Display** element, and the supported camera types are Analog Camera and IP Camera.

- **Analog Camera**

  An Analog Camera only supports CVBS signals in PAL or NTSC format. The analog camera connector is a BNC connector. The input format of CVBS signal is different from that of AHD, TVI, and CVI signals.

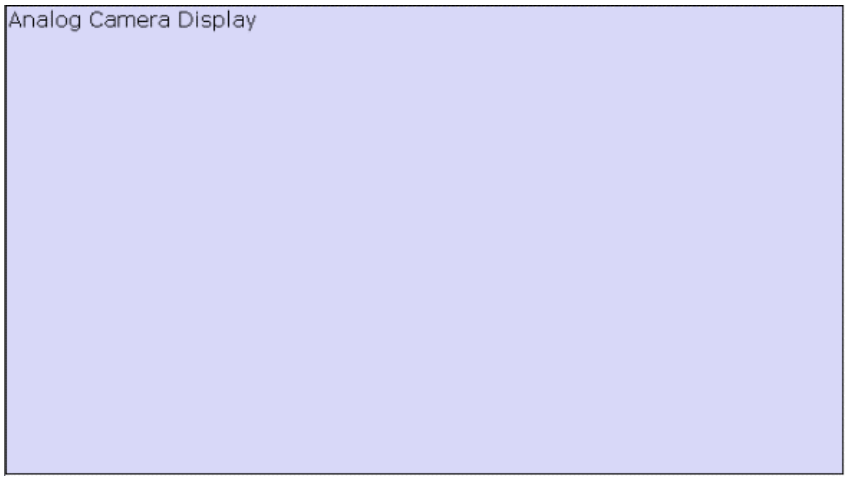
  - PAL (Phase Alternation Line) is a phase alternating scan. Generally, PAL uses a bandwidth of 8 MHz and the color signal transmission can tolerate a larger phase error.
  - NTSC (National Television System Committee) is the earliest developed television system. Generally, NTSC uses a bandwidth of 6 MHz and covers the widest range of colors.
  - CVBS (Composite Video Baseband Signal) is commonly known as the visual signal of the AV connector. CVBS is the original video signal format that hasn't been processed. A CVBS signal cable includes luminance (Y), chrominance (C), horizontal synchronization (Hsync), and vertical synchronization (Vsync).

- **IP Camera**

  IP Camera combines traditional cameras and network technologies. It is a digital device based on network transmission and is equipped with a network output interface through which it can use the Ethernet for remote network connection. The IP Camera supported by the HMI only provides Real Time Streaming Protocol (RTSP). RTSP is an application layer protocol in the TCP/IP protocol system, which is designed for entertainment and communication systems to control streaming media servers.

  The following describes how to use the Analog Camera type **Camera Display** element.

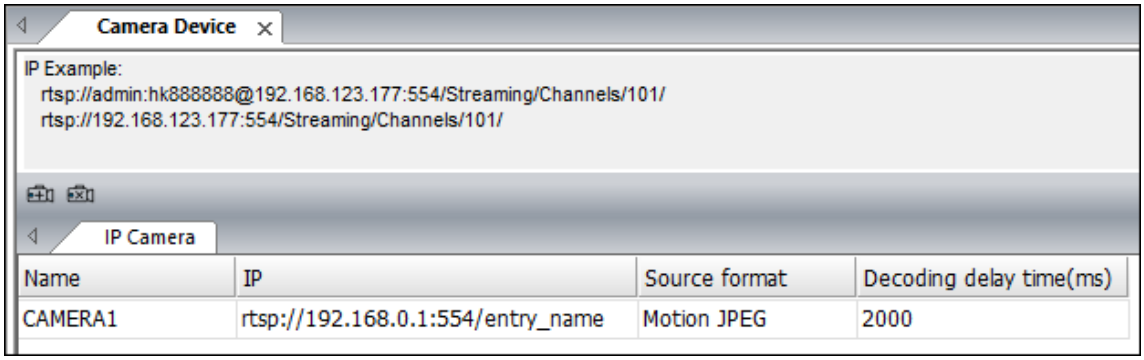
  1. Create a **Camera Display** element on the screen and set the **Camera Type** to **ANALOG CAMERA**, **Camera Name** to **CH1**, and **Show Size** to **640x360**.

  2. Compile and download the screen to the HMI.
  3. After the download is complete, you can see the direction of the video recorded by the Analog Camera displayed on the **Camera Display** element.

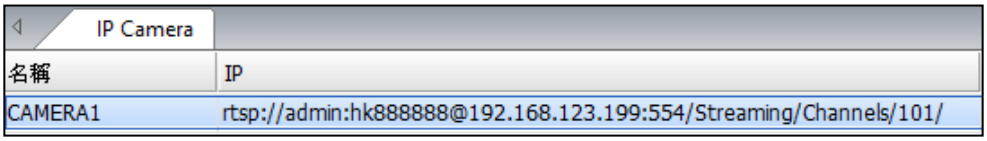
The following describes how to use the IP Camera.

1. Go to **Project** > **Other Setting** > **Camera Device**.
2. Click to add a camera in the **Camera Device** page.
3. Modify the IP address of the camera. The default IP is rtsp://192.168.0.1:554/entry_name.

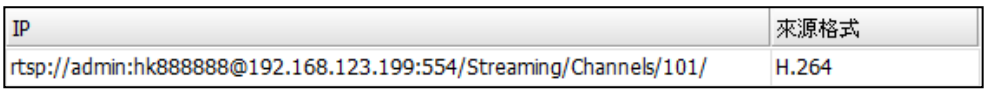


The IP Example above is the account, password, IP address, Port, and entry name used by HIKVISION camera. You can refer to the following example to modify the IP address of the corresponding IP camera.
rtsp://admin:hk888888@192.168.123.177:554/Streaming/Channels/101/

Account	admin
Password	hk888888
IP Address	192.168.123.177
Port	554
Entry name	Streaming/Channels/101/ Streaming/Channels/102/

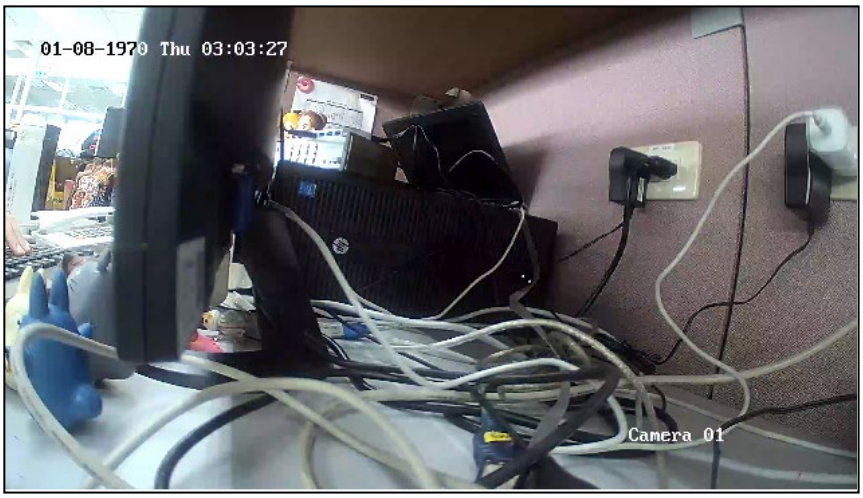


4. Select the **Source format**. The default is **H.264**.



5. Create a **Camera Display** element on the screen and set the **Camera Type** to **IP CAMERA**, **Camera Name** to **CAMERA1** and **Show Size** to **640x360**.

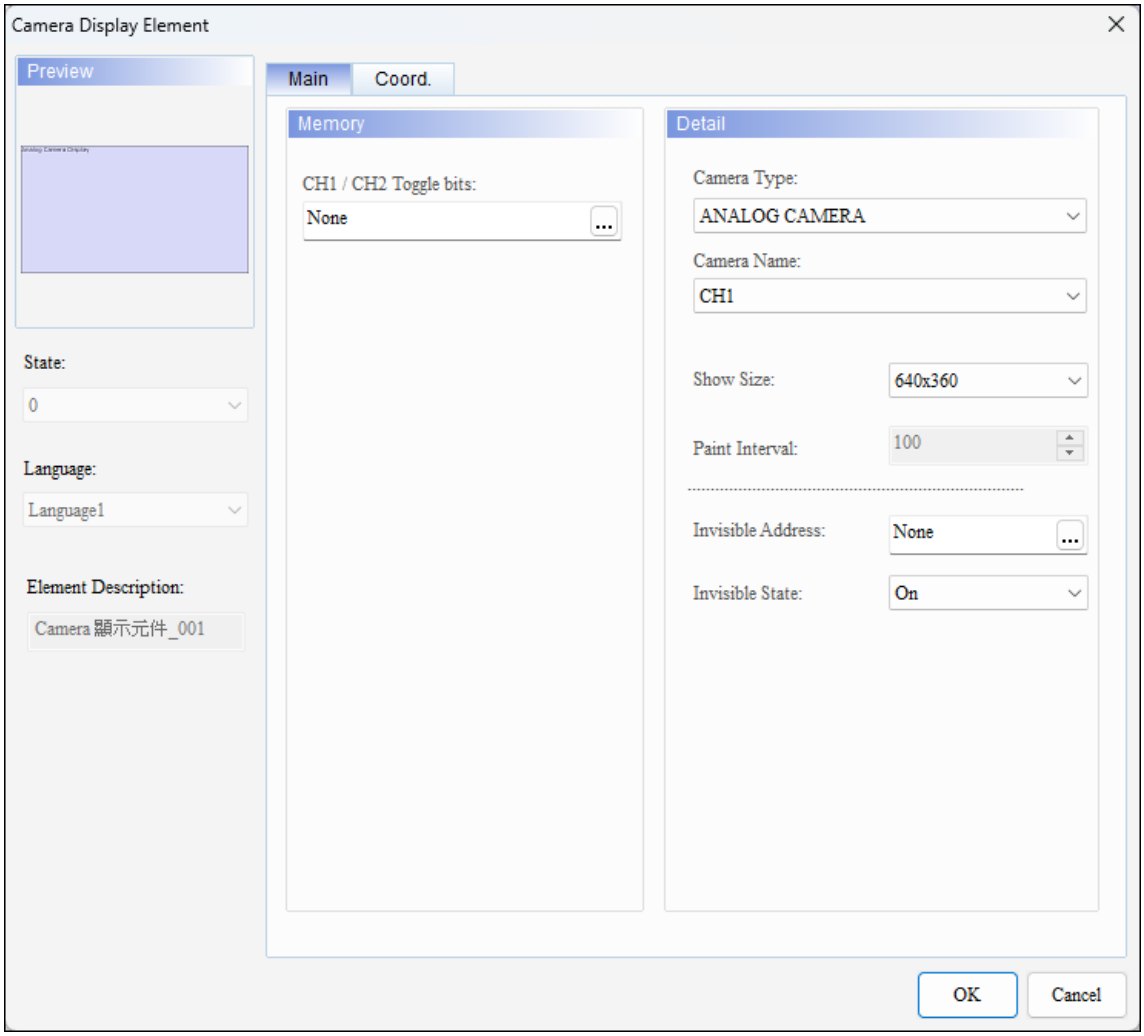
6. Compile and download the screen to the HMI.
7. After the download is complete, you can see the direction of the video recorded by the IP Camera displayed on the **Camera Display** element.



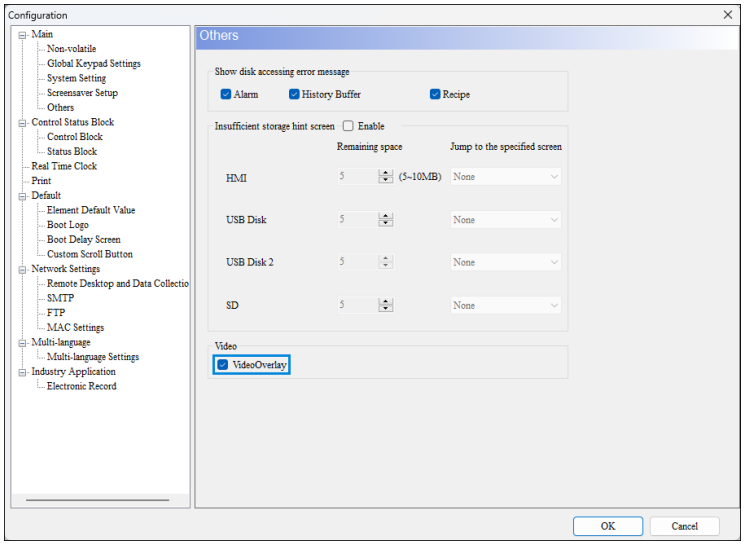
The following table lists the properties of the **Camera Display** element with their description.

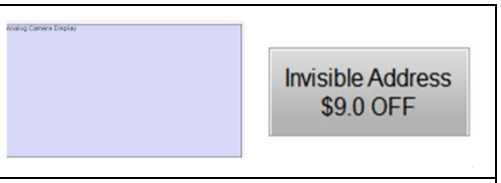
Note: Since the **Camera Display** element is shared between Analog Camera and IP Camera, refer to the settings for the Analog Camera when using the IP Camera.

- **Main**

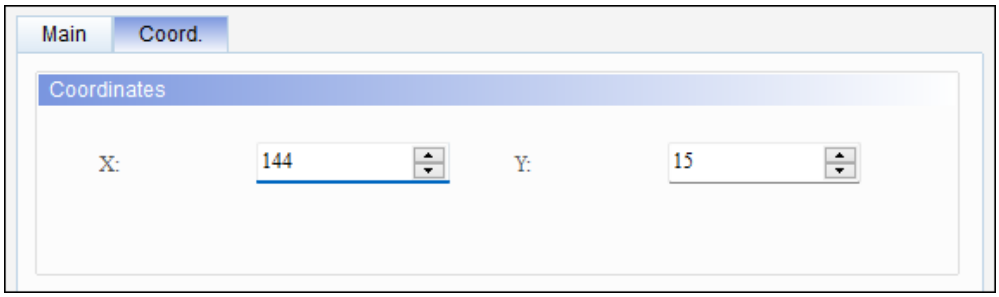


Legend	Setting	Description
❶	CH1/CH2 Toggle bits	Set the channel of the Analog camera to CH1 or CH2. When the toggle bit is set to ON, the set channel will be immediately switched to another channel.
❷	Camera Type	Select the type of camera.

Legend	Setting	Description
❸	Camera Name	The name varies depending on the selected camera type. • When the **Camera Type** is set to ANALOG CAMERA, the available options are CH1 and CH2. Connect a corresponding BNC connector to the CH1 or CH2 port on the back of the HMI. • When the **Camera Type** is set to IP CAMERA, the available options are CAMERA1 and CAMERA2. Use a webcam and set its IP address to the same network segment as that of the HMI.
❹	Show Size	Select the display resolution of the element.
❺	Paint Interval	Select or enter the capture time for drawing the image onto the element in milliseconds (ms). This setting is only available when the **VideoOverlay** checkbox is selected. 

<table>
<tr><th>Legend</th><th>Setting</th><th>Description</th></tr>
<tr><td>⑥</td><td>Invisible Address Invisible State</td><td>
The Invisible Address hides an element on the HMI and must be used with Invisible State. If Invisible State is set to On, the Invisible Address hides the element. Instead, if Invisible State is set to Off, the Invisible Address hides the element.
The following is an example of using invisible address.
1. Create a button element on the screen and set its address to $9.0, then set the Invisible Address of the element to hide to $9.0.
2. To hide the element, you must click the $9.0 button.
<table>
<tr><td rowspan="2">Invisible State is On</td><td> 
</td></tr>
<tr><td>
</td></tr>
<tr><td rowspan="2">Invisible State is Off</td><td>
</td></tr>
<tr><td>
</td></tr>
</table>
</td></tr>
</table>

- **Coord.**

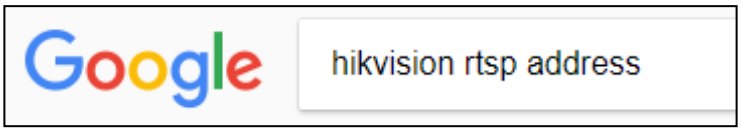


Set the XY coordinates on the upper left of the element.

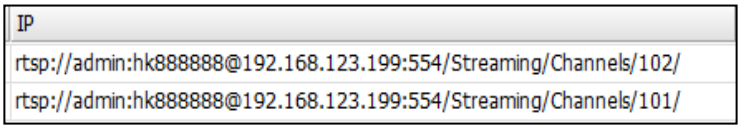
How to open the **Camera Device** page?

- Click **Project** > **Other Setting** > **Camera Device** on the toolbar.

The following table lists the settings in the **Camera Device** page with their description.

<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td> </td><td>Click to add a camera.</td></tr>
<tr><td> </td><td>Click to delete a camera.</td></tr>
<tr><td>Name</td><td>Enter a name for the camera.</td></tr>
<tr><td>IP</td><td>Enter the IP address of the camera.
The format of the IP address is fixed. Modify the content according to the network camera settings. The IP address format is as follows.
<table>
<tr><td>Account</td><td>Enter the account according to the settings of the network camera.</td></tr>
<tr><td>Password</td><td>Enter the password according to the settings of the network camera.</td></tr>
<tr><td>IP address</td><td>Enter the IP address according to the settings of the network camera.</td></tr>
<tr><td>Port</td><td>The default is 554.</td></tr>
<tr><td>Entry name</td><td>The entry name varies from different brands. Enter “Brand name rtsp address” in the field of the search engine such as “HIKVISION rtsp address”.
 

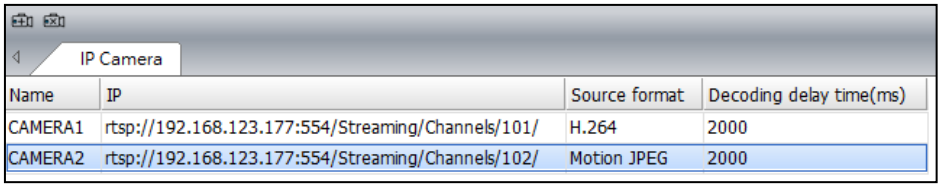
The search results are as follows.
Main Stream
rtsp://192.168.1.100:554/Streaming/Channels/101/
rtsp://admin:examplepass123!@192.168.1.100:554/Streaming/Channels/101/

Sub Stream
rtsp://192.168.1.100:554/Streaming/Channels/102/
rtsp://admin:examplepass123!@192.168.1.100:554/Streaming/Channels/102
101 indicates the Main Stream and 102 indicates the Sub Stream of HIKVISION. The contents of the IP field can be entered as follows.
 
</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Source format</td><td>
Select the corresponding Source format according to the settings of the network camera.

Take the brand HIKVISON for example. Go to IP address “http://192.168.123.199/doc/page/login.asp?_1534130000127” and enter the Account and Password to log in.
<table>
<tr><td>Account</td><td>admin</td></tr>
<tr><td>Password</td><td>hk888888</td></tr>
</table>

Go to Configuration > Video / Audio to check the Video Encoding. For HIKIVISION, the Main Stream supports the coding formats H.264 and H.265, and the Sub Stream supports the coding formats H.264, H.265, and Motion JPEG. Since the HMI supports only Motion JPEG and H.264, if users select the Main Stream (101), the Source format of the HMI should be set to H.264. If users select the Sub Stream (102), the Source format of the HMI should be set to Motion JPEG.

The HMI settings are as follows.


</td></tr>
<tr><td>Decoding delay time</td><td>Select or enter the decoding delay time.</td></tr>
</table>

Event Trigger

The **Event Trigger** function is mainly used with the **Camera Display** element. After the conditions set for the **Event Trigger** function are met, the content captured by the camera at that time can be archived into an mpeg4 file.

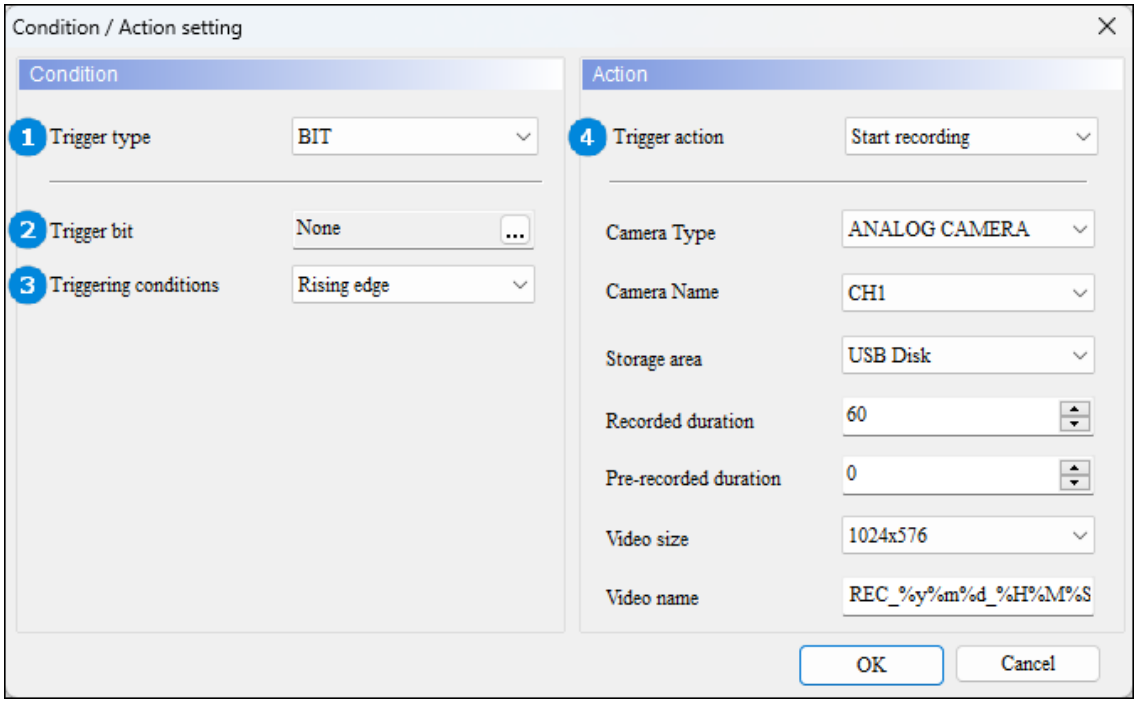
How to open the **Event Trigger** page?

- Click **Project** > **Other Setting** > **Event Trigger** on the toolbar.

The following table lists the settings in the **Event Trigger** page with their description.

Setting	Description
	Click to add a trigger event.
	Click to delete a trigger event.
No.	Displays the number of the triggered event.
Condition	Displays the condition of the triggered event.
Action	Displays the action of the triggered event.

The following table lists the setting in the **Condition / Action** setting dialog with their description.



Legend	Setting	Description
1	Trigger type	Only bit triggering is available.
2	Trigger bit	Set the address of the trigger bit, which can be an external controller memory address or an Internal Memory address.
3	Triggering conditions	Select the triggering condition. • The **Rising edge** indicates that the Bit changes from On to Off. • The **Falling edge** indicates that the Bit changes from Off to On.

Legend	Setting	Description
④	Trigger action	Select the trigger action. • **Start recording** ○ **Camera Type**: Select the type of camera. ○ **Camera Name**: Select the name of camera. The name varies depending on the selected camera type. If select IP CAMERA, only the created IP Camera name can be selected. Go to **Project** > **Other Setting** > **Camera Device** to create the name for the IP Camera. ○ **Storage area**: Select the area to store the videos. If select **MEDIA**, the video will be stored in the HMI and only 2G of recorded videos can be stored. If select **USB Disk** or **SD**, the videos will be stored based on the selected capacity. The supported format is FAT32. ○ **Recorded duration**: Select or enter the recorded duration. When meeting the trigger bit condition, the camera records for the set duration. ○ **Pre-recorded duration**: Select or enter the pre-recorded duration. ○ **Video size**: Select the size of the video file. If select 640x480 and the duration is 120 seconds, one file will take up 30 MB. ○ **Video name**: Enter the name of the video file. • **Stop recording**: Select the camera type and name to stop recording. • **Goto screen**: Set the change screen. If select, • **Take Picture**: The settings are the same as those for **Start recording**, but the **Recorded duration** and **Pre-recorded duration** cannot be set.

Sound Settings

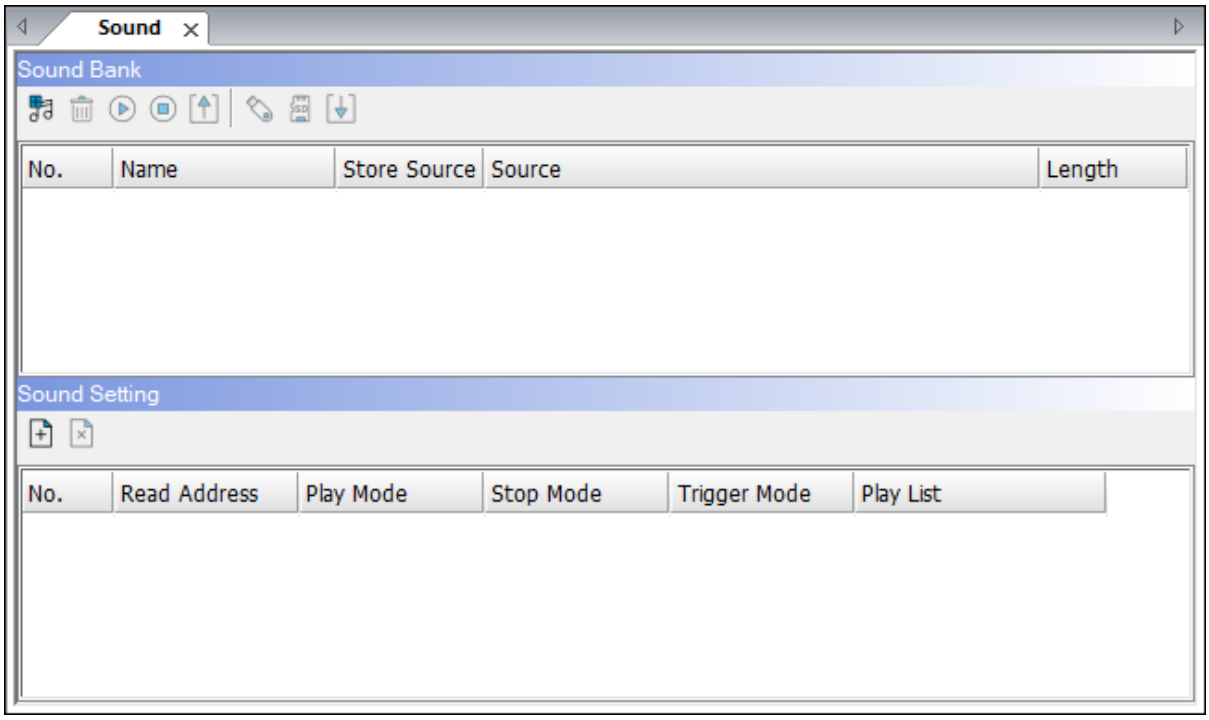
If there is an error, on-site operators can be informed through the **Sound Settings** function. This function can be triggered by the Bit address of the audio file and can be used with the values in the Word register to specify different audio files. The supported audio file formats are MP3 and WAV.

Note: The currently supported models are DOP-107IV, DOP-107EG, DOP-108IG, DOP-110IG, DOP-110IS as well as DOP-112MX, DOP-112WX, DOP-115MX, and DOP-115WX.

How to open the **Sound** setting page?

- Click **Project** > **Other Setting** > **Sound Settings** on the toolbar.

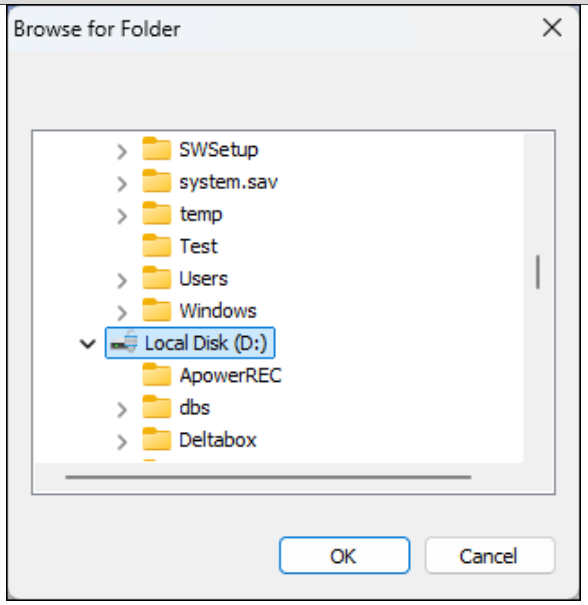
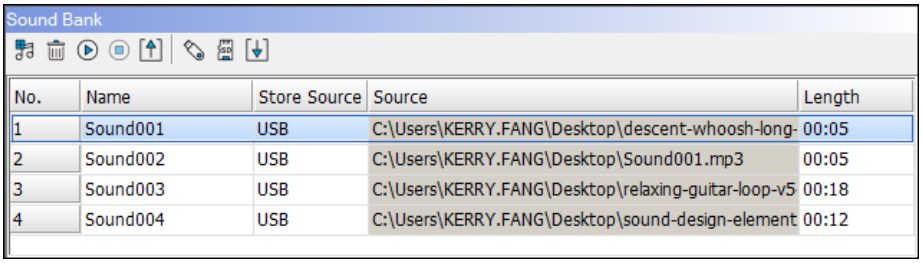
The following table lists the settings in the **Sound** setting page with their description.



- **Sound Bank**

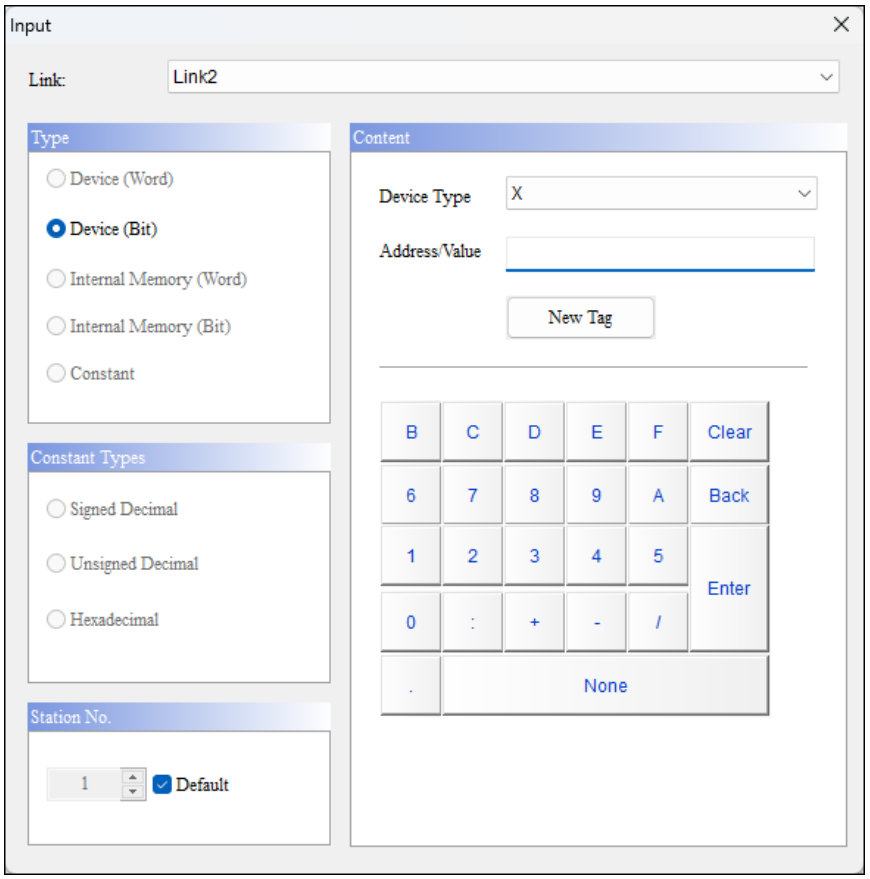
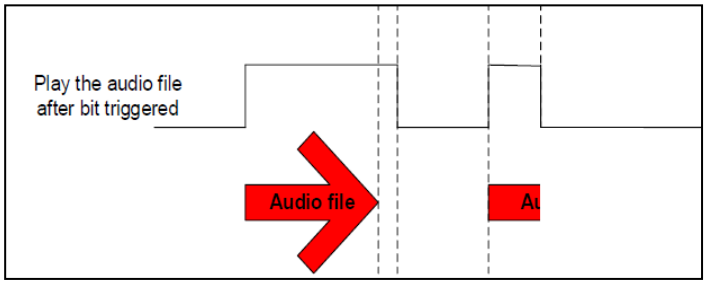
  The **Sound Bank** pane allows you to add, delete, or export sound files.

Setting	Description
	Click to add an audio file. Up to 1000 (0~999) audio files can be added.
	Click to delete the selected audio file. To delete multiple audio files, hold the **Shift** or **Ctrl** key and click the files to be deleted, and then click this icon.
	Click to play the selected audio file and check whether it can be played properly.
	Click to stop the currently playing audio file.
	Click to export the audio file. **Note**: The audio file must be saved to the root directory.

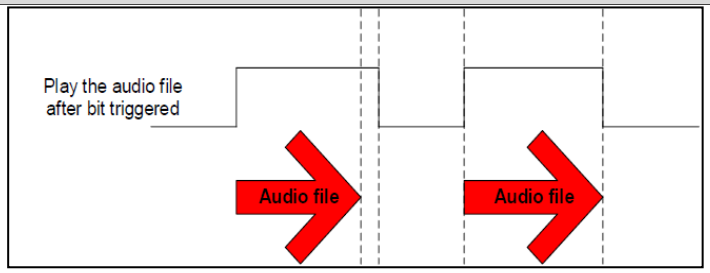
Setting	Description
	
 	Click to quickly change the storage device of single or multiple audio files to USB or SD. The selection method for changing multiple audio files is the same as the selection method for deleting multiple audio files.
No.	The **No.** field is a major reference value when playing audio files. When the playlist's read mode is set to **Address**, the software will determine which audio file to play based on the set memory address. 
Name	The filename is displayed as **Sound** by default, and the first audio file has a filename starting with 001, and so on.
Store Source	Select the location to store the audio files.
Source	Displays the address of the added audio file. You can change the source audio file, i.e., replace the soundtrack.
Length	Displays the length of the audio file.

- **Sound Setting**

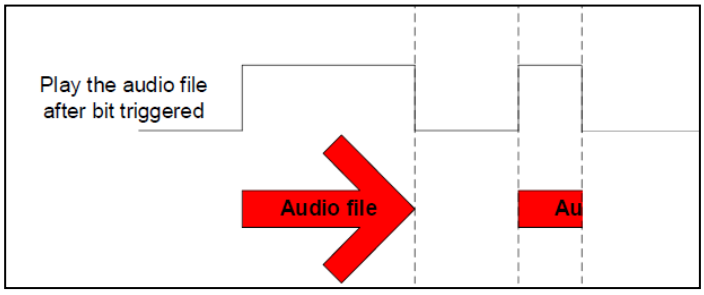
The **Sound Setting** pane allows you to set the play, stop, and trigger modes of audio files.

Setting	Description
	Click to add an audio setting item.
	Click to delete an audio setting item.
No.	Up to 512 audio setting items can be set.
Read Address	The read address can only be set as Bit to trigger and play audio files 
Play Mode / Stop Mode	• **Play**: The sound plays immediately after the Bit is triggered. ○ If the **Stop Mode** is set to **Stop**, no matter the soundtrack is playing or not, it stops immediately.  ○ If the **Stop Mode** is set to **Stop after playing**, the soundtrack is played to the end and stopped.

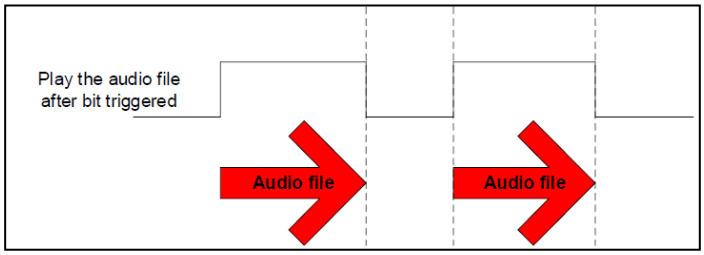
Setting	Description



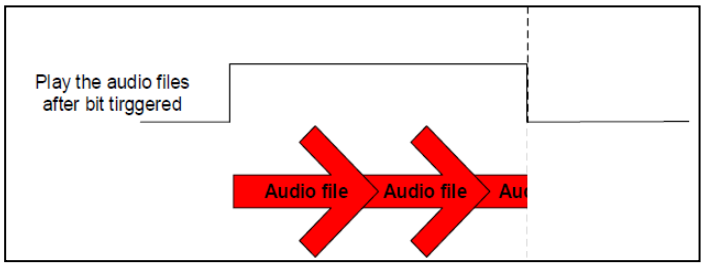
- **Cleared automatically after playing**: The audio file clears the Bit after the soundtrack is played. If the same address is selected for triggering and playing two soundtracks, this Bit is cleared after these soundtracks finish playing.
  - **Stop Mode**: Stop



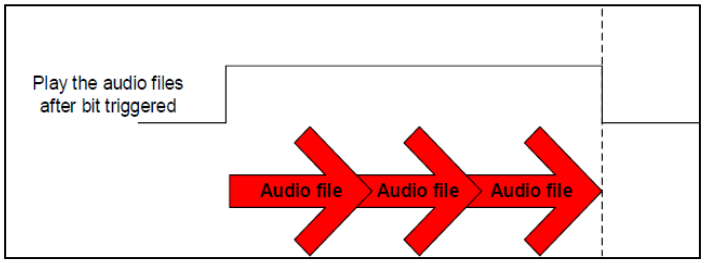
  - **Stop Mode**: Stop after playing

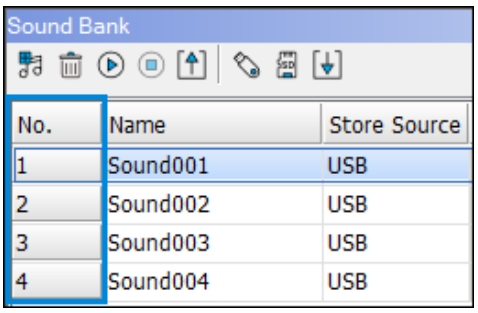


- **Repeat play**
  - **Stop Mode**: Stop



  - **Stop Mode**: Stop after playing



<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Trigger Mode</td><td>Select whether to trigger the soundtrack when the button is ON or OFF.</td></tr>
<tr><td>Play List</td><td>Each set trigger address supports up to 100 audio files.
<table>
<tr><td>Mode</td><td>Select the mode according to the added audio file.
File: Select the audio file from the Sound Bank to play directly and add it to the Play List. When the read Bit address of the Sound Setting is triggered, the selected audio file is played.
Address: Specify a register address. When the trigger condition is met, the HMI reads the input values of the register and plays the audio file corresponding to the No. in the Sound Bank. 
</td></tr>
<tr><td>Name</td><td>Displays the name of the audio file.</td></tr>
<tr><td>Source</td><td>Displays the source paths of the audio files and cannot be edited.
To change the source, modify the Source in the Sound Bank.</td></tr>
</table></td></tr>
</table>

MODBUS TCP/COM Mapping Table

Delta HMI can be used as a Modbus Slave for external Modbus Server devices to transmit data. Modbus communication supports TCP/IP and Serial Port connections. You can configure the HMI address to the corresponding Modbus address through **Modbus TCP/COM mapping table** function.

How to open **Modbus TCP/COM mapping table**?

- On the toolbar, click **Project** > **Other Settings** > **Modbus TCP/COM mapping table**.

The following table lists the Modbus range limitation of TCP/IP and COM connection.

Connection	Data Type	Maximum Length	Modbus Range
TCP/IP	Coils	65536	00001(0x0000) ~ 65536(0xFFFF)
	Registers	65536	00001(0x0000) ~ 65536(0xFFFF)
COM	Coils	2048	00001(0x0000) ~ 02048(0x07FF)
	Registers	20000	40001(0xA410) ~ 60000(0xEA5F)

The following describes the setting methods of Modbus Slave (TCP/IP) and Modbus Slave (COM).

- **Modbus Slave (TCP/IP)**

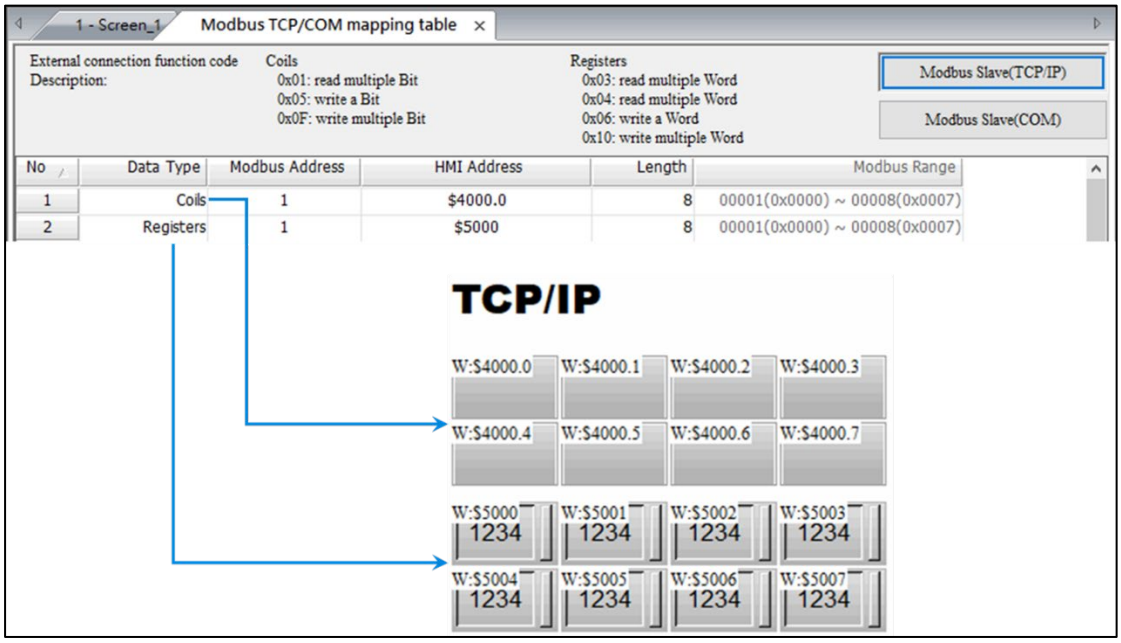
  HMI with Ethernet ports will have the Modbus Slave function enabled by default. You can configure the Modbus address corresponding to the HMI address through the **Modbus TCP/COM mapping table**.

  Example Settings

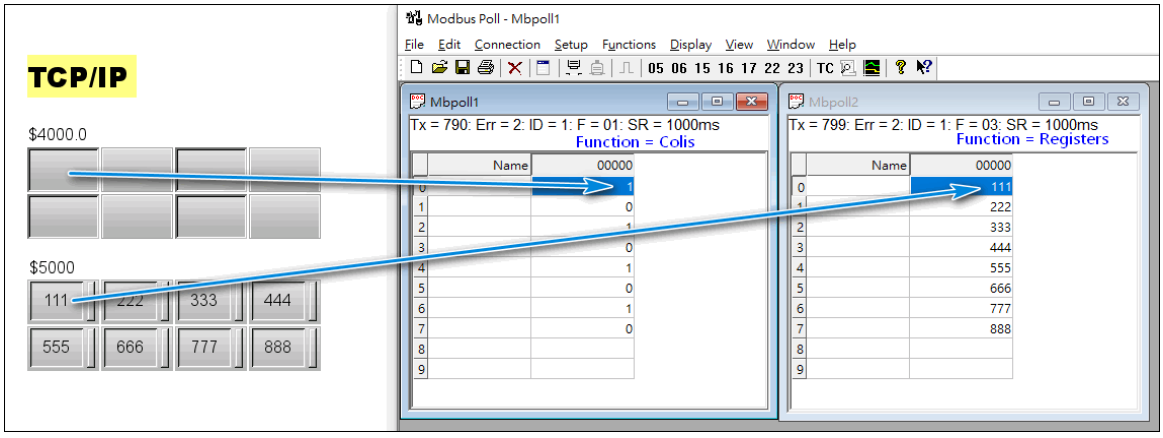
  1. In the **Modbus TCP/COM mapping table**, configure the mapping table.
     - Set the HMI address to $4000.0, corresponding to the Modbus address of **Coils** function code 0x00000.
     - Set the HMI address to $5000, corresponding to the Modbus address of **Registers** function code 0x00000.

2. Create elements on the screen and set the HMI address corresponding to Modbus, and then download the project to the HMI.

   The Modbus Master device is connected with TCP/IP and set connected HMI IP. Change values of elements on the HMI, and the HMI transmits data through Modbus.



The following figure shows the result that **Modbus Poll** is used as the Modbus Master and communicated with the HMI.



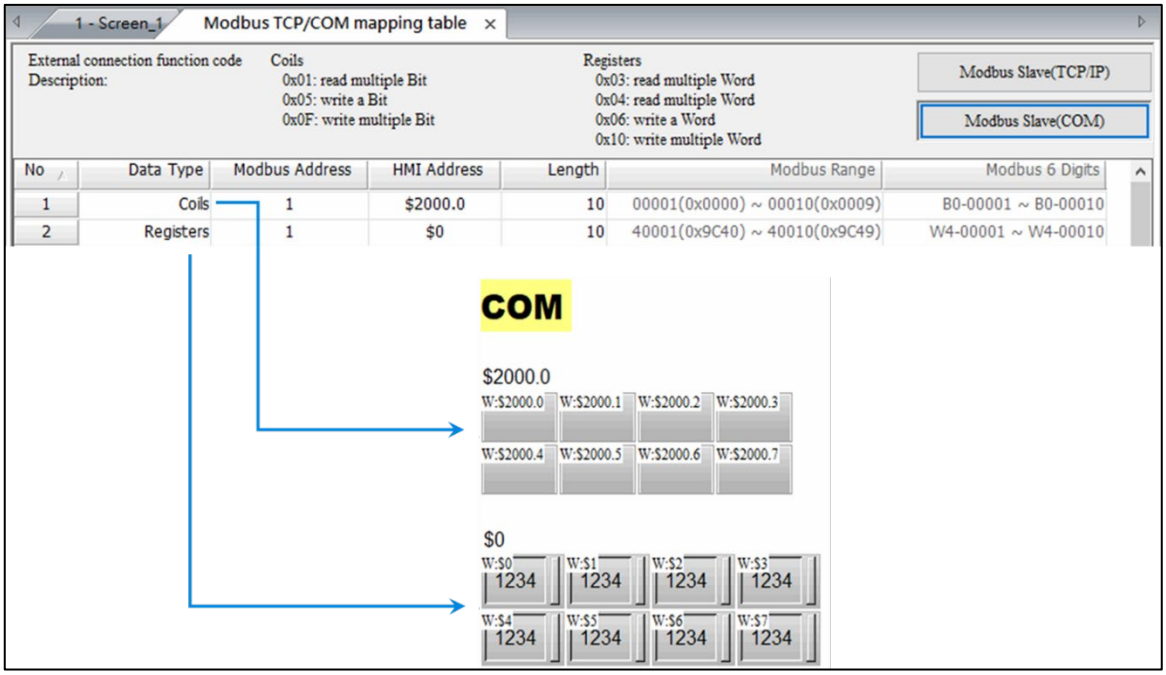
- **Modbus Slave (COM)**

  Follow the steps to set the **Modbus Slave (COM)** mapping.

  1. In the **Communication Settings** dialog, select **COM2**.

2. Select **Modbus** for **Manufacturers** and **RTU (Slave)** for **Series** to perform the Modus communication of Serial Port.
3. In **the Modbus TCP/COM mapping table**, click **Modbus Slave (COM)** to configure the corresponding relationship between the HMI address and Modbus address.

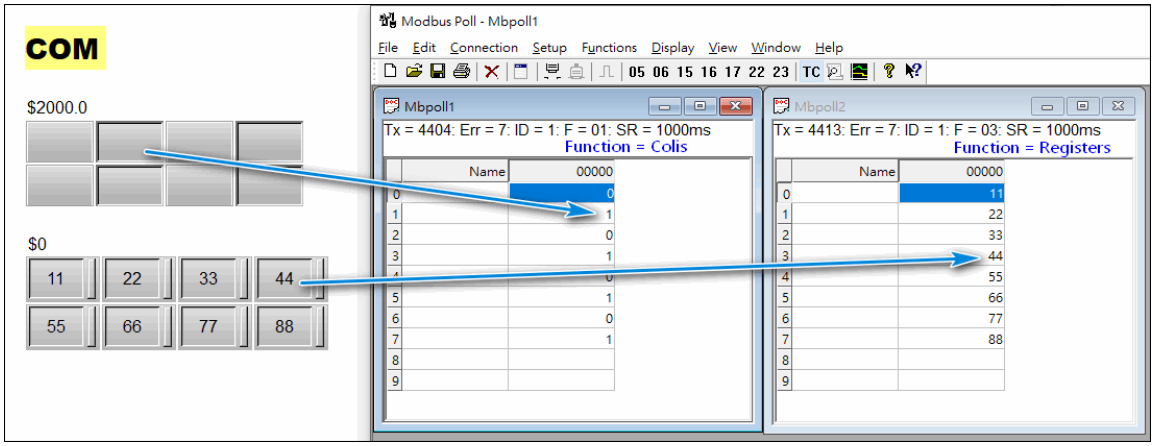
Example Settings



1. In the **Modbus TCP/COM mapping table**, configure the mapping table.
   - Set the HMI address to $2000.0, corresponding to the Modbus address of **Coils** function code 00001.
   - Set the HMI address to $0, corresponding to the Modbus address of **Registers** function code 40001.
2. Create elements on the screen and set the HMI address corresponding to Modbus, and then download the project to the HMI.

   The Modbus Master device is connected through the Serial Port. Change values of elements on the HMI, and the HMI transmits data through Modbus.

The following figure shows the result that **Modbus Poll** is used as the Modbus Master and communicated with the HMI.

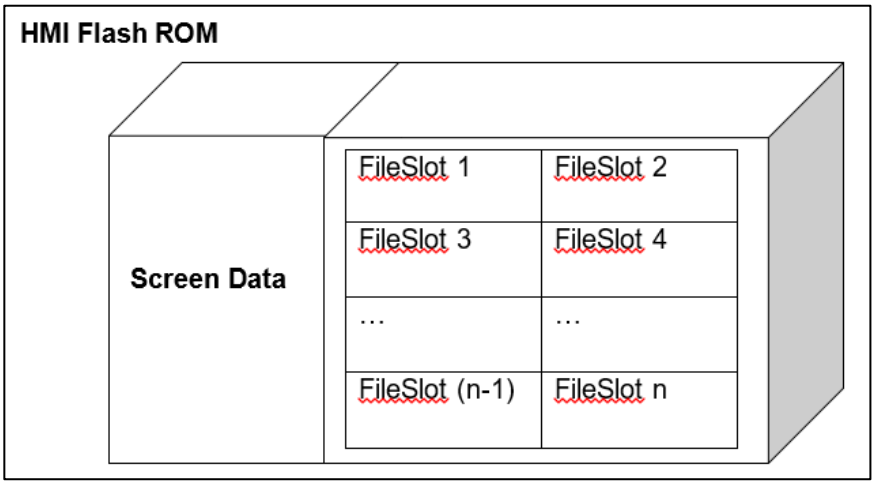

COM
$2000.0
$0
11
22
33
44
55
66
77
88
Modbus Poll - Mbpoll1
File Edit Connection Setup Functions Display View Window Help
05 06 15 16 17 22 23 TC
Mbpoll1
Tx = 4404: Err = 7: ID = 1: F = 01: SR = 1000ms
Function = Colis
Name 00000
0 0
1 1
2 0
3 1
5 1
6 0
7 1
Mbpoll2
Tx = 4413: Err = 7: ID = 1: F = 03: SR = 1000ms
Function = Registers
Name 00000
0 11
1 22
2 33
3 44
4 55
5 66
6 77
7 88

FileSlot Management

The **FileSlot Management** feature enables flexible planning for saving and using large amounts of data. In the **FileSlot Management** dialog, set the size and content of each FileSlot, and then use the macro command (FileSlotRead) to read the file content into the register for use. You can also store register data within these files. For example, in products with different robotic arm programs, these programs can be stored in separate FileSlots and loaded from their respective FileSlot to realize rapid switching between product programs.

The usage rules of FileSlot are as follows.

- After downloading a FileSlot to the HMI, the HMI's ROM memory will reserve the required space for the FileSlot.



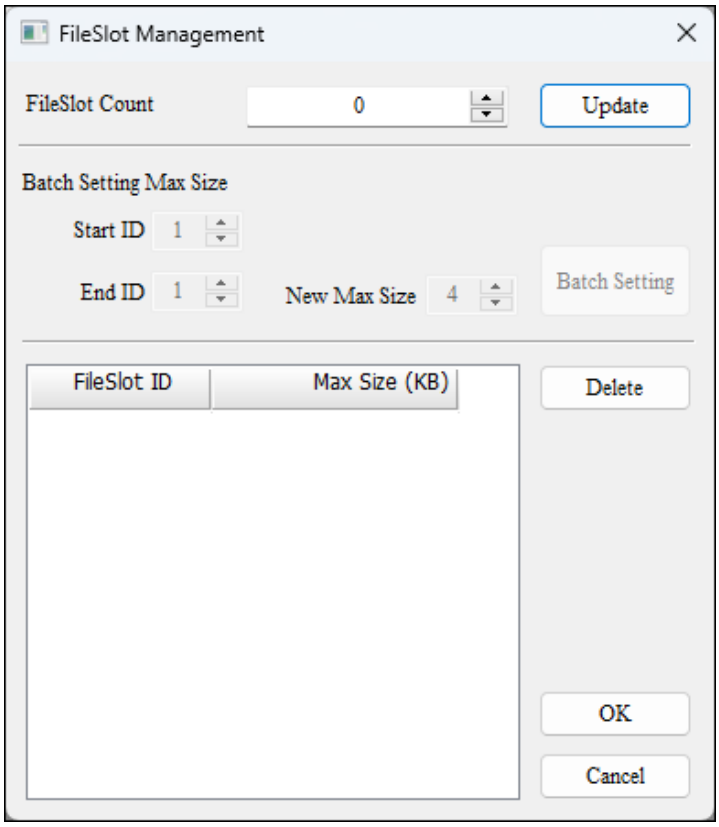
- For efficiency and data retention purposes, when a screen with FileSlot settings is downloaded to the HMI, the HMI only reserves the memory space for the FileSlot; actual files must be generated using file macros (FileSlotWrite, ImportFileSlot).
- To clear FileSlot data from the HMI, you must either perform **Format**, **Reset HMI**, or remove the macro command (FileSlotRemove).
- If the HMI already has FileSlot data and the downloaded FileSlot data is fewer in number, the excess FileSlot data will be cleared during the screen download process. For example, if 5 FileSlot data are downloaded, but the HMI originally has 10, FileSlots 6 through 10 will be cleared.

- When using the **File Copy** function of the HMI system to copy the screen data files or auto update data files created in DIAScreen to an HMI containing FileSlot data, the above-mentioned rules apply: the FileSlot data will be cleared only under this condition.
- Since FileSlot data remains unchanged during screen updates, it is recommended to perform **Format** or **Reset** before redesigning the screen layout or adjusting FileSlot file size.
- FileSlot data can be exported through the **File Copy** function in the HMI system. The exported files are stored in the folder HMI-000\FILESLOT and can be imported into other HMIs. If the target HMI already contains FileSlot data, it will be overwritten.

How to open the **FileSlot Management** dialog?

- On the toolbar, click **Project** > **Other Setting** > **FileSlot Management**.

The following table lists the settings in the **FileSlot Management** dialog with their description.

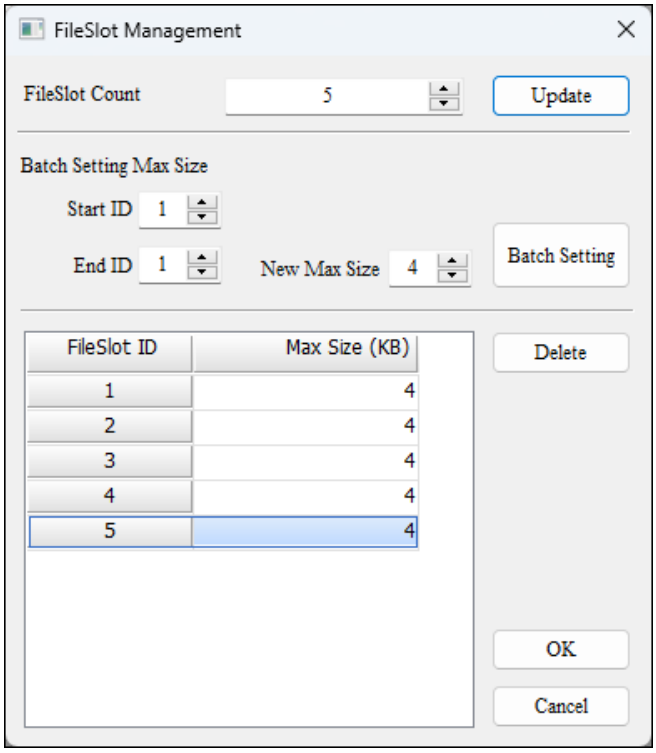


Function	Description
FileSlot Count	Select or enter the desired number of FileSlots; a maximum of 1024 FileSlots can be entered.
Update	After entering the total number of FileSlots, click **Update** to allocate the specified number of FileSlots.
Batch Setting Max Size	Select or enter the maximum file size for FileSlot, then click **Batch Setting** to apply **New Max Size** to the FileSlots with specified FileSlot IDs.
Max Size	Individually set the maximum file size for each FileSlot. The range value is 4~10,240 KB.
Delete	Click to delete the selected FileSlot ID. After deletion, the FileSlot ID will be automatically replaced.

Follow these steps to configure FileSlot.

To configure FileSlot

1. On the toolbar, click **Project** > **Other Setting** > **FileSlot Management**.
2. In the **FileSlot Management** dialog, set **FileSlot Count** to **5**, and then click **Update** to display the set number of FileSlots.
3. Set **New Max Size** for each FileSlot, then click **OK**.

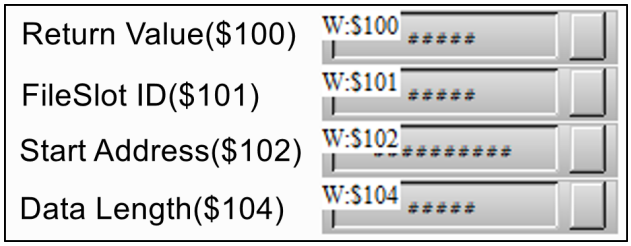


4. Create two **Momentary Button** elements on the screen and set their **Write Address** to **$50.0** and **$50.1**.
5. In **On Macro**, enter the following instructions respectively:

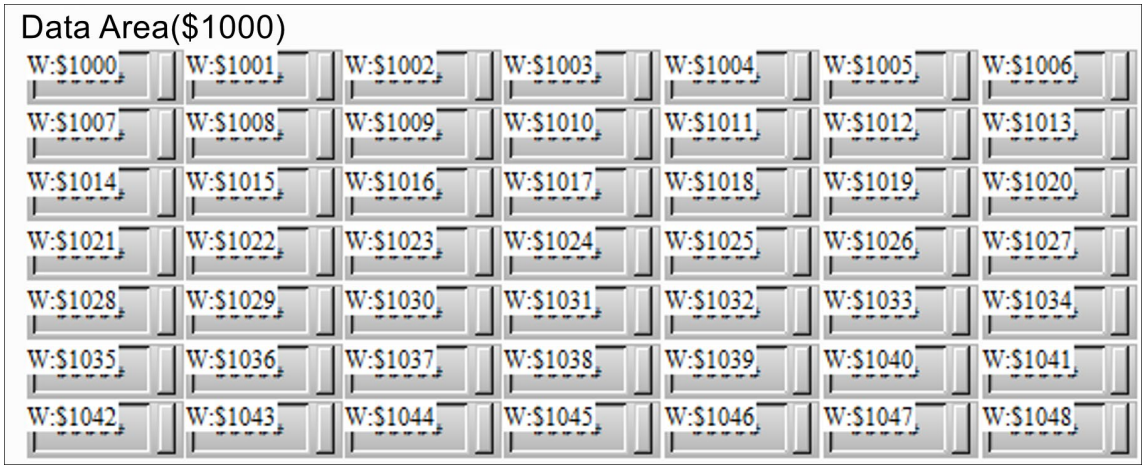
   $100 = FileSlotRead($101, $1000, $102, $104)

   $100 = FileSlotWrite($101, $1000, $102, $104)

6. Create **Numeric Entry** elements on the screen that correspond to the macro instruction parameters.



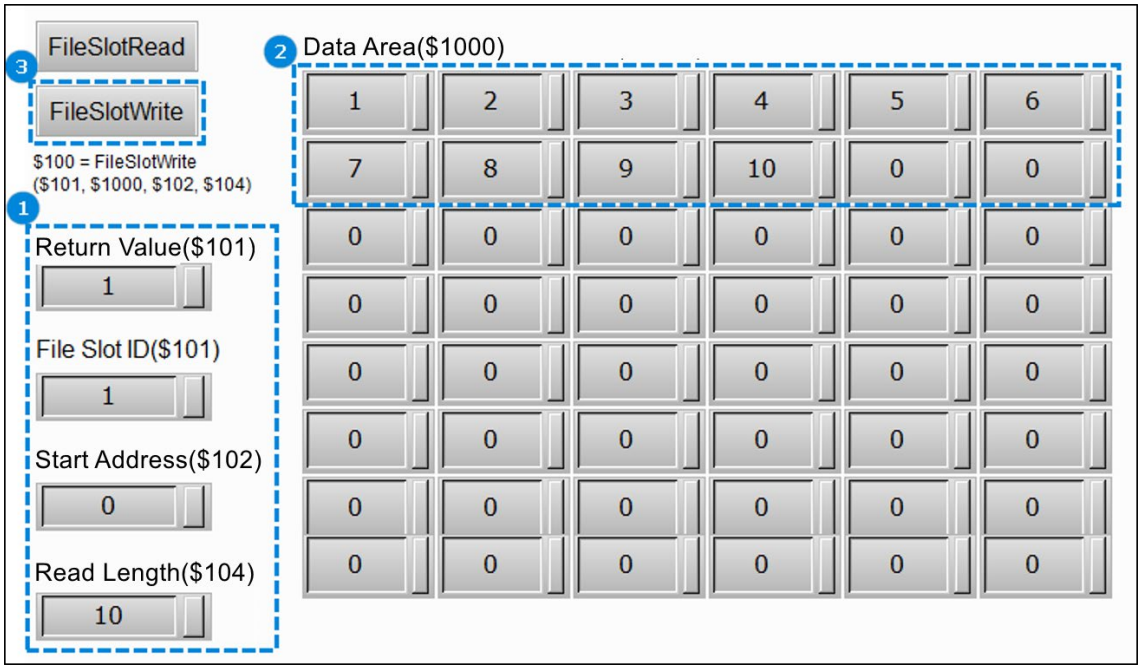
7. Create **Numeric Entry** elements on the screen to read the FileSlot file content and sequentially set their **Write Address** from **$1000** to **$1048**.



8. Compile and download the screen to the HMI.

The execution result is as follows.

Use the FileSlotWrite macro to write data from the data area into a specified FileSlot file. As illustrated below, write 10 Word type address from the data area into the FileSlot with ID = 1. After writing, the data can be read back using the FileSlotRead macro.


FileSlotRead
FileSlotWrite
$100 = FileSlotWrite
($101, $1000, $102, $104)
Return Value($101)
1
File Slot ID($101)
1
Start Address($102)
0
Read Length($104)
10
Data Area($1000)
1 2 3 4 5 6
7 8 9 10 0 0
0 0 0 0 0 0
0 0 0 0 0 0
0 0 0 0 0 0
0 0 0 0 0 0
0 0 0 0 0 0
0 0 0 0 0 0

EIP Data Exchange Table

The HMI supports Ethernet / IP connection. You can import EDS (Electronic Datasheet) files or manually enter CIP parameters to establish connection data and configure the data exchange between the HMI internal memory and network devices through the data exchange table.

Note:

- Only available for DOP-300 series HMI.
- A maximum of 6 data exchange tables are supported and the same device supports up to 4 data exchange tables.

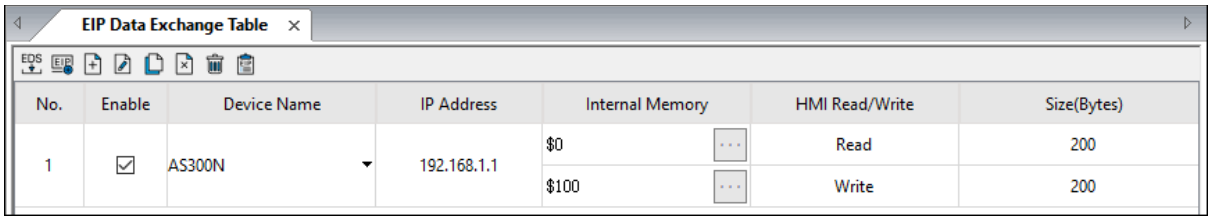
How to open the **EIP Data Exchange Table** configuration page?

- In the **Project** pane, double-click the EIP Data Exchange Table.

The following table lists the settings on the toolbar in the **EIP Data Exchange Table** configuration page and their descriptions.

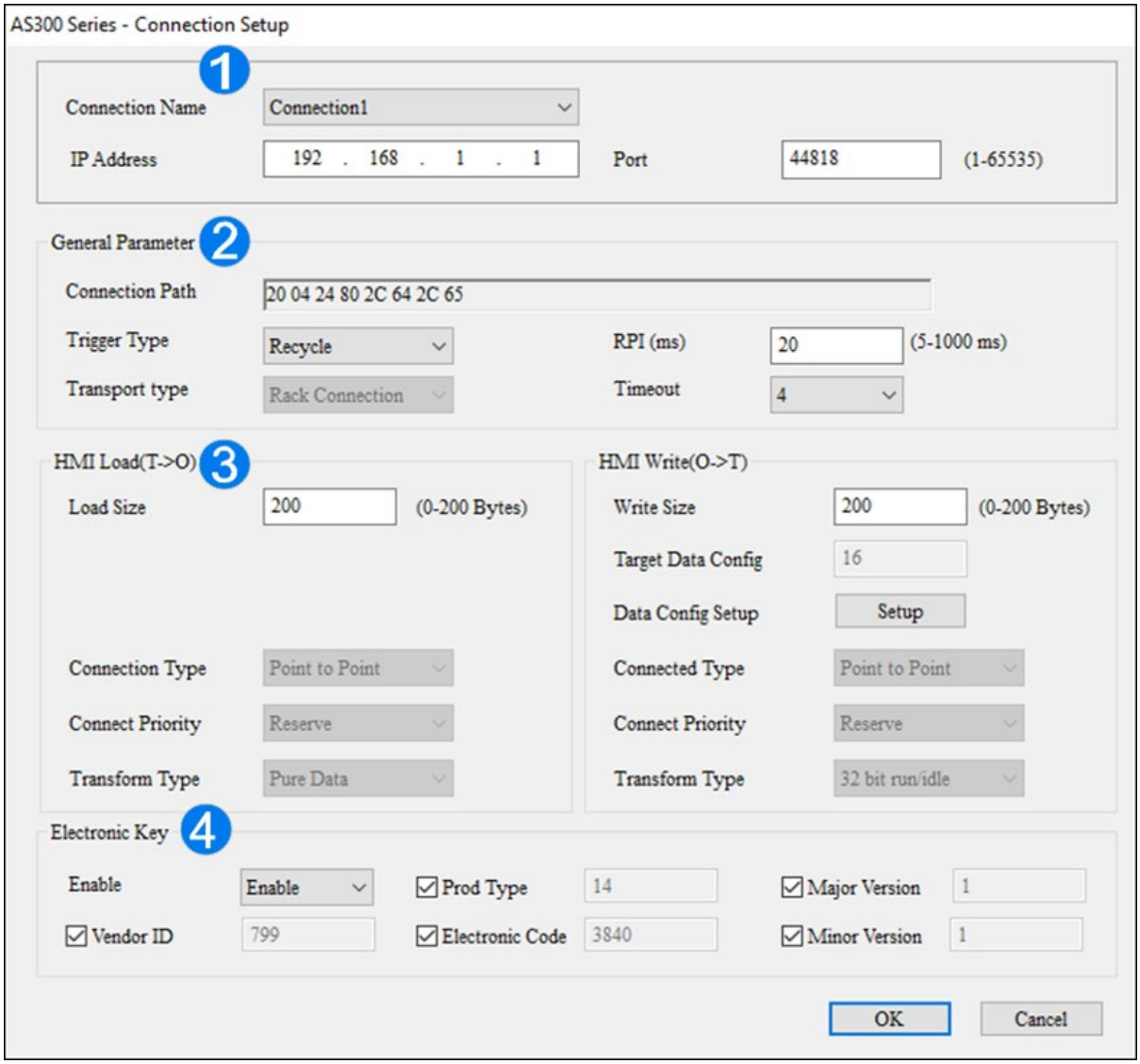
Setting	Description
	Import the EDS file of the connected device and set the connection parameters.
	Manually add an EIP device. Establish a device for connection by entering CIP parameters.
	After adding an EIP device, click this icon to add a data exchange table.
	After selecting one row of the data exchange table, click this icon to edit the connection parameters.
	After selecting one row of the data exchange table, click this icon to copy the selected row.
	After selecting one row of the data exchange table, click this icon to delete the selected row.
	Click to delete all data exchange tables.
	Click to open the **Device List** dialog. In the **Device List** dialog, select the newly added or imported EIP devices to delete. **Note**: When a device is deleted, its corresponding data exchange table is also deleted.

The following table lists each field setting in the **EIP Data Exchange Table** configuration page and their descriptions.



Setting	Description
No.	Displays the line numbers of the data exchange table.
Enable	Select to enable the data exchange of the row. If not selected, the data exchange configuration will not be loaded to the HMI.
Device Name	Select the name of the connected device. This name is defined in an EDS file or when you manually add an EIP device.
IP Address	Displays the IP address of the connected device. It can be configured in the **Connection Setup** dialog. **Note**: The same IP cannot be set for different devices.
Internal Memory	Set the internal memory address of the HMI for data exchange. **Note**: The read / write addresses of the same data exchange table cannot be the same.
HMI Read / Write	Displays the read / write mark for the HMI.
Size (Bytes)	Displays the size of data exchange. It can be configured in the **Connection Setup** dialog.

The following table lists the settings in the **Connection Setup** dialog of the device imported through EDS and their descriptions.



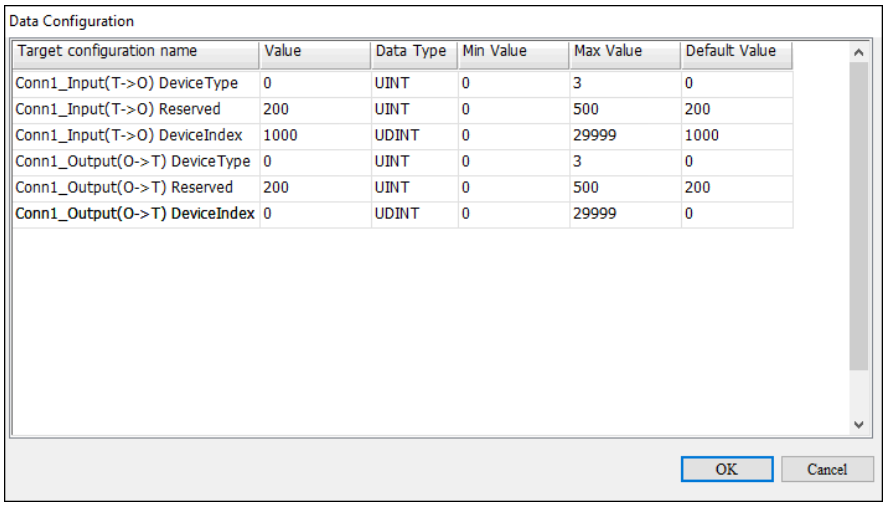
1 Connection settings

Setting	Description
Connection Name	Select the name of the connected device for data exchange. **Note**: When the same IP address is set for the same device, the same connection name cannot be selected.
IP Address	Enter the IP address of the device.
Port	Enter the device port. The range value is 1–65525, and the default value is **44818**.

2 General Parameters

Setting	Description
Connection Path	Displays the connection path.
Trigger Type	Select the trigger type of the connection.
Transport Type	This setting is available only when you manually add an EIP.
RPI (ms)	Enter the RPI of the connection. The range value is 5–1000, and the default value is **20**.
Timeout	Select a timeout for the connection. The default value is **4**.

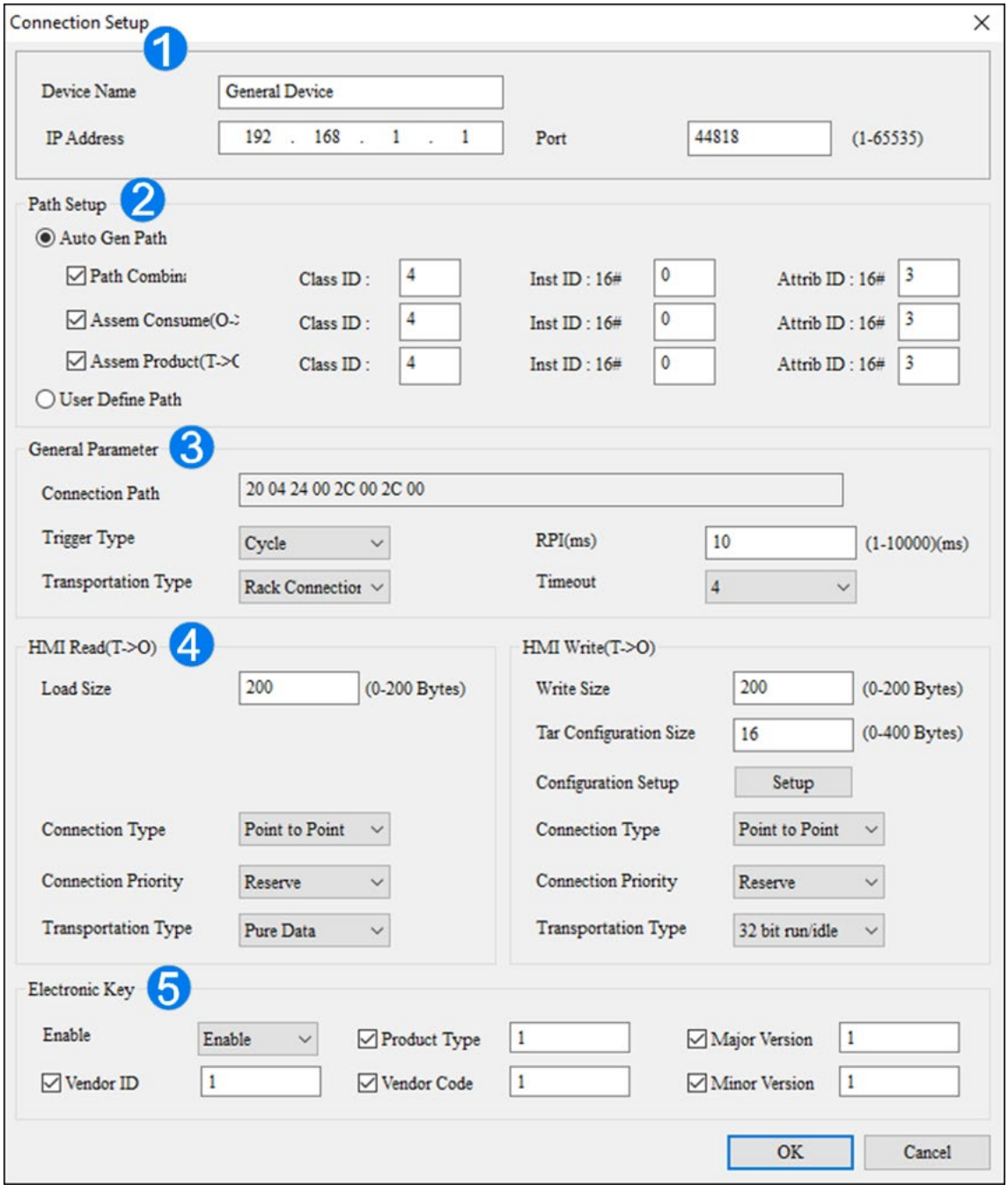
❸ HMI Read / Write

Setting	Description
Load Size	Enter the read size. The range value is 0–200 (bytes), and the default value is **200** (bytes).
Write Size	Enter the write size. The range value is 0–200 (bytes), and the default value is **200** (bytes).
Tar Configuration Size	The setting is not available.
Configuration Setup	Click **Setup** to configure the connection data of the target device. In the **Data Configuration** dialog, set the data parameters in the **Value** field. 
Connection Type	The setting is not available.
Connection Priority	The setting is not available.
Transform Type	The setting is not available.

④ Electronic Key

Setting	Description
Enable	Select the items to match. **Note**: The parameter setting of the electronic key is available when you select the **Enable** checkbox in the **Enable** field.

The following table lists the settings in the **Connection Setup** dialog by manually establishing an EIP device and their descriptions.



1 Connection settings

Setting	Description
Device Name	Enter the name of the device.
IP Address	Enter the IP address of the device.
Port	Enter the device port. The range value is 1–65525, and the default value is **44818**.

2 Path Setup

Setting	Description
Auto Gen Path	Enter the CIP parameters, and the **Connection Path** will be automatically generated according to the set parameters.
User Define Path	Define the **Connection Path** in the **General Parameter** area.

3 General Parameter

Setting	Description
Connection Path	This setting is available only when the **User Define Path** is selected.
Trigger Type	Select the trigger type of the connection.
Transportation Type	Select the transportation type of the connection.
RPI (ms)	Enter the RPI of the connection. The range value is 1–10000, and the default value is **10**.
Timeout	Select a timeout for the connection. The default value is **4**.

4 HMI Read / Write

Setting	Description
Load Size	Enter the read size. The range value is 0–200 (bytes), and the default value is **200** (bytes).
Write Size	Enter the write size. The range value is 0–200 (bytes), and the default value is **200** (bytes).
Tar Configuration Size	Enter the size of the target configuration. The range value is 0–400 (bytes), and the default value is **16** (bytes).

Setting	Description
Configuration Setup	Click **Setup** to configure the connection data of the target device. In the **Data Configuration** dialog, set the data parameters in the **Value** field.
Connection Type	Select the connection type.
Connection Priority	Select a connection priority.
Transportation Type	Select the transportation format.

⑤ Electronic Key

Setting	Description
Enable	Select the items to match and enter the parameters. **Note**: The parameter setting of the electronic key is available when the **Enable** is selected in the **Enable** field.

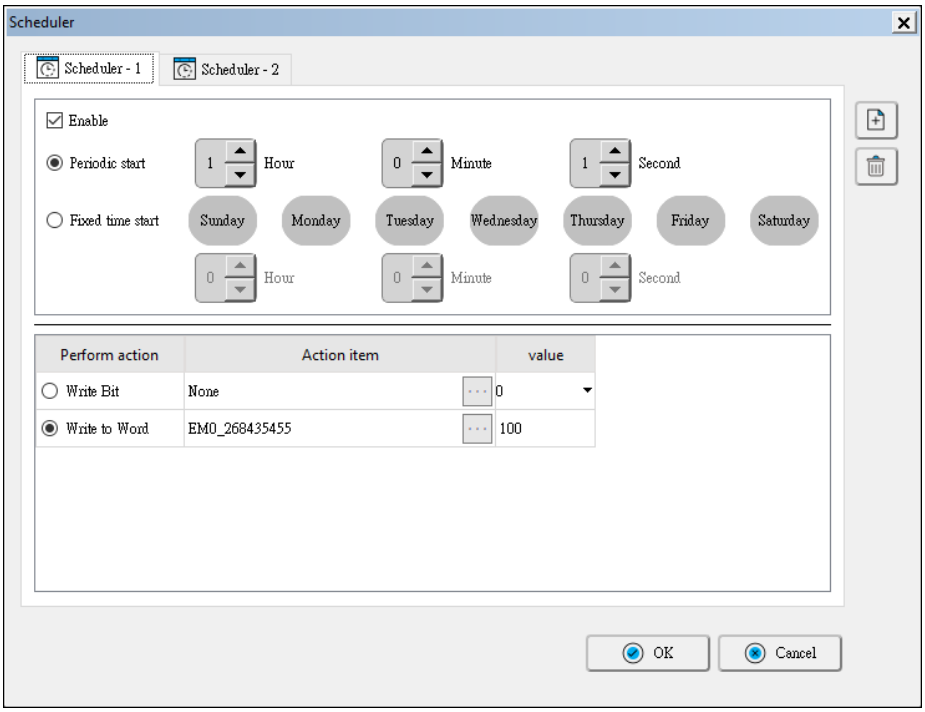
Scheduler

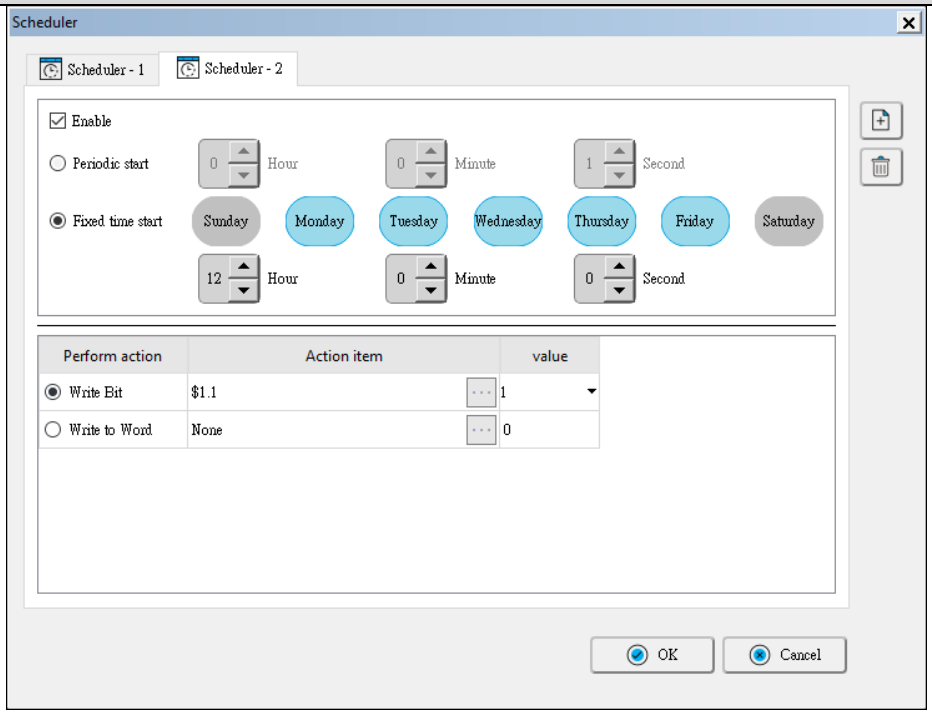
You can use the Scheduler to periodically set the writing value for the address.

How to open the **Scheduler**?

- Click **Project** > **Other Settings** > **Scheduler** on the toolbar.

The following table lists the settings in the Scheduler dialog with their description.

Setting	Description
Enable	The scheduler settings are available after selecting this setting.
Periodic start	Select to start the schedule periodically. The following figure is an example of writing the value 100 to the Word address EM0_268435455 every hour. 
Fixed time start	Select to start the schedule at a fixed time. The following figure is an example of writing the value 1 to the Bit address $1.1 at a fixed time every week.

Setting	Description
	
	Click to add the scheduler. **Note**: Up to 3 schedulers can be added.
	Click to delete the scheduler.

Installment

The installment function allows customers to configure the device in stages by paying in installments to solve trial use and funding issues. A trial is also provided for customers. When the trial expires, it will lock the HMI screen to prevent further use.

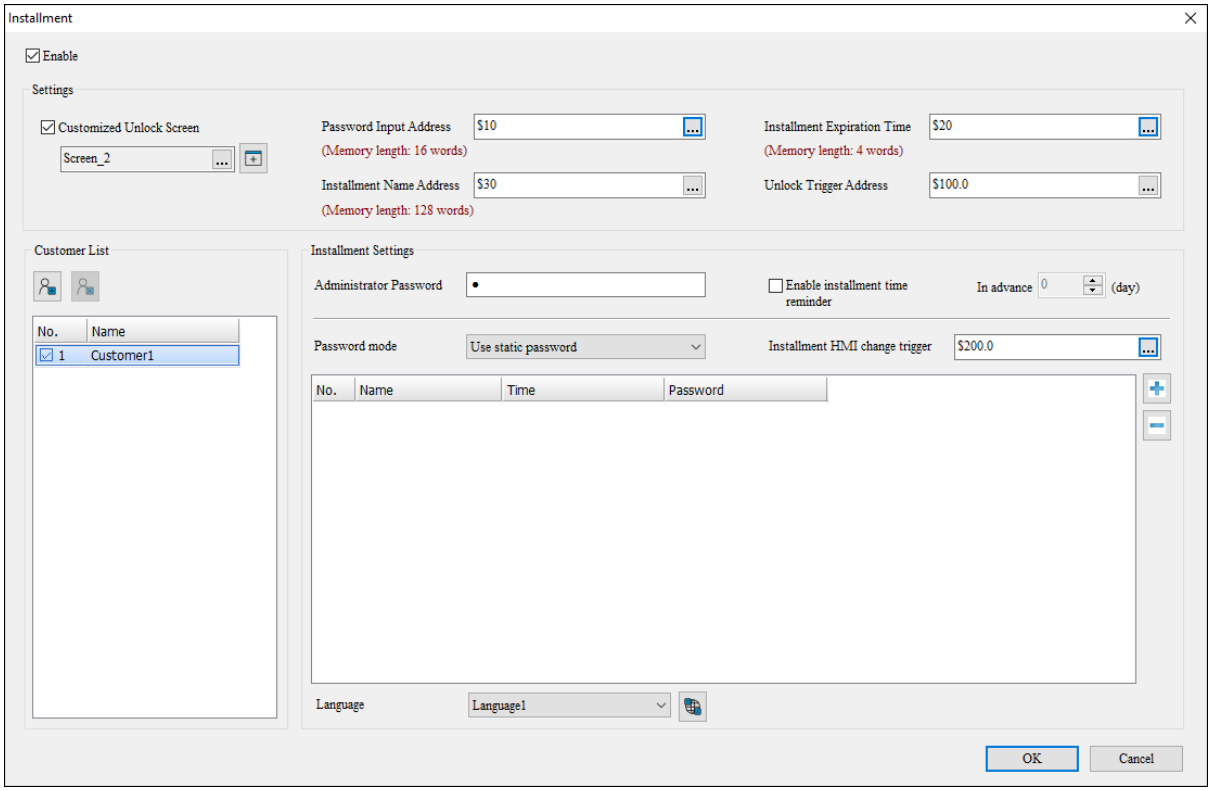
Note:

- If external memory addresses, such as controller addresses, are used for the installment settings, it is important to ensure that the device is communicating properly and functioning correctly to avoid errors occurring when executing the installment function.
- The installment function does not support online simulation and offline simulation.

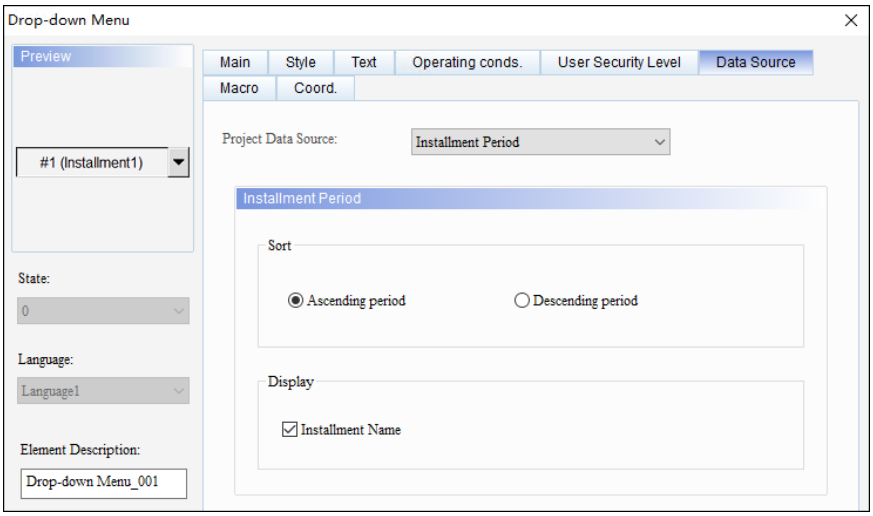
How to open the **installment** function?

- Click **Project** > **Other Setting** > **Installment** on the toolbar.

The following table lists the settings in the **Installment** dialog with the descriptions.



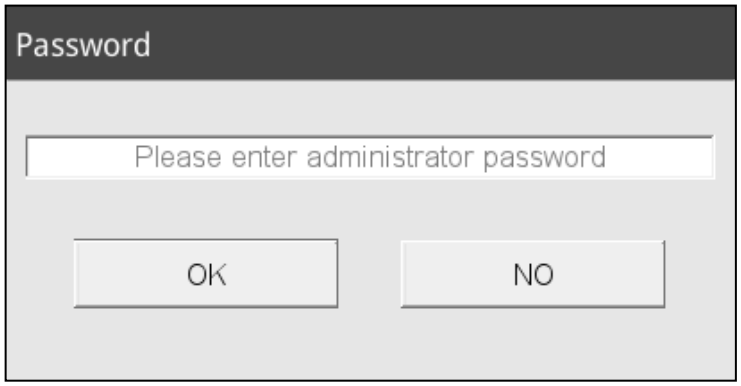
❶ Settings

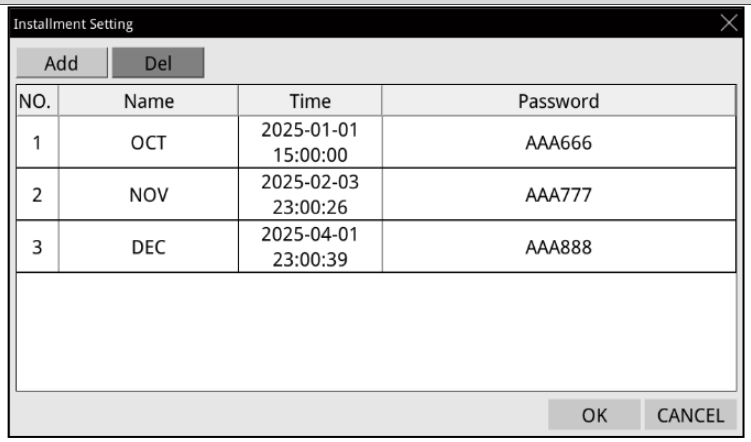
Settings	Description
Customized Unlock Screen	If selected, the existing sub-screen is available for you to select which can be used as the unlock screen. You can also click to add the example unlock screen. **Note**: The **Unlock Screen** must be configured with a **Drop-down Menu** element, and the **Project Data Source** must be selected as **Installment Period**, so that you can select the installment to be unlocked when performing unlocking. 
Password Input Address	Set the password input address. Corresponding to the address set for the **Character Entry** element on the unlock screen.
Installment Name Address	Set the installment name address. The **Character Display** or **Multi-language Input** element can be configured on the screen to show the name of the current unlocked installment. **Note**: If the installment name is set with characters other English and numbers, use the **Multi-language Input** element to display it.
Installment Expiration Time	Set the installment expiration time address. The **Date Display**, **Time Display**, and **Weekday Display** elements can be configured on the screen to show the time of the next installment.
Unlock Trigger Address	Set the unlock trigger address. Corresponding to the unlock button configured on the unlock screen.

❷ Customer List

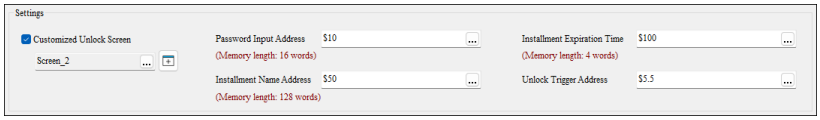
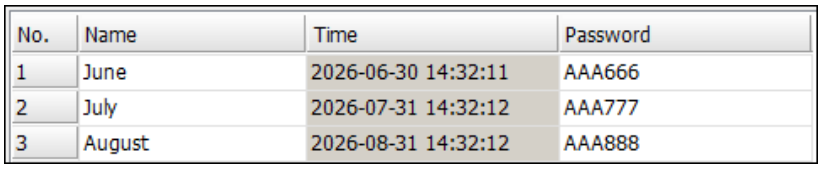
Settings	Description
	Click to add new customer. Up to 10 customers can be added and the names must not be the same. Select the customer number, the customer installment information will be applied when downloading the project.
	After selecting the customer number, click this icon to delete.

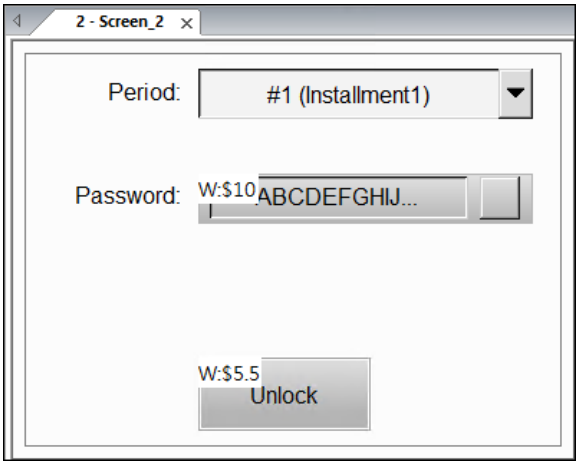
❸ Installment Settings

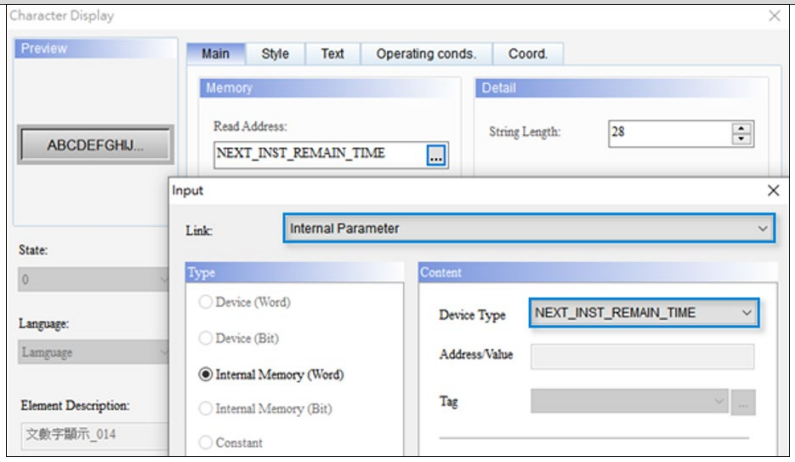
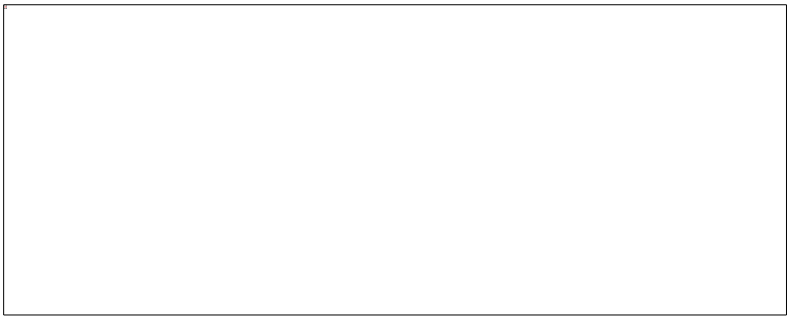
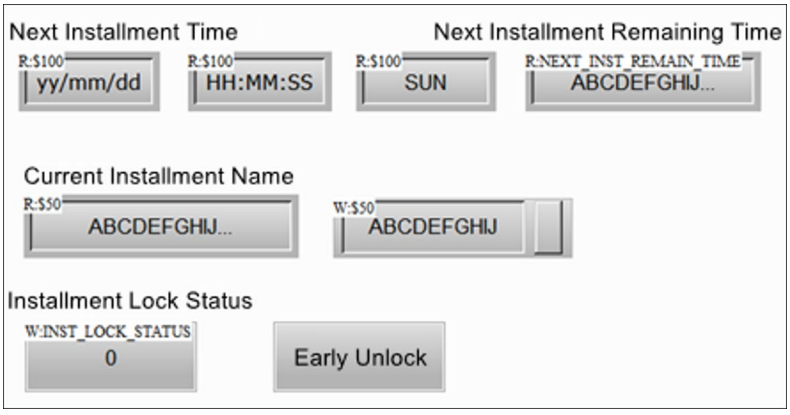
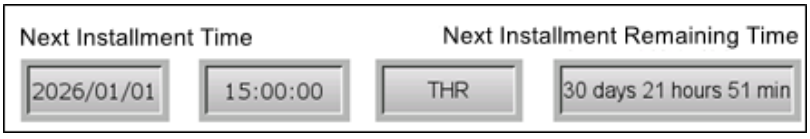
Settings	Description
Administrator Password	Set the administrator password. Enter this password on the HMI to unlock all installments.
Enable installment time reminder	If selected, you can configure the number of days to set expiration reminders. The range value is 0–15 and the default value is **1**.
Password mode	Select the password mode. • Select **Use static password**, the software will use a fixed password for each installment. • Select **Use variable password**, you can set an internal memory address and customize the **Password Length**, making it easy to set the password through macros or LUA.
Installment HMI change trigger	Set the HMI change trigger address to change the installment content on the HMI. • Trigger this address on the HMI and enter the installment setting **manager password** in the pop-up **Password** dialog.  • Displays the **Installment Settings** dialog. Add / delete the installment setting in this dialog.

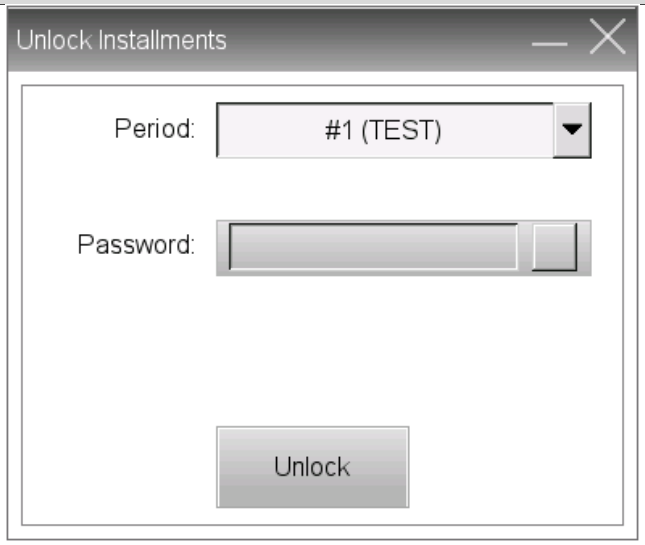
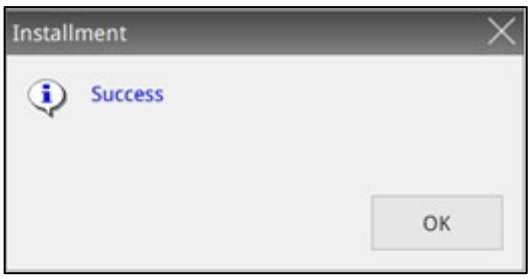
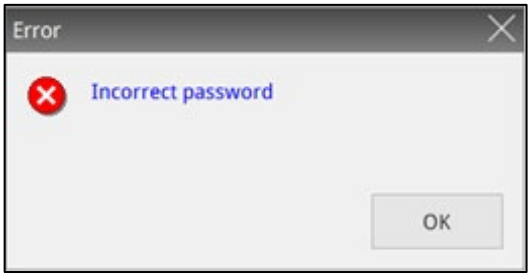
Settings	Description
	
	Click to add a new installment, up to 32 installments can be added. **Note**: • The installment password supports the characters 09, !, $, #, %, a~z, and A~Z. • For the same customer, the installment password cannot be the same (including the administrator password).
	Click to delete the selected installment(s).
Language	The installment name can be set to different languages with the **Multi-language** function.

The following table lists examples of the **Installment** setting.

Steps	Description
Setting the Installment	1. Open the Installment dialog, and then select the **Enable** checkbox. 2. Select the **Customized Unlock Screen** checkbox, and then click to add the unlock screen. 3. Set the control address.  4. Add the installment setting in the setting table. 
Setting the Unlock Screen	1. Set the **Password Input Address** to $10 for the **Character Input** element used for password input.

Steps	Description
	2. Set the **Unlock Trigger Address** to $5.5 for the unlock button. 
Setting the HMI Screen	Configure elements for operating installment and reading installment status on the screen. 1. Create **Date Display**, **Time Display** and **Week Display** elements and set the **Installment Expiration Time** to $100 to display the next installment time. 2. Create a **Character Display** element and set the **Installment Name Address** to $50 to display the name of the current installment. **Note**: If the installment name is set with characters other English and numbers, use the **Multi-language Input** element to display it. 3. Create an **Enable Next Installment** button to early unlock the installment setting before the set installment time is reached. 4. Create a **Character Display** element and set the **Read Address** to Internal Parameter “NEXT_INST_REMAIN_TIME” to display the remaining time until the next payment. **Note**: The **String Length** must be set to at least 23 characters to display the time completely.

Steps	Description
	 5. Create a **Multistate Indicator** element and set the **Read Address** to Internal Parameter “INST_LOCK_STATUS” to read the unlock status of the installment. Status 0 indicates that the installment is unlocked, while Status 1 indicates the installment is waiting to be unlocked.  
Unlocking the installment	Perform the unlock function on the HMI. 1. If the set installment time is not reached, the next installment time and remaining time will be displayed.  2. When the installment time is up or the **Enable Next Installment** button is pressed on the HMI screen, the **Unlock Installments** screen displays.

Steps	Description
	 3. In the **Unlock Installments** screen, choose the period to be unlocked, and then enter the password of the period. 4. After entering the password, click the **Unlock** button. ○ If the password is entered correctly, a message displays and prompts you that the unlock is successfully.  ○ If the password is entered incorrectly, a message displays and prompts you that the unlock is failed. 

- **Keep Installment Data when Downloading**

  To ensure the installment unlock record will not be cleared, you can go to **General** > **Configuration** > **General** > **System Setting** and select **Keep Installment payment settings after screen download** checkbox to not to update the installment information after downloading the screen.

- **System Time Setting Restriction**

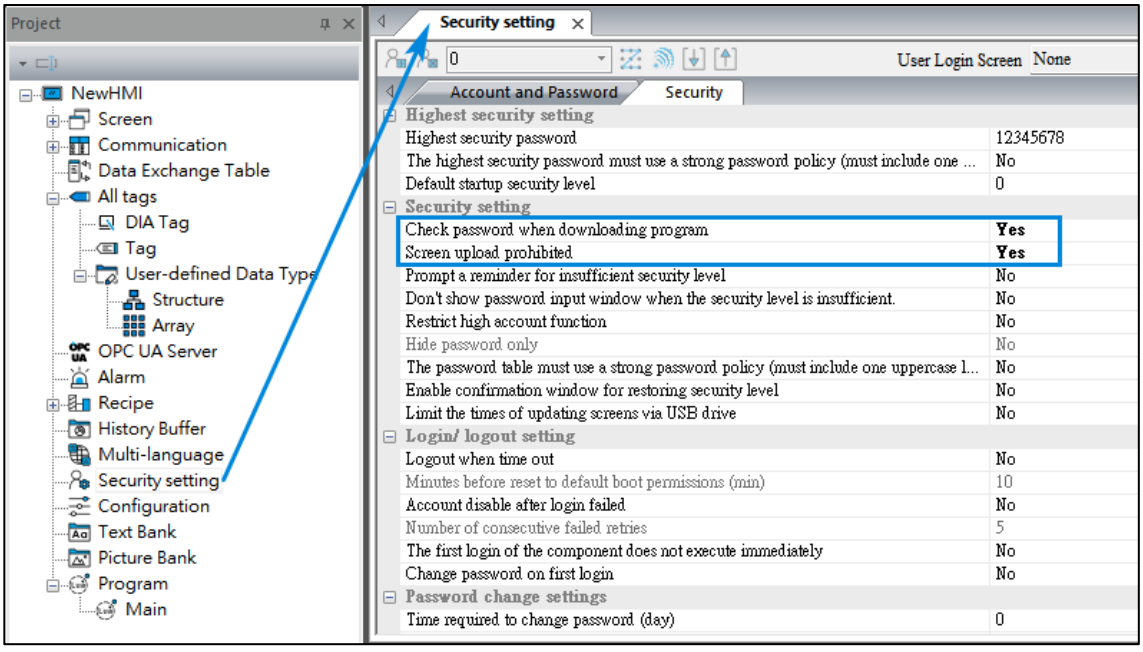
  When the **Installment** function is enabled, the related functions of system time setting will be disabled, including LUA instructions, macros, NTP, system parameters, etc to prevent the device time setting from tampering during installment, causing the installment to fail to execute as scheduled. If system time calibration is required, you can use a project without installment settings or perform a factory reset, and then download the project with installment settings.

  Note: If there is an abnormality with the device battery, the system time may be restored to the default value of 1970/01/01, and the HMI will be locked and cannot be used.

- **Project Protection**

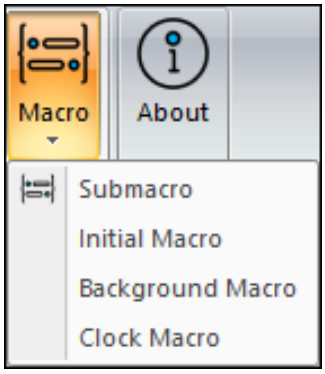
  The **Installment** function involves the financial transactions, the projects must be well-protected and set the protection mechanism, such as **Check password when downloading program**, **Screen upload prohibited**. Project files, download screen executable files, automatic screen update files, etc. must not be easily provided to outsiders.

  In the **Project** pane, double-click **Security setting**, and then set **Check password when downloading program** and **Screen upload prohibited** to **Yes** in the **Security** tab to enable the protection mechanism.



Macro

The options include Submacro, Initial Macro, Background Macro, and Clock Macro.



The following lists the cross-reference table of each function.

Function	Section
Submacro	Refer to Submacro.
Initial Macro	Refer to Initial Macro.
Background Macro	Refer to Background Macro.
Clock Macro	Refer to Clock Macro.

Types of Macro

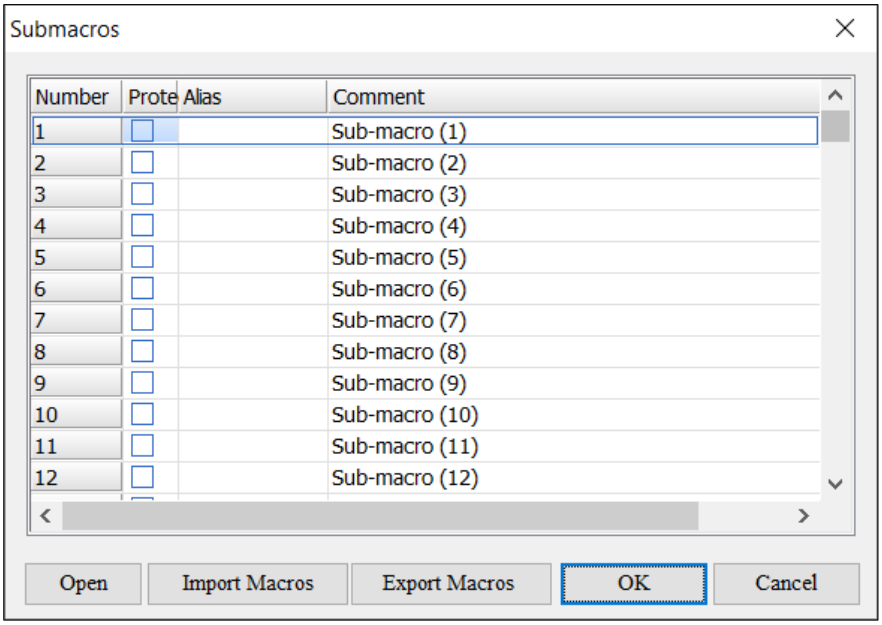
Macro consists of independent commands processing programs written by users. 512 lines of commands can be edited in each type of Macro. The main features of each type of Macro are shown in the following table.

Types of Macro	Description
Submacro	• **Submacro** provides 512 submacros, and 512 lines of commands can be programmed in each submacro. • **Submacro** is similar to subroutines in programming languages. Write highly repetitive execution programs or functions into submacros, and then call the submacros when needed.
Initial Macro	The **Initial Macro** is the first macro to be executed after the HMI is started, and it is only executed once.

Types of Macro	Description
Background Macro	The **Background Macro** executes repeatedly during the operation of the man–machine. One or several lines of programs execute at a time (not completed at one time). After the last line of the program is executed, it executes again from the beginning.
Clock Macro	Execute the macro continuously on the screen. If users have set the **Screen Open Macro**, the system first executes the **Screen Open Macro** and then executes **Screen Cycle Macro**.

Submacro

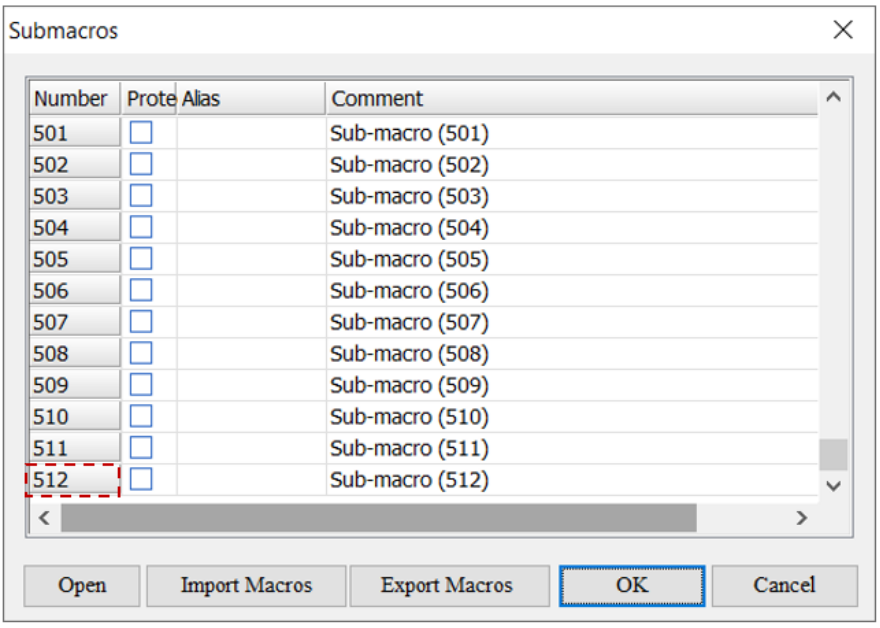
Go to **Project** > **Macro** > **Submacro** to set Submacro.



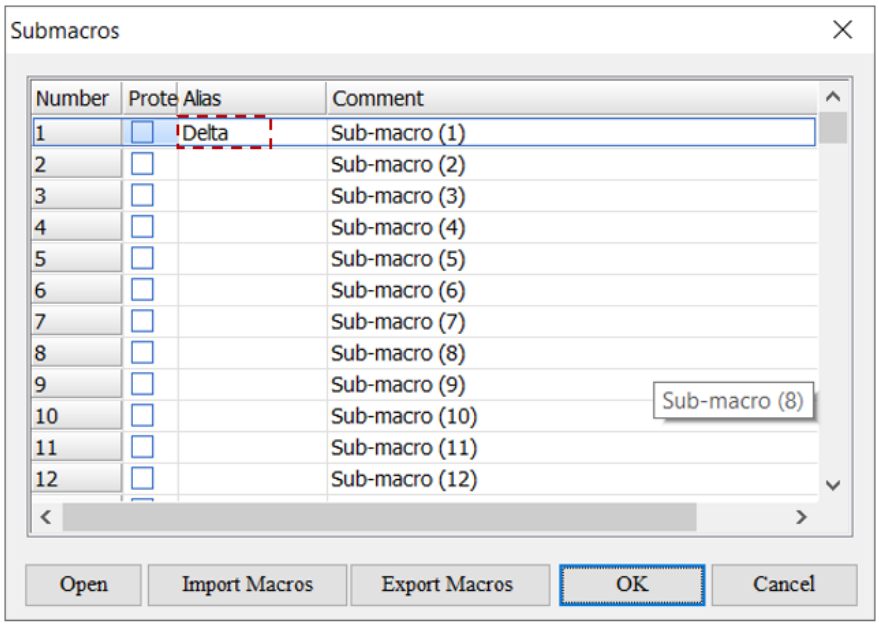
The **Submacro** provides 512 submacro programs with numbers 1 ~ 512 respectively. Submacros are similar to subroutines in programming languages where users can write highly repetitive actions or functions. In addition, users can call the submacros when needed, which saves the time for writing the macros and makes it easier for maintenance.

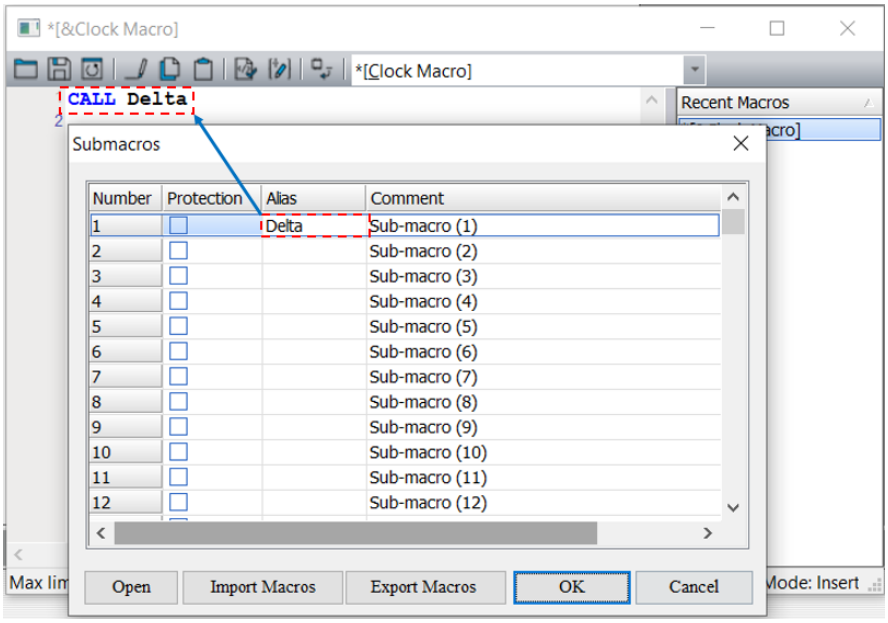
Note:

The action of calling submacros in the **Submacro** should not exceed six levels.

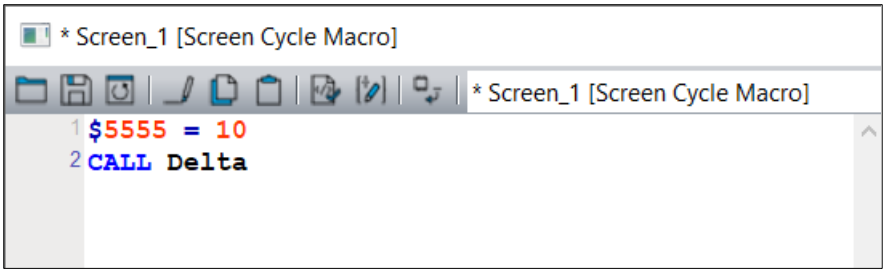


To call a **Submacro**, call its number directly or name it in the **Alias** column and call it alias. Submacro names support character and Chinese input for up to 64 characters.

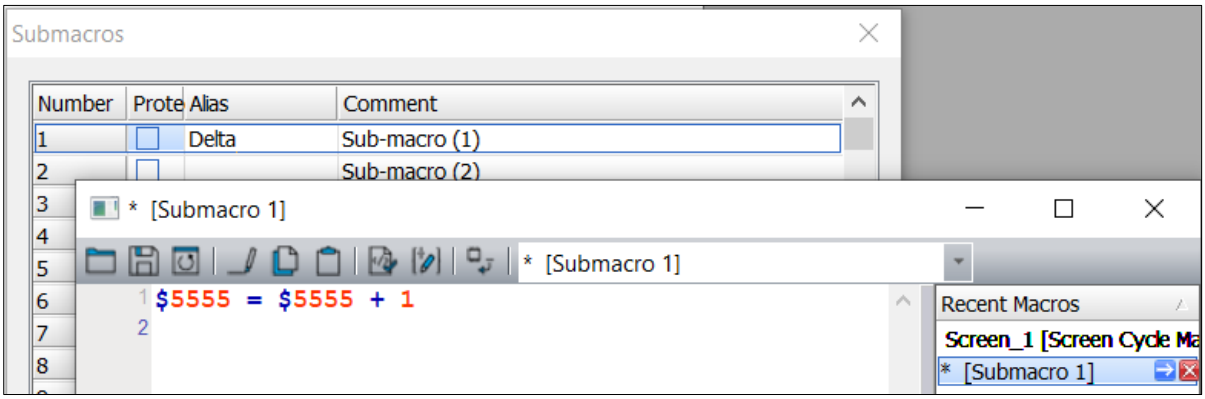


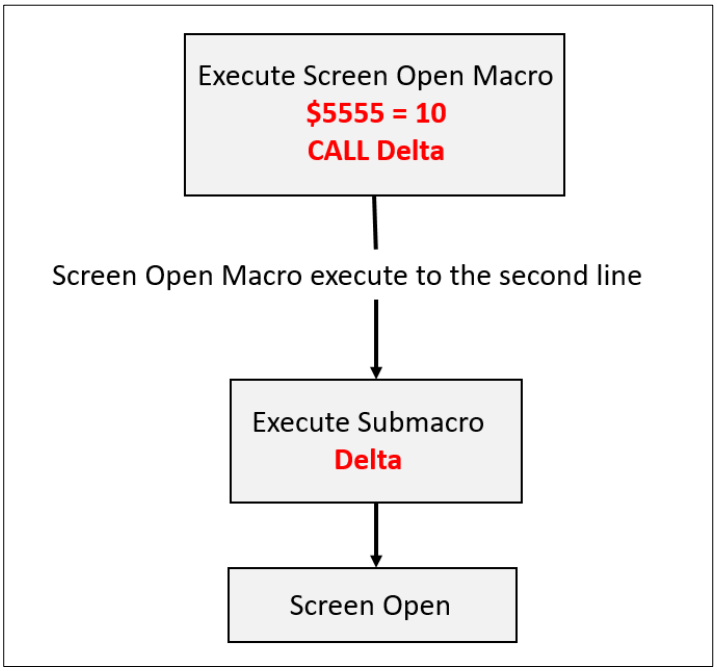


The screen opens the macro content written in the macro.

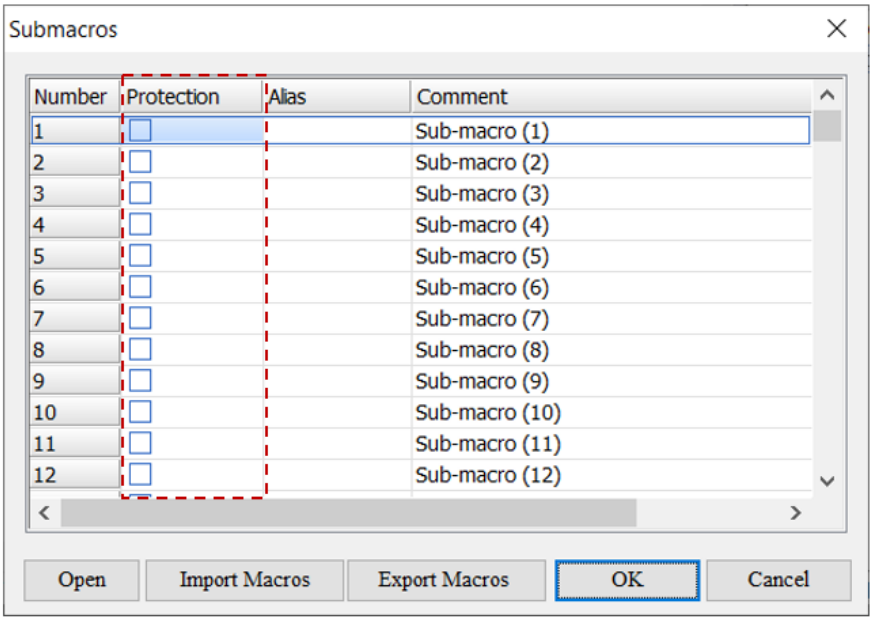


The macro content written in the **Submacros**.

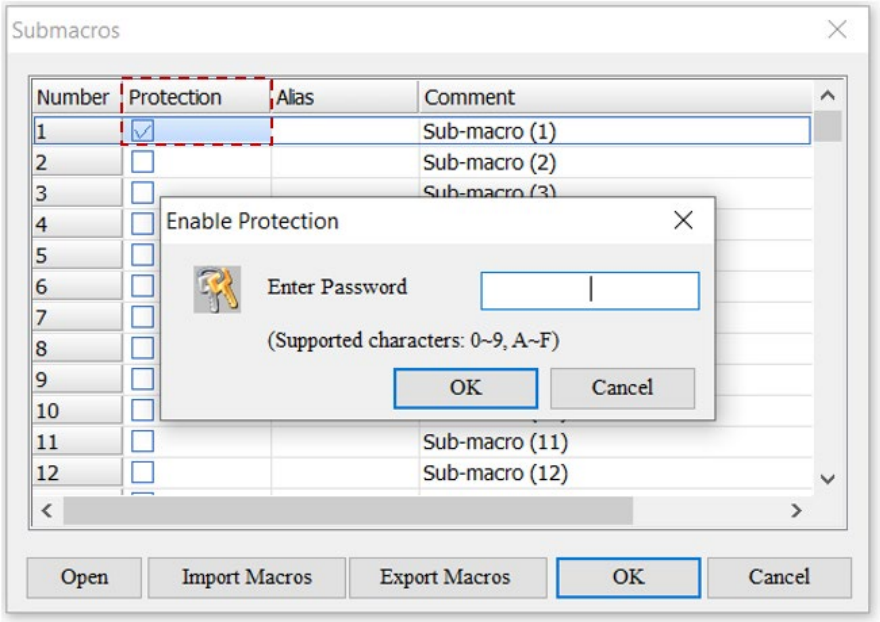




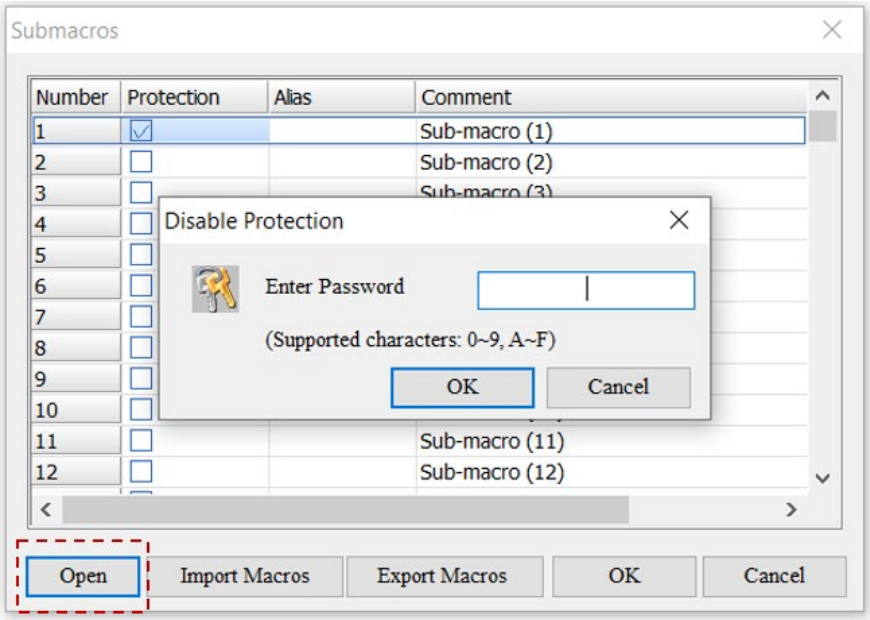
Submacro also provides password protection, which can be encrypted for each sub-macro.



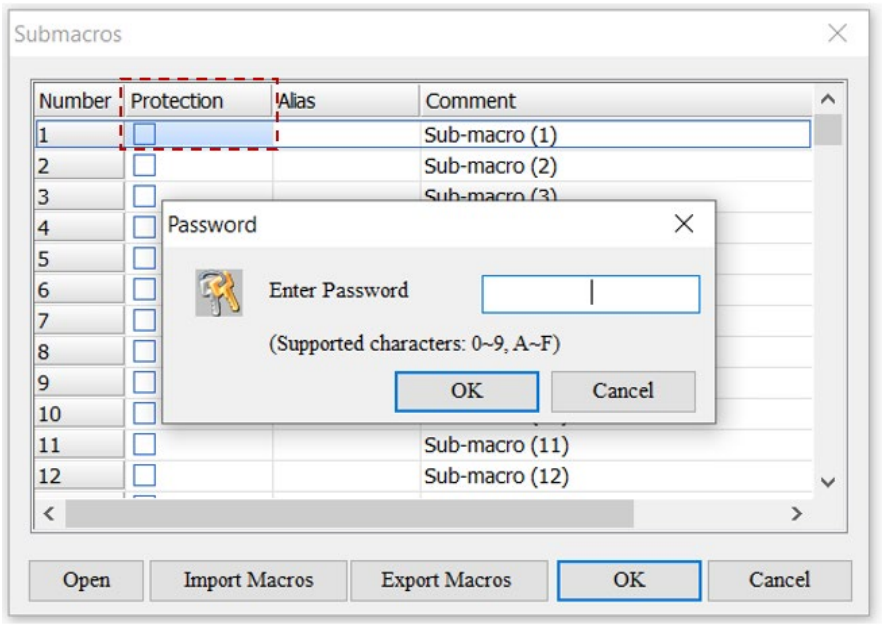
When the checkbox of **Protection** is selected, users are immediately asked to enter a set of passwords.



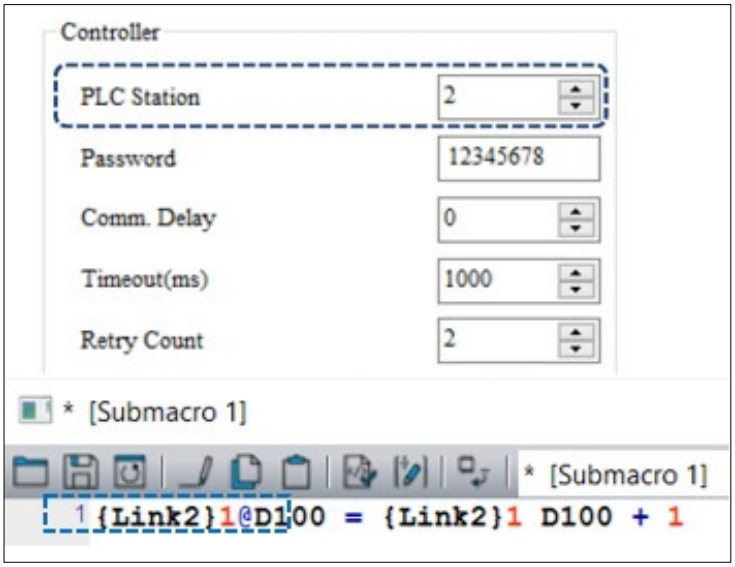
After the sub-macro number 1 is encrypted, the next time users want to open it, they must enter the password to enter the sub-macro number 1 to edit the macro command.



When the protection is unchecked, users should enter the protection password of the sub-macro number 1 which is set newly.

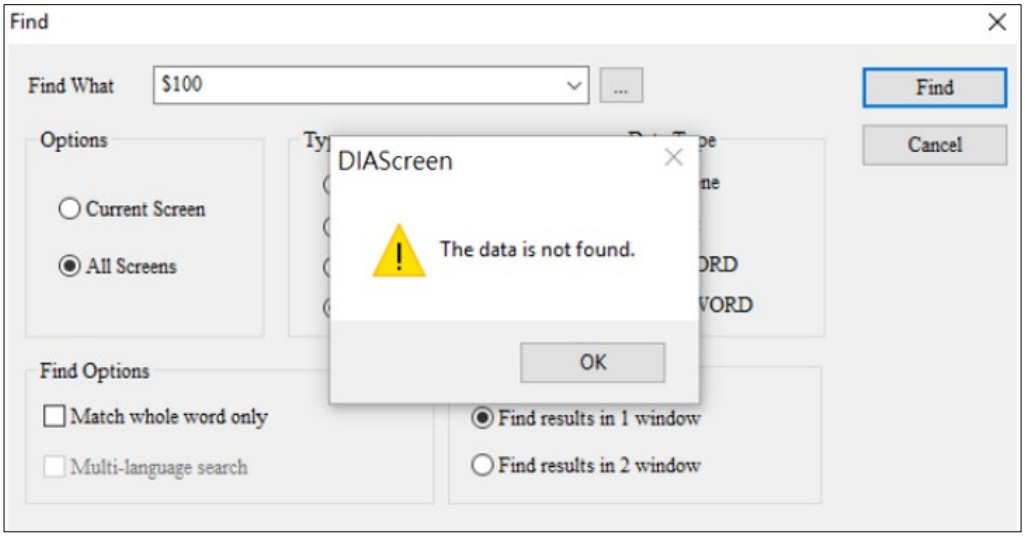


When the sub-macros are password protected, the affected functions are as follows.

(1)	• If there is a communication station number in a protected submacro, the function of changing station number will be invalid. • If the station number in the protected submacro is 1, change the **PLC Station** to 2 and then enter the protected submacro. Users will find that the station number must be 1 and will not be changed. 

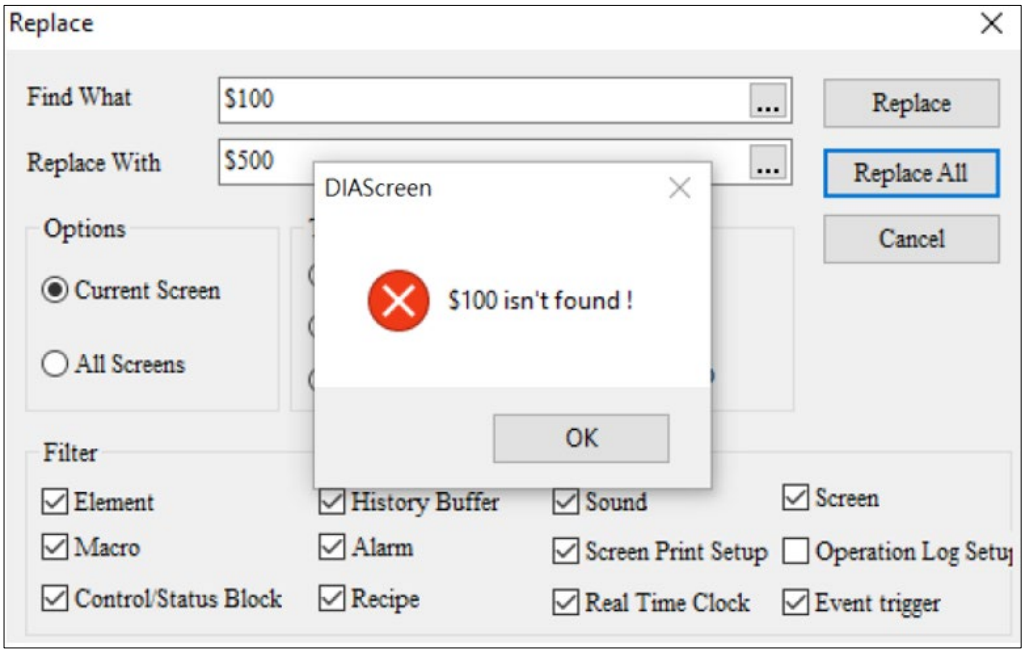
(2)

- If $100 is set in the protected sub-macro, $100 cannot be found when executing in the **Find** window.



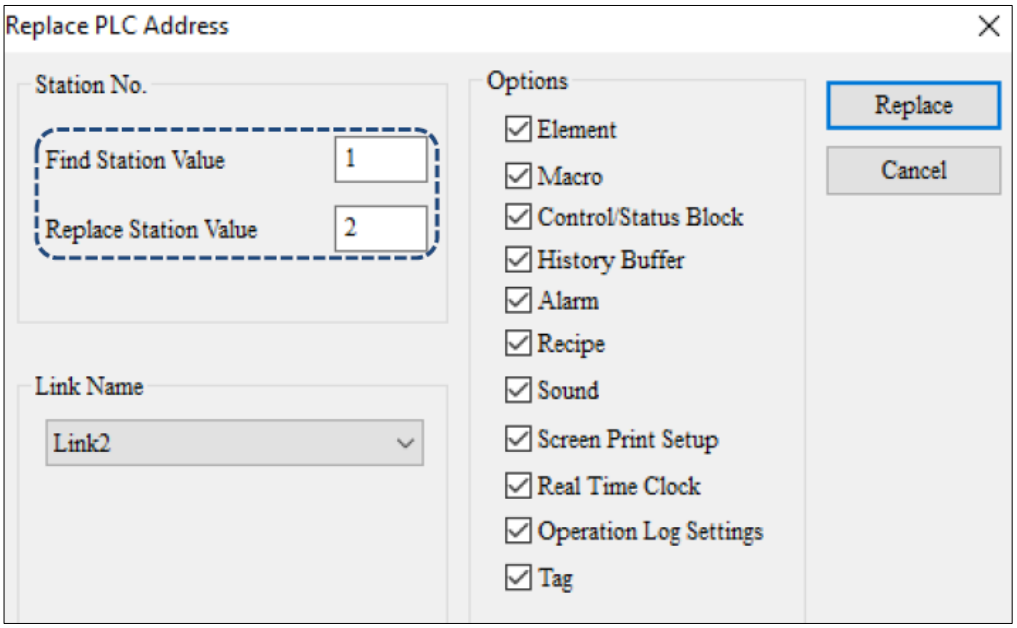
(3)

- If $100 is set in the protected sub-macro, when executing in the **Replace** window, $100 cannot be replaced with $500.



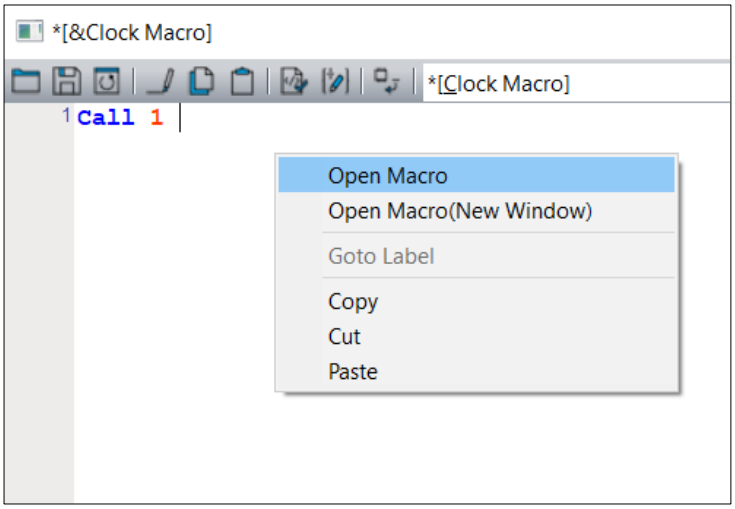
(4)

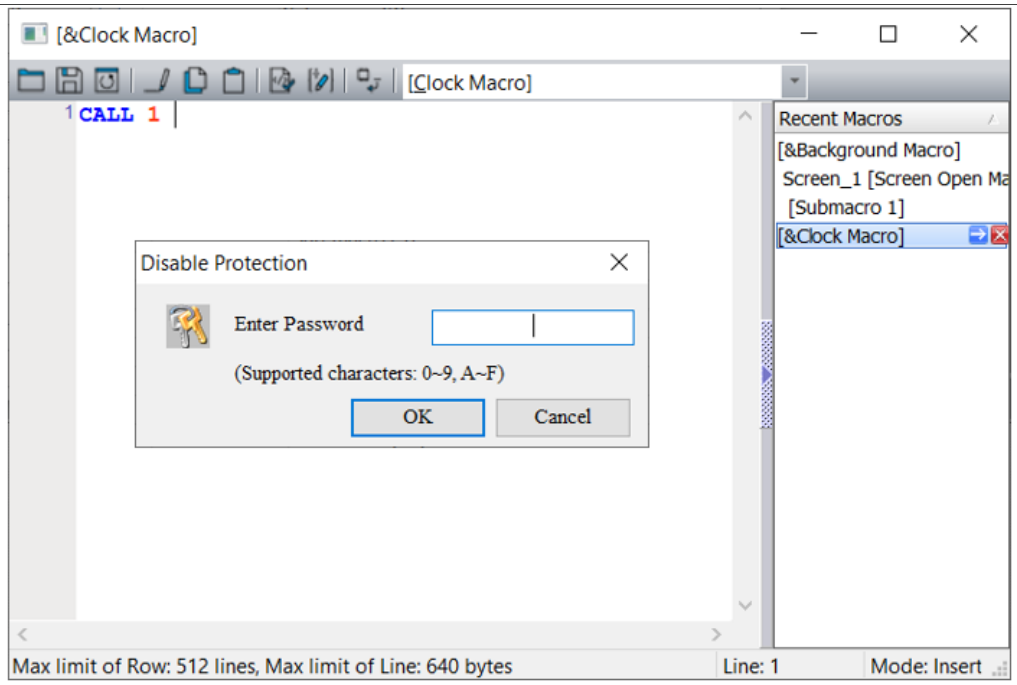
- If the Station Value in the protected submacro is 1, it cannot be replaced by 2 when the **Replace Station Value** number (Replace PLC Address) function is executed.



(5)

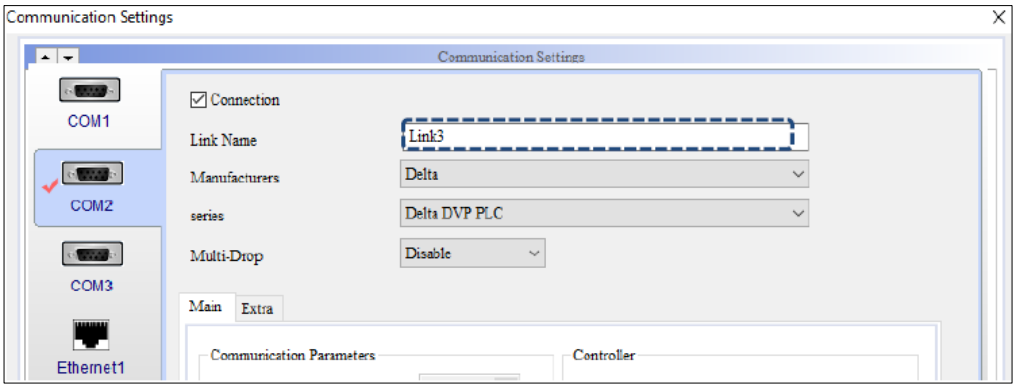
- When there is a **Call** command in the macro, right-click and select the **Open Macro** option. Users will be asked to enter the password if the submacro is password-protected.

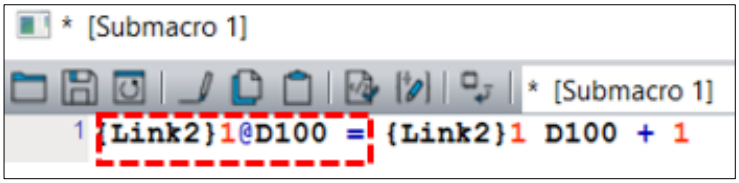




(6)

- Change the **Link Name** to Link3, and Link2 cannot be changed to Link3 in the protected submacro.

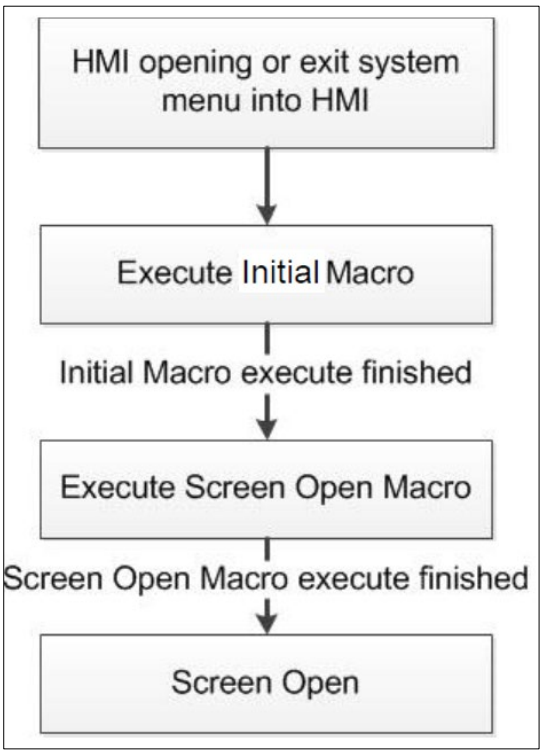




Initial Macro

Go to **Project** > **Macro** > **Initial Macro** to set the Initial Macro.

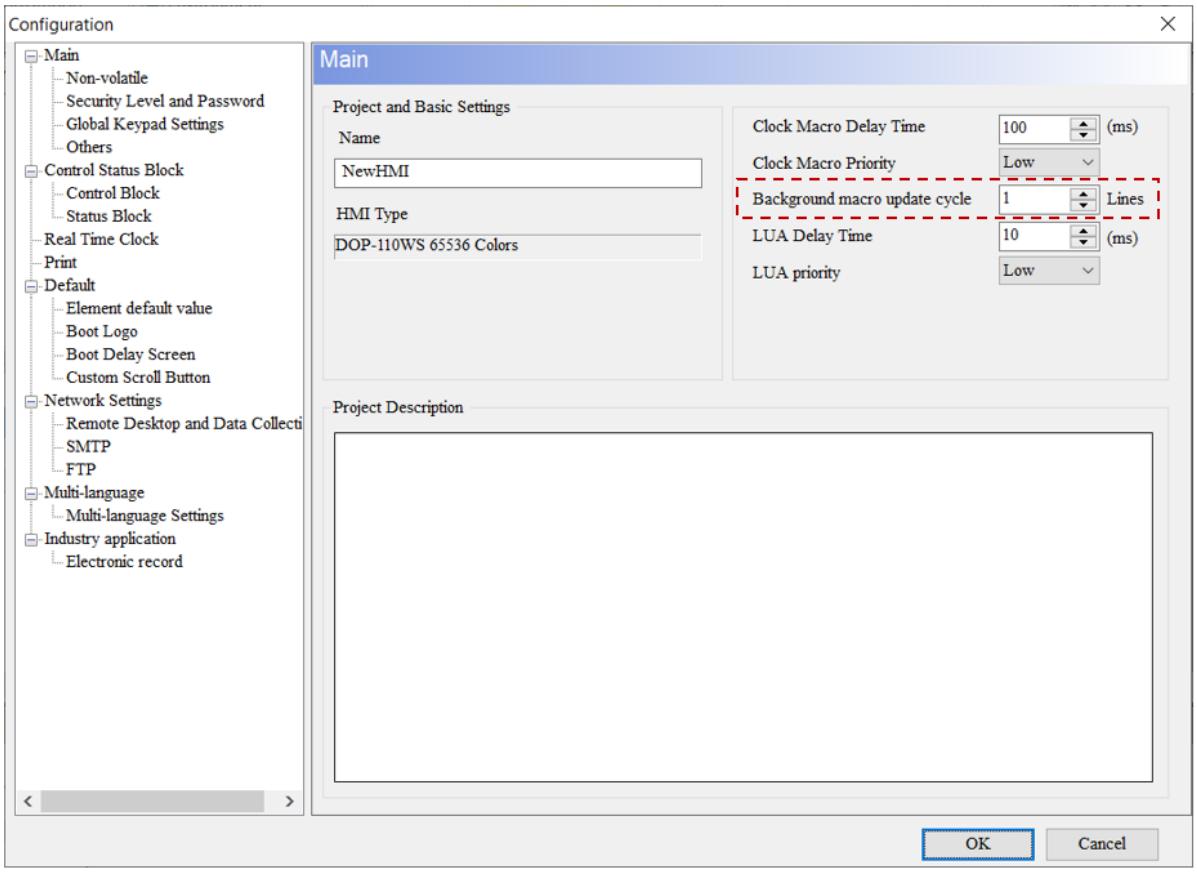
Initial Macro is the first macro to be executed when the HMI starts up, so users can write the initial settings required for the whole HMI program in the Initial Macro.



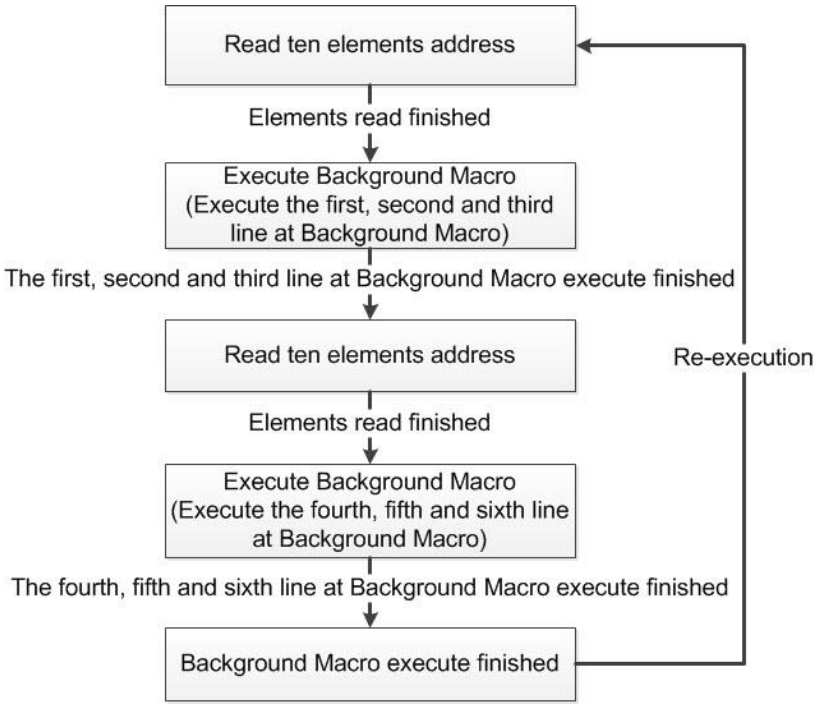
Background Macro

Go to **Project** > **Macro** > **Background Macro** to set Background macro.

The **Background Macro** is a program that is executed repeatedly during the HMI operation. The **Background Macro** is executed in the manner of one line or several lines at a time (instead of finishing with one execution). When the last line is executed, this macro starts all over again. To define the number of lines for each Background Macro execution, go to **General** > **Configuration** to set the **Background macro update cycle** with a maximum of 512 lines.



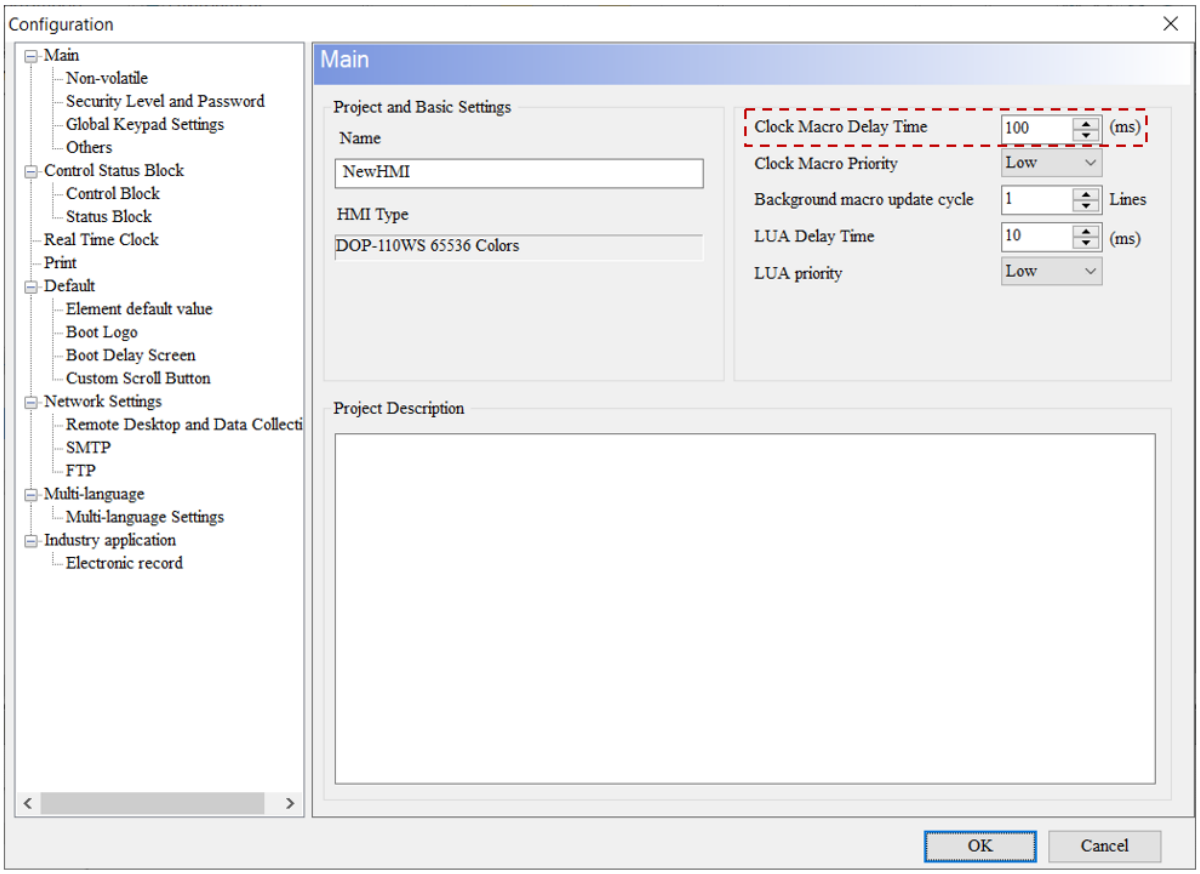
Suppose that 10 elements are created on the HMI screen, enter 6 lines of macro commands into the **Background Macro** and set the **Background macro update cycle** to 3 lines, then the execution process of **Background Macro** is shown as follows.



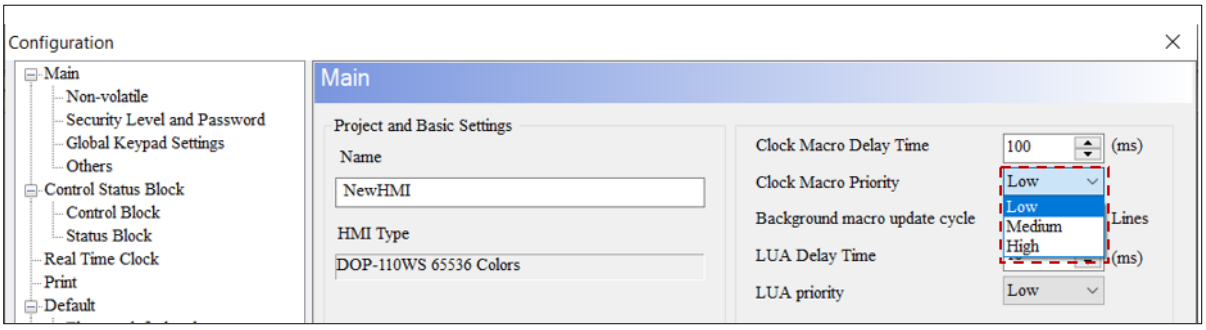
Clock Macro

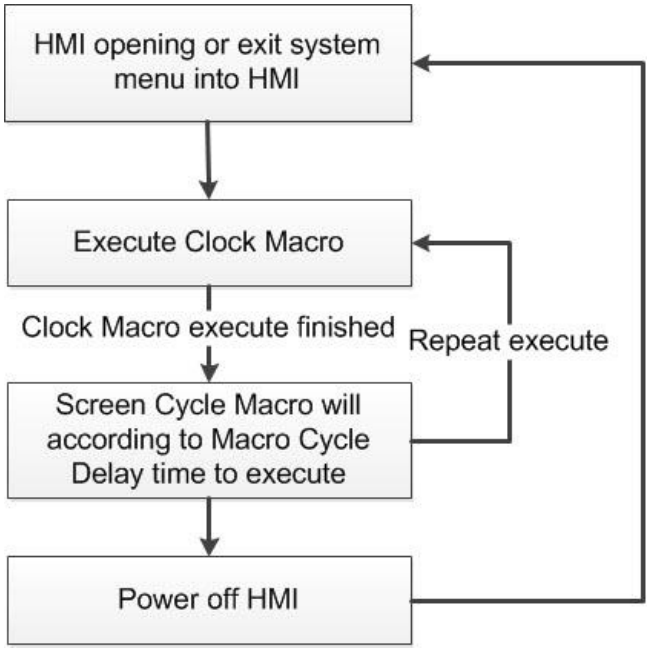
Go to **Project** > **Macro** > **Clock Macro** to set Clock Macro.

Clock Macro executes repeatedly in the process of the HMI operation. Unlike **Background Macro**, **Clock Macro** executes its commands all at once rather than one line or several lines at a time. Similar to **Screen Cycle Macro**, the **Clock Macro** is executed repeatedly according to the set macro delay time. Go to **General** > **Configuration** to set the **Clock Macro Delay Time**. At the end of each **Clock Macro** execution, it resumes its execution according to the set delay time. The default value is **100** ms and the maximum is **65535** ms.



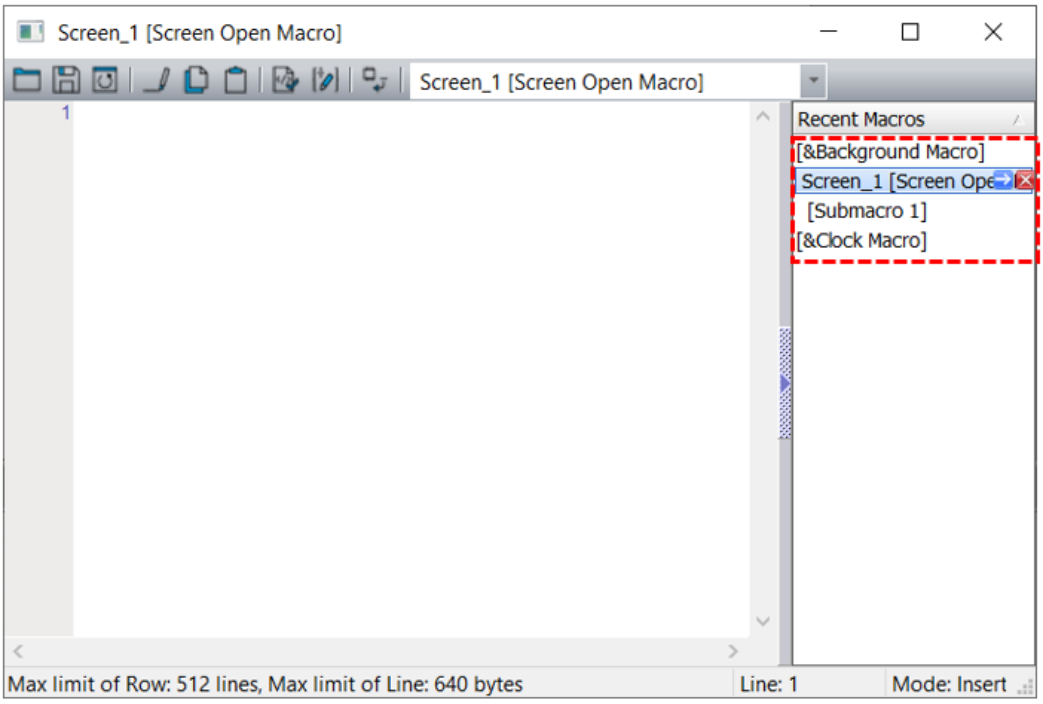
Clock Macro also provides users with the function to prioritize the execution of the Clock Macro, which provides the selections of **Low**, **Medium**, and **High**. Set the **Clock Macro Priority**, and when the priority is higher, it ensures the **Clock Macro Delay Time** is more accurate.



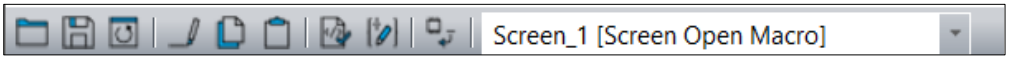


Macro Edit Window

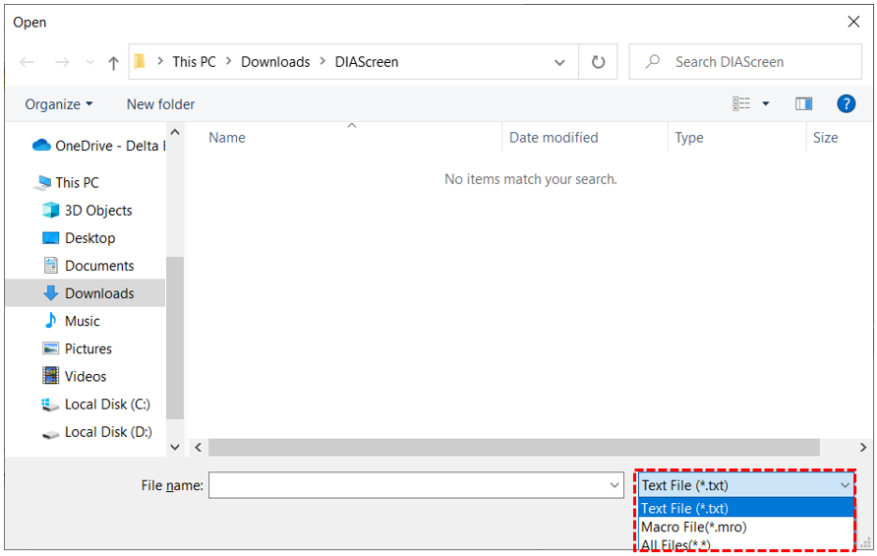
Open the macro screen window to start editing the macros. The maximum lines for each type of macros are **512** lines and the maximum words is **640** bytes or **640** words. The **Recent Macros** on the right side of the **Macro Edit Window** can only record a maximum of 10 recently opened macros. In case of more than 10 records, the first macro will be closed, and a new macro will be added. If the first macro record is updated before being closed, the system prompts users to save the first macro before adding the new macro.

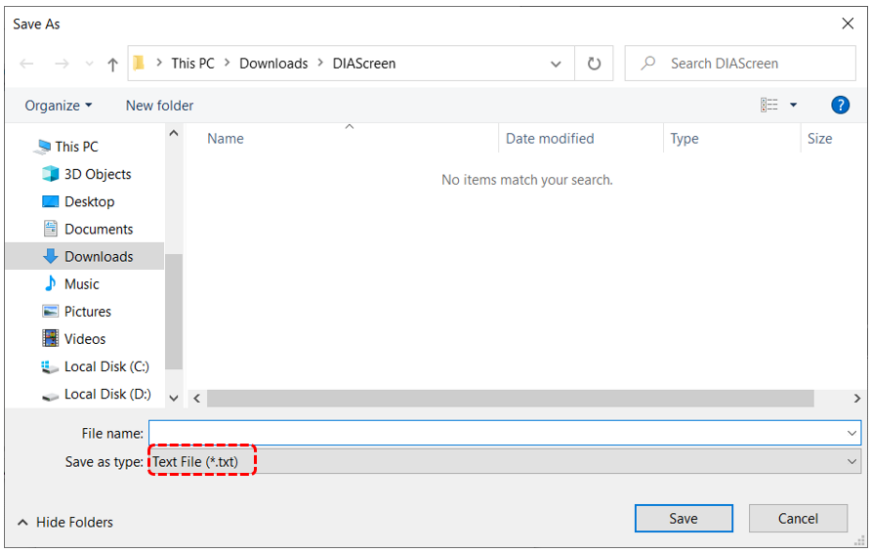
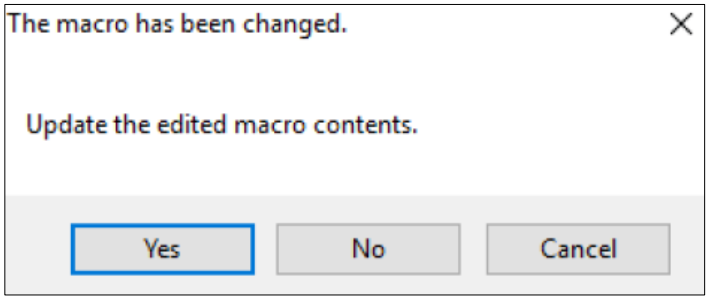


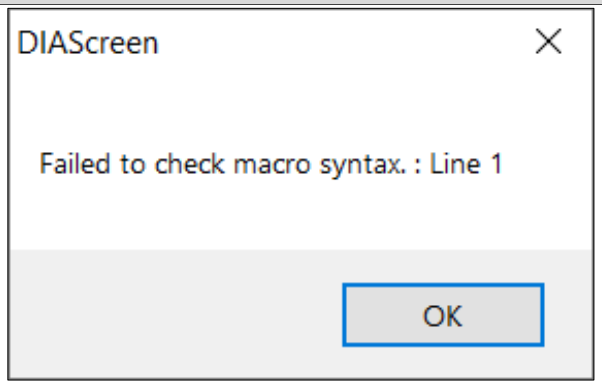
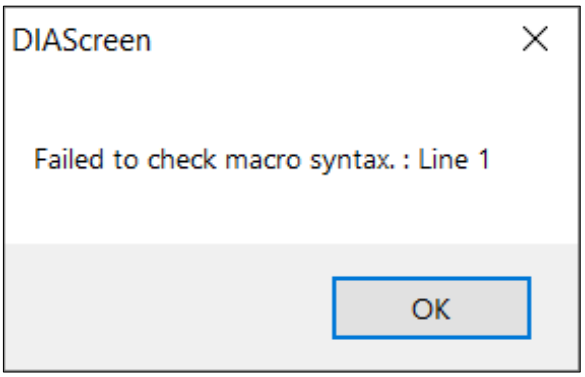
Each macro provides a macro toolbar to assist users in planning and writing macro commands.

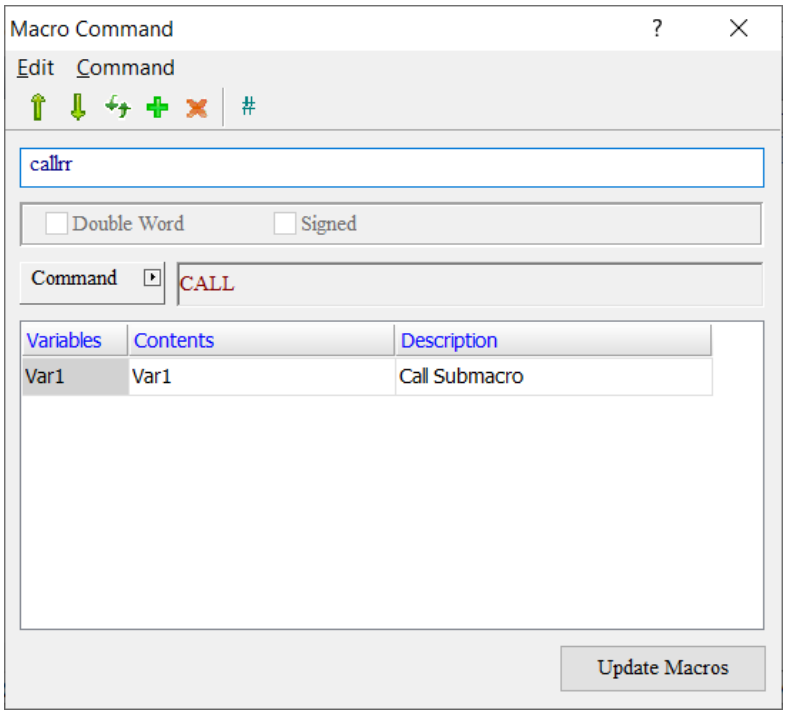
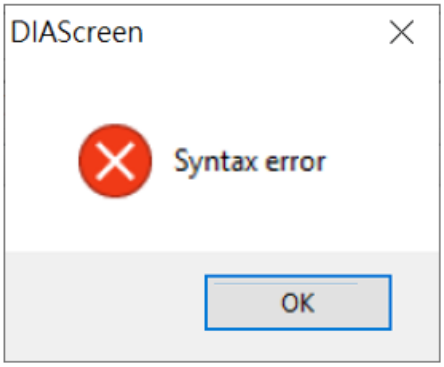


The following table describes the function options in the macro toolbar.

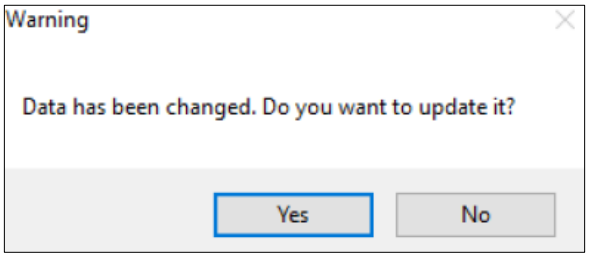
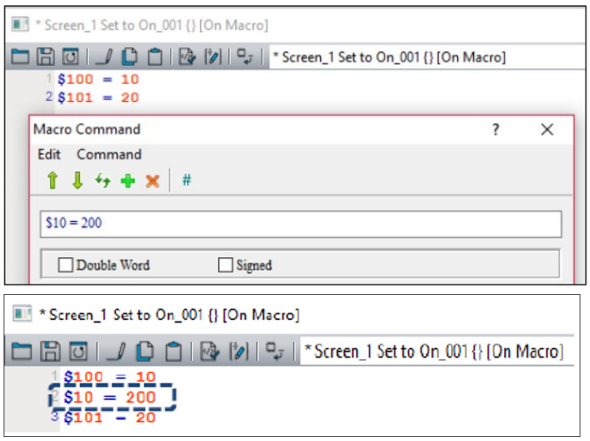
Macro Toolbar functions		
Icon	**Function**	**Description**
	Open	The **Open** function is equal to the action of Import. DIAScreen provides two formats: txt and mro. Import edited macros to reduce the time for repeated editing. 

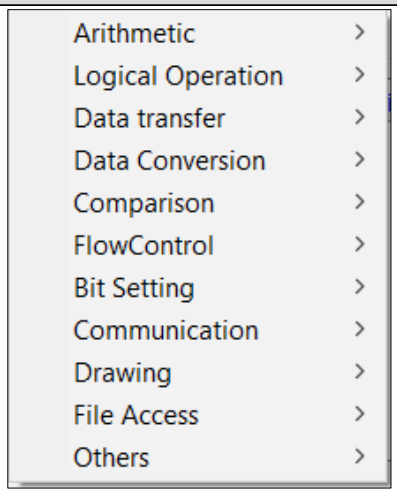
Macro Toolbar functions		
Icon	**Function**	**Description**
	Save	The **Save** function is equal to the action of export. The software only provides the txt format for saving. Save the edited macros for backup or for use of other screens. 
	Update	• The **Update** function is to update the modified macro contents and check the macro syntax as well. If users close the macro edit window without executing , the software will inform users that the macro has been changed.  • If is executed, the current syntax will be checked. If there are syntax errors, the software will show the following message.

Macro Toolbar functions		
Icon	**Function**	**Description**
		
	Cut	Operations of **Cut**, **Copy**, and **Paste** are the same as those of Office. Users can also execute Cut, Copy, and Paste with the keyboard shortcuts (Cut: Ctrl + X; Copy: Ctrl + C; Paste: Ctrl + V).
	Copy	
	Paste	
	Syntax check	The **Syntax check** function is used to make sure that macro commands are correct. An error message will pop up if there is any syntax error.  **Note:** The **Syntax check** function is not equivalent to macro compilation. Users need to execute the compile function to compile macros.

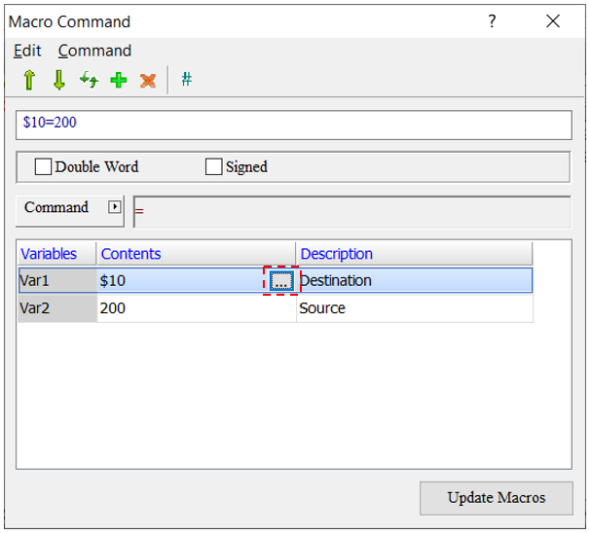
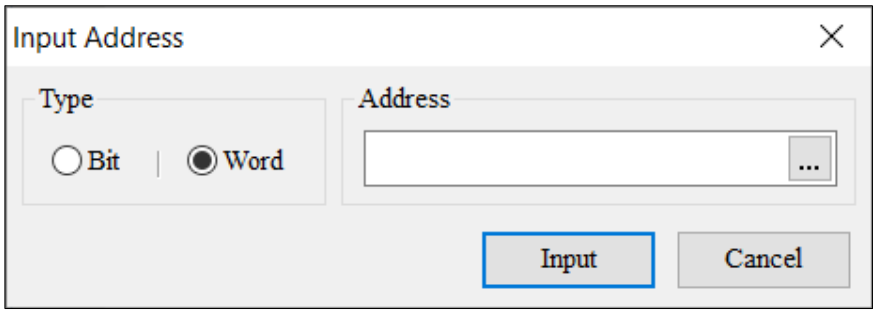
<table>
<tr><th colspan="4">Macro Toolbar functions</th></tr>
<tr><th>Icon</th><th>Function</th><th colspan="2">Description</th></tr>
<tr><td></td><td>Macro Wizard</td><td colspan="2">The Macro Wizard function provides convenient and easy input of macro commands, which are less error than manually entering macro commands.
 
</td></tr>
<tr><td></td><td></td><td>Edit</td><td>Up
Down</td><td>• The cursor can be moved Up or Down.
• Use the Up and Down functions to select which line of macro to move to.</td></tr>
<tr><td></td><td></td><td></td><td>Update</td><td>Execute the Update Button and check whether the syntax of macro command is correct. If there are Syntax error, an error message will pop up.
 

If users change data and close the Macro Wizard window without updating it, DIAScreen displays a message asking if</td></tr>
</table>

Macro Toolbar functions			
Icon	**Function**	**Description**	
			users want to update the data as it has been changed. 
		Insert	**Insert** (add) a line of macro command. The new macro command will be inserted above the macro command of which the cursor has selected and the original one will be moved down. 
		Delete	Delete the macro command of which the cursor has selected. If there are other macro commands following the deleted one, they will be moved up automatically. If the cursor has selected a line without a macro command, the delete action is invalid.
		Comment	The function of **Comment** is to help users manage macros, improve readability, and easy maintenance. Type the symbol "#" in the macro edit screen or use the **Edit** > **Comment** on the **Macro Wizard** or click #, and then write comments or macro programs. The comments written will not be executed in macros.
		Command	• **Command** includes all functions of macro commands which are shown as follows:

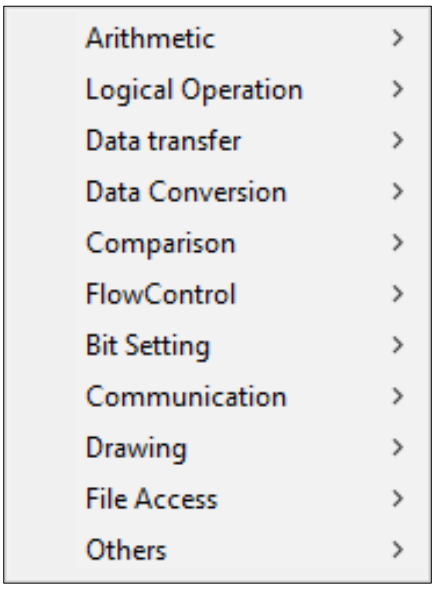
<table>
<tr><th colspan="3">Macro Toolbar functions</th></tr>
<tr><th>Icon</th><th>Function</th><th>Description</th></tr>
<tr><td></td><td></td><td>
<table>
<tr><td></td><td> 

For a detailed description of macro command functions, refer to Macro Commands.</td></tr>
<tr><td>Double Word</td><td>Macro commands provide 32 bits and signed number operations. If users select the checkbox of Signed, the command is operated with signed number; if users do not select this checkbox, the command is operated with unsigned number. If users select the Double Word, the command is operated with 32 bits; if users do not select this checkbox, the command is operated with 16 bits.
<table>
<tr><td>Unsigned</td><td>Unsigned number</td></tr>
<tr><td>Signed</td><td>Signed number</td></tr>
<tr><td>Word</td><td>16-bit data</td></tr>
<tr><td>DW (Double Word)</td><td>32-bit data</td></tr>
</table></td></tr>
<tr><td>Signed</td><td>Note: If the Macro Command is set to Double Word, two registers are occupied for each memory address in the command.</td></tr>
<tr><td>Command</td><td>Command has the same function as the Command of the Macro Wizard window, both are functions of selecting the macro command to be used. For details, refer to Macro Commands.</td></tr>
</table>
</td></tr>
</table>

Macro Toolbar functions		
Icon	**Function**	**Description**
	Variable	Click … to set parameters if the variables are required in macro commands. 
	Input Address	Enter the PLC memory address to be used in the macro through the **Input Address** function to prevent entering the wrong address. 

Macro Commands

Macro commands include Arithmetic, Logical Operation, Data transfer, Data Conversion, Comparison, FlowControl, Bit Setting, Communication, Drawing, File Access, and Others.



The following table lists the Macro commands.

Macro Type	Command	Expression	Description
Arithmetic	+	Var1 = Var2 + Var3	Addition
	-	Var1 = Var2 - Var3	Subtraction
	*	Var1 = Var2 * Var3	Multiplication
	/	Var1 = Var2 / Var3	Division
	%	Var1 = Var2 % Var3	Remainder
	+-*/	Var1 = Var2	Arithmetic
	MUL64	Var1 = MUL64(Var2, Var3) (Signed DW)	64-Bit Multiplication
	ADDSUMW	Var1 = ADDSUMW(Var2, Var3)	Accumulative
	FADD	Var1 = FADD(Var2, Var3) (Signed DW)	Floating-point number addition
	FSUB	Var1 = FSUB(Var2, Var3) (Signed DW)	Floating-point number subtraction
	FMUL	Var1 = FMUL(Var2, Var3) (Signed DW)	Floating point number multiplication
	FDIV	Var1 = FDIV(Var2, Var3) (Signed DW)	Floating-point number division
	FMOD	Var1 = FMOD(Var2, Var3) (Signed DW)	Floating-point number remainder
	QADD	Var1 = QADD(Var2, Var3)	Quadruple word addition

Macro Type	Command	Expression	Description
	QSUB	Var1 = QSUB(Var2, Var3)	Quadruple Subtraction
	QMUL	Var1 = QMUL(Var2, Var3)	Quadruple multiplication
	QDIV	Var1 = QDIV(Var2, Var3)	Quadruple division operation
	QMOD	Var1 = QMOD(Var2, Var3)	Quadruple word remainder
	SIN	Var1 = SIN(Var2) (Signed DW)	Sine function
	COS	Var1 = COS(Var2) (Signed DW)	Cosine function
	TAN	Var1 = TAN(Var2) (Signed DW)	Tangent function
	COT	Var1 = COT(Var2) (Signed DW)	Cotangent function
	SEC	Var1 = SEC(Var2) (Signed DW)	secant function
	CSC	Var1 = CSC(Var2) (Signed DW)	Cosecant function
	FSQRT	Var1 = FSQRT(Var2) (Signed DW)	Floating-point number square root
	FCOS	Var1 = FCOS(Var2) (Signed DW)	Floating-point number cosine function
	FSIN	Var1 = FSIN(Var2) (Signed DW)	Floating-point number sine function
	FATAN2	Var1 = FATAN2(Var2,Var3) (Signed DW)	Floating-point inverse tangent function
Logical Operation	|	Var1 = Var2 | Var3	OR logical operation
	&&	Var1 = Var2 && Var3	AND logical operation
	^	Var1 = Var2 ^ Var3	XOR logical operation
	NOT	Var1 = NOT Var2	NOT logical operation

Macro Type	Command	Expression	Description
	<<	Var1 = Var2 << Var3	SHL (shift left) logical operation
	>>	Var1 = Var2 >> Var3	SHR (shift right) logical operation
Data Transfer	MOV	Var1 = Var2	Data Specified Operand
	BMOV	BMOV(Var1, Var2, Var3)	Copy Block
	ArrayCopy	Var1 = ArrayCopy(Var2, Var3, Var4, Var5, Var6)	Copy Array
	FILL	FILL(Var1, Var2, Var3)	Fill Block
	FILLASC	FILLASC(Var1, " ")	Convert text to ASCII value
	STRCAT	Var1 = STRCAT(Var2, Var3, Var4)	Connection string
	FMOV	Var1 = FMOV(Var2) (Signed DW)	Specify floating-point number data
	QMOV	Var1 = QMOV(Var2)	Quadruple word data designation
Data Conversion	BCD	Var1 = BCD(Var2)	Convert decimal values to BCD values
	BIN	Var1 = BIN(Var2)	Convert BCD values to decimal values
	TODWORD	Var1 = TODWORD(Var2)	Convert values from WORD to DOUBLE WORD
	TOWORD	Var1 = TOWORD(Var2, Var3)	Convert value from BYTE to WORD
	TOBYTE	Var1 = TOBYTE(Var2, Var3)	Convert values from Word to Byte
	SWAP	SWAP(Var1, Var2, Var3)	Swap high and low bytes of Word
	XCHG	XCHG(Var1, Var2, Var3)	Exchange value data
	MAX	Var1 = MAX(Var2, Var3)	Get maximum
	MIN	Var1 = MIN(Var2, Var3)	Get minimum

Macro Type	Command	Expression		Description
	TOHEX	Var1 = TOHEX(Var2)		Convert 4 ASCII characters to Word hexadecimal integers
	TOASC	Var1 = TOASC(Var2)		Convert a hexadecimal integer of Word to 4 Word ASCII characters
	FCNV	Var1 = FCNV(Var2) (Signed DW)		Convert an integer to a floating-point number
	ICNV	Var1 = ICNV(Var2) (Signed DW)		Convert float to integer
	SPRINTF	Var1 = SPRINTF(Var2, "%u", Var4)		Format String
Compare	IF…THEN GOTO	IF ==	IF Var1 == Var2 THEN GOTO LABEL Var3	If... ,then execute according to the specified label name
		IF !=	IF Var1 != Var2 THEN GOTO LABEL Var3	
		IF >	IF Var1 > Var2 THEN GOTO LABEL Var3	
		IF >=	IF Var1 >= Var2 THEN GOTO LABEL Var3	
		IF <	IF Var1 < Var2 THEN GOTO LABEL Var3	
		IF <=	IF Var1 <= Var2 THEN GOTO LABEL Var3	
		IF AND == 0	IF (Var1 && Var2) == 0 THEN GOTO LABEL Var3	
		IF AND != 0	IF (Var1 && Var2) != 0 THEN GOTO LABEL Var3	

Macro Type	Command	Expression		Description
		IF == ON	IF Var1 == ON THEN GOTO LABEL Var2	
		IF == OFF	IF Var1 == OFF THEN GOTO LABEL Var2	
		IFB == ON	IFB Var1 == ON THEN GOTO LABEL Var2	
		IFB == OFF	IFB Var1 == OFF THEN GOTO LABEL Var2	
	IF…THEN CALL	IF == CALL	IF Var1 == Var2 THEN CALL Var3	If... call submacro
		IF != CALL	IF Var1 != Var2 THEN CALL Var3	
		IF > CALL	IF Var1 > Var2 THEN CALL Var3	
		IF >= CALL	IF Var1 >= Var2 THEN CALL Var3	
		IF < CALL	IF Var1 < Var2 THEN CALL Var3	
		IF <= CALL	IF Var1 <= Var2 THEN CALL Var3	
		IF AND == 0 CALL	IF (Var1 && Var2) == 0 THEN CALL Var3	
		IF AND != 0 CALL	IF (Var1 && Var2) != 0 THEN CALL Var3	
		IF == ON CALL	IF Var1 == ON THEN CALL Var2	
		IF == OFF CALL	IF Var1 == OFF THEN CALL Var2	

Macro Type	Command	Expression		Description
	IF…	IF ==	IF Var1 == Var2	Logical comparison
		IF !=	IF Var1 != Var2	
		IF >	IF Var1 > Var2	
		IF >=	IF Var1 >= Var2	
		IF <	IF Var1 < Var2	
		IF <=	IF Var1 <= Var2	
		IF AND == 0	IF (Var1 && Var2) == 0	
		IF AND != 0	IF (Var1 && Var2) != 0	
		IF == ON	IF Var1 == ON	
		IF == OFF	IF Var1 == OFF	
	ELSEIF…	ELSEIF ==	ELSEIF Var1 == Var2	Logical comparison
		ELSEIF !=	ELSEIF Var1 != Var2	
		ELSEIF >	ELSEIF Var1 > Var2	
		ELSEIF >=	ELSEIF Var1 >= Var2	
		ELSEIF <	ELSEIF Var1 < Var2	
		ELSEIF <=	ELSEIF Var1 <= Var2	
		ELSEIF AND == 0	ELSEIF (Var1 && Var2) == 0	
		ELSEIF AND != 0	ELSEIF (Var1 && Var2) != 0	
		ELSEIF == ON	ELSEIF Var1 == ON	
		ELSEIF == OFF	ELSEIF Var1 == OFF	
	ELSE	ELSE		Logical comparison
	ENDIF	ENDIF		Logical comparison
	FCMP	Var1 = FCMP(Var2, Var3) (Signed DW)		Comparison of floating-point value

Macro Type	Command	Expression	Description
	QCMP	Var1 = QCMP(Var2, Var3) (Signed DW)	Quadruple Numerical Comparison
FlowControl	GOTO	GOTO LABEL Var1	Go to a label unconditionally (LABEL)
	LABEL	LABEL Var1	Label
	CALL	CALL Var1	Call Submacro
	RET	RET	Exit Submacro
	FOR	FOR Var1	Program Loop
	NEXT	NEXT	
	END	END	End Macro Program
Bit setting	BITON	BITON Var1	Set the bit to ON
	BITOFF	BITOFF Var1	Set the bit to OFF
	BITNOT	BITNOT Var1	Inverse bit (ON→OFF，OFF→ON)
	GETB	Var1 = GETB Var2	Get bit value
Communic-ation	INITCOM	Var1 = INITCOM(Var2, Var3, Var4, Var5, Var6, Var7, Var8)	COM PORT initialization
	ADDSUM	Var1 = ADDSUM(Var2, Var3)	Calculate CHECKSUM by addition
	XORSUM	Var1 = XORSUM(Var2, Var3)	Calculate CHECKSUM by XOR
	PUTCHARS	Var1 = PUTCHARS(Var2, Var3, Var4)	Export characters via the communication port
	GETCHARS	Var1 = GETCHARS(Var2, Var3, Var4)	Get characters via the communication port
	SELECTCOM	SELECTCOM(Var1)	Select COM Port
	CLEARCOMBUFFER	CLEARCOMBUFFER(Var1, Var2)	Clear buffer of Com Port
	CHRCHKSUM	Var1 = CHRCHKSUM("Var2", Var3, Var4)	Calculate the length and checksum of string

Macro Type	Command	Expression	Description
	LOCKCOM	Var1 = LOCKCOM(Var2, Var3)	Lock COM Port
	UNLOCKCOM	UNLOCKCOM(Var1)	Unlock COM Port
	STATIONCHK	Var1 = STATIONCHK(Var2, Var3)	Check COM Port communication state
	STATIONON	STATIONON(Var1, Var2)	Station On
	STATIONOFF	STATIONOFF(Var1, Var2)	Station Off
	IPON	Var1 = IPON(Var2, Var3, Var4, Var5, Var6)	Enable IP address
	IPOFF	Var1 = IPOFF(Var2, Var3, Var4, Var5, Var6)	Disable IP address
	IPCHANGE	Var1 = IPCHANGE(Var2, Var3, Var4, Var5, Var6, Var7)	Change the IP address of the connecting controller and the communication port
	COMLINKSTATUS	Var1 = COMLINKSTATUS(Var2)	Communication port status
	NETLINKSTATUS	Var1 = NETLINKSTATUS(Var2)	Internet connection status
	CLOSECOM	Var1 = CLOSECOM(Var2)	Close port
Drawing	RECTANGLE	RECTANGLE(Var1)	Rectangle
	LINE	LINE(Var1)	Line
	POINT	POINT(Var1)	Point
	CIRCLE	CIRCLE(Var1)	Circle
File Access	FileSlotRead	Var1 = FileSlotRead(Var2, Var3, Var4, Var5)	Read file
	FileSlotWrite	Var1 = FileSlotWrite(Var2, Var3, Var4, Var5)	Write file
	FileSlotRemove	Var1 = FileSlotRemove(Var2)	Remove file
	FileSlotGetLength	Var1 = FileSlotGetLength(Var2, Var3)	Read length of file

Macro Type	Command	Expression	Description
	FileSlotExport	Var1 = FileSlotEXPORT(Var2, Var3, Var4, Var5)	Export file
	FileSlotImport	Var1 = FileSlotIMPORT(Var2, Var3, Var4, Var5)	Import file
	FileSlotGetName	Var1 = FileSlotGetName(Var2, Var3, Var4)	Get file name
	FileSlotSetName	Var1 = FileSlotSetName(Var2, Var3, Var4)	Set file name
	FileSlotGetID	Var1 = FileSlotGetID(Var2, Var3, Var4)	Get file id
Others	Time Tick	Var1 = TIMETICK	Get the time from the start of the system up to now
	GETLASTERROR	Var1 = GETLASTERROR	Get the error value of the previous command
	Comment	#	Comment
	Delay	Delay(Var1)	Delay
	GETSYSTEMTIME	Var1 = GETSYSTEMTIME	Get system time
	SETSYSTEMTIME	SETSYSTEMTIME(Var1)	Set system time
	GETHISTORY	Var1 = GETHISTORY(Var2, Var3, Var4, Var5, Var6)	Get history data
	EXPORT	EXPORT	EXPORT(Var1)
	EXRCP16	EXRCP16	Var1 = EXRCP16(Var2, Var3)
	IMRCP16	IMRCP16	Var1 = IMRCP16(Var2, Var3)
	EXRCP32	Var1 = EXRCP32(Var2, Var3)	Export 32-bit Recipe
	IMRCP32	Var1 = IMRCP32(Var2, Var3)	Import 32 bit Recipe

Macro Type	Command	Expression	Description
	EXENRCP	Var1 = EXENRCP(Var2, Var3)	Export Enhanced Recipe
	IMENRCP	Var1 = IMENRCP(Var2, Var3)	Import Enhanced Recipe
	EXHISTORY	Var1 = EXHISTORY(Var2, Var3, Var4)	Export history data
	EXALARM	Var1 = EXALARM(Var2, Var3)	Export alert data
	EXALARMGROUP	Var1 = EXALARMGROUP(Var2, Var3, Var4)	Export alert assignment group information
	DISKFORMAT	Var1 = DISKFORMAT(Var2)	Disk format
	BMPCAPTURE	Var1 = BMPCAPTURE(Var2)	Capture screen
	PLCDOWNLOAD	Var1 = PLCDOWNLOAD(Var2, Var3, Var4, Var5, Var6)	Download DVP or ISP file to PLC via the HMI
	OPENSCREEN	OPENSCREEN(Var1)	• Open the screen • Cannot be used in Screen On Macro/Screen Off Macro/Screen Cycle Macro
	CLOSESUBSCREEN	CLOSESUBSCREEN(Var1)	• Close the sub-screen • Cannot be used in Screen On Macro/Screen Off Macro/Screen Cycle Macro
	GetCircleCenter	Var1 = GetCircleCenter(Var2, Var3)	Calculate the coordinates of the center of the circle
	VAR	VAR Var1	Variable
Industrial application	ECAM	Var1 = ECAM(Var2, Var3, Var4, Var5)	

Macro Type	Command	Expression	Description
	RSVP	Var1 = RSVP(Var2, Var3, Var4)	
	WSVP	Var1 = WSVP(Var2, Var3, Var4)	
	SAVEMOTIONINSLOG	Var1 = SAVEMOTIONINSLOG(Var2, Var3, Var4)	Record Motion command operation

Arithmetic

Arithmetic includes integer and floating-point operations, and the macro commands are described as follows.

+
–
*
/
%
+-*/
MUL64
ADDSUMW

FADD
FSUB
FMUL
FDIV
FMOD

QADD
QSUB
QMUL
QDIV
QMOD

SIN
COS
TAN
COT
SEC
CSC

FSQRT
FCOS
FSIN
FATAN2

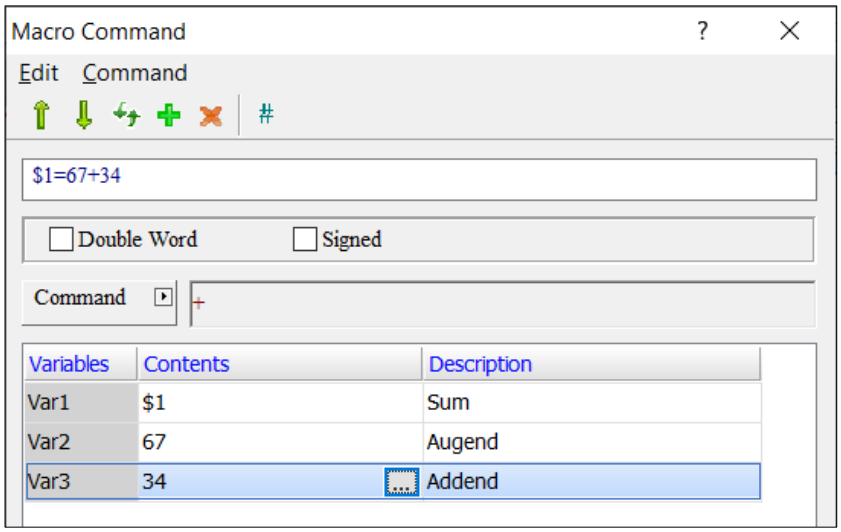
- **+ (Addition)**

Expression	Meaning of Variable		Note
Var1 = Var2 + Var3 (W) Var1 = Var2 + Var3 (DW) Var1 = Var2 + Var3 (Signed W) Var1 = Var2 + Var3 (Signed DW)	Var1	Sum	W: Word DW: Double Word Signed: Number
	Var2	Augend	
	Var3	Addend	
	Description of Action		
	Add Var2 and Var3, and put the result in Var1.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V
Var3	V	V	V

Example

- Var1 is the Internal Memory address, and Var2 and Var3 are constants.



- When the command is $1= 67 + 34, an addition operation will be executed to put the result obtained by adding 67 and 34 into $1, so $1 = 101.

- **- (Subtraction Operation)**

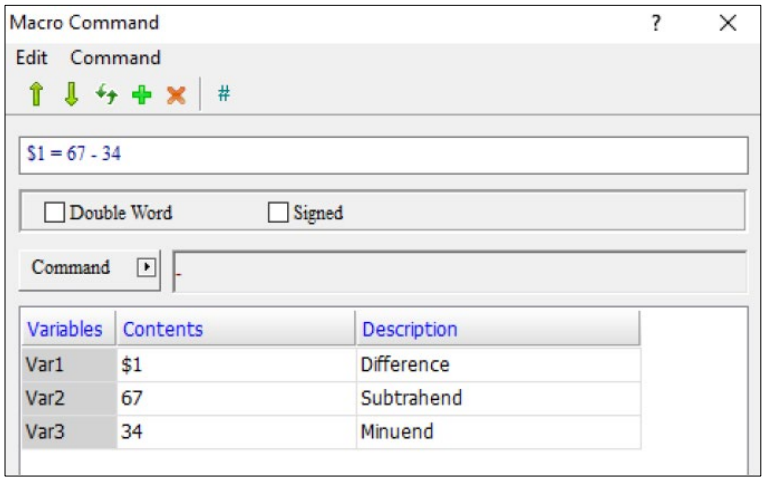
Expression	Meaning of Variable		Note
Var1 = Var2 - Var3 (W)	Var1	Difference	W: Word

Expression	Meaning of Variable		Note
Var1 = Var2 - Var3 (DW) Var1 = Var2 - Var3 (Signed W) Var1 = Var2 - Var3 (Signed DW)	Var2	Subtrahend	DW: Double Word Signed: Number
	Var3	Minuend	
	Description of Action		
	Subtract Var2 from Var3 and put the result in Var1.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V
Var3	V	V	V

Example

- Var1 is the Internal Memory address, and Var2 and Var3 are constants.



- When the command is $1 = 67 - 34, the subtraction operation is executed and the result of 67 – 34 is put in $1, so $1 = 33.

- *** (Multiplication)**

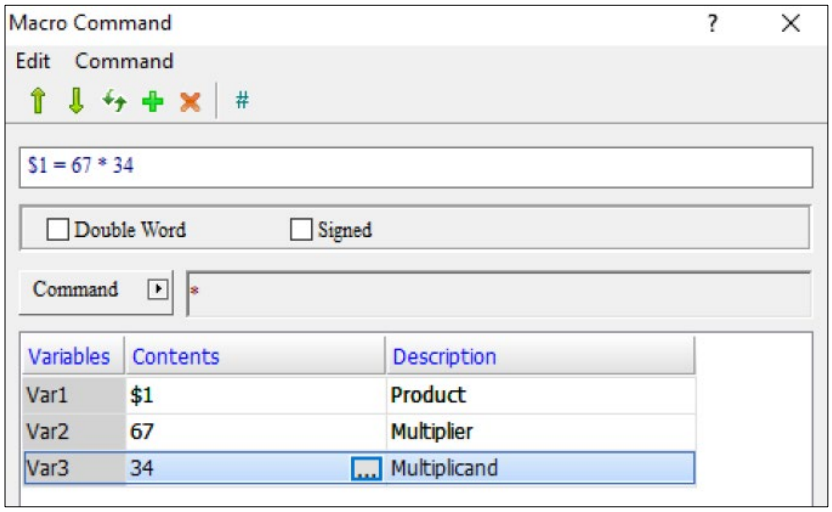
Expression	Meaning of Variable		Note
Var1 = Var2 * Var3 (W) Var1 = Var2 * Var3 (DW) Var1 = Var2 * Var3 (Signed W) Var1 = Var2 * Var3 (Signed DW)	Var1	Product	W: Word DW: Double Word Signed: Number
	Var2	Multiplier	
	Var3	Multiplicand	

Expression	Meaning of Variable	Note
	Description of Action	
	Multiply Var2 by Var3 and put the result in Var1.	

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V
Var3	V	V	V

Example

- Var1 is the Internal Memory address, and Var2 and Var3 are constants.



- When the command is $1 = 67 * 34, the multiplication operation is executed and the result of 67 * 34 is put in $1, so $1 = 2278.

- **/ (Division)**

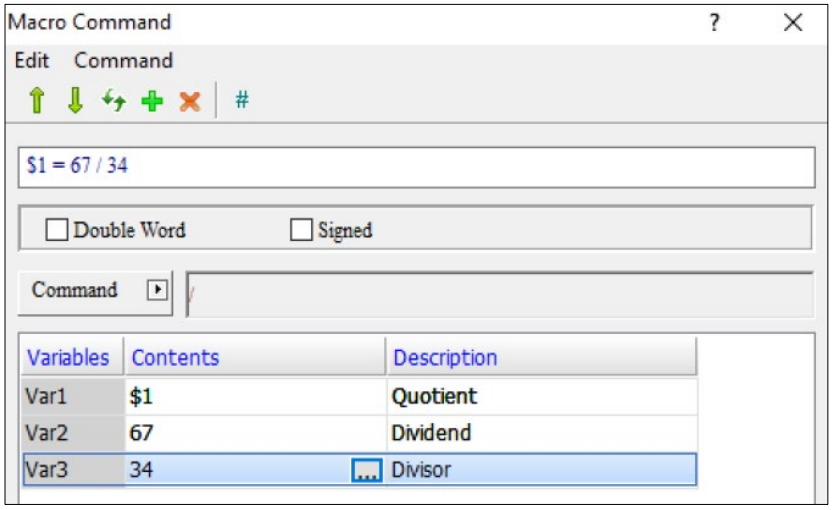
Expression	Meaning of Variable		Note
Var1 = Var2 / Var3 (W) Var1 = Var2 / Var3 (DW) Var1 = Var2 / Var3 (Signed W) Var1 = Var2 / Var3 (Signed DW)	Var1	Quotient	W: Word DW: Double Word Signed: Number
	Var2	Dividend	
	Var3	Divisor	
	Description of Action		
	Divide Var2 by Var3 and put the result in Var1.		

Expression	Meaning of Variable	Note
Note: Var3 cannot be 0.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V
Var3	V	V	V

Example

- Var1 is the Internal Memory address, and Var2 and Var3 are constants.



- When the command is $1 = 67 / 34, the division operation is executed and the result of 67 / 34 is put in $1, so $1 = 1.

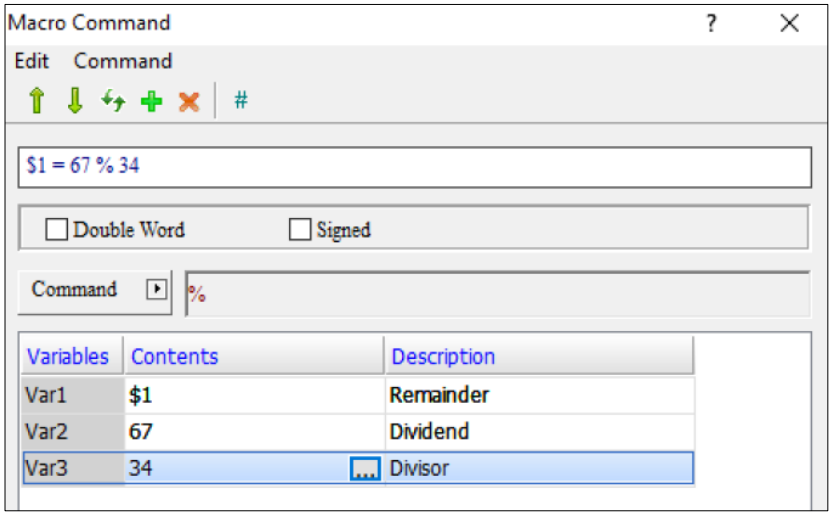
- **% (Remainder)**

Expression	Meaning of Variable		Note
Var1 = Var2 % Var3 (W) Var1 = Var2 % Var3 (DW) Var1 = Var2 % Var3 (Signed W) Var1 = Var2 % Var3 (Signed DW)	Var1	Remainder	W: Word DW: Double Word Signed: Number
	Var2	Dividend	
	Var3	Divisor	
	Description of Action		
	Divide Var2 by Var3 and put the remainder in Var1.		
Note: Var 3 cannot be 0.			

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V
Var3	V	V	V

Example

- Var1 is the Internal Memory address, and Var2 and Var3 are constants.



- When the command is $1 = 67 % 34, the remainder operation is executed and the result of 67 % 34 is put in $1, so $1 = 33.

- **MUL64 (64-bit Multiplication)**

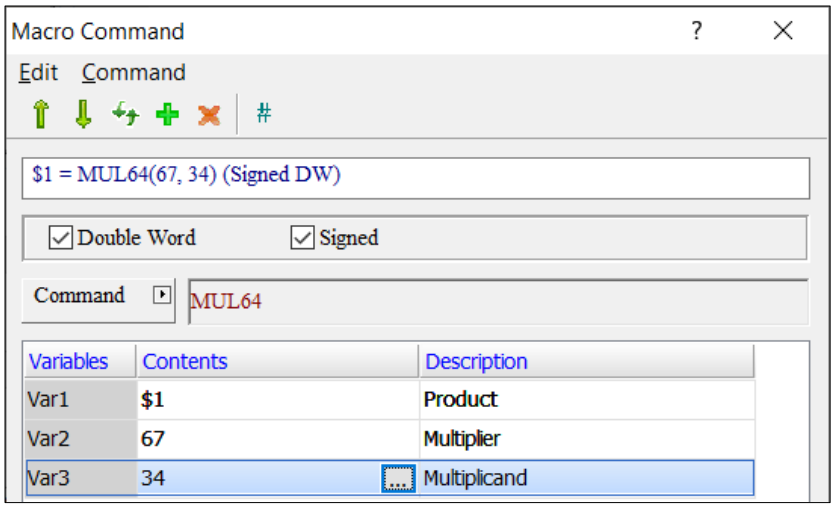
Expression	Meaning of Variable		Note
Var1 = MUL64(Var2, Var3) (W) Var1 = MUL64(Var2, Var3) (DW) Var1 = MUL64(Var2, Var3) (Signed W) Var1 = MUL64(Var2, Var3) (Signed DW)	Var1	Product	W: Word DW: Double Word Signed: Number
	Var2	Multiplier	
	Var3	Multiplicand	
	Description of Action		
	Multiple Var3 by Var2, and put the result in Var1.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	

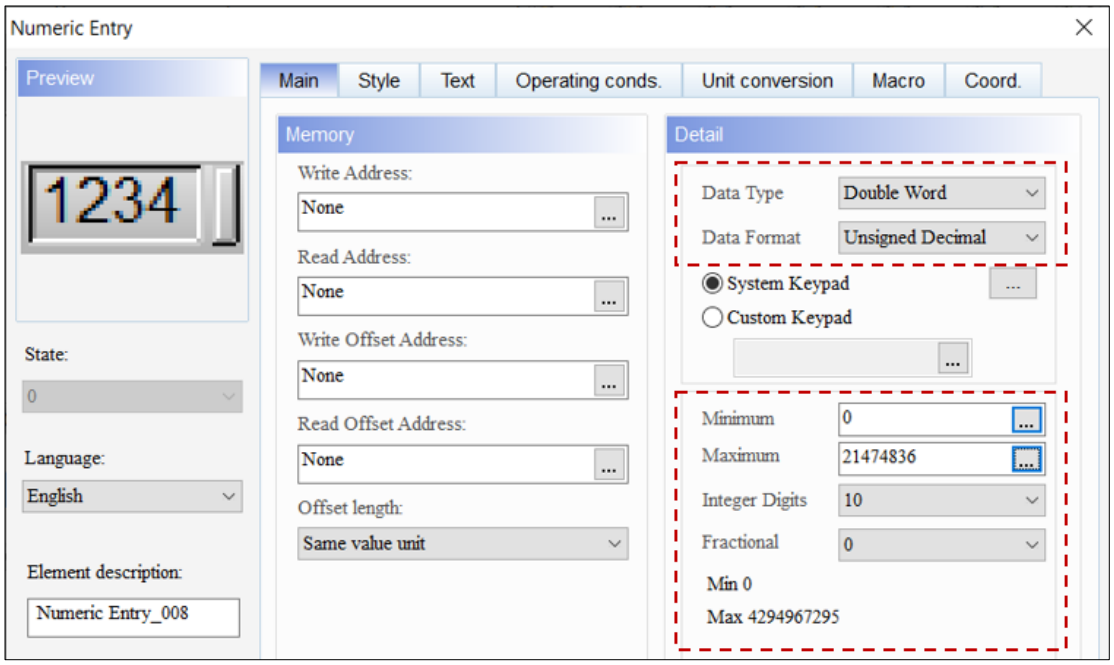
Var2	V	V	V
Var3	V	V	V

Example

- Var1 is the Internal Memory address, and Var2 and Var3 are constants.



- If the checkbox of **Double Word** is selected, set the **Data Type** of the elements to **Double Word** so as to save the correct value.



- When the command is $1 = MUL64 (67, 34), the MUL64 bit multiplication is executed and the result of 67 multiplied by 34 is put in $1, so $1 = 2278.

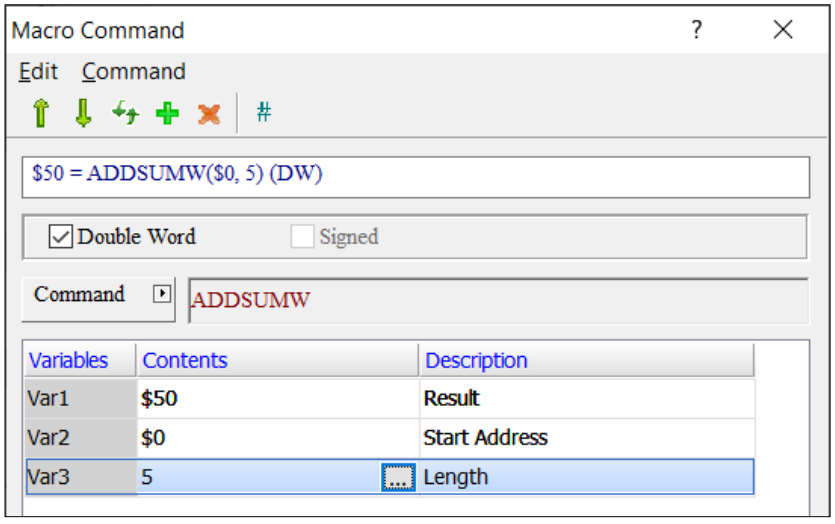
- **ADDSUMW (Accumulation)**

<table>
<tr><th>Expression</th><th colspan="2">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="5">Var1 = ADDSUMW(Var2, Var3) (W)
Var1 = ADDSUMW(Var2, Var3) (DW)</td><td>Var1</td><td>Result</td><td rowspan="5">W: Word
DW: Double Word</td></tr>
<tr><td>Var2</td><td>Start address</td></tr>
<tr><td>Var3</td><td>Length</td></tr>
<tr><td colspan="2">Description of Action</td></tr>
<tr><td colspan="2">Add up Var3 length addresses continuously from Var2 start address and put the result in Var1.</td></tr>
<tr><td colspan="4">Note: If Double Word is selected, the Start Address increments by 2 to Length; if Word is selected, it increments by 1.</td></tr>
</table>

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V
Var3	V	V	V

Example

- Var1 and Var2 are the Internal Memory addresses, Var3 is a constant, and the checkbox of Double Word is selected.
- Memory address increases by 2 addresses from $0 to 5 lengths: $0, $2, $4, $6, & $8.



Example
• Enter $0 = 1, $2 = 2, $4 = 3, $6 = 4, and $8 = 5, and put the accumulated value in $50, so $50 = 15.

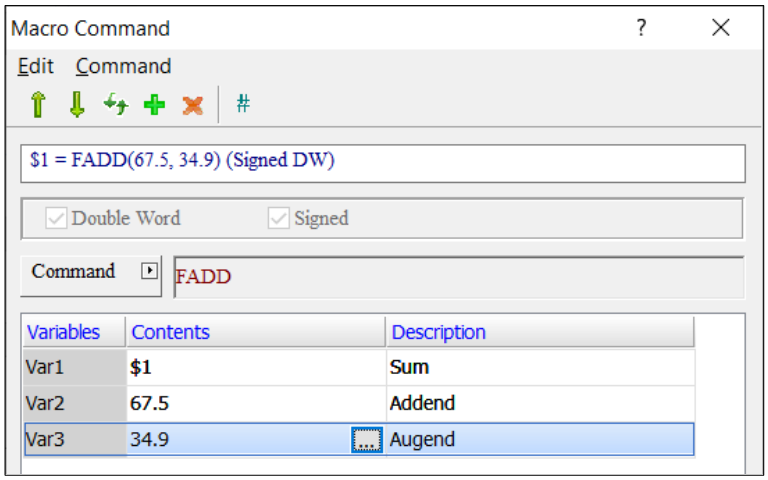
- **FADD (Floating-Point Number Addition)**

Expression	Meaning of Variable		Note
Var1 = FADD(Var2, Var3) (Signed DW)	Var1	Sum	DW: Double Word Signed: Number
	Var2	Addend	
	Var3	Augend	
	Description of Action		
	Add Var2 and Var3 and put the result in Var1.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V
Var3	V	V	V

Example

- Var1 is the Internal Memory address, and Var2 and Var3 are constants.



- When the command is $1 = FADD (67.5, 34.9), the floating-point number addition operation is executed and the result of 67.5 plus 34.9 is put in $1, so $1 = 102.4.

- **FSUB (Floating-Point Number Subtraction)**

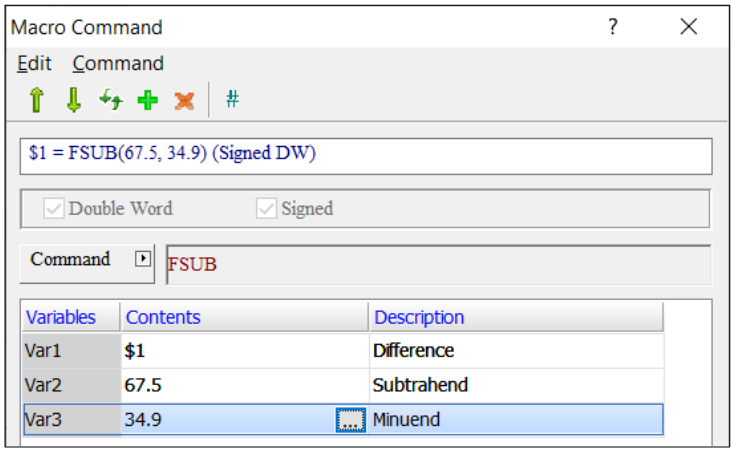
Expression	Meaning of Variable		Note
Var1 = FSUB(Var2, Var3) (Signed DW)	Var1	Difference	DW: Double Word Signed: Number
	Var2	Subtrahend	

Expression	Meaning of Variable		Note
	Var3	Minuend	
	Description of Action		
	Subtract Var2 from Var3 and put the result in Var1.		

Variable	Type		
	Internal Memory	PLC Register	constant
Var1	V	V	
Var2	V	V	V
Var3	V	V	V

Example

- Var1 is the Internal Memory address, and Var2 and Var3 are constants.



- When the command is $1 = FSUB (67.5, 34.9), the floating-point number subtraction operation is executed and the result of 67.5 minus 34.9 is put in $1, so $1 = 32.6.

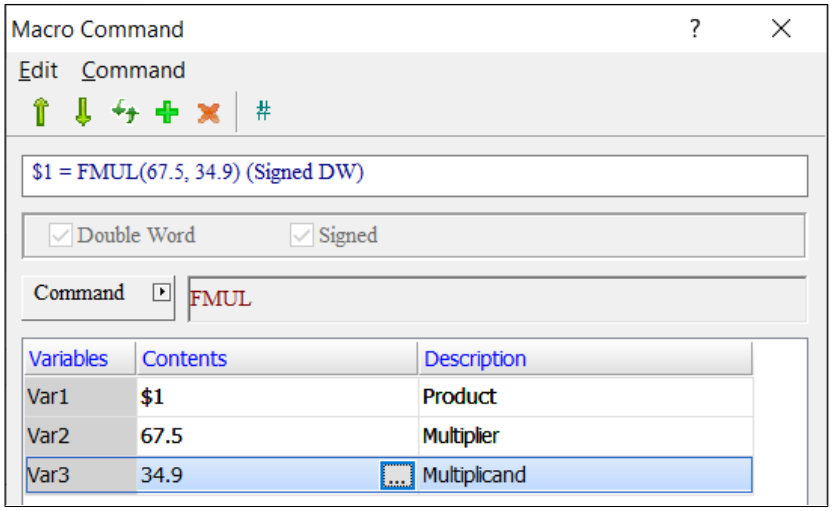
- **FMUL (Floating Point Number Multiplication)**

Expression	Meaning of Variable		Note
Var1 = FMUL(Var2, Var3) (Signed DW)	Var1	Product	DW: Double Word Signed: Number
	Var2	Multiplier	
	Var3	Multiplicand	
	Description of Action		
	Multiply Var3 by Var2 and put the result in Var1.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V
Var3	V	V	V

Example

- Var1 is the Internal Memory address, and Var2 and Var3 are constants.



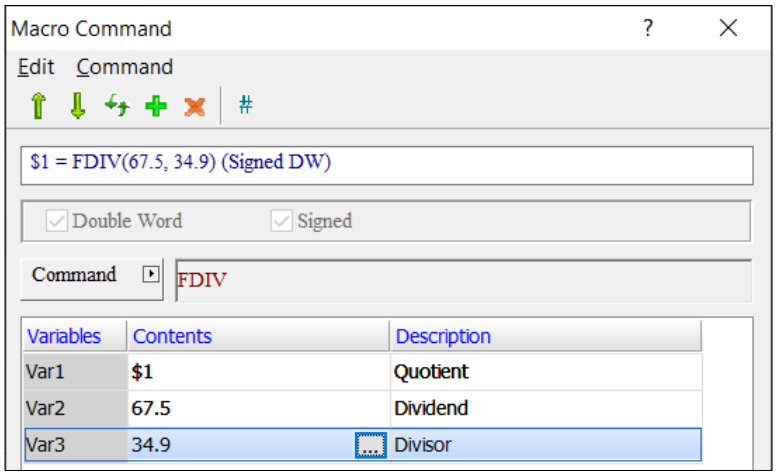
- When the command is $1 = FMUL (67.5, 34.9), the floating-point number multiplication is executed and the result of 67.5 multiplied by 34.9 is put in $1, so $1 = 2355.75.

- **FDIV (Floating Point Number Division)**

Expression	Meaning of Variable		Note
Var1 = FDIV(Var2, Var3) (Signed DW)	Var1	Quotient	DW: Double Word Signed: Signed number
	Var2	Dividend	
	Var3	Divisor	
	Description of Action		
	Divide Var2 by Var3 and put the result in Var1.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V

Variable	Type		
	Internal Memory	PLC Register	Constant
Var3	V	V	V

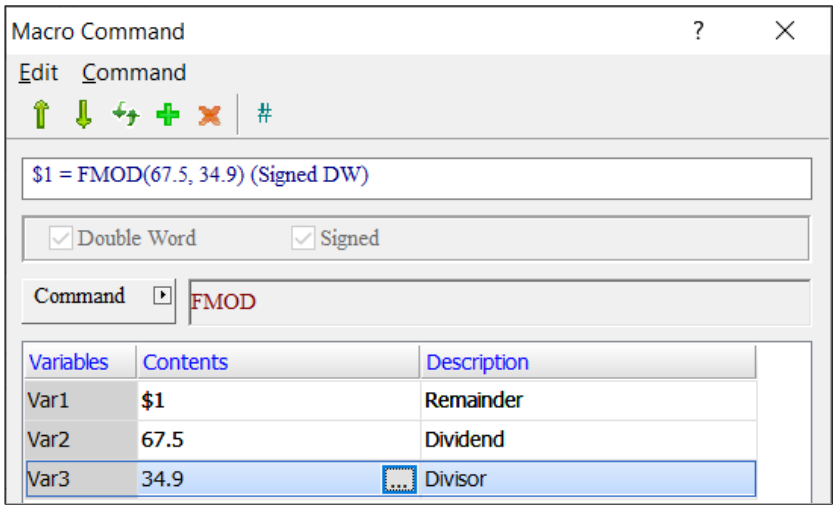
Example
• Var1 is the Internal Memory address, and Var2 and Var3 are constants.  • When the command is $1 = FDIV (67.5, 34.9), the floating-point number division operation is executed and the result of 67.5 divided by 34.9 is put in $1, so $1 = 1.934.

- **FMOD (Floating-Point Number Remainder)**

Expression	Meaning of Variable		Note
Var1 = FMOD(Var2, Var3) (Signed DW)	Var1	Remainder	DW: Double Word Signed: Signed number
	Var2	Dividend	
	Var3	Divisor	
	Description of Action		
	Divide Var2 by Var3 and put the remainder in Var1.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V
Var3	V	V	V
Example			

- Var1 is the Internal Memory address, and Var2 and Var3 are constants.



- When the command is $1 = FMOD (67.5, 34.9), the floating-point number remainder operation is executed and the remainder of 67.5 divided by 34.9 is put in $1, so $1 = 32.6.

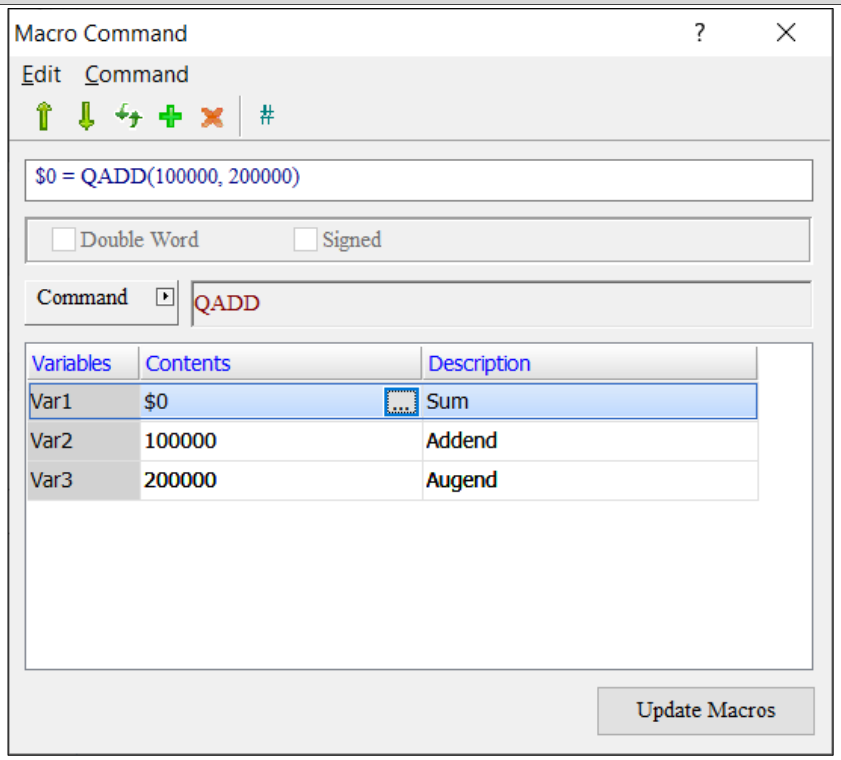
- **QADD (Quadruple Word Addition)**

Expression	Meaning of Variables		Note
Var1 = QADD(Var2, Var3)	Var1	Sum	Q: Quad Word
	Var2	Addend	
	Var3	Augend	
	Description of Action		
	The sum obtained by dividing Var2 by Var3 is placed in Var1.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V
Var3	V	V	V

Example
• Var1 is the Internal Memory address, and Var2 and Var3 are constants.

Example

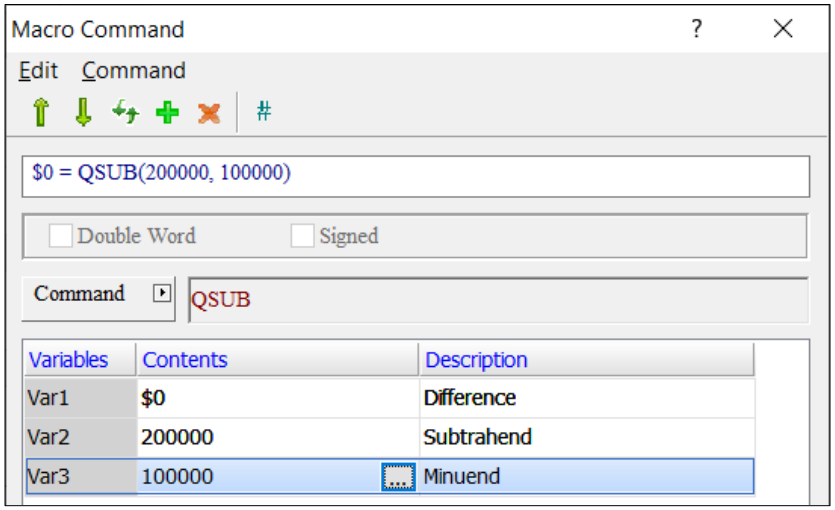


- When the command is $0 = QADD (100000, 200000), it will execute quadruple word addition operation and put the result obtained by adding 100000 and 200000 into $0, so $0 = 300000.

- **QSUB (Quadruple Subtraction)**

Expression	Meaning of Variables		Note
Var1 = QSUB(Var2, Var3)	Var1	Difference	Q: Quad Word
	Var2	Subtrahend	
	Var3	Minuend	
	Description of Action		
	Var2 is subtracted from Var3, and the result is placed in Var1.		

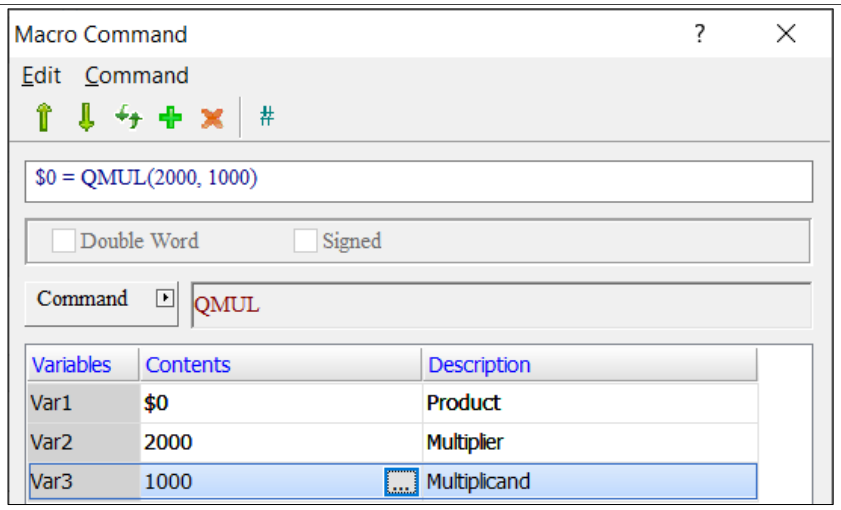
Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V
Var3	V	V	V

Example
• Var1 is the Internal Memory address, and Var2 and Var3 are constants.    • When the command is $0 = QSUB (200000, 100000), the floating-point subtraction operation will be executed to put the result obtained by subtracting 100000 from 200000 into $0, so $0 = 100000.

- **QMUL (Quad Word Multiplication)**

Expression	Meaning of Variable		Note
Var1 = QMUL(Var2, Var3)	Var1	Product	Q: Quad Word
	Var2	Multiplier	
	Var3	Multiplicand	
	Description of Action		
	Var2 is multiplied by Var3, and the result is placed in Var1.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V
Var3	V	V	V
Example			
• Var1 is the Internal Memory address, and Var2 and Var3 are constants.			



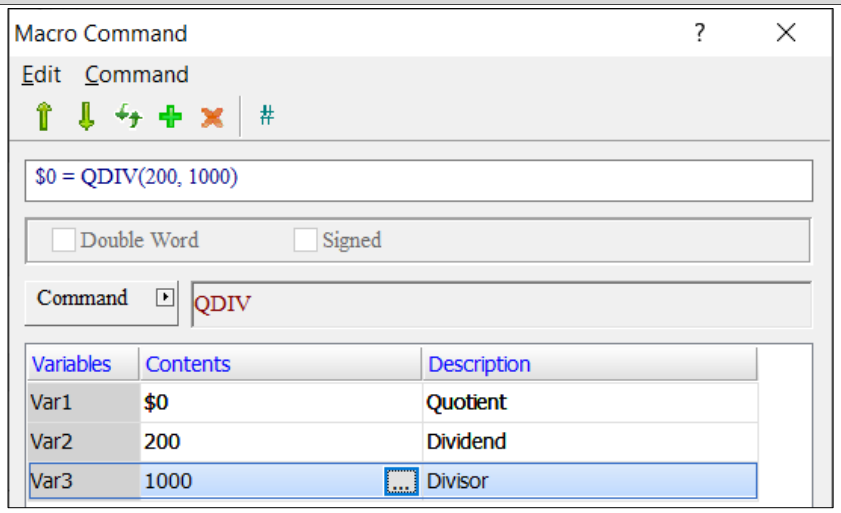
- When the command is $0 = QMUL (2000, 1000), it will perform floating-point multiplication and put the result obtained by multiplying 2000 by 1000 into $0, so $0 = 2000000.

- **QDIV (Quad Word Division)**

Expression	Meaning of Variable		Note
Var1 = QDIV(Var2, Var3)	Var1	Quotient	Q: Quad Word
	Var2	Dividend	
	Var3	Divisor	
	Description of Action		
	Var2 is divided by Var3, and the result is placed in Var1.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V
Var3	V	V	V

Example
• Var1 is the Internal Memory address, and Var2 and Var3 are constants.

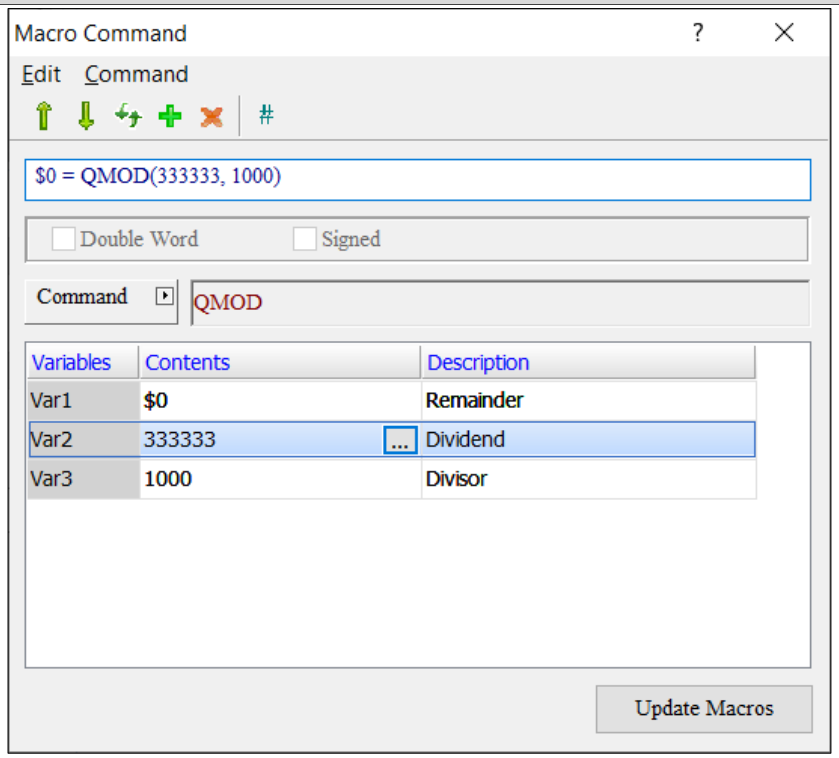
Example
 • When the command is $0 = QDIV (200, 1000), the floating-point number division operation will be executed and the result obtained by dividing 200 by 1000 will be put into $0, so $0 = 0.2.

- **Q MOD (Quadruple Word Remainder Operation)**

Expression	Meaning of Variable		Note
Var1 = QMOD(Var2, Var3)	Var1	Remainder	Q: Quad Word
	Var2	Dividend	
	Var3	Divisor	
	Description of Action		
	The remainder obtained by dividing Var2 and Var3 is put into Var1.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V
Var3	V	V	V

Example
• Var1 is the Internal Memory address, and Var2 and Var3 are constants.

Example
 • When the command is $0 = QMOD (333333, 1000), the floating-point remainder operation will be executed to put the remainder obtained by dividing 333333 by 1000 into $0, so $0 = 333.

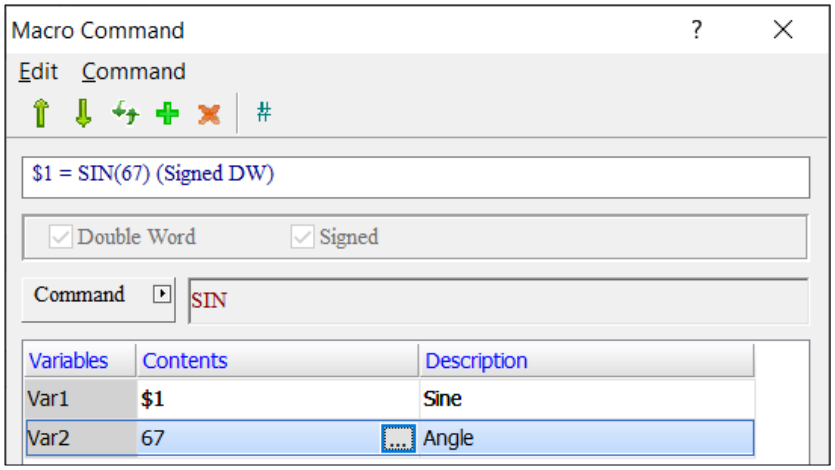
- **SIN (Sine function)**

Expression	Meaning of Variable		Comment
Var1 = SIN(Var2) (Signed DW)	Var1	Sine	DW: Double Word Signed: Number
	Var2	Angle	
	Description of Action		
	Sine operation is executed for the value input in Var2, and the result is put in Var1.		

Note:

1. The Data format of Var1 must be Floating.
2. The Data format of Var2 must be Signed Decimal, and the number of decimal places cannot be set.

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		V

Example
• Var1 is the Internal Memory address and Var2 is a constant.    • When the command is $1 = SIN (67), the input angle of 67 is used for sine operation and the result is put in $1, so $1 = 0.921.

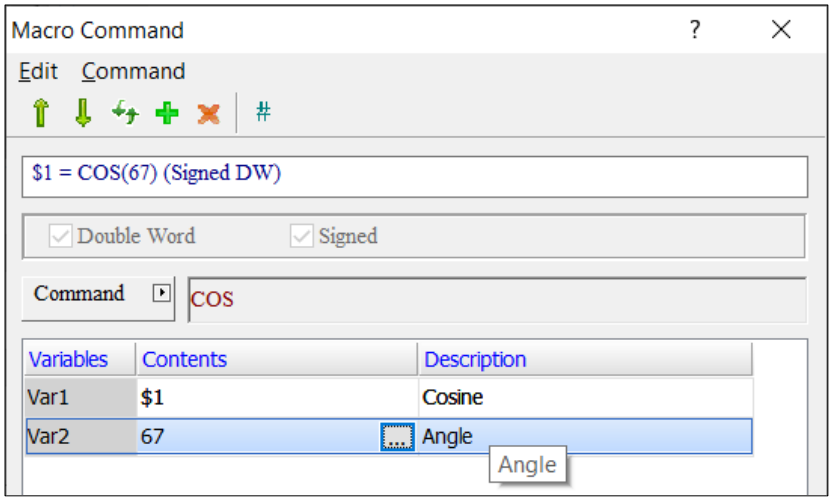
- **COS (Cosine function)**

Expression	Meaning of Variable		Comment
Var1 = COS(Var2) (Signed DW)	Var1	Cosine	DW: Double Word Signed: Signed number
	Var2	Angle	
	Description of Action		
	Cosine operation is executed for the value input in Var2, and the result is put in Var1.		

Note:

1. The Data Format for Var1 must be Floating.
2. The Data Format for Var2 must be Signed Decimal, and no decimal digits can be set.

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		V

Example
• Var1 is the Internal Memory address and Var2 is a constant.  • When the command is $1 = COS (67), the input angle of 67 is used for cosine operation and the result is put in $1, so $1 = 0.391.

- **TAN (Tangent function)**

Expression	Meaning of Variable		Note
Var1 = TAN(Var2) (Signed DW)	Var1	Tangent	DW: Double Word Signed: Signed number
	Var2	Angle	
	Description of Action		
	Tangent operation is executed for the value input in Var2, and the result is put in Var1.		

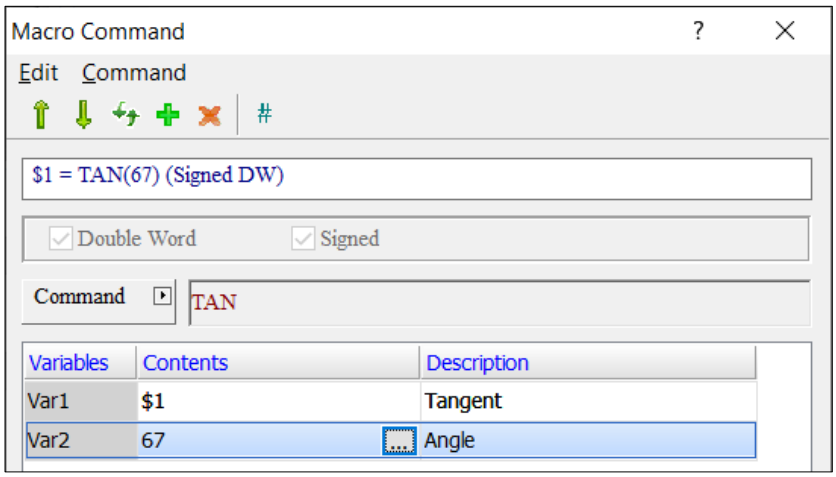
Note:

1. The Data Format for Var1 must be Floating.
2. The Data Format for Var2 must be Signed Decimal, and no decimal digits can be set.

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		V

Example

- Var1 is the Internal Memory address and Var2 is a constant.



- When the command is $1 = TAN (67), the input angle of 67 is used for tangent operation and the result is put in $1, so $1 = 2.356.

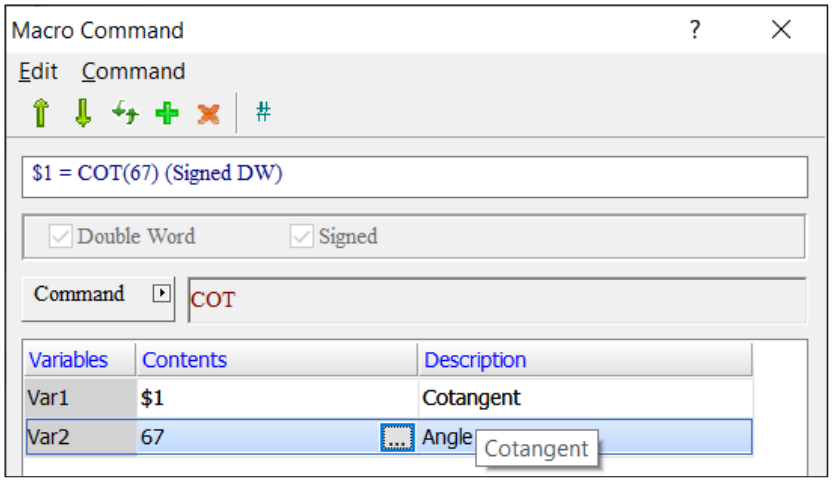
- **COT (Cotangent function)**

Expression	Meaning of Variable		Note
Var1 = COT(Var2) (Signed DW)	Var1	Cotangent	DW: Double Word Signed: Signed number
	Var2	Angle	
	Description of Action		
	Cotangent operation is executed for the value input in Var2. and the result is put in Var1.		

Note:

1. The Data Format for Var1 must be Floating.
2. The Data Format for Var2 must be Signed Decimal, and no decimal digits can be set.

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		V

Example
• Var1 is the Internal Memory address and Var2 is a constant.    • When the command is $1 = COT (67), the input angle of 67 is used for cotangent operation and the result is put in $1, so $1 = 0.424.

- **SEC (Secant function)**

Expression	Meaning of Variable		Note
Var1 = SEC(Var2) (Signed DW)	Var1	Secant	DW: Double Word Signed: Signed number
	Var2	Angle	
	Description of Action		
	Secant operation is executed for the value input in Var2, and the result is put in Var1.		

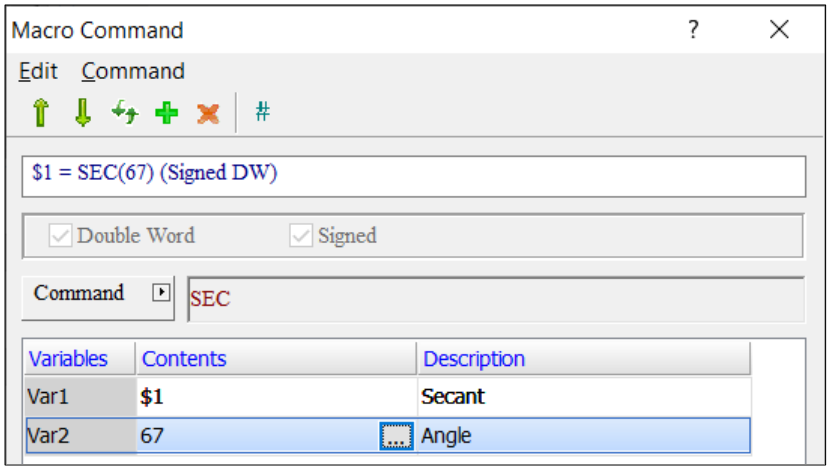
Note:

1. The Data Format for Var1 must be Floating.
2. The Data Format for Var2 must be Signed Decimal, and no decimal digits can be set.

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		V

Example

- Var1 is the Internal Memory address and Var2 is a constant.



- When the command is $1 = SEC (67), the input angle of 67 is used for secant operation and the result is put in $1, so $1 = 2.559.

- **CSC (Cosecant function)**

Expression	Meaning of Variable		Note
Var1 = CSC(Var2) (Signed DW)	Var1	Cosecant	DW: Double Word Signed: Signed number
	Var2	Angle	
	Description of Action		
	Cosecant operation is executed for the value input in Var2, and the result is put in Var1.		

Note:

1. The Data Format for Var1 must be Floating
2. The Data Format for Var2 must be Signed Decimal, and no decimal digits can be set.

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		V

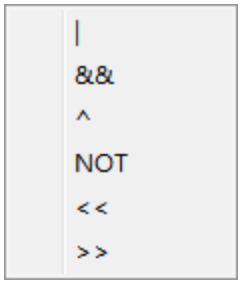
Example

- Var1 is the Internal Memory address and Var2 is a constant.

- When the command is $1 = CSC (67), the input angle of 67 is used for cosecant operation and the result is put in $1, so $1 = 1.086.

Logical Operation

Logic Operation contains six operators which convert numerical values to binary 0 and 1 representations, and then conduct |, &&, ^, NOT, < <, and > > operations. The macro commands are detailed as follows.



- **| (OR Logical Operation)**

Or Logical Operation Feature	
Expression	Result
0 | 0	0

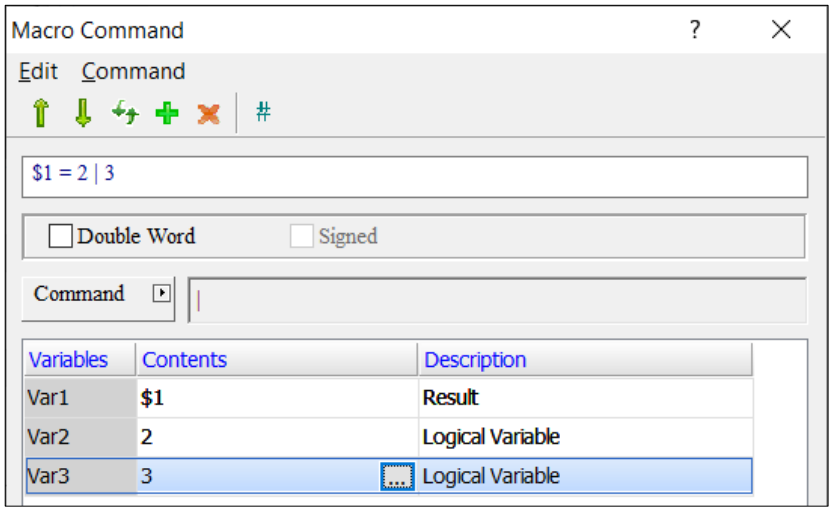
Or Logical Operation Feature	
Expression	Result
0 | 1	1
1 | 0	1
1 | 1	1

Expression	Meaning of Variable		Note
Var1 = Var2 | Var3 (W) Var1 = Var2 | Var3 (DW)	Var1	Result	W: Word DW: Double Word
	Var2	Logical Variable	
	Var3	Logical Variable	
	Description of Action		
	Execute OR operation on Var2 and Var3, and put the result in Var1.		

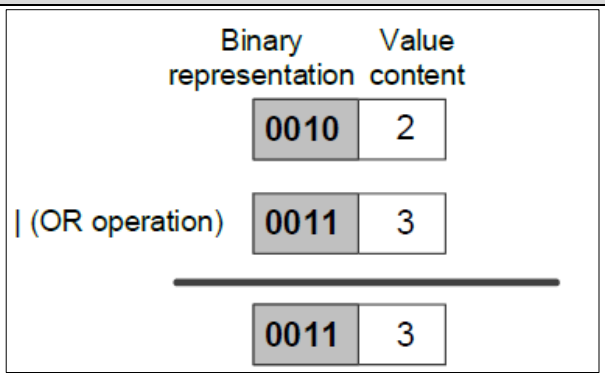
Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V
Var3	V	V	V

Example

- Var1 is the Internal Memory address, and Var2 and Var3 are Constant.



- After converting the values of Var2 and Var3 into binary representations (2 = 0010 and 3 = 0011), execute OR logical operation on 0010 and 0011, and the result is 0011 which is also 3.

Example


- **&& (AND Logical Operation)**

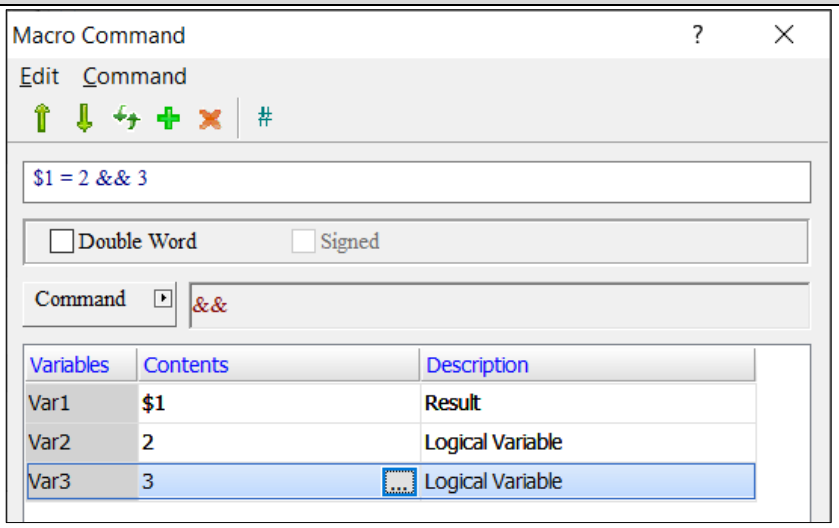
AND Logical Operation feature	
Expression	**Result**
0 && 0	0
0 && 1	0
1 && 0	0
1 && 1	1

Expression	Meaning of Variable		Note
Var1 = Var2 && Var3 (W) Var1 = Var2 && Var3 (DW)	Var1	Result	W: Word DW: Double Word
	Var2	Logical Variable	
	Var3	Logical Variable	
	Description of Action		
	Execute AND operation on Var2 and Var3, and put the result in Var1.		

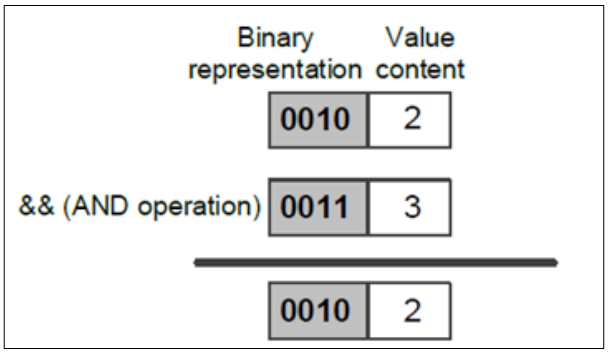
Variable	Type		
	Internal Memory	**PLC Register**	**Constant**
Var1	V	V	
Var2	V	V	V
Var3	V	V	V

Example
• Var1 is the Internal Memory address, and Var2 and Var3 are constants

Example



- After converting the values of Var2 and Var3 into binary representations (2 = 0010 and 3 = 0011), execute AND logical operation on 0010 and 0010, and the result is 0010 which is also 2.



- **^ (XOR Logical Operation)**

XOR Logical Operation feature	
Expression	Result
0 ^ 0	0
0 ^ 1	1
1 ^ 0	1
1 ^ 1	0

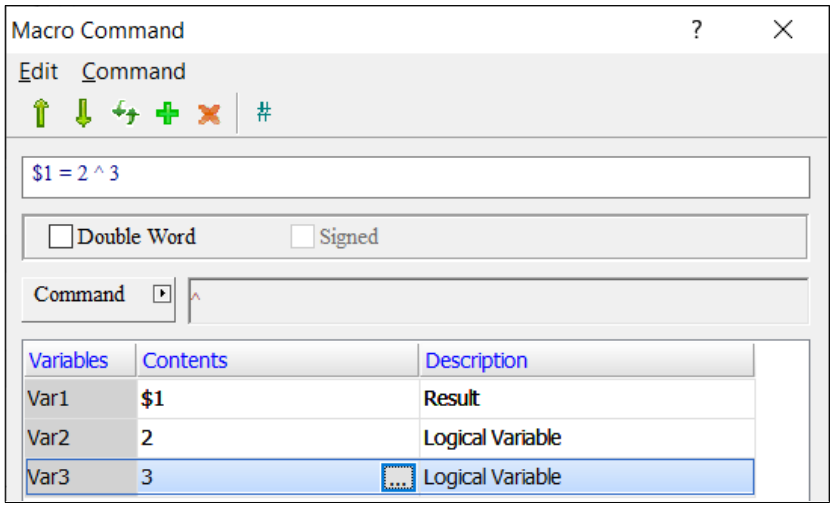
Expression	Meaning of Variable		Note
Var1 = Var2 ^ Var3 (W) Var1 = Var2 ^ Var3 (DW)	Var1	Result	W: Word DW: Double Word
	Var2	Logical Variable	
	Var3	Logical Variable	

Expression	Meaning of Variable	Note
	Description of Action	
	Execute XOR operation on Var2 and Var3 and put the result in Var1.	

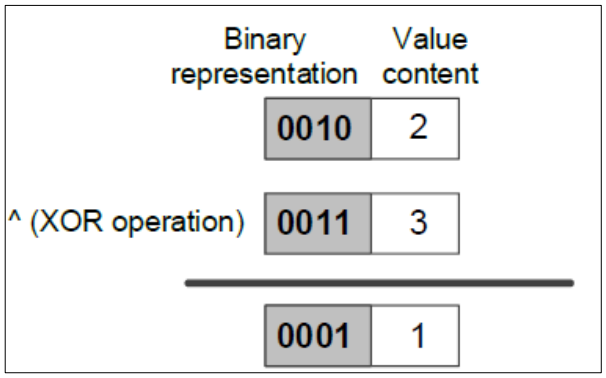
Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V
Var3	V	V	V

Example

- Var1 is the Internal Memory address, and Var2 and Var3 are constants



- After converting the values of Var2 and Var3 into binary representations (2 = 0010 and 3 = 0011), execute XOR logical operation on 0010 and 0011, and the result is 0001 which is also 1.



- **NOT (NOT Logical Operation)**

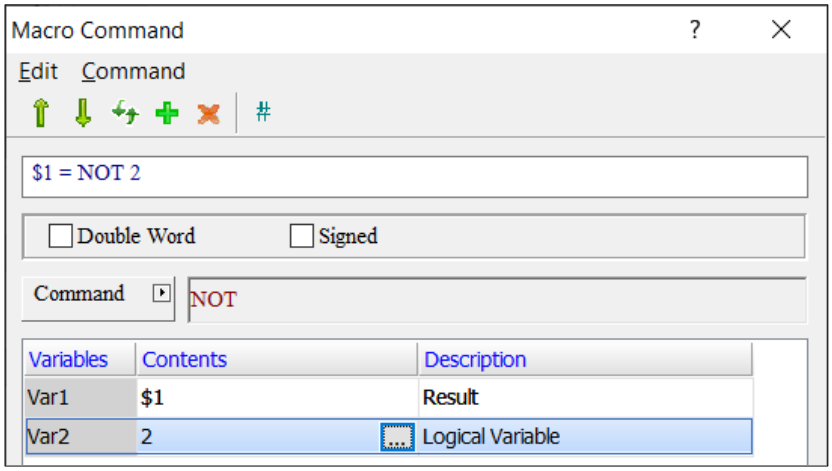
NOT Logical Operation feature	
Expression	**Result**
NOT 0	1
NOT 1	0

Expression	Meaning of Variable		Note
Var1 = NOT Var2 (W) Var1 = NOT Var2 (DW) Var1 = NOT Var2 (Signed W) Var1 = NOT Var2 (Signed DW)	Var1	Result	W: Word DW: Double Word Signed: Signed number
	Var2	Logical Variable	
	Description of Action		
	Execute NOT operation on Var2 and put the result in Var1.		

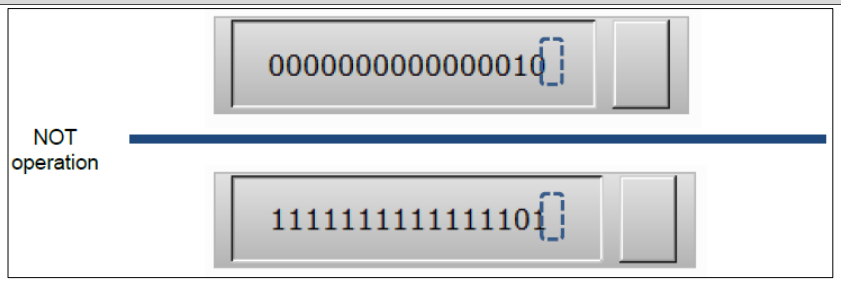
Variable	Type		
	Internal Memory	**PLC Register**	**Constant**
Var1	V	V	
Var2	V	V	V

Example

- Var1 is the Internal Memory address and Var2 is a constant



- After converting the value of Var2 into binary representation (2 = 0000000000000010), execute NOT 2 logical operation on 0000000000000010, and the result is 1111111111111101.

Example


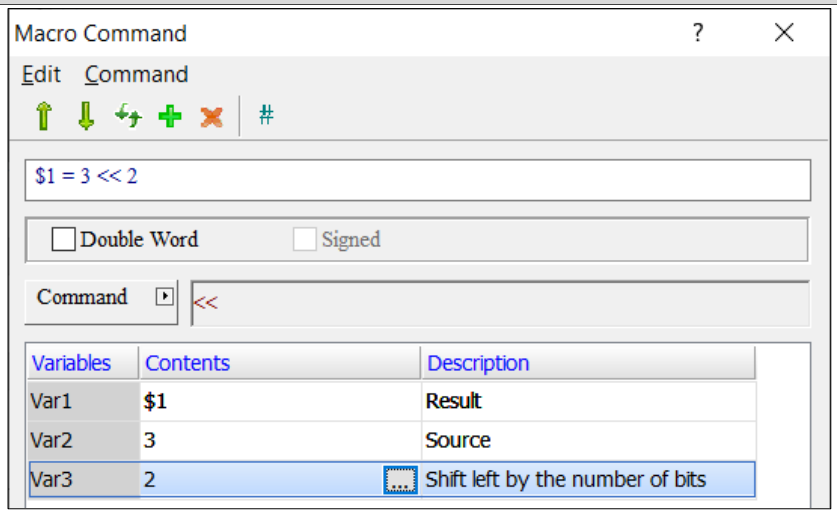
- **<< (SHL Left Shift Logical Operation)**

Expression	Meaning of Variable		Note
Var1 = Var2 << Var3 (W) Var1 = Var2 << Var3 (DW)	Var1	Result	W: Word DW: Double Word
	Var2	Source	
	Var3	Shift left by the number of bits	
	Description of Action		
	Move the number of bits of Source Var2 to the left by Var3 and put the result in Var1.		

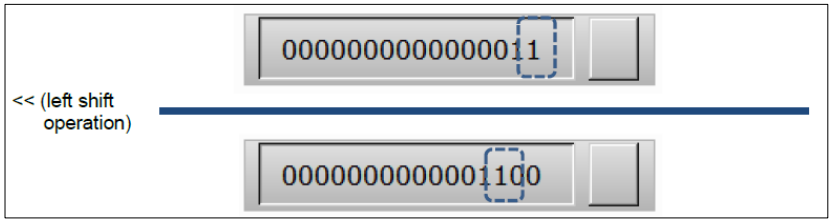
Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V
Var3	V	V	V

Example
• Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.

Example



- After converting the value of Var2 into binary representation (3 = 0000000000000011), execute the logical operation of << left shift by 2 bits on 0000000000000011, and the result is 0000000000001100.



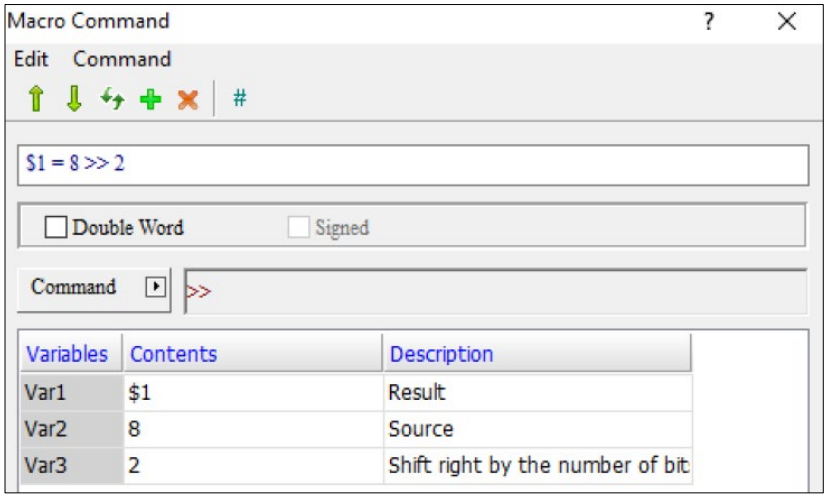
- **>> (SHR Right Shift Logical Operation)**

Expression	Meaning of Variable		Note
Var1 = Var2 >> Var3 (W) Var1 = Var2 >> Var3 (DW)	Var1	Result	W: Word DW: Double Word
	Var2	Source	
	Var3	Shift right by the number of bits	
	Description of Action		
	Move the number of bits of Source Var2 to the right by Var3 and put the result in Var1.		

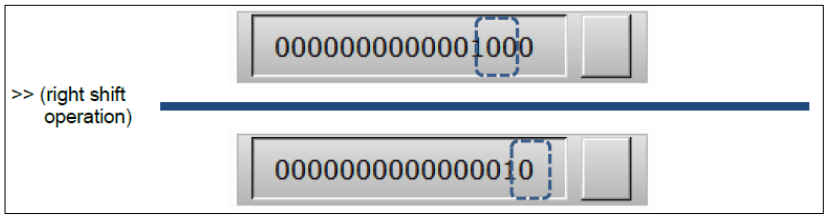
Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V
Var3	V	V	V

Example

- Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.

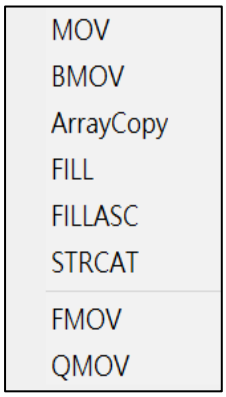


- After converting the value of Var2 into binary representation (8 = 0000000000001000), execute the logical operation of >> right shift by 2 bits on 0000000000001000, and the result is 0000000000000010.



Data Transfer

Data transfer includes the following transfer commands, which are detailed as follows.



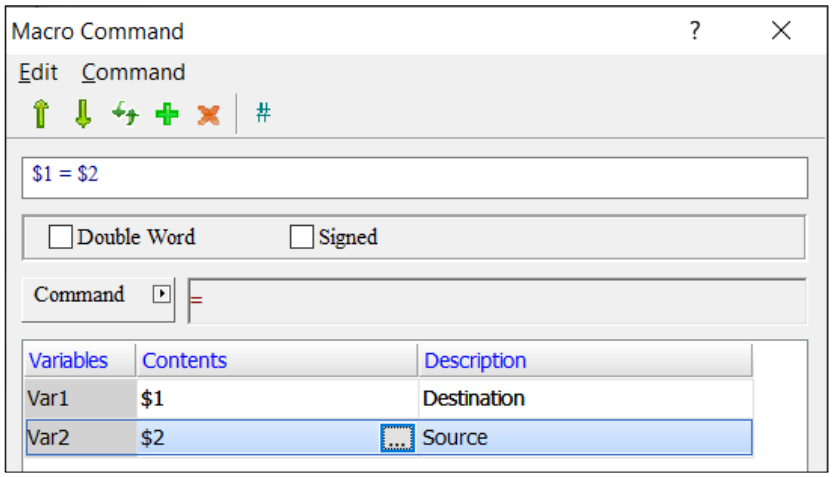
- **MOV (Data Specified Operand)**

Expression	Content		Note
Var1 = Var2 (W) Var1 = Var2 (DW) Var1 = Var2 (Signed W) Var1 = Var2 (Signed DW)	Var1	Destination	W: Word DW: Double Word Signed: Signed number
	Var2	Source	
	Description of Action		
	Copy the source of Var2 to the Var1 destination, and Var2's source will not be changed due to the data specify command.		

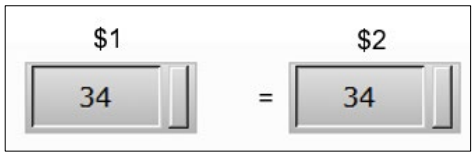
Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V

Example

- Var1 and Var2 are Internal Memory addresses.



- When the command $1 = $2 is executed, the Input value of $2 will be moved to $1. If the Input value of $2 is 34, $1 is 34.



- **BMOV (Copy Block)**

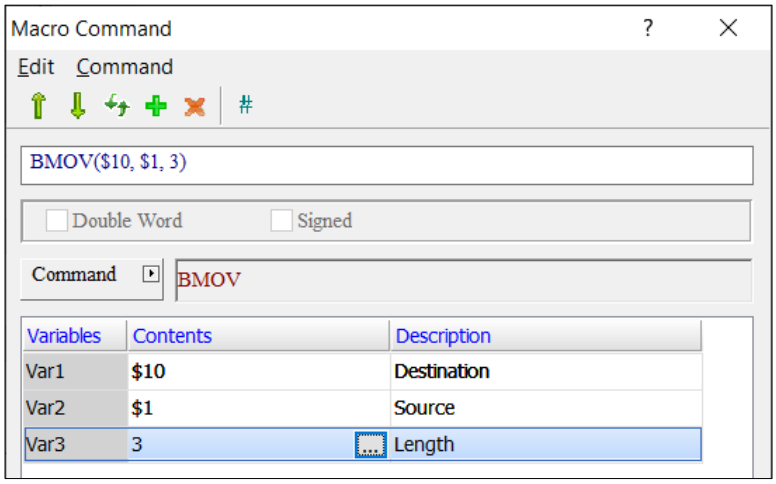
Expression	Meaning of Variable		Note
BMOV(Var1, Var2, Var3)(W)	Var1	Destination	W: Word

Expression	Meaning of Variable		Note
	Var2	Source	
	Var3	Length (Word)	
	Description of Action		
	Copy Var3 data lengths from Var2 source to Var1 destination.		

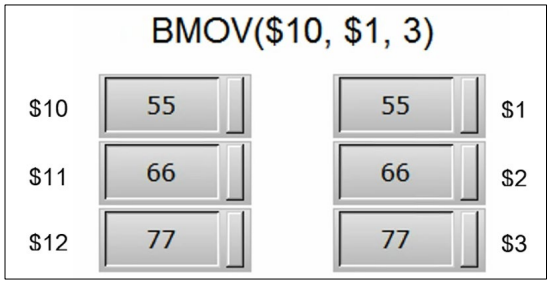
Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	
Var3	V		V

Example

- Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.



- Get the data of the three addresses starting from $1 and move the data to $10, so the Input value in $1, $2, and $3 are transferred to $10, $11, and $12 respectively.



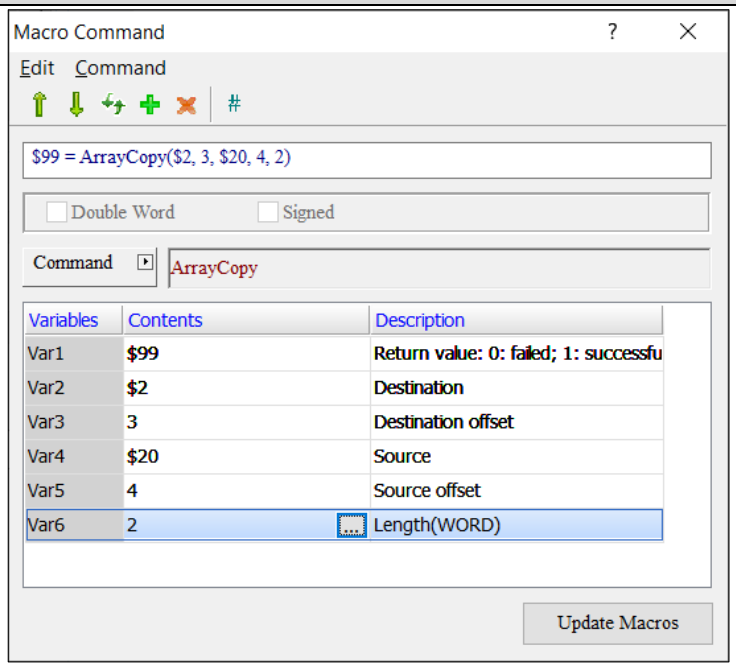
- **ArrayCopy (Copy Array)**

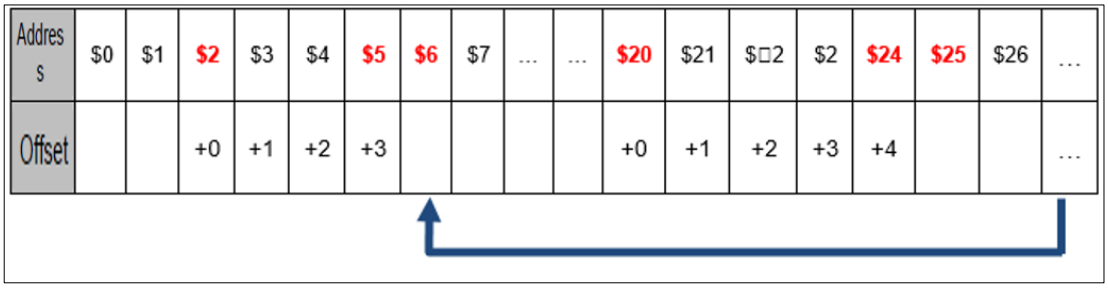
Expression	Meaning of Variable		
Var1 = ArrayCopy(Var2, Var3,Var4, Var5, Var6)	Var1	Return Value	
		Failed	0
		Succeeded	1
	Var2	Destination	
	Var3	Destination offset	
	Var4	Source	
	Var5	Source offset	
	Var6	Length (Word)	
	Description of Action		
	Copy a Continuous address data to another address.		

Variable	Type			
	Internal Memory	PLC Register	String	Constant
Var1	V			
Var2	V	V		
Var3	V	V		V
Var4	V	V		
Var5	V	V		V
Var6	V	V		V

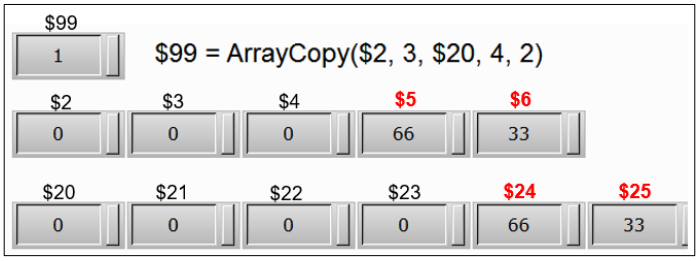
Example
• Var1, Var2, and Var4 are Internal Memory addresses, and Var3 and Var5 are constants.

Example





- Copy the offset (4, address is $24) specified by the source address ($20) to the offset (3, address is $5) specified by the destination address ($2) and move 2 Words in length according to the start address determined by the offset, as shown in the following figure.



- **FILL (Fill Block)**

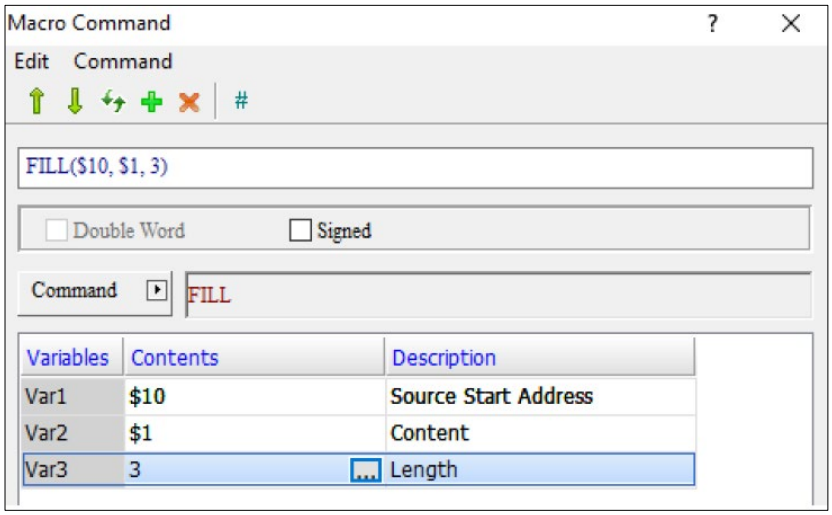
Expression	Meaning of Variable		Note
FILL(Var1, Var2, Var3) (W) FILL(Var1, Var2, Var3) (Signed W)	Var1	Destination Start Address	W: Word
	Var2	Source	
	Var3	Length	

	Description of Action	
	Store values of Var2 to the Var1 start address in sequence. The total length is Var3.	

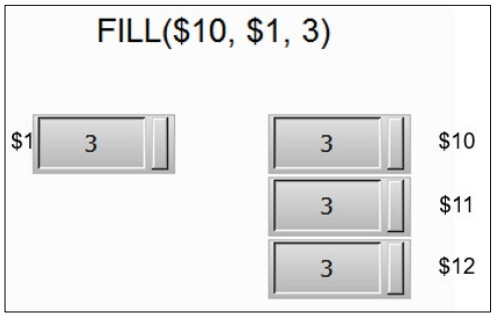
Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V
Var3	V	V	V

Example

- Var1 is the Internal Memory address, and Var2 and Var3 are constants.



- Save the input value of $1 for 3 data lengths in sequence to $10, $11, and $12.



- **FILLASC (Convert Text to ASCII Values)**

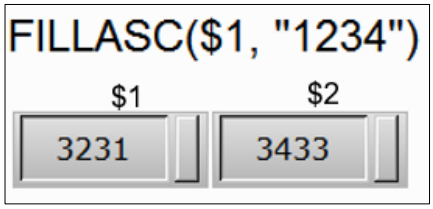
Expression	Meaning of Variable		Note
FILLASC(Var1, "Var2") (W)	Var1	Destination address for storing string	W: Word
	Var2	String	
	Description of Action		
	Convert each character of Var2 string into an ASCII value and put it in Var1 address.		

Variable	Type			
	Internal Memory	PLC Register	Constant	String
Var1	V	V		
Var2				V

Example

- Var1 is the Internal Memory address and Var2 is a string.

- The results after operation are $1 = 3231H and $2 = 3433H.



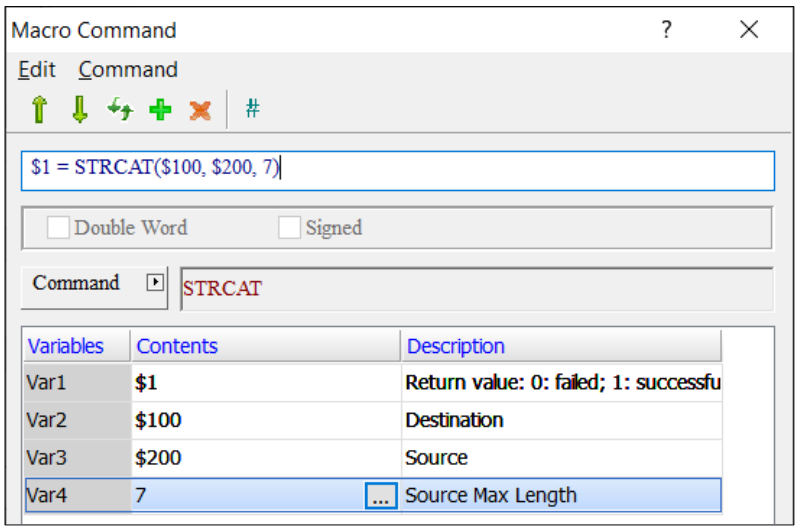
- **STRCAT (Connection String)**

<table>
<tr><th>Expression</th><th colspan="3">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="7">Var1 =
STRCAT(Var2,Var3,Var4)</td><td rowspan="3">Var1</td><td colspan="2">Return value</td><td rowspan="7"></td></tr>
<tr><td>Failed</td><td>0</td></tr>
<tr><td>Succeeded</td><td>1</td></tr>
<tr><td>Var2</td><td colspan="2">Destination</td></tr>
<tr><td>Var3</td><td colspan="2">Source</td></tr>
<tr><td>Var4</td><td colspan="2">Maximum length of the destination string (unit: Char)</td></tr>
<tr><td colspan="3">Description of Action</td></tr>
<tr><td colspan="3">Connect the source address string to the destination address string.</td></tr>
</table>

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		
Var3	V		
Var4	V		V

Example

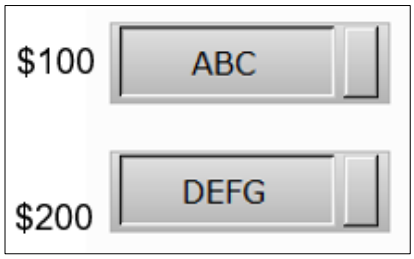
- Var1 is the Internal Memory address, Var2 and Var3 are strings, and Var4 is a constant.



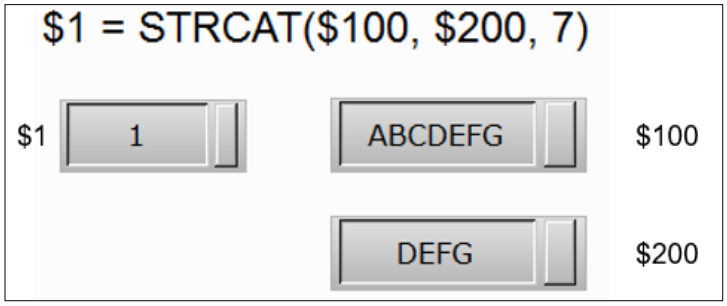
- Create a **Numeric Entry** element with the address as $1 and create **Character Entry** elements of $100 and $200.

Example

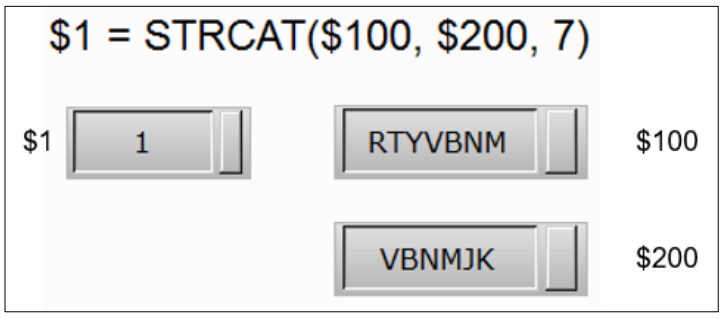
- After execution, enter the following values.



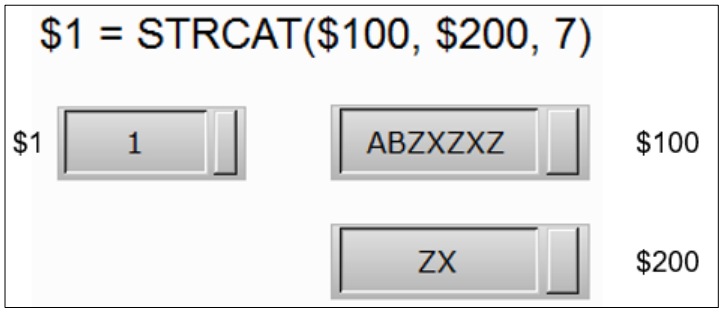
- After operation the results are $100 = ABCDEFG and $1 = 1.



- A length of 7 indicates a maximum length of 7 characters in $100. If the connection string is more than 7 characters, only 7 characters will be displayed.



- If the total length is less than 7 characters, it will be added to 7 characters for display.



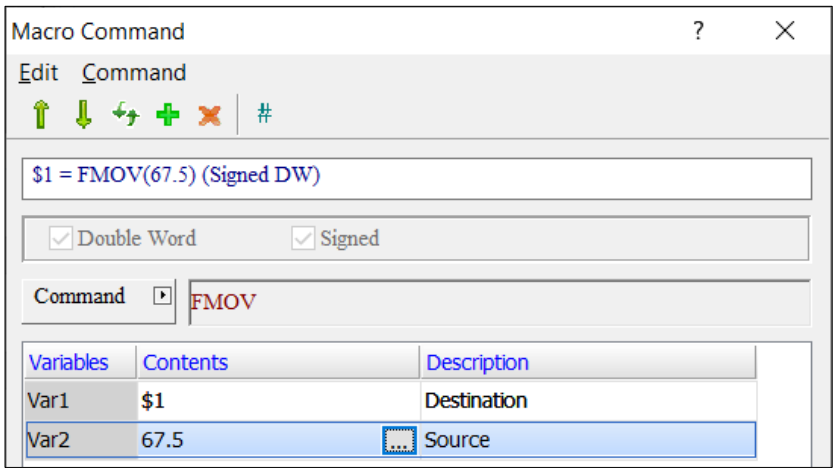
- **FMOV (Specify Floating-Point Number Data)**

Expression	Meaning of Variable		Note
Var1 = FMOV(Var2) (Signed DW)	Var1	Destination	DW: Double Word Signed: Signed number
	Var2	Source	
	Description of Action		
	Copy the floating-point number data of Var2 to Var1. The source data of Var2 will not be changed by the floating-point number data command.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V

Example

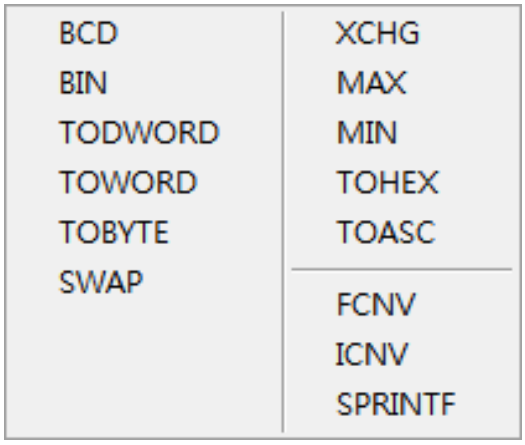
- Var1 is the Internal Memory address and Var2 is a constant.



- Store the floating-point number data of 67.5 to $1, so $1 = 67.5.

Data Conversion

Data Conversion includes conversion of Data Format, Maximum and Minimum, and value data swap commands, which are explained in detail as follows.



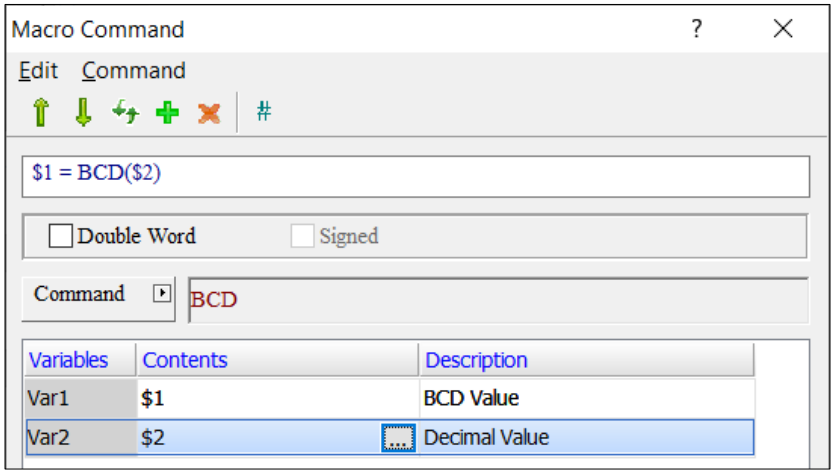
- **BCD (Convert Decimal Values to BCD values)**

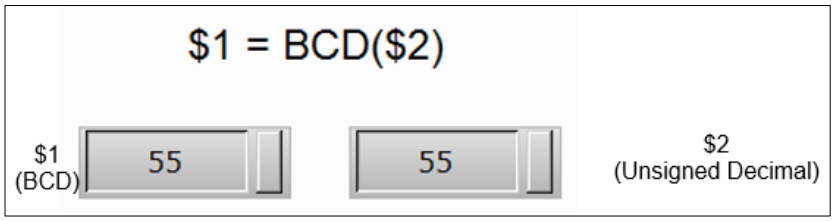
Expression	Meaning of Variable		Note
Var1 = BCD(Var2) (W) Var1 = BCD(Var2) (DW)	Var1	BCD Value	W: Word DW: Double Word
	Var2	Decimal Value	
	Description of Action		
	Convert the decimal value in Var2 to the BCD format value and put it in Var1.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		

Example

- Var1 and Var2 are Internal Memory addresses.

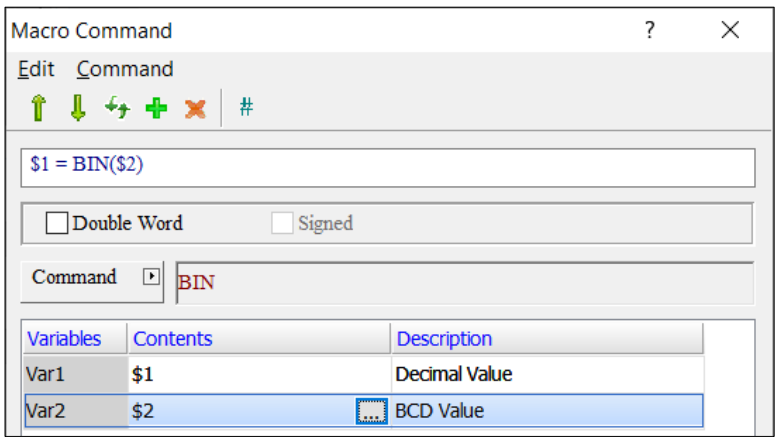


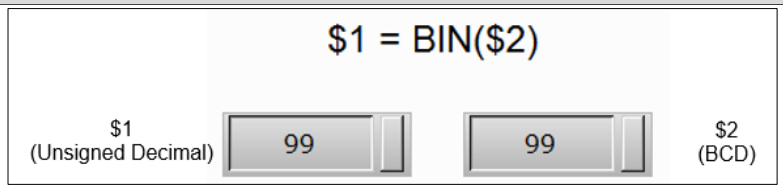
Example
• Convert the decimal value of $2 to BCD and store it in $1. 

- **BIN (Convert BCD Values to Decimal Values)**

Expression	Meaning of Variable		Note
Var1 = BIN(Var2) (W) Var1 = BIN(Var2) (DW)	Var1	Decimal Value	W: Word DW: Double Word
	Var2	BCD Value	
	Description of Action		
	Convert the BCD format value in Var2 to a decimal value and put it in Var1.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		

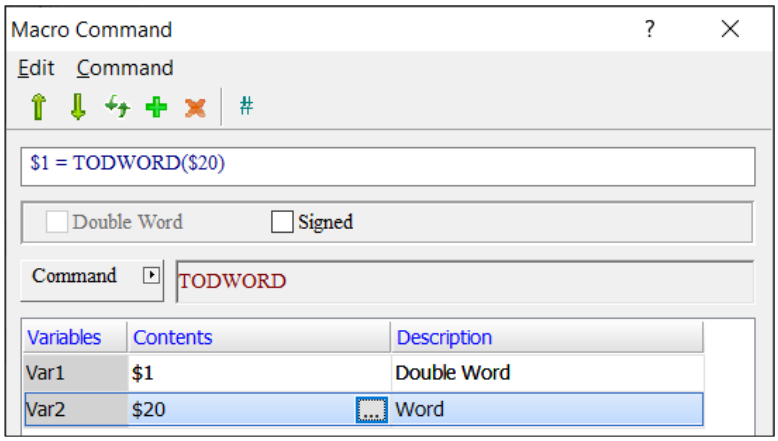
Example
• Var1 and Var2 are Internal Memory addresses.  • Convert the BCD value format of $2 to decimal value format and save it in $1.

Example


- **TODWORD (Convert Values from Word to Double Word)**

Expression	Meaning of Variable		Note
Var1 = TODWORD(Var2) (W) Var1 = TODWORD(Var2) (Signed W)	Var1	Double Word value	W: Word Signed: Signed number
	Var2	Word value	
	Description of Action		
	Convert the Word format value in Var2 to the Double Word value and put it in Var1.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		

Example
• Var1 and Var2 are Internal Memory addresses.  • Convert the Word value of $20 to Double Word value and store it in $1. Because the value has been converted to Double Word format, it occupies two addresses, $1 and $2

- **TOWORD (Convert Values from Byte to Word)**

Expression	Meaning of Variable		Note
Var1 = TOWORD(Var2, Var3) (W)	Var1	Word value	W: Word
	Var2	Source Start Address	
	Var3	Length	
	Description of Action		
	Starting from Var2 start address, convert a continuous Var3 bytes to Word value and save the result in Var1. Add 0 for the high byte of each Var1.		

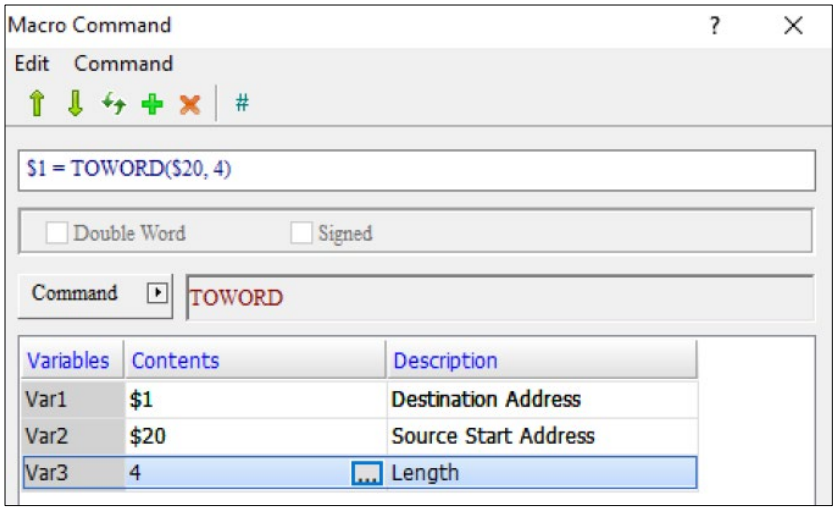
Note:

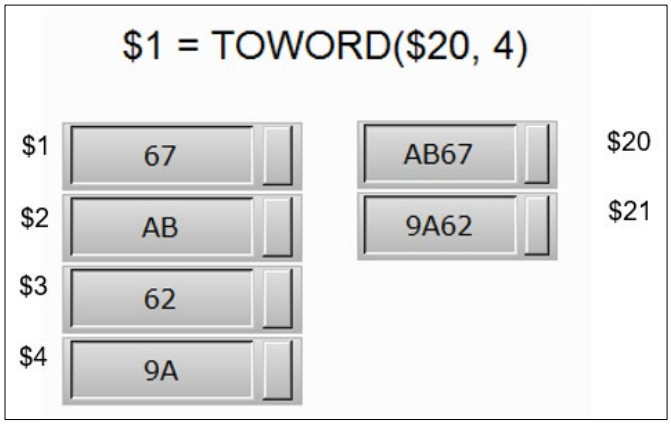
1. Since the unit of Var2 is Word, each Word of Var2 can be converted into 2 Words.
2. After data conversion, the high and low bytes of Word will be swapped.

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		
Var3	V		V

Example

- Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.



Example
• Convert a continuous 4 bytes from $20 to Word value and put it in $1. • The Data Format of $1 and $20 is set to Hex. • Suppose $20 = AB67H and $21 = 9A62H, read 4 bytes of data in $20 with the TOWORD command and put them in $1, $2, $3, and $4. Accordingly, the data obtained are $1 = 67H, $2 = ABH, $3 = 62H, and $4 = 9AH. 

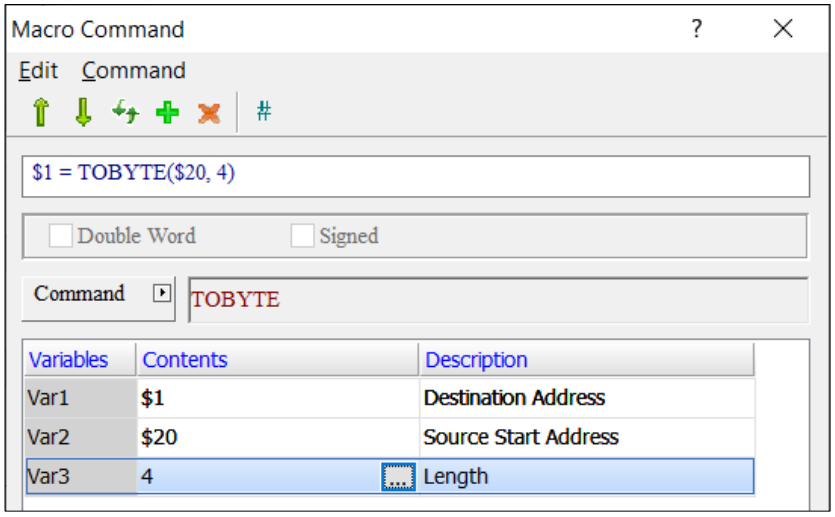
- **TOBYTE (Convert Values from Word to Byte)**

Expression	Meaning of Variable		Note
Var1 = TOBYTE(Var2, Var3) (W)	Var1	BYTE Value	W: Word
	Var2	Source Start Address	
	Var3	Length	
	Description of Action		
	Starting from the low byte of Var2, convert a continuous Var3 Words to byte values and ignore the high byte of Var2. The result is saved in Var1.		
Note: After data conversion, the high and low bytes of Word will be swapped.			

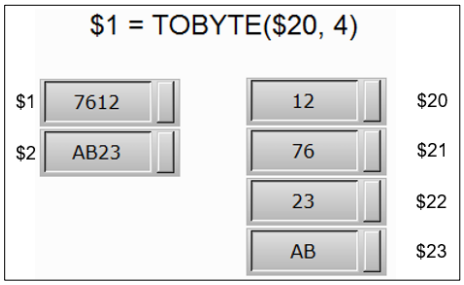
Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		
Var3	V		V

Example

- Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.



- Convert a continuous 4 Words from $20 to byte value and put it in $1.
- The Data format of $1 and $20 is set to Hex.
- Suppose $20 = 12H, $21 = 76H, $22 = 23H, and $23 = ABH, read 4 Words of data in $20 with the
- TOBYTE command and put them in $1 and $2. Accordingly, the data obtained are $1 = 7612H and $2 = AB23H.



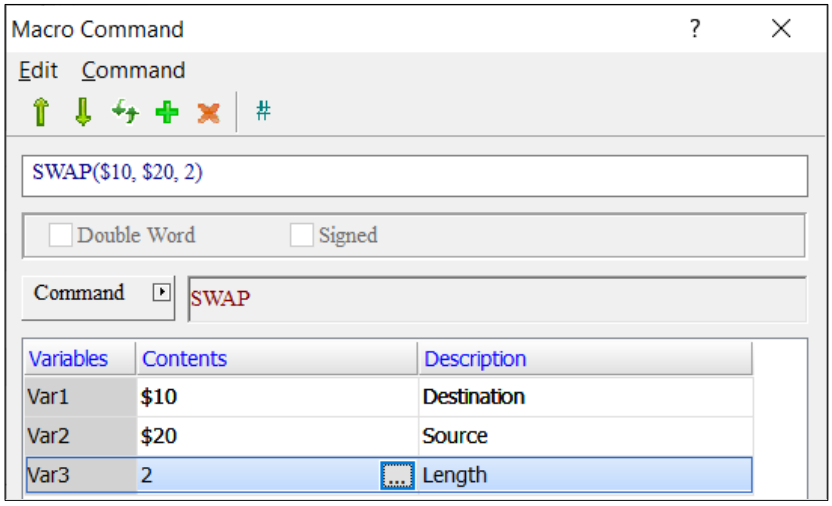
- SWAP (Swap High and Low Bytes of Word)

Expression	Meaning of Variable		Note
SWAP(Var1, Var2, Var3) (W)	Var1	Destination Start Address	W: Word
	Var2	Source Start Address	
	Var3	Length	
	Description of Action		
	Convert Var3 lengths from Var2 start address and swap high and low bytes of Var2 Word data to Var1.		

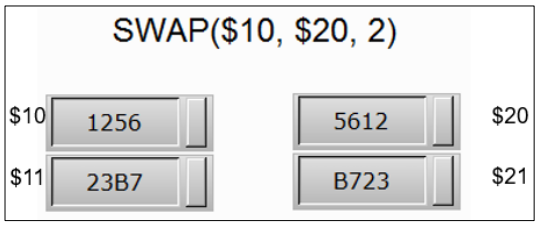
Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		
Var3	V		V

Example

- Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.



- Swap high and low bytes of the value in $20 and put them in $10. Exchange 2 data lengths of values.
- The Data format of $10 and $20 is set to Hex.
- Suppose $20 = 5612H and $21 = B723H, swap high and low bytes of the data in $20 with the
- SWAP command and put them in $10 and $11. Accordingly, the data obtained are $10 = 1256H and $11 = 23B7H.



- XCHG (Exchange Value Data)

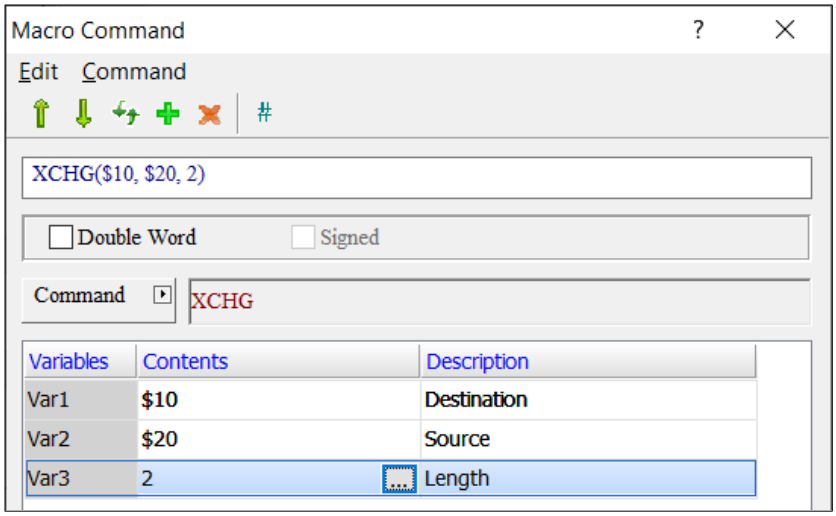
Expression	Meaning of Variable		Note
XCHG(Var1, Var2, Var3) (W) XCHG(Var1, Var2, Var3) (DW)	Var1	Destination Start Address	W: Word DW: Double Word
	Var2	Source Start	

Expression	Meaning of Variable		Note
		Address	
	Var3	Length	
	Description of Action		
	Exchange Var2 start address data to Var1 destination start address according to Var3 lengths.		

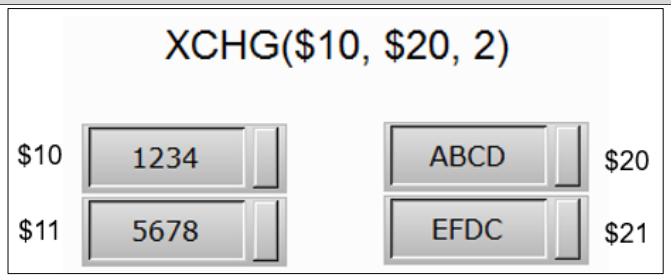
Variable	Type		
	Internal Memory	**PLC Register**	**Constant**
Var1	V		
Var2	V		
Var3	V		V

Example

- Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.



- Exchange 2 data lengths of values in $20 and $10.
- The Data Formats of $10 and $27 is set to Hex.
- Suppose $20 = 1234H, $21 = 5678H, $10 = ABCDH, and $11 = EFDCH, exchange the data in $20 and $21 with the data in $10 and $11 with the XCHG command. Accordingly, the data obtained are $20 = ABCDH, $21 = EFDCH, $10 = 1234H, and $11 = 5678H.

Example


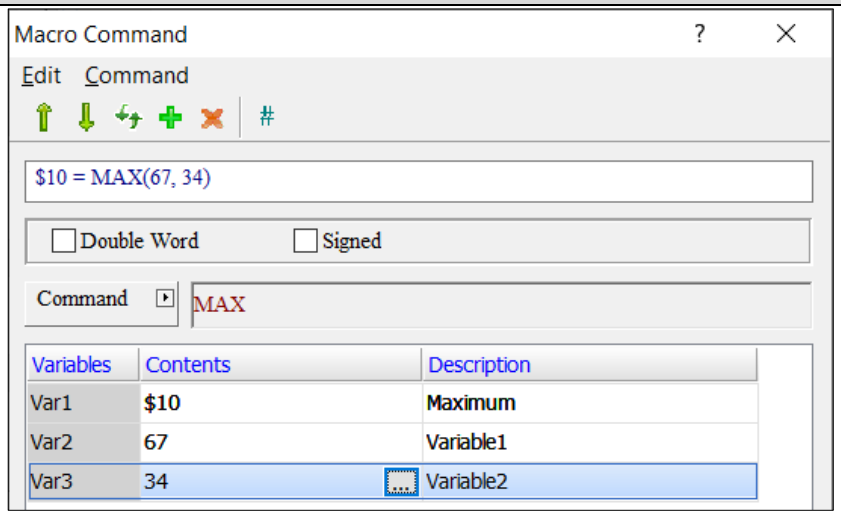
- **MAX (Get Maximum)**

Expression	Meaning of Variable		Note
Var1 = MAX(Var2, Var3) (W) Var1 = MAX(Var2, Var3) (DW) Var1 = MAX(Var2, Var3) (Signed W) Var1 = MAX(Var2, Var3) (Signed DW)	Var1	Maximum	W: Word DW: Double Word
	Var2	Variable1	
	Var3	Variable2	
	Description of Action		
	Take the maximum value of Var2 and Var3 and put it in Var1.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		V
Var3	V		V

Example
• Var1 is the Internal Memory address, and Var2 and Var3 are constants

Example

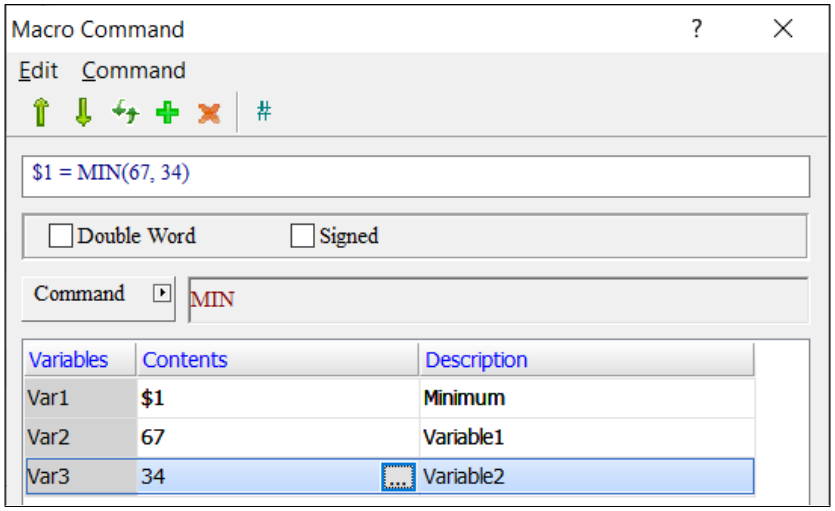


- Enter the Maximum value of 67 and 34 in $1, so $1 = 67.

- **MIN (Get Minimum)**

Expression	Meaning of Variable		Note
Var1 = MIN(Var2, Var3) (W) Var1 = MIN(Var2, Var3) (DW) Var1 = MIN(Var2, Var3) (Signed W) Var1 = MIN(Var2, Var3) (Signed DW)	Var1	Minimum	W: Word DW: Double Word
	Var2	Variable1	
	Var3	Variable2	
	Description of Action		
	Take the minimum value of Var2 and Var3 and put it in Var1.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		V
Var3	V		V

Example
• Var1 is the Internal Memory address, and Var2 and Var3 are constants.  • Put the Minimum value of 67 and 34 in $1, so $1 = 34.

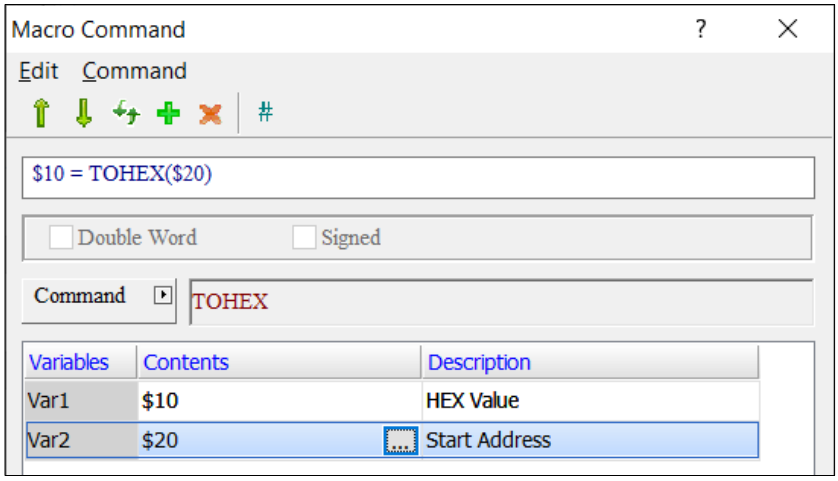
- **TOHEX (Convert 4 ASCII Characters to HEX)**

Expression	Meaning of Variable		Note
Var1 = TOHEX(Var2) (W)	Var1	HEX Value	W: Word
	Var2	ASCII Start Address	
	Description of Action		
	Convert 4 ASCII characters of WORD to HEX from Var2 and put it in Var1.		

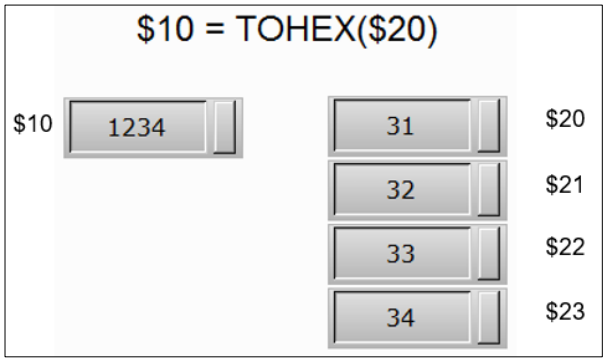
Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		

Example

- Var1 and Var2 are Internal Memory addresses.



- Convert 4 consecutive ASCII characters of Word from the Start Address of $20 to HEX value and put it in $10.
- The Data Format of $10 and $20 is set to Hex.
- Suppose $20 = 31H, $21 = 32H, $22 = 33H, and $23 = 34H, convert ASCII characters in $20, $21, $22, and $23 to HEX with the TOHEX command and put it in $10. Accordingly, the data obtained is $10 = 1234H.



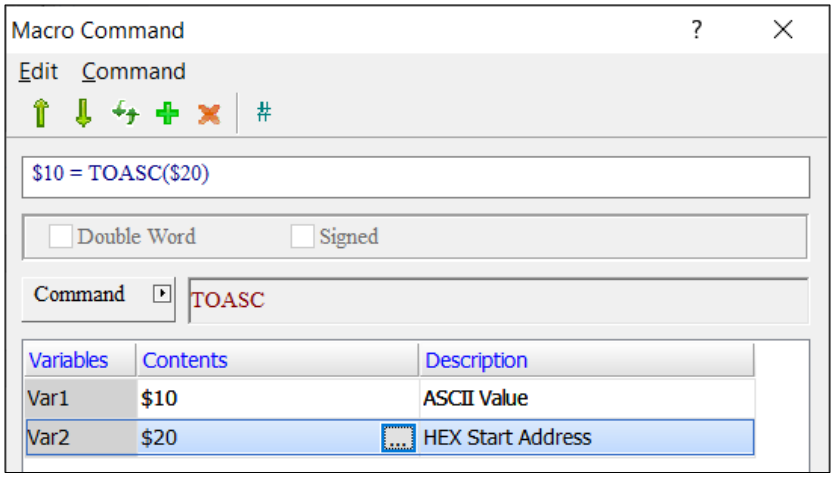
- **TOASC (Convert HEX to 4 ASCII Characters of Word)**

Expression	Meaning of Variable		Note
Var1 = TOASC(Var2) (W)	Var1	ASCII Value	W: Word
	Var2	HEX Start Address	
	Description of Action		
	Convert HEX from the address of Var2 to 4 ASCII characters of WORD and put them in Var1.		

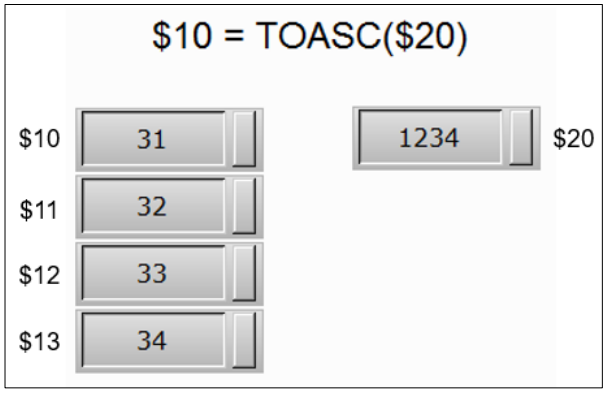
Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		

Example

- Var1 and Var2 are Internal Memory addresses.



- Convert HEX value of $20 into 4 consecutive ASCII characters of Word and put them in $10.
- The Data format of $10 and $20 is set to Hex.
- Suppose $20 = 1234H, convert HEX in $20 into ASCII characters with the TOASC command and put them in $10, $11, $12, and $13. Accordingly, the data obtained are $10 = 31H, $11 = 32H, $12 = 33H, and $13 = 34H.



- **FCNV (Convert Integer to Floating-Point Number)**

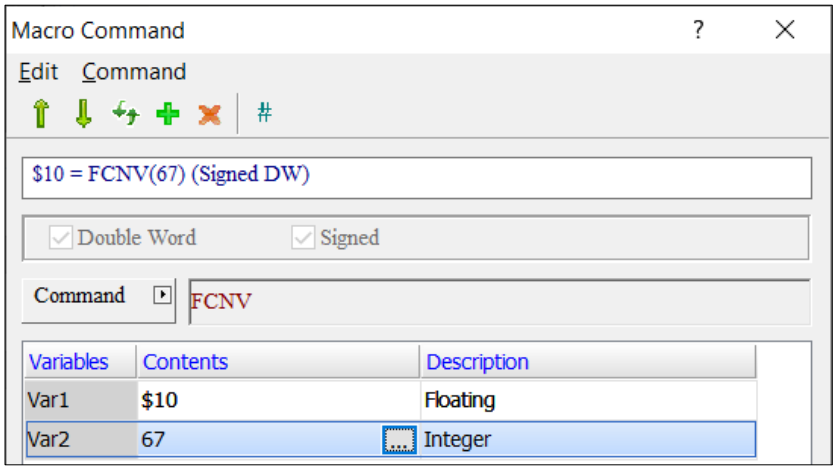
Expression	Meaning of Variable		Note
Var1 = FCNV(Var2) (Signed DW)	Var1	Floating	DW: Double Word Signed: Signed number
	Var2	Integer	

Expression	Meaning of Variable	Note
	Description of Action	
	Convert the integer in Var2 to floating-point number and put it in Var1.	

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		V

Example

- Var1 and Var2 are Internal Memory addresses.



- Convert the integer value of 67 to a floating-point value and put it in $10.
- The Data format set for $10 is Floating and the Data Type is Double Word.
- The result after the HMI operation is $10 = 67.0.

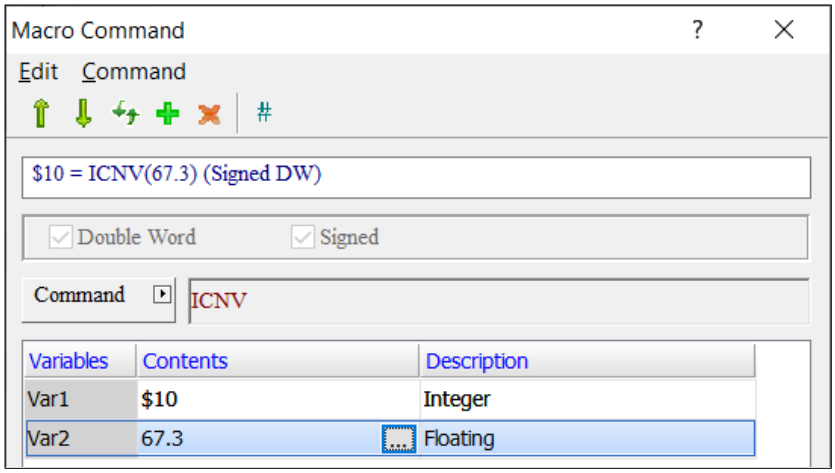
- **ICNV (Convert Floating-Point Number to Integer)**

Expression	Meaning of Variable		Note
Var1 = ICNV(Var2) (Signed DW)	Var1	Integer	DW: Double Word Signed: Signed number
	Var2	Floating	
	Description of Action		
	Convert the floating-point number in Var2 to integer and put it in Var1.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		V

Example

- Var1 and Var2 are Internal Memory addresses.



- Convert the floating-point value of 67.3 into an integer value and put it in $10.
- The Data format set for $10 is Unsigned Decimal, and the Data Type is Word.
- After the HMI operation, the result is $10 = 67.

- **SPRINTF (Format String)**

Expression	Meaning of Variable			Note
Var1 = SPRINTF (Var2, "%u", Var4) (DW) Var1 = SPRINTF (Var2, "%u", Var4, …, Var23) (DW) (Note2)	Var1	Return value		DW = Double Word
		Failed	0	
		Succeeded	1	
	Var2	Enter destination address of string		
	Var3	Format description string (Note 1)		
	Var4	Value 1		
	…	…		
	Var23	Value 20		
	Description of Action			

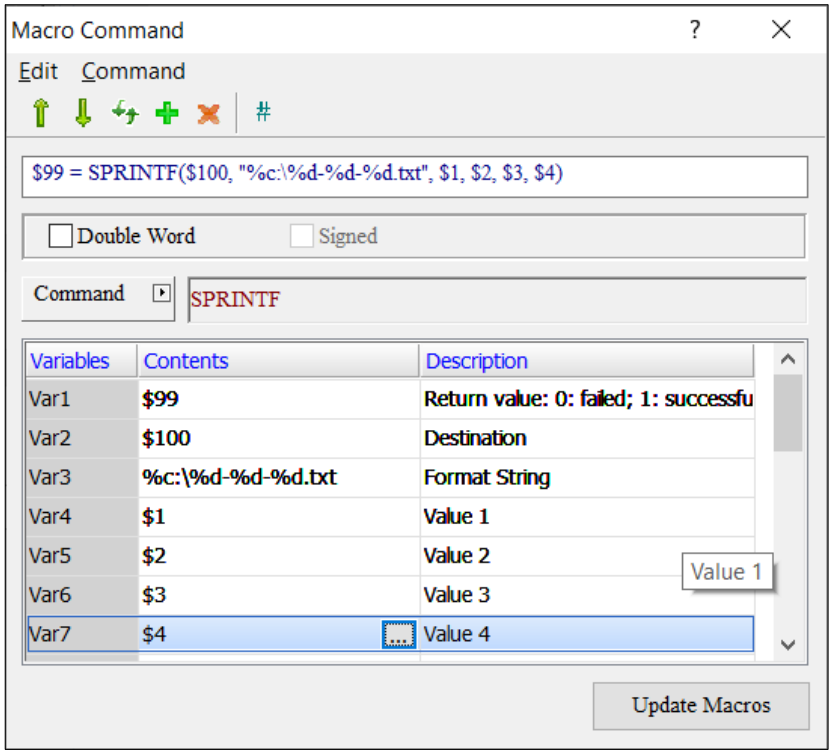
Expression	Meaning of Variable	Note
	Connect the values according to Var2 format description string and fill in the destination address.	

Note:

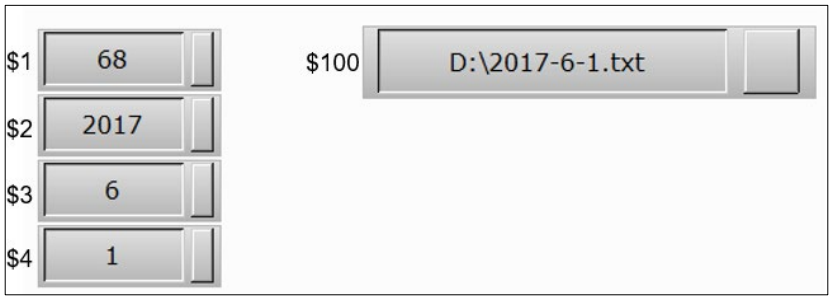
1. The supporting string formats are as follows:
   %d: signed integer
   %u: unsigned integer
   %c: ASCII character
   %x: hexadecimal value
2. Support up to 20 numeric variables.
3. The number of the parameters of variable conversion in the string contents should be consistent with the number of values.

Variable	Type			
	Internal Memory	PLC Register	String	Constant
Var1	V			
Var2	V	V		
Var3			V	
Var4	V	V		V
…	…	…		…
Var23	V	V		V

Example

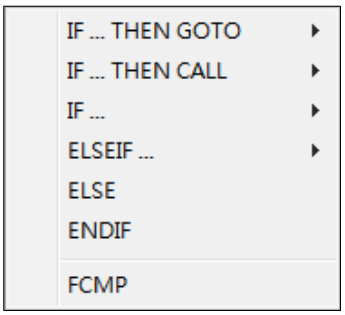


- Create a **Character Entry** element, set the address as $100, and set the length as 20.
- Create **Numeric Entry** elements of $1, $2, $3, and $4.
- After downloading the screen to the HMI, enter $1 = 68 (representing D in ASCII code), $2 = 2017, $3 = 6, and $4 = 1 on the HMI. The results are as follows.

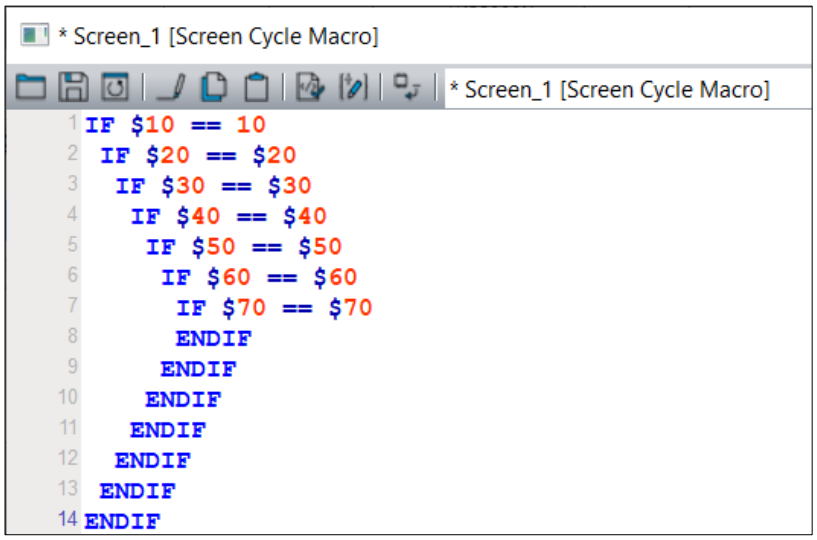


Comparison

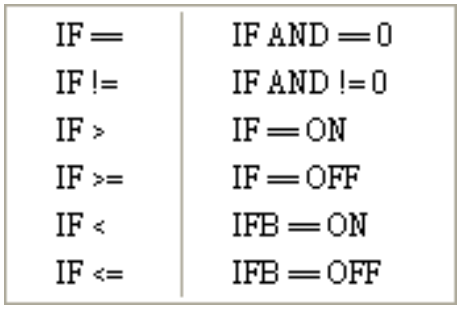
Comparison contains IF... THEN GOTO, IF... THEN CALL, IF…, ELSEIF… and other comparison commands, as detailed as follows.



- The nested structure of IF... supports up to 7 layers



- IF... THEN GOTO (if..., then execute according to the specified label name)



The following describes the 12 commands of the IF... THEN GOTO macro.

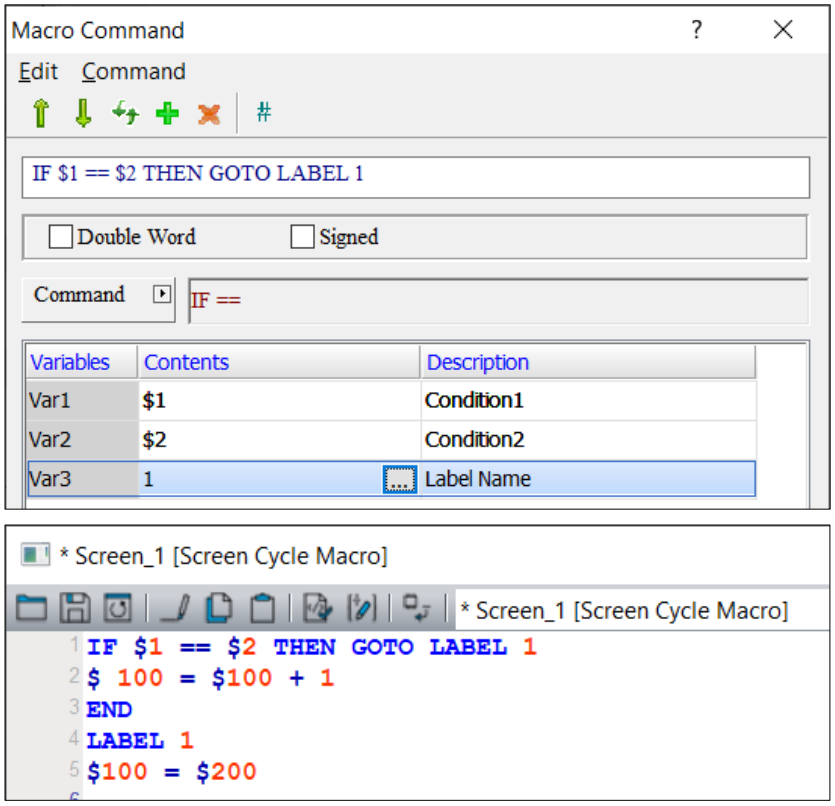
(1) IF ==			
Expression	**Meaning of Variable**		**Note**
IF Var1 == Var2 THEN GOTO LABEL Var3 (W) IF Var1 == Var2 THEN GOTO LABEL Var3 (DW) IF Var1 == Var2 THEN GOTO LABEL	Var1	Condition1	W: Word DW: Double Word Signed: Signed number
	Var2	Condition2	
	Var3	Label Name	
	Description of Action		

(1) IF ==		
Expression	**Meaning of Variable**	**Note**
Var3 (Signed W) IF Var1 == Var2 THEN GOTO LABEL Var3 (Signed DW)	If Condition1 equals Condition2, then GOTO executes LABEL Var3.	

Variable	Type		
	Internal Memory	**PLC Register**	**Constant**
Var1	V		V
Var2	V		V
Var3			V

Example

- Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.



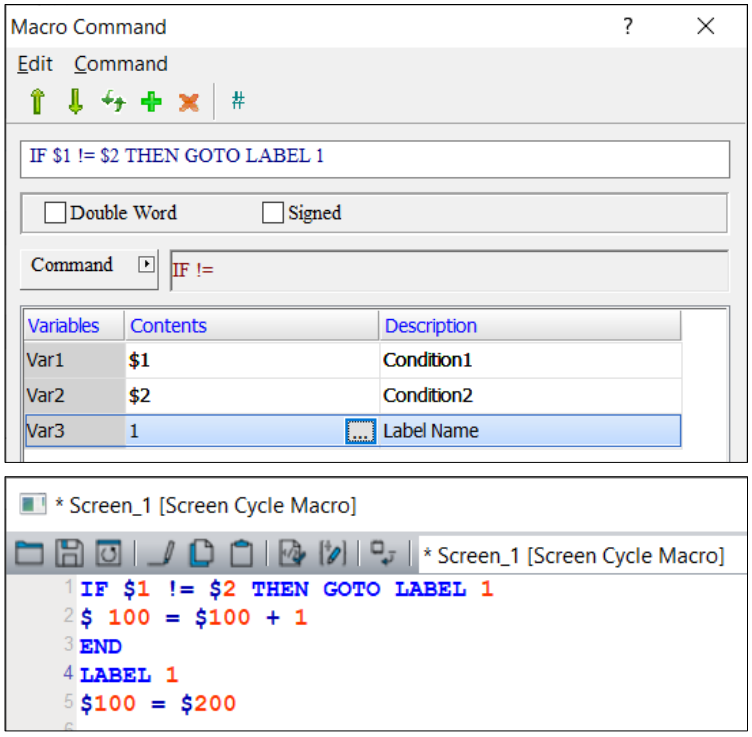
- If the value of $1 equals $2, execute LABEL1 ($100 = $200); if $1 does not equal $2, then execute $100 = $100 + 1.

(2) IF !=			
Expression	**Meaning of Variable**		**Note**
IF Var1 != Var2 THEN GOTO LABEL Var3 (W) IF Var1 != Var2 THEN GOTO LABEL Var3 (DW) IF Var1 != Var2 THEN GOTO LABEL Var3 (Signed W) IF Var1 != Var2 THEN GOTO LABEL Var3 (Signed DW)	Var1	Condition1	W: Word DW: Double Word Signed: Signed number
	Var2	Condition2	
	Var3	Label Name	
	Description of Action		
	If Condition1 does not equal Condition2, then GOTO executes LABEL Var3.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		V
Var2	V		V
Var3			V

Example

- Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.



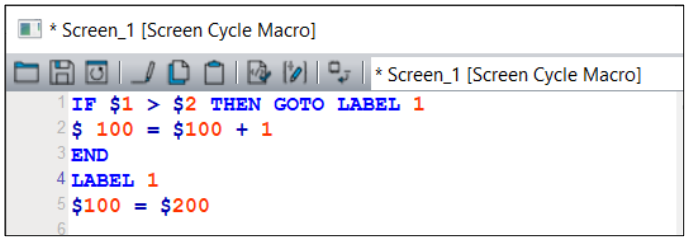
- If the value of $1 does not equal $2, execute LABEL1 ($100 = $200); if $1 equals $2, then execute $100 = $100 + 1.

(3) IF >			
Expression	**Meaning of Variable**		**Note**
IF Var1 > Var2 THEN GOTO LABEL Var3 (W) IF Var1 > Var2 THEN GOTO LABEL Var3 (DW) IF Var1 > Var2 THEN GOTO LABEL Var3 (Signed W) IF Var1 > Var2 THEN GOTO LABEL Var3 (Signed DW)	Var1	Condition1	W: Word DW: Double Word Signed: Signed number
	Var2	Condition2	
	Var3	Label index name	
	Description of Action		
	If Condition1 is greater than Condition2, then GOTO executes LABEL Var3.		

Variable	Type		
	Internal Memory	**PLC Register**	**Constant**
Var1	V		V
Var2	V		V
Var3			V

Example

- Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.



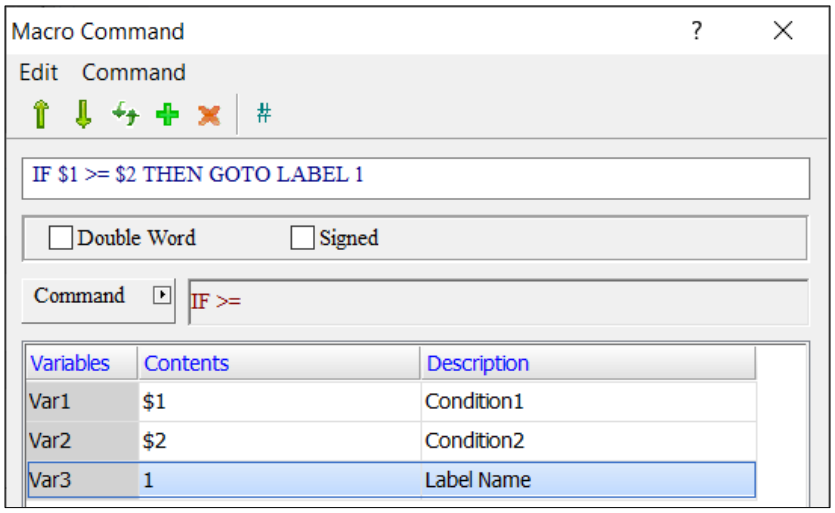
- If the value of $1 is greater than $2, execute LABEL1 ($100 = $200); if $1 is less than or equal to $2, then execute $100 = $100 + 1.

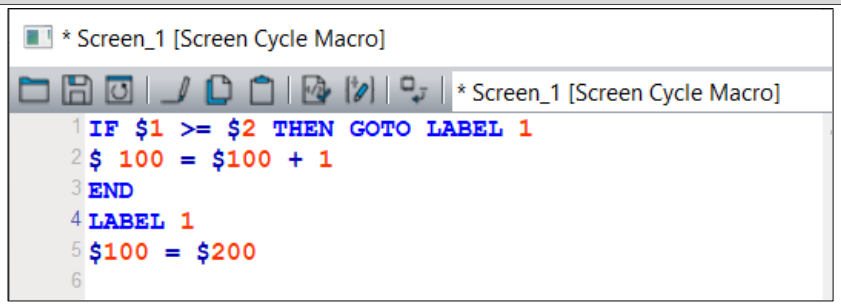
(4) IF >=			
Expression	**Meaning of Variable**		**Note**
IF Var1 >= Var2 THEN GOTO LABEL Var3 (W) IF Var1 >= Var2 THEN GOTO LABEL Var3 (DW) IF Var1 >= Var2 THEN GOTO LABEL Var3 (Signed W) IF Var1 >= Var2 THEN GOTO LABEL Var3 (Signed DW)	Var1	Condition1	W: Word DW: Double Word Signed: Signed number
	Var2	Condition2	
	Var3	Label name	
	Description of Action		
	If Condition1 is greater than or equal to Condition 2, then GOTO executes LABEL Var3.		

Variable	Type		
	Internal Memory	**PLC Register**	**Constant**
Var1	V		V
Var2	V		V
Var3			V

Example

- Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.



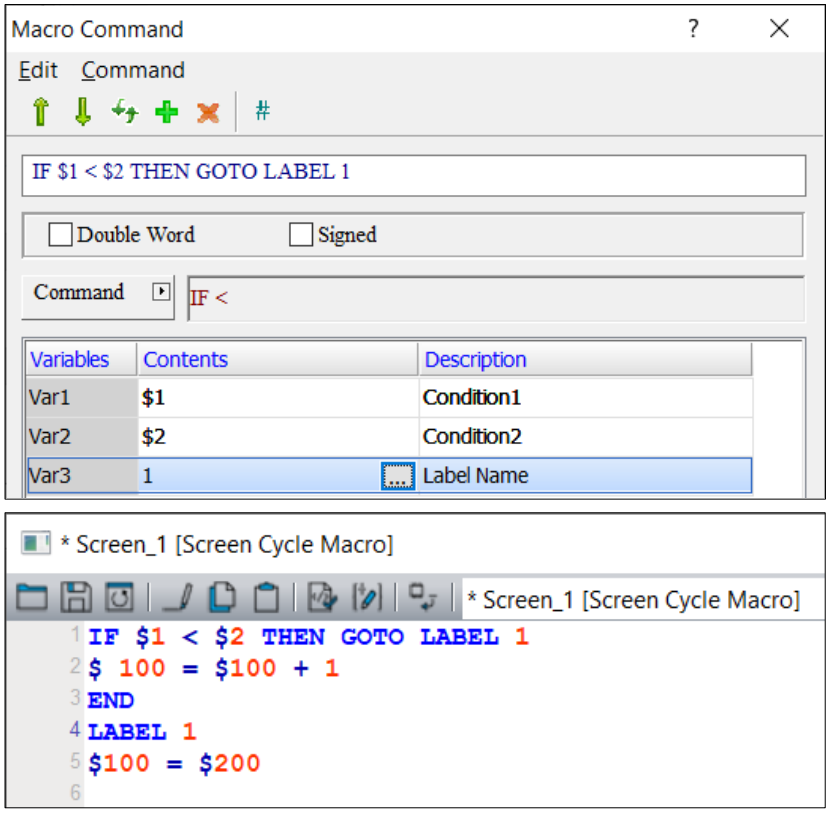
Example
 • If the value of $1 is greater than or equal to $2, execute LABEL1 ($100 = $200); if $1 is less than $2, then execute $100 = $100 + 1.

(5) IF <			
Expression	**Meaning of Variable**		**Note**
IF Var1 < Var2 THEN GOTO LABEL Var3 (W) IF Var1 < Var2 THEN GOTO LABEL Var3 (DW) IF Var1 < Var2 THEN GOTO LABEL Var3 (Signed W) IF Var1 < Var2 THEN GOTO LABEL Var3 (Signed DW)	Var1	Condition1	W: Word DW: Double Word Signed: Signed number
	Var2	Condition2	
	Var3	Label name	
	Description of Action		
	If Condition1 is less than Condition2, then GOTO executes LABEL Var3.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		V
Var2	V		V
Var3			V

Example

- Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.



- If the value of $1 is less than $2, execute LABEL1 ($100 = $200); If $1 is greater than or equal to $2, execute $100 = $100 + 1.

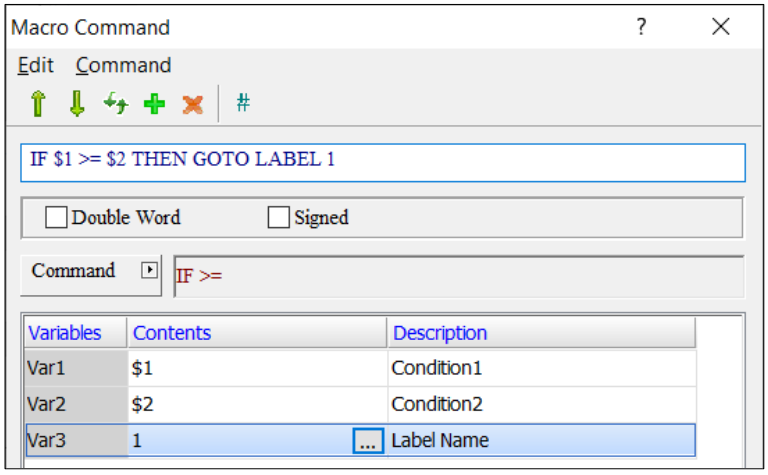
(6) IF <=

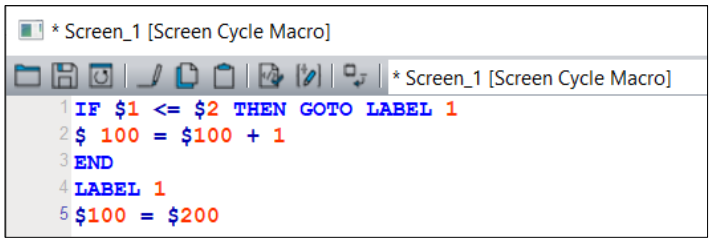
Expression	Meaning of Variable		Note
IF Var1 <= Var2 THEN GOTO LABEL Var3 (W) IF Var1 <= Var2 THEN GOTO LABEL Var3 (DW) IF Var1 <= Var2 THEN GOTO LABEL Var3 (Signed W) IF Var1 <= Var2 THEN GOTO LABEL Var3 (Signed DW)	Var1	Condition1	W: Word DW: Double Word Signed: Signed number
	Var2	Condition2	
	Var3	Label name	
	Description of Action		
	If Condition1 is less than or equal to Condition2, then GOTO executes LABEL Var3.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		V
Var2	V		V
Var3			V

Example

- Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.





- If the value of $1 is less than or equal to $2, execute LABEL1 ($100 = $200); if $1 is greater than $2, then execute $100 = $100 + 1.

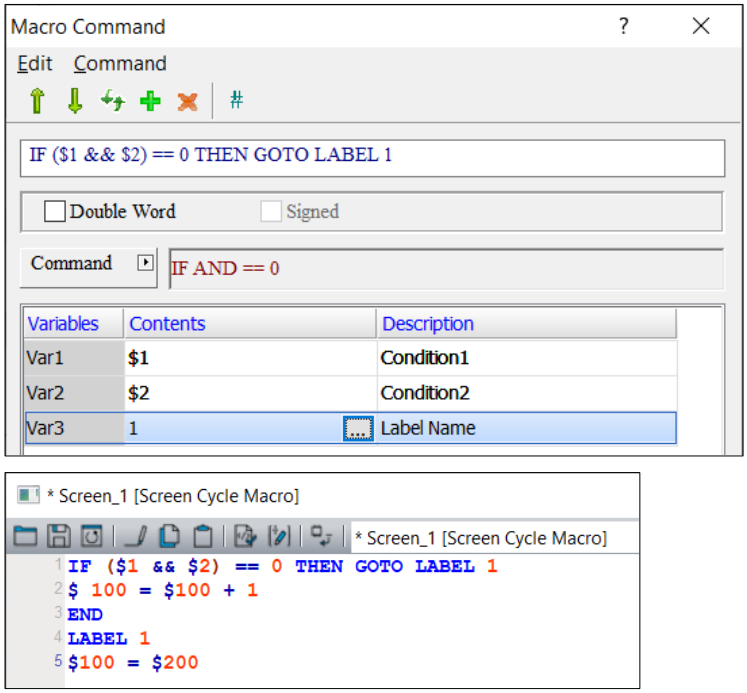
(7) IF AND == 0

Expression	Meaning of Variable		Note
IF (Var1 && Var2) == 0 THEN GOTO LABEL Var3 (W) IF (Var1 && Var2) == 0 THEN GOTO LABEL Var3 (DW)	Var1	Condition1	W: Word DW: Double Word
	Var2	Condition2	
	Var3	Label name	
	Description of Action		
	If the result of AND operation on Condition1 and Condition2 is 0, then GOTO executes LABEL Var3.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		V
Var2	V		V
Var3			V

Example

- Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.



- If the result of AND operation on $1 and $2 is 0, execute LABEL1 ($100 = $200); if the result of AND operation on $1 and $2 is not 0, then execute $100 = $100 + 1.

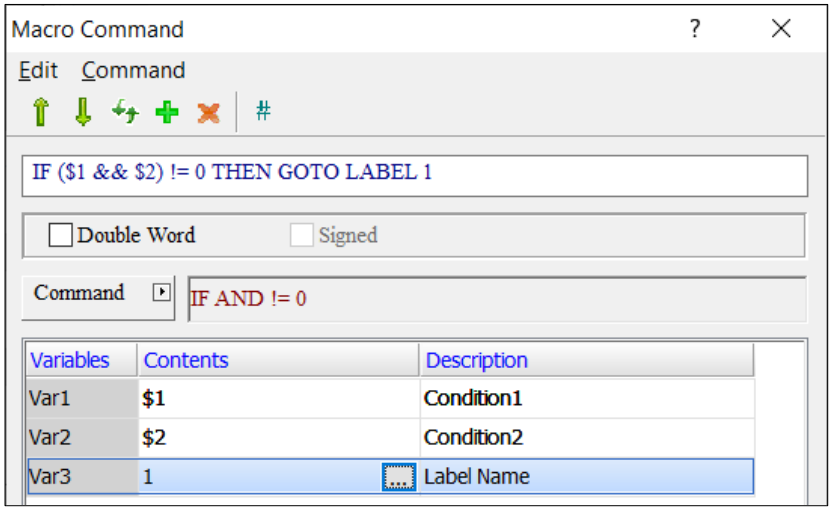
(8) IF AND != 0

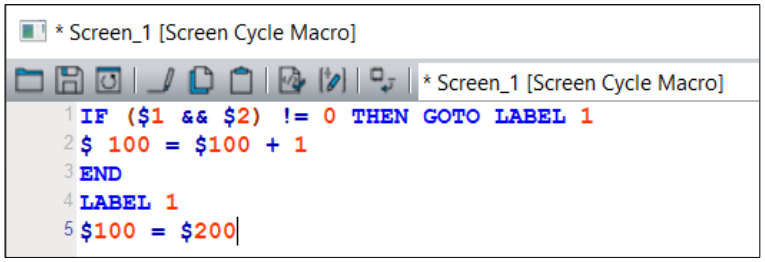
Expression	Meaning of Variable		Note
IF (Var1 && Var2) != 0 THEN GOTO LABEL Var3 (W) IF (Var1 && Var2) != 0 THEN GOTO LABEL Var3 (DW)	Var1	Condition1	W: Word DW: Double Word
	Var2	Condition2	
	Var3	Label name	
	Description of Action		
	If the result after AND operation on Condition1 and Condition2 is not 0, then GOTO executes LABEL Var3.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		V
Var2	V		V
Var3			V

Example

- Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.





- If the result of AND operation on $1 and $2 is not 0, execute LABEL1 ($100 = $200); if the result of AND operation on $1 and $2 is 0, then execute $100 = $100 + 1.

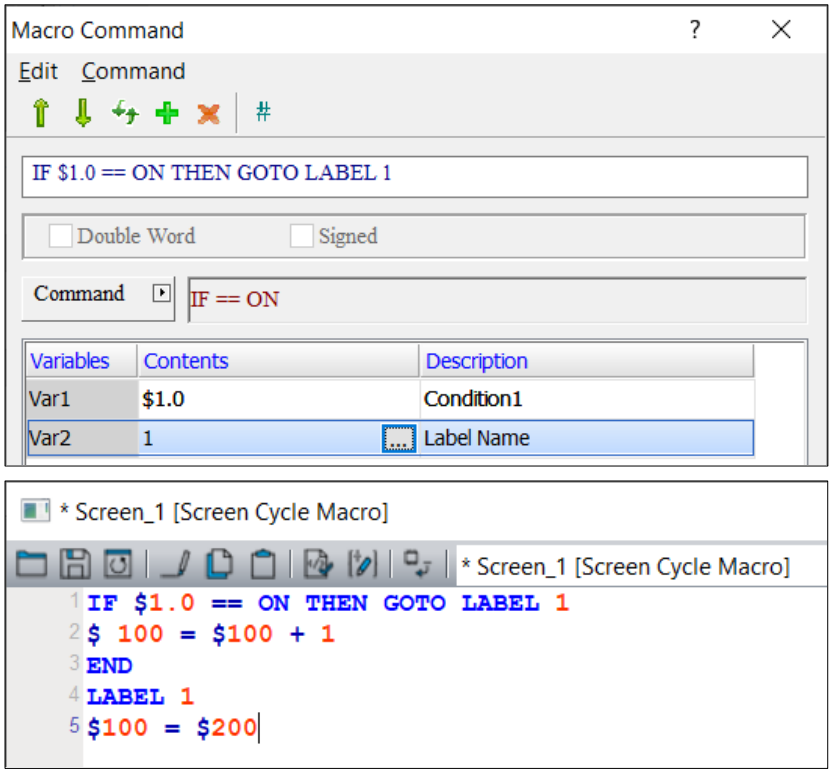
(9) IF == ON

Expression	Meaning of Variable		Note
IF Var1 == ON THEN GOTO LABEL Var2 (W)	Var1	Condition1	W: Word
	Var2	Label name	
	Description of Action		
	If Condition1 is ON, then GOTO executes LABEL Var2.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V (can only be Bit)		
Var2			V

Example

- Var1 is the Internal Memory address and Var2 is a constant.



- If $1.0 is ON, execute LABEL1 ($100 = $200); if $1.0 is not ON, then execute $100 = $100 + 1.

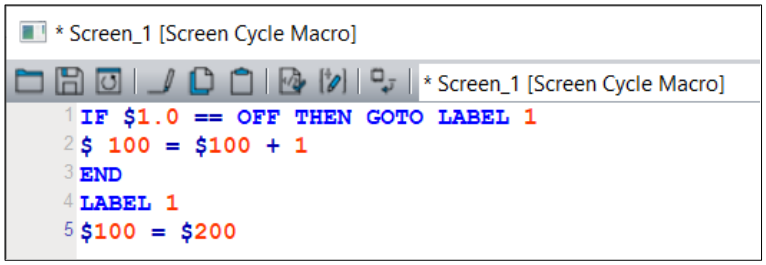
(10) IF == OFF

Expression	Meaning of Variable		Note
IF Var1 == OFF THEN GOTO LABEL Var2 (W)	Var1	Condition1	W: Word
	Var2	Label name	
	Description of Action		
	If Condition1 is OFF, then GOTO executes LABEL Var2.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V (can only be Bit)		
Var2			V

Example

- Var1 is the Internal Memory address and Var2 is a constant.



- If $1.0 is OFF, execute LABEL1 ($100 = $200); if $1.0 is not OFF, then execute $100 = $100 + 1.

(11) IFB == ON

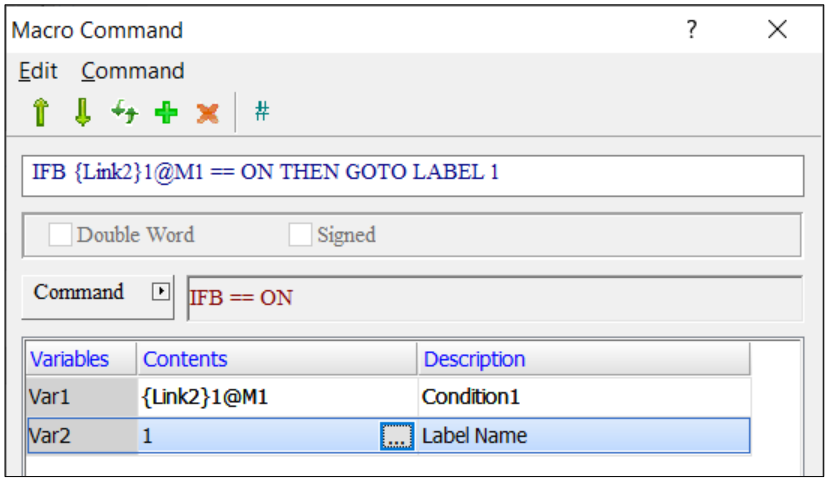
Expression	Meaning of Variable		Note
IFB Var1 == ON THEN GOTO LABEL Var2 (W)	Var1	Condition1	W: Word
	Var2	Label name	
	Description of Action		
	If Condition1 is ON, then GOTO executes LABEL Var2.		

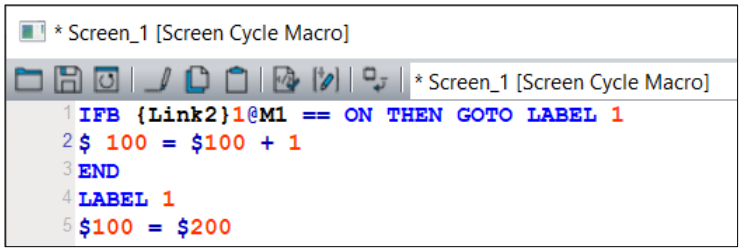
Note: For the IFB == ON command, its Bit address of Var1 can support the setting of external PLC Register.

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V (can only be Bit)	V (can only be Bit)	
Var2			V

Example

- Var1 is the address of PLC Register and Var2 is a constant.





- If M1 is ON, execute LABEL1 ($100 = $200); if M1 is not ON, then execute $100 = $100 + 1.

(12) IFB == OFF

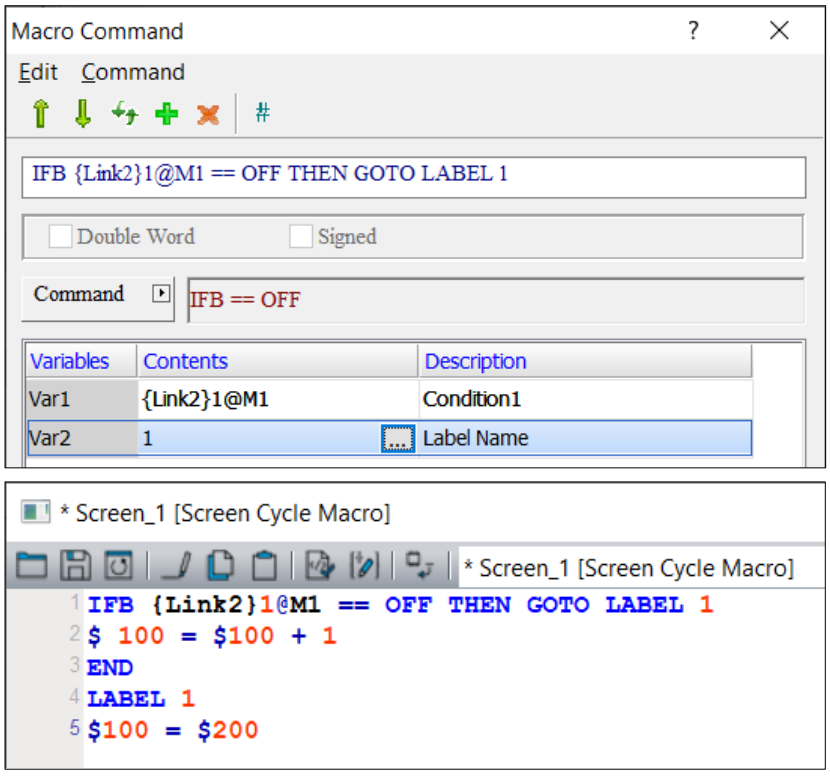
Expression	Meaning of Variable		Note
IFB Var1 == OFF THEN GOTO LABEL Var2 (W)	Var1	Condition1	W: Word
	Var2	Label name	
	Description of Action		
	If Condition 1 is OFF, then GOTO executes LABEL Var2.		

Note: For the IFB == OFF command, its Bit address of Var1 can support the setting of external PLC Register.

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V (can only be Bit)	V (can only be Bit)	
Var2			V

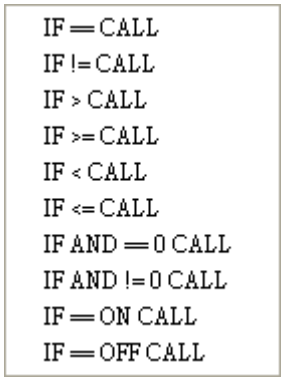
Example

- Var1 is the address of PLC Register and Var2 is a constant.



- If M1 is OFF, execute LABEL1 ($100 = $200); if M1 is not OFF, then execute $100 = $100 + 1.

- IF… THEN CALL (If…call submacro)



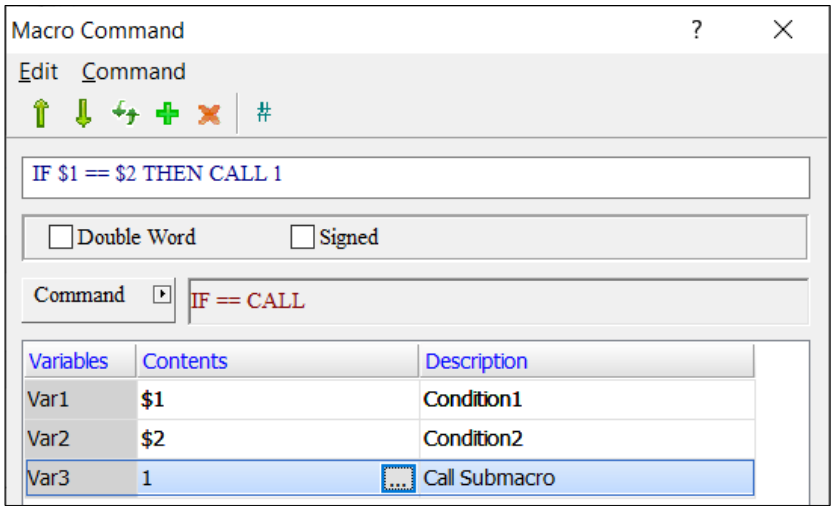
The following describes the 10 commands of the IF... THEN CALL macro.

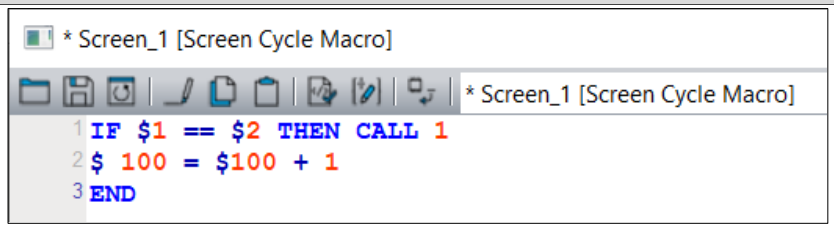
(1) IF ==			
Expression	**Meaning of Variable**		**Note**
IF Var1 == Var2 THEN CALL Var3 (W) IF Var1 == Var2 THEN CALL Var3 (DW) IF Var1 == Var2 THEN CALL Var3 (Signed W) IF Var1 == Var2 THEN CALL Var3 (Signed DW)	Var1	Condition1	W: Word DW: Double Word Signed: Signed number
	Var2	Condition2	
	Var3	Label name	
	Description of Action		
	If Condition1 equals Condition2, then call Submacro Var3.		

Variable	Type		
	Internal Memory	**PLC Register**	**Constant**
Var1	V		V
Var2	V		V
Var3			V

Example

- Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.



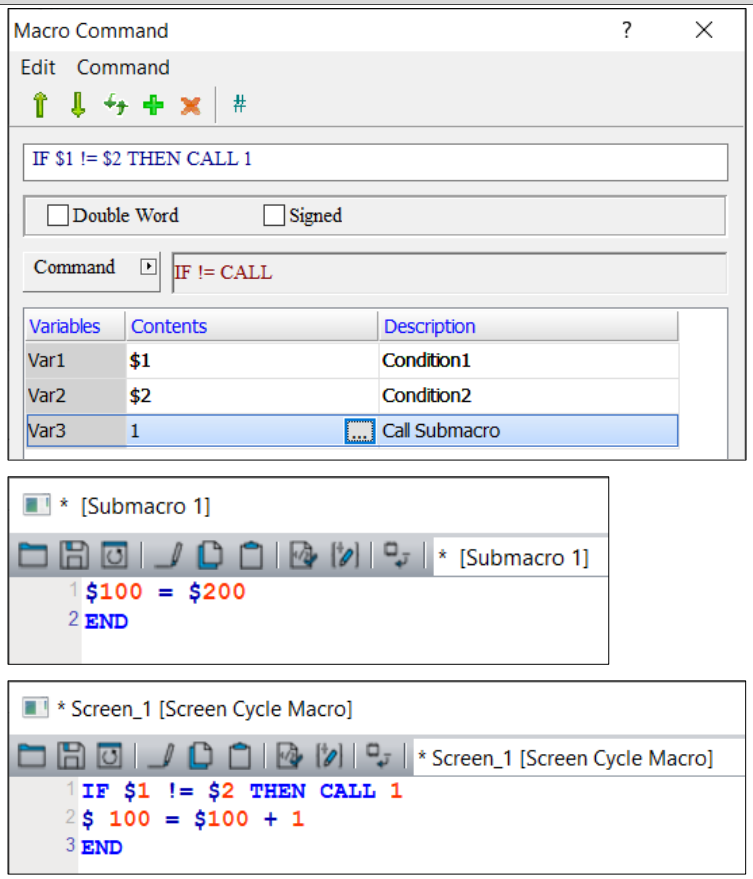
Example
 • If the value of $1 equals $2, call Submacro 1 ($100 = $200); if $1 does not equal $2, then execute $100 = $100 + 1.

(2) IF !=			
Expression	**Meaning of Variable**		**Note**
IF Var1 != Var2 THEN CALL Var3 (W) IF Var1 != Var2 THEN CALL Var3 (DW) IF Var1 != Var2 THEN CALL Var3 (Signed W) IF Var1 != Var2 THEN CALL Var3 (Signed DW)	Var1	Condition1	W: Word DW: Double Word Signed: Signed number
	Var2	Condition2	
	Var3	Label name	
	Description of Action		
	If Condition1 does not equal Condition2, call Submacro Var3.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		V
Var2	V		V
Var3			V

Example
• Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.

Example



- If the value of $1 equals $2, call Submacro 1 ($100 = $200); if $1 does not equal $2, then execute $100 = $100 + 1.

(3) IF >

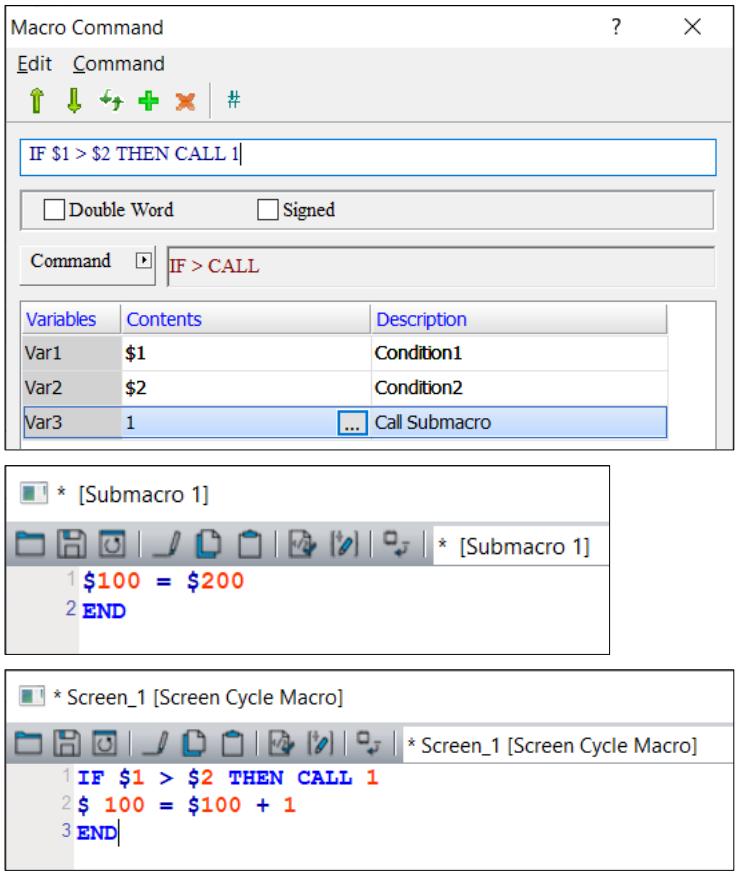
Expression	Meaning of Variable		Note
IF Var1 > Var2 THEN CALL Var3 (W) IF Var1 > Var2 THEN CALL Var3 (DW) IF Var1 > Var2 THEN CALL Var3 (Signed W) IF Var1 > Var2 THEN CALL Var3 (Signed DW)	Var1	Condition1	W: Word DW: Double Word Signed: Signed number
	Var2	Condition2	
	Var3	Label name	
	Description of Action		
	If Condition1 does not equal Condition2, call Submacro Var3.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		V
Var2	V		V

Variable	Type		
	Internal Memory	PLC Register	Constant
Var3			V

Example

- Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.



- If the value of $1 is greater than $2, call Submacro 1 ($100 = $200); if $1 is less than or equal to $2, then execute $100 = $100 + 1.

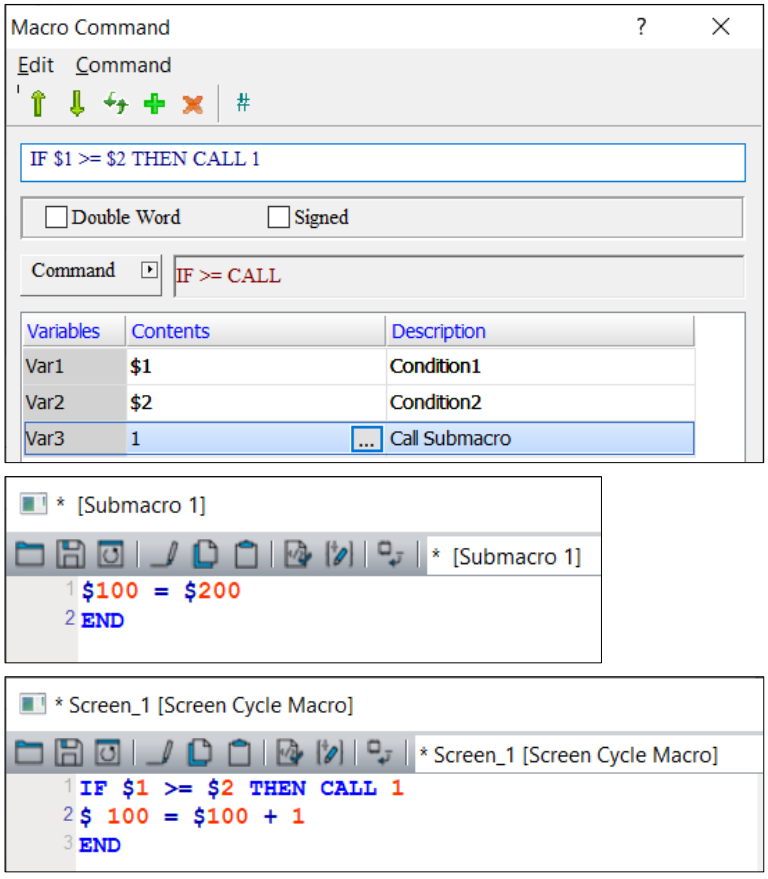
(4) IF >=

Expression	Meaning of Variable		Note
IF Var1 >= Var2 THEN CALL Var3 (W) IF Var1 >= Var2 THEN CALL Var3 (DW) IF Var1 >= Var2 THEN CALL Var3 (Signed W) IF Var1 >= Var2 THEN CALL Var3 (Signed DW)	Var1	Condition1	W: Word DW: Double Word Signed: Signed number
	Var2	Condition2	
	Var3	Label name	
	Description of Action		
	If Condition1 is greater than or equal to Condition2, call Submacro Var3.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		V
Var2	V		V
Var3			V

Example

- Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.



- If the value of $1 is greater than or equal to $2, call Submacro 1 ($100 = $200); if $1 is less than $2, then execute $100 = $100 + 1.

(5) IF <

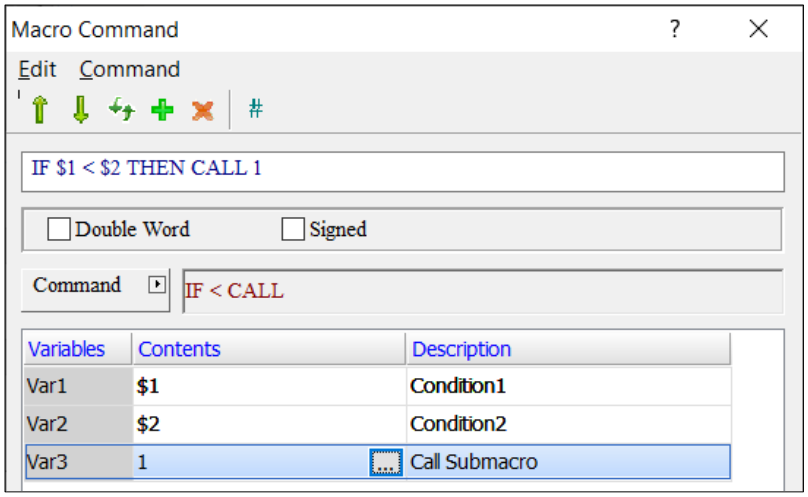
Expression	Meaning of Variable		Note
IF Var1 < Var2 THEN CALL Var3 (W) IF Var1 < Var2 THEN CALL Var3 (DW) IF Var1 < Var2 THEN CALL	Var1	Condition1	W: Word DW: Double Word Signed: Signed number
	Var2	Condition2	
	Var3	Label name	
	Description of Action		

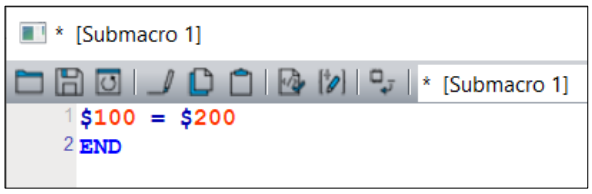
(5) IF <		
Expression	**Meaning of Variable**	**Note**
Var3 (Signed W) IF Var1 < Var2 THEN CALL Var3 (Signed DW)	If Condition1 is less than Condition2, call Submacro Var3.	

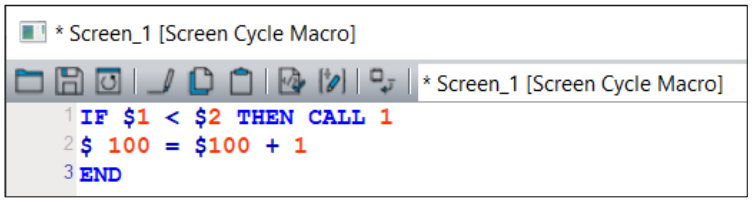
Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		V
Var2	V		V
Var3			V

Example

- Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.







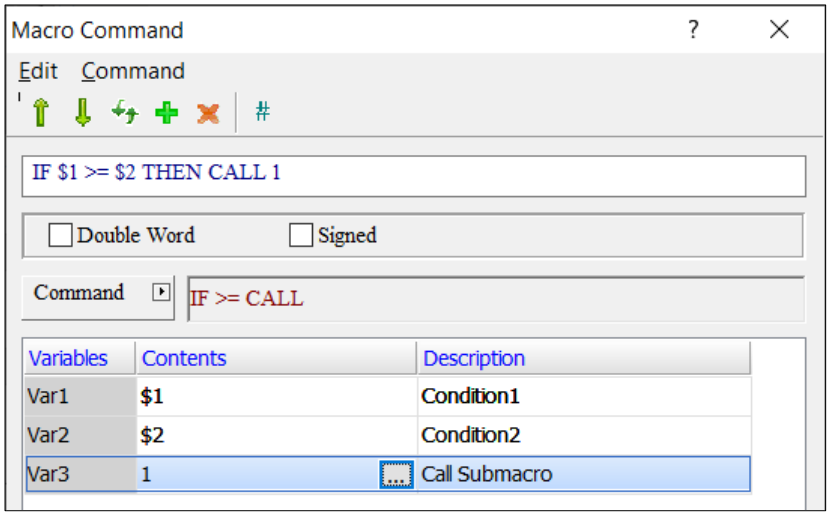
- If the value of $1 is less than $2, call Submacro 1 ($100 = $200); if $1 is greater than or equal to $2, then execute $100 = $100 + 1.

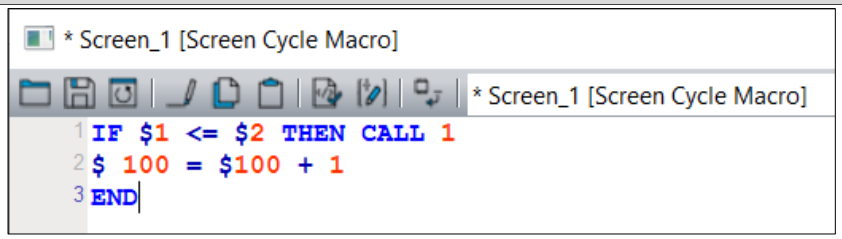
(6) IF <=			
Expression	**Meaning of Variable**		**Note**
IF Var1 <= Var2 THEN CALL Var3 (W) IF Var1 <= Var2 THEN CALL Var3 (DW) IF Var1 <= Var2 THEN CALL Var3 (Signed W) IF Var1 <= Var2 THEN CALL Var3 (Signed DW)	Var1	Condition1	W: Word DW: Double Word Signed: Signed number
	Var2	Condition2	
	Var3	Label name	
	Description of Action		
	If Condition1 is less than or equal to Condition2, call Submacro Var3.		

Variable	Type		
	Internal Memory	**PLC Register**	**Constant**
Var1	V		V
Var2	V		V
Var3			V

Example

- Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.



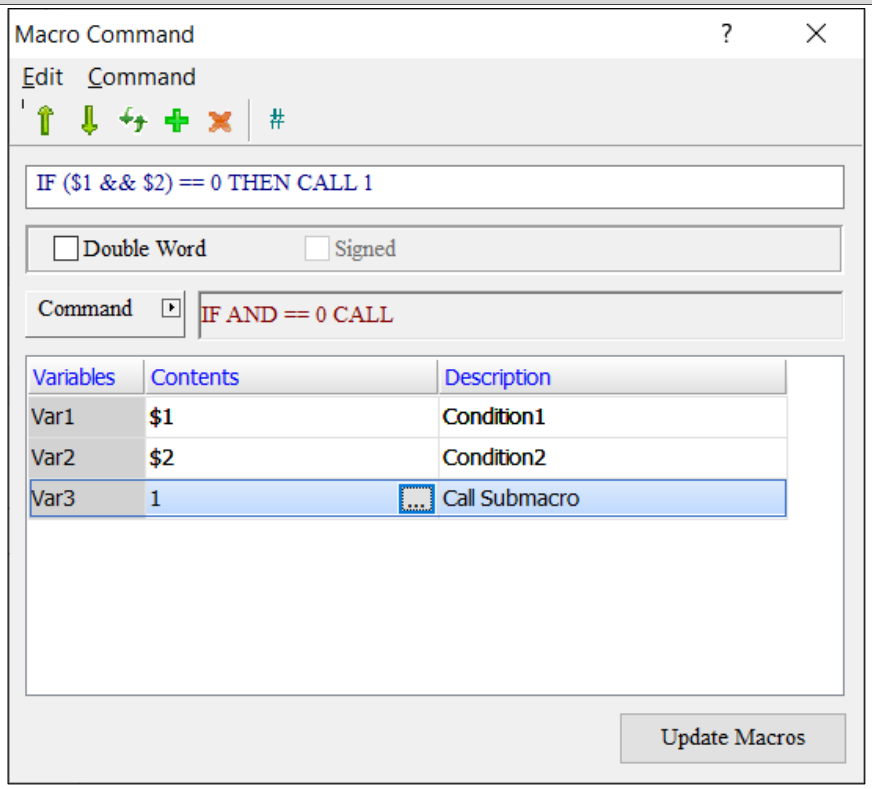
Example
 • If the value of $1 is less than or equal to $2, call Submacro 1 ($100 = $200); if $1 is greater than $2, then execute $100 = $100 + 1.

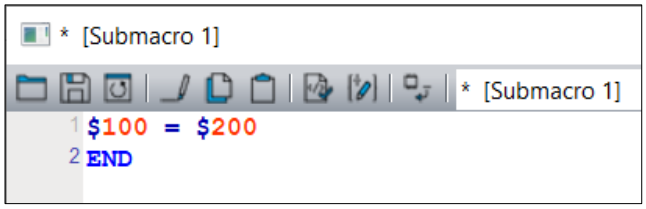
(7) IF AND == 0			
Expression	**Meaning of Variable**		**Note**
IF (Var1 && Var2) == 0 THEN CALL Var3 (W) IF (Var1 && Var2) == 0 THEN CALL Var3 (DW)	Var1	Condition1	W: Word DW: Double Word
	Var2	Condition2	
	Var3	Label name	
	Description of Action		
	If the result of the AND operation on Condition1 and Condition2 is 0, call Submacro Var3.		

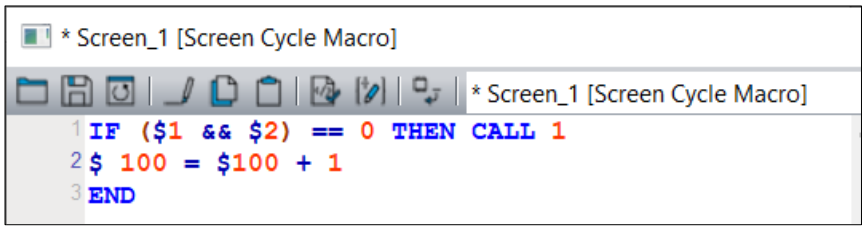
Variable	Type		
	Internal Memory	**PLC Register**	**Constant**
Var1	V		V
Var2	V		V
Var3			V

Example
• Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.

Example







- If the result of the AND operation on $1 and $2 is 0, call Submacro 1 ($100 = $200); if the result of the AND operation on $1 and $2 is not 0, then execute $100 = $100 + 1.

(8) IF AND != 0

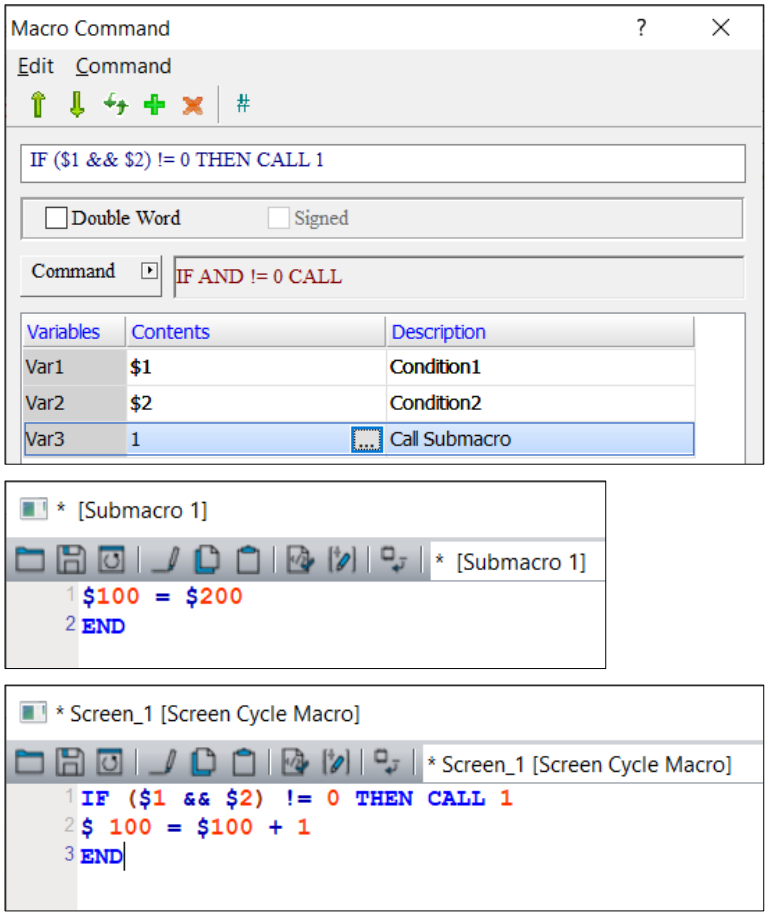
Expression	Meaning of Variable		Note
IF (Var1 && Var2) != 0 THEN CALL Var3 (W) IF (Var1 && Var2) != 0 THEN CALL Var3 (DW)	Var1	Condition1	W: Word DW: Double Word
	Var2	Condition2	
	Var3	Label name	
	Description of Action		

(8) IF AND != 0		
Expression	**Meaning of Variable**	**Note**
	If the result of the AND operation on Condition1 and Condition2 is not 0, call Submacro Var3.	

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		V
Var2	V		V
Var3			V

Example

- Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.



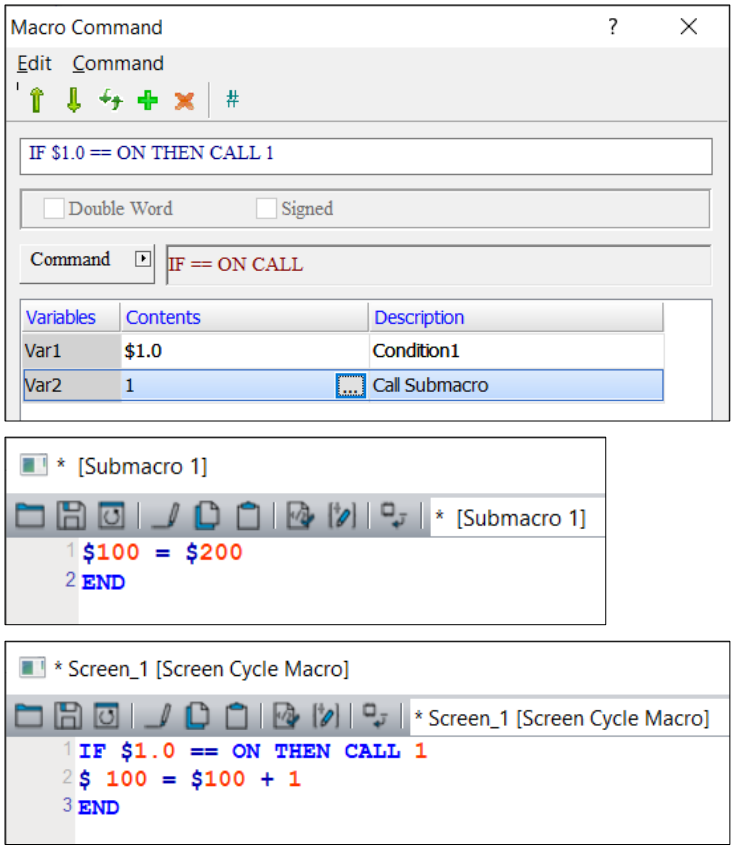
- If the result of the AND operation on $1 and $2 is not 0, call Submacro 1 ($100 = $200); if the result of the AND operation on $1 and $2 is 0, then execute $100 = $100 + 1.

(9) IF == ON			
Expression	**Meaning of Variable**		**Note**
IF Var1 == ON THEN CALL Var2 (W)	Var1	Condition1	W: Word
	Var2	Label name	
	Description of Action		
	If Condition1 is ON, call Submacro Var2.		

Variable	**Type**		
	Internal Memory	**PLC Register**	**Constant**
Var1	V (can only be bit)		
Var2			V

Example

- Var1 is the Internal Memory address and Var2 is a constant.



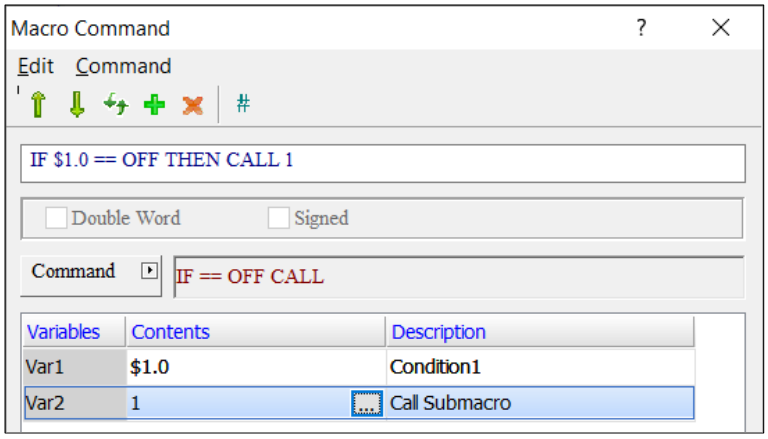
- If $1.0 is ON, call Submacro 1 ($100 = $200); if $1.0 is not ON, then execute $100 = $100 + 1.

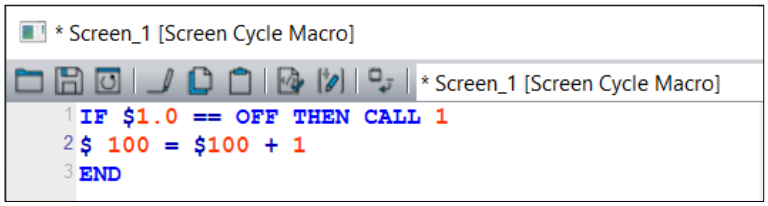
(10) IF == OFF			
Expression	**Meaning of Variable**		**Note**
IF Var1 == OFF THEN CALL Var2 (W)	Var1	Condition1	W: Word
	Var2	Label name	
	Description of Action		
	If Condition1 is OFF, call Submacro Var2.		

Variable	Type		
	Internal Memory	**PLC Register**	**Constant**
Var1	V (can only be Bit)		
Var2			V

Example

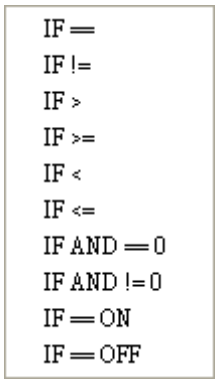
- Var1 is the Internal Memory address and Var2 is a constant.





- If $1.0 is OFF, call Submacro 1 ($100 = $200); if $1.0 is not OFF, then execute $100 = $100 + 1.

- **IF… (If…)**



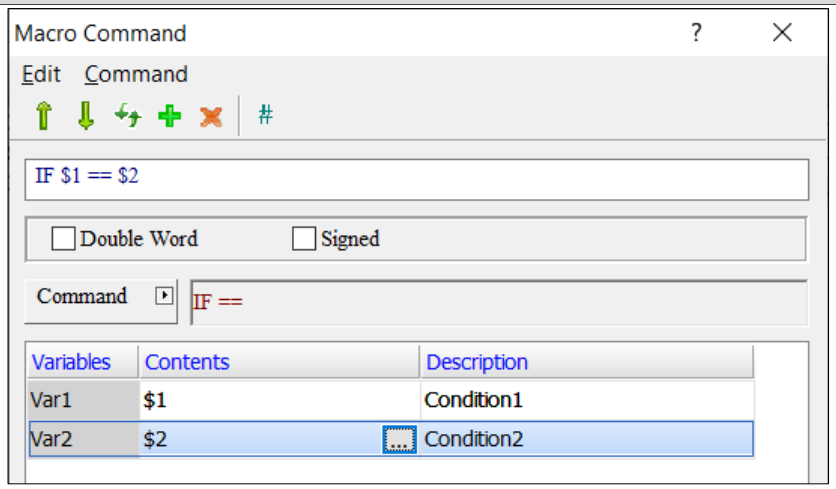
The following introduces 10 commands of the IF… macro.

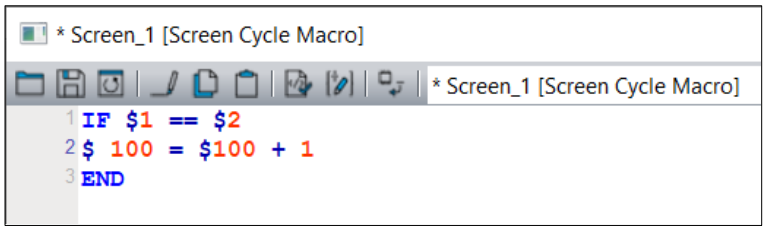
(1) IF ==			
Expression	**Meaning of Variable**		**Note**
IF Var1 == Var2 (W) IF Var1 == Var2 (DW) IF Var1 == Var2 (Signed W) IF Var1 == Var2 (Signed DW)	Var1	Condition1	W: Word DW: Double Word Signed: Signed number
	Var2	Condition2	
	Description of Action		
	If Condition1 equals Condition2, execute...		
Note: The IF macro command must be used with ENDIF, otherwise an error message will pop up while compiling.			

Variable	**Type**		
	Internal Memory	**PLC Register**	**Constant**
Var1	V		V
Var2	V		V
Var3			V

Example
• Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.

Example





- If the value of $1 equals $2, execute $100 = $100 + 1; if $1 is greater than or less than $2, then $100 = $100 + 1 will not be executed.

(2) IF !=

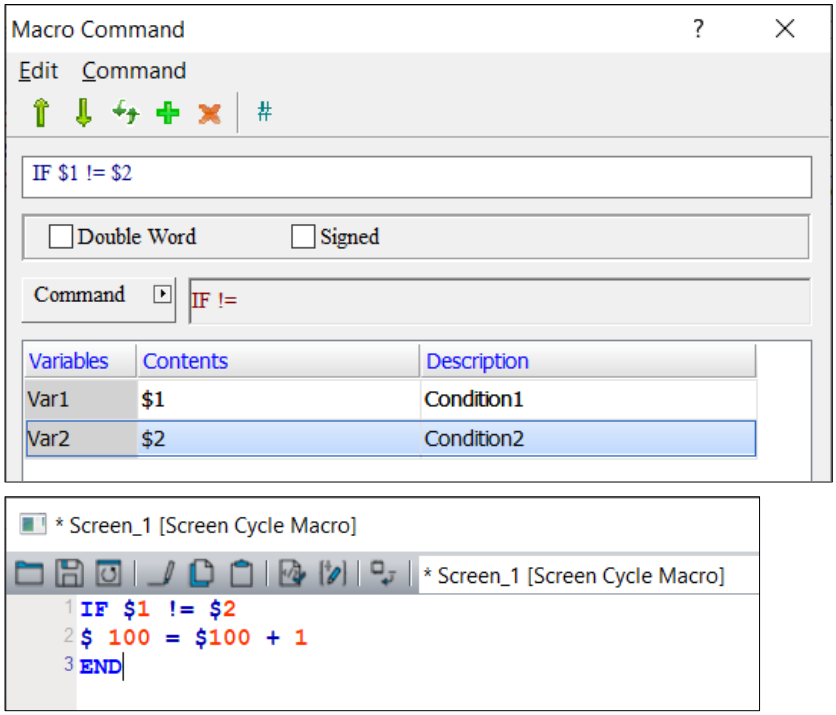
Expression	Meaning of Variable		Note
IF Var1 != Var2 (W) IF Var1 != Var2 (DW) IF Var1 != Var2 (Signed W) IF Var1 != Var2 (Signed DW)	Var1	Condition1	W: Word DW: Double Word Signed: Signed number
	Var2	Condition2	
	Description of Action		
	If Condition1 does not equal Condition2, execute…		

Note: The IF macro command must be used with ENDIF, otherwise an error message will pop up while compiling.

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		V
Var2	V		V
Var3			V

Example

- Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.

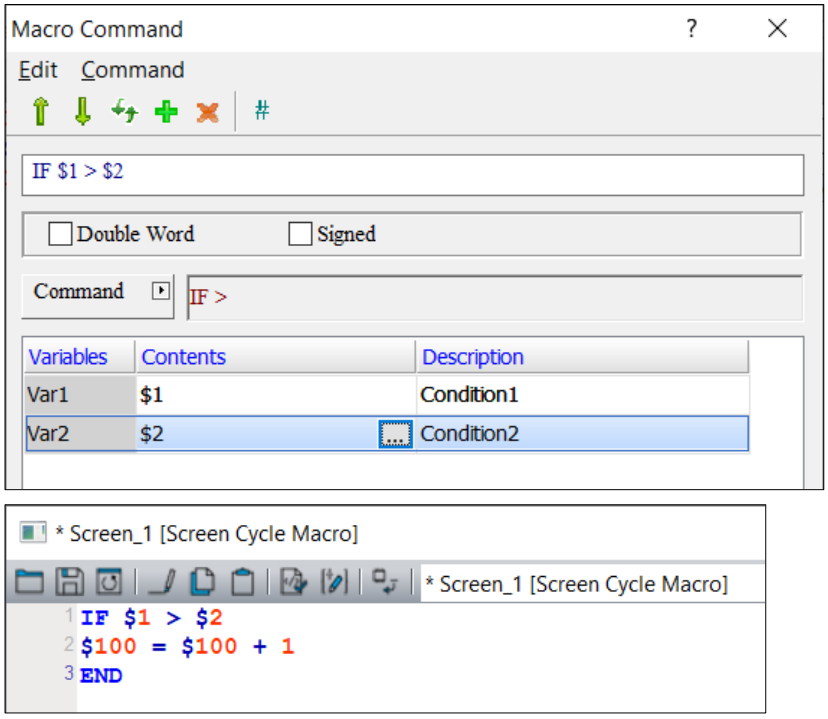


- If the value of $1 does not equal $2, execute $100 = $100 + 1; if $1 is greater than or less than $2, then $100 = $100 + 1 will not be executed.

(3) IF >

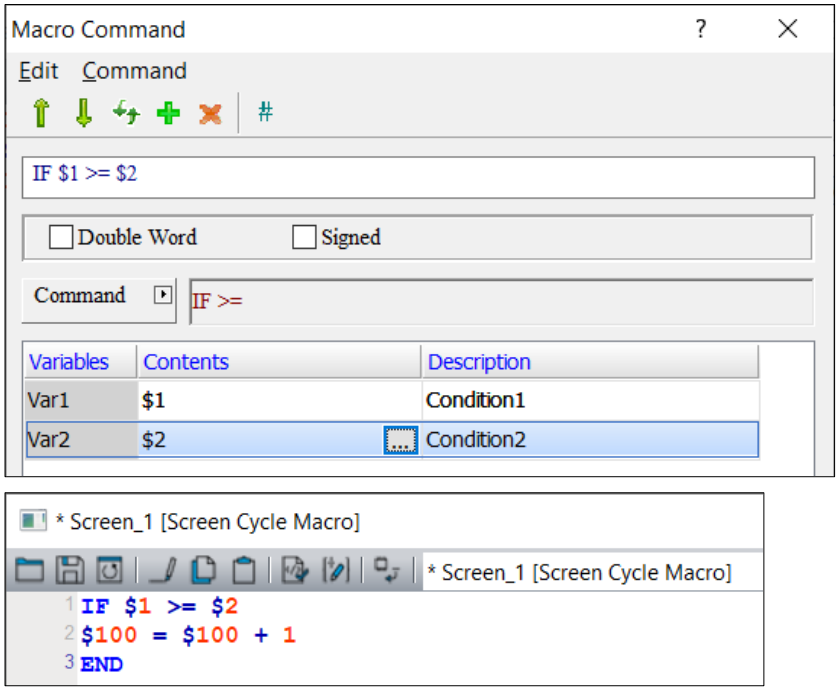
Expression	Meaning of Variable		Note
IF Var1 > Var2 (W) IF Var1 > Var2 (DW) IF Var1 > Var2 (Signed W) IF Var1 > Var2 (Signed DW)	Var1	Condition1	W: Word DW: Double Word Signed: Signed number
	Var2	Condition2	
	Description of Action		
	If Condition1 is greater than Condition2, execute…		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		V
Var2	V		V
Var3			V

Example
• Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.    • If the value of $1 is greater than $2, execute $100 = $100 + 1; if $1 is less than or equal to $2, then $100 = $100 + 1 will not be executed.

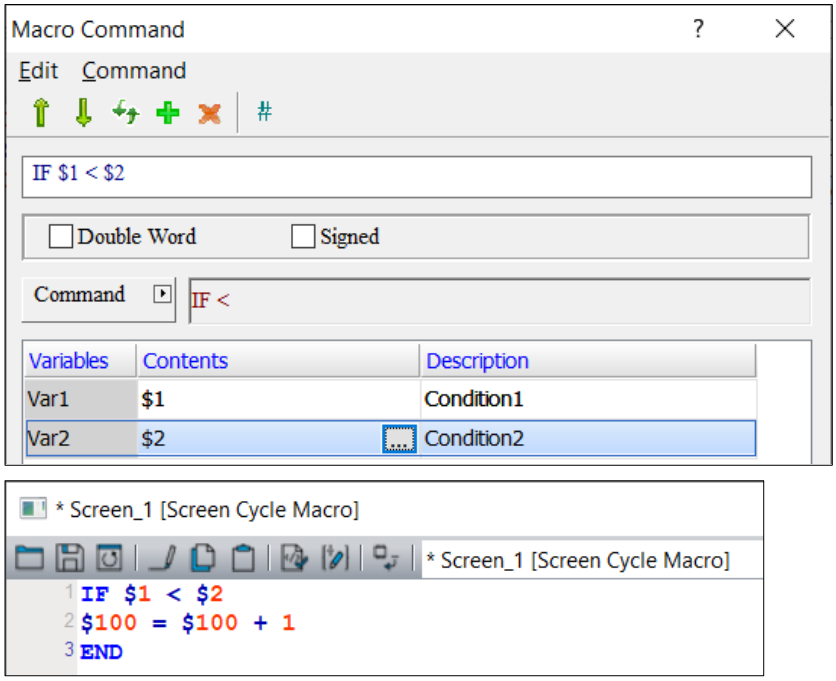
(4) IF >=			
Expression	**Meaning of Variable**		**Note**
IF Var1 >= Var2 (W) IF Var1 >= Var2 (DW) IF Var1 >= Var2 (Signed W) IF Var1 >= Var2 (Signed DW)	Var1	Condition1	W: Word DW: Double Word Signed: Signed number
	Var2	Condition2	
	Description of Action		
	If Condition1 is greater than or equal to Condition2, execute…		
Note: The IF macro command must be used with ENDIF, otherwise an error message will pop up while compiling.			

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		V
Var2	V		V
Var3			V

Example
• Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.    • If the value of $1 is greater than or equal to $2, execute $100 = $100 + 1; if $1 is less than $2, then $100 = $100 + 1 will not be executed.

(5) IF <			
Expression	**Meaning of Variable**		**Note**
IF Var1 < Var2 (W) IF Var1 < Var2 (DW) IF Var1 < Var2 (Signed W) IF Var1 < Var2 (Signed DW)	Var1	Condition1	W: Word DW: Double Word Signed: Signed number
	Var2	Condition2	
	Description of Action		
	If Condition1 is less than Condition2, execute…		
Note: The IF macro command must be used with ENDIF, otherwise an error message will pop up while compiling.			

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		V
Var2	V		V
Var3			V

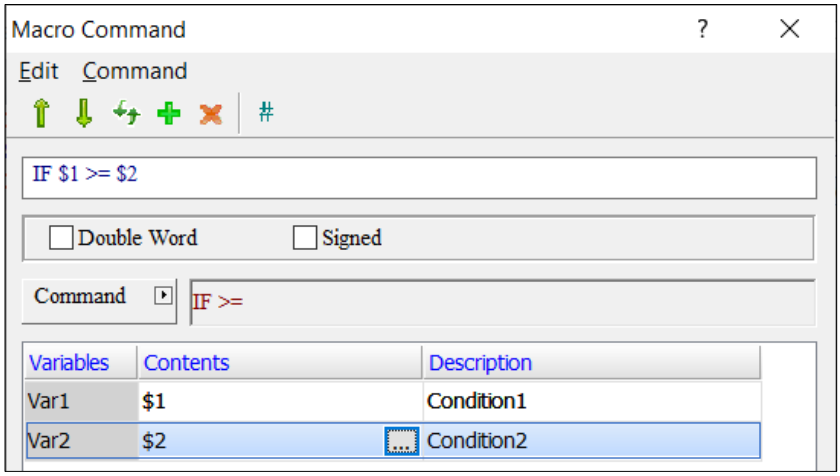
Example
• Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.    • If the value of $1 is less than $2, execute $100 = $100 + 1; if $1 is greater than or equal to $2, then $100 = $100 + 1 will not be executed.

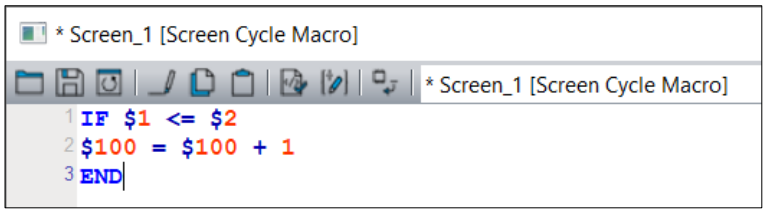
(6) IF <=			
Expression	**Meaning of Variable**		**Note**
IF Var1 <= Var2 (W) IF Var1 <= Var2 (DW) IF Var1 <= Var2 (Signed W) IF Var1 <= Var2 (Signed DW)	Var1	Condition1	W: Word DW: Double Word Signed: Signed number
	Var2	Condition2	
	Description of Action		
	If Condition1 is less than or equal to Condition2, execute…		
Note: IF macro command must be used with ENDIF, otherwise an error message will pop up while compiling.			

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		V
Var2	V		V
Var3			V

Example

- Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.





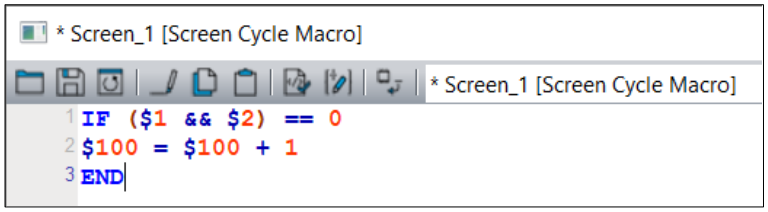
- If the value of $1 is less than or equal to $2, execute $100 = $100 + 1; if $1 is greater than $2, then $100 = $100 + 1 will not be executed.

(7) IF AND == 0			
Expression	**Meaning of Variable**		**Note**
IF (Var1 && Var2) == 0 (W) IF (Var1 && Var2) == 0 (DW)	Var1	Condition1	W: Word DW: Double Word
	Var2	Condition2	
	Description of Action		
	If the result of the AND operation on Condition1 and Condition2 is 0, execute…		

Variable	Type		
	Internal Memory	**PLC Register**	**Constant**
Var1	V		V
Var2	V		V

Example

- Var1 and Var2 are Internal Memory addresses.



- If the result of the AND operation on $1 and $2 is 0, execute $100 = $100 + 1; if the result of the AND operation on $1 and $2 is not 0, then $100 = $100 + 1 will not be executed.

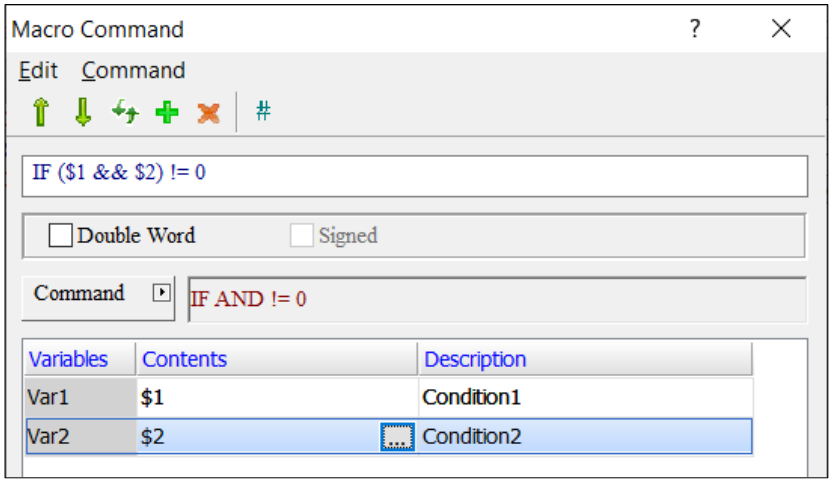
(8) IF AND != 0

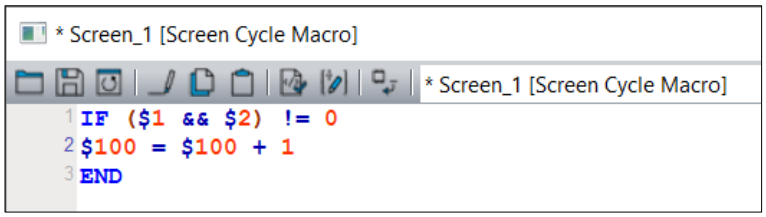
Expression	Meaning of Variable		Note
IF (Var1 && Var2) != 0 (W) IF (Var1 && Var2) != 0 (DW)	Var1	Condition1	W: Word DW: Double Word
	Var2	Condition2	
	Description of Action		
	If the result of the AND operation on Condition1 and Condition2 is not 0, execute…		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		V
Var2	V		V

Example

- Var1 and Var2 are Internal Memory addresses.

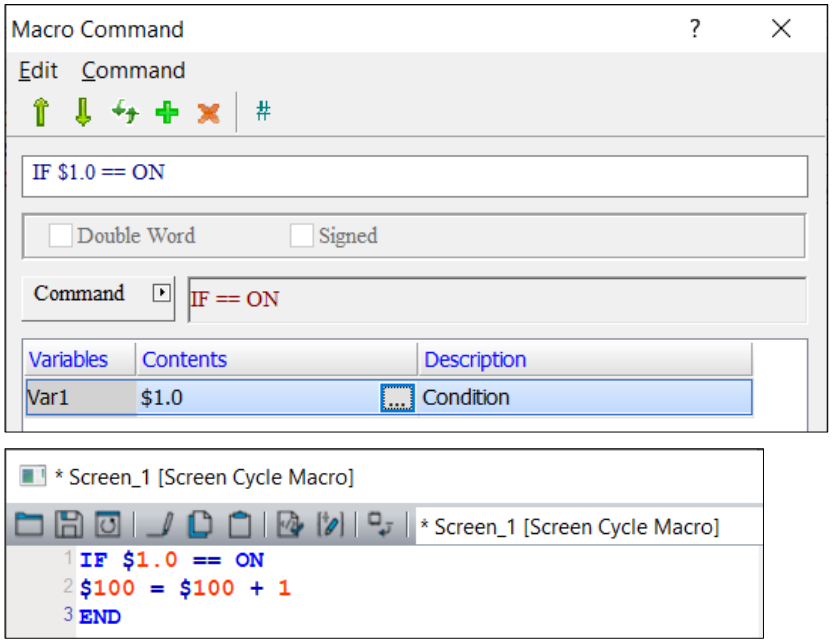




- If the result of the AND operation on $1 and $2 is not 0, execute $100 = $100 + 1; if the result of the AND operation on $1 and $2 is 0, then $100 = $100 + 1 will not be executed.

(9) IF == ON			
Expression	**Meaning of Variable**		**Note**
IF Var1 == ON (W)	Var1	Condition1	W: Word
	Description of Action		
	If Condition1 is ON, execute…		

Variable	**Type**		
	Internal Memory	**PLC Register**	**Constant**
Var1	V (can only be Bit)		

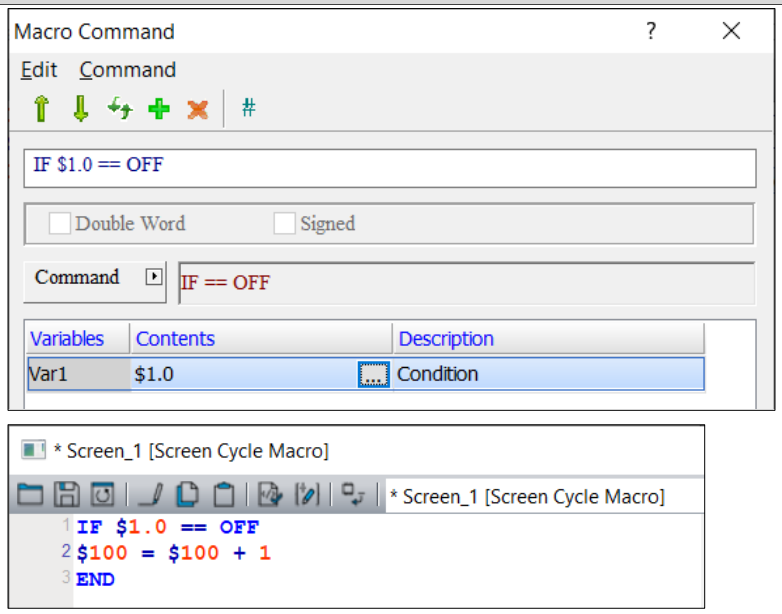
Example
• Var1 is the Internal Memory address.    • If $1.0 is ON, execute $100 = $100 + 1; if $1.0 is not ON, then $100 = $100 + 1 will not be executed.

(10) IF == OFF			
Expression	**Meaning of Variable**		**Note**
IF Var1 == OFF (W)	Var1	Condition1	W: Word
	Description of Action		
	If Condition1 is OFF, execute…		

Variable	Type		
	Internal Memory	**PLC Register**	**Constant**
Var1	V (can only be Bit)		

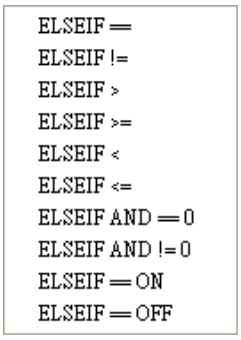
Example
• Var1 is the Internal Memory address.

Example



- If $1.0 is OFF, execute $100 = $100 + 1; if $1.0 is not OFF, then $100 = $100 + 1 will not be executed.

- ELSEIF… (else if…)



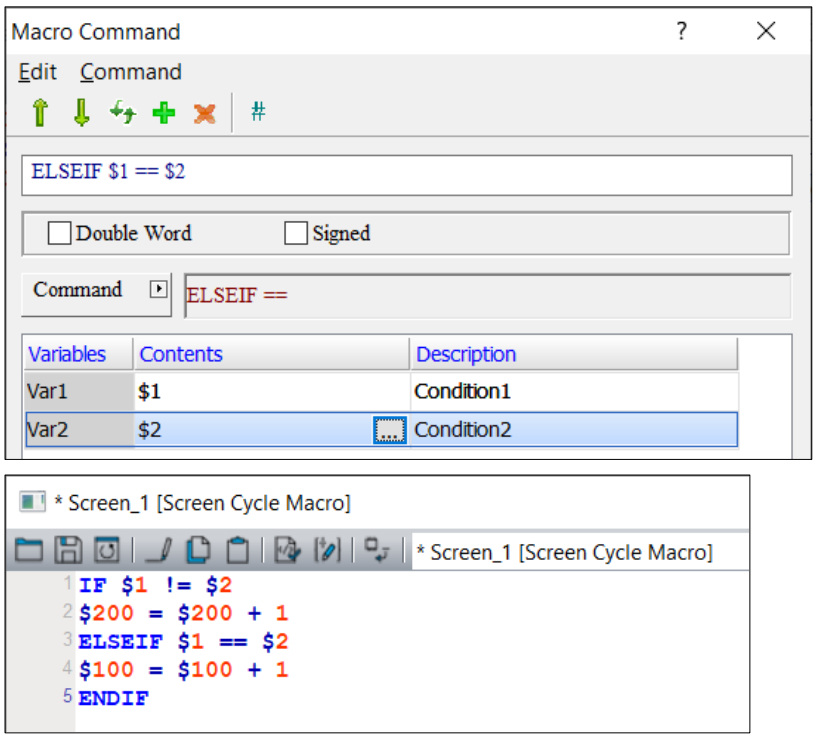
The following introduces 10 commands of ELSEIF… macro.

(1) ELSEIF ==			
Expression	**Meaning of Variable**		**Note**
ELSEIF Var1 == Var2 (W) ELSEIF Var1 == Var2 (DW) ELSEIF Var1 == Var2 (Signed W) ELSEIF Var1 == Var2 (Signed DW)	Var 1	Condition1	W: Word DW: Double Word Signed: Signed number
	Var 2	Condition2	
	Description of Action		
	Else if Condition1 equals Condition2, execute...		
Note: ELSEIF macro command must be used with IF...ENDIF, otherwise an error message will pop up while compiling.			

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		V
Var2	V		V

Example

- Var1 and Var2 are Internal Memory addresses.



- If the value of $1 does not equal $2, execute $200 = $200 + 1; else if $1 equals $2, then execute $100 = $100 + 1.

(2) ELSEIF !=

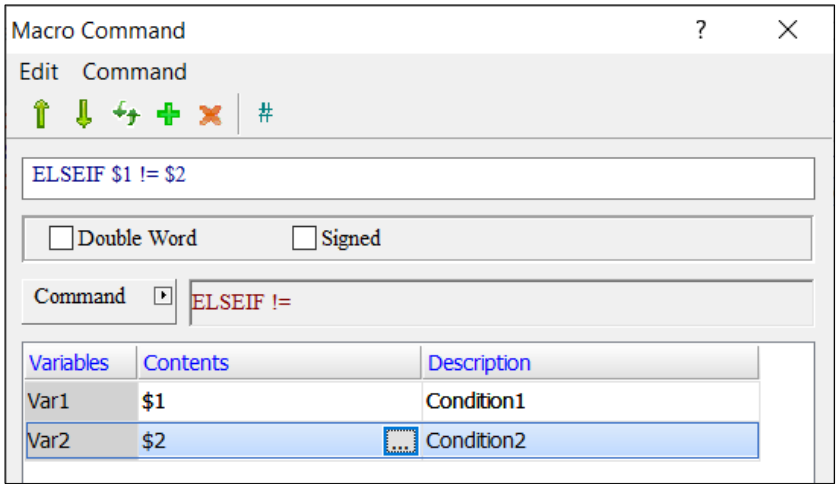
Expression	Meaning of Variable		Note
ELSEIF Var1 != Var2 (W) ELSEIF Var1 != Var2 (DW) ELSEIF Var1 != Var2 (Signed W) ELSEIF Var1 != Var2 (Signed DW)	Var1	Condition1	W: Word DW: Double Word Signed: Signed number
	Var2	Condition2	
	Description of Action		
	Else if Condition1 does not equal Condition2, execute…		

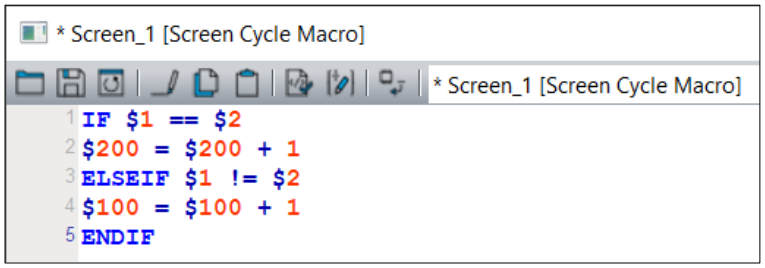
Note: ELSEIF macro command must be used with IF...ENDIF, otherwise an error message will pop up while compiling.

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		V
Var2	V		V

Example

- Var1 and Var2 are Internal Memory addresses.





- If the value of $1 equals $2, execute $200 = $200 + 1; else if $1 does not equal $2, then execute $100 = $100 + 1.

(3) ELSEIF >

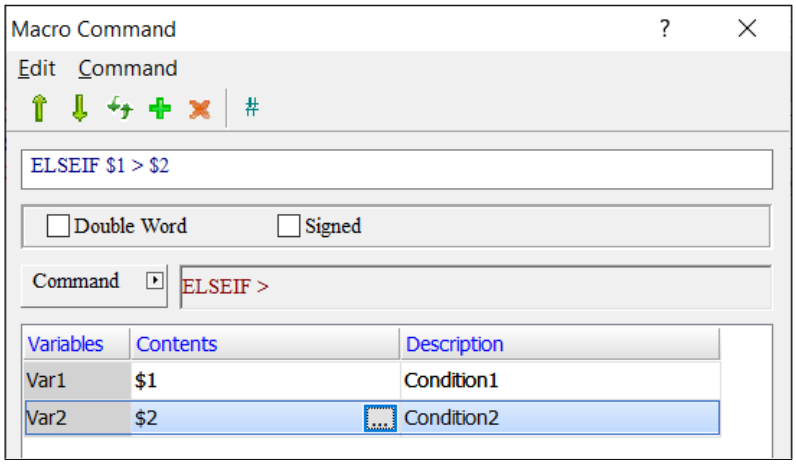
Expression	Meaning of Variable		Note
ELSEIF Var1 > Var2 (W) ELSEIF Var1 > Var2 (DW) ELSEIF Var1 > Var2 (Signed W) ELSEIF Var1 > Var2 (Signed DW)	Var1	Condition1	W: Word DW: Double Word Signed: Signed number
	Var2	Condition2	
	Description of Action		
	Else if Condition1 is greater than Condition2, execute...		

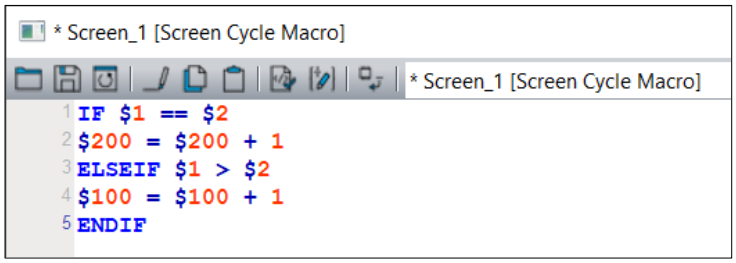
Note: ELSEIF macro command must be used with IF...ENDIF, otherwise an error message will pop up while compiling.

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		V
Var2	V		V

Example

- Var1 and Var2 are Internal Memory addresses.





- If the value of $1 equals $2, execute $200 = $200 + 1; else if $1 is greater than $2, then execute $100 = $100 + 1.

(4) ELSEIF >=

Expression	Meaning of Variable		Note
ELSEIF Var1 >= Var2 (W) ELSEIF Var1 >= Var2 (DW) ELSEIF Var1 >= Var2 (Signed W) ELSEIF Var1 >= Var2 (Signed DW)	Var1	Condition1	W: Word DW: Double Word Signed: Signed number
	Var2	Condition2	
	Description of Action		
	Else if Condition1 is greater than or equal to Condition2, execute…		

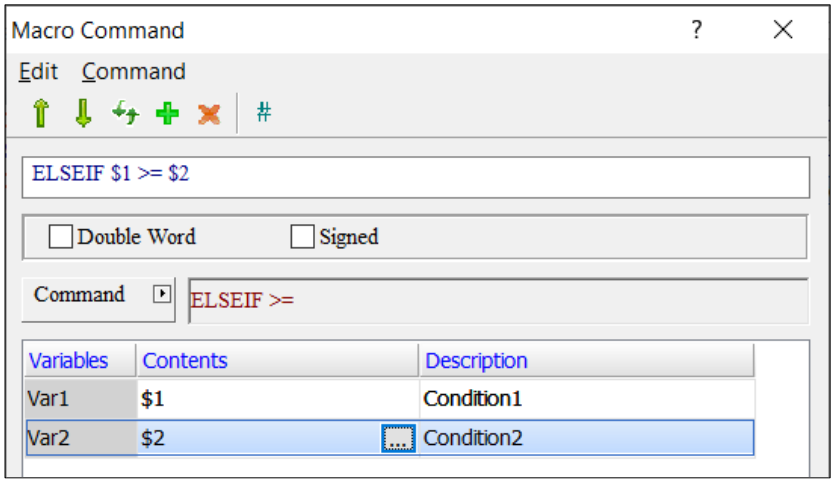
(4) ELSEIF >=

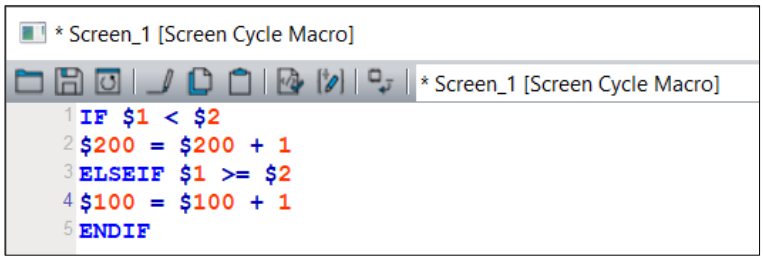
Note: ELSEIF macro command must be used with IF...ENDIF, otherwise an error message will pop up while compiling.

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		V
Var2	V		V

Example

- Var1 and Var2 are Internal Memory addresses.





- If the value of $1 is less than $2, execute $200 = $200 + 1; else if $1 is greater than or equal to $2, then execute $100 = $100 + 1.

(5) ELSEIF <

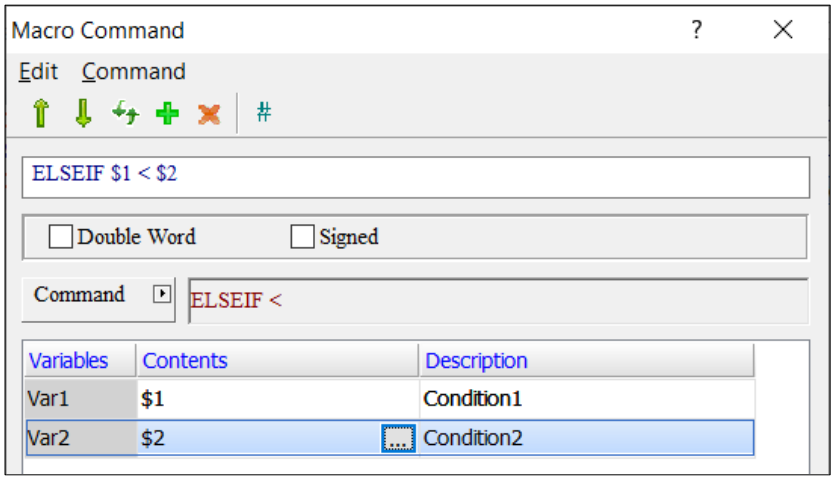
Expression	Meaning of Variable		Note
ELSEIF Var1 < Var2 (W) ELSEIF Var1 < Var2 (DW) ELSEIF Var1 < Var2 (Signed W) ELSEIF Var1 < Var2 (Signed	Var1	Condition1	W: Word DW: Double Word Signed: Signed number
	Var2	Condition2	
	Description of Action		

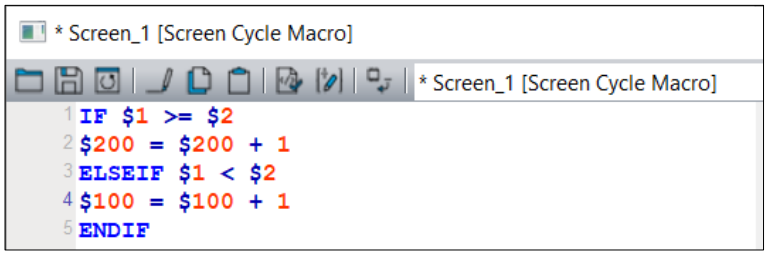
(5) ELSEIF <		
Expression	**Meaning of Variable**	**Note**
DW)	Else if Condition1 is less than Condition2, execute…	
Note: ELSEIF macro command must be used with IF...ENDIF, otherwise an error message will pop up while compiling.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		V
Var2	V		V

Example

- Var1 and Var2 are Internal Memory addresses.





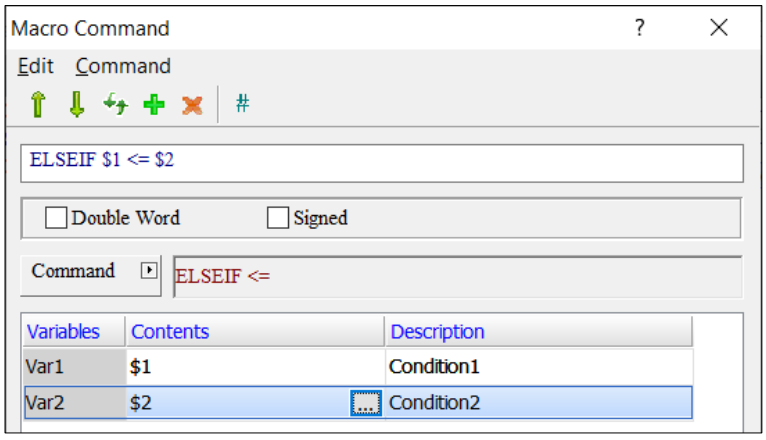
- If the value of $1 is greater than or equal to $2, execute $200 = $200 + 1; else if $1 is less than $2, then execute $100 = $100 + 1.

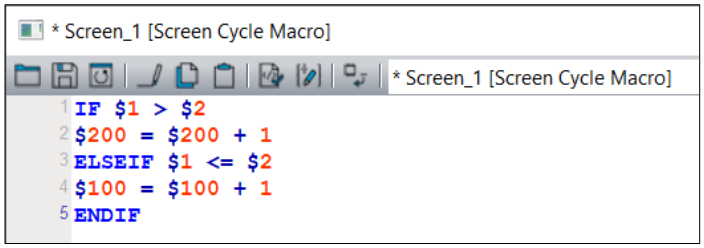
(6) ELSEIF <=			
Expression	**Meaning of Variable**		**Note**
ELSEIF Var1 <= Var2 (W) ELSEIF Var1 <= Var2 (DW) ELSEIF Var1 <= Var2 (Signed W) ELSEIF Var1 <= Var2 (Signed DW)	Var1	Condition1	W: Word DW: Double Word Signed: Signed number
	Var2	Condition2	
	Description of Action		
	Else if Condition1 is less than or equal to Condition2, execute…		
Note: ELSEIF macro command must be used with IF...ENDIF, otherwise an error message will pop up while compiling.			

Variable	Type		
	Internal Memory	**PLC Register**	**Constant**
Var1	◎		◎
Var2	◎		◎

Example

- Var1 and Var2 are Internal Memory addresses.





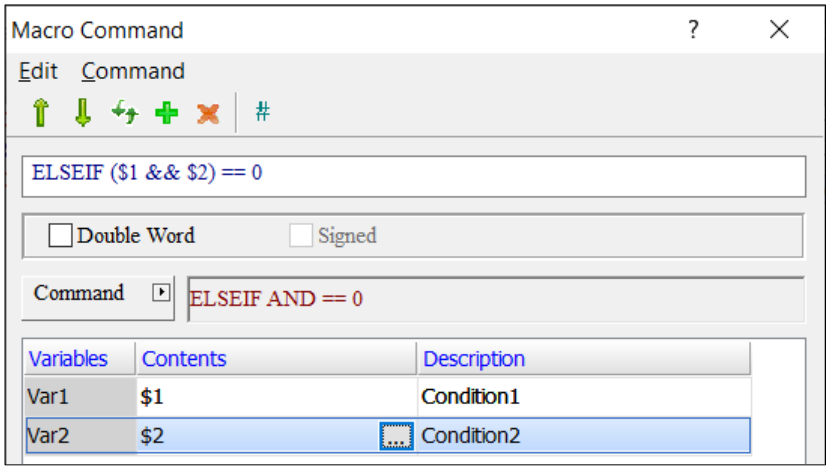
- If the value of $1 is greater than $2, execute $200 = $200 + 1; else if $1 is less than or equal to $2, then execute $100 = $100 + 1.

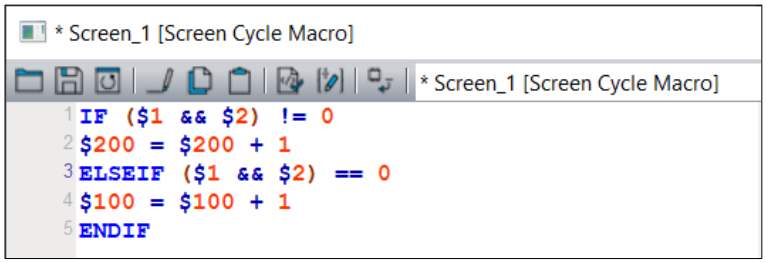
(7) ELSEIF AND == 0			
Expression	**Meaning of Variable**		**Note**
ELSEIF (Var1 && Var2) == 0 (W) ELSEIF (Var1 && Var2) == 0 (DW)	Var1	Condition1	W: Word DW: Double Word
	Var2	Condition2	
	Description of Action		
	Else if the result of the AND operation on Condition1 and Condition2 is 0, execute…		

Variable	Type		
	Internal Memory	**PLC Register**	**Constant**
Var1	V		V
Var2	V		V

Example

- Var1 and Var2 are Internal Memory addresses.





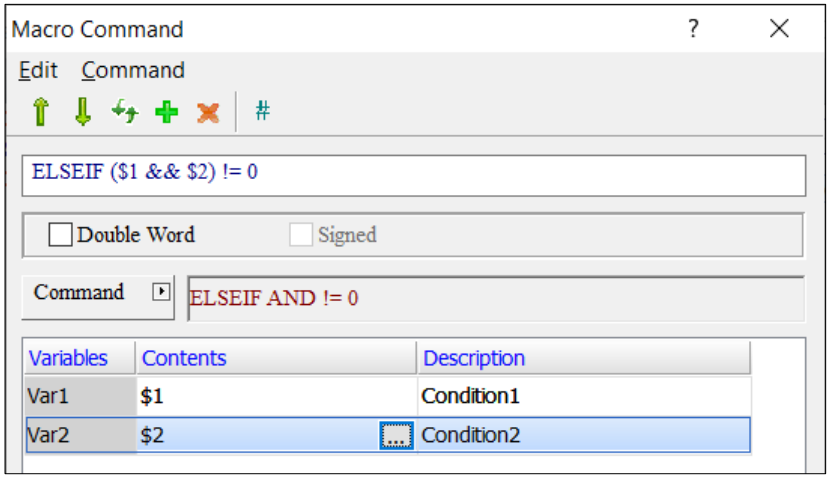
- If the result of the AND operation on $1 and $2 is not 0, execute $200 = $200 + 1; else if the result of the AND operation on $1 and $2 is 0, then execute $100 = $100 + 1.

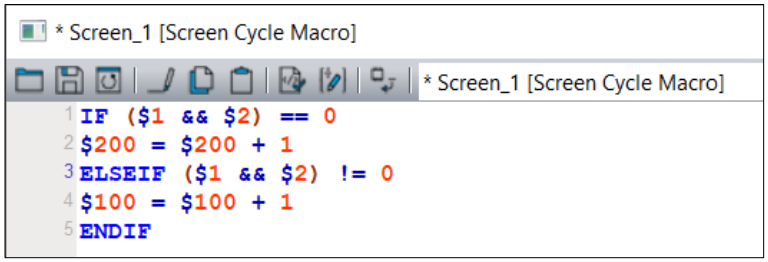
(8) ELSEIF AND != 0			
Expression	**Meaning of Variable**		**Note**
ELSEIF (Var1 && Var2) != 0 (W) ELSEIF (Var1 && Var2) != 0 (DW)	Var1	Condition1	W: Word DW: Double Word
	Var2	Condition2	
	Description of Action		
	Else if the result of the AND operation on Condition1 and Condition2 is not 0, execute…		

Variable	Type		
	Internal Memory	**PLC Register**	**Constant**
Var1	V		V
Var2	V		V

Example

- Var1 and Var2 are Internal Memory addresses.





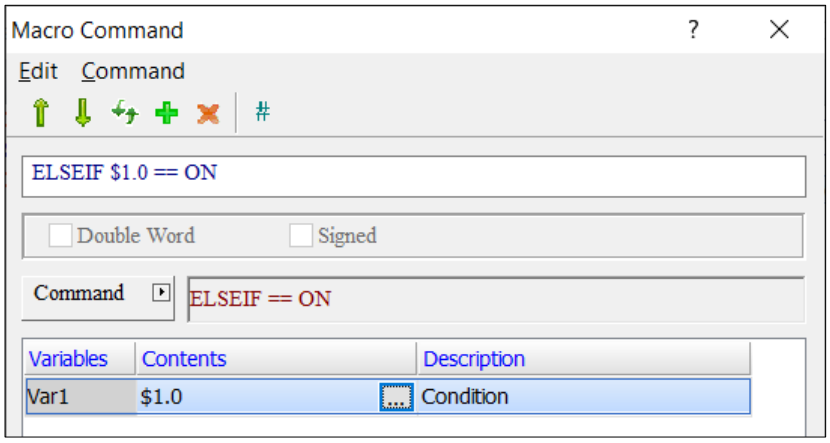
- If the result of the AND operation on $1 and $2 is 0, execute $200 = $200 + 1; else if the result of the AND operation on $1 and $2 is not 0, then execute $100 = $100 + 1.

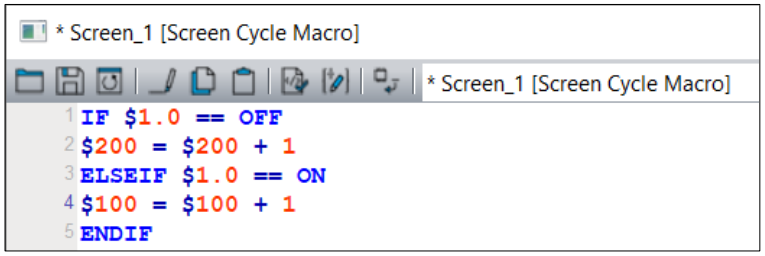
(9) ELSEIF == ON			
Expression	**Meaning of Variable**		**Note**
ELSEIF Var1 == ON (W)	Var1	Condition1	W: Word
	Description of Action		
	Else if Condition1 is ON, execute…		

Variable	**Type**		
	Internal Memory	**PLC Register**	**Constant**
Var1	V (can only be Bit)		

Example

- Var1 is the Internal Memory address.





- If $1.0 is OFF, execute $200 = $200 + 1; else if $1.0 is ON, then execute $100 = $100 + 1.

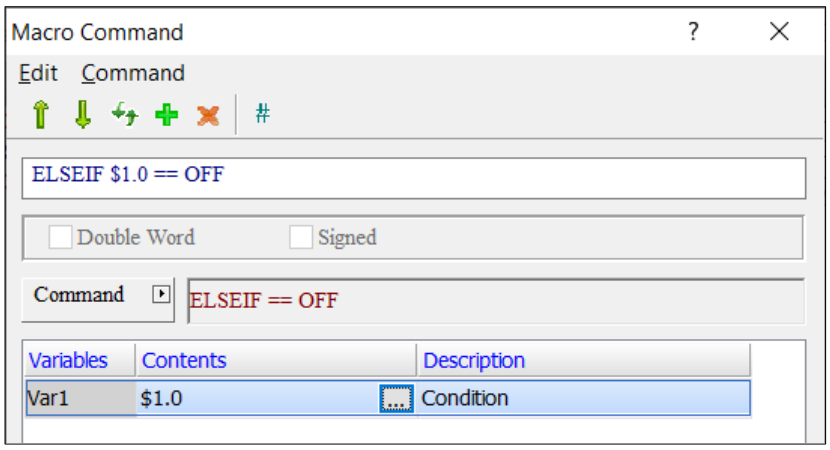
(10) ELSEIF == OFF			
Expression	**Meaning of Variable**		**Note**
ELSEIF Var1 == OFF (W)	Var1	Condition1	W: Word
	Description of Action		

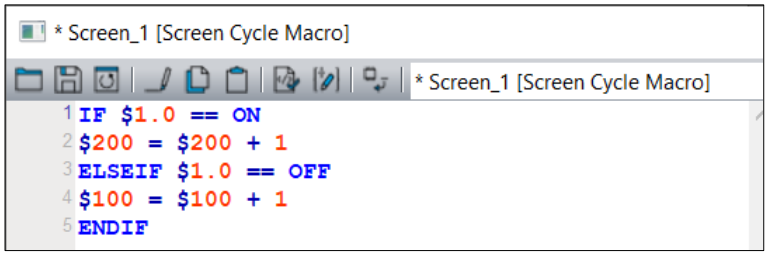
(10) ELSEIF == OFF		
Expression	**Meaning of Variable**	**Note**
	Else if Condition1 is OFF, execute…	

Variable	Type		
	Internal Memory	**PLC Register**	**Constant**
Var1	V (can only be Bit)		

Example

- Var1 is the Internal Memory address.

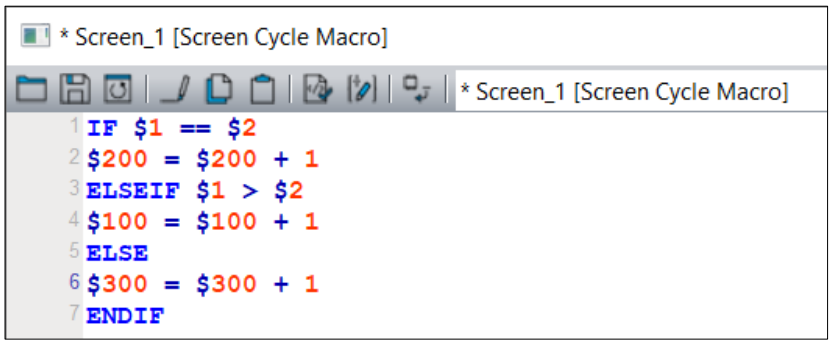




- If $1.0 is ON, execute $200 = $200 + 1; else if $1.0 is OFF, then execute $100 = $100 + 1.

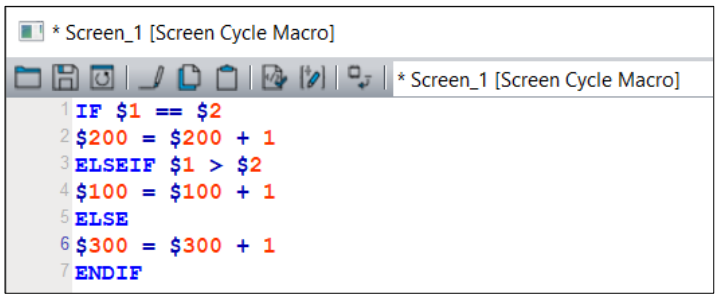
- **ELSE**

  The ELSE command is mainly used to execute other programs when conditions of IF... or ELSEIF are not met. ELSE must be used together with IF… and ENDIF, otherwise the software will prompt a syntax error message while compiling



- **ENDIF**

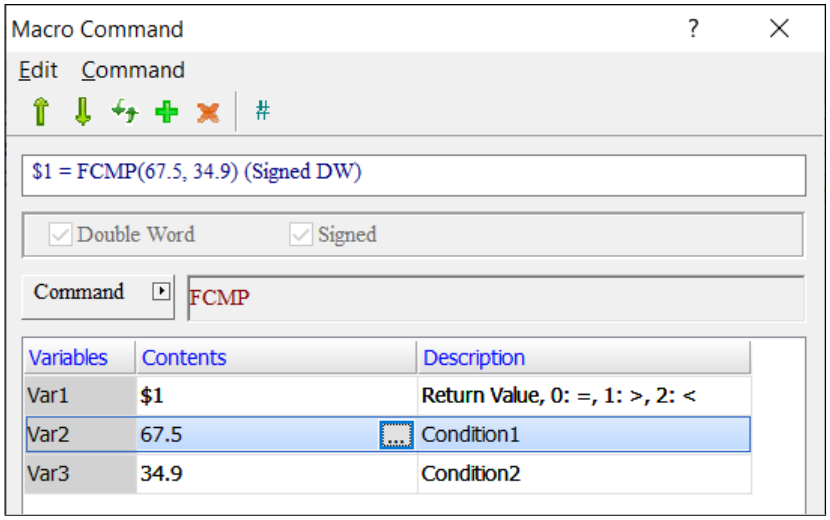
  ENDIF is mainly used together with IF..., ELSE, and ELSEIF... commands.



- **FCMP (Comparison of Floating-point Value)**

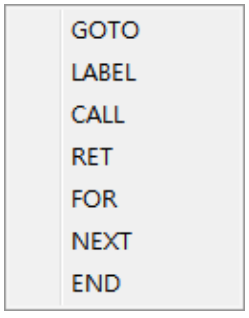
<table>
<tr><th>Expression</th><th colspan="3">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="8">Var1 = FCMP(Var2, Var3)
(Signed DW)</td><td>Var1</td><td colspan="2">Return value of comparison result</td><td rowspan="8">DW: Double Word
Signed: Signed number</td></tr>
<tr><td></td><td>=</td><td>0</td></tr>
<tr><td></td><td>></td><td>1</td></tr>
<tr><td></td><td><</td><td>2</td></tr>
<tr><td>Var2</td><td colspan="2">Condition1</td></tr>
<tr><td>Var3</td><td colspan="2">Condition2</td></tr>
<tr><td colspan="3">Description of Action</td></tr>
<tr><td colspan="3">Compare Var2 and Var3, and put the result in Var1.</td></tr>
</table>

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		V
Var3	V		V

Example
• Var1 is the Internal Memory address, and Var2 and Var3 are constants.  • Compare the floating-point values of 67.5 and 34.9. 67.5 > 34.9, and the return value is 1, so $1 = 1.

FlowControl

FlowControl includes commands such as GOTO, LABEL, CALL, RET, FOR, NEXT, END, etc., to control the execution process while writing macro programs. It is explained as follows.



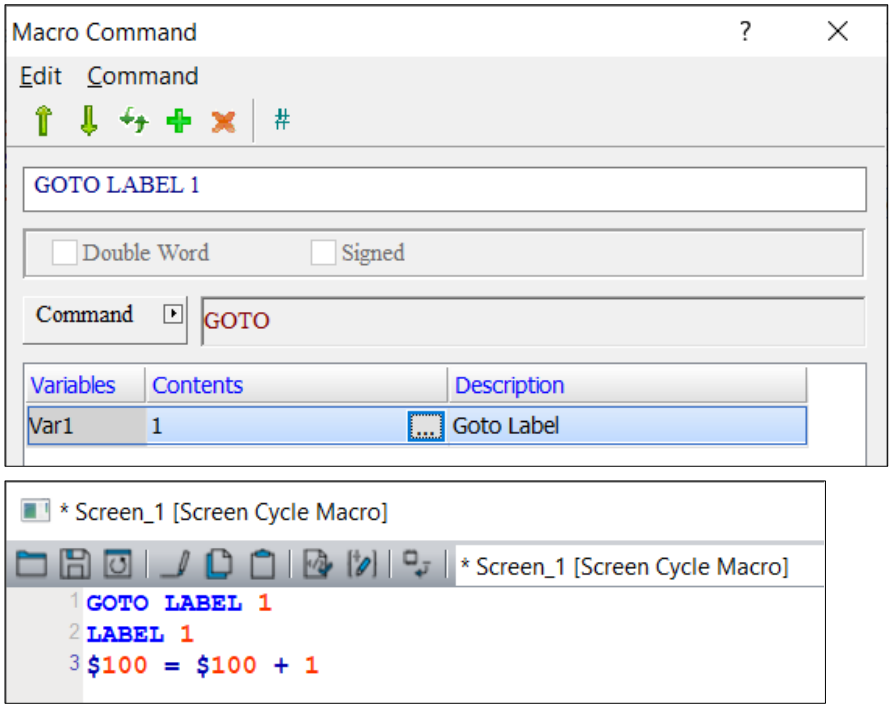
- GOTO LABEL (unconditionally execute a label name)

Expression	Meaning of Variable		Note
GOTO LABEL Var1 (W)	Var1	Goto Label	W: Word
	Description of Action		
	Directly execute the specified label name.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1			V

Example

- Var1 can only be a constant.

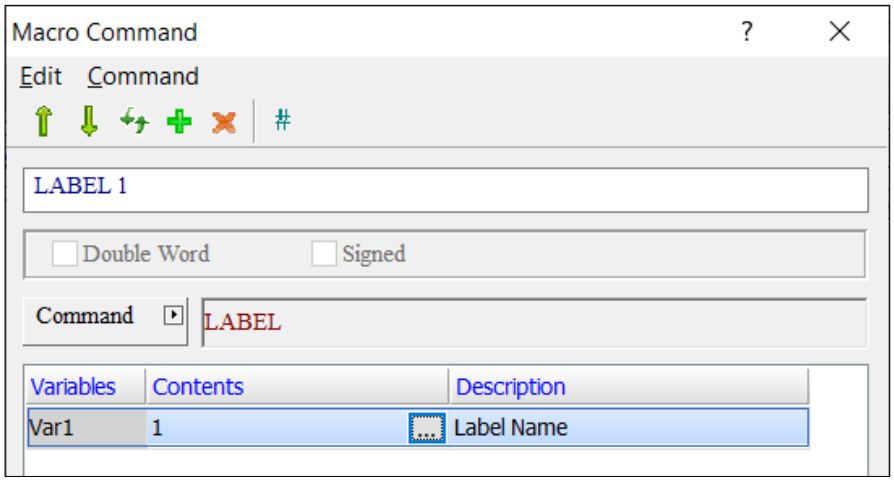


- Execute LABEL 1 directly. The command of LABEL 1 is $100 = $100 + 1.

- LABEL (Label name)

Expression	Meaning of Variable		Note
LABEL Var1 (W)	Var1	Label name	W: Word
	Description of Action		
	Specify the label name.		
Note: The same label name shall not be used in the same macro.			

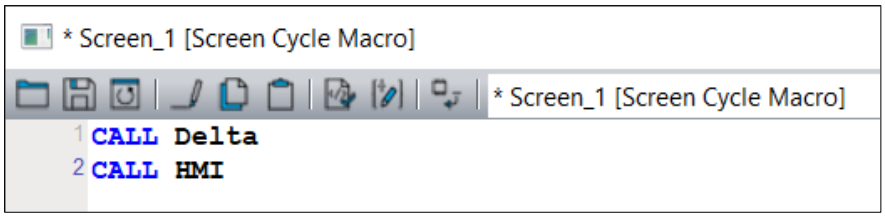
Variable	Type		
	Internal Memory	PLC Register	Constant
Var1			V

Example
• Var1 can only be a constant. 

- CALL (Call Submacro)

Expression	Meaning of Variable		Note
CALL Var1 (W)	Var1	Submacro number (1 - 512)	W: Word
	Description of Action		
	Specify the names of submacros.		

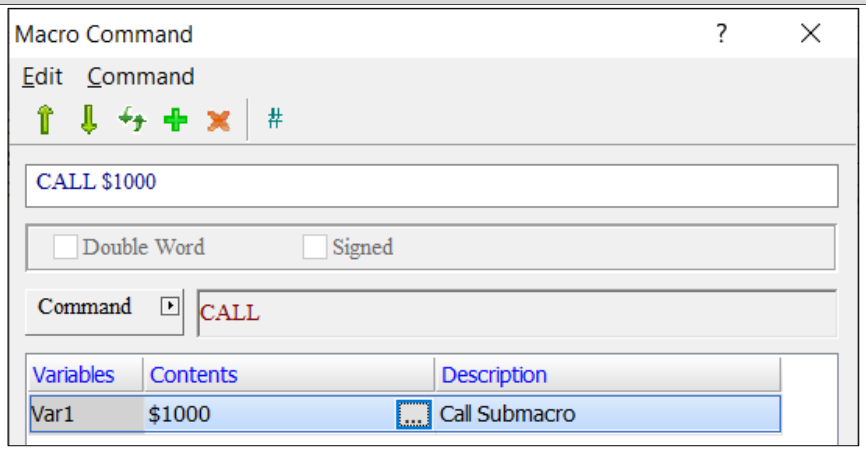
Note: Var1 can support the input of Chinese and English names. If users want to input Chinese and English names, enter its macro alias manually. The Macro Wizard only supports the input of submacro numbers.



Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		V

Example
• Var1 is the Internal Memory address.

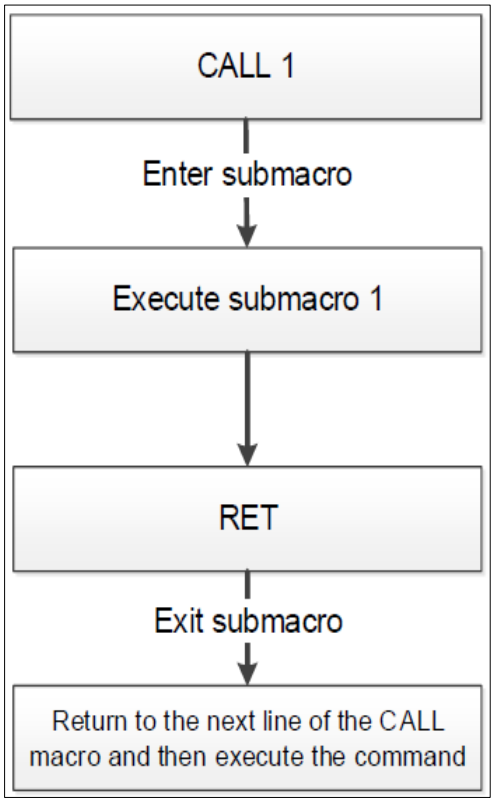
Example

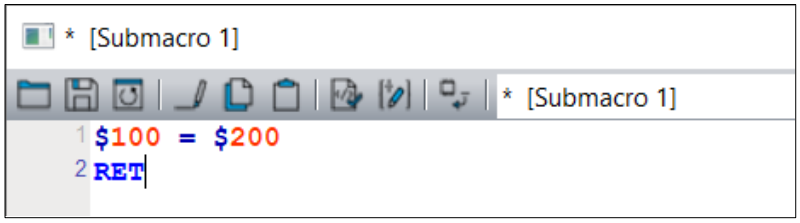
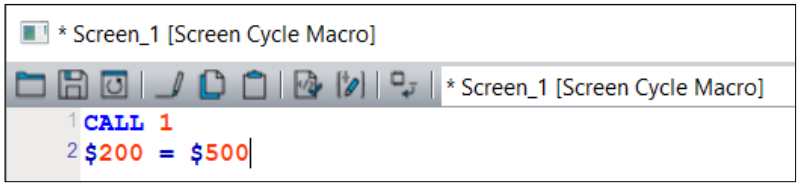


- Execute the submacro commands by entering submacro numbers via the Internal Memory address $1000 (**Numeric Entry** element).

- RET (Exit submacro)

Expression	Description of Action	Note
RET	Exit submacro and return to the next line of the CALL submacro and then execute the command.	RET should be added at the end of the submacro, and must be used together with the CALL command.



Example
• The RET command must be written at the end of the submacro.    • Return and execute the next line of the CALL macro command after exiting the submacro. 

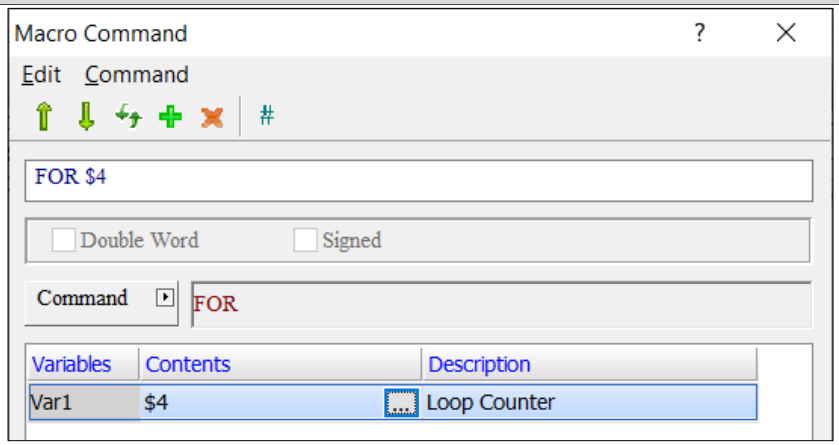
- **FOR, NEXT (Program Loop)**

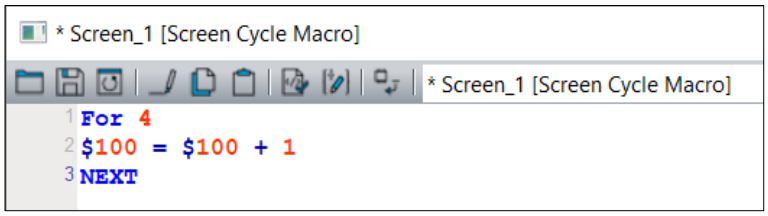
Expression	Meaning of Variable		Note
FOR Var1 (W)	Var1	Loop Counter	W: Word
	Description of Action		
	Execute the statement for Var1 times continuously.		
Expression	**Description of Action**		**Note**
NEXT	It must be used together with the FOR command.		
Note: The multilayer loops can be used to support up to 10 layers.			

Memory Usage			
Variable	**Internal Memory**	**PLC Register**	**Constant**
Var1	V		V

Example
• Var1 is a constant.

Example

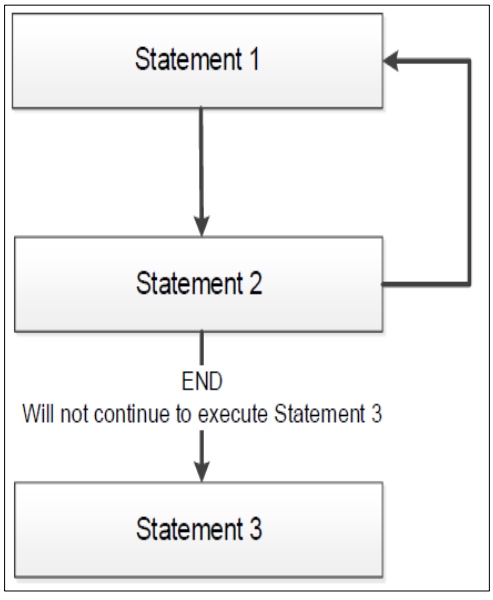




- FOR 4 means that the $100 = $100 + 1 command is executed for four times, so the result obtained is 4.

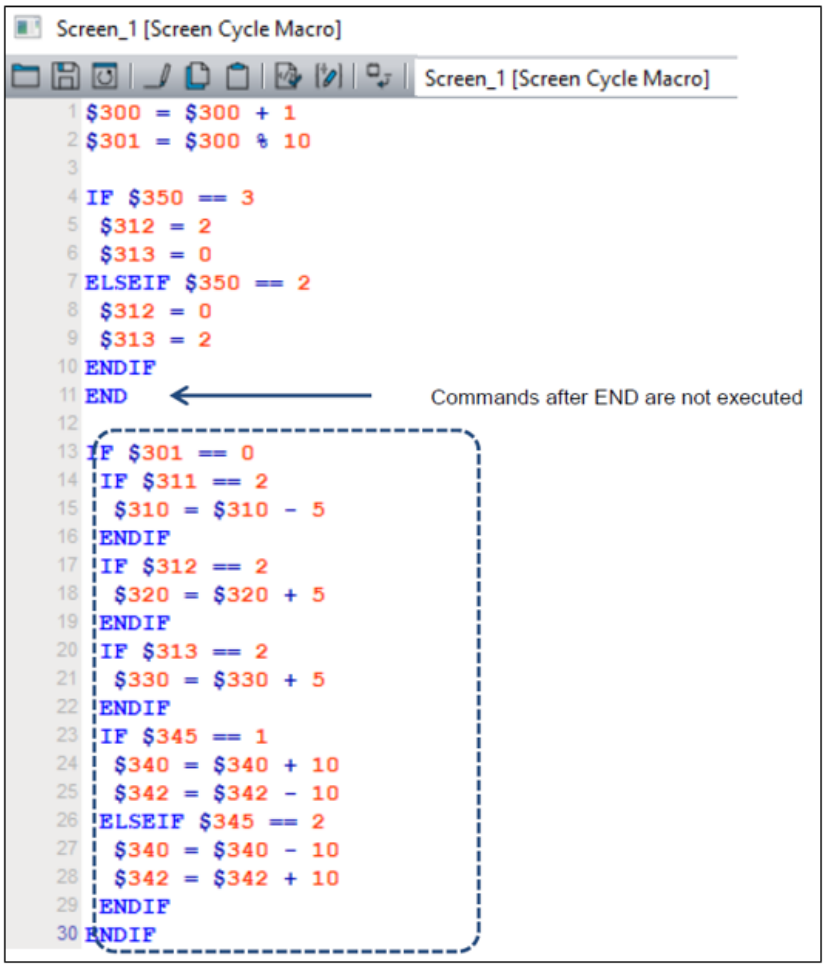
- **END (End Macro Programs)**

Expression	Description of Action	Note
END	End macro programs.	Adding END to the submacro means the program will not return to the original macro command to continue executing the next line.

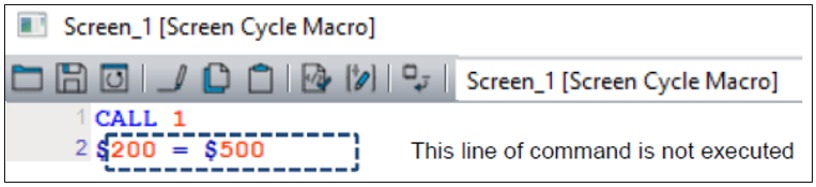


Example

- The commands after END are not executed.

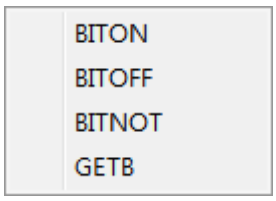


- When the END command is written to the end of a submacro, it means the program will not return to execute the previous macro command.



Bit Setting

Bit Setting includes BITON, BITOFF, BITNOT, GETB and other commands, which allow users to set the On / Off state of bit, inverse bit, and get the value represented by the bit. The details are provided as follows.



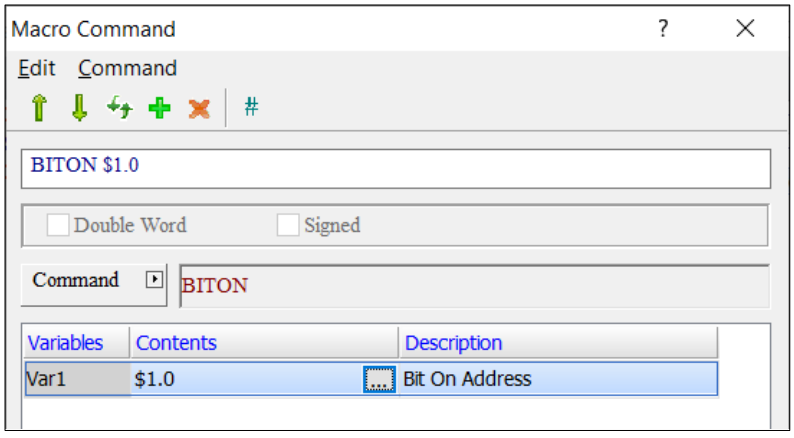
- **BITON (Set Bit as On)**

Expression	Meaning of Variable		Note
BITON Var1 (W)	Var1	Bit setting	W: Word
	Description of Action		
	Set Var1 to ON.		

Variable	Type		
	Internal Memory	**PLC Register**	**Constant**
Var1	V (can only be bit)	V (can only be bit)	

Example

- Var1 is the Internal Memory address.



- Create a **Maintained Button** and set the Write Address to $1.0. When **BITON** $1.0 is executed, the result is as follows.

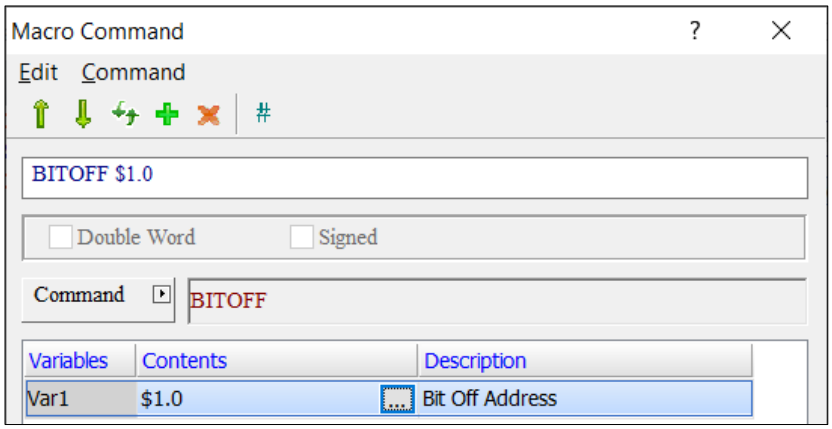
- **BITOFF (Set bit as Off)**

Expression	Meaning of Variable		Note
BITOFF Var1 (W)	Var1	Bit setting	W: Word
	Description of Action		
	Set Var1 to Off.		

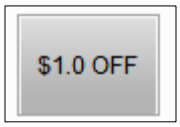
Variable	Type		
	Internal Memory	**PLC Register**	**Constant**
Var1	V (can only be bit)	V (can only be bit)	

Example

- Var1 is the Internal Memory address.



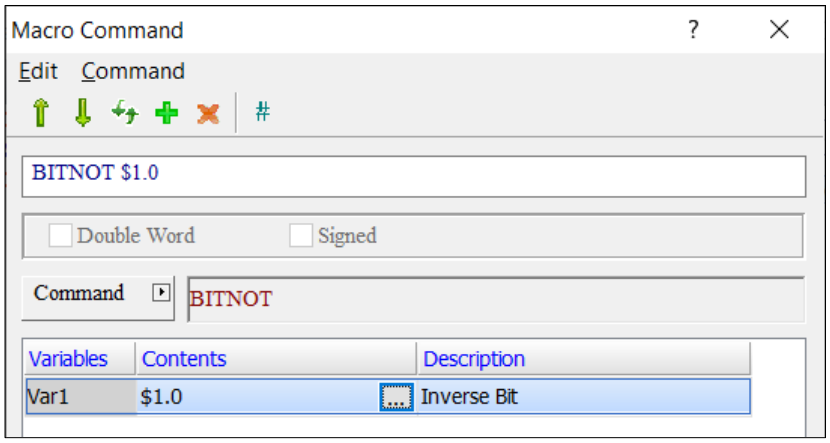
- Create a **Maintained Button** and set the Write Address to $1.0. When BITOFF $1.0 is executed, the result is as follows.



- **BITNOT (Inverse Bit, ON→OFF, OFF→ON)**

Expression	Meaning of Variable		Note
BITNOT Var1 (W)	Var1	Bit Setting	W: Word
	Description of Action		
	Set Var1 bit from On to Off and from Off to On.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V (can only be Bit)	V (can only be Bit)	

Example
• Var1 is the Internal Memory address.    • Create a **Maintained Button** and set the **Write Address** to $1.0. When BITNOT $1.0 is executed, the actions of the **Maintained Button** will be continuously switched from On to Off and from Off to On.

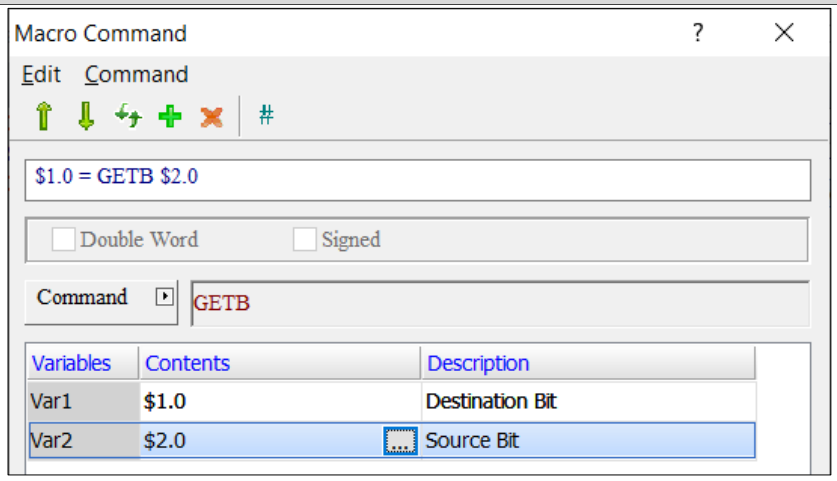
- **GETB (Get Bit Value)**

Expression	Meaning of Variable		Note
(Var1) = GETB (Var2) (W)	Var1	Bit setting	W: Word
	Description of Action		
	Put the Var2 bit value in Var1.		

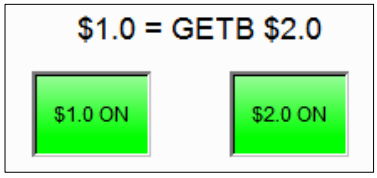
Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V (can only be bit)	V (can only be bit)	
Var2	V (can only be bit)	V (can only be bit)	

Example
• Var1 and Var2 are Internal Memory.

Example

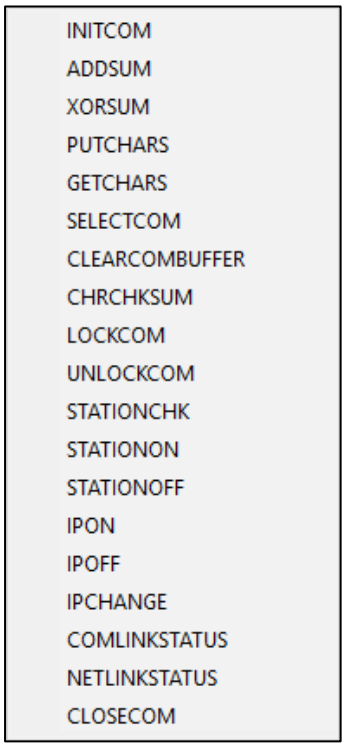


- Set $1.0 and $2.0 as **Set to On Button** elements. When executing $1.0 = GETB $2.0, press the $2.0 **Button**, then $1.0 will be triggered to ON.



Communication

The Communication macro command provides several macros related to COM Port and network IP control, as detailed as follows.



- **INITCOM (COM Port Initialization)**

<table>
<tr><th>Expression</th><th colspan="3">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="11">Var1 = INITCOM(Var2, Var3, Var4, Var5, Var6, Var7, Var8) (W)</td><td rowspan="3">Var1</td><td colspan="2">Return value</td><td rowspan="11">W: Word</td></tr>
<tr><td>Failed</td><td>0</td></tr>
<tr><td>Succeeded</td><td>1</td></tr>
<tr><td>Var2</td><td colspan="2">COM Port</td></tr>
<tr><td>Var3</td><td colspan="2">Interface</td></tr>
<tr><td>Var4</td><td colspan="2">Data Bits</td></tr>
<tr><td>Var5</td><td colspan="2">Parity Bits</td></tr>
<tr><td>Var6</td><td colspan="2">Stop Bits</td></tr>
<tr><td>Var7</td><td colspan="2">Baud Rate</td></tr>
<tr><td>Var8</td><td colspan="2">Flow Control</td></tr>
<tr><td colspan="3">Description of Action</td></tr>
<tr><td colspan="3">The initialization of COM Port is used to open the communication port, set the communication protocol (Var2 - Var8), and put the return value of the initialization result in Var1.</td></tr>
<tr><td colspan="5">Note: INITCOM command can only be issued once. If it is used again, the command is invalid.</td></tr>
</table>

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		

Parameter Setting			
Variable	**Option**	**Option Content**	**Corresponding Code**
Var2	COM Port	COM1	0
		COM2	1
		COM3	2
Var3	Interface	RS232	0
		RS422	1
		RS485	2

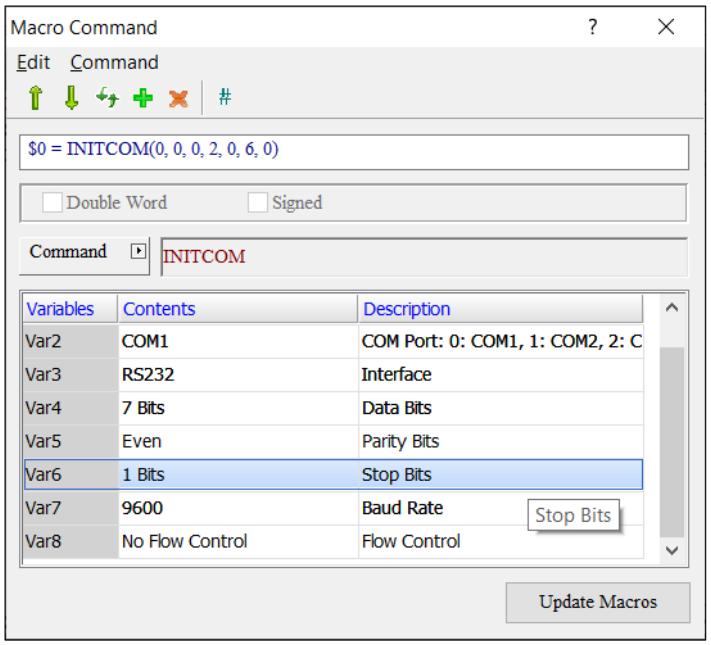
Parameter Setting			
Variable	**Option**	**Option Content**	**Corresponding Code**
Var4	Data Bits	7 Bits	0
		8 Bits	1
Var5	Parity Bits	None	0
		Old	1
		Even	2
Var6	Stop Bits	1 Bits	0
		2 Bits	1
Var7	Baud Rate	300	0
		600	1
		900	2
		1200	3
		2400	4
		4800	5
		9600	6
		14400	7
		19200	8
		28800	9
		38400	10
		57600	11
		115200	12
Var8	Flow Control	No Flow Control	0
		CTS RTS Flow Control	1
		DTR DSR Flow Control	2
		Xon Xoff Flow Control	3

Notes for FlowControl	
No FlowControl	FlowControl is not set.

Notes for FlowControl		
FlowControl		When using the serial port to transmit data, due to the new transmission processing technology such as real-time compression and debugging, the speed and accuracy of the communication are greatly improved, but the speed of data transmission between the computer and the HMI is greater than its real Data transmission speed. In order to ensure the security and integrity of data during transmission, it is necessary to have the control of transmission flow.
FlowControl	CTS / RTS FlowControl	FlowControl for the hardware, which is achieved by the electrical pulse wave generated by the hardware flowing to internal modem or to external modem through a connecting cable.
	DSR / DTR FlowControl	FlowControl for the hardware, which is achieved by a cable directly connecting the computer and the HMI.
	X On / X Off FlowControl	FlowControl for the software, which is only used in 2400 bps modem. It is achieved by adding the control code generated by software to the data in transmission.

Example

- Var1 is the Internal Memory address.



- After executing the INITCOM command, 0 or 1 will be returned to $1 for failure or success, respectively.

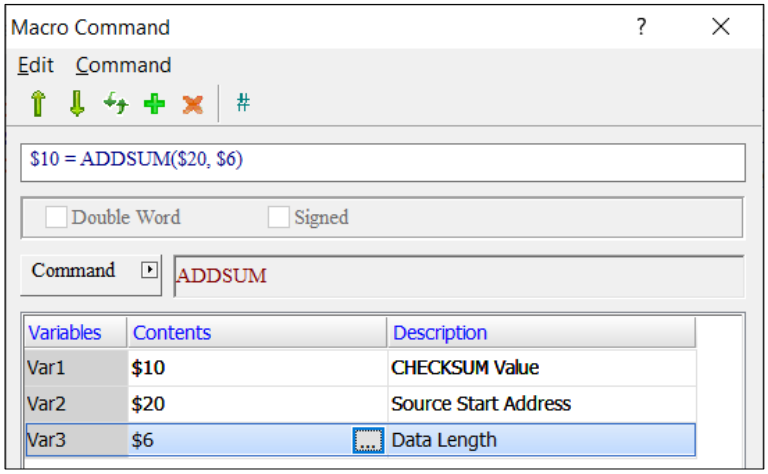
- **ADDSUM (Get CHECKSUM through Addition)**

<table>
<tr><th>Expression</th><th colspan="2">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="5">Var1 = ADDSUM(Var2, Var3) (W)</td><td>Var1</td><td>CHECKSUM Value</td><td rowspan="5">W: Word</td></tr>
<tr><td>Var2</td><td>Source Start Address</td></tr>
<tr><td>Var3</td><td>Data Length</td></tr>
<tr><td colspan="2">Description of Action</td></tr>
<tr><td colspan="2">Get CHECKSUM through addition. Var1 is the CHECKSUM value after calculation, Var2 is the start address of the data to be calculated, and Var3 is the length of the data.</td></tr>
<tr><td colspan="4">NOTE: The value of CHECKSUM calculated by ADDSUM is based on Byte. If the length of data is 6, it must be divided by 2, so the actual length is 3.</td></tr>
</table>

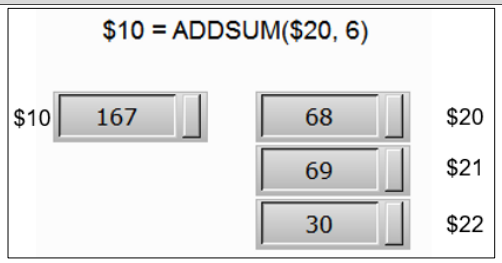
Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		
Var3	V		V

Example

- Var1 and Var2 are Internal Memory addresses, and Var3 is a constant.



- Conduct addition operation for 3 (6 / 2 = 3) consecutive data length from the start address of $20 and put the value result in $10. The expression is $20 + $21 + $22 = $10.

Example


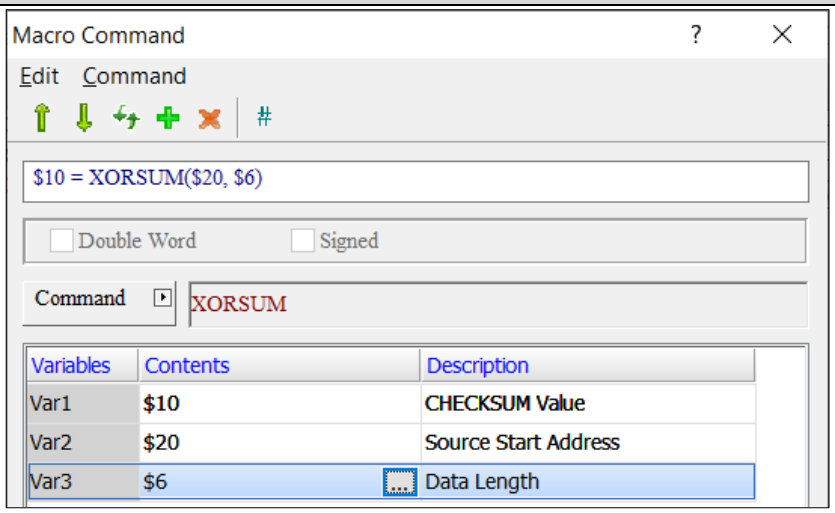
- **XORSUM (Get CHECKSUM through XOR)**

Expression	Meaning of Variable		Note
Var1 = XORSUM(Var2, Var3) (W)	Var1	CHECKSUM Value	W: Word
	Var2	Source Start Address	
	Var3	Data Length	
	Description of Action		
	Get CHECKSUM through addition. Var1 is the CHECKSUM value after calculation, Var2 is the start address of the data to be calculated, and Var3 is the length of the data.		
Note: The value of CHECKSUM calculated by XORSUM is based on Byte. If the length of data is 6, it has to be divided by 2, so the actual length is 3.			

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		
Var3	V		V

Example
• Var1 and Var2 are Internal Memory addresses, and Var3 is a constant

Example



- Conduct XOR operation for 3 (6 / 2 = 3) consecutive data length from the start address of $20 and put the value result in $10.

- **PUTCHARS (Output Characters via the Communication Port)**

<table>
<tr><th>Expression</th><th colspan="3">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="7">Var1 = PUTCHARS(Var2, Var3, Var4) (W)</td><td rowspan="3">Var1</td><td colspan="2">Return value</td><td rowspan="7">W: Word</td></tr>
<tr><td>Failed</td><td>0</td></tr>
<tr><td>Succeeded</td><td>1</td></tr>
<tr><td>Var2</td><td colspan="2">Source Start Address</td></tr>
<tr><td>Var3</td><td colspan="2">Data Length</td></tr>
<tr><td>Var4</td><td colspan="2">Communication Time</td></tr>
<tr><td colspan="3">Description of Action</td></tr>
<tr><td colspan="4">By the selected communication port, output characters of Var3 data length to Var2 start address within the required Var4 communication time, and put the return value in Var1.</td></tr>
</table>

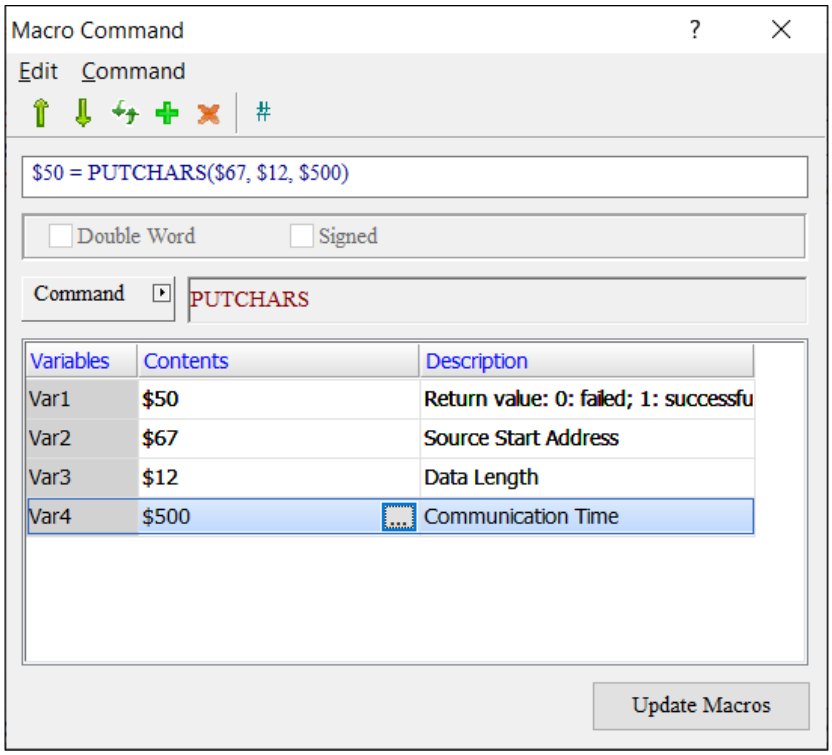
Note:

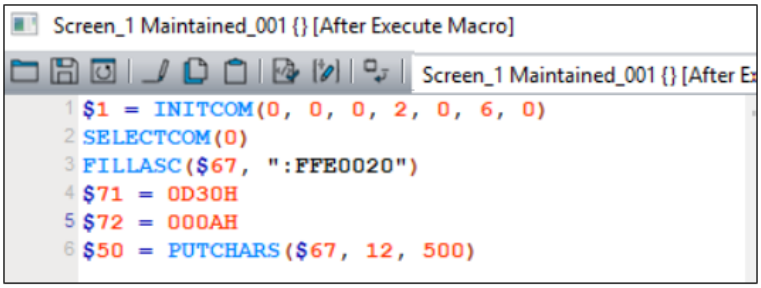
3. The PUTCHARS command must be used together with INITCOM and SELECTCOM.
4. The unit of Var3 is Byte.
5. The unit of Var4 is ms, which means to complete the macro execution within a specific time and end the execution when the time is up, so as to avoid delaying the execution of macros afterwards.

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		
Var3	V		V
Var4	V		V

Example

- Var1 and Var2 are Internal Memory addresses, and Var3 and Var4 are constants.





- In the case of $50 = PUTCHARS ($67, 12, 500), its action is to send 12 bytes (6 words) data and write them to $67. If 12 bytes cannot be sent within 500 ms, the program will exit the macro command when the time is up and write 0 to $50; if 12 bytes are sent successfully, it will exit this command immediately and write 1 to $50.

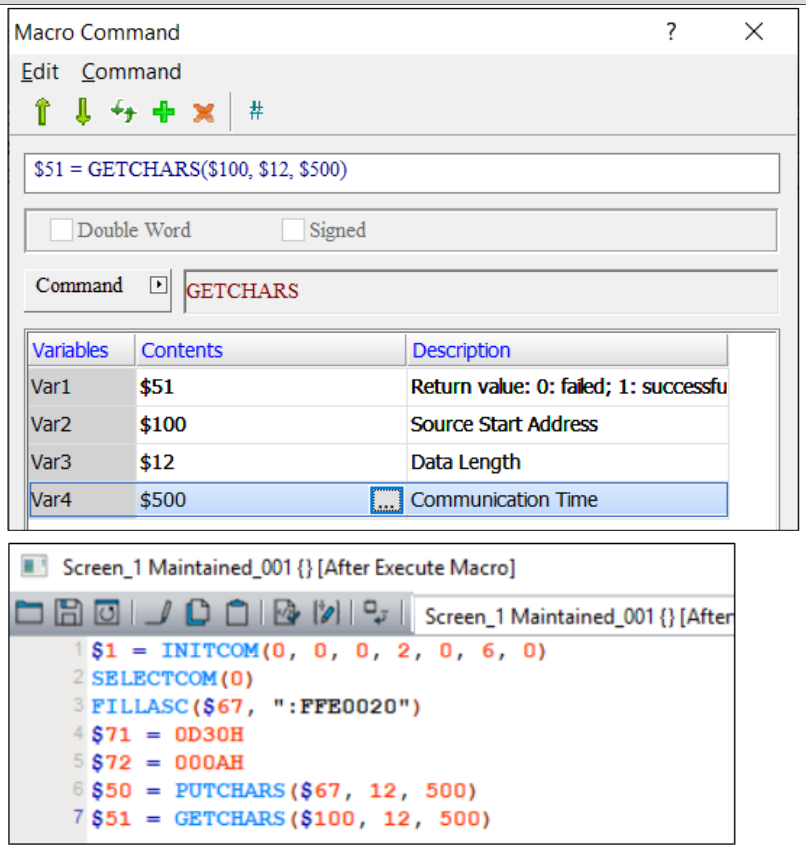
- **GETCHARS (Get Characters via the COM Port)**

<table>
<tr><th>Expression</th><th colspan="3">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="8">Var1 = GETCHARS(Var2, Var3, Var4) (W)</td><td rowspan="3">Var1</td><td colspan="2">Return value</td><td rowspan="8">W: Word</td></tr>
<tr><td>Failed</td><td>0</td></tr>
<tr><td>Succeeded</td><td>1</td></tr>
<tr><td>Var2</td><td colspan="2">Source Start Address</td></tr>
<tr><td>Var3</td><td colspan="2">Data Length</td></tr>
<tr><td>Var4</td><td colspan="2">Communication Time</td></tr>
<tr><td colspan="3">Description of Action</td></tr>
<tr><td colspan="3">By the selected communication port, obtain characters of Var3 data length to Var2 start address within the required Var4 communication time, and put the return value in Var1.</td></tr>
<tr><td colspan="5">Note:
1. The GETCHARS command must be used together with INITCOM and SELECTCOM.
2. The unit of Var3 is Byte.
3. The unit of Var4 is ms, which means to complete the macro execution within a specific time and end the execution when the time is up, so as to avoid delaying the execution of macros afterwards.</td></tr>
</table>

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		
Var3	V		V
Var4	V		V

Example
• Var1 and Var2 are Internal Memory addresses, and Var3 and Var4 are constants.

Example



- In the case of $51 = GETCHARS ($100, 12, 500), its action is to collect 12 bytes (6 words) data and write them to $100. If 12 bytes cannot be collected within 500 ms, the program will exit the macro command when the time is up and write 0 to $51; if 12 bytes are collected successfully, it will exit this command immediately and write 1 to $51.

- **SELECTCOM (Select COM Port)**

<table>
<tr><th>Expression</th><th colspan="3">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="4">SELECTCOM(Var1)
(W)</td><td rowspan="3">Var1</td><td>COM1</td><td>0</td><td rowspan="4">W: Word</td></tr>
<tr><td>COM2</td><td>1</td></tr>
<tr><td>COM3</td><td>2</td></tr>
<tr><td colspan="3">Description of Action</td></tr>
<tr><td colspan="4">Select the communication port.</td></tr>
</table>

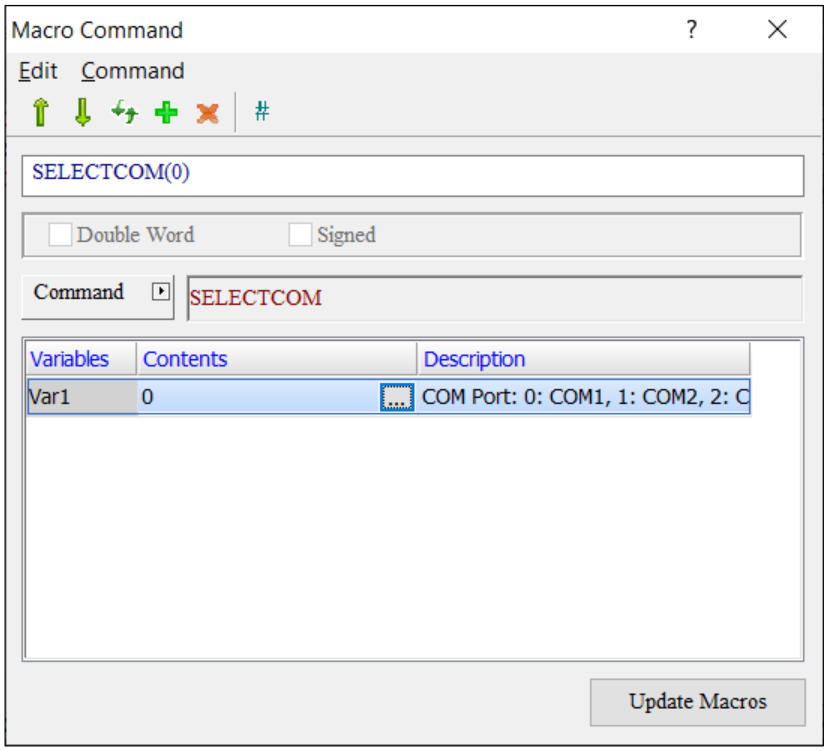
Note:

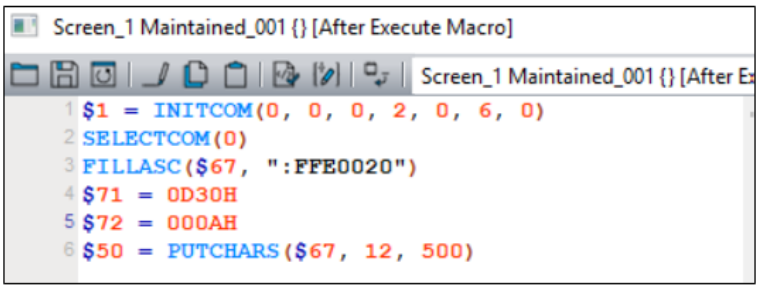
1. The SELECTCOM command must be used together with INITCOM.
2. The specified COM Port cannot be the same as the one used by the system. The selected COM Port will process relevant communication commands, so the SELECTCOM command in different macros will not support or interfere with each other.

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1			V

Example

- Var1 can only be a constant.





- **CLEARCOMBUFFER (Clear Buffer of COM Port)**

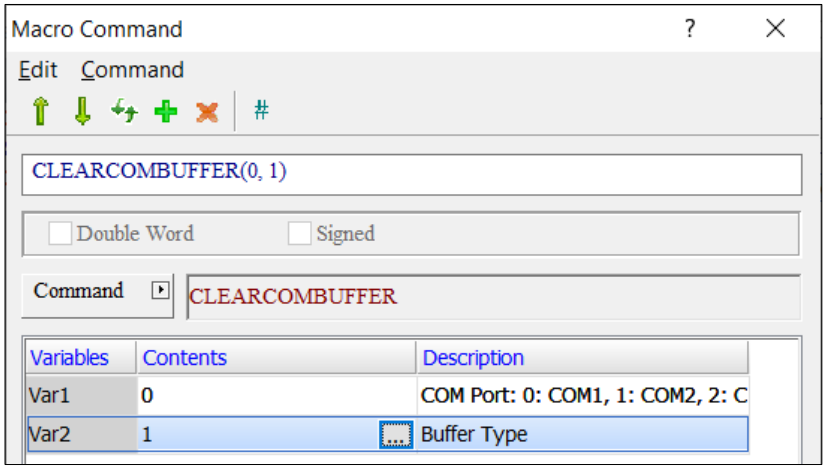
Expression	Meaning of Variable			Note
CLEARCOMBUFFER(Var1, Var2) (W)	Var1	COM1	0	W:Word
		COM2	1	
		COM3	2	
	Var2	Receive Buffer	0	

Expression	Meaning of Variable			Note
		Transmit Buffer	1	
	Description of Action			
	Clear buffer of Var1 COM Port.			

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1			V
Var2			V

Example

- Var1 and Var2 can only be constants.



- **CHRCHKSUM (Calculate the Length and CHECKSUM Value of the String)**

Expression	Meaning of Variable				Note
Var1 = CHRCHKSUM("Var2", Var3, Var4) (W)	Var1	String Length			W: Word
	Var2	Input String			
	Var3	Memory address for storing strings			
	Var4	Select the display format for CHECKSUM result	1 BYTE	1	
			2 BYTES (WORD)	2	
	Description of Action				

	Calculate string length and CHECKSUM value and put them in Var1.	

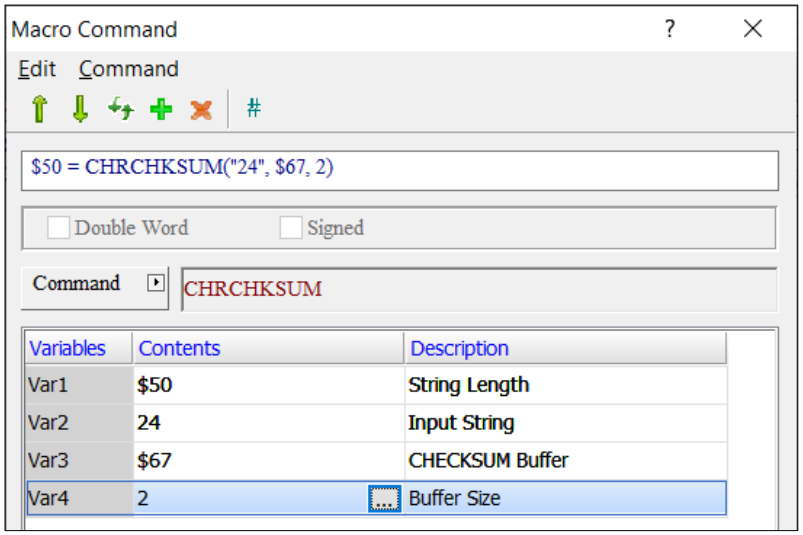
Note:

1. The string length of Var1 varies according to the format set by Var4.
2. If the input string is "345", Var4 is set as 2, the result value of the string length of Var1 is 5. On the contrary, if Var4 is set as 1, the result value is 4. (Unit based on Byte.)

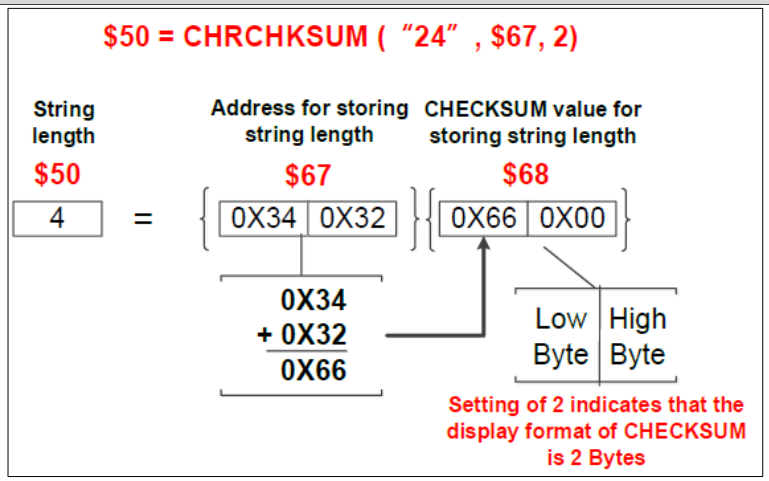
Variable	Type			
	Internal Memory	PLC Register	String	Constant
Var1	V			
Var2			V	
Var3	V			
Var4				V (can only input 1 and 2)

Example

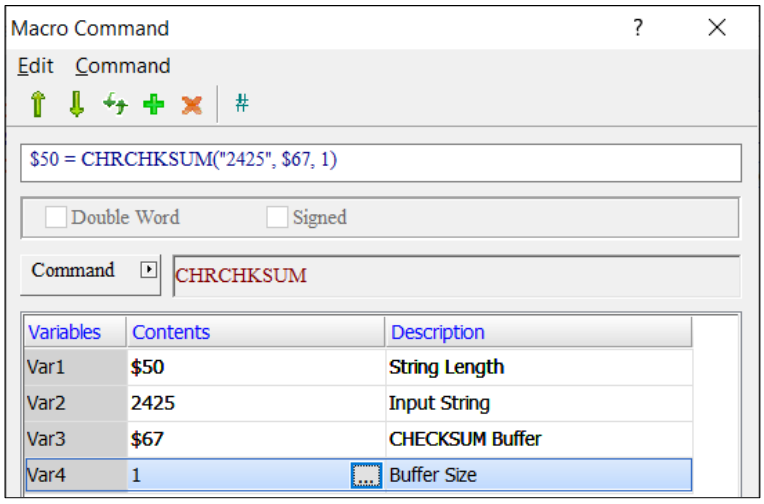
- Example 1

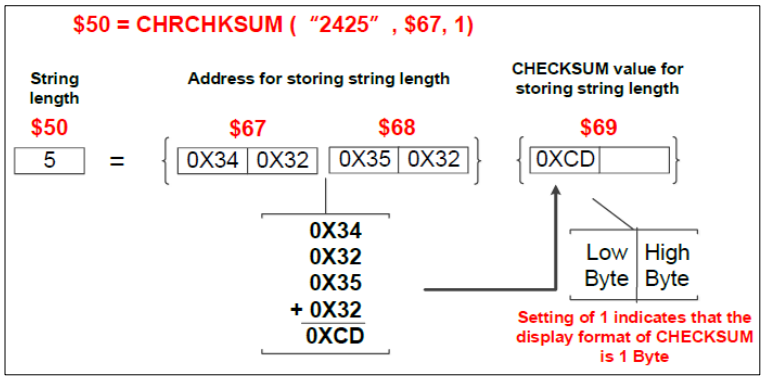


Example



- Example 2





- **LOCKCOM / UNLOCKCOM (Lock COM Port / Unlock COM Port)**

Expression	Meaning of Variable			Note
Var1 = LOCKCOM(Var2, Var3) (W)	Var1	Return value		W: Word
		Failed	0	
		Succeeded	1	

<table>
<tr><th>Expression</th><th colspan="3">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="5"></td><td rowspan="3">Var2</td><td>COM1</td><td>0</td><td rowspan="5"></td></tr>
<tr><td>COM2</td><td>1</td></tr>
<tr><td>COM3</td><td>2</td></tr>
<tr><td>Var3</td><td colspan="2">Time Out</td></tr>
<tr><td colspan="3">Description of Action</td></tr>
<tr><td colspan="4">Lock COM port.</td></tr>
<tr><td rowspan="5">UNLOCKCOM(Var1) (W)</td><td rowspan="3">Var1</td><td>COM1</td><td>0</td><td rowspan="5"></td></tr>
<tr><td>COM2</td><td>1</td></tr>
<tr><td>COM3</td><td>2</td></tr>
<tr><td colspan="3">Description of Action</td></tr>
<tr><td colspan="4">Unlock COM port.</td></tr>
<tr><td colspan="5">Note:
1. If the LOCKCOM command is set to infinite waiting (i.e. Var3 = 0), it means LOCKCOM will be executed twice in the same macro which will cause the HMI unable to respond.
2. If there are communication commands in more than one macro at the same time, it may cause interference of the communication data. To prevent this problem, LOCKCOM and UNLOCKCOM can be added before and after the communication commands. This can avoid communication interruption and execution of other communication actions in the same period of time to ensure the integrity of communication contents.
3. Use LOCKCOM and UNLOCKCOM together, otherwise the HMI cannot execute the download action.
4. The unit of Var3 Time Out value is ms.</td></tr>
</table>

Variable	LOCKCOM command type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2			V
Var3			V

Variable	UNLOCKCOM command type		
	Internal Memory	PLC Register	Constant
Var1			V

The following are examples of the proper, improper, and incorrect usages of the LOCKCOM / UNLOCKCOM macro commands.

LOCKCOM / UNLOCKCOM example (Proper use)

Background Macro	Element On Macro	Screen Cycle Macro
$50 = LOCKCOM(0,500) $51 = PUTCHARS($67, 3, 300) UNLOCKCOM(0)	$50 = LOCKCOM(0,500) $51 = GETCHARS($67, 3, 300) UNLOCKCOM(0)	$50 = LOCKCOM(0,500) $51 = PUTCHARS($67, 3, 300) UNLOCKCOM(0)

Suppose that communication commands are currently executed in three macros, when the **Background Macro** executes LOCKCOM (0, 500) first, then COM 1 is locked. As a result, LOCKCOM (0, 500) in **Element On Macro** and **Screen Cycle Macro** will stop, which will not be executed until Background macro executes UNLOCKCOM (0) to unlock COM 1. This action can avoid data interference or receiving error.

LOCKCOM / UNLOCKCOM example (Improper use)

Element On Macro	Screen Cycle Macro
$51 = GETCHARS($67, 3, 300)	$50 = LOCKCOM(0,500) $51 = PUTCHARS($67, 3, 300) UNLOCKCOM(0)

Suppose that communication commands are currently executed in two macros, when **Screen Cycle Macro** executes LOCKCOM (0, 500) first, COM 1 is locked. However, because **Element On Macro** is not locked by LOCKCOM, the GETCHARS command can still be executed, meaning it does not need to wait until the **Screen Cycle Macro** executes the UNLOCKCOM command. This will result in data interference and errors, so please avoid the above usage.

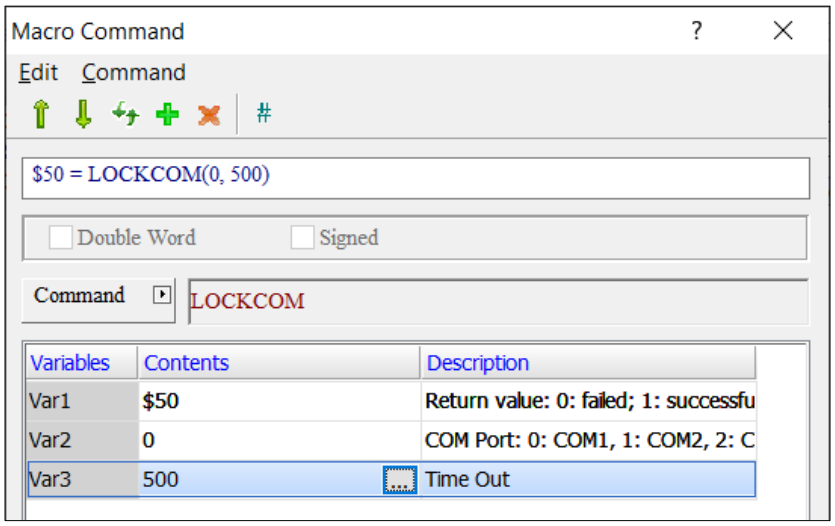
LOCKCOM / UNLOCKCOM example (Incorrect use)

Background Macro	Element On Macro
$50 = LOCKCOM(0, 500) $51 = PUTCHARS($67, 3, 300)	UNLOCKCOM(0)

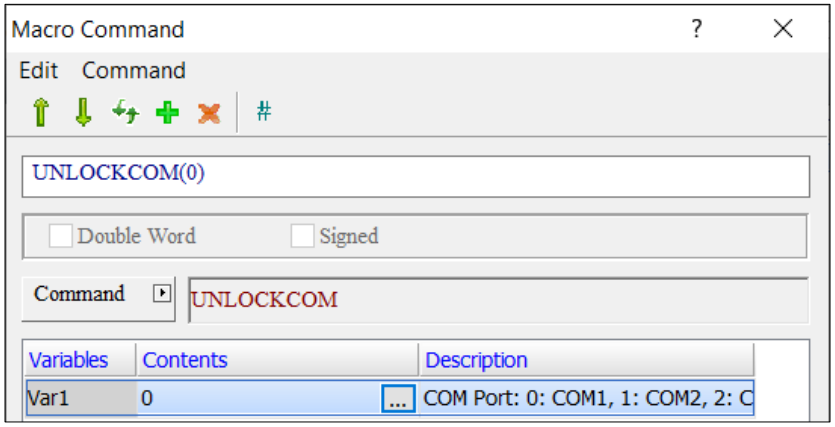
Suppose users lock COM Port in the **Background Macro** and transmits data through COM Port, but users cannot unlock COM Port in **Element On Macro**. This means that the two commands of locking COM Port and unlocking COM Port cannot be written separately.

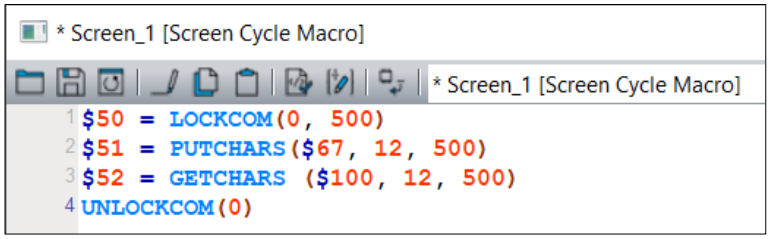
Example

- Var1 is the Internal Memory address, and Var2 and Var3 can only be constants.



- Var1 can only be a constant.





- **STATIONCHK (Check COM Connection Status)**

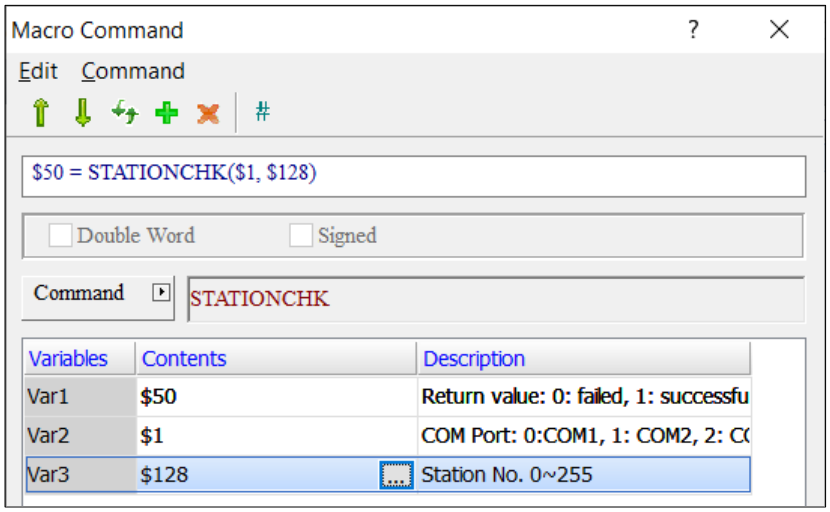
Expression	Meaning of Variable			Note
Var1 = STATIONCHK(Var2, Var3) (W)	Var1	Return value		W: Word
		Failed	0	
		Succeeded	1	
	Var2	COM1	0	

Expression	Meaning of Variable			Note
		COM2	1	
		COM3	2	
	Var3	Station No.		
	Description of Action			
	Check COM connection status.			
Note: This command reads Internal Memory parameters without increasing the HMI communication.				

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		V
Var3	V		V

Example

- Var1 is the Internal Memory address, and Var2 and Var3 are constants.



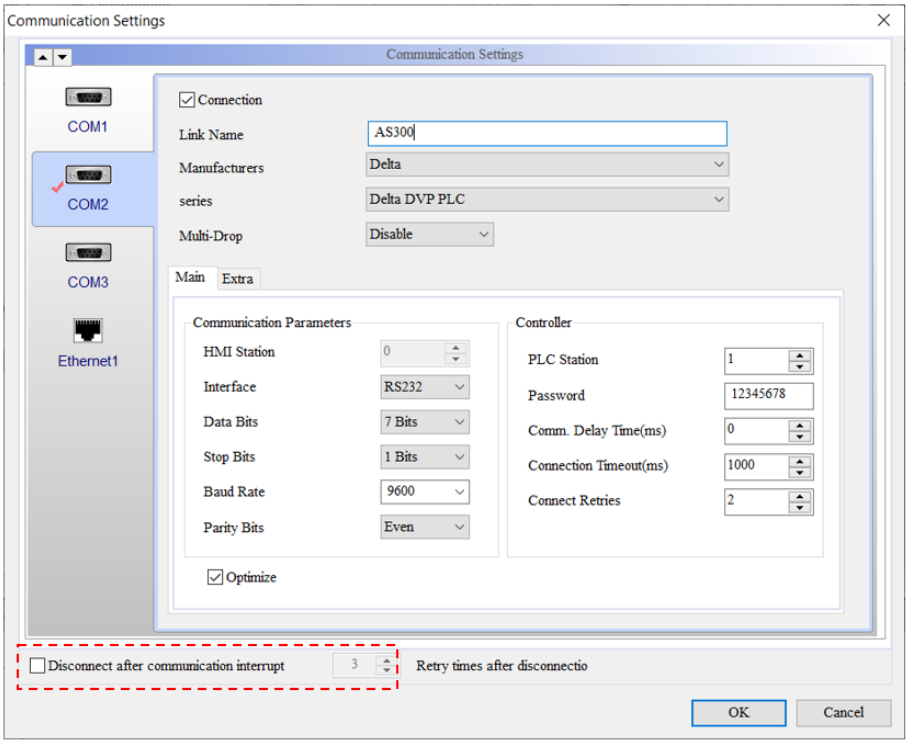
- After executing this **Macro**, if the communication between the HMI COM2 and the PLC of Station No. 128 is normal, the return value is $50 = 1; if the communication is abnormal, the return value is $50 = 0.

- **STATIONON (Station ON)**

Expression	Meaning of Variable			Note
STATIONON(Var1, Var2) (W)	Var1	COM1	0	W: Word
		COM2	1	

Expression	Meaning of Variable			Note
		COM3	2	
	Var2	Station No.		
	Description of Action			
	Enable Station No. Var2 of COM Var1 and the HMI can communicate with the station controller.			

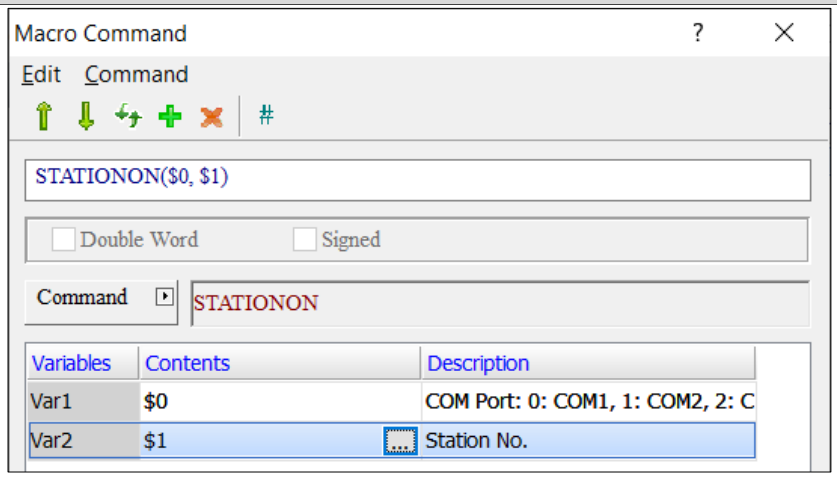
Note: The STATIONON macro and the Disconnect after communication interrupt of **General** > **Communication Settings** cannot be used at the same time.



Variable	Type		
	Internal Memory	**PLC Register**	**Constant**
Var1	V		V
Var2	V		V

Example
• Var1 and Var2 are constants. • Enable Station No. 1 on COM 1.

Example

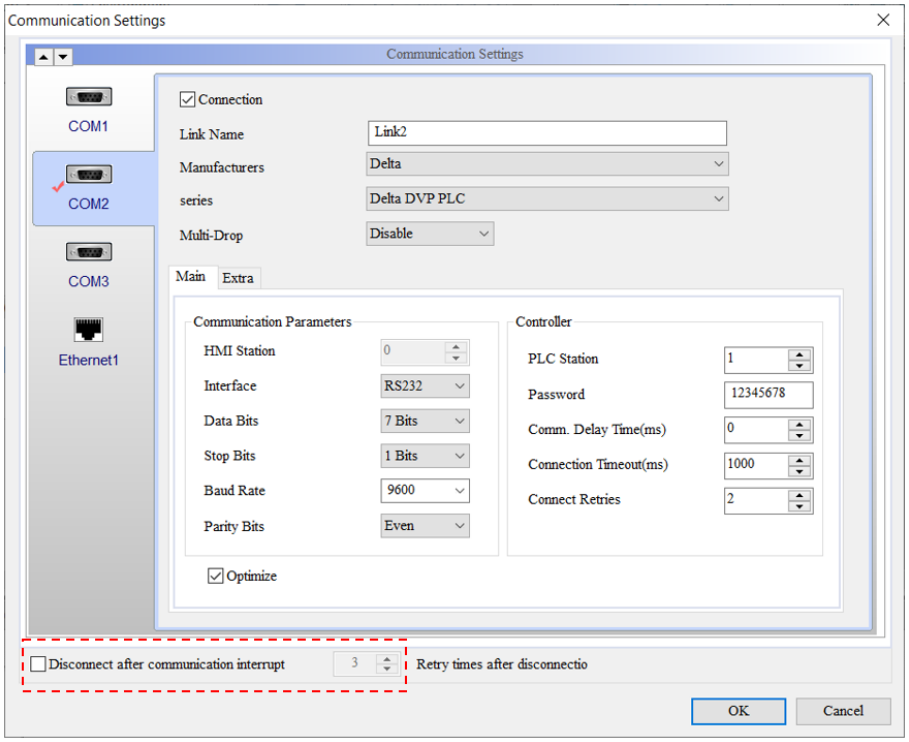


- **STATIONON (Station OFF)**

Expression	Meaning of Variable			Note
STATIONOFF(Var1, Var2) (W)	Var1	COM1	0	W: Word
		COM2	1	
		COM3	2	
	Var2	Station No.		
	Description of Action			
	Disable Station No. Var2 of COM Var1, and the HMI cannot communicate with the station controller.			

Note: The STATIONOFF macro and the Disconnect after communication interrupt of **General** > **Communication Settings** cannot be used at the same time.

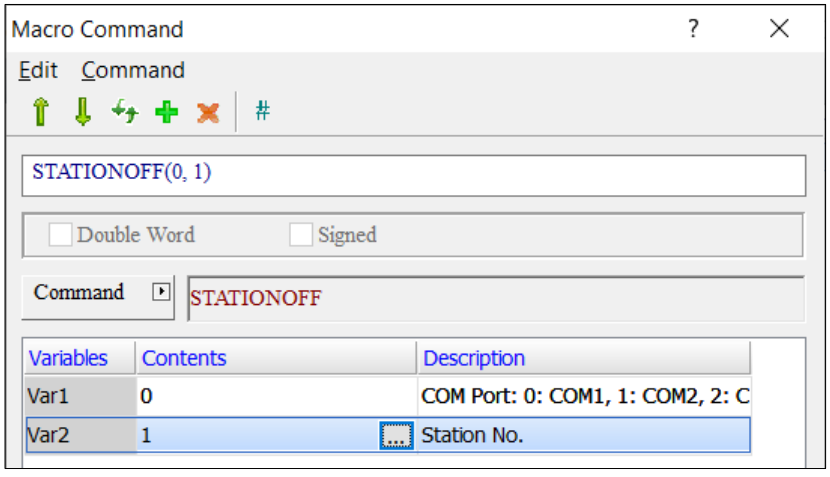
Expression	Meaning of Variable	Note



Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		V
Var2	V		V

Example

- Var1 and Var2 are constants.
- Disable Station No. 1 on COM 1.

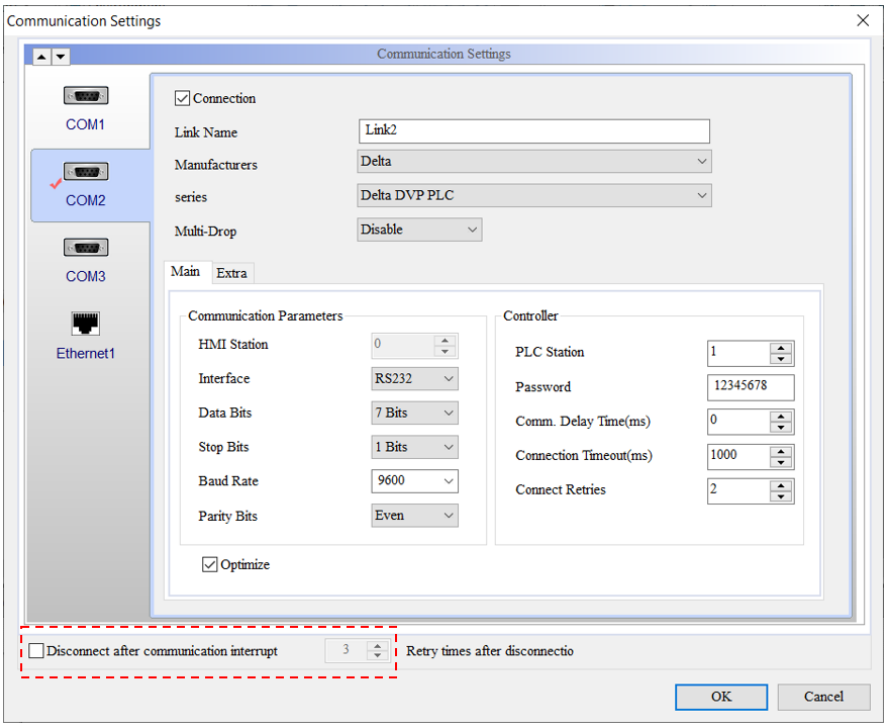


- **IPON (Activate IP address)**

<table>
<tr><th>Expression</th><th colspan="3">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="9">Var1 = IPON(Var2,Var3,Var4, Var5,Var6) (W)</td><td rowspan="3">Var1</td><td colspan="2">Return value</td><td rowspan="9">W: Word</td></tr>
<tr><td>Failed</td><td>0</td></tr>
<tr><td>Succeeded</td><td>1</td></tr>
<tr><td>Var2</td><td colspan="2">IP1</td></tr>
<tr><td>Var3</td><td colspan="2">IP2</td></tr>
<tr><td>Var4</td><td colspan="2">IP3</td></tr>
<tr><td>Var5</td><td colspan="2">IP4</td></tr>
<tr><td>Var6</td><td colspan="2">Port</td></tr>
<tr><td colspan="3">Description of Action</td></tr>
<tr><td colspan="4">Activate IP Var2, Var3, Var4, Var5, and Port Var6, and the HMI can communicate with the station controller.</td></tr>
</table>

Note:

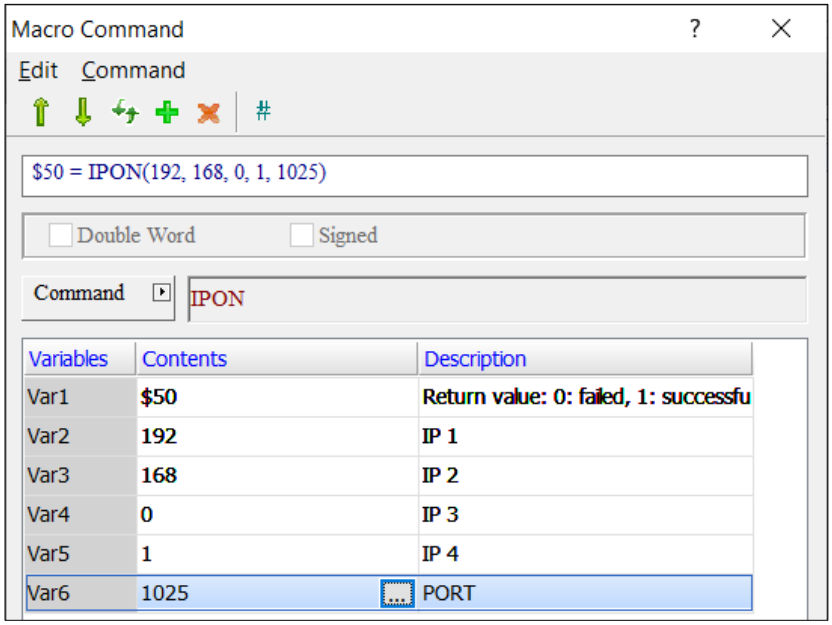
1. Using Var6 is not necessary. All ports under this IP will be activated when it is not in use.
2. The IPON macro and the Disconnect after communication interrupt of ***General*** > ***Communication Settings*** cannot be used at the same time.



Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V	V	V
Var3	V	V	V
Var4	V	V	V
Var5	V	V	V
Var6	V	V	V

Example

- Var1 is the Internal Memory address, and Var2 to Var6 are constants. Activate IP 192.168.0.1 Port: 1025.



- If users are not using Var6, enter $50 = IPON (192, 168, 0, 1), and all ports of IP 192.168.0.1 will be activated after the macro operation.

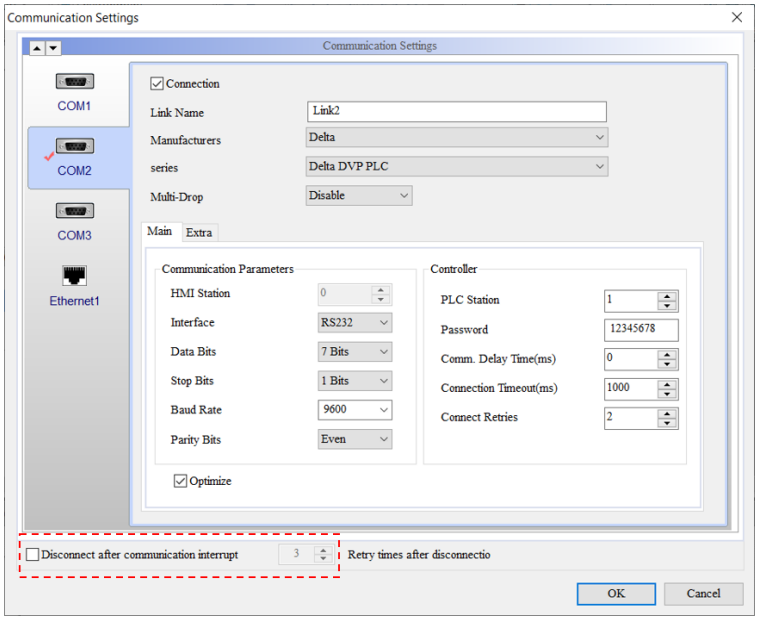
- **IPOFF (Disable IP address)**

Expression	Meaning of Variable			Note
Var1 = IPOFF(Var2, Var3,Var4,Var5,Var6) (W)	Var1	Return value		W: Word
		Failed	0	
		Succeeded	1	
	Var2	IP1		
	Var3	IP2		

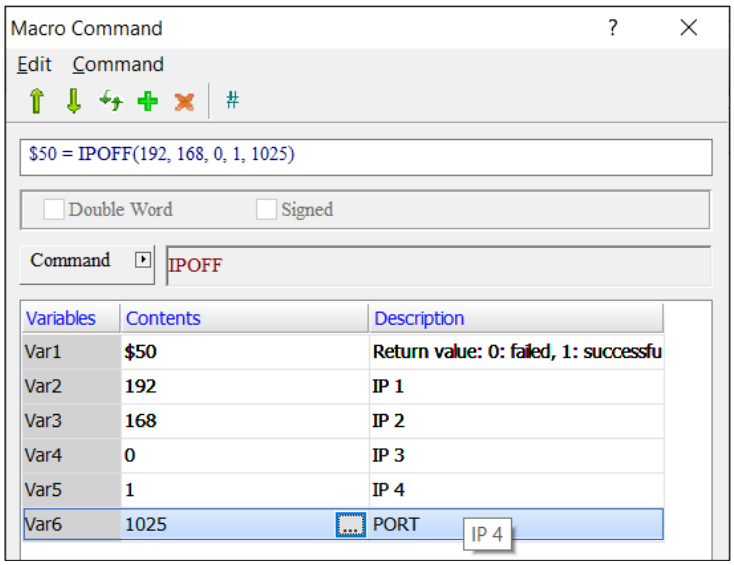
Expression	Meaning of Variable		Note
	Var4	IP3	
	Var5	IP4	
	Var6	Port	
	Description of Action		
	Disable IP Var2, Var3, Var4, Var5, and Port Var6, and the HMI cannot communicate with the station controller.		

Note:

1. Using Var6 is not necessary. All ports under this IP will be disabled when it is not in use.
2. The IPOFF macro and the Disconnect after communication interrupt of ***General*** > ***Communication Setting*** cannot be used at the same time.

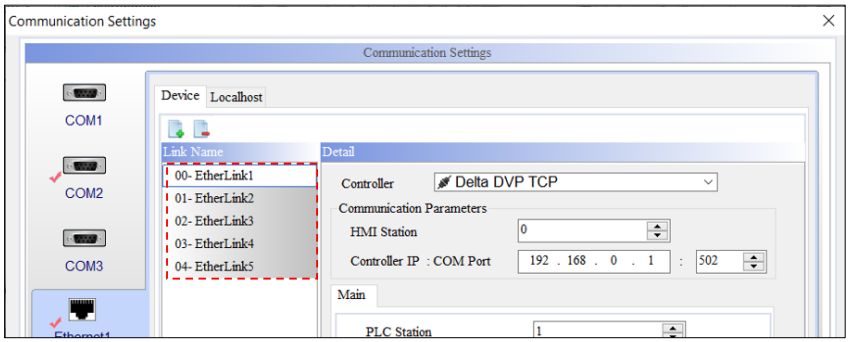


Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V	V	V
Var3	V	V	V
Var4	V	V	V
Var5	V	V	V
Var6	V	V	V

Example
• Var1 is the Internal Memory address, and Var2 to Var6 are constants. Disable IP 192.168.0.1 Port: 1025.    • If users are not using Var6, enter $50 = IPON (192, 168, 0, 1), and all ports in IP 192.168.0.1 will be disabled after the macro operation.

- **IPCHANGE (Change the Connection IP and Communication Port of the Connection Controller)**

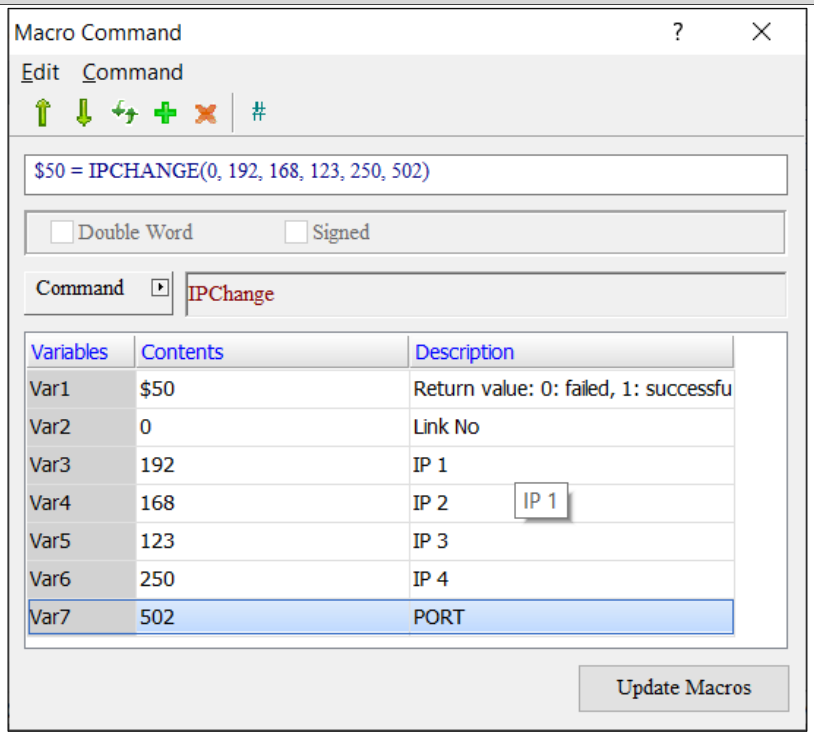
<table>
<tr><th>Expression</th><th colspan="3">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="10">Var1 = IPCHANGE(Var2, Var3, Var4, Var5, Var6, Var7) (W)</td><td rowspan="3">Var1</td><td colspan="2">Return value</td><td rowspan="10">W: Word</td></tr>
<tr><td>Failed</td><td>0</td></tr>
<tr><td>Succeeded</td><td>1</td></tr>
<tr><td>Var2</td><td colspan="2">Link No</td></tr>
<tr><td>Var3</td><td colspan="2">IP1</td></tr>
<tr><td>Var4</td><td colspan="2">IP2</td></tr>
<tr><td>Var5</td><td colspan="2">IP3</td></tr>
<tr><td>Var6</td><td colspan="2">IP4</td></tr>
<tr><td>Var7</td><td colspan="2">Port</td></tr>
<tr><td colspan="3">Description of Action</td></tr>
<tr><td colspan="3">Change the IP setting of PLC connection, Var3, Var4, Var5, Var6, and Port Var7, and the HMI can dynamically change the information of Link for the HMI to reconnect with another PLC.</td></tr>
</table>

Expression	Meaning of Variable	Note
Note: Link No. starts from 0. 		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V	V	V
Var3	V	V	V
Var4	V	V	V
Var5	V	V	V
Var6	V	V	V
Var7	V		

Example
• Var1 is the Internal Memory address, and Var2 to Var7 are constants.

Example



- Change the IP of PLC connection to 192.168.123.250 Port: 502.

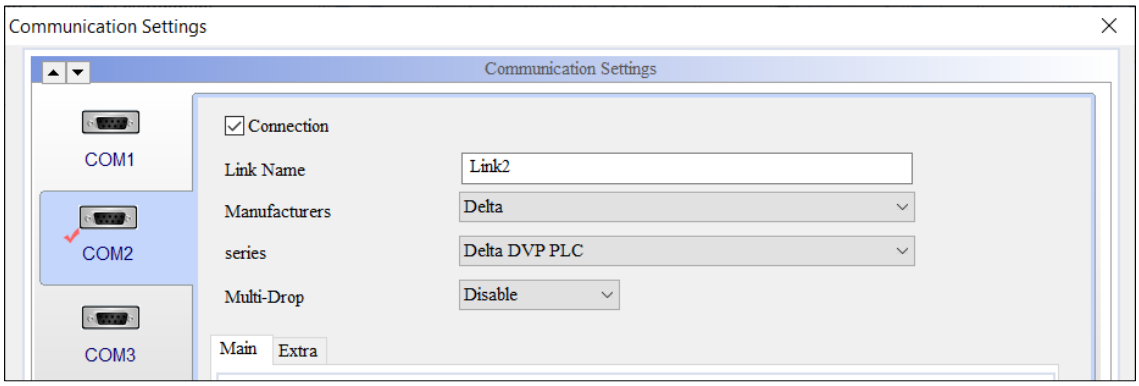
- **COMLINKSTATUS (Communication Port Connection Status)**

<table>
<tr><th>Expression</th><th colspan="3">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="5">Var1 = COMLINKSTATUS(Var2)</td><td>Var1</td><td colspan="2">Communication error code. Return 0 when the communication is normal.</td><td rowspan="5"></td></tr>
<tr><td rowspan="3">Var2</td><td>COM1</td><td>0</td></tr>
<tr><td>COM2</td><td>1</td></tr>
<tr><td>COM3</td><td>2</td></tr>
<tr><td colspan="3">Description of Action</td></tr>
<tr><td colspan="3">Return the error code of COM Var2 to Var1</td></tr>
</table>

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V

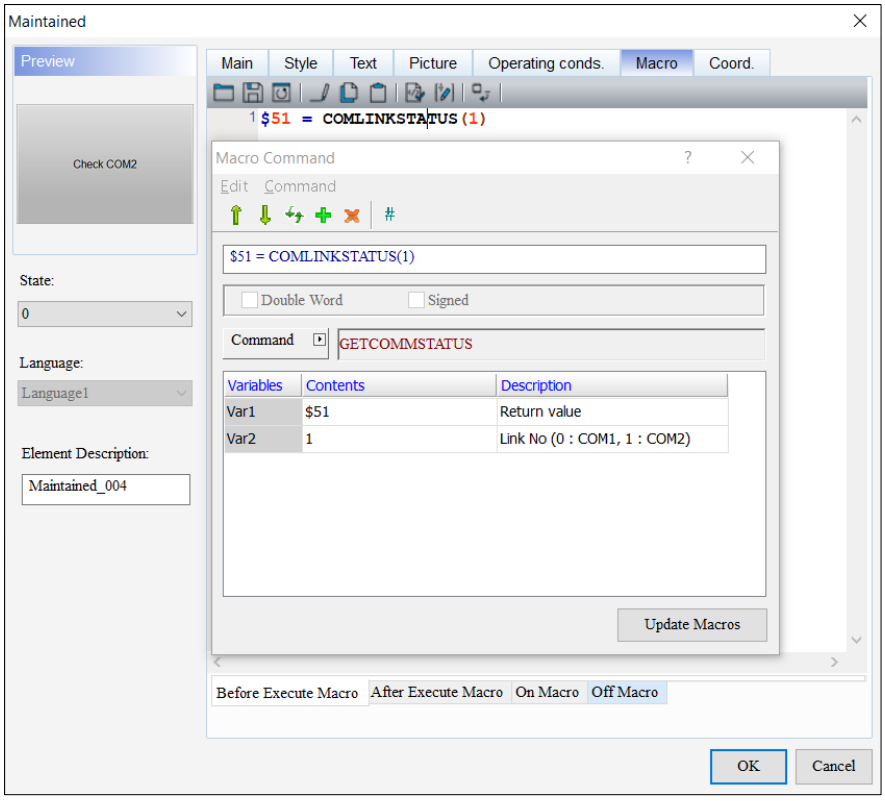
Example

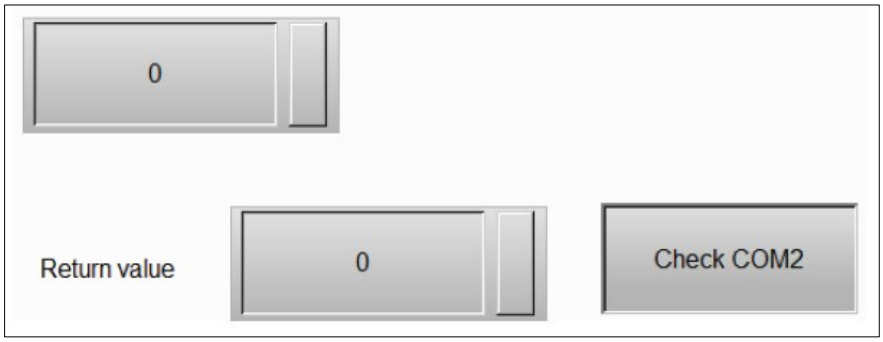
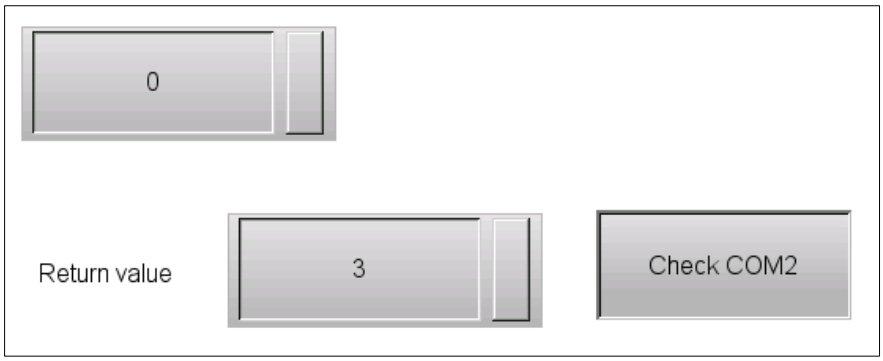
Step 1: In the **Communication Settings**, set the controller **Series** to Delta DVP PLC.



Step 2: Create two **Numeric Entry** elements on the HMI screen with the addresses as $51 and {Link2}1@D0 and the data formats as Hexadecimal.

Step 3: Create a **Maintained Button**, set the address to $100.1, and add the **On Macro** to detect the communication status of COM2.

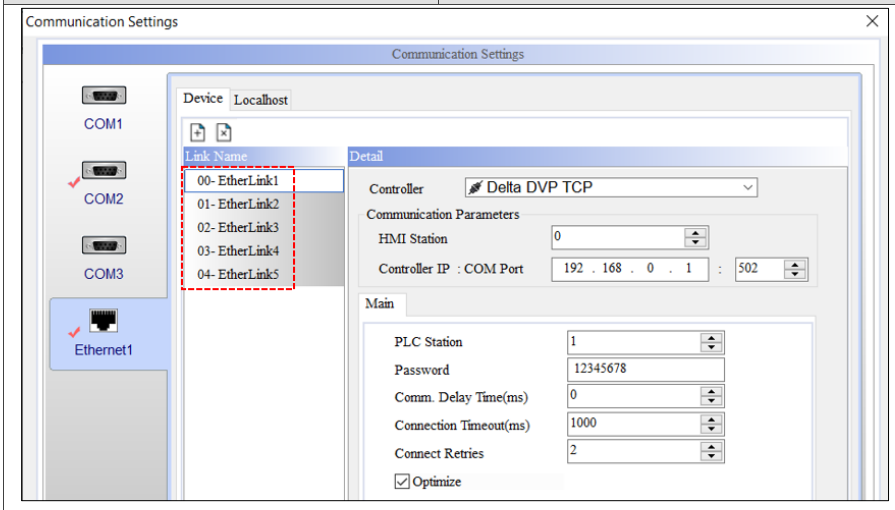


Example
Step 4: Download the project to the HMI. Under normal communication, the return value is 0 when users execute the **Maintained Button**.  **Step 5**: Remove the communication cable. When the HMI displays Communication error 3 and users execute the **Maintained Button**, the return value is 3. 

- NETLINKSTATUS (Network Connection Status)

Expression	Meaning of Variable		Note
Var1 = NETLINKSTATUS(Var2)	Var1	Return value	-
		Communication error code. When the return value is 0, it means the communication is normal.	
	Var2	Link No	
	Description of Action		
	Return the error code of Var2 of the current Link No. to Var 1.		
Note: The Link No. starts from 0.			

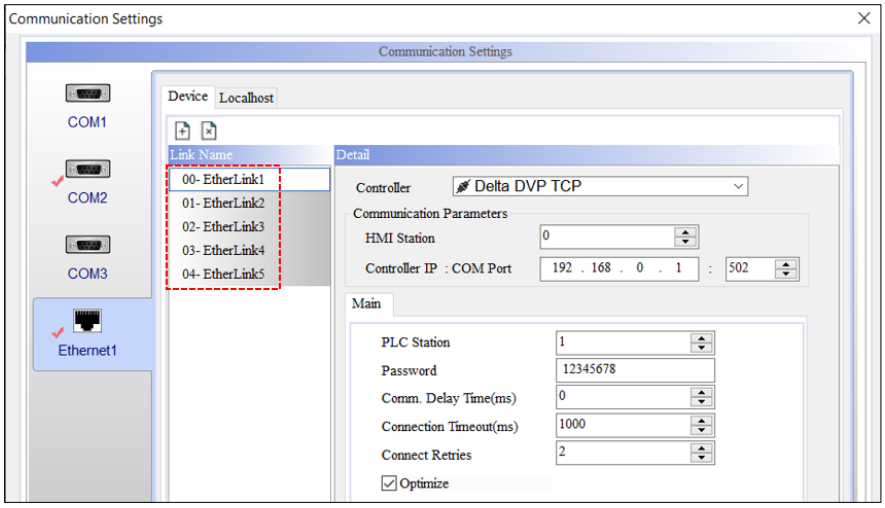
Expression	Meaning of Variable	Note



Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V

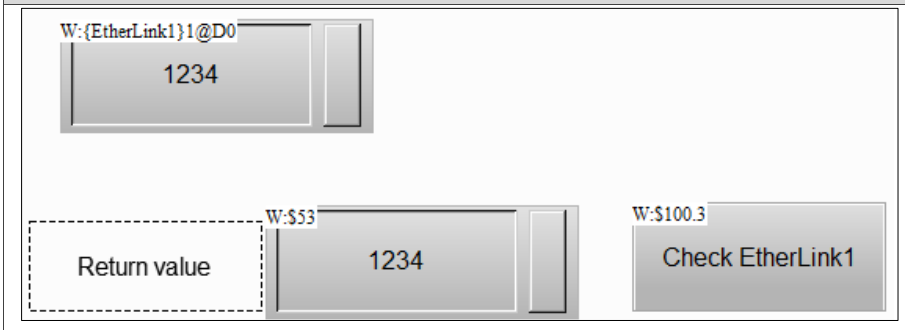
Example

Step 1: In the **Communication Settings**, set the **Controller** to Delta DVP TCP/IP.

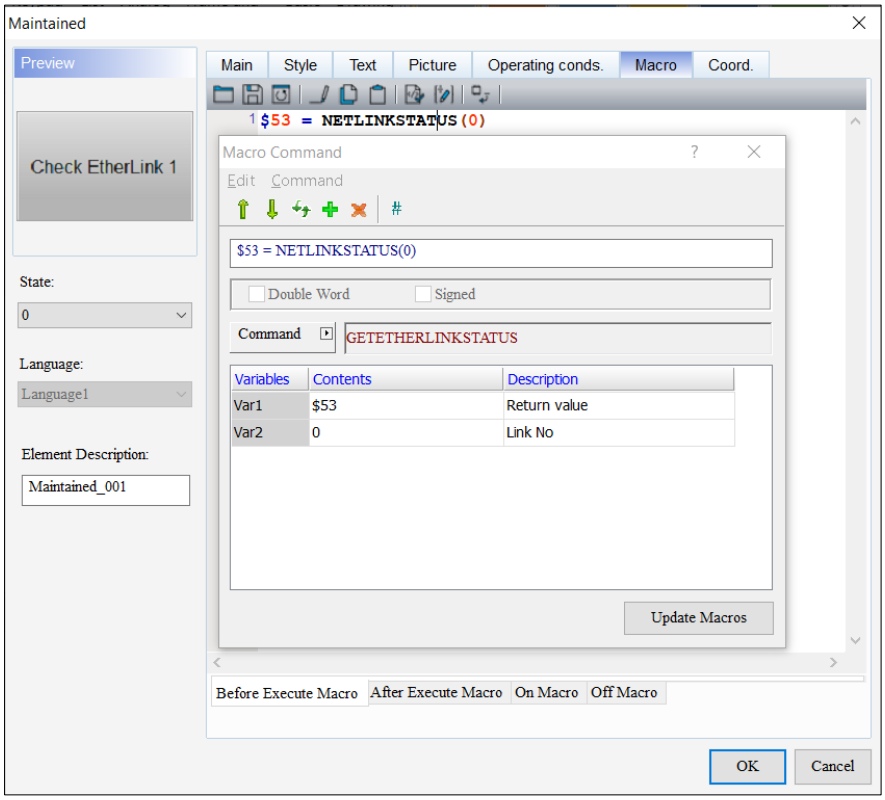


Step 2: Create two **Numeric Entry** elements on the HMI screen with the addresses as $53 and {EtherLink1}1@D0 and the data formats as Hexadecimal.

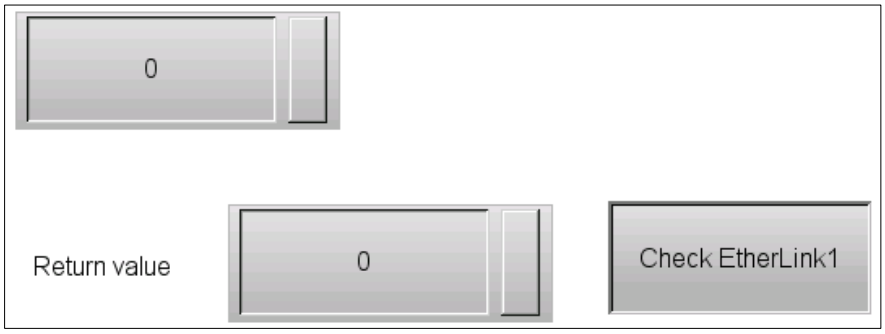
Example



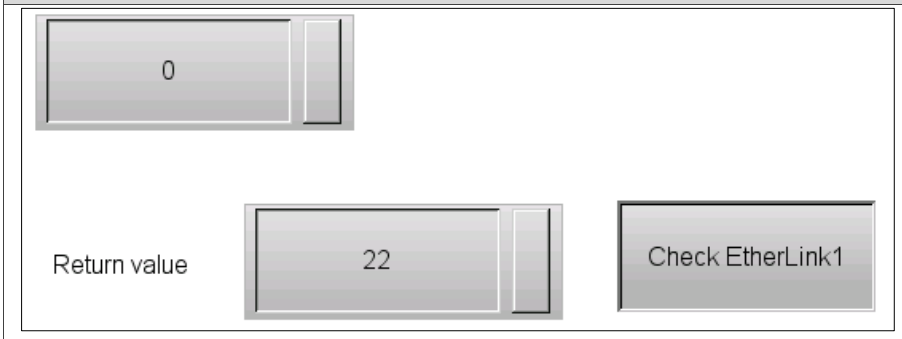
Step 3: Create a **Maintained Button**, set the address to $100.3, and add the **On Macro** to detect the communication status of Link No.0.



Step 4: Download the project to the HMI. Under normal communication, the return value is 0 when users execute the **Maintained Button**.



Step 5: Remove the network cable. When the HMI displays TCP reading error 22 and users execute the **Maintained Button**, the return value is 22.

Example


- **CLOSECOM (Disable the Communication Port)**

Expression	Meaning of Variable			Note
Var1 = CLOSECOM(Var2)	Var1	Return value		
		Failed	0	
		Succeeded	1	
	Var2	COM1	0	
		COM2	1	
		COM3	2	
	Description of Action			
	Disable the communication port and return the value indicating success or failure to Var1.			

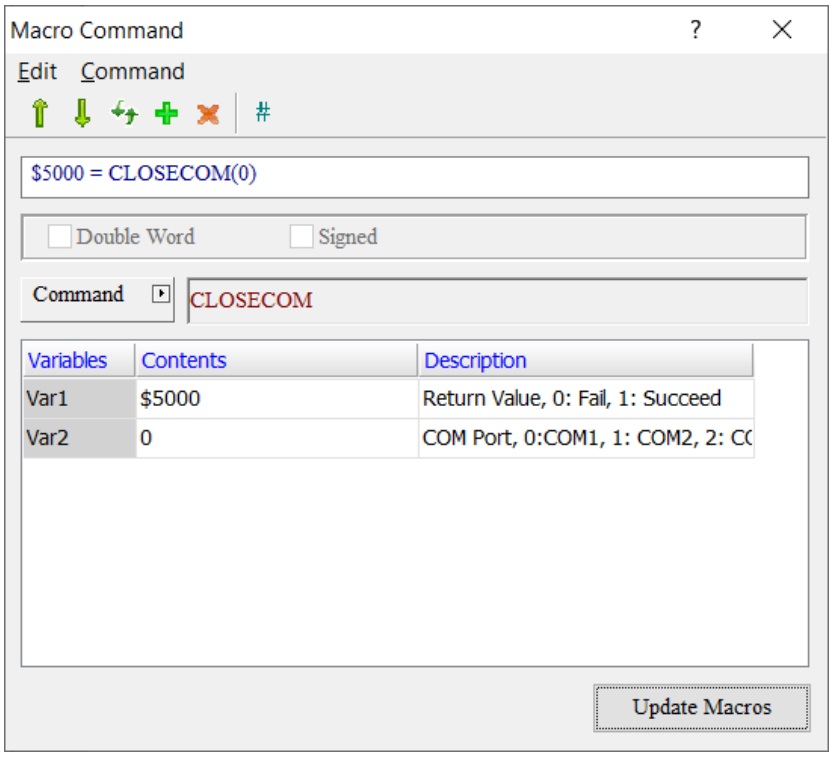
Note:

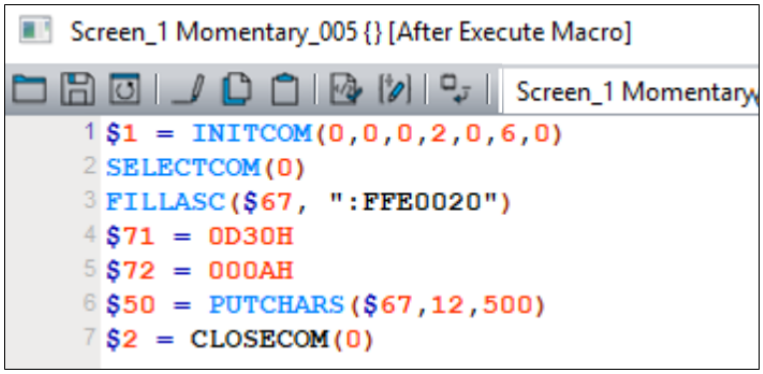
1. The CLOSECOM command must be used with INITCOM.
2. The communication port that users specify must be different from the port used for the system. All communication commands are processed based on the selected communication ports, so there are no interference or reference between the specified communication port commands used in different macros.
3. If users need to change to another communication, each time they complete the INITCOM communication, execute the CLOSECOM macro to disable the current COM communication. Next, execute the INITCOM macro for the communication parameters to be used next time.

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	V
Var2	V	V	V

Example

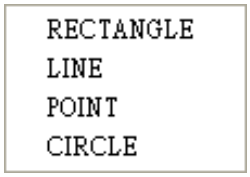
- Var1 and Var2 can be external memory, Internal Memory, or constants.





Drawing

Drawing includes commands such as RECTANGLE, LINE, POINT, CIRCLE, etc., which allow users to draw figures. The commands are described in detail as follows.



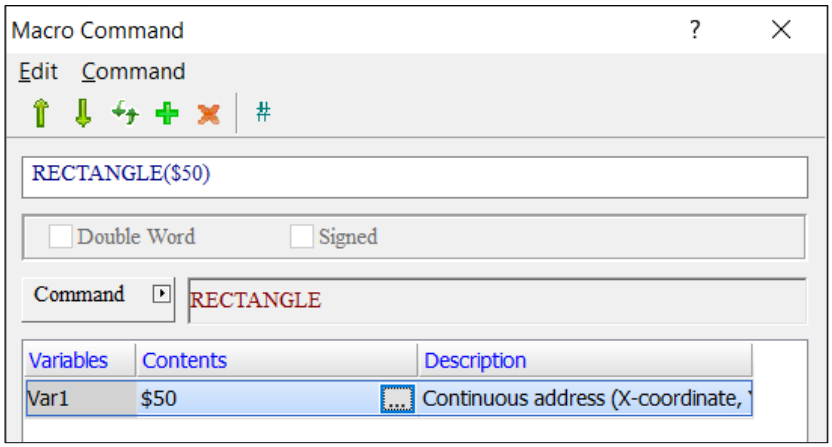
- **RECTANGLE (Rectangle)**

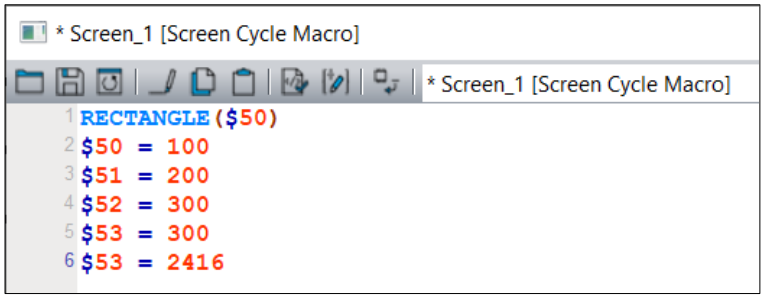
Expression	Meaning of Variable		Note
RECTANGLE(Var1) (W)	Var1	Upper-left X-coordinate	W: Word
	Var1 + 1	Upper-left Y-coordinate	
	Var1 + 2	Rectangle width	
	Var1 + 3	Rectangle height	
	Var1 + 4	Rectangle color	
	Description of Action		
	Draw a rectangle with continuous addresses.		

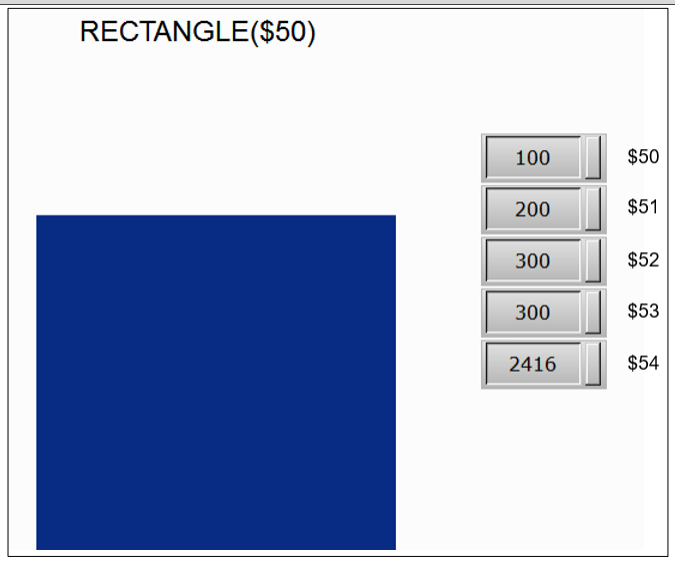
Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		

Example

- Var1 is the Internal Memory address.





Example


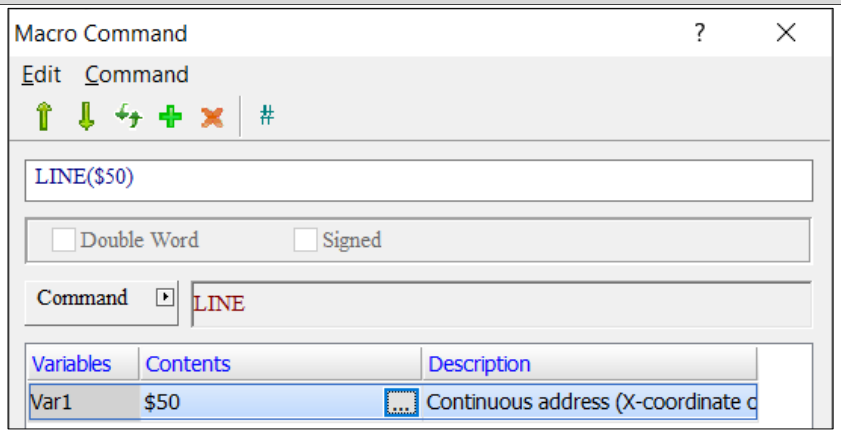
- **LINE (Line)**

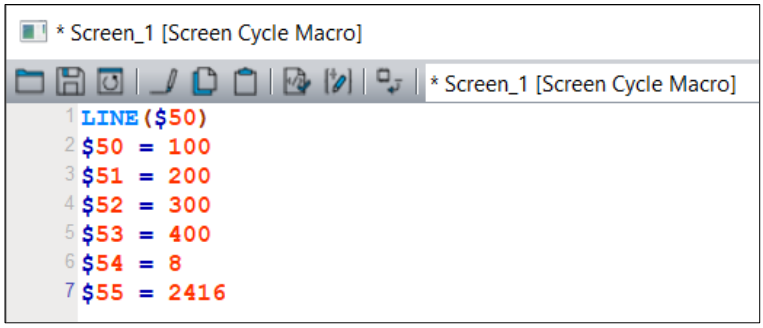
Expression	Meaning of Variable		Note
LINE(Var1) (W)	Var1	X-Coordinate of starting point	
	Var1 + 1	Y-Coordinate of starting point	W: Word
	Var1 + 2	X-Coordinate of end point	
	Var1 + 3	Y-Coordinate of end point	
	Var1 + 4	Line width	
	Var1 + 5	Line color	
	Description of Action		
	Draw a line with continuous addresses.		

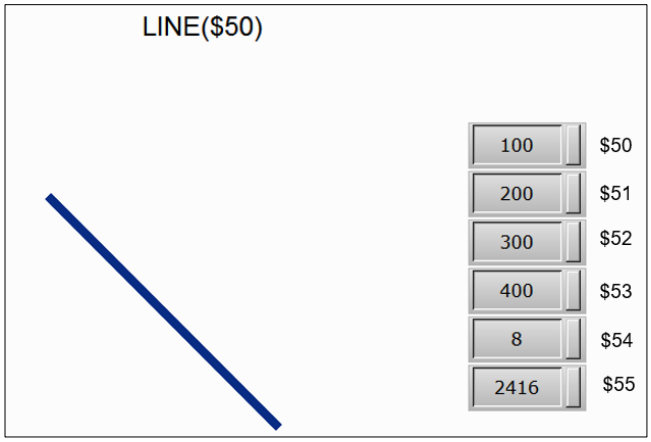
Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		

Example
• Var1 is the Internal Memory address.

Example







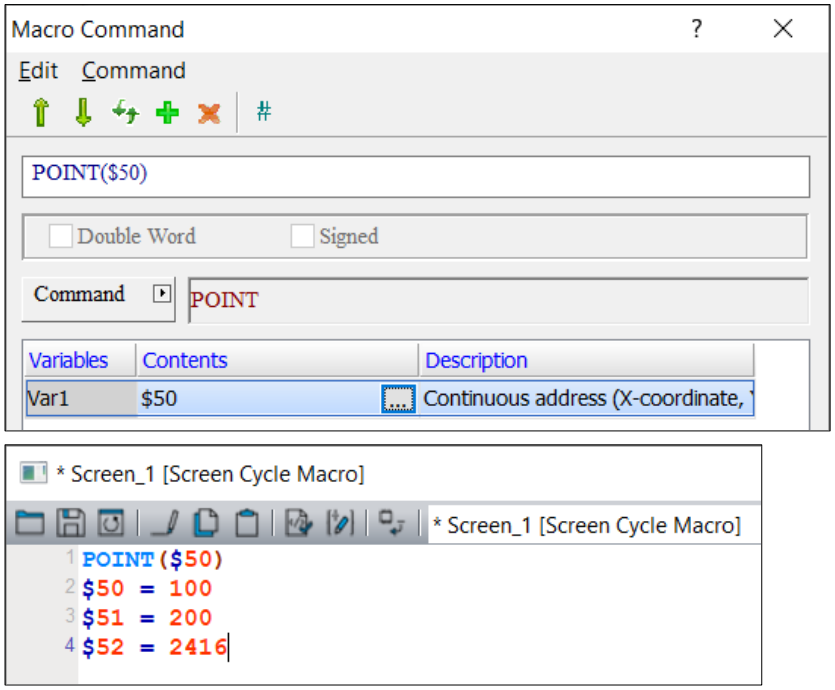
- **POINT (Point)**

Expression	Meaning of Variable		Note
POINT(Var1) (W)	Var1	X-Coordinate	W: Word
	Var1 + 1	Y-Coordinate	
	Var1 + 2	Point color	
	Description of Action		
	Draw a point with continuous addresses.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		

Example

- Var1 is the Internal Memory address.



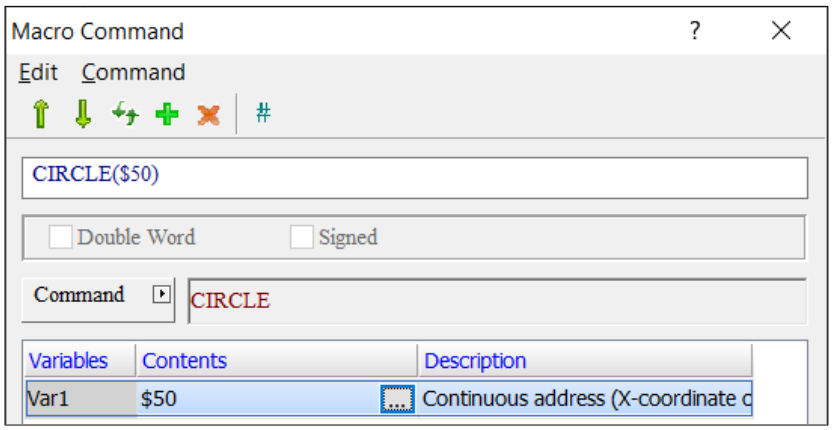
- **CIRCLE (Circle)**

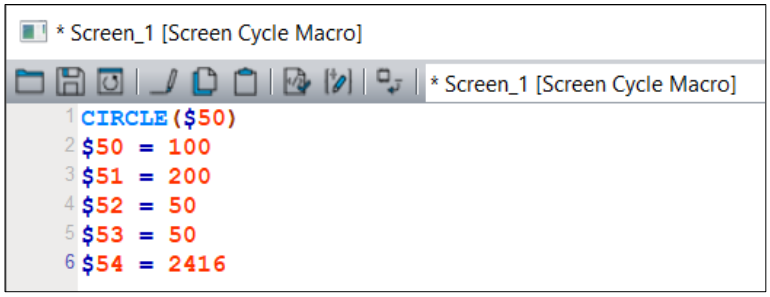
Expression	Meaning of Variable		Note
CIRCLE(Var1) (W)	Var1	X-Coordinate at the center of a circle	W: Word
	Var1 + 1	Y-Coordinate at the center of a circle	
	Var1 + 2	Circle length	
	Var1 + 3	Circle width	
	Var1 + 4	Circle color	
	Description of Action		
	Draw a circle with continuous addresses.		

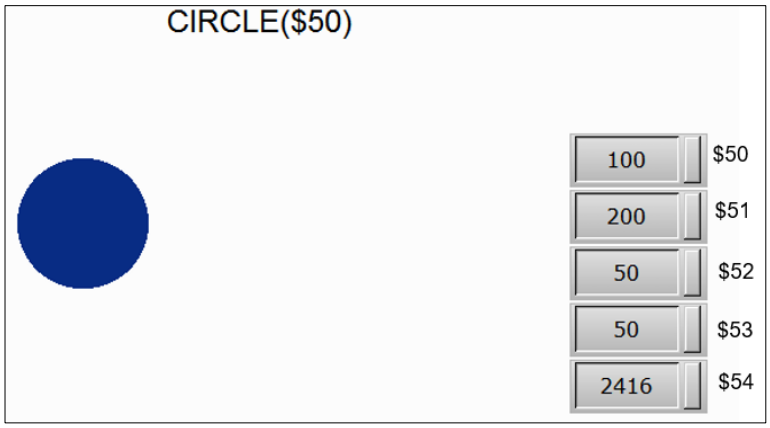
Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		

Example

- Var1 is the Internal Memory address.

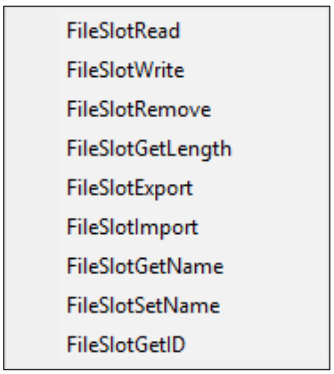






File Access

File Access includes FileSlotRead, FileSlotWrite, FileSlotRemove, FileSlotGetLength, FileSlotExport, FileSlotImport and other FileSlot related commands. The commands are described in detail as follows.



- **FileSlotRead (Read the file)**

<table>
<tr><th>Expression</th><th colspan="3">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="8">Var1 = FileSlotRead (Var2, Var3, Var4, Var5) (W)</td><td rowspan="3">Var1</td><td colspan="2">Return value</td><td rowspan="8">W: Word</td></tr>
<tr><td>Failed</td><td>0</td></tr>
<tr><td>Succeeded</td><td>1</td></tr>
<tr><td>Var2</td><td colspan="2">FileSlot ID</td></tr>
<tr><td>Var3</td><td colspan="2">Destination</td></tr>
<tr><td>Var4</td><td colspan="2">FileSlot content start address (DW)</td></tr>
<tr><td>Var5</td><td colspan="2">Word data length</td></tr>
<tr><td colspan="3">Description of Action</td></tr>
<tr><td colspan="4">Read FileSlot contents of Var2 for Var5 Word data starting from Var4 to Var3 destination address and return the results to Var1.</td></tr>
<tr><td colspan="5">Note: If the specified FileSlot file does not exist yet, use the FileSlotWrite command to create the file.</td></tr>
</table>

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V	V	V
Var3	V	V	
Var4	V	V	V
Var5	V	V	V

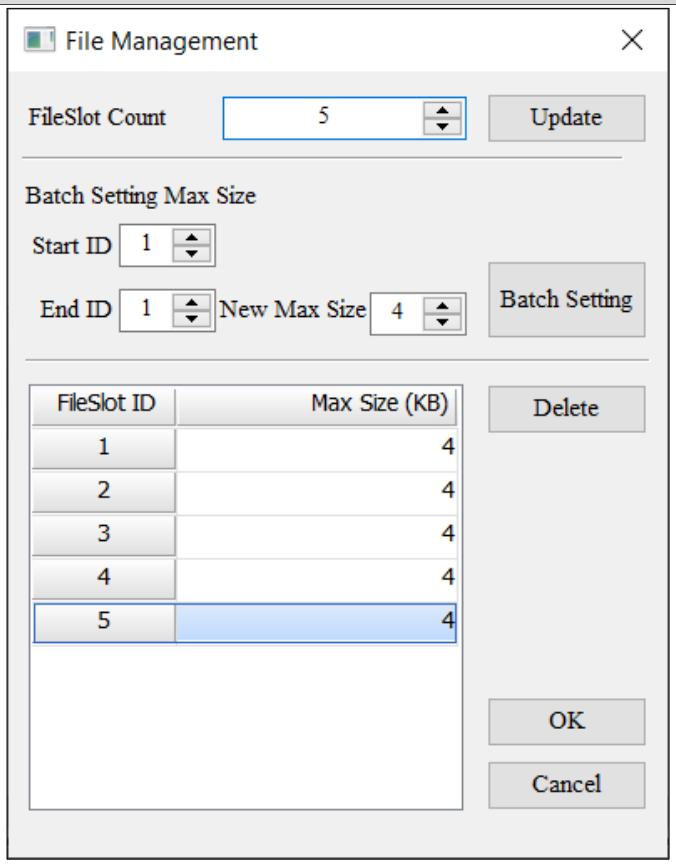
- **FileSlotWrite (Write the file)**

<table>
<tr><th>Expression</th><th colspan="3">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="8">Var1 = FileSlotWrite (Var2, Var3, Var4, Var5) (W)</td><td rowspan="3">Var1</td><td colspan="2">Return value</td><td rowspan="8">W: Word</td></tr>
<tr><td>Failed</td><td>0</td></tr>
<tr><td>Succeeded</td><td>1</td></tr>
<tr><td>Var2</td><td colspan="2">FileSlot ID</td></tr>
<tr><td>Var3</td><td colspan="2">Source</td></tr>
<tr><td>Var4</td><td colspan="2">FileSlot content start address (DW)</td></tr>
<tr><td>Var5</td><td colspan="2">Word data length</td></tr>
<tr><td colspan="3">Description of Action</td></tr>
<tr><td colspan="4">Read Var5 Word data starting from Var3, write in Var2 FileSlot from Var4 start address, and return the results to Var1.</td></tr>
</table>

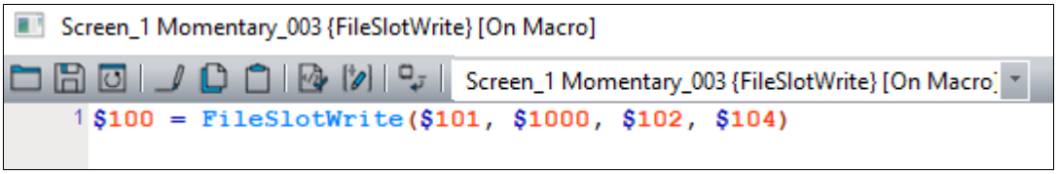
Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V	V	V
Var3	V	V	
Var4	V	V	V
Var5	V	V	V

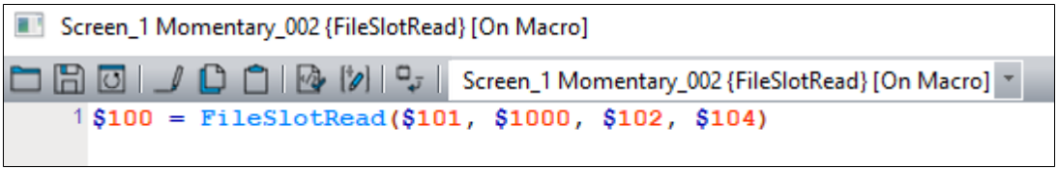
Example
• Users must use FileSlotWrite to create a file before using FileSlotRead to read the data. The procedures are as follows. 1. Go to **Project** > **Other Setting** > **Fileslot File Management** to set FileSlot.

Example



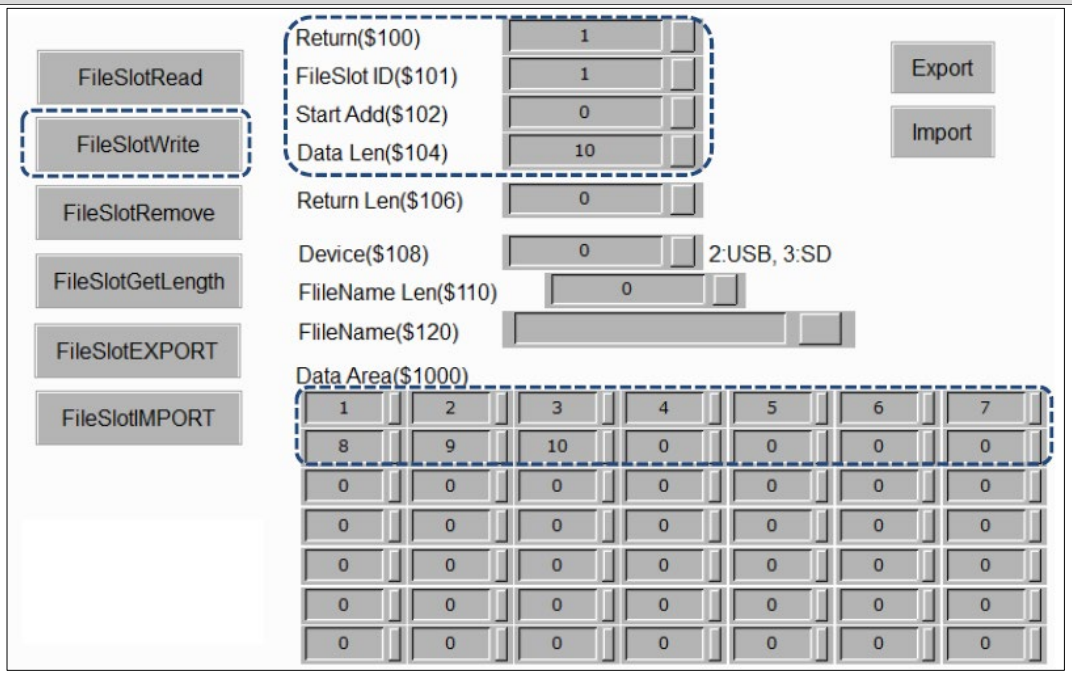
2. Create two **Momentary Button** elements and write the **On Macro**.



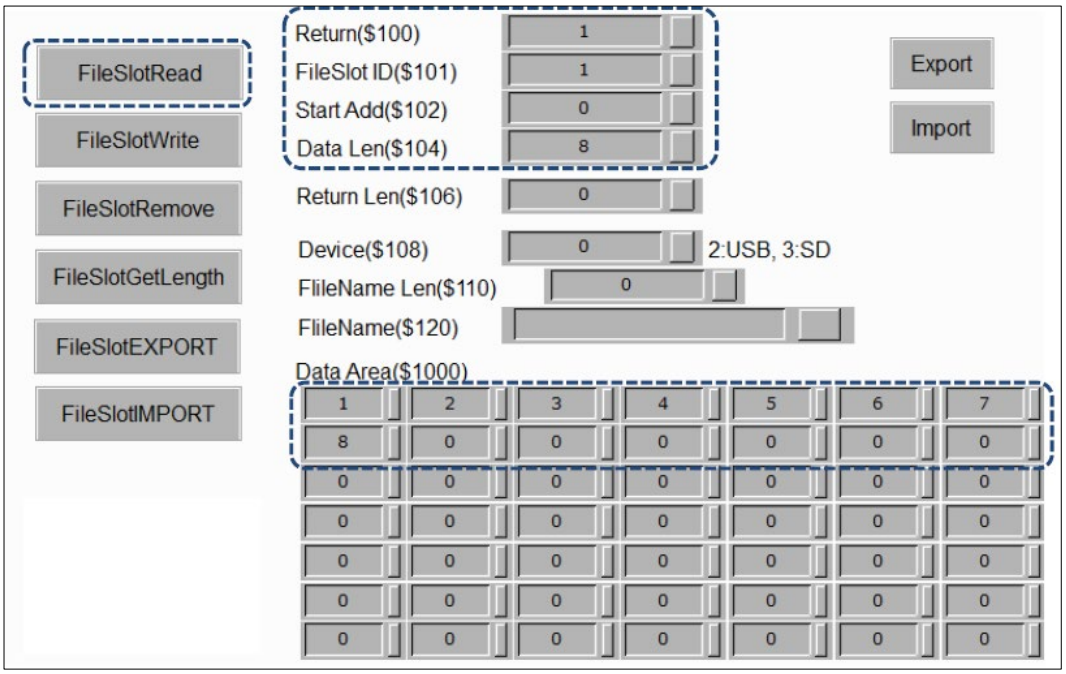


3. After compiling the screen and downloading it to the HMI, edit the data of 10 Words with $1000 as the start address. Set $101 as 1, $102 as 0, and $104 as 10. By triggering FileSlotWrite, the data of $1,000 - $1009 will be written to the FileSlot ID 1 starting from the 0th address.

Example



4. Reset the data of $1000 - $1009 to zero, and then set $104 as 8. By triggering **FileSlotRead**, the data of 8 Words starting from the 0th address of the FileSlot ID 1 will be written to $1000 to $1007.

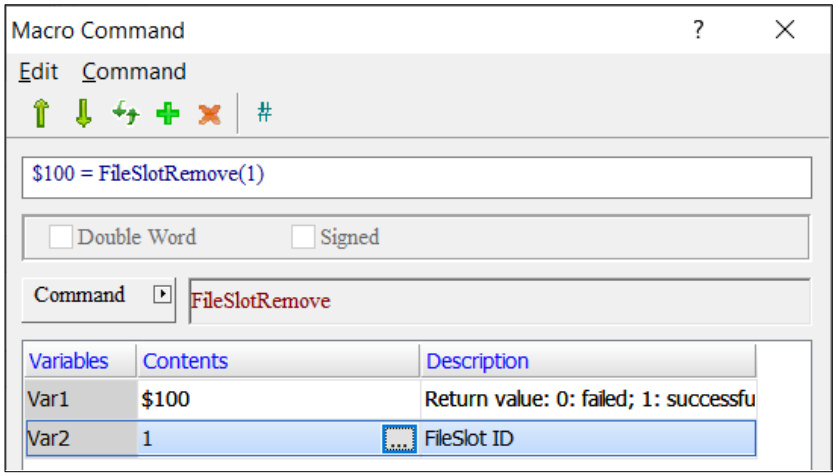


- FileSlotRemove (Remove the file)

Expression	Meaning of Variable			Note
Var1 = FileSlotRemove (Var2) (W)	Var1	Return value		W: Word
		Failed	0	

Expression	Meaning of Variable			Note
		Succeeded	1	
	Var2	FileSlot ID		
	Description of Action			
	Remove the FileSlot of Var2 and return the result to Var1.			

Variable	Type		
	Internal Memory	**PLC Register**	**Constant**
Var1	V		
Var2	V	V	V

Example
• Var1 is the Internal Memory address and Var2 is a constant. • Remove FileSlot ID 1 (Var2) and put the return value in $100 (Var1). 

- FileSlotGetLength (Read the file length)

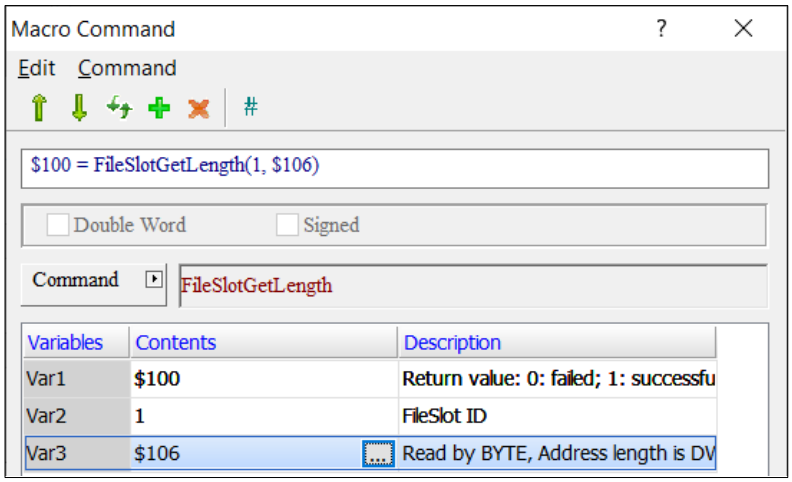
Expression	Meaning of Variable			Note
Var1 = FileSlotGetLength (Var2, Var3) (W)	Var1	Return value		W: Word
		Failed	0	
		Succeeded	1	
	Var2	FileSlot ID		
	Var3	FileSlot LengthReturn value(DW)		
	Description of Action			

Expression	Meaning of Variable	Note
	Store the length of the Var2 FileSlot to Var3 and return the result to Var1.	
Note: Unit of read length is Byte.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V	V	V
Var3	V	V	

Example

- Var1 and Var3 are Internal Memory addresses, and Var2 is a constant.
- Get the length of FileSlot ID 1 (Var2) and save it to $106, and put the return value in $100 (Var1). If the FileSlot length is 10 words, the value returned to $106 is 20 (Byte).



- FileSlotExport (Export the file)

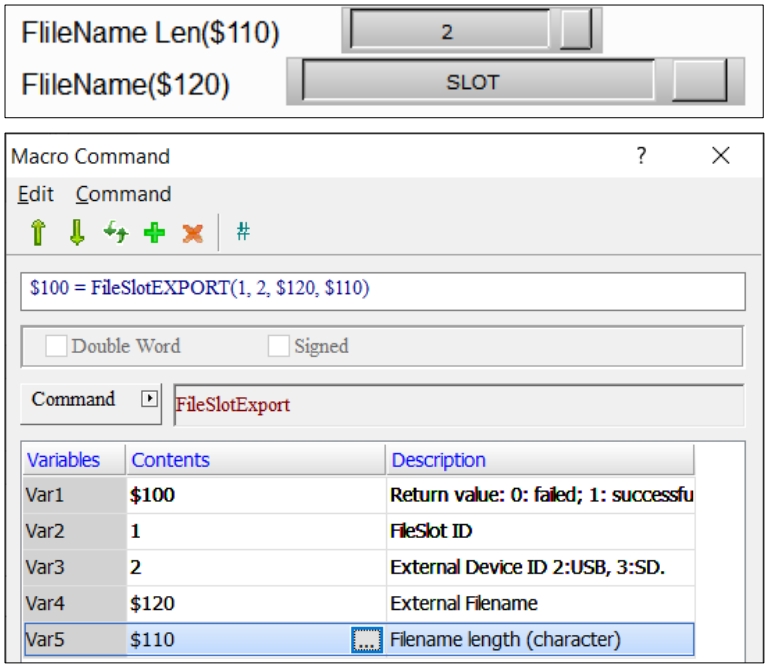
Expression	Meaning of Variable				Note
Var1 = FileSlotExport (Var2, Var3, Var4, Var5) (W)	Var1	Return value			W: Word
		Failed	0		
		Succeeded	1		
	Var2	FileSlot ID			
	Var3	File export device	USB Disk	2	
			SD Card	3	

Expression	Meaning of Variable		Note
	Var4	Name of the file exported	
	Var5	Name length of the file exported	
	Description of Action		
	Export the Var2 FileSlot to the external storage device Var3, name the file as Var4, and return the result to Var1.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V	V	V
Var3	V	V	V
Var4	V	V	
Var5	V	V	V

Example

- Var1, Var4, and Var5 are Internal Memory addresses, and Var2 and Var3 are constants.
- Export FileSlot ID 1 (Var2) to USB Disk (Var3), which name length of the file is 2 (Var5) and file name is Slot (Var4) and put the return value in $100 (Var1).



- FileSlotImport (Import the file)

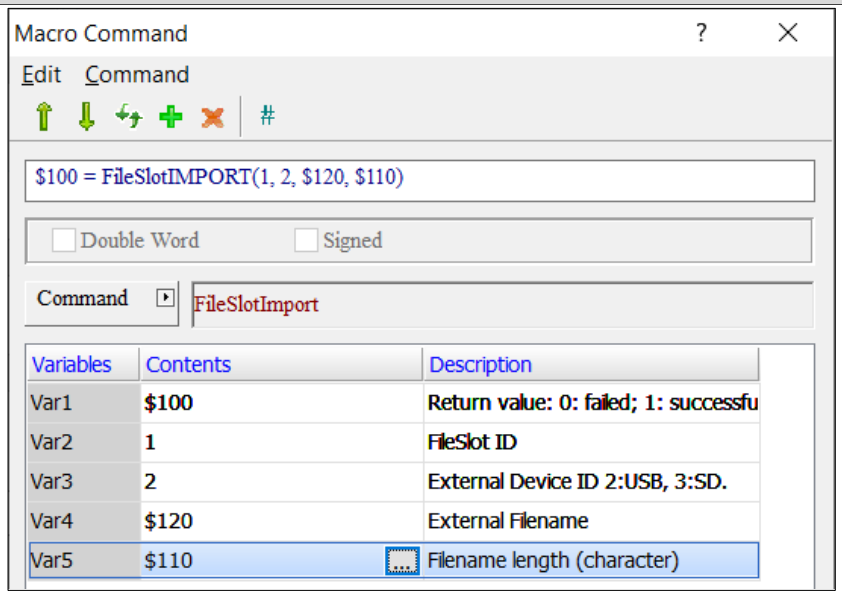
<table>
<tr><th>Expression</th><th colspan="4">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="8">Var1 = FileSlotImport (Var2, Var3, Var4, Var5) (W)</td><td rowspan="3">Var1</td><td colspan="3">Return value</td><td rowspan="8">W: Word</td></tr>
<tr><td>Failed</td><td colspan="2">0</td></tr>
<tr><td>Succeeded</td><td colspan="2">1</td></tr>
<tr><td>Var2</td><td colspan="3">FileSlot ID</td></tr>
<tr><td rowspan="2">Var3</td><td rowspan="2">Device to export the file</td><td>USB Disk</td><td>2</td></tr>
<tr><td>SD Card</td><td>3</td></tr>
<tr><td>Var4</td><td colspan="3">Name of the file imported</td></tr>
<tr><td>Var5</td><td colspan="3">Length of the filename to be imported</td></tr>
<tr><td colspan="5">Description of Action</td></tr>
<tr><td colspan="5">Import the file named Var4 in the external storage device Var3 to the Var2 FileSlot and send the return value to Var1.</td></tr>
</table>

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V	V	V
Var3	V	V	V
Var4	V	V	
Var5	V	V	V

Example

- Var1, Var4, and Var5 are Internal Memory addresses, and Var2 and Var3 are constants.
- Import the file named Slot (Var4) with the length of 2 (Var5) in USB Disk (Var3) to the FileSlot ID 1 (Var2) and put the return value in $100 (Var1).

Example



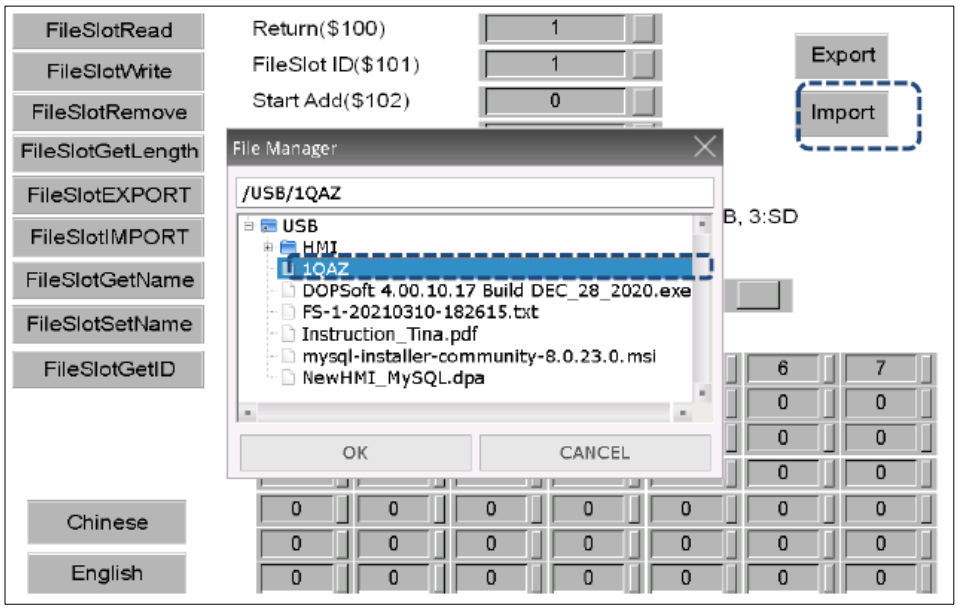
- FileSlotGetName (Get the filename)

Expression	Meaning of Variable			Note
Var1 = FileSlotGetName(Var2, Var3, Var4)	Var1	Return value		
		Failed	0	
		Succeeded	1	
	Var2	FileSlot ID		
	Var3	Filename		
	Var4	Length of the filename (read in units of Byte)		
	Description of Action			
	Store the filename of Var2 (FileSlot ID) referencing the length specified by Var4 and return the result to Var1.			

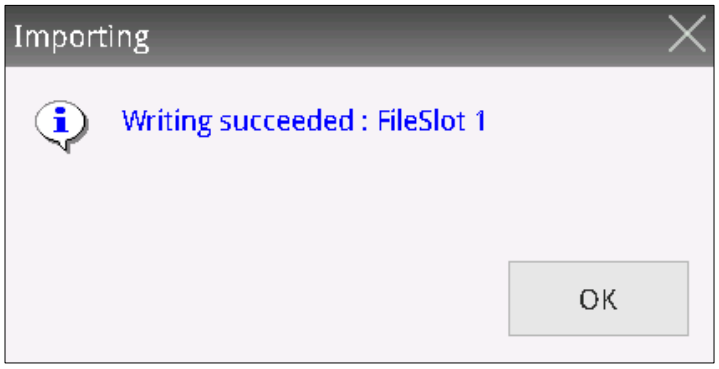
Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V	V	V
Var3	V	V	
Var4	V	V	V

Example

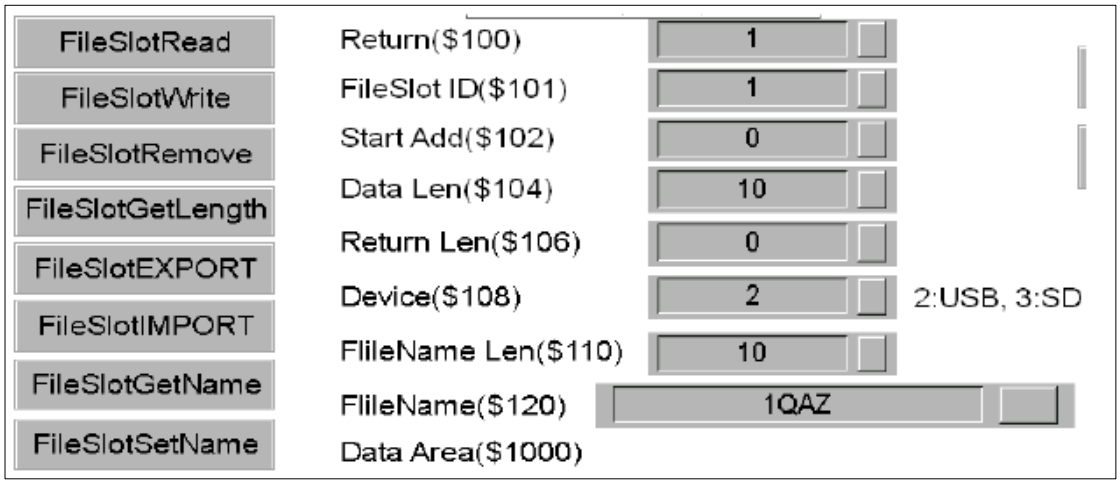
- Import the FileSlot file, and then get the filename through the imported FileSlot ID.



- When the **Import** is successful, the following message appears.



- Execute the **FileSlotGetName** command to get the filename of the file that is just imported.



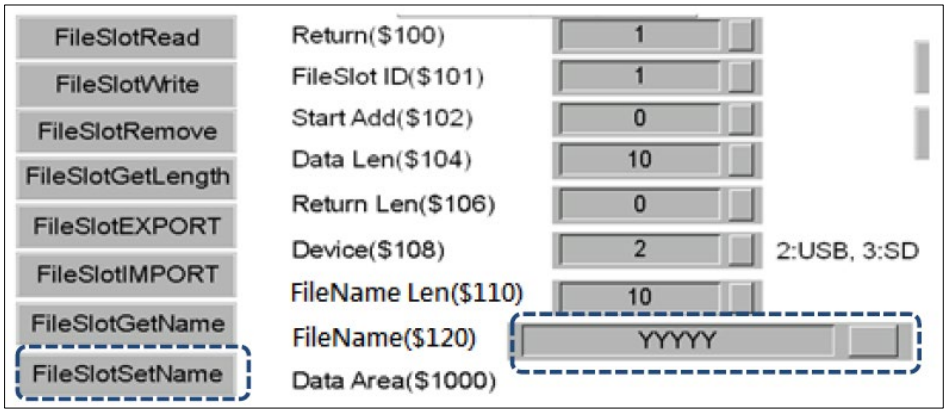
- **FileSlotSetName (Set the filename)**

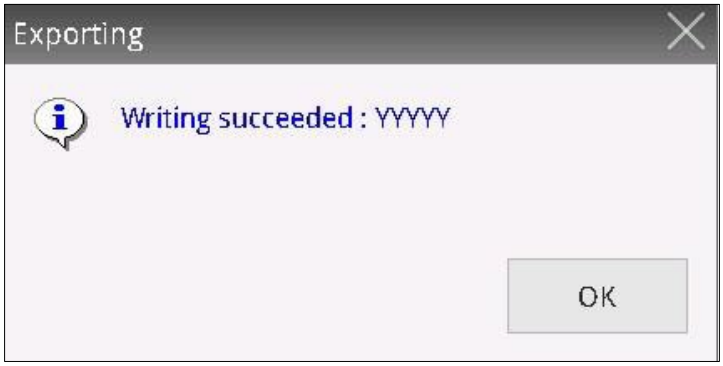
<table>
<tr><th>Expression</th><th colspan="3">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="7">Var1 = FileSlotSetNaSme(Var2, Var3, Var4)</td><td rowspan="3">Var1</td><td colspan="2">Return value</td><td rowspan="7"></td></tr>
<tr><td>Failed</td><td>0</td></tr>
<tr><td>Succeeded</td><td>1</td></tr>
<tr><td>Var2</td><td colspan="2">FileSlot ID</td></tr>
<tr><td>Var3</td><td colspan="2">Filename</td></tr>
<tr><td>Var4</td><td colspan="2">Length of the filename (read in units of Byte)</td></tr>
<tr><td colspan="3">Description of Action</td></tr>
<tr><td colspan="4">Set the filename of Var2 (FileSlot ID) to Var3 referencing the length specified by Var4 and then return the result to Var1.</td></tr>
</table>

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V	V	V
Var3	V	V	
Var4	V	V	V

Example

This command modifies the name of the imported FileSlot file.



Example
After users modified the filename, **Export** the file to the USB Disk or SD Card. 

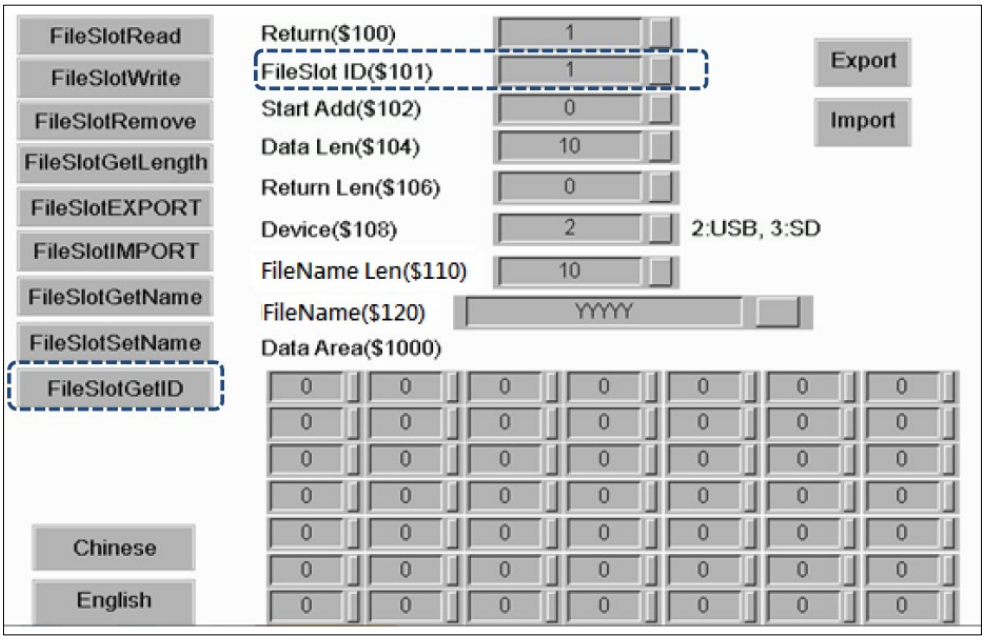
- **FileSlotGetID (Get the file ID)**

Expression	Meaning of Variable			Note
Var1 = FileSlotGetID (Var2, Var3, Var 4) (W)	Var1	Return value		
		Failed	0	
		Succeeded	1	
	Var2	Filename		
	Var3	Length of the filename (read in units of Byte)		
	Var4	FileSlot ID		
	Description of Action			
	Refer to the length of the filename specified by Var3, store the FileSlot ID of the filename specified by Var2 to Var4, and return the result to Var1.			

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V	V	V
Var3	V	V	V
Var4	V	V	

Example

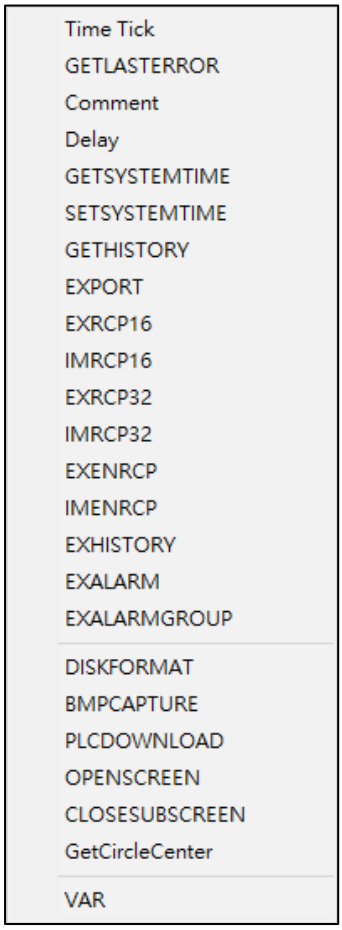
Use the filename of the FileSlot file to get the corresponding FileSlot ID.



Others

Others include Time Tick, GETLASTERROR, Comment, Delay, GETSYSTEMTIME, SETSYSTEMTIME, GETHISTORY, EXPORT, EXRCP, IMRCP, EXENRCP, IMENRCP, EXHISTORY, EXALARM, EXALARMGROUP, DISKFORMAT, BMPCAPTURE, PLCDOWNLOAD, OPENSCREEN, CLOSESUBSCREEN, GetCircleCenter, VAR, and other commands.

The commands are described in detail as follows.

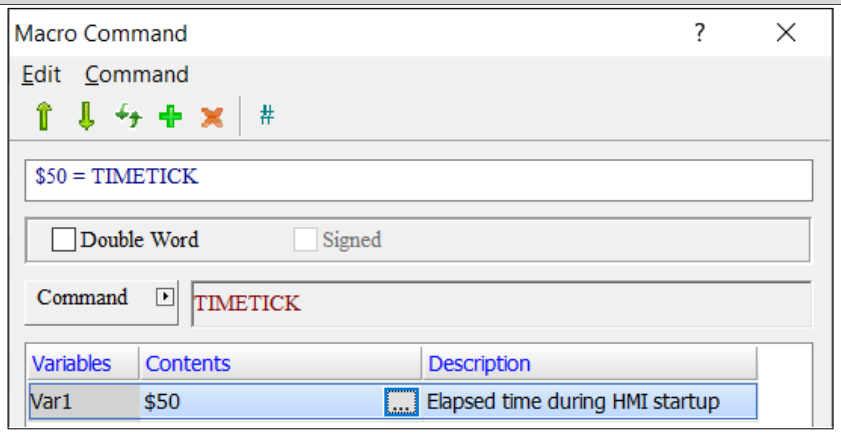


- **Time Tick (get the time from the start of the system up to now)**

Expression	Meaning of Variable		Note
Var1 = TIMETICK (W) Var1 = TIMETICK (DW)	Var1	Elapsed time during HMI startup	W: Word DW: Double Word
	Description of Action		
	Get the time from the start of the system up to now and put it in Var1 (unit: ms).		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		

Example
• Var1 is the Internal Memory address. • Put elapsed time during the HMI startup in $50.

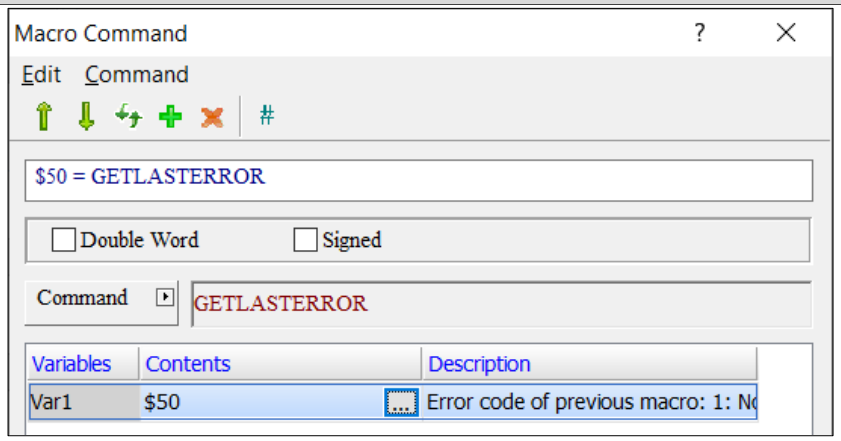
Example


- **GETLASTERROR (Get Error Value of Previous Command)**

Expression	Meaning of Variable		Note
Var1 = GETLASTERROR (W) Var1 = GETLASTERROR (DW) Var1 = GETLASTERROR (Signed W) Var1 = GETLASTERROR (Signed DW)	Var1	Error code of previous macro 1: Succeeded Negative value: error (Refer to Macro Error Codes)	W: Word DW: Double Word Signed: Signed number
	Description of Action		
	Get the error value of the previous macro command and put the result in Var1.		
Note: This command must follow a macro command with an error to obtain its error value.			

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		

Example
• Var1 is the Internal Memory address. • Get the result of the error value of the previous macro command and put it in $50.

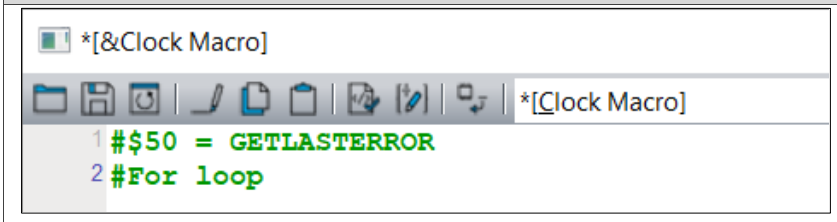
Example


- **COMMENT (Comments)**

Expression	Meaning of Variable		Note
#Var1 (W)	Var1	Contents of command	W: Word
	Description of Action		
	Comment Var1.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		

Example
Add # before the command to make a comment on the command. # can also be used for describing the purpose of a macro.

Example


- **Delay (Delay)**

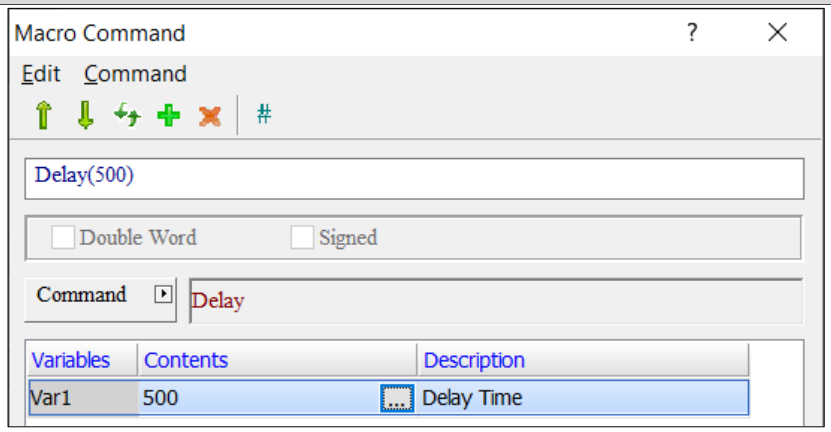
Expression	Meaning of Variable		Note
Delay(Var1) (W)	Var1	Delay time	W: Word
	Description of Action		
	Delay the time of Var1 before executing commands of the next line. The unit is ms.		

Note:

1. Due to multitasking of the HMI, a system delay may occur by default. If users set this Delay command, the delay time may increase because of the busy operations of the system, and the command will not be executed in advance.
2. Excessive Delay setting will result in slow response of the HMI.
3. When the Delay command is executed, the HMI will suspend all actions until the Delay time is over.

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		V

Example
• Var1 is a constant. • Set the Delay time to 500 ms.

Example


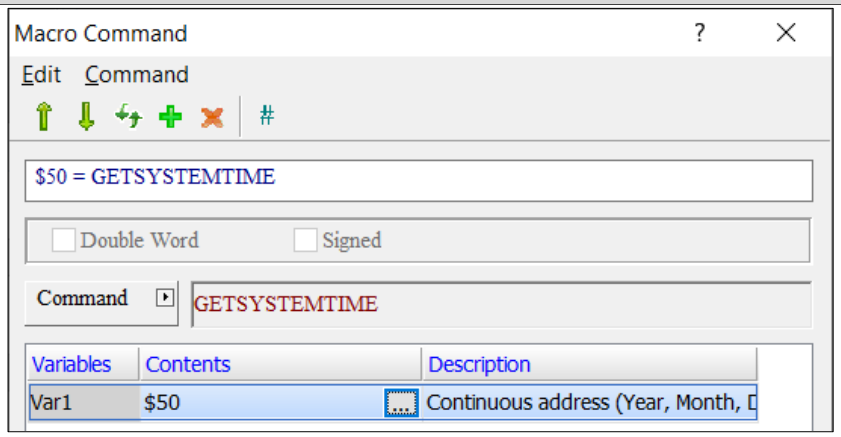
- **GETSYSTEMTIME (Get System Time)**

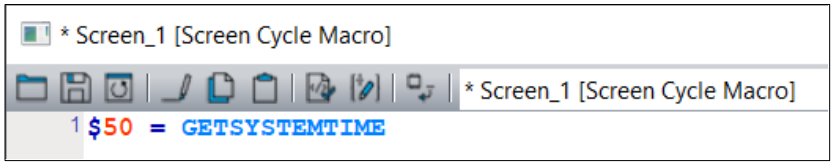
Expression	Meaning of Variable		Note
Var1 = GETSYSTEMTIME (W)	Var1	Year	W: Word
	Var1 + 1	Month	
	Var1 + 2	Day	
	Var1 + 3	Week	
	Var1 + 4	Hour	
	Var1 + 5	Minute	
	Var1 + 6	Second	
	Description of Action		
	Get the system time from Var1 to Var7 for 7 consecutive Words address.		

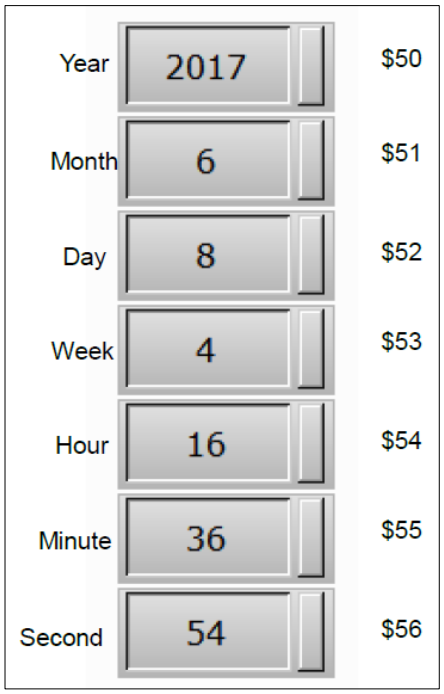
Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		

Example
• Var1 is the Internal Memory address. • Put the current system time in $50 to $56.

Example







- **SETSYSTEMTIME (Set System Time)**

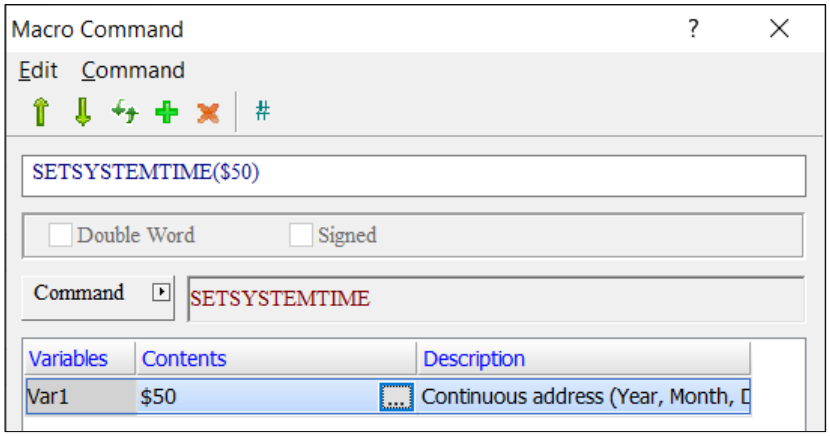
Expression	Meaning of Variable		Note
Var1 = SETSYSTEMTIME (W)	Var1	Year	W: Word
	Var1 + 1	Month	
	Var1 + 2	Day	
	Var1 + 3	Week	
	Var1 + 4	Hour	

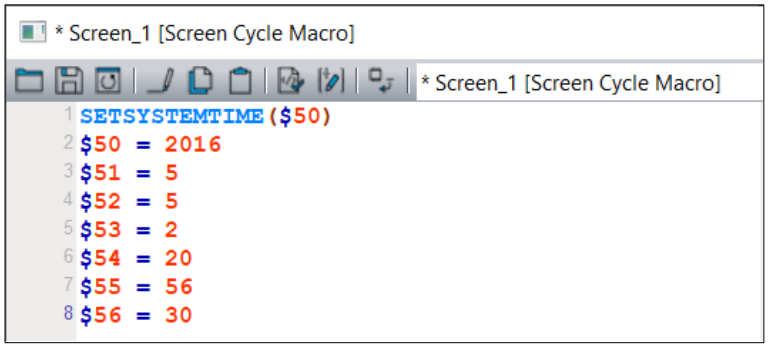
Expression	Meaning of Variable		Note
	Var1 + 5	Minute	
	Var1 + 6	Second	
	Description of Action		
	Set the system time from Var1 to Var7 for 7 consecutive Words address.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		

Example

- Var1 is the Internal Memory address.
- Set the current system time and put it in $50 to $56.





Example

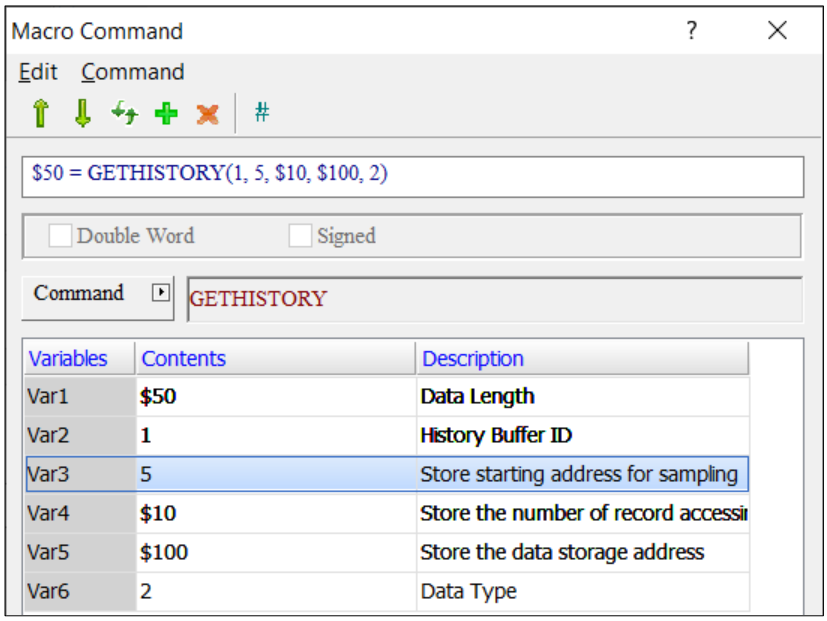
- **GETHISTORY (Get History Data)**

Expression	Meaning of Variable			Note
Var1 = GETHISTORY(Var2, Var3, Var4, Var5, Var6) (W)	Var1	Store Data Length		W: Word
	Var2	History Buffer ID		
	Var3	Read starting address for sampling		
	Var4	Read points		
	Var5	Data storage address		
	Var6	Data Type	Data	0
			Time	1
			Date and Time	2
	Description of Action			
	Get history data.			
Note: Double Word is recommended to be used for Var1, Var3, and Var4. If the continuous address of Word is used, data may be overwritten, and the result may be affected.				

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		V
Var3	V		V
Var4	V		V
Var5	V	V	
Var6	V		V

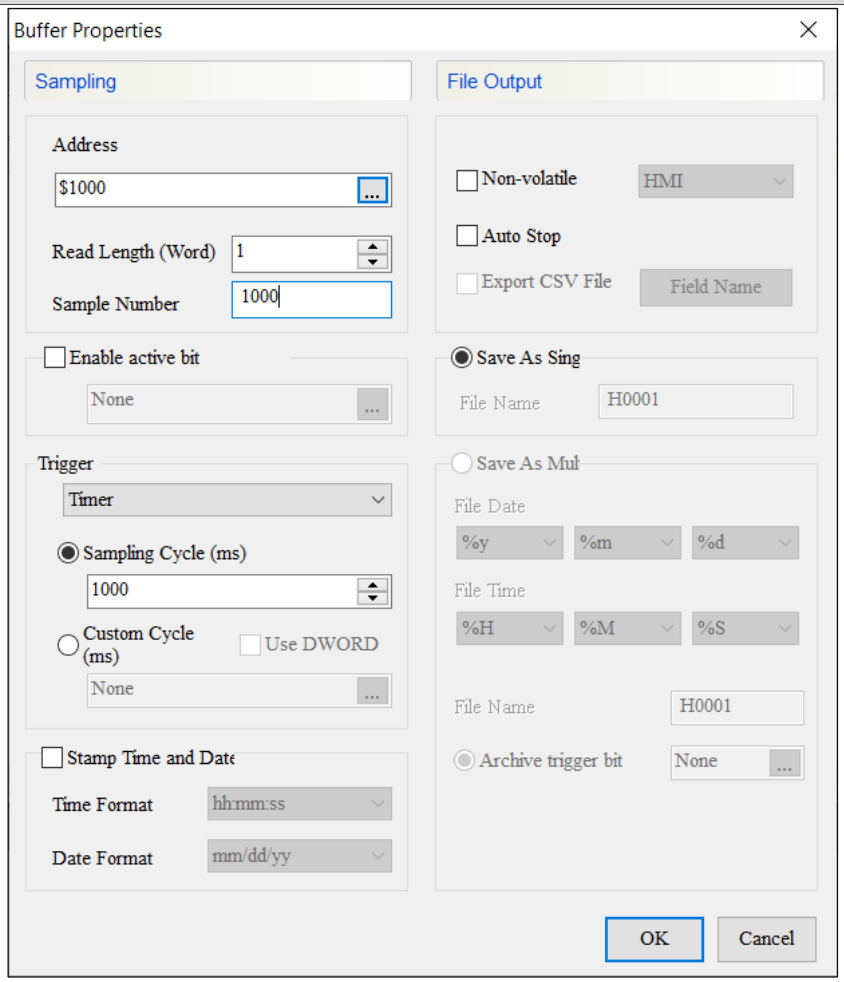
Example

- Var1 and Var5 are Internal Memory addresses, and Var2, Var3, Var4, and Var6 are constants.
- Set the History Buffer ID as 1 (Var2), sample from the fifth data (Var2) to the tenth data (Var3), put the data type (including time and data) set as 2 (Var6) in the continuous address (Var5) of $100, and finally put the obtained Data Length in $50 (Var1).

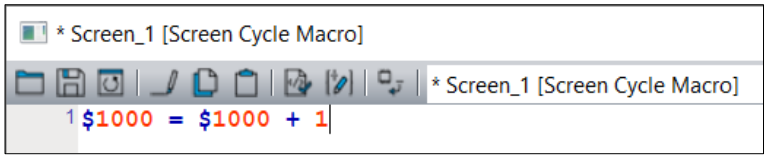


- Set the **Read Address** $1000 as the history buffer **Sampling** data **Address**.

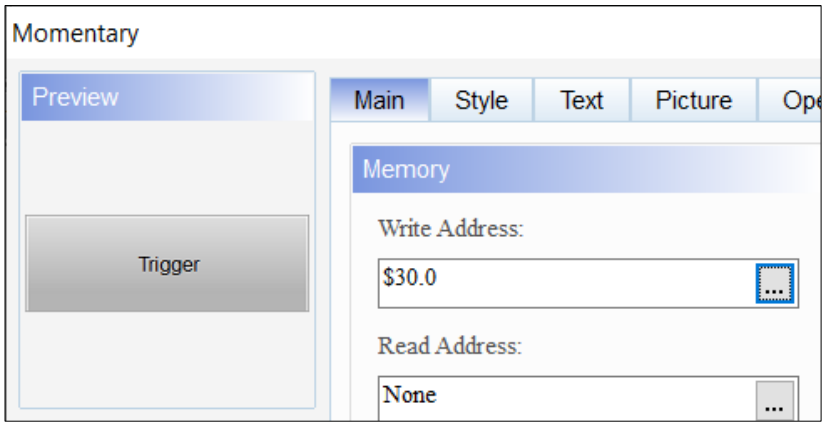
Example



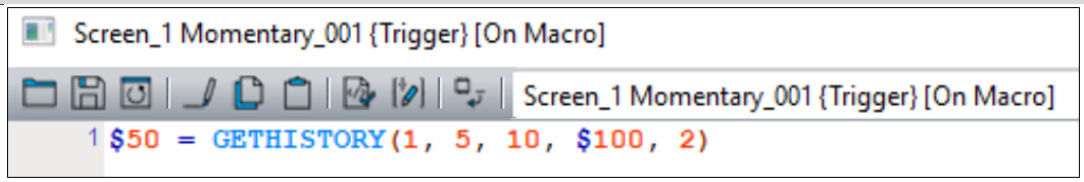
- Edit the **Screen Cycle Macro** to add up the history data of $1000.



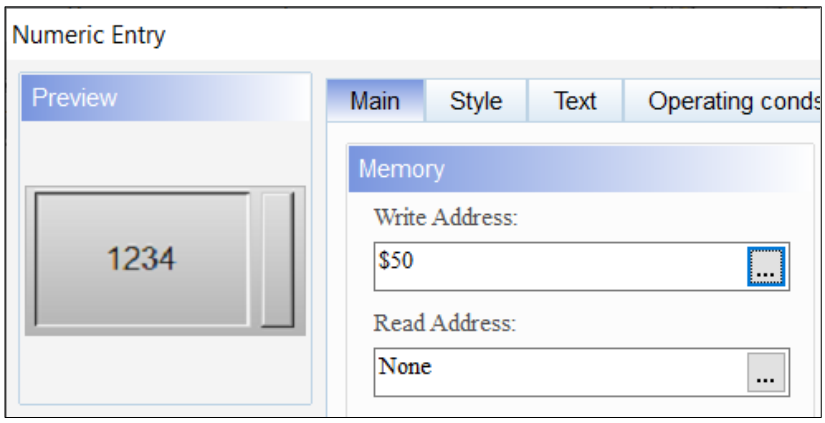
- Create a **Momentary Button**, edit the **On Macro**, and set the GETHISTORY command as follows.



Example



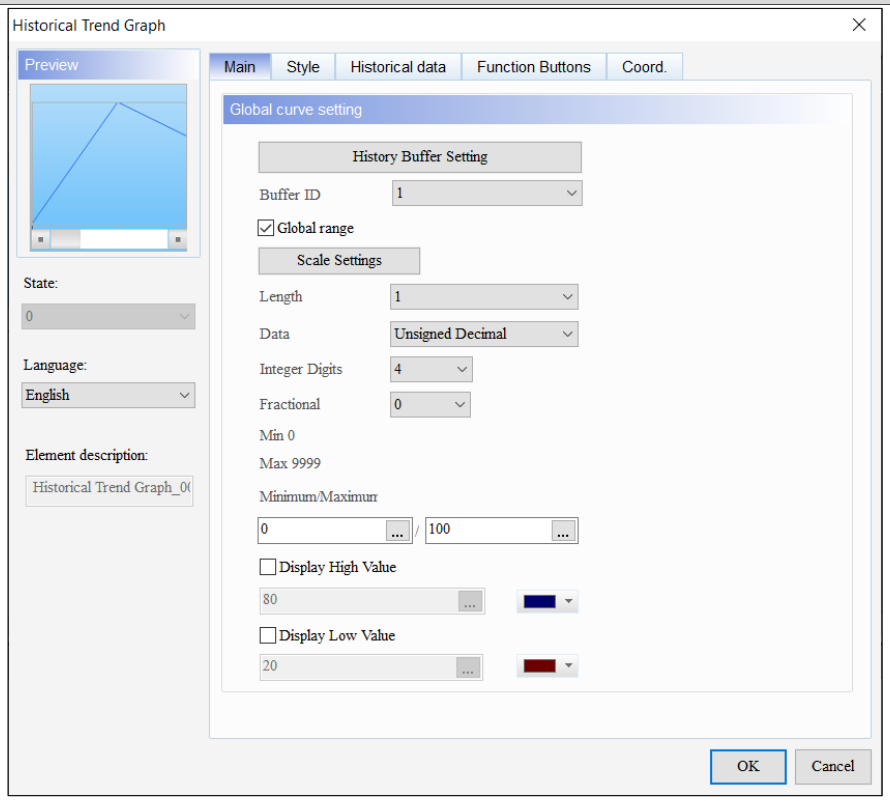
- Create a **Numeric Entry** element and set the **Write Address** as $50.

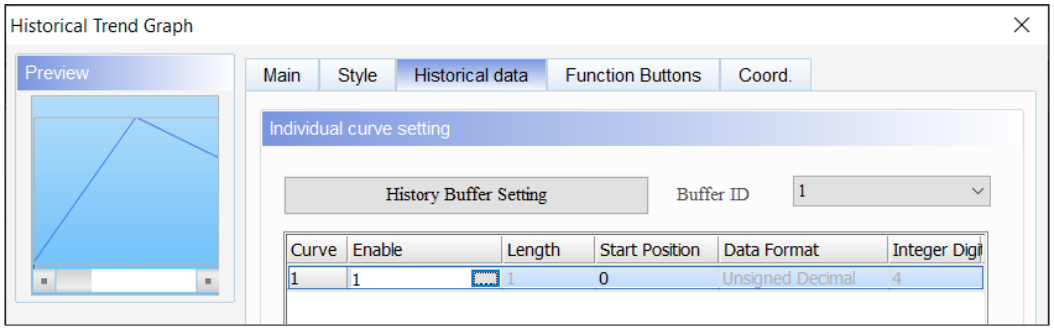


- Create **Numeric Entry** elements of $100 to $134, as shown as follows.

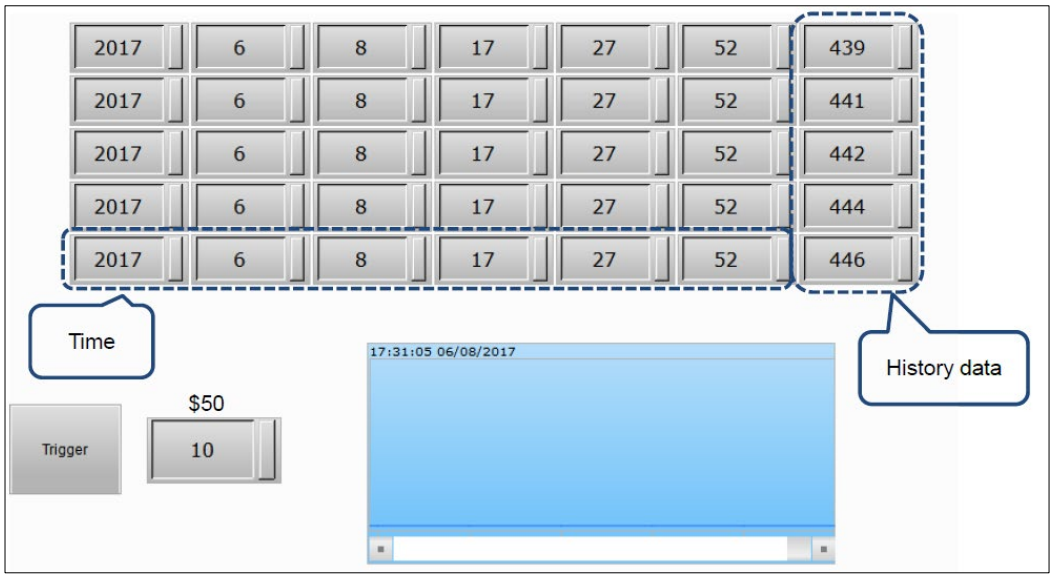
- Create a **Historical Trend Graph** and enable it.

Example





- After compiling the screen, the execution result is as follows:



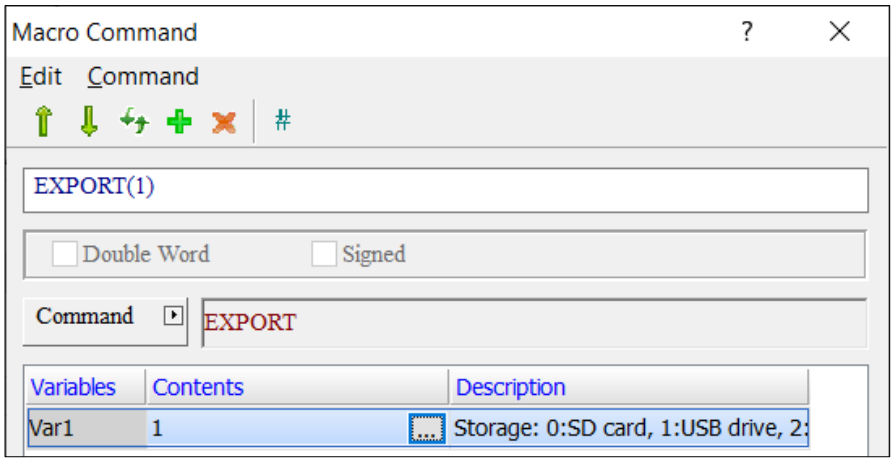
- **EXPORT (Export Data)**

Expression	Meaning of Variable				Note
EXPORT(Var1) (W)	Var1	Data export device	SD Card	0	W: Word
			USB Disk	1	
			Printer	2	
	Description of Action				
	Export and print historical and alarm data to external storage device Var1.				

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		V

Example

- Var1 is a constant.
- Export data to the USB Disk.



- **EXRCP16 / EXRCP32 (Export 16-bit Recipe / Export 32-bit Recipe)**

Expression	Meaning of Variable			Note
Var1 = EXRCP16(Var2, Var3) (W) Var1 = EXRCP32(Var2, Var3) (W)	Var1	Return value		W: Word
		Failed	0	
		Succeeded	1	
	Var2	File name of the exported 16-bit Recipe / 32-bit Recipe		

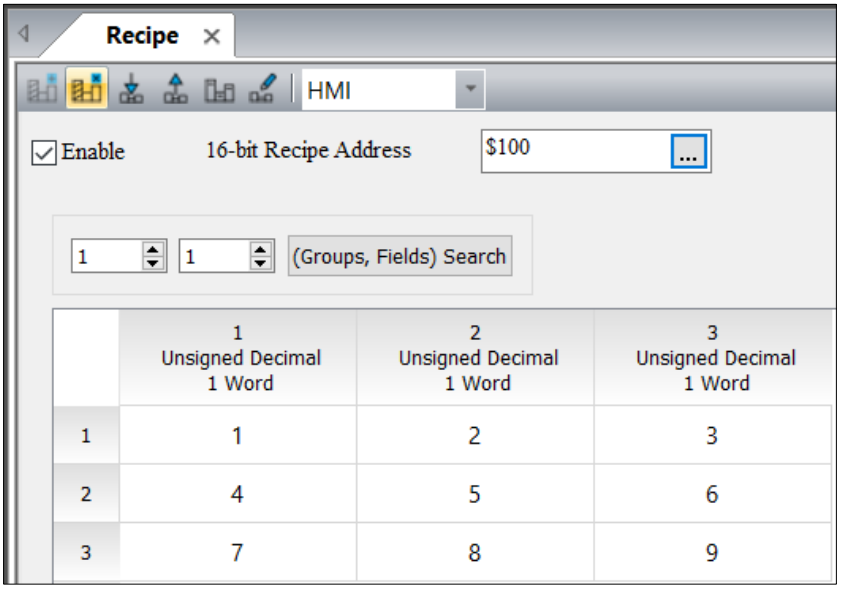
<table>
<tr><th>Expression</th><th colspan="4">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="3"></td><td rowspan="2">Var3</td><td rowspan="2">Recipe export storage device</td><td>USB Disk</td><td>2</td><td rowspan="3"></td></tr>
<tr><td>SD Card</td><td>3</td></tr>
<tr><td colspan="4">Description of Action</td></tr>
<tr><td colspan="5">Export the 16-bit Recipe / 32-bit Recipe and store in Var3 and return the result to Var1.</td></tr>
<tr><td colspan="6">Note: The exported 16-bit and 32-bit Recipe files will be stored in the root directory of the external storage device.</td></tr>
</table>

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	
Var3	V	V	V

Example

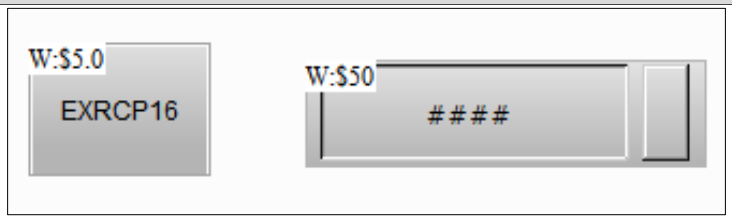
- The following example shows the export command of 16-bit Recipe, which is the same as that of 32-bit Recipe.
- Export the16-bit Recipe data to the USB Disk, and the file name is Delta. The steps are as follows:

1. Go to **Data Management** > **Recipe** and set the recipe data.

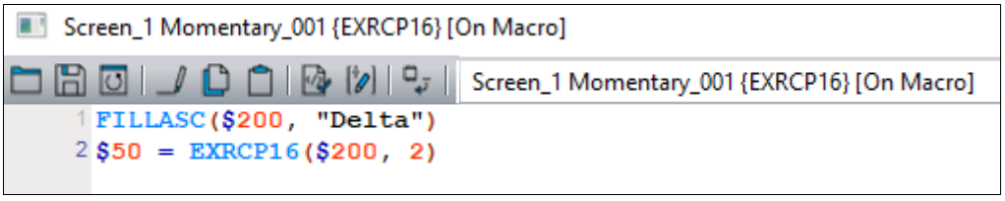


2. Create a **Momentary Button** ($5.0) and a **Numeric Entry** element ($50).

Example



3. Go to the **Momentary Button** and write the **On Macro** as follows. Put the ***Delta*** string in $200 and store the data in the USB Disk by exporting 16-bit Recipe, and the file name is Delta.



4. After compiling the screen and downloading the recipe data to the HMI, trigger the $5.0 **Button** and $50 shows 1, representing successful action, and then the HMI exports the 16-Bit Recipe data to the ***USB Disk***.

Delta.csv

	A	B	C
1	RCP16-1.0		
2			
3	3	3	
4	1	2	3
5	4	5	6
6	7	8	9

- **IMRCP16 / IMRCP32 (Import 16-bit Recipe / Import 32-bit Recipe)**

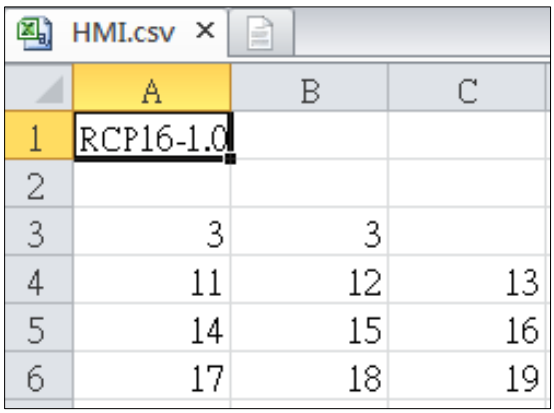
Expression	Meaning of Variable				Note
Var1 = IMRCP16(Var2, Var3) (W) Var1 = IMRCP32(Var2, Var3) (W)	Var1	Return value			W: Word
		Failed	0		
		Succeeded	1		
	Var2	File name of the imported 16-bit Recipe / 32-bit Recipe			
	Var3	Recipe	USB	2	

Expression	Meaning of Variable				Note
		import storage device	Disk		
			SD Card	3	
	Description of Action				
	Import the 16-bit Recipe / 32-bit Recipe from Var3 to the HMI and return the result to Var1.				

Variable	Type		
	Internal Memory	**PLC Register**	**Constant**
Var1	V	V	
Var2	V	V	
Var3	V	V	V

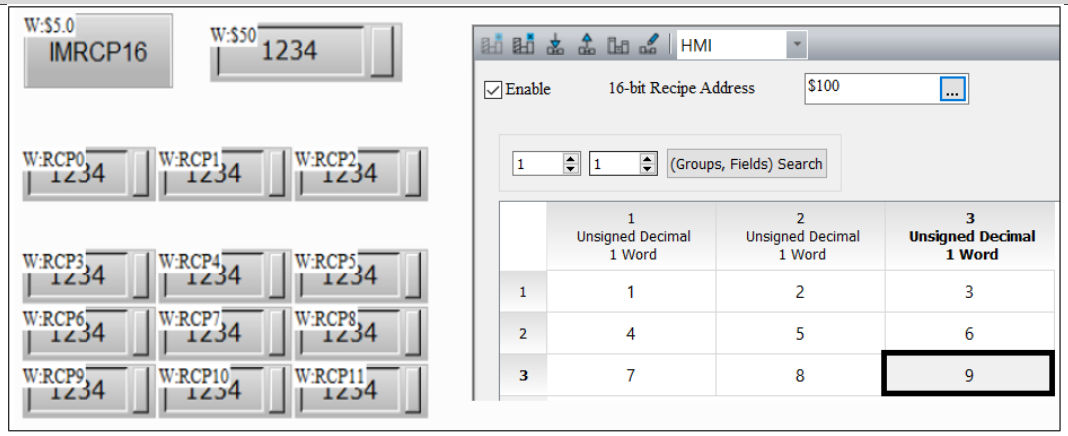
Example

- The following example shows the import command of 16-bit Recipe, which is the same as that of 32-bit Recipe.
- Import 16-Bit Recipe data from the ***USB Disk*** to the ***HMI***, and the file name is HMI. The steps are as follows:
  1. Refer to the following figure, use Excel to make a recipe file called HMI.csv and store it to the USB Disk.

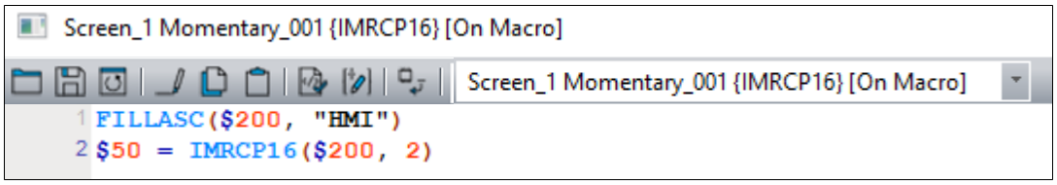


  2. Create a **Momentary Button** ($5.0), a **Numeric Entry** element ($50), and recipe addresses RCP0 to RCP11. This is the default recipe content of the software.

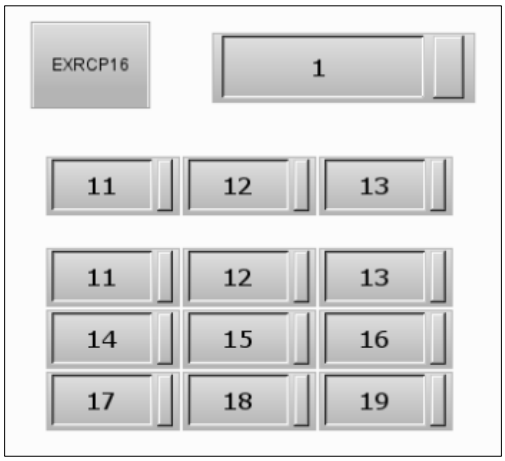
Example



3. Go to the **Momentary Button** and write the **On Macro** as follows. Put the ***HMI*** string in $200 and **Import** 16-Bit Recipe data from the USB Disk.



4. After compiling the screen and downloading the recipe data to the HMI, trigger the $5.0 **Button** and $50 shows 1, representing successful action, and then the 16-bit Recipe data is imported to the HMI. The recipe data of the HMI is then changed to the HMI recipe file.



- **EXENRCP (Export Enhanced Recipe)**

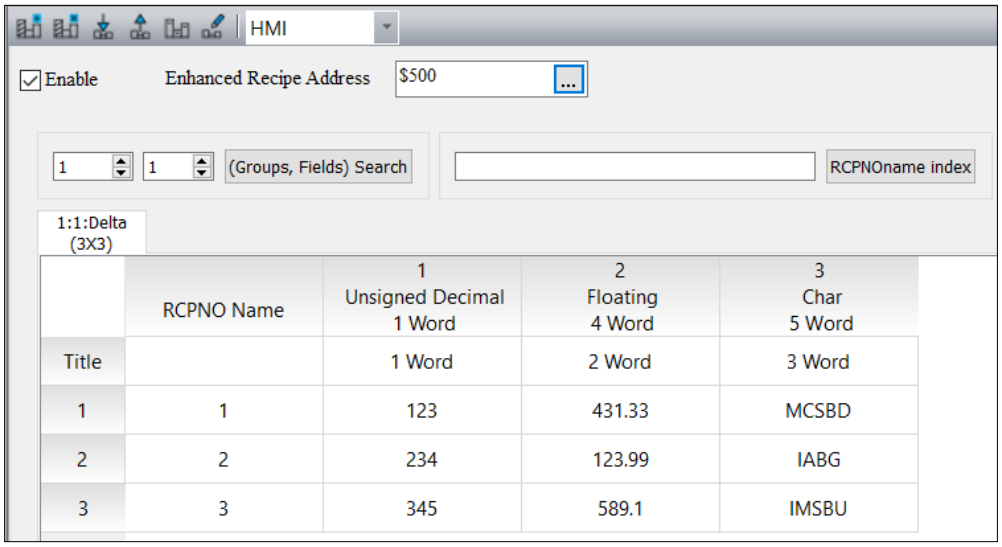
Expression	Meaning of Variable			Note
Var1 = EXENRCP(Var2, Var3) (W)	Var1	Return value		W: Word
		Failed	0	
		Succeeded	1	
	Var2	File name of the exported enhanced recipe		

<table>
<tr><th>Expression</th><th colspan="4">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="3"></td><td rowspan="2">Var3</td><td rowspan="2">Recipe export storage device</td><td>USB Disk</td><td>2</td><td rowspan="3"></td></tr>
<tr><td>SD Card</td><td>3</td></tr>
<tr><td colspan="4">Description of Action</td></tr>
<tr><td colspan="5">Export the enhanced recipe and store it in Var3 and return the result to Var1.</td></tr>
<tr><td colspan="6">Note: The exported enhanced recipe file will be stored in the root directory of the external storage device.</td></tr>
</table>

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	
Var3	V	V	V

Example

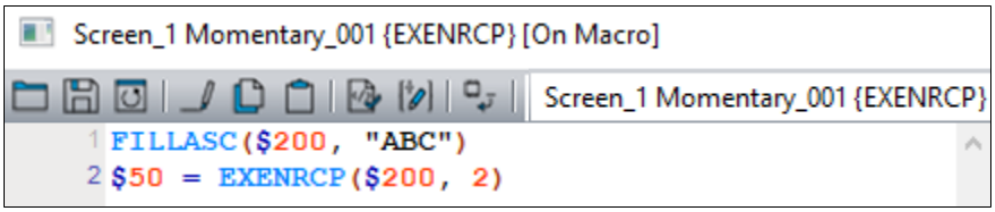
- Export the enhanced recipe data to the USB Disk, and the file name is ABC. The steps are as follows:
  1. Go to **Data Management** > **Enhanced Recipe** and set the recipe data.



  2. Create a **Momentary Button** ($6.0) and a **Numeric Entry** element ($50).

Example

3. Go to the **Momentary Button** to write the **On Macro** as follows. Put the ABC string in $200 and store the data in the USB Disk by exporting the enhanced recipe, and the file name is ABC.



4. After compiling the screen and downloading the recipe data to the HMI, trigger the $5.0 **Button** and $50 shows 1, representing successful action, and then the **HMI** exports the enhanced recipe data to the **USB Disk**.

ABC.csv

	A	B	C	D
1	ENRCP-2.0			
2	Delta	3	3	
3	2	1	0	0
4	5	2	3	2
5	8	5	0	0
6	123	431.33	MCSBD	
7	234	123.99	IABG	
8	345	589.1	IMSBU	

- For each **Data Format** and the corresponding defined value, refer to the following table.

Data Format	DIAScreen
BCD	0
Signed Decimal	1
Unsigned Decimal	2
Hexadecimal	3
Binary	4
Floating	5
Char	8

- Each cell of the exported CSV file is explained as follows:

Example

	A	B	C	D
1	ENRCP-2.0			
2	Delta	3	3	
3	2	1	0	0
4	5	2	3	2
5	8	5	0	0
6	123	431.33	MCSBD	
7	234	123.99	IABG	
8	345	589.1	IMSBU	

Excel cell	Definition	Content
A-1	Version	ENRCP-2.0
A-2	Enhanced recipe group name	Delta
B-2	Column number	3
C-2	Group number	3
A-3	Data Format of Column 1	2 (Unsigned Decimal)
B-3	Data Length of Column 1	1 (Word)
C-3	Integer Digits of Column 1	0
D-3	Fractional (Digits) of Column 1	0
A-4	Data Format of Column 2	5 (Floating)
B-4	Data Length of Column 2	2 (Word)
C-4	Data Length of Column 2	3
D-4	Fractional (Digits) of Column 2	2
A-5	Data Format of Column 3	8 (Char)
B-5	Data Length of Column 3	5 (Word)
C-5	Integer Digits of Column 3	0
D-5	Fractional (Digits) of Column 3	0

- **IMENRCP (Import Enhanced Recipe)**

<table>
<tr><th>Expression</th><th colspan="4">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="6">Var1 = IMENRCP(Var2, Var3)
(W)</td><td rowspan="3">Var1</td><td colspan="3">Return value</td><td rowspan="6">W: Word</td></tr>
<tr><td>Failed</td><td colspan="2">0</td></tr>
<tr><td>Succeeded</td><td colspan="2">1</td></tr>
<tr><td>Var2</td><td colspan="3">File name of the imported enhanced recipe</td></tr>
<tr><td rowspan="2">Var3</td><td rowspan="2">Recipe import storage device</td><td>USB Disk</td><td>2</td></tr>
<tr><td>SD Card</td><td>3</td></tr>
<tr><td colspan="5">Description of Action</td></tr>
<tr><td colspan="5">Import the enhanced recipe from Var3 to the HMI and return the result to Var1</td></tr>
</table>

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	
Var3	V	V	V

Example

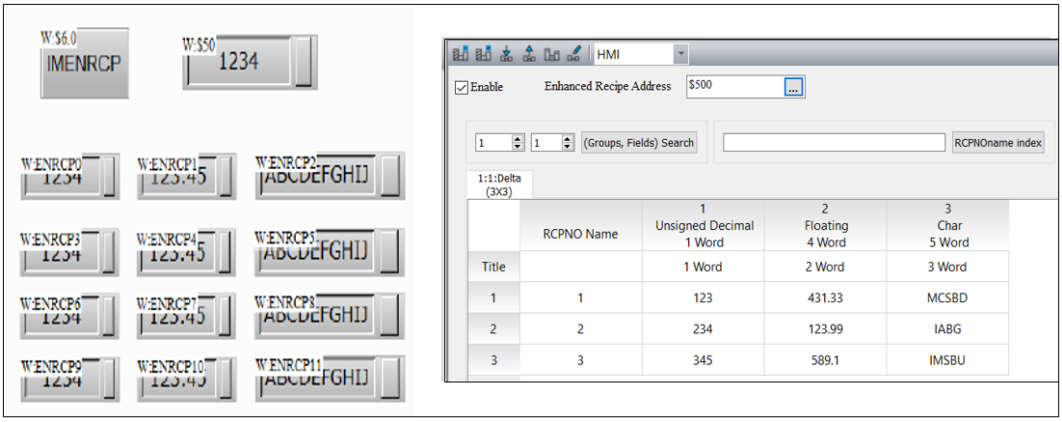
- **Import** the enhanced recipe from the ***USB Disk*** to the HMI, and the file name is **DEF**. The steps are as follows:
  1. Refer to the following figure, use Excel to make a recipe file called **DEF.csv** and store it to the **USB Disk**.

DEF.csv

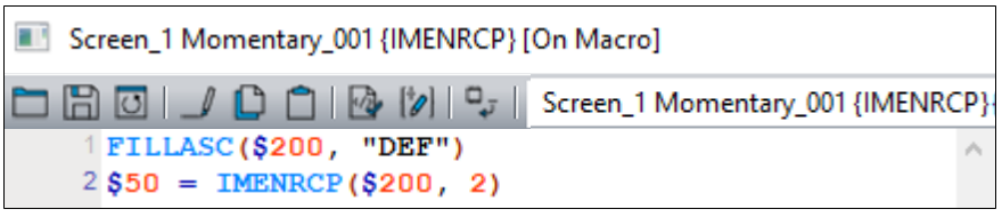
	A	B	C	D
1	ENRCP-2.0			
2	Delta	3	3	
3	2	1	0	0
4	5	2	3	2
5	8	5	0	0
6	1123	31.33	HMI	
7	1234	23.99	DOP-B	
8	1345	89.1	DOP-W	

  2. Create a **Momentary Button** ($6.0), a **Numeric Entry** element ($50), and recipe addresses RCP0 to RCP11.

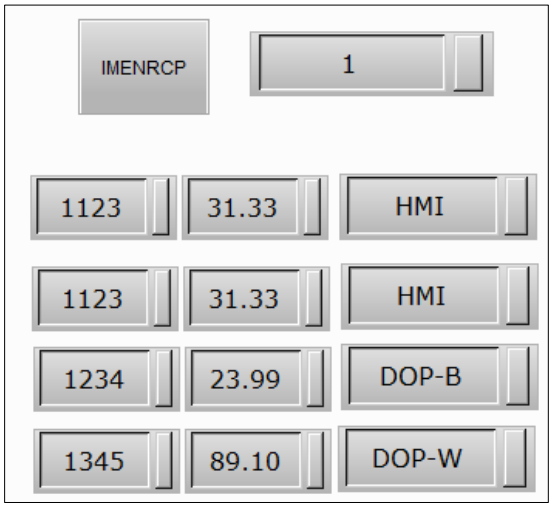
Example



3. Go to the **Momentary Button** to write the **On Macro** as follows. Put the DEF string in $100 and import the enhanced recipe from the USB Disk.



4. After compiling the screen and downloading the recipe data to the HMI, trigger the $5.0 **Button** and $50 shows 1, representing the successful action, and then the enhanced recipe data is imported to the HMI. The recipe data of the HMI is then changed to the DEF recipe file.



- **EXHISTORY (Export History Data)**

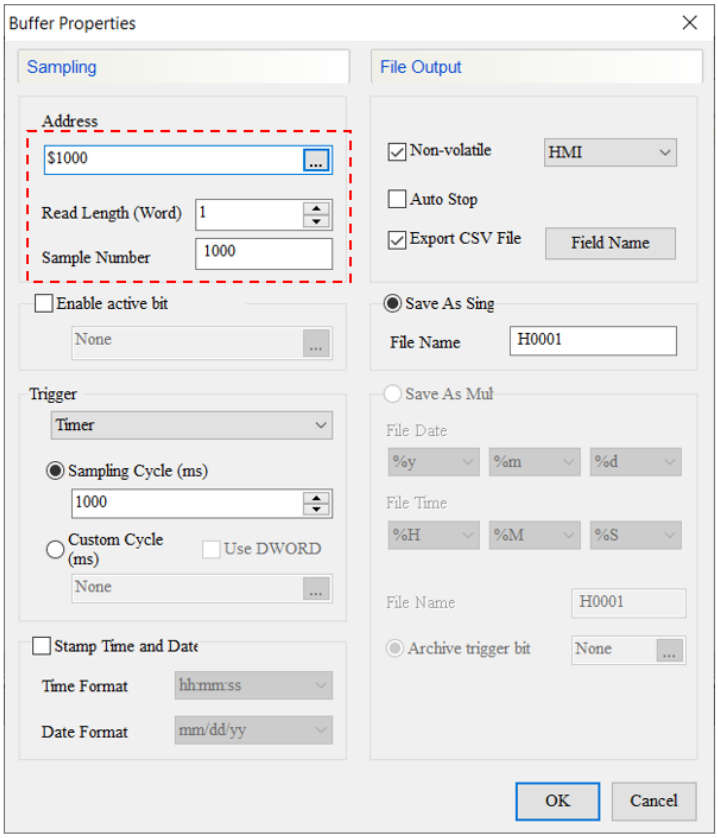
Expression	Meaning of Variable			Note
Var1 = EXHISTORY(Var2, Var3, Var4)	Var1	Return value		-
		Failed	0	
		Succeeded	1	

<table>
<tr><th>Expression</th><th colspan="4">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="5"></td><td>Var2</td><td colspan="3">History Buffer ID</td><td rowspan="5"></td></tr>
<tr><td>Var3</td><td colspan="3">Exported file name</td></tr>
<tr><td rowspan="2">Var4</td><td rowspan="2">External storage device</td><td>USB Disk</td><td>2</td></tr>
<tr><td>SD Card</td><td>3</td></tr>
<tr><td colspan="4">Description of Action</td></tr>
<tr><td colspan="4">Export history data to an external storage device.</td></tr>
<tr><td colspan="6">Note:
1. Users must select the checkbox for Export CSV File in the Buffer Properties setting. If not, execution of this macro will only export the .dat file.
2. When users input 0 to the History Buffer ID, it means to export all history buffers. If 3 history buffers are opened, 3 files will be exported after execution. The file names are Export file name 1.csv, Export file name 2.csv, and Export file name 3.csv respectively. When users input a non-0 number, it represents that the specified ID of history buffer will be exported. After execution, one file will be exported with the name of Export file name.csv
3. File extension .csv will be added automatically after exporting. The file name cannot contain characters such as \, /,:, *, ?, ”, <, >, and | x00 indicates the end of the file name string.</td></tr>
</table>

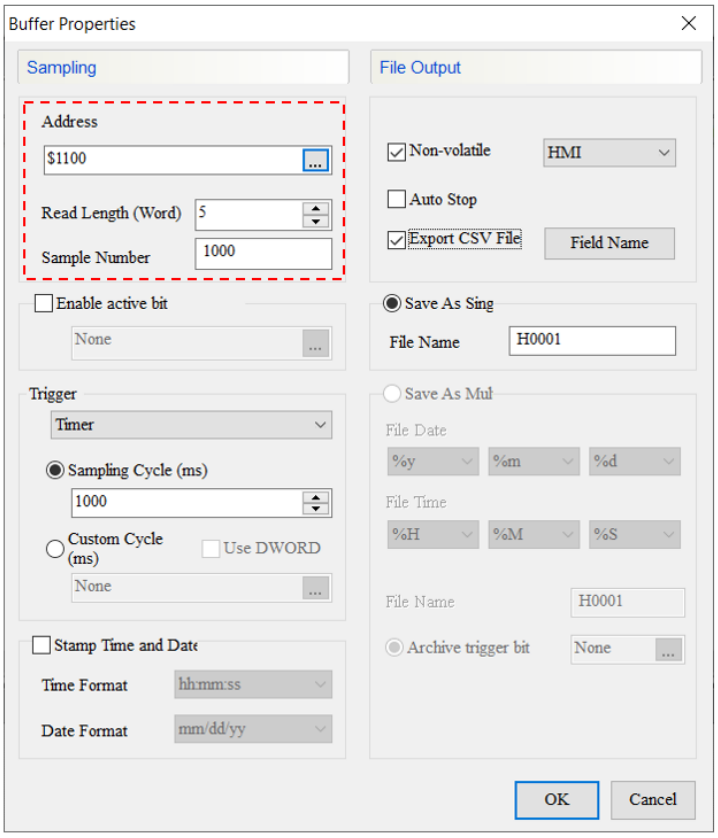
Variable	Type			
	Internal Memory	PLC Register	String	Constant
Var1	V	V		
Var2	V	V		V
Var3	V	V		
Var4	V	V		V

Example
Step 1: Set three history buffers as follows: 1. For History data buffer 1, set the **Address** to $1000, **Read Length (Word)** to 10, and select the checkbox of **Export CSV File**.

Example

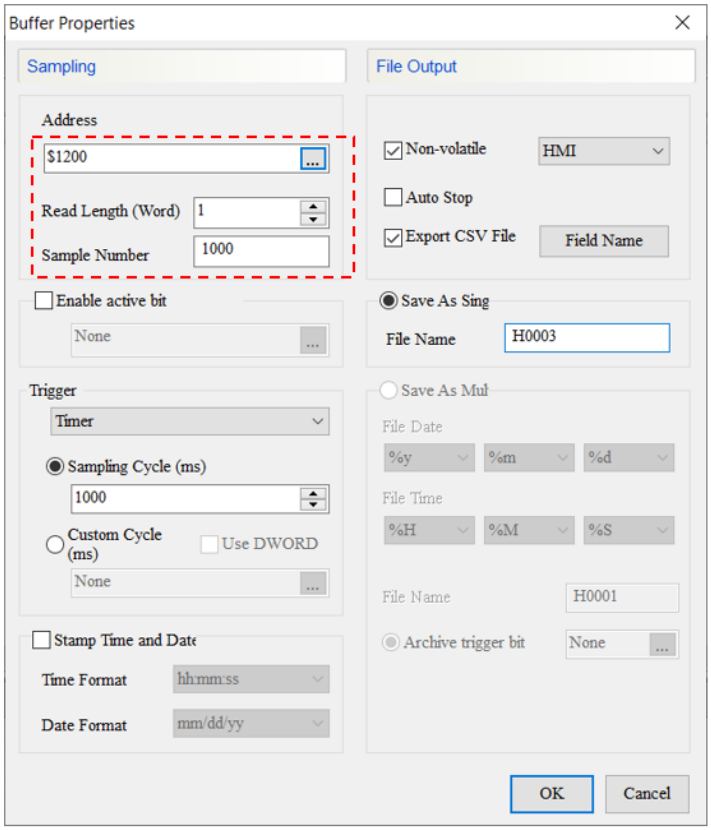


2. For History data buffer 2, set the **Address** to $1100, **Read Length (Word)** to 5, and select the checkbox of **Export CSV File**.

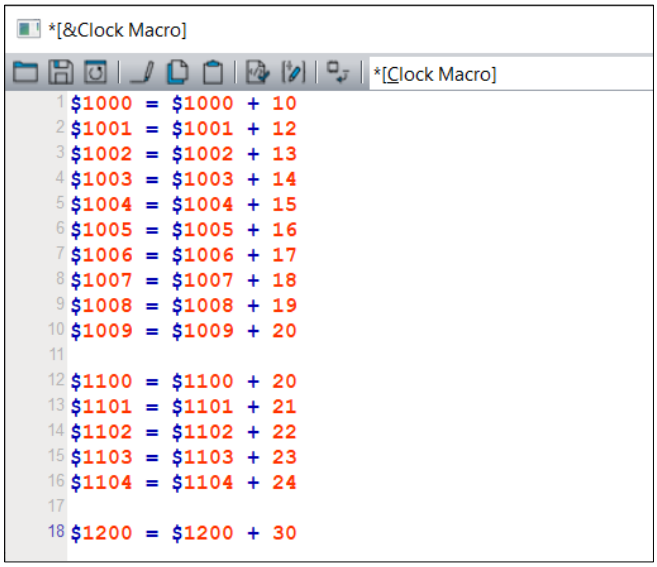


Example

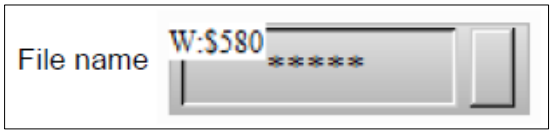
3. For History data buffer 3, set the **Address** to $1200, **Read Length (Word)** to 1, and select the checkbox of **Export CSV File**.



Step 2: Add up history data in the **Clock Macro**.

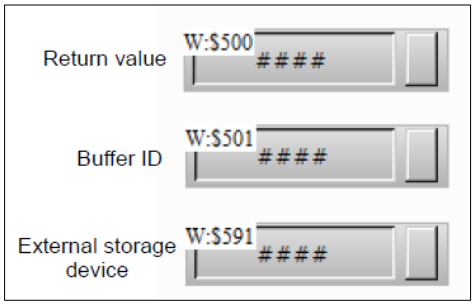


Step 3: Create a **Character Entry** element on the screen with the address as $580 and string length as 5.

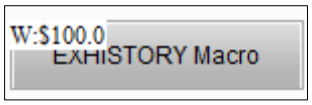


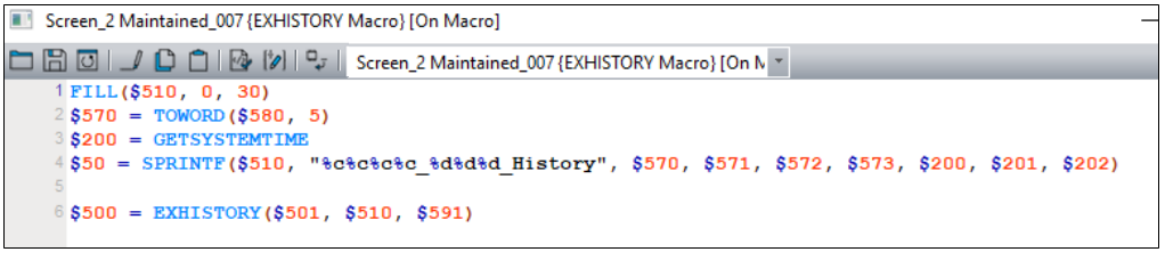
Example

Step 4: Create 3 **Numeric Entry** elements on the screen with the addresses of $500, $591, and $501.



Step 5: Create a **Maintained Button** element on the screen with the address of $100.0, and add the On Macro.

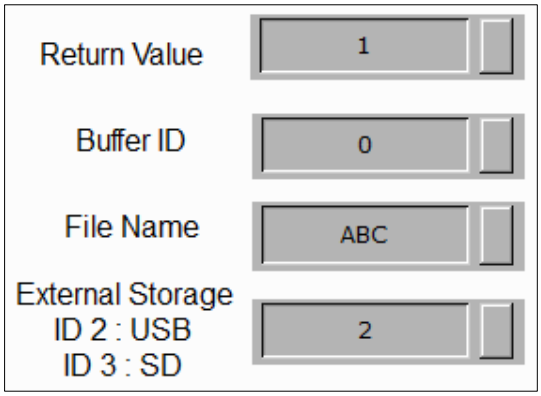


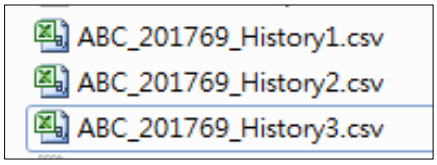


- Macro commands description:
- Line 1: Clear $510 - $539.
- Line 2: Convert the unit of $580 (file name string) from byte to word.
- Line 3: Move the system time to $200 - $206 (year, month, day, week, hour, minute, second).
- Line 4: Add _year, month, day and _History to three characters of device name to form a continuous string and assign it to the $510 start address.
- Line 6: Export the history data to the specified external device and file name.

Step 6: Download the editing screen to the HMI and insert the **USB Disk** into the HMI.

Step 7: Enter the device name as ABC on the screen and select 2 to use the USB Disk for the external storage device. Set the **Buffer ID** as 0 (export all) and press the EXHISTORY Button. The **Return Value** becomes 1 after the action.



Example
Step 8: After the **USB Disk** is removed, the files exported to the USB Disk are as follows: 

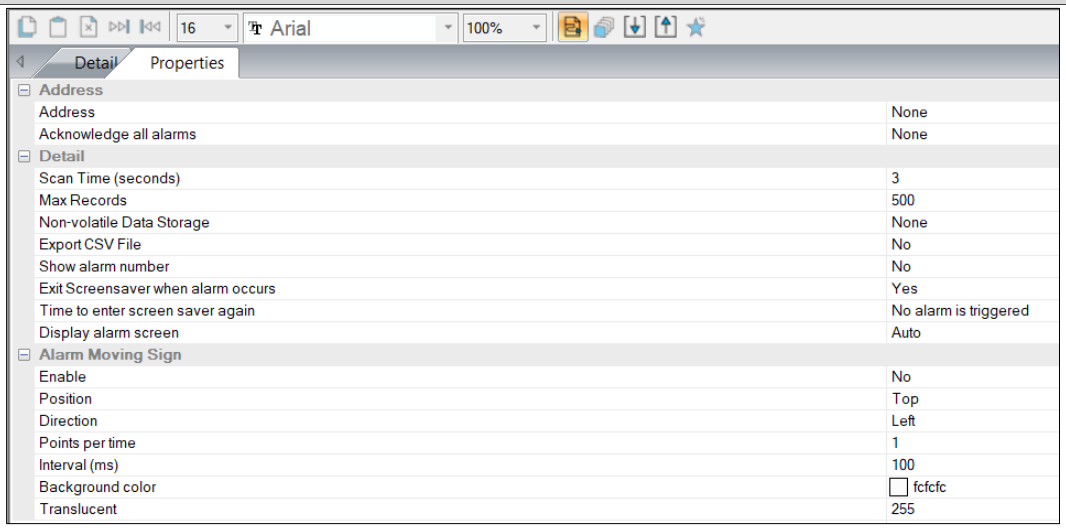
- **EXALARM (Export Alarm Data)**

<table>
<tr><th>Expression</th><th colspan="5">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="7">Var1 = EXALARM(Var2, Var3)</td><td rowspan="3">Var1</td><td colspan="4">Return value</td><td rowspan="7"></td></tr>
<tr><td colspan="2">Failed</td><td colspan="2">0</td></tr>
<tr><td colspan="2">Succeeded</td><td colspan="2">1</td></tr>
<tr><td>Var2</td><td colspan="4">Exported file name</td></tr>
<tr><td rowspan="2">Var3</td><td rowspan="2">External storage device</td><td>USB Disk</td><td>2</td></tr>
<tr><td>SD Card</td><td>3</td></tr>
<tr><td colspan="5">Description of Action</td></tr>
<tr><td colspan="5">Export alarm data to external storage devices.</td></tr>
<tr><td colspan="7">Note:
1. Users must select the checkbox of Export CSV File in the alarm settings.
2. File extension ".csv" will be added automatically after exporting. The file name cannot contain characters such as \, /, :, *, ?, ”, <, >, and |.x00 indicates the end of the file name string.</td></tr>
</table>

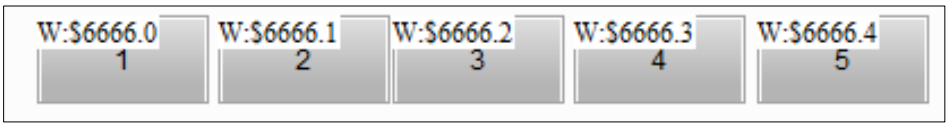
Variable	Type			
	Internal Memory	PLC Register	String	Constant
Var1	V	V		
Var2	V	V		
Var3	V	V		V

Example
• The steps are as follows: 1. The alarm settings are as follows. Set the **Address** as $6666 and set the **Export CSV File** to **Yes**.

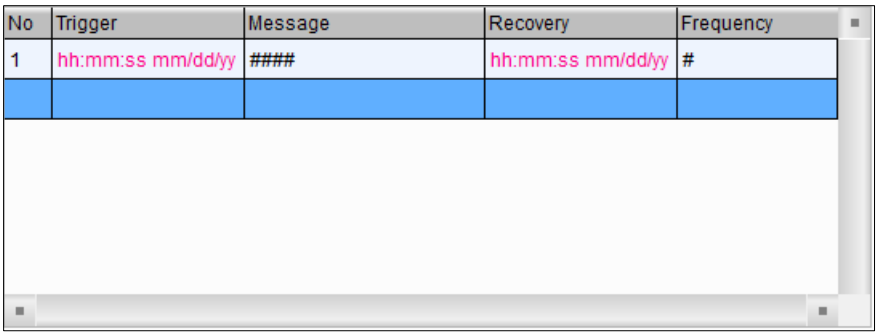
Example



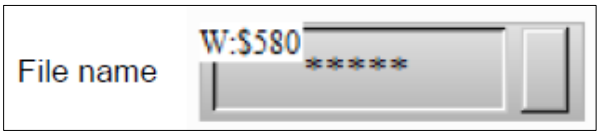
2. Create **Maintained Button** elements that trigger alarm bits on the screen to trigger alarms. Set the addresses as $6666.0, $6666.1, $6666.2, $6666.3, and $6666.4 in sequence.



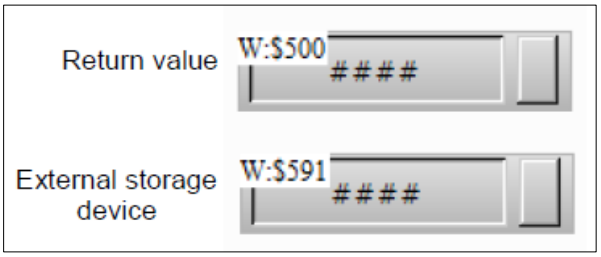
3. Create an **Alarm History Table** element on the screen to display the current history alarm.



4. Create a **Character Entry** element on the screen with the address as $580 and string length of 5.

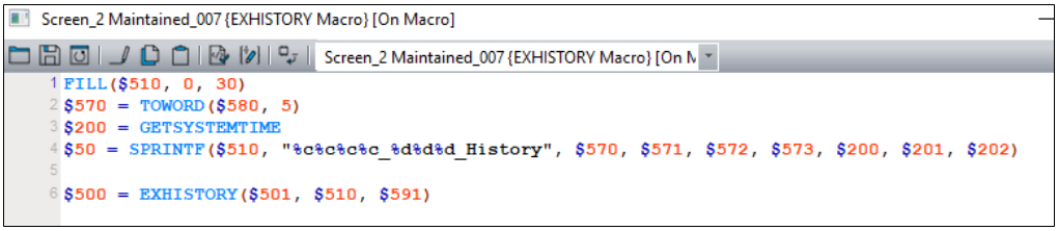


5. Create two **Numeric Entry** elements on the screen with the addresses as $500 and $591.



Example

6. Create a **Maintained Button** element on the screen with the address as $100.1, and add **On Macro**.



The macro commands are described as follows:

Line 1: Clear $510 - $539.

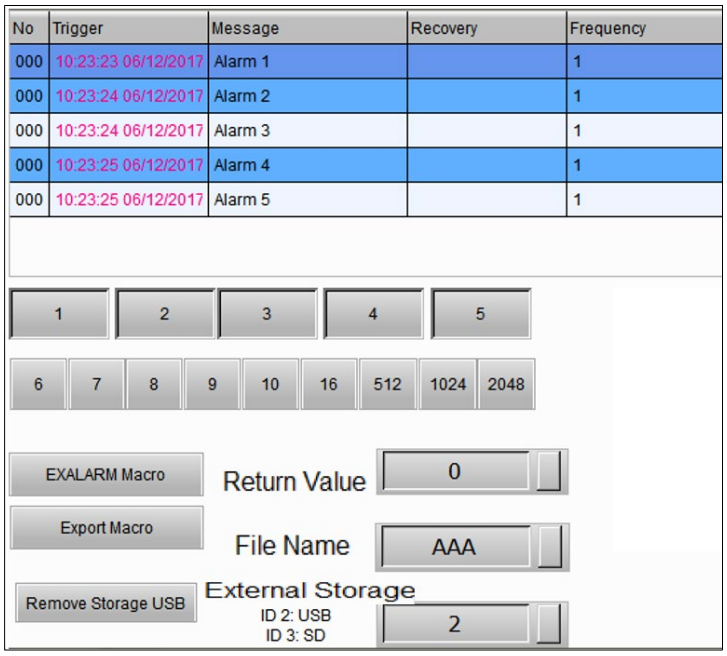
Line 2: Convert the unit of $580 (device name string) from byte to word.

Line 3: Move the system time to $200 - $206 (year, month, day, week, hour, minute, second).

Line 4: Add "_year, month, day" and "_Alarm" to "three characters of device name" to form a continuous string and assign to the $510 start address.

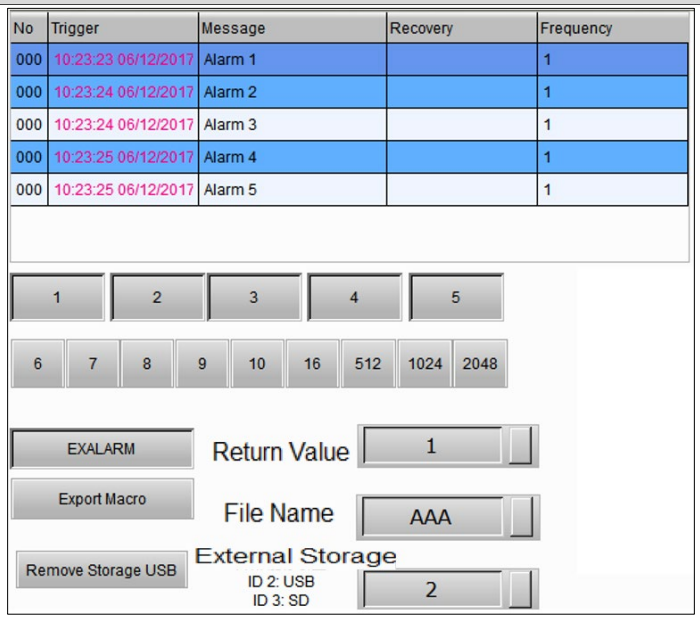
Line 6: Export the history alarm to the specified external device and file name.

7. Download the editing screen to the HMI and insert the USB Disk into the HMI.
8. Trigger the alarm and enter the device name as **AAA** on the screen and select 2 to use the **USB Disk** for the external storage device.

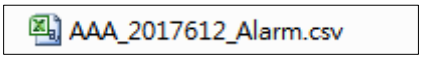


9. Press the **EXALARM Macro Button** and the **Return Value** becomes 1 after the action.

Example



10. After the USB Disk is removed, the file exported to the **USB Disk** is as follows:



- **EXALARMGROUP (Export Group Data Specified by Alarm)**

<table>
<tr><th>Expression</th><th colspan="4">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="7">Var1 = EXALARMGROUP(Var2, Var3, Var4)</td><td rowspan="3">Var1</td><td colspan="3">Return value</td><td rowspan="7"></td></tr>
<tr><td colspan="2">Failed</td><td>0</td></tr>
<tr><td colspan="2">Succeeded</td><td>1</td></tr>
<tr><td>Var2</td><td colspan="3">Alarm group number</td></tr>
<tr><td>Var3</td><td colspan="3">Exported file name</td></tr>
<tr><td rowspan="2">Var4</td><td rowspan="2">External storage device</td><td>USB Disk</td><td>2</td></tr>
<tr><td>SD Card</td><td>3</td></tr>
<tr><td colspan="5">Description of Action</td></tr>
<tr><td colspan="5">Export the alarm data to the external storage device in accordance with the specified group.</td></tr>
</table>

NOTE:

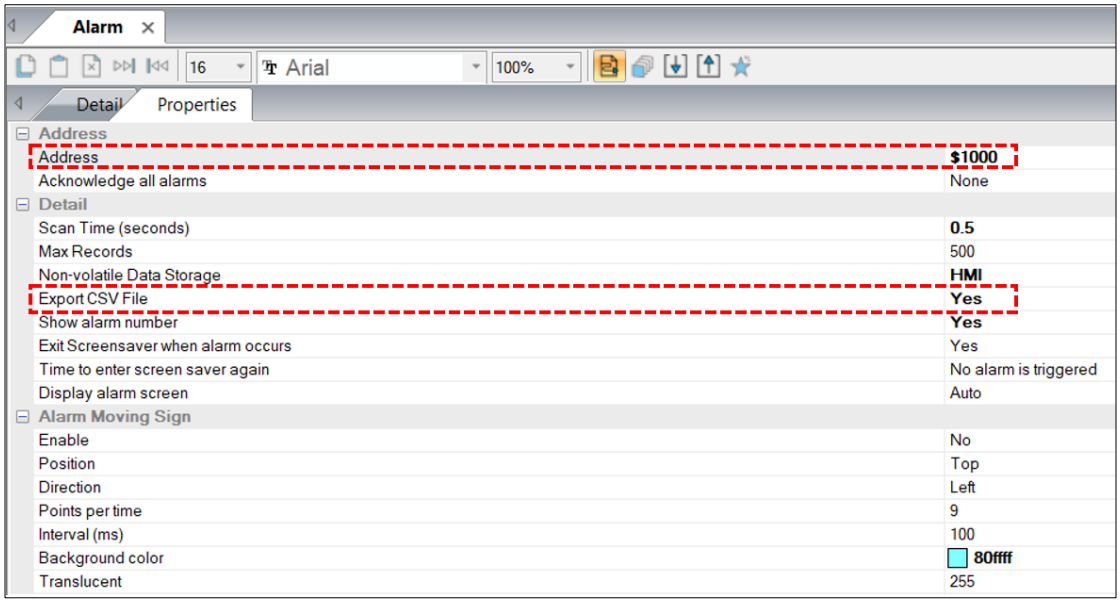
1. Users must select checkbox of Export CSV File in the alarm settings.
2. File extension of "- group number" and ".csv " will be added automatically after exporting. The file name cannot contain characters such as \, /, :, *, ?, ”, <, >, and |.x00 indicates the end of the file name string.

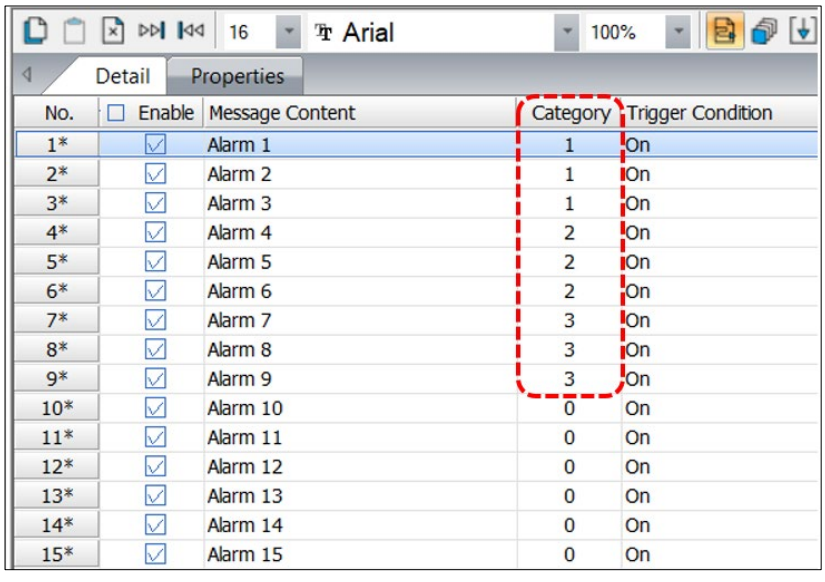
Variable	Type			
	Internal Memory	PLC Register	String	Constant
Var1	V	V		
Var2	V	V		V
Var3	V	V		
Var4	V	V		V

Example

The steps are as follows:

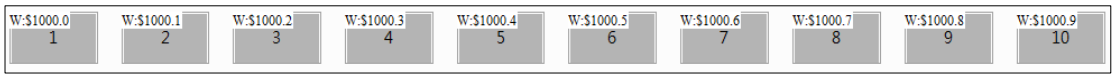
1. The alarm settings are as follows. Set the **Address** to $1000 and set the **Export CSV File** to Yes. Set Alarms 1 ~ 3 as Group 1, Alarms 4 ~ 6 as Group 2, and Alarms 7 ~ 9 as Group 3.



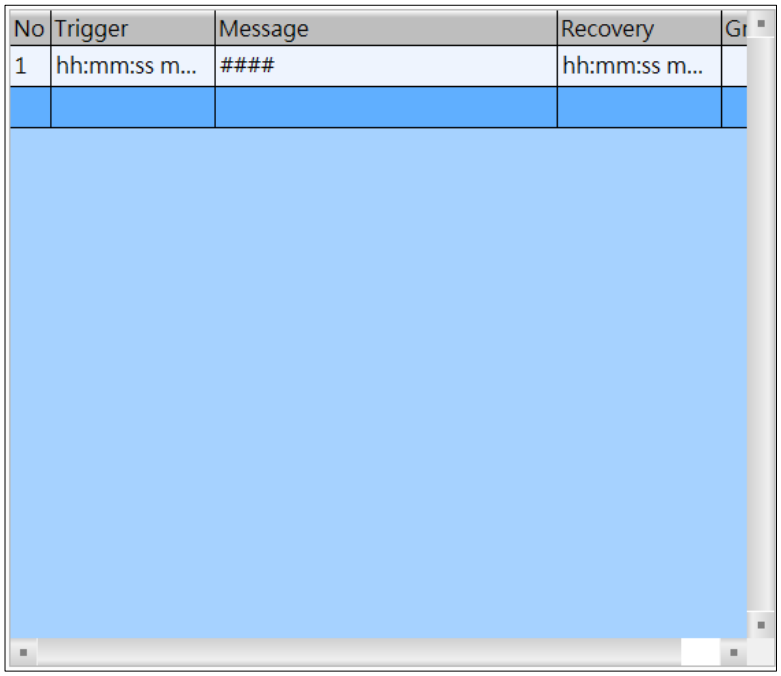


Example

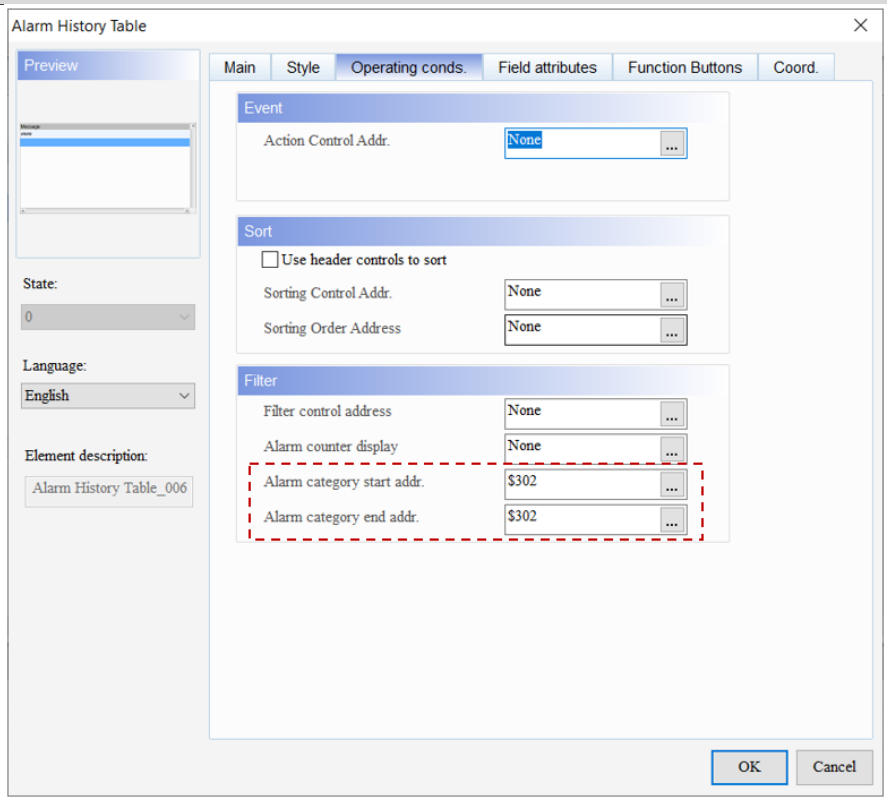
2. Create **Maintained Button** elements that trigger alarm bits on the screen to trigger alarms. Set the Write addresses as $1000.0, $1000.1, $1000.2 to $1000.9 in sequence.



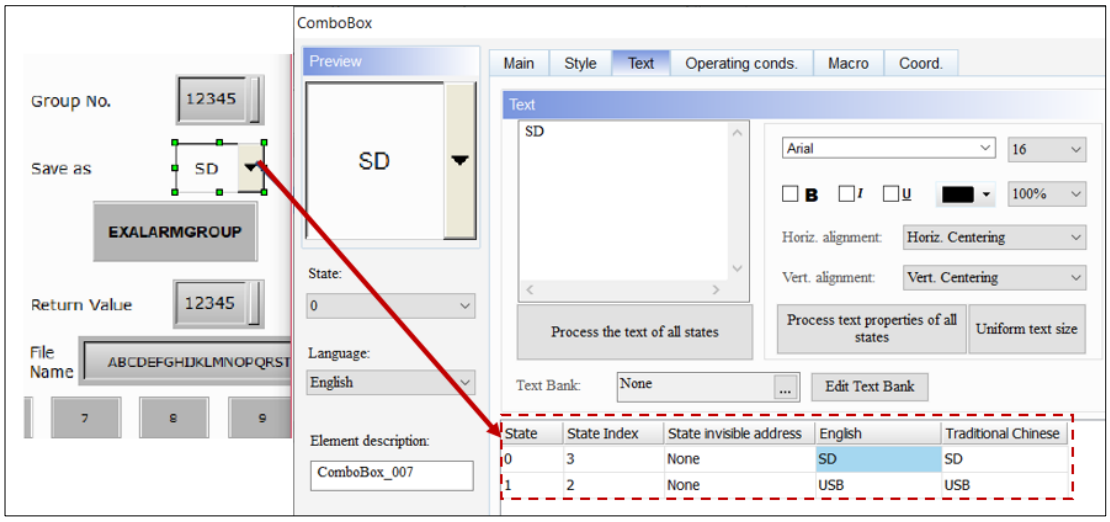
3. Create an **Alarm History Table** element on the screen to display the current history alarm and set the group number as variable and the read address as $302.



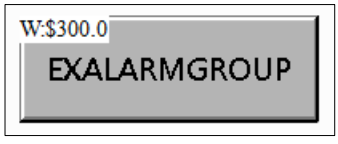
Example



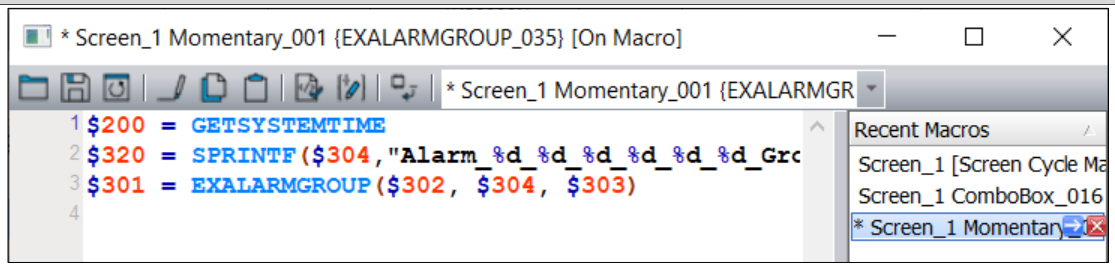
4. Create a **Character Display** element on the screen with the address as $304 and the string length as 20; Create 2 **Numeric Entry** elements with the addresses as $302 and $301. Create a **ComboBox** with the address as $303. The total number of **State** is 2, the **State Index** of State 1 is 3, the text is SD, the **State Index** of State 2 is 2, and the **Text** is USB.



5. Create a **Maintained Button** element on the screen with the address as $300.0, and add the **On Macro**.



Example



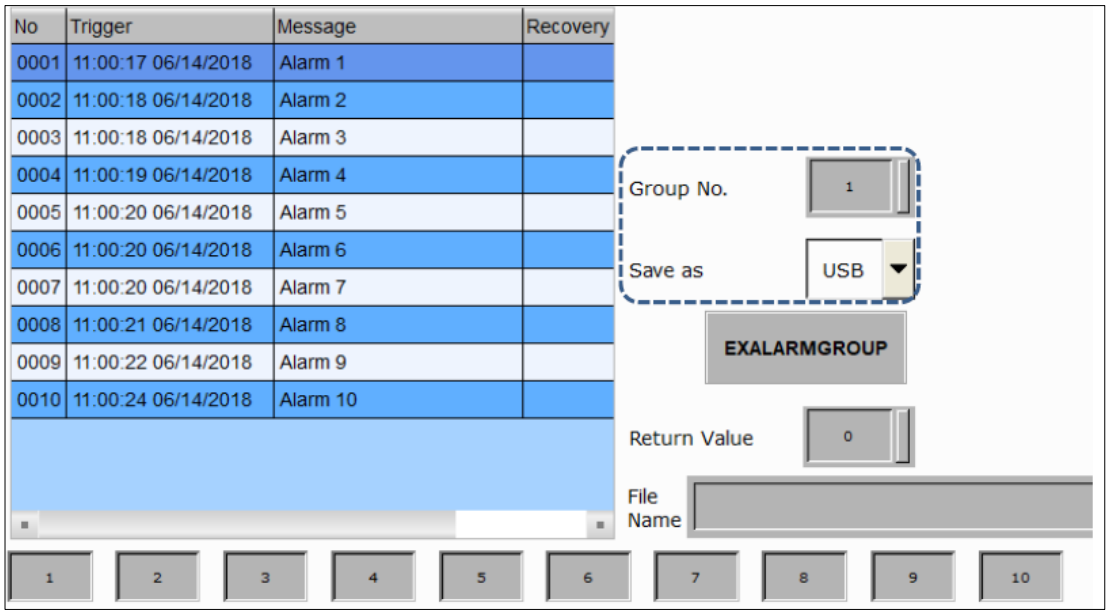
The macro commands are described as follows:

Line 1: Record the system time to $200 - $206 (year, month, day, week, hour, minute, second).

Line 2: Add _year, month, day, hour, minute, second and _Group to Alarm to form a continuous string and assign to the $304 start address as file name.

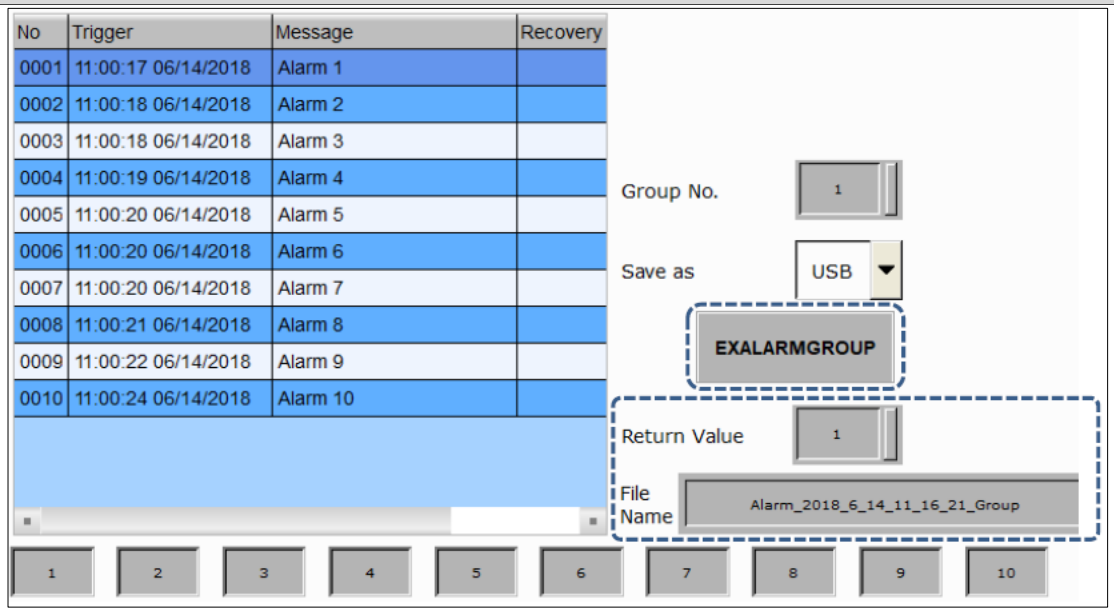
Line 3: Export history alarm to the specified external device according to the specified group number and file name.

6. Download the editing screen to the HMI and insert the USB Disk into the HMI.
7. Trigger Alarm 1 to Alarm 10, enter group number **1** on the screen, and select USB for the external storage device.

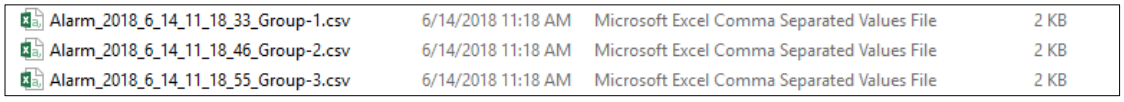


8. Press the **EXALARMGROUP Button** and the file name will be displayed, and the return value will become 1 after the action.

Example



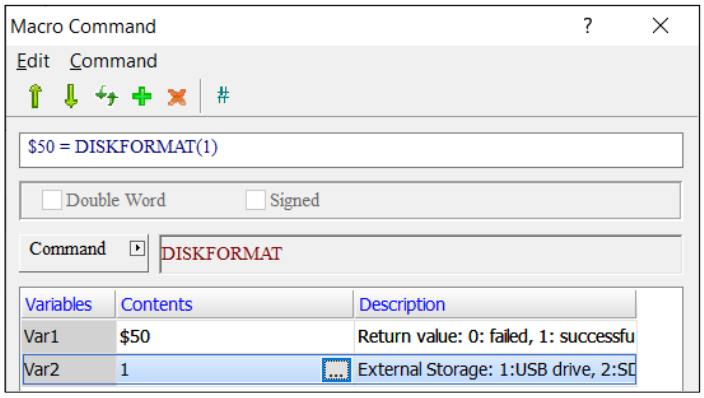
9. Then, set group numbers as 2 and 3 as in Step 8, and press the **EXALARMGROUP Button** respectively.
10. After the ***USB Disk*** is removed, the files exported to the USB Disk are as follows.



- **DISKFORMAT (Disk Format)**

<table>
<tr><th>Expression</th><th colspan="4">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="5">Var1 = DISKFORMAT(Var2) (W)</td><td rowspan="3">Var1</td><td colspan="3">Return value</td><td rowspan="5">W: Word</td></tr>
<tr><td>Failed</td><td colspan="2">0</td></tr>
<tr><td>Succeeded</td><td colspan="2">1</td></tr>
<tr><td rowspan="2">Var2</td><td rowspan="2">External storage device</td><td>USB Disk</td><td>1</td></tr>
<tr><td>SD Card</td><td>2</td></tr>
<tr><td colspan="5">Description of Action</td></tr>
<tr><td colspan="5">Select Var2 device to be formatted and return the result value to Var1.</td></tr>
</table>

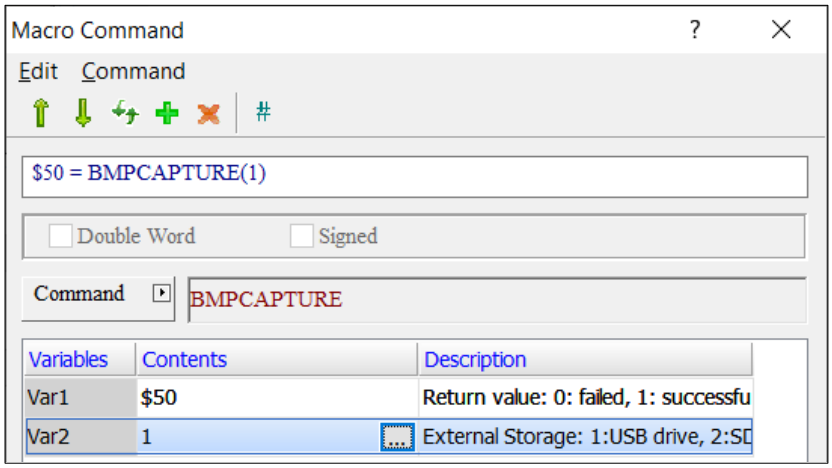
Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V

Example
• Var1 is the Internal Memory address and Var2 is a constant. • Format the USB Disk and put the Return value in $50. 

- **BMPCAPTURE (Screen Capture)**

<table>
<tr><th>Expression</th><th colspan="4">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="6">Var1 = BMPCAPTURE(Var2)
(W)</td><td rowspan="3">Var1</td><td colspan="3">Return value</td><td rowspan="6">W: Word</td></tr>
<tr><td>Failed</td><td colspan="2">0</td></tr>
<tr><td>Succeeded</td><td colspan="2">1</td></tr>
<tr><td rowspan="2">Var2</td><td rowspan="2">External storage device</td><td>USB Disk</td><td>1</td></tr>
<tr><td>SD Card</td><td>2</td></tr>
<tr><td colspan="4">Description of Action</td></tr>
<tr><td colspan="4">Store the captured screen file in the Var2 device and return the result value to Var1.</td></tr>
<tr><td colspan="6">Note:
1. All formats of the exported files are .BMP.
2. The exported path is the folder with the current date in the root directory of the external storage device, and the current screen file is saved in hour, minute and second.</td></tr>
</table>

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V

Example
• Var1 is the Internal Memory address and Var2 is a constant. • Save the captured screen to the USB Disk and put the Return value in $50. 

- **PLCDOWNLOAD (PLC File Download)**

<table>
<tr><th>Expression</th><th colspan="4">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="11">Var1 = PLCDOWNLOAD(Var2, Var3, Var4, Var5, Var6) (W)</td><td rowspan="3">Var1</td><td colspan="3">Return value</td><td rowspan="11">W: Word</td></tr>
<tr><td>Failed</td><td colspan="2">0</td></tr>
<tr><td>Succeeded</td><td colspan="2">1</td></tr>
<tr><td rowspan="3">Var2</td><td rowspan="3">COM Port</td><td>COM1</td><td>0</td></tr>
<tr><td>COM2</td><td>1</td></tr>
<tr><td>COM3</td><td>2</td></tr>
<tr><td>Var3</td><td colspan="3">PLCStation No.</td></tr>
<tr><td>Var4</td><td colspan="3">DELTA PLC file name (i.e. delta.dvp, delta.isp)</td></tr>
<tr><td rowspan="2">Var5</td><td rowspan="2">External storage device</td><td>USB Disk</td><td>1</td></tr>
<tr><td>SD Card</td><td>2</td></tr>
<tr><td>Var6</td><td colspan="3">PLC Password</td></tr>
<tr><td></td><td colspan="4">Description of Action</td><td></td></tr>
<tr><td></td><td colspan="4">Download PLC file to PLC.</td><td></td></tr>
</table>

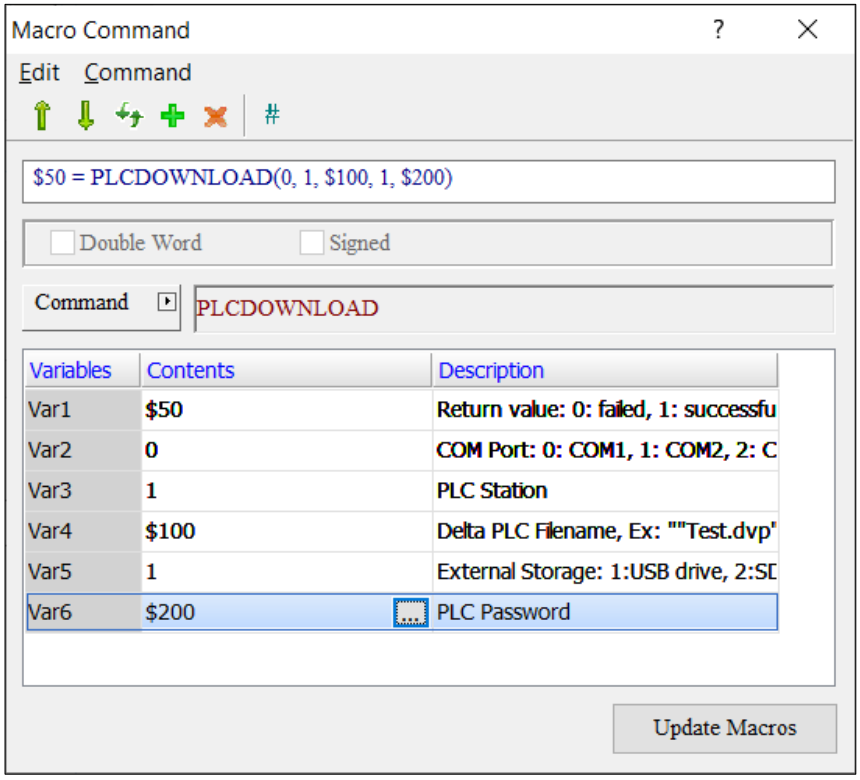
Note:

1. Only support Delta PLC.
2. Support .dvp and .isp file formats.
3. Use **Character Entry** elements for Var6 password.

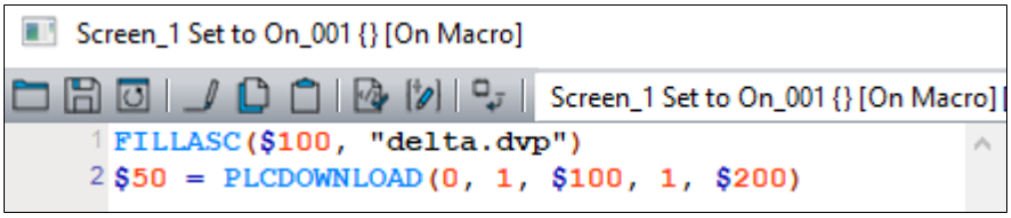
Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	
Var2	V	V	V
Var3	V	V	V
Var4	V	V	
Var5	V	V	V
Var6	V	V	

Example

- Store the DVP or ISP file to be downloaded to the USB Disk or SD Card and put the Return value in $50.

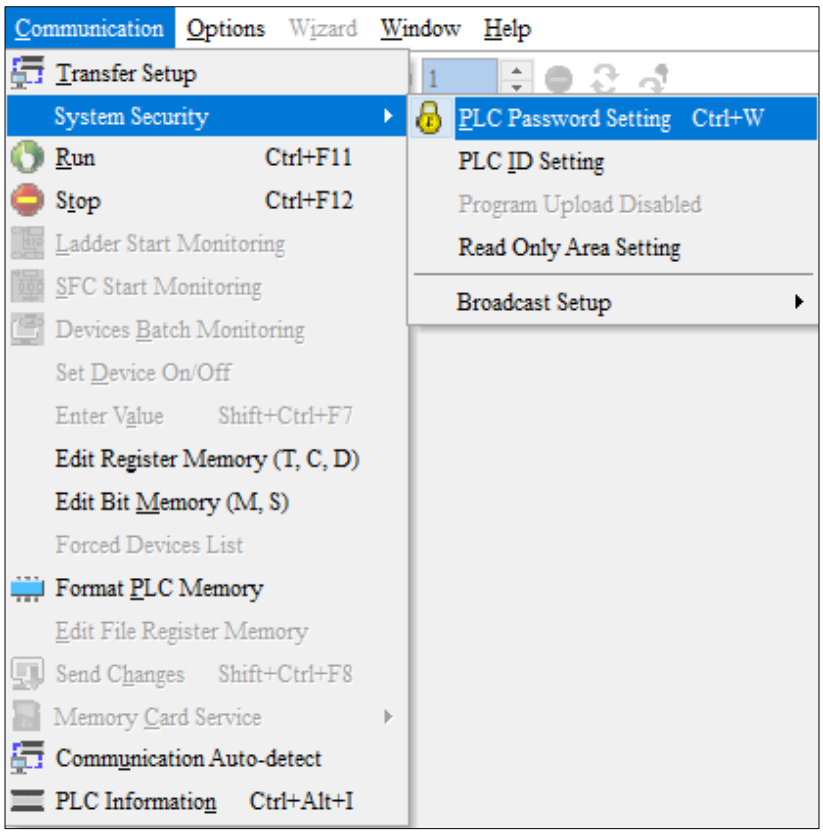


- Var4 is the PLC file name. First, the name string of the file to be downloaded must be put in a register address with the **FILLASC** command provided by the software. Then fill in the register address $100 with Var4 of **PLCDOWNLOAD**.



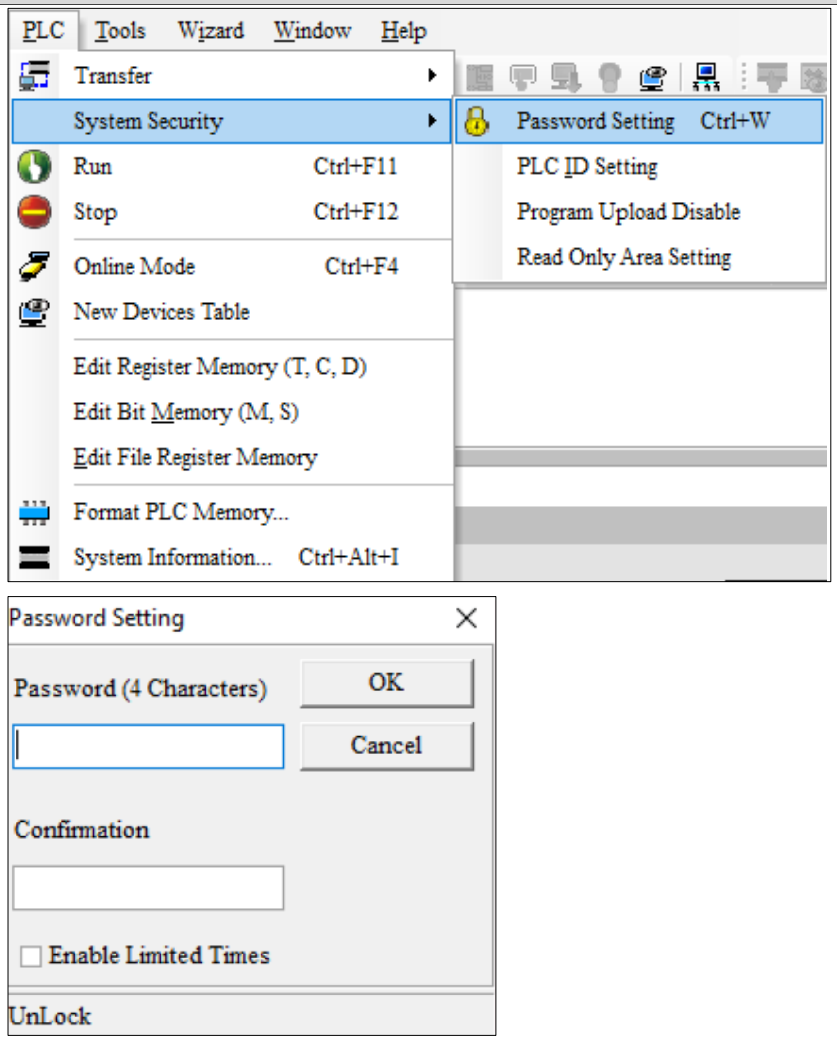
Example

- The PLC password of Var6 must be set by going to **System Security** > **Password Setting** of the WPLSoft and ISPSoft. After the setting is finished, enter the password with the **Character Entry** element of the HMI, then the PLC files can be downloaded to the PLC.
- WPLSoft password setting.



- ISPSoft password setting.

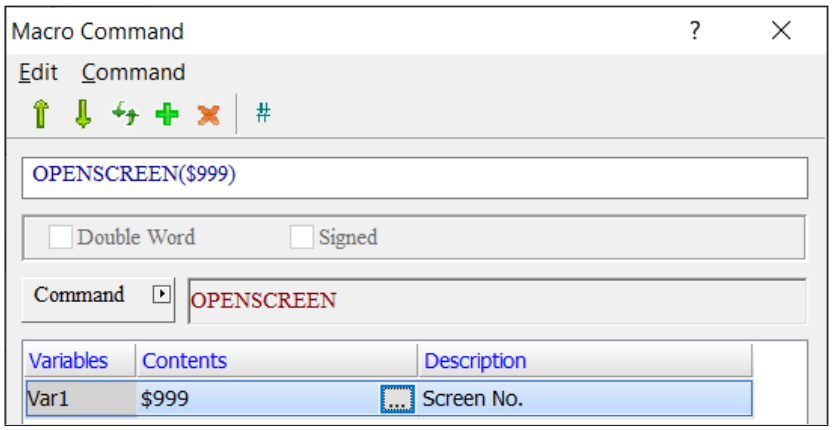
Example



- OPENSCREEN

Expression	Meaning of Variable		Note
OPENSCREEN(Var1) (W)	Var1	Screen number	W: Word
	Description of Action		
	Open the screen number specified by Var1.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	V

Example
• Var1 is the Internal Memory address. When $999 = 2, the macro switches the screen to Screen 2.    • This **Macro** does not support Screen Cycle Macro, Screen Open Macro, and Screen Close Macro.

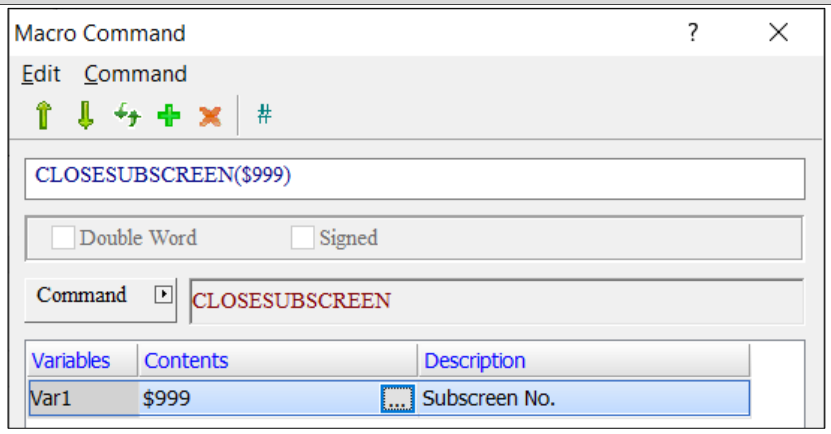
- CLOSESUBSCREEN

Expression	Meaning of Variable		Note
CLOSESUBSCREEN(Var1) (W)	Var1	Subscreen number	W: Word
	Description of Action		
	Close the subscreen number specified by Var1.		

Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V	V	V

Example
• Var1 is the Internal Memory address. When $999 = 2, the macro closes Subscreen No. 2.

Example



- This macro does not support Screen Cycle Macro, Screen Open Macro, and Screen Close Macro.

- **GetCircleCenter (Calculate Coordinates of the center of a circle)**

<table>
<tr><th>Expression</th><th colspan="2">Meaning of Variable</th><th>Note</th></tr>
<tr><td rowspan="6">Var1 = GetCircleCenter (Var2, Var3)</td><td rowspan="3">Var1</td><td>Return value</td><td rowspan="6">DW: Double Word</td></tr>
<tr><td>Failed</td><td>0</td></tr>
<tr><td>Succeeded</td><td>1</td></tr>
<tr><td>Var2</td><td colspan="2">Input coordinates of the 3 points (Note 1)</td></tr>
<tr><td>Var3</td><td colspan="2">Center coordinates after calculation (Note 2)</td></tr>
<tr><td colspan="3">Description of Action</td></tr>
<tr><td colspan="4">Enter the calculated coordinates of 3 points to calculate the center coordinates. (Note 3)</td></tr>
</table>

Note:

1. Input coordinates of the 3 points:
   Set the 3 points as P1(x1, y1), P2(x2, y2), and P3(x3, y3), and the length of each point is Signed DW.
   If Var2 is $100:
   Save LOW WORD of x1 in $100, HIGH WORD in $101
   Save LOW WORD of y1 in $102, HIGH WORD in $103
   Save LOW WORD of x2 in $104, HIGH WORD in $105
   Save LOW WORD of y2 in $106, HIGH WORD in $107

Expression	Meaning of Variable	Note

Save LOW WORD of x3 in $108, HIGH WORD in $109

Save LOW WORD of y3 in $110, HIGH WORD in $111

2. Center coordinates after calculation:

   Set the center coordinates as P4(x4, y4), and the length of each point is Signed DW.

   If Var 3 is $200:

   Save LOW WORD of x4 in $200, HIGH WORD in $201

   Save LOW WORD of y4 in $202, HIGH WORD in $203

3. Formulas:

   $x = \triangle x/\triangle$

   $y = \triangle y/\triangle$

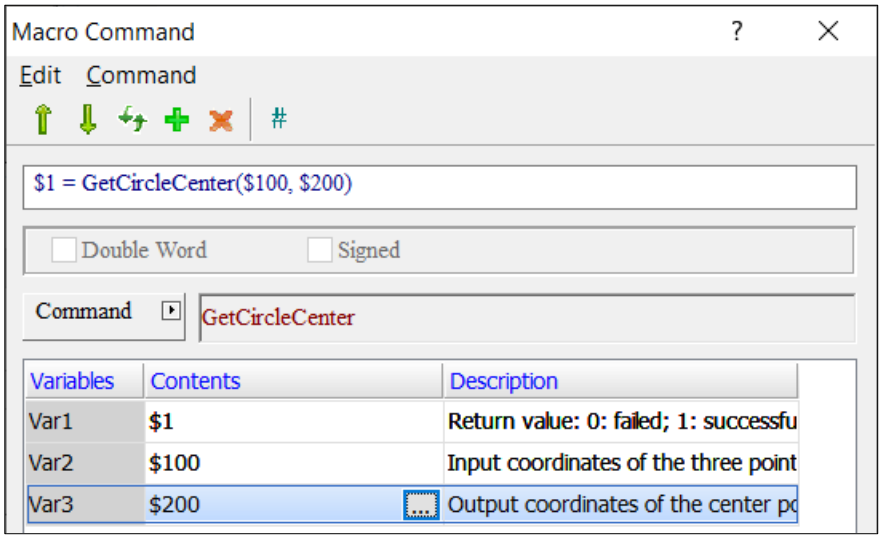
   where $\triangle = 2(xa - xb) * (yc - yb) - 2(ya - yb) * (xc - xb)$

   $\triangle x = (yc - yb) * (xa^2 + ya^2 - xb^2 - yb^2) - (ya - yb) * (xc^2 + yc^2 - xb^2 - yb^2)$

   $\triangle y = (xa - xb) * (xc^2 + yc^2 - xb^2 - yb^2) - (xc - xb) * (xa^2 + ya^2 - xb^2 - yb^2)$

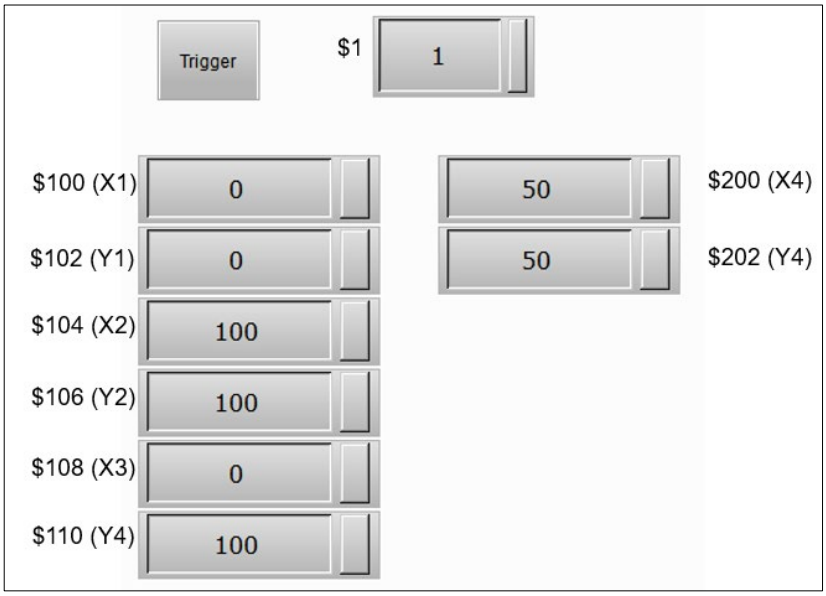
Variable	Type		
	Internal Memory	PLC Register	Constant
Var1	V		
Var2	V		
Var3	V		

Example



The steps are as follows:

1. Create a **Numeric Display** element as $1 and the Data Type is Word.
2. Create **Numeric Entry** elements of $100, $102, $104, $106, $108, and $110, and the Data Type is Double Word.

Example
Enter the following values: $100 = 0 $102 = 0 $104 = 100 $106 = 100 $108 = 0 $110 = 100 3. Create **Numeric Display** elements of $200 and $202, and the **Data Type** is Double Word. After executing the macro, the screen is as follows 

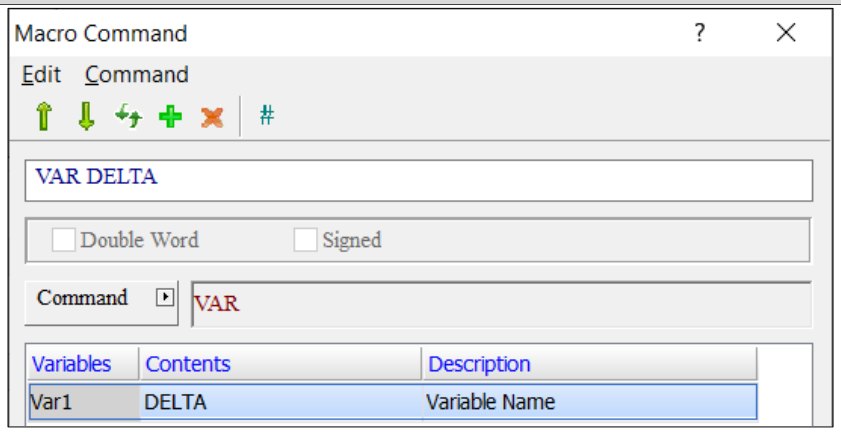
- **VAR**

Expression	Meaning of Variable		Note
VAR Var1 (W)	Var1	Variable Name	W: Word
	Description of Action		
	Specify a name as a global variable.		

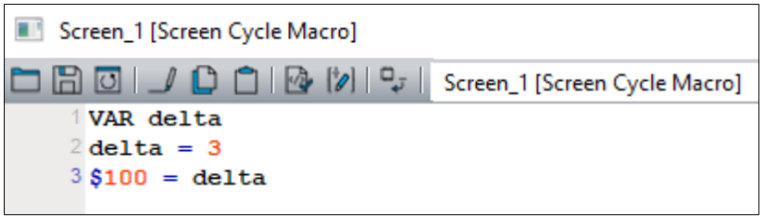
Variable	Type			
	Internal Memory	**PLC Register**	**String**	**Constant**
Var1			V	

Example
• Var1 is a word string. Declare DELTA as a variable.

Example



- Set the value of the variable “delta” as 3, then move the value to $100, and execute $100 = 3.



Macro Error Codes

Error codes will be listed in the output field for easier troubleshooting while compiling the macros. If users miss out a command, the software will also prompt error messages to notify syntax errors.

- **Error message during editing**

Code	Code Name	Cause of Error
-91	Local variable is not allowed	Using local variables is not allowed when parameters of continuous addresses are used in the macro. Example: Parameter 1 of FILL.
-92	Elementary arithmetic error	The variable format is incorrect when the codes of elementary arithmetic are edited.

Code	Code Name	Cause of Error
-93	Submacro does not exist	This Error message indicates that the submacro users are calling does not exist. For example, when users write a line of CALL 5, which means the submacro 5 is to be called, but users did not edit the submacro 5 (could be an input error or users forgot to edit the corresponding submacro), this message serves as a reminder to prevent unpredictable errors.
-94	Number of labels exceeds the maximum	The number of labels exceeds the maximum of 65536.
-95	Flow control setting error	A non-existing flow control setting is used.
-96	Calling a null address	Memory configuration failure causes the macro to use memory address 0 to process the data.
-97	IF exceeds the maximum	The maximum layer of the nested IF statement is 6.
-98	Lack of ENDIF	When the IF statement is used, users must put the ENDIF at the end, so the macro nested within and outside the IF statement can be identified.
-99	IF ELSE execution error	This warning is sent when the IF ELSE statement is edited with its code not matching the IF ELSE logic.
-100	LABEL cannot be found	This error message indicates that the LABEL requested by GOTO cannot be found.
-101	Recursion occurred	This error message indicates the occurrence of recursion which mostly occurs in submacros. The reason is that if a submacro calls itself, whether directly or indirectly, it is the occurrence of recursion. Technically, submacros cannot be recursively edited, but if this is unavoidable, use GOTO or For (infinite loop) instead.
-102	More than 10 nested FOR are used	This error message indicates that more than 10 nested FOR commands have been used. The purpose is to prevent excessive use of nested FOR and insufficient memories. Use GOTO or IF when necessary.
-103	Non-existing submacro is called	The submacro called is not within the range of 1 – 512 (i.e. CALL 0 or CALL 513).
-104	The number of NEXT is less than the number of FOR	This error message indicates that the number of NEXT operand does not match the number of FOR operand. Because FOR and NEXT need to be used in pairs, if a NEXT is missing, an error will occur while the program is running.

Code	Code Name	Cause of Error
-105	The number of FOR is less than NEXT	This error message indicates that the number of NEXT operand does not match the number of FOR operand. Because FOR and NEXT need to be used in pairs, if there is one extra NEXT, an error will occur while the program is running.
-106	Repeated LABEL	This error message indicates that a LABEL is set repeatedly in the same macro. This means GOTO will generate two different results. In order to avoid unpredictable mistakes, this editing error message appears to remind users.
-107	There is RET in macro	This error message indicates that the use of RET in a macro. Since RET is designed for the submacro to return to the next command called, which means the program is not finished. END should be used to end a program if it is necessary to mark the end of a macro.
-109	Incorrect address format in macro	This error message indicates that the input address format in the macro is incorrect.
-110	Recipe address input error	This error message indicates that the input recipe address in the macro is incorrect. The recipe address does not exist.
-111	Recipe address input error	This error message indicates that the input recipe address in the macro is incorrect. The recipe address exceeds the range.
-112	Incorrect COM setting in macro	This error message indicates that the COM setting in the macro is incorrect; the COM port does not allow flow control.
-113	Incorrect COM setting in macro	This error message indicates that the COM setting in the macro is incorrect; the COM port is occupied.
-114	Incorrect COM setting in macro	This error message indicates that the COM setting in the macro is incorrect. Users must set both COM2 and COM3 to RS485. Otherwise, set COM2 to RS232 and COM3 to RS422.
-115	Macro command error	This error message indicates that the command in the macro is incorrect or the called submacro is incorrect.
-117	Disk formatting error	This error message indicates that the device code in the macro is incorrect. It is possible that the parameter is set to an unsupported device (such as an SD card).
-118	Unsupported device is used in macro	This error message indicates that the macro has used the command not supported by the device.

Code	Code Name	Cause of Error
-119	Repeat variable statement	This error message indicates that the macro includes repeat variable statements.

- **Macro Error Messages in the HMI**

  The GETLASTERROR macro command can be used to read an error message on the HMI, but if a new correct command is executed before the error message is read, the error message will be changed. The execution of a macro will not change the error messages of other macros.

Code	Code Name	Cause of error
-1	Program End	The program has ended.
-2	Encoding error	This macro command does not exist.
-3	Lock error	Unable to lock the specified COM Port.
-10	GOTO error	There is a GOTO error in the macro.
-11	Stack overflow	This error message indicates that users have called too many submacros, so the stacks are not enough for use since many different macros are executing at the same time. This is a protection to avoid memory insufficiency.
-12	Empty submacro	This is a call submacro error. Since the command of CALL can call the submacro ID represented by the value in the Internal Memory of CALL, if users enter the value of this memory address while there is no corresponding submacro to call, this error message occurs.
-13	Data Read Error	Data read error may be an error of the Internal Memory data, but it mostly represents data read error of the controller.
-14	Data Write Error	Data write error may be an error of the Internal Memory data, but it mostly represents data write error of the controller.
-15	Divisor is 0	The divisor is 0 during the division or remainder operation.
-16	Data Process Error with BCD format	An error occurred while executing the BCD macro command.
-17	Data Process Error while converting ASCII to HEX	An error occurred while executing the TOHEX macro.

Code	Code Name	Cause of error
-18	NEXT OFFSET Processing Error	The macro data error results in errors while executing the NEXT macro command.
-19	Character Command Error	An error occurred while executing FILLASC.
-20	Data process error with BIN format	An error occurred while executing the BIN command.
-21	Submacro Data Error	The macro data error results in errors while calling submacros.
-22	FOR loop has OFFSET error	The macro data error results in errors while executing the FOR macro command.
-23	INITIAL Error	An error occurred while executing the INICOM macro command.
-24	Memory allocation error	The HMI memory is insufficient to execute macros.
-25	COM Port error	An error occurred in COM Port resulting in execution failure of the COM Port related communication macros.
-26	Print Port error	Print Port selection error when printing.
-27	Read value error	An error occurred when macros read parameter data that are out of range.
-28	IF ELSE ENDIF Error	An error occurred while executing the IF ELSE ENDIF macro command.
-29	Pen Width Setting Error	An error occurred while setting the pen width for the drawing macro.
-30	History Data Error	An error occurred while executing the GETHISTORY macro.
-31	Export Option Error	An error occurred while executing the EXPORT macro.
-32	Disk Reading Error	An error occurred in the external or internal disk resulting in execution error in the associated macros (EXPORT, DISKFORMAT...).
-33	Print Error	An error occurred while the macro is executing printing.
-34	Stack Overflow in IF ELSE ENDIF.	An error occurred due to stack overflow when the IF ELSE ENDIF macro command is executing.
-35	Password Error	Password confirmation is required when executing the macro, and the password entered is incorrect.

Code	Code Name	Cause of error
-36	Password Lock Error	Password confirmation is required when executing the macro, and users have reached the password attempt limit.
-37	ID Code Identification Error	ID password confirmation is required when executing the macro, and the ID password entered is incorrect.
-38	Syntax Error	Syntax error occurred after downloading the PLC program.
-39	Connection Failure/No Response	The connection fails or does not respond when the PLC program is downloaded.
-66	History ID Error	History buffer ID error.
-67	Identified String Error	Unrecognizable RL program string.
-69	Identified Encoding Error	An error occurs when analyzing the RL program.
-70	Variable 1 Error	The first variable in the macro command is incorrect.
-71	Variable 2 Error	The second variable in the macro command is incorrect.
-72	Variable 3 Error	The third variable in the macro command is incorrect.
-73	Field Language Reading Error	Unable to read the RL program or G-code.
-74	Field Language Writing Error	Unable to write the RL program or G-code.
-75	Field Language Execution Error	Unable to execute the RL program or G-code.
-76	Subprogram ID Error	Incorrect Subprogram ID of the RL program.
-77	Subprogram Size Error	Incorrect subprogram size of the RL program.
-78	Exceed the Maximum Number of Program Lines	Number of lines of the RL program exceeds the maximum allowable number.
-79	G Code Compilation Error	G code compilation error.
-80	Exceeding the Range when Converting Double to INT	The value in the data type of Double exceeds the range for the value in int type.
-82	Local Variable Configuration Error	For example, Var1 and Var2 of the BMOV command cannot use the local variables.
-83	Restricted Macro	The current execution is not allowed because there is a restricted macro command. For

Code	Code Name	Cause of error
	Command	example, executing the OpenScreen macro in the Screen Cycle macro.
-85	Temporary Memory Space Error	The specified file space is too small.

- **PLC related file error description, including file formats of DVP and ISP**

Code	Code Name	Cause of error
-40	This file name is not supported	The file name to be opened by the macro execution is not supported.
-41	This version is not supported	The file version to be opened by the macro execution is not supported.
-42	File Open Error	There is an action to open file when the macro is executed, and the action failed.
-43	File Handle Error	Handle error in the file opened by the macro execution.
-44	File Read Error	The file opened by the macro execution cannot be read properly.
-45	File Seek Error	The content of the file opened by the macro execution cannot be moved properly.
-46	File Write Error	The file opened by the macro execution cannot be written properly.
-47	File Removal Error	There is an action to remove file when the macro is executed, and the action failed.
-48	File Rename Error	There is an action to rename the file when the macro is executed, and the action failed.
-49	File Length Error	A file length error is found when the macro is executed.
-50	File Data Error	A file data error is found when the macro is executed.

- **File Slot file Error Description**

Code	Code Name	Cause of Error
-51	File Slot Macro ID Error	The ID for the macro execution to open does not exist.
-52	File Slot Macro Handle Error	Handle error in the file opened by the macro execution.

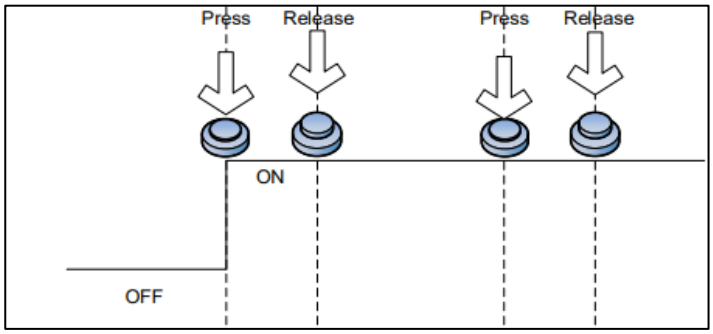
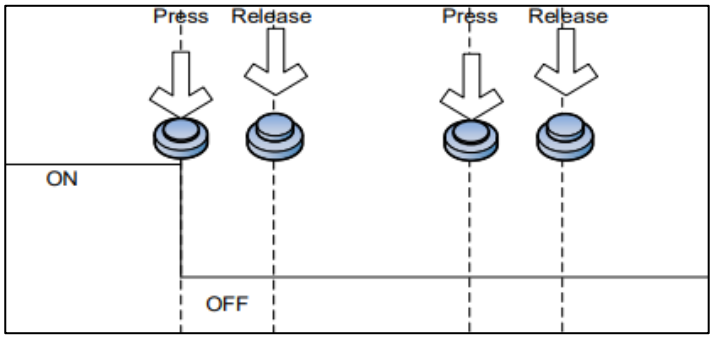
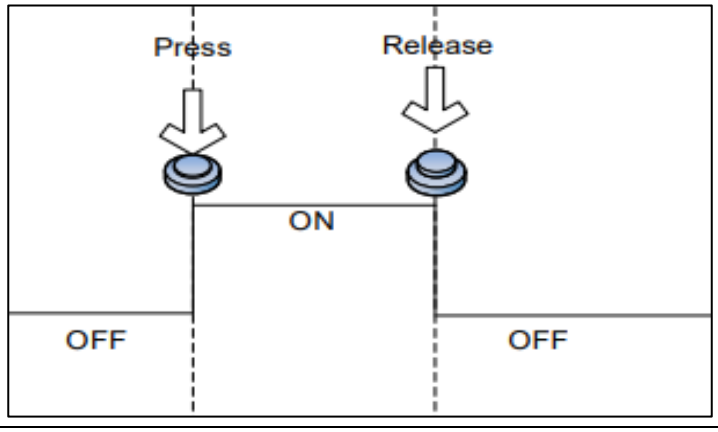
Code	Code Name	Cause of Error
-53	File Slot Macro Read Error	The file opened by the macro execution cannot be read properly.
-54	File Slot Macro Seek Error	The file opened by the macro execution cannot be moved properly.
-55	File Slot Macro Write Error	The file opened by the macro execution cannot be written properly.
-56	File Slot Macro Removal Error	There is an action to remove file when the macro is executed, and the action failed.
-57	File Slot Macro Length Error	A file length error is found when the macro is executed.
-58	File Slot Macro Export Error	The destination device assigned for the file export is incorrect when the macro is executed.
-59	File Slot Export Error	The macro execution fails to export the file.
-61	File Slot Import Error	The macro execution fails to import the file.
-62	The file corresponding to the File Slot does not exist	The FileSlot file read or exported by the macro execution does not exist.
-63	Failed to Read the file from External Disk	The external file to be imported by the macro execution does not exist.
-64	Failed to Copy the file	The macro execution fails to copy the file when performing file import or export.
-84	Failed to Read the File Slot filename	The macro execution fails to read the FileSlot filename.
-86	Failed to set the File Slot filename	The macro execution fails to set the FileSlot filename.
-87	Failed to search for the File Slot ID	The macro execution fails to search for the selected FileSlot ID.

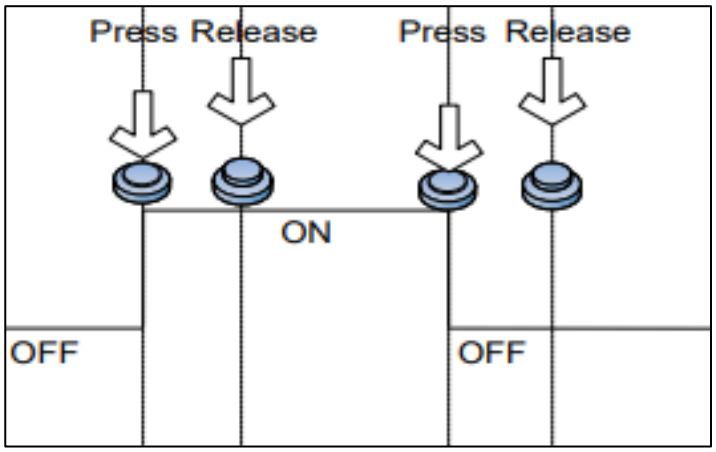
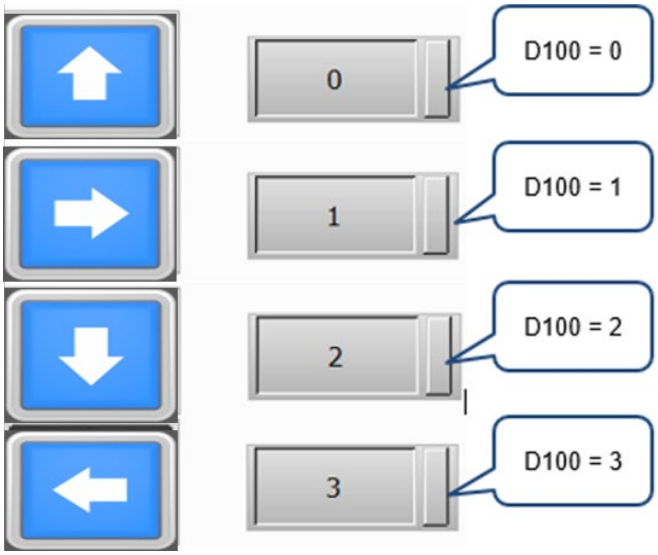
Element

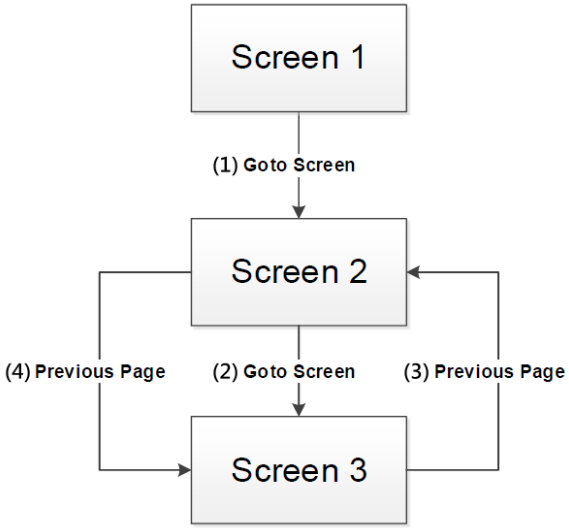
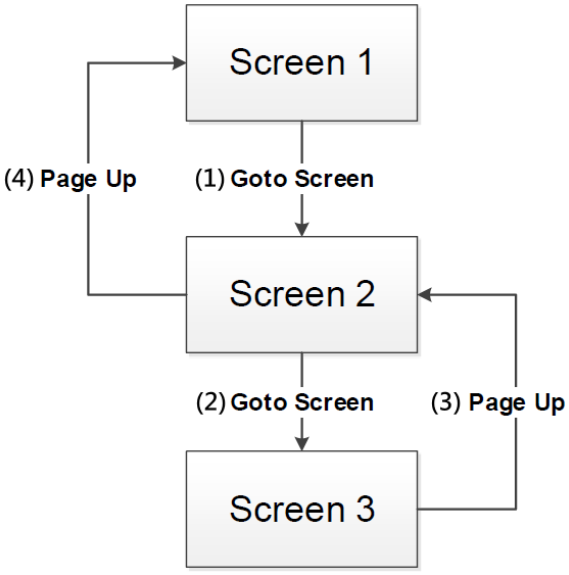
Button

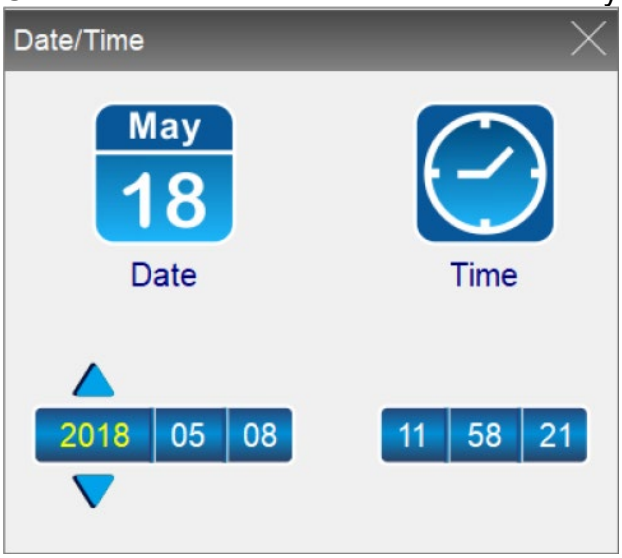
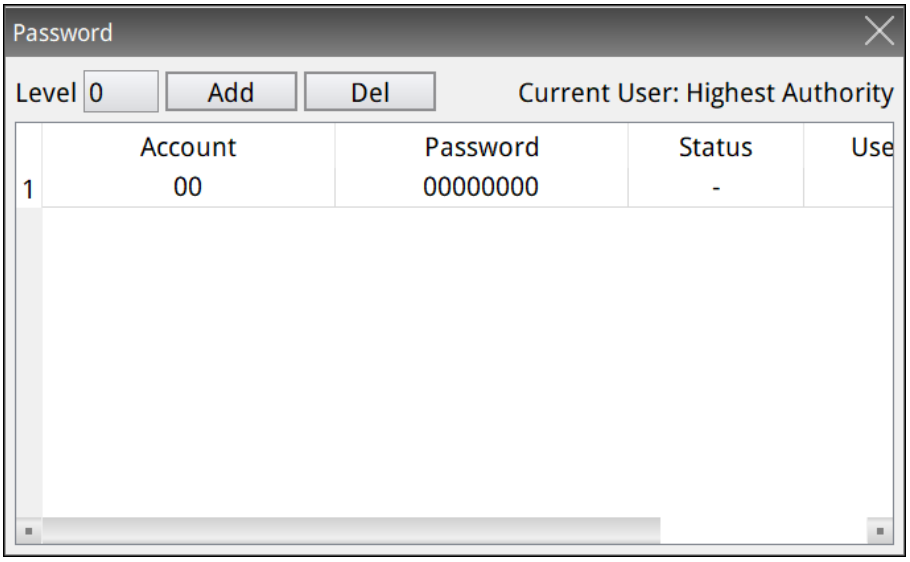
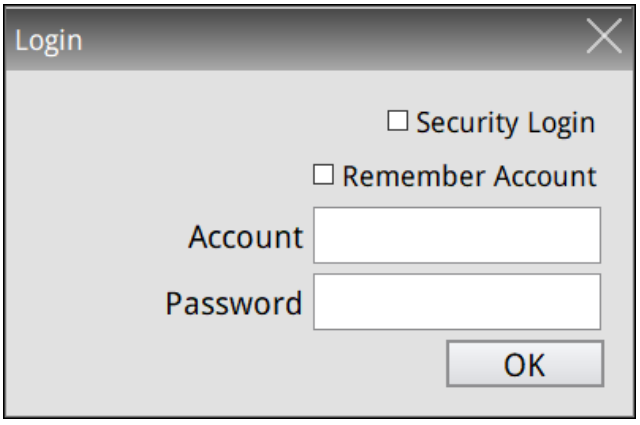
DIAScreen provides multiple button elements, allowing the operations for memory address assignment, system setting change and file processing triggering.

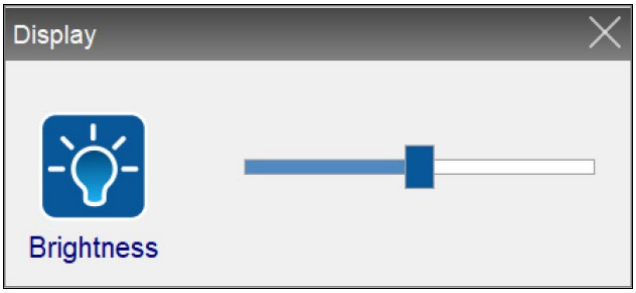
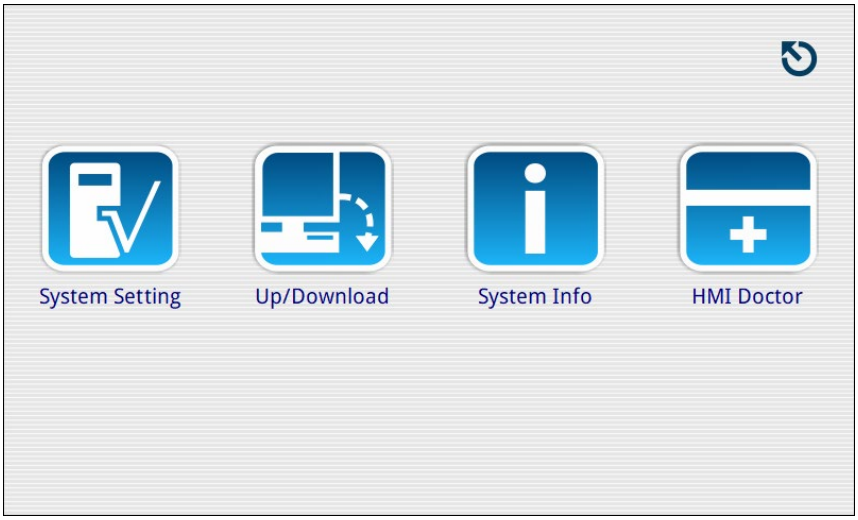
The following table lists the function of each button and their description.

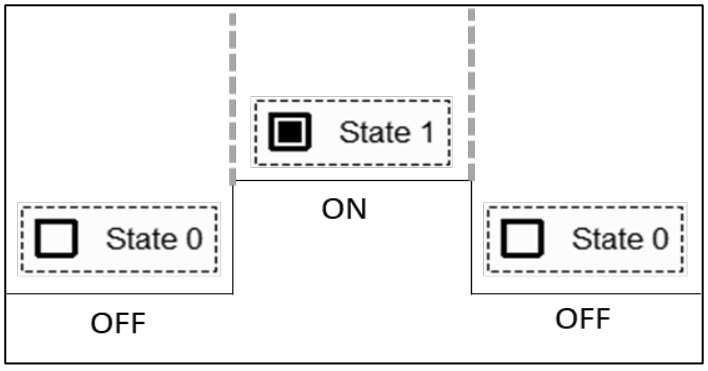
Button	Description
Set to ON	Press this button to set the specified memory address to ON; the state remains ON when pressing it again. 
Set to OFF	Press this button to set the specified memory address to OFF, the state remains OFF when pressing it again. 
Momentary	Press this button to reverse the specified memory address values. The memory address recovers to its original state when the button is released. To keep reversing the memory address values, press the button continuously. 

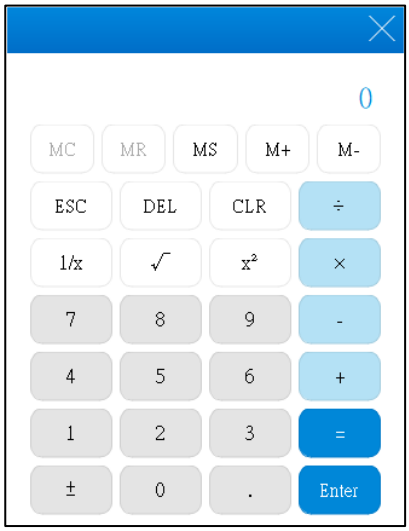
Button	Description
Maintained	Press this button to reverse the specified memory address values. To reverse the current memory address state again, press the button again. 
Multistate	Used to display multiple different state pictures or state values. When pressing this button, the button state switches according to the set **Sequence**. The following figure shows the values of the button state and memory address when **Sequence** is set to **Next State**. 
Set Value	Press this button, and a system keypad pops-up. Enter the value on the keypad to change the memory address value.
Set Constant	Press this button, and the HMI changes the memory address to the specified constant value.
Increment Decrement	Press the **Increment** or **Decrement** button, the HMI changes the memory address values to the set **Increase/Decrease**. If the increased or decreased value exceeds the set upper or lower limit, the memory address values are not changed when the **Increment** or **Decrement** button is pressed again.
Goto Screen	Used to switch the HMI display screen, the options are:

Button	Description
	• **Goto Screen**: The HMI switches to the set screen. You can select the specified screen or set the internal memory address to specify the screen to switch on the HMI. • **Previous Page**: The HMI switches to the previously displayed screen.  • **Page Up**: The HMI records the screen switching sequence. Click this button to switch to the previous recorded page. 

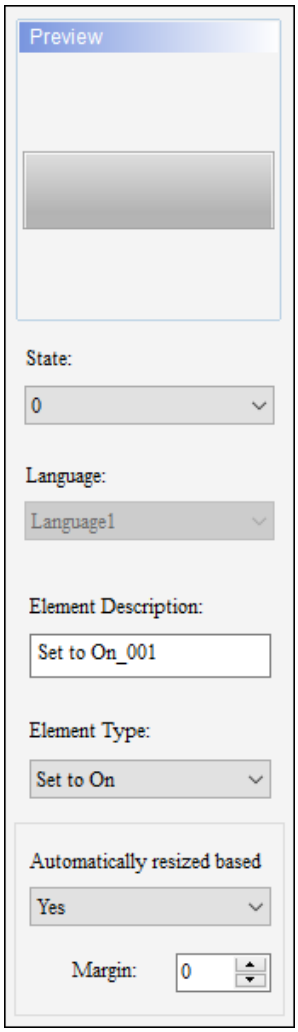
Button	Description
System Date and Time	Used to set the date and time of the HMI system on the HMI. 
Password Table	Used to open the password table of the HMI. The HMI displays the password table under the current user level. 
Enter Password	Click this button to display the **Login** dialog. 

Button	Description
Contrast Brightness	Used to adjust the displayed brightness of the HMI. 
Set Low Security	Used to set the user security level to the lowest.
System Menu	Click this button to enter the System Menu of the HMI. 
Print Output	When the HMI project has set up a printer, the print action can be executed.
Report List	Used to export the alarm records as a CSV file. For operation instructions, refer to Report List.
Template Output	Used to export the alarm records, operation logs, recipe data, and data history records as a PDF file and store them in an external storage device. For operation instructions, refer to Template Output.
Screen Capture	Used to capture the current HMI screen and store it in an external storage device. The captured screen is stored in the “screen_capture” folder of the device.
Remove Storage	To avoid data loss of the storage device connected to the HMI, remove the storage device before turning off the HMI, replacing or removing the storage device.
Import/Export Recipe	Used to import recipe data from the storage device or export recipe data as a CSV file and store it on the device.
Calibration	Used to perform 5-point calibration.

Button	Description
Language Change	Used to switch the display language of the element on the HMI quickly. Go to **General** > **Configuration** > **Multi-language Settings**. **Note**: This setting does not support the system language switching.
Import/Export FileSlot	Used to read / write the specified FileSlot ID on the HMI. For operation instructions, refer to Import / Export FileSlot.
Multiple Actions	Provides multiple button actions. You can configure multiple actions to perform when operating the button. For operation instructions, refer to Multiple Actions.
Motion Actuator	Used to call external programs. **Note**: This function supports only AX-8 and IMP series.
Checkbox	Click this button to reverse the specified memory address values. Click this button again to restore the memory address state to its original state. 
Select File	Used to read the file path on the HMI. After clicking this button, the **Open** window appears. The file path will be returned to the file name address after the file is selected. **Note**: This function supports only AX-8 and IMP series.
Import/Export Contact Information	Used to import Contact.csv to the HMI from the storage device or export the contact data on the HMI to contact.csv.
Enable Next Installment	Click this button to display an unlock screen to set the unlock installment.
Timezone Setup	Used to set the time zone of the HMI. **Note**: This function supports only AX-8 series Linux model.
Import/Export Account	Used to import accounts.csv from the storage device to the password table of the HMI or export the password table as accounts.csv.zip. **Note**: The exported file must be encrypted, and the password must be at least 8 characters and include uppercase/lowercase letters, numbers, and symbols (!, $, %, ^, &).

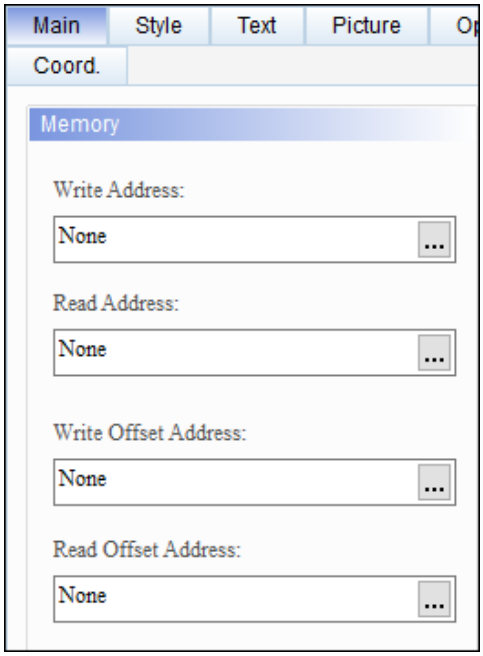
Button	Description
Calculator	Click this button to show calculator on the HMI. 

Common Properties



Setting	Description
State	Select the display state of the Preview image, and the **Style** tab will apply the settings according to the selected state.
Language	Select the display language of the Preview image.
Element Description	Enter the element description. The description will be displayed in the **Description** field of the Operation Log Table to identify the performed actions.
Element Type	Select the element type. This setting supports only the button change for the same type.
Automatically resized based text	If set to **Yes**, when the set text length exceeds the element size, the element is automatically resized according to the **Margin** to show the complete text. **Note**: This setting supports the element enlargement function, the string is shortened, and the element will not become smaller.

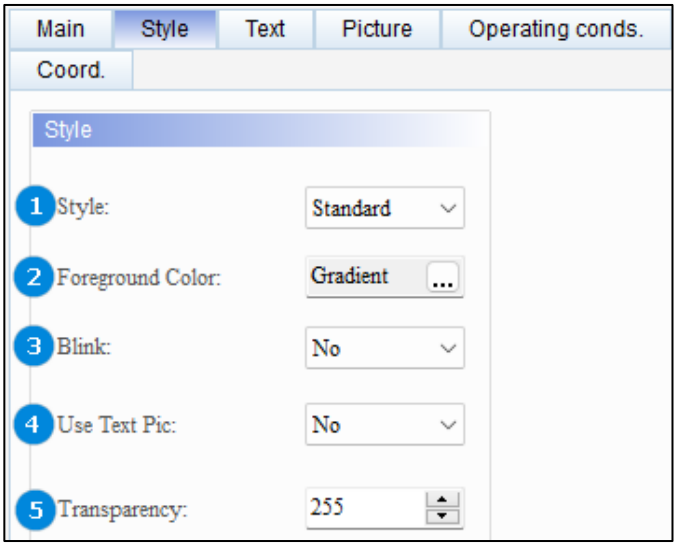
- **General**
  - **Memory**



Setting	Description
Write / Read Address	Refer Address Settings to set the write / read address of the element. If only the **Write Address** is set but the **Read Address** is not set, the HMI will consider the read address and write address to be the same.

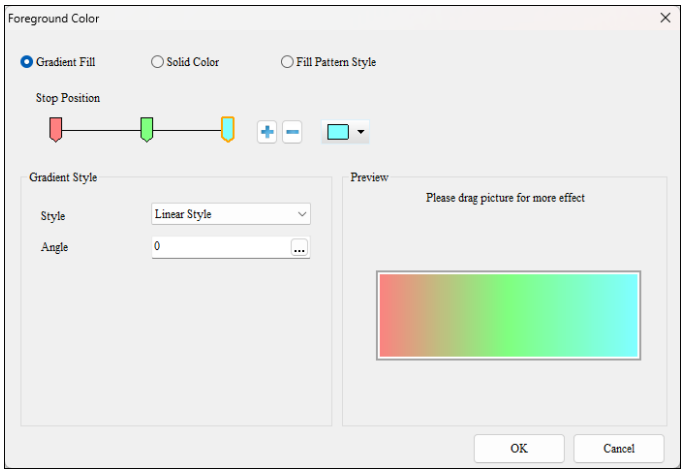
Setting	Description
Write / Read Offset Address	The HMI will configure the value to the corresponding write / read address according to the set Offset Address. After setting the offset address, the actual read and write address are as follows. • The actual write address of the element: Write Address + Value in Write Offset Address × Element [Data Type]. • The actual read address of the element: Read Address + Value in Read Offset Address × Element [Data Type]

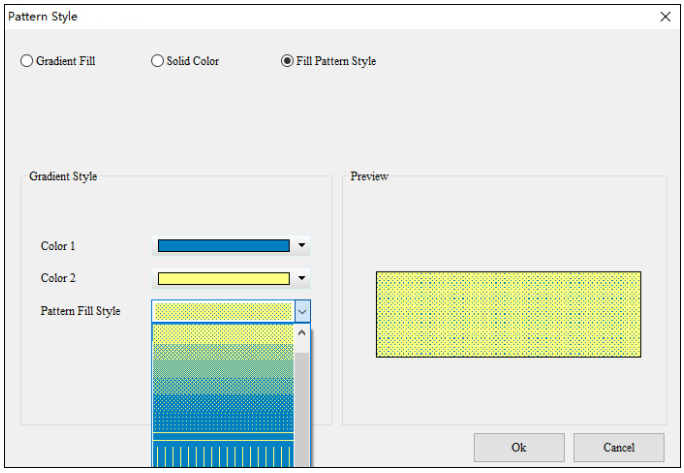
- **Style**



After selecting a state, modify the following settings and apply them to the state.

Legend	Setting	Description
1	Style	Select the appearance of the element. The default value is **Standard**. **Standard** / **Raised** / **Round** / **Invisible**
2	Foreground Color	Select the color of the element. • **Gradient Fill:** Set the gradient style by setting the current angle or center point. You can drag the mouse on the preview image to set the

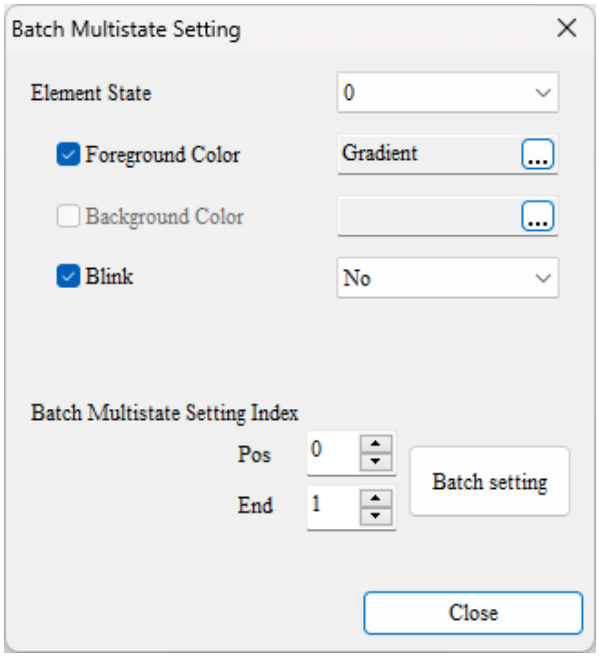
<table>
<tr><th>Legend</th><th>Setting</th><th>Description</th></tr>
<tr><td></td><td></td><td>current angle or center point, or set the memory address to dynamically configure it on the HMI. You can also set the stop point colors to present the effect of multi-color gradients.
 

Note: Up to 5 stop points can be set.
• Solid Color: Select a single-color fill.
• Fill Pattern Style: Select a two-color combination configuration and select a pattern fill style.
 
</td></tr>
<tr><td>3</td><td>Blink</td><td>Choose whether to enable the blink prompt. The blink color is opposite to the foreground color of the status.</td></tr>
<tr><td>4</td><td>Use Text Pic</td><td>Choose whether to use a text picture. This setting can be compatible with the effect of the text display of the DOP-B project. The effects are as follows.
<table>
<tr><th>Use Text Pic is set to Yes</th><th>Use Text Pic is set to No</th></tr>
<tr><td>HELLO</td><td>HELLO</td></tr>
</table></td></tr>
</table>

Legend	Setting	Description
		Note: When using DIAScreen to open the DOP-B project, this setting will be set to **Yes** by default; if adding a new DOP-100 project, this setting will be set to **No** by default.
⑤	Transparency	Select or enter the transparency value of the element. The range value is 50—255, and the default value is **255**. The smaller the value, the higher the transparency of the element.

- **Batch Multistate Setting**

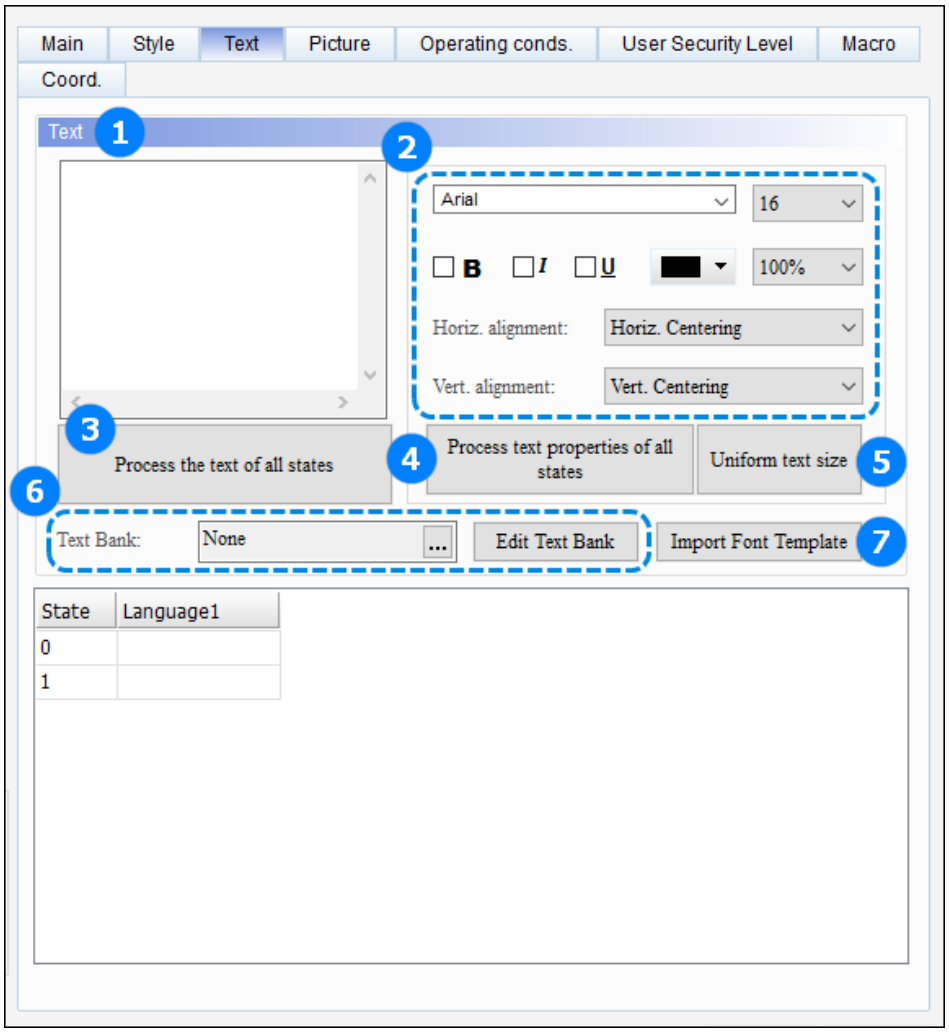
  Note: The setting is supported in **Bit Setting Button** and **Multistate**.



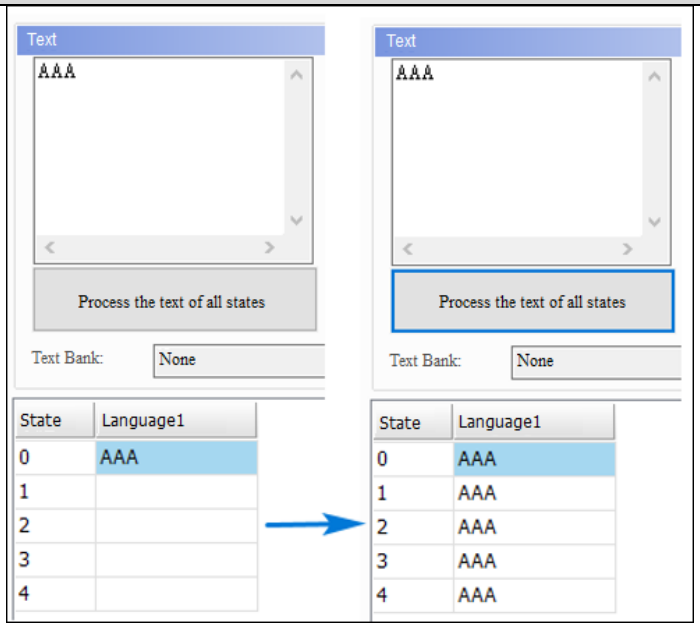
Follow these steps to configure the **Batch Multistate Setting**.

1. Select the **Element State**.
2. Select the properties checkbox to apply and set **Batch Multistate Setting Index**. For example, set the **Foreground Color** of state 0~5 in batch to **Gradient**, and then click **Batch setting**.

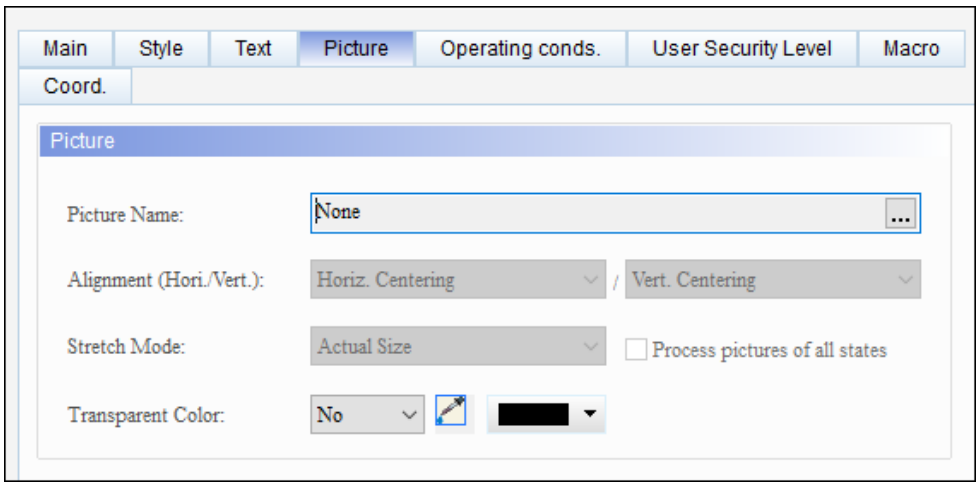
- **Text**

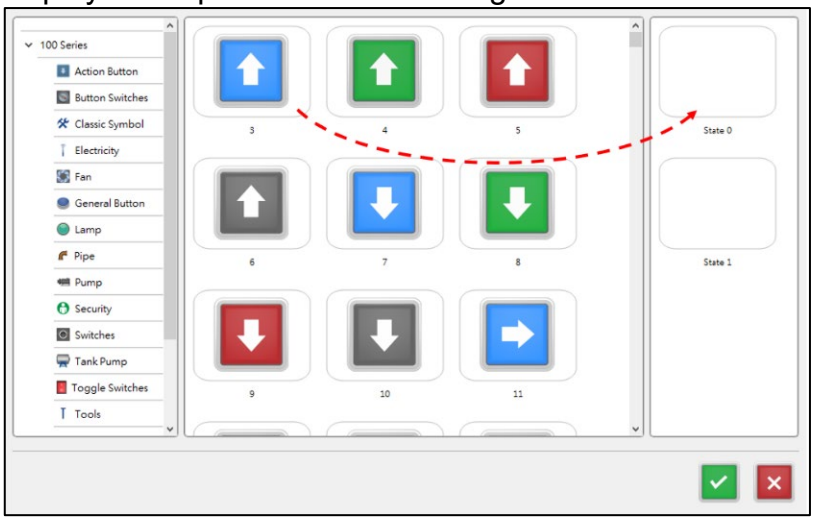


Legend	Setting	Description
1	Text	Enter the text message to display in the text box. Select the state / language table below to directly set the text displayed by that state / language. Enter the text message to be displayed in the text box. Click the element on the screen and press **Space** to edit the text content of the current state / language.
2	Text-related properties	Set the text properties.
3	Process the text of all states	Click to apply the selected text contents of the state to those of other states.

Legend	Setting	Description
		
❹	Process text properties of all states	Click to apply the selected text properties of the state to those of other states.
❺	Uniform text size	Changes the text size in all languages with one click according to the currently selected state.
❻	Text Bank	Applies the text in the **Text Bank** to the content of each state / language. You can also click **Edit Text Bank** to edit the text content in the **Text Bank** editor.
❼	Import Font Template	Click to open the **Font Template** dialog, and then select the template and import the text properties of the selected state.

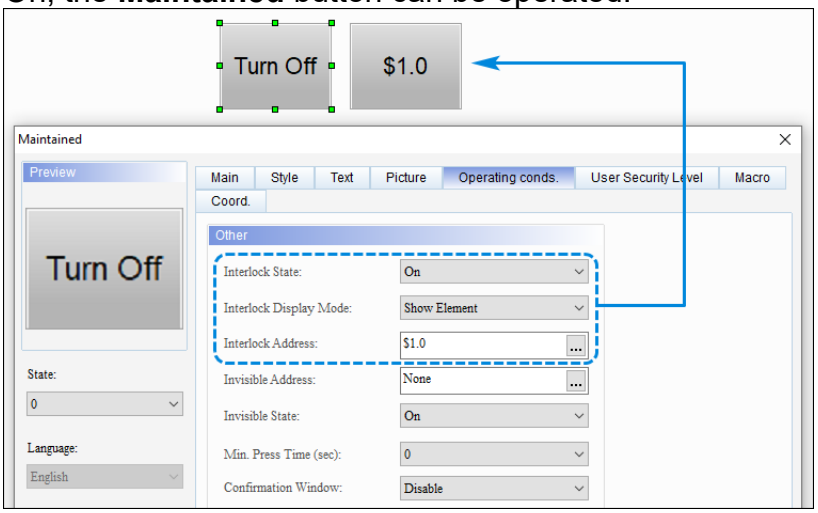
- **Picture**

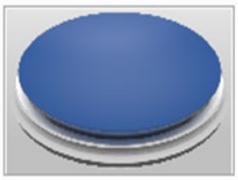
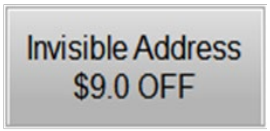


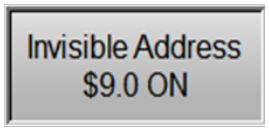
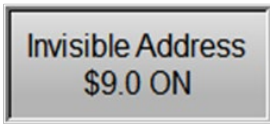
<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Picture Name</td><td>Enter the picture name. The default value is None.To set the picture display, click ... to select the picture to display in the picture bank and drag it to each state icon.
 

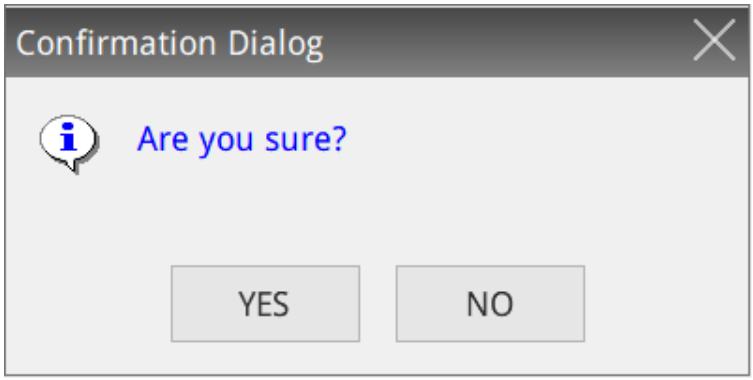
To configure other pictures, refer to Picture Bank.</td></tr>
<tr><td>Alignment</td><td>Select the alignment of the picture.</td></tr>
<tr><td>Stretch Mode</td><td>Select the stretch mode of the picture.
<table>
<tr><th>Stretch All</th><th>Stretch 1:1</th><th>Actual Size</th></tr>
<tr><td>The picture fills the full element display area.
 </td><td>The picture displays in 1:1 size based on the element width and length.
 </td><td>The picture is displayed in the display area of the element in its actual size.
 </td></tr>
</table></td></tr>
<tr><td>Process pictures of all states</td><td>If an element has multiple state values, and its picture may not fill the full display area of the element. Select this checkbox to process all pictures settings</td></tr>
<tr><td>Transparent Color</td><td>Choose whether to specify a color in the picture as a transparent color. If set to Yes, you can select colors or use the color picker to select one of the colors on the picture to specify it as a transparent color, that is, the foreground color. The following figures show the effect of specifying white as a transparent color.
</td></tr>
</table>

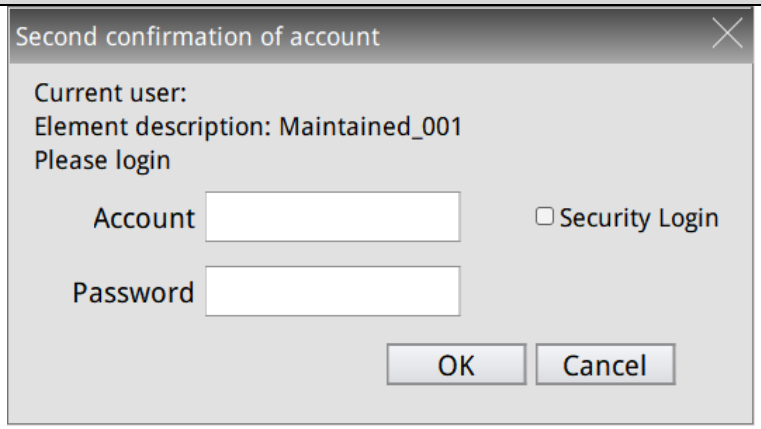
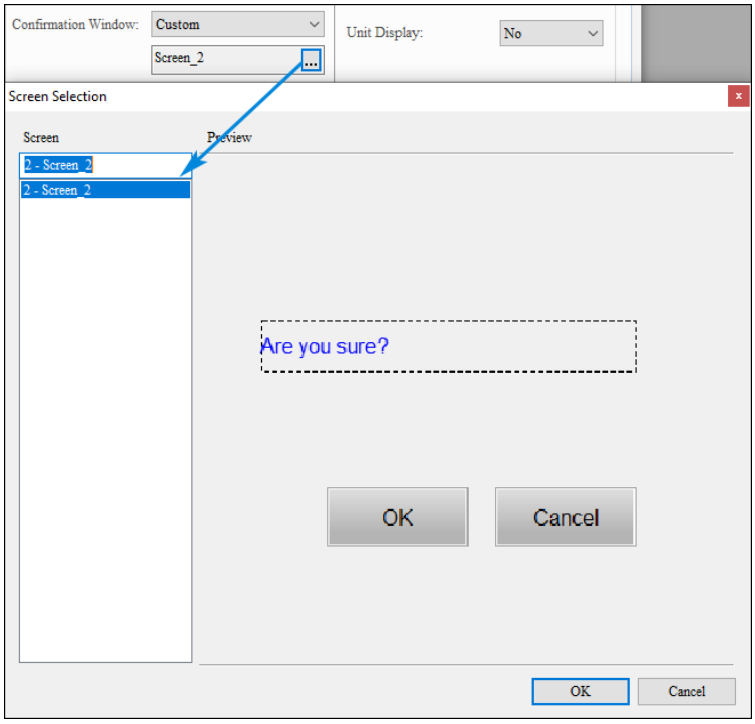
- **Operating Conds.**

<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Interlock State</td><td rowspan="3">The Interlock State enables to operation of a certain element and must be used with Interlock Address. If the Interlock State is set to Off, the Interlock Address enables to operation of a certain element. Instead, if the Interlock State is set to On, the Interlock Address enables to operation of a certain element. As shown in the following figure, when the Interlock Address $1.0 is On, the Maintained button can be operated.
 
</td></tr>
<tr><td>Interlock Display Mode</td></tr>
<tr><td>Interlock Address</td></tr>
<tr><td>Interlock Address</td><td>The effects of Interlock Display Mode are as follows.
<table>
<tr><th>Show Element</th><th>Show Prohibition Symbol</th></tr>
<tr><td> </td><td> </td></tr>
</table></td></tr>
<tr><td>Invisible Address</td><td>The Invisible Address hides an element on the HMI and must be used with Invisible State. If Invisible State is set to On, the Invisible Address hides the element. Instead, if Invisible State is set to Off, the Invisible Address hides the element.
The following is an example of using invisible address.
1. Create a button element on the screen and set its address to $9.0, then set the Invisible Address of the element to hide to $9.0.
2. To hide the element, you must click the $9.0 button.</td></tr>
</table>

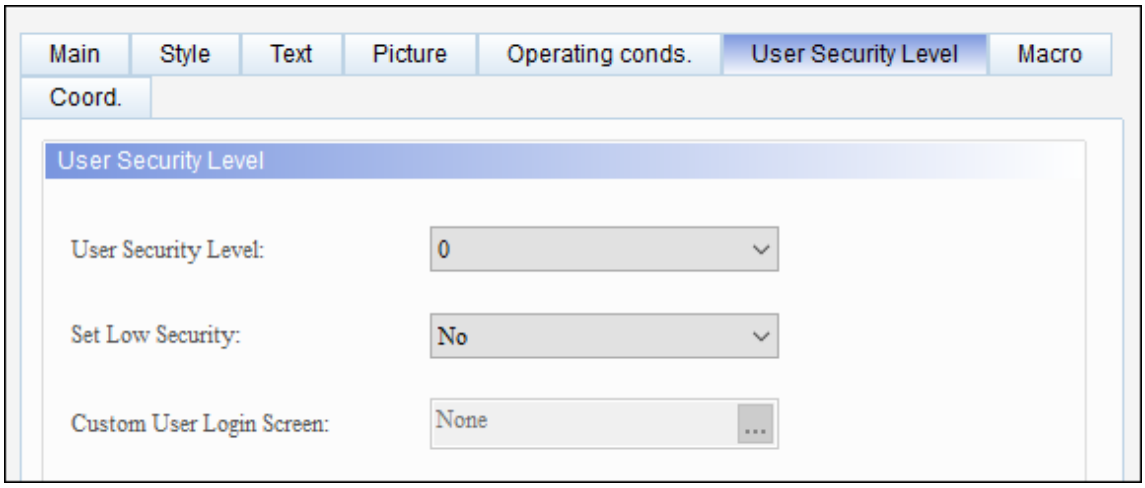
<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Invisible State</td><td>
<table>
<tr><td>Invisible State is On</td><td> 
 

Element disappear
 
</td></tr>
<tr><td>Invisible State is Off</td><td>
 

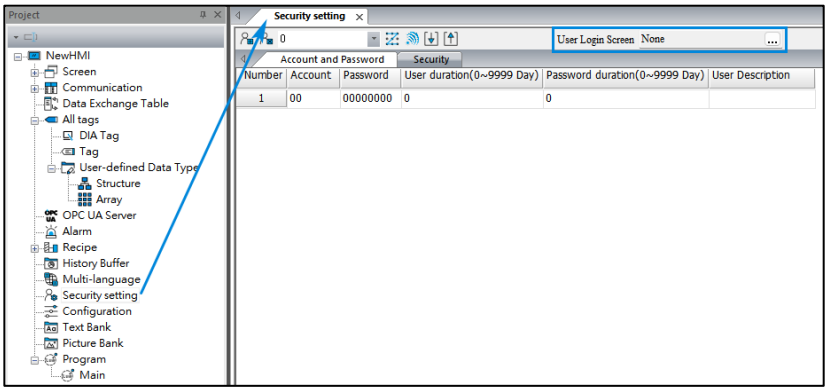
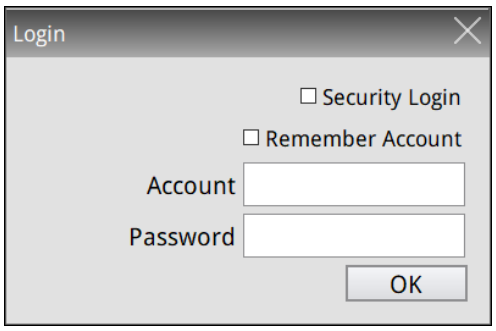
Element disappear
</td></tr>
</table>
</td></tr>
<tr><td>Min. Press Time</td><td>Select the time the element acts after it is pressed. The element will only activate after being pressed for a set time to prevent malfunction. The range value is 0–10.</td></tr>
<tr><td>Confirmation Window</td><td>Select the confirm window type.
Disable: After pressing the element, the button action will be performed without displaying the dialog.
Confirmation: On the HMI, pressing the element on will pop up the Confirmation Dialog, and the button action will be performed only after confirmation in the dialog.
 
Second confirmation of account: Pressing the element will display the Second confirmation of account dialog, and you need to enter the account and password to perform the button action.</td></tr>
</table>

Setting	Description
	 • **Customize**: Select the custom keypad dialog. You can go to **General** > **New Screen** > **Confirmation Window Screen** to add a Confirmation dialog screen. 
Modifier + Hot Key	Select the modifier and hot key. You can execute buttons through the shortcut keys when using an external keyboard.
Ignore Operation Log	Choose whether to enable **Ignore Operation Log**. When the operation log settings are enabled, the HMI will record the operations of all written elements. If **Ignore Operation Log** of the element is set to **Yes**, the element action will not be recorded in the operation log.

- **User Security Level**

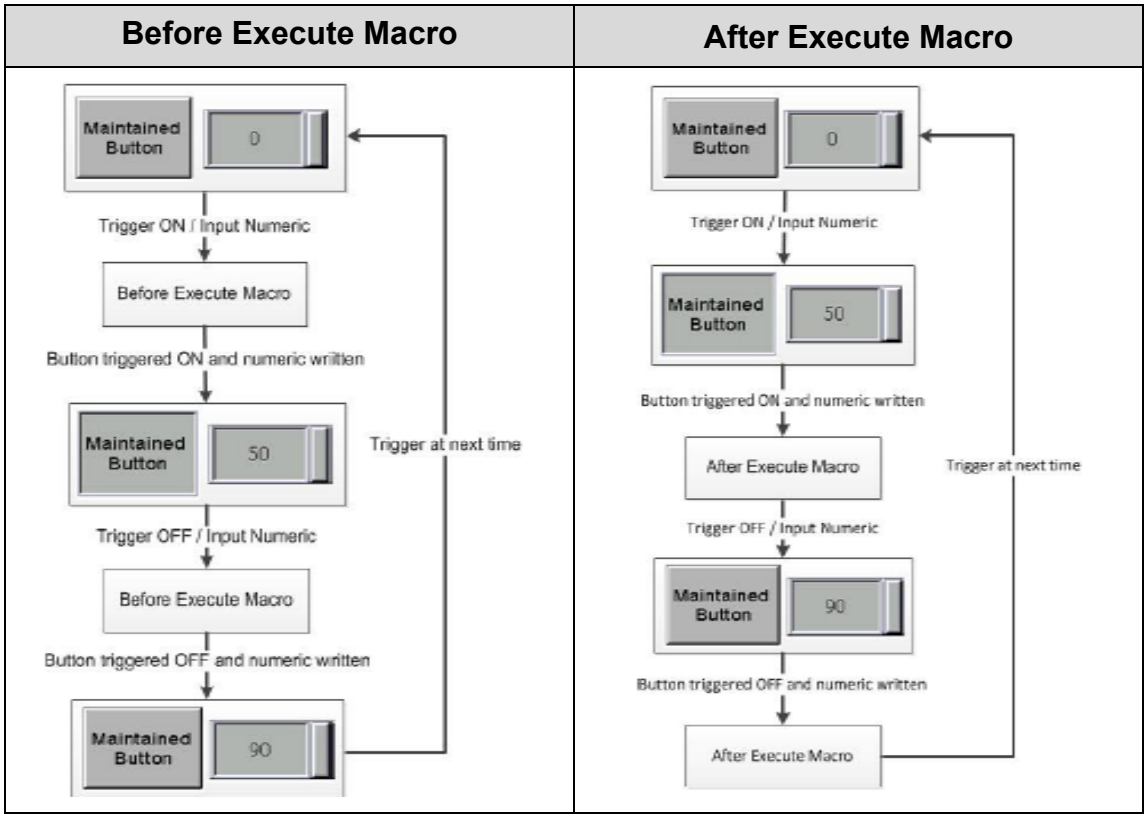


Setting	Description
User Security Level	Select the user security level. If the current security level is insufficient, the **Login** dialog pops up when the element is clicked on the HMI. The element action is performed if the login security level is equivalent to or higher than the set **User Security Level**. **Note**: When set to **0**, the element action can be performed without login.
Set Low Security	Choose whether to set the security level to the lowest security level after entering the security level. If set to **Yes**, the HMI will automatically set the security level to the lowest (Level 0) after performing the element action. Every time the element is triggered, the **Login** dialog will pop up, and you can perform the element action after logging in.

Setting	Description
Custom User Login Screen	Set the custom login screen. When **User Security Level** is not set to **0**, the custom login screen can be selected. When triggering the element and the user security level is required the set screen pops up. • On the toolbar, click **General** > **New Screen** > **User Login Screen** > **Login/Logout Screen**. • If the element does not have a specified login screen, you can go to **Project** > **Security Setting**. Set the **User Login Screen** of each security level on the **Security Settings** page. When the element sets this security level, the login screen configured in this page is displayed.  • If the element or account does not set the login screen, the default login screen is displayed. 

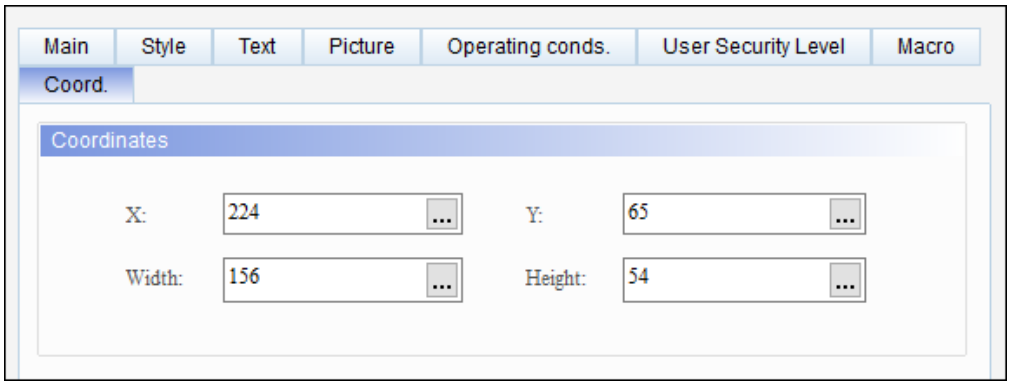
- **Macro**

  After clicking the element on the HMI, the HMI executes **Before Execute Macro**, then executes the button action, and finally executes **After Execute Macro**.



Note: If the **Button** states are not changed with the button touch (using external controller commands or other macros and programs instead), the HMI does not execute the **Macro** commands.

- **Coord.**



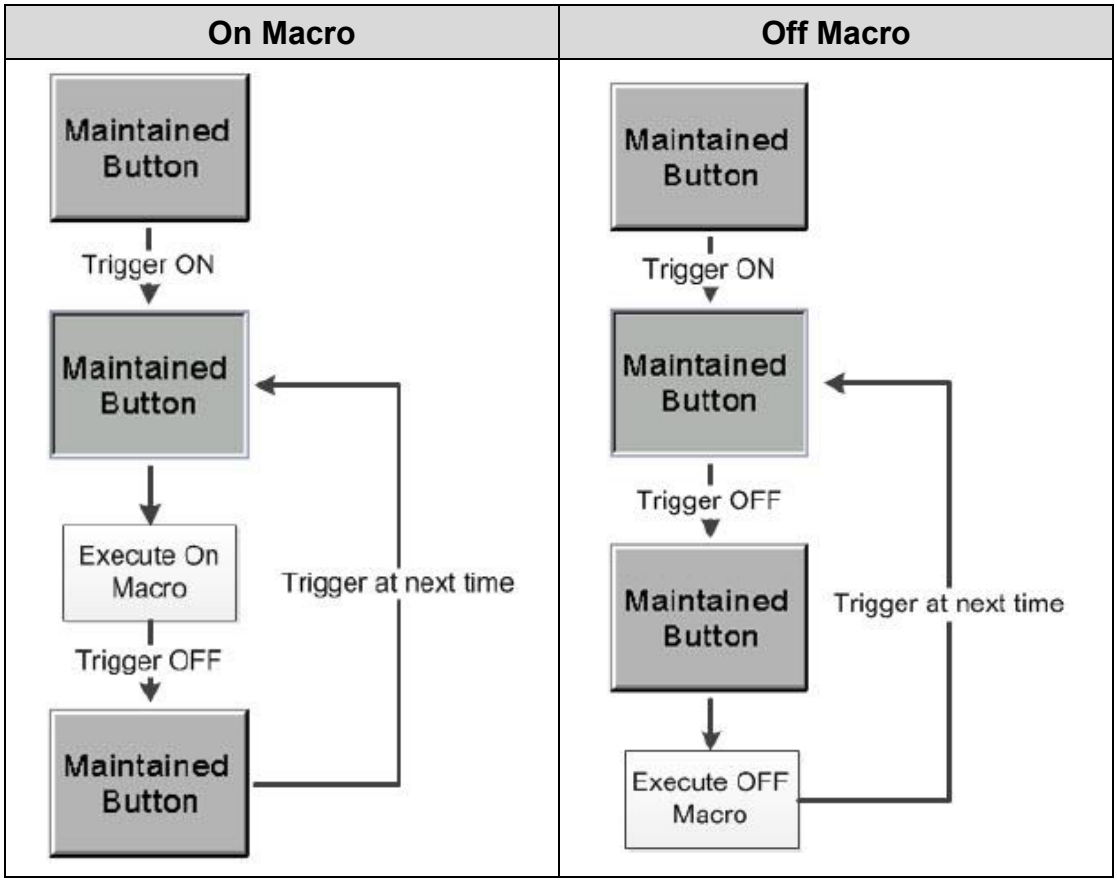
Set the XY coordinates, width, and height on the upper left of the element. You can enter a constant as fixed position or memory address to dynamically configure the position of the element on the HMI.

Bit Setting Button

The Bit setting button contains Set to ON, Set to OFF, Momentary, Maintained, Checkbox, and so on. The type of these elements can be changed by setting **Element Type**. The following describes the special properties of the Bit settings button.

- **Macro**
  - **On Macro / Off Macro**

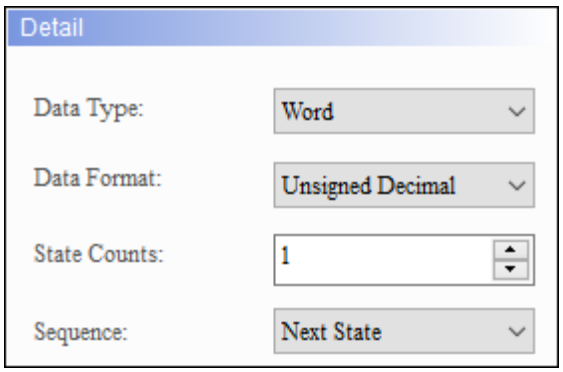
    When switching the state to On by pressing the button element, the HMI executes the On Macro commands. When switching the state to Off by pressing the button element, the HMI executes the Off Macro commands. Each time the ON / OFF state is triggered, the On Macro / Off Macro will only be executed once and will not be executed repeatedly.



Multistate

The following table lists the special properties of the **Multistate** button and their description.

- **Main – Numeric value settings**



Setting	Description
Data Type	Select the unit for state switching
Data Format	Select the data format. **Note**: This setting is only available when **Data Type** is set to Word.
State Counts	Select or enter the state counts according to the selected Data Type. • Bit: 2 states can be set. • Word: 1~256 states can be set. • LSB: 16 states can be set. • LSB (Support State 0): 17 states can be set. **Note**: LSB means that the register data is first converted into binary data, and then the current object state is defined by using the lowest non-zero bit. For example, decimal value 3 converted to binary is 0000000000000011, and the lowest non-zero bit is Bit 0, State is 1.
Sequence	Select the switching sequence for the state of the multistate button. • Next State: When switching states, the state is switched in ascending order. • Previous State: When switching states, the states is switched in descending order.

Numeric Value Setting Button

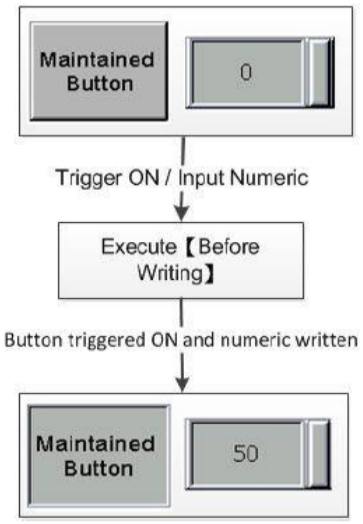
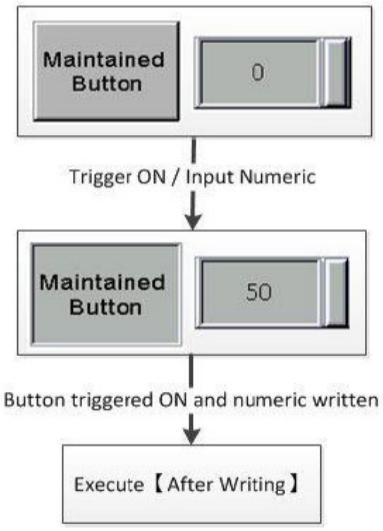
The numeric value setting button contains Set Values, Set Constant, Increment, Decrement, and so on. The following table lists the special properties of the numeric value setting button.

- **Main – Numeric value settings**

Setting	Description
Data Type	Select the data type. The available options are Word, Double Word, and Quad Word.
Data Format	Select the data format. The options vary from the data type settings, including BCD, Signed BCD, Signed Decimal, Unsigned Decimal, Hexadecimal, Binary, and Floating.
Value	Enter a valid range value according to the selected data type and data format.

- **Operating conds. – Trigger Mode / Trigger Address**

The following table list the trigger mode and trigger flowchart.

Trigger Before Writing	Trigger After Writing
The trigger address is set to ON first, and the value changes. 	The value is changed first, and the trigger address is set to ON. 

Note: Since the trigger function only sets the set trigger address to ON, you need to set the trigger address to OFF if you want to trigger it again.

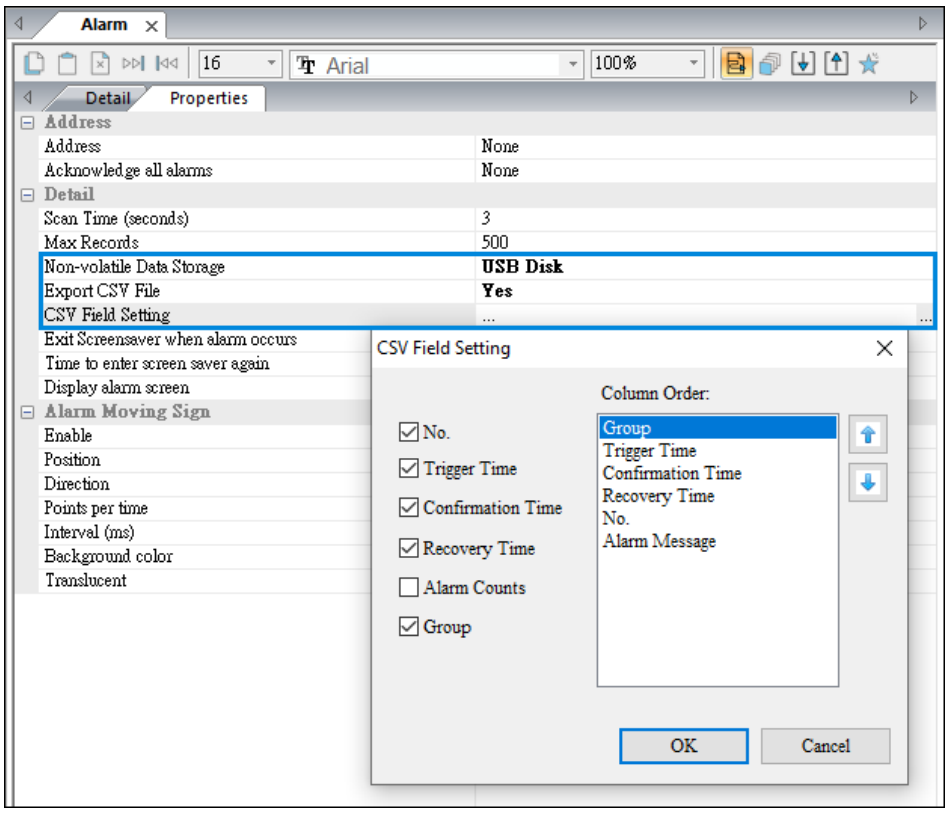
Report List

The alarm data can be exported as CSV file and stored in external devices through the **Report List** element.

Follow these steps to set the report list.

To set the report list

1. In **Project**, double-click **Alarm**.
2. In the **Alarm** settings page, select the storage location of non-volatile, and then select **Yes** for **Export CSV File**.
3. Configure the **CSV Field Setting**. In the **CSV Field Setting** dialog, configure the content and order of the field to display.



4. Create a **Report List** element on the screen.
5. Click the element on the HMI, then export the alarm information as a CSV file and store it to the set report device.

Note: The report file storage path is `\HMI\HMI-000\@HMI0000\HistoryOp\CSV`.

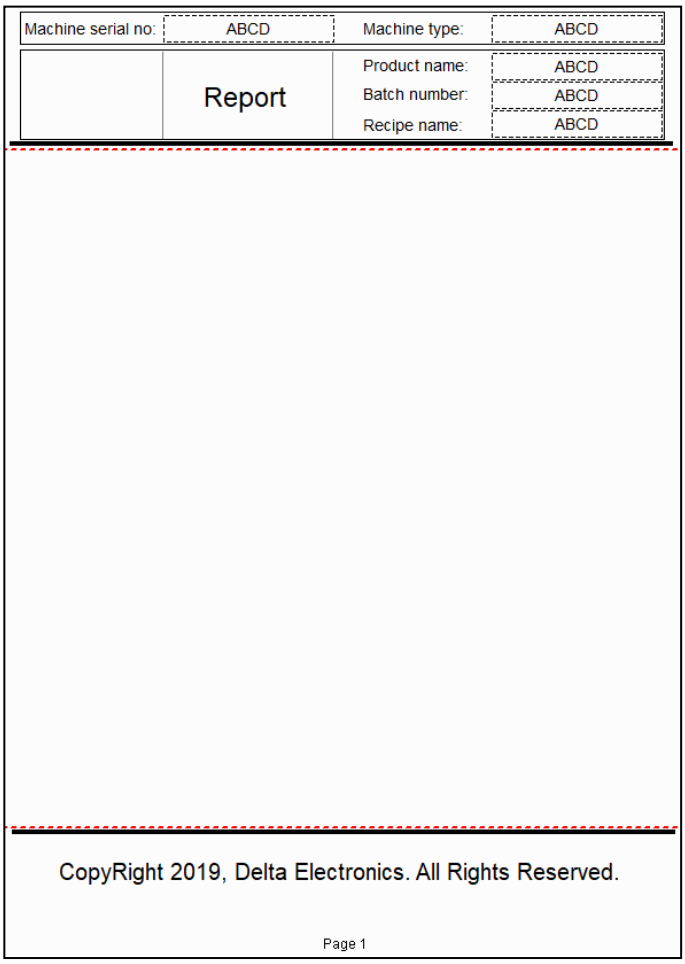
Template Output

Alarm records, operational records, recipe data and data history records can be exported as PDF files and stored in external storage devices through the **Template Output** element.

Follow these steps to set the template screen.

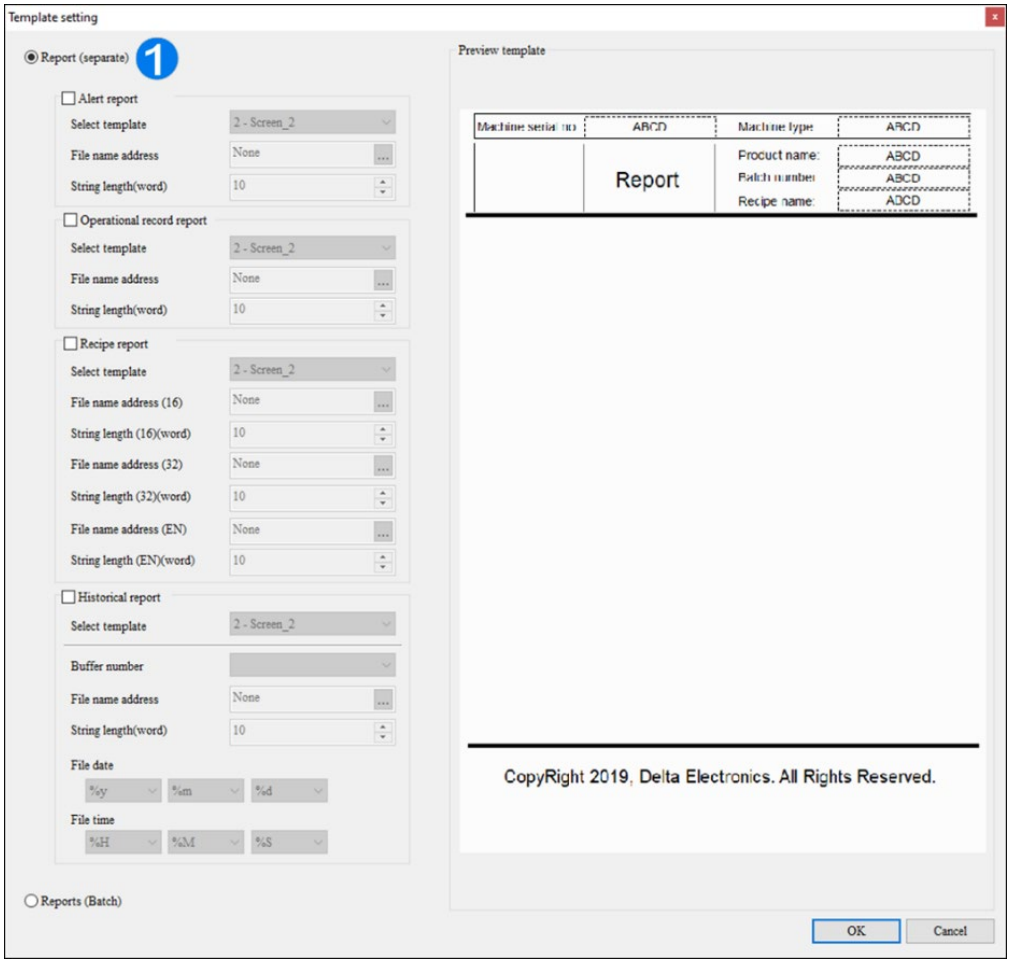
To set the template screen

1. Create a template screen. Do one of the following:
   - In the menu bar, click **General** > **New Screen** > **Template Screen**.
   - Right-click the blank space on the **Screen Management Window** and select **New Screen**. Select **Template Screen** for **Screen Type** and click **OK**.
2. In the template screen, you can customize the content displayed at the header and footer of the page. The blank space between two red dotted lines is the content of the report.



3. In the **Screen property setting** dialog, set the display style of the page number in the **Template setting** area.
4. Create a **Template Output** element on the screen.
5. In the **Template Output** property dialog, select the **Report Device**.

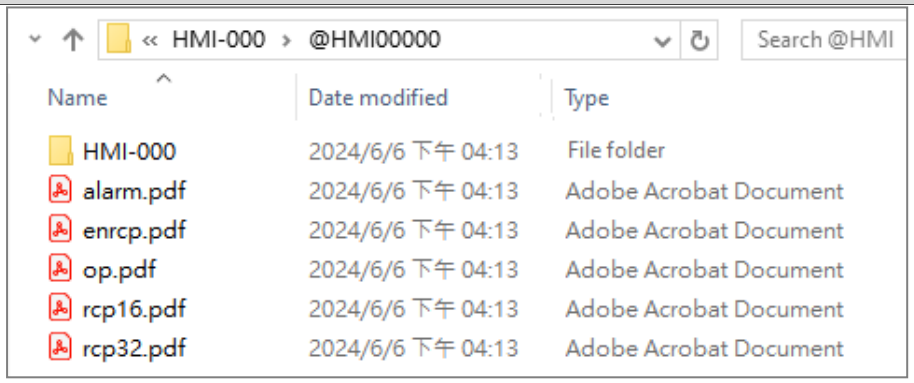
The following table lists the settings in the **Template setting** dialog with their descriptions.

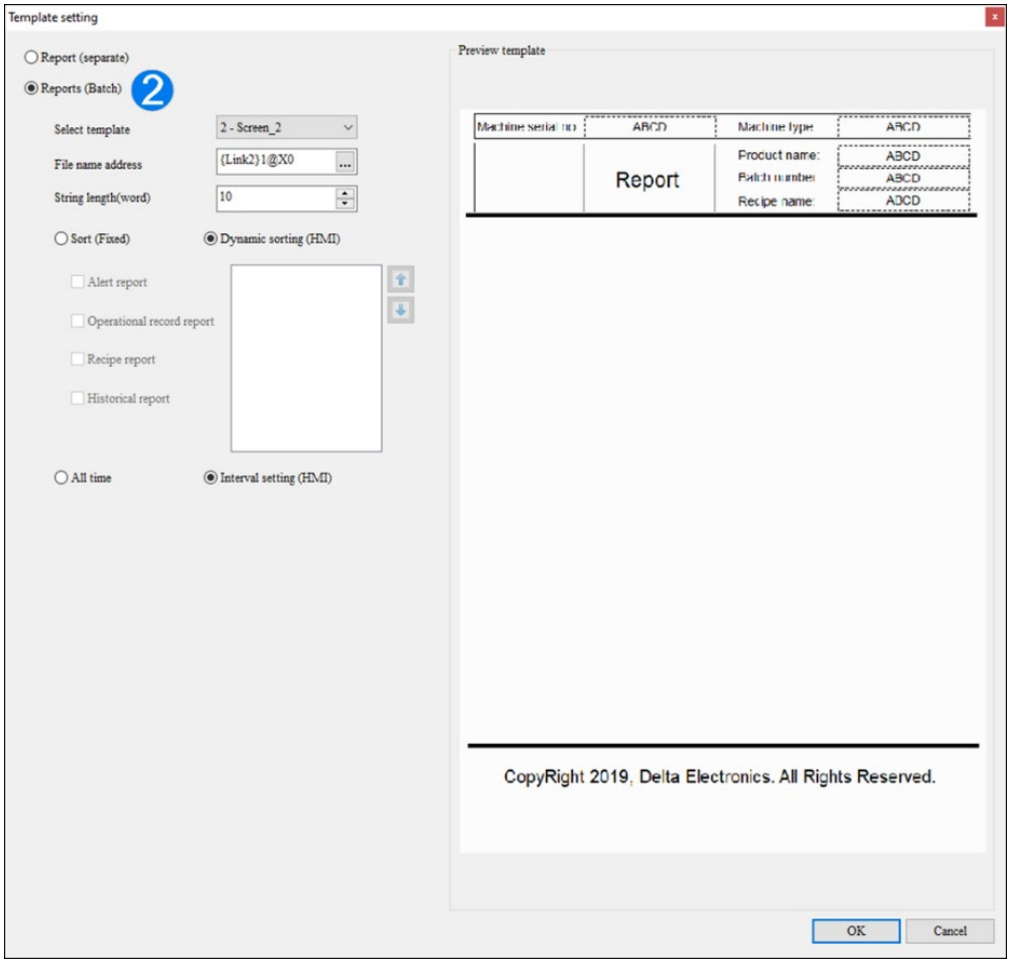


1 Report (separate)

After creating report output of each type, PDF files will be generated respectively.

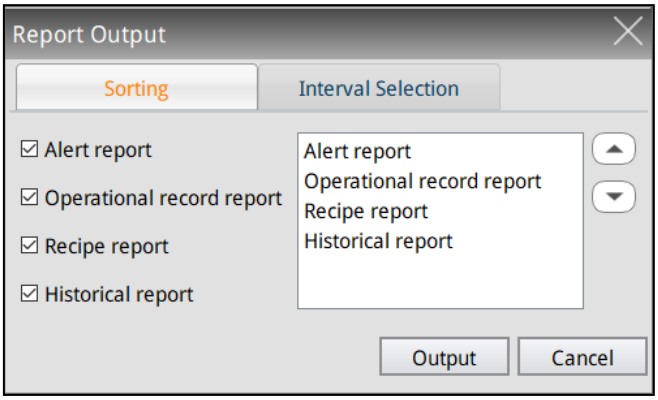
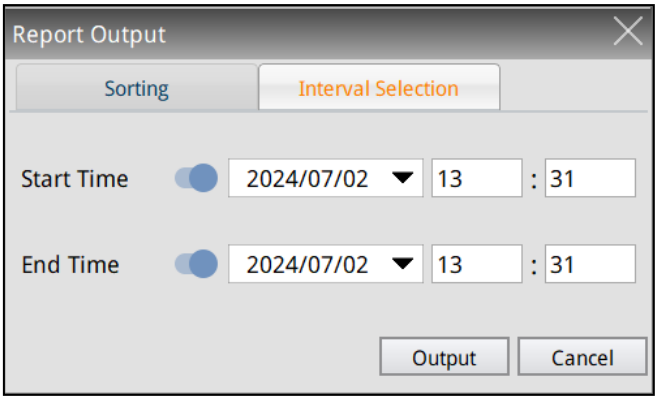
Setting	Description
Select template	Select the template you want to apply.
File name address	Set the profile name address. If not set, the report file gets exported with the default file name. **Note**: The report file storage path is `\HMI\HMI-000\@HMI0000`.

Setting	Description
	
String length	Select or enter the string length for the file name.
File date	Select the date display format for the file. You can also set it to None to not to export the file date. **Note**: This setting is only available for historical reports.
File time	Select the time display format for the file. You can also set it to None to not to export the file time. **Note**: This setting is only available for historical reports.



② Reports (Batch)

After customizing the display order of each type of report and exporting the reports, they will be consolidated as a single PDF file.

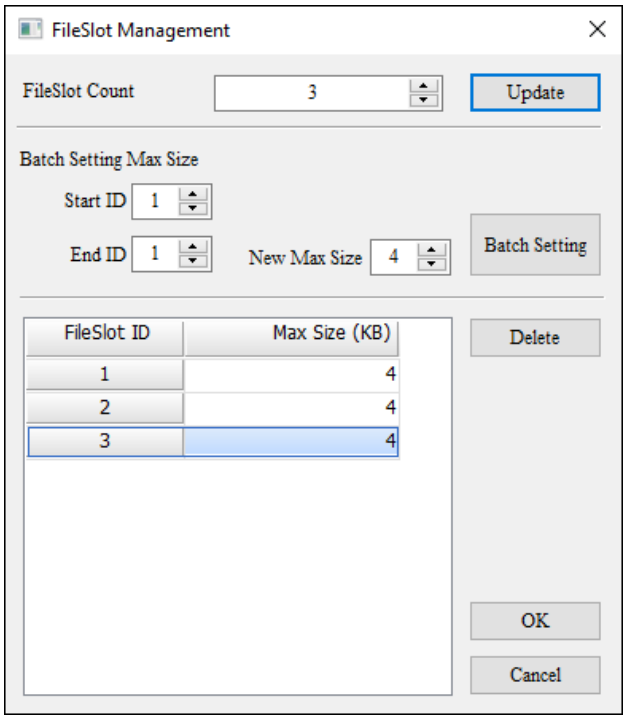
<table>
<tr><th>Setting</th><th colspan="2">Description</th></tr>
<tr><td>Select template</td><td colspan="2">Select the template to apply.</td></tr>
<tr><td>File name address</td><td colspan="2">Set the file name address. If not set, the report file will be output with the default file name.
Note: The default report file storage path is <code>\HMI\HMI-000\@HMI0000\batch.pdf</code>.</td></tr>
<tr><td>String length</td><td colspan="2">Select or enter the string length for the file name.</td></tr>
<tr><td rowspan="2">Sort</td><td>Sort
(Fixed)</td><td>Select the output type of report and click to sort the order in which the report is displayed.</td></tr>
<tr><td>Dynamic sorting (HMI)</td><td>Select to sort the display order of reports on the HMI screen.
 
</td></tr>
<tr><td rowspan="2">Time</td><td>All time</td><td>Select to output reports of all recorded data.</td></tr>
<tr><td>Interval setting (HMI)</td><td>Select to specify the time interval for the output data on the HMI screen.
 
</td></tr>
</table>

Import/Export FileSlot

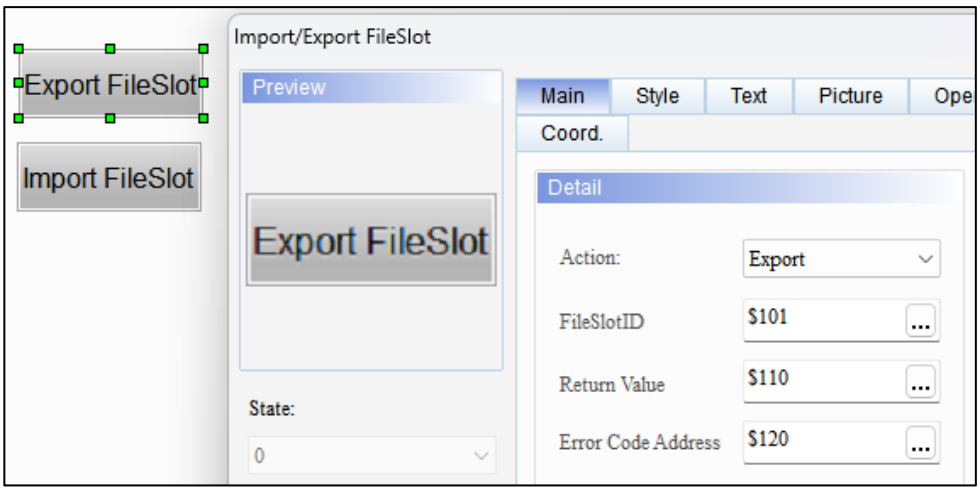
The following steps describes how to use the **Import/Export FileSlot** button.

To use the Import/Export FileSlot button

1. In the menu bar, click **Project** > **Other Setting** > **FileSlot Management**, and then set FileSlot.

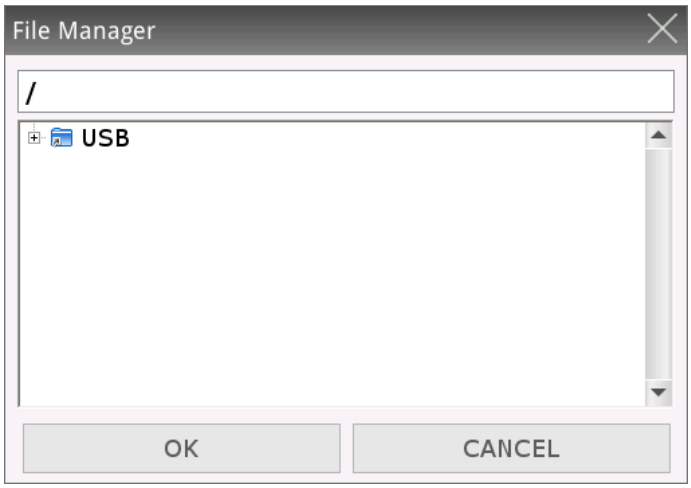


2. Create two **Import/Export FileSlot** buttons on the screen, select **Import** and **Export** for Action separately.



3. Set **FileSlotID** as memory address or constant.

4. Click the **Import/Export FileSlot** button on the HMI, and then select the file to read or write in the **File Manager** dialog.



Multiple Actions

The **Multiple Actions** button is used to configure the button action of Set ON, Set OFF, Maintained, Multistate, Set Constant, Increment, Decrement, Goto Screen, Set Low Security, System Menu, Report List, Screen Capture, Remove Storage, Import/Export Recipe, Language Change, and Delay.

Note:

- Up to 32 actions for each press, release, and long press can be set, that is, one **Multiple Actions** button can perform up to 32*3 actions.
- The System Menu can only be the last action.
- One **Multiple Actions** button can only have one page change action, including Goto Screen and Previous Page.
- If the button action is set with a Macro, the execution of the Macro is invalid.

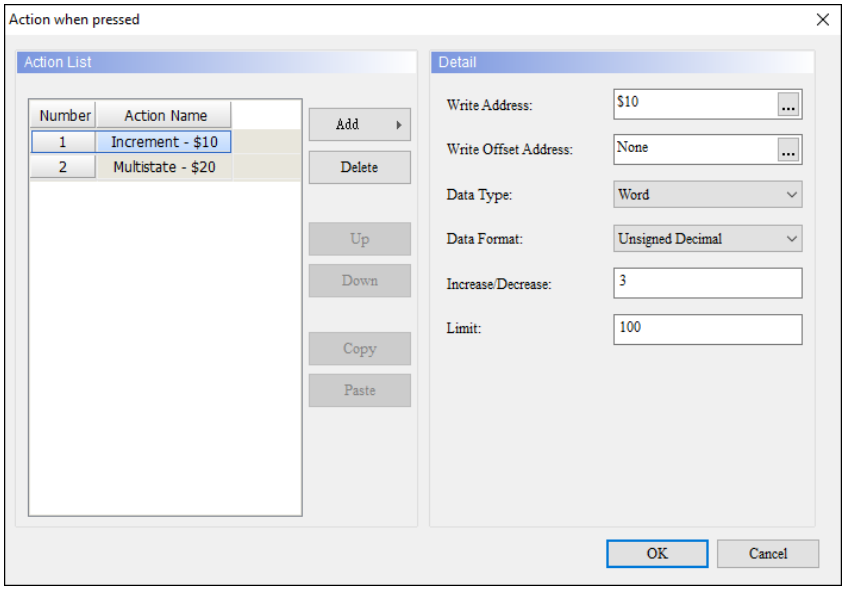
The following steps describe how to use the **Multiple Actions** button.

To use the Multiple Actions button

1. Create a **Multiple Actions** button on the screen.

2. Double-click the element and set the **Action when pressed** in the element properties dialog. In the **Action when pressed** dialog, add two actions, for example, Increment and Multistate.

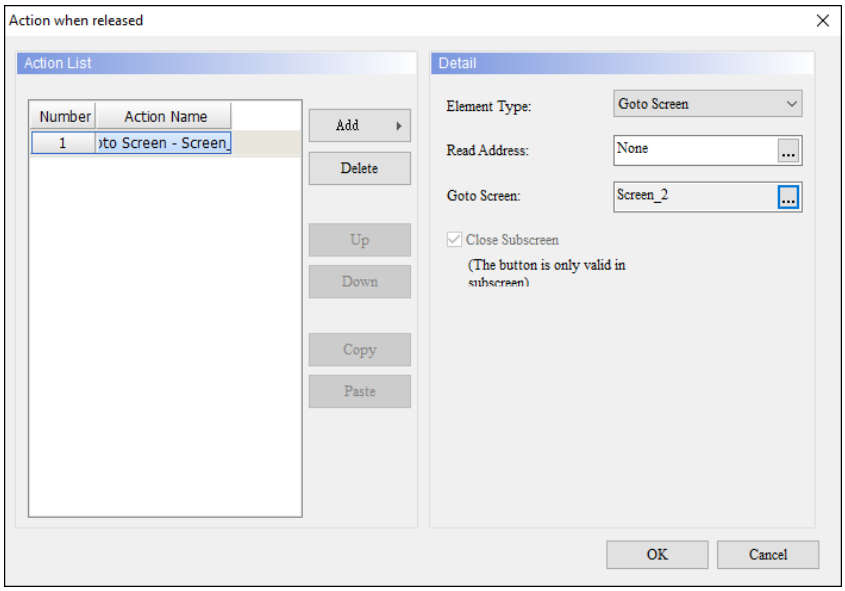
   - For **Increment** action, set the **Write Address** to **$10** and **Increase** to **3** for Increment.
   - For **Multistate** action, set the **Write Address** to **$20** and **State Counts** to **3**.



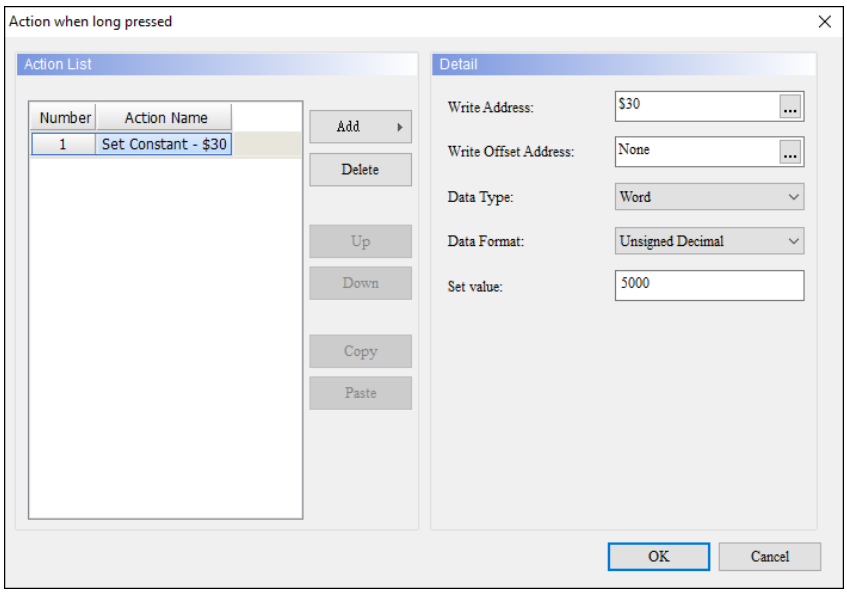
3. Set the **Action when released**. In the **Action when released** dialog, add one action, for example, Goto Screen.

   - For **Goto Screen** action, set **Goto Screen** to **Screen_2**.

     Note: You must first add a screen in the **Screen Management Window**.



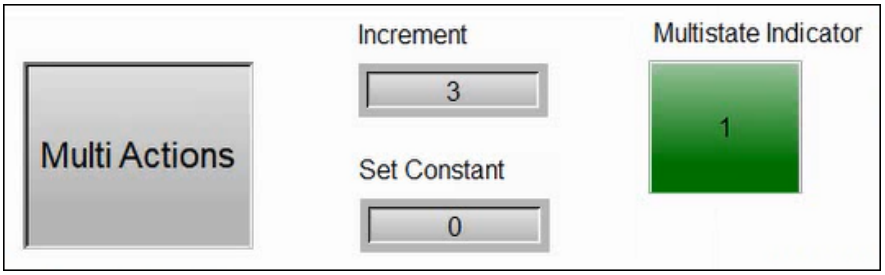
4. Set the **Action when long pressed**. In the **Action when long pressed** dialog, add one action, for example, Set Constant.
   - For **Set Constant** action, set the **Write Address** to **$30** and the **Set Value** to **5000**.



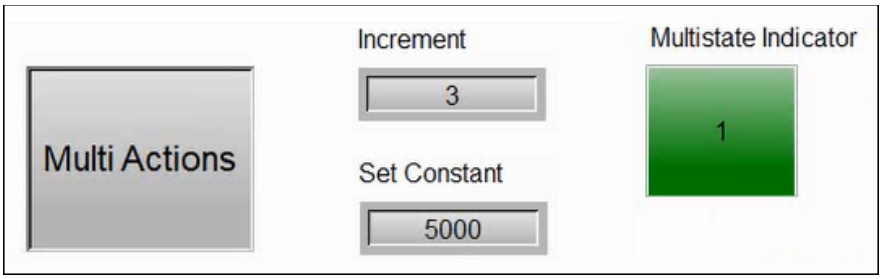
5. Set the **Long pressed time** to **3**.
6. Create 2 **Numeric Display** elements on the screen and set the **Read Address** to **$10** and **$30** separately.
7. Create a **Multistate Indicator** element on the screen and set the **Data Type** to **Word** and the **Read Address** to **$20**, and then set the foreground color for two states.

The results are as follows.

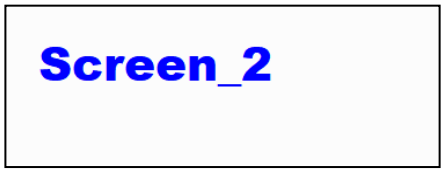
1. Press the **Multistate** button, and the element performs the **Action when pressed**.



2. Press the button for 3 seconds, and the element performs the **Action when long pressed.**



3. Release the button, and the element performs the **Action when released.**



Meter

The **Meter** element is used to display the measurement value of the set memory address. You can set whether the value exceeds an upper or lower limit, as well as set a target value. DIAScreen provides four different styles of meter elements.

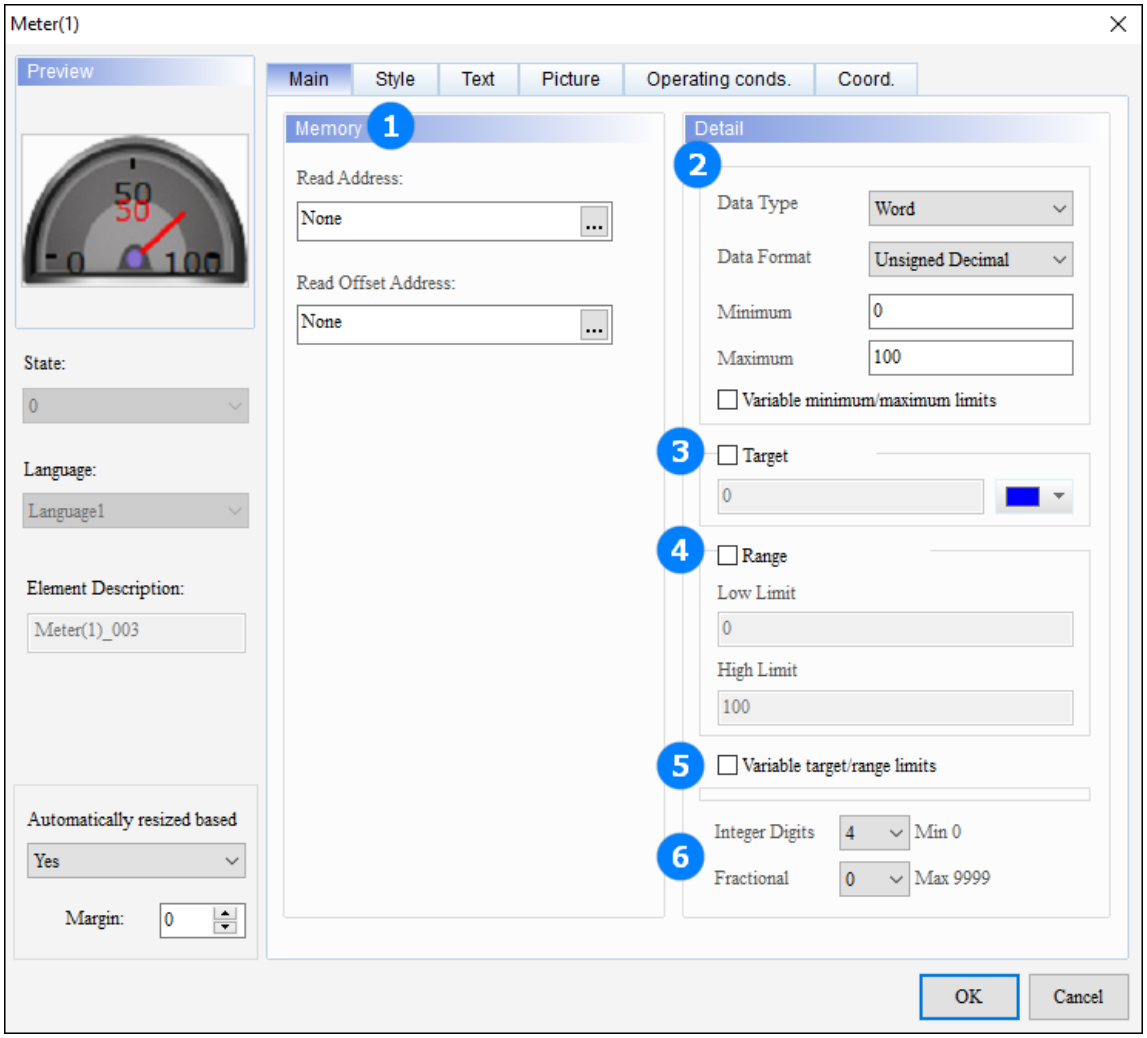
Meter(1)	Meter(2)	Meter(3)	Meter(4)

Common Properties

If **Automatically resized based** is set to **Yes**, when the set text length exceeds the element size, the element is automatically resized according to the **Margin** to show the complete text.

Note: This setting supports the element enlargement function, the string is shortened, and the element will not become smaller.

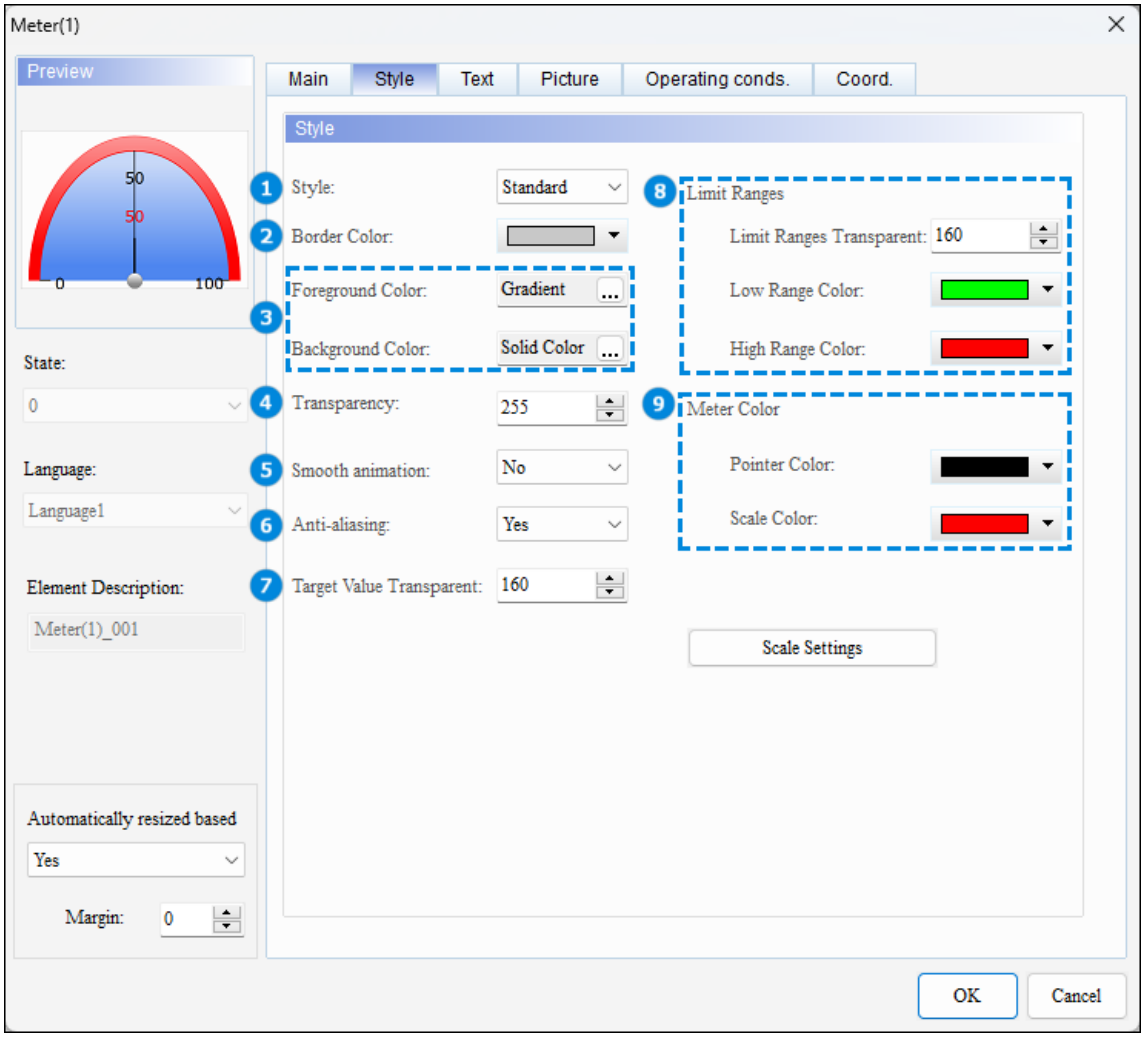
- **Main**

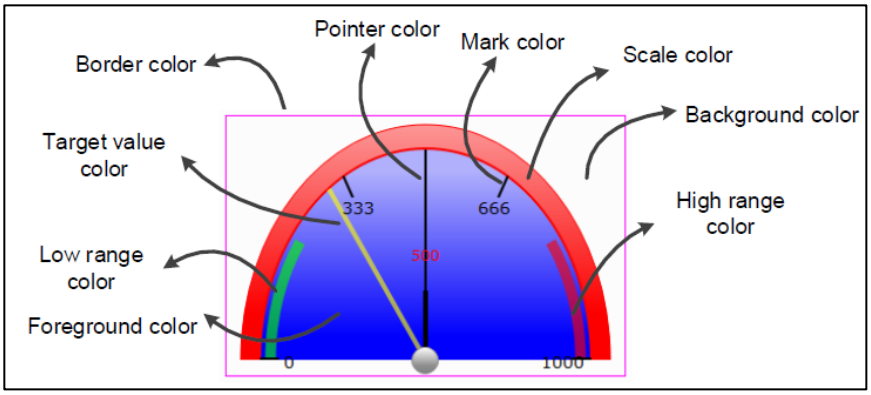


Legend	Setting	Description
1	Read Address	You can select the **Internal Memory** address or controller register address, and the **Input** memory **Type** can only be **Word**.
	Read Offset Address	The HMI will configure the value to the corresponding read address according to the set Offset Address. After setting the offset address, the actual read address is as follows. Actual read address of the element: Read Address + Value in Read Offset Address × Element [Data Type]
2	Numeric Value Settings	• **Data Type**: Select the data type. • **Data Format**: Select the data format. • **Minimum / Maximum**: Enter the valid range value according to the selected **Data Type** and **Data Format**. • **Variable minimum / maximum limits**: If selected, you can define the minimum and

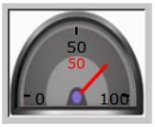
Legend	Setting	Description
		maximum value through the corresponding value of the memory address.
❸	Target	Select and enter the target value. You can also select its display color. If the **Variable target/range limits** checkbox is not selected, only constant values can be entered for the **Target**.
❹	Range	Select and enter the **Low Limit** and **High Limit**. If the **Variable target/range limits** checkbox is not selected, only constant values can be entered for the **Low Limit** and **High Limit**.
❺	Variable target/range limits	If selected, you can define memory addresses to dynamically change the display target value as well as the low and high limit values.
❻	Integer Digits Fractional	Select the valid display digits according to the selected **Data Type**. **Note**: The **Fractional** here is display format.

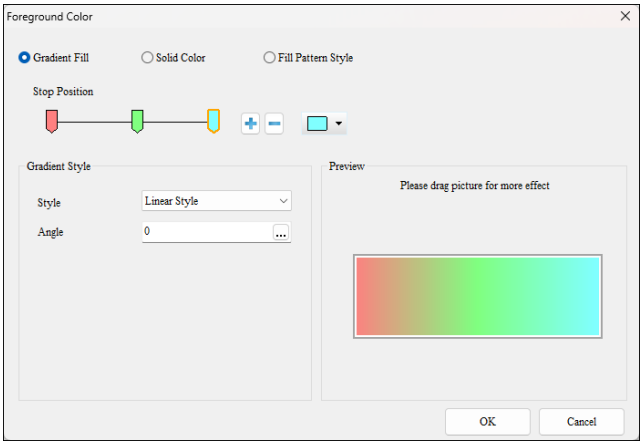
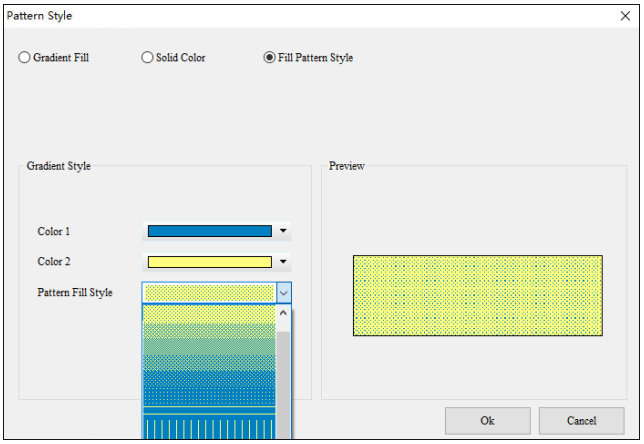
- **Style**

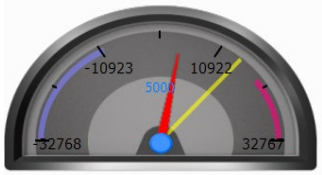
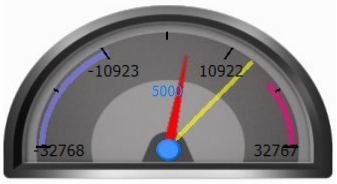
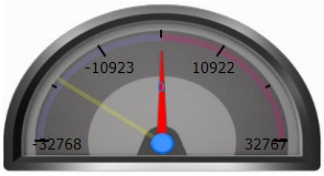
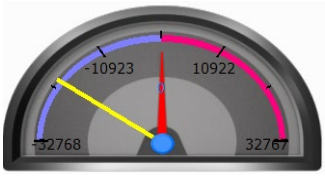




Legend	Setting	Description
1	Style	Select the appearance of the element. The default value is **Standard**.

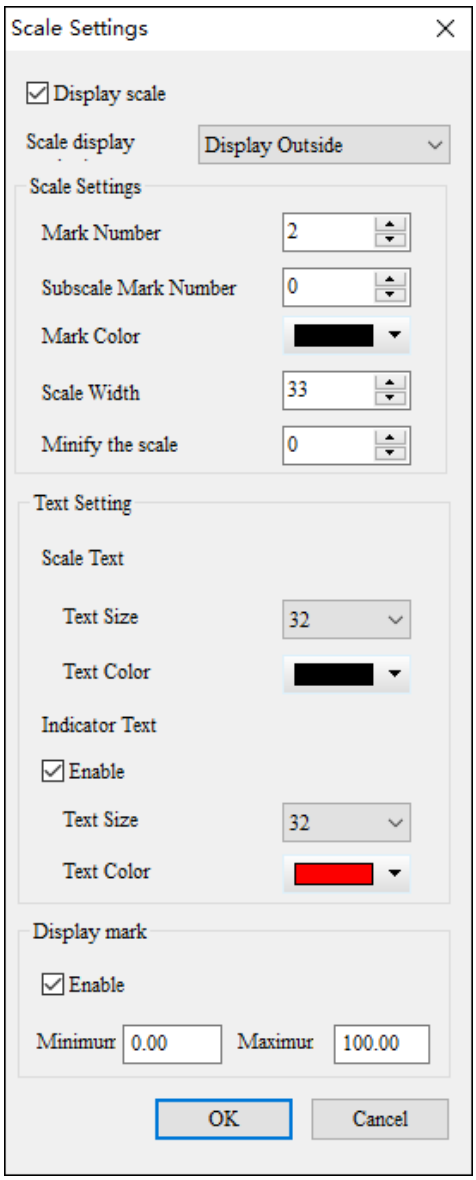
Standard	Raised	Sunken	Transparent
			

Legend	Setting	Description
②	Border Color	Select the border color of the element.
③	Foreground Color Background Color	Set the foreground and background colors of the element. • **Gradient Fill:** Set the gradient style by setting the current angle or center point. You can drag the mouse on the preview image to set the current angle or center point, or set the memory address to dynamically configure it on the HMI. You can also set the stop point colors to present the effect of multi-color gradients.  **Note**: Up to 5 stop points can be set. • **Solid Color**: Select a single-color fill. • **Fill Pattern Style**: Select a two-color combination configuration and select a pattern fill style.  **Note**: The **Foreground Color** is only available when the **Picture Bank Mode** is selected in the **Picture** tab.

<table>
<tr><th>Legend</th><th>Setting</th><th>Description</th></tr>
<tr><td>④</td><td>Transparency</td><td>Select or enter the transparency value of the element. The range value is 50–255, and the default value is 255.
The smaller the value, the higher the transparency of the element.</td></tr>
<tr><td>⑤</td><td>Smooth animation</td><td>Choose whether to enable the smooth animation function.
If select Yes, the element display becomes smoother.</td></tr>
<tr><td>⑥</td><td>Anti-aliasing</td><td>Select whether to enable anti-aliasing effect.
If select Yes, the element border displays smooth.
<table>
<tr><th>Yes</th><th>No</th></tr>
<tr><td> 
</td><td> 
</td></tr>
</table></td></tr>
<tr><td>⑦</td><td>Target Value Transparent</td><td>Select or enter the transparency value of limit ranges and target of the element. The range value is 50–255, and the default value is 255. The less the value, the higher the transparency.
<table>
<tr><th>Transparency is set to 50</th><th>Transparency is set to 255</th></tr>
<tr><td> 
</td><td> 
</td></tr>
</table>
Note: The effect of Target Value Transparent is visible only when the Target checkbox is selected.</td></tr>
<tr><td>⑧</td><td>Limit Ranges</td><td>• Limit Ranges Transparent: The range value is 50–255 and the default value is 255. The less the value, the higher the transparency.
• Low / Hight Range Color:
Note: The effect of Limit Ranges is visible only when the Range checkbox is selected.</td></tr>
</table>

Legend	Setting	Description
⑨	Limit Ranges	• **Limit Ranges Transparent**: The range value is 50–255 and the default value is **255**. The less the value, the higher the transparency. • **Low / Hight Range Color**: **Note**: The effect of **Limit Ranges** is visible only when the **Range** checkbox is selected.

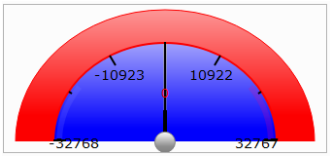
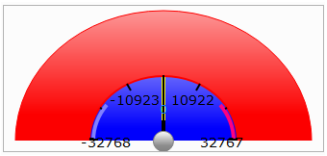
- **Scale Settings**



Setting	Description
Display scale	Select to display the scale on the element.
Scale display	Select the way to display the scale on the element.

<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Scale Settings</td><td>

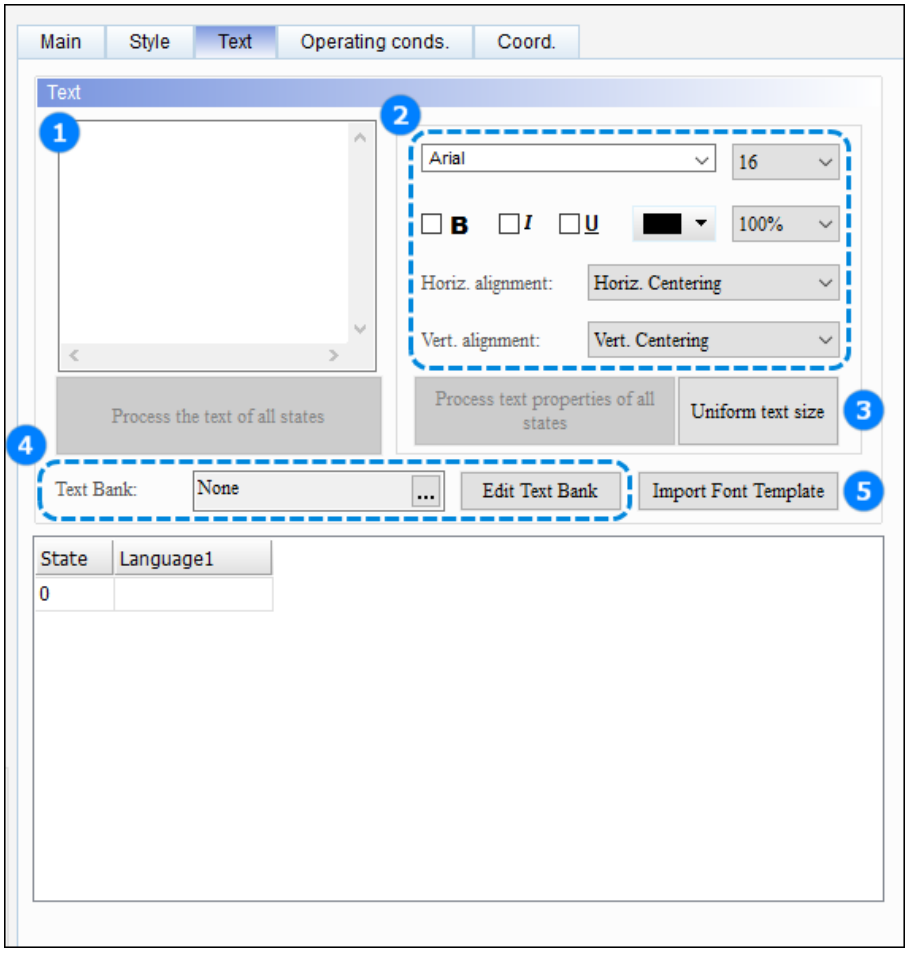
Mark Number: Select or enter the mark number. The range value is 1–10.
Subscale Mark Number: Select or enter the subscale mark number. The range value is 0–99.
Mark Color: Select the color of the mark.
Scale Width: Select or enter the width of the scale.
Note: This setting is only available when the Picture Bank Mode is selected in the Picture tab and the Scale display is Display Outside.
Minify the scale: Select or enter the range of scale minification. The greater the value, the wider the distance between the scale and the ruler. The range value is 0–8.
Note: This setting is only available when the Picture Bank Mode is selected in the Picture tab.

<table>
<tr><th>Minify the scale is set to 3</th><th>Minify the scale is set to 8</th></tr>
<tr><td> 
</td><td> 
</td></tr>
</table>
</td></tr>
<tr><td>Text Setting</td><td>

Scale Text: Set the size and color of the scale text.
Indicator Text: Select the Enable checkbox to display the read value of the element and set the size and color of the text.

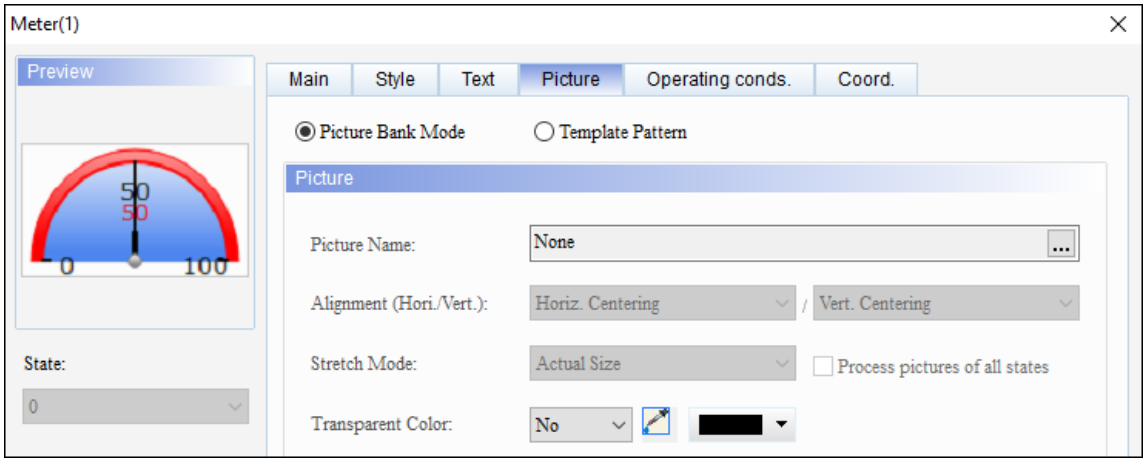
</td></tr>
<tr><td>Display mark</td><td>Select the Enable checkbox, and the scale value will display based on the set Maximum / Minimum. The range value is –32768~32767 and support entering two decimal places.</td></tr>
</table>

- **Text**

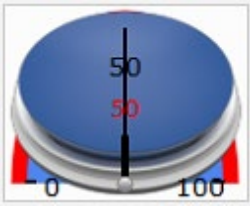
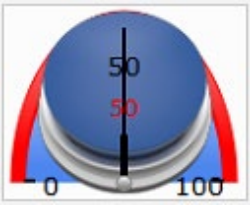
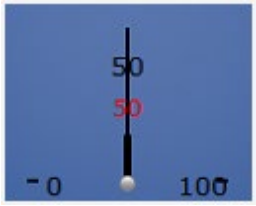


Legend	Setting	Description
1	Text	Enter the text message to display in the text box. Select the state / language table below to directly set the text displayed by that state / language. Enter the text message to be displayed in the text box. Click the element on the screen and press **Space** to edit the text content of the current state / language.
2	Text Property	Set the text properties.
3	Uniform text size	Changes the text size in all languages with one click according to the currently selected state.
4	Text Bank	Applies the text in the **Text Bank** to the content of each state / language. You can also click **Edit Text Bank** to edit the text content in the **Text Bank** editor.
5	Import Font Template	Click to open the **Font Template** dialog, and then select the template and import the text properties of the selected state.

- **Picture**
  - **Picture Bank Mode**: You can select the picture from the picture bank and apply it to the meter, pointer, and meter center.
  - **Template Pattern**: Click **Start the Template Wizard** and select the built-in template.



Setting	Description
Picture Name	Enter the picture name. The default value is **None**.To set the picture display, click ... to select the picture to display in the picture bank and drag it to each state icon. To configure other pictures, refer to Picture Bank.
Alignment	Select the alignment of the picture.

<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Stretch Mode</td><td>Select the stretch mode of the picture.
<table>
<tr><th>Stretch All</th><th>Stretch 1:1</th><th>Actual Size</th></tr>
<tr><td>The picture fills the full element display area.
  </td><td>The picture displays in 1:1 size based on the element width and length.
  </td><td>The picture is displayed in the display area of the element in its actual size.
  </td></tr>
</table></td></tr>
<tr><td>Transparent Color</td><td>Choose whether to specify a color in the picture as a transparent color. If set to Yes, you can select colors or use the color picker to select one of the colors on the picture to specify it as a transparent color, that is, the foreground color. The following figures show the effect of specifying white as a transparent color.
 </td></tr>
</table>

- **Operating Conds.**

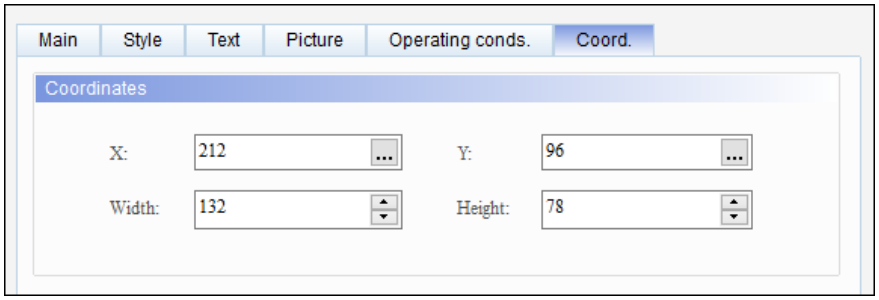
  The **Invisible Address** displays or hides the element on the HMI. If the **Invisible State** is set to **On**, the element will be hidden when the **Invisible Address** is ON. Instead, if the **Invisible State** is set to **Off**, the element will be hidden when the **Invisible Address** is OFF.

  The following is an example of using invisible address.

  1. Create a button element on the screen and set its **Read Address** to $9.0, then set the **Invisible Address** of the element to hide to $9.0.
  2. To hide the element, you must first click the $9.0 button.

Invisible State is On	Invisible Address $9.0 OFF Element disappear Invisible Address $9.0 ON
Invisible State is Off	Invisible Address $9.0 ON Element disappear Invisible Address $9.0 OFF

- **Coord.**

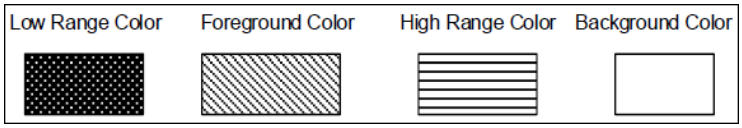


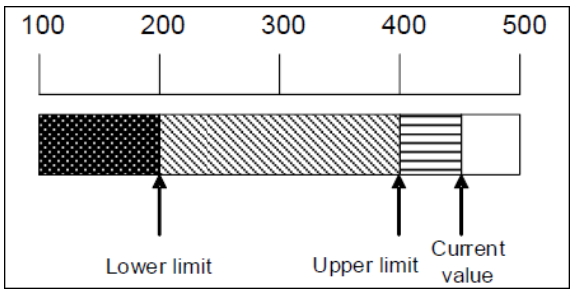
Set the XY coordinates, width, and height on the upper left of the element. You can enter a constant as fixed position or memory address to dynamically configure the position of the element on the HMI.

Bar

- **Normal**

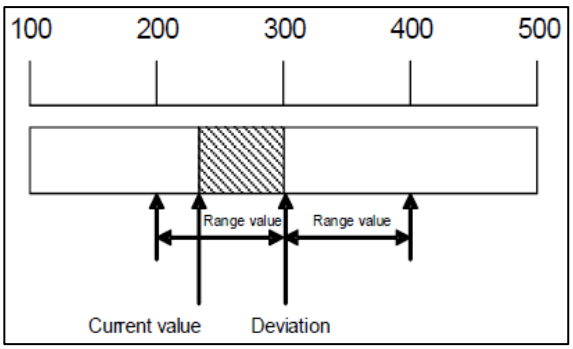
  The **Normal Bar** element displays the values corresponding to the read addresses in the form of a bar from the minimum value to the maximum value on the HMI. You can also set the colors for the lower limit, upper limit, and target values to identify them.





- **Differential**

  The **Differential Bar** element displays the deviation between the value of the **Read Address** and the set target value on the HMI.



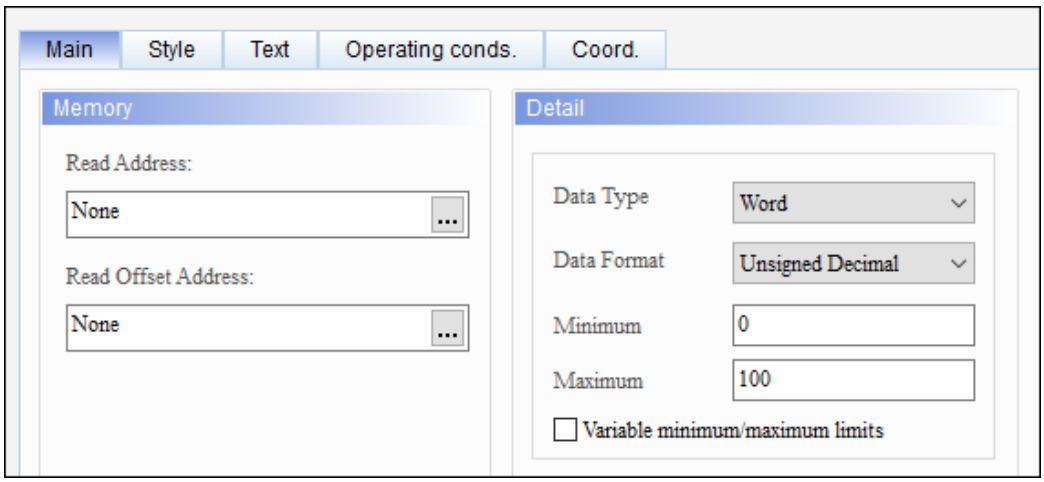
The bar element of DIAScreen provides memory addresses with customizable target values and lower and upper limit values, making the application more flexible and meeting users' requirements.

Common Properties

If **Automatically resized based** is set to **Yes**, when the set text length exceeds the element size, the element is automatically resized according to the **Margin** to show the complete text.

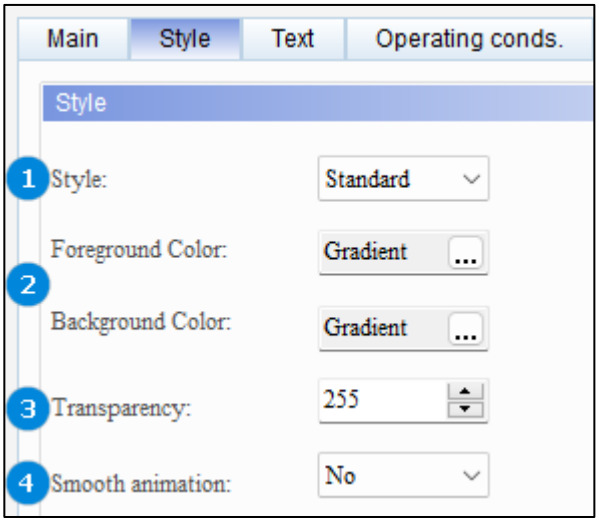
Note: This setting supports the element enlargement function, the string is shortened, and the element will not become smaller.

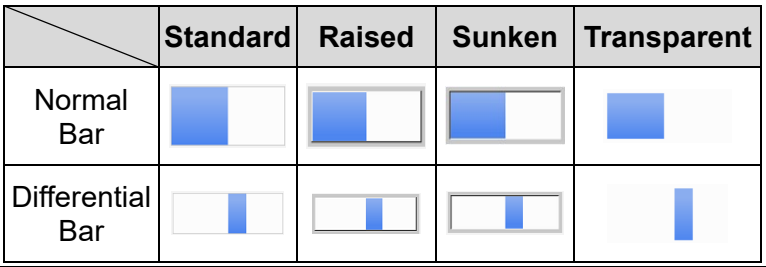
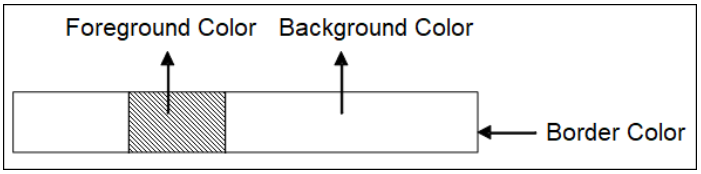
- **Main**

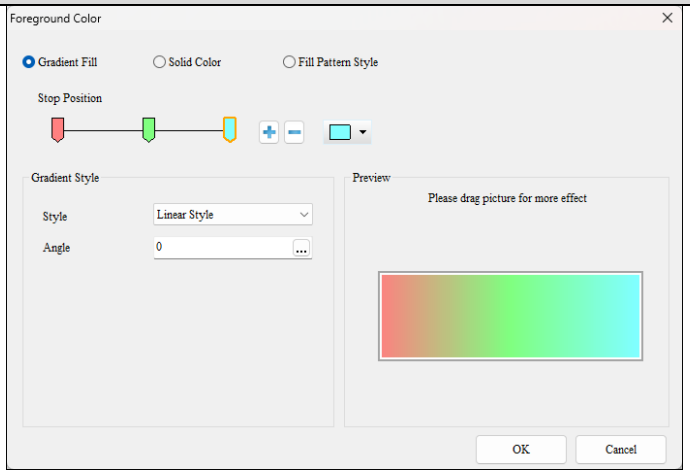
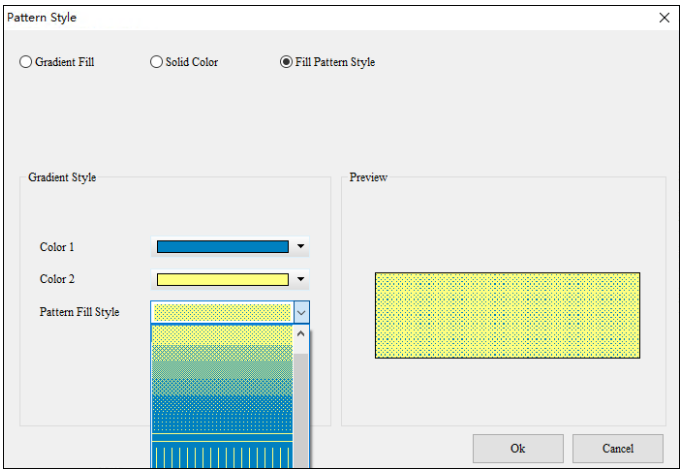


Setting	Description
Read Address	Set the **Internal Memory** address or controller register address. **Note**: The **Input** memory **Type** must be Word.
Read Offset Address	The HMI will configure the value to the corresponding write / read address according to the set Offset Address. After setting the offset address, the actual read and write address are as follows. Actual read address of the element: Read Address + Value in Read Offset Address × Element [Data Type]
Data Type	Select the data type. The available options are Word and Double Word.
Data Format	Select the data format. The available options are BCD, Signed BCD, Signed Decimal, Unsigned Decimal, and Hexadecimal.
Minimum / Maximum	Enter the valid range value according to the selected **Data Type** and **Data Format**.
Variable minimum/maximum limits	If selected, you can define the minimum and maximum value through the corresponding value of the memory address.

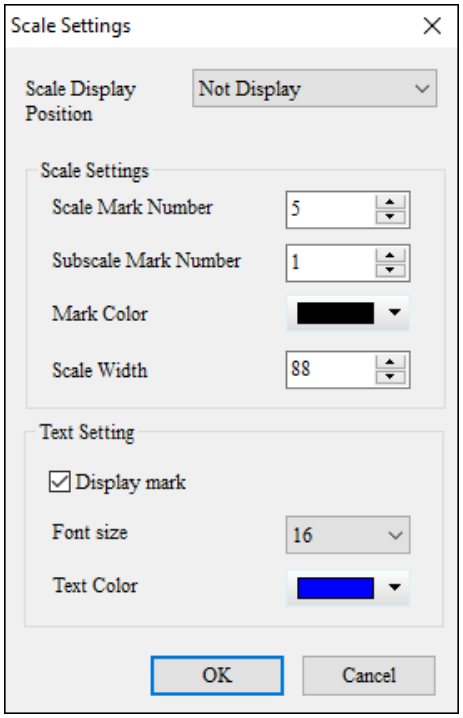
- **Style**



Legend	Setting	Description
1	Style	Select the appearance of the element. The available options are Standard, Raised, Sunken, and Transparent. The default value is **Standard**. 
2	Foreground Color / Background Colo	Select the foreground and background colors of the element.  • **Gradient Fill:** Set the gradient style by setting the current angle or center point. You can drag the mouse on the preview image to set the current angle or center point, or set the memory address to dynamically configure it on the HMI. You can also set the stop point colors to present the effect of multi-color gradients.

Legend	Setting	Description
		 **Note**: Up to 5 stop points can be set. • **Solid Color**: Select a single-color fill. • **Fill Pattern Style**: Select a two-color combination configuration and select a pattern fill style. 
❸	Transparency	Select or enter the transparency value of the element. The range value is 50—255 and the default value is **255**. The lower the value is, the higher the transparency is.
❹	Smooth animation	Select whether to enable the **Smooth animation** function. When this function is enabled, the element display becomes smoother.

- **Style – Scale Settings**



Setting	Description
Scale Display Position	Select the position to display scale on the element according to **Direction**. • Scale can be displayed on the top or bottom when **Direction** is **Left** or **Right**. • Scale can be displayed on the left or right when **Direction** is **Top** or **Bottom**.
Scale Settings	Set the mark number, color, width of the scale. The range value of **Scale Mark Number** is 1–99.
Text Setting	Select the **Display mark** checkbox to display the mark and set the text size and color.

- **Operating conds.**

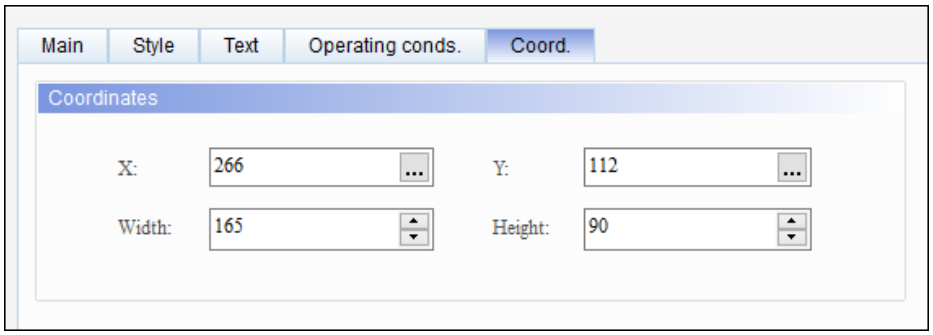
  The **Invisible Address** displays of hides the element on the HMI. If **Invisible State** is set to **On**, the **Invisible Address** hides the element. Instead, if the **Invisible State** is set to **Off**, the **Invisible Address** hides the element.

  The following is an example of using invisible address.

3. Create a button element on the screen and set its **Read Address** to **$9.0**, then set the **Invisible Address** of the element to hide to **$9.0**.
4. To hide the element, you must first click the $9.0 button.

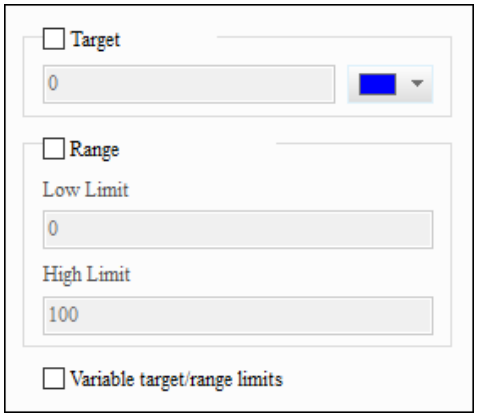
Invisible State is On	
Invisible State is Off	

- **Coord.**



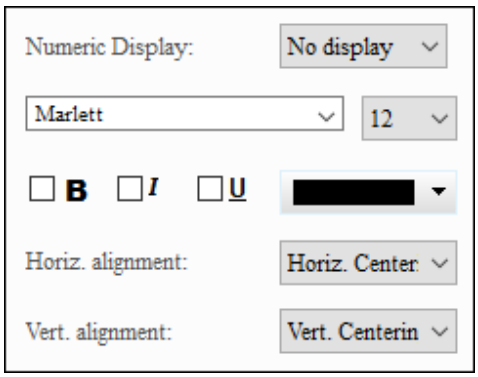
Set the XY coordinates, width, and height on the upper left of the element. You can enter a constant as fixed position or memory address to dynamically configure the position of the element on the HMI.

Normal Bar Properties

- **Main**
  - **Detail**
    - 

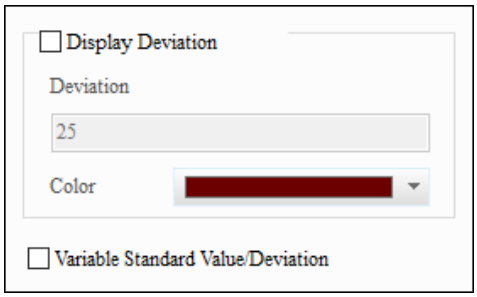
Setting	Description
Target	Select and enter the target value. You can also specify the display color. If the **Variable target / range limits** checkbox is not selected, enter only a constant value.
Range	Select and enter the **Low Limit** and **High Limit**. If the **Variable target / range limits** checkbox is not selected, only constant values can be entered for the **Low Limit** and **High Limit**.
Variable target / range limits	If selected, you can define memory addresses to dynamically change the display target value as well as the low and high limit values.

  - **Style**
    - **Numeric Display**



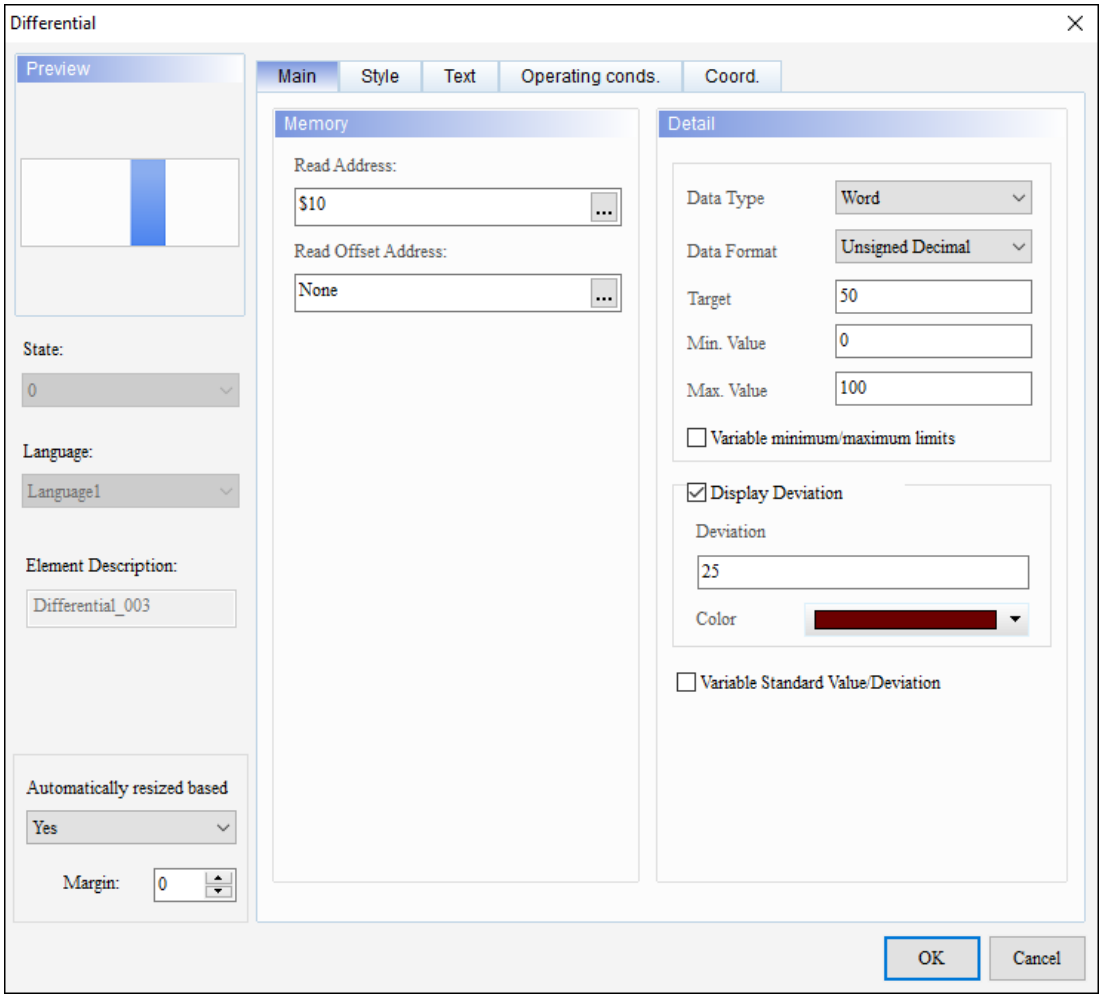
The available options are No display, Value, and Percentage. The default value is **No display**. If you select **Value** or **Percentage**, you can set the display style and alignment type.

Differential Bar Properties

- **Main**
  - **Detail**
  - 

  - Select and enter the **Deviation**. You can also set the display color for deviation values within the range. If the **Variable target / range limits** checkbox is not selected, only constant values can be entered to define the deviation value of the **Differential Bar** element.
  - The following describes how to use the **Differential** element.

1. Create a **Differential** element on the screen, and then set the **Read Address** to **$10** and select the **Display Deviation** checkbox.

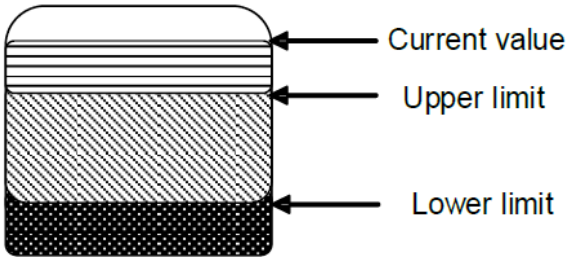


2. Create a **Numeric Entry** element on the screen, and then set the **Write Address** to **$10**.
3. Compile and download the screen to the HMI.

- The result is as follows.
- Enter 60 for the **Numeric Entry** element. The current value is within 25 of the target values and is displayed in the deviation color.

- Enter 20 for the **Numeric Entry** element. The current value is not within 25 of the target values and is displayed in the foreground color.

Pipe

Pipe	Description
Pipe(1) Pipe(2)	Read the value corresponding to the memory address, and display on the HMI in different pipe style. You can also set different color to distinguish its lower limit, upper limit, and target value. 
Pipe(3) Pipe (4) Pipe (5)	Used for connecting other pipe elements. These pipe elements have no parameters for memory address or numeric value setting. You can only set the displaying pipe diameter and rotation angle.
Pipe (6) Pipe (7)	Used as a water pipe diagram to set the water flow direction by the memory address.

Common Properties

Note: Pipe(3) / Pipe(4) / Pipe(5) has no text settings.

- **Operating conds.**

  The **Invisible Address** displays or hides the element on the HMI. If the **Invisible State** is set to **On**, the **Invisible Address** hides the element. Instead, if the **Invisible State** is set to **Off**, the **Invisible Address** hides the element.

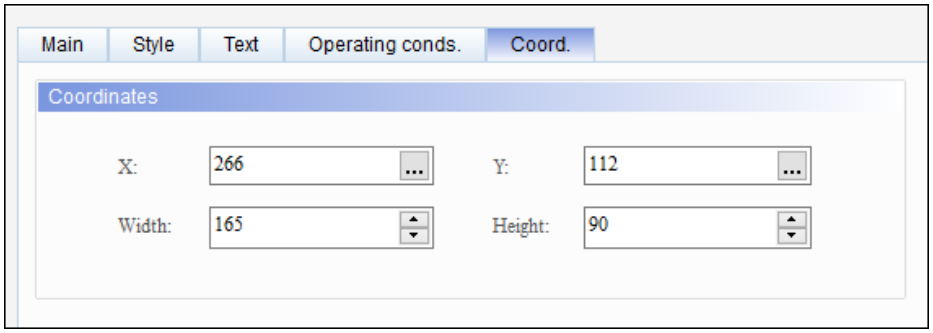
  The following is an example of using invisible address.

  1. Create a button element on the screen and set its **Read Address** to $9.0, then set the **Invisible Address** of the element to hide to $9.0.
  2. To hide the element, you must first click the $9.0 button.

Invisible State is On	

	Element disappear	Invisible Address $9.0 ON
Invisible State is Off		Invisible Address $9.0 ON
	Element disappear	Invisible Address $9.0 OFF

- **Coord.**



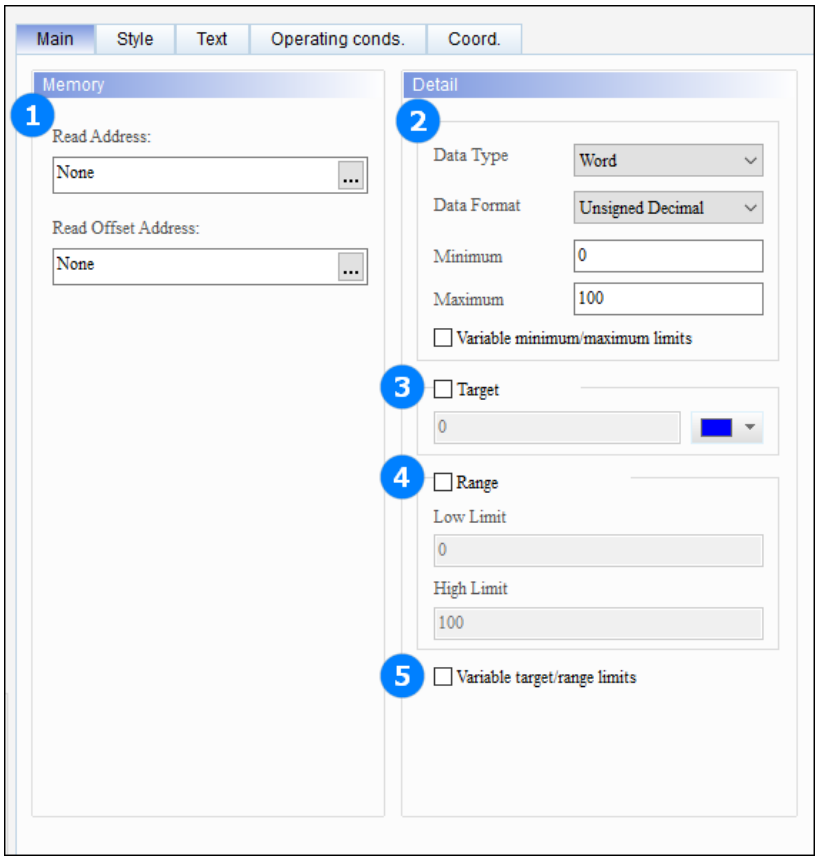
Set the XY coordinates, width, and height on the upper left of the element. You can enter a constant as fixed position or memory address to dynamically configure the position of the element on the HMI.

Pipe(1) / Pipe(2)

If **Automatically resized based** is set to **Yes**, when the set text length exceeds the element size, the element is automatically resized according to the **Margin** to show the complete text.

Note: This setting supports the element enlargement function, the string is shortened, and the element will not become smaller.

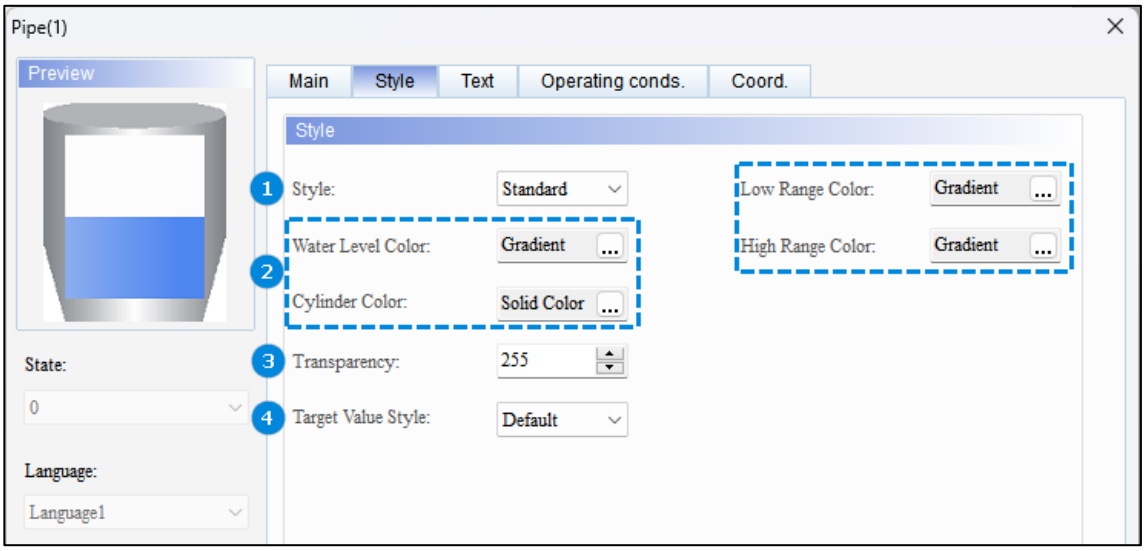
- **Main**



Legend	Setting	Description
1	Read Address	You can select the **Internal Memory** address or controller register address, and the **Input** memory **Type** can only be **Word**.
	Read Offset Address	The HMI will configure the value to the corresponding read address according to the set Offset Address. After setting the offset address, the actual read address is as follows. The actual read address of the element: Read Address + Value in Read Offset Address × Element [Data Type]
2	Numeric Value Settings	• **Data Type**: Select the data type. • **Data Format**: Select the data format. • **Minimum / Maximum**: Enter the valid range value according to the selected **Data Type** and **Data Format**. • **Variable minimum / maximum limits**: If selected, you can define the minimum and maximum value through the corresponding value of the memory address.

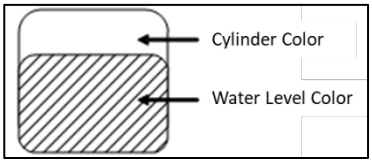
Legend	Setting	Description
❸	Target	Select and enter the target value. You can also select its display color. If the **Variable target/range limits** checkbox is not selected, only constant values can be entered for the **Target**.
❹	Range	Select and enter the **Low Limit** and **High Limit**. If the **Variable target/range limits** checkbox is not selected, only constant values can be entered for the **Low Limit** and **High Limit**.
❺	Variable target / range limits	If selected, you can define memory addresses to dynamically change the display target value as well as the low and high limit values.

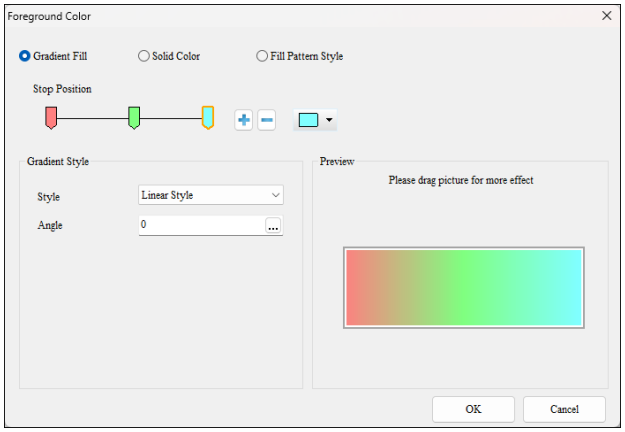
- **Style**

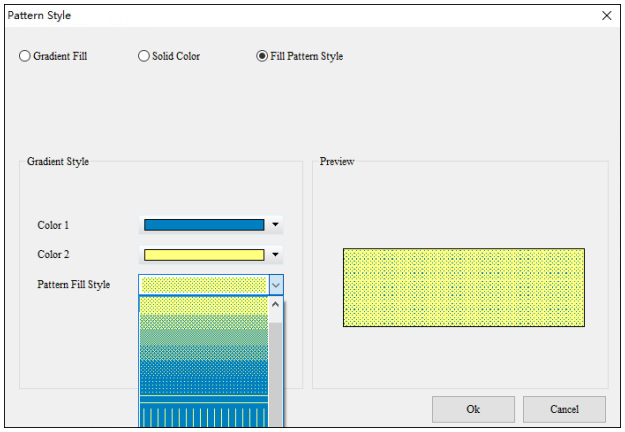
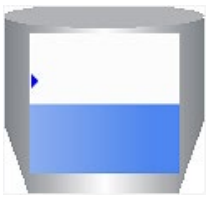


Legend	Setting	Description
❶	Style	Select the appearance of the element. The default value is **Standard**.

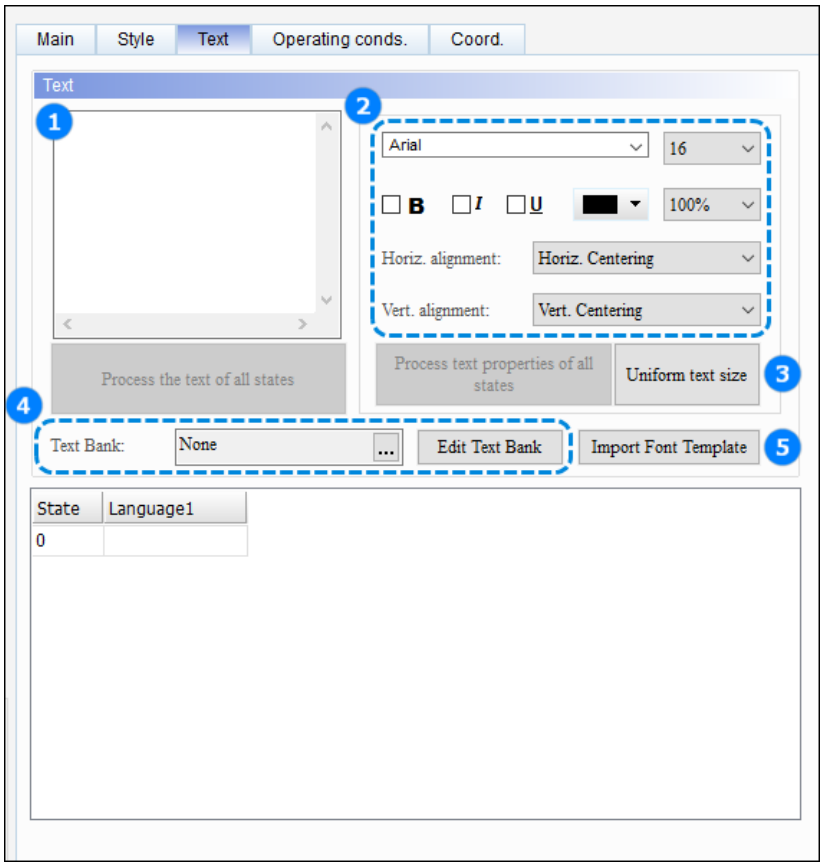
Standard	Rotation 180

Legend	Setting	Description
❷	Water Level Color Cylinder Color Low Range Color High Range Color	Select the color of the element and the low range and high range colors to be displayed on the HMI.  **Note**: When the **Range** enable, if the reading value on the HMI is less than the low limit value, the foreground color will display the low range color; if the reading value is greater than the high limit value, the foreground color will display the high range color. • **Gradient Fill:** Set the gradient style by setting the current angle or center point. You can drag the mouse on the preview image to set the current angle or center point, or set the memory address to dynamically configure it on the HMI. You

<table>
<tr><th>Legend</th><th>Setting</th><th>Description</th></tr>
<tr><td></td><td></td><td>can also set the stop point colors to present the effect of multi-color gradients.
 

Note: Up to 5 stop points can be set.
• Solid Color: Select a single-color fill.
• Fill Pattern Style: Select a two-color combination configuration and select a pattern fill style.
 
</td></tr>
<tr><td>3</td><td>Transparency</td><td>Select or enter the transparency of the element. The range value is 50–255 and the default value is 255.
The smaller the value, the higher the transparency of the element.</td></tr>
<tr><td>4</td><td>Target Value Style</td><td>Select the target value style of the element.
<table><tr><th>Default</th><th>Style 1</th></tr><tr><td> </td><td>  </td></tr></table>Note: The Target Value Style is visible only when the Target checkbox is selected.</td></tr>
</table>

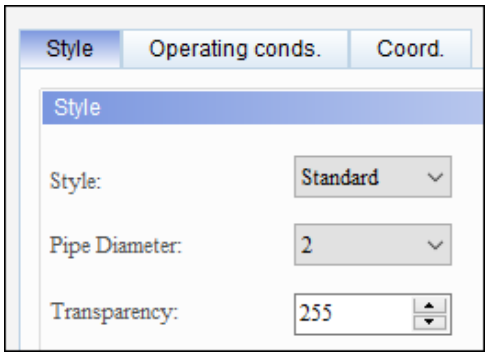
- **Text**



Legend	Setting	Description
1	Text	Enter the text message to display in the text box. Select the state / language table below to directly set the text displayed by that state / language. Enter the text message to be displayed in the text box. Click the element on the screen and press **Space** to edit the text content of the current state / language.
2	Text Property	Set the text properties.
3	Uniform text size	Changes the text size in all languages with one click according to the currently selected state.
4	Text Bank	Applies the text in the **Text Bank** to the content of each state / language. You can also click **Edit Text Bank** to edit the text content in the **Text Bank** editor.
5	Import Font Template	Click to open the **Font Template** dialog, and then select the template and import the text properties of the selected state.

Pipe(3) / Pipe(4) / Pipe(5)

- **Style**



Setting	Description
Style	Select the appearance of the element. The default value is **Standard**. **Note**: This setting is only available for Pipe(4) / Pipe(5).
Pipe Diameter	Select the pipe diameter size of the element.
Transparency	Select the transparency of the element. The range value is 50–255 and the default value is **255**. The smaller the value, the higher the transparency of the element.

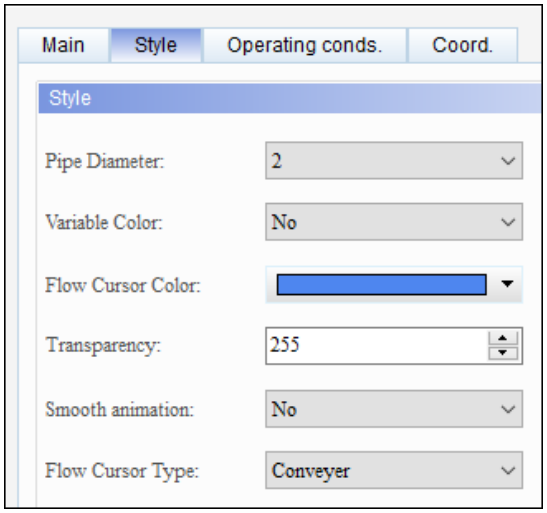
Pipe(6) / Pipe(7)

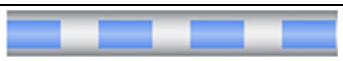
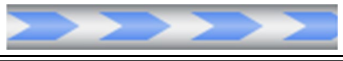
- **Main**

Set the water flow direction.

Memory Address Value	Pipe(6)	Pipe(7)
Read Address = 1	Right → Left	Top → Bottom
Read Address = 2	Left → Right	Bottom → Top

- **Style**



Setting	Description
Pipe Diameter	Select the pipe diameter size of the element.
Variable Color	Choose whether the color can be changed. If select **Yes**, you can dynamically change the flow cursor color through the memory address on the HMI. The range value is 0–65535.
Flow Cursor Color	Select the color of the flow cursor. When the **Variable Color** is set to **Yes**, you can set the memory address, and the color can be configured on the HMI according to the set address value.
Transparency	Select the transparency of the element. The range value is 50–255 and the default value is **255**. The smaller the value, the higher the transparency of the element.
Smooth animation	Choose whether to enable the smooth animation function. If select **Yes**, the element display becomes smoother.
Flow Cursor Type	Select the type of flow cursor. The effects of each type are as follows. Conveyer  Bubble Arrow 

Flow Block

Through the flow block component, the flow block can be set on the line segment, and the flow effect will be displayed according to the set direction speed.

Follow these steps to draw a flow block.

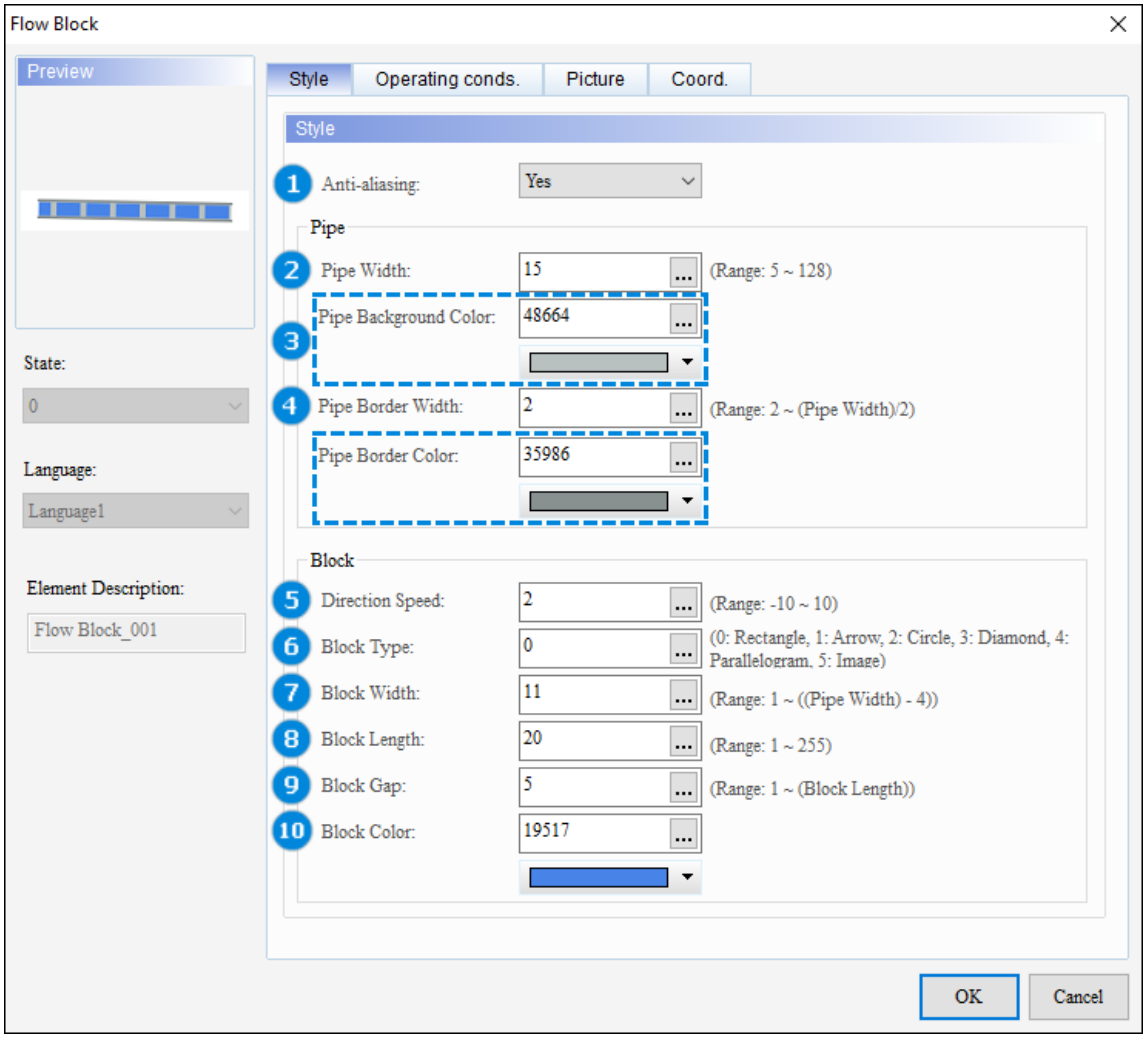
To draw a flow block

1. Click **Element** > **Pipe** > **Flow Block** on the toolbar.
2. Drag the mouse on the screen and left-click to determine the polyline of the pipe. Then, right-click to complete the drawing of the flow block.

Note: The angle of the pipeline cannot be an acute angle, and the minimum angle is 90 degrees.

The following table lists the special properties of the **Flow Block** element with their description.

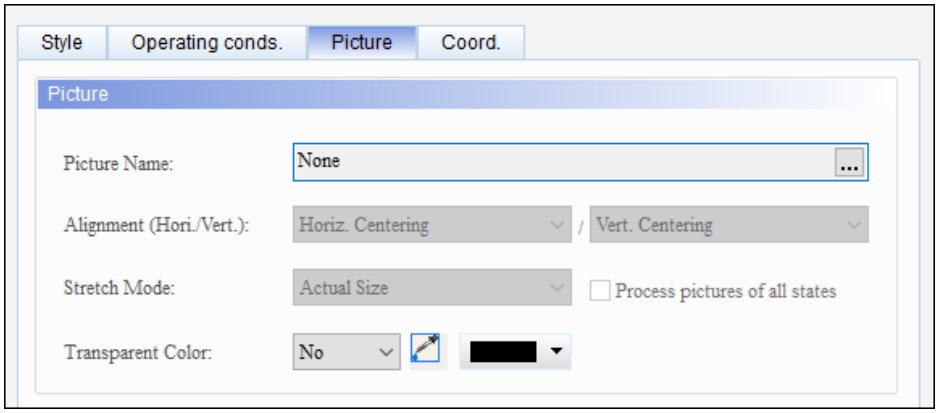
- **Style**



Legend	Setting	Description
❶	Anti-aliasing	Choose whether to enable anti-aliasing effect. If select **Yes**, the element border displays smooth without aliasing. **Yes**: **No**:
❷	Pipe Width	Set the width of the pipe, and the range value is 5–128.
❸	Pipe Background Color Pipe Border Color	Select the background and border color of the pipe. If the color is dynamically set by the HMI, the range value of the color is 0–65535.
❹	Pipe Border Width	Set the border width of the pipe, and the range value is 2–**Pipe Width**/2.
❺	Direction Speed	Set the direction speed. Take the direction in which the flow block is drawn as forward, and enter a negative number to reverse the flow. The range value is –10~10。
❻	Block Type	Set the type of the block. The numerical values are defined as follows: • 1: Rectangle • 2: Arrow • 3: Circle • 4: Parallelogram • 5: Image. Displays the image configured in the **Picture** tab. **Note**: If the address value on the HMI is set to more than 5, the block will be displayed as a rectangle.
❼	Block Width	Set the width of the block, and the range value is 1–(**Pipe Width**-4)。
❽	Block Length	Set the length of the block, and the range value is 1–255.
❾	Block Gap	Set the gap of the block, and the range value is 1–**Block Length**.
❿	Block Color	Select color of the block. If the color is dynamically set by the HMI, then the range value of the color is 0–65535.

Note: You can set the memory address for the above properties to dynamically change the property value on the HMI.

- **Picture**



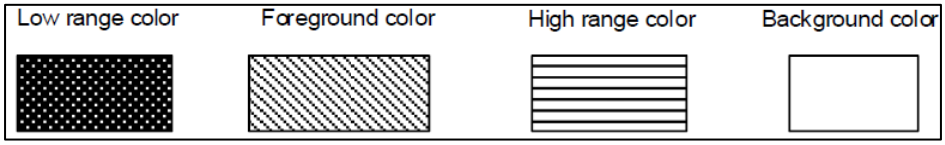
Note: The picture will only be displayed on the element when the **Block Type** is set to 5.

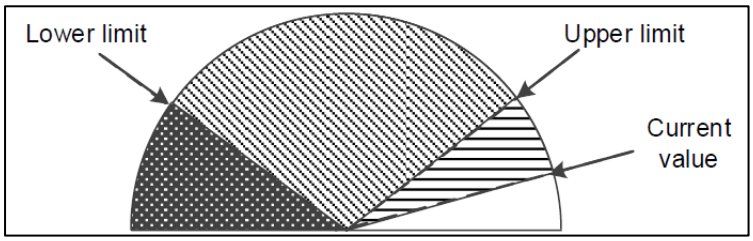
Setting	Description
Picture Name	Enter the picture name. The default value is **None**.To set the picture display, click ... to select the picture to display in the picture bank and drag it to each state icon. To configure other pictures, refer to Picture Bank.
Alignment	Select the alignment of the picture.

<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Stretch Mode</td><td>Select the stretch mode of the picture.
<table>
<tr><th>Stretch All</th><th>Stretch 1:1</th><th>Actual Size</th></tr>
<tr><td>The picture fills the full element display area.
 </td><td>The picture displays in 1:1 size based on the element width and length.
 </td><td>The picture is displayed in the display area of the element in its actual size.
 </td></tr>
</table></td></tr>
<tr><td>Process pictures of all states</td><td>If an element has multiple state values, and its picture may not fill the full display area of the element. Select this checkbox to process all pictures settings</td></tr>
<tr><td>Transparent Color</td><td>Choose whether to specify a color in the picture as a transparent color. If set to Yes, you can select colors or use the color picker to select one of the colors on the picture to specify it as a transparent color, that is, the foreground color. The following figures show the effect of specifying white as a transparent color.
</td></tr>
</table>

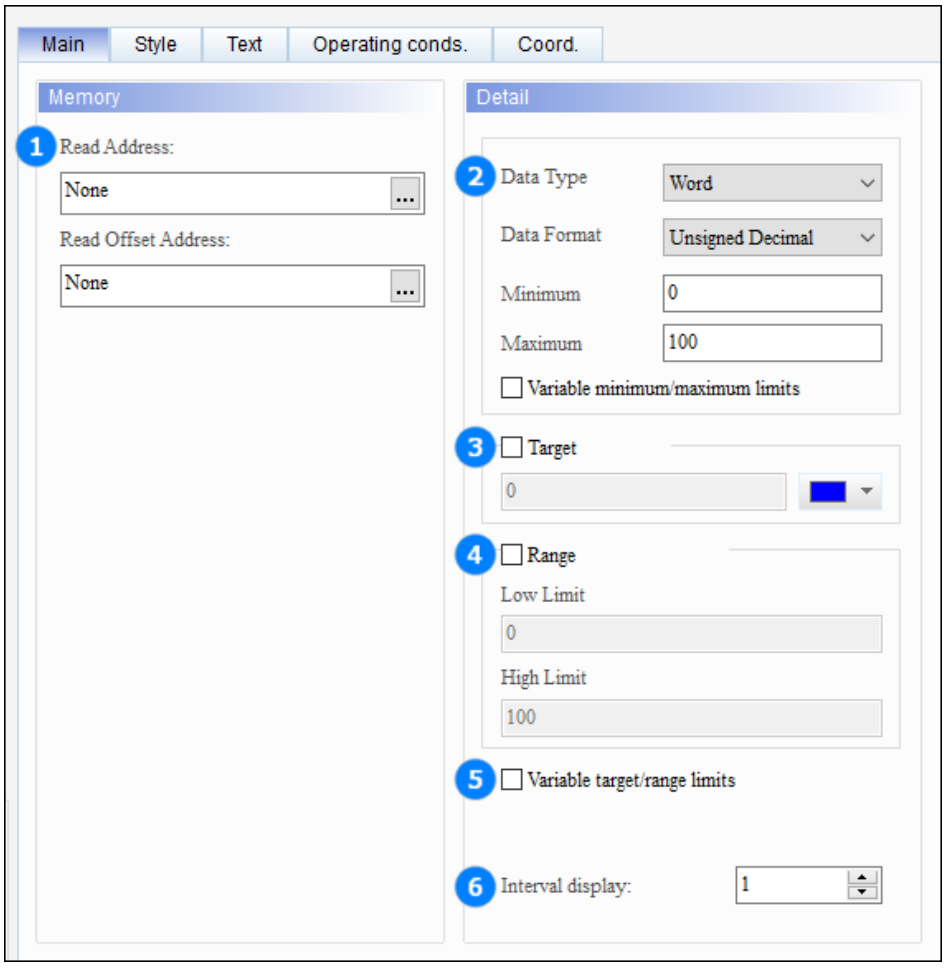
Pie

The **Pie** element displays the values corresponding to the read addresses in different pie styles on the HMI. You can also set the colors for the lower limit, upper limit, and target values to identify them.



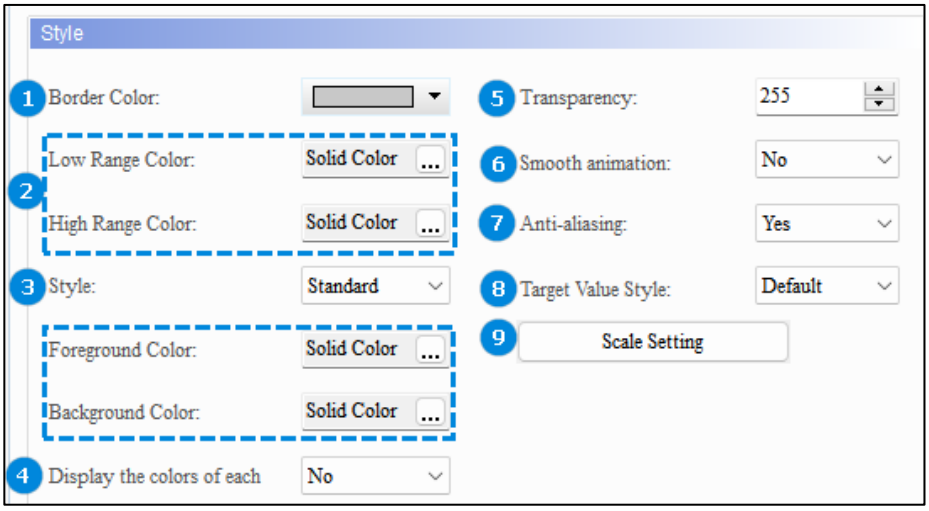


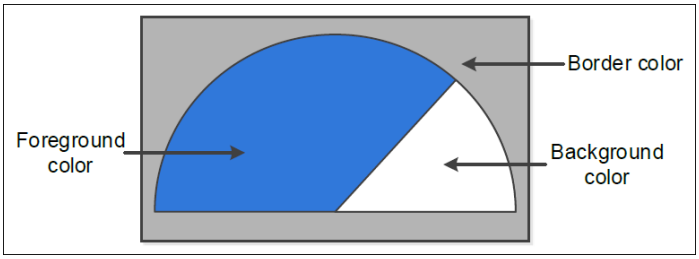
Common Properties

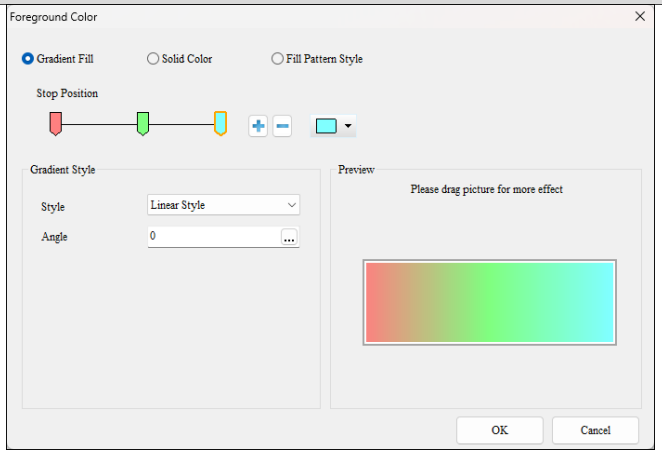


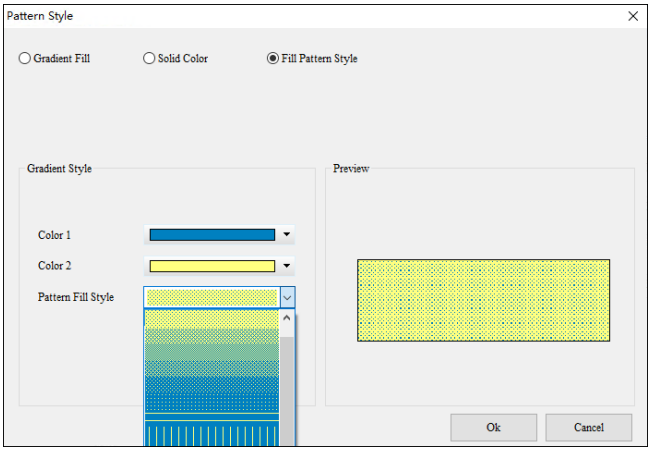
Legend	Function	Description
❶	Read Address	You can select the **Internal Memory** address or controller register address, and the **Input** memory **Type** can only be **Word**.
	Read Offset Address	The HMI will configure the value to the corresponding read address according to the set Offset Address. After setting the offset address, the actual read address is as follows. The actual read address of the element: Read Address + Value in Read Offset Address × Element [Data Type]
❷	Data Settings	• **Data Type**: Select the data type. • **Data Format**: Select the data format. • **Minimum / Maximum**: Enter the valid range value according to the set **Data Type** and **Data Format**. • **Variable minimum / maximum limits**: If selected, you can define the minimum and maximum value through the corresponding value of the memory address.
❸	Target	Select and enter the target value. You can also select its display color. If the **Variable target / range limits** checkbox is not selected, only constant values can be entered for the **Target**.
❹	Range	Select and enter the **Low Limit** and **High Limit**. If the **Variable target/range limits** checkbox is not selected, only constant values can be entered for the **Low Limit** and **High Limit**.
❺	Variable target / range limits	If selected, you can define memory addresses to dynamically change the display target value as well as the low and high limit values.
❻	Interval display	Select or enter the interval display value. The range value is 0–100. To have an effect, the value displayed in the interval needs to be divided by the difference between the maximum value and the minimum value. For example, if the minimum value is 0, the maximum value is 100, the interval display value is 20, no scale is displayed when the value is 0~19; the scale is 20 when the value is 20~39; the scale is 40 when the value is 40~59, and so on.

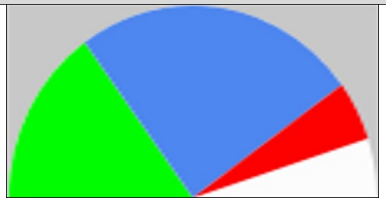
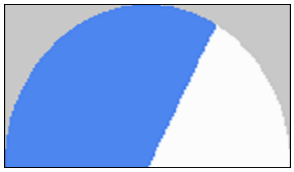
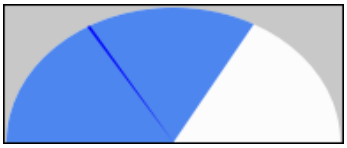
- **Style**

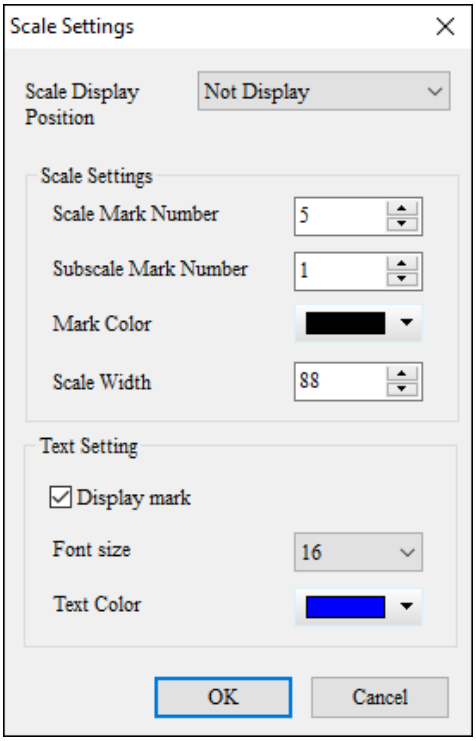


Legend	Function	Description
1	Border Color	Select the border color of the element. 
2	Low Range Color High Range Color Foreground Color Background Color	Select the color of the element and the low range and high range colors to be displayed on the HMI. **Note**: When the **Range** enable, if the reading value on the HMI is less than the low limit value, the foreground color will display the low range color; if the reading value is greater than the high limit value, the foreground color will display the high range color. • **Gradient Fill:** Set the gradient style by setting the current angle or center point. You can drag the mouse on the preview image to set the current angle or center point, or set the memory address to dynamically configure it on the HMI. You can also set the stop point colors to present the effect of multi-color gradients.

<table>
<tr><th>Legend</th><th>Function</th><th>Description</th></tr>
<tr><td></td><td></td><td> 

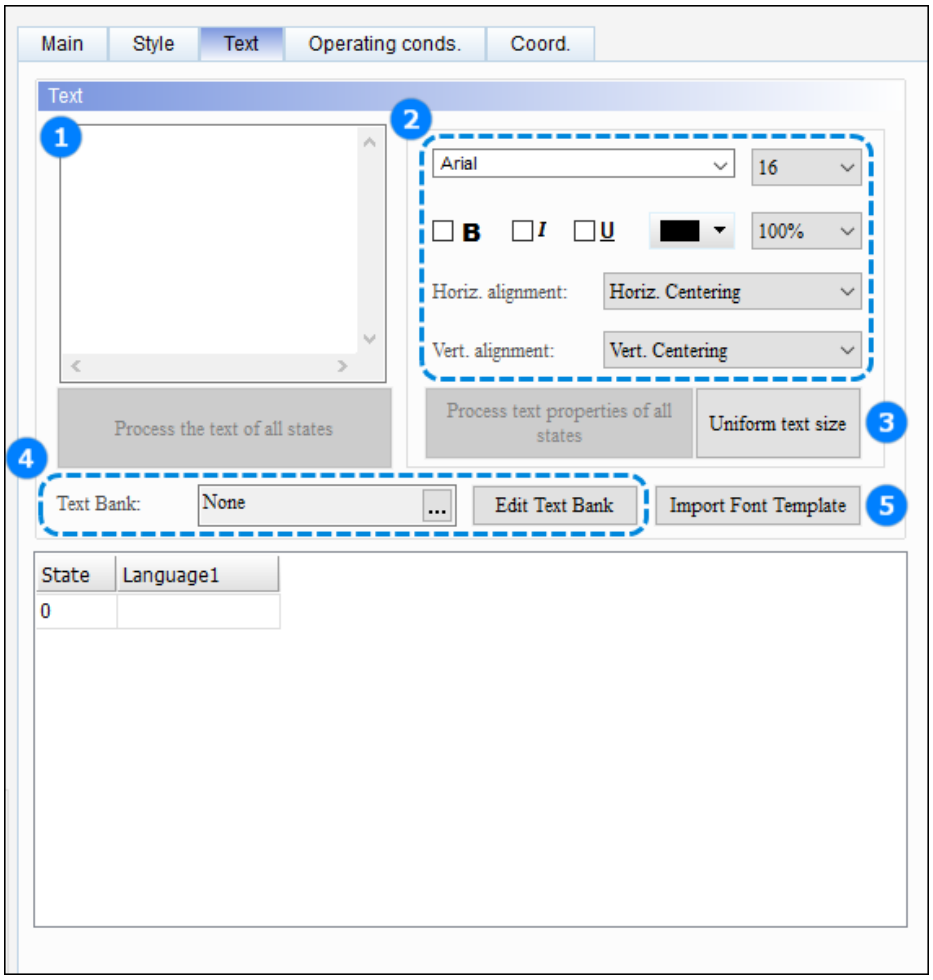
Note: Up to 5 stop points can be set.
• Solid Color: Select a single-color fill.
• Fill Pattern Style: Select a two-color combination configuration and select a pattern fill style.

</td></tr>
<tr><td>3</td><td>Style</td><td>Select the appearance of the element. The default value is Standard.
<table>
<tr><th>Standard</th><th>Raised</th><th>Sunken</th><th>Transparent</th></tr>
<tr><td> </td><td> </td><td> </td><td> </td></tr>
</table></td></tr>
<tr><td>4</td><td>Display the colors of each</td><td>• If select No, no matter what the value is, only the Foreground Color will be displayed.
• If select Yes, with the Low Range Color and Range functions, it can represent different colors of each stage.
The following is an example of setting Display the colors of each for Pie(1) element.</td></tr>
</table>

<table>
<tr><th>Legend</th><th>Function</th><th>Description</th></tr>
<tr><td></td><td></td><td>  </td></tr>
<tr><td>⑤</td><td>Transparency</td><td>Select or enter the transparency value of the element. The range value is 50–255 and the default value is 255.
The lower the value is, the higher the transparency is.</td></tr>
<tr><td>⑥</td><td>Smooth animation</td><td>Select whether to enable the Smooth Animation function.
When this function is enabled, the element display becomes smoother.</td></tr>
<tr><td>⑦</td><td>Anti-aliasing</td><td>Select whether to enable the Anti-aliasing function.
When this function is enabled, the display of the elements will be more delicate, and there will be no aliasing.
<table><tr><th>Yes</th><th>No</th></tr><tr><td> </td><td>  </td></tr></table></td></tr>
<tr><td>⑧</td><td>Target Value Style</td><td>Select the style of the target value. The available options are Default and Style 1, and the default value is Default.
<table><tr><th>Default</th><th>Style 1</th></tr><tr><td>  </td><td> </td></tr></table>Note: The effect of this setting is visible only when the Target checkbox is selected.</td></tr>
</table>

<table>
<tr><th>Legend</th><th>Function</th><th>Description</th></tr>
<tr><td>⑨</td><td>Scale Settings</td><td>


<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Scale Display Position</td><td>Select the position to display scale on the element according to Direction.</td></tr>
<tr><td>Scale Settings</td><td>Set the mark number, color, width of the scale. The range value of Scale Mark Number is 1–99.</td></tr>
<tr><td>Text Setting</td><td>Select the Display mark checkbox to display the mark and set the text size and color.</td></tr>
</table>
</td></tr>
</table>

- **Text**



Legend	Setting	Description
1	Text	Enter the text message to display in the text box. Select the state / language table below to directly set the text displayed by that state / language. Enter the text message to be displayed in the text box. Click the element on the screen and press **Space** to edit the text content of the current state / language.
2	Text Property	Set the text properties.
3	Uniform text size	Changes the text size in all languages with one click according to the currently selected state.
4	Text Bank	Applies the text in the **Text Bank** to the content of each state / language. You can also click **Edit Text Bank** to edit the text content in the **Text Bank** editor.
5	Import Font Template	Click to open the **Font Template** dialog, and then select the template and import the text properties of the selected state.

- **Operating conds.**

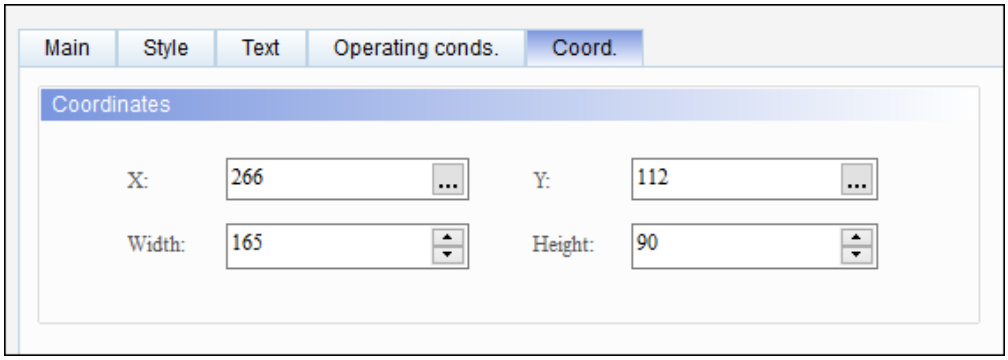
  The **Invisible Address** displays or hides the element on the HMI. If the **Invisible State** is set to **On**, the element will be hidden when the **Invisible Address** is ON. Instead, if the **Invisible State** is set to **Off**, the element will be hidden when the **Invisible Address** is OFF.

  The following is an example of using invisible address.

  1. Create a button element on the screen and set its **Read Address** to $9.0, then set the **Invisible Address** of the element to hide to $9.0.
  2. To hide the element, you must first click the $9.0 button.

<table>
<tr><td rowspan="2">Invisible State is On</td><td>
</td></tr>
<tr><td>Element disappear
Invisible Address $9.0 ON</td></tr>
<tr><td rowspan="2">Invisible State is Off</td><td>
</td></tr>
<tr><td>Element disappear
Invisible Address $9.0 OFF</td></tr>
</table>

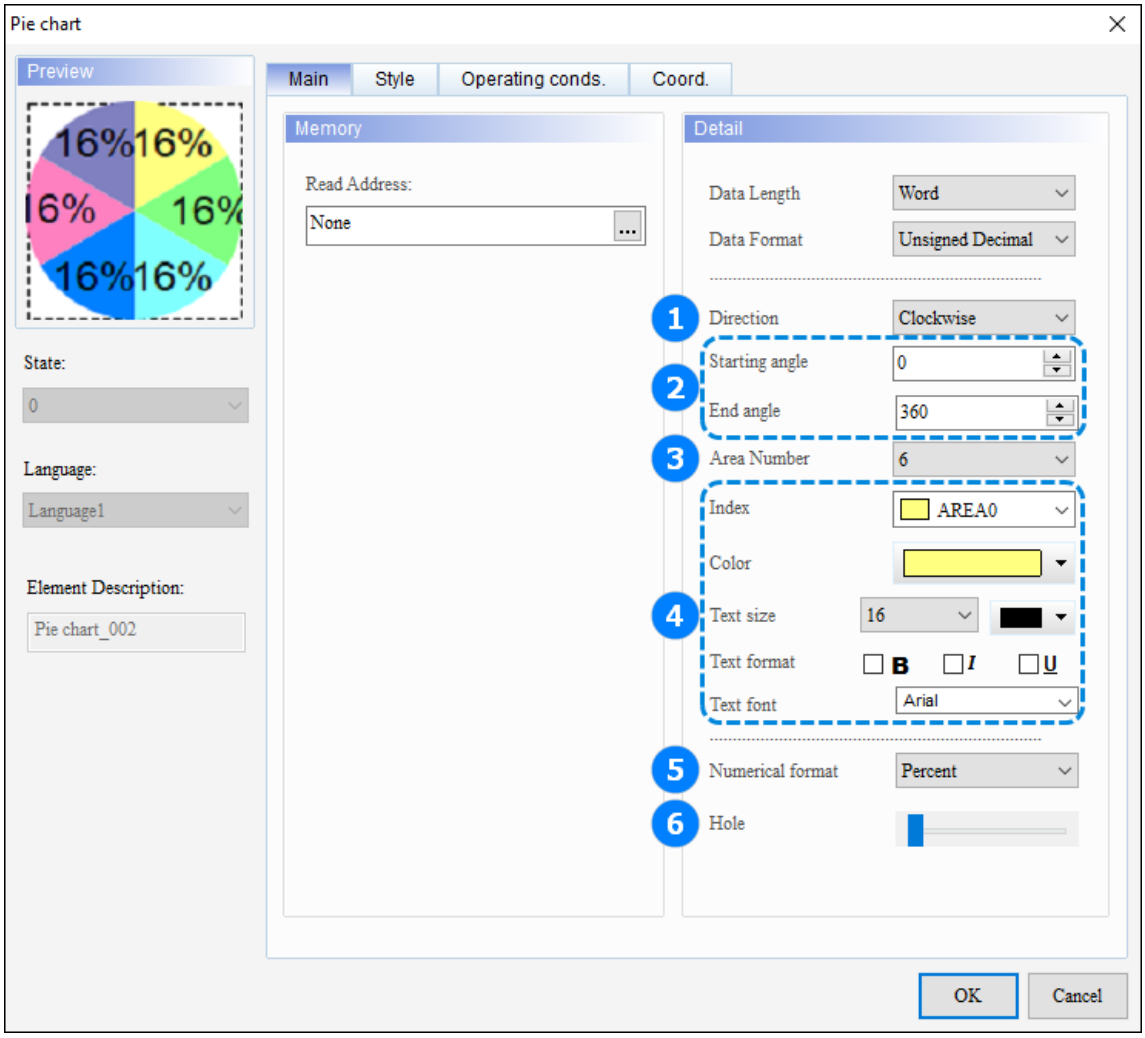
- **Coord.**



Set the XY coordinates, width, and height on the upper left of the element. You can enter a constant as fixed position or memory address to dynamically configure the position of the element on the HMI.

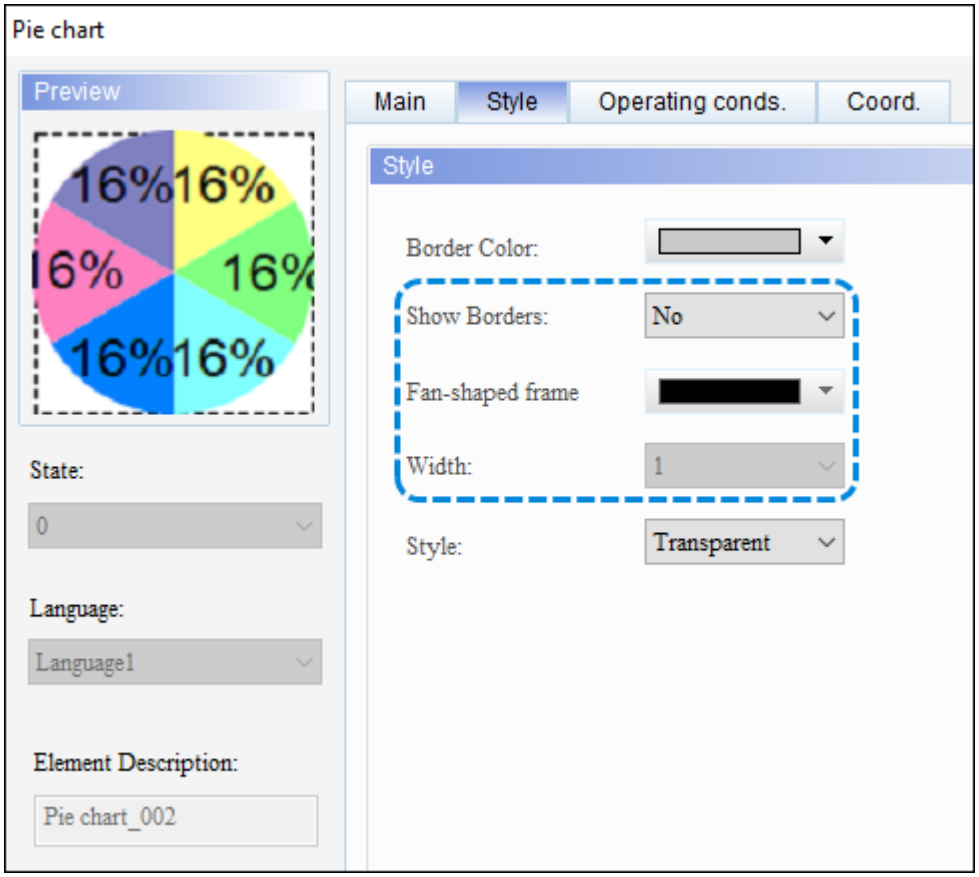
Pie Chart

- **Main**



Legend	Setting	Description
❶	Direction	Select the display direction of the pie chart area.
❷	Starting angle End angle	Select or enter the starting angle and end angle of the pie chart.
❸	Area Number	Select the size of the area number. The range value is 2–15.
❹	Area Color / Text Settings	Select the color of the area and set the text properties. **Note**: Area settings are only available after selecting an **Area Number**.
❺	Numerical format	Select the numerical format.
❻	Hole	Drag the slider to change the hole size.

- **Style**



The area border settings are only available when **Show Borders** is set to **Yes**.

Indicator

The Indicator element displays the state of the address. You can specify the data type or data range to configure the different styles of the state.

Common Properties

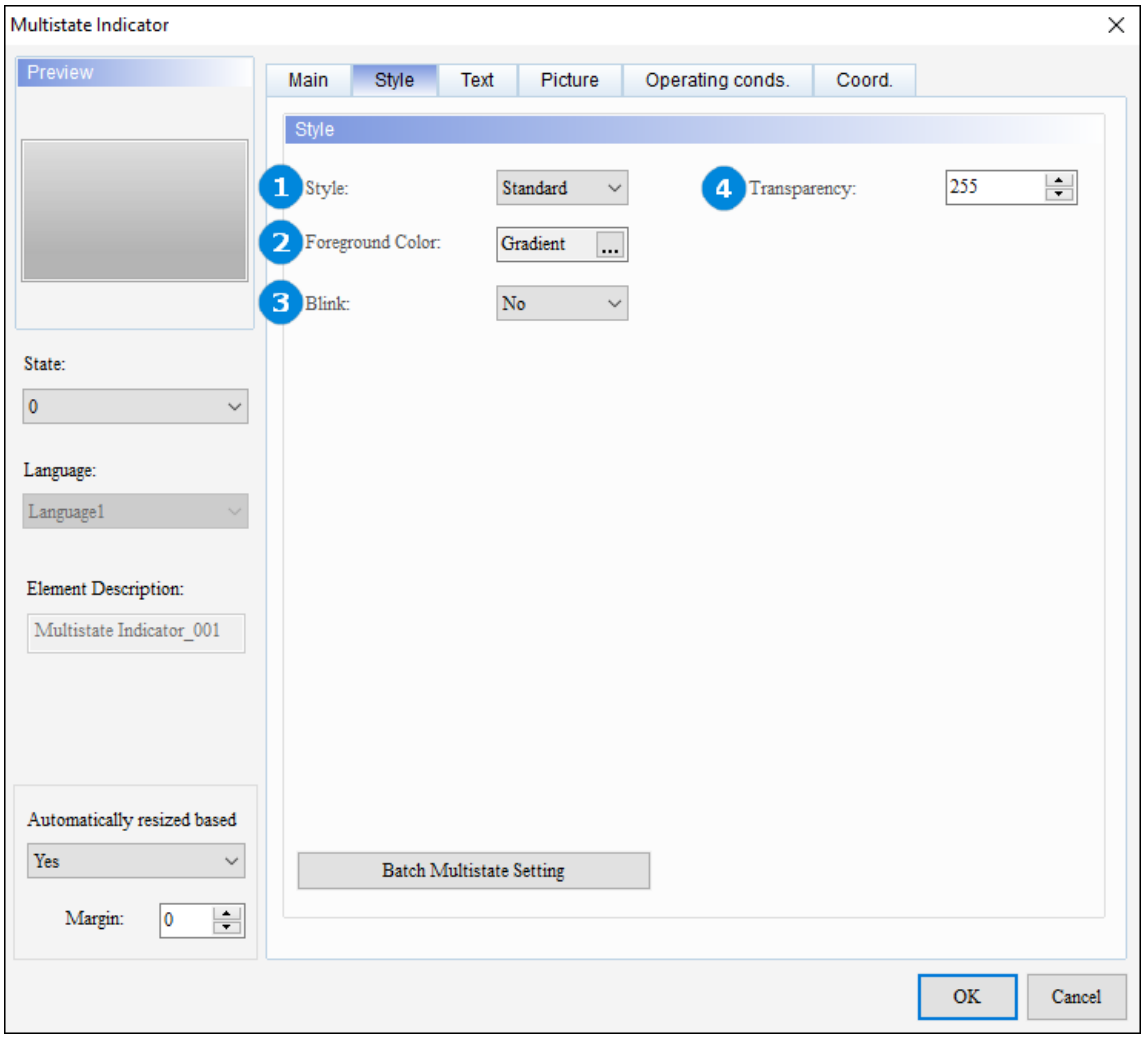
If **Automatically resized based** is set to **Yes**, when the set text length exceeds the element size, the element is automatically resized according to the **Margin** to show the complete text.

Note: This setting supports the element enlargement function, the string is shortened, and the element will not become smaller.

- **Main**
  - **Read Address**: You can select the **Internal Memory** address or controller register address, and the **Input** memory **Type** can only be **Word**.
  - **Read Offset Address**: The HMI will configure the value to the corresponding read address according to the set Offset Address. After setting the offset address, the actual read address is as follows.

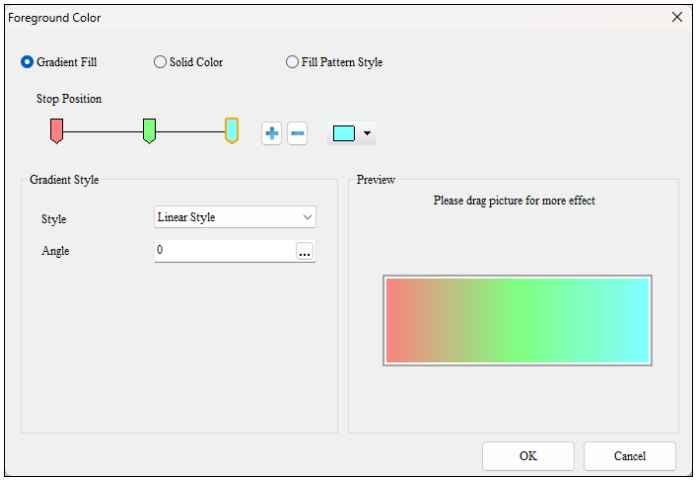
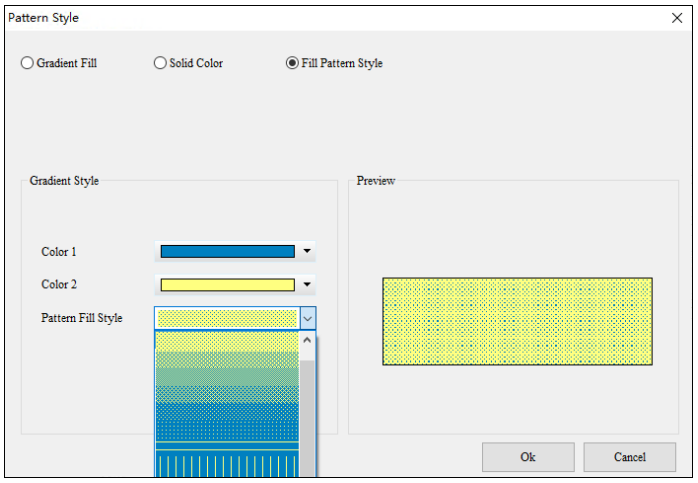
    The actual read address of the element: Read Address + Value in Read Offset Address × Element [Data Type]

- **Style**

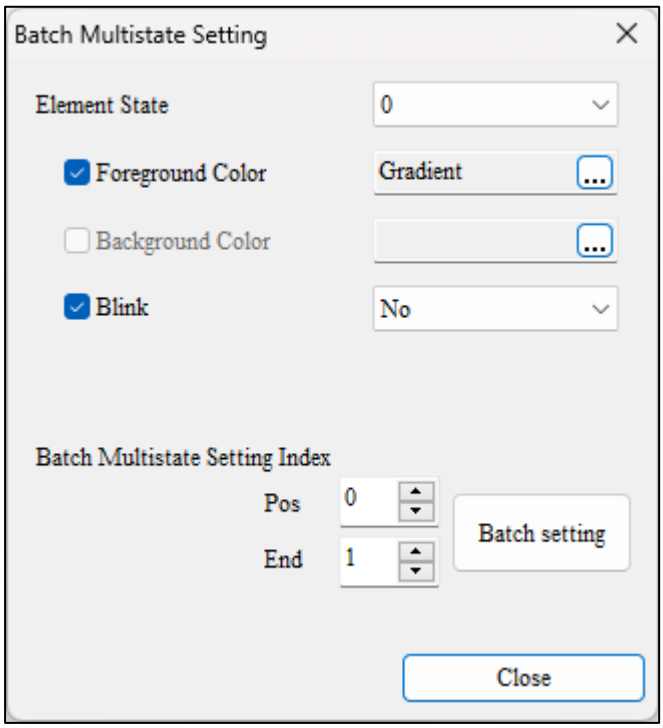


After selecting a state, modify the following settings and apply them to the state.

Legend	Setting	Description
1	Style	Select the appearance of the element. The default value is Standard. Standard / Raised / Round / Invisible
2	Foreground Color	Select the color of the element. • **Gradient Fill:** Set the gradient style by setting the current angle or center point. You can drag the mouse on the preview image to set the current angle or center point, or set the memory address to dynamically configure it on the HMI.

Legend	Setting	Description
		You can also set the stop point colors to present the effect of multi-color gradients.  **Note**: Up to 5 stop points can be set. • **Solid Color**: Select a single-color fill. • **Fill Pattern Style**: Select a two-color combination configuration and select a pattern fill style. 
③	Blink	Choose whether to enable the blink prompt. The blink color is opposite to the foreground color of the status.
④	Transparency	Select or enter the transparency value of the element. The range value is 50–255, and the default value is **255**. The smaller the value, the higher the transparency of the element.

- **Batch Multistate Setting**

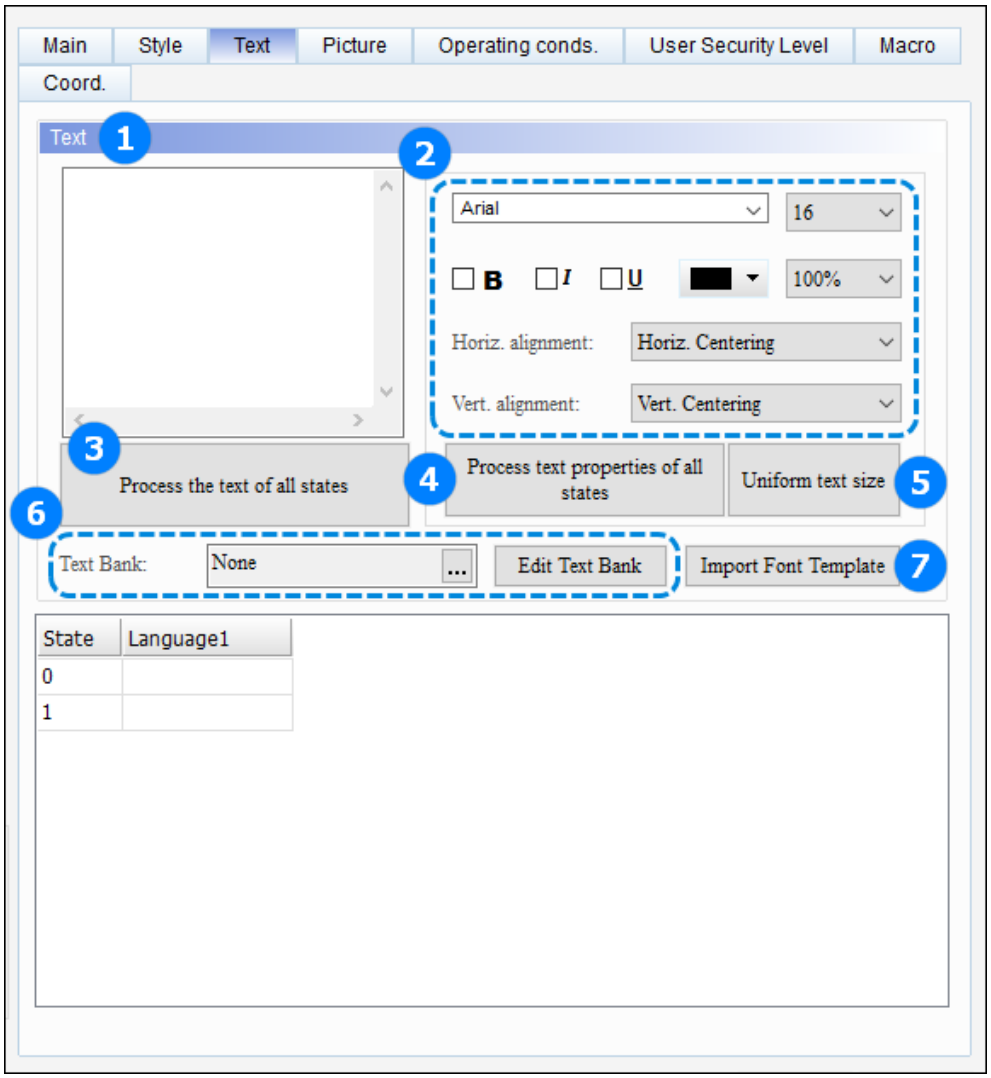


Follow these steps to configure the **Batch Multistate Setting**.

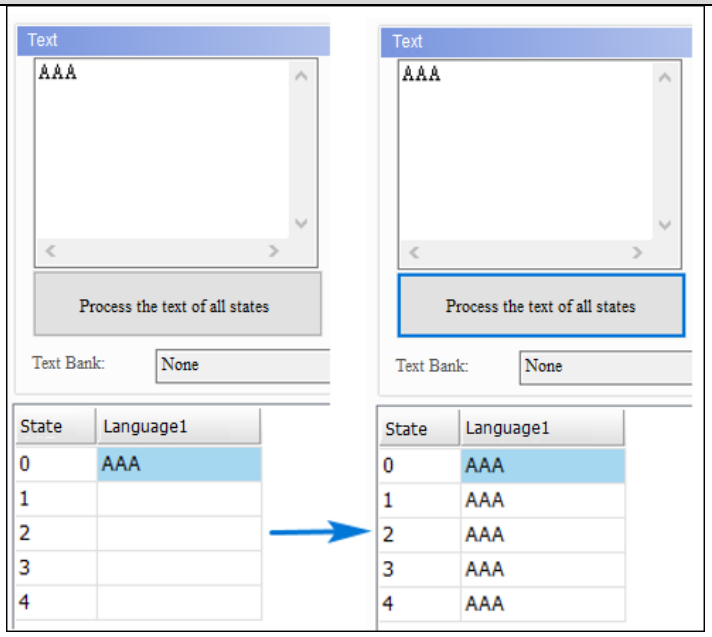
Configure the Batch Multistate Setting

1. Select the **Element State**.
2. Select the properties checkbox to apply and set **Batch Multistate Setting Index**. For example, set the **Foreground Color** of state 0~state 5 in batch to **Gradient**, and then click **Batch setting.**

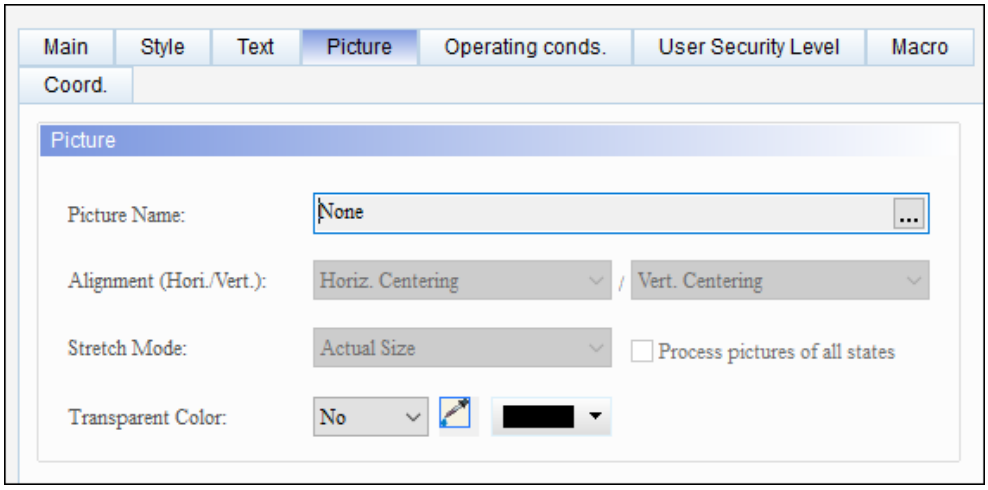
- **Text**



Legend	Setting	Description
❶	Text	Enter the text message to display in the text box. Select the state / language table below to directly set the text displayed by that state / language. Enter the text message to be displayed in the text box. Click the element on the screen and press **Space** to edit the text content of the current state / language.
❷	Text-related properties	Set the text properties.
❸	Process the text of all states	Click to apply the selected text contents of the state to those of other states.

Legend	Setting	Description
		
❹	Process text properties of all states	Click to apply the selected text properties of the state to those of other states.
❺	Uniform text size	Changes the text size in all languages with one click according to the currently selected state.
❻	Text Bank	Applies the text in the **Text Bank** to the content of each state / language. You can also click **Edit Text Bank** to edit the text content in the **Text Bank** editor.
❼	Import Font Template	Click to open the **Font Template** dialog, and then select the template and import the text properties of the selected state.

- **Picture**



<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Picture Name</td><td>Enter the picture name. The default value is None.To set the picture display, click ... to select the picture to display in the picture bank and drag it to each state icon.

To configure other pictures, refer to Picture Bank.</td></tr>
<tr><td>Alignment</td><td>Select the alignment of the picture.</td></tr>
<tr><td>Stretch Mode</td><td>Select the stretch mode of the picture.
<table>
<tr><th>Stretch All</th><th>Stretch 1:1</th><th>Actual Size</th></tr>
<tr><td>The picture fills the full element display area.
 </td><td>The picture displays in 1:1 size based on the element width and length.
 </td><td>The picture is displayed in the display area of the element in its actual size.
 </td></tr>
</table></td></tr>
<tr><td>Process pictures of all states</td><td>If an element has multiple state values, and its picture may not fill the full display area of the element. Select this checkbox to process all pictures settings</td></tr>
</table>

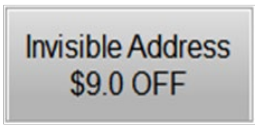
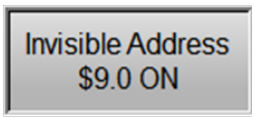
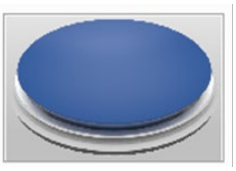
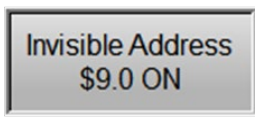
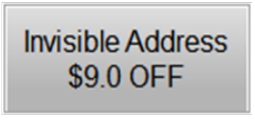
Setting	Description
Transparent Color	Choose whether to specify a color in the picture as a transparent color. If set to **Yes**, you can select colors or use the color picker to select one of the colors on the picture to specify it as a transparent color, that is, the foreground color. The following figures show the effect of specifying white as a transparent color.

- **Operating conds.**

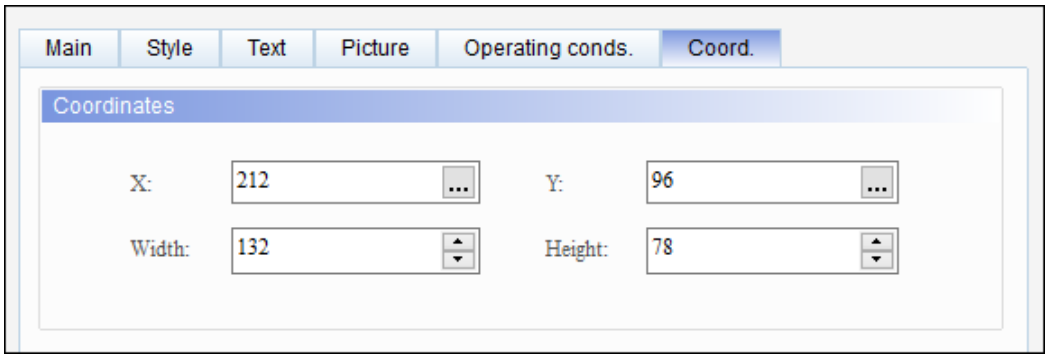
  The **Invisible Address** displays or hides the element on the HMI. If the **Invisible State** is set to **On**, the element will be hidden when the **Invisible Address** is ON. Instead, if the **Invisible State** is set to **Off**, the element will be hidden when the **Invisible Address** is OFF.

  The following is an example of using invisible address.

3. Create a button element on the screen and set its **Read Address** to $9.0, then set the **Invisible Address** of the element to hide to $9.0.
4. To hide the element, you must first click the $9.0 button.

Invisible State is On		
	Element disappear	
Invisible State is Off		
	Element disappear	

- **Coord.**



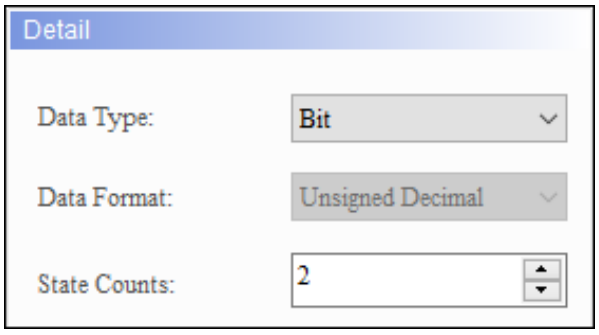
Set the XY coordinates, width, and height on the upper left of the element. You can enter a constant as fixed position or memory address to dynamically configure the position of the element on the HMI.

Multistate Indicator

The **Multistate Indicator** element displays the state of an address. When the state changes, the element displays the style of that state.

The following table lists the special properties of the Multistate Indicator element with their description.

- **Main – Detail**



Setting	Description
Data Type	Select the unit for state switching
Data Format	Select the data format. **Note**: This setting is only available when **Data Type** is set to Word.
State Counts	Select or enter the state counts according to the selected Data Type.

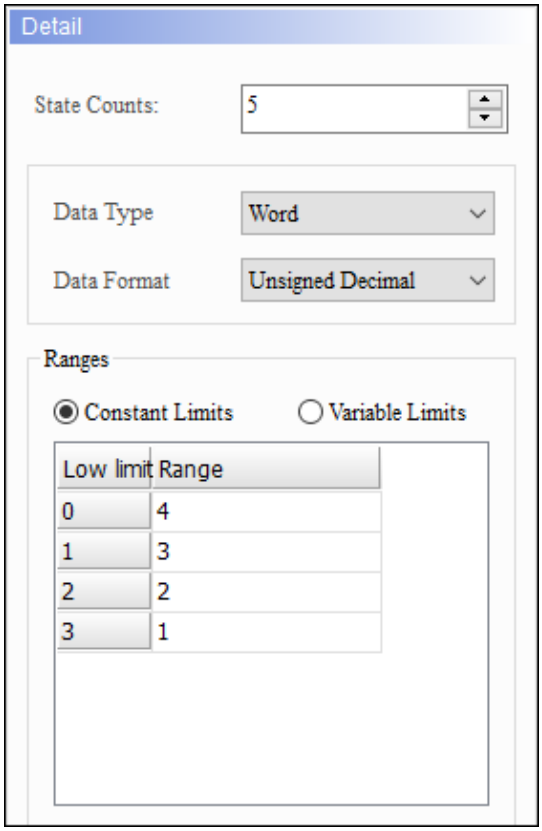
Setting	Description
	• Bit: 2 states can be set. • Word: 1~256 states can be set. • LSB: 16 states can be set. • LSB (Support State 0): 17 states can be set. **Note**: LSB means that the register data is first converted into binary data, and then the current object state is defined by using the lowest non-zero bit. For example, decimal value 3 converted to binary is 0000000000000011, and the lowest non-zero bit is Bit 0, State is 1.

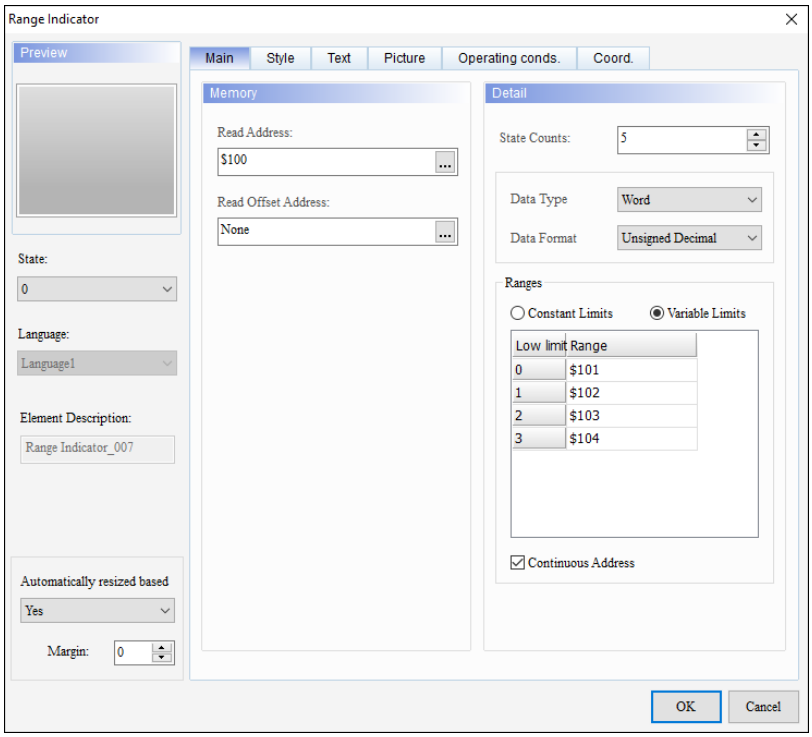
Range Indicator

The **Range Indicator** element reads the memory address to display the state of the specified range.

The following table lists the special properties of the Range Indicator element with their description.

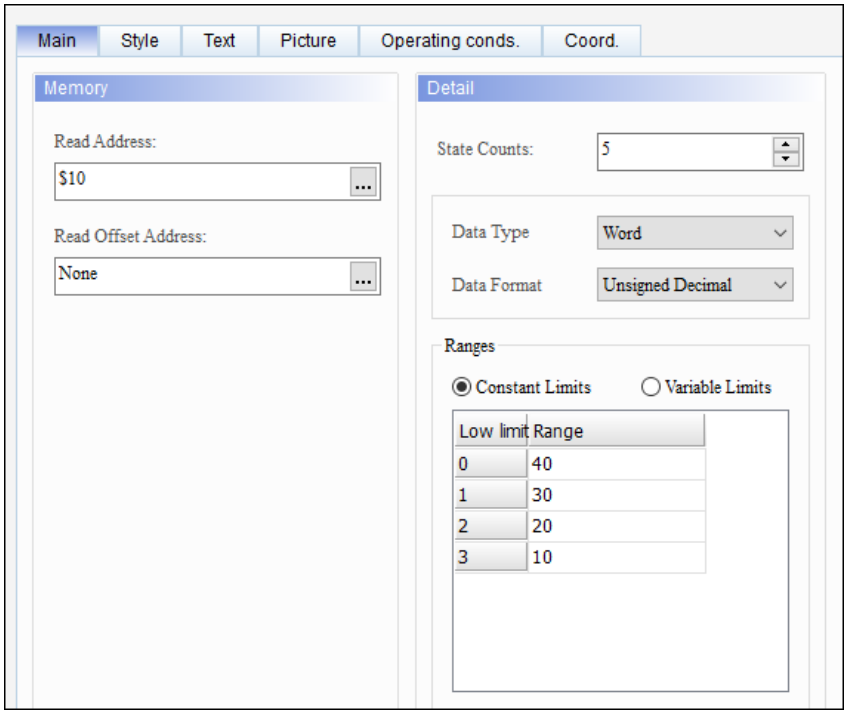
- **Main – Detail**

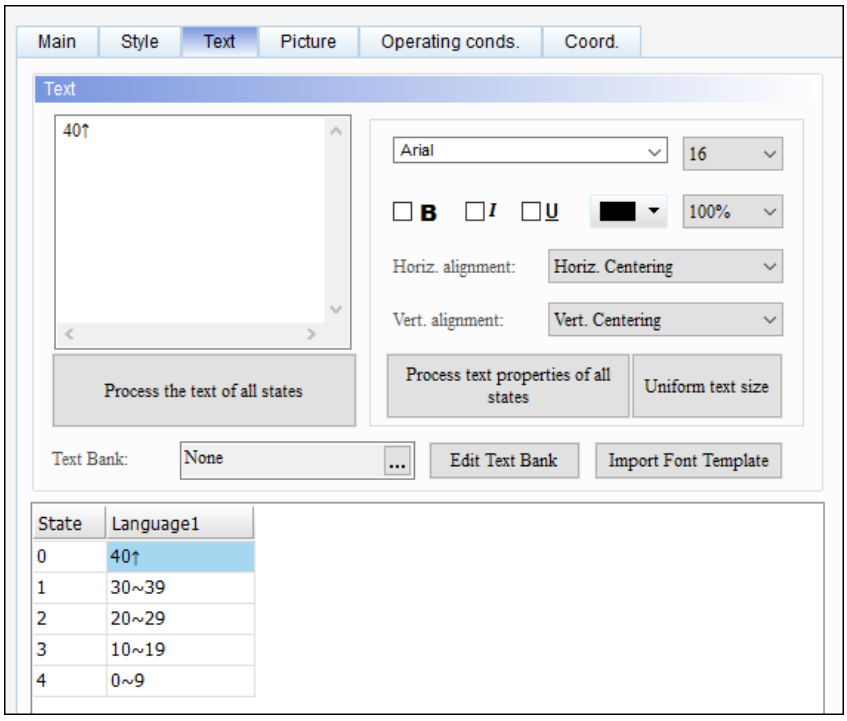


Setting	Description
State Counts	Select or enter the state counts. The range value is 1–256.
Data Type	Select the data type.
Data Format	Select the data format.
Ranges	Choose **Constant Limits** or **Variable Limits**. • **Constant Limits**: The value entered is a constant and the range is from maximum value to minimum value from top to bottom. • **Variable Limits**: The value entered can be determined by the memory address, and you can set whether to use the continuous address to read the value. ○ If the **Continuous Address** checkbox is selected, the read memory address will be used as the starting value to continuously configure each state.  ○ If the **Continuous Address** checkbox is not selected, you can configure the read address of each state.

The following describes how to use the **Range Indicator** element.

1. Create a **Range Indicator** element on the screen.
2. Double-click the element, and then set the range and text of each state in the property dialog of the element.





The execution result is as follows.

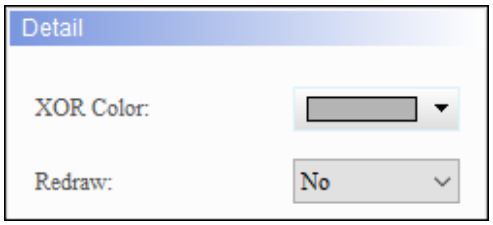
Enter the value to the **Numeric Entry** element, and the Range Indicator displays the different styles of the state.

Simple Indicator

The **Simple Indicator** provides ON and OFF states. When switching the state of the read address, the element color will change according to the set XOR color.

The following table lists the special properties of the **Simple Indicator** element with their description.

- **Main**



Setting	Description
XOR Color	Select the XOR Color of the element.
Redraw	Choose whether to redraw. If select **Yes**, when overlapping the element on top of the dynamic element, the data can be read successfully.

The following describes how to use the **Simple Indicator** element.

1. Create a **Range Indicator** element on the screen and set the **Read Address** to **$10**, and then set the different **XOR Color** for each state.
2. Create two Simple Indicator elements and set the Read Address to **$0.0**, and then set **Yes** and **No** for **Redraw** separately.
3. Overlap two **Simple Indicator** elements on top of the **Range Indicator** element.

The execution result is as follows.

Set $0.0 of the **Simple Indicator** to On, and the element status is as shown on the left; modifying the element value of $10 can make the **Range Indicator** change to a different color. Then, after modifying the value again, the color of the **Simple Indicator** configured with redraw setting will change as shown on the right.

Data Display

DIAScreen provides multiple **Data Display** elements. Set the read address and select the element to display according to the read data format. The following table lists the description of each element.

Element	Description
Numeric Display	Used to display the read address values.
Character Display	Used to display the character. The HMI converts the read address value to ASCII code to display on the **Character Display** element.
Data Display	Used to display the date, time, and week of the HMI system. You can also set timestamp through **Read Address** to display the date and time, and the **Week Display** element can be edited using the **Multi-language** function.
Time Display	
Week Display	
General Message Display	Used to display the text message of different states and read the text of the corresponding state according to the set address.
Moving Sign	Used to display the text message of the state to toggle and adjust the display direction, moving points, and interval of the text.
QR code display	Used to convert the string of read address to QR code to display on the HMI.
Barcode	Used to generate Barcode and must be used with the **Barcode Input** element.

Common Properties

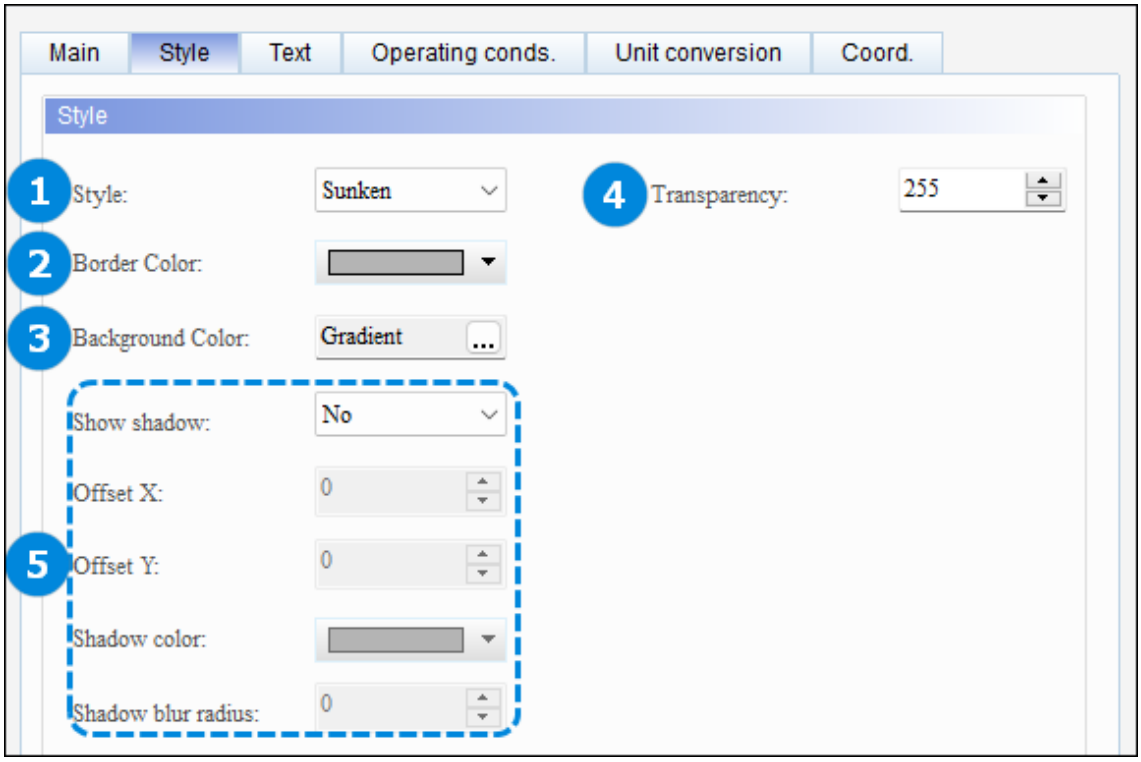
If **Automatically resized based on text** is set to **Yes**, when the set text length exceeds the element size, the element is automatically resized according to the **Margin** to show the complete text.

Note: This setting supports the element enlargement function, the string is shortened, and the element will not become smaller.

- **Main**

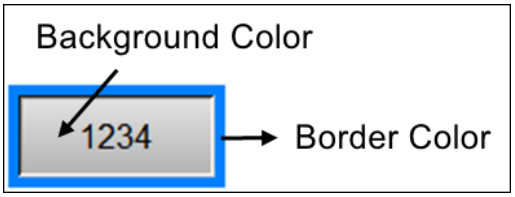
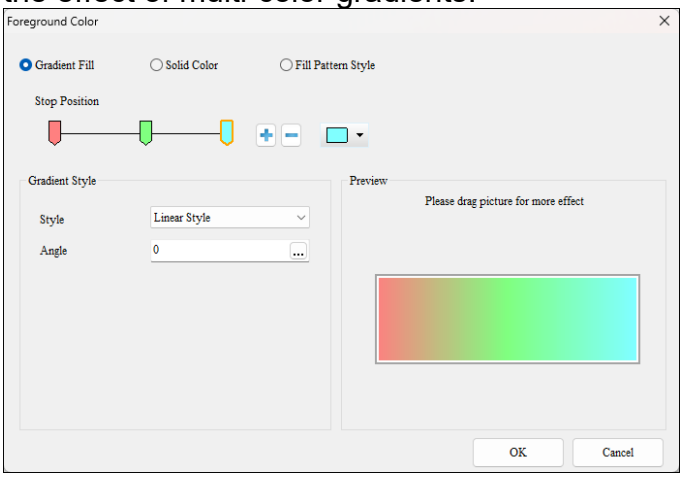
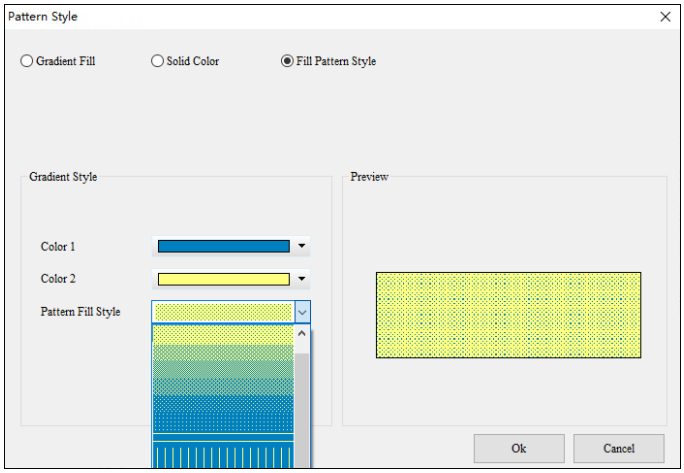
Setting	Description
Read Address	You can select the **Internal Memory** address or controller register address, and the **Input** memory **Type** can only be **Word**.
Read Offset Address	The HMI will configure the value to the corresponding read address according to the set Offset Address. After setting the offset address, the actual read address is as follows. The actual read address of the element: Read Address + Value in Read Offset Address × Element [Data Type] **Note**: The **Offset length** can only be set through the **Numeric Display** element.

- **Style**



Legend	Setting	Description
1	Style	Select the appearance of the element. The default value is **Standard**.

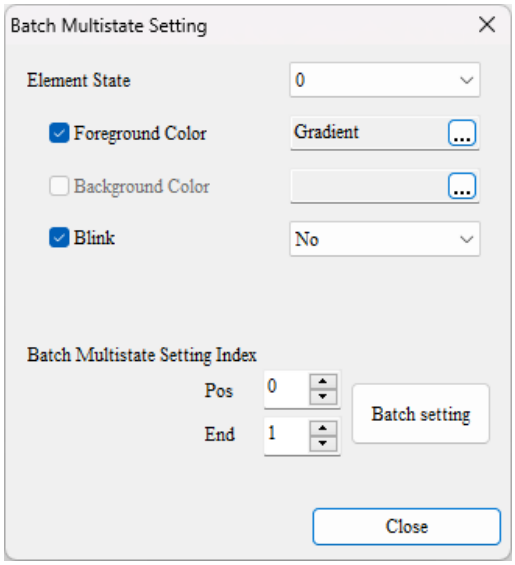
Standard	Raised	Sunken	Transparent
1234	1234	1234	1234

Legend	Setting	Description
②	Border Color	Select the display border color of the element. 
③	Background Color	Select the color of the element. • **Gradient Fill:** Set the gradient style by setting the current angle or center point. You can drag the mouse on the preview image to set the current angle or center point, or set the memory address to dynamically configure it on the HMI. You can also set the stop point colors to present the effect of multi-color gradients.  **Note**: Up to 5 stop points can be set. • **Solid Color**: Select a single-color fill. • **Fill Pattern Style**: Select a two-color combination configuration and select a pattern fill style. 

Legend	Setting	Description
❹	Transparency	Select or enter the transparency of the element. The range value is 50–255 and the default value is **255**. The smaller the value, the higher the transparency of the element.
❺	Shadow Settings	• **Show shadow**: Choose whether to show shadow. The shadow settings are available only when **Show shadow** is set to **Yes**. • **Offset X**: Select or enter the X offset value. The range value is an integer between –100~100. If enter a negative number, the shadow appears on the left side of the text, and if enter a positive number, the shadow appears on the right side. • **Offset Y**: Select or enter the Y offset value. The range value is an integer between –100~100. If enter a negative number, the shadow appears above the text, and if enter a positive number, the shadow appears as shown in the following figure. • **Shadow color**: Set the color of the text shadow. • **Shadow blur radius**: Select or enter the shadow blur radius. The range value is an integer between 0~100. The greater the value, the more blurred the shadow. **Note**: These settings only apply to the **Numeric Display** and **Character Display** element.

- **Batch Multistate Setting**

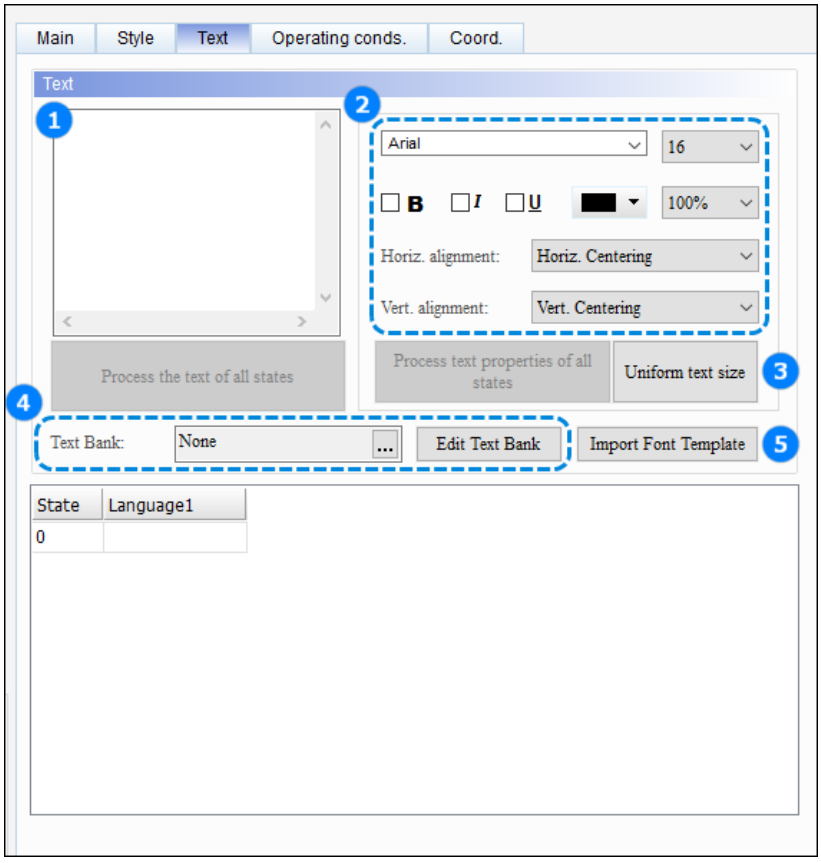
  Note: The setting is supported in **Week Display** and **Message Display** elements.



Follow these steps to configure the **Batch Multistate Setting**.

1. Select the **Element State**.
2. Select the properties checkbox to apply and set **Batch Multistate Setting Index**. For example, set the **Foreground Color** of state 0~state 5 in batch to **Gradient**, and then click **Batch setting**.

- **Text**



Note: The properties of this tab only applies to the **General Message Display, Moving Sign**, and **Week Display** element. Other **Dada Display** elements do not have multi-state settings.

Legend	Setting	Description
❶	Text	Enter the text message to display in the text box. Select the state / language table below to directly set the text displayed by that state / language. Enter the text message to be displayed in the text box. Click the element on the screen and press **Space** to edit the text content of the current state / language.
❷	Text Property	Set the text properties.
❸	Uniform text size	Changes the text size in all languages with one click according to the currently selected state.
❹	Text Bank	Applies the text in the **Text Bank** to the content of each state / language. You can also click **Edit Text Bank** to edit the text content in the **Text Bank** editor.
❺	Import Font Template	Click to open the **Font Template** dialog, and then select the template and import the text properties of the selected state.

- **Operating conds.**

The **Invisible Address** displays or hides the element on the HMI. If the **Invisible State** is set to **On**, the element will be hidden when the **Invisible Address** is ON. Instead, if the **Invisible State** is set to **Off**, the element will be hidden when the **Invisible Address** is OFF.

The following is an example of using invisible address.

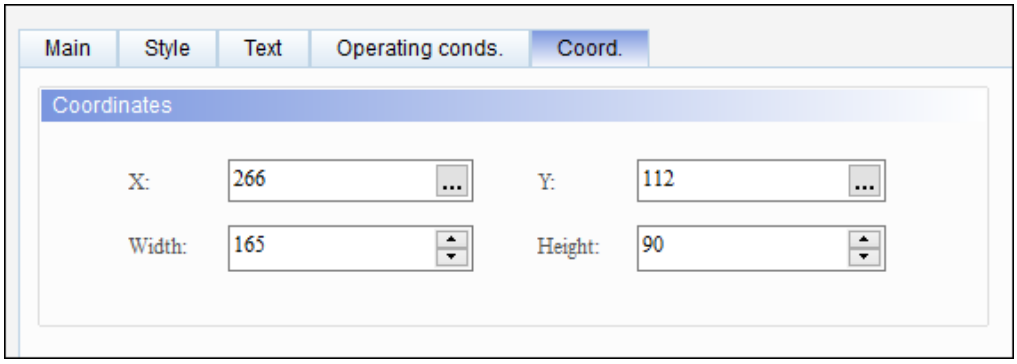
1. Create a button element on the screen and set its **Read Address** to $9.0, then set the **Invisible Address** of the element to hide to $9.0.
2. To hide the element, you must first click the $9.0 button.

Invisible State is On	1234	Invisible Address $9.0 ON
	Element disappear	Invisible Address $9.0 ON

Invisible State is Off	1234	Invisible Address $9.0 OFF
	Element disappear	Invisible Address $9.0 OFF

Note: This property only applies to the **Numerica Display**, **Character Display** and **General Message Display** element.

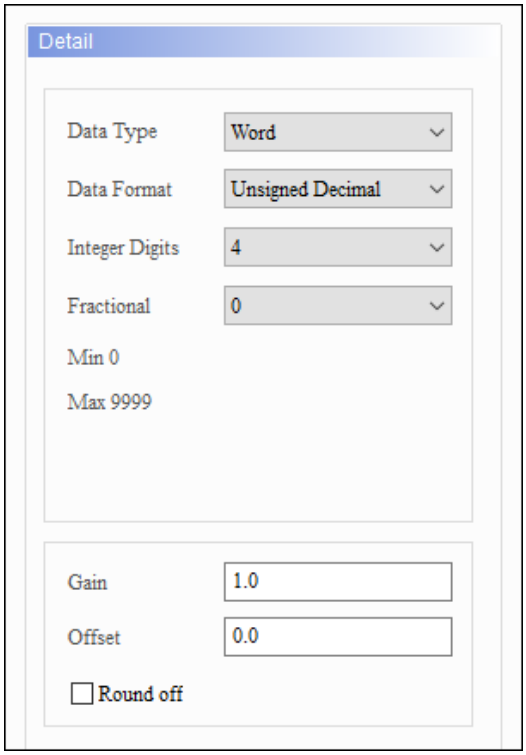
- **Coord.**



Set the XY coordinates, width, and height on the upper left of the element. You can enter a constant as fixed position or memory address to dynamically configure the position of the element on the HMI.

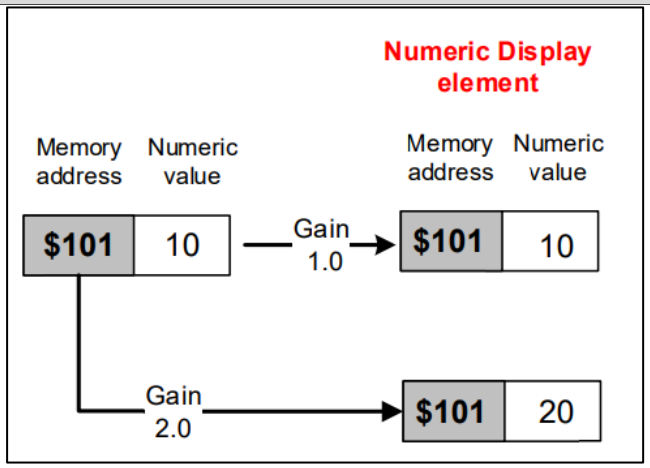
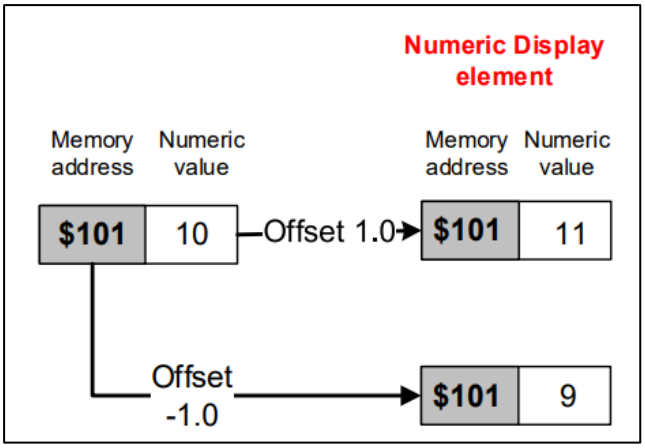
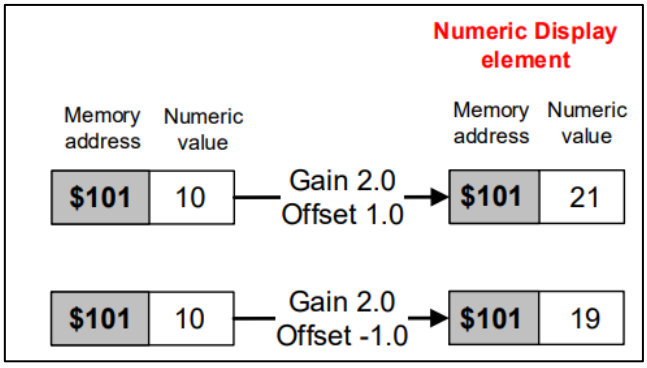
Numeric Display

- **Main**
  - **Detail**

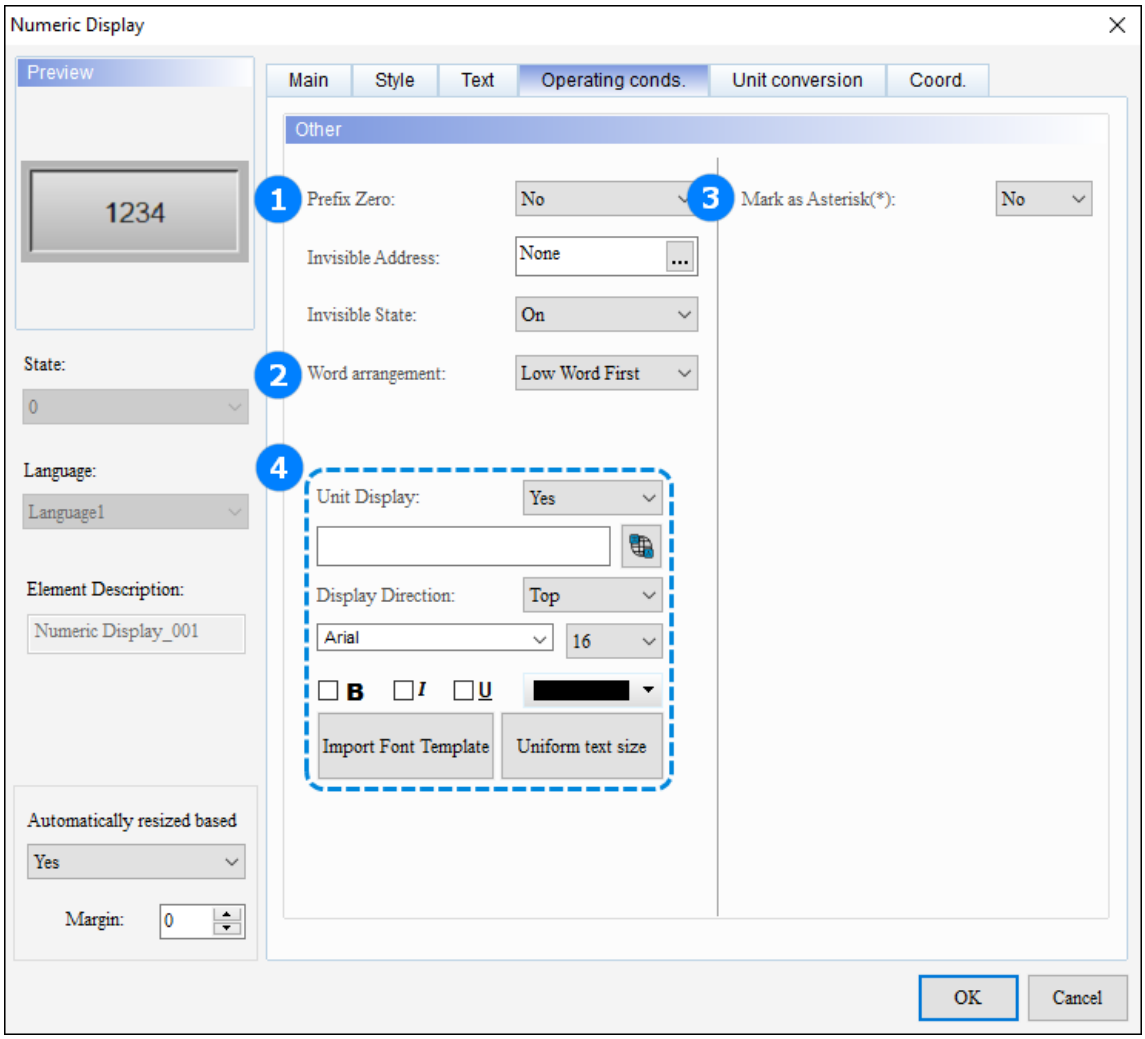


Setting	Description
Data Type	Select the data type.
Data Format	Select the data format. **Note**: Floating format is available only when the **Data type** is **Double Word** and **Quad Word**.
Integer Digits / Fractional	Select the valid display digits according to the data type. **Note**: The number of decimal places here is not a real decimal value, but the display format. Only when Floating is selected for the **Data Format**, it is a real decimal.
Gain / Offset	Enter the gain value and offset value. • The formula for **Gain** and **Offset** is y = (a)x + (b). • The **Numeric Display** element multiplies the actual value of the register by the set **Gain** value, and then display it on the HMI screen, and the gain is preset to 1.0. If the gain is set to 2.0, when the element reads the register value of 10, the actual value displayed on the element is 20.

y	a	x	b
Element display value	Gain	Actual Register Value	Offset

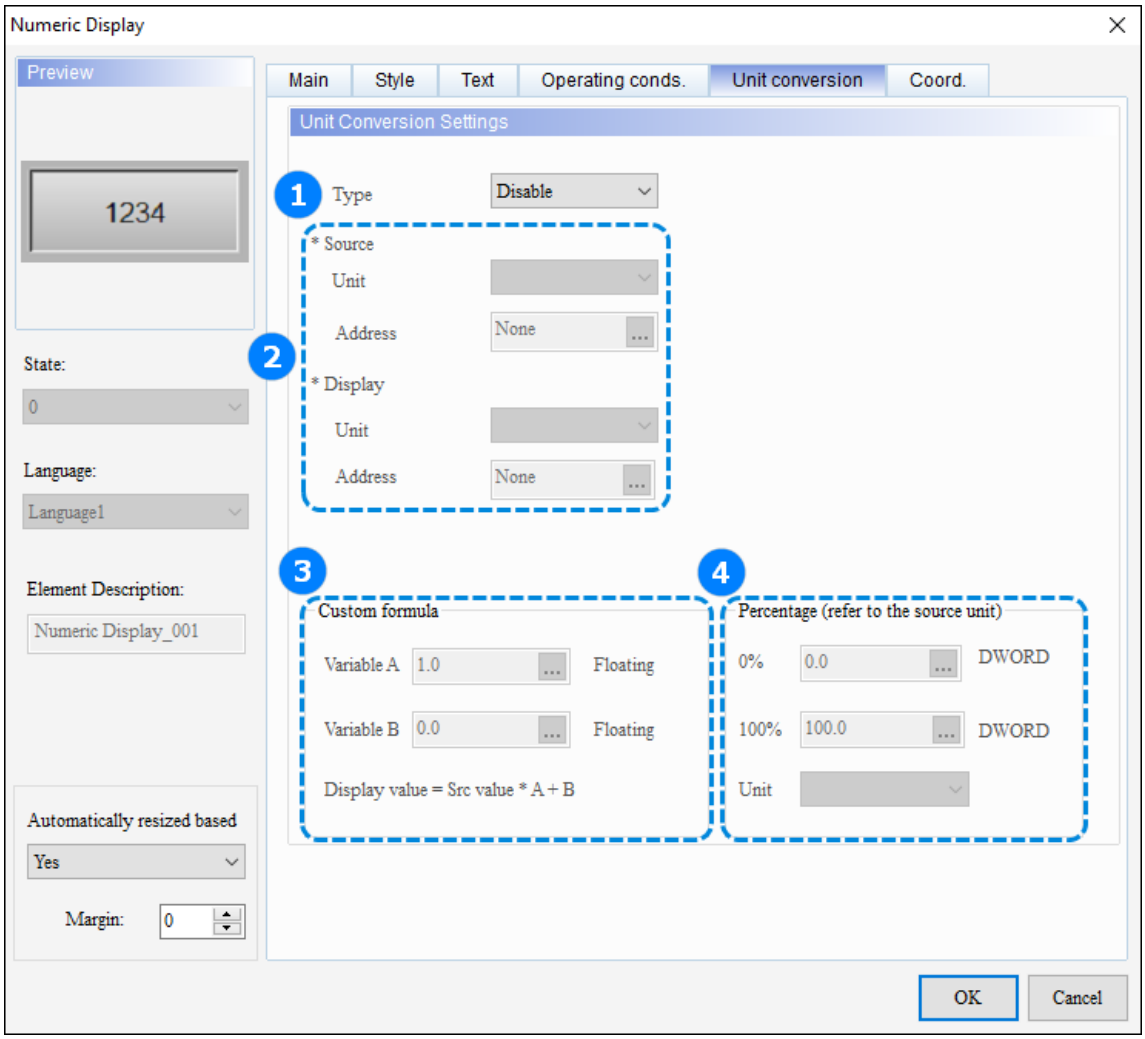
Setting	Description
	 • The **Numeric Display** element adds the actual value of the register to the set **Offset** value, and then displays it on the HMI. The **Offset** is preset to 0.0. If the **Offset** is set to 1.0, when the element reads the register value of 10, the value actually displayed on the element is 11; on the contrary, when the **Offset** is set to –1.0, when the element reads the register value to 10, then the value actually displayed on the element is 9.  • The following is a schematic figure of setting **Gain** 2.0, **Offset** 1.0 and **Gain** 2.0 and **Offset** –1.0.  • If users check **Round off**, the calculation results are rounded off before displayed on the element.

- **Operating conds.**



Legend	Setting	Description
1	Prefix Zero	Choose whether the integer digits are filled in with zeros. If the integer digits are 5 digits and the read value is 50, when selecting **Yes** for **Prefix Zero**, the **Numeric Display** element displays as follows.
2	Word arrangement	Select the word arrangement to swap high and low characters.
3	Unit Display	Choose whether to display the unit. Select **Yes** to set the text content, display direction, and relevant text properties.
4	Mark as Asterisk(*)	Select **Yes**, the read value will be displayed with a * symbol on the element.

- **Unit conversion**



Legend	Setting	Description
1	Type	Select the type to convert. • If **Disable** is selected, the value does not need unit conversion. • If **Custom formula** is selected, enter **Variable A** and **Variable B** in the **Custom formula** area, and the element value will be converted and displayed using the formula “**Display value = Syn value * A + B**”.

<table>
<tr><th>Legend</th><th>Setting</th><th>Description</th></tr>
<tr><td>②</td><td>Source Display</td><td>
Select the unit of source and display according to the set Type.

The Percentage settings are available when selecting % or Using the code for the Unit. When the Percentage area is available, you need to set the values corresponding to 0% and 100%, and set the unit according to the unit of the Source.
When selecting Using the code for the Unit, the conversion unit can be dynamically specified on the HMI through the memory address. The following table lists the corresponding code of each unit.

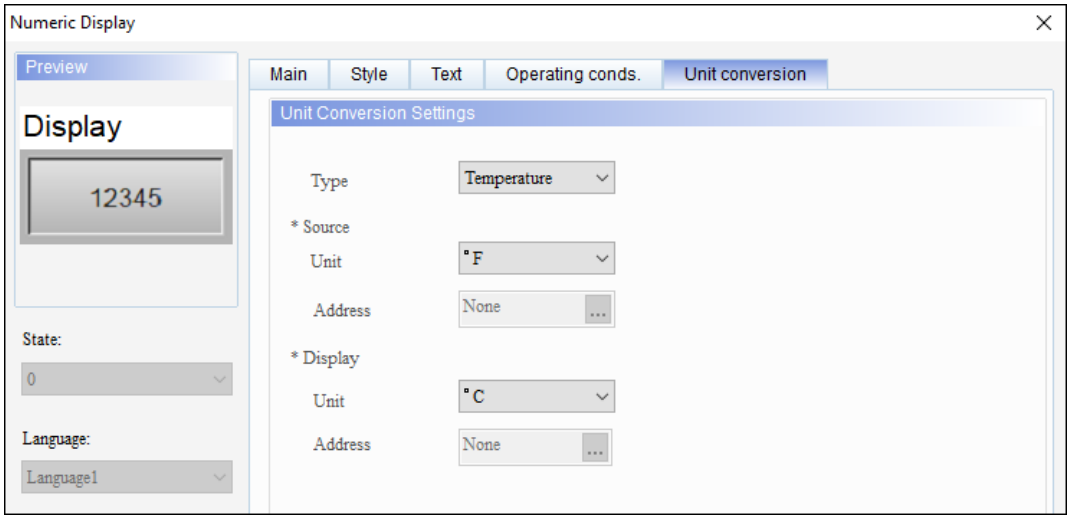
<table>
<tr><th>Type</th><th>Unit</th><th>Code</th></tr>
<tr><td rowspan="3">Speed</td><td>mm/sec</td><td>101</td></tr>
<tr><td>inch/sec</td><td>102</td></tr>
<tr><td>%</td><td>700</td></tr>
<tr><td rowspan="5">Pressure</td><td>kg/cm^2</td><td>201</td></tr>
<tr><td>bar</td><td>202</td></tr>
<tr><td>MPa</td><td>203</td></tr>
<tr><td>psi</td><td>204</td></tr>
<tr><td>%</td><td>700</td></tr>
<tr><td rowspan="3">Location</td><td>mm</td><td>301</td></tr>
<tr><td>inch</td><td>302</td></tr>
<tr><td>%</td><td>700</td></tr>
<tr><td rowspan="3">Temperature</td><td>°F</td><td>401</td></tr>
<tr><td>°C</td><td>402</td></tr>
<tr><td>%</td><td>700</td></tr>
<tr><td rowspan="5">Weight</td><td>T</td><td>501</td></tr>
<tr><td>kN</td><td>502</td></tr>
<tr><td>g</td><td>503</td></tr>
<tr><td>oz</td><td>504</td></tr>
<tr><td>%</td><td>700</td></tr>
<tr><td rowspan="4">Volume</td><td>L</td><td>601</td></tr>
<tr><td>ml</td><td>602</td></tr>
<tr><td>kL</td><td>603</td></tr>
<tr><td>%</td><td>700</td></tr>
</table>
Note: When selecting Using the code for Source and Display, do not use the same address.
</td></tr>
</table>

Legend	Setting	Description
❸	Custom formula	This setting is available only when the **Type** is set to **Custom formula**. The element value will be converted and displayed using the formula "**Display value = Syn value * A + B**" and the unit is Floating.
❹	Percentage	This setting is only available when the unit of the **Source** or **Display** is set to **%** or **Using the code**. The **Unit** synchronizes to the unit of the **Source**, but if the unit of the **Source** is selected as **%** or **Using the code**, you can select to convert all units of the type.

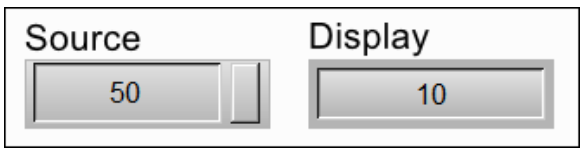
The following describes an example of using **Unit Conversion**.

- **Fixed Unit**

1. Create a **Numeric Display** element on the screen as **Source**, and then set its **Read Address** to **$10**.
2. Create a **Numeric Display** element on the screen as **Display**, and then set its **Read Address** to **$10**, and then configure unit conversion.

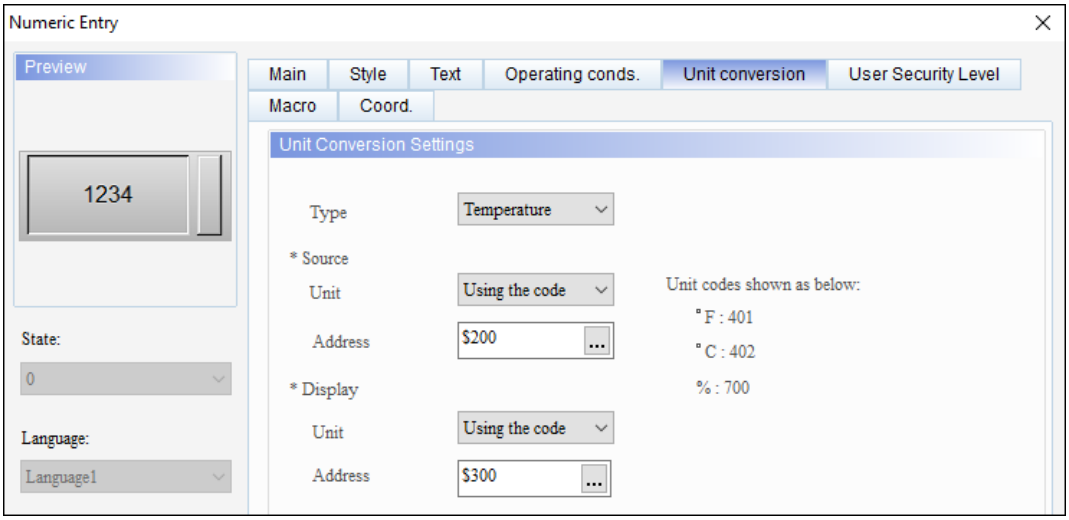


3. Perform an offline simulation, and then enter 50 (°F) for **Source**. **Display** will show 10 (°C).



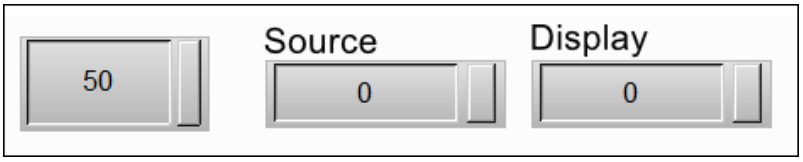
- **Using the code**

1. Create a **Numeric Entry** element on the screen, and then set the **Write Address** to $100 and configure the unit conversion.

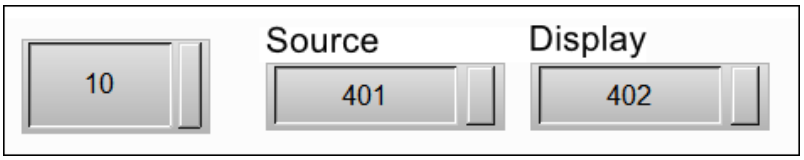


2. Create two more **Numeric Entry** elements, and then set the **Write Address** to **$200** and **$300** separately.

3. Perform an offline simulation, and then enter 50 for the **Numeric Entry** element of $100.



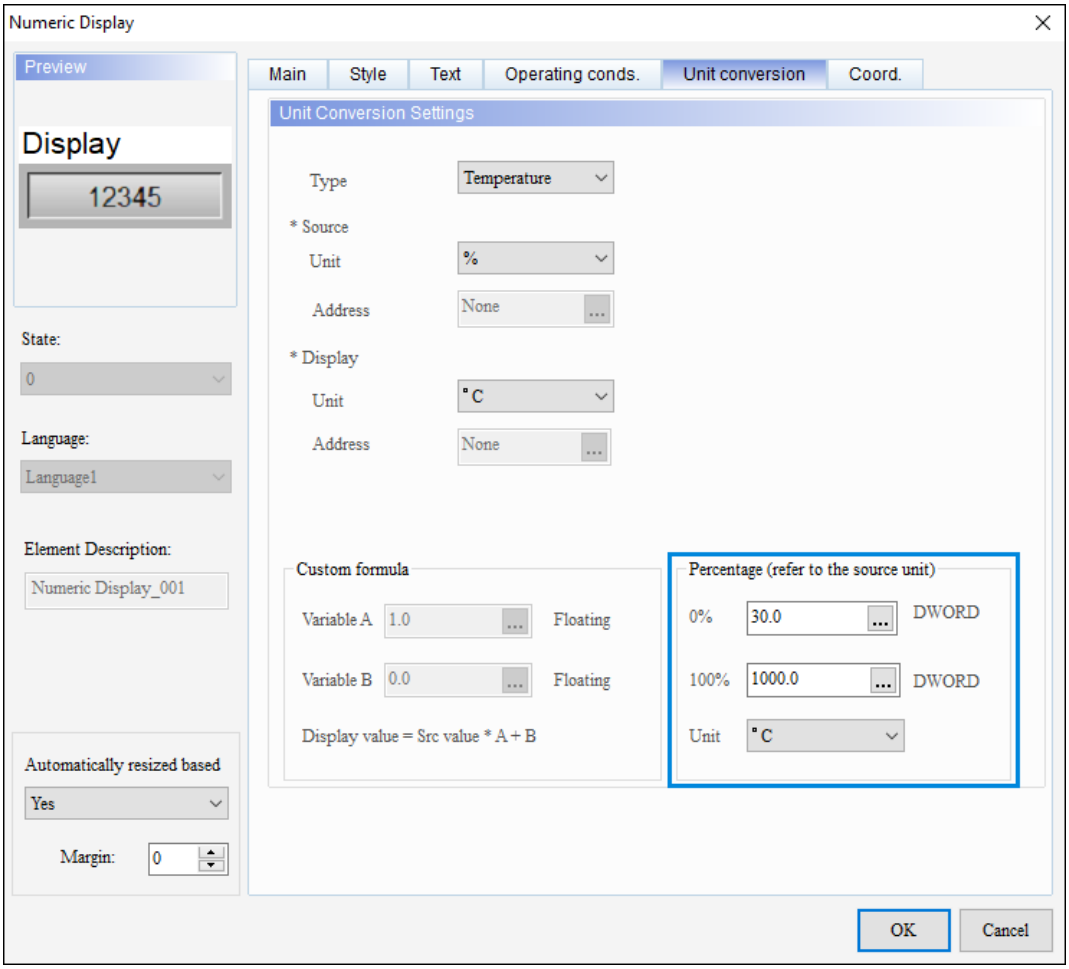
4. Enter 401 (°F) for **Source** and 402 (°C) for **Display**, and the **Numeric Entry** element of $100 will have its unit converted according to the code.



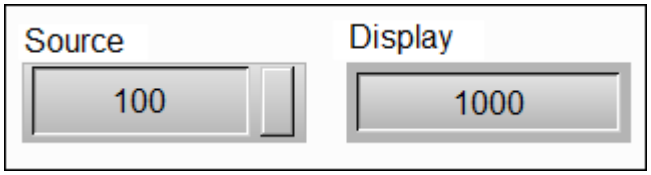
- **Percentage**

1. Create a **Numeric Entry** element on the screen as **Source**, and then set the **Write Address** to **$20**.

2. Create a **Numeric Display** element as **Display**, and then set the **Write Address** to **$20** and configure the unit conversion.



3. Perform an offline simulation, and then enter 100 for **Source**. **Display** will show the value 1000 corresponding to 100%.



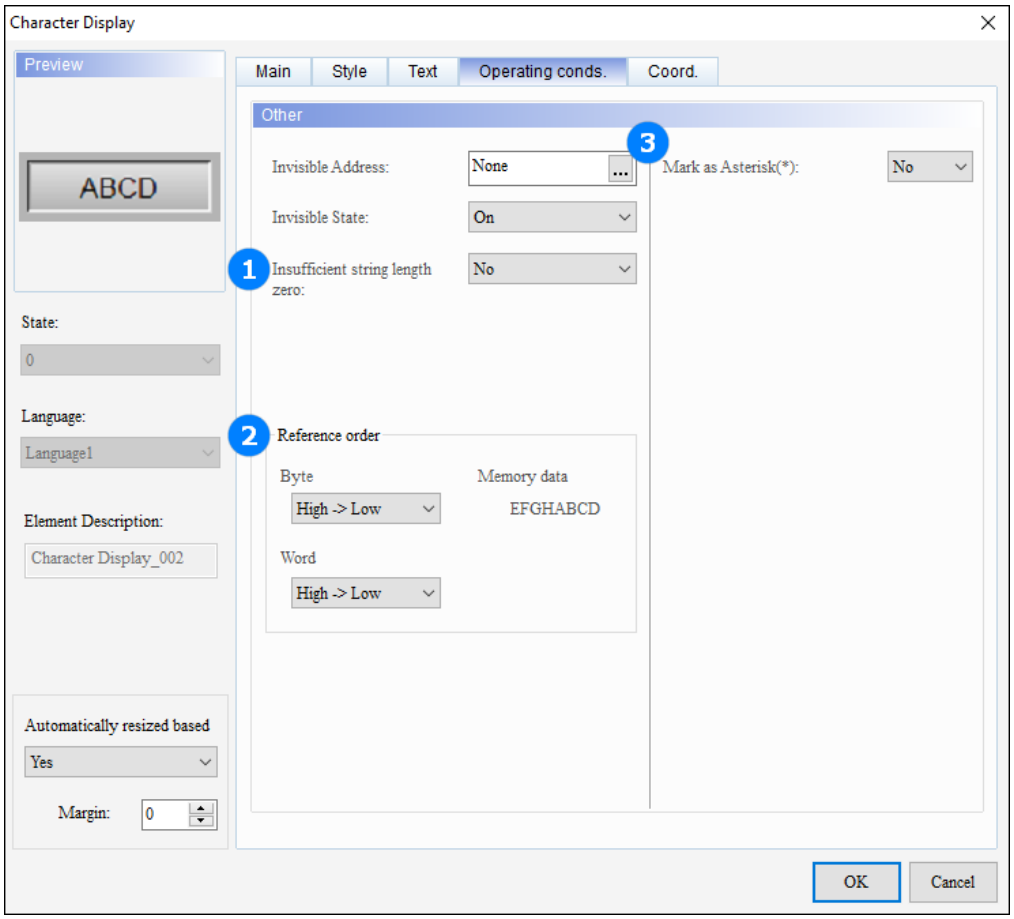
Character Display

The **Character Display** element reads the Byte data, and the data type of the **Numeric Entry** element is Word, which consists of double byte, so one Word represents two bytes. The value read by the **Character Display** element must perform a swapping of high and low bytes, and this result will be the actual display characters. For example, if $0 is 4241 (Hexadecimal), it is displayed as “AB” after the high and low bytes are swapped.

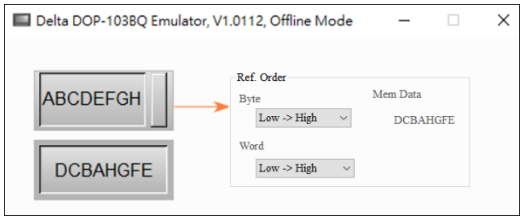
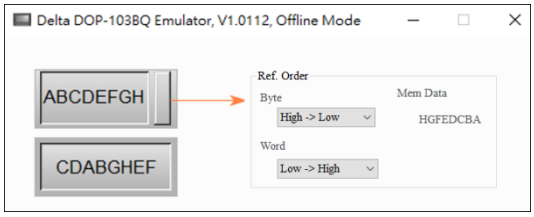
Unsigned Decimal	Hexadecimal	Character
65	41	A
66	42	B
67	43	C

Note: A **Character Display** element supports up to 256 characters.

- **Operating conds.**



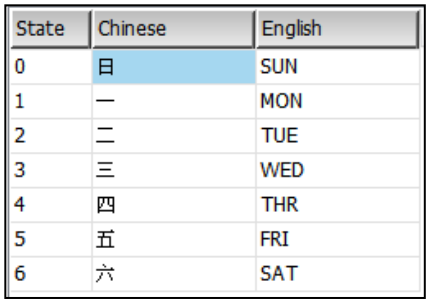
Legend	Setting	Description
1	Insufficient string length zero	Choose whether to fill in with zero when the string length is insufficient. When the length of the entered string is less than the set **String Length**, the remaining characters are filled in with zeros for display. For example, when the **String Length** is set to 10, but only 5 characters are entered, the following characters will be filled in with zeros.

<table>
<tr><th>Legend</th><th>Setting</th><th>Description</th></tr>
<tr><td>❷</td><td>Reference order</td><td>Select the display mode of Read Address. DIAScreen provides two swap modes: ByteSwap and WordSwap. Both can be used at the same time, but use ByteSwap first and then WordSwap.
The following is the display result of using two swap modes.
<table>
<tr><td>Reference order Byte is Low→High</td><td>Enter ABCDEFGH in the Character Entry element, and the result of writing to the memory after reading the swap by the Character Display element is DCBAHGFE.
 
</td></tr>
<tr><td>Reference order Word is High→Low</td><td>Enter ABCDEFGH in the Character Entry element, and the result of writing to the memory after reading the swap by the Character Display element is CDABGHEF.
 
</td></tr>
</table></td></tr>
<tr><td>❸</td><td>Mark as Asterisk(*)</td><td>Choose whether to display the * symbol. Select Yes and the read value will be displayed on the element with an * symbol.</td></tr>
</table>

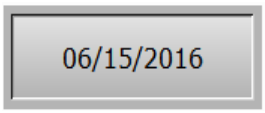
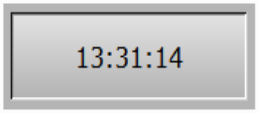
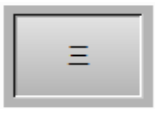
Date / Time / Week Display

The following table lists the description of the **Date Display**, **Time Display**, and **Week Display** elements.

Element	Description
Date Display	Select the date display format. The default value is **mm/dd/yy**.
Time Display	Select the time display format. The default value is **HH:MM:SS**.

Element	Description
Week Display	If multi-languages are configured, you can edit them in the **Text** tab of the element properties dialog to present the effects. 

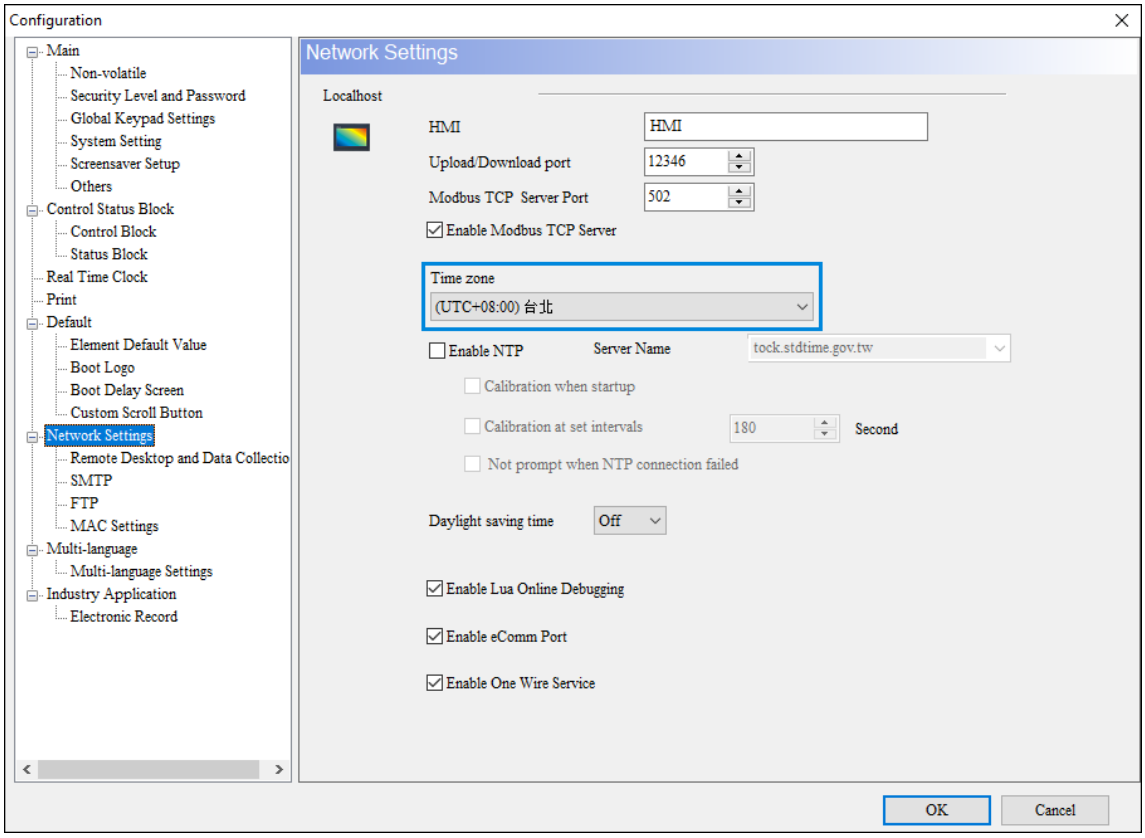
After compiling and downloading the screen to the HMI, the HMI screen is displayed as follows.

Date Display	Time Display	Week Display	
		Chinese	English
			

- **Main – Read Address**

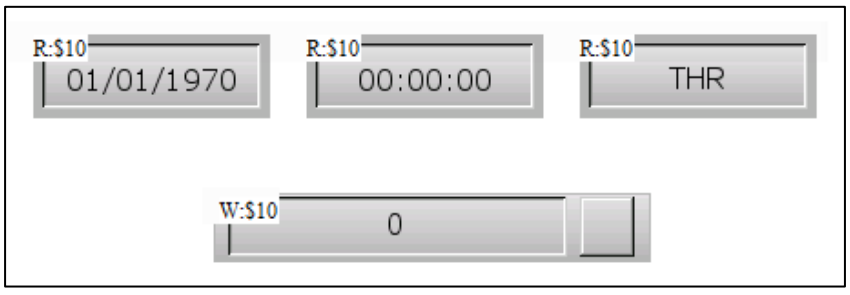
  If the **Read Address** is set to **None**, the system time of the HMI will be displayed; if a memory address is set, the corresponding time will be displayed as a timestamp. Timestamp has the following rules:

  - Start time: 1970/01/01
  - Standard type: in **seconds**
  - Time zone setting: Go to **General** > **Configuration** > **Network Settings** > **Time zone** to set the time zone of the HMI time.



The following describes an example of using the **Date Display** element to display timestamp.

1. Create **Date Display**, **Time Display**, and **Week Display** elements on the screen, and then set the **Read Address** to $10.
2. Create a **Numeric Entry** element, and then set the **Write Address** to $10.
3. Compile and download the screen to the HMI. Enter 0 for the **Numeric Entry** element, and the time displays as 01/01/1970 00:00 (GTM +00:00).



4. Enter 86400 for the **Numeric Entry** element, and the time displays as 01/02/1970 00:00 (GTM +00:00).

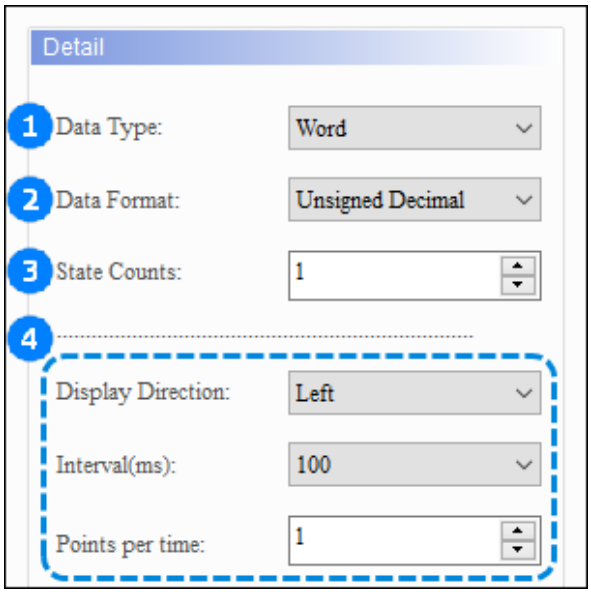
Note:

- The timestamp is in seconds. It is suggested to set an address that supports the **Quad Word** data type.
- For timestamp conversion, refer to epochconverter.com/.

Message Display

The following table lists the special properties of the **General Message Display** and **Moving Sign** elements with their description.

- **Main – Numeric Settings**

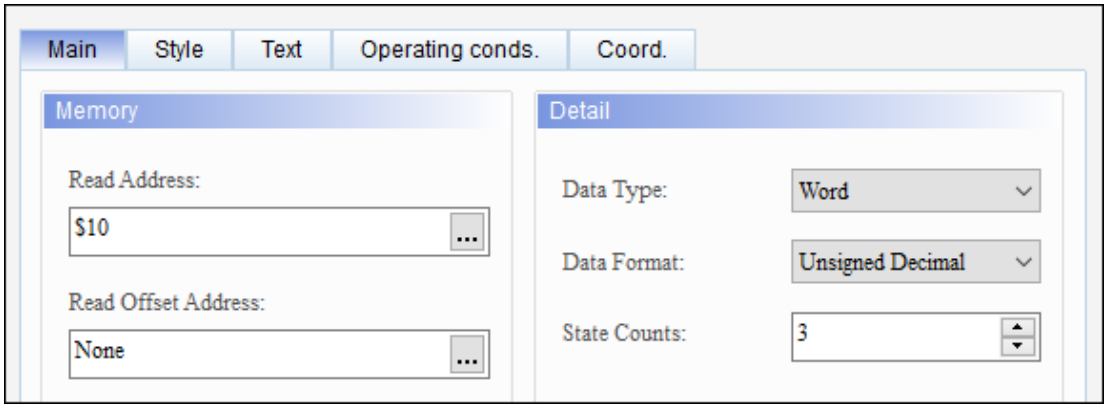


Legend	Setting	Description
1	Data Type	Select the data type for state switching.
2	Data Format	Select the data format. **Note**: This setting is only available when the **Data Type** is Word.
3	State Counts	Select or enter the state counts according to the selected **Data Type**. • Bit: 2 states can be set. • Word: 1~1000 states can be set for the **General Message Display** element; 1~256 states can be set for the **Moving Sign** element. • LSB: 16 states can be set. • LSB (Support State 0): 17 states can be set.

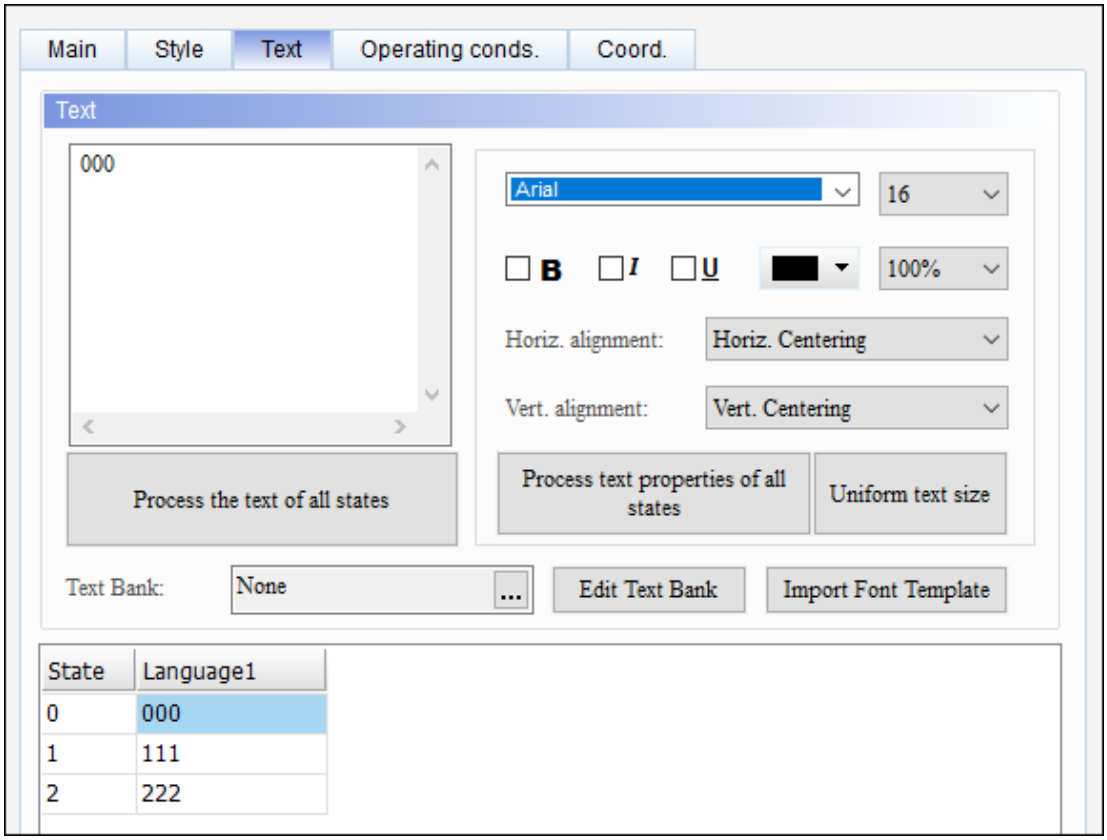
Legend	Setting	Description
		Note: L LSB means that the register data is first converted into binary data, and then the current object state is defined by using the lowest non-zero bit. For example, decimal value 3 converted to binary is 0000000000000011, and the lowest non-zero bit is Bit 0, State is 1.
❹	Text Display Settings	• Select the **Display Direction** of the text on the element. • The **Moving Sign** element updates the display according to the selected **Interval** and **Points per time**. 1~50 pixels can be set for **Points per time**. **Note**: These settings only apply to the **Moving Sign** element.

The following describes an example of using the **Message Display** element.

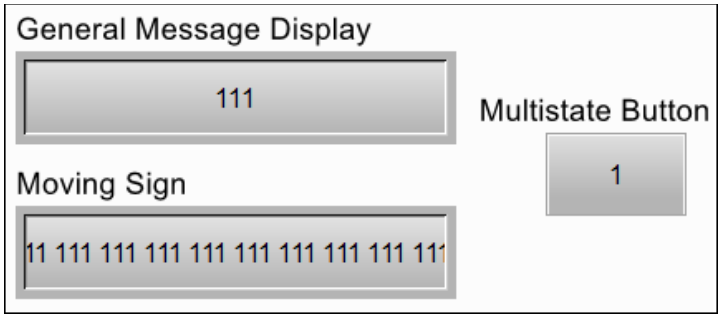
1. Create **General Message Display**, **Moving Sign**, and **Multistate** elements on the screen, and then set the **Read Address** to **$10**, the **Data Type** to **Word**, and the **State Counts** to **3**.



2. Enter the message of states 0~2 in the **Text** tab.



3. Perform an offline simulation, and then click the **Multistate** button. The **Message Display** elements will display the corresponding text according to the state.



QR Code Display

QR code (Quick Response Code) is a square two-dimensional barcode (matrix barcode) comprising only black and white colors. There are three position markers (a square in a grid) on the three corners of the QR code, and another position marker is at a position near the lower right corner. These markers are for identification and positioning, so there is no need to carefully align the code while scanning. The scanning software decodes the code correctly regardless of angle and direction. The data types stored in a QR code include numbers, letters, binary data, and Kanji

Use the **QR code display** element to convert the string into a QR code and display it on the HMI, making it easy to be scanned to obtain the string or website address.

The following table lists the special properties of the **QR code display** element with their description.

- **Main – Detail**

<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>String Length</td><td>Select or enter the string length, and up to 256 characters can be entered.</td></tr>
<tr><td>Error correction</td><td>Select the error correction level. The available options are LOW, MEDIUM, QUARTILE, and HIGH, based on the error tolerance capabilities.
The higher the error correction level, the less data can be stored. In other words, if there are graphics or pictures in the middle of the QR code, when the graphics or pictures are large, a higher error correction level is required to prevent data errors; If there are no graphics or pictures in the middle, select LOW for error correction.
The data restoration rate for each error level is as follows.
<table>
<tr><th>Error Correction Level</th><th>Data Restoration Rate</th></tr>
<tr><td>LOW</td><td>7%</td></tr>
<tr><td>MEDIUM</td><td>15%</td></tr>
<tr><td>QUARTILE</td><td>25%</td></tr>
<tr><td>HIGH</td><td>30%</td></tr>
</table>
</td></tr>
</table>

The following describes an example of using the **QR Code Display** element.

1. Create **QR Code Display** and **Character Display** elements on the screen, and then set the **Read Address** and **Write Address** to **$100** and the **String Length** to **30**.
2. Perform an offline simulation, and then enter www.deltaww.com for the **Character Entry** element, and you can use your mobile phone to scan the QR code to enter the Delta website.

Barcode Code

The following table lists the special properties of the **Barcode Code** element with their description.

- **Main – Settings**

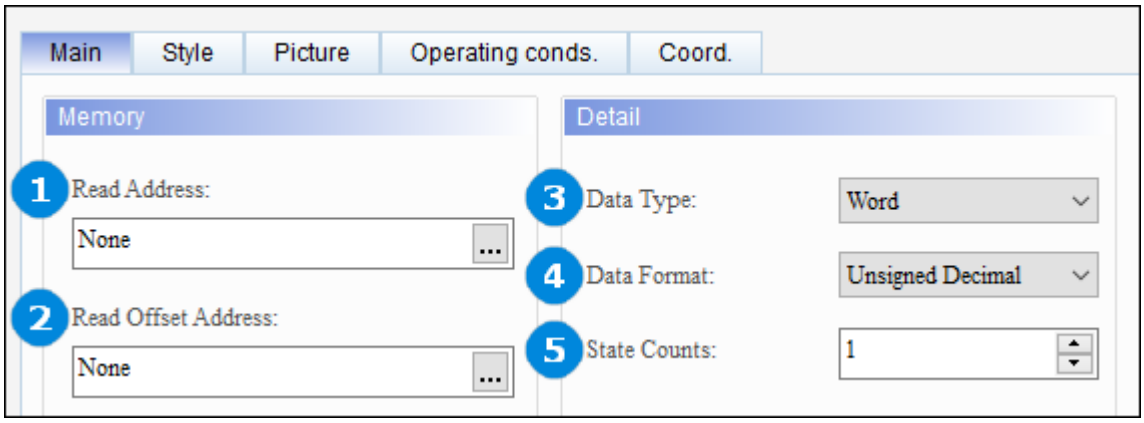
Setting	Description
Encoded Mode	The available options are EAN13 and CODE128. EAN13 CODE128
Encoded Length	Select or enter the encoded length. • If select EAN13, the length supports up to 12. • If select CODE128, the length supports up to 48.
Encoded Text Size	Select the display size of the number on the Barcode.

Graphic

The **Graphic element** is used to display graphics in different formats, and the appearance of the graphic display can be controlled through the memory address.

Common Properties

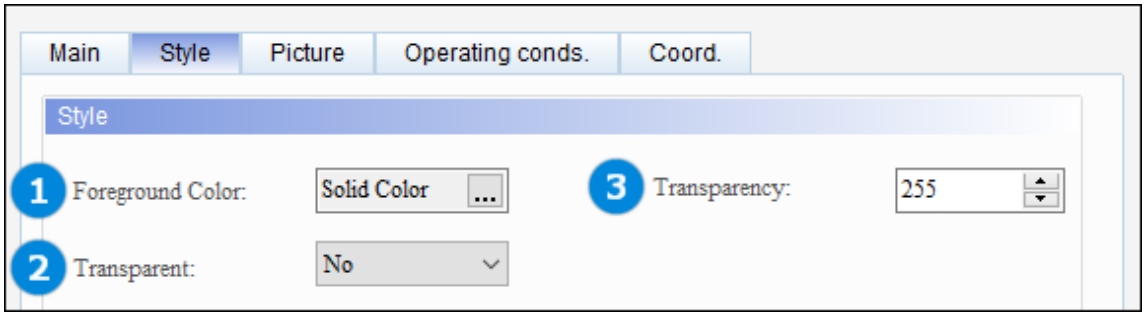
- **Main**



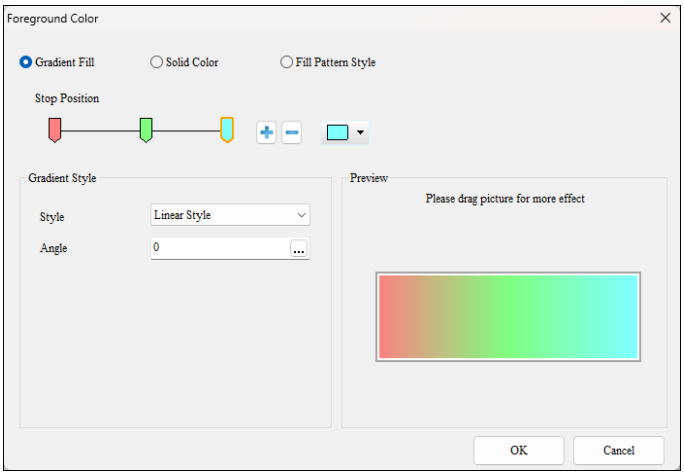
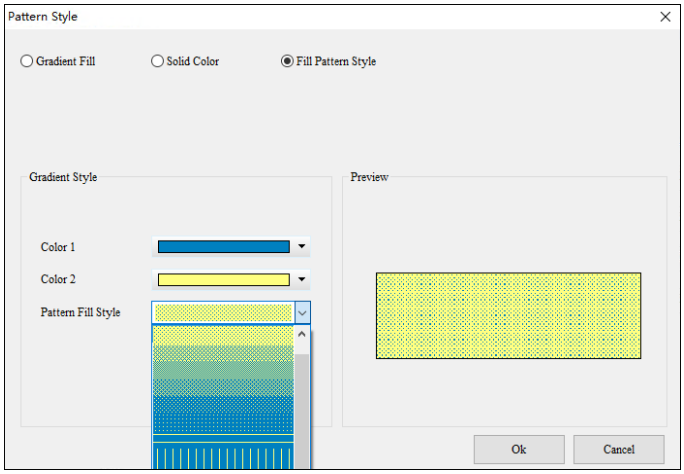
Legend	Setting	Description
1	Read Address	You can select the **Internal Memory** address or controller register address, and the **Input** memory **Type** can only be **Word**.
2	Read Offset Address	The HMI will configure the value to the corresponding read address according to the set Offset Address. After setting the offset address, the actual read address is as follows. The actual read address of the element: Read Address + Value in Read Offset Address × Element [Data Type]
3	Data Type	Select the data type for state switching.
4	Data Format	Select the data format. **Note**: This setting is only available when the **Data Type** is Word.

Legend	Setting	Description
⑤	State Counts	Select or enter the state counts according to the selected **Data Type**. • Bit: 2 states can be set. • Word: 1~256 states can be set. • LSB: 16 states can be set. • LSB (Support State 0): 17 states can be set. **Note**: LSB means that the register data is first converted into binary data, and then the current object state is defined by using the lowest non-zero bit. For example, decimal value 3 converted to binary is 0000000000000011, and the lowest non-zero bit is Bit 0, State is 1.

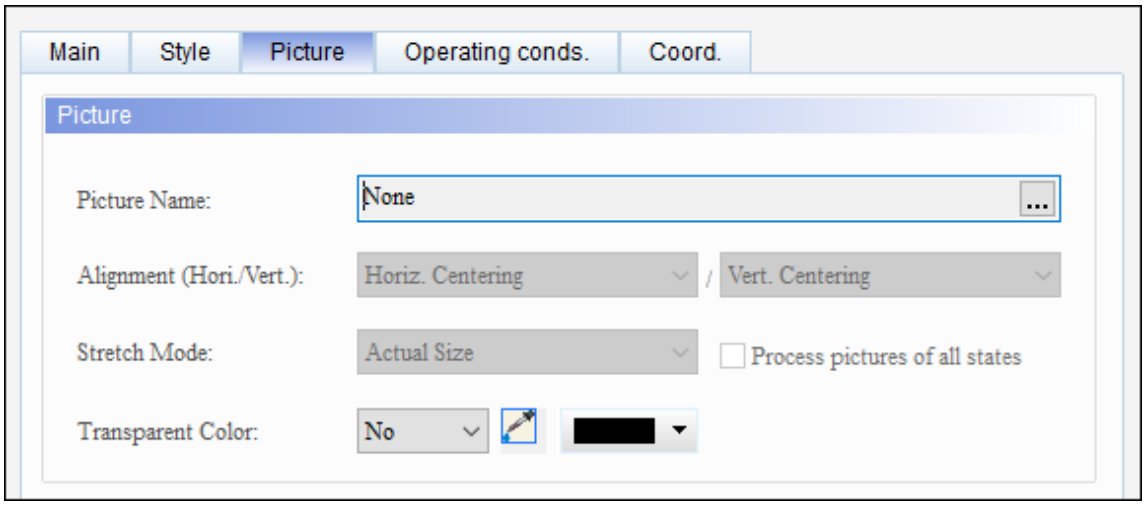
- **Style**



After selecting a state, modify the following settings and apply them to the state.

Legend	Setting	Description
❶	Foreground Color	Select the color of the element. • **Gradient Fill:** Set the gradient style by setting the current angle or center point. You can drag the mouse on the preview image to set the current angle or center point, or set the memory address to dynamically configure it on the HMI. You can also set the stop point colors to present the effect of multi-color gradients.  **Note**: Up to 5 stop points can be set. • **Solid Color**: Select a single-color fill. • **Fill Pattern Style**: Select a two-color combination configuration and select a pattern fill style. 
❷	Transparent	Choose whether the element displays the transparency.
❸	Transparency	Select or enter the transparency value of the element. The range value is 50–255, and the default value is **255**. The smaller the value, the higher the transparency of the element.

- **Picture**



Note: This tab setting only applies to the **State Graphic** and **GIF Viewer** elements.

Setting	Description
Picture Name	Enter the picture name. The default value is **None**.To set the picture display, click ... to select the picture to display in the picture bank and drag it to each state icon. To configure other pictures, refer to Picture Bank.
Alignment	Select the alignment of the picture.

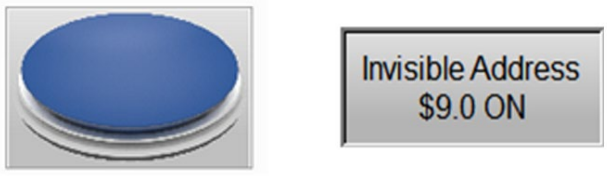
<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Stretch Mode</td><td>Select the stretch mode of the picture.
<table>
<tr><th>Stretch All</th><th>Stretch 1:1</th><th>Actual Size</th></tr>
<tr><td>The picture fills the full element display area.
 </td><td>The picture displays in 1:1 size based on the element width and length.
 </td><td>The picture is displayed in the display area of the element in its actual size.
 </td></tr>
</table></td></tr>
<tr><td>Process pictures of all states</td><td>If an element has multiple state values, and its picture may not fill the full display area of the element. Select this checkbox to process all pictures settings</td></tr>
<tr><td>Transparent Color</td><td>Choose whether to specify a color in the picture as a transparent color. If set to Yes, you can select colors or use the color picker to select one of the colors on the picture to specify it as a transparent color, that is, the foreground color. The following figures show the effect of specifying white as a transparent color.
</td></tr>
</table>

- **Operating conds.**

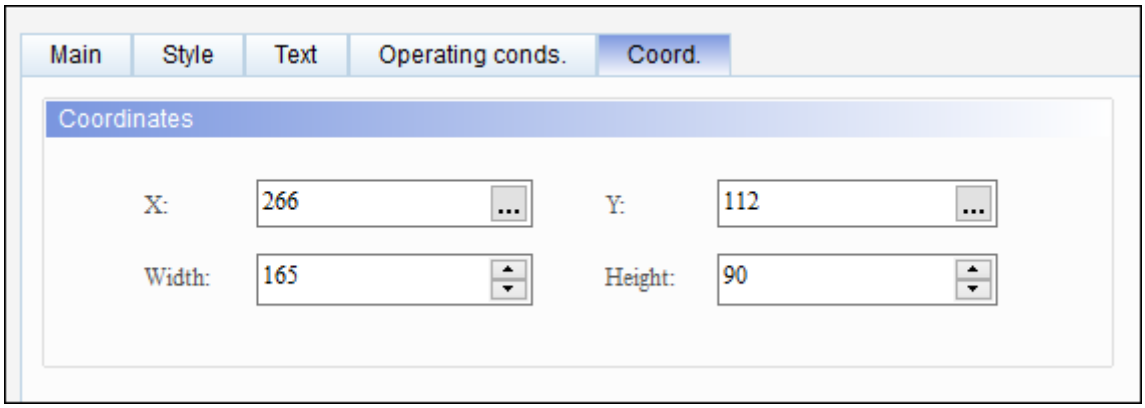
  The **Invisible Address** displays or hides the element on the HMI. If the **Invisible State** is set to **On**, the element will be hidden when the **Invisible Address** is ON. Instead, if the **Invisible State** is set to **Off**, the element will be hidden when the **Invisible Address** is OFF.

  The following is an example of using invisible address.

  1. Create a button element on the screen and set its **Read Address** to $9.0, then set the **Invisible Address** of the element to hide to $9.0.
  2. To hide the element, you must first click the $9.0 button.

Invisible State is On	
Invisible State is Off	

- **Coord.**



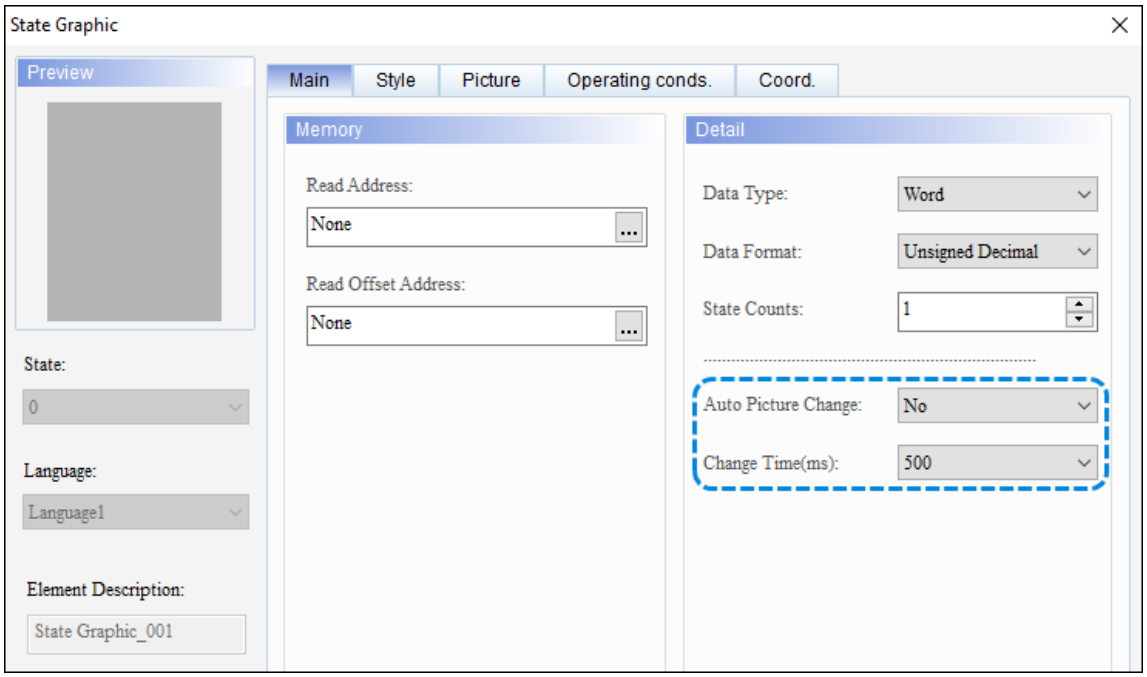
Set the XY coordinates, width, and height on the upper left of the element. You can enter a constant as fixed position or memory address to dynamically configure the position of the element on the HMI.

State Graphic

Use the **State Graphic** element to set multiple states of pictures and display the corresponding state graphic through the value read by the **Read Address**.

The following table lists the special properties of the **State Graphic** element with their description.

- **Main**



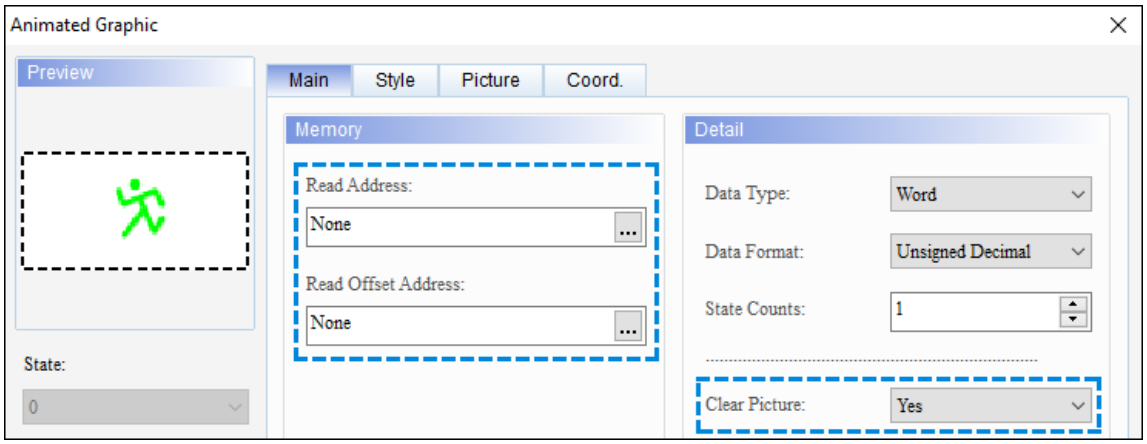
Setting	Description
Auto Picture Change	Choose whether to change the picture automatically. • If select **No**, the corresponding state graphic will be displayed according to the value of the **Read Address**. • If select **Yes**, the state graphic will be changed according to **Auto Picture Change** in order when the value of the **Read Address** is not 0. • If select **Variation**, the **Read Address+1** will be used to control whether to automatically change pictures.
Change Time	Select the picture change time, the unit is in millimeters.

Animated Graphic

Use the **Animated Graphic** element to set multiple state graphics or import GIF files to display the animation effects.

The following table lists the special properties of the **Animated Graphic** element with their description.

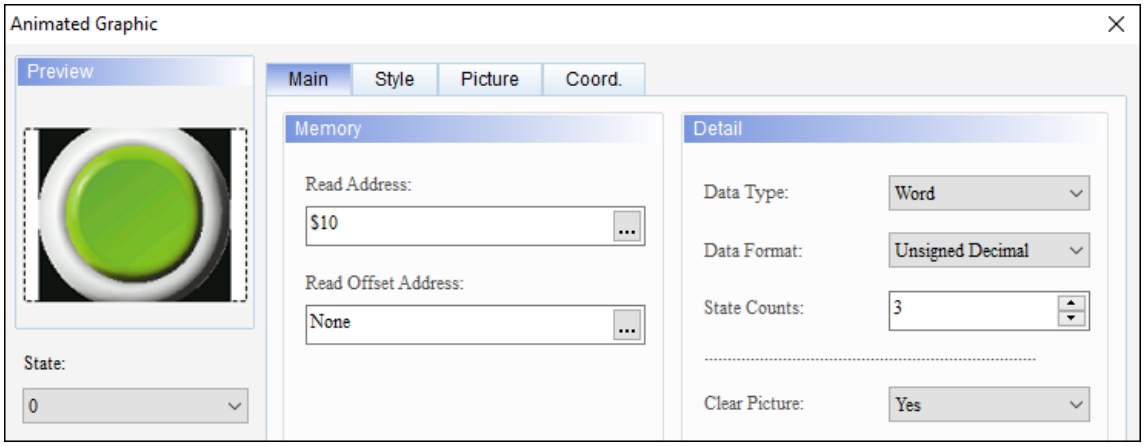
- **Main**



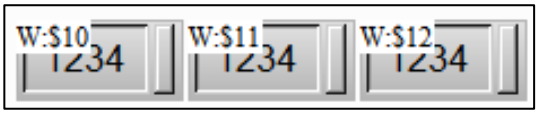
Setting	Description
Read Address	The **Animated Graphic** element switches according to the value of the **Read Address**. • **Read Address+1** is the position where the horizontal axis (X) of the **Animated Graphic** element moves. • **Read Address+2** is the position where the vertical axis (Y) of the **Animated Graphic** element moves.
Clear Picture	Choose whether to clear the pictures. • If select **Yes**, the picture of the previous state will be cleared. • If select **No**, the HMI will remain the pictures of all states.

The following steps describe how to use the **Animated Graphic** element.

1. Create an **Animated Graphic** element on the screen, then set the **Read Address** to **$10** and the **State Counts** to **3**, and configure pictures for three states.

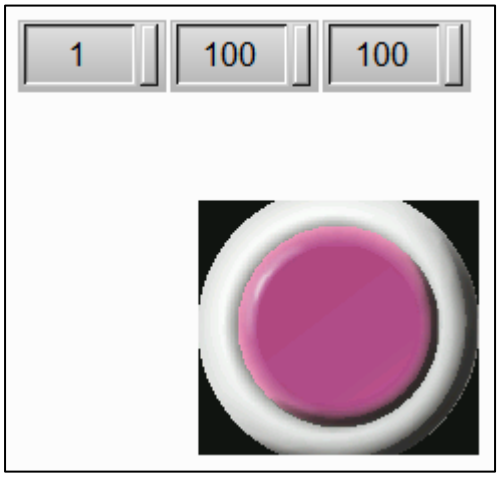


2. Create 3 more **Numeric Entry** elements, and then set the **Write Address** to **$10**, **$11**, and **$12** separately.

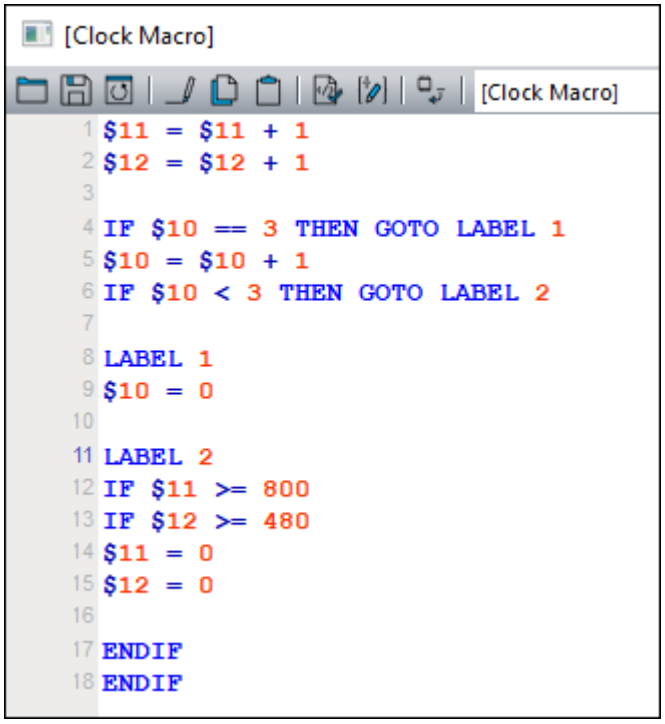


The execution result is as follows.

Enter 1 for the **Numeric Entry** element of $10, and the **Animated Graphic** element displays the style of the state 1; enter 100 for the **Numeric Entry** element of $11 and $12, and the **Animated Graphic** element is displayed at the XY position of (100,100).



You can go to **Project** > **Clock Macro** to update the address for the fixed time to display the animation effect. The Clock Macro is written as follows.



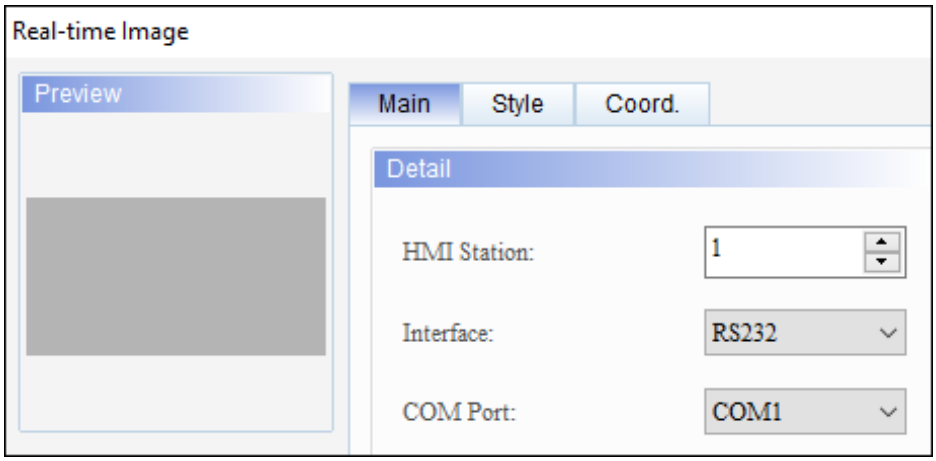
Note: Go to **General** > **Configuration** > **Main** to set the **Clock Macro Delay Time**.

Real-time Image

DIAScreen provides a transfer tool, TestTransfer, which can transfer the images from the computer to the HMI and display them on the **Real-time Image** element.

The following table lists the special properties of the **Real-time Image** element with their description.

- **Main**



Setting	Description
HMI Station	Select or enter the station number provided for RS485. The range value is 1–255.
Interface	Select the interface to connect the HMI via the COM port.
COM Port	Select the COM port. The available options are COM and LAN.

The following steps describe how to use the **Real-time** element.

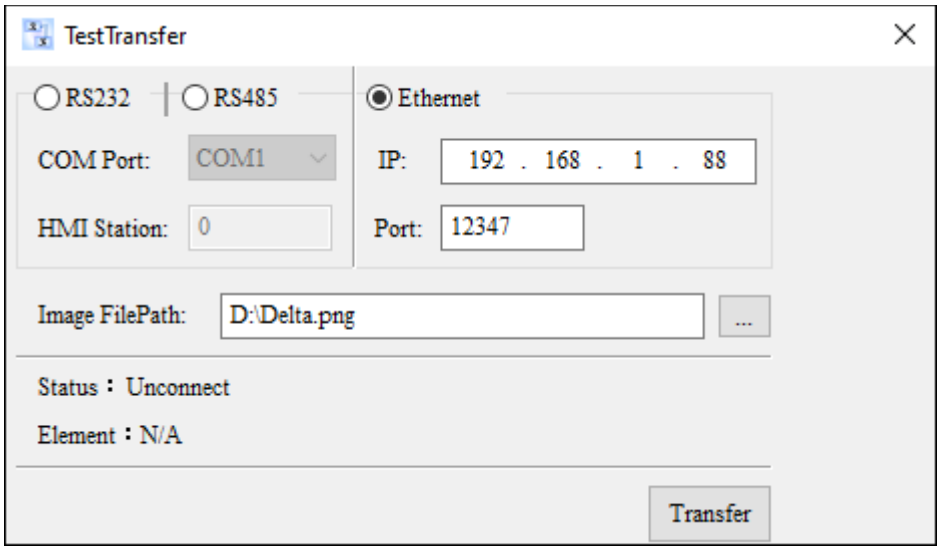
1. Create a **Real-time** element on the screen and set **LAN** for the **Interface**, and then download the screen to the HMI.
2. Enable **TestTransfer.exe** under the installation directory of DIAScreen.

   The default installation path: `C:\Program Files (x86)\Delta Industrial Automation\DIAStudio\DIAScreen (version)\Utility\ImgTrans\TestTransfer.exe`

The execution result is as follows.

In the **TestTransfer** dialog, select **Ethernet** and enter the **IP** of the HMI.

In the **Image FilePaht** field, select the path of the image to transfer, and then click **Transfer**.

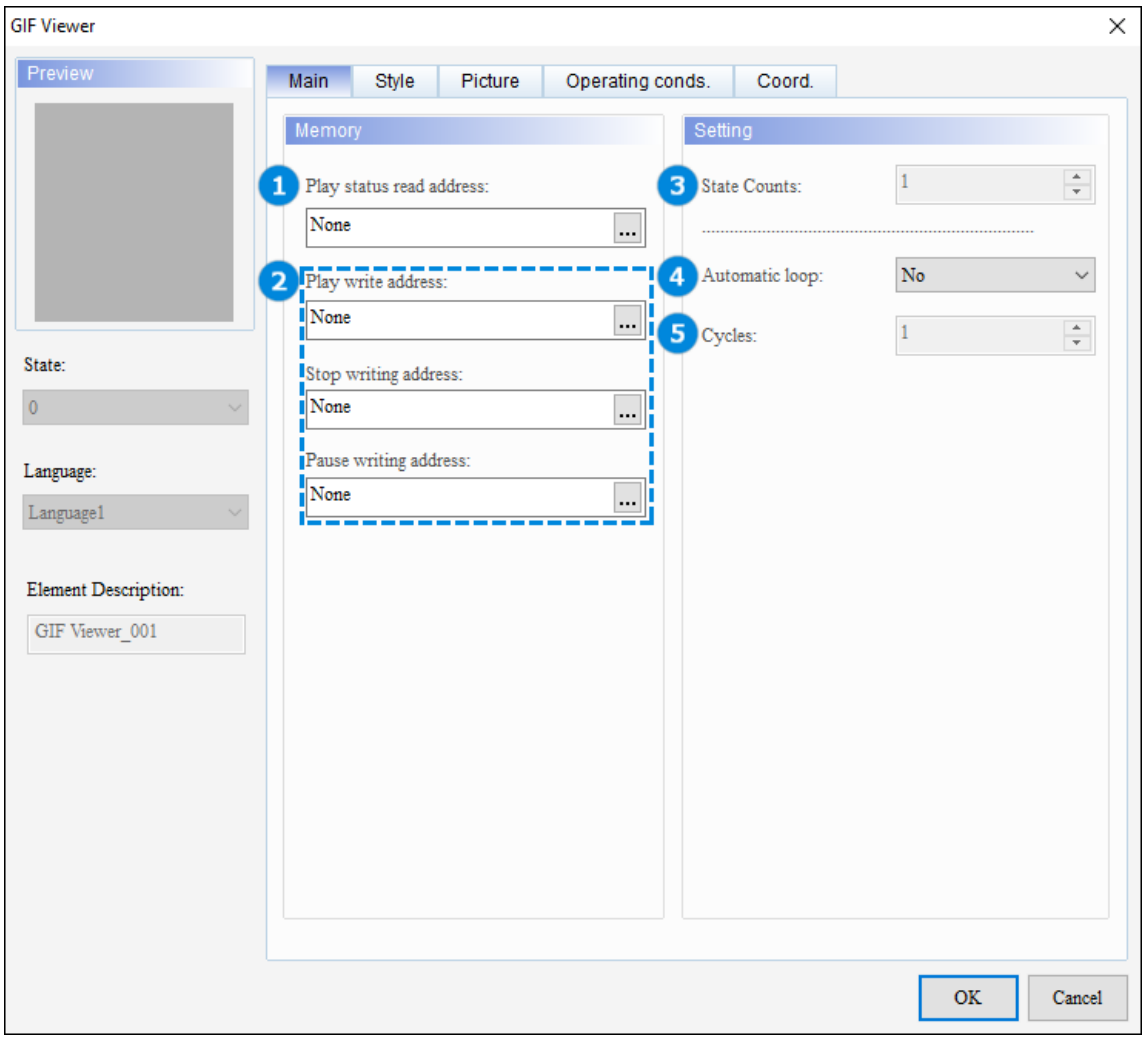


The **Real-time Image** element on the HMI displays the transferred image.

GIF Viewer

The following table lists the special properties of the **GIF Viewer** element with their description.

- **Main**



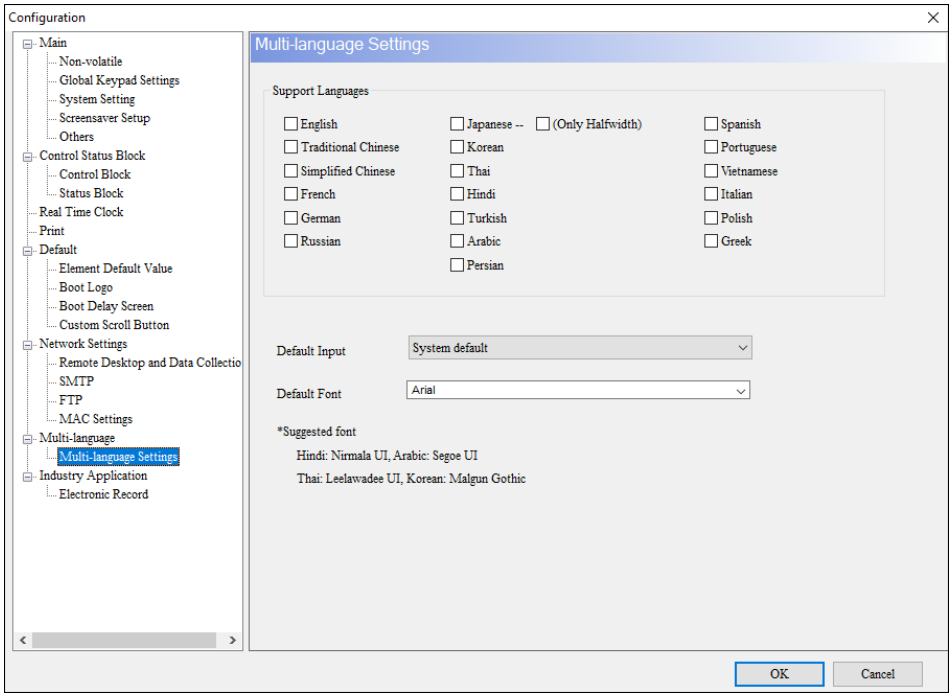
Legend	Setting	Description
1	Play status read address	Set the address to read the GIF status. • 0: Stop playing • 1: Pause • 2: Playing
2	Control Address	Set the control address for GIF execution. When the control address on the HMI is On, the play, stop, and pause actions will be performed. **Note**: The control address has no effect when the **Automatic loop** is set to **Yes**.

Legend	Setting	Description
❸	State Counts	This setting is only available when the **Automatic loop** is set to **Yes**. The range value is 1–256.
❹	Automatic loop	Choose whether to loop automatically. If select **No**, the GIF action can be controlled through the control address.
❺	Cycles	This setting is only available when the **Automatic loop** is set to **Yes**. The range value is 1–200.

Input

Use the **Input** element to write values to the memory address or read the memory address to display values.

The following table lists the description of each **Input** element.

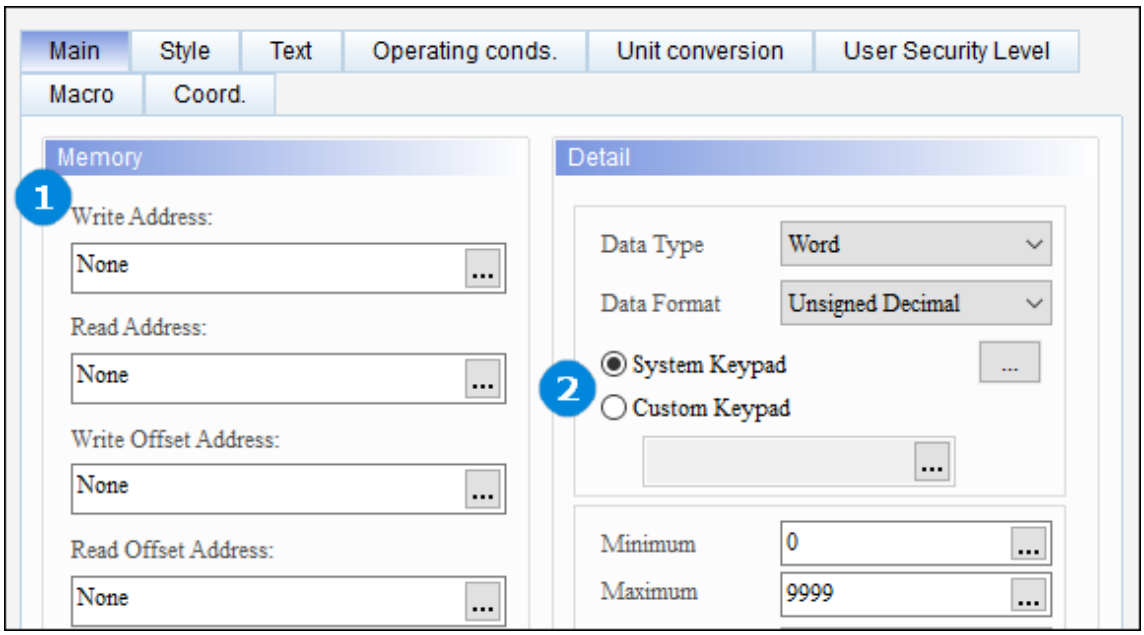
Element	Description
Numeric Entry	Uses the keypad to write a value to the memory address.
Character Entry	Uses an ASCII keypad to write ASCII code characters to the memory address.
Barcode Input	Uses an external input device to scan the barcode and write the read characters to the memory address in ASCII code. **Note**: Refer to Device Communication for the Barcode Reader setting.
Multi-language Input	Go to **General** > **Configuration** > **Multi-language Settings** and select the input keypad to use on the HMI.  **Note**: The Multi-language Input does not support **On-line Simulation** and **Off-line Simulation**.
Multi-line Text Input	Used to view and edit the text files and G-code files. Supports multi-language display.

Common Properties

If **Automatically resized based** is set to **Yes**, when the set text length exceeds the element size, the element is automatically resized according to the **Margin** to show the complete text.

Note: This setting supports the element enlargement function, the string is shortened, and the element will not become smaller.

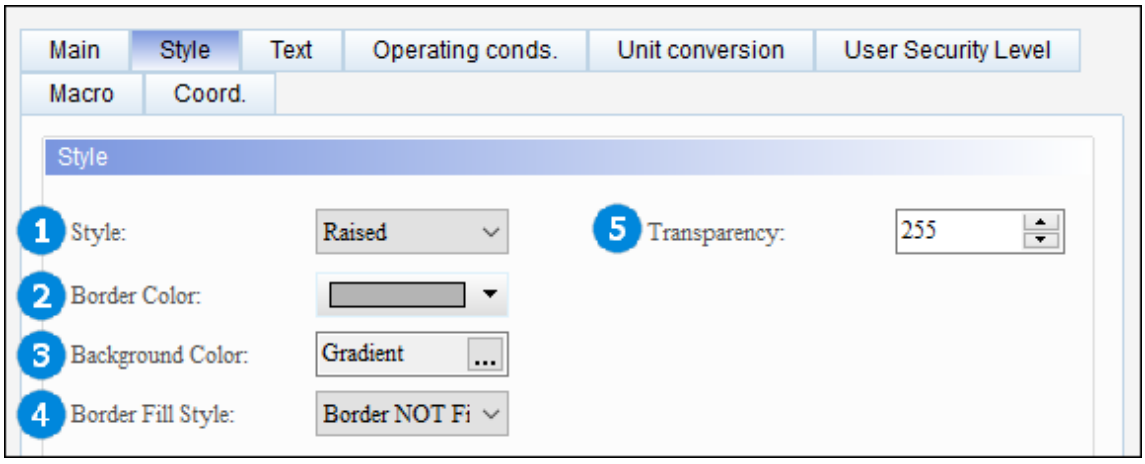
- **Main**



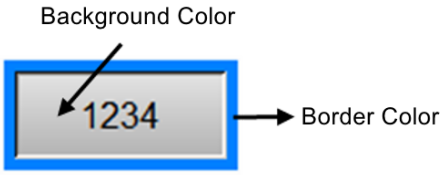
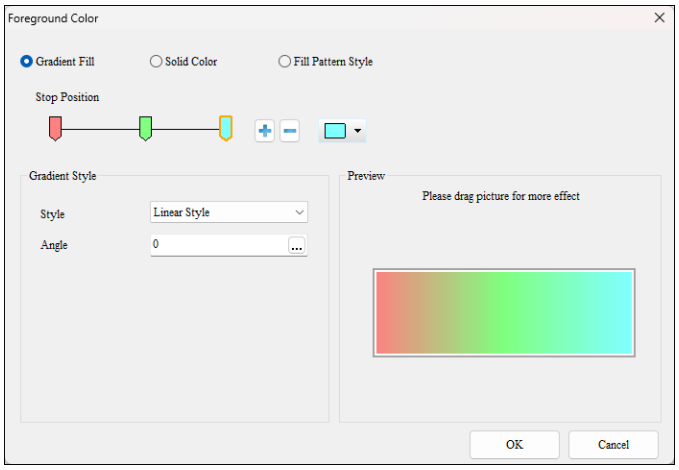
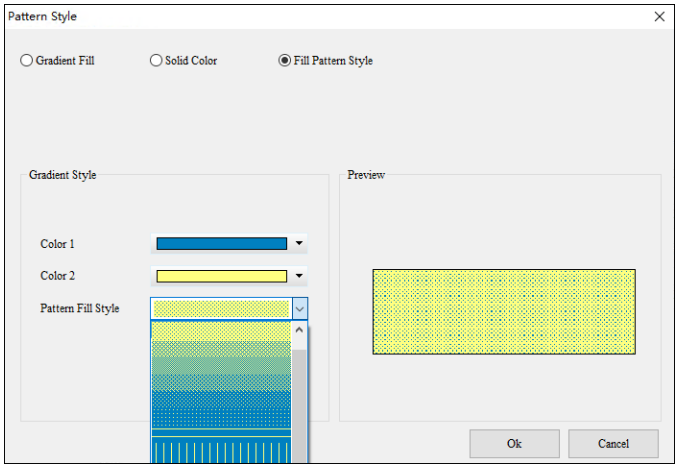
Legend	Setting	Description
1	Write / Read Address	You can select the **Internal Memory** address or controller register address, and the **Input** memory **Type** can only be **Word**. **Note**: If the write memory address is set but the read memory address is not set, the HMI will regard the read memory address and the write memory address as the same.
	Write / Read Offset Address	The HMI will configure the value to the corresponding read address according to the set Offset Address. After setting the offset address, the actual read address is as follows. • The actual write address of the element: Write Address + Value in Write Offset Address × Element [Data Type]

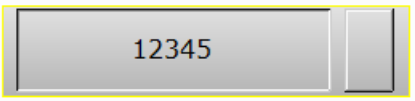
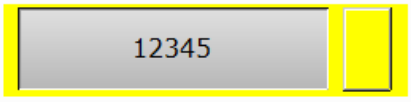
Legend	Setting	Description
		• The actual read address of the element: Read Address + Value in Read Offset Address × Element [Data Type] **Note**: The **Offset Length** can be selected in the **Numeric Input** element.
②	Keypad	elect the keypad type. • **System Keypad**: Open the **System Keypad Layout** dialog to edit the keypad style to display on the HMI. Click to resize the keypad and double-click the text area on the keypad to modify the text. You can also adjust the text size, font, color, and background color with the toolbar. **Note**: The **Numeric Entry** element uses keypads with different number systems depending on the **Data Format**. • **Custom Keypad**: Select the created keypad screen.

- **Style**

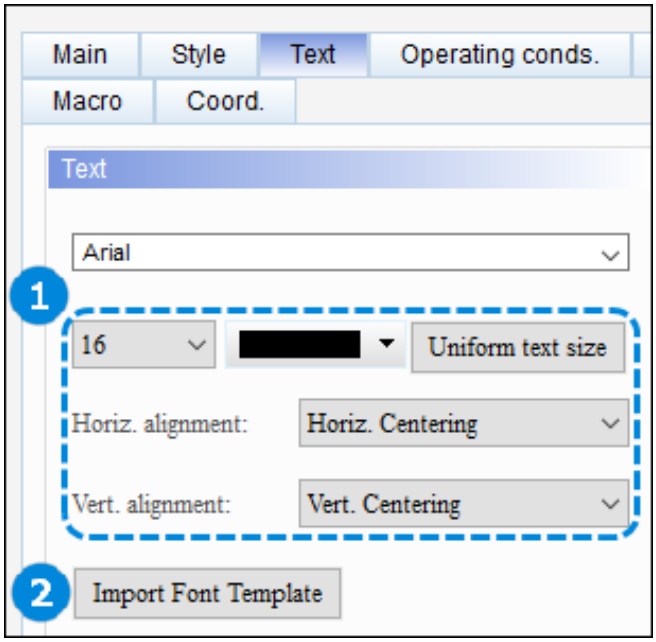


Legend	Setting	Description
①	Style	Select the appearance of the element. The default value is **Raised**. **Standard** **Raised** **Sunken** **Transparent** 1234 1234 1234 1234

Legend	Setting	Description
❷	Border Color	Select the display color of the border. 
❸	Background Color	Select the color of the element. • **Gradient Fill:** Set the gradient style by setting the current angle or center point. You can drag the mouse on the preview image to set the current angle or center point, or set the memory address to dynamically configure it on the HMI. You can also set the stop point colors to present the effect of multi-color gradients.  **Note**: Up to 5 stop points can be set. • **Solid Color**: Select a single-color fill. • **Fill Pattern Style**: Select a two-color combination configuration and select a pattern fill style. 

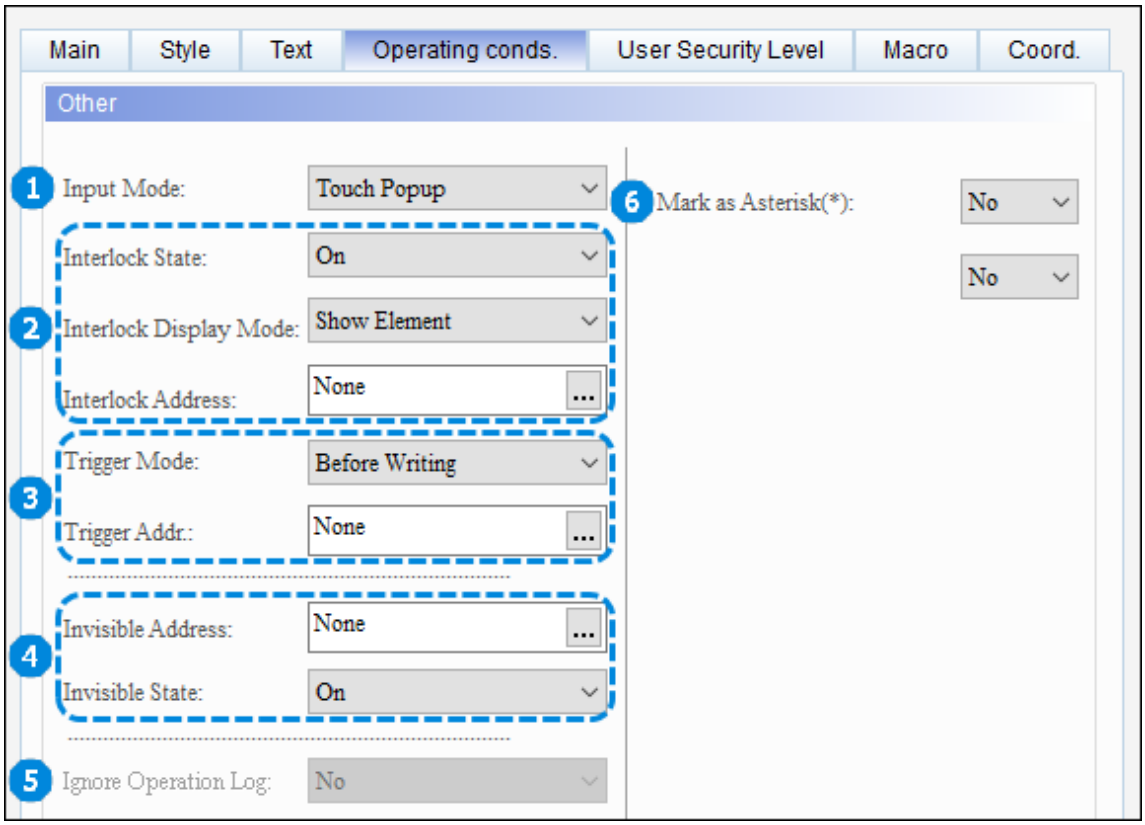
Legend	Setting	Description
4	Border Fill Style	Select the fill style of the element. **Border NOT Fill**  **Border Fill** 
5	Transparency	Select or enter the transparency of the element. The range value is 50–255 and the default value is **255**. The smaller the value, the higher the transparency of the element.

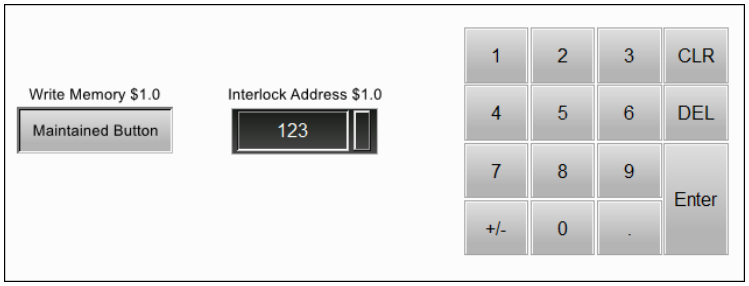
- **Text**

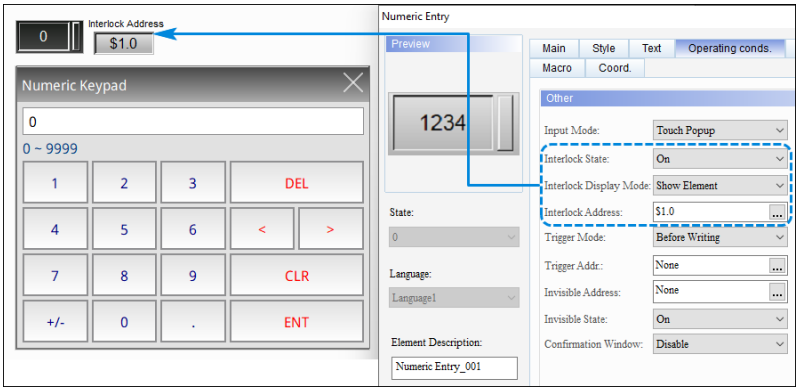


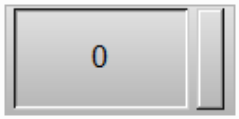
Legend	Setting	Description
1	Text Property	Set the text properties.
2	Import Font Template	Click to open the **Font Template** dialog, and then select the template and import the text properties of the selected state.

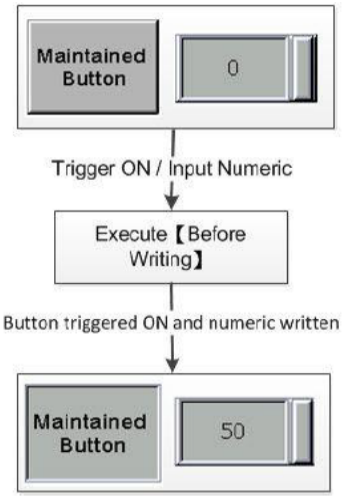
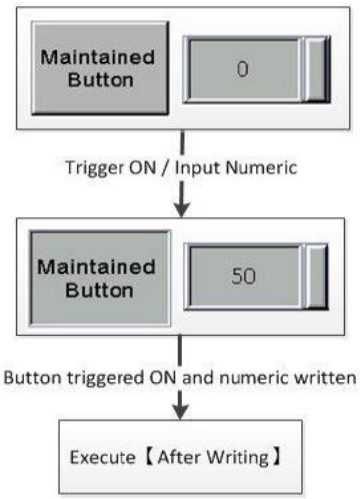
- **Operating conds.**

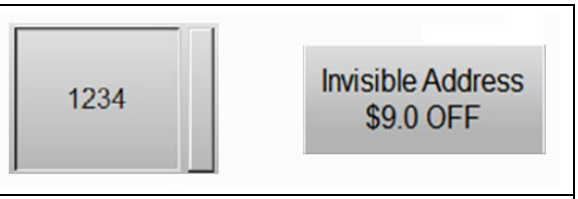


Legend	Setting	Description
1	Input Mode	Select the input mode. • **Touch Popup**: After pressing the input element on the HMI, the keypad dialog will pop up. • **Active Non-Popup**: This mode must be used with **Interlock Address** to enable the input. After the Interlock Address is triggered, the input element will flash, and you can enter the value through the keypad.  • **Touch Non-Popup**: After pressing the input element on the HMI, the element will flash, and you can enter the value through the keypad.

<table>
<tr><th>Legend</th><th>Setting</th><th>Description</th></tr>
<tr><td rowspan="3">❷</td><td>Interlock State</td><td>The Interlock State enables to operation of a certain element and must be used with Interlock Address. If the Interlock State is set to Off, the Interlock Address enables to operation of a certain element. Instead, if the Interlock State is set to On, the Interlock Address enables to operation of a certain element. As shown in the following figure, when the Interlock Address $1.0 is On, the Numeric Entry element can be operated.
 

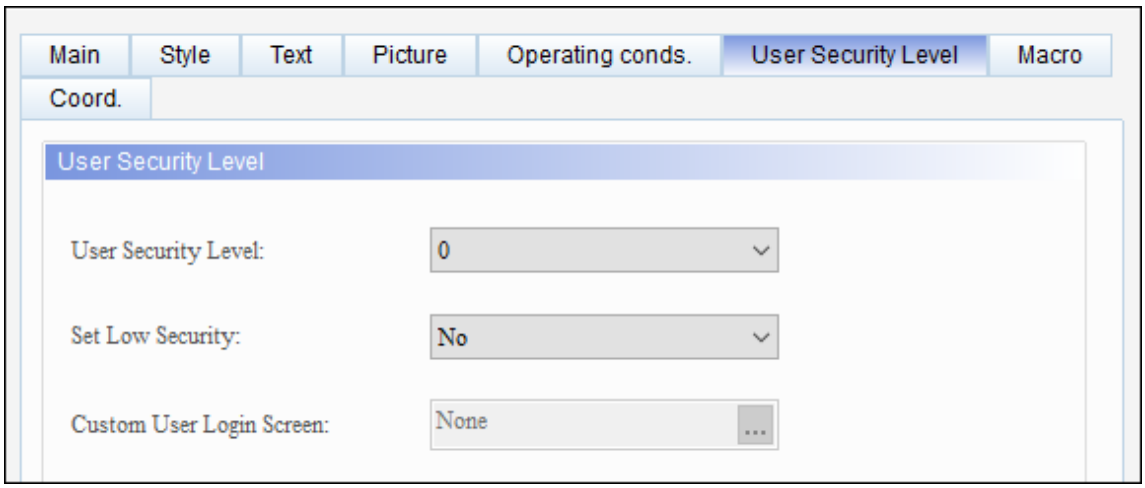
The effects of Interlock Display Mode are as follows.
<table>
<tr><th>Show Element</th><th>Show Prohibition Symbol</th></tr>
<tr><td> 
</td><td>
</td></tr>
</table></td></tr>
<tr><td>Interlock Display Mode</td><td></td></tr>
<tr><td>Interlock Address</td><td></td></tr>
</table>

<table>
<tr><th>Legend</th><th>Setting</th><th>Description</th></tr>
<tr><td>❸</td><td>Trigger Mode
Trigger Addr.</td><td>Select the trigger mode.
<table>
<tr><th>Trigger Before Writing</th><th>Trigger After Writing</th></tr>
<tr><td>The trigger address is set to ON first, and the value changes.
 
</td><td>The value is changed first, and the trigger address is set to ON.
 
</td></tr>
</table>
Note: Since the trigger function only sets the set trigger address to ON, you need to set the trigger address to OFF if you want to trigger it again.</td></tr>
</table>

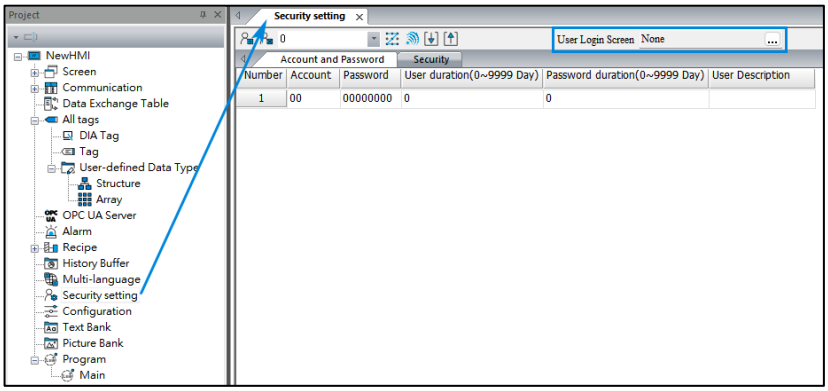
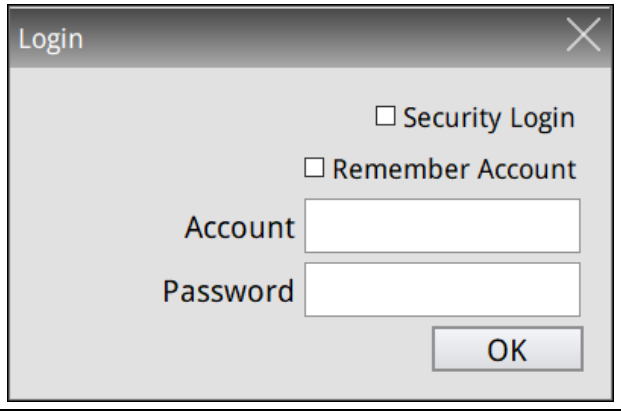
<table>
<tr><th>Legend</th><th>Setting</th><th>Description</th></tr>
<tr><td>④</td><td>Invisible Address Invisible State</td><td>
The Invisible Address hides an element on the HMI and must be used with Invisible State. If Invisible State is set to On, the Invisible Address hides the element. Instead, if Invisible State is set to Off, the Invisible Address hides the element.
The following is an example of using invisible address.
1. Create a button element on the screen and set its address to $9.0, then set the Invisible Address of the element to hide to $9.0.
2. To hide the element, you must click the $9.0 button.
<table>
<tr><td rowspan="2">Invisible State is ON</td><td> 
</td></tr>
<tr><td>
</td></tr>
<tr><td rowspan="2">Invisible State is OFF</td><td>
</td></tr>
<tr><td>
</td></tr>
</table>
</td></tr>
<tr><td>⑤</td><td>Ignore Operation Log</td><td>Choose whether to enable Ignore Operation Log.
When the operation log settings are enabled, the HMI will record the operations of all written element. If Ignore Operation Log of the element is set to Yes, the element action will not be recorded in the operation log.</td></tr>
<tr><td>⑥</td><td>Mark as Asterisk (*)</td><td>Select Yes, the read value will be displayed with a * symbol on the element.</td></tr>
</table>

Legend	Setting	Description

- **User Security Level**

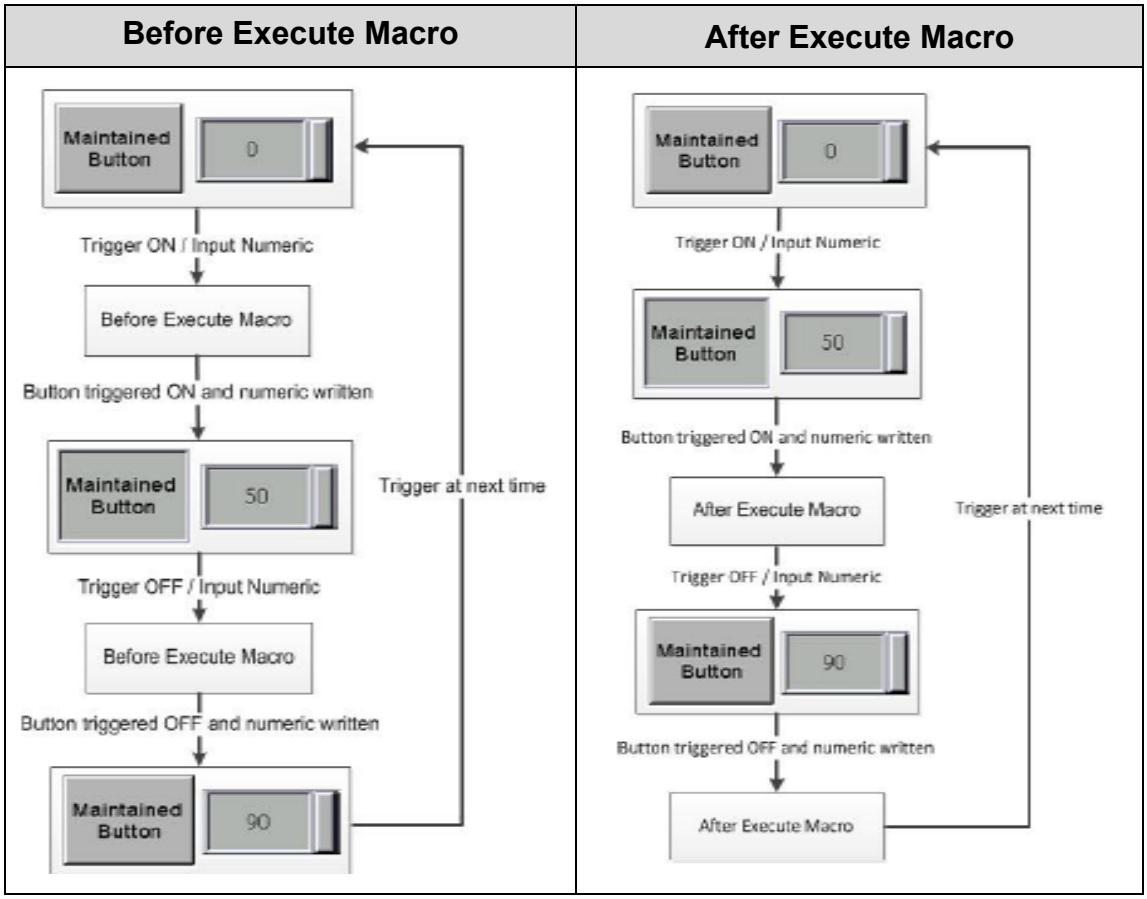


Setting	Description
User Security Level	Select the user security level. If the current security level is insufficient, the **Login** dialog pops up when the element is clicked on the HMI. The element action is performed if the login security level is equivalent to or higher than the set **User Security Level**. **Note**: When set to **0**, the element action can be performed without login.
Set Low Security	Choose whether to set the security level to the lowest security level after entering the security level. If set to **Yes**, the HMI will automatically set the security level to the lowest (Level 0) after performing the element action. Every time the element is triggered, the **Login** dialog will pop up, and you can perform the element action after logging in.

Setting	Description
Custom User Login Screen	Set the custom login screen. When **User Security Level** is not set to **0**, the custom login screen can be selected. When triggering the element and the user security level is required the set screen pops up. • On the toolbar, click **General** > **New Screen** > **User Login Screen** > **Login/Logout Screen**. • If the element does not have a specified login screen, you can go to **Project** > **Security Settings**. Set the **User Login Screen** of each security level in the **Security Settings** page. When the element sets this security level, the login screen configured in this page is displayed.  • If the element or account does not set the login screen, the default login screen is displayed. 

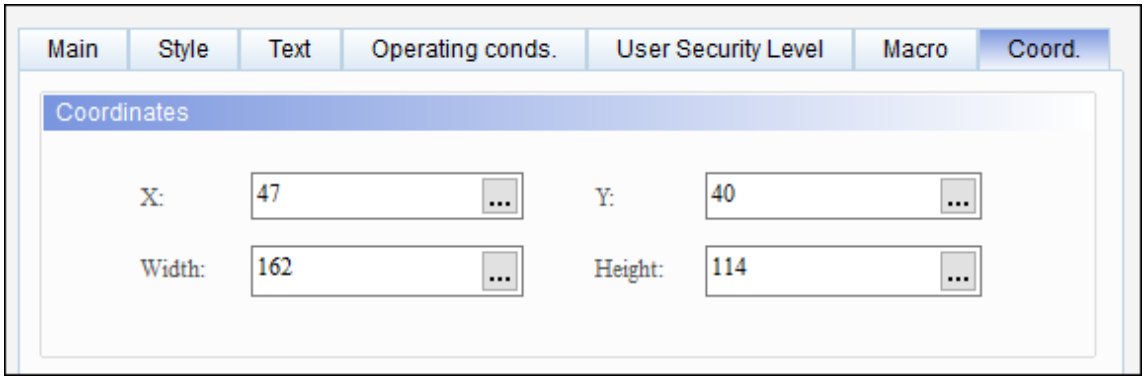
- **Macro**

After clicking the element on the HMI, the HMI executes **Before Execute Macro**, then executes the button action, and finally executes **After Execute Macro**.



Note: If the **Button** states are not changed with the button touch (using external controller commands or other macros and programs instead), the HMI does not execute the **Macro** commands.

- **Coord.**

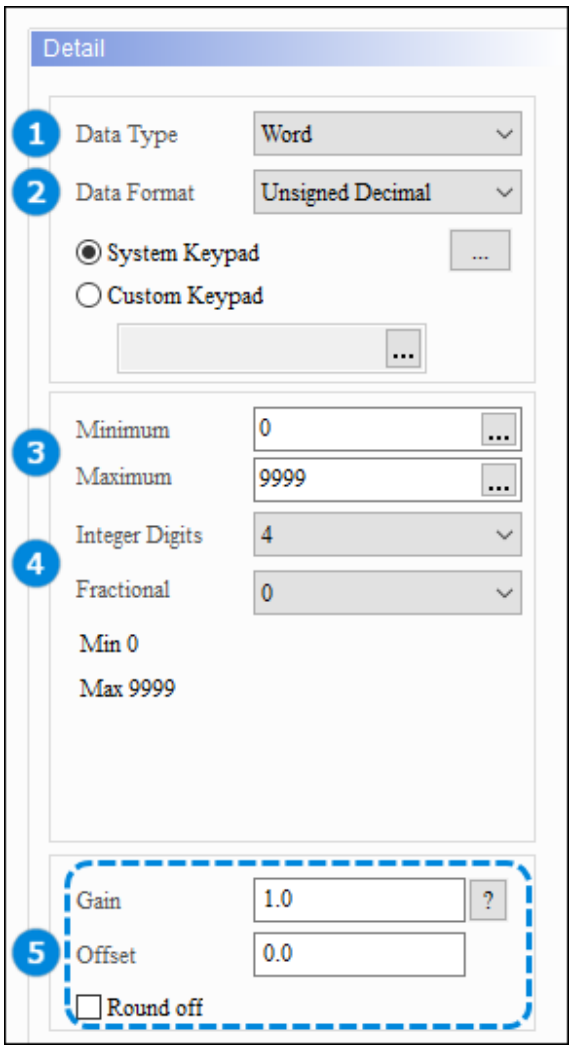


Set the XY coordinates, width, and height on the upper left of the element. You can enter a constant as fixed position or memory address to dynamically configure the position of the element on the HMI.

Numeric Entry

The following table lists the special properties of the **Numeric Entry** element with their description.

- **Main – Numeric Setting**

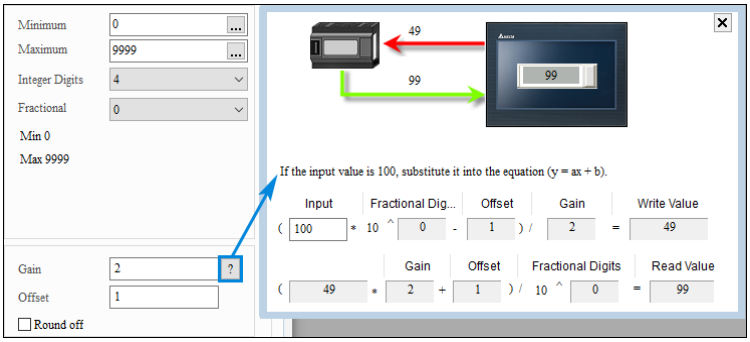


Legend	Setting	Description
1	Data Type	Select the data type. The available options are Word, Double Word, and Quad Word.
2	Data Format	Select the data format. The options vary from the data type settings, including BCD, Signed BCD, Signed Decimal, Unsigned Decimal, Hexadecimal, Binary, and Floating. **Note**: **Note**: Floating format is available only when the **Data type** is **Double Word** and **Quad Word**.

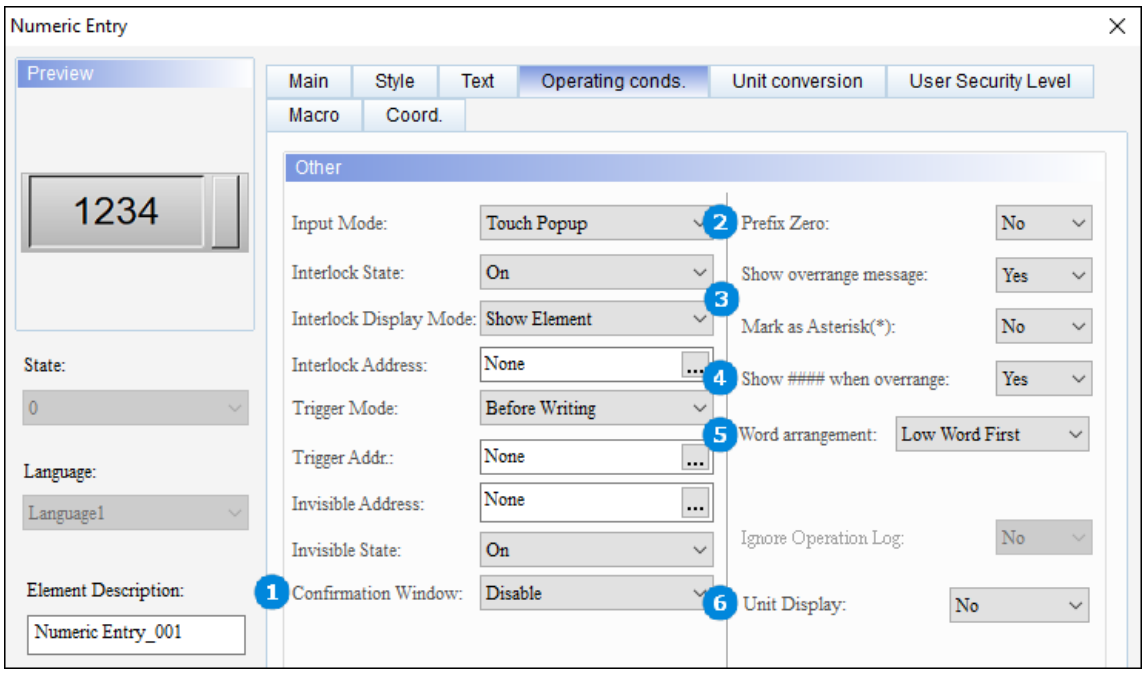
<table>
<tr><th>Legend</th><th>Setting</th><th>Description</th></tr>
<tr><td>③</td><td>Minimum
Maximum</td><td>The valid range values for minimum and maximum values vary depending on the selected Data Type and Data Format.
<table>
<tr><th>Data Type</th><th>Data Format</th><th>Valid range</th></tr>
<tr><td rowspan="6">Word</td><td>BCD</td><td>0~9999</td></tr>
<tr><td>Signed BCD</td><td>–999~+9999</td></tr>
<tr><td>Signed Decimal</td><td>–32768~+32767</td></tr>
<tr><td>Unsigned Decimal</td><td>0~65535</td></tr>
<tr><td>Hexadecimal</td><td>0~0xFFFF</td></tr>
<tr><td>Binary</td><td>0~0xFFFF</td></tr>
<tr><td rowspan="7">Double Word</td><td>BCD</td><td>0~99999999</td></tr>
<tr><td>Signed BCD</td><td>–9999999~+99999999</td></tr>
<tr><td>Signed Decimal</td><td>–2147483648~+2147483647</td></tr>
<tr><td>Unsigned Decimal</td><td>0~4294967295</td></tr>
<tr><td>Hexadecimal</td><td>0~0xFFFFFFFF</td></tr>
<tr><td>Binary</td><td>0~0xFFFFFFFF</td></tr>
<tr><td>Floating</td><td>0~9999999</td></tr>
<tr><td rowspan="7">Quad Word</td><td>BCD</td><td>0~9999999999999999</td></tr>
<tr><td>Signed BCD</td><td>–9999999999999999~
+9999999999999999</td></tr>
<tr><td>Signed Decimal</td><td>–9223372036854775808~
+9223372036854775807</td></tr>
<tr><td>Unsigned Decimal</td><td>0~18446744073709551615</td></tr>
<tr><td>Hexadecimal</td><td>0~0xffffffffffffffff</td></tr>
<tr><td>Binary</td><td>0~0xffffffffffff`fff</td></tr>
<tr><td>Floating</td><td>0~999999999999999</td></tr>
</table></td></tr>
<tr><td>④</td><td>Integer Digits
Fractional</td><td>Select the valid display digits according to the data type.
Note: The number of decimal places here is not a real decimal value, but the display format. Only when Floating is selected for the Data Format, it is a real decimal.</td></tr>
</table>

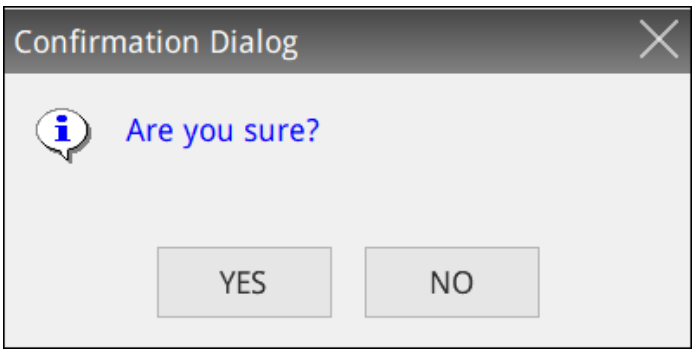
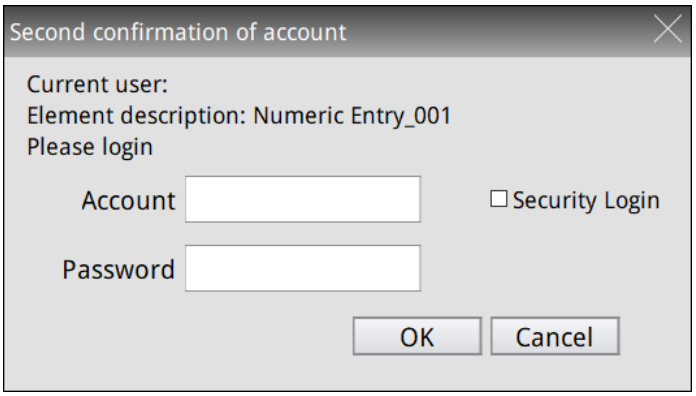
Legend	Setting	Description
5	Gain / Offset	Enter the gain value and offset value. • The formula for **Gain** and **Offset** is y = (a)x + (b). • Click ? to open calculation dialog. Enter the value in the **Input** field, and the **Write Value** / **Read Value** will be displayed according to the calculation method. • If select the **Round off** checkbox, the calculation results will be rounded off and displayed on the **Numeric Entry** element. • If the set **Gain** or **Offset** value is decimal, set the **Data Format** to **Floating**.

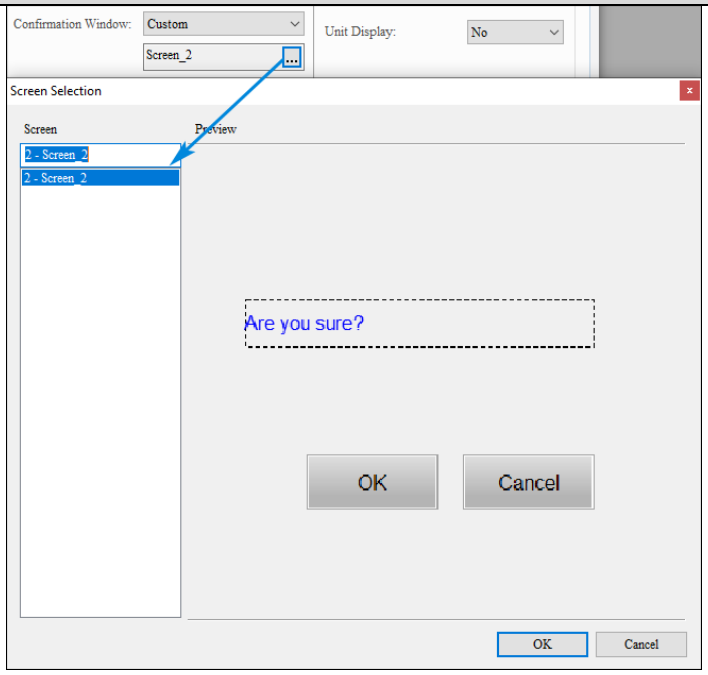
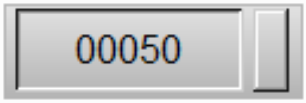
y	a	x	b
Result	Gain value	Input value	Offset value



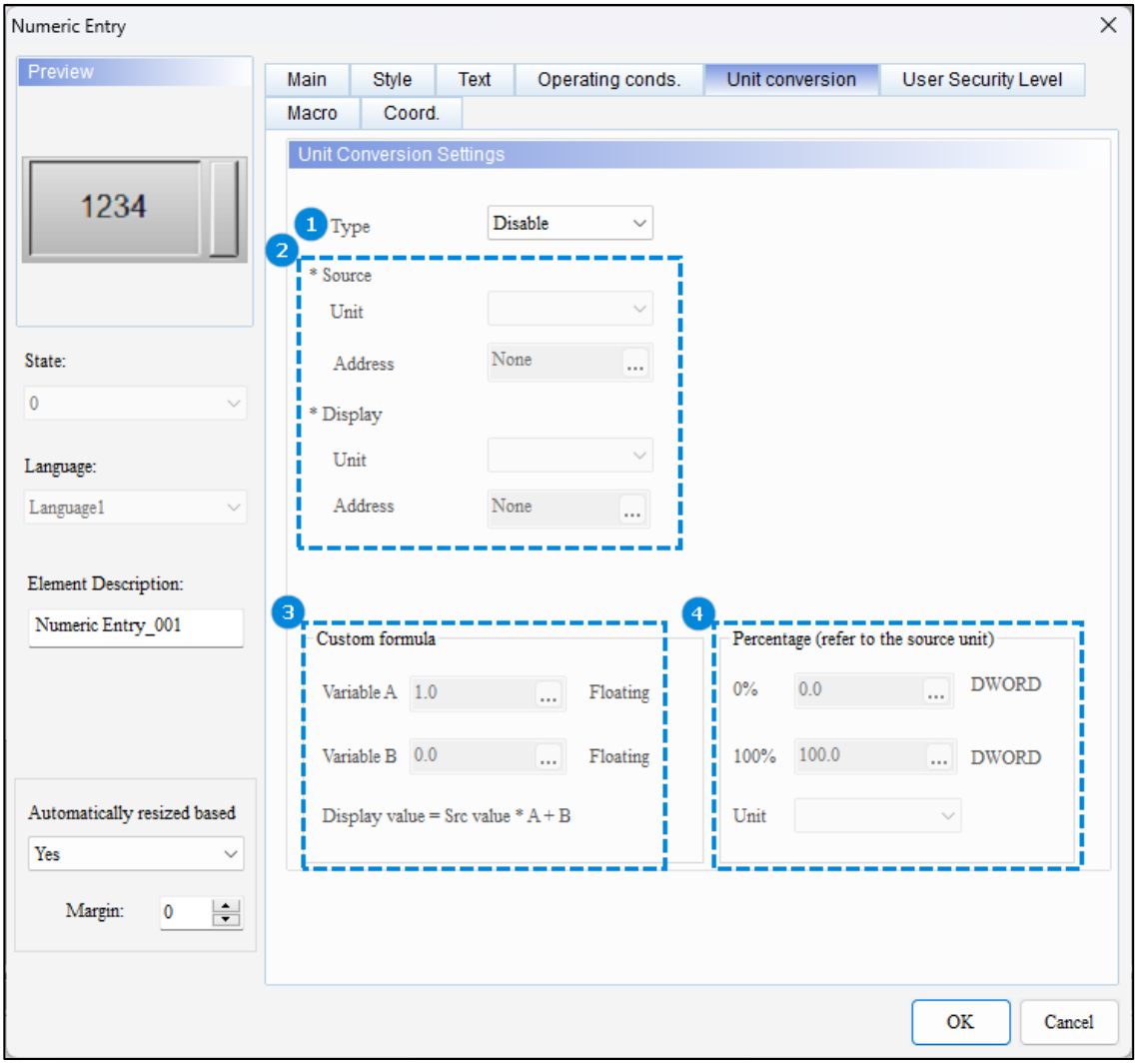
- **Operating conds.**



Legend	Setting	Description
❶	Confirmation Window	Select the type of confirmation dialog. • **Disable**: After pressing the element, the element action will be performed without displaying the dialog. • **Confirmation**: On the HMI, pressing the element on will pop up the **Confirmation Dialog**, and the element action will be performed only after confirmation in the dialog.  • **Second confirmation account**: Pressing the element will display the **Second confirmation of account** dialog, and you need to enter the account and password to perform the button action.  • **Custom**: Select the custom keypad dialog. You can go to **General** > **New Screen** > **Confirmation Window Screen** to add a Confirmation dialog screen.

Legend	Setting	Description
		
❷	Prefix Zero	Choose whether the integer digits are filled in with zeros. If the integer digits are 5 digits and the read value is 50, when selecting **Yes** for **Prefix Zero**, the **Numeric Display** element displays as follows. 
❸	Show overrange message	Choose whether to show overrange message when the input value exceeds the valid range.
❹	Show #### when overrange	Choose whether to show #### when the value range is not within the limited minimum and maximum values.
❺	Word arrangement	Select the word arrangement to swap high and low characters.
❻	Unit Display	Choose whether to display the unit. Select **Yes** to set the text content, display direction, and relevant text properties.

- **Unit conversion**



Legend	Setting	Description
1	Type	Select the type to convert. • If **Disable** is selected, the value does not need unit conversion. • If **Custom formula** is selected, enter **Variable A** and **Variable B** in the **Custom formula** area, and the element value will be converted and displayed using the formula “Display value = Syn value * A + B”.
2	Source Display	Select the unit of source and display according to the set Type. • The **Percentage** settings are available when selecting **%** or **Using the code** for the **Unit**. When the **Percentage** area is available, you need to set the values corresponding to **0%**

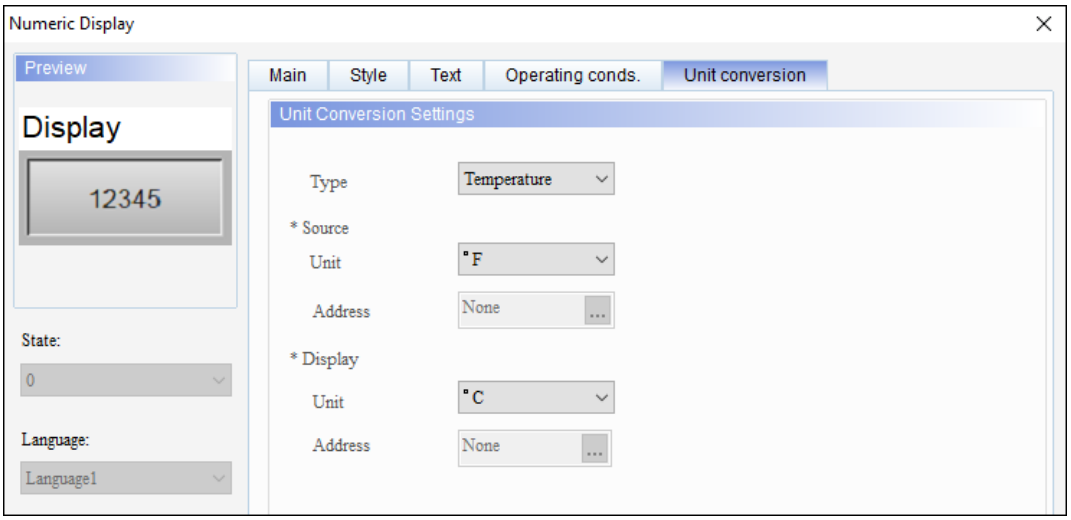
<table>
<tr><th>Legend</th><th>Setting</th><th>Description</th></tr>
<tr><td></td><td></td><td>and 100%, and set the unit according to the unit of the Source.
• When selecting Using the code for the Unit, the conversion unit can be dynamically specified on the HMI through the memory address. The following table lists the corresponding code of each unit.
<table>
<tr><th>Type</th><th>Unit</th><th>Code</th></tr>
<tr><td rowspan="3">Speed</td><td>mm/sec</td><td>101</td></tr>
<tr><td>inch/sec</td><td>102</td></tr>
<tr><td>%</td><td>700</td></tr>
<tr><td rowspan="5">Pressure</td><td>kg/cm^2</td><td>201</td></tr>
<tr><td>bar</td><td>202</td></tr>
<tr><td>MPa</td><td>203</td></tr>
<tr><td>psi</td><td>204</td></tr>
<tr><td>%</td><td>700</td></tr>
<tr><td rowspan="3">Location</td><td>mm</td><td>301</td></tr>
<tr><td>inch</td><td>302</td></tr>
<tr><td>%</td><td>700</td></tr>
<tr><td rowspan="3">Temperature</td><td>°F</td><td>401</td></tr>
<tr><td>°C</td><td>402</td></tr>
<tr><td>%</td><td>700</td></tr>
<tr><td rowspan="5">Weight</td><td>T</td><td>501</td></tr>
<tr><td>kN</td><td>502</td></tr>
<tr><td>g</td><td>503</td></tr>
<tr><td>oz</td><td>504</td></tr>
<tr><td>%</td><td>700</td></tr>
<tr><td rowspan="4">Volume</td><td>L</td><td>601</td></tr>
<tr><td>ml</td><td>602</td></tr>
<tr><td>kL</td><td>603</td></tr>
<tr><td>%</td><td>700</td></tr>
</table>
Note: When selecting Using the code for Source and Display, do not use the same address.</td></tr>
<tr><td>❸</td><td>Custom formula</td><td>This setting is available only when the Type is set to Custom formula. The element value will be converted and displayed using the formula “Display value = Syn value * A + B” and the unit is Floating.</td></tr>
</table>

Legend	Setting	Description
④	Percentage	This setting is only available when the unit of the **Source** or **Display** is set to **%** or **Using the code**. The **Unit** synchronizes to the unit of the **Source**, but if the unit of the **Source** is selected as **%** or **Using the code**, you can select to convert all units of the type.

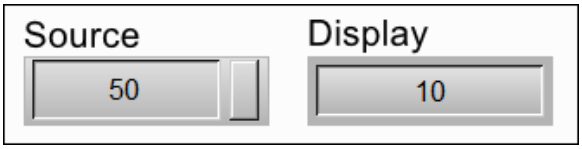
The following describes an example of using **Unit Conversion**.

- **Fixed Unit**

4. Create a **Numeric Entry** element on the screen as **Source**, and then set its **Write Address** to **$10**.
5. Create a **Numeric Display** element on the screen as **Display**, and then set its **Read Address** to **$10**, and then configure unit conversion.

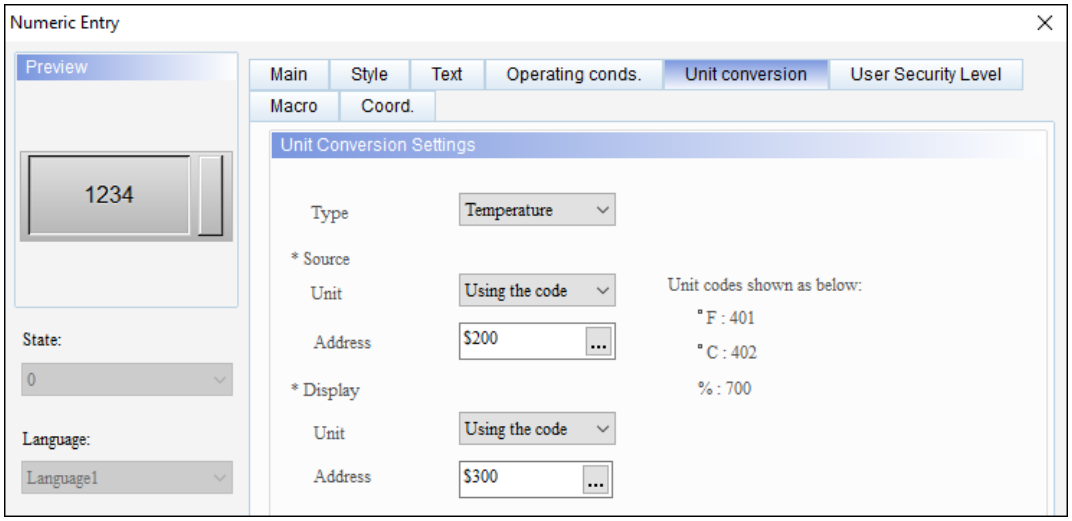


6. Perform an offline simulation, and then enter 50 (°F) for **Source**. **Display** will show 10 (°C).



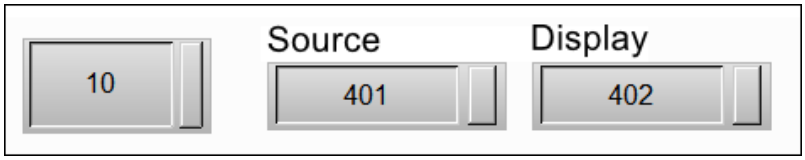
- **Using the code**

1. Create a **Numeric Entry** element on the screen, and then set the **Write Address** to $100 and configure the unit conversion.



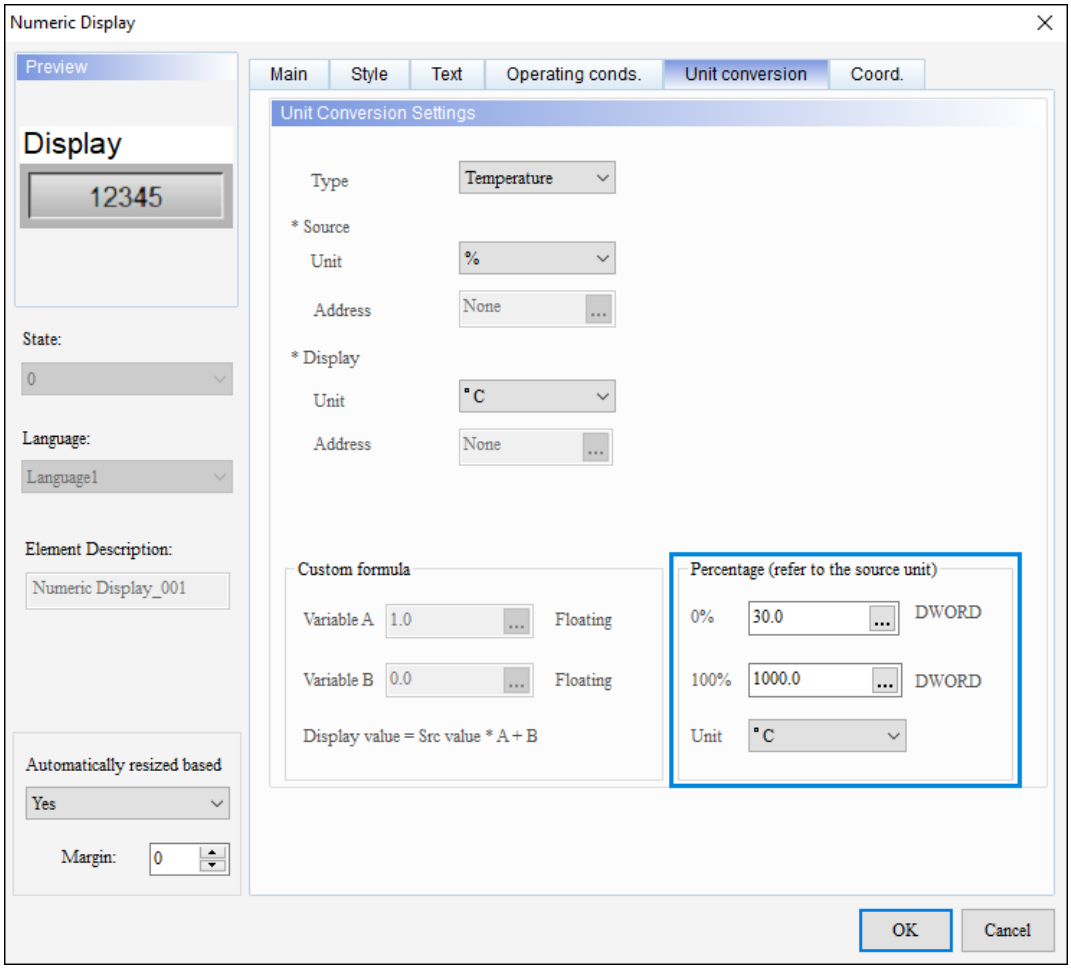
2. Create two more **Numeric Entry** elements, and then set the **Write Address** to **$200** and **$300** separately.
3. Perform an offline simulation, and then enter 50 for the **Numeric Entry** element of $100.

4. Enter 401 (°F) for **Source** and 402 (°C) for **Display**, and the **Numeric Entry** element of $100 will have its unit converted according to the code.

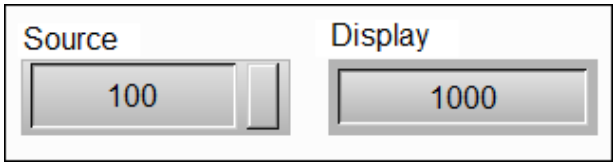


- **Percentage**

1. Create a **Numeric Entry** element on the screen as **Source**, and then set the **Write Address** to **$20**.
2. Create a **Numeric Display** element as **Display**, and then set the **Write Address** to **$20** and configure the unit conversion.

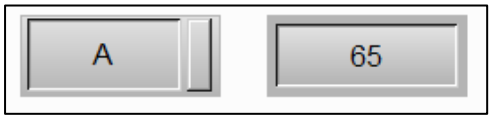


3. Perform an offline simulation, and then enter 100 for **Source**. **Display** will show the value 1000 corresponding to 100%.



Character Entry

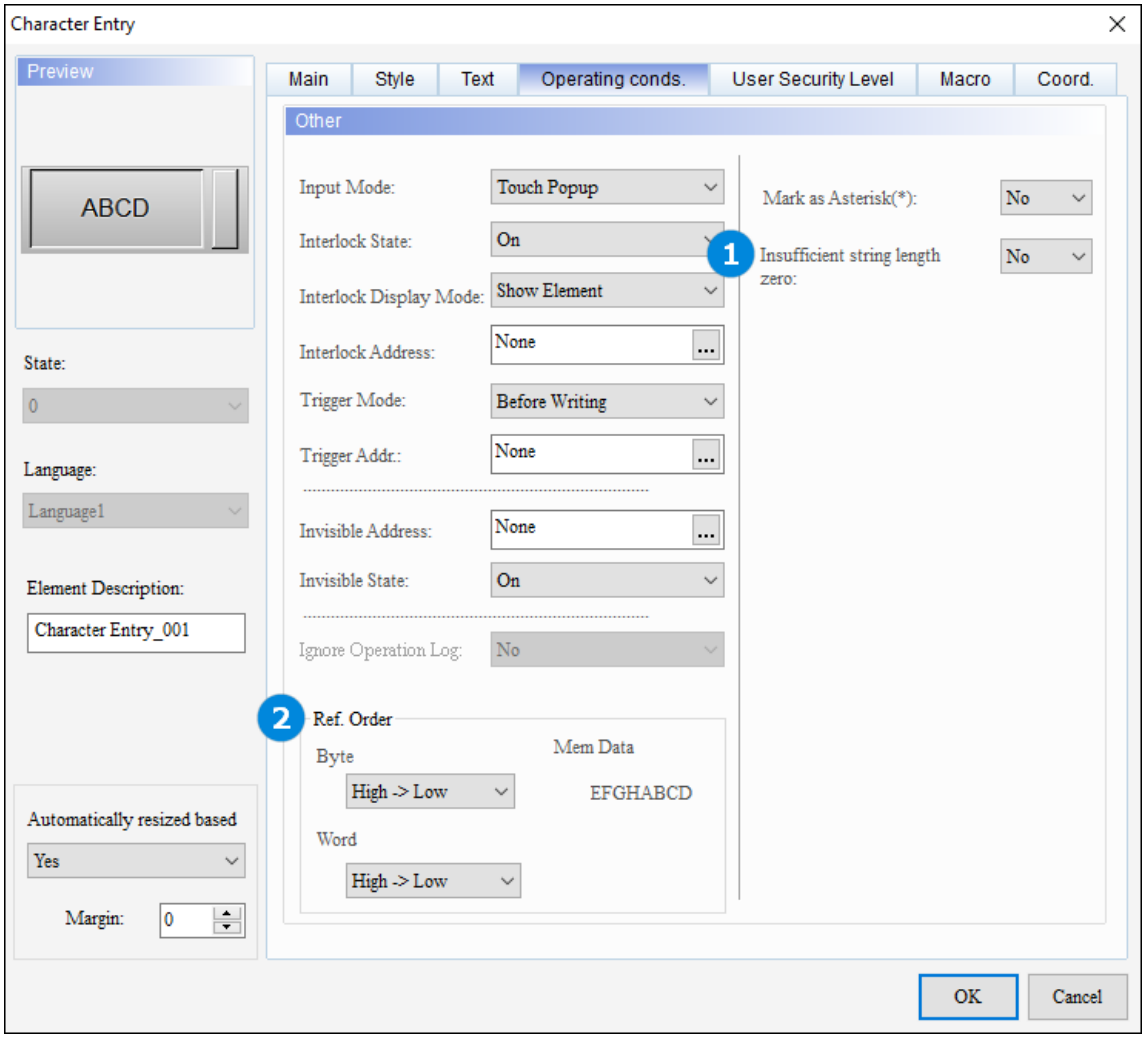
The **Character Entry** element is an entry element supporting ASCII code input. The following table lists the character conversion corresponding to the **Data Format**. When entering A on the **Character Entry** element, the **Numeric Entry** element will read the address value and display 65 (Unsigned Decimal).

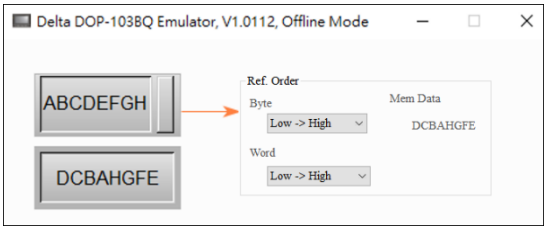
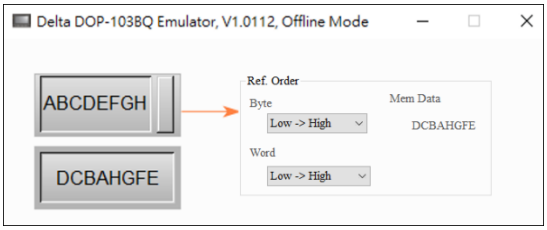


Unsigned Decimal	Hexadecimal	Character
65	41	A
66	42	B
67	43	C

Note: A **Character Entry** element supports up to 512 characters.

- **Operating conds.**

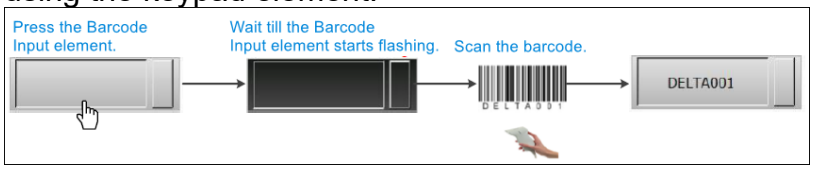
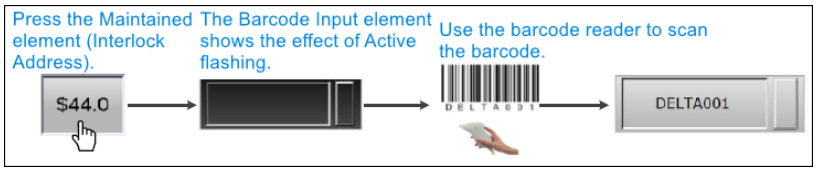
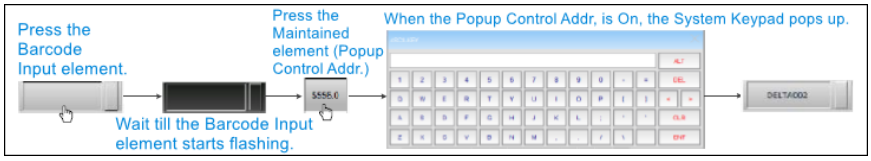


<table>
<tr><th>Legend</th><th>Setting</th><th>Description</th></tr>
<tr><td>1</td><td>Insufficient string length zero</td><td>Choose whether to fill in with zero when the string length is insufficient.
When the length of the entered string is less than the set String Length, the remaining characters are filled in with zeros for display. For example, when the String Length is set to 10, but only 5 characters are entered, the following characters will be filled in with zeros.
</td></tr>
<tr><td>2</td><td>Reference order</td><td>Select the display mode of Read Address. DIAScreen provides two swap modes: ByteSwap and WordSwap. Both can be used at the same time, but use ByteSwap first and then WordSwap.
The following is the display result of using two swap modes.
<table>
<tr><td>Reference order Byte is Low→High</td><td>Enter ABCDEFGH in the Character Entry element, and the result of writing to the memory after reading the swap by the Character Display element is DCBAHGFE.
 
</td></tr>
<tr><td>Reference order Word is High→Low</td><td>Enter ABCDEFGH in the Character Entry element, and the result of writing to the memory after reading the swap by the Character Display element is CDABGHEF.
 
</td></tr>
</table></td></tr>
</table>

Barcode Input

The following table lists the special properties of the **Barcode Input** element with their description.

- **Operating conds.**

Setting	Description
Input Mode	Select the input mode. • **Touch:** After pressing the **Barcode Input** element on the HMI, the element will flash, and you can scan the barcode through the external device or input the string using the keypad element.  • **Active**: This mode must be used with the **Interlock Address** to enable the input. After the **Input Mode is triggered**, the **Barcode Input** element will flash, and you can scan the barcode through the external device or input the string using the keypad element. 
Popup Control Addr.	Set whether to display the system keypad dialog. Set the **Popup Control Addr.** to On, and the keypad dialog appears after pressing the **Barcode Input** element. When the System Keypad appears, the **Popup Control Addr.** becomes Off. To enter the barcode again, the **Popup Control Addr.** must be set to On again.  **Note**: This setting is valid only when the **Input Mode** is set to **Touch**.

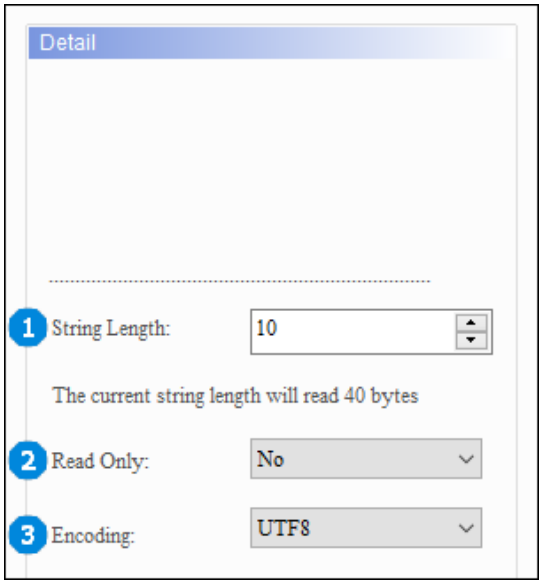
Note: A **Barcode Input** element supports up to 512 characters.

Multi-language Input

- **Note**: Some models support read-only of the **Multi-language Input** element.

- The following table lists the special properties of the **Multi-language Input** element with their description.

- **Main**

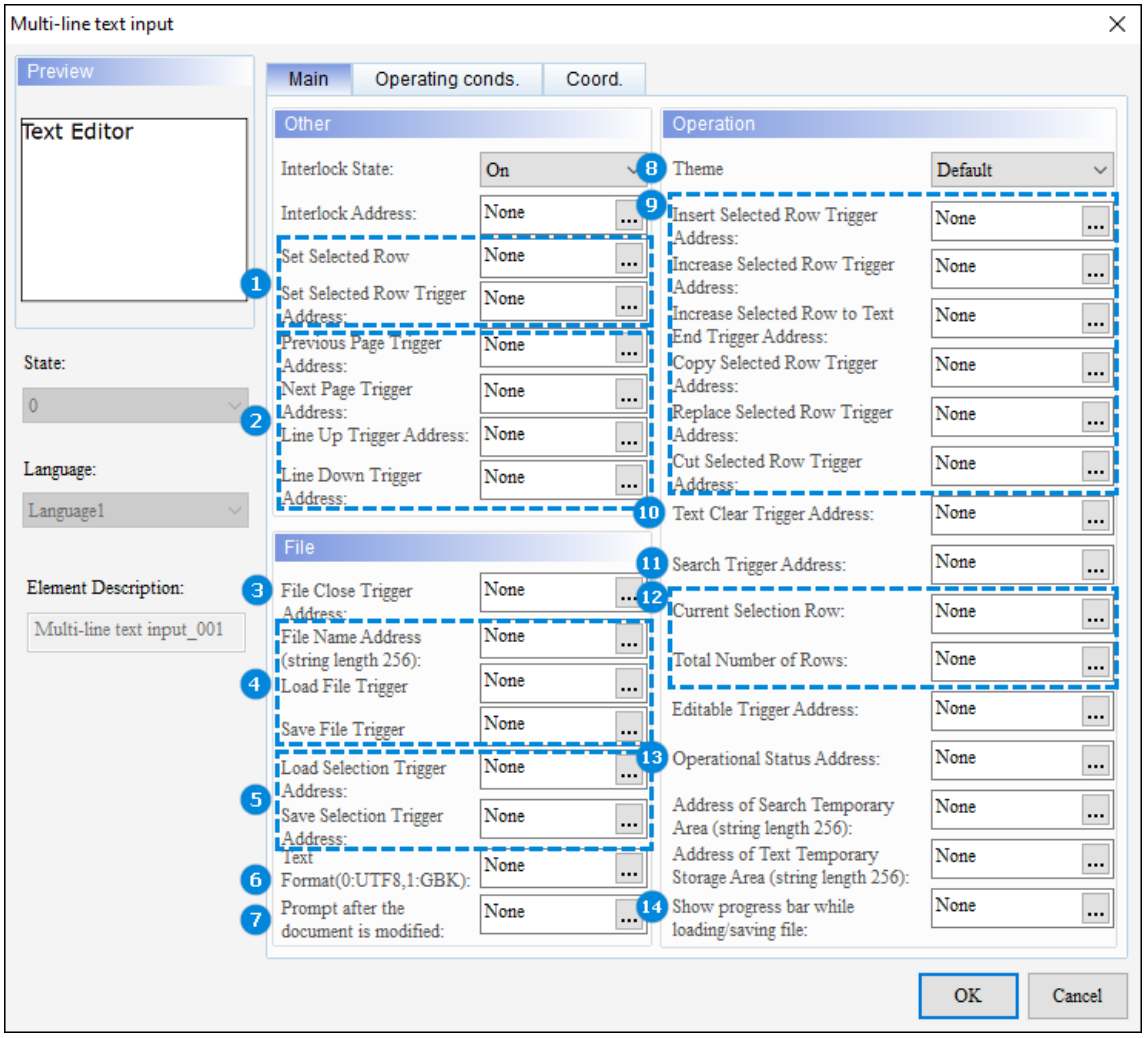


Legend	Setting	Description
1	String Length	Select the length of the string, and the range value is 1–512
2	Read Only	Choose whether the element is read-only. The default value is **No**. select **Yes**, the element can only read data
3	Encoding	Select the encoding format of the text.

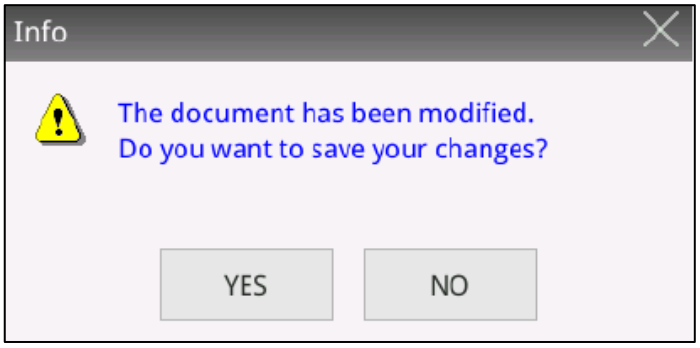
Multi-line Text Input

The following table lists the special properties of the **Multi-line Text Input** element with their description.

- **Main**



Legend	Setting	Description
❶	Set Selected Row Address	Specify the line to select with the **Set Select Row Address**. Then set the **Set Selected Row Trigger** to On, and the element displays the selected item. For example, set the selected row to 8 and set the **Selected Row Trigger** to On, and the text data specifies row 8.
	Set Selected Row Trigger	1 中文測試 2 중국어 시험 3 中国語テスト 4 Čínský test 5 Китайский тест 6 중국어 시험 7 Éditeur de texte 8 Kínverska próf̈ið
❷	Previous Page Trigger Address	Switch when the trigger address is On, and the address is automatically cleared after completion.

Legend	Setting	Description
	Next Page Trigger Address	**Note**: When there is only a page of text content, switching to the **Previous** page will select the first row, and switching to the Next page will select the last row.
	Line Up Trigger Address	
	Line Down Trigger Address	
❸	File Close Trigger Address	When the trigger address is On, the currently opened text file can be closed.
❹	File Name Address	Enter the file path in the **File Name Address** field. The following actions can be executed: • **Load File Trigger**: After loading the text file, the **Multi-line Text Input** element displays the text content. • **Save File Trigger**: Store the current text content in the file name path. **Note**: The file name length supports up to 256 characters.
	Load File Trigger	
	Save File Trigger	
❺	Load Selection Trigger Address	When the trigger address is On, the file browsing dialog will open, and you can select the file to load or save. **Note**: Write the selected file path to the **File Name Address** after the execution.
	Save Selection Trigger Address	
❻	Text format	Set the encoding format of the text. The values are defined as follows: • 0: UTF8 • 1: GBK
❼	Prompt after document has been modified	Set whether to pop up a prompt message after modifying the file. The values are defined as follows: • 0: Do not prompt • 1: Pop up the prompt dialog 

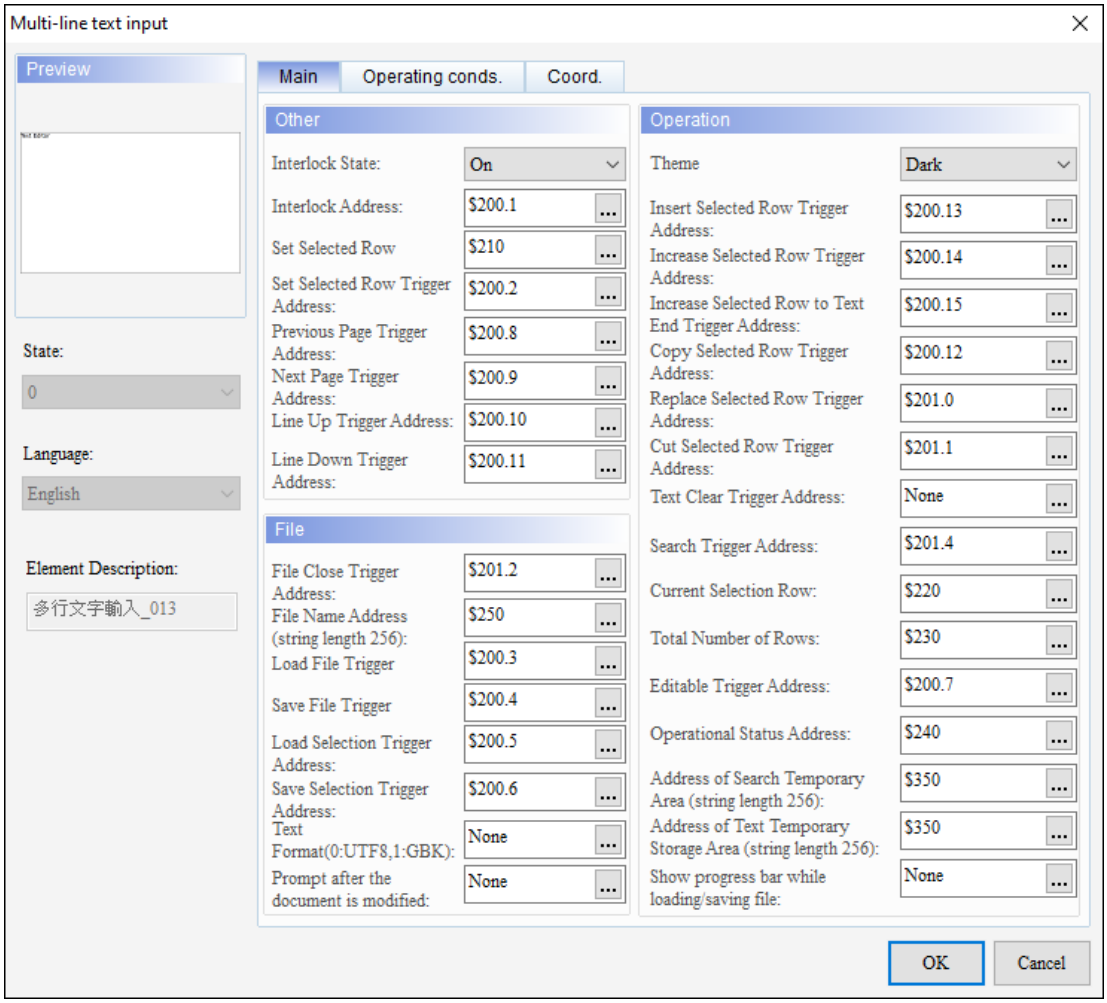
Legend	Setting	Description
⑧	Theme	Select the theme to display on the element. **Note**: **GCode** theme loads the .NC format file belonged to G-code and the text displays in color.
⑨	Insert Selected Row Trigger Address	After setting the **Editable Trigger Address** to On, select and enter text at the starting address of the text temporary area by the **Selected Row Address**, and then set the trigger address to On. • **Insert**: Insert the content of the text temporary area into the previous row of the selected row. • **Increase**: Add the content of the text temporary area to the previous row of the selected row. • **Increase to the end**: Add the content of the text temporary area to the end of the text row. • **Copy**: Copy the text of the selected row to the text temporary area. • **Replace**: Replace the text of the selected row with the content of the text temporary area. • **Cut**: Cut the text of the selected row and paste it into the text temporary area. **Note**: The trigger address is automatically cleared after the action is completed.
	Increase Selected Row Trigger Address	
	Increase Selected Row to Text End Trigger Address	
	Copy Selected Row Trigger Address	
	Replace Selected Row Trigger Address	
	Cut Selected Row Trigger Address	
⑩	Text Clear Trigger Address	When the trigger address is On, all text data on the element is cleared.
⑪	Search Trigger Address	Enter keywords to be researched into the **Address of Search Temporary Area** and set the **Search Trigger Address** to On, and the corresponding text row can be selected.
	Address of Search Temporary Area	
⑫	Current Selection Row	Displays the number of rows of text currently selected.
	Total Number of Rows	Displays the total number of rows in the currently opened file.
⑬	Show progress bar while loading / saving file	Set whether to display a progress bar when processing control files. • 0: Not show • 1: Show

- **Operating conds.**

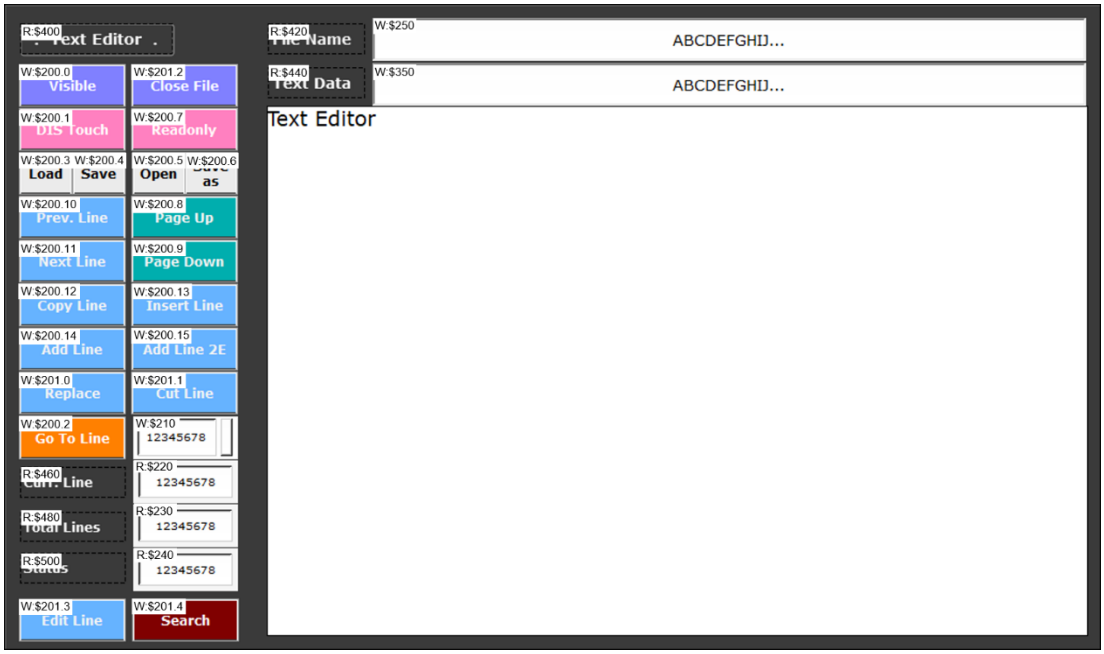
  Editable Trigger Address: When the trigger address is On, the multi-language input keypad pops up, and you can edit the selected rows of text.

  The following describes the example of using the **Multi-line Text Input** element.

1. Create a **Multi-line Text Input** element on the screen, and then configure the address corresponding to each property in the element properties dialog.

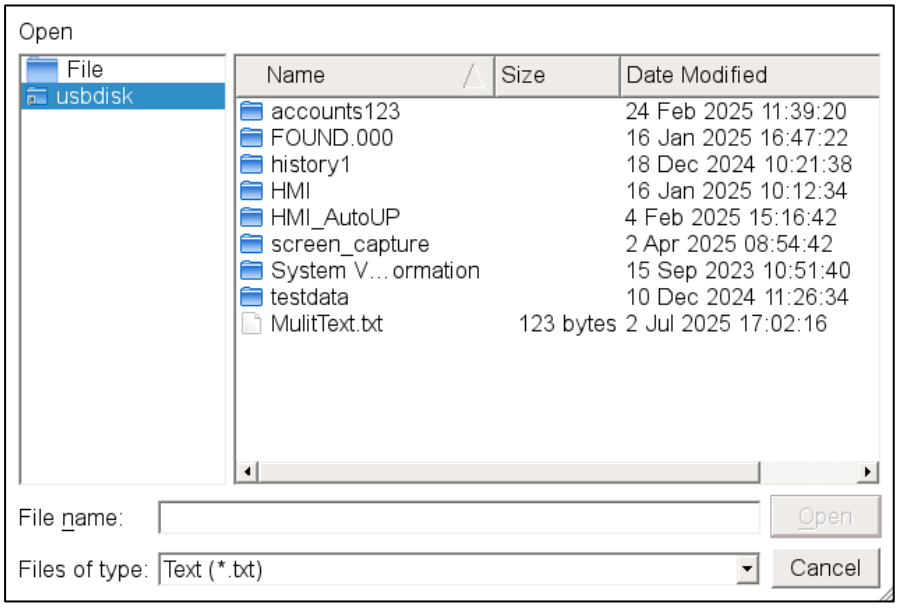


2. Create the control elements corresponding to the used memory address. Refer to the following figure for screen planning.

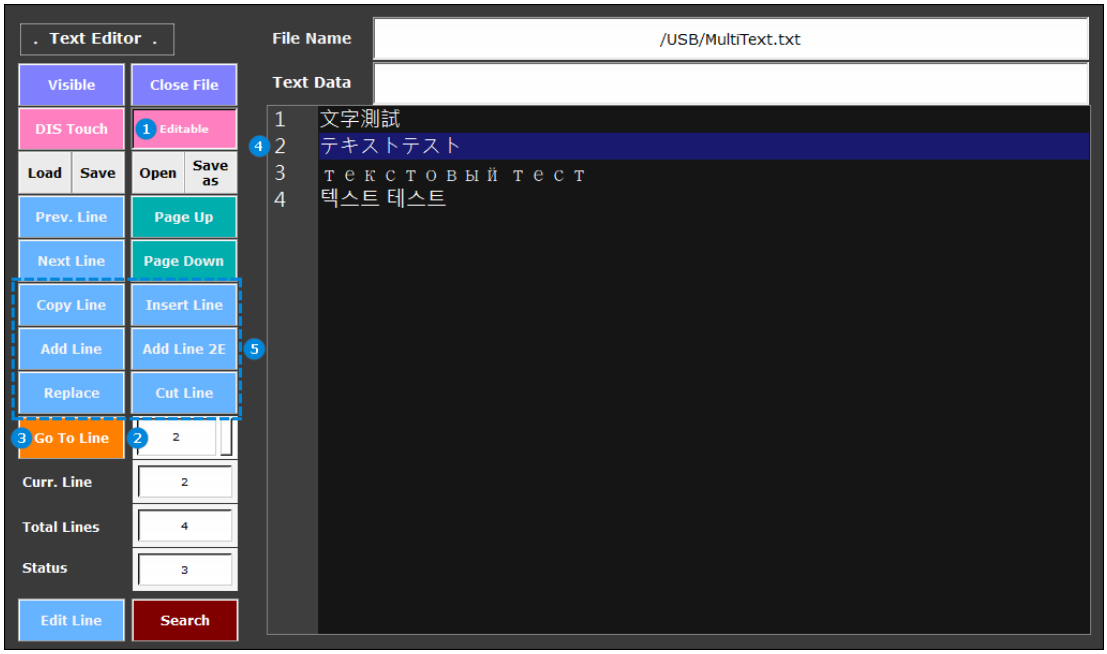


3. Compile and download the screen to the HMI, and then save the TXT file to be read to the USB and insert it into the HMI.

4. Press **Open** on the HMI screen to load the file trigger bit. Select the file in the pop-up file browse dialog to load the **Multi-line Text Input** element.



5. Press **Readonly** to edit the trigger bit. After entering the edit mode, press **Go To Line** to enter the number of rows, and then press the **Set Selected Row**. The element will select the text, and you can perform operations such as copying and cutting.



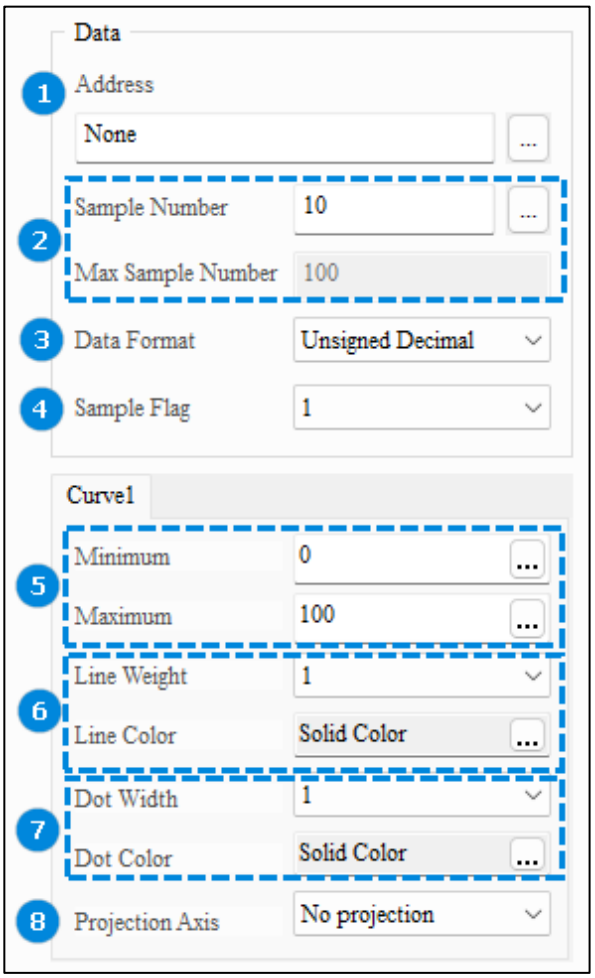
6. To add text, you can enter text in the **Text Data** (**Address of Text Temporary Storage Area**) field to add, insert, or replace the text to write into the element.
7. Press **Edit Line** to edit the selected row trigger address. You can change the text content on the popup multi-language input keypad.
8. Press **Save** or **Save as** to save the text content of the element to the file.

Curve

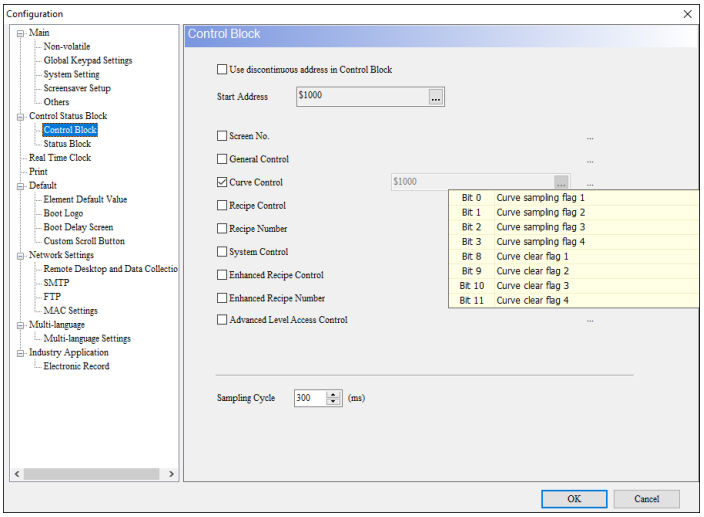
The **Curve** reads values continuously according to the read address and draws a curve.

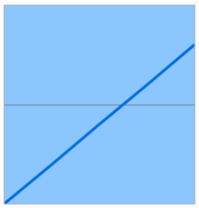
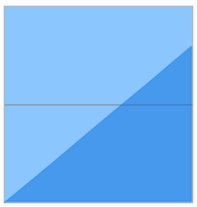
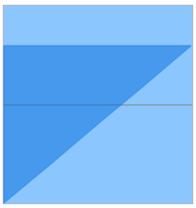
Common Properties

- **Main**

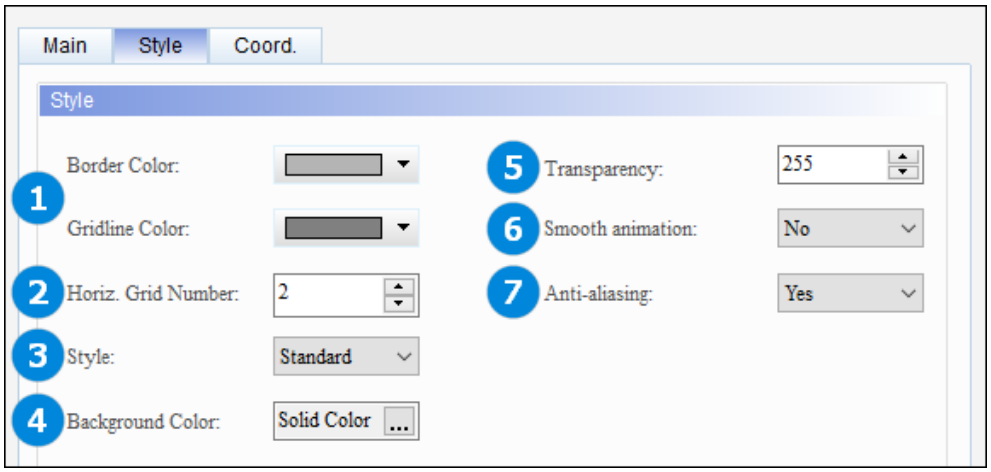


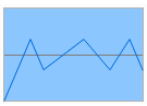
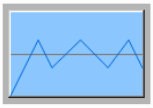
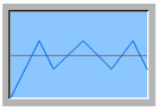
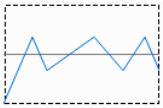
Legend	Setting	Description
1	Address	You can select the **Internal Memory** address or controller register address, and the **Input** memory **Type** can only be **Word**. **Note**: For the **X-Y Chart**, the horizontal and vertical reading addresses can be set respectively.

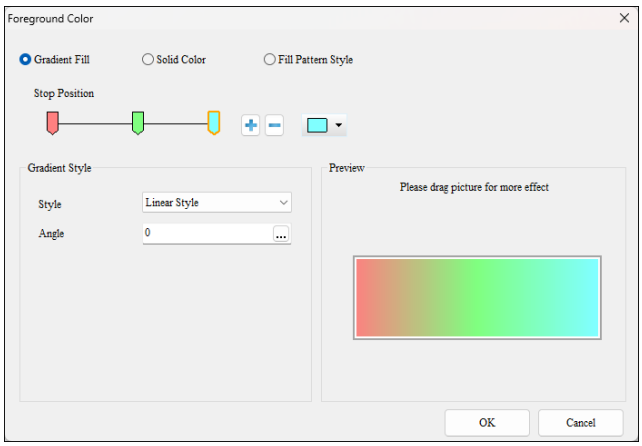
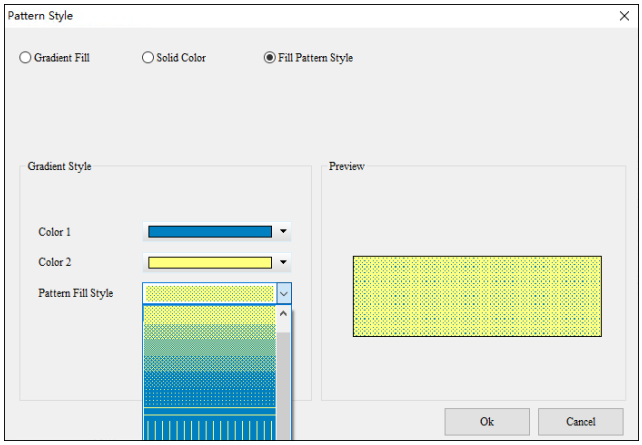
Legend	Setting	Description
②	Sample Number Max Sample Number	Set the sample number and maximum sample number. The size of **Sample Number** depends on the width and **Style** of the element. • When the **Style** is set to **Standard**, the maximum value of the **Sample Number** is the element width. • When the **Style** is set to **Raise** or **Sunken**, since the border width is 7pt, the maximum value of the **Sample Number** is the element width minus 7. **Note**: The Max Sample Number is only available when the Sample Number is set as address. The maximum value of the **Max Sample Number** also depends on the width of the element.
③	Data Format	Select the data format. **Note**: Only the **X-Y Chart** can choose Word or Double Word.
④	Sample Flag	Select the sample flag. This setting corresponds to the setting of the **Curve Control** which can be set in **General** > **Configuration** > **Control Block**.  **Note**: The **Curve Control** is a share setting for **Trend Graph** and **X-Y Chart**.

<table>
<tr><th>Legend</th><th>Setting</th><th>Description</th></tr>
<tr><td>❺</td><td>Minimum
Maximum</td><td>Set the valid range value according to the selected Datat Type and Datat Format. You can also dynamically set the minimum and maximum values through the memory address on the HMI.
<table>
<tr><th>Data Type</th><th>Data Format</th><th>Data Valid Range</th></tr>
<tr><td rowspan="5">Word</td><td>BCD</td><td>0~9999</td></tr>
<tr><td>Signed BCD</td><td>–999~+9999</td></tr>
<tr><td>Signed Decimal</td><td>–32768~+32767</td></tr>
<tr><td>Unsigned Decimal</td><td>0~65535</td></tr>
<tr><td>Hexadecimal</td><td>0~0xFFFF</td></tr>
<tr><td rowspan="5">Double Word</td><td>BCD</td><td>0~99999999</td></tr>
<tr><td>Signed BCD</td><td>–9999999~
+99999999</td></tr>
<tr><td>Signed Decimal</td><td>–2147483648~
+2147483647</td></tr>
<tr><td>Unsigned Decimal</td><td>0~4294967295</td></tr>
<tr><td>Hexadecimal</td><td>0~0xFFFFFFFF</td></tr>
</table></td></tr>
<tr><td>❻</td><td>Line Weight
Line Color</td><td>Select the width and color of the line.
Note: If you set the Projection Axis, the line color can only be set to Solid Color.</td></tr>
<tr><td>❼</td><td>Dot Weight
Dot Color</td><td>Select the width and color of the dot.
Note: The setting is only supported in Trend Graph and Curve Input elements.</td></tr>
<tr><td>❽</td><td>Projection Axis</td><td>Select the projection axis of the curve.
<table>
<tr><th>No projection</th><th>X-axis projection</th><th>Y-axis projection</th></tr>
<tr><td>  </td><td>  </td><td>  </td></tr>
</table></td></tr>
</table>

- **Style**

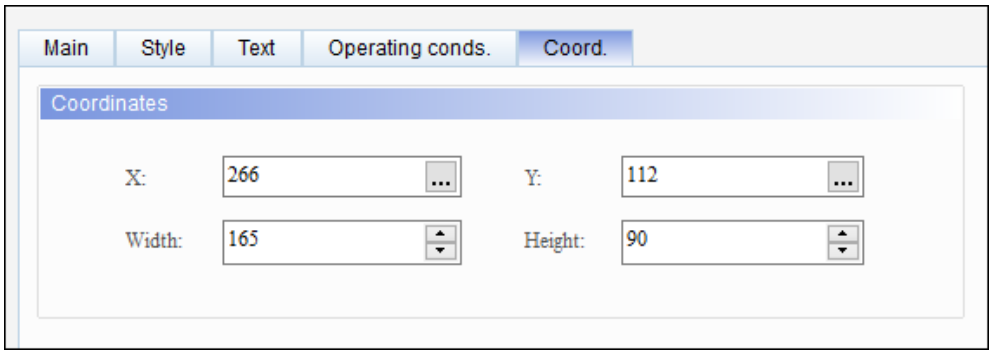


Legend	Setting	Description
1	Border Gridline Color	Select the border and gridline color of the element.
2	Horiz. Grid Number	Select or enter the gridline number of the trend graph. Support up to 50 grids, and the default value is **2**, meaning there is one grid line dividing the **Trend Graph** element into two zones. **Note**: This setting is applicable for the **Trend Graph** and **Curve Input** elements.
3	Style	Select the appearance of the element. The default value is **Standard**. **Standard**  **Raised**  **Sunken**  **Transparent** 

Legend	Setting	Description
④	Background Color	Select the color of the element. • **Gradient Fill:** Set the gradient style by setting the current angle or center point. You can drag the mouse on the preview image to set the current angle or center point, or set the memory address to dynamically configure it on the HMI. You can also set the stop point colors to present the effect of multi-color gradients.  **Note**: Up to 5 stop points can be set. • **Solid Color**: **Select** a single-color fill. • **Fill Pattern Style**: Select a two-color combination configuration and select a pattern fill style. 
⑤	Transparency	Select or enter the transparency of the element. The range value is 50–255 and the default value is **255**. The smaller the value, the higher the transparency of the element.
⑥	Smooth animation	Choose whether to enable the smooth animation function. If select **Yes**, the element display becomes smoother.

Legend	Setting	Description
❼	Anti-aliasing	Choose whether to enable anti-aliasing effect. If select **Yes**, the element border displays smooth.

- **Coord.**

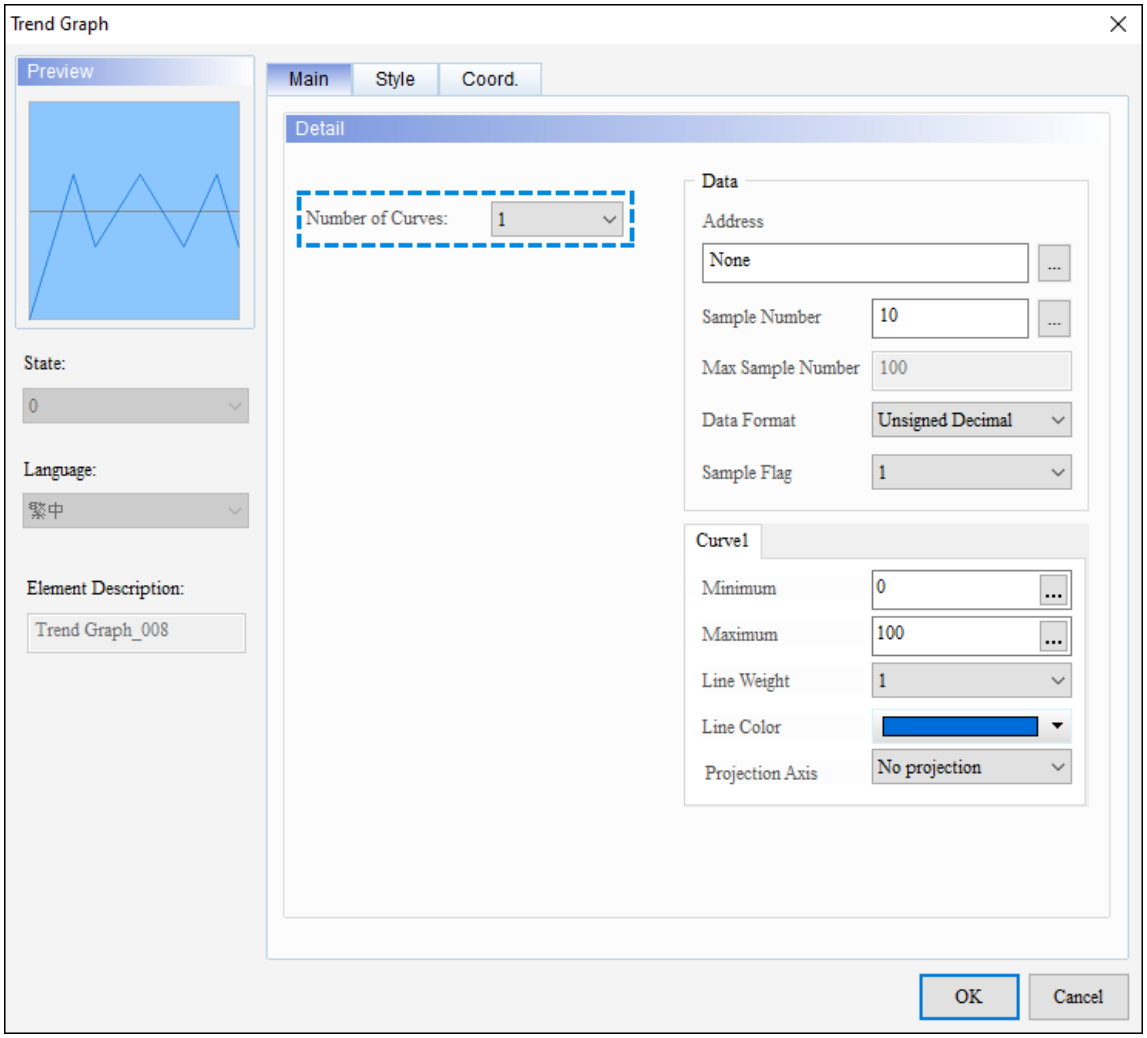


Set the XY coordinates, width, and height on the upper left of the element.

Trend Graph

The following table lists the special properties of the **Trend Graph** element with their description.

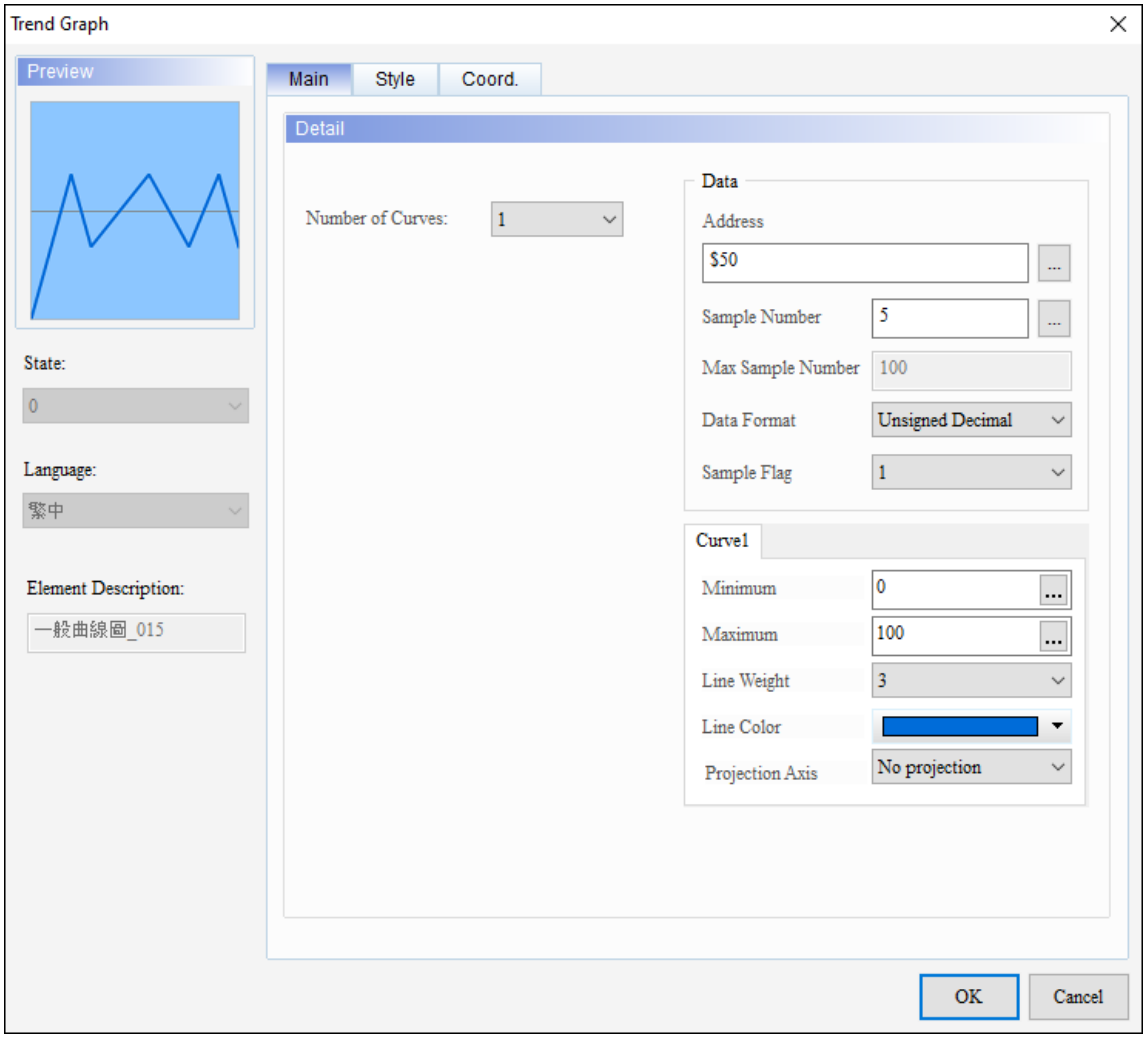
- **Main**



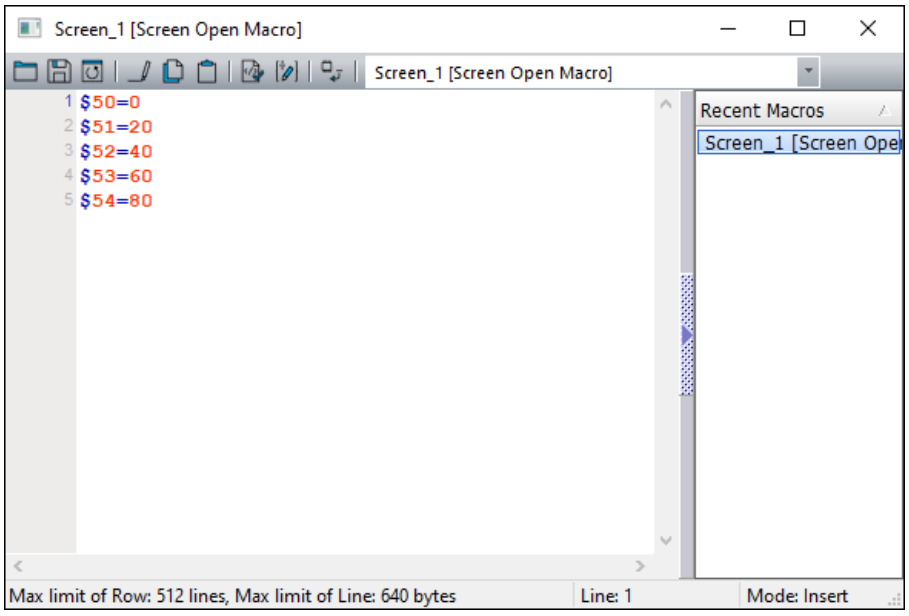
Setting	Description
Number of Curves	Select the number of curves. Up to 12 curves can be set. You can set the properties of curves respectively in the lower right of the element properties dialog.

The following describes how to use the **Trend Graph** element.

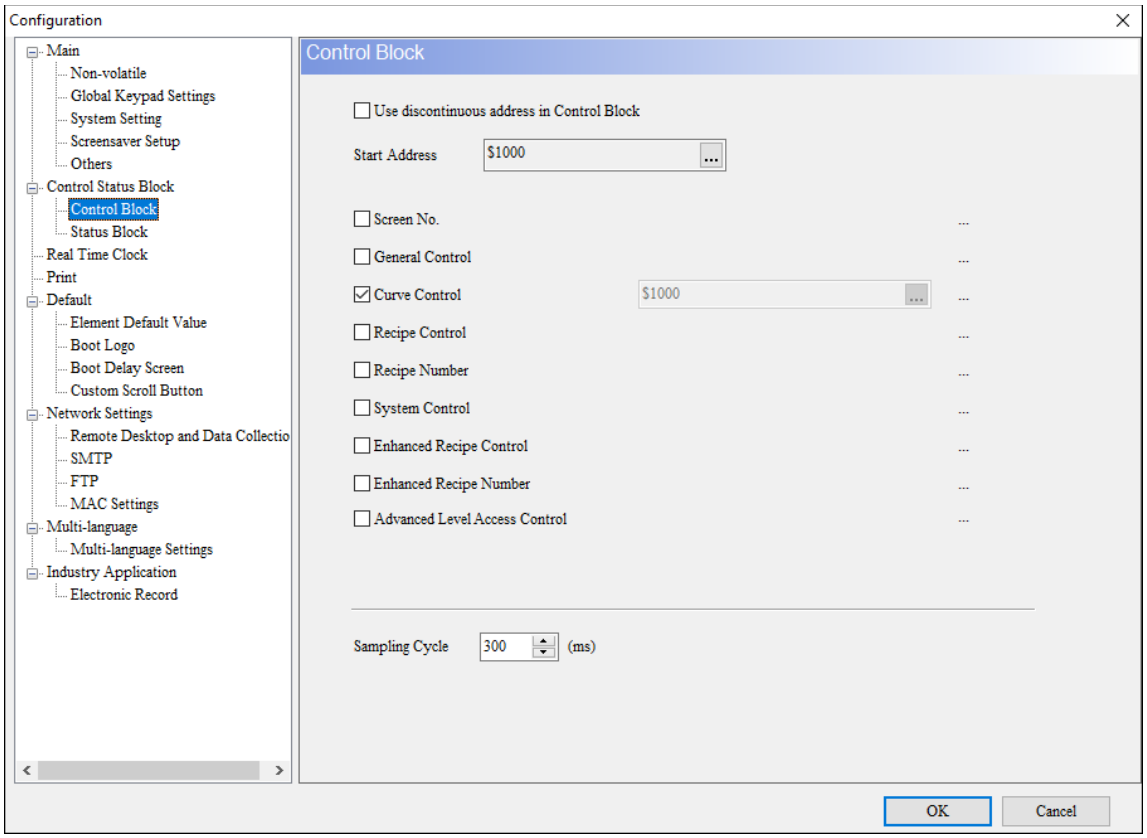
1. Create a **Trend Graph** element on the screen, and then set the **Read Address** to **$50** and the **Sample Number** to **5**.



2. Create 5 **Numeric Entry** elements, set the **Write Address** to **$50~$54** separately, and then give these addresses an initial value in the **Screen Open Macro** dialog.



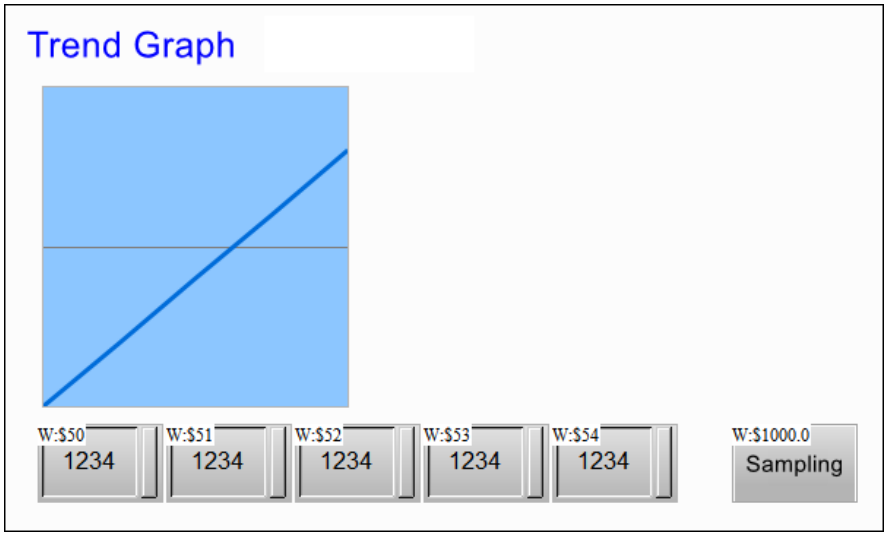
3. Go to **General** > **Configuration** > **Control Block** and set the address of the **Curve Control** to **$1000**.



4. Create a **Momentary** button element on the screen, and then set the **Write Address** to **$1000.0**.

The execution result is as follows.

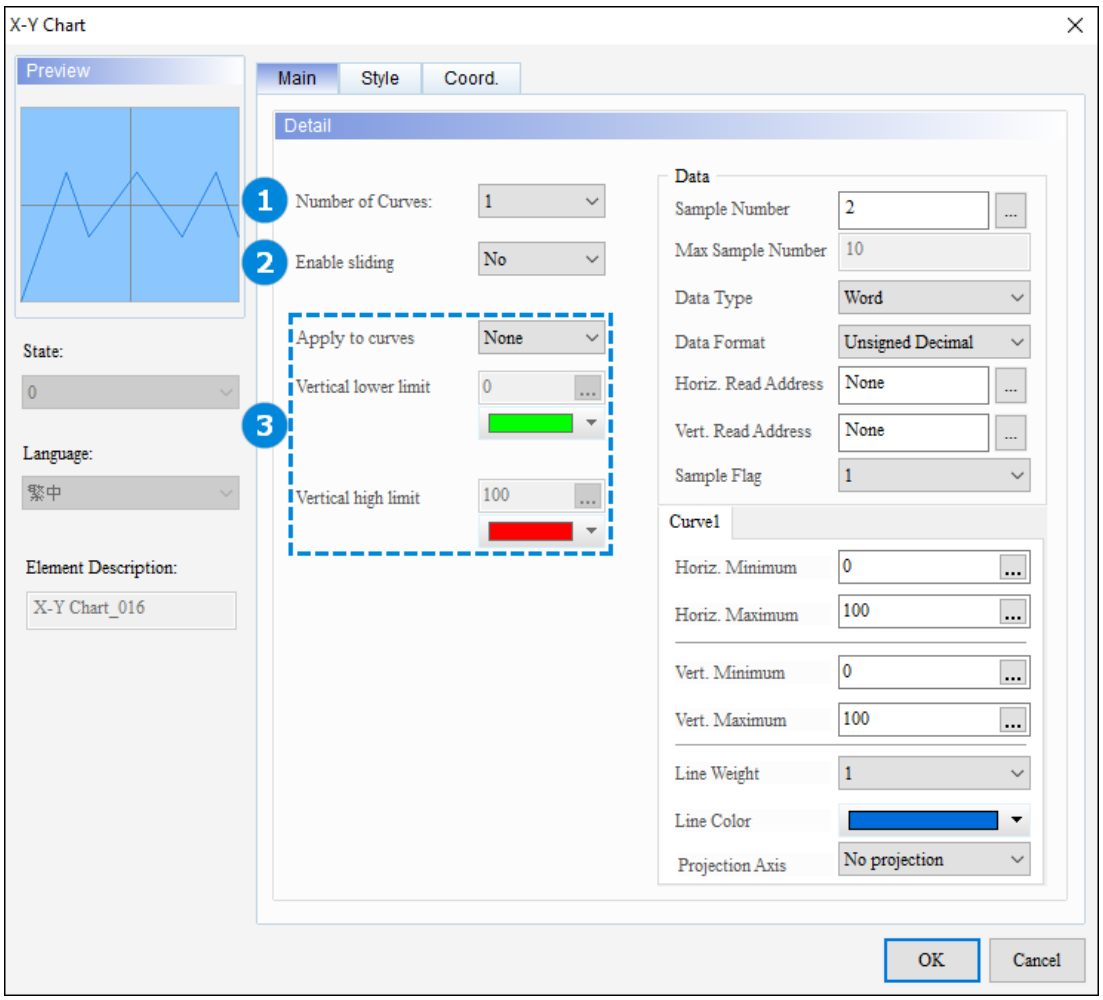
Press the **Sampling** button on the HMI, the **Trend Graph** will display each sampling point.



X-Y Chart

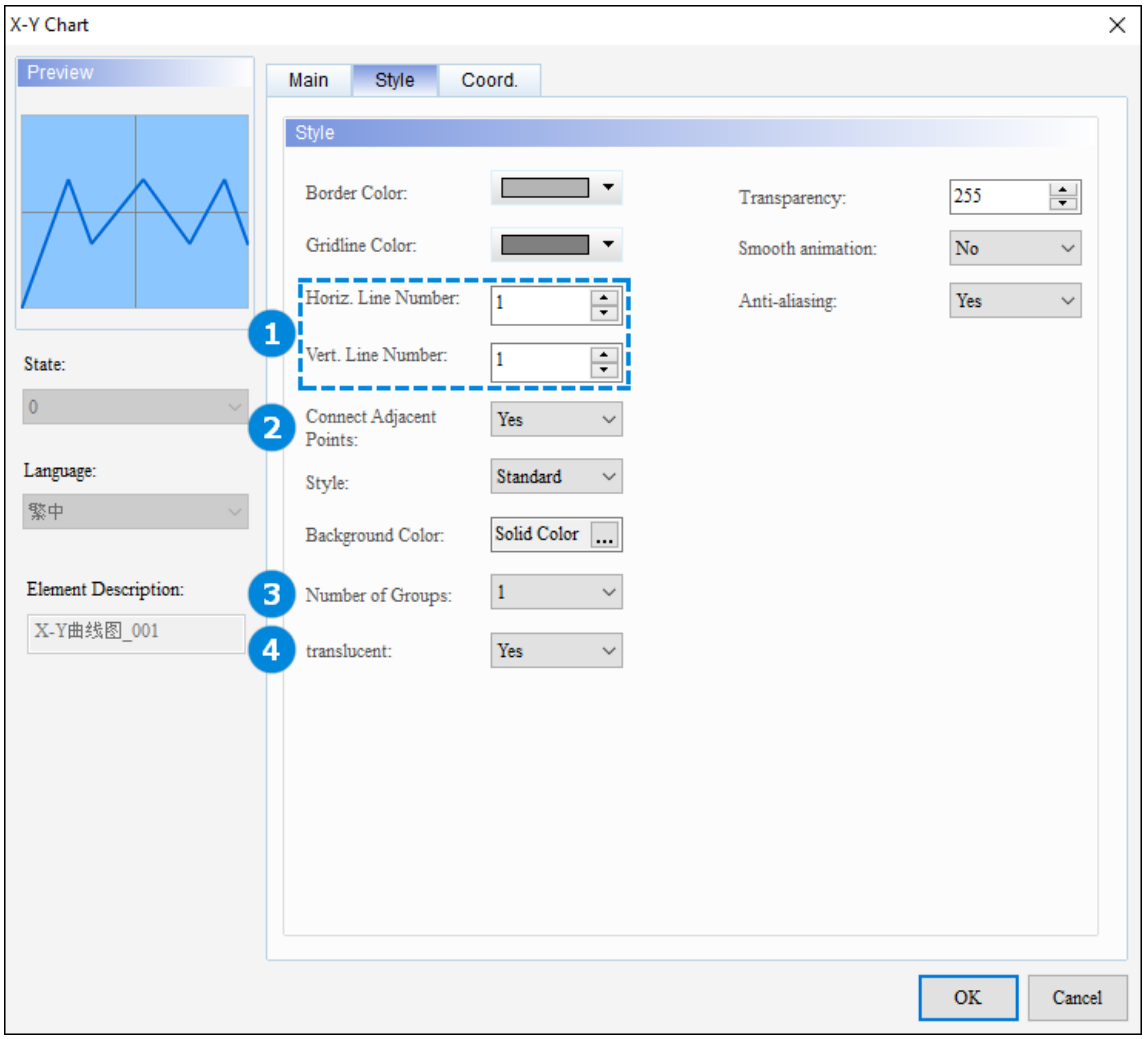
The following table lists the special properties of the **X-Y Chart** element with their description.

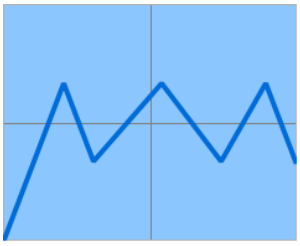
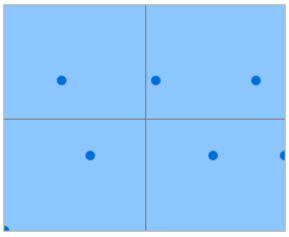
- **Main**

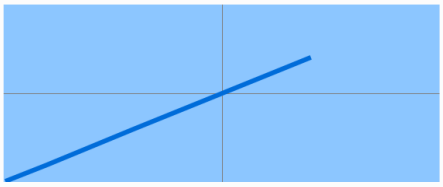
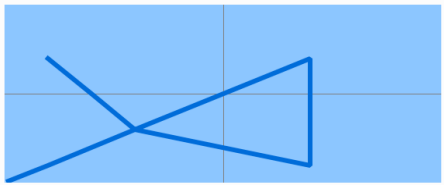
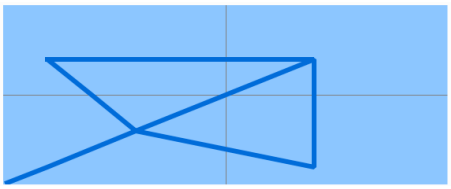
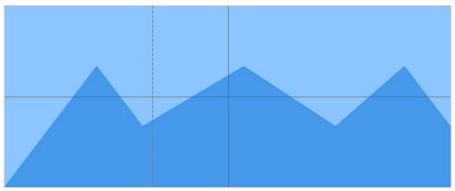
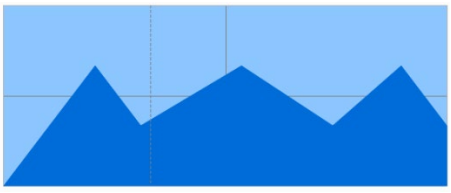


Legend	Setting	Description
❶	Number of Curves	Select the number of curves. Up to 8 curves can be set. You can set the properties of curves respective in the lower right of the element properties dialog.
❷	Enable sliding	Choose whether to enable sliding. If select **Yes**, after pressing the **X-Y Chart** on the HMI, the **X-Y Chart** will display the sampling value at the pressed position.
❸	High limit / Lower limit	Select the curve to which the upper and lower limits apply, and then set its upper and lower limits and display color.

- **Style**

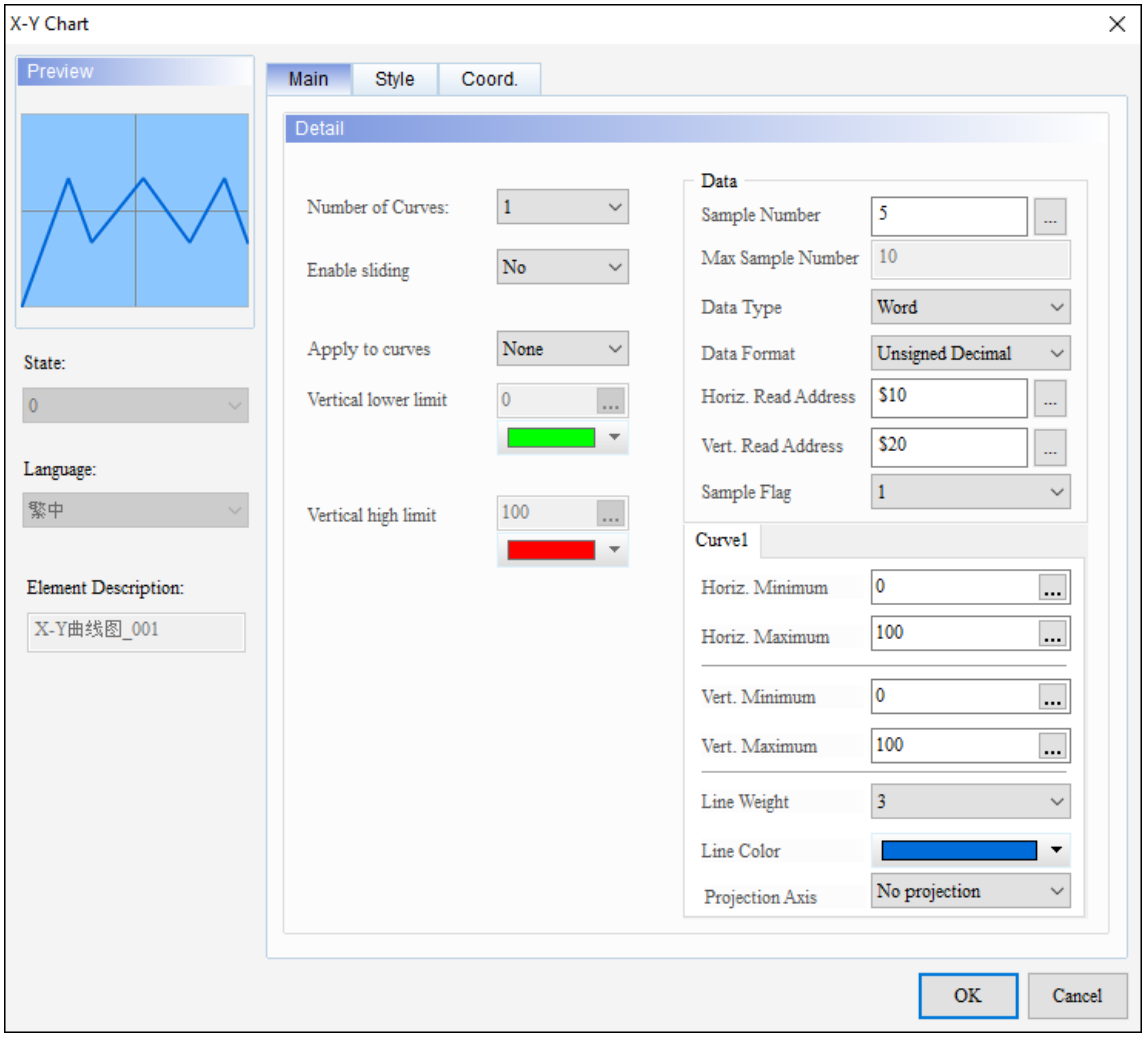


Legend	Setting	Description
1	Horiz. Line Number Vert. Line Number	Set the number of horizontal and vertical line to be displayed on the element. Support up to 99 lines.
2	Connect Adjacent Points	Choose whether to connect the sampling points. **Yes**:  **No**: 

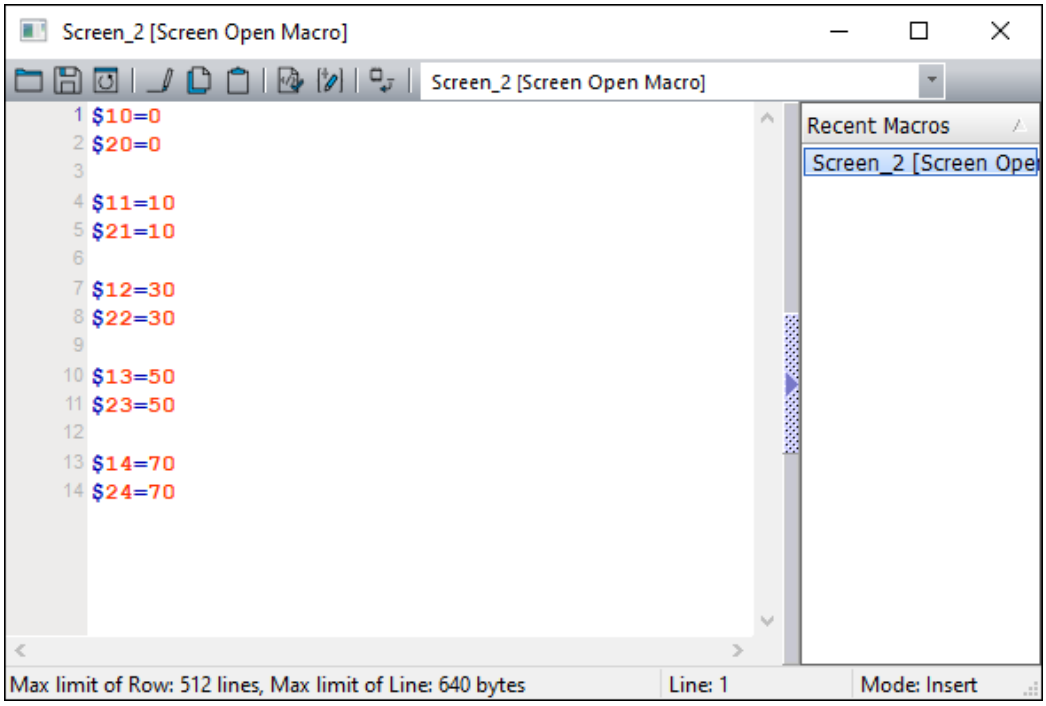
<table>
<tr><th>Legend</th><th>Setting</th><th>Description</th></tr>
<tr><td>❸</td><td>Number of Groups</td><td>Select the number of groups to be displayed on the element. Up to 10 groups can be selected.
Each time a sample is taken, a piece of data is recorded. The following table is an example of displaying 3 groups of statuses.
<table>
<tr><th>Group</th><th>State</th></tr>
<tr><td>1</td><td>  </td></tr>
<tr><td>2</td><td>  </td></tr>
<tr><td>3</td><td>  </td></tr>
</table></td></tr>
<tr><td>❹</td><td>Translucent</td><td>When setting the Projection Axis, select whether the graph is displayed in translucent.
<table>
<tr><td>Yes</td><td>  </td></tr>
<tr><td>No</td><td>  </td></tr>
</table></td></tr>
</table>

The following describes how to use the **X-Y Chart** element.

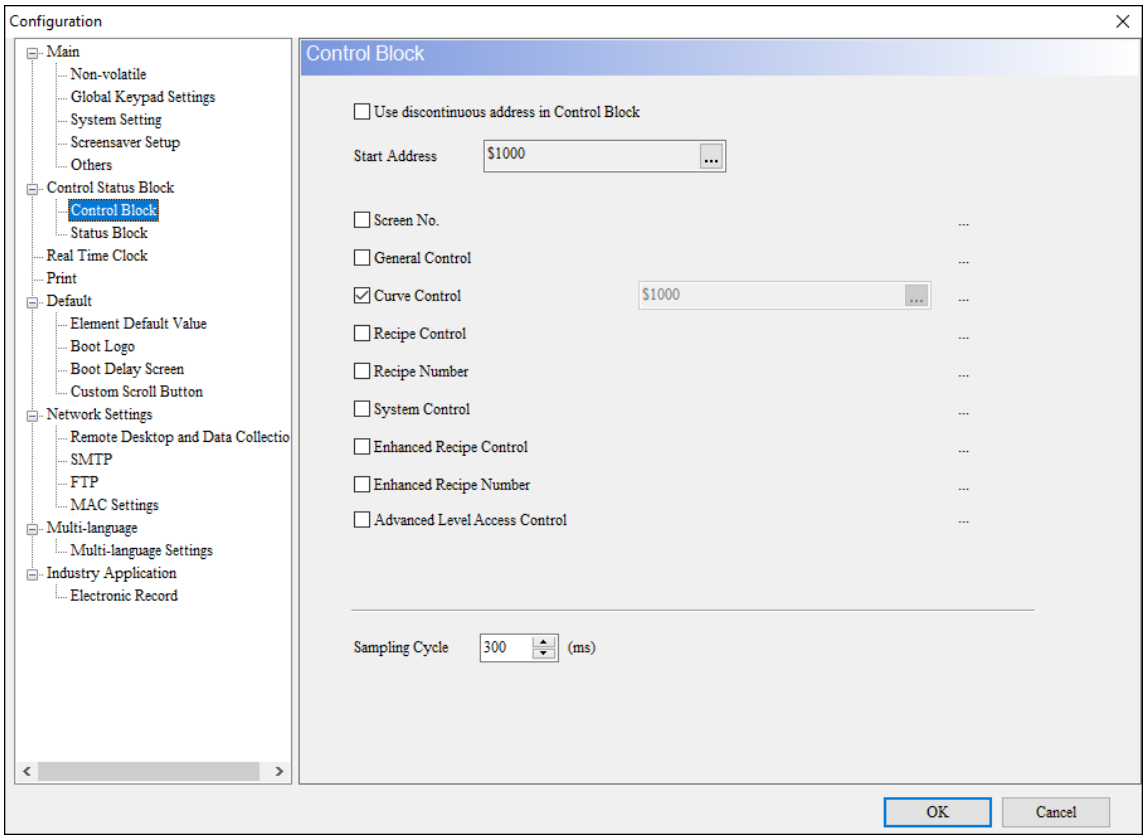
1. Create an **X-Y Chart** element, set the **Horiz. Read Address** to **$10** and the **Vert. Read Address** to **$20**, and then the **Sample Number** to **5**.



2. Create 10 **Numeric Entry** elements, set the **Write Address** to **$10~$14** and **$20~$24** separately, and then give these addresses an initial value in the **Screen Open Macro** dialog.



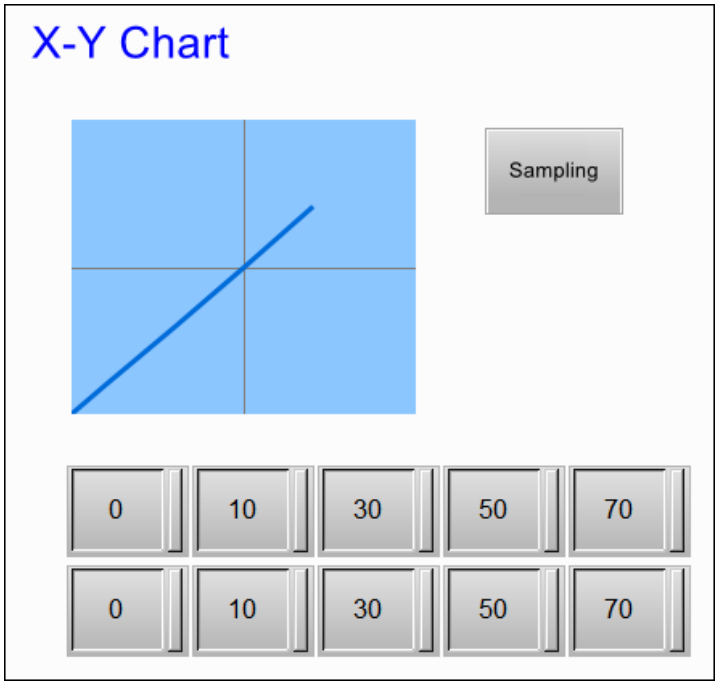
3. Go to **General** > **Configuration** > **Control Block** to set the address of the **Curve Control** to **$1000**.



4. Create a **Momentary** button element on the screen, and then set the **Write Address** to **$1000.0** 。

The execution result is as follows.

Press the **Sampling** button on the HMI, and the **X-Y Chart** will display each sampling point.

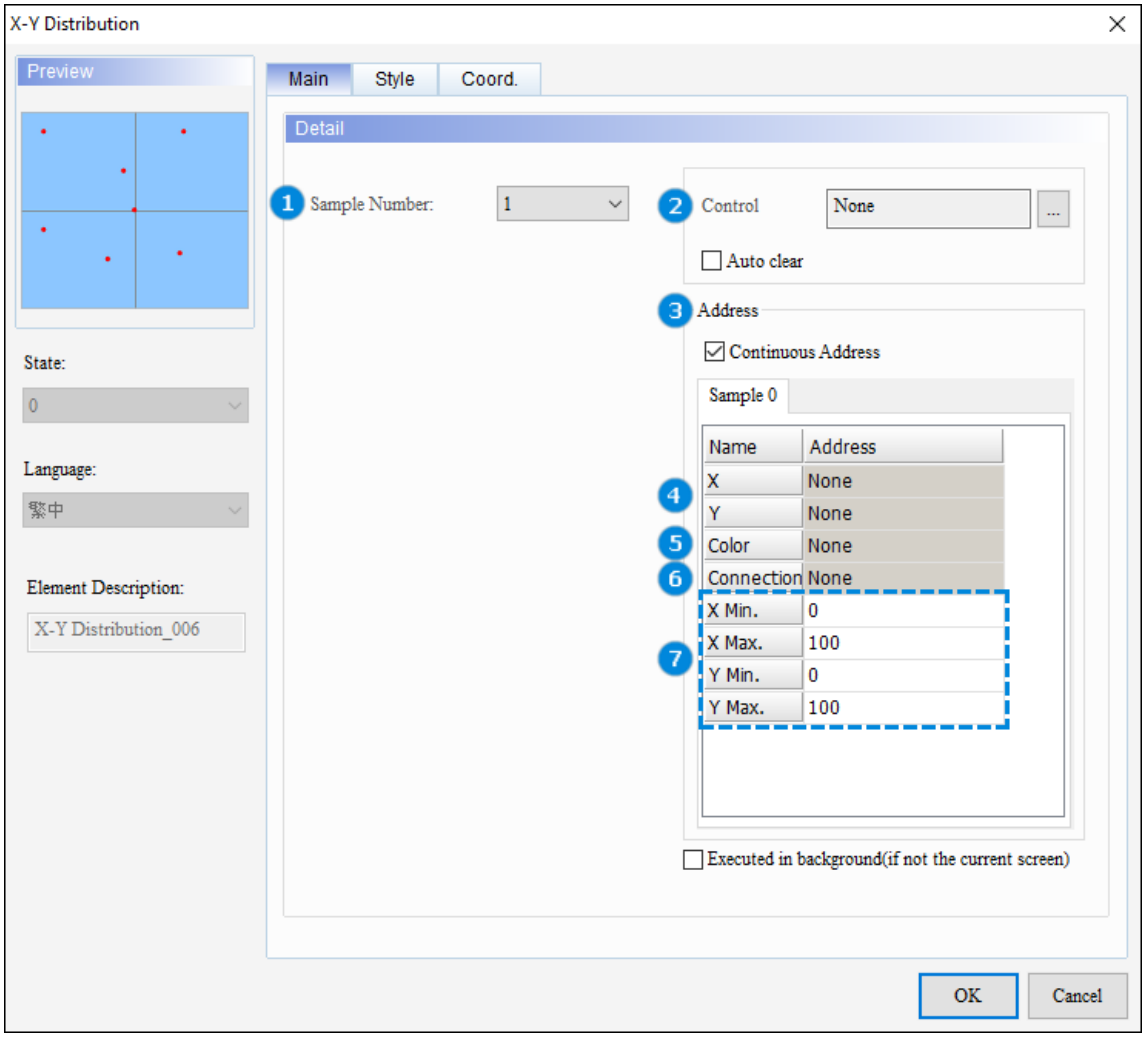


X-Y Distribution

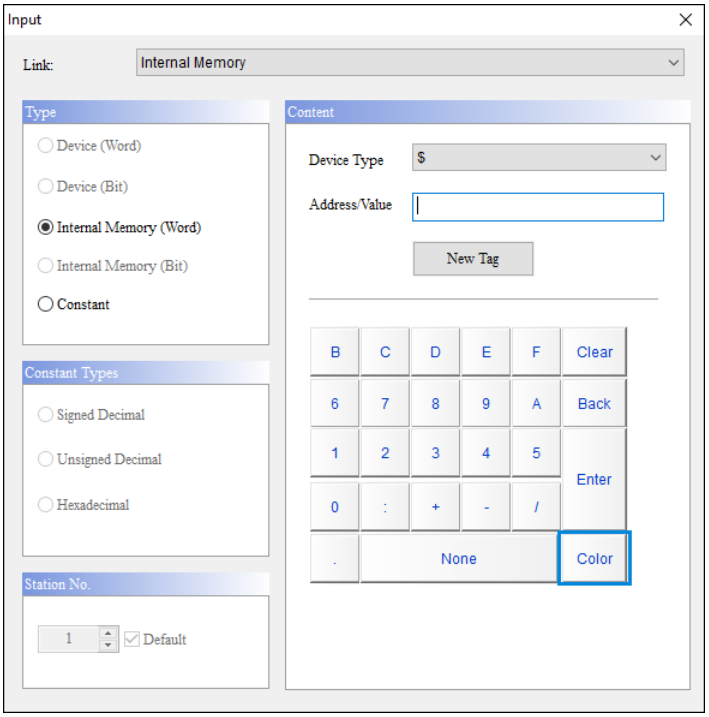
You can use the **X-Y Distribution** to respectively set the read address of the X coordinate and Y coordinate to draw sampling points at the corresponding positions on the element.

The following table lists the special properties of the **X-Y Distribution** element with their description.

- **Main**

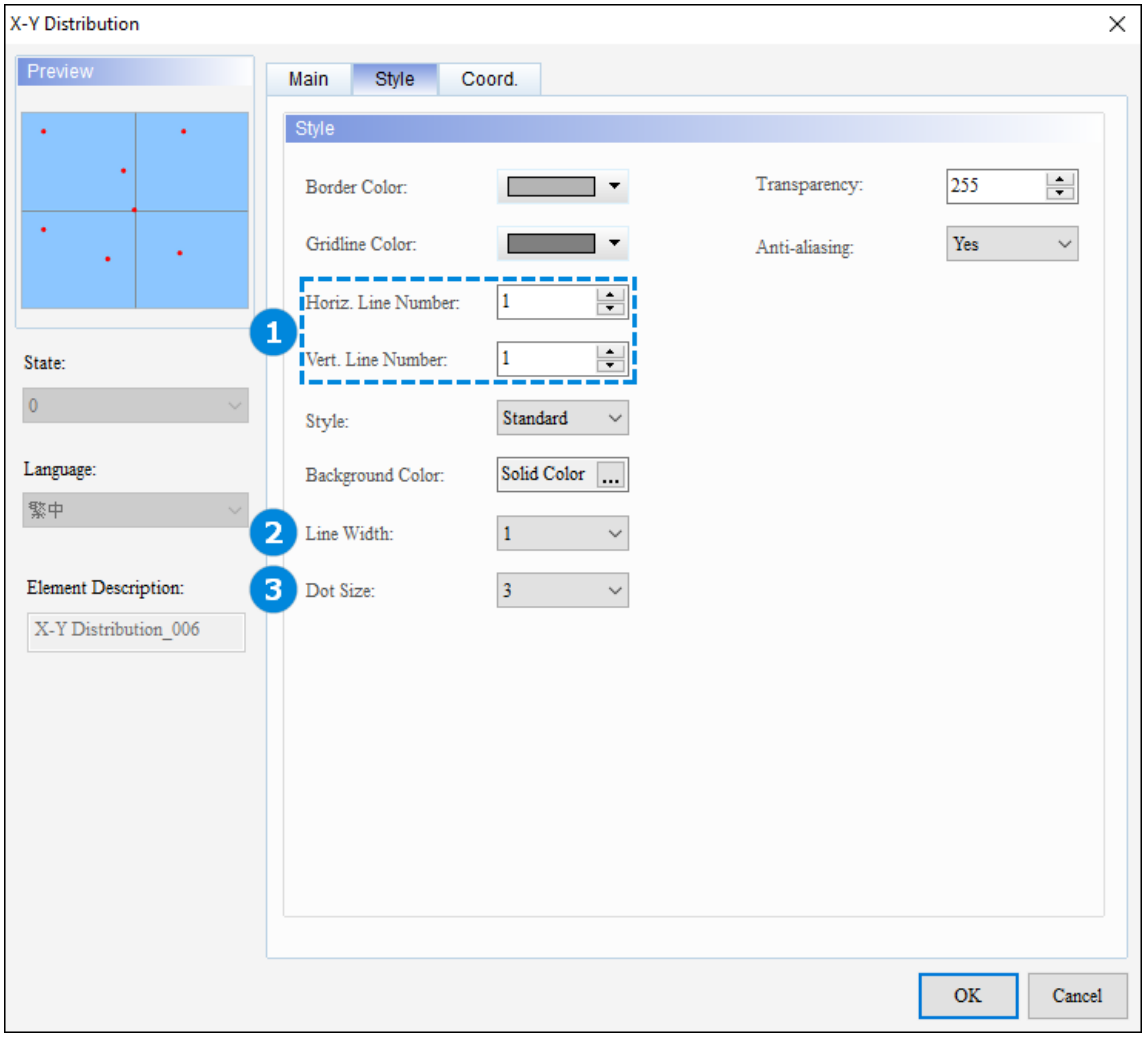


Legend	Setting	Description
1	Sample Number	Select the sample number. Up to 4 numbers of sampling can be set. You can set the properties of sampling respectively in the lower right of the element properties dialog.
2	Control	Set the control address for sampling. • Set the control flag address drawn by the element. Trigger Bit 0 to draw the sampling points and trigger Bit 1 to clear the sampling points. • Only one point will be drawn each time you trigger the Curve Control flag. To draw the second point, set this flag off and then trigger it again. Select the **Auto clear** checkbox to have the HMI automatically clear the flag.
3	Continuous Address	• If the **Continuous Address** checkbox is selected, **X**, **Y**, **Color**, and **Connection** are

Legend	Setting	Description
		continuous addresses, continuing the set X address, and the Y address is read-only. For example, when X address is set to $1000, the addresses of Y, Color, and Connection are $1001 to $1003. The address of Color or Connection can only be set as a constant value. • If the Continuous Address checkbox is not selected, you can set the address of **X**, **Y**, **Color**, and **Connection** individually. The address of Color and Connection can also be set as a constant value.
4	X / Y	Set the read address of X and Y coordinates.
5	Color	Set the read address of the color. You can also set it as a constant value, or directly click **Color** to select the sampling color. 
6	Connection	Set the read address or constant value of the connection. • 0: No connection. • 1: Connect the previous sampling point.
7	X Min. X Max. Y Min. Y Max.	Set the minimum and maximum values of the X / Y. The range value is 0–65535.

Legend	Setting	Description
⑧	Executed in background	If select, the **X-Y Distribution** can also execute sampling when you are operating on other screens. For example, if Screen 1 has the **X-Y Distribution** and Screen 2 does not have, after selecting this setting, execute sampling on Screen 2 and then switch back to Screen 1, the **X-Y Distribution** chart will continue to complete the sampling. **Note**: If **Auto clear** is not selected, you must execute sampling on other screens before setting the sampling flag to Off.

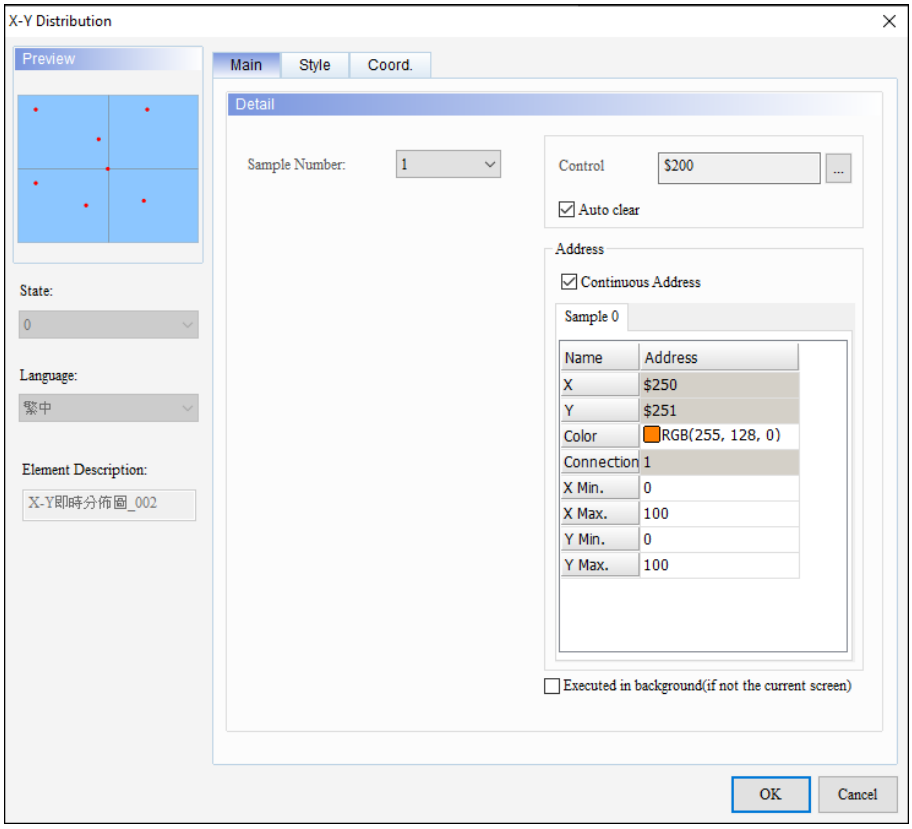
- **Style**



Legend	Setting	Description
❶	Horiz. Line Number Vert. Line Number	Select or enter the number of horizontal lines and vertical lines to be displayed on the element. Support up to 99 lines.
❷	Line Width	Select the width of the connection line. The range value is 1–8. **Note**: The line width takes effect when the **Connection** value is 1.
❸	Dot Size	Select the width of the sampling point. The range value is 1–8.

The following describes how to use the **X-Y Distribution** element.

1. Create a **X-Y Distribution** element on the screen, and then set the **Control**, **X / Y**, **Color**, and **Connection** addresses.

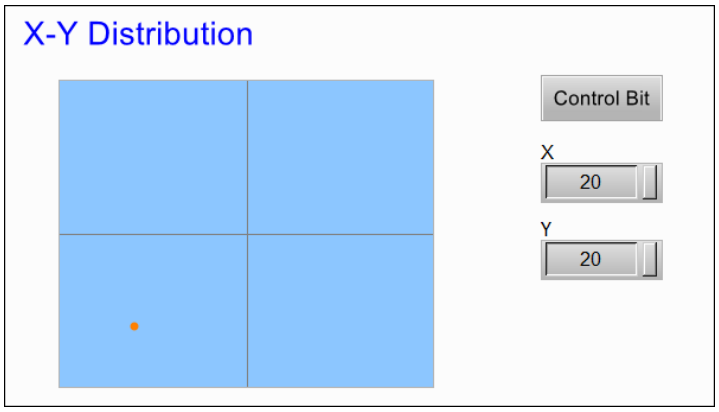


2. Create a **Momentary** button element, and then set the **Write Address** to **$200.0**.
3. Create 2 **Numeric Entry** element, and then set the **Write Address** to **$250** and **$251**.

4. Compile and download the screen to the HMI.

The execution result is as follows.

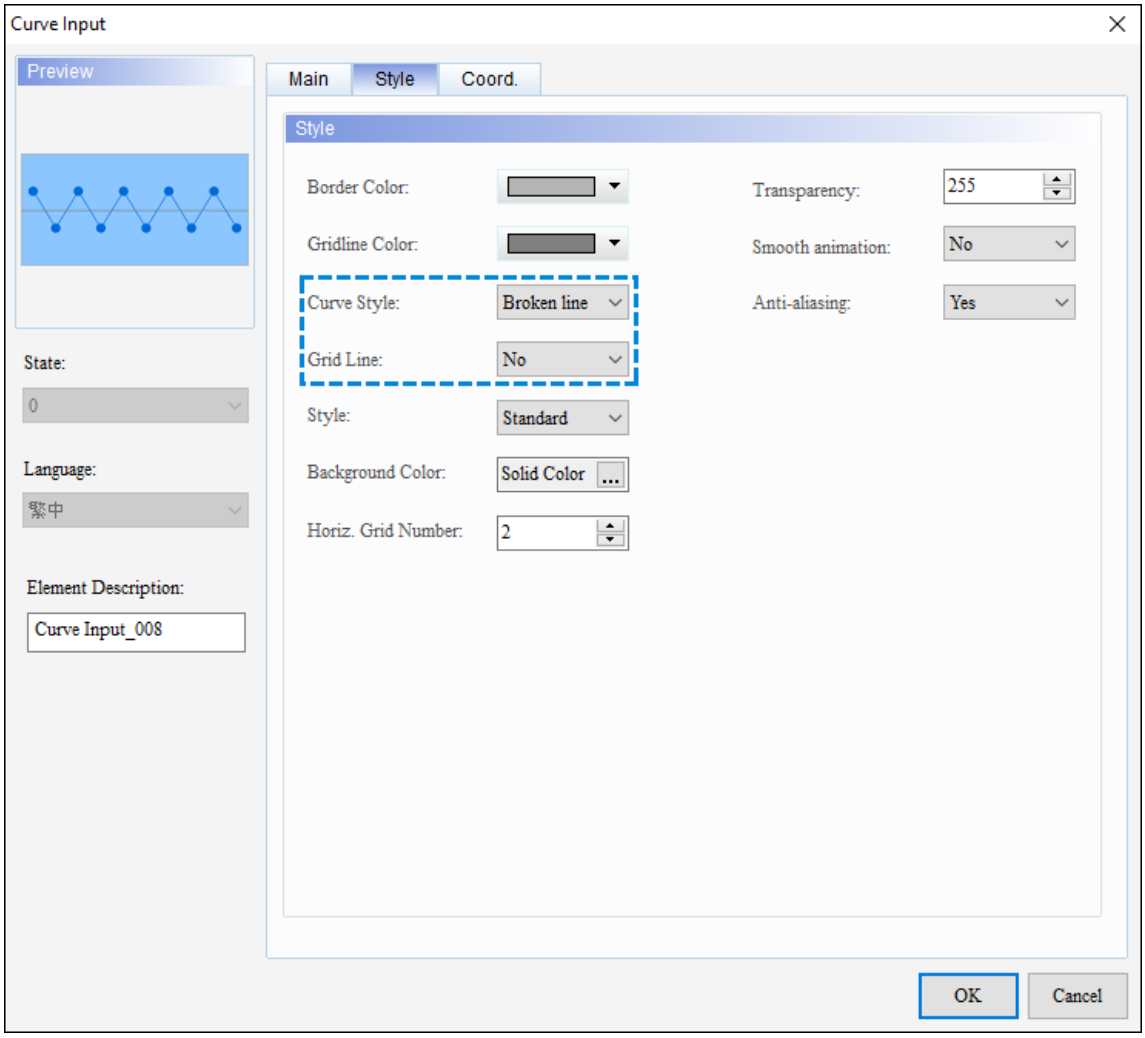
Enter X and Y values to the **Numeric Entry** element on the HMI screen, and then press the **Control Bit** button. The **X-Y Distribution** draws the corresponding sampling points.

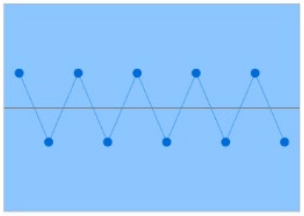
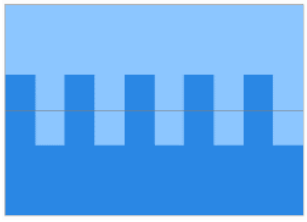


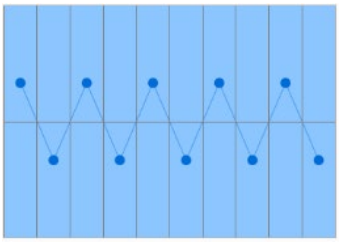
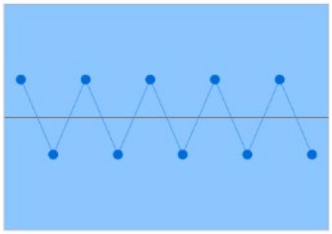
Curve Input

The following table lists the special properties of the **Curve Input** element with their description.

- **Style**

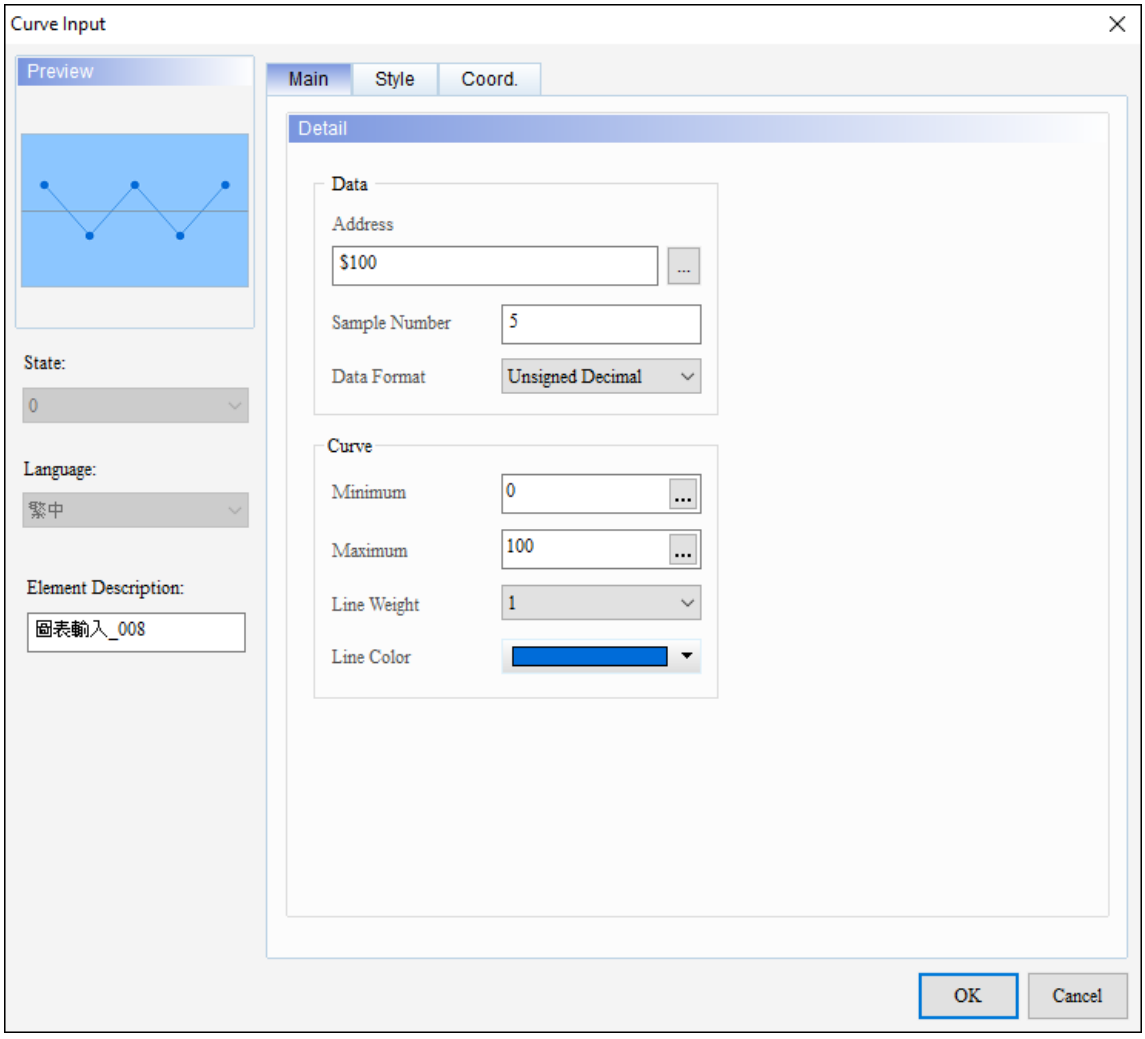


Legend	Setting	Description
❶	Curve Style	Select the curve style. **Broken line**:  **Block graphs**: 

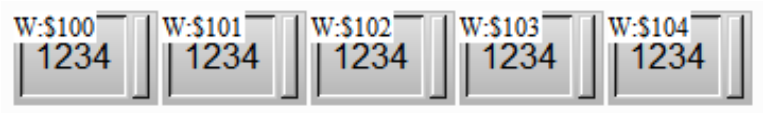
Legend	Setting	Description
②	Grid Line	Choose whether to display the grid line on the graph. **Yes**:  **No**: 

The following describes how to use the **Curve Input** element.

1. Create a **Curve Input** element, and then set the **Read Address** to **$100** and the **Sample Number** to **5**.



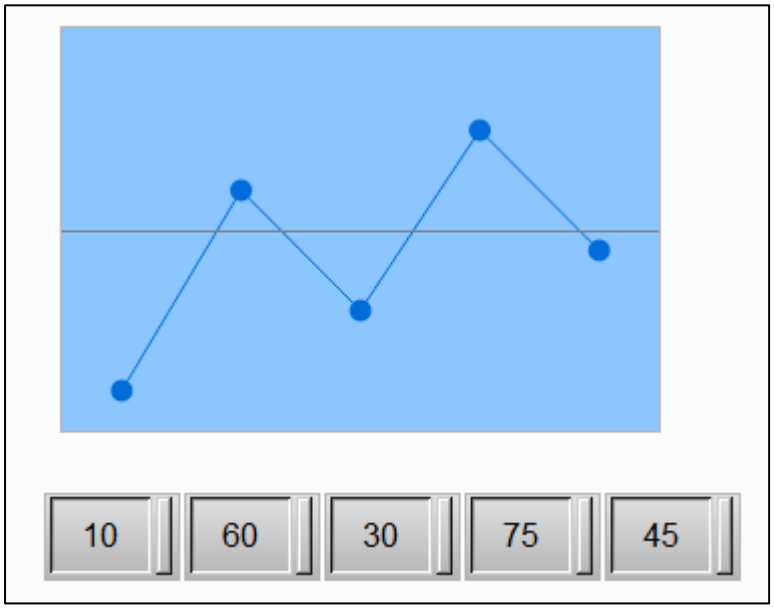
2. Create 5 **Numeric Entry** elements, set the **Write Address** to **$100~$104** separately.



3. Compile and download the screen to the HMI.

The execution result is as follows.

Enter the sampling value to the **Numeric Entry** element on the HMI, and the **Curve Input** element displays the curves.



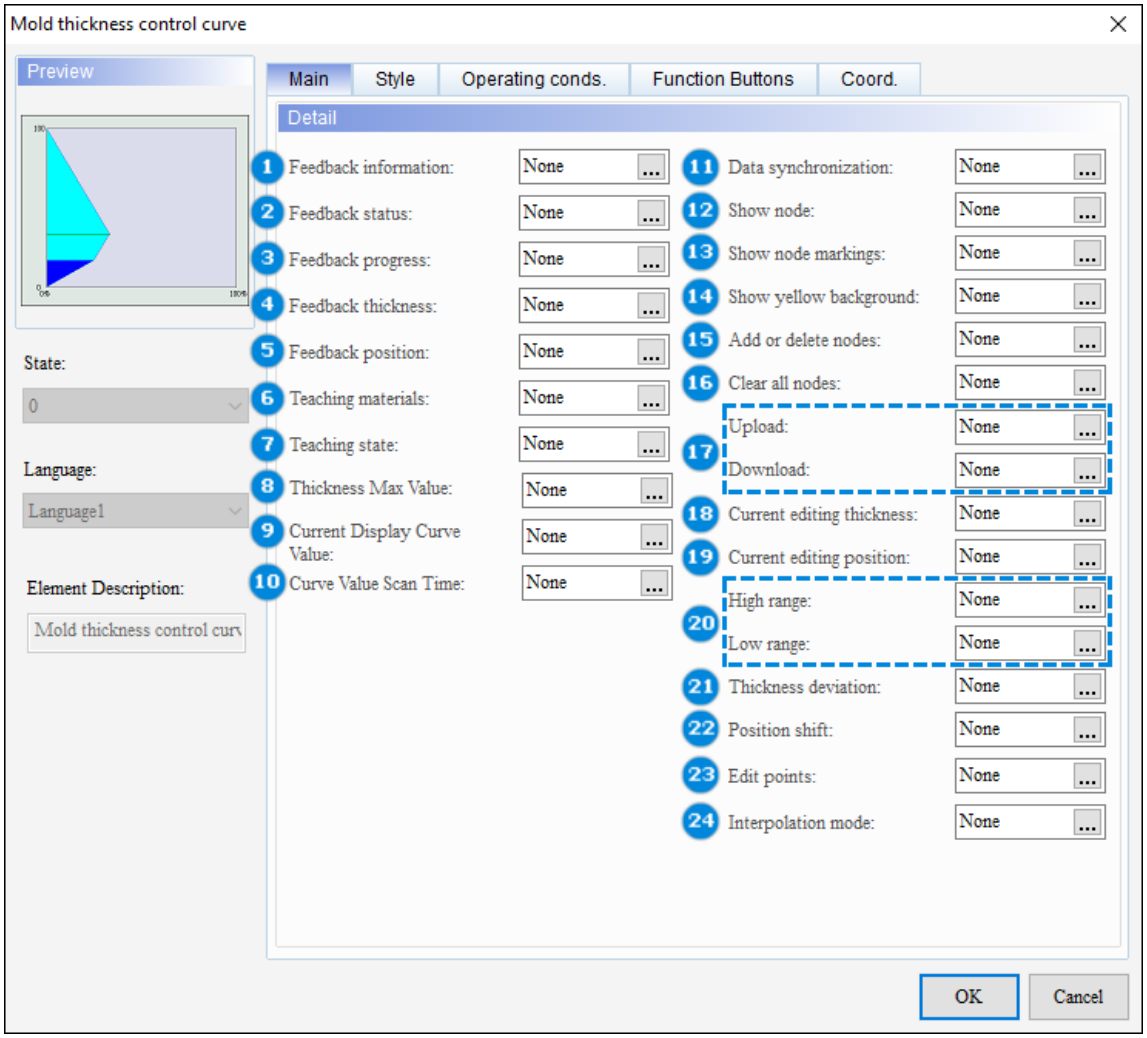
Mold Thickness Control Curve

The **Mold Thickness Control Curve** element is used to edit a curve on the HMI and store the data at the address specified by users. The software then draws the feedback line according to the status points data provided by users. After importing the data, the software will recalculate the curve according to the current parameter.

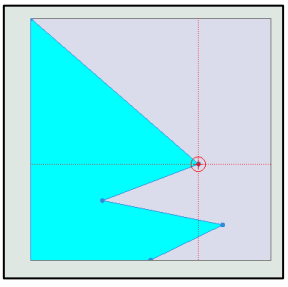
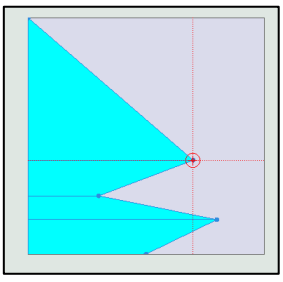
Note: This function is only supported in the AX series and IMP series.

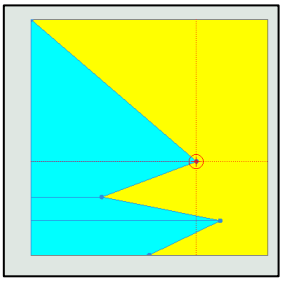
The following table lists the special properties of the **Mold Thickness Control Curve** element with their description.

- **Main**



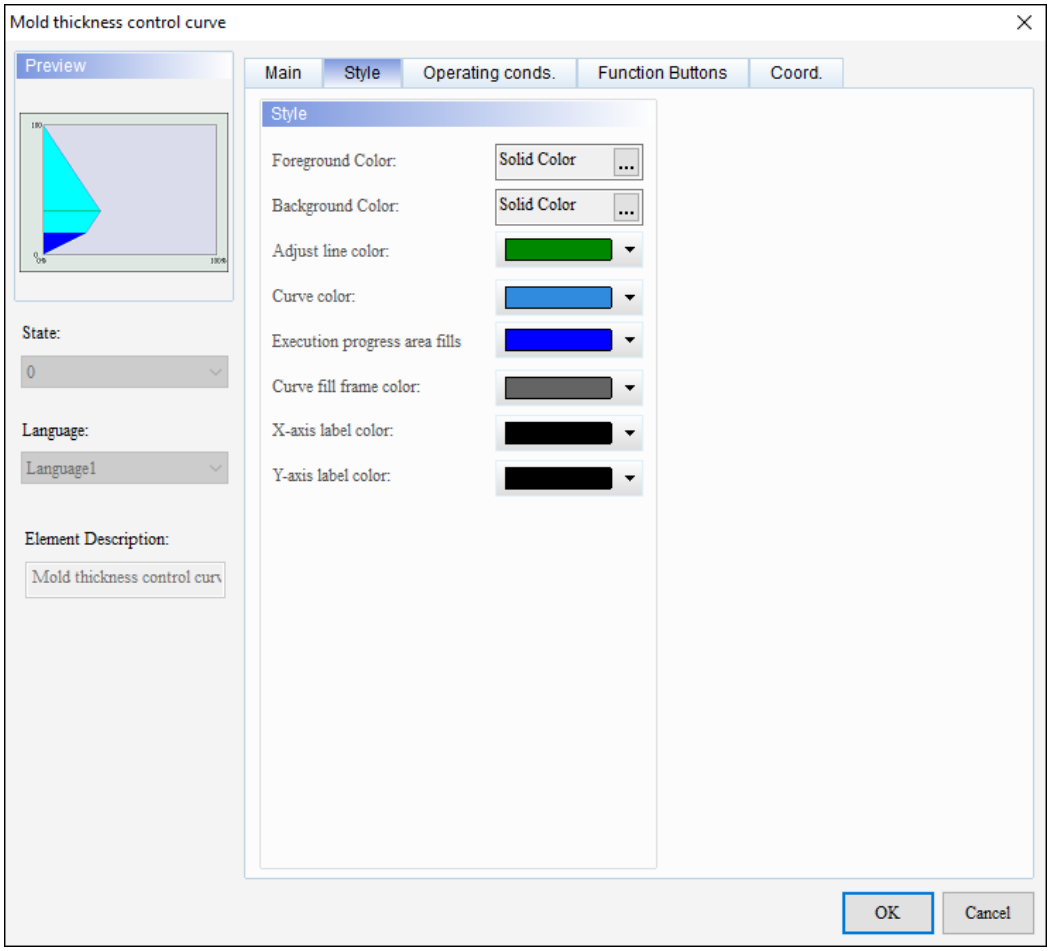
Legend	Setting	Description
1	Feedback information	Set the number of feedback information according to the edited points.

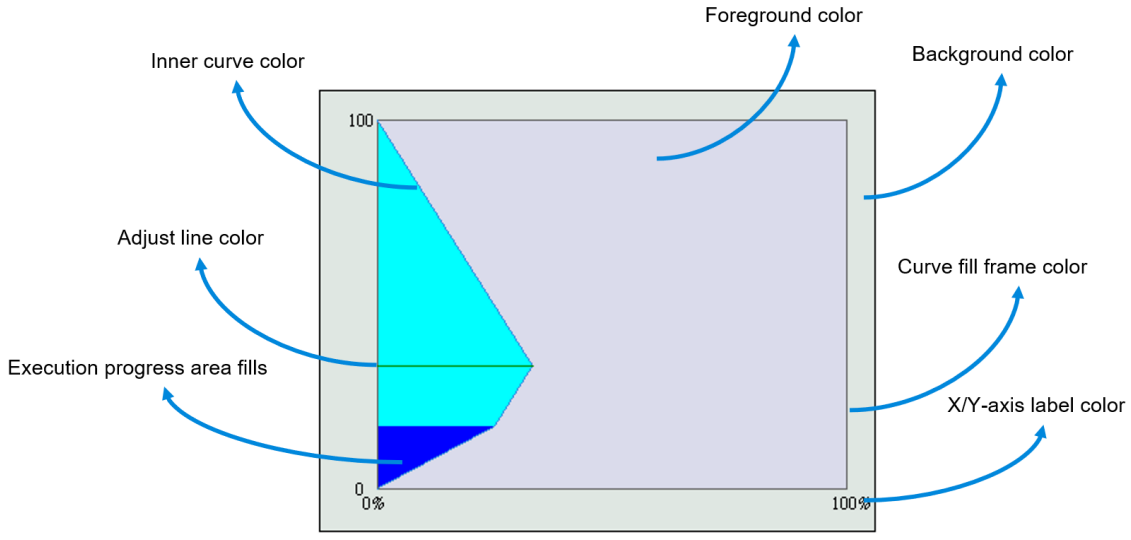
Legend	Setting	Description
②	Feedback status	Set the memory address to read the feedback status. The status contains Disable, Start, Pause, and Reset.
③	Feedback progress	Set the memory address to read the feedback progress. This setting is read-only.
④	Feedback thickness	Set the memory address to read the feedback thickness.
⑤	Feedback position	Set the memory address to read the feedback position.
⑥	Teaching material	Set the curve data stored on the HMI.
⑦	Teaching state	Set the state of curve data stored on the HMI. The state contains In action and Completed.
⑧	Thickness Max Value	Set the memory address to read the maximum value of thickness.
⑨	Current Display Curve Value	Set the memory address to record the curve value on the current display screen. The data format of each numerical value is Word, and the maximum number of records is 400.
⑩	Curve Value Scan Time	Set the memory address to control the value recording interval when the same controls are triggered.
⑪	Data synchronization	Set the synchronous data when the same element is on different screens.
⑫	Show node	Set to show all nodes set by users. 
⑬	Show node markings	Set to show the connection between nodes and the Y axis. 

Legend	Setting	Description
⑭	Show yellow background	Set to show a yellow background. It also means that the curve is being edited. 
⑮	Add or delete nodes	After confirming the current position and thickness, nodes can be added or deleted.
⑯	Clear all nodes	Clear all nodes on the curve, and the curve will also be cleared at the same time.
⑰	Upload Download	• Upload: Upload data from the **Teaching materials** to the element. • Download: Download the data to the **Teaching materials** after editing the curve from the element.
⑱	Current editing thickness	The currently edited thickness.
⑲	Current editing position	The currently edited position.
⑳	High range Low range	• High range: Set the highest point of the node in the curve, and other nodes will use this point as the reference point. When the node reaches 0 (%), the graph will not change, and it can be greater than 100 (%). • Low range: Set the lowest point of the node in the curve, and other nodes will use this point as the reference point. When the node reaches 0 (%), the graph will not change, and it can be greater than 100 (%).
㉑	Thickness deviation	Set to synchronously increase or decrease the thickness of all nodes. When the node reaches 0 (%), the graph will not change, it can be greater than 100 (%).
㉒	Position shift	Set to shift the curve upward or downward.
㉓	Edit points	Set the edit point. The available edit points are 100, 200, and 400.

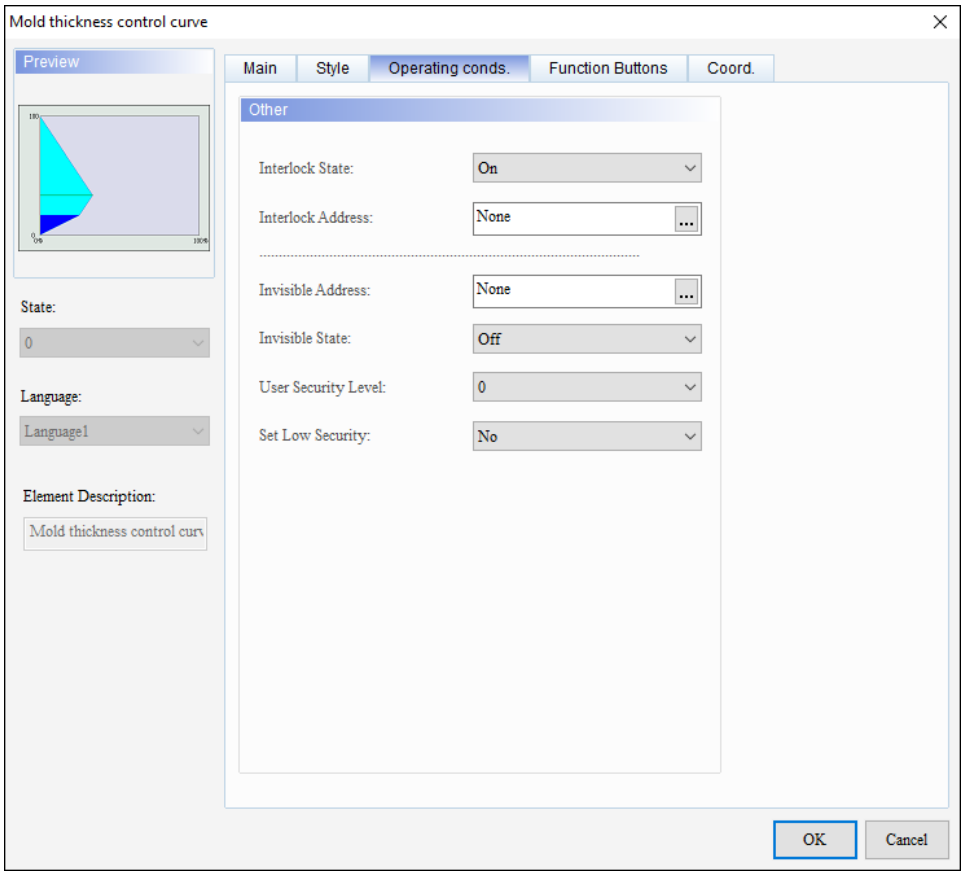
Legend	Setting	Description
24	Interpolation mode	• 0: Linear curve • 1: Spline curve

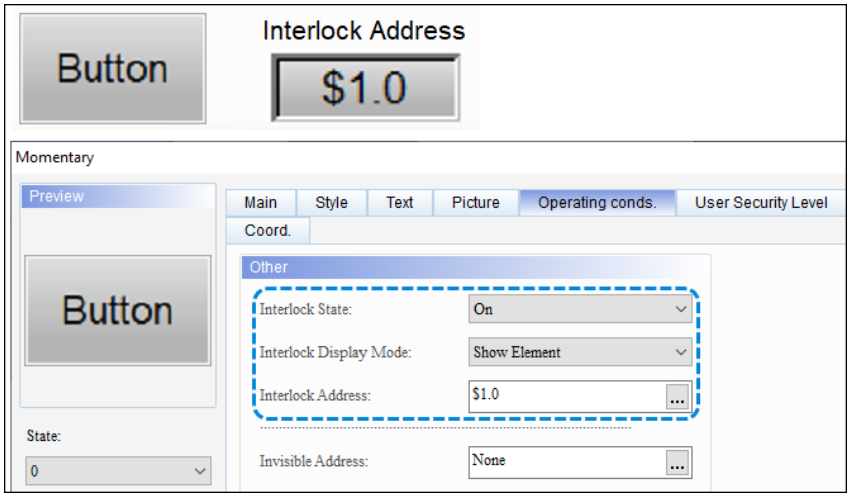
- **Style**

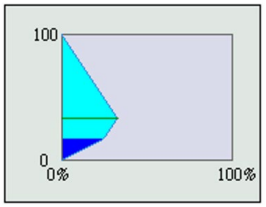




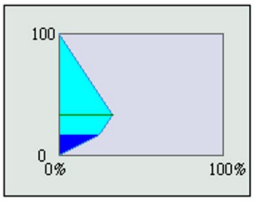
- **Operating conds.**



Setting	Description
Interlock State	The **Interlock State** enables to operation of a certain element and must be used with **Interlock Address**. If the **Interlock State** is set to **Off**, the **Interlock Address** enables to operation of a certain element. Instead, if the **Interlock State** is set to **On**, the **Interlock Address** enables to operation of a certain element. As shown in the following figure, when the Interlock Address $1.0 is On, the **Maintained** button can be operated.
Interlock Address	

<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Invisible Address
Invisible State</td><td>
The Invisible Address displays or hides the element on the HMI. If the Invisible State is set to On, the element will be hidden when the Invisible Address is ON. Instead, if the Invisible State is set to Off, the element will be hidden when the Invisible Address is OFF.
The following is an example of using invisible address.
1. Create a button element on the screen and set its Read Address to $9.0, then set the Invisible Address of the element to hide to $9.0.
2. To hide the element, you must first click the $9.0 button.
<table>
<tr><td>Invisible State is On</td><td> 

Invisible Address
$9.0 OFF

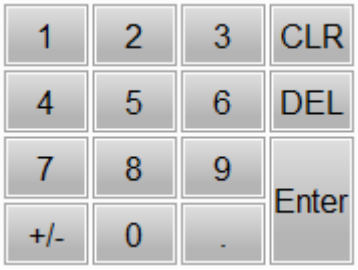
Element disappear
Invisible Address
$9.0 ON</td></tr>
<tr><td>Invisible State is Off</td><td> 

Invisible Address
$9.0 ON

Element disappear
Invisible Address
$9.0 OFF</td></tr>
</table>
</td></tr>
<tr><td>User Security Level</td><td>Select the user security level.
When pressing the element on the HMI, the password enter dialog will appear if the current security level is not sufficient. Log in with higher user security level, the element action will be performed.
Note: When the user security level is set to 0, you do not need to log in to operate.</td></tr>
<tr><td>Set Low Security</td><td>Choose whether to set low security after entering the security.
If set to Yes, the HMI automatically sets the security level to the lowest (level 0) after the element is triggered.
Afterwards, every time the element is triggered, the login screen will pop up, and the action will be executed only after logging in.</td></tr>
</table>

Keypad

DIAScreen provides three types of **Keypad** elements for configuring the **Keypad Screen**. If you select the **Custom Keypad** in the **Entry** element, you can set a custom keypad screen. After clicking the **Entry** element, the corresponding keypad will appear.

Keypad	Description
Keypad(1)	A Decimal format keypad. 
Keypad(2)	A Hexadecimal format keypad.
Keypad(3)	An ASCII format keypad.

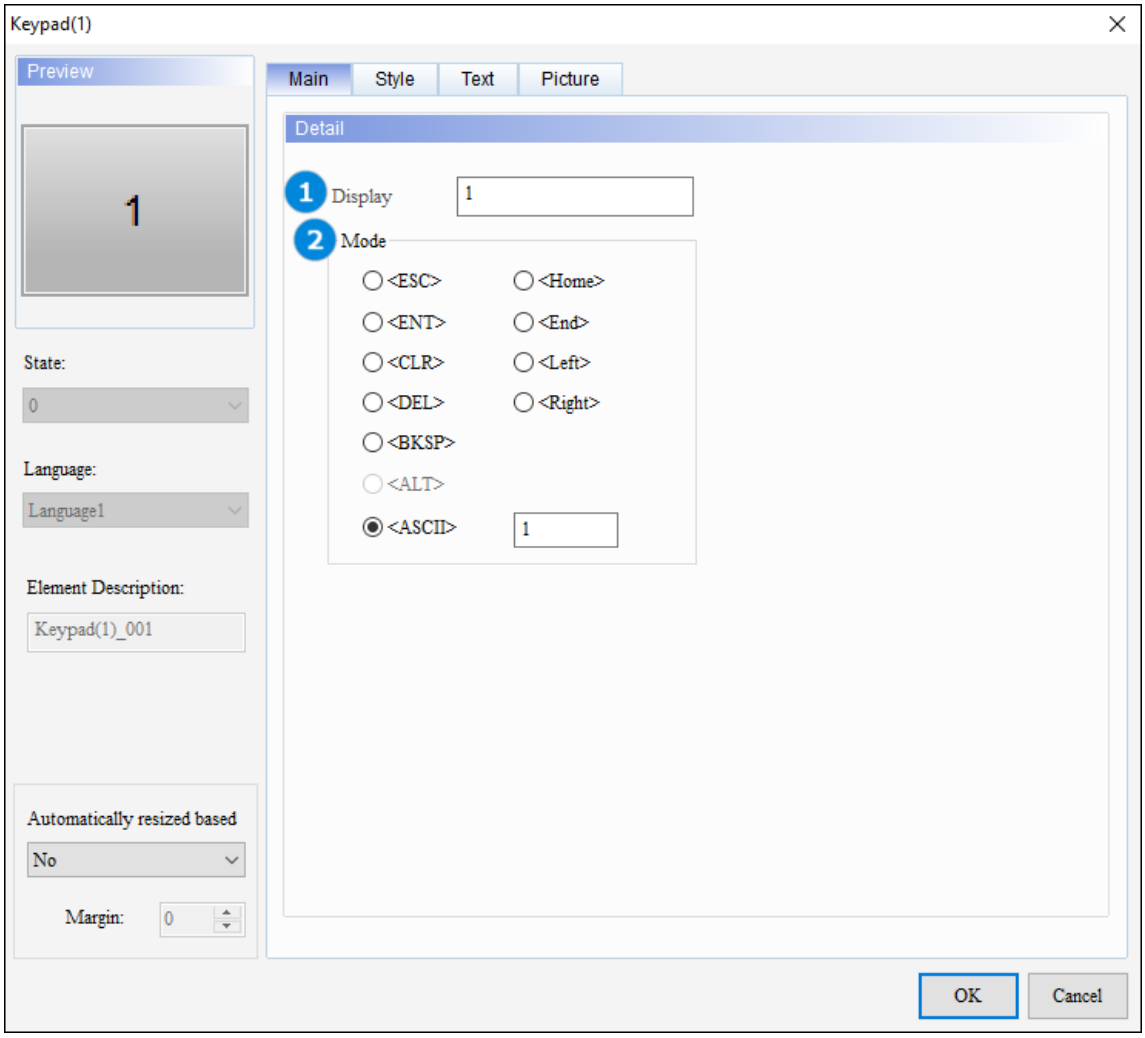
The **Keypad** element is a grouped element. You can directly double-click the element to open its properties dialog and make settings. To rearrange the position of the buttons, right-click the element and select **UnGroup**, and the position of the buttons can be changed.

Common Properties

If **Automatically resized based** is set to **Yes**, when the set text length exceeds the element size, the element is automatically resized according to the **Margin** to show the complete text.

Note: This setting supports the element enlargement function, the string is shortened, and the element will not become smaller.

- **Main**

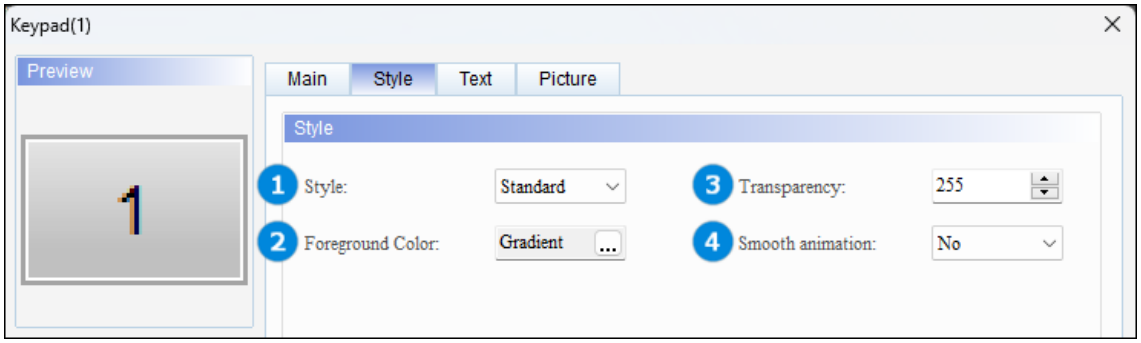


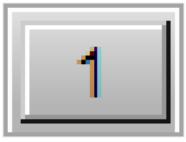
Legend	Setting	Description
1	Display	Enter the display text for the keypad buttons.
2	Mode	elect the execution mode of the keypad buttons.

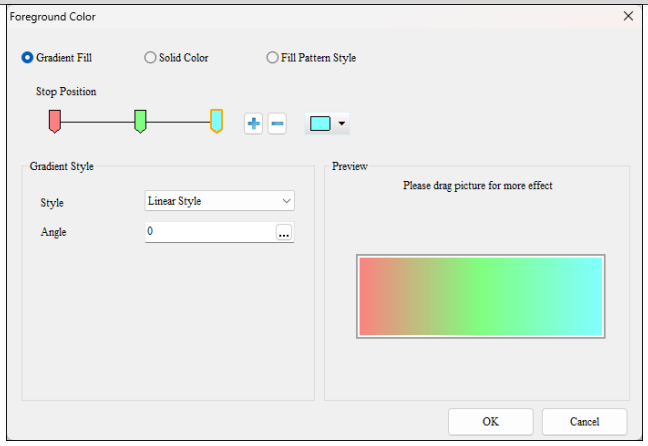
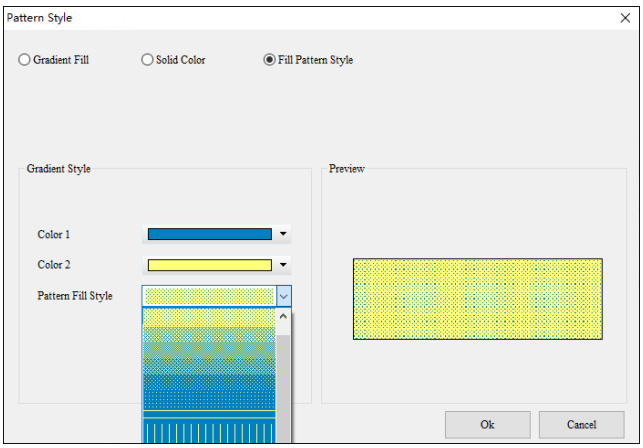
Mode	Description
[ESC]	Cancel the entry. If the **Keypad** element is on a sub-screen, the sub-screen will also be closed when executing ESC.

Legend	Setting	Description
		[ENT] Input the entry. [CLR] Clear a string of characters. [DEL] Delete a single character after the input cursor. [BKSP] Delete a single character before the input cursor. [Home] Move the input cursor to the beginning of that line. [End] Move the input cursor to the end of that line. [Left] Move the input cursor to the left by one character. [Right] Move the input cursor to the right by one character. [ASCII] Specify the input code.

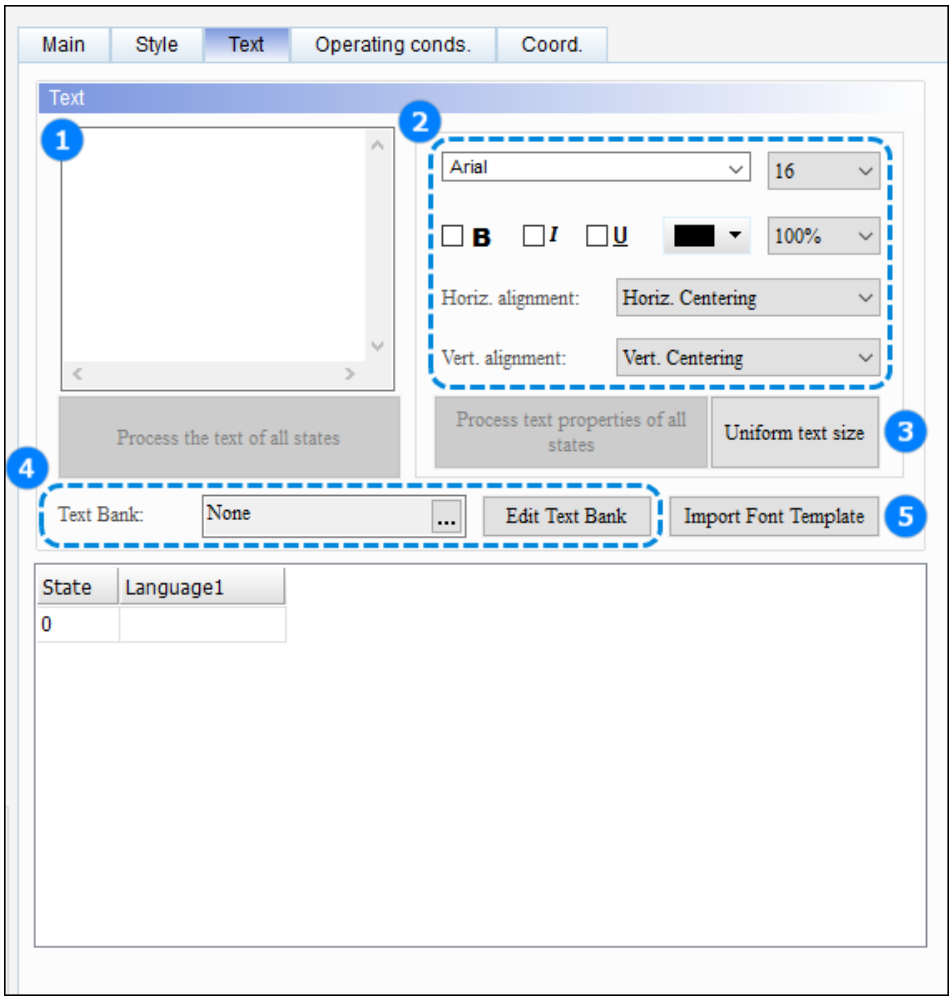
- **Style**



Legend	Setting	Description
1	Style	Select the appearance of the element. The default value is **Standard**. **Standard** **Raised** 
2	Foreground Color	Select the color of the element. • **Gradient Fill:** Set the gradient style by setting the current angle or center point. You can drag the mouse on the preview image to set the current angle or center point, or set the memory address to dynamically configure it on the HMI. You can also set the stop point colors to present the effect of multi-color gradients.

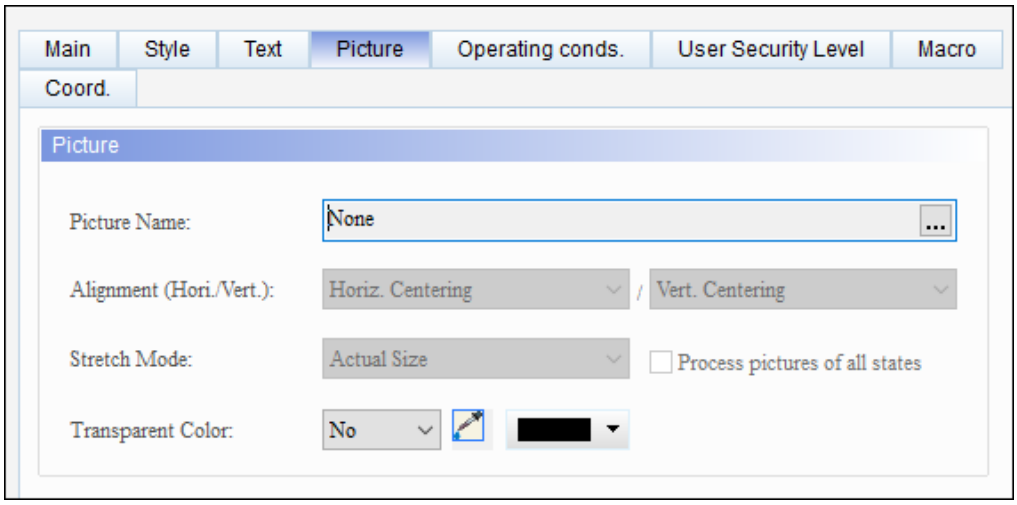
Legend	Setting	Description
		 **Note**: Up to 5 stop points can be set. • **Solid Color**: Select a single-color fill. • **Fill Pattern Style**: Select a two-color combination configuration and select a pattern fill style. 
③	Transparency	Select or enter the transparency of the element. The range value is 50–255 and the default value is **255**. The smaller the value, the higher the transparency of the element.
④	Smooth animation	Choose whether to enable the smooth animation function. If select **Yes**, the element display becomes smoother.

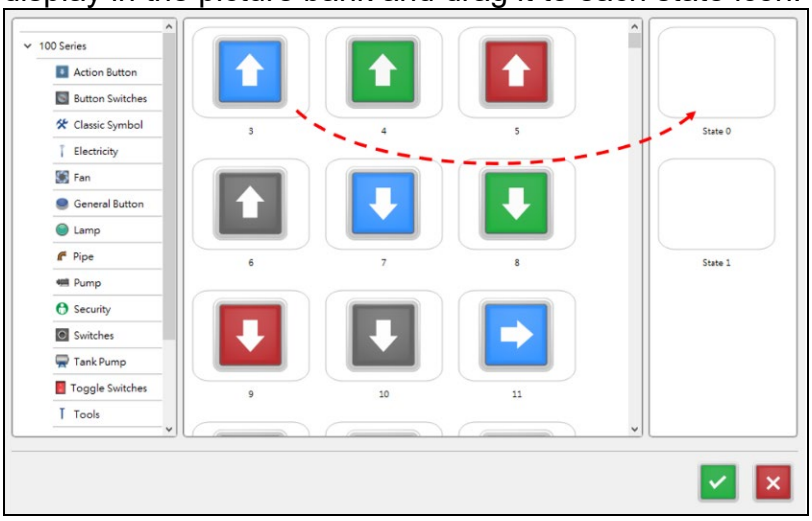
- **Text**



Legend	Setting	Description
1	Text	Enter the text message to display in the text box. Select the state / language table below to directly set the text displayed by that state / language. Enter the text message to be displayed in the text box. Click the element on the screen and press **Space** to edit the text content of the current state / language.
2	Text Property	Set the text properties.
3	Uniform text size	Changes the text size in all languages with one click according to the currently selected state.
4	Text Bank	Applies the text in the **Text Bank** to the content of each state / language. You can also click **Edit Text Bank** to edit the text content in the **Text Bank** editor.
5	Import Font Template	Click to open the **Font Template** dialog, and then select the template and import the text properties of the selected state.

- **Style**



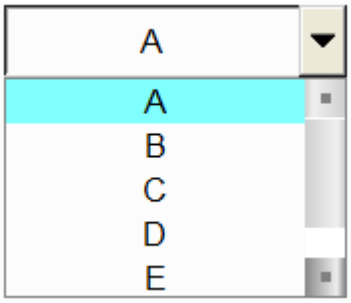
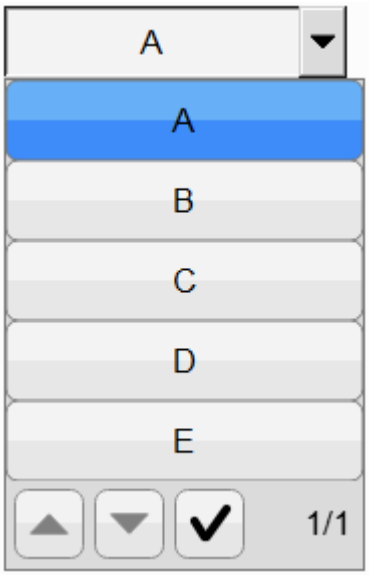
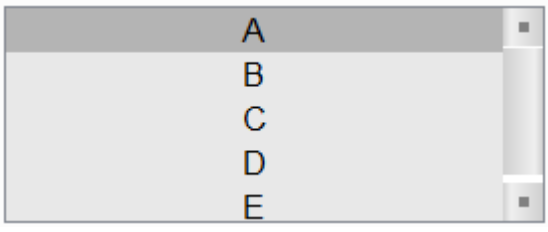
Setting	Description
Picture Name	Enter the picture name. The default value is **None**.To set the picture display, click [...] to select the picture to display in the picture bank and drag it to each state icon.  To configure other pictures, refer to Picture Bank.
Alignment	Select the alignment of the picture.
Stretch Mode	Select the stretch mode of the picture.

Stretch All	Stretch 1:1	Actual Size
The picture fills the full element display area. 	The picture displays in 1:1 size based on the element width and length. 	The picture is displayed in the display area of the element in its actual size.

Setting	Description
Process pictures of all states	If an element has multiple state values, and its picture may not fill the full display area of the element. Select this checkbox to process all pictures settings
Transparent Color	Choose whether to specify a color in the picture as a transparent color. If set to **Yes**, you can select colors or use the color picker to select one of the colors on the picture to specify it as a transparent color, that is, the foreground color. The following figures show the effect of specifying white as a transparent color.

List

Use different types of List elements to switch corresponding memory status. The following table lists the description of each List element.

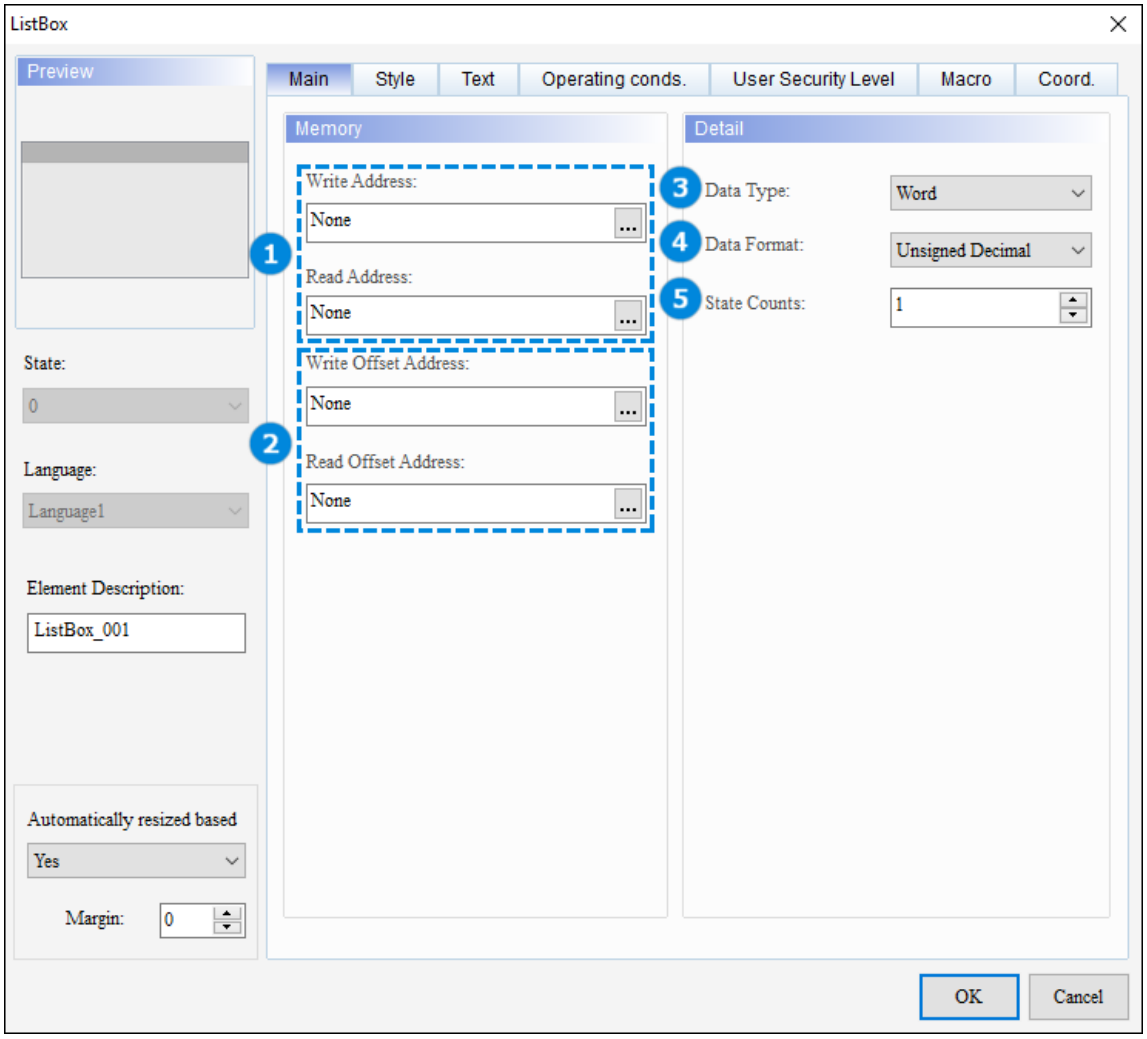
List	Description
ComboBox	The items of the list are displayed in a drop-down list. After selecting an item on the HMI, write the status value to the corresponding memory address. 
Drop-down Menu	Set the number of display item in the drop-down menu. On the HMI, you can use the up and down buttons to switch list pages. After selecting the item, press ✔ to write the status value to the corresponding memory address. 
ListBox	The items of the list are directly displayed on the element. After selecting an item on the HMI, write the status value to the corresponding memory address. 

Common Properties

- **Main**

  If **Automatically resized based** is set to **Yes**, when the set text length exceeds the element size, the element is automatically resized according to the **Margin** to show the complete text.

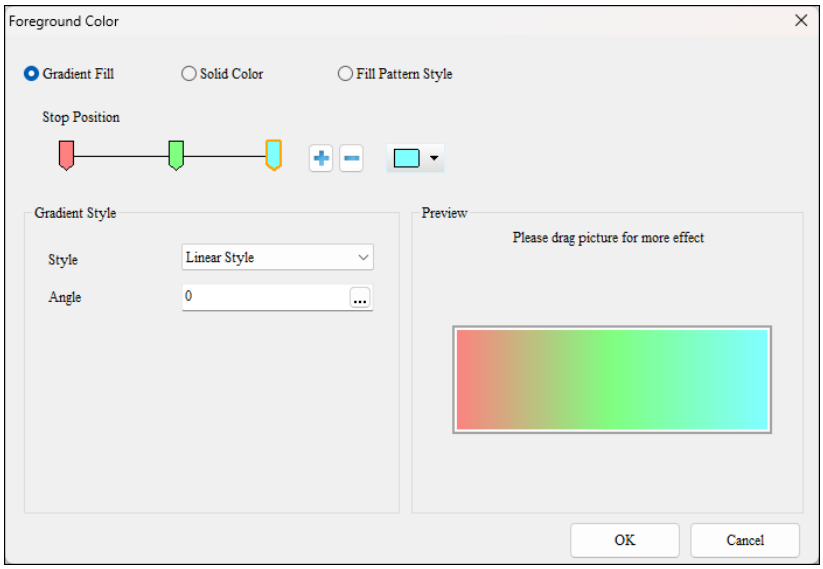
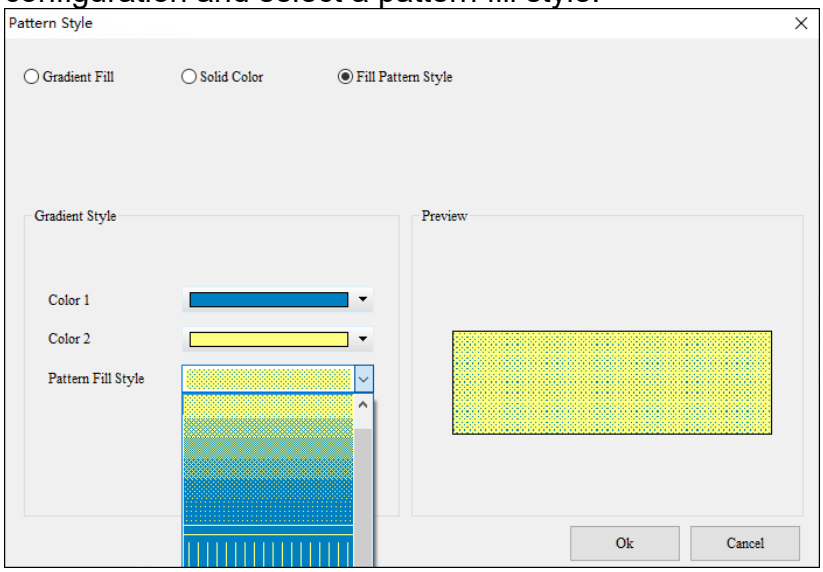
  Note: This setting supports the element enlargement function, the string is shortened, and the element will not become smaller.



Note: The following property descriptions only apply to the ComboBox, Drop-down Menu, and **ListBox** elements.

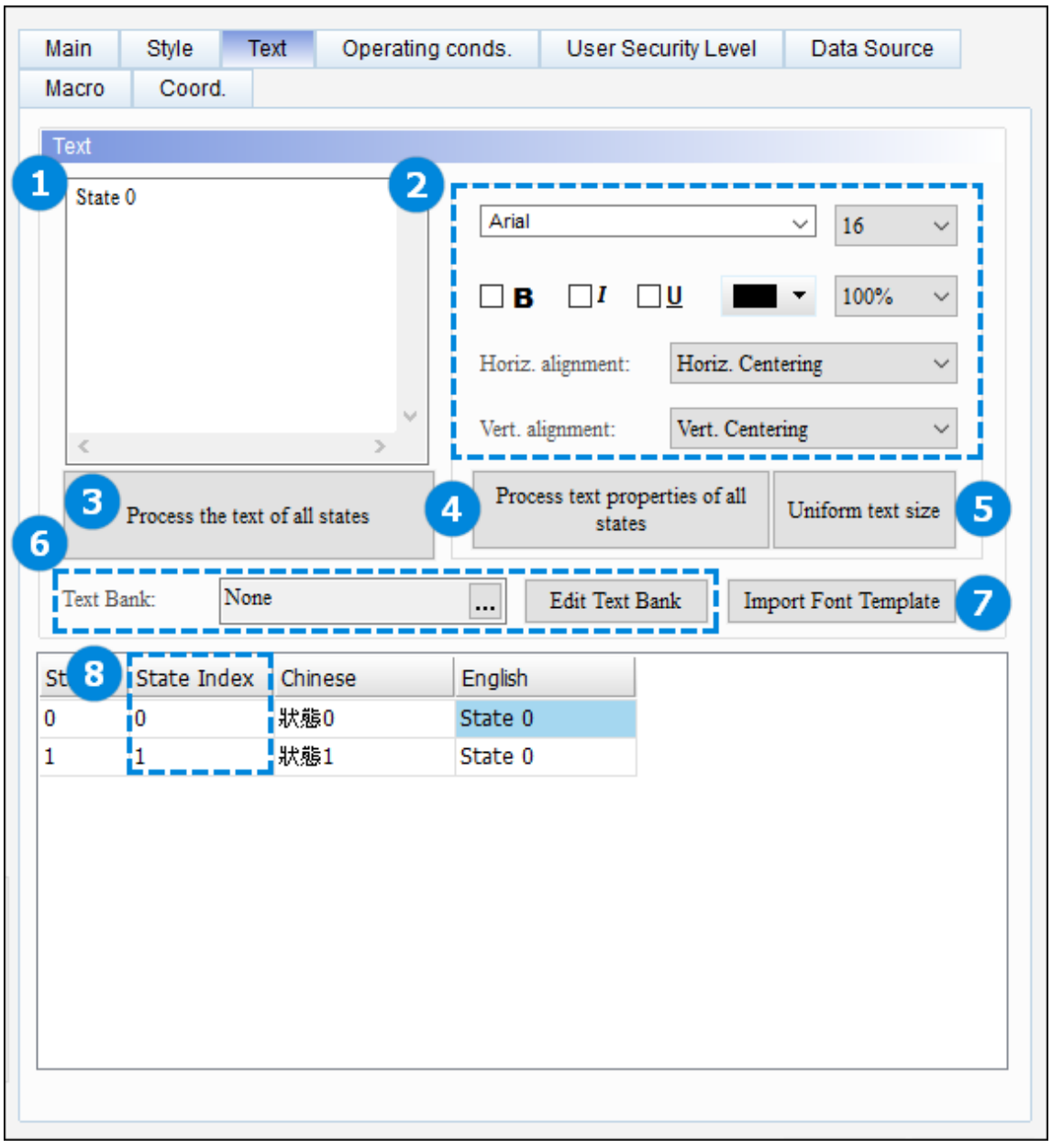
Legend	Setting	Description
❶	Write / Read Address	You can select the **Internal Memory** address or controller register address, and the **Input** memory **Type** can only be **Word**. **Note**: If the write memory address is set but the read memory address is not set, the HMI will regard the read memory address and the write memory address as the same.
❷	Write / Read Offset Address	The HMI will configure the value to the corresponding read address according to the set Offset Address. After setting the offset address, the actual read address is as follows. • The actual write address of the element: Write Address + Value in Write Offset Address × Element [Data Type] • The actual read address of the element: Read Address + Value in Read Offset Address × Element [Data Type]
❸	Data Type	Select the unit for state switching
❹	Data Format	Select the data format. **Note**: This setting is only available when **Data Type** is set to Word.
❺	State Counts	Select or enter the state counts according to the selected Data Type. • Bit: 2 states can be set. • Word: 1~256 states can be set. • LSB: 16 states can be set. • LSB (Support State 0): 17 states can be set. **Note**: LSB means that the register data is first converted into binary data, and then the current object state is defined by using the lowest non-zero bit. For example, decimal value 3 converted to binary is 0000000000000011, and the lowest non-zero bit is Bit 0, State is 1.

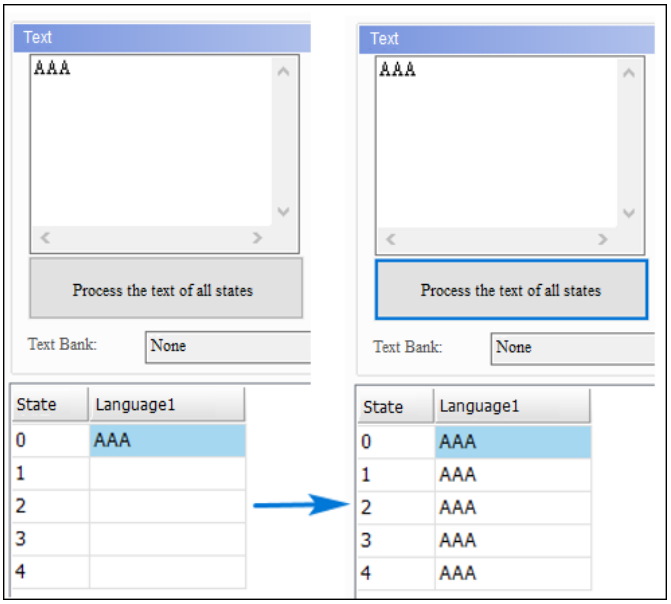
- **Style**

Setting	Description
Border Color	Select the border color of the element.
Foreground Color	Select the color of the element. • **Gradient Fill:** Set the gradient style by setting the current angle or center point. You can drag the mouse on the preview image to set the current angle or center point, or set the memory address to dynamically configure it on the HMI. You can also set the stop point colors to present the effect of multi-color gradients.  **Note**: Up to 5 stop points can be set. • **Solid Color**: Select a single-color fill. **Fill Pattern Style**: Select a two-color combination configuration and select a pattern fill style. 

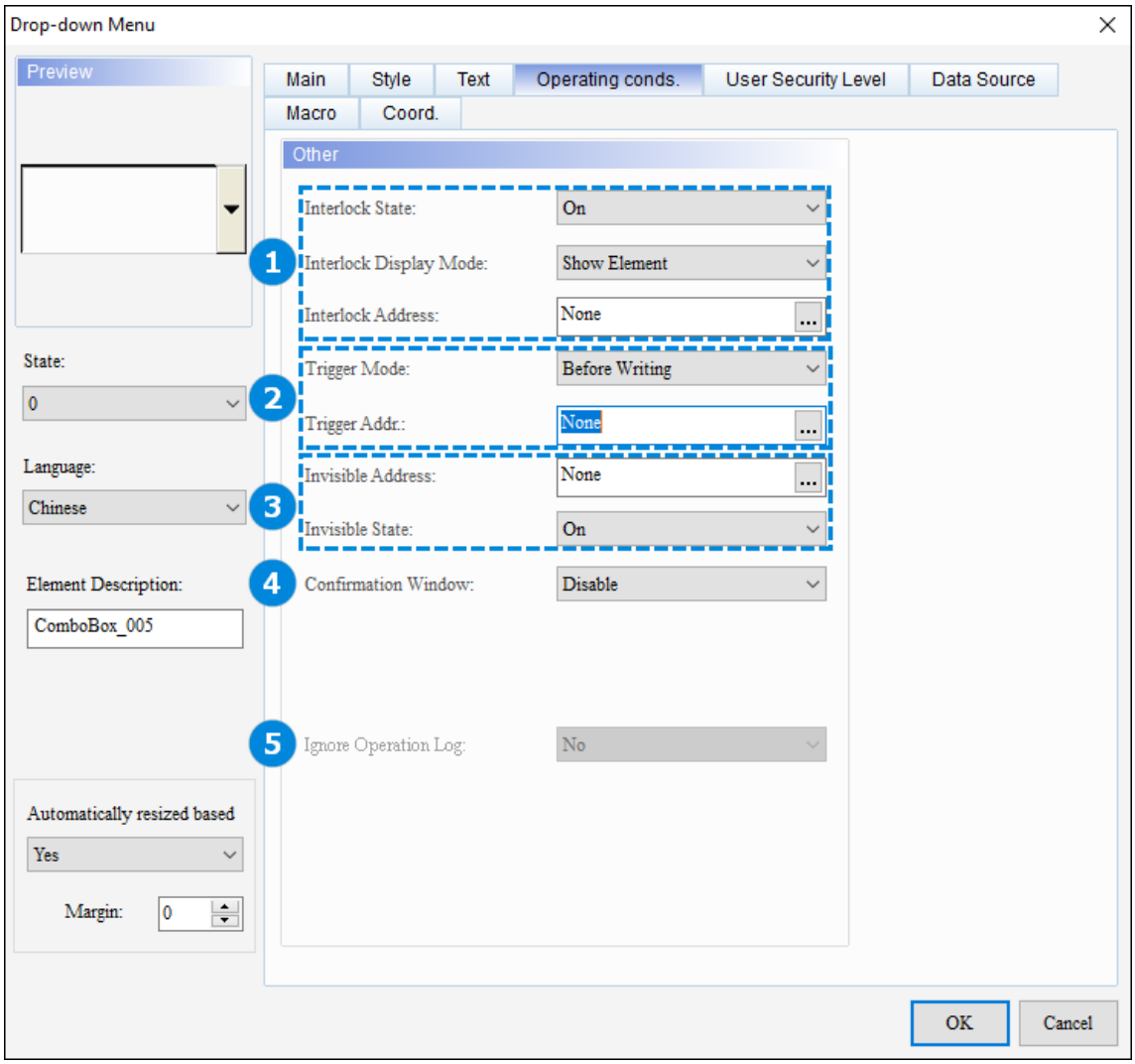
Setting	Description
Blink	Choose whether to enable the blink prompt. The blink color is opposite to the foreground color of the status.
Transparency	Select or enter the transparency value of the element. The range value is 50–255, and the default value is **255**.

- **Text**

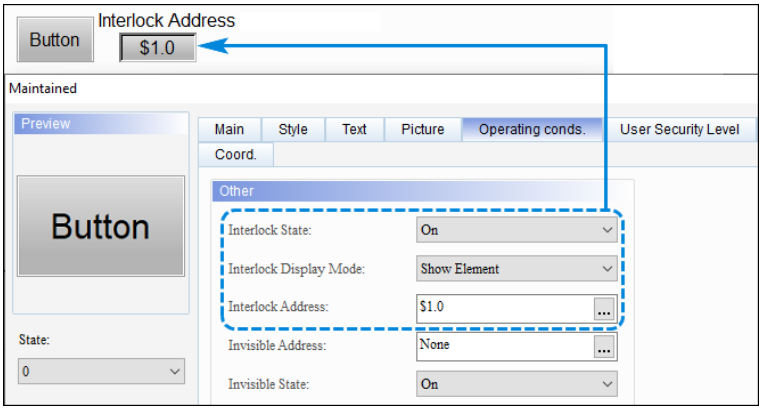


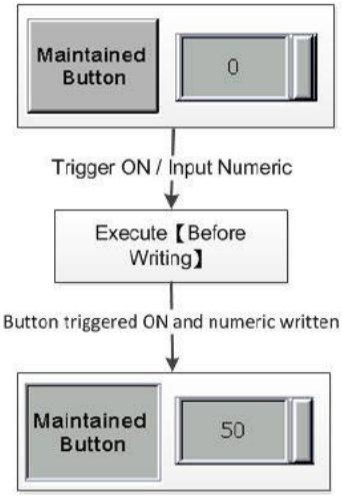
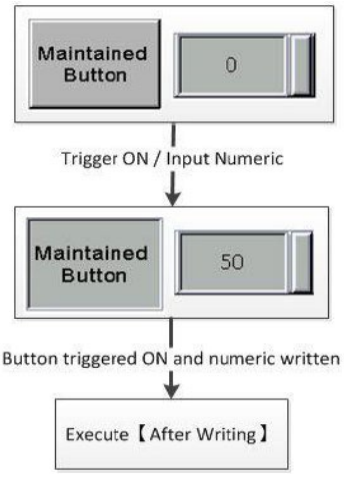
Legend	Setting	Description
❶	Text	Enter the text message to display in the text box. Select the state / language table below to directly set the text displayed by that state / language. Enter the text message to be displayed in the text box. Click the element on the screen and press **Space** to edit the text content of the current state / language.
❷	Text-related properties	Set the text properties.
❸	Process the text of all states	Click to apply the selected text contents of the state to those of other states. 
❹	Process text properties of all states	Click to apply the selected text properties of the state to those of other states.
❺	Uniform text size	Changes the text size in all languages with one click according to the currently selected state.
❻	Text Bank	Applies the text in the **Text Bank** to the content of each state / language. You can also click **Edit Text Bank** to edit the text content in the **Text Bank** editor.
❼	Import Font Template	Click to open the **Font Template** dialog, and then select the template and import the text properties of the selected state.
❽	State Index	Select the address value corresponding to the selected state, and the range value is 0–65535.

- **Operating conds.**



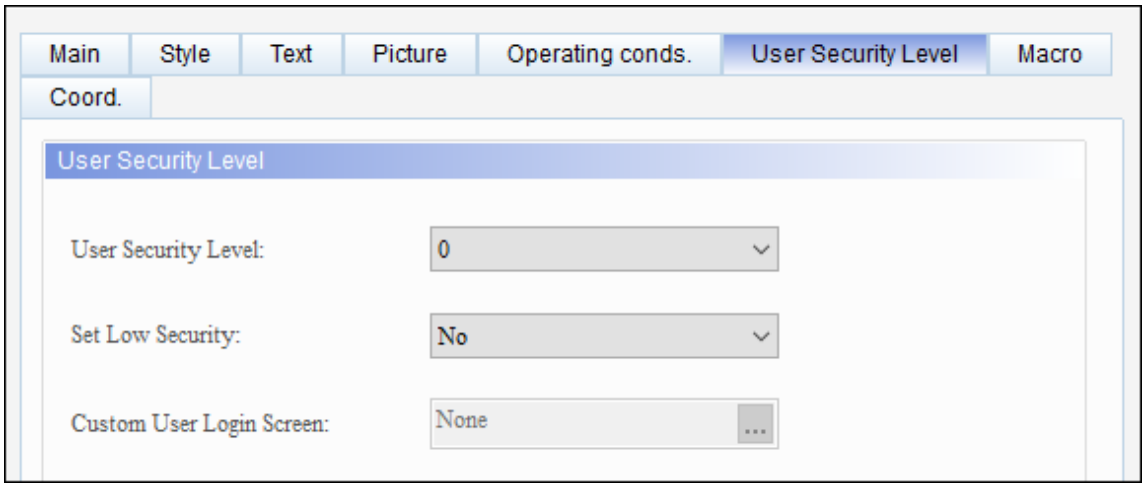
Legend	Setting	Description
1	Interlock State Interlock Address	The **Interlock State** enables to operation of a certain element and must be used with **Interlock Address**. If the **Interlock State** is set to **Off**, the **Interlock Address** enables to operation of a certain element.

<table>
<tr><th>Legend</th><th>Setting</th><th>Description</th></tr>
<tr><td></td><td>Interlock Display Mode</td><td>Instead, if the Interlock State is set to On, the Interlock Address enables to operation of a certain element. As shown in the following figure, when the Interlock Address $1.0 is On, the Maintained button can be operated.
 

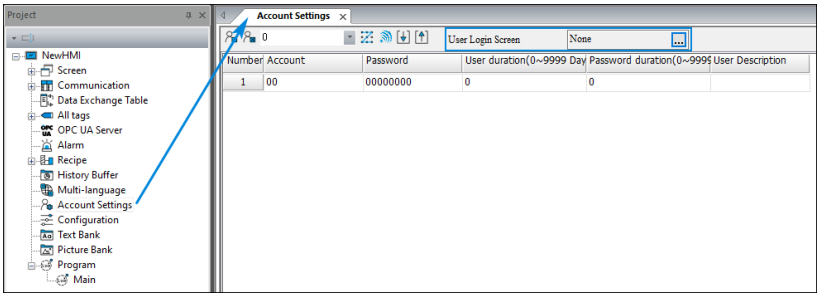
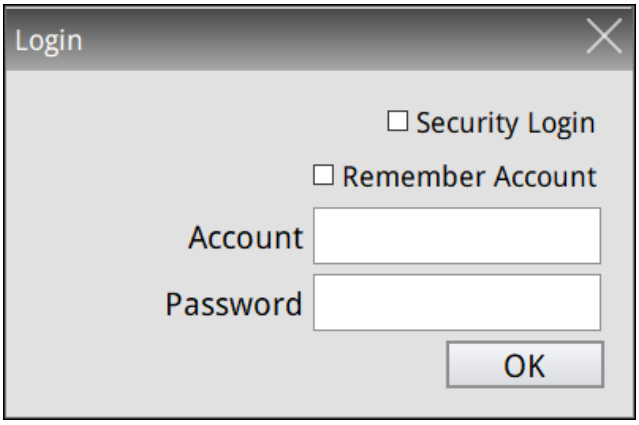
The effects of Interlock Display Mode are as follows.
<table>
<tr><th>Show Element</th><th>Show Prohibition Symbol</th></tr>
<tr><td> </td><td> </td></tr>
</table></td></tr>
<tr><td>❷</td><td>Trigger Mode
Trigger Addr.</td><td>Select the trigger mode.
<table>
<tr><th>Trigger Before Writing</th><th>Trigger After Writing</th></tr>
<tr><td>The trigger address is set to ON first, and the value changes.
 
</td><td>The value is changed first, and the trigger address is set to ON.
 
</td></tr>
</table>
Note: Since the trigger function only sets the set trigger address to ON, you need to set the trigger address to OFF if you want to trigger it again.</td></tr>
</table>

<table>
<tr><th>Legend</th><th>Setting</th><th>Description</th></tr>
<tr><td rowspan="2">❸</td><td>Invisible Address</td><td>The Invisible Address hides an element on the HMI and must be used with Invisible State. If Invisible State is set to On, the Invisible Address hides the element. Instead, if Invisible State is set to Off, the Invisible Address hides the element.
The following is an example of using invisible address.
1. Create a button element on the screen and set its address to $9.0, then set the Invisible Address of the element to hide to $9.0.
2. To hide the element, you must click the $9.0 button.</td></tr>
<tr><td>Invisible State</td><td>
<table>
<tr><td>Invisible State is ON</td><td>
</td></tr>
<tr><td>Invisible State is OFF</td><td>
</td></tr>
</table>
</td></tr>
<tr><td>❹</td><td>Confirmation Window</td><td>Select the confirm window type. If set to Confirmation, the Confirmation Dialog appears to confirm whether to perform the button action.
</td></tr>
<tr><td>❺</td><td>Ignore Operation Log</td><td>Choose whether to enable Ignore Operation Log.
When the operation log settings are enabled, the HMI will record the operations of all written element. If Ignore Operation Log of the element is set to Yes, the element action will not be recorded in the operation log.</td></tr>
</table>

- **User Security Level**

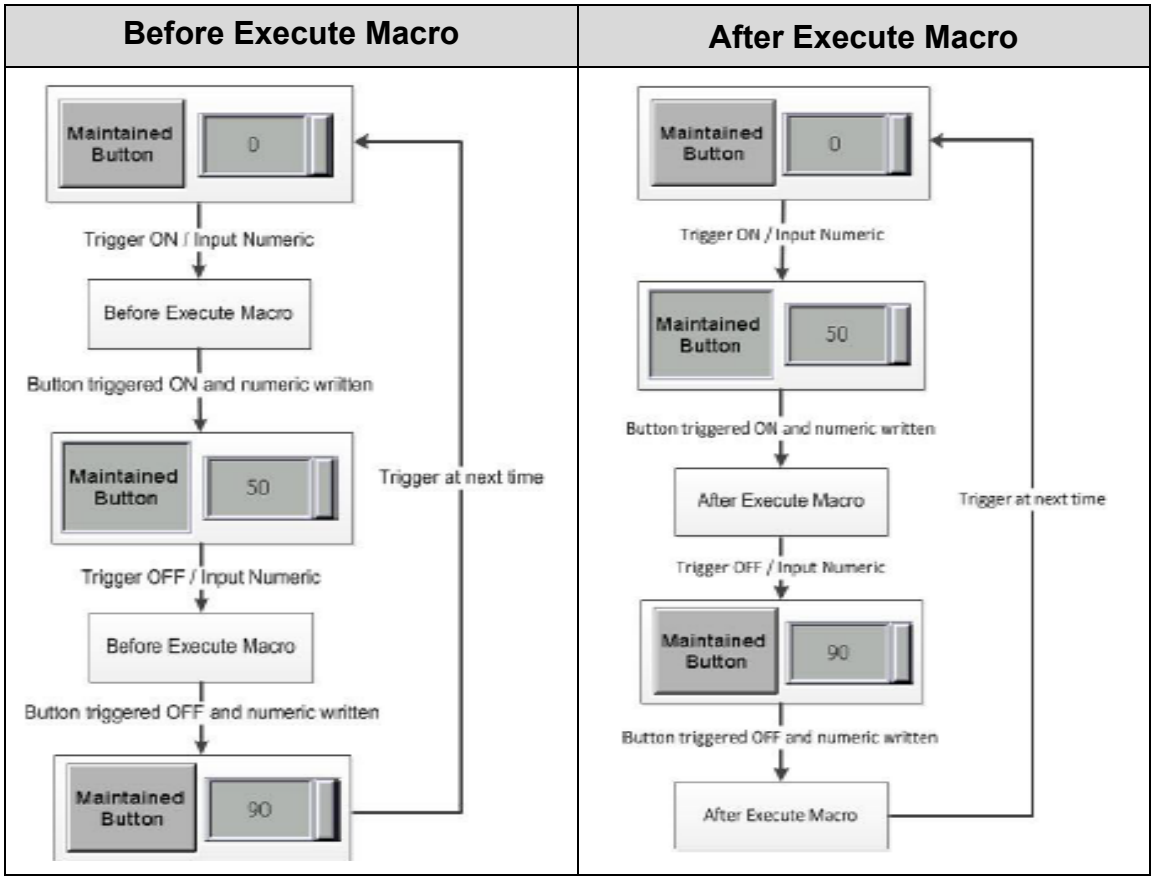


Setting	Description
User Security Level	Select the user security level. If the current security level is insufficient, the **Login** dialog pops up when the element is clicked on the HMI. The element action is performed if the login security level is equivalent to or higher than the set **User Security Level**. **Note**: When set to **0**, the element action can be performed without login.
Set Low Security	Choose whether to set the security level to the lowest security level after entering the security level. If set to **Yes**, the HMI will automatically set the security level to the lowest (Level 0) after performing the element action. Every time the element is triggered, the **Login** dialog will pop up, and you can perform the element action after logging in.

Setting	Description
Custom User Login Screen	Set the custom login screen. When **User Security Level** is not set to **0**, the custom login screen can be selected. When triggering the element and the user security level is required the set screen pops up. • On the toolbar, click **General** > **New Screen** > **User Login Screen** > **Login/Logout Screen**. • If the element does not have a specified login screen, you can go to **Project** > **Account Settings**. Set the **User Login Screen** of each security level in the **Account Settings** page. When the element sets this security level, the login screen configured in this page is displayed.  • If the element or account does not set the login screen, the default login screen is displayed. 

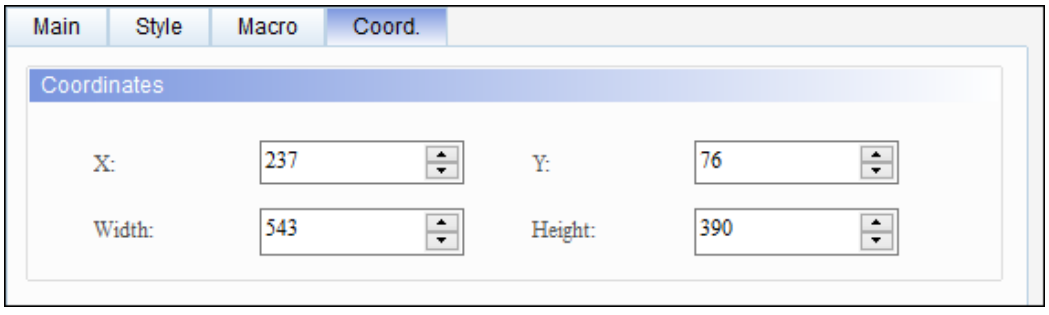
- **Macro**

After clicking the element on the HMI, the HMI executes **Before Execute Macro**, then executes the button action, and finally executes **After Execute Macro**.



Note: If the **Button** states are not changed with the button touch (using external controller commands or other macros and programs instead), the HMI does not execute the **Macro** commands.

- **Coord.**



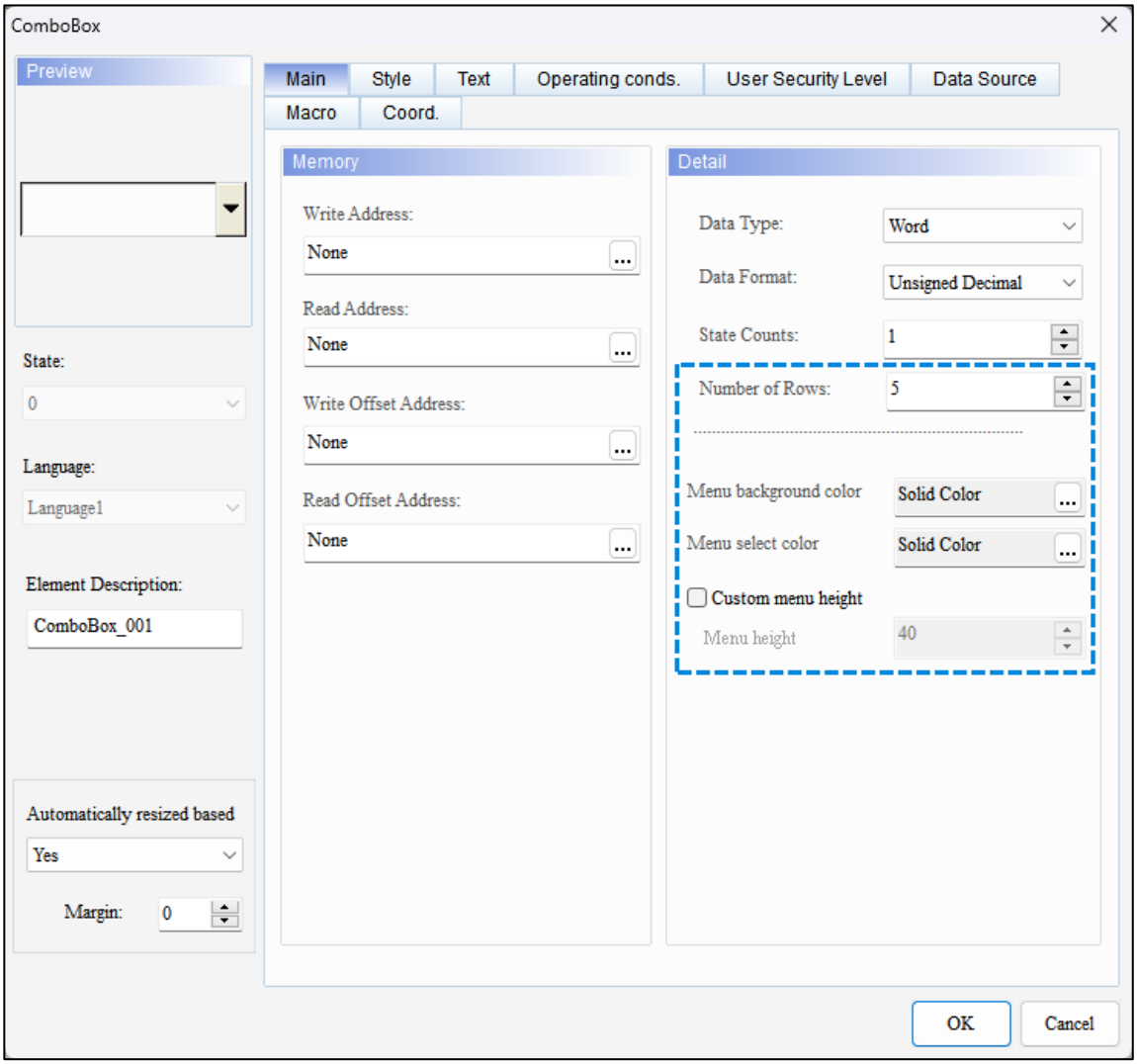
Select or enter the XY coordinates, width, and height on the upper left of the element.

ComboBox

The **ComboBox** element displays multiple state messages and functions of the same category in the same combo box, allowing you to select the desired option for execution.

The following table lists the properties of the **ComboBox** element with their description.

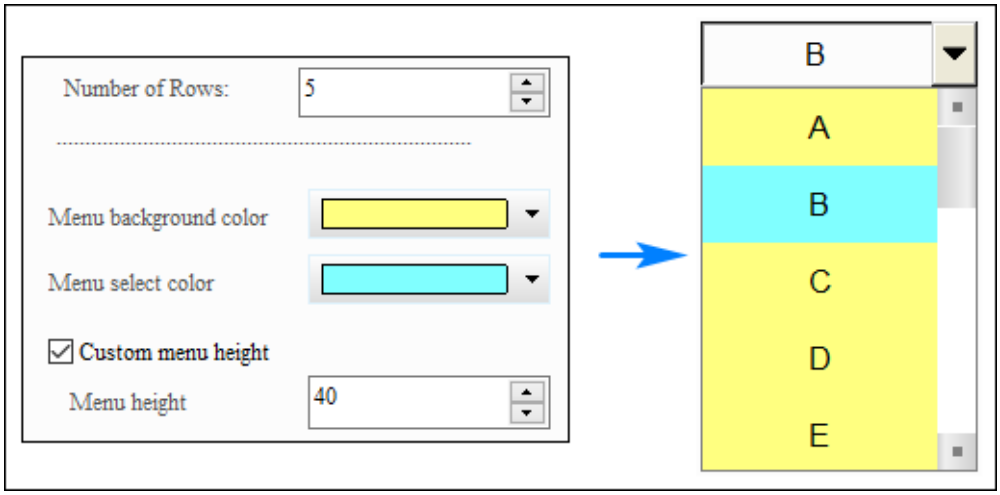
- **Main**



Setting	Description
Number of Rows	Select or enter the number of rows the combo box displays when the **ComboBox** element on the HMI is pressed. The range value is 5–15, and the default value is **5**.

Setting	Description
Menu background color Menu select color	Select the background color of the combo box and the color of the selected item.
Menu height	Select the **Custom menu height** checkbox, and then select or enter an item height for the combo box. The maximum height of the menu is screen **Height**. The default value is **40**.

The following figure demonstrates the menu settings of the **ComboBox** element and its effect on the HMI.

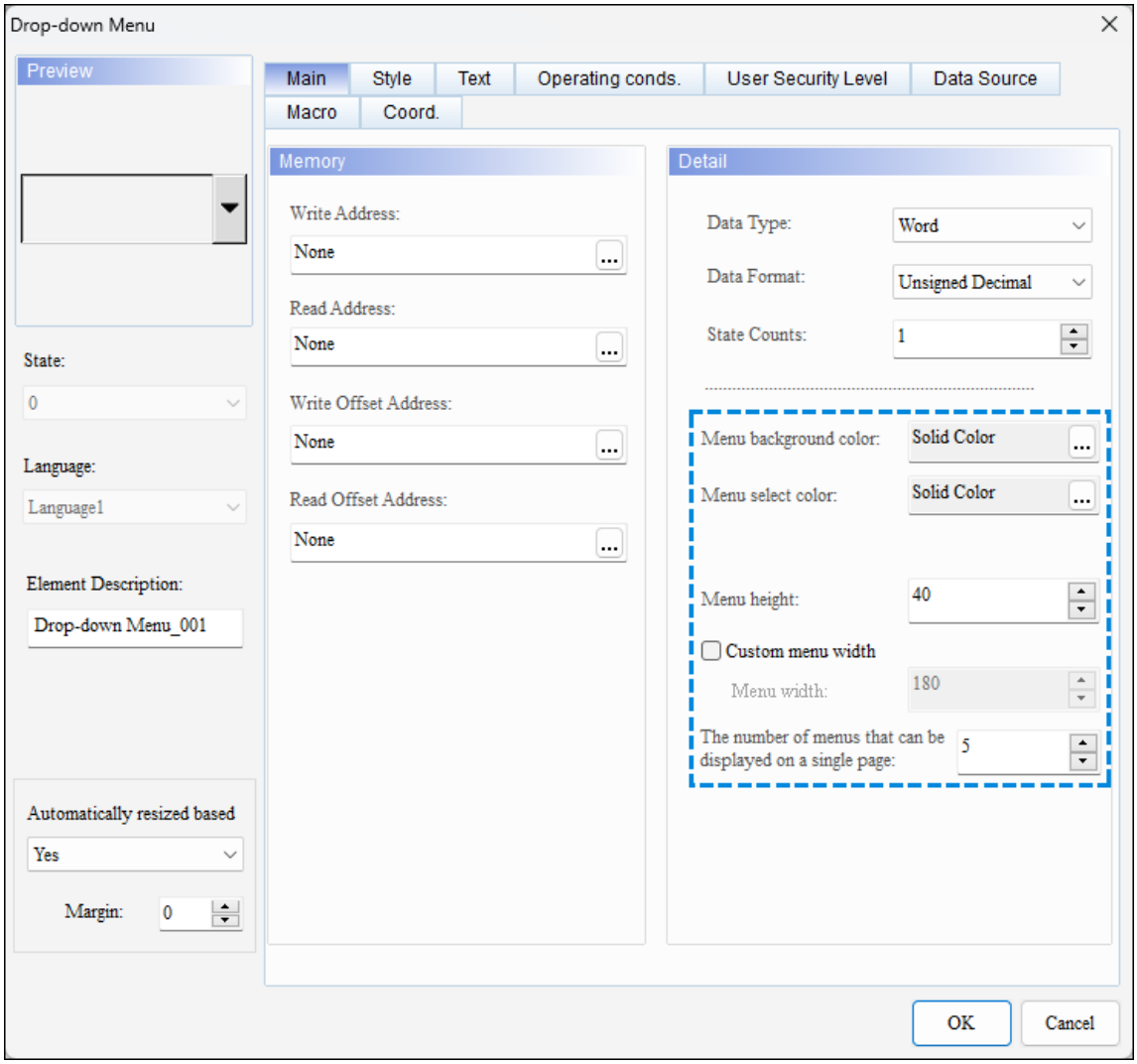


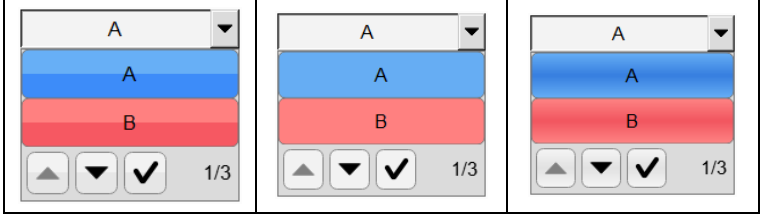
Drop-down Menu

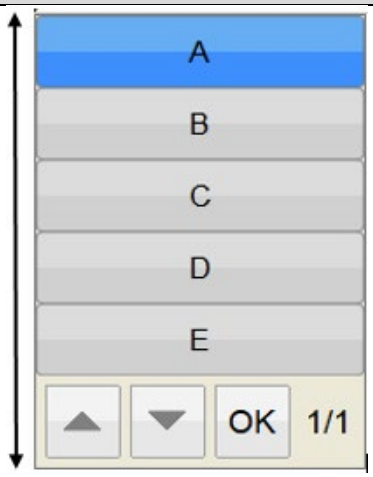
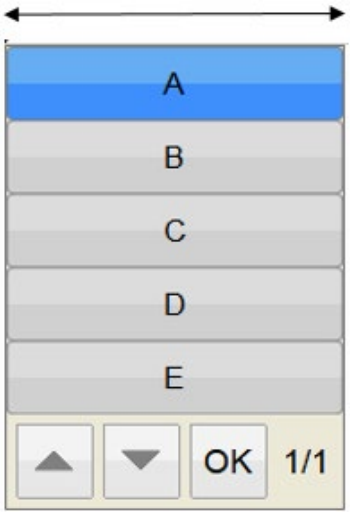
The **Drop-down Menu** element displays multiple state messages, allowing you to select the desired options for execution. You must click ☑ after selecting to complete the selection.

The following table lists the special properties of the **Drop-down Menu** element with their description.

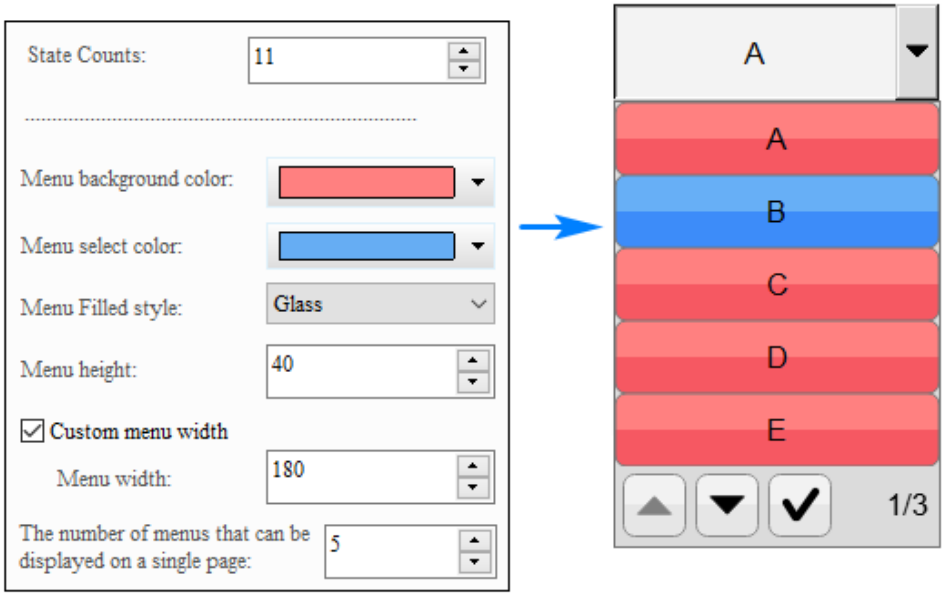
- **Main**



Setting	Description
Menu background color Menu select color	Select the background color of the combo box and the color of the selected item. **Glass** / **Solid Color** / **Gradient** 
Menu height	Select or enter the height of the menu, which must not exceed the screen **Height**.

Setting	Description
	
Menu width	Select the **Custom menu width** checkbox, and then select or enter the width of the menu, which must not exceed the screen **Width**. 
The number of menus that can be displayed on a single page	Select or enter the number of items that can be displayed on a single menu page.

The following figure demonstrates the menu settings of the **Drop-down Menu** element and its effect on the HMI.

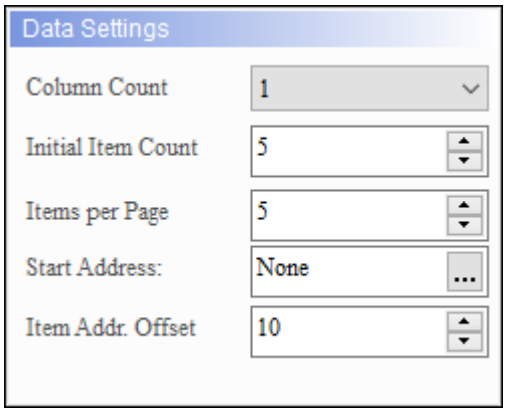


GridBox

The **GridBox** element displays the custom data content in grid format.

The following table lists the special properties of the **GridBox** element with their description.

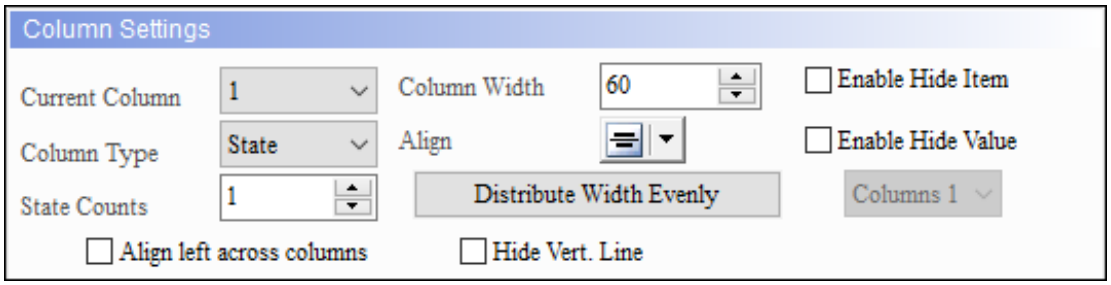
- **Main**
  - **Data Settings**



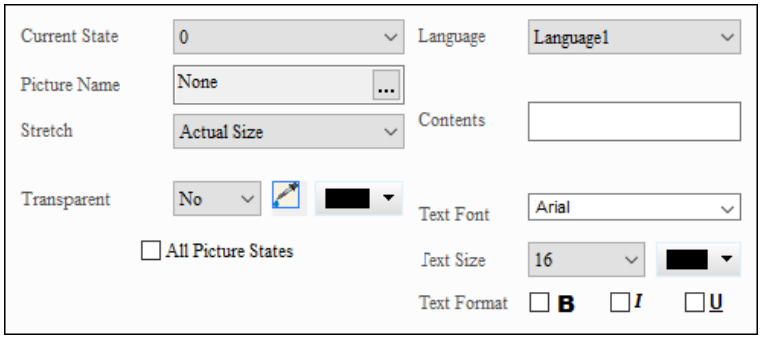
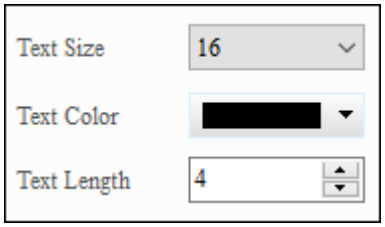
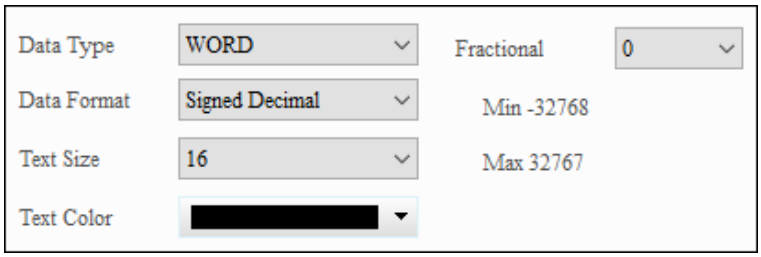
Setting	Description
Column Count	Select the number of columns. Up to 32 columns can be set. **Note**: If the **Column Count** is greater than the **Item Addr. Offest**, a warning message appears, reminding you that addresses overlap.

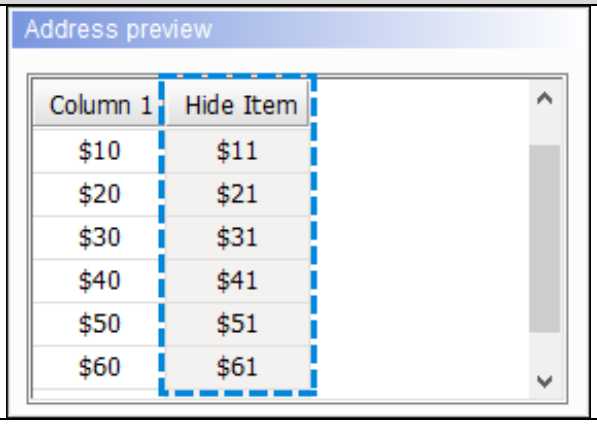
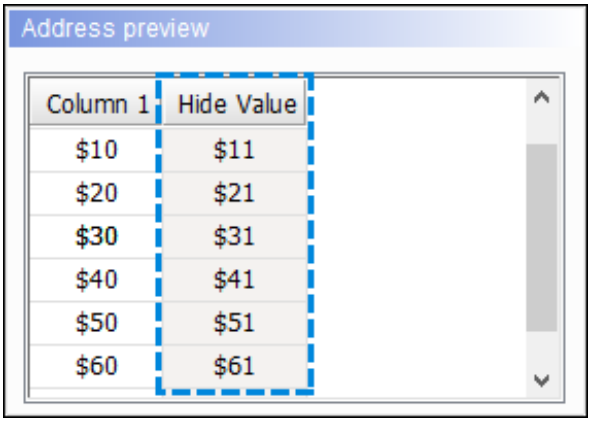
Setting	Description
Initial Item Count	Select or enter the initial item count of the element. Up to 1000 items can be set.
Items per Page	Select or enter the number of items that can be displayed on a single page of the element. Up to 100 items can be set.
Start Address	Set the starting address for reading. You can select the **Internal Memory** address or controller register address, and the **Input** memory **Type** can only be **Word**. After setting the address, you can view the read address corresponding to each column in the **Address preview** area.
Item Addr. Offset	Select or enter the offset of the address between items. Up to 10000 can be set. **Note**: If the **Item Addr. Offest** is greater than the **Column Count**, a warning message appears, reminding you that addresses overlap.

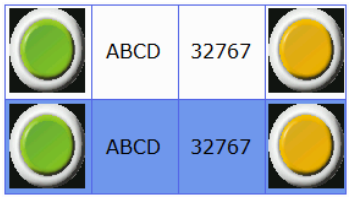
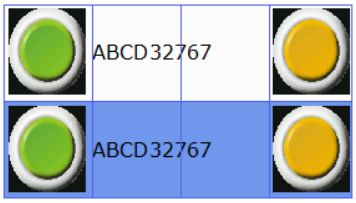
- **Column Settings**



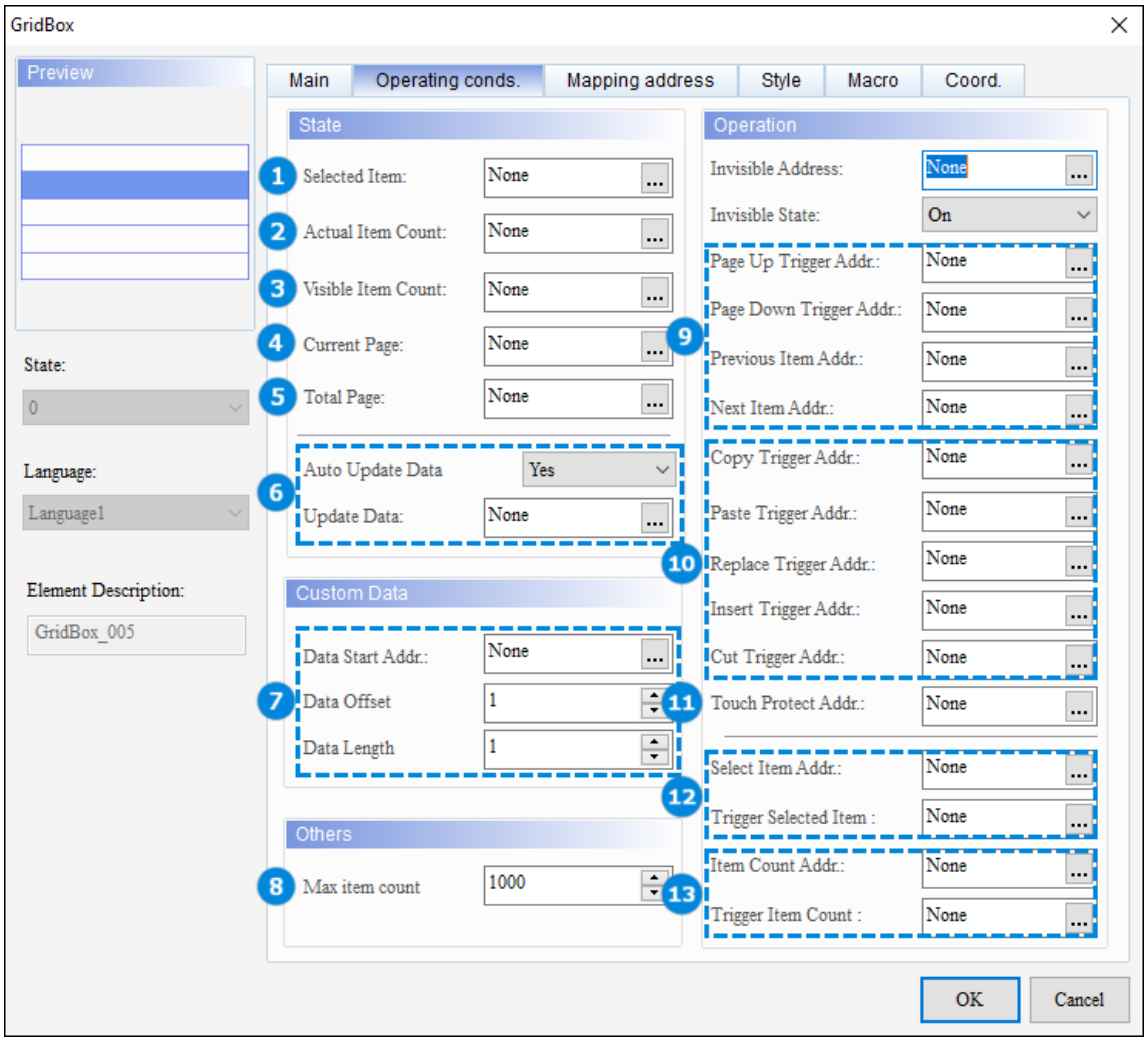
Setting	Description
Current Column	Select the column to edit and configure the column setting, and then apply them to the column.

Setting	Description
Column Type	Select the type of the column. • **State**: Displays the state. After selecting the state, set the properties of the picture and text.  • **String**: Displays the string. You can set the properties of the text.  • **Numeric**: Displays the numeric. You can set the data format and the properties of the text. 
State Counts	Select or enter the state counts to display. Up to 256 states can be set. **Note**: This setting is only available when the **Column Type** is set to **State**.
Column Width	Select or enter the width of the currently edited column. Up to 65535 can be set. You can also click **Distribute Width Evenly** to automatically adjust the width.
Align	Select the alignment style for the cell contents.
Enable Hide Item	After enabling this setting, the **Address preview** area automatically displays the column and corresponding address of the hidden item. When the address of a hidden item is not 0, this item will not be displayed on the element.

<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td></td><td> 
</td></tr>
<tr><td>Enable Hide Value</td><td>After enabling this setting and selecting the hidden column, the Address preview area displays the column and corresponding address of the hidden value.
 

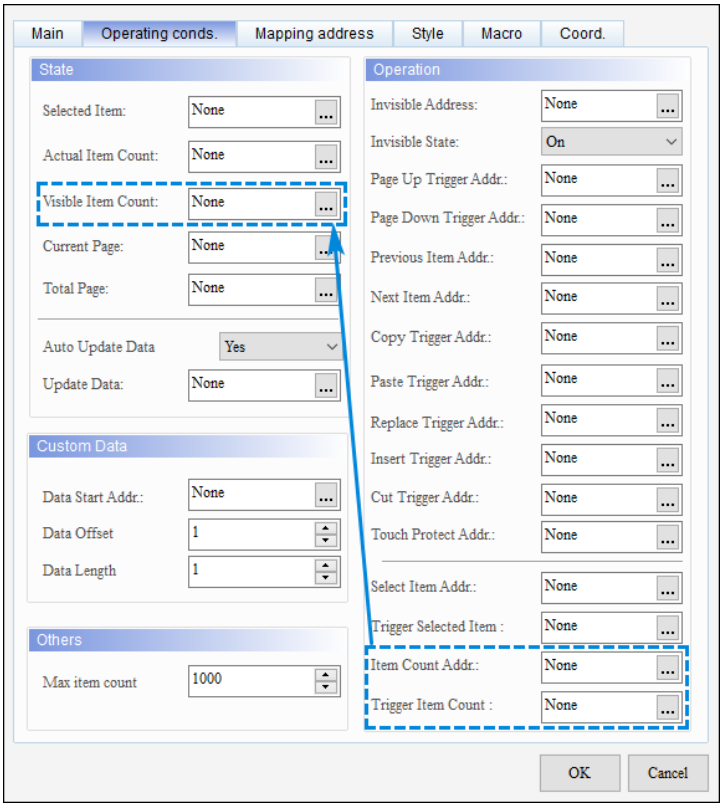
• Each bit of the address in the Hide Value column determines whether to display the corresponding column. For example, the first column data of the item will not be displayed when bit 0 is On; the second column data of the item will not be displayed when bit 1 is On, and so forth.</td></tr>
<tr><td>Align left across columns</td><td>Enable this setting to merge the cells and left-align the content. Before merging, you must enable Align left across columns for the columns to merge.
<table>
<tr><th>Align left across columns not enabled</th><th>Align left across columns enabled</th></tr>
<tr><td> 
</td><td> 
</td></tr>
</table></td></tr>
<tr><td>Hide Vert. Line</td><td>Enable this setting to hide the right separator line for the selected columns in the table.</td></tr>
</table>

- **Operating conds.**

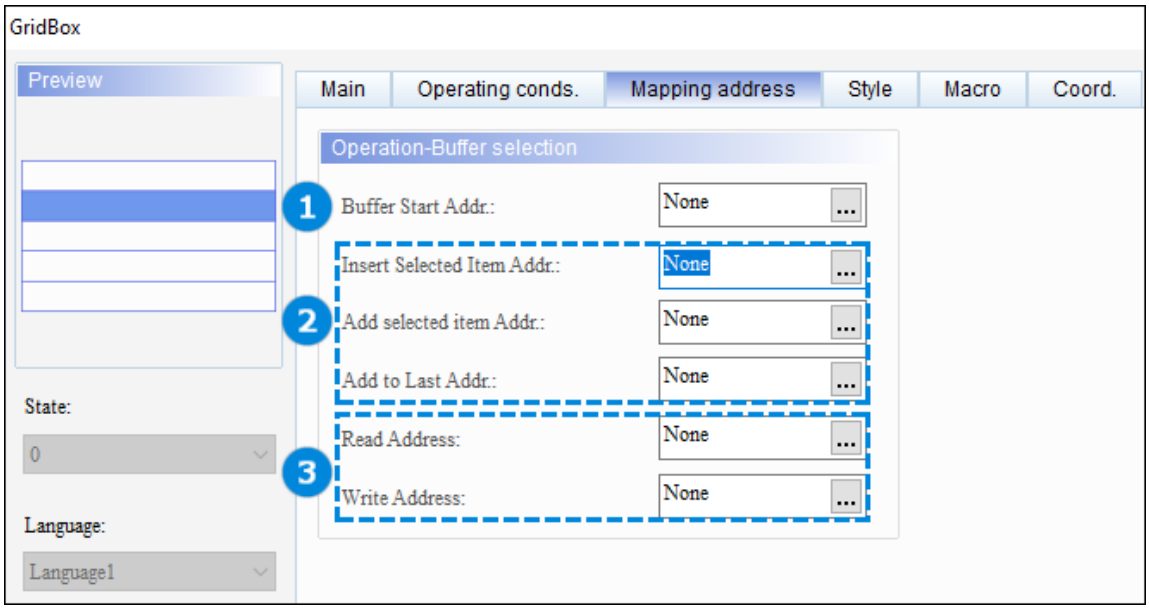


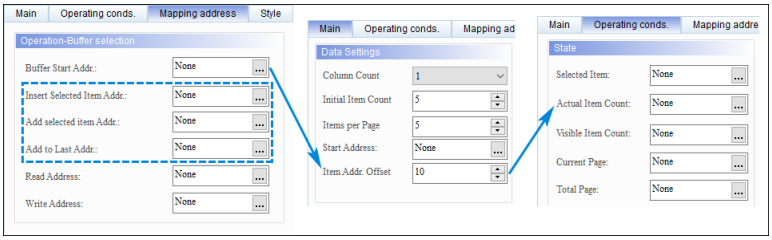
Legend	Setting	Description
❶	Selected Item	Set the memory address for reading the selected item number. Click the item in the GridBox on the HMI, and the **Selected Item** address displays the item number.
❷	Actual Item Count	Set the memory address for reading the total number of items displayed on the GridBox.
❸	Visible Item Count	Set the memory address for reading the total numbers of visible item after deducted the total hidden items on the GridBox.
❹	Current Page	Set the memory address for reading the currently displayed page of the GridBox.
❺	Total Page	Set the memory address of the total number of pages to read all visible items in the GridBox.

Legend	Setting	Description
⑥	Auto Update Data	Choose whether to automatically update the data of the current page and set the address for updating the data. If the **Auto Update Data** is set to **Yes**, the element will automatically update the data; if the **Auto Update Data** is set to **No**, you must trigger the **Update Data** to On to update the data.
	Update Data	
⑦	Custom Data	Customize another data area whose data will change as the element is edited.
⑧	Max item count	Used to set the maximum number of items that can be added to the Gridbox. The maximum value is 1000.
⑨	Page Up Trigger	Switch when the trigger address is On. The address will be automatically cleared after the action is completed.
	Page Down Trigger	
	Previous Item Addr.	
	Next Item Addr.	
⑩	Copy Trigger Addr.	Specify the selected item through the **Selected Item Addr.** and trigger the **Trigger Selected Item** to On, and then execute the following actions after triggering the trigger address to On. • **Copy**: Copy the currently selected item. • **Paste**: Paste the copied item after the row of selected item. • **Replace**: Replace the content of the currently selected item with the content of copied item. • **Insert**: Insert the blank item into the next row of the selected item. • **Cut**: Cut the currently selected items. **Note**: The trigger address is automatically cleared after the action is completed.
	Paste Trigger Addr.	
	Replace Trigger Addr.	
	Insert Trigger Addr.	
	Cut Trigger Addr.	
⑪	Touch Protect Addr.	When set to On, the **GridBox** element cannot be touched.
⑫	Select Item Addr.	Specify the selected item through the **Selected Item Addr.**, and then trigger the **Trigger Selected Item** to On, and the element displays the selected item.
	Trigger Selected Item	

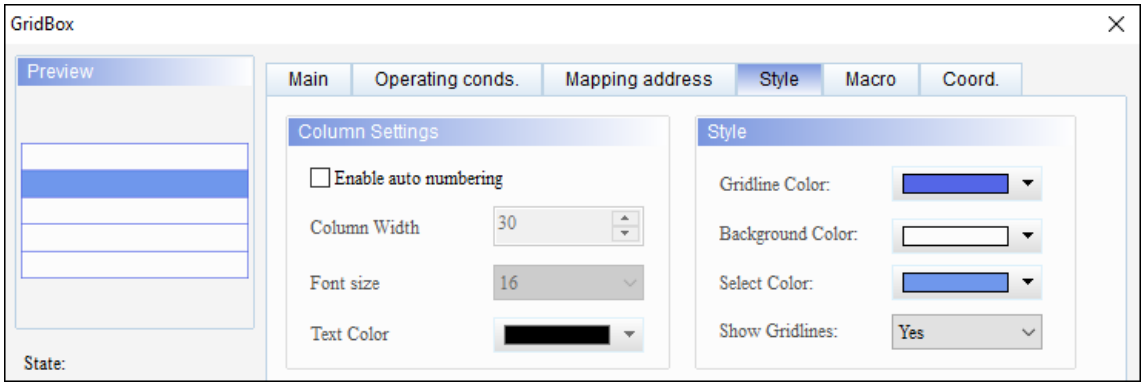
Legend	Setting	Description
⑬	Item Count Addr.	Specify the visible item through the **Item Count Addr.**, and then trigger the **Trigger Item Count** to On, and the **Visible Item Count** updates. 

- **Mapping address**



Legend	Setting	Description
❶	Buffer Start Addr.	Set the starting address to temporarily store data.
❷	Insert Selected Item Addr.	When the bit is set to On, the address of the **Item Addr. Offset** will be increased starting from the **Buffer Start Addr**. Execute the following actions to increase the **Actual Item Count** by 1. If the **Buffer Start Addr**. is $10 and the **Item Addr. Offset** is 10, then the selected item to be inserted or added to write to the address of $20. • **Insert selected item**: Insert the **Selected Item**. • **Add selected item**: Add the item after the **Selected Item**. • **Add to last**: Add to the last item.  **Note**: If the **Actual Item Count** reaches the **Max item count**, the actions cannot be executed.
	Add selected item Addr.	
	Add to Last Addr.	
❸	Read Address	When the **Write Address** bit is On, the **Selected Item** will be written to the **Buffer Start Addr.** or read from the Buffer Start Addr, and the element content does not change.
	Write Address	

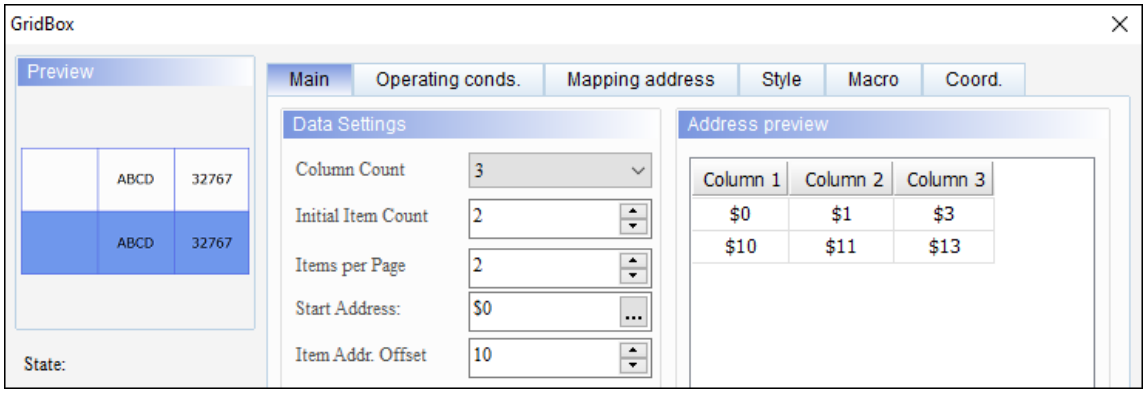
- **Style**



Setting	Description
Column Settings	Select the **Enable auto numbering** checkbox to add the numbering column on the GridBox and set the column width, text size, and text color.
Style	Select the gridline color, background color, and selection color of the element, and choose whether to display gridlines.

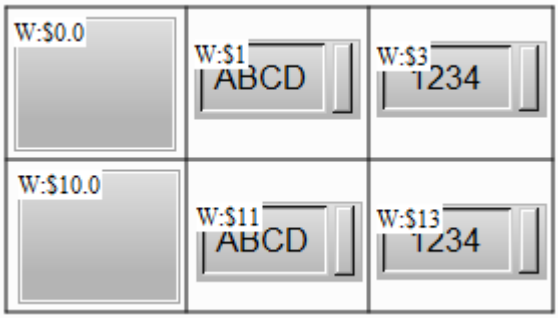
The following describes how to use the **GridBox** element.

1. Create a **GridBox** element on the screen, and then set the **Column Count** to **3**, **Items per Page** to **2**, and **Start Address** to **$10**.



2. Respectively set the different types for 3 columns. For example, set Column 1 to **State**, Column 2 to **String**, and Column 3 to **Numeric**.

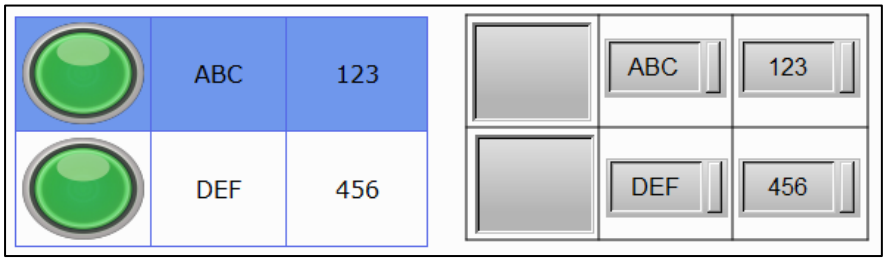
3. Create control elements such as **Maintained**, **Character Entry**, and **Numeric Entry** elements on the screen, and set the corresponding addresses.



4. Compile and download the screen to the HMI.

The execution result is as follows.

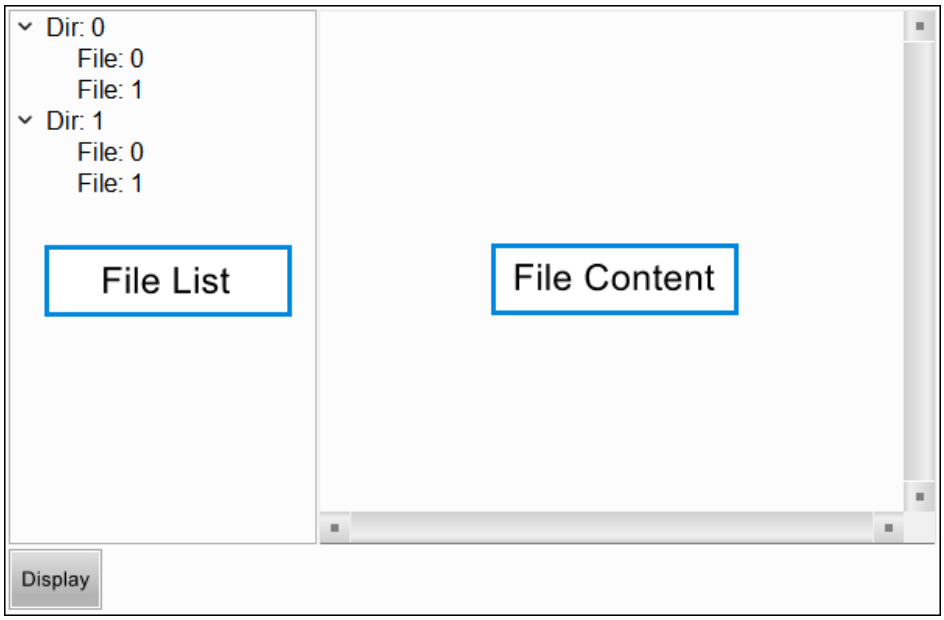
When entering a value through a control element, the GridBox's columns display the corresponding value.



PDF Viewer / Text Viewer

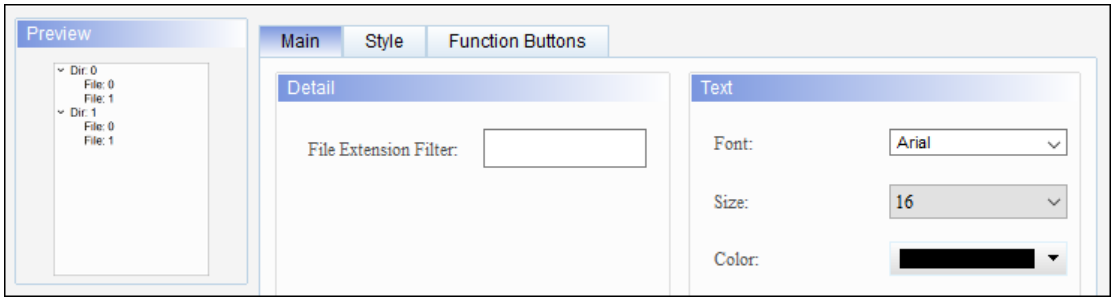
Use the Viewer element to read PDF, TXT, or CSV files on the HMI and store the file to an external storage device, then insert it into the HMI. The Viewer element will display the file content.

The Viewer element has the **File List** and **File Content** area.



The following respectively lists the properties of the **File List** and **File Content** with their description.

- **File List**
  - **Main**

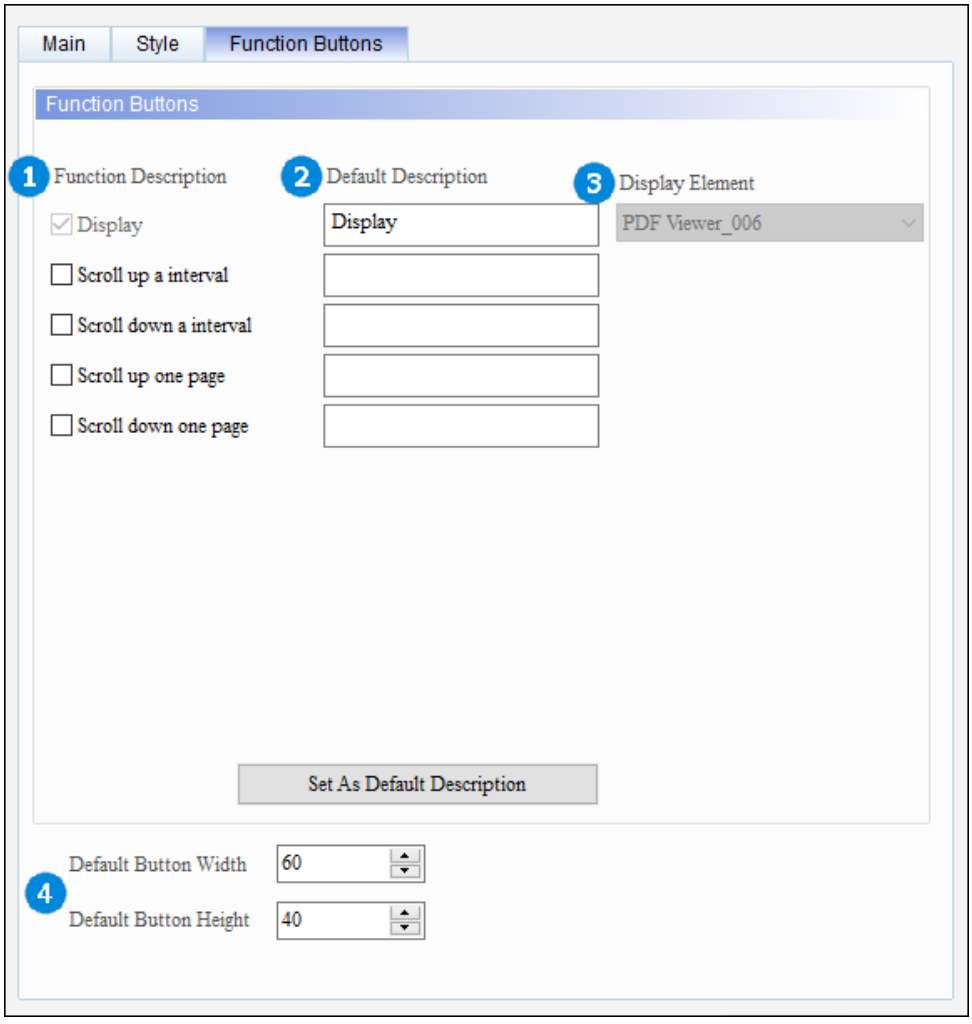


Setting	Description
File Extension Filter	Enter the file extension to filter.
Text	Set the text font, size, and color of the text.

- **Style**

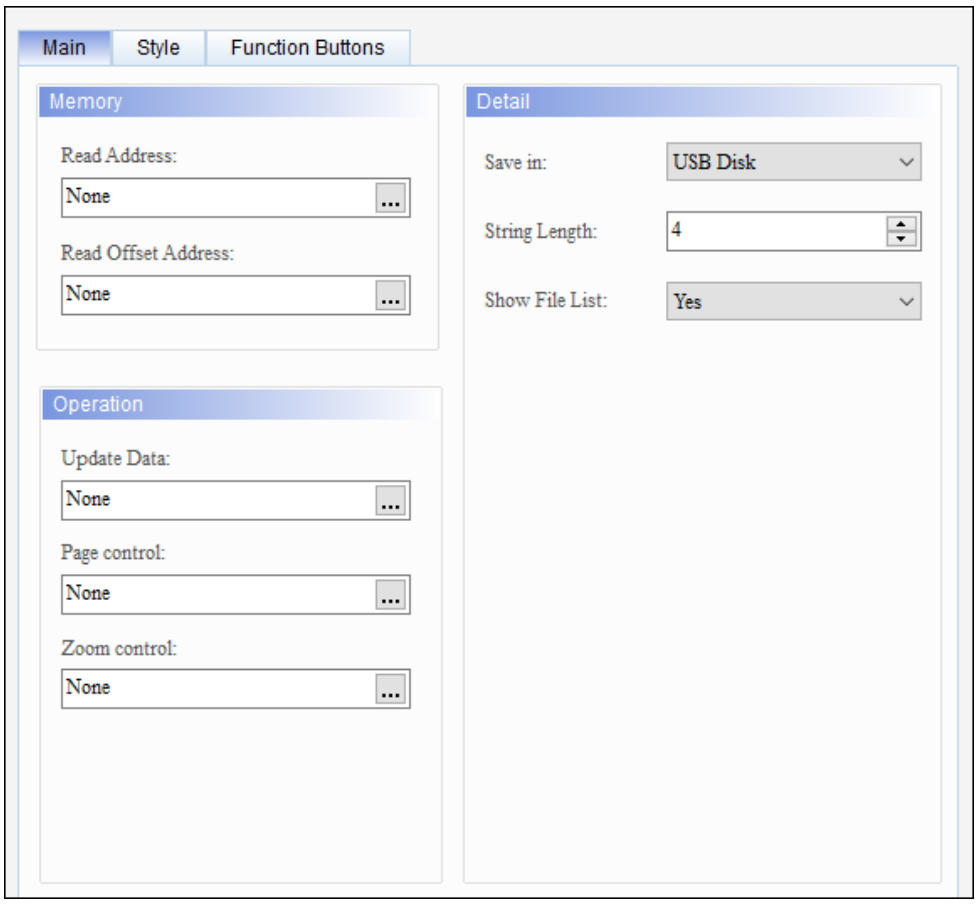
  Set the border color, background color, transparency, and smooth animation of the viewer area.

- **Function Buttons**



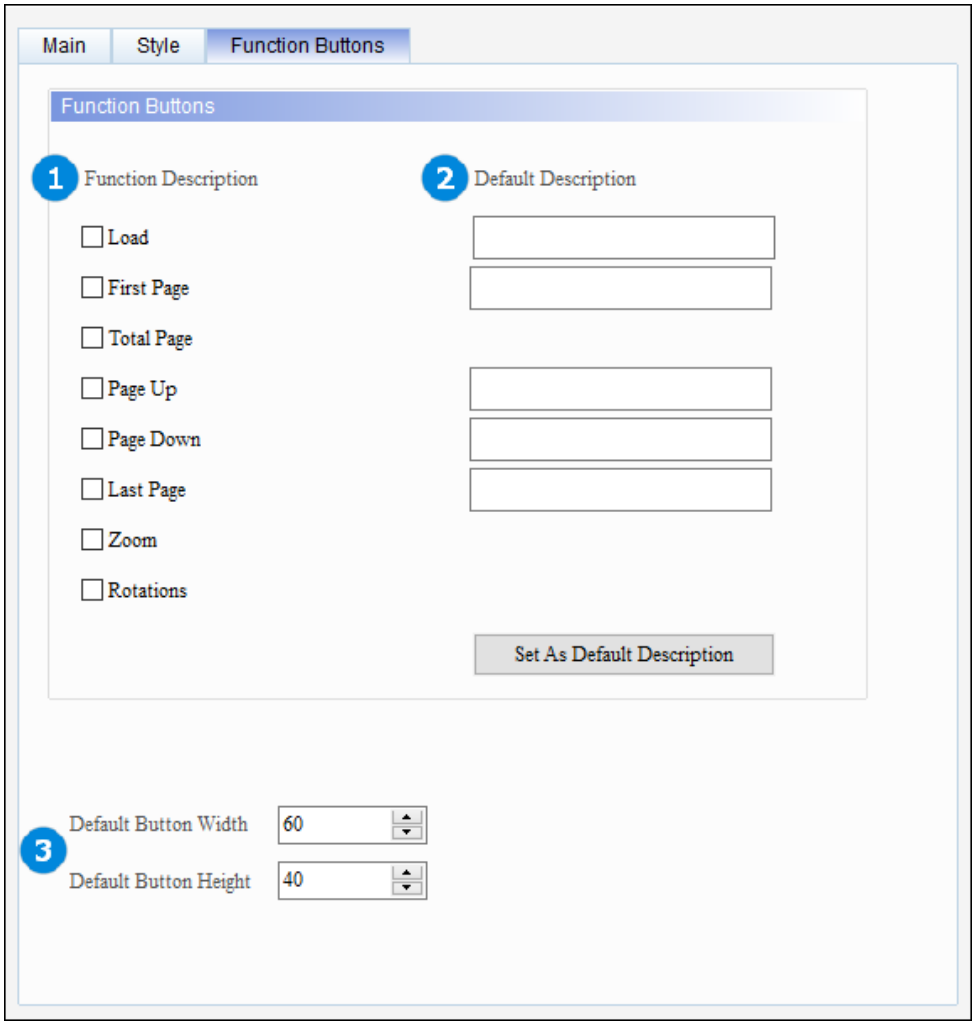
Legend	Setting	Description
❶	Function Description	Select the function button to use for the **File List**. • **Display**: Used for the **Display Element** and cannot be disabled. Select a file in the **File List** area of the element on the HMI screen, and then press the **Display** button, and the file will be displayed in the **File Content** area.
❷	Default Description	Enter the text to display on the function button.
❸	Display Element	Used to display the file and cannot be set.
❸	Default Button Width Default Button Height	Select or enter the height and width of the function button. After changing the height and width, you can update the button size through the **Rearrange Function Buttons** from the element context menu.

- **File Content**
  - **Main**



<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Memory</td><td>Set the memory address of the file path.</td></tr>
<tr><td>Operation</td><td>Set the address for the Update Data. When the address is On. the display content is update.
Text Viewer control address
Goto addr.: Set the address to go to the specified row.
Goto trigger addr.: Set the address to go to the specified row when the bit is On.
PDF Viewer control address
Page control: Set the address to specify the display page of the file
Zoom control: Set the display zoom ratio address of the control file.
<table><tr><th>State Value</th><th>Display Ratio</th></tr><tr><td>0</td><td>Full page dislapy</td></tr><tr><td>1</td><td>50%</td></tr><tr><td>2</td><td>75%</td></tr><tr><td>3</td><td>100%</td></tr><tr><td>4</td><td>125%</td></tr><tr><td>5</td><td>150%</td></tr><tr><td>6</td><td>200%</td></tr></table></td></tr>
<tr><td>Save in</td><td>Select the device to store the file.</td></tr>
<tr><td>String Length</td><td>Select or enter the string length of the file path.</td></tr>
<tr><td>Show File List</td><td>Choose whether to display the file list.</td></tr>
</table>

- **Function Buttons**



Legend	Setting	Description
1	Function Description	Select the function buttons used in the **File Content**. • **Load**: After pressing the **Load** button on the HMI, the **File Content** area displays the file path of **Read Address**. • **Total / Zoom / Rotations**: The special buttons for the **PDF Viewer**. You can switch the displayed page and control the displayed zoom ratio and direction.
2	Default Description	Enter the text to display on the function button.
3	Default Button Width Default Button Height	Select or enter the height and width of the function buttons. After changing the height or width, you can update the button size through the **Rearrange Function Buttons** function in the context menu.

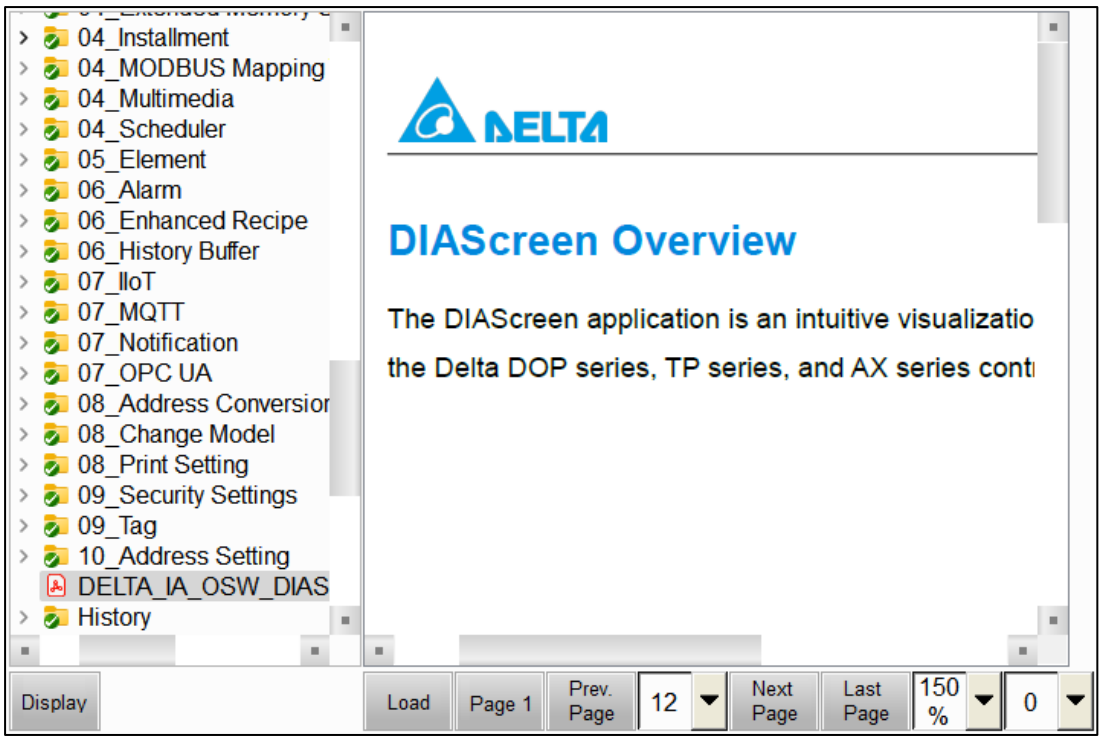
Note:

- The **Text Viewer** only supports UTF-8 code text. If the file format is .CSV, go to **General** > **Environment** to set the **CSV Separator**.
- Only one viewer element per screen is supported.

The following describes an example of viewing different files.

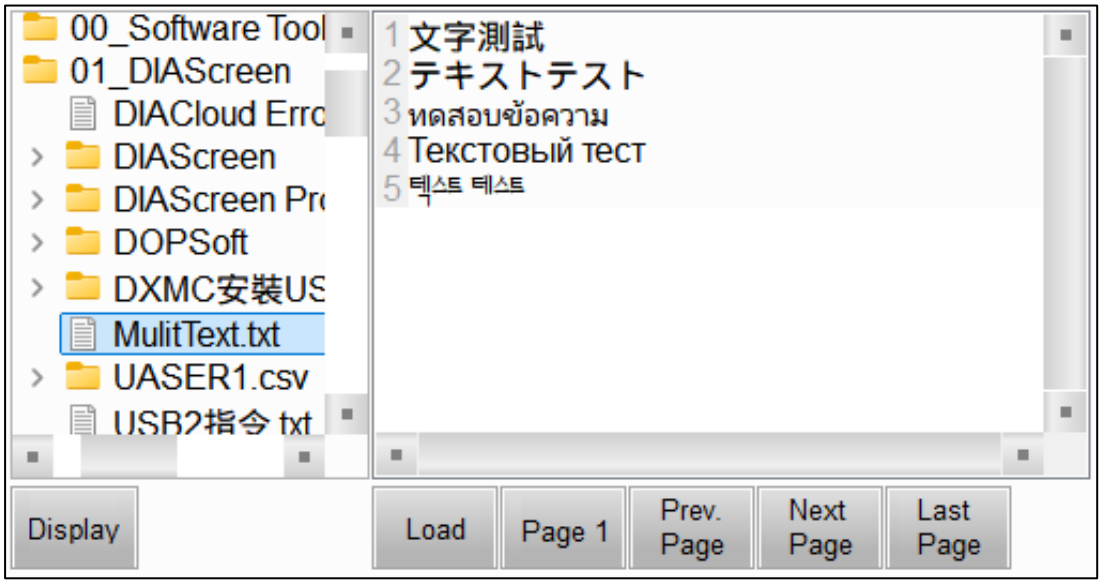
- **PDF file**

  1. Create a **PDF Viewer** element on the screen and download the screen to the HMI.
  2. Select a PDF file in the **File List** area of the element, and then press the **Display** button, and the **File Content** area displays the file preview.



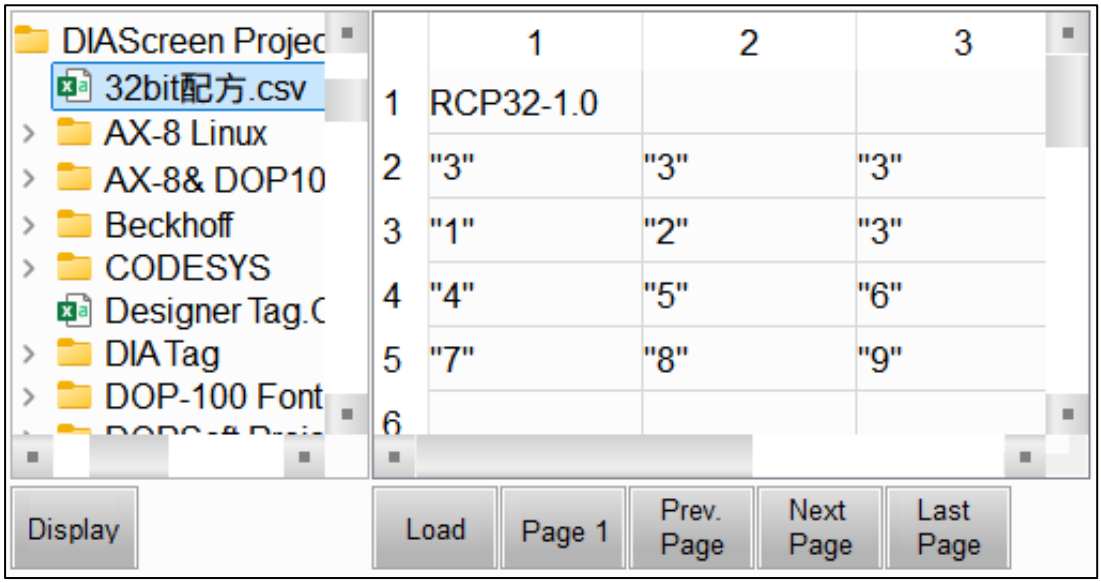
- **TXT file**

  1. Create a **Text Viewer** element on the screen and download the screen to the HMI.
  2. Select the TXT file in the **File List** area of the element, and then press the **Display** button, and the **File Content** area displays the file preview.



- **CSV file**

1. Create a **Text Viewer** element on the screen and enter the **File extension Filter** as CSV and download the screen to the HMI.
2. Select the CSV file in the File List area of the element, and then press the Display button, and the **File Content** area displays the file preview.



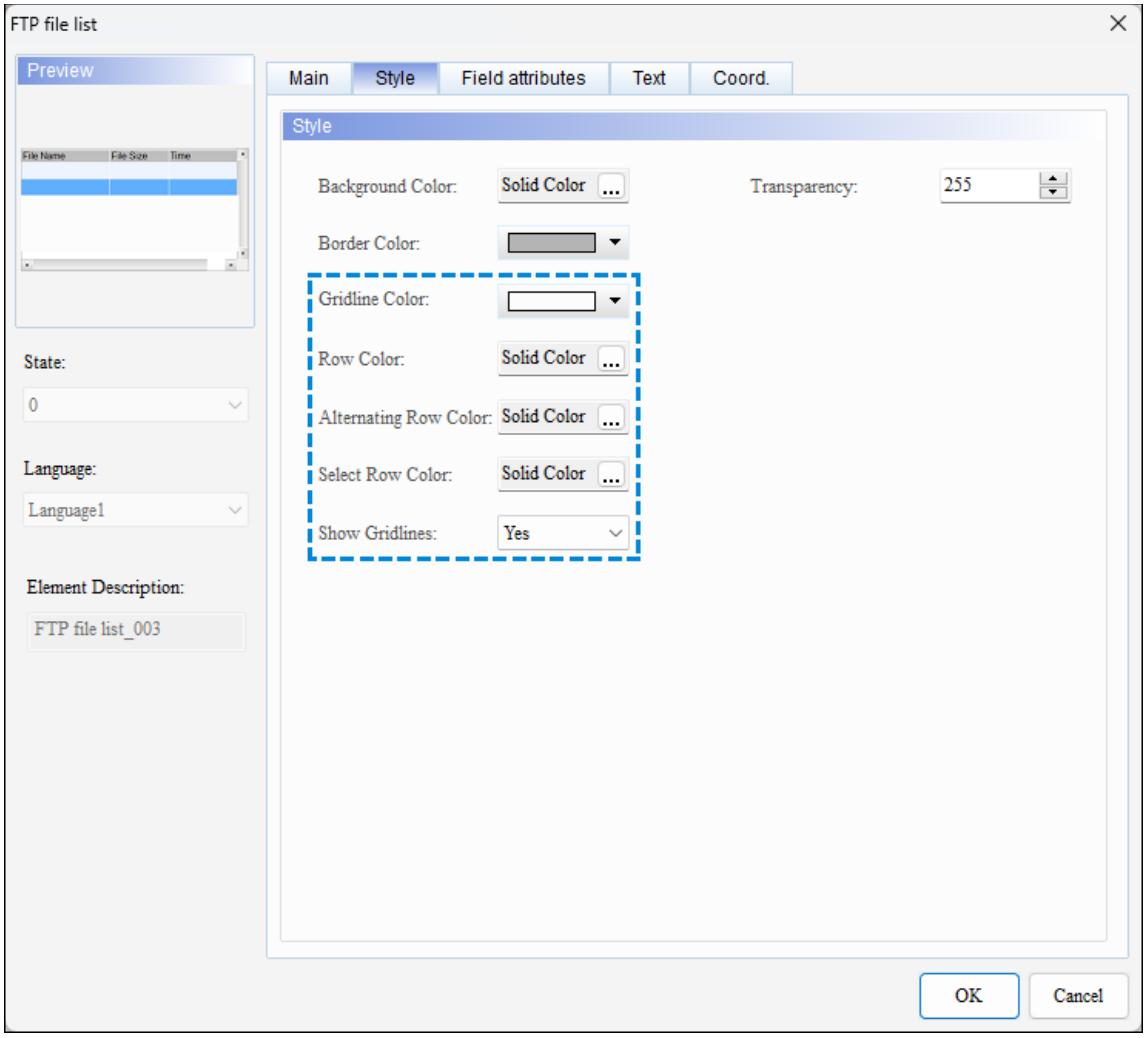
FTP file list

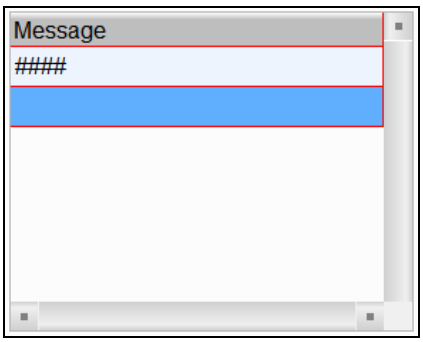
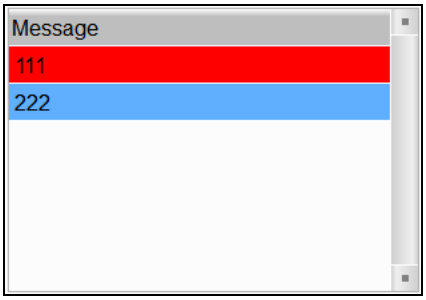
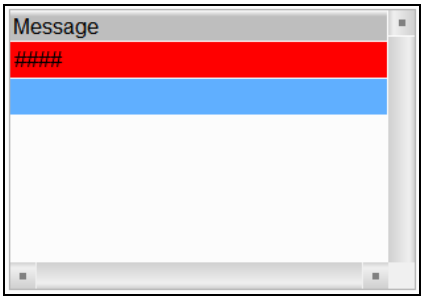
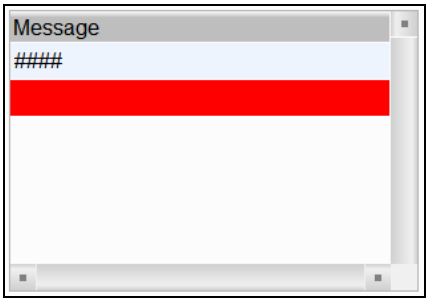
The following table lists the special properties of the **FTP file list** element with their description.

- **Main**

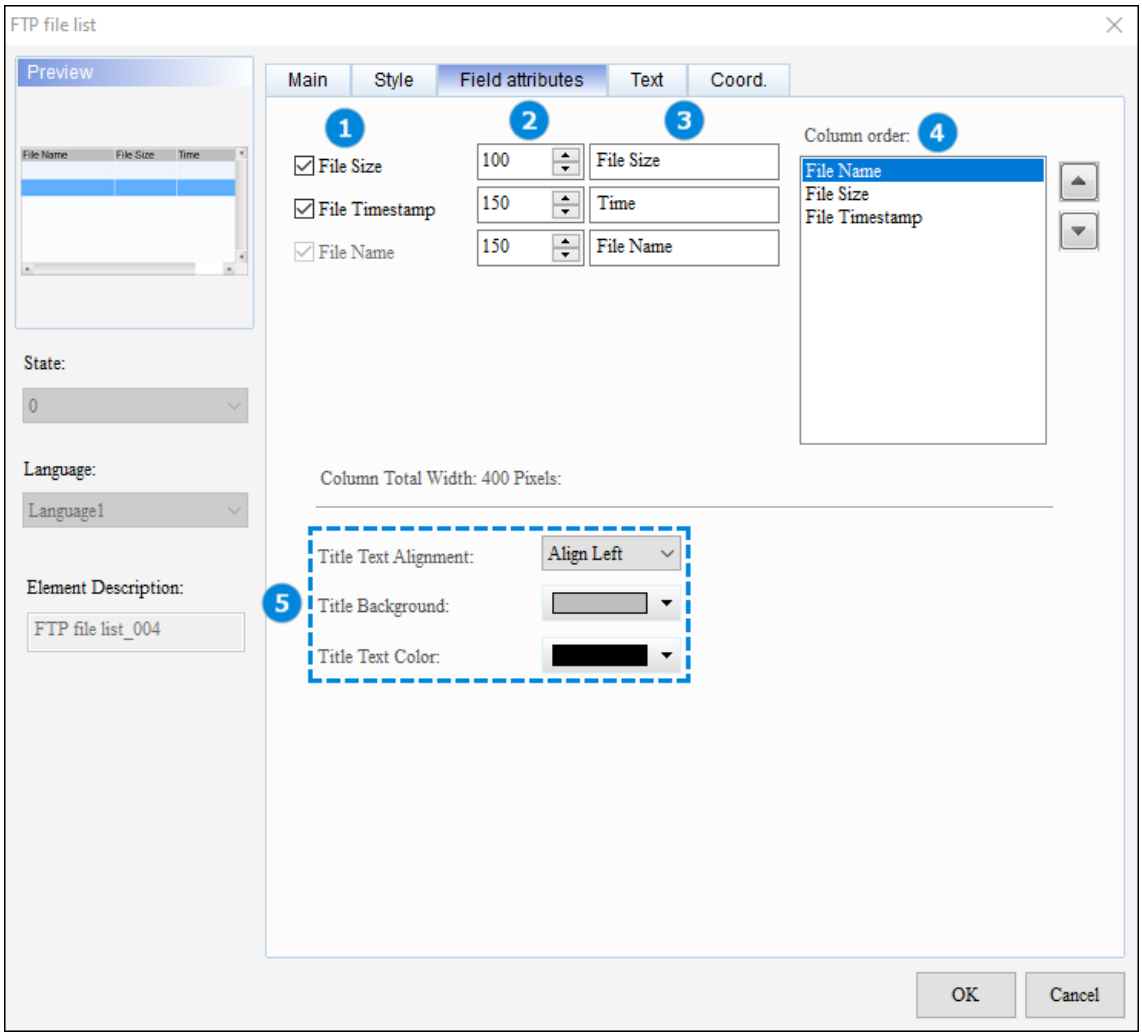
Setting	Description
Switch to the directory effective bit by clicking	Set the interlock address. When this bit is On, double-click the item to enter to the directory.
Prev page trigger bit	Set the trigger bit. When this bit is On, the element displays the switching page. After the action is completed, the bit will be automatically cleared.
Next page trigger bit	

- **Style**



Setting	Description
Gridline Color / Show Gridlines	Set the gridline color of the element. The default is white. When set to red, it displays as follows. **Note**: The **Gridline Color** setting is valid only when you select **Yes** for **Show Gridlines**. 
Select Row Color	Set the color of the selected row of data when operating the element on the HMI. The default is blue. When set to red, it displays as follows. 
Row Color	Set the color to display for each row of the alarm. The default is light blue. When set to red, it displays as follows. 
Alternating Row Color	Set the color for the alternating row of the alarm. The default is blue. When set to red, it displays as follows. 

- **Field Properties**



Legend	Setting	Description	
1	Column Display	Select the column to display on the element.	
2	Column Width	Select or enter the width for each column.	
3	Column title	Enter the title for each column.	
4	Column order	After selecting the column to display, click to adjust the display order of the column.	
5	Title bar	Text Alignment	Select the text of the title bar aligns to the left, center, or right.
		Background	Select the background color of the title bar.
		Text Color	Select the text color of the title bar.

- **Text**

Setting	Description
Text Property	Set the text properties.
Import Font Template	Click to open the **Font Template** dialog, and then select the template and import the text properties of the selected state.

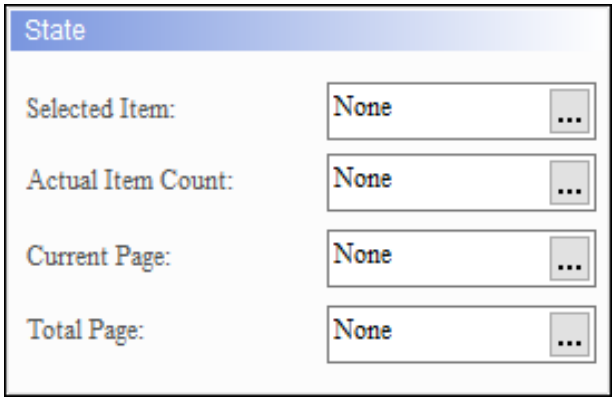
Text List

The **Text List** element reads the content of the File Slot and supports multiple text edit function.

Note: This function only applies to the DOP-107H Hand-Held model.

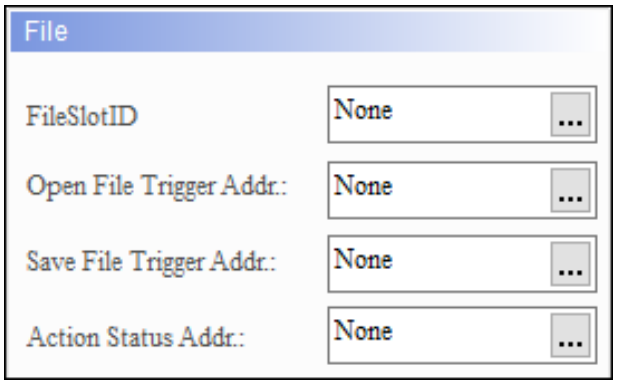
The following table lists the special properties of the **Text List** with their description.

- **Main**
  - **Data Settings**



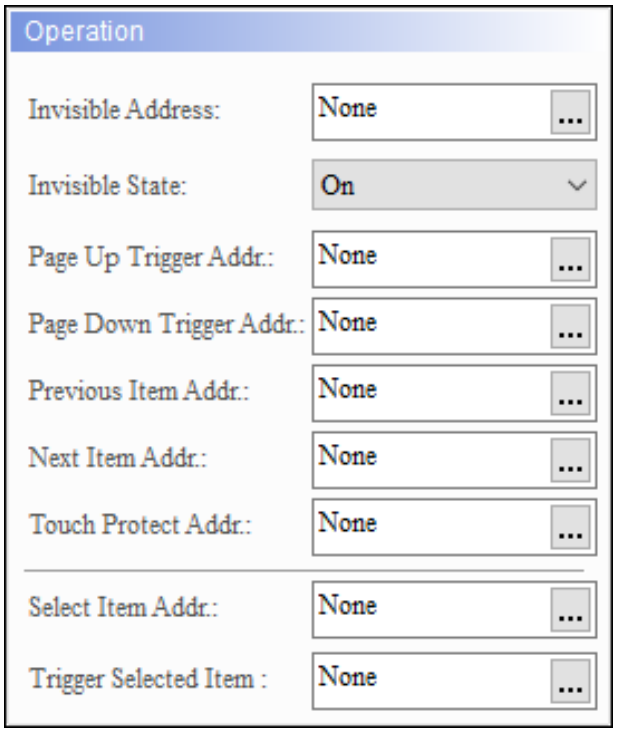
Setting	Description
Selected Item	Set the number of rows of text currently selected.
Actual Item Count	Set the total number of rows in the opened file.
Current Page	Set the currently displayed page. When the total number of rows of file data is greater than the display range of the element, the data will be displayed in pages.
Total Page	Set the total number of pages in the opened file.

- **File**



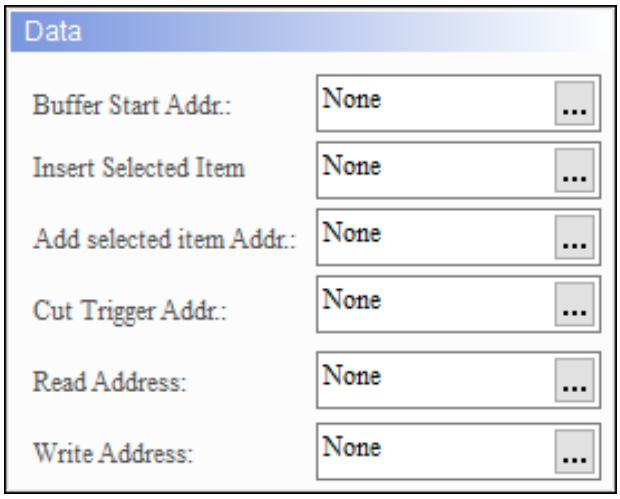
<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>FileSlot ID</td><td rowspan="3">Go to Project > Other Settings > FileSlot Management to configure FileSol, and then perform the following operations after specifying FileSlotID on the HMI.
Open File Trigger Addr.: After triggering the Open File Trigger Addr. to On, open the specified FileSlot content.
Save File Trigger Addr.: After triggering the Save File Trigger Addr. to On, save the current content to the file corresponding to the specified FileSlotID.
Note: The address is automatically cleared after completion.</td></tr>
<tr><td>Open File Trigger Addr.</td></tr>
<tr><td>Save File Trigger Addr.</td></tr>
<tr><td>Action Status Addr.</td><td>Obtain the execution status of the current Text List.
<table>
<tr><th>State Value</th><th>Description</th></tr>
<tr><td>0</td><td>Zero state</td></tr>
<tr><td>1</td><td>Row number does not exist</td></tr>
<tr><td>3</td><td>File does not exist</td></tr>
<tr><td>5</td><td>Failed to open file</td></tr>
<tr><td>6</td><td>Failed to write file</td></tr>
<tr><td>7</td><td>Triggered Open File Trigger Addr. or Save File Trigger Addr. without specifying FileSlotID</td></tr>
<tr><td>13</td><td>Element is touch-protected</td></tr>
<tr><td>14</td><td>Element is invisible</td></tr>
<tr><td>16</td><td>Exceeded the maximum file size (Maximum: 20 MB)</td></tr>
</table></td></tr>
</table>

- **Operation**



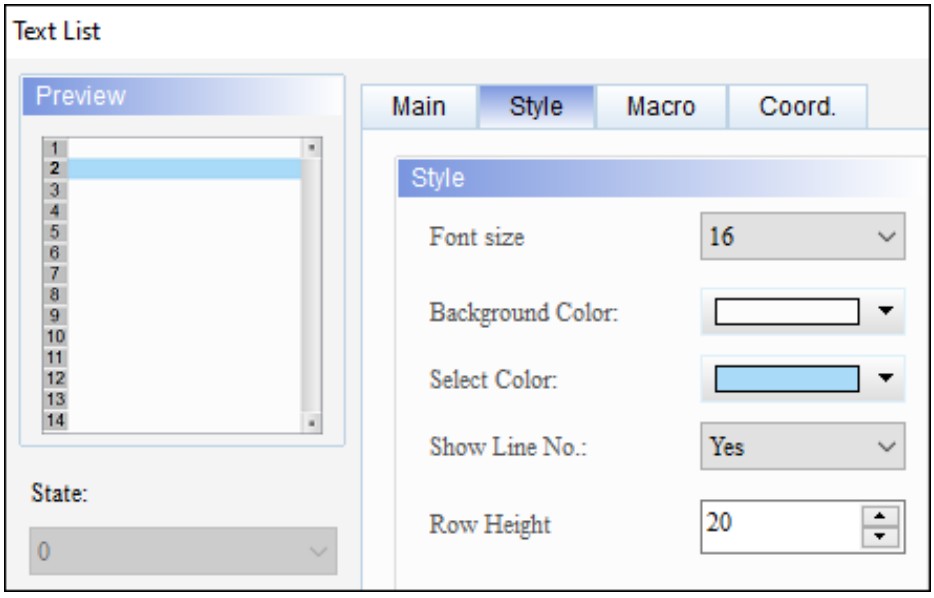
Setting	Description
Page Up Trigger	Switching when the trigger address is On, and the address is automatically cleared after completion.
Page Down Trigger	
Previous Item Addr.	
Next Item Addr.	
Touch Protect Addr.	When the **Touch Protect Addr**. is triggered On, the Text List element cannot be touched. Other related operations are invalid, and the **Action Status Addr.** will be set to 13.
Select Item Addr.	Specify the item by setting the **Select Item Addr.** and trigger the **Trigger Selected Item**, and the element will select the specified item.
Trigger Selected Item	

- **Data**



Setting	Description
Buffer Start Addr.	Specify the item through the **Select Item Addr.** and trigger **Trigger Selected Item**, enter then perform the following operations after entering text at the Buffer Start Addr. and triggering the control bit. • **Insert**: Insert the content of the **Buffer Start Addr.** into the previous row of the selected item. • **Add**: Add the content of the **Buffer Start Addr.** to the previous row of the selected item. • **Cut**: Cut the text of the selected item and paste it into the **Buffer Start Addr.** • **Read**: Copy the text of the selected item to the **Buffer Start Addr.** • **Write**: Replace the text of the selected item with the content of the **Buffer Start Addr.** **Note**: The trigger address is automatically cleared after the action is completed.
Insert Selected Item	
Add selected item	
Cut Trigger Addr.	
Read Address	
Write Address	

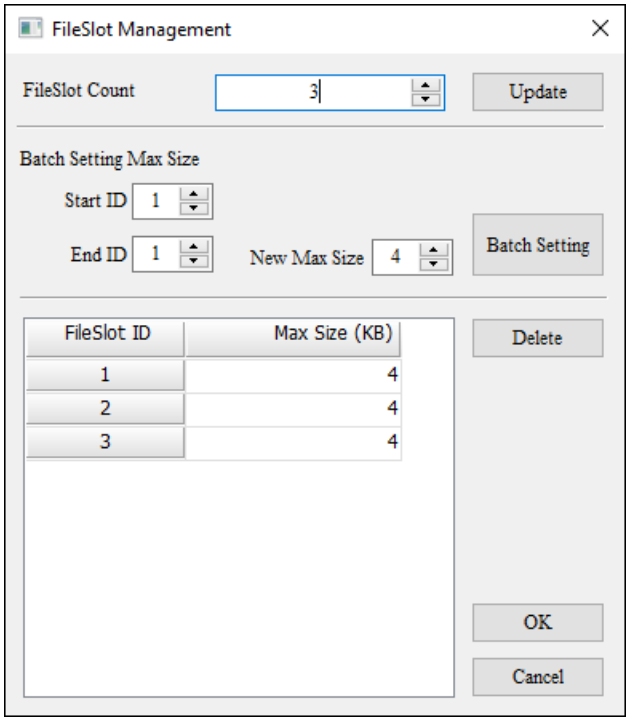
- **Style**



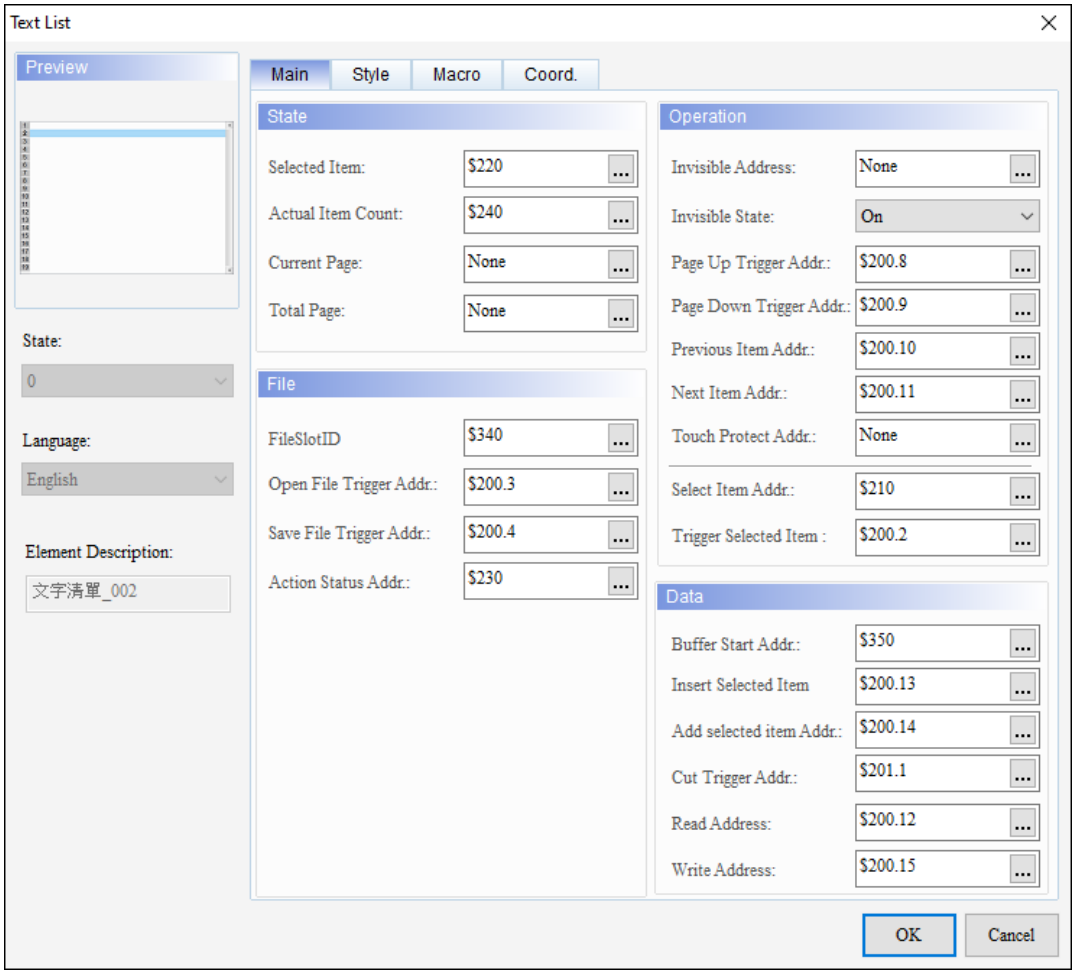
Setting	Description
Font size	Select the font size of the element.
Background Color	Select the background color of the element.
Select Color	Select the color of the selected item.
Show Line No.	Choose whether to display the line number.
Row Height	Select the row height of each row. The range value is 1–256, and the default value is **20**.

The following describes how to use the **Text List** element.

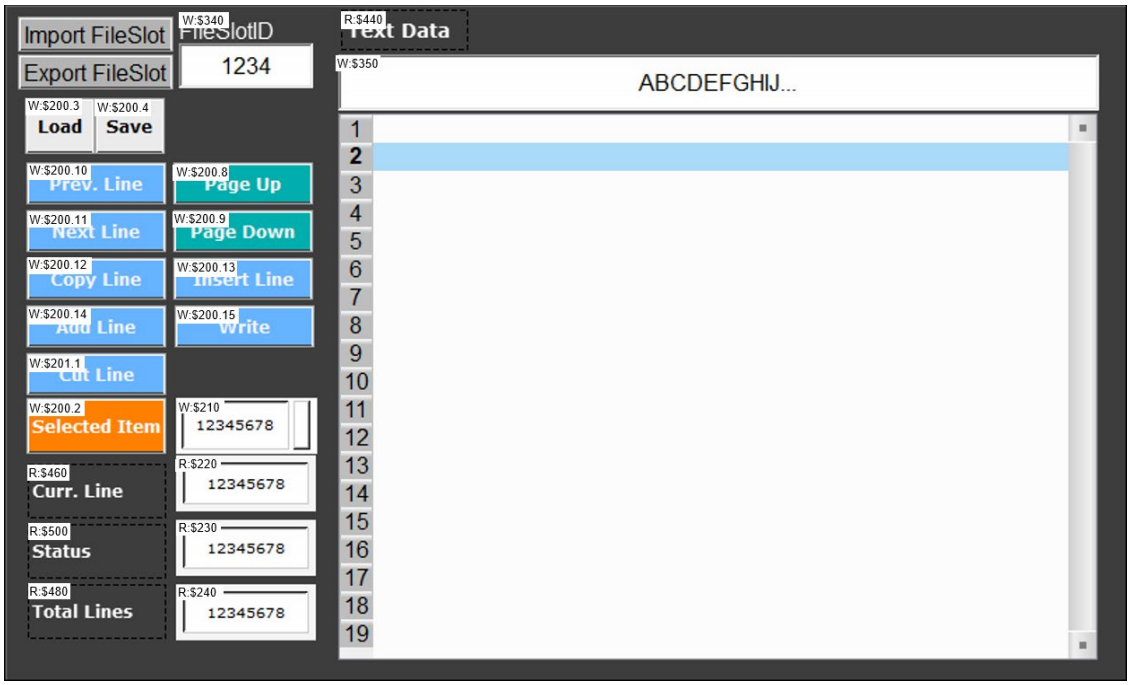
1. Click **Project** > **Other Setting** > **FileSlot Management**, and then add 3 FileSolt in the **FileSlot Management** dialog.



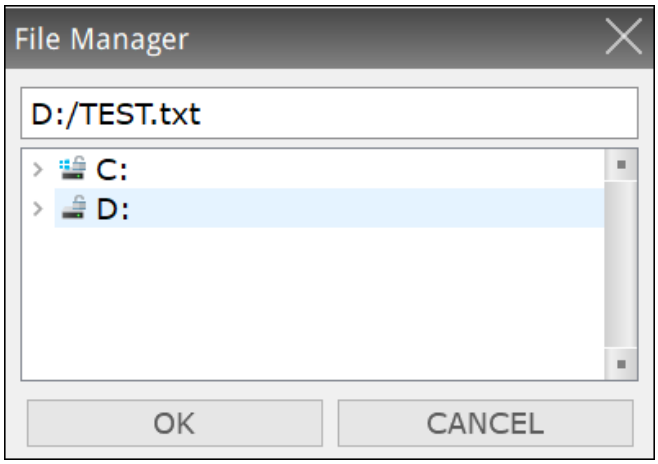
2. Create a **Text List** element on the screen, and then configure the address corresponding to each property in the element properties dialog.



3. Create the control elements corresponding to the used memory address. Refer to the following figure for screen planning.

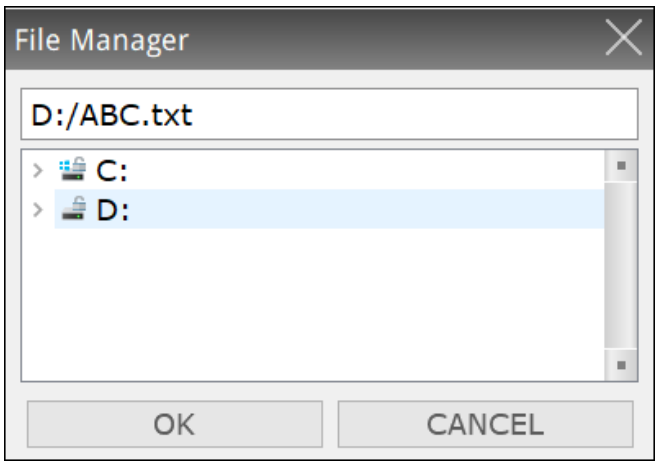


4. Create an **Import / Export FileSlot** button element and set the **FileSlotID** to **$340**.
5. Compile and download the screen to the HMI, and then save the TXT file to import to the USB and insert it to the HMI.
6. Specify the FileSlot ID on the HMI, and then click **Import FileSolt**.
7. Select or enter the TXT file path to import in the **File Manager** dialog, and then write the file into FileSlot.



8. Press the **Load** button to load FileSlot data to the **Text List** element.
9. After specifying the item, enter text in the **Text Data** (**Buffer Start Address**) field, you can perform operations such as copying and cutting, and then write the text into the element.

10. After the edition is completed, press the **Save** button to write the text content into FileSlot.
11. Press the **Export FileSolt** button to select or enter the file path to export in the **File Manager** dialog, and then write the FileSlot data into the file.

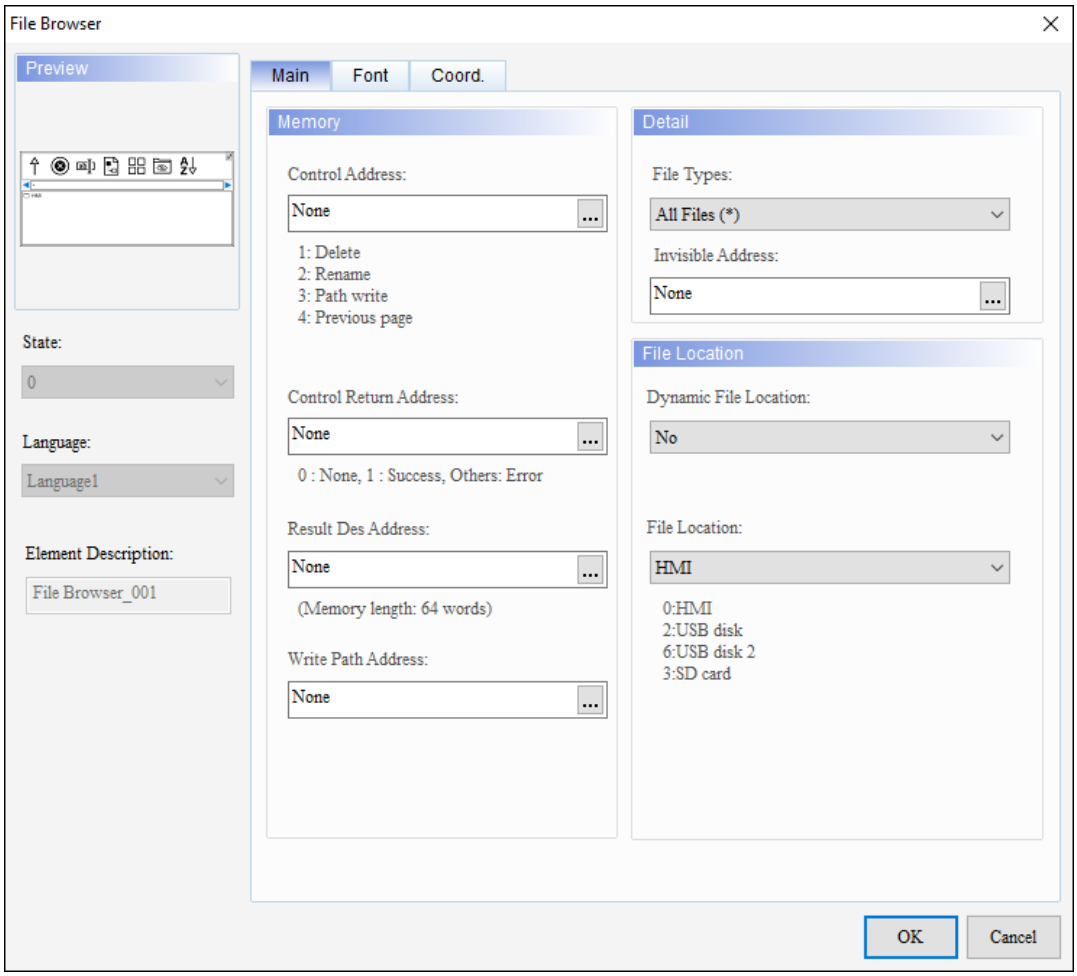


File Browser

You can browse the stored device files on HMI, and rename and delete files through the **File Browser** element.

The following table lists the properties of the **File Browser** element with their descriptions.

- **Main**



  - **Memory**

<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Control Address</td><td>Set the address to trigger actions. The following table lists the corresponding actions of the state value.
<table>
<tr><th>State Value</th><th>Action</th></tr>
<tr><td>1</td><td>File deletion</td></tr>
<tr><td>2</td><td>File rename</td></tr>
<tr><td>3</td><td>Path write</td></tr>
<tr><td>4</td><td>Switch to the previous page</td></tr>
</table></td></tr>
<tr><td>Control Return Address

Result Describes Address</td><td>Set the return address after the action is completed, the status code and description of the return are as follows:
<table>
<tr><th>Action</th><th>Status</th><th>Description</th></tr>
<tr><td>No action</td><td>0</td><td>-</td></tr>
<tr><td>Succeed</td><td>1</td><td>-</td></tr>
<tr><td>File deletion</td><td>–1</td><td>Failed to delete the file</td></tr>
<tr><td rowspan="5">File rename</td><td>–2</td><td>File name is too long
Note: The maximum length of the file name is 255 characters.</td></tr>
<tr><td>–3</td><td>File does not exist</td></tr>
<tr><td>–4</td><td>The file name cannot contain the following characters: \/:*?\"<>|</td></tr>
<tr><td>–5</td><td>The name of the file has not been changed</td></tr>
<tr><td>–6</td><td>The same file name exists</td></tr>
<tr><td rowspan="2">Path write</td><td>–7</td><td>The path does not exist</td></tr>
<tr><td>–8</td><td>The path write address is not set</td></tr>
</table>
Note:
• The data type of Control Return Address is signed decimal.
• If the HMI language is non-English, the result description should be displayed using a Multi-language Input element.</td></tr>
</table>

Setting	Description
Write Path address	Do one of the following to write path address: • Select the file on the HMI and click on the toolbar of the element to write the path. • Write value 3 for **Control Address**, and then write the current file path for **Write Path Address**. Write the file path to this address, and the file browser redirects to the specified file. **Note**: • The file path does not contain the root directory, but the path after the address is written to the file location. • Set this parameter using the **Character Display** or the **Multi-language Input** element.

- **Detail**

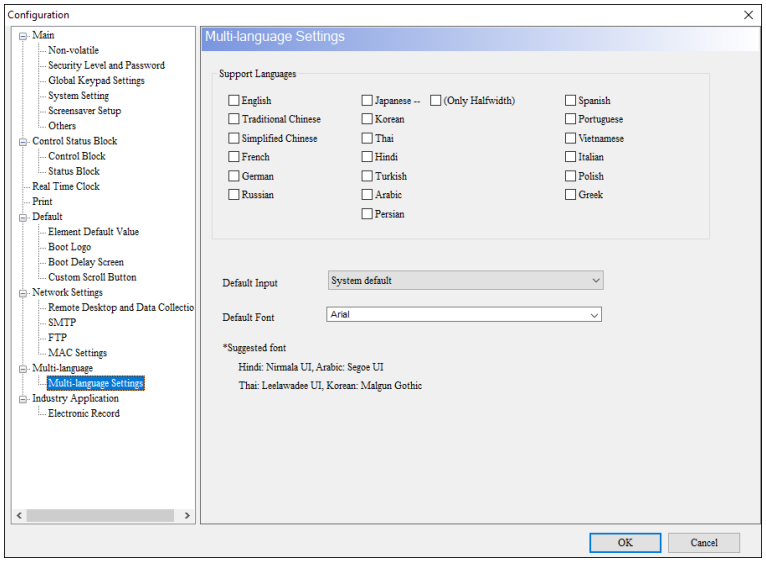
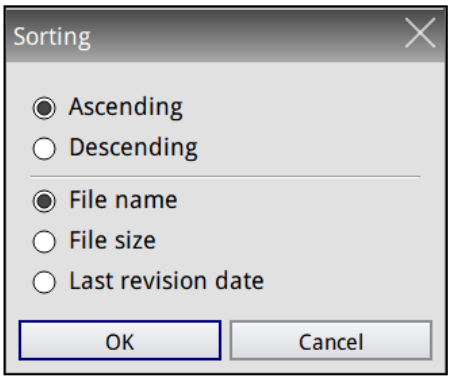
Function	Description
File Types	Select the type of file to display.
Invisible Address	Set the invisible address. When the invisible bit is On, the element is not visible to the HMI.

- **File Location**

Function	Description
Dynamic File Location	Choose whether to set the file location dynamically. Select **Yes** to select by the root directory of the address file.
File Location	Choose the root directory to which the path is written.

The following table lists the functions on the toolbar of the **File Browser** element with their description.

Icon	Function	Description
	Previous	Click to switch to the previous level of directory.

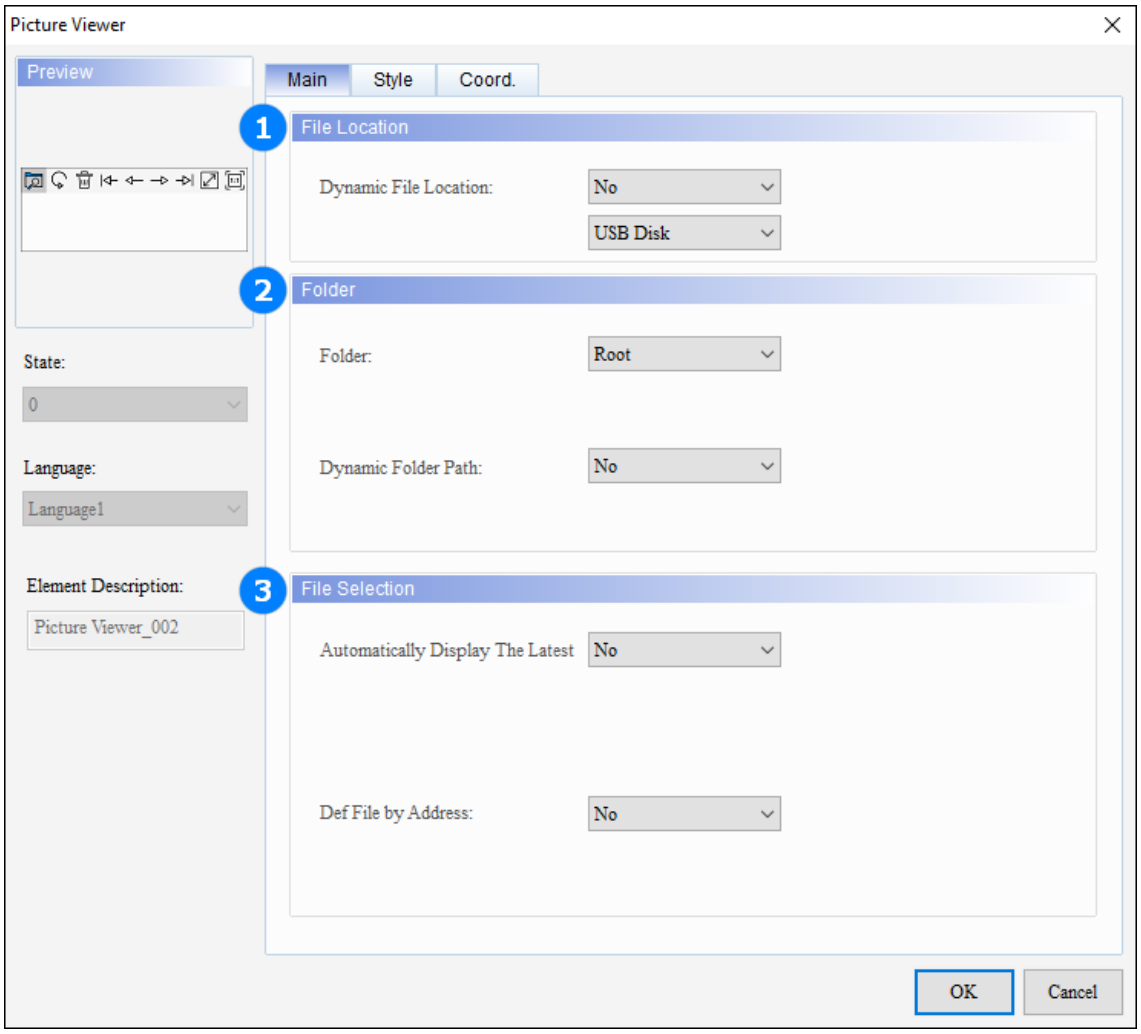
Icon	Function	Description
	File deletion	Click to delete the currently selected file.
	File rename	After selecting a file, click this icon to enter the file name to change the name of the selected file. **Note**: If you want to input words other than English or numbers, check the languages in the **Configuration** dialog. 
	Path write	After selecting the file, click this icon to display the full path of the file at the **Write Path Address**. **Note**: If you don't select a file, clicking this button has no effect.
	Display toggle	Click to display files as icons or manifests.
	Folder switching	Click to display or not display the folder.
	File sorting	Click to set how the files are sorted. 

Picture Viewer

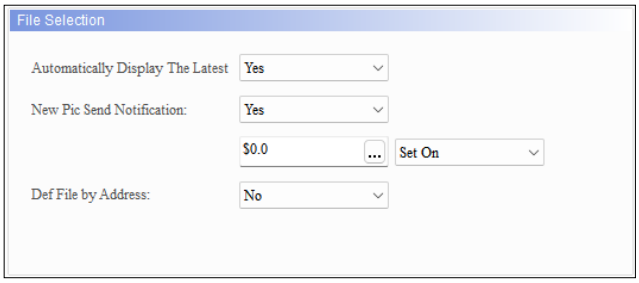
You can select files and view pictures on the HMI through the **Picture Viewer** element.

The following lists the properties of the **Picture Viewer** element and their descriptions.

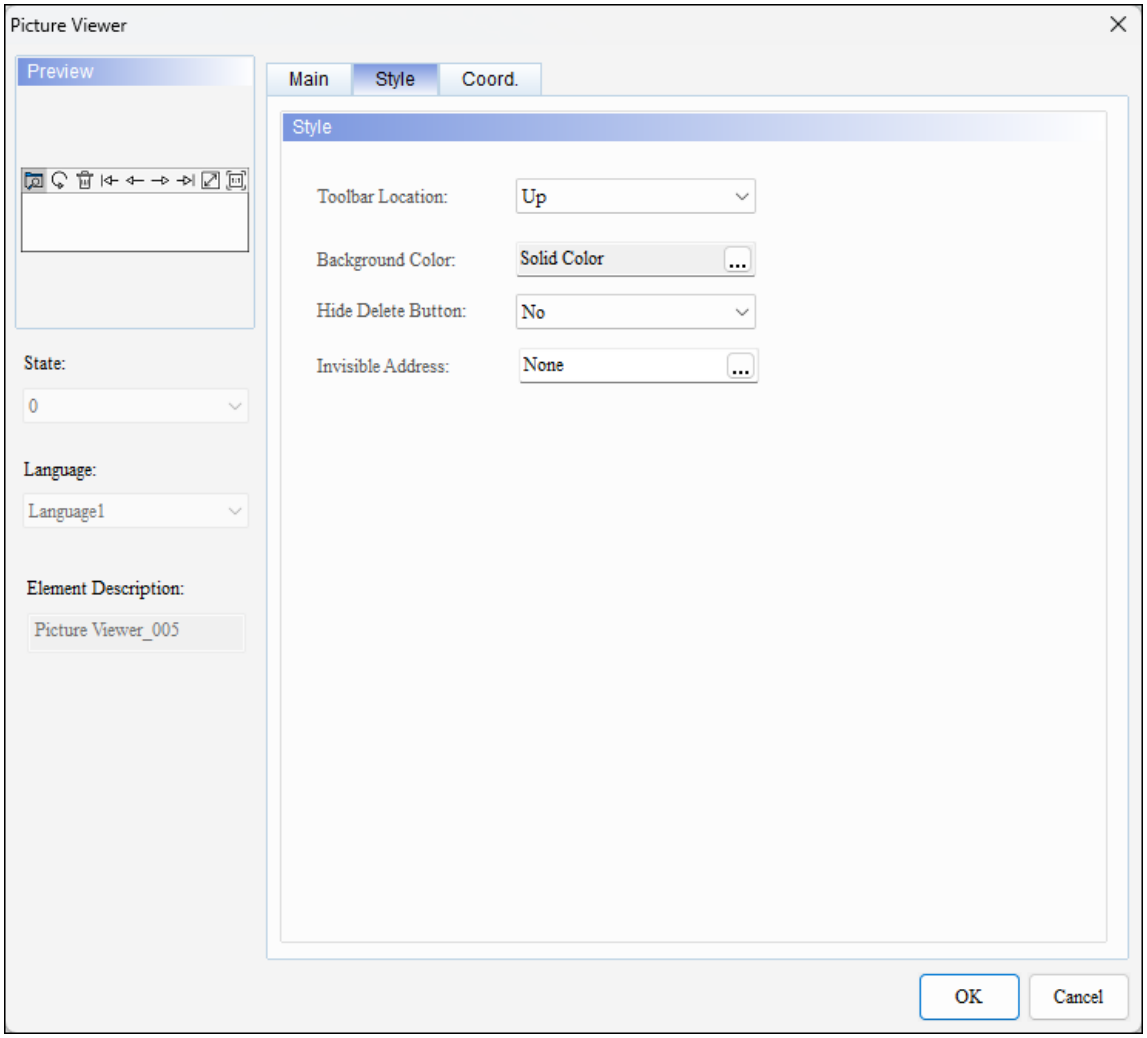
- **Main**



Legend	Function	Description
1	File Location	Select the location to store the files. • If **Dynamic File Location** is set to **No**, the storage device can be specified for reading. • If **Dynamic File Location** is set to **Yes**, the storage device can be dynamically specified on the HMI. The value 2 is USB disk, 6 is USB disk 2, 3 is SD card.

Legend	Function	Description
②	Folder	Set the folders to display in the File Browser. • **Root**: The root directory of the specified **File Location**. • **Capture Picture**: The screen_capture folder under the specified **File Location**. • **Define By User**: The specified folder under the specified **File Location**. If **Dynamic Folder Path** is set to **Yes**, the folder to display can be dynamically specified on the HMI. **Note**: You can only specify internal memory address of the HMI for **Dynamic Folder Path**.
③	File Selection	• Choose whether to automatically display the latest pictures. Select **Yes** to select whether to send a notification when the picture is updated, and specify the memory address of the notification.  • Choose whether to specify the file by the address. Select **Yes** to specify the file name of the picture by the address on the HMI to display the picture on the element.

- **Style**



Setting	Description
Toolbar Location	Select the location to which the toolbar is displayed on the element.
Background Color	Select the background color of the element.
Hide Delete Button	Select whether to hide the delete button.
Invisible Address	Set the invisible address. When the invisible bit is On, the element is not visible to the HMI.

Note:

- Only one **Picture Viewer** element can be configured for a screen, and only for a general screen.

- The picture will not be loaded by default when the HMI starts, and the picture will only be displayed if you enter the **File Browser** element and click on the picture file, or specify the file name by the file address.

The following table lists the functions on the toolbar of the **Picture Viewer** element with their descriptions.

Icon	Function	Description
	File browsing	Click to toggle to **File Browser**. The directory under the set **File Location** is displayed by default. For the functions on the toolbar of the **File Browser**, refer to File Browser.
	Refresh	Click to reload the currently displayed picture.
	Delete	Click to delete the currently displayed picture.
	The first picture	Displays the pictures in alphabetical order by file name.
	Previous picture	
	The next picture	
	The last picture	
	Maintain proportions	Click to maintain the picture's aspect ratio. The ratio scales according to the size of the element.
	Actual size	Click to display the original size of the picture.

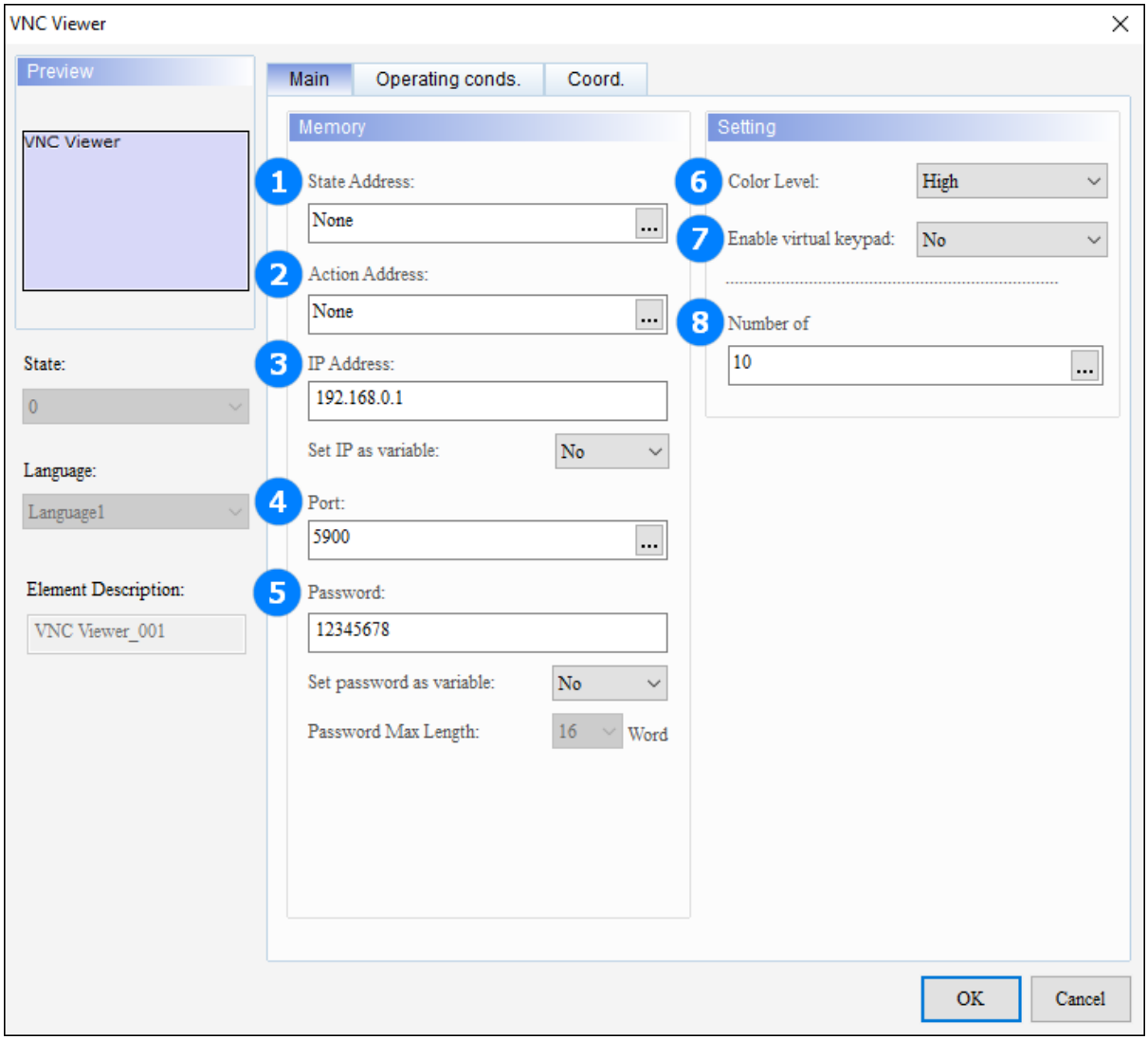
Frame and Multimedia

VNC Viewer

VNC (Virtual Network Computing) uses the RFB protocol to achieve screen sharing and remote operation. It can transmit keyboard and mouse movements and real-time images through the network. The **VNC Viewer** element of the HMI receives and controls the display device operated by the server.

Note: This function is only available for DOP-300 series models.

The following table lists the settings in the property dialog of the VNC Viewer element with the description.



<table>
<tr><th>Legend</th><th>Function</th><th>Description</th></tr>
<tr><td>1</td><td>State Address</td><td>Set the State Address in Word type.
Obtain the current connection state through the State Address values. The following table lists the state values with the corresponding description.<table><tr><th>State</th><th>Description</th></tr><tr><td>0</td><td>Stopped</td></tr><tr><td>1</td><td>Connecting</td></tr><tr><td>2</td><td>Connect failed</td></tr><tr><td>3</td><td>Authentication error</td></tr><tr><td>4</td><td>Server disconnect</td></tr><tr><td>5</td><td>Security error</td></tr><tr><td>6</td><td>Attempting to reconnect</td></tr><tr><td>7</td><td>Reconnect failed</td></tr><tr><td>8</td><td>Reconnect successful</td></tr></table></td></tr>
<tr><td>2</td><td>Action Address</td><td>Set the Action Address in Word type.
Send the action to the VNC device through the Action Address. The following table lists the state values with the corresponding description.<table><tr><th>State</th><th>Description</th></tr><tr><td>0</td><td>None</td></tr><tr><td>1</td><td>Start</td></tr><tr><td>2</td><td>Stop</td></tr><tr><td>3</td><td>Update</td></tr></table></td></tr>
<tr><td>3</td><td>IP Address</td><td>Set the IP address for the VNC server.
When selecting Yes for Set IP as variable, you can dynamically change the IP address on the HMI. Use the 4 Word addresses read by the IP starting address continuously as the set IP address.</td></tr>
<tr><td>4</td><td>Port</td><td>Set the Port for the VNC server. The default value is 5900.</td></tr>
</table>

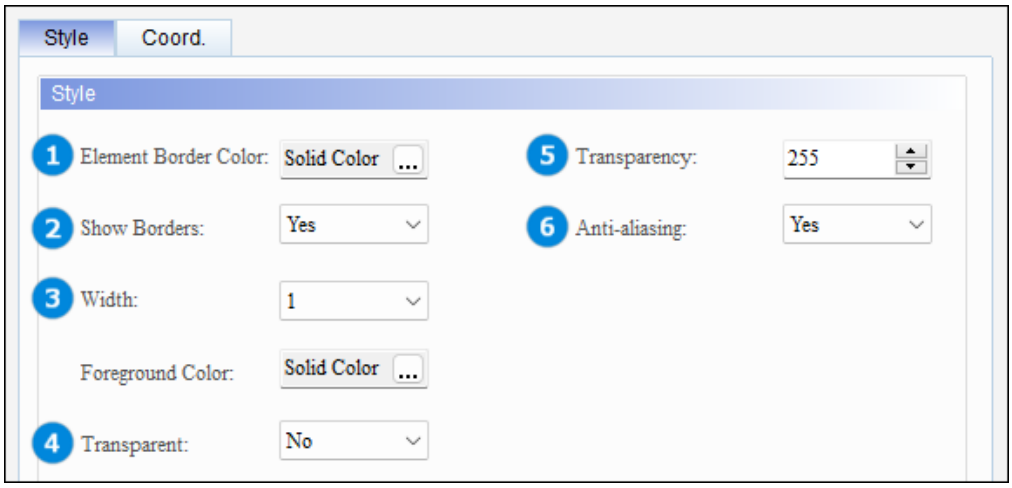
Legend	Function	Description
⑤	Password	When selecting **Yes** for **Set password as variable**, you can dynamically change the VNC connection password on the HMI. Use the set **Password Max Length**, read by the set starting address, as the password. Password can consist of numbers and letters.
⑥	Color Level	Set the display color for different bits. The available options are High (32 Bit), Middle (16 Bit), and Low (8 Bit). The default value is **High**.
⑦	Enable virtual keypad	If select **Yes**, the **Keyboard** button appears at the bottom right corner of the element after clicking the **VNC Viewer** element on the HMI. Click the **Keypad** button to call out the ASCII keypad.
⑧	Number of	• If the connection to the server is disconnected, a message “Attempting to reconnect to VNC server” appears. When the connection is not restored after the set retries are reached, it will be automatically aborted and a message “Disconnected from VNC server” will appear. • If set to 0, it means the number of retries is unlimited. The maximum number is 65535, and the interval between each retry is 3 seconds.

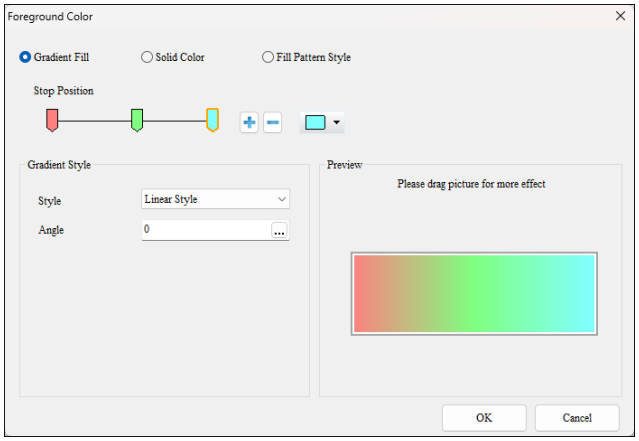
Basic Shape

Basic shape elements include a variety of geometric shapes, and you can set their colors and borders.

Common Properties

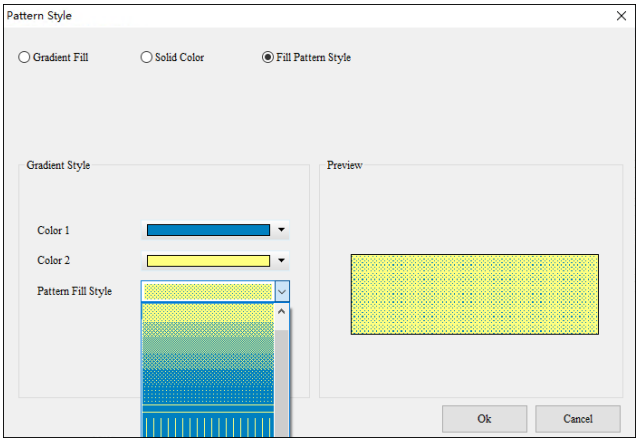
- **Style**

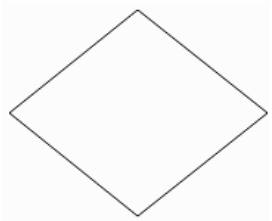
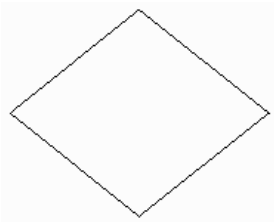


Legend	Setting	Description
1	Element Border Color Foreground Color	Set the display color of the element. • **Gradient Fill:** Set the gradient style by setting the current angle or center point. You can drag the mouse on the preview image to set the current angle or center point, or set the memory address to dynamically configure it on the HMI. You can also set the stop point colors to present the effect of multi-color gradients.  **Note**: Up to 5 stop points can be set.

<table>
<tr><th>Legend</th><th>Setting</th><th>Description</th></tr>
<tr><td></td><td></td><td>

Solid Color: Select a single-color fill.
Fill Pattern Style: Select a two-color combination configuration and select a pattern fill style.



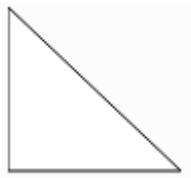
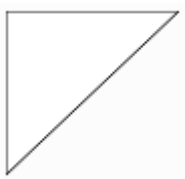
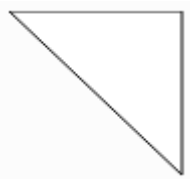
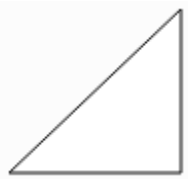
</td></tr>
<tr><td>2</td><td>Show Borders</td><td>Select whether to display the element border.</td></tr>
<tr><td>3</td><td>Width</td><td>Select the border width of the element, and the range value is 1~8.</td></tr>
<tr><td>4</td><td>Transparent</td><td>Select whether to display the element in a transparent. The default value is No.
If set to Yes, the setting of the Foreground Color does not have effect.</td></tr>
<tr><td>5</td><td>Transparency</td><td>Select or enter the transparency value of the element. The range value is 50–255, and the default value is 255.
The smaller the value, the higher the transparency of the element.</td></tr>
<tr><td>6</td><td>Anti-aliasing</td><td>Select whether to enable the Anti-aliasing effect.
If set to Yes, the display of the elements will be more delicate, and there will be no aliasing.
<table>
<tr><th>Yes</th><th>No</th></tr>
<tr><td>  </td><td>  </td></tr>
</table>
</td></tr>
</table>

- **Coord.**

Set the XY coordinates, width, and height on the upper left of the element.

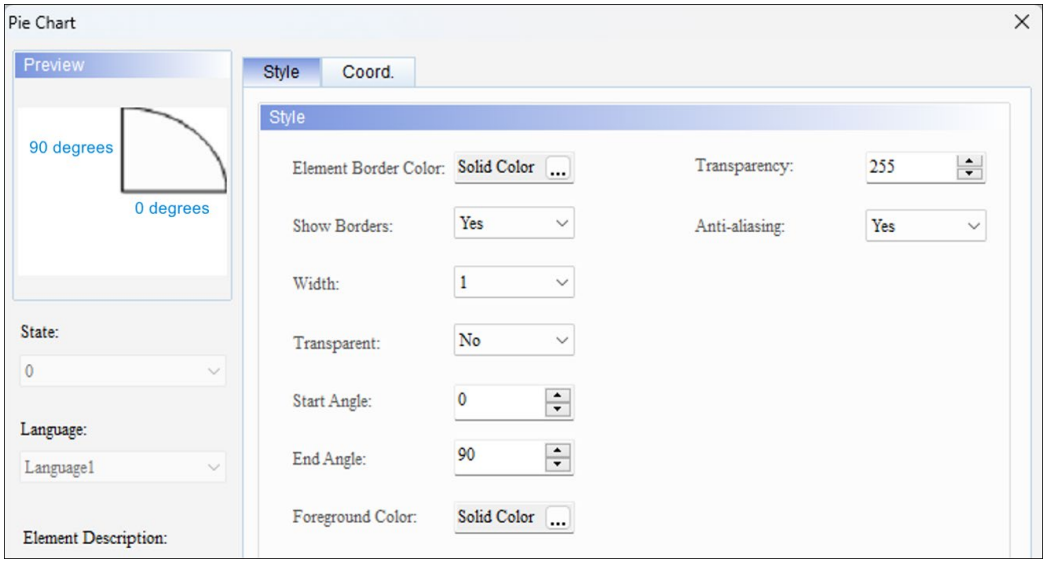
Right Triangle

Select the **Style** of the element. The available options are Standard, Rotation 90, Rotation 180, and Rotation 270. The following table lists the effect of each style.

Standard	Rotation 90	Rotation 180	Rotation 270
			

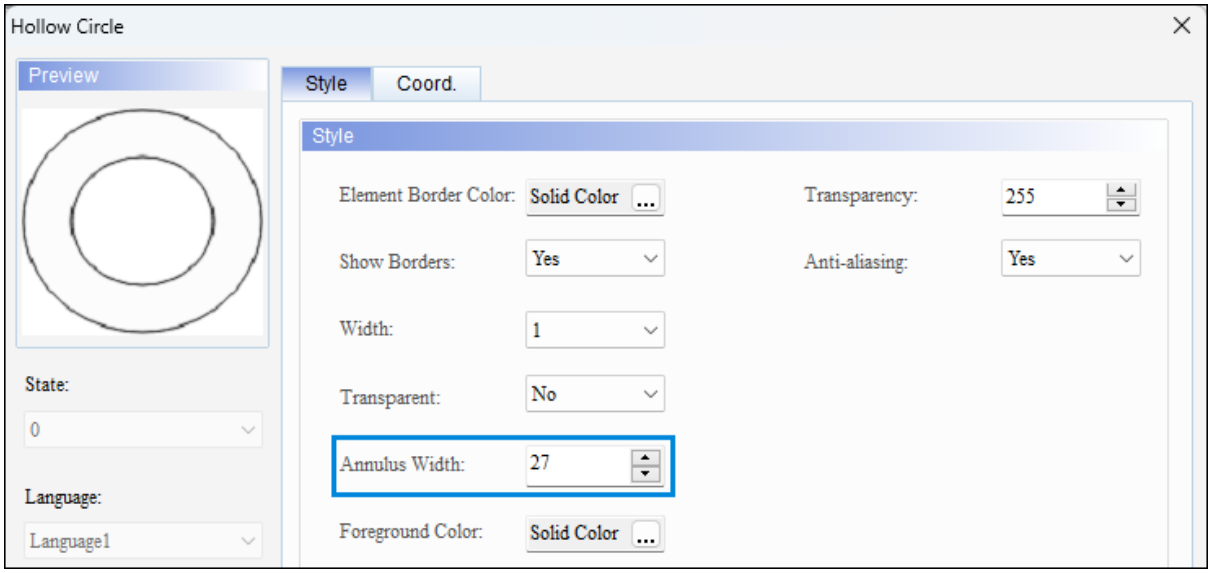
Pie Chart / Arc

Set the **Start Angle** and **End Angle** to determine the opening angle of the shape. Take the three o'clock direction as 0 degrees, and draw the shape counterclockwise.



Hollow Circle / Stop Circle

Set the **Annulus Width** between the two circles of the element. The maximum value is the minimum width or height divided by 2.



Drawing

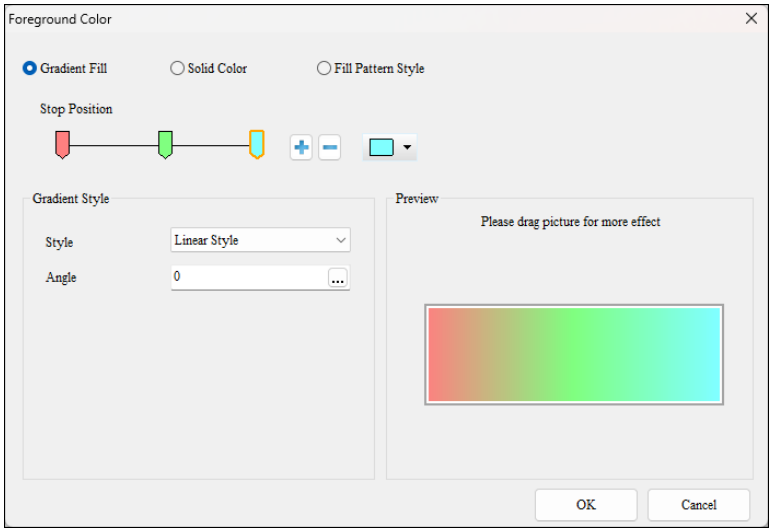
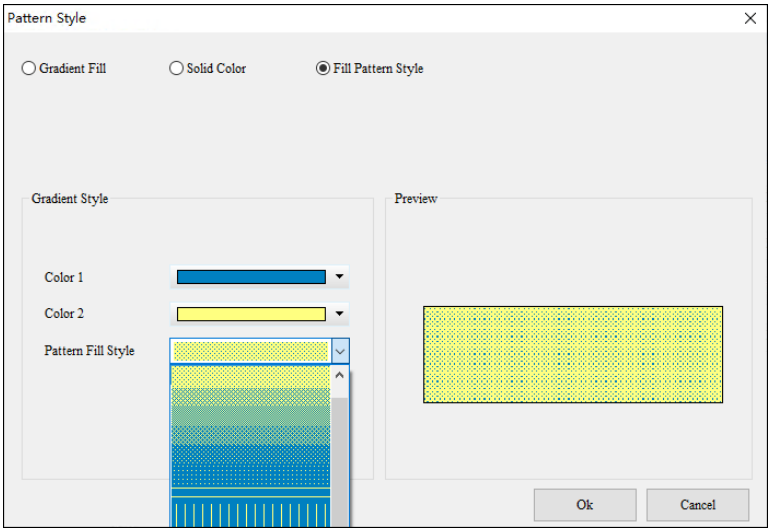
The **Drawing** elements contain multiple shape elements and elements for auxiliary marks. You can control the display position, color, and size of the element through the memory address.

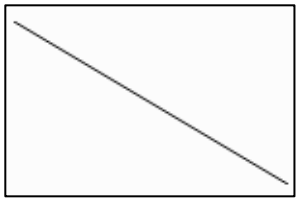
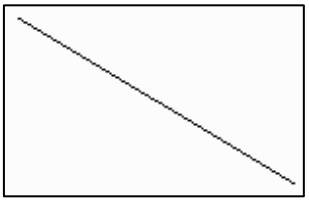
Common Properties

- **Main**

Setting	Description
Read Address	You can select the **Internal Memory** address or controller register address, and the **Input** memory **Type** can only be **Word**. **Note**: This memory address is used to control the display style of the element on the HMI.
Read Offset Address	The HMI will configure the value to the corresponding read address according to the set Offset Address. After setting the offset address, the actual read address is as follows. The actual read address of the element: Read Address + Value in Read Offset Address × Element [Data Type]
Data Format	Select the data format, and the default value is **Unsigned Decimal**.

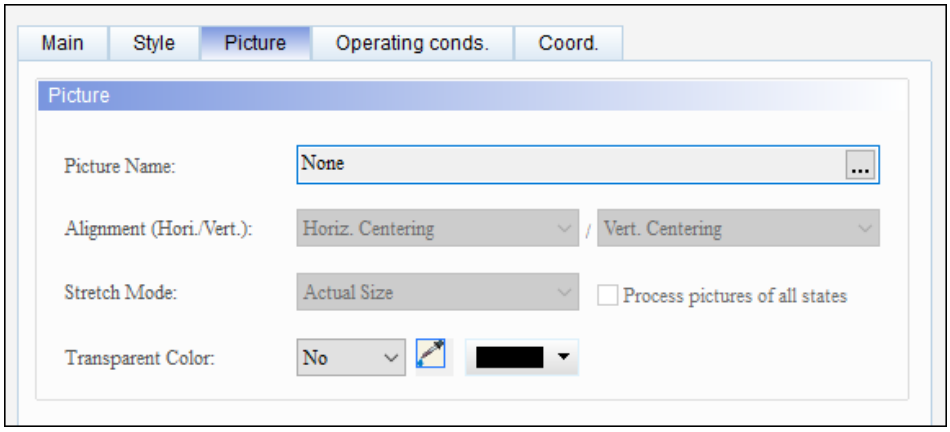
- **Style**

Setting	Description
Foreground Color	Set the display color of the element. • **Gradient Fill:** Set the gradient style by setting the current angle or center point. You can drag the mouse on the preview image to set the current angle or center point, or set the memory address to dynamically configure it on the HMI. You can also set the stop point colors to present the effect of multi-color gradients.  **Note**: Up to 5 stop points can be set. • **Solid Color**: Select a single-color fill. • **Fill Pattern Style**: Select a two-color combination configuration and select a pattern fill style. 

<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Blink</td><td>Select whether to enable the blink prompt, and the default value is No.
If select Yes, the dynamic element will blink when its corresponding Read Address value is greater than 1; the element will not blink when the value is 0.
If select No, the element will not blink.
Note: This property is only applied to the Line, Rectangle, Circle elements.</td></tr>
<tr><td>Transparent</td><td>Selet whether the element is displayed in transparent, and the default value is No.
If select Yes, the setting of the Foreground Color is invalid.
Note: This property is only applied to the Rectangle, Circle, and Polygon elements.</td></tr>
<tr><td>Transparency</td><td>Select or enter the transparency of the element. The range value is 50–255 and the default value is 255.
The smaller the value, the higher the transparency of the element.</td></tr>
<tr><td>Anti-aliasing</td><td>Select whether to enable the Anti-aliasing function. If select Yes, the display of the elements will be more delicate, and there will be no aliasing.<table><tr><th>Yes</th><th>No</th></tr><tr><td>  </td><td>  </td></tr></table>Note: This setting only has effect when the Round Corner Radius of the Rectangular element is not 0.</td></tr>
</table>

- **Picture**

Note: The Picture tab is only applied to the **Rectangle**, **Circle**, and **Polygon** elements.



Setting	Description
Picture Name	Enter the picture name. The default value is **None**.To set the picture display, click [...] to select the picture to display in the picture bank and drag it to each state icon. To configure other pictures, refer to Picture Bank.
Alignment	Select the alignment of the picture.
Stretch Mode	Select the stretch mode of the picture.

Stretch All	Stretch 1:1	Actual Size
The picture fills the full element display area. 	The picture displays in 1:1 size based on the element width and length. 	The picture is displayed in the display area of the element in its actual size.

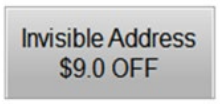
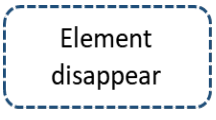
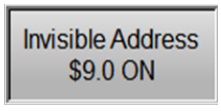
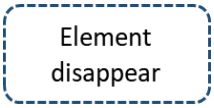
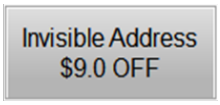
Setting	Description
Transparent Color	Select whether to specify a color in the picture as a transparent color. If set to **Yes**, you can select colors or use the color picker to select one of the colors on the picture to specify it as a transparent color, that is, the foreground color. The following figures show the effect of specifying white as a transparent color.

- **Operating Conds.**

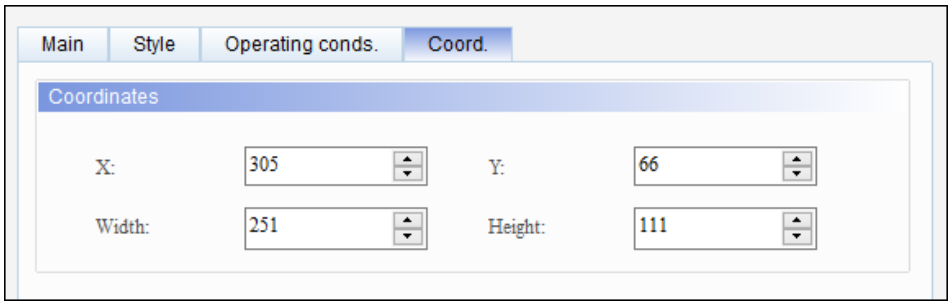
  The **Invisible Address** displays or hides the element on the HMI. If the **Invisible State** is set to **On**, the **Invisible Address** hides the element. Instead, if the **Invisible State** is set to **Off**, the **Invisible Address** hides the element.

  The following is an example of using invisible address.

  1. Create a button element on the screen and set its **Read Address** to $9.0, then set the **Invisible Address** of the element to hide to $9.0.
  2. To hide the element, you must first click the $9.0 button.

Invisible State is On		
		
Invisible State is Off		
		

- **Coord.**



Set the XY coordinates, width, and height on the upper left of the element.

Line

Select the **Line** element, press and hod the mouse button and drag, and then release the mouse button to complete the line drawing. The position where the mouse button is pressed is the starting point and the position where the mouse button is released is the end point.

The following table lists the special properties of the **Line** element with their description.

- **Main**

Setting	Description
Variable Position	Select whether the position can be changed. If select **Yes**, the position of the line on the HMI can be controlled by the memory address. • **Read Address** is regarded as the X coordinate of the starting point of the line. • **Read Address+1** is regarded as the Y coordinate of the starting point of the line.
Variable Color	Select whether the color can be changed. If select **Yes**, the color of the line can be dynamically changed on the HMI through the memory address. The range value is 0~65535. If the **Variable Position** is **Yes**, the line color is **Read Address+2**. **Note**: This setting has no effect when the **Element Line Color** is a multi-color setting such as **Gradient Fill** or **Fill Pattern Style**.

- **Style**

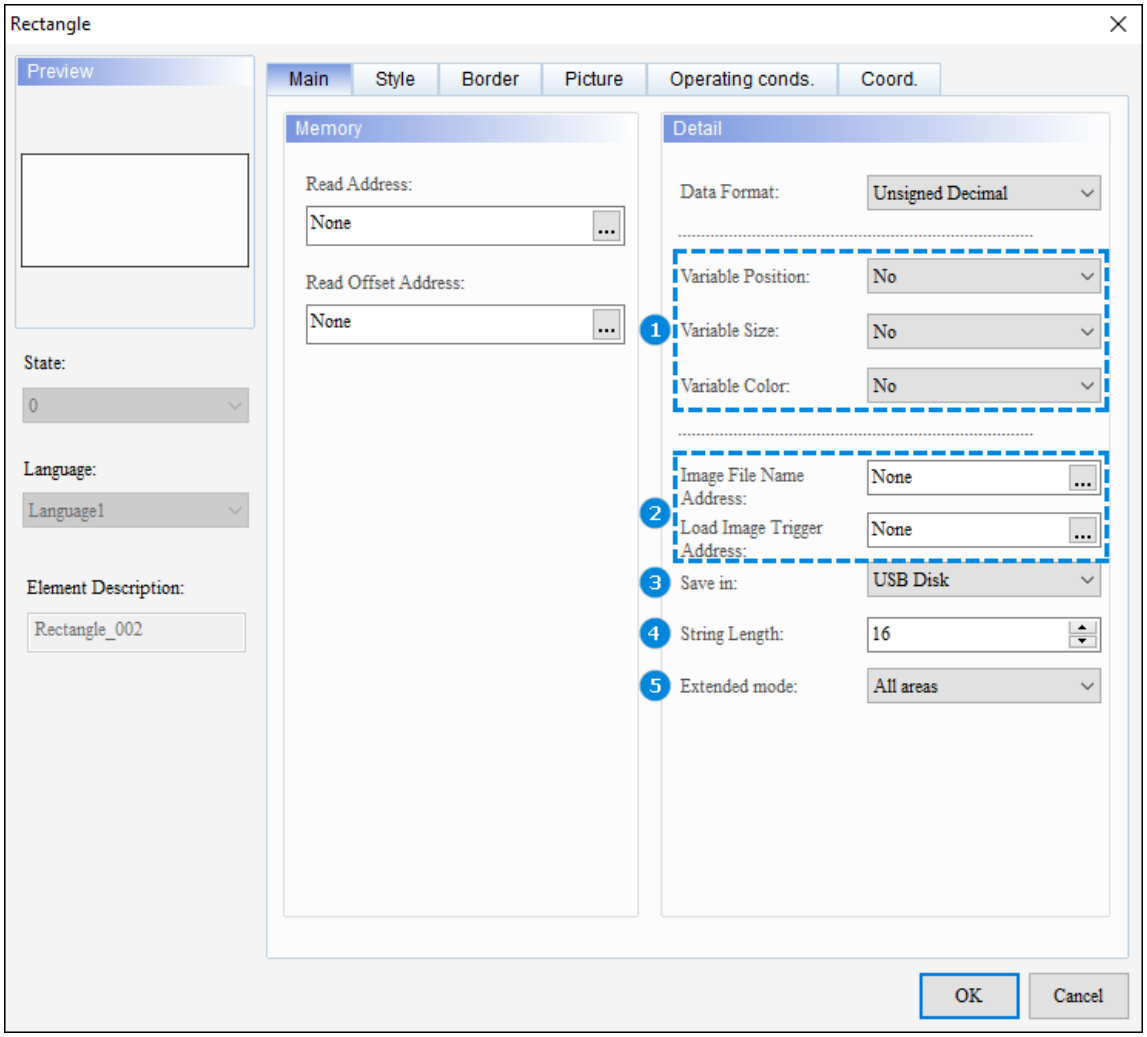
Setting	Description
Line Style	Select the display style of the line.
Line Width	Select the width of the line, the range value is 1~8.

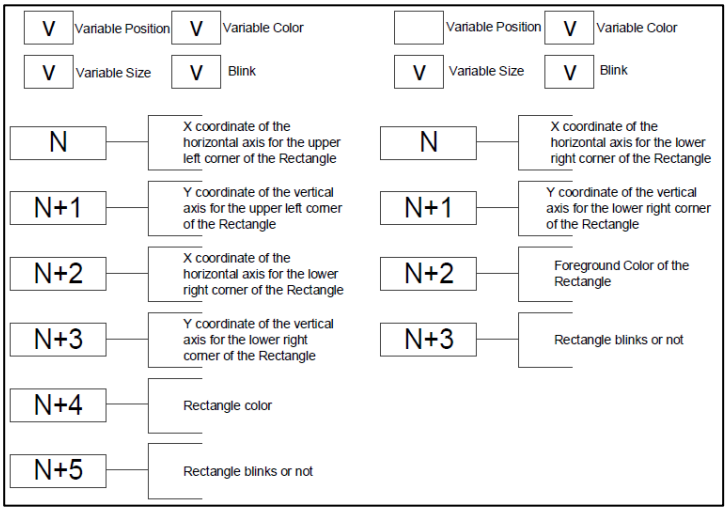
Rectangle

The **Rectangle** element provides various filled styles and border configurations and supports importing pictures from the picture bank. Use the set read address to control the moving position, color, size, and blinking of the rectangle element.

The following table lists the special properties of the **Rectangle** element with their description.

- **Main**

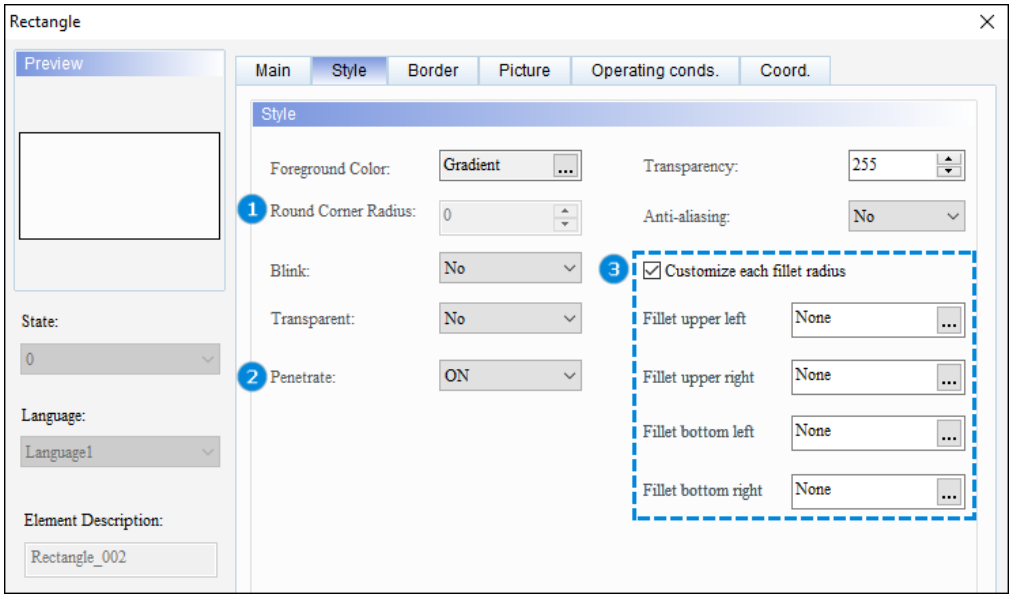


<table>
<tr><th>Legend</th><th>Setting</th><th>Description</th></tr>
<tr><td rowspan="3">1</td><td>Variable Position</td><td rowspan="3">Select whether the position, size, color can be changed. If select Yes, the position, size, color of the element can be dynamically changed on the HMI.
Read Address is regarded as the X coordinate of the left upper corner of the element.
Read Address+1 is regarded as the Y coordinate of the left upper corner of the element.
Read Address+2 is regarded as the width of the element.
Read Address+3 is regarded as the height of the element.
Read Address+4 is regarded as the color of the element, and the range value is 0~65535。
Read Address+5 determines whether the element blinks or not.
The above addresses are configured with all variables set to Yes. If one of the addresses is set to No, the memory address will be automatically substituted.
 

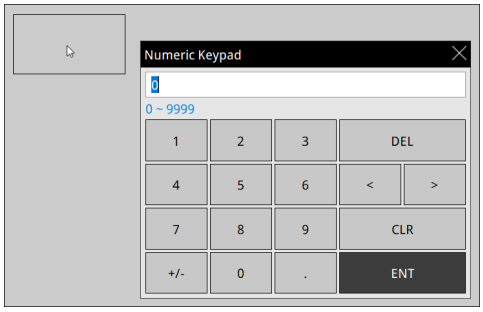
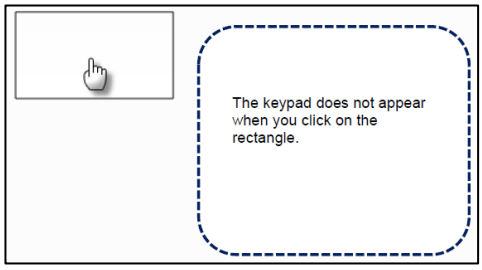
Note: The Variable Color has no effect when the Foreground Color is a multi-color setting such as Gradient Fill or Fill Pattern Style.</td></tr>
<tr><td>Variable Size</td></tr>
<tr><td>Variable Color</td></tr>
<tr><td rowspan="2">2</td><td>Image File Name Address</td><td rowspan="2">Specify the image path through Image file name address, and then set Trigger to load the image address to On, the element displays the specified item.</td></tr>
<tr><td>Load Image Trigger address</td></tr>
</table>

Legend	Setting	Description
❸	Save in	Select the stored type. The default value is **USB Disk**. By storing the image file in a USB Disk, USB Disk 2 or SD Card, the image file can be read on the HMI. Currently supports BMP, JPEG, PNG, and TIFF file formats.
❹	String Length	Select or enter the length of the file name to read. The default value is **16**.
❺	Extended mode	Select the extended mode. The default value is **All areas**. • **All areas**: The picture fills the full element display area. • **Keep the ratio**: The picture displays in 1:1 size based on the element width and length • **Actual Size**: The picture is displayed in the display area of the element in its actual size.

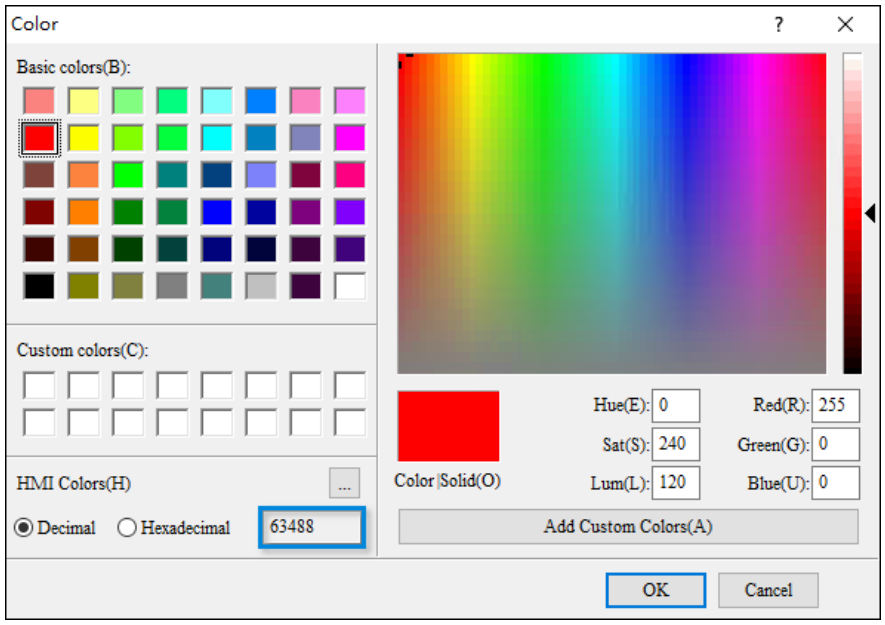
- **Style**



Legend	Setting	Description
❶	Round Corner Radius	Select or enter the round corner radius The size of this round corner radius is determined by the width and height of the **Rectangle** element. Take the smaller value among width/2 and height/2 of the **Rectangle** element. This is the maximum value that can be set for the radius.

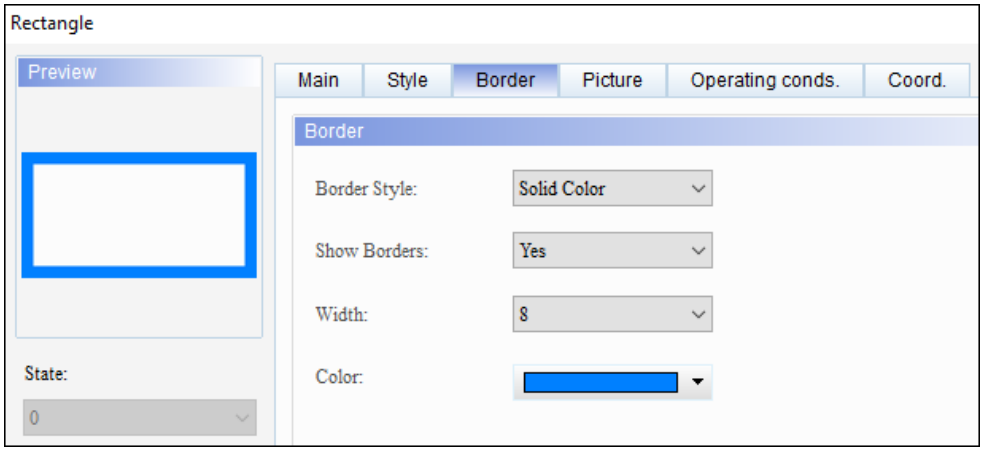
<table>
<tr><th>Legend</th><th>Setting</th><th>Description</th></tr>
<tr><td>②</td><td>Penetrate</td><td>Select whether the element that overlaps with the Rectangle element can be clicked. The default value is ON.
Example Operation
1. Create a Numeric Entry element.
2. Create one more Numeric Entry element and make it overlaps on top of the Numeric Entry element.
<table>
<tr><th>Setting</th><th>Illustration</th></tr>
<tr><td>Penetrate ON</td><td>The Numeric Entry element under Rectangle element can be clicked.
 
</td></tr>
<tr><td>Penetrate OFF</td><td>The Numeric Entry element under Rectangle element cannot be clicked.
 
</td></tr>
</table>
</td></tr>
<tr><td>③</td><td>Customize each fillet radius</td><td>Select the Customize each fillet radius checkbox to set the fillet radius for the vertices of the rectangle, or set memory address to dynamically set the fillet radius on the HMI.</td></tr>
</table>

- **Border**

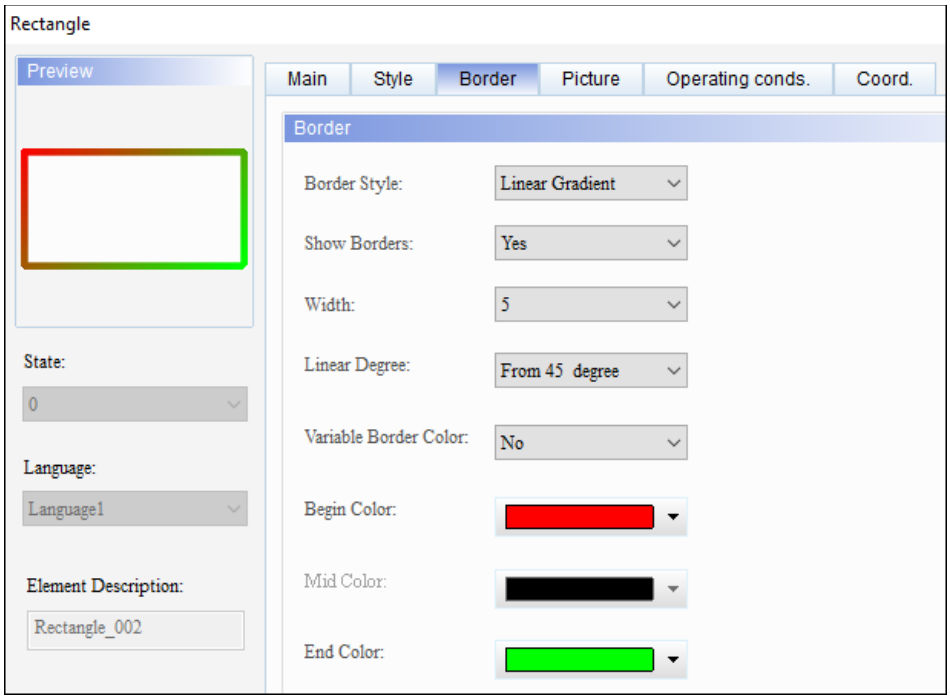
Setting	Description
Show Borders	Select whether to display the border.
Width	Select the width of the border, and the range value is 1~8.
Color	Select the color of the border, and the range value is 0~65535. The value entered in the address corresponds to the color as shown in the following figure. 

The following describes the properties and effects corresponding to each **Border Style**.

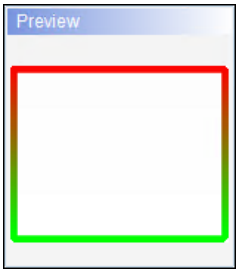
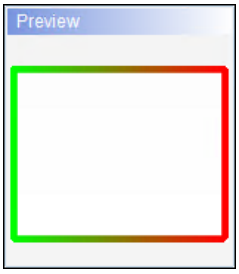
- **Fixed**: Displays the border in singe color.



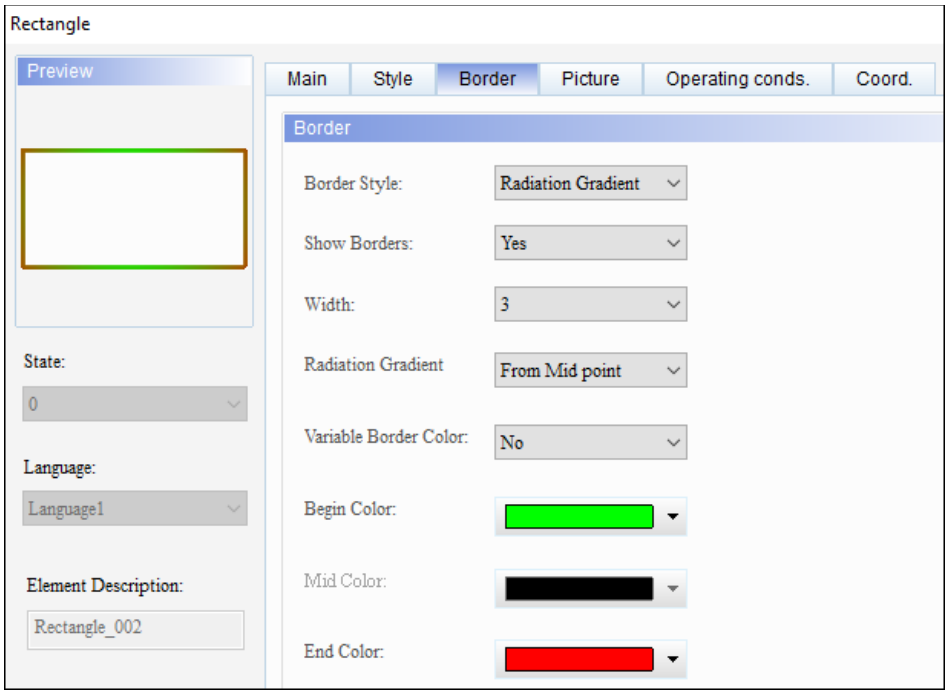
- **Linear Gradient**: Two colors can be set to present a gradient effect according to the angle.



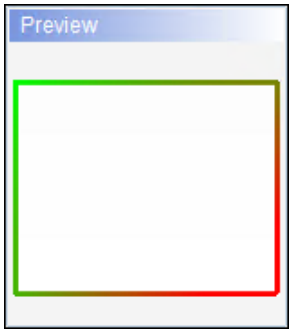
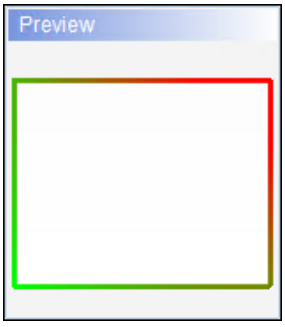
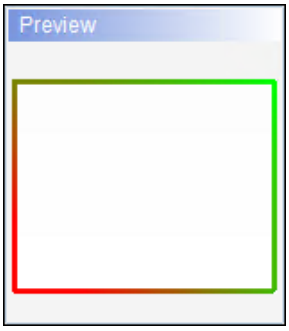
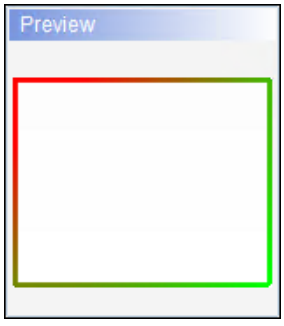
Transition from **Begin Color** to **End Color** according to the angle.

From 0 degree	From 45 degree
From 90 degree	**From 180 degree**
	

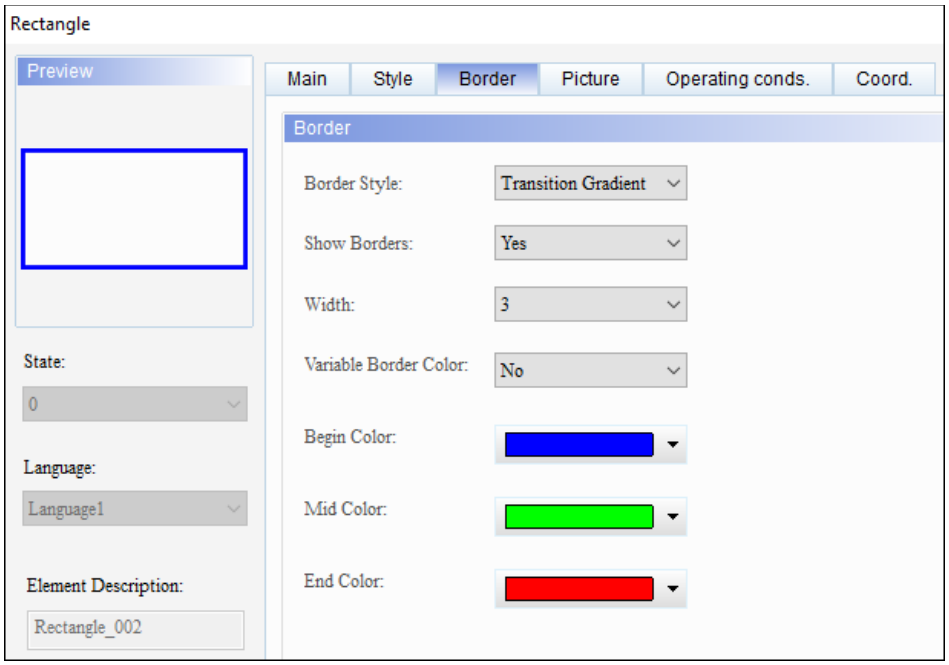
- **Radiation Gradient**: Two colors can be set to present a gradient effect from the mid-point.



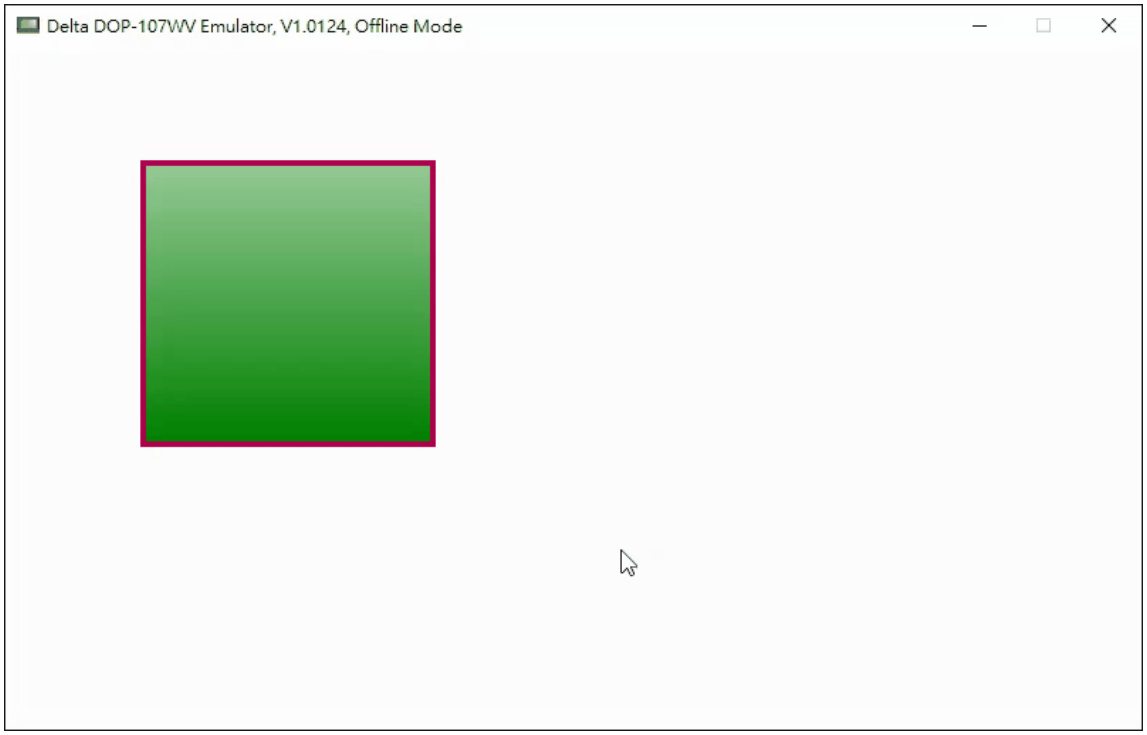
Begin Color radiates from the mid-point of the gradient to the **End Color**.

From Top-Left cor.	From Bot-Left cor.
	
From Top-Right cor.	**From Bot-Right cor.**
	

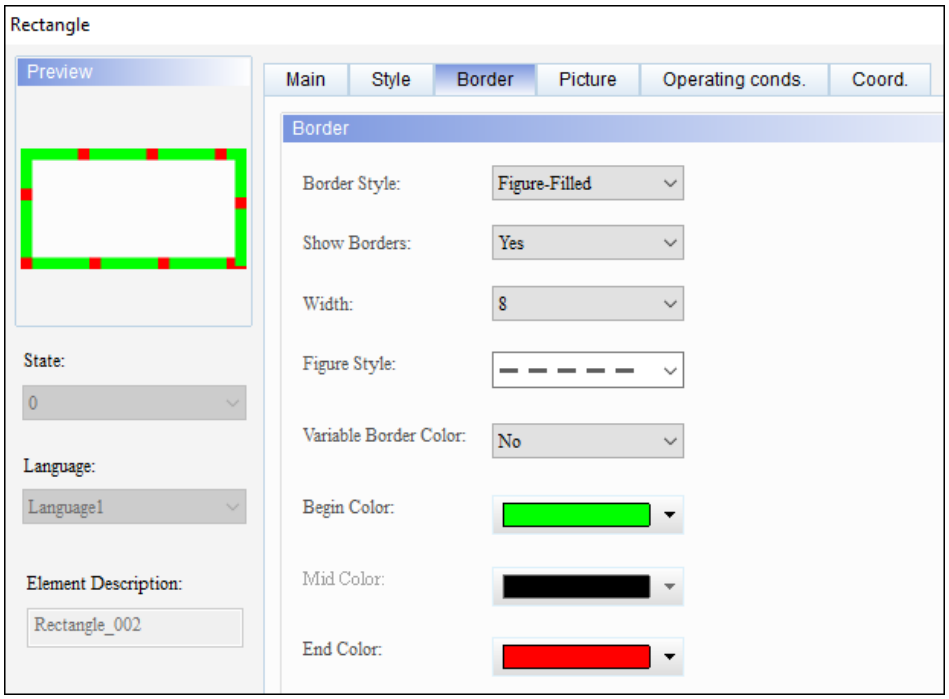
- **Transition Gradient**: Three colors can be set to cyclically present the gradient effect on the HMI.



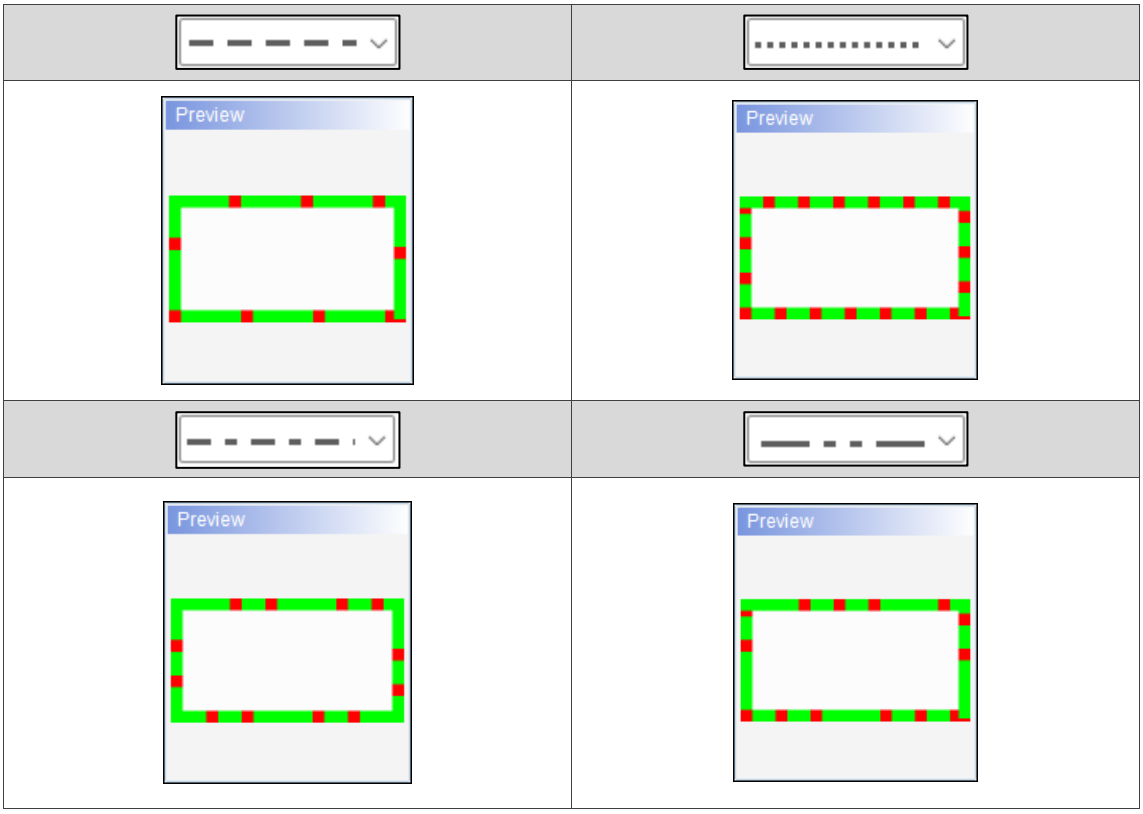
The HMI will dynamically cycle according to **Begin Color**, **Mid Color**, and **End Color**.



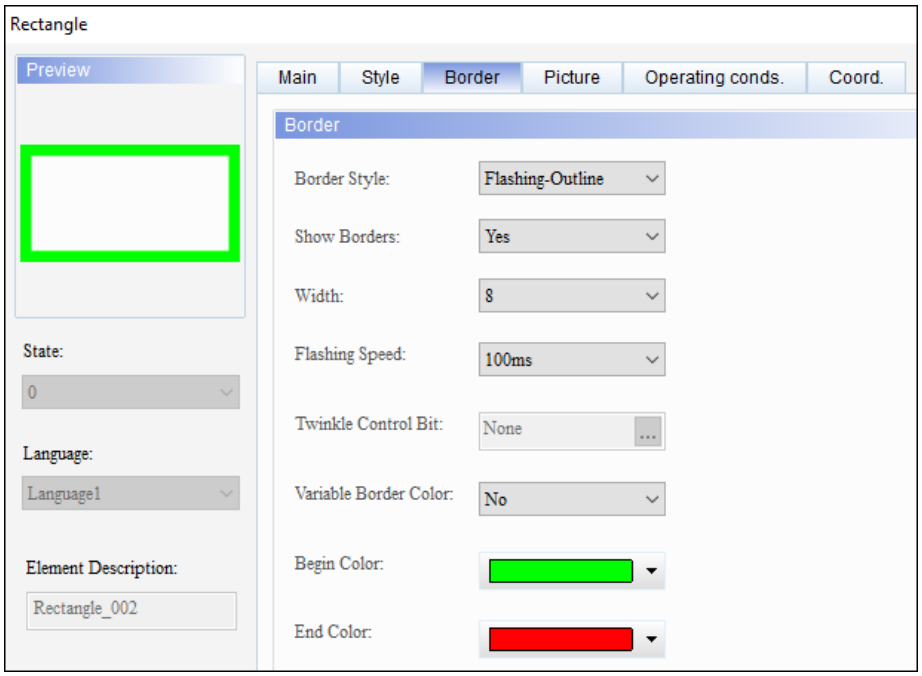
- **Figure-Filled**



Uses **Begin Color** as the line background color and **End Color** as the dotted line color. The following table demonstrates the effects of using different figure style.



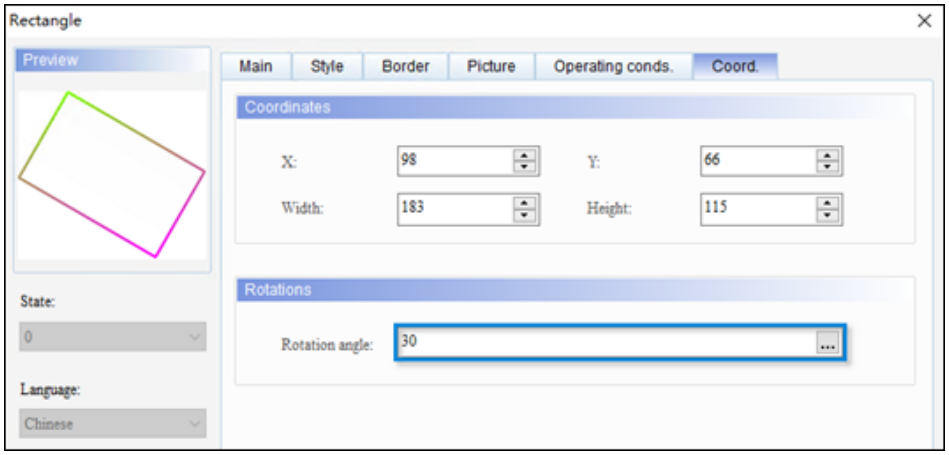
- **Flashing-Outline**



Setting	Description
Flashing Speed	Select the flashing speed. The **Begin Color** and **End Color** flash alternately according to the **Flashing Speed**.
Twinkle Control Bit	This function is only available when the **Flashing Speed** is set to **Custom**. When the control bit is OFF, the border color displays as **Begin Color**; when the control bit is ON, the border color displays as **End Color**.

- **Coord.**

You can set **Rotation angle** as a variable or a constant, which is a value for clockwise rotation.

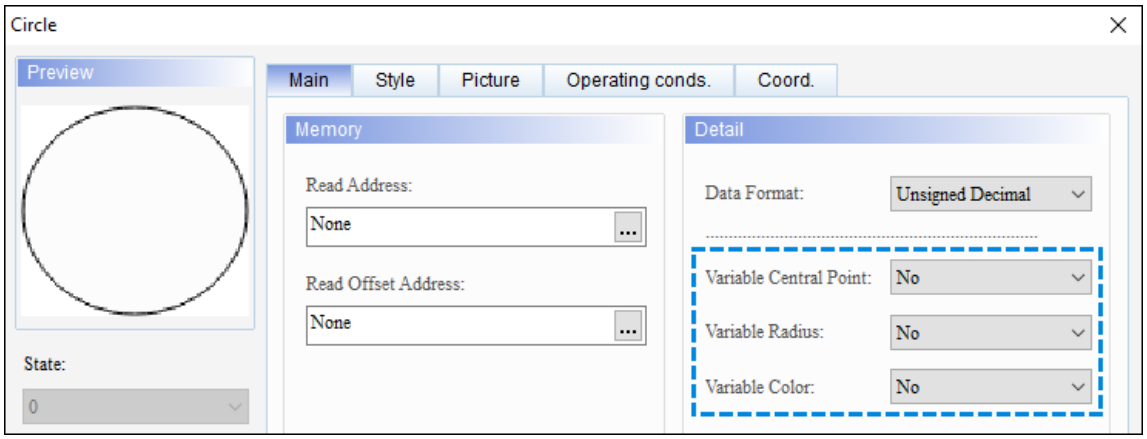


Circle

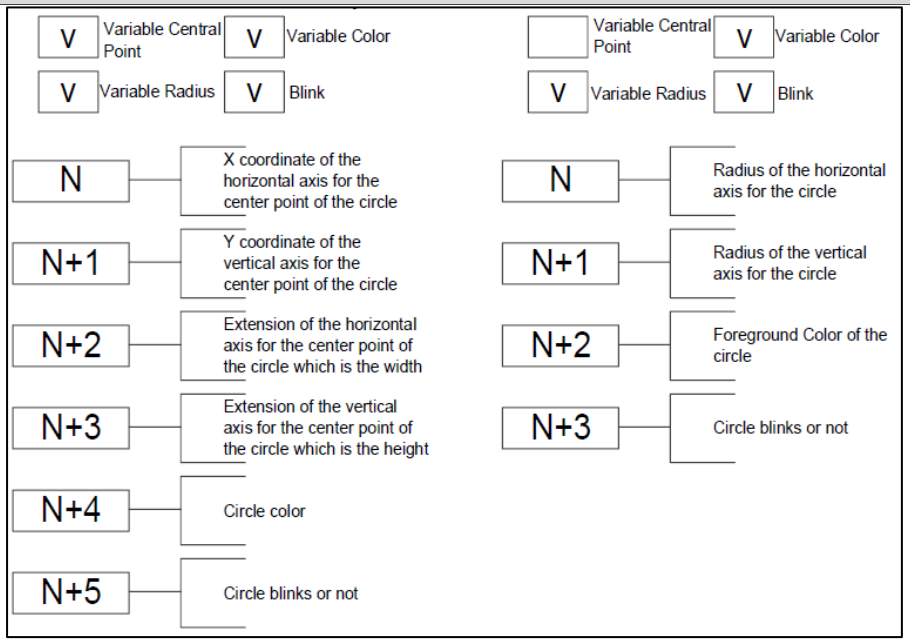
Press and drag the mouse to form a circle or oval through the **Circle** element. You can use the picture bank to import the picture, or use the set read address to control the color, size, central point, radius and blinking of the circle.

The following table list the special properties of the **Circle** element with their description.

- **Main**



Setting	Description
Variable Central Point	Select whether the color, center point, and radius can be changed. If select **Yes**, the position, size, color of the element can be dynamically changed on the HMI. • **Read Address** is regarded as the X coordinate for the center point of the circle. • **Read Address+1** is regarded as the Y coordinate for the center point of the circle. • **Read Address+2** is regarded as the extension of the horizontal axis for the center point of the **Circle** element which is the width. • **Read Address+3** is regarded as the extension of the vertical axis for the center point of the **Circle** element which is the height. • **Read Address+4** is regarded as the color of the circle, and the range value is 0~65535。 • **Read Address+5** determines whether the circle blinks or not. The above addresses are configured with all variables set to **Yes**. If one of the addresses is set to **No**, the memory address will be automatically substituted.
Variable Radius	
Variable Color	

Setting	Description
	 **Note**: The **Variable Color** has no effect when the **Foreground Color** is a multi-color setting such as **Gradient Fill** or **Fill Pattern Style**.

- **Style**

Setting	Description
Element Border Color	Select the color of the border.
Show Borders	Select whether to display the border.
Width	Select the width of the border, and the range value is 1~8.

Polygon

Select the Polygon element, click and drag the mouse to determine the vertices of the polygon, and then right-click to complete the polygon drawing.

The following table list the special properties of the **Polygon** element with their description.

- **Style**

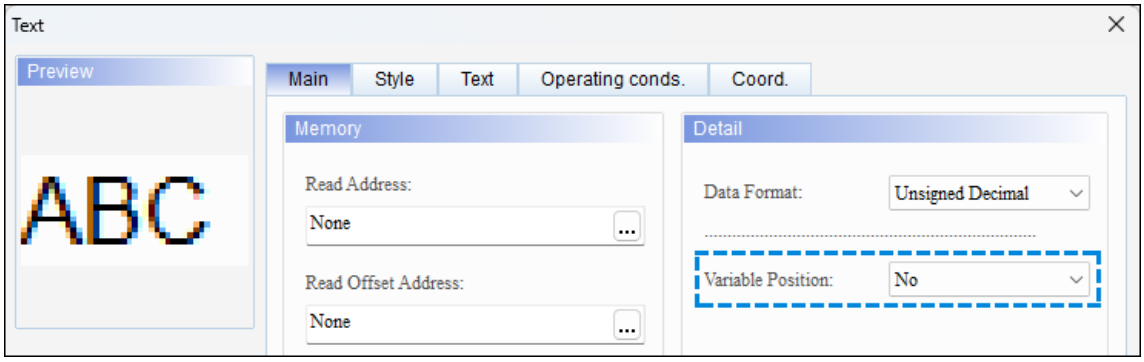
Setting	Description
Element Border Color	Select the color of the border.
Show Borders	Select whether to display the border.
Width	Select the width of the border, and the range value is 1~8.

Static Text

Enter text through the **Static Text** element and display it on the screen.

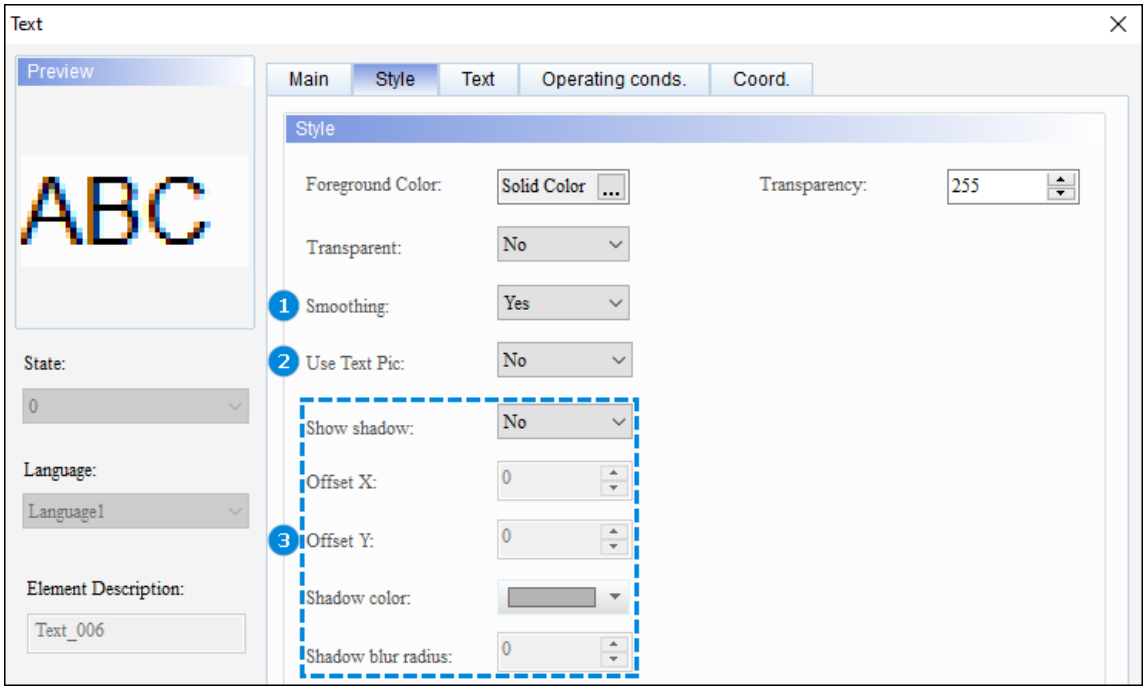
The following table lists the special properties of the **Static Text** element with their description.

- **Main**

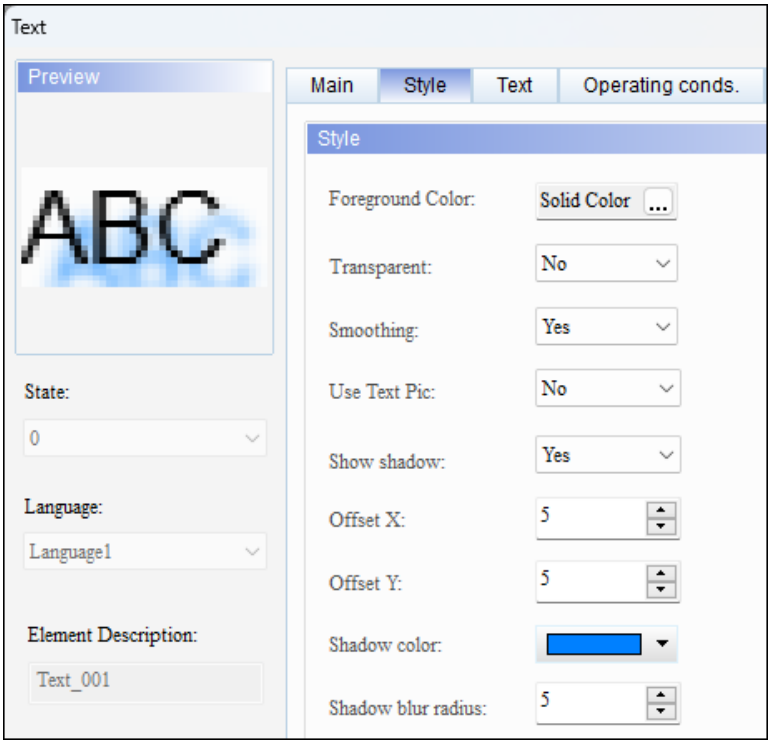


Setting	Description
Variable Position	Select whether the position can be changed. If select **Yes**, the position of the line on the HMI can be controlled by the memory address. • **Read Address** is regarded as the X coordinate of the left upper corner of the element. • **Read Address+1** is regarded as the Y coordinate of the left upper corner of the element.

- **Style**

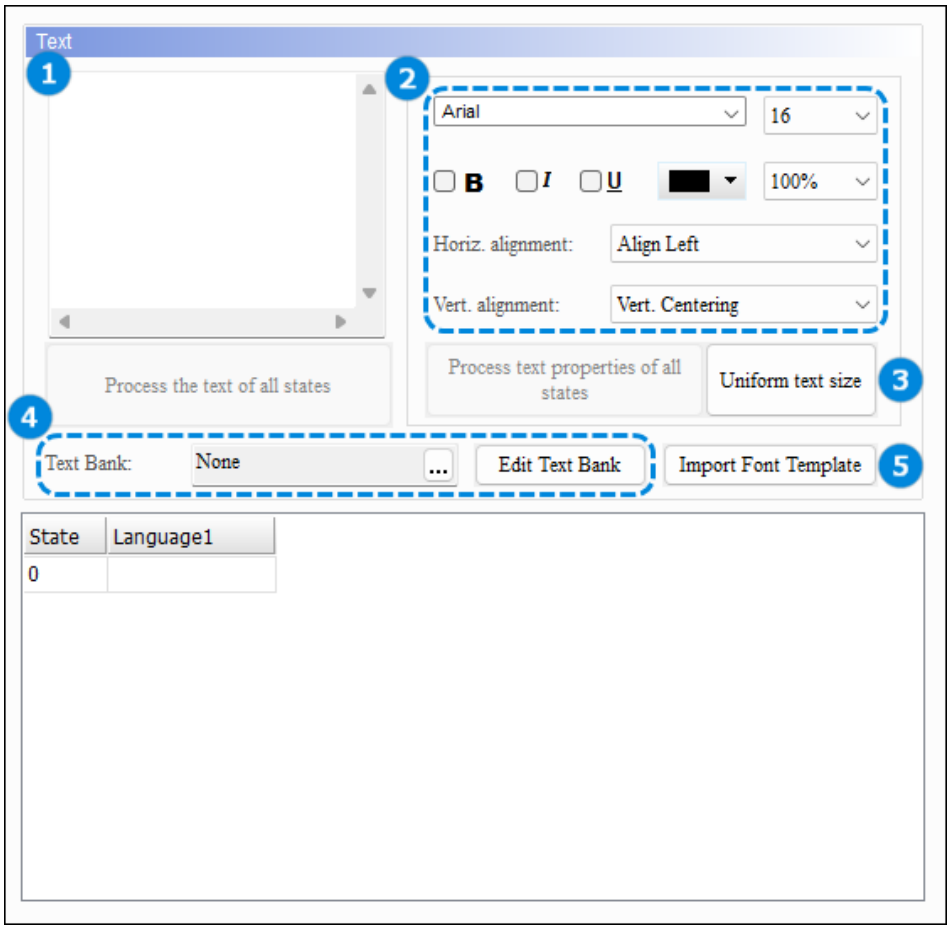


Legend	Setting	Description
1	Smoothing	Select whether to open the smoothing effect. If select **Yes**, the font is displayed without jagged edges and is smoother. Smoothing Yes: Smoothing No:
2	Use Text Pic	Select whether to use a text picture. This setting can be compatible with the effect of the text display of the DOP-B project. The effects are as follows. **Use Text Pic is set to Yes**: **Use Text Pic is set to No**: **Note**: When using DIAScreen to open the DOP-B project, this setting will be set to **Yes** by default; if adding a new DOP-100 project, this setting will be set to **No** by default.

<table>
<tr><th>Legend</th><th>Setting</th><th>Description</th></tr>
<tr><td>❸</td><td>Show shadow</td><td>
Select whether to display the shadow.
If select Yes, the shadow setting can be adjusted.
<table>
<tr><td>Offset X</td><td>Enter the X offset value, which is an integer in the range of –100~100. Enter a negative value, and the shadow will be displayed on the left of the text; enter a positive value, and the shadow will be displayed on the right of the text.</td></tr>
<tr><td>Offset Y</td><td>Enter the Y offset value, which is an integer in the range of –100~100. Enter a negative value, and the shadow will be displayed above the text; enter a positive value, and the shadow will be displayed below the text.</td></tr>
<tr><td>Shadow color</td><td>Set the color of the text shadow.</td></tr>
<tr><td>Shadow blur radius</td><td>Enter the bur radius of the shadow, which is an integer in the range of 0~100. The larger the value, the more blurred the shadow is.</td></tr>
</table>


</td></tr>
</table>

- **Text**



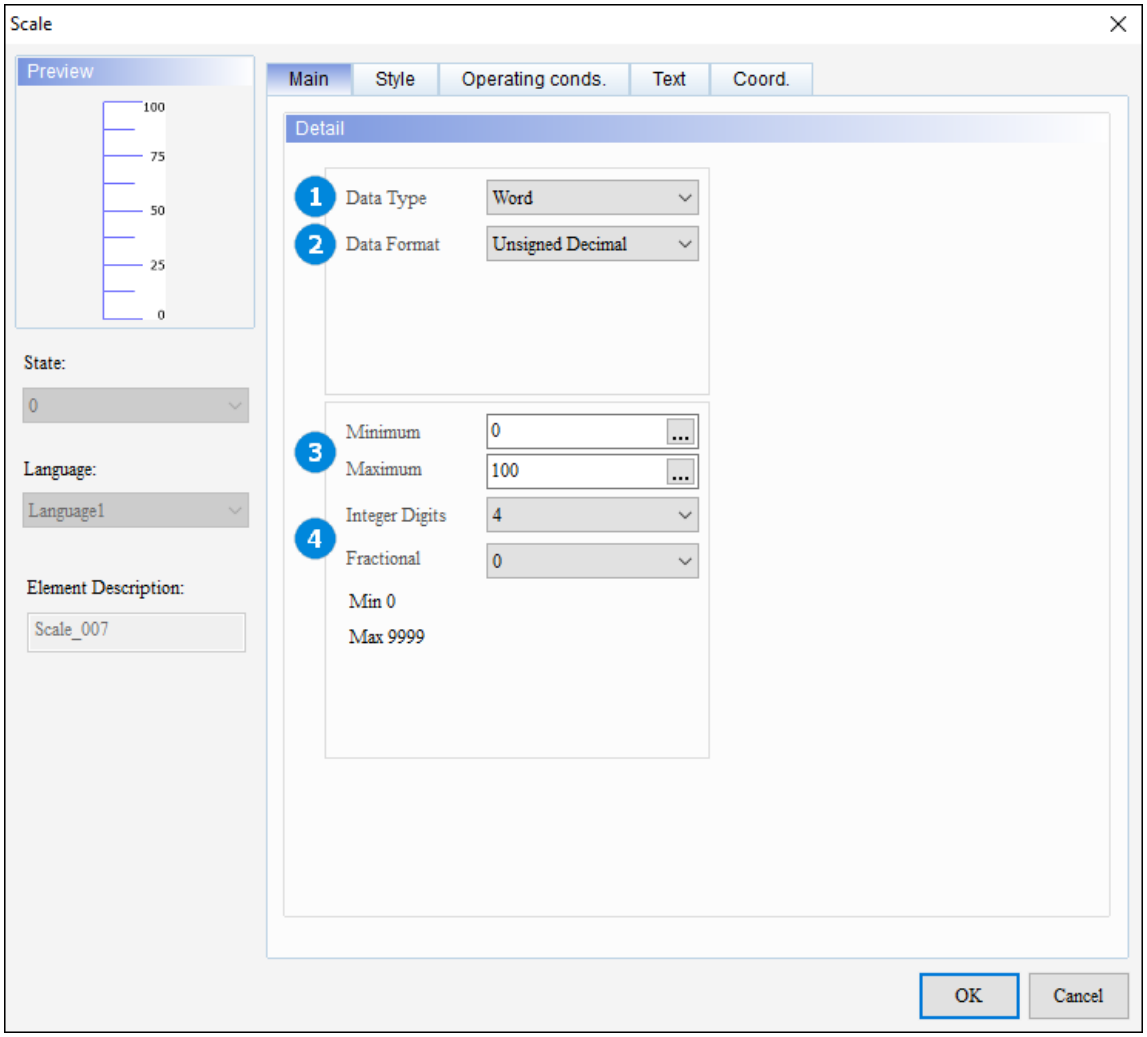
Legend	Setting	Description
1	Text	Enter the text message to display in the text box. Select the state / language table below to directly set the text displayed by that state / language. Enter the text message to be displayed in the text box. Click the element on the screen and press **Space** to edit the text content of the current state / language.
2	Text-related properties	Set the text properties.
3	Uniform text size	Changes the text size in all languages with one click according to the currently selected state.
4	Text Bank	Applies the text in the **Text Bank** to the content of each state / language. You can also click **Edit Text Bank** to edit the text content in the **Text Bank** editor.
5	Import Font Template	Click to open the **Font Template** dialog, and then select the template and import the text properties of the selected state.

Scale

Set the scale and mark value through the Scale element.

The following table lists the special properties of the **Scale** element with their description.

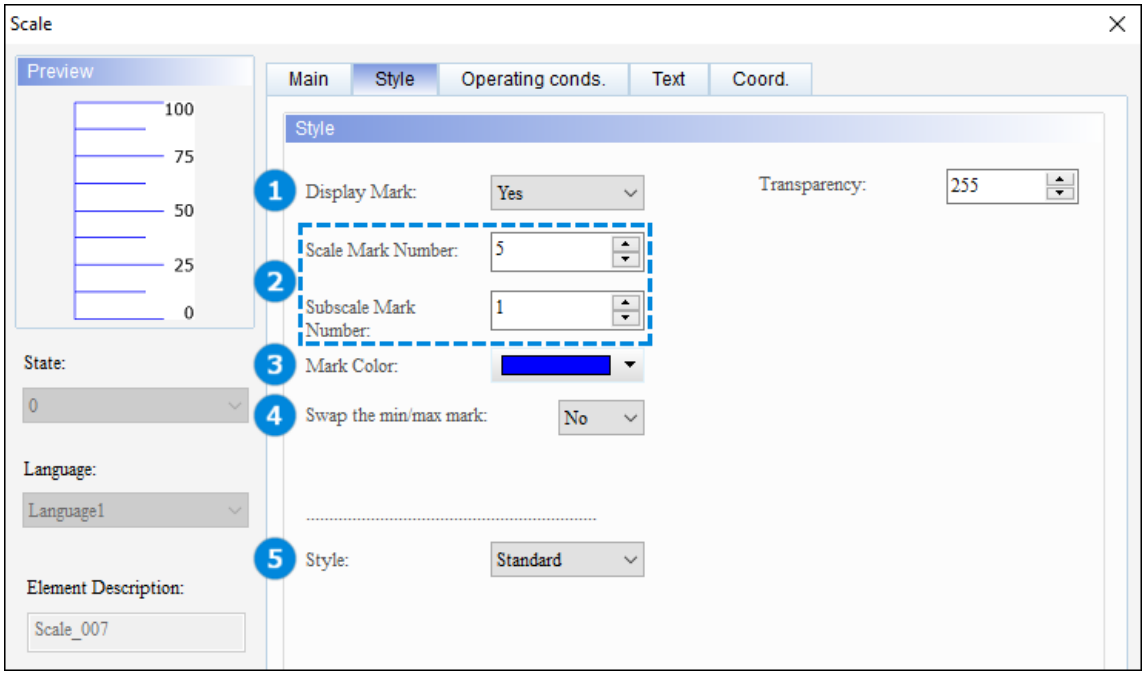
- **Main**

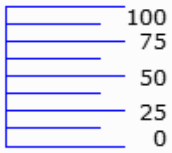
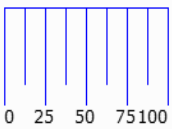
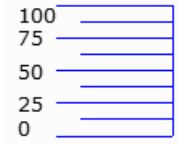


Legend	Setting	Description
1	Data Type	Select the data type. The available options are Word and Double Word.
2	Data Format	Select the data format. The available options are BCD, Signed Decimal, Unsigned Decimal, and Floating. **Note**: Floating format is available only when the **Data type** is **Double Word** and **Quad Word**.

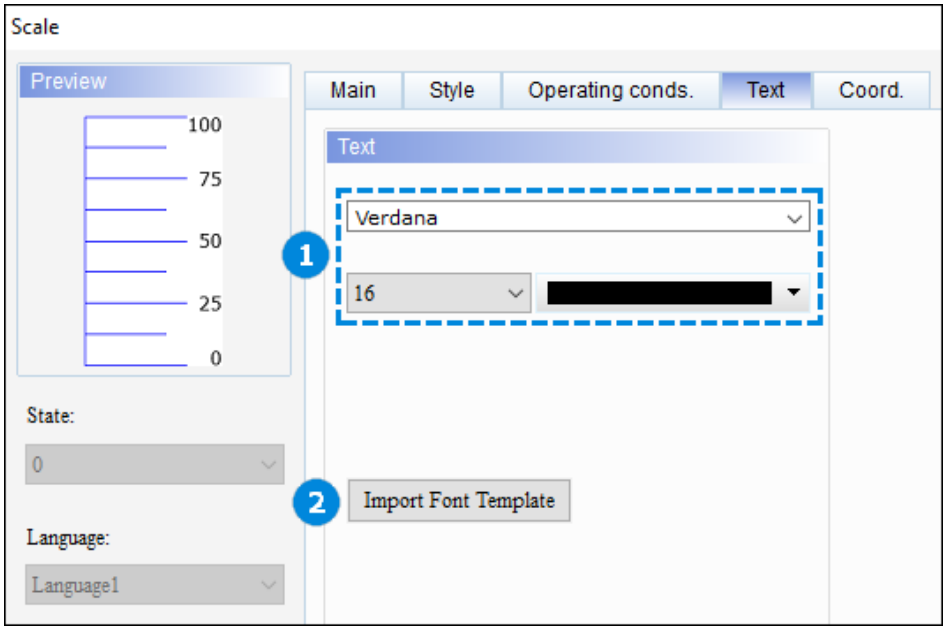
<table>
<tr><th>Legend</th><th>Setting</th><th>Description</th></tr>
<tr><td>3</td><td>Minimum
Maximum</td><td>The valid range values for minimum and maximum values vary depending on the selected Data Type and Data Format.
<table>
<tr><th>Data Type</th><th>Data Format</th><th>Valid Range</th></tr>
<tr><td rowspan="3">Word</td><td>BCD</td><td>0~9999</td></tr>
<tr><td>Signed Decimal</td><td>–32768~+32767</td></tr>
<tr><td>Unsigned Decimal</td><td>0~65535</td></tr>
<tr><td rowspan="4">Double Word</td><td>BCD</td><td>0~99999999</td></tr>
<tr><td>Signed Decimal</td><td>–2147483648~+2147483647</td></tr>
<tr><td>Unsigned Decimal</td><td>0~4294967295</td></tr>
<tr><td>Floating</td><td>0~9999999</td></tr>
</table></td></tr>
<tr><td>4</td><td>Integer
Digits
Fractional</td><td>Select the valid display digits according to the data type.
Note: The number of decimal places here is not a real decimal value, but the display format. Only when Floating is selected for the Data Format, it is a real decimal.</td></tr>
</table>

- **Style**



Legend	Setting	Description
❶	Display Mark	Select whether to display the mark. ○ If select **Yes**, the mark number will be displayed on the scale. ○ If select **No**, only the mark is displayed on the scale.
❷	Scale Mark Number Subscale Mark Number	Select or enter the scale mark number and subscale mark number, and the range value is 1–9.
❸	Mark Color	Select the display color of the marks.
❹	Swap the min/max mark	Select whether to swap the min/max value order of marks.
❺	Style	Select the style of the element. **Standard**  **Rotation 90**  **Rotation 180**  **Rotation 270**

- **Text**



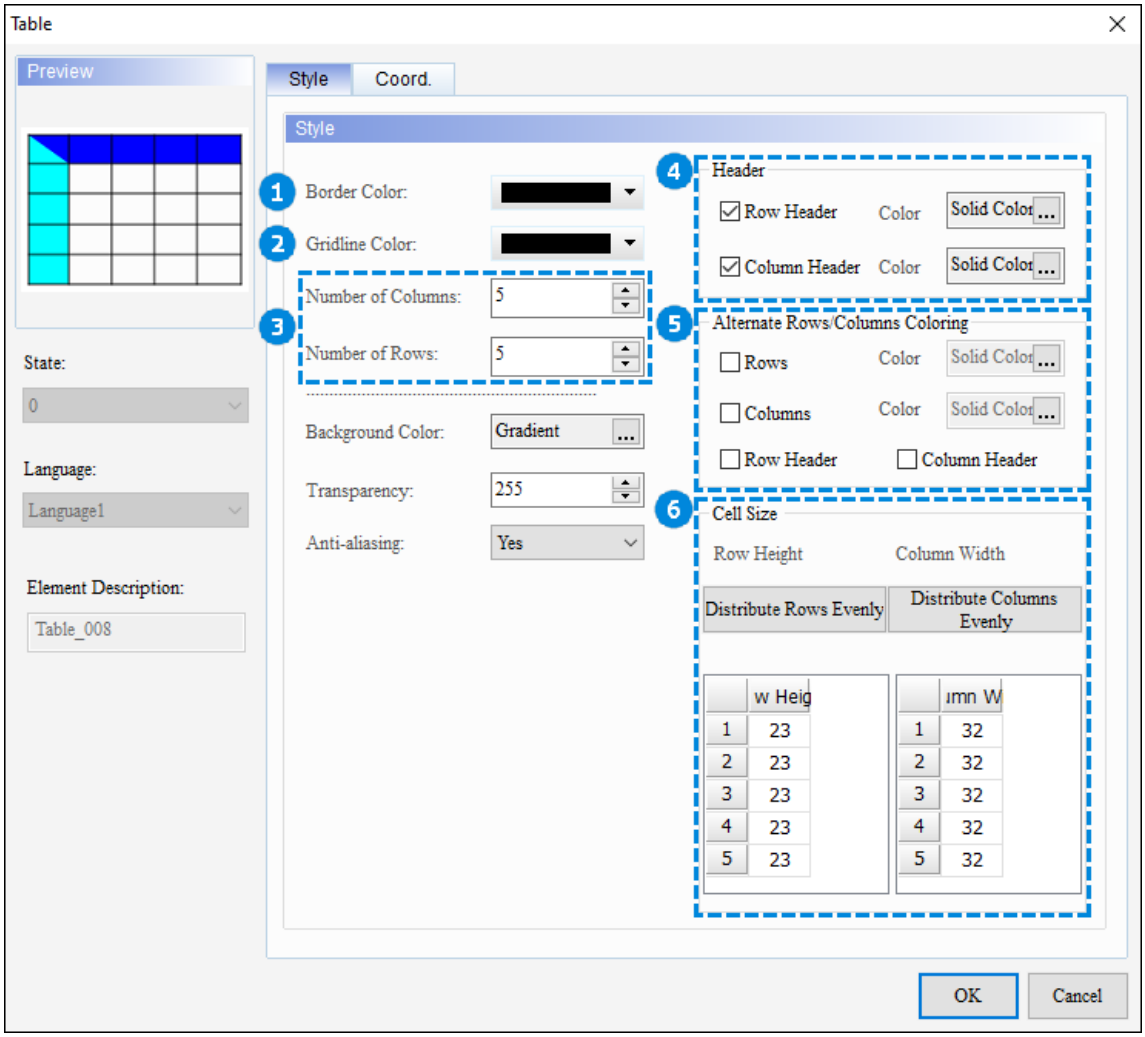
Legend	Setting	Description
1	Text Property	Set the text properties of the scale.
2	Import Font Template	Click to open the **Font Template** dialog, and then select the template and import the text properties of the selected state.

Table

Set the number of rows and columns and header style through the **Table** element. This function is same as the function of Microsoft Office.

The following table lists the special properties of the **Table** element with their description.

- **Style**



Legend	Setting	Description
❶	Border Color	Select the color of the border.
❷	Gridline Color	Select the color of the gridline.
❸	Number of Columns	Select or enter the number of columns, and the range value is 1–99.
❹	Header	Select to display the header and select the display color of the header.
❺	Alternate Rows/Columns Coloring	Select to display alternate rows and select the color of the columns.
❻	Cell Size	Set the size of each table column or distribute rows or columns evenly. The sum of column heights cannot exceed the element height, and the sum of row heights cannot exceed the element width.

Alarm

Alarm Setting

You can set the properties to be used by the alarm elements, such as read address, sampling cycle, maximum records, non-volatile data storage, and alarm moving sign, and can also edit the alarm message contents to be displayed. The Bit or Word addresses can also be configured to trigger alarms through the continuous alarm address and non-continuous alarm address functions.

The content of the alarm message also supports dynamic changes. You can add %d1 and %f1 to the alarm message, and then enter the value through the monitoring address function. When the alarm is triggered, the alarm message with the current value can be displayed immediately.

The formula provided by the software calculates all the alarm relevant data edited. Then, the set non-volatile memory (HMI, USB Disk, and SD) saves these calculation results. For data saved in the HMI, the alarm data size is subject to change based on the HMI model, you can see the specifications for non-volatile memory in the HMI installation manual. For data saved in the external storage devices (USB Disk or SD Card), the alarm data size is determined by the external storage devices.

The alarm formulas are applicable to the **Alarm History Table** and **Alarm Frequency Table**. The CSV file size is determined by the message (length) entered, there is no formula for the CSV file.

The following describes the formulas for the **Alarm History Table** and **Alarm Frequency Table**.

1. Alarm History Table

$$\{[6\ Bytes(a)+2\ Bytes(b)]\times \mathrm{N}(c)\}+6\ Bytes(d)=\text{Actual file size}\ Bytes$$

a	Time and Date Data
b	Alarm Data
c	Scan Points
d	Data File Header

2. Alarm Frequency Table

$$2\ Bytes(a) \times N(b) = \text{Actual file size } Bytes$$

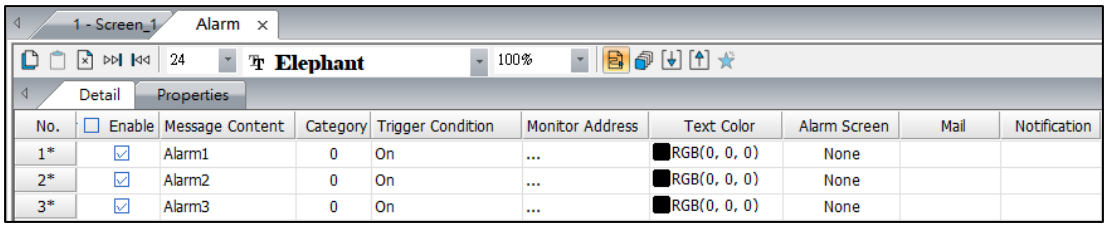
a	Alarm Frequency Data
b	Alarm Records

How to open the **Alarm** setting page?

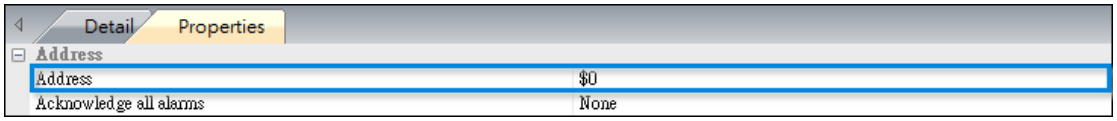
- In the **Project** pane, double click **Alarm**.
- On the toolbar, click **Data Management** > **Alarm Settings**.

Use the following the steps to set an alarm.

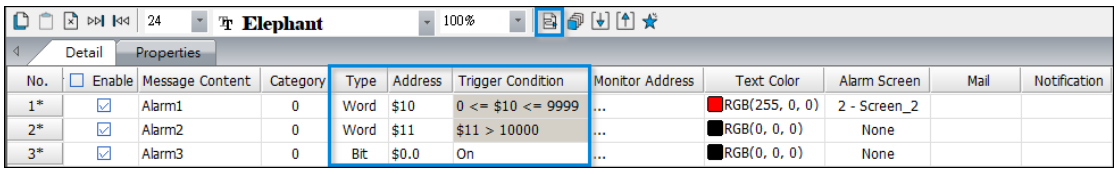
1. Click **Data Management** > **Alarm Settings** on the toolbar.
2. In the **Alarm** setting page, set the alarm for continuous alarm address and non-continuous alarm address.
   - **Continuous Alarm address**: Add 3 alarms.



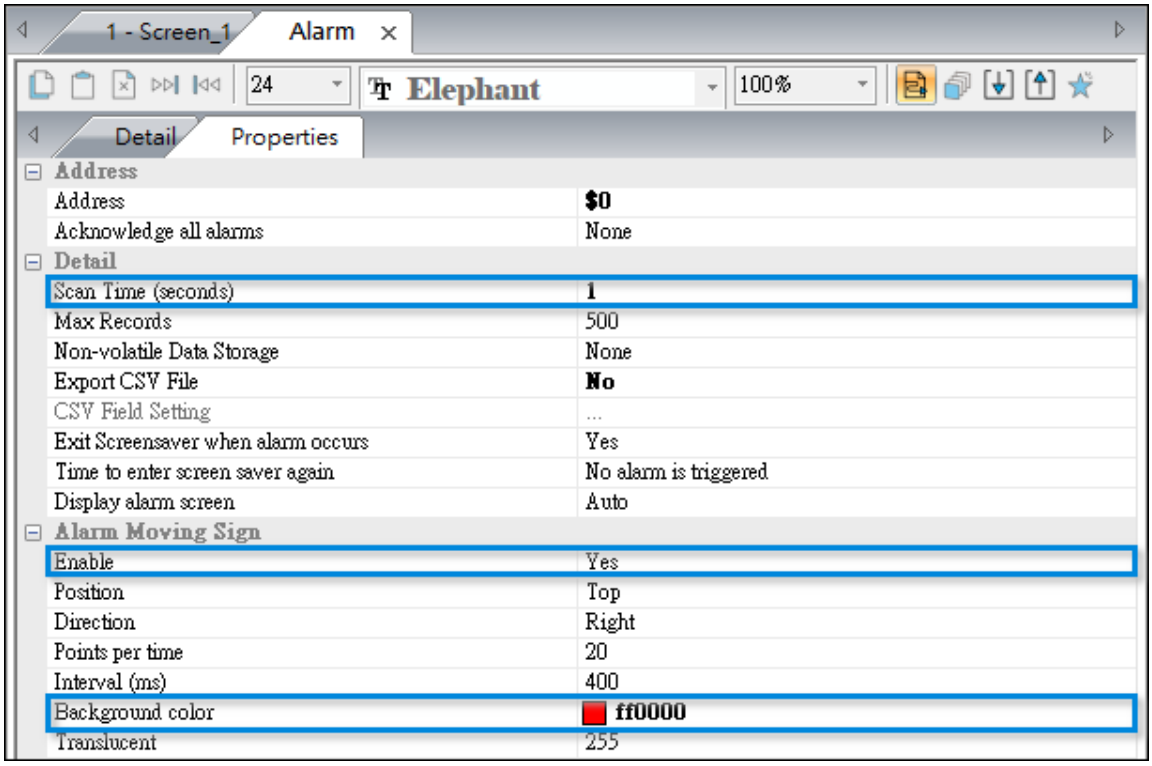
Set the alarm Address in the Properties tab, as shown below.



   - **Non-continuous Alarm address**: Click on the toolbar and set the **Type**, **Address**, and **Trigger Condition**.

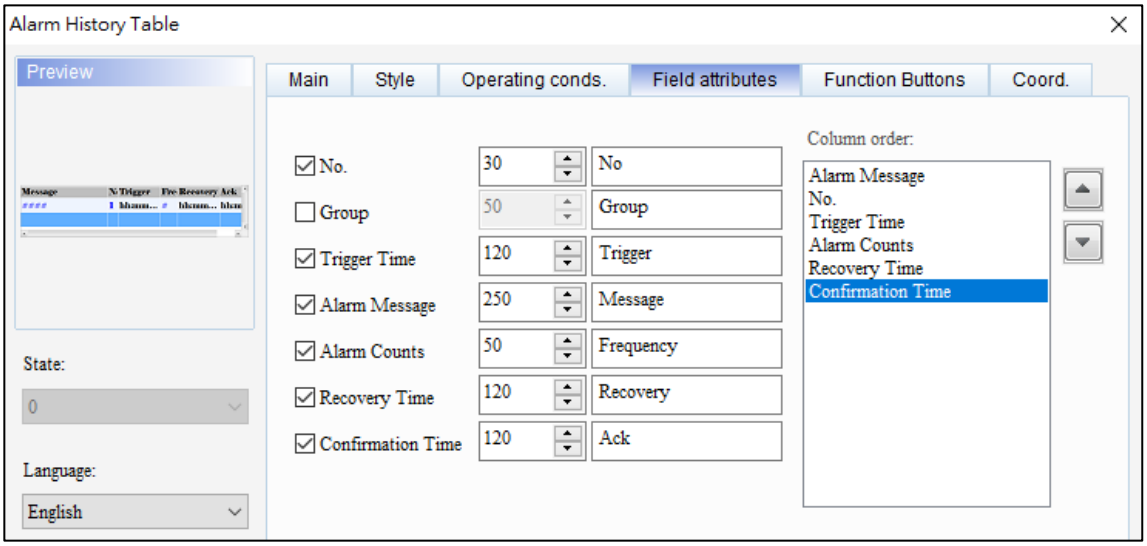


3. In the **Alarm** setting page, set the **Scan Time (seconds)** and **Alarm Moving Sign** in the **Properties** tab.



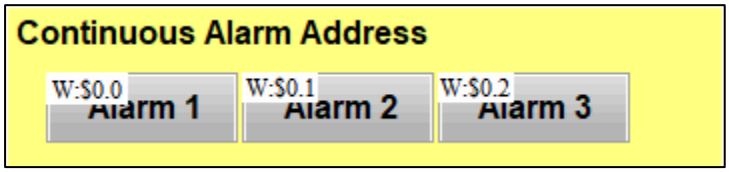
4. Click **Data Management** > **Alarm** on the toolbar and select **Alarm History Table**, and add it to the screen.

5. In the **Alarm History Table** element, select the items to display in the **Field attributes** tab.

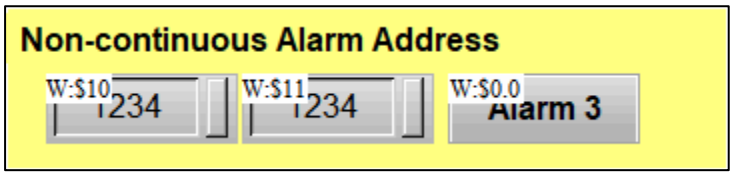


6. Create elements on the screen to trigger alarms.

   - **Continuous Alarm address**: Create 3 **Maintained Button** elements with addresses set to $0.0, $0.1, and $0.2.

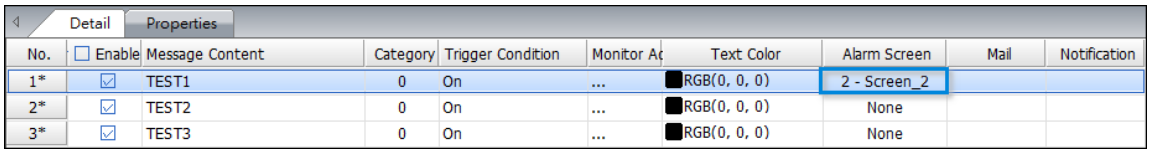


   - **Non-continuous Alarm address**: Create 2 **Numeric Entry** elements to trigger Word type alarms and a **Button** element to trigger Bit type alarms.



7. Click **General** > **New Screen** > **Subscreen** on the toolbar.

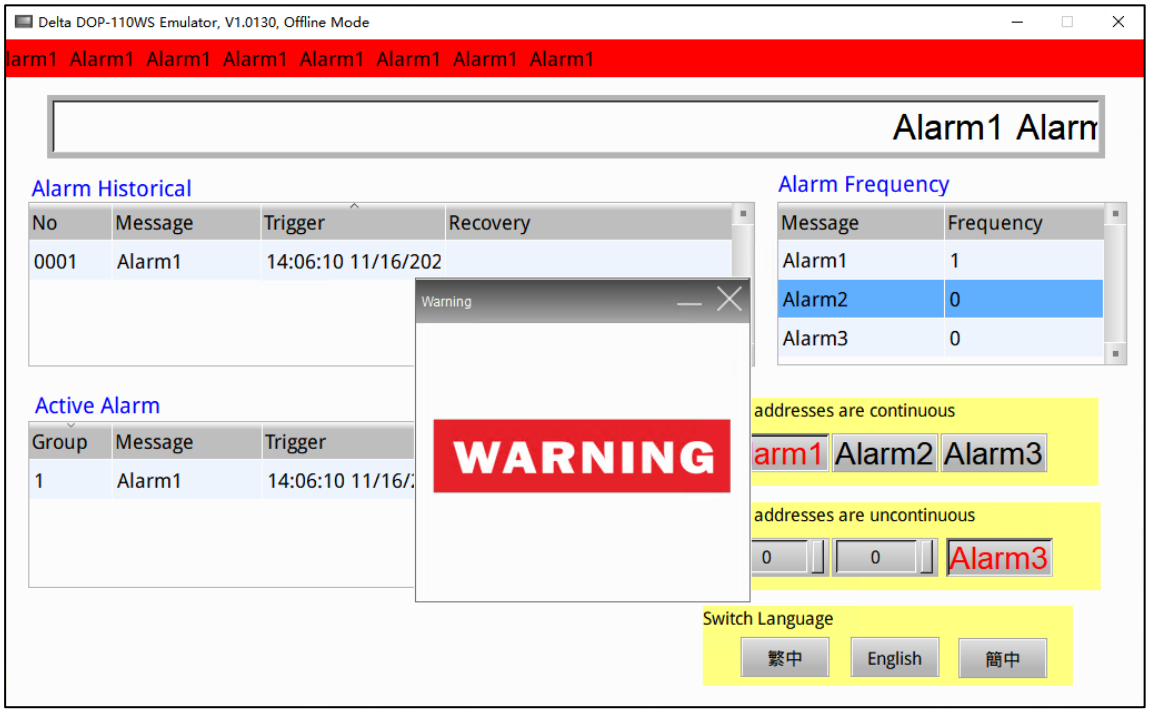
8. In the **Alarm** setting page, set the **Alarm Screen** of the number 1 item to the subscreen you have created.



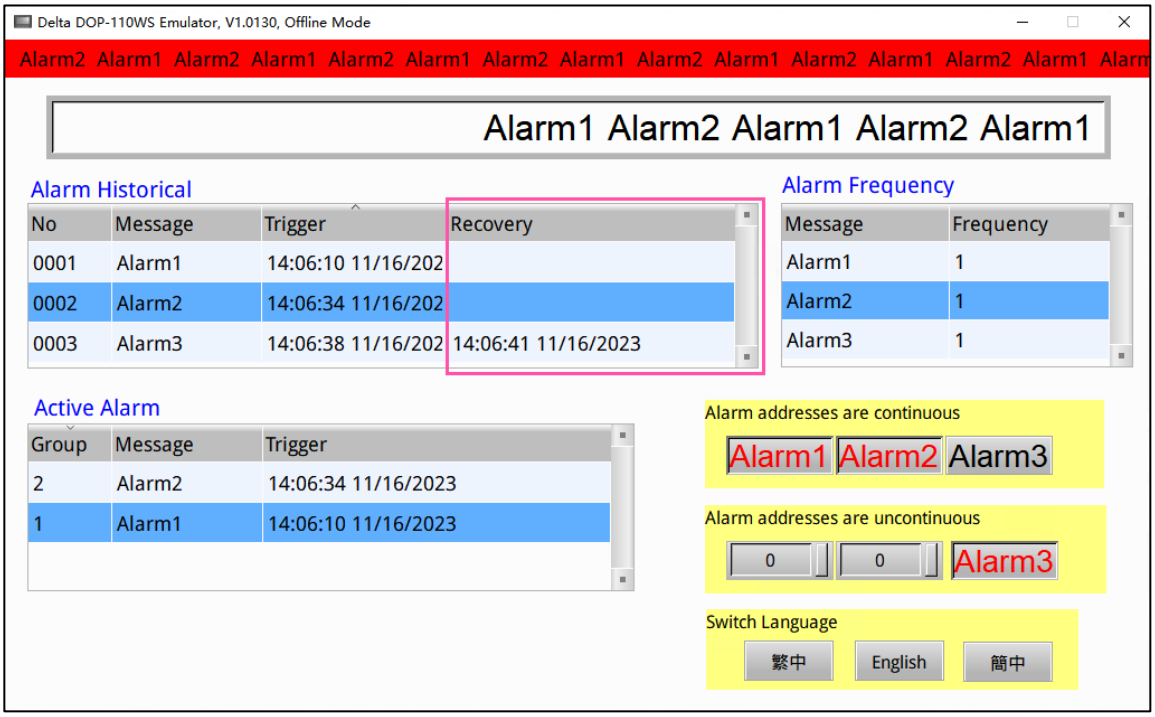
9. Click **Project** > **Compile All** on the toolbar and download the screen to the HMI.

   The following example shows the behavior of continuous alarm address on the HMI.

   Since Alarm 1 has an alarm screen set up, when clicking the button with the address $0.0 will trigger Alarm 1 and pop up an alarm dialog.

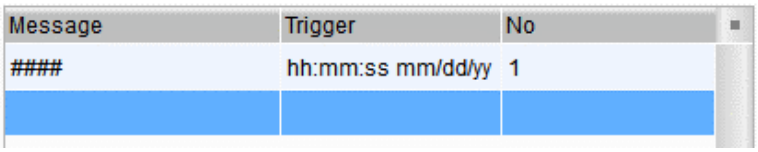
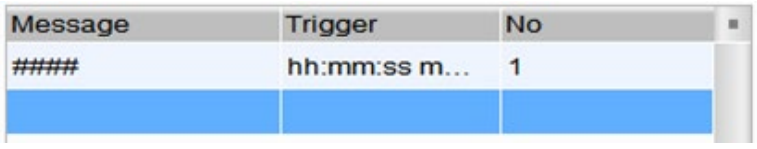
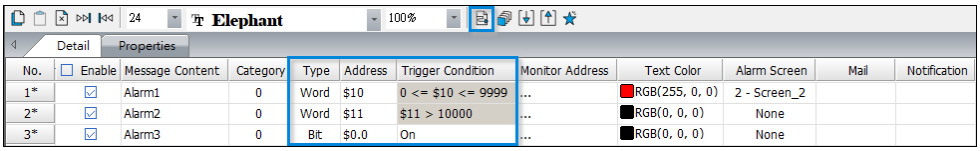


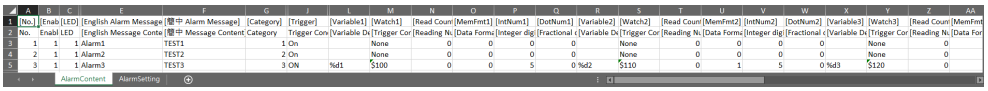
When the alarm is recovered, the **Recovery** field of the **Alarm History Table** displays the recovery time.



Alarm Settings Toolbar

The following table lists the settings on the toolbar in the **Alarm** setting page and their descriptions.

<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td> </td><td>Click to copy the selected alarm items.
Support single and multiple copy functions.
Press Ctrl to select the alarm item to copy, and press Shift to select a range of alarm items to copy.</td></tr>
<tr><td> </td><td>Click to paste the selected alarm items.
Supports single and multiple paste functions.
Note: You must first copy the alarm items.</td></tr>
<tr><td> </td><td>After selecting the alarm items, click this icon to delete. The settings of the item will return to default settings.</td></tr>
<tr><td> </td><td>Click to display the next 2048 entries.</td></tr>
<tr><td> </td><td>Click to display the previous 2048 entries.</td></tr>
<tr><td>Alarm Font Size</td><td>Set the text size for displaying alarm message on the screen.</td></tr>
<tr><td>Alarm Font</td><td>Set the font for displaying alarm message on the screen.</td></tr>
<tr><td>Alarm Font Ratio</td><td>Set the scaling ratio for display alarm message on the screen.
The value range is 33%–200%, and the default value is 100%.
<table>
<tr><th>Ratio</th><th>Result</th></tr>
<tr><td>100%</td><td> 
</td></tr>
<tr><td>150%</td><td> 
</td></tr>
</table></td></tr>
<tr><td> </td><td>Click to switch between Continuous Alarm Address and Non-continuous Alarm Address.
Set the alarm address in the Alarm setting page, and the alarm can be triggered using Bit or Word addresses.
 
</td></tr>
</table>

Setting	Description
	Click to quickly set alarm categories in batches. In the **Alarm Category Settings** dialog, set the **Starting Alarm Number**, **Ending Alarm Number** and **Category ID**, click **Batch Setting**. The alarms and categories from the start number to the end number will be set to the specified numbers. • The category number range is 0~4095.
	Click to import the alarm data file. Supports .xlsx, .alm, and .ini formats. **Note**: .alm and .ini are alarm configuration files exported by the old software.
	Click to export the edited alarm data file. Supports .xlsx format. The exported .xlsx file contains AlarmContent and AlarmSetting. You can edit the contents of each field and alarm properties of the alarm in the alarm data file. 
	Click to optimize the reading of non-continuous alarm addresses and speed up the alarm reading and writing.

	A	B	C	D
1	[Language]	[Font]	[Size]	[Ratio]
2		Font	Size	Ratio
3	中文	Delius	20	100
4	English	Elephant	24	100
5	簡中	Comic Sans MS	28	100
6				
7	Alarm Setting	Address		
8	Address	Address	$0	
9	Scan Time	Scan Time (seconds)	3.000000	
10	Max Records	Max Records	500	
11	Hold	Enable non-volatile mem	1	
12	Hold Place	Non-volatile Data Storag	0	
13	CSV	Export CSV File	1	
14	CSV Column Setting	CSV Field Setting	0|1|3|4|2|	
15	Exit Screen Saver	Exit Screensaver when a	1	
16	Time To Enter Screen Sa	Time to enter screen sav	0	
17	Screen Display Mode	Display alarm screen	0	
18	Continue Address	Continuous alarm addres	1	
19				
20	Alarm Moving Sign	Alarm Moving Sign		
21	Enable	Enable	1	
22	Position	Position	0	
23	Direction	Direction	1	
24	Moving Points	Points per time	20	
25	Interval	Interval (ms)	400	
26	BackgroundColor	Background color	RGB(213,213,213)	
27	Opacity	Translucent	255	
28	AckAllAddress	Acknowledge all alarms	None	

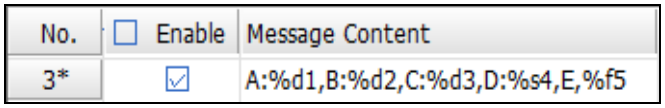
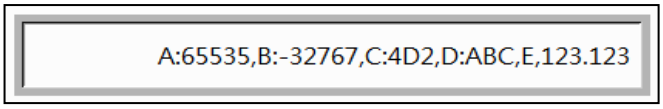
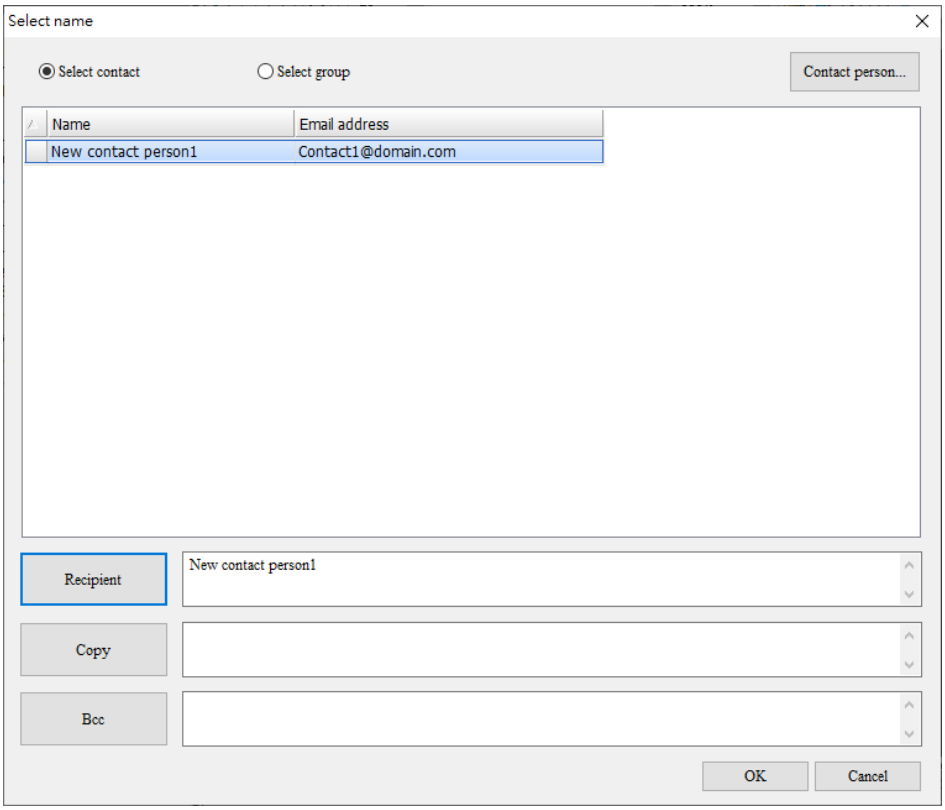
AlarmContent AlarmSetting

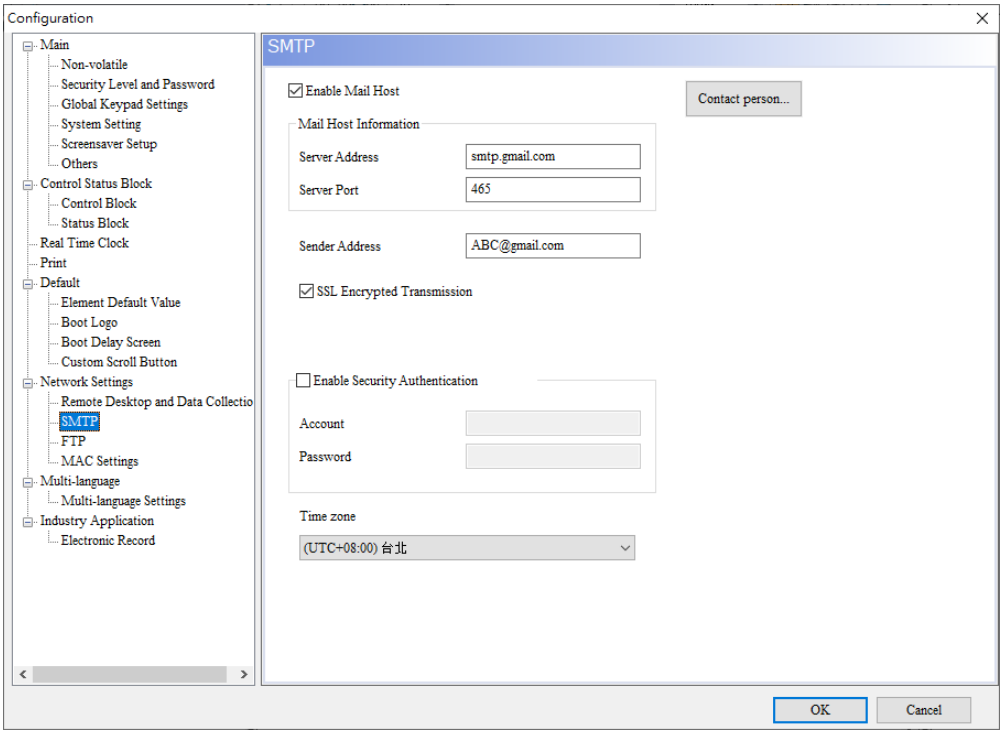
Alarm Message Settings

The following table lists the setting in the **Detail** tab in the **Alarm** setting page.

Setting	Description
No.	The number of alarm messages, and supports up to 20480 alarms.
Enable	Enable / disable the alarm. • If selected, run according to the set alarm rules after downloading. When you change any alarm setting field, the alarm is automatically enabled. • If not selected, the alarm content will not be checked when compiling the rule, and the alarm rule will not be executed when downloading to the HMI.
Message Content	Set the content of the alarm message to be displayed. • If you want to dynamically display message content, you can use it with the **Monitor Address** field. • If you want to set the message content displayed in different languages, you can go to **Project** pane > **Multi-language** to add a new language, and then go to **General** > **Language** drop-down menu to swtich.
Category	Set the category of the alarm number. On the toolbar, click to quickly set the category numbers. After the setting is complete, you can set the display order in the alarm elements or display the alarm category according to the set range. The value range is 0–4095.
Type	Set the trigger type of the alarm as Word or Bit. **Note**: You must switch the alarm address to non-continuous.
Address	Set the trigger address of the alarm. **Note**: You must switch the alarm address to non-continuous.

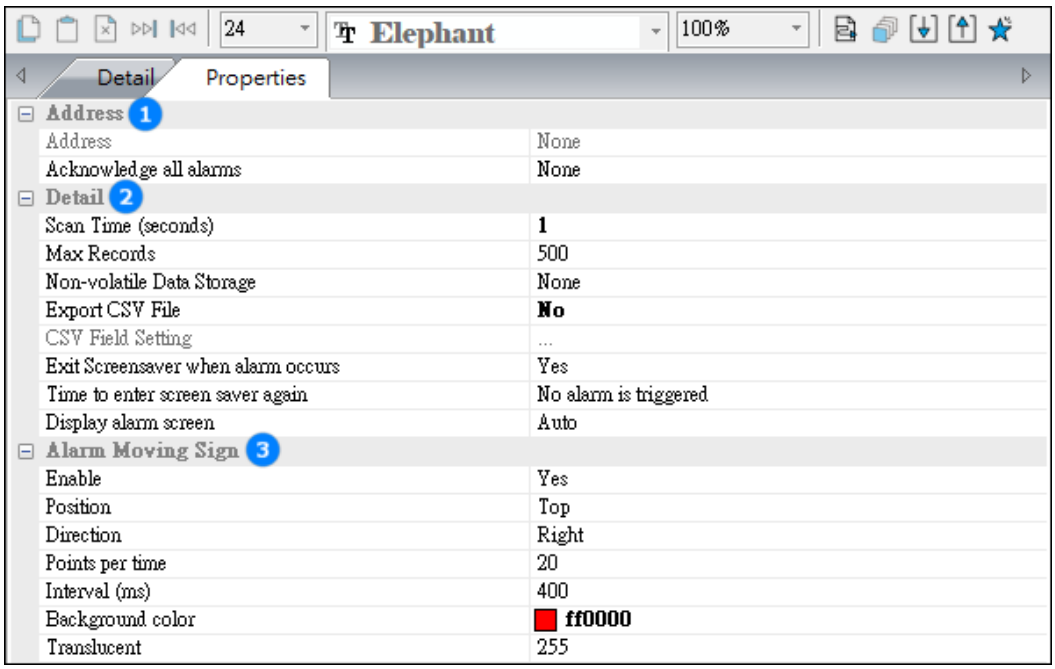
Trigger Condition	• **Continuous Alarm Address**: Set the **Trigger Condition** as On or Off.
	  ○ Select **On** to trigger an alarm when the Bit is On.
	  ○ Select **Off** to trigger an alarm when the Bit is Off.
	• **Non-continuous Alarm Address**: Establish the Trigger conditions according to the set trigger type.
	  ○ Select **Bit** for **Trigger Type**, the **Trigger Condition** can be configured as On or Off.
	  ○ Select **Word** for **Trigger Type**, you can configure the trigger condition range in the **Trigger Condition** field.

<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Monitor Address</td><td>Configure the settings for dynamic display of alarm message. The alarm message can display the current value of the monitor address.
The supported formats include %d (positive integer), %f (floating-point number), and %s (Alphanumeric). Besides, one alarm can set up to 8 monitoring addresses.
Enter the alarm message to be displayed in the Message Content field, and add the set Monitor Address such as %dx or %fx where x indicates 1~8.
<table>
<tr><td>Alarm message setting</td><td> 
</td></tr>
<tr><td>Result</td><td> 
</td></tr>
</table></td></tr>
<tr><td>Text Color</td><td>Set the text color of the alarm message displayed by the alarm element or the alarm moving sign when the alarm is triggered.</td></tr>
<tr><td>Alarm Screen</td><td>Set whether to show the specified alarm screen when the alarm is triggered. The default value is None.</td></tr>
<tr><td>Mail</td><td>Set the mail information. When an alarm occurs, the Mail function sends an e-mail to relevant recipients.
Click Recipient, Copy or Bcc in the Select Name dialog and select Select contact to select the recipient's email address in the contact list.
 
</td></tr>
</table>

Setting	Description
	Note: E-mail sending requires SMTP settings. Go to **General** > **Configuration** > **SMTP** to configure the mail address. 

Alarm Property Settings

The following table lists the settings in the **Properties** tab in the **Alarm** setting page.



1 Address

Setting	Description
Address	Set the Internal Memory or the controller register address to read addresses. **Note**: The **Address** field is available only the alarm address is switched to continuous.
Acknowledge all alarms	Set the Internal Memory address to acknowledge all alarms. When the alarm is triggered, the **Alarm History Table** will record the alarm acknowledge time.

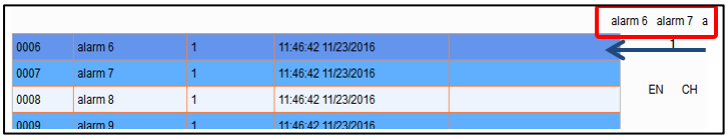
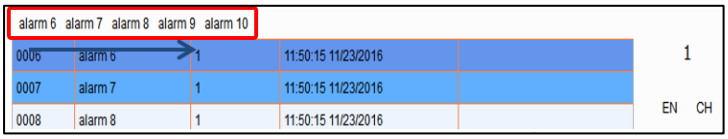
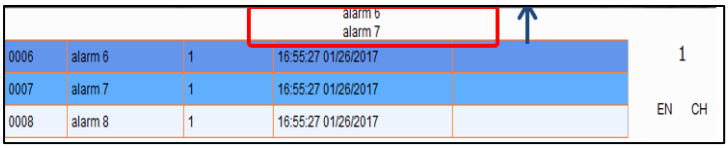
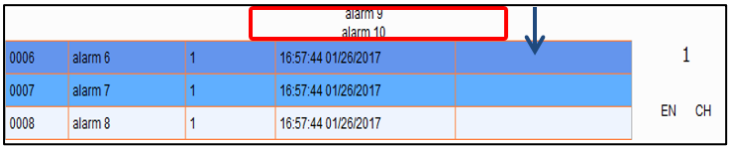
② Detail

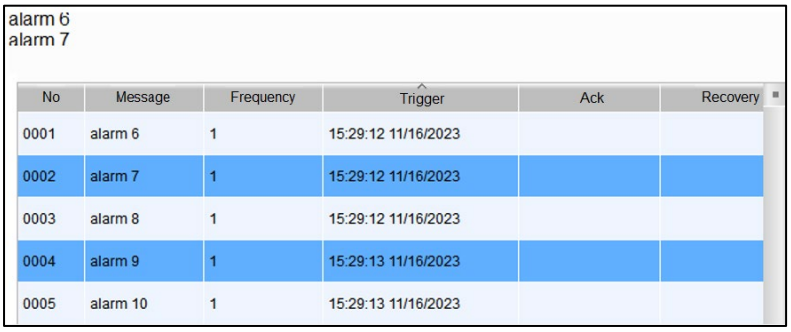
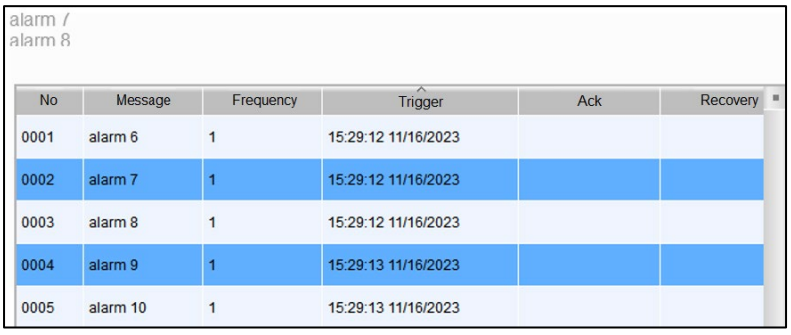
Setting	Description
Scan Time (seconds)	Set the time when the alarm is updated. The value range is 0.5–10, and the default value is **3**.
Max Records	Set the max records for scan point. When the recorded scan point reaches the maximum, the record starts from 1 and overwrites the previous data**.** The value range is 1–9,999, and the default value is **500**.
Non-volatile Data Storage	Set the storage location when the power is off. The storage location varies depending on the HMI model, including None, HMI, USB Disk, USB Disk 2, and SD. • If select **HMI**, when the HMI is powered off, the data is saved in the HMI SRAM. • If **Export CSV File** is set to **Yes**, set **Non-volatile Data Storage** to **USB Disk** or **SD**.
Export CSV File	Set as **Yes** to save the alarm data as CSV files in the external storage devices (USB Disk or SD Card).
CSV Field Setting	Select the column items for the exported CSV file, and you can adjust the column order. Use the **Print Output** button element to print the historical alarm content into a CSV file according to the CSV Field Setting and store it in the external storage devices. **Note**: This setting is available only the **Export CSV File** is set to **Yes**.
Exit Screensaver when alarm occurs / Time to enter screen saver again	• These functions are used with the screen saver. • Set **Exit Screensaver when alarm occurs** to **Yes**: When an alarm occurs, the screensaver screen will stop immediately. if the screensaver screen is not set, the HMI will not enter backlight mode. The following table lists the timing to enter the screensaver screen again.

<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td></td><td>
<table>
<tr><th>Timing</th><th>Description</th></tr>
<tr><td>No alarm is triggered on the HMI</td><td>As long as no alarm is triggered, the HMI can enter the screensaver mode.</td></tr>
<tr><td>No alarm is triggered during the screen saver waiting time</td><td>• As long as no alarm is triggered during the screen saver waiting time, the HMI can enter the screensaver mode.
• The screen saver waiting time can be set in Configuration > Screensaver Setup.</td></tr>
</table>
• Set Exit Screensaver when alarm occurs to No: The HMI exits the screen saver when the alarm is triggered for the first time. After that, whether the alarm is cleared or not, the HMI enters the screensaver mode according to the set time.
• Screensaver screen creation: Create a normal screen and go to General > Screensaver Setup. In the Screensaver Setup dialog, drag the screen from All Screens to Screensaver Screen.
</td></tr>
<tr><td>Display alarm screen</td><td>Set to display the alarm screen automatically or manually.
• Set to Auto: The HMI displays the alarm screen as soon as the alarm with a set alarm screen is triggered.
• Set to Manual: It must be controlled with the Alarm History Table element. The triggering methods are as follows:
○ In the element property dialog, set Action Control Addr. in the Operating conds. tab. When the value is set to 2, the alarm screen displays.
○ In the element property dialog, select the Trigger alarm screen checkbox in the Function Buttons tab to display the alarm screen.</td></tr>
</table>

③ Alarm Moving Sign

Setting	Description
Enable	Set whether to display the alarm moving sign. The default is No. • Set to **Yes**: When the alarm is triggered, no matter which page you operate on the HMI, an alarm moving sign shows at the specified position on the screen. • Set to **No**: When the alarm is triggered, alarm messages are not displayed.

<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Position</td><td>Set the display position. The default is Top.
Set to Top: When the alarm is triggered, the alarm message is displayed at the top of the HMI screen.
Set to Bottom: When the alarm is triggered, the alarm message is displayed at the bottom of the HMI screen.</td></tr>
<tr><td>Direction</td><td>
<table>
<tr><th>Direction</th><th>Result</th></tr>
<tr><td>Left</td><td> 
</td></tr>
<tr><td>Right</td><td> 
</td></tr>
<tr><td>Up</td><td> 
</td></tr>
<tr><td>Down</td><td> 
</td></tr>
</table>
</td></tr>
<tr><td>Points per time</td><td>Set the points per time. The greater the number, the greater the distance each time the text moves. The value range is 1–50 (pixel).</td></tr>
<tr><td>Interval (ms)</td><td>Set how often the alarm moving sign moves. The value range is 1–50.
Note: The moving distance is determined by the setting of Points per time.</td></tr>
<tr><td>Background color</td><td>Set the background color of the alarm moving sign.</td></tr>
</table>

<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Translucent</td><td>Set the transparency level of the alarm moving sign. The value range is 0–255, and the default value is 255.
<table>
<tr><th>Value</th><th>Result</th></tr>
<tr><td>255</td><td> 
</td></tr>
<tr><td>100</td><td> 
</td></tr>
</table></td></tr>
</table>

Alarm Element

You can create alarm element to display alarm status. Alarm elements including Alarm History Table, Active Alarm List, Alarm Frequency Table,Alarm Moving Sign, and Gantt Chart. This section introduces the properties of each alarm element.

Common Properties

- **Main – Enable group number filtering**

  You can select the group number for **Category** in the **Detail** in the alarm setting page.

  In the alarm element property dialog, select the **Enable group number filtering** checkbox in the **Main** tab. On the HMI, the element will display the alarm group according to the specified number. The value of the group number can be a variable or constant.

  Note: When the **Group** is 0, all alarms are displayed.

- **Operating conds. – Filter**

  In the alarm element property dialog, you can use the filter function to configure the items displayed on the alarm element in the **Operating conds.** tab.

  The following table lists the settings in the **Filter** area in the Operating conds. tab and their descriptions.

<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Filter control address</td><td>Set the filter control address to filter the specified items.
<table>
<tr><th>Value</th><th>Action description</th></tr>
<tr><td>0</td><td>Default status, displays all triggered alarms.</td></tr>
<tr><td>1</td><td>Hides the alarms with both Recovery Time and Acknowledge Time.</td></tr>
<tr><td>2</td><td>Hides the alarms with Recovery Time.</td></tr>
<tr><td>3</td><td>Hides the alarms with Recovery Time or Acknowledgement Time.</td></tr>
<tr><td>4</td><td>Hides the alarms with Acknowledgement Time.</td></tr>
<tr><td>5</td><td>• This setting must be used with Alarm counter display.
• The alarm display count refers to the value of Alarm counter display. When the alarm display count is smaller than this value, the alarm is hidden.</td></tr>
<tr><td>6</td><td>• This setting must be used with Alarm category start addr. and Alarm category end addr.
• When the alarm category number is not within the range set by Alarm category start addr. and Alarm category end addr., the alarm is hidden.</td></tr>
<tr><td>9</td><td>• This setting must be used with Alarm number start addr. and Alarm number end addr.
• When the alarm number is not within the range set by Alarm number start addr. and Alarm number end addr., the alarm is hidden.</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Alarm counter display</td><td>

This setting is applicable to the Alarm History Table element and the must be used with Filter control address.
When Filter control address is set to 5, enter the value of the alarm count.

<table>
<tr><th>Example</th><th>Description</th></tr>
<tr><td>Alarm element has alarms with alarm counts 1, 2, and 3 times</td><td>

Set to 1 for Alarm counter display, the Alarm History Table element displays the alarms with 1 or more alarm counts.
Set to 2 for Alarm counter display, the Alarm History Table element displays the alarms with 2 or more alarm counts.
Set to 3 for Alarm counter display, the Alarm History Table element displays the alarms with 3 or more alarm counts.

</td></tr>
</table>
</td></tr>
<tr><td>Alarm category start addr./ Alarm category end addr.</td><td>
When Filter control address is set to 6, enter the alarm category number.
<table>
<tr><th>Example</th><th>Description</th></tr>
<tr><td>Alarm element has alarms with alarm category numbers 1 and 5</td><td>

Set to 1 for Alarm category start addr. and 3 for Alarm category end addr., only the alarms with alarm category number 1 are displayed on the alarm element.
Set to 1 for Alarm category start addr. and 5 for Alarm category end addr., the alarms with alarm category numbers 1 and 5 are displayed on the alarm element.

</td></tr>
</table>
</td></tr>
<tr><td>Alarm number start addr./ Alarm number end addr.</td><td>
When Filter control address is set to 9, enter the alarm category number.
<table>
<tr><th>Example</th><th>Description</th></tr>
<tr><td>Alarm element has alarms with alarm numbers 1~5
•</td><td>

Set to 1 for Alarm number start addr. and 3 for Alarm number end addr., only the alarms with alarm number 1~3 are displayed on the alarm element.

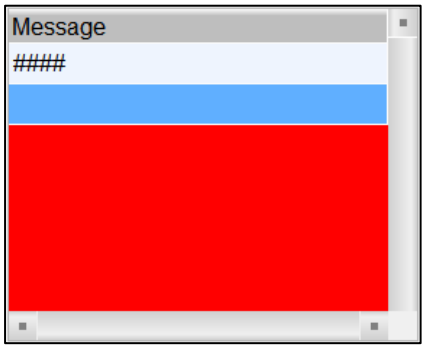
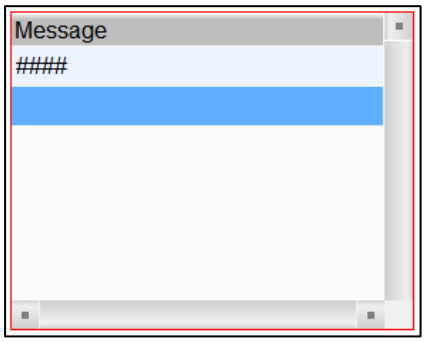
</td></tr>
</table>
</td></tr>
</table>

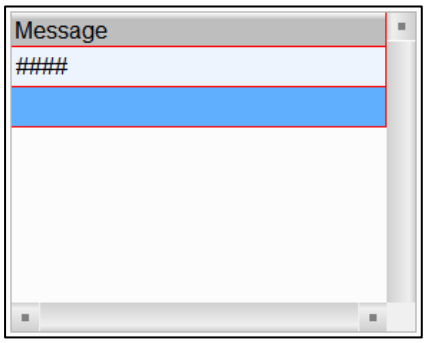
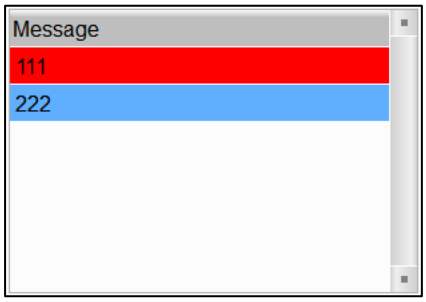
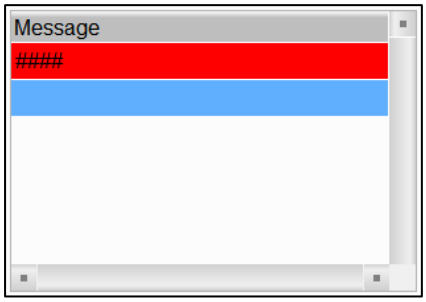
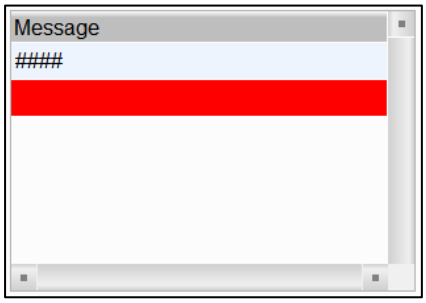
- **Operating conds. – Control**

  In the alarm element property dialog, you can set the scroll control address in the **Operating conds**. tab and switch the alarm table display page on the HMI through the memory address. The scroll control supports 16-bit Unsigned, 16-bit signed, 32-bit Unsigned, and 32-bit signed data formats.

- **Style**

  The display style of Alarm elements on the HMI can be set.

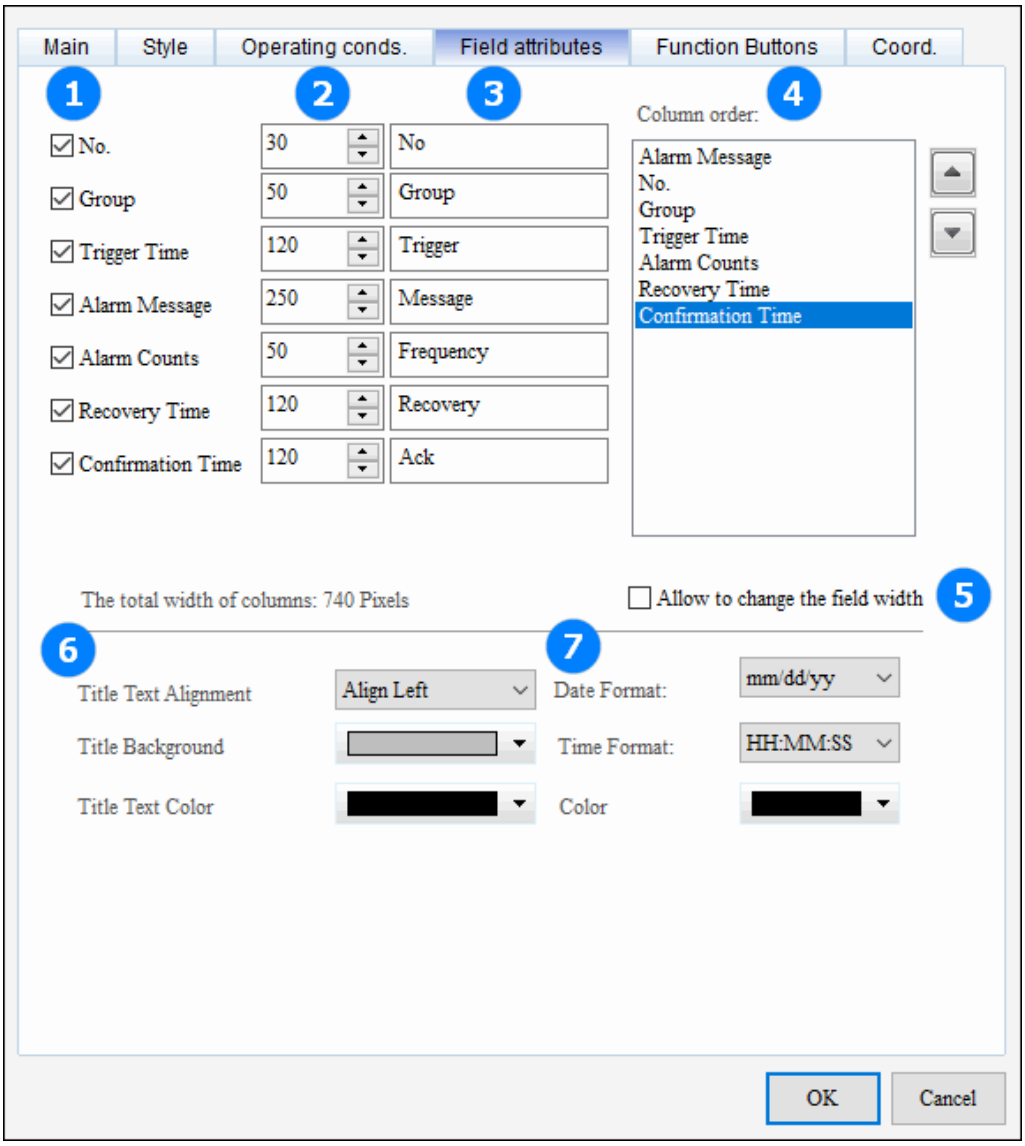
Setting	Description
Background Color	Set the background color of the element. The default is white. When set to red, it displays as follows. 
Border Color	Set the border color of the element. The default is gray. When set to red, it displays as follows. 

Setting	Description
Gridline Color / Show Gridlines	Set the gridline color of the element. The default is white. When set to red, it displays as follows. **Note**: The **Gridline Color** setting is valid only when you select **Yes** for **Show Gridlines**. 
Select Row Color	Set the color of the selected row of data when operating the element on the HMI. The default is blue. When set to red, it displays as follows. 
Row Color	Set the color to display for each row of the alarm. The default is light blue. When set to red, it displays as follows. 
Alternating Row Color	Set the color for the alternating row of the alarm. The default is blue. When set to red, it displays as follows. 

Setting	Description
Transparency	Set the transparency of the element. The range value is 50–255, and the default value is **255**. The smaller the value, the more transparent the element is.

- **Field attributes**

  You can set the field items to be displayed in the alarm table in the **Field attributes** tab.



Legend	Setting	Description
1	Column display	Select the columns to be displayed in the element.
2	Column width	Select or enter the width for each column.

Legend	Setting	Description	
❸	Column title	Enter the titles for each column.	
❹	Column order	After selecting the columns to be displayed, click to adjust the column displaying order.	
❺	Allow to change the field width	Select to drag the field on the HMI to change the displaying width.	
❻	Title bar	Title Text Alignment	Select the title bar text as left, center or right.
		Title Background	Select the background color of the title bar.
		Title Text Color	Set the text color of the title bar.
❼	Date and time	Date Format	Set the display date format.
		Time Format	Select the display time format.
		Color	Select the display color of date and time. The default is black.

- **Function Buttons**

  You can add function buttons in the **Function Buttons** tab of the alarm element properties dialog to control the display of the alarm table.

Setting	Description
Function Description	Select to add function buttons.
Default Description	Set the description of the function button. Click **Set As Default Description** to insert the default strings to the input fields. You can switch between the different languages to define the default text.
Default Button Width	Set the width of the function buttons. The value range is 25–100, and the default value is **60**. After adjusting the width, right-click the element on the screen and select **Rearrange Function Buttons**, all function buttons display according to the settings.
Default Button Height	Set the width of the function buttons. The value range is 25–100, and the default value is **40**. After adjusting the height, right-click the element on the screen and select **Rearrange Function Buttons**, all function buttons display according to the settings.

Alarm History Table Properties

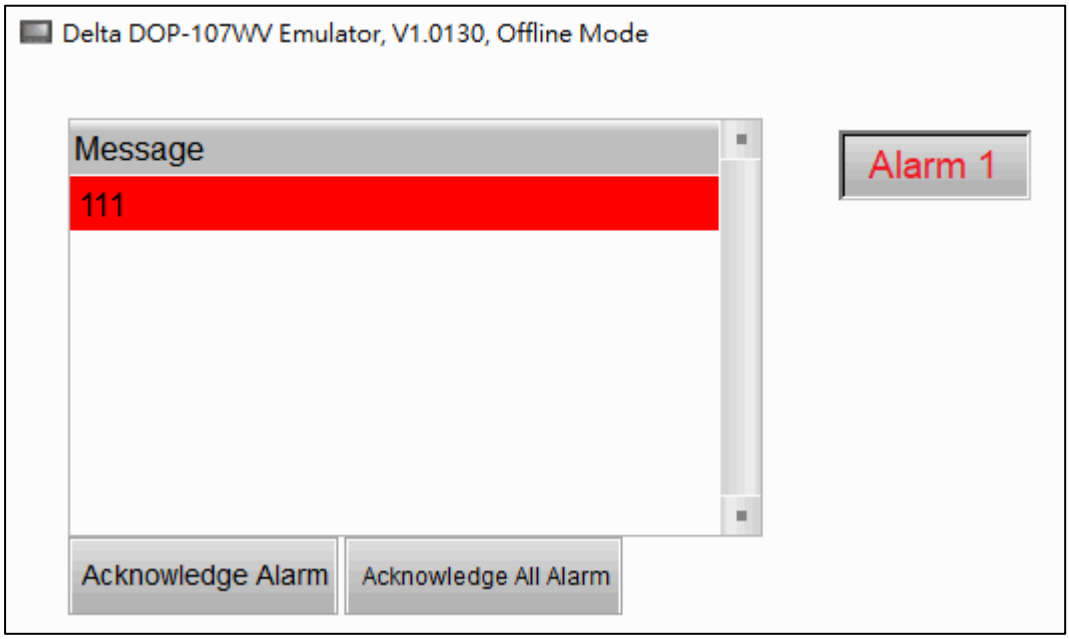
- **Style**

  You can set **Trigger Column Color**, **Confirm Column Color**, and **Recovery Column Color** after selecting **Color Mode** as **Alarm Status**. When an alarm is triggered on the HMI, colors display according to different statuses.

  Example Setting

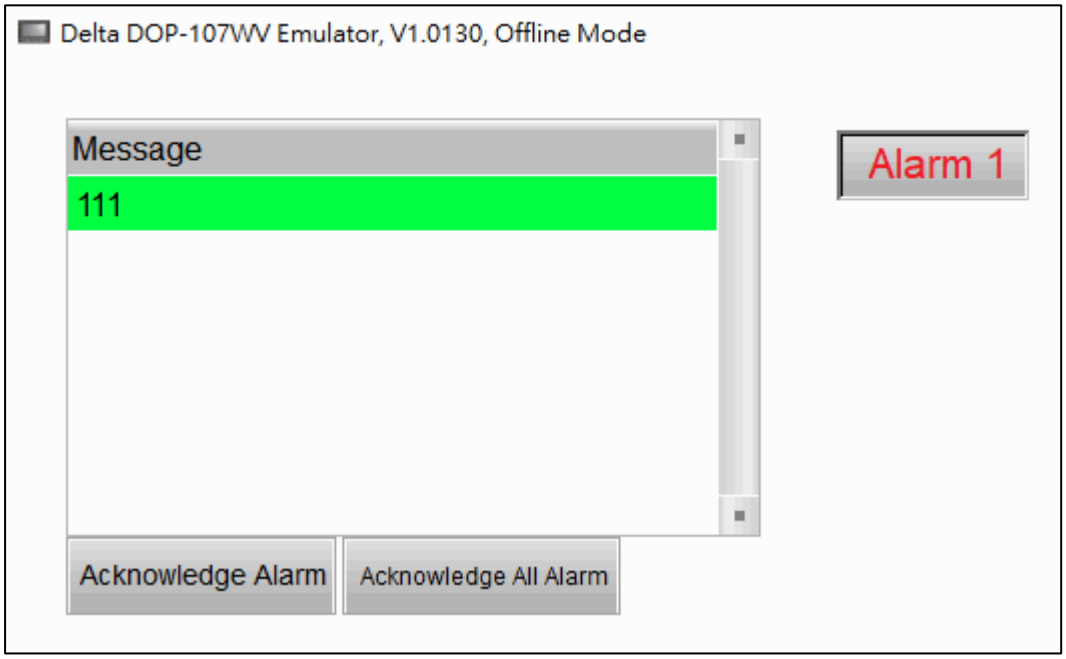
  1. Select the **Color Mode**.
  2. Trigger an alarm on the HMI.

     The trigger column is displayed in red.

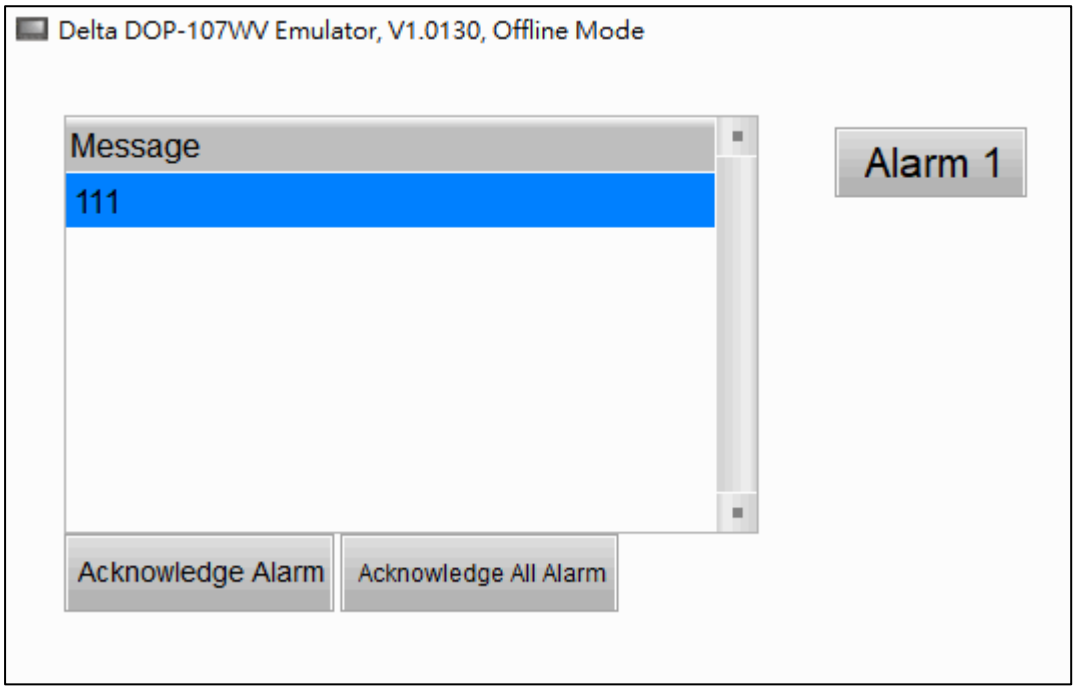


  3. Click the **Acknowledge Alarm** or **Acknowledge All Alarm** button on the Alarm History Table.

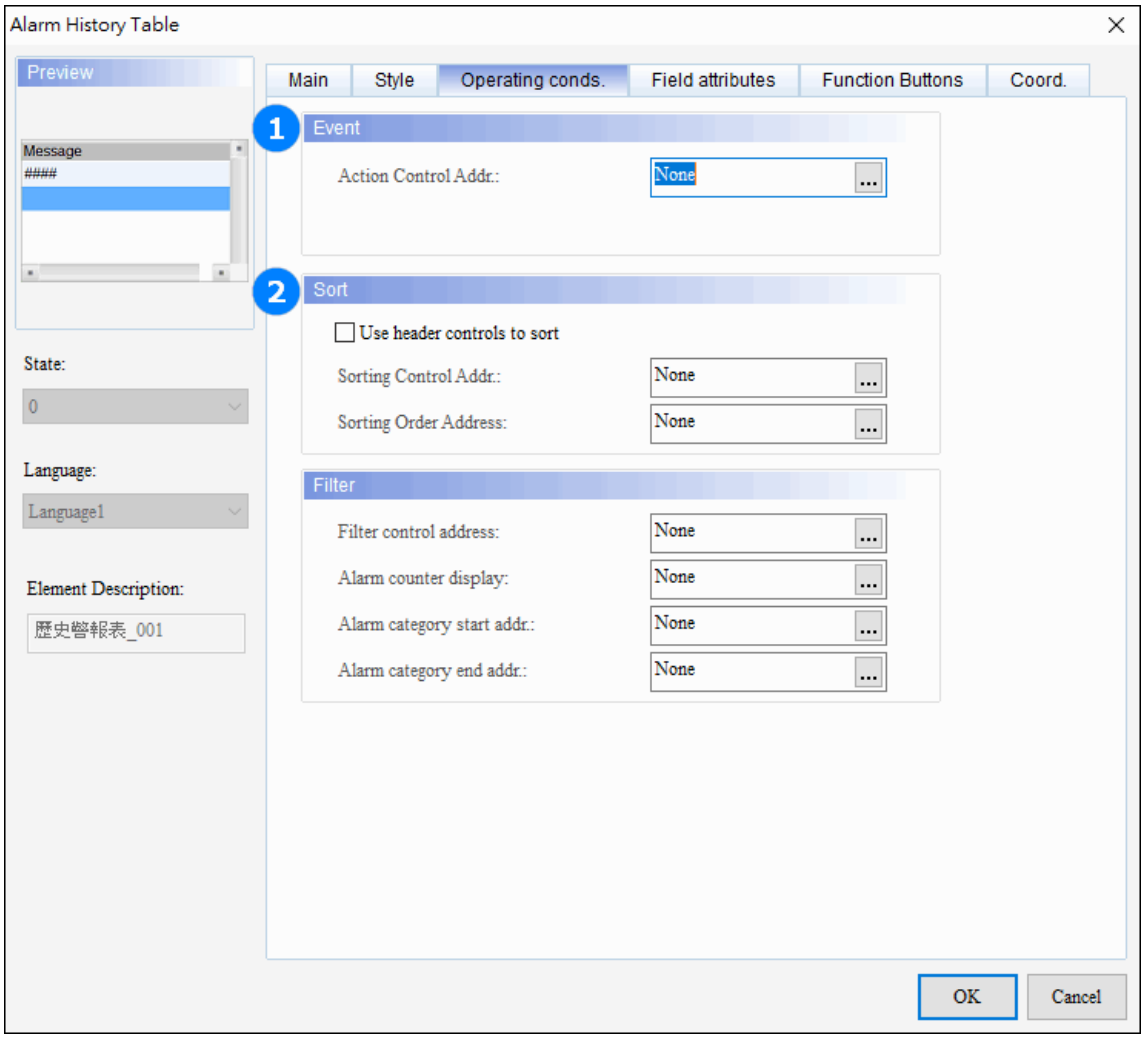
     The confirm column is displayed in green.



When an alarm is recovered, the recovery column is displayed in blue.



- **Operating conds.**



1 Event

Setting	Description
Action Control Addr.	Set the action control address to change screens or acknowledge the alarms for specified alarms.

Value	Description
0	Default status, no action.
1	Acknowledge for selected alarms in the **Alarm History Table element**.
2	If the selected alarm in the **Alarm History Table element** has a set alarm screen which is set to display manually, the alarm screen displays when the value is 2.

❷ Sort

<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Use header controls to sort</td><td>If selected, click the header of the Alarm History Table element to sort the alarms in ascending or descending order. The Sorting Control Addr. and Sorting Order Address settings are not available.
Note: The sorting of the Message column is not supported.</td></tr>
<tr><td>Sorting Control Addr.</td><td>Set Sort Control Addr. for the specified item.
<table>
<tr><th>Value</th><th>Description</th></tr>
<tr><td>0</td><td>Default status, no action.</td></tr>
<tr><td>1</td><td>Sort by Trigger Time.</td></tr>
<tr><td>2</td><td>Sort by Acknowledge Time.</td></tr>
<tr><td>3</td><td>Sort by Recovery Time.</td></tr>
<tr><td>4</td><td>Sort by alarm count.</td></tr>
<tr><td>5</td><td>Sort by alarm category.</td></tr>
</table></td></tr>
<tr><td>Sorting Order Address</td><td>Set the sorting order address with the Sorting Control Addr. value to determine the ascending or descending order. For example, if the Sorting Control Addr. is set to 1 and the Sorting Order Address is set to 0, the Trigger Time is sorted in ascending order.
<table>
<tr><th>Value</th><th>Description</th></tr>
<tr><td>0</td><td>Sort in ascending order.</td></tr>
<tr><td>1</td><td>Sort in descending order.</td></tr>
</table></td></tr>
</table>

Active Alarm List / Alarm Frequency Table Properties

- **Operating conds.**

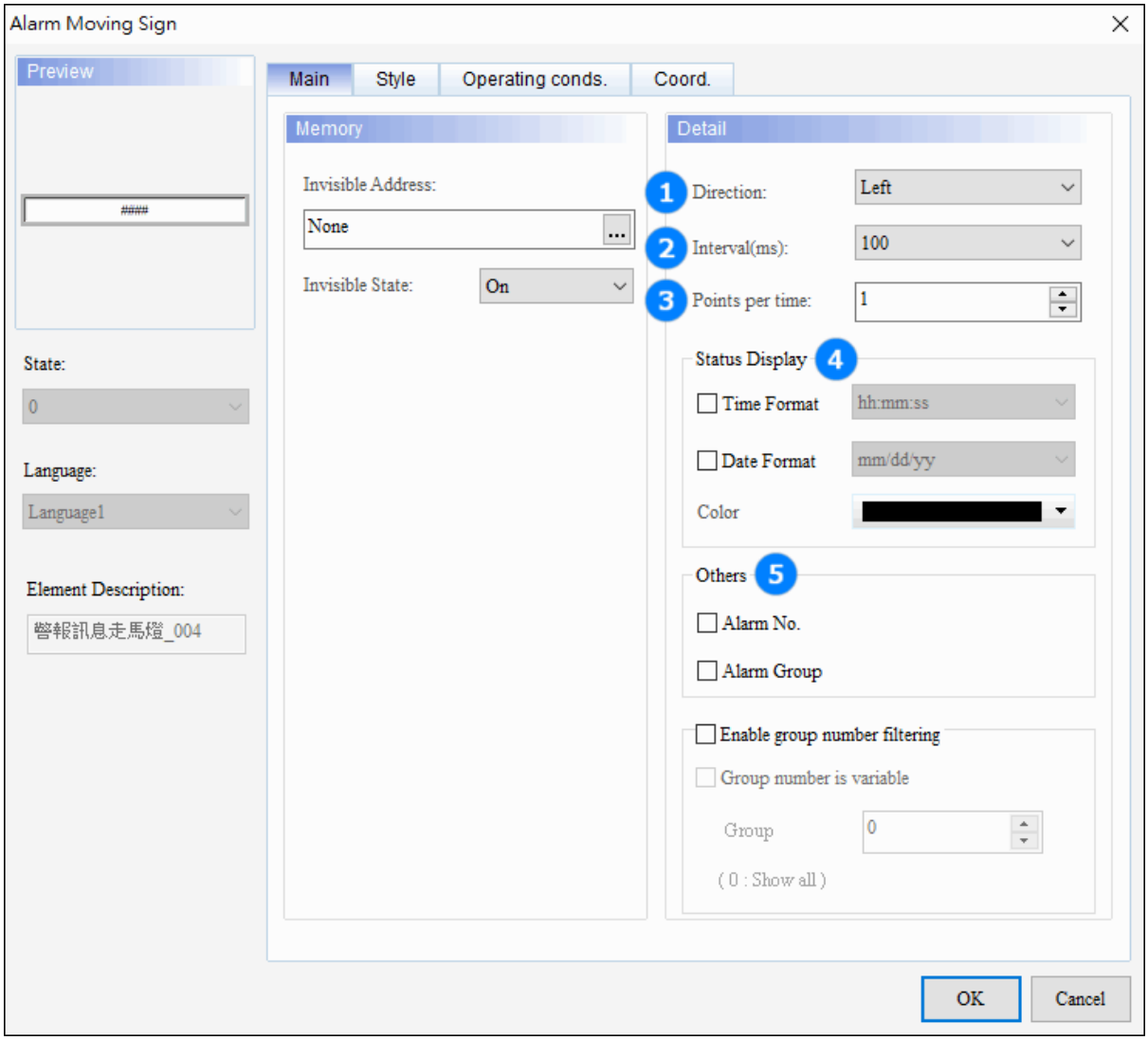
  Select the fields you want to display in the **Field attributes** tab, and then select **Use header controls to sort** checkbox in the **Operating conds.** tab, the **Default sort** field can be selected.

Alarm Moving Sign Properties

The **Alarm Moving Sign** element records the alarm number, time and date when an alarm is triggered. You can also set the time interval and moving distance according to your needs.

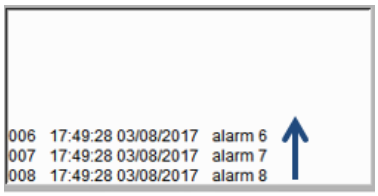
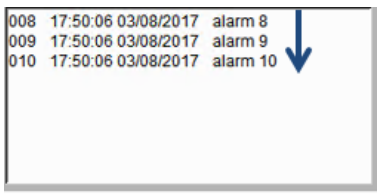
In the **Main** tab, you can set the content of the Alarm Moving Sign.

The following table lists the properties in the **Main** tab in the **Alarm Moving Sign** element and their descriptions.



❶ Direction

Setting	Description
Direction	Select the display direction. The default value is Left.

Left	
Right	
Up	
Down	

❷ Interval

Setting	Description
Interval (ms)	Set how often the message moves. The default value is **100**.

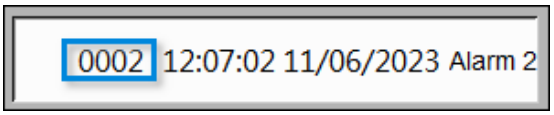
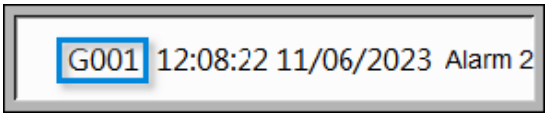
❸ Points per time

Setting	Description
Points per time	Set the distance each time text moves. The greater the number, the greater the distance each time the text moves. The value range is 1–50, and the default value is **1**.

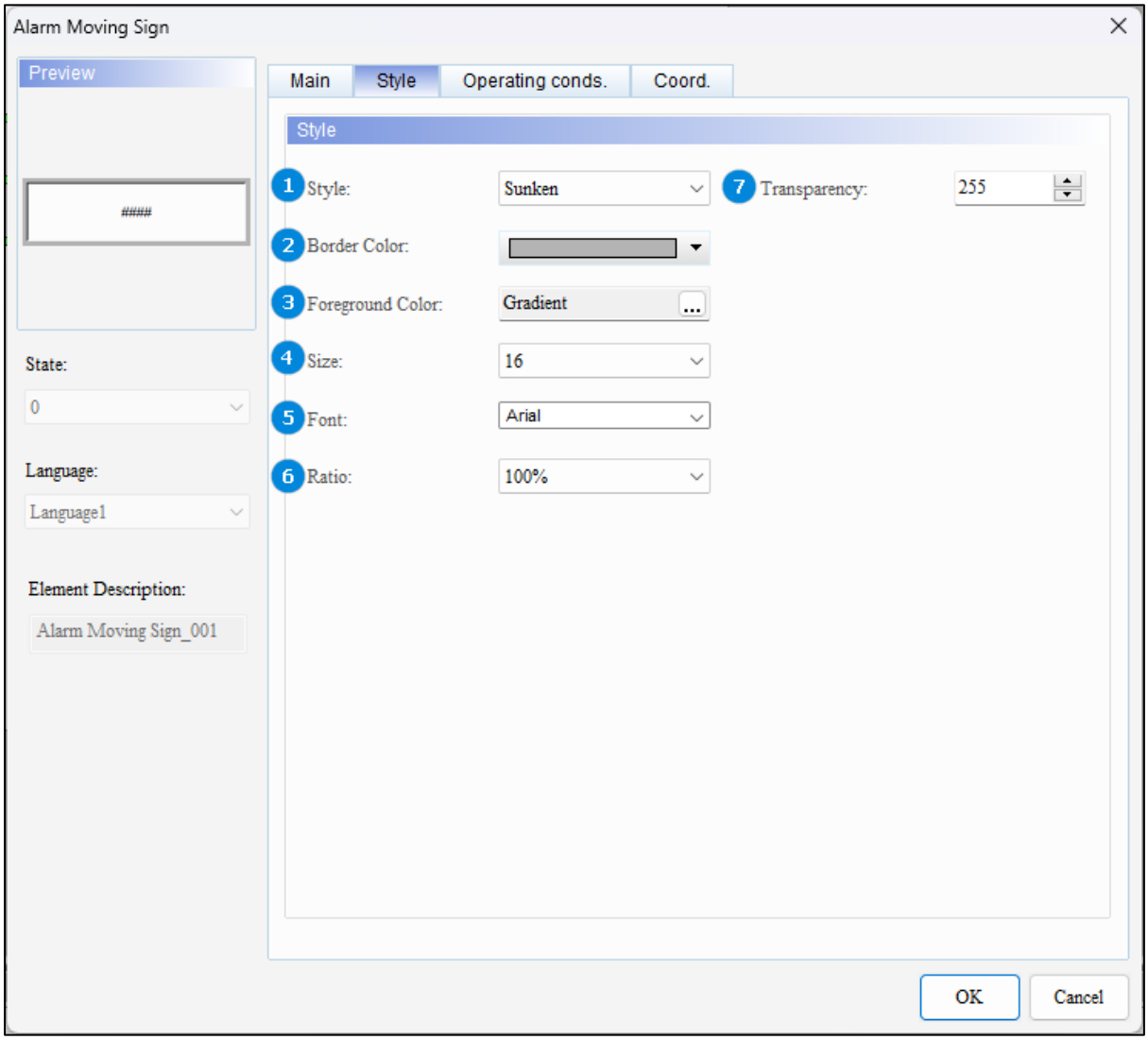
❹ Status Display

Setting	Description
Time format	Select the time format. The default value is hh:mm:ss.
Date format	Select the date format. The default value is mm/dd/yy.
Display Color	Select the display color of the time and date.

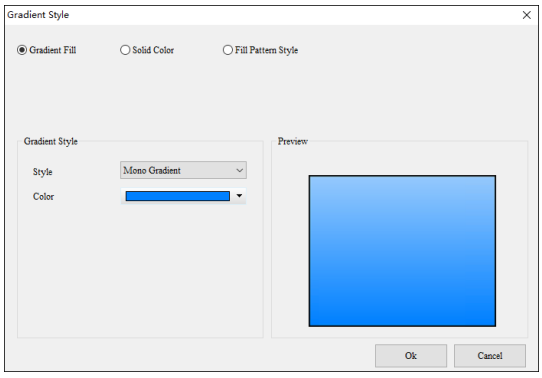
❺ Other displays

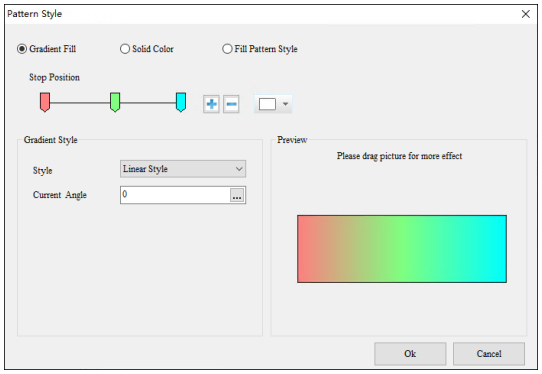
Item	Description
Alarm No.	If selected, the **Alarm Moving Sign** element displays the alarm number when an alarm is triggered. 
Alarm Group	If selected, the **Alarm Moving Sign** element displays the alarm category when an alarm is triggered. 

In the **Style** tab of the property dialog, you can set the display style of the **Alarm Moving Sign** element.

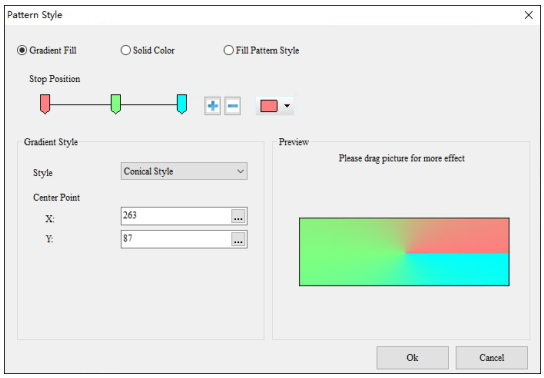
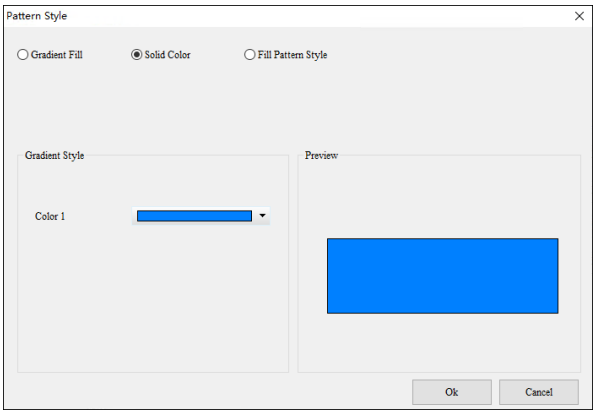


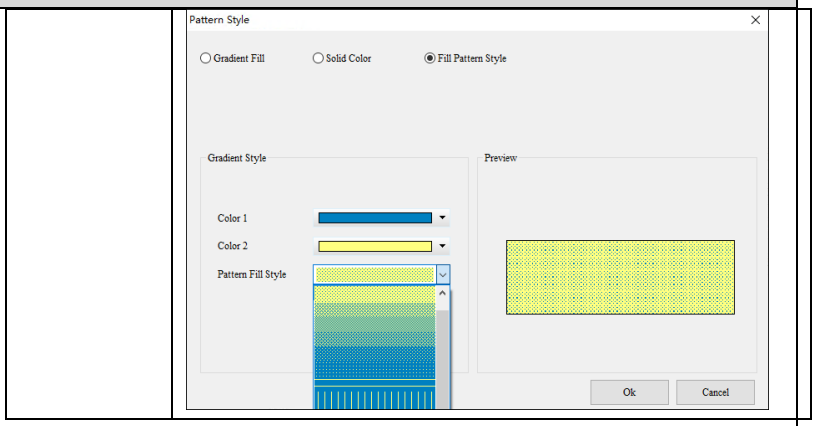
Legend	Setting	Description
1	Style	Set the appearance of the element. The options include Standard, Raised, Sunken, and Transparent. Standard: #### Raised: #### Sunken: #### Transparent: ####
2	Border Color	Set the border color of the element. The default is gray. When set to red, it displays as follows. ####

<table>
<tr><th>Legend</th><th>Setting</th><th>Description</th></tr>
<tr><td>③</td><td>Foreground Color</td><td>Select the Foreground Color of the element. The default value is Gradient. The following table lists the functions in the Gradient Fill view with their description.
<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Gradient Style</td><td>Set the Gradient Style by setting the Current Angle or Center Point. Set the Current Angle or Center Point by dragging the mouse on the preview image, or set the memory address for dynamic configuration on the HMI. Supports adding gradient stop points and setting stop point colors to present the effect of multi-color gradients.
Note: Up to 5 stop points can be set.
• Mono Gradient
The gradient color automatically configured by the software does not provide stop point configuration and is compatible with the behavior of DOPSoft software.
 

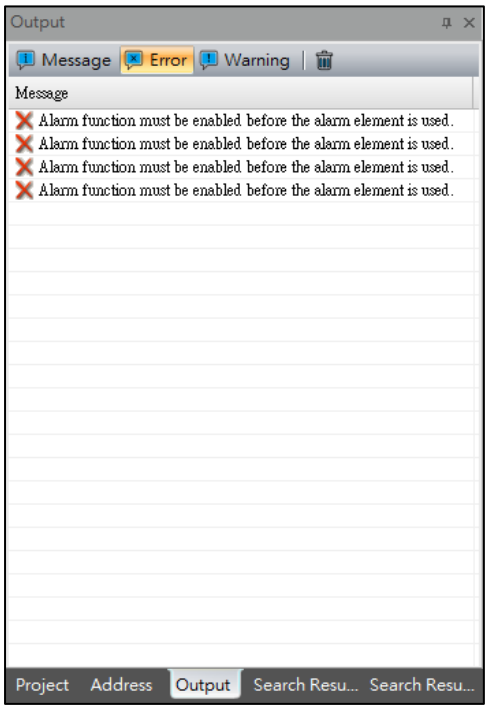
• Linear Style
 

• Radiation Gradient</td></tr>
</table></td></tr>
</table>

Legend	Setting	Description	
			 • **Conical Style** 
		Solid Color	Set a single color fill, the same as the **Foreground Color** setting. 
		Fill Pattern Style	Provides a variety of Fill Pattern styles and two-color combinations to present pattern fill effects.

Legend	Setting	Description
		 **Note**: If select **Transparent** for Style, the set **Foreground Color** will not be applied.
4	Size	Set the text size of the alarm message to be displayed.
5	Font	Set the font of the alarm message to be displayed. You can switch between different languages to set the font settings corresponding to the language.
6	Ratio	Set the scaling ratio of the alarm message to be displayed.
7	Transparency	Set or enter the transparency value. The smaller the value, the higher the transparency of the element. The value range is 50–255, and the default value is **255**.

Note: When an alarm element is created and the alarm reading address is not set, the error message will be displayed in the **Output** pane after compilation.



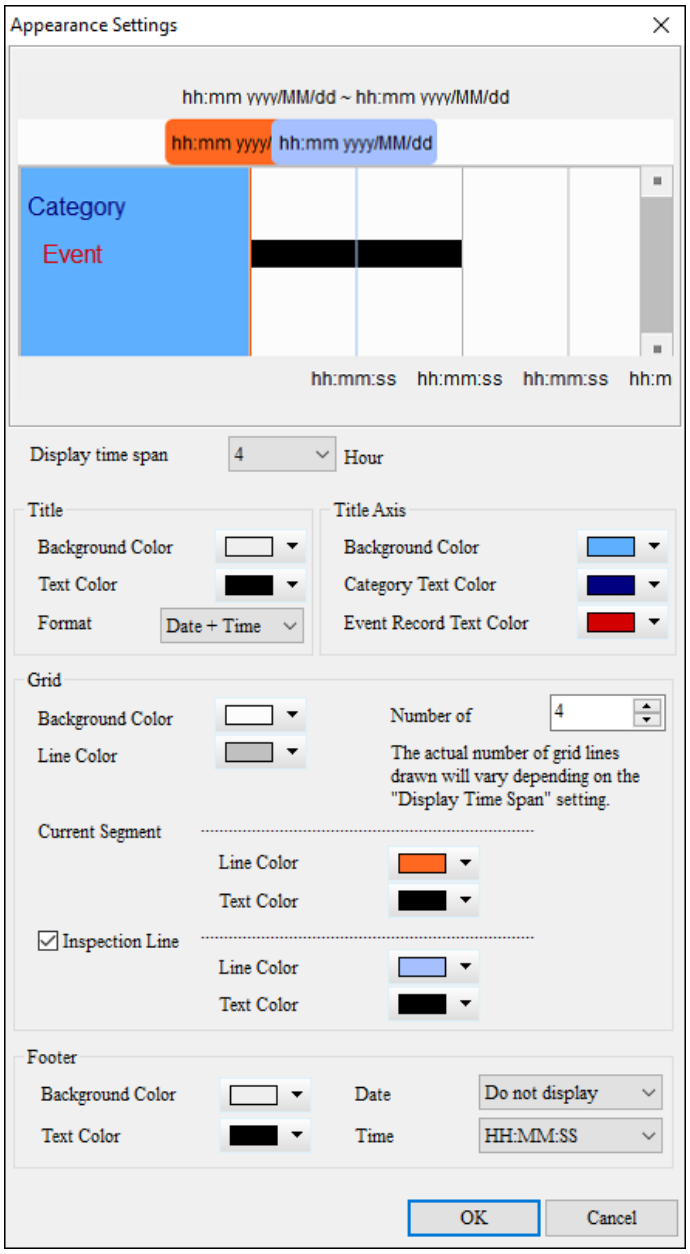
Gantt Chart Properties

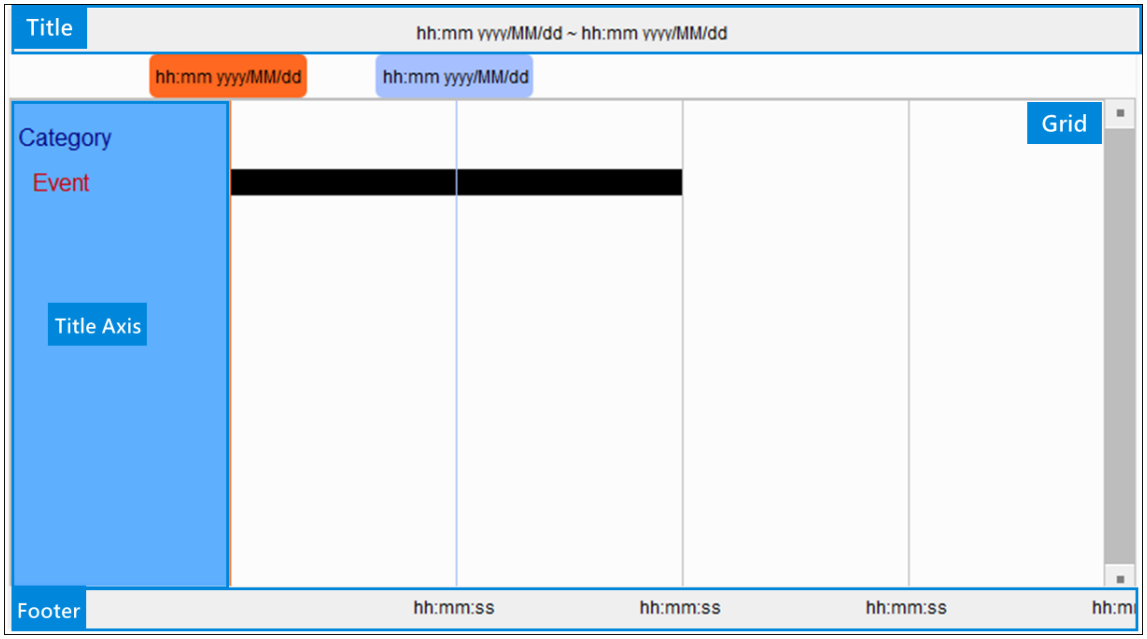
The **Gantt Chart** element displays the alarm trigger time to restore time. You can filter the alarm information to view using

Note: This element is only available for DOP-300 series model, and one screen can only have one Gantt Chart element.

- **Style**

  The following table lists the settings in the **Appearance Settings** dialog with their description.





Setting	Description
Display time span	Set the time range to display on the Gantt chart.
Title	Set the background color, text color, and date / time format of the title. **Note**: The format setting synchronously applies to **Current Segment** and **Inspection Line**.
Title Axis	Set the background color, category text color, and event record text color of the title axis.
Grid	Set the background, line color, and number of grids.
Current segment	Set the line color and text color.
Inspection Line	
Footer	Set the background color, text color, and date and time format.

Notification Function

WeChat Push Notification Function

This section describes how to send an alarm message to the WeChat App through the push notification function when an alarm occurs on the HMI.

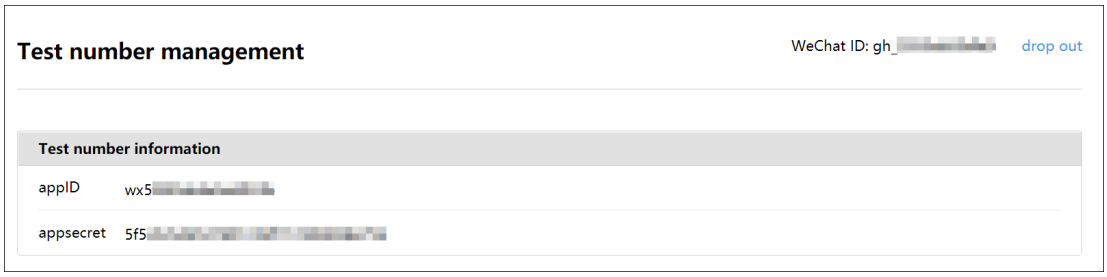
Prerequisites

- WeChat account is registered.

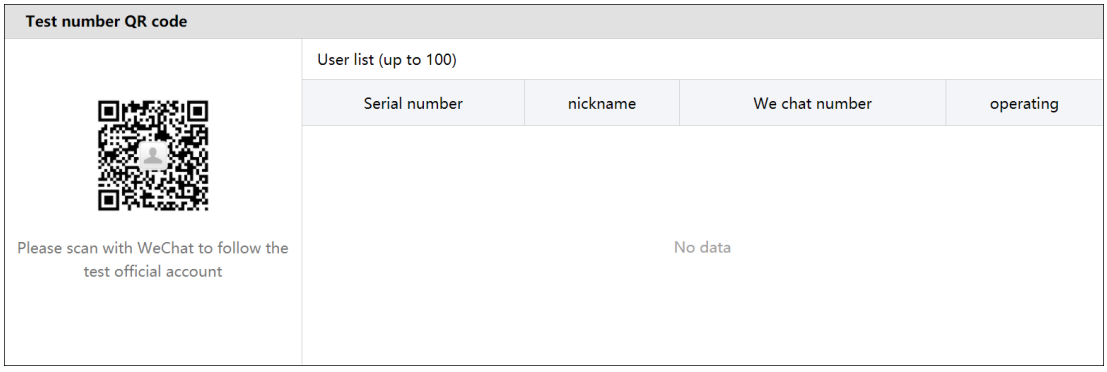
Follow these steps to set up WeChat push notification function.

To set up WeChat push notification function

1. Go to mp.weixin.qq.com/debug/cgi-bin/sandbox?t=sandbox/login and log in to your existing WeChat account (WeChat Public Platform).
   a. After logging in, enter **appID** and **appsecret** in the **Test number information** area for subsequent use in DIAScreen.

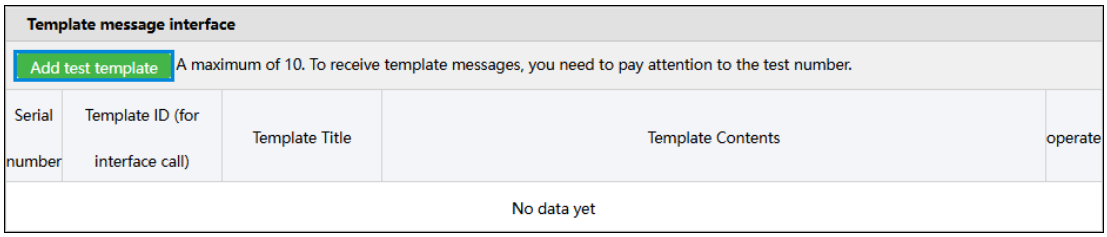


   b. In **Test number QR code** area, scan the QR code to track the test official account to receive notifications. After tracking, the user information appears in **User list**, and the **WeChat ID** must be recorded for subsequent use in DIAScreen.

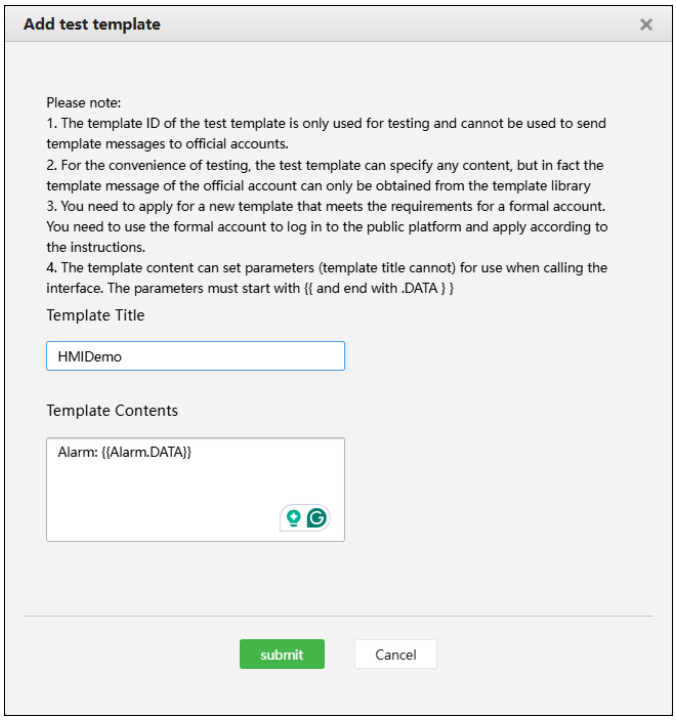


2. Add new test template.

   a. In the **Template message interface** area, click **Add test template**.



   b. In the **Add test template** dialog, enter **Template Title** and **Template Contents** according to your need.



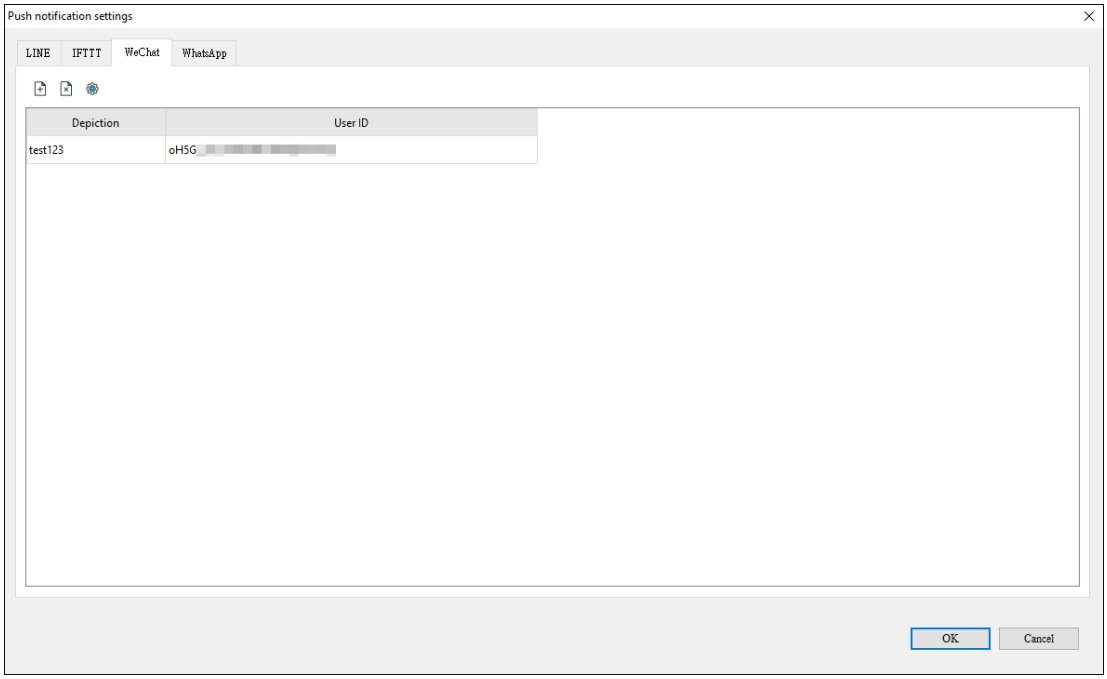
   Note: The template content must contain the {{Alarm.DATA}} string. This {{Alarm.DATA}} string will automatically import the DIAScreen alarm message content.

   c. After the addition is completed, the module information will be displayed in the list. Record **Template ID** for subsequent DIAScreen use.

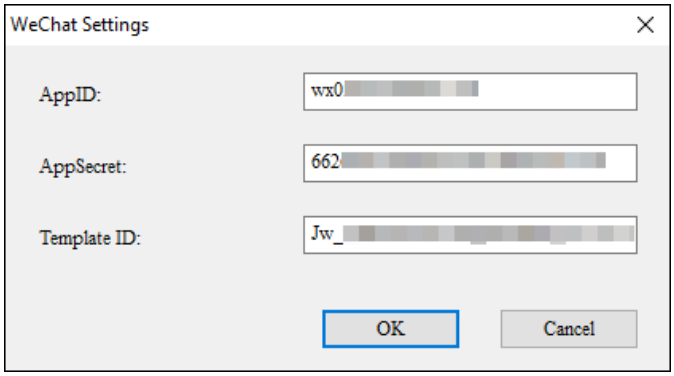
3. Create a project in DIAScreen.

   a. After the project is created, go to **IIoT** > **Notifications**.

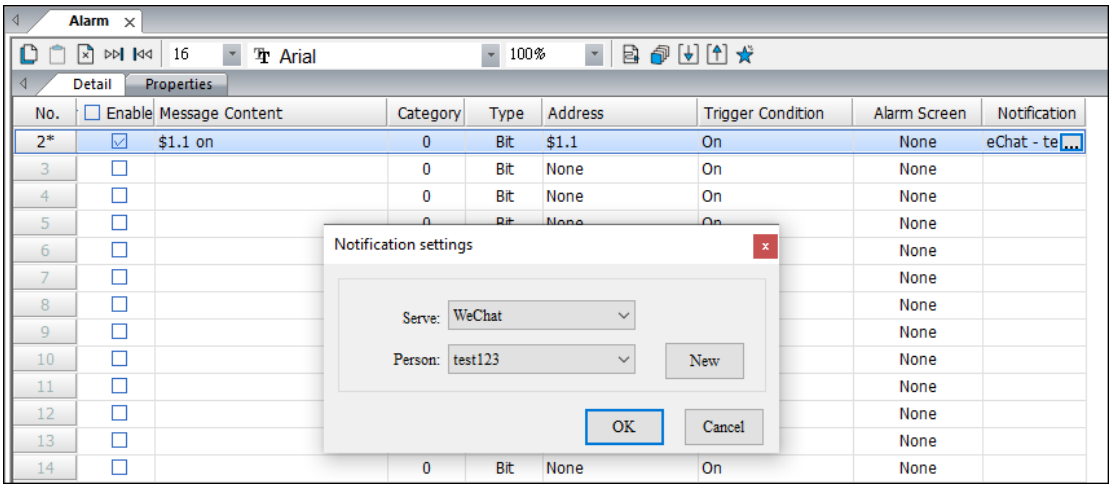
   b. In the **WeChat** tab of the **Push notification settings** dialog, click to add a user, and then enter the **WeChat ID** obtained from the **WeChat Public Platform** in the **User ID** field and enter the **Description** according to your need.



   c. Click , and then enter **AppID**, **AppSecret**, and **Template ID** in the **WeChat Settings** dialog.

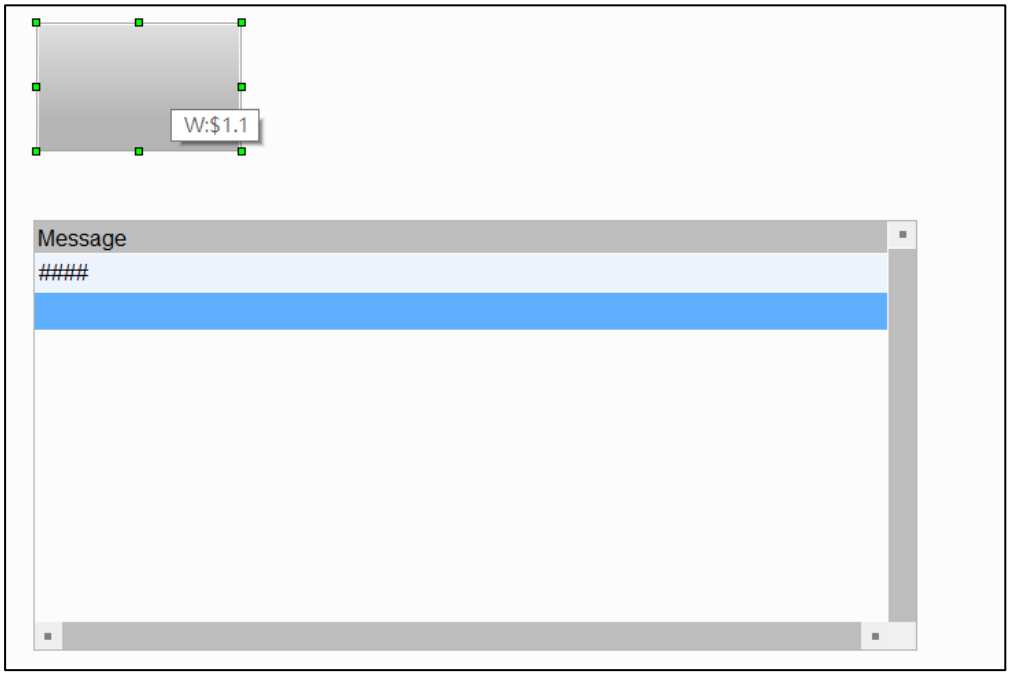


d. Go to **Data Management** > **Alarm Settings**, set up alarm in the **Detail** tab of the **Alarm** page, and then set **Notification**.



e. Create an **Alarm History Table** element and a **Maintained Button** element on the screen.

f. Set up **Alarm History Table** element and set **Write Address** of the **Maintained Button** element to $1.1 (for triggering the alarm).

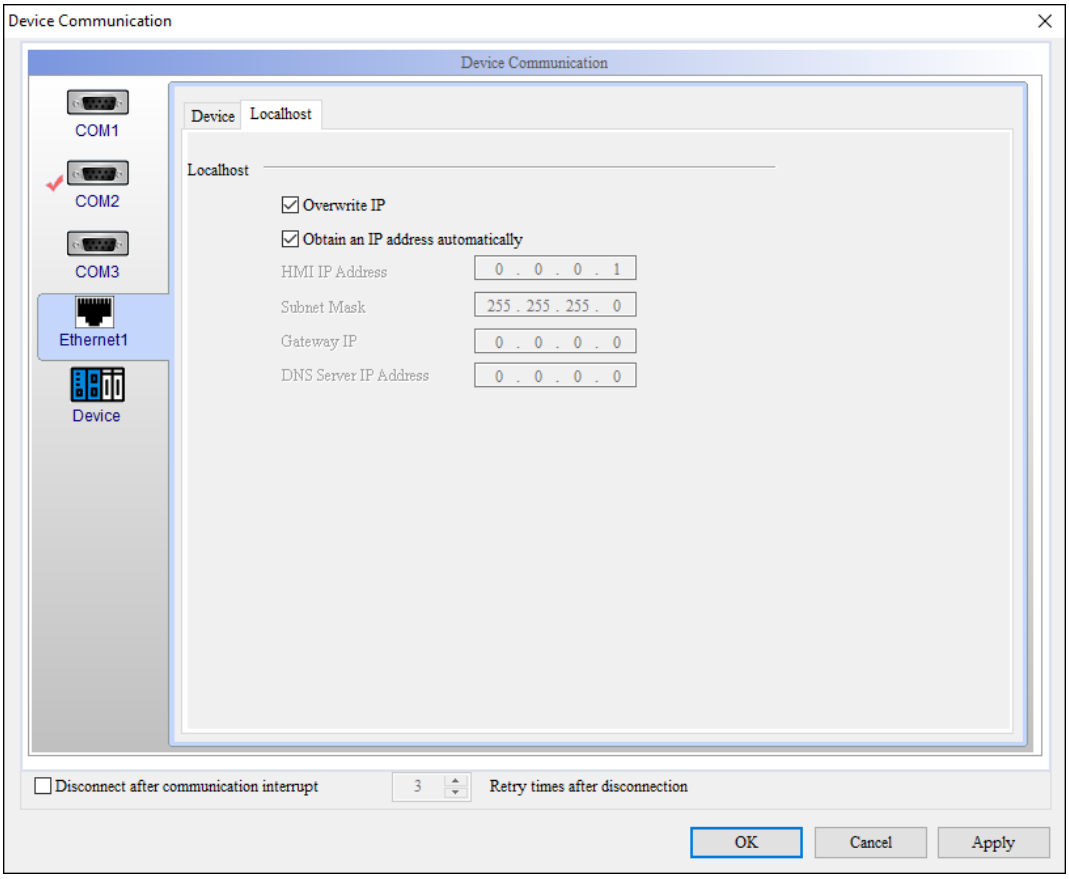


4. Connect to a network.

- **DOP-100 series models (without Wi-Fi Dongle)**

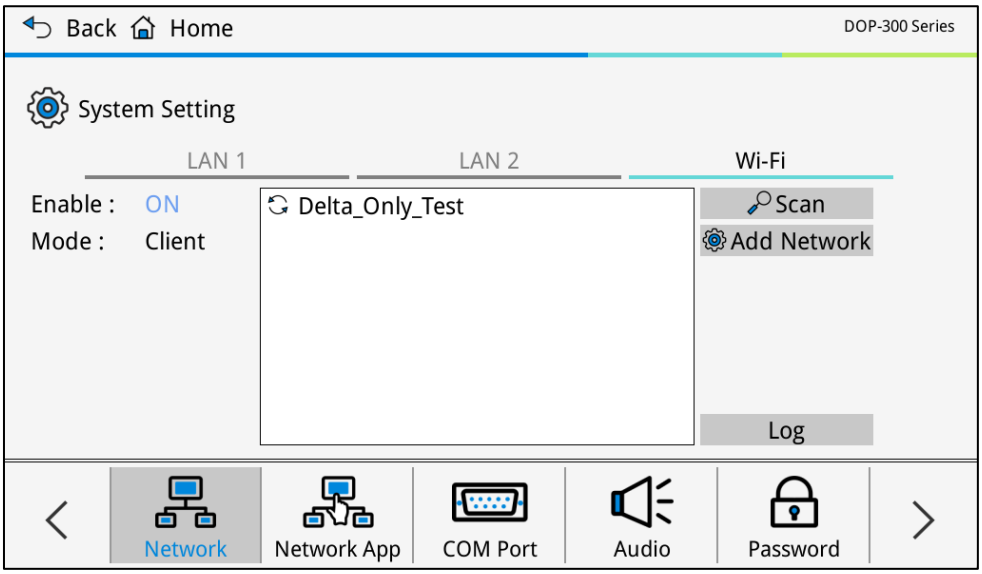
  a. In DIAScreen, go to **General** > **Device Communication**.

  b. In the **Ethernet 1** page of the **Device Communication** dialog, select **Overwrite IP** and **Obtain an IP address automatically** checkboxes in the **Localhost** tab.

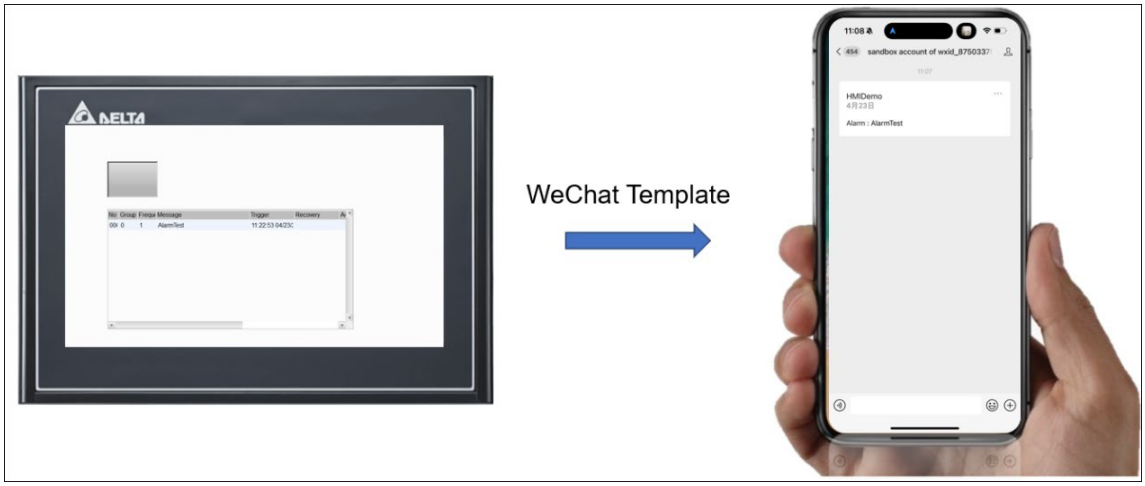


- **DOP-300 series models (connect to Wi-Fi Dongle)**

  a. On the HMI screen, go to **System Menu** > **System Setting** > **Network** > **Wi-Fi**, and connect to Wi-Fi.



5. After completing the above settings, download the screen to the HMI and connect the HMI to the external network. After the alarm is triggered, you can view the corresponding alarm message on WeChat App.



Note: If WeChat App does not receive push messages, confirm the network connection of the HMI or go to the system directory to confirm whether the network IP has been assigned.

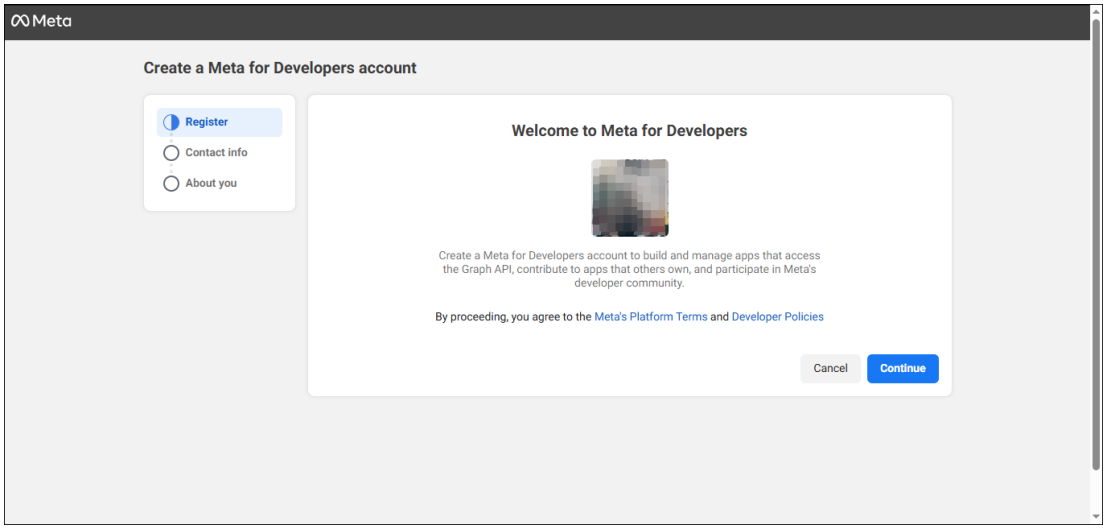
WhatsApp Push Notification Function

This section describes how to send an alarm message to the WhatsApp through the push notification function when an alarm occurs on the HMI.

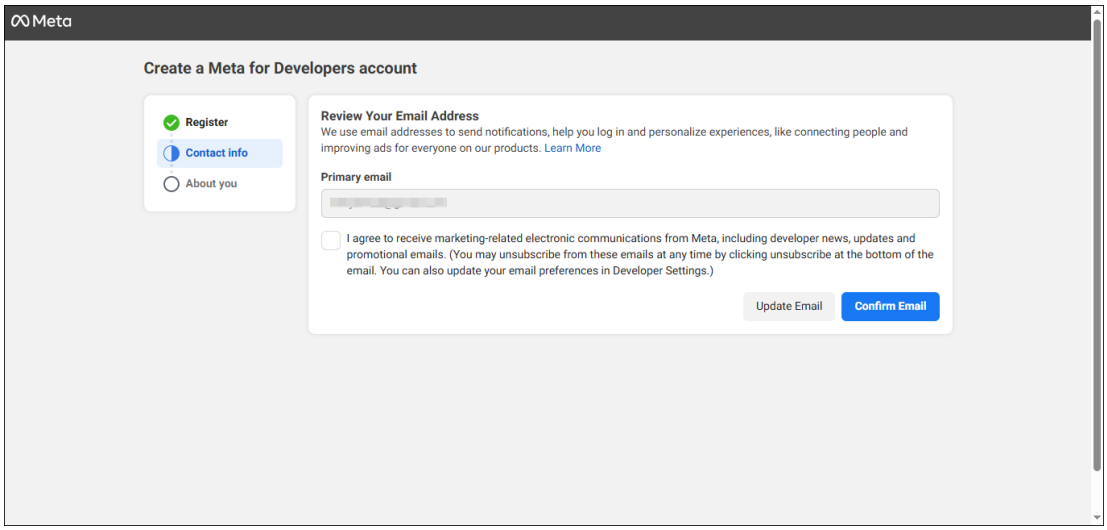
Follow these steps to set up the WhatsApp push notification function.

To set up WhatsApp push notifications function

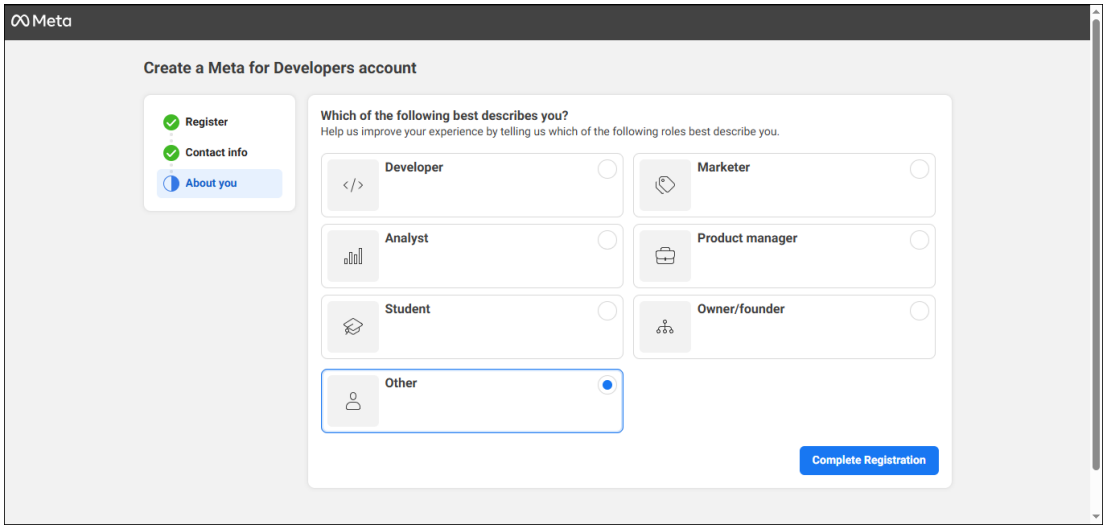
1. Go to developers.facebook.com/async/registration to register a Meta account and create a Developer account.



a. Confirm your e-mail.



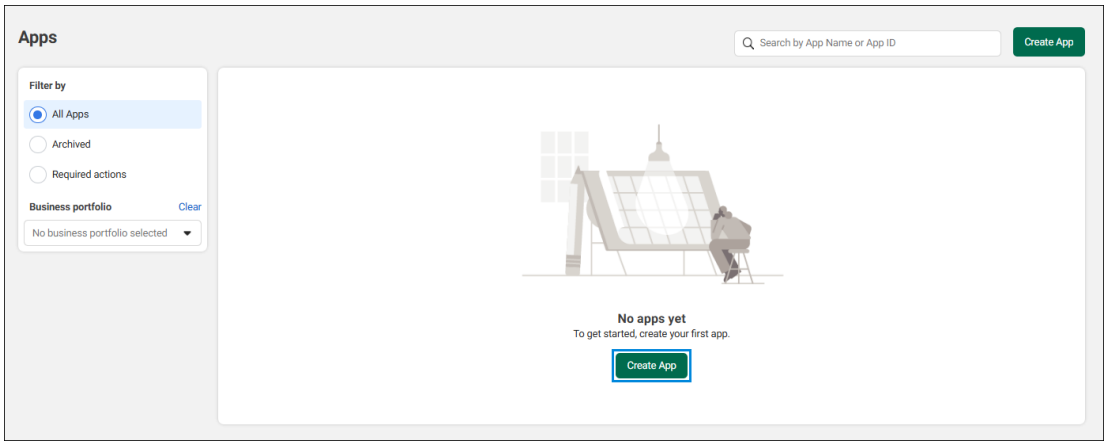
b. Confirm your registration.



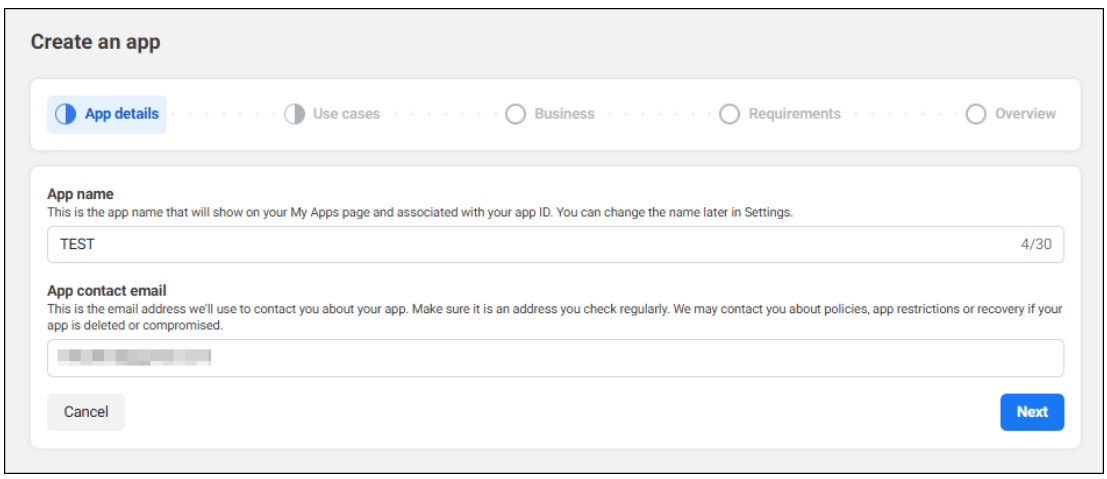
2. Create an application.

   a. After signing up, go to developers.facebook.com/apps/?show_reminder=true to create an app.

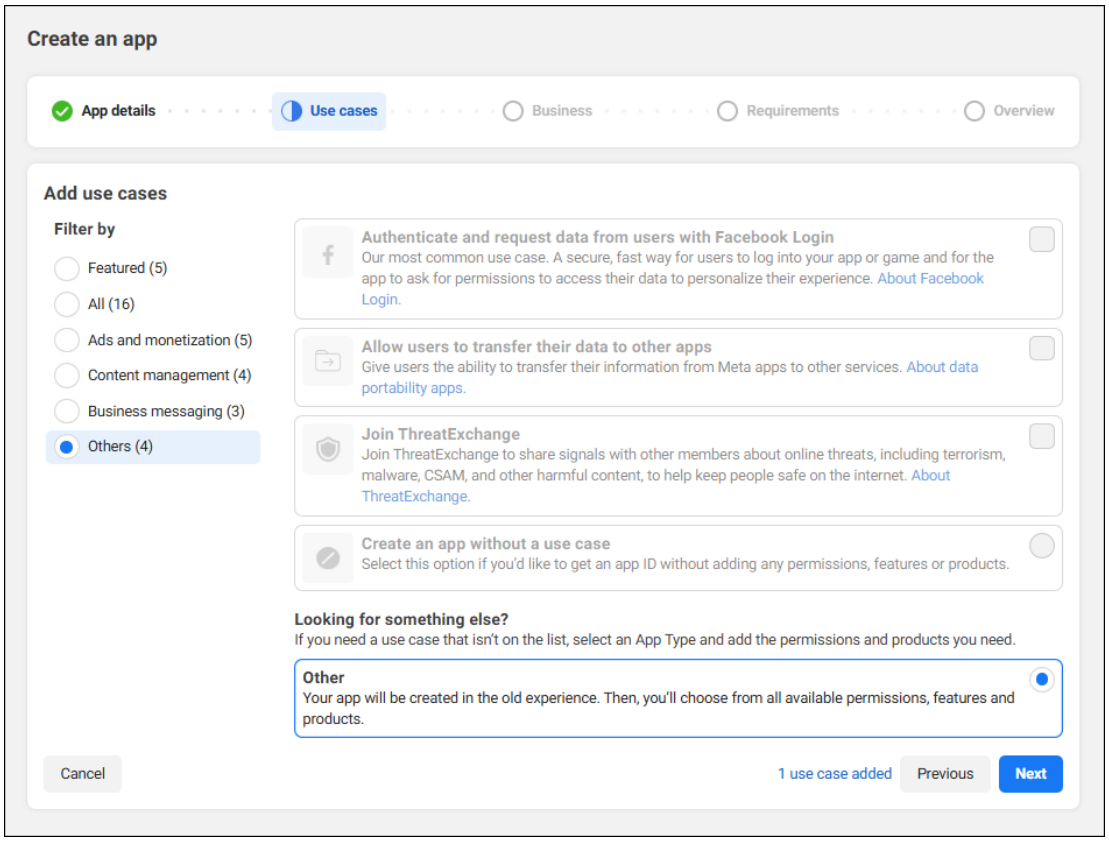
   b. In **Apps**, click **Create App**.



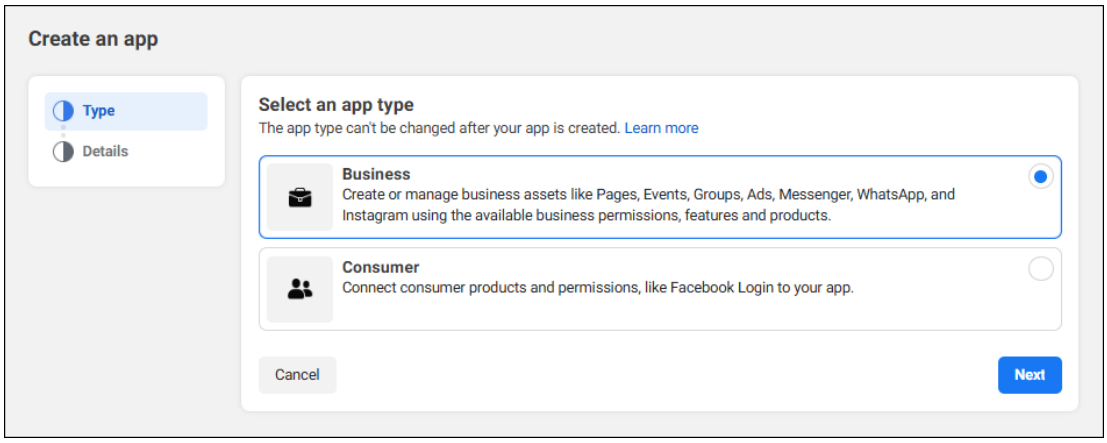
   c. In the **Create an app** dialog, enter **App name** and **App contact email**.



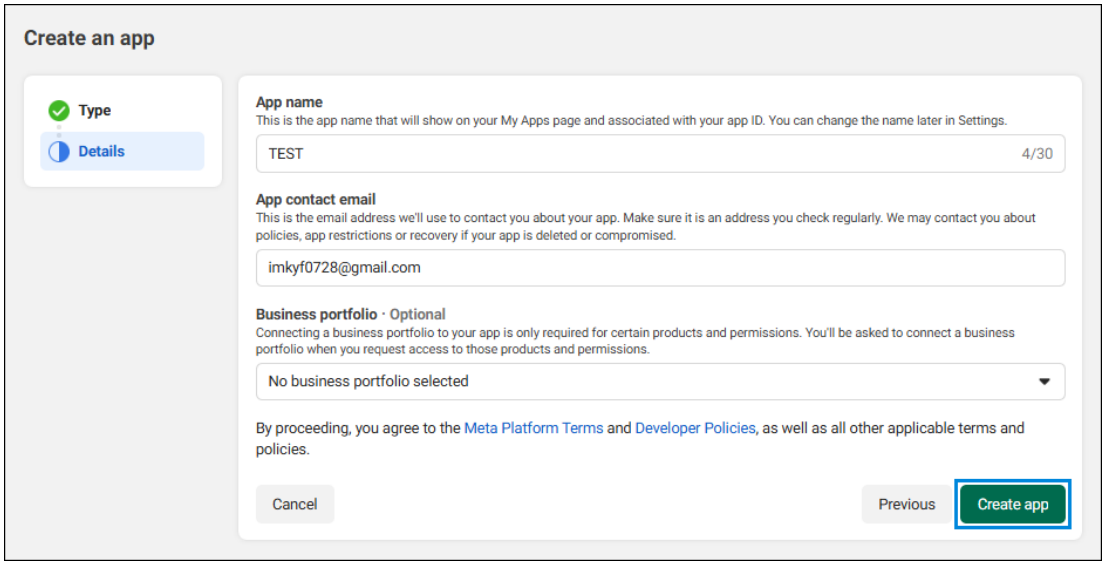
d. For **Use cases**, select **Others**.



e. Select **Business** in **Select an app type**.

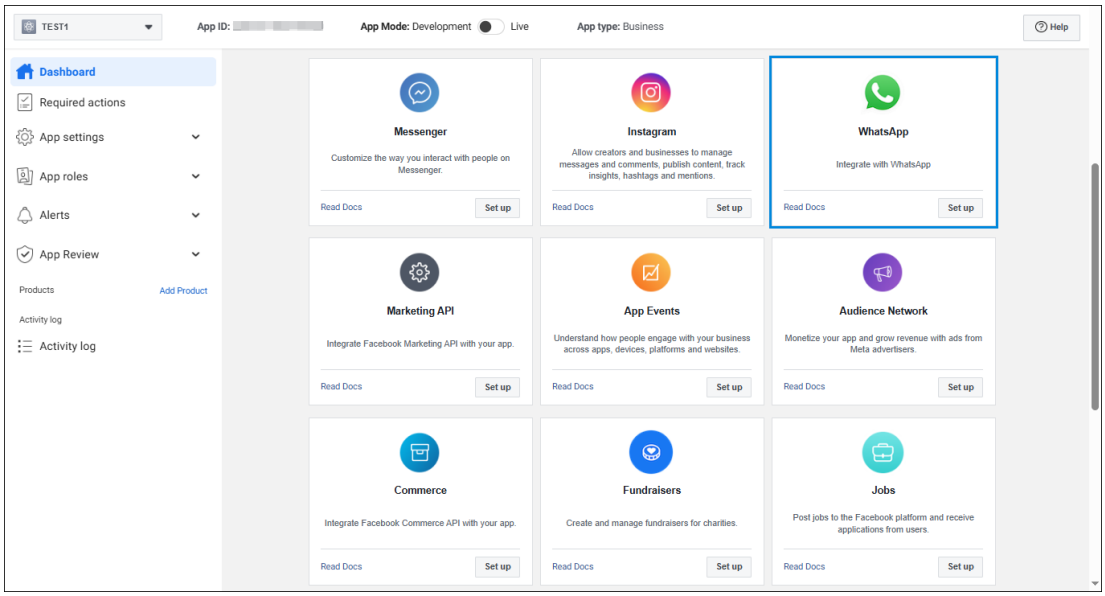


f. Click **Create app**.

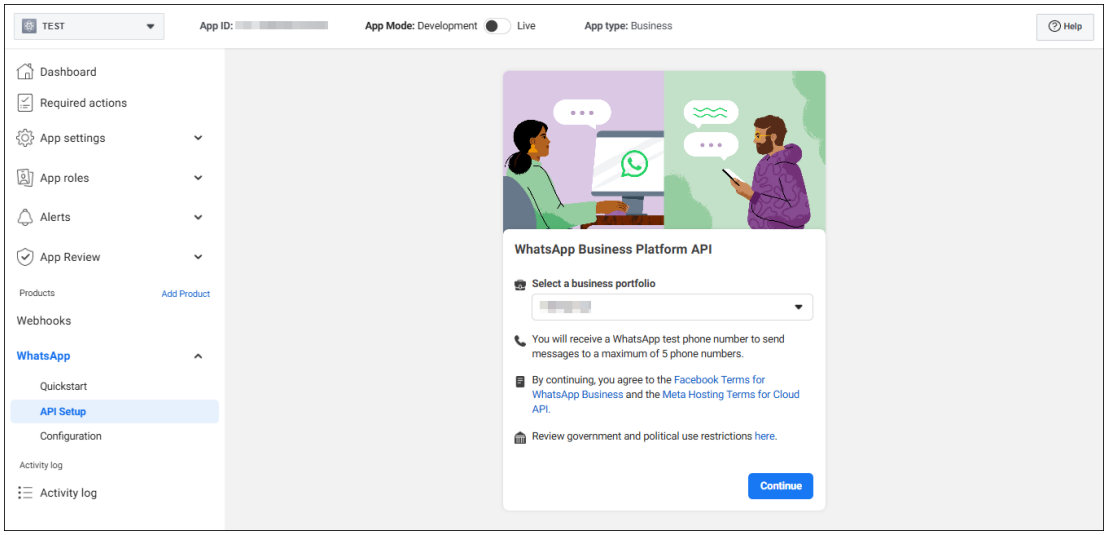


3. Setting WhatsApp.

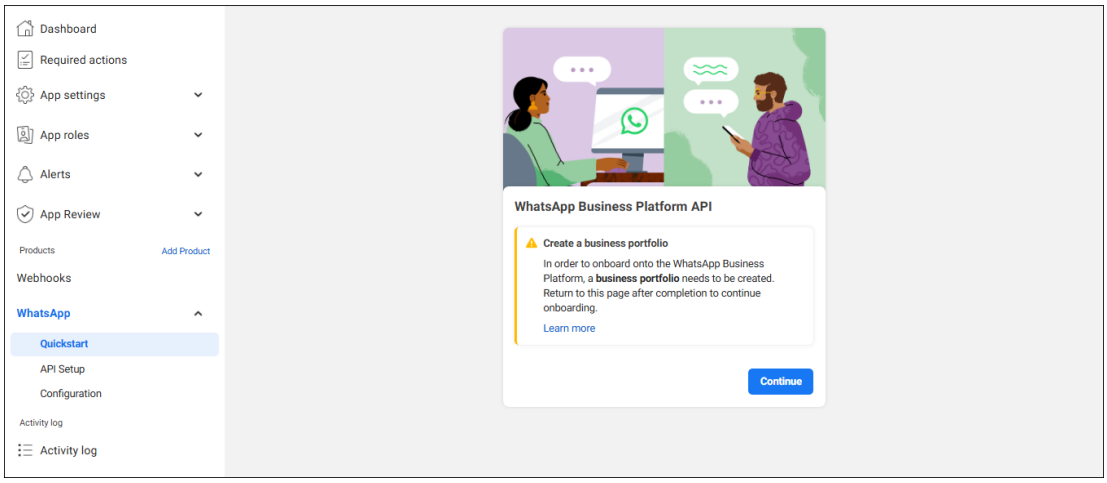
a. In **Meta App Dashboard**, click **WhatsApp**.

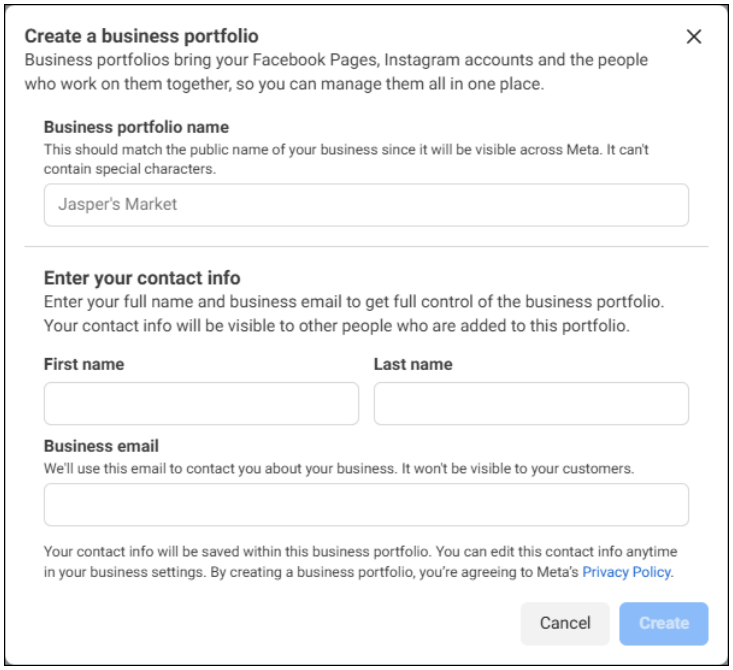


b. Select a business portfolio in **WhatsApp Business Platform API** and click **Continue**.

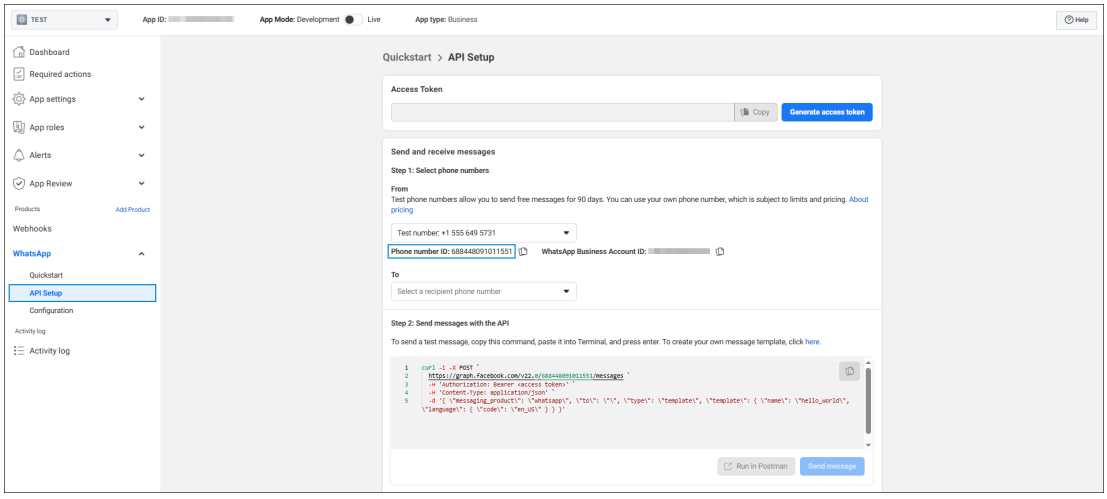


Note: If there is no business portfolio, the screen will be displayed as follows. Click **Continue** to create a business portfolio.





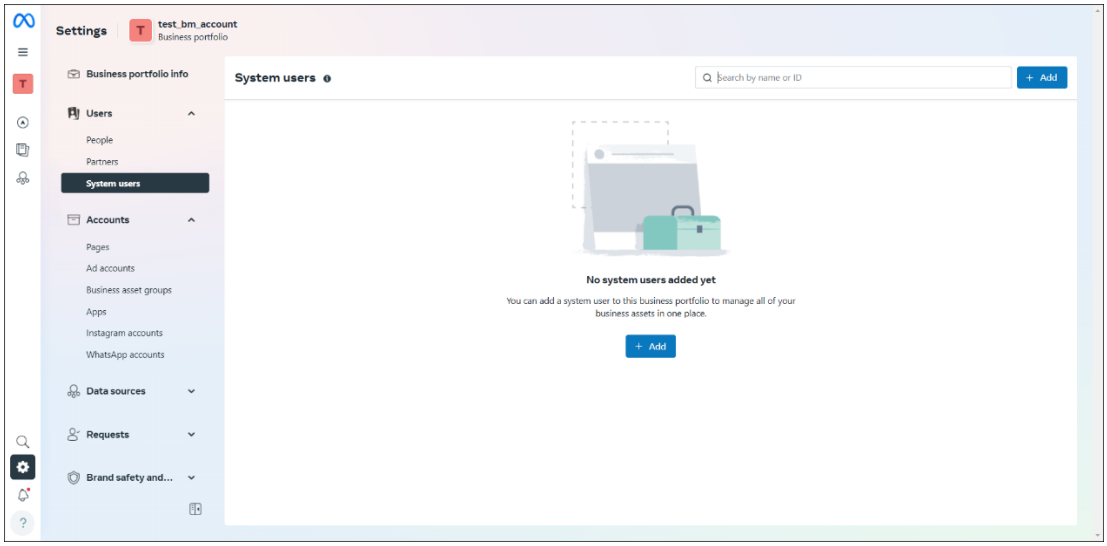
c. After the creation is complete, go to the **API Setup** page and record the obtained **Phone Number ID** for subsequent use in DIAScreen.



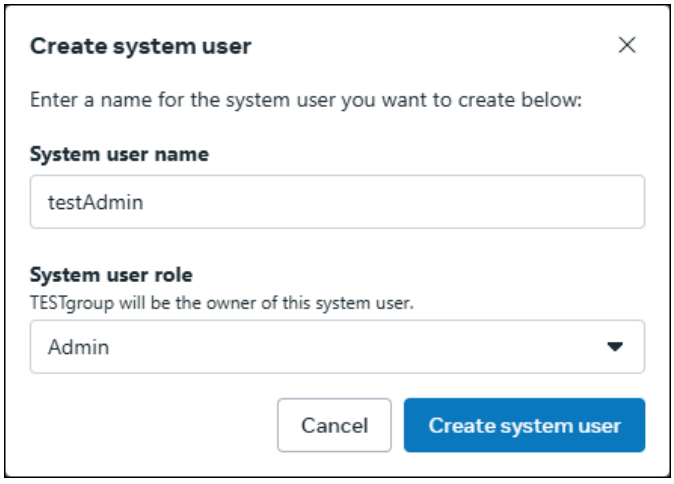
4. Generate token.

a. Go to business.facebook.com/settings/ and log in to Meta Business Suite.

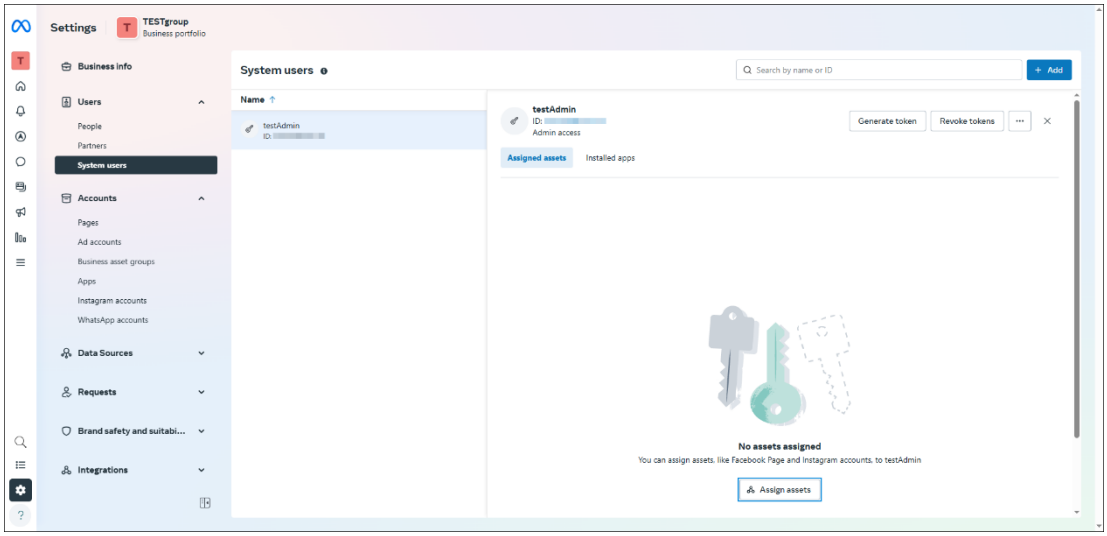
b. In the **System users** tab, click **Add**.

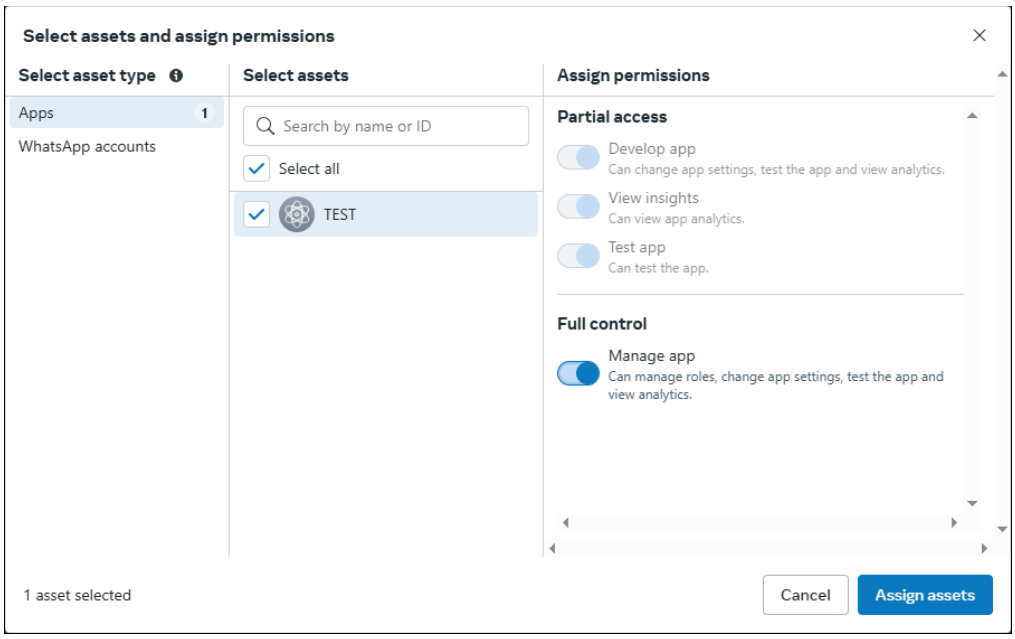


c. In the **Create system user** dialog, enter **System user name** and set **System user role** to Admin.

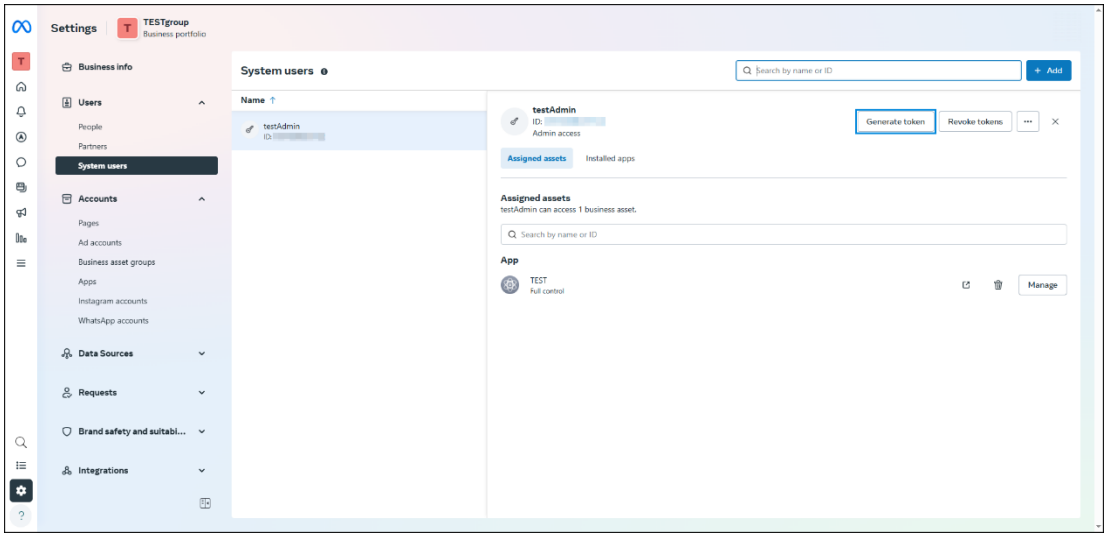


d. Click **Assigned assets**, then assign the corresponding application in the **Select assets and assign permissions** dialog and grant full control.

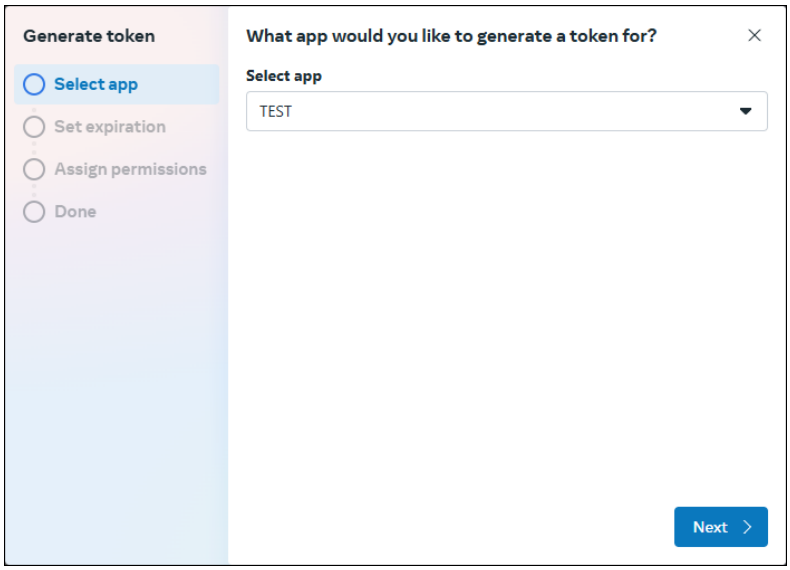


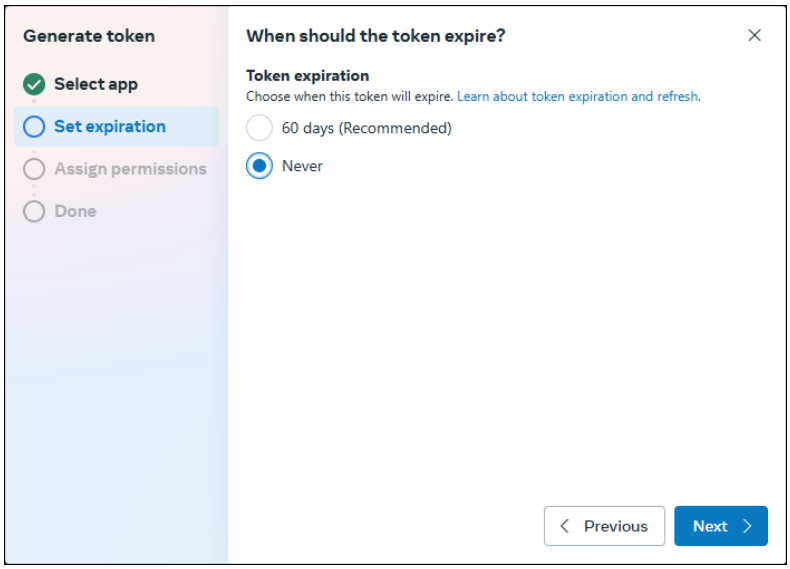


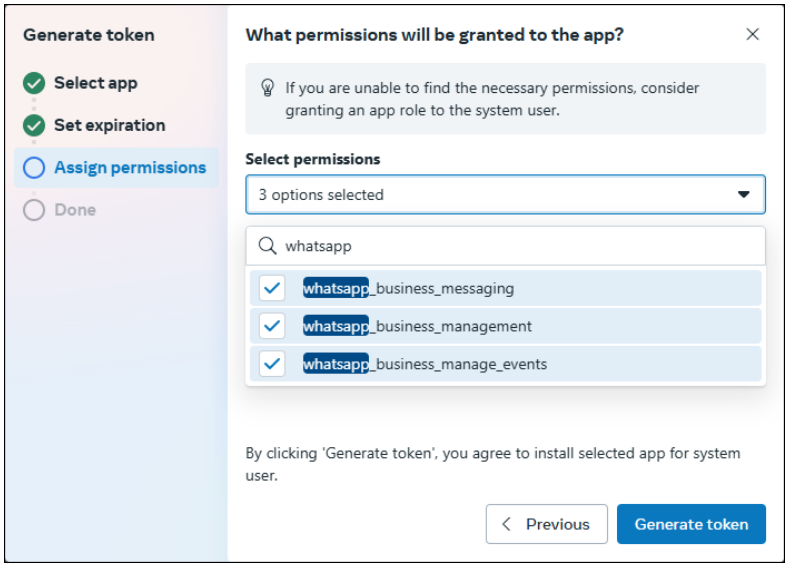
e. After completing the above settings, click **Generate token**.



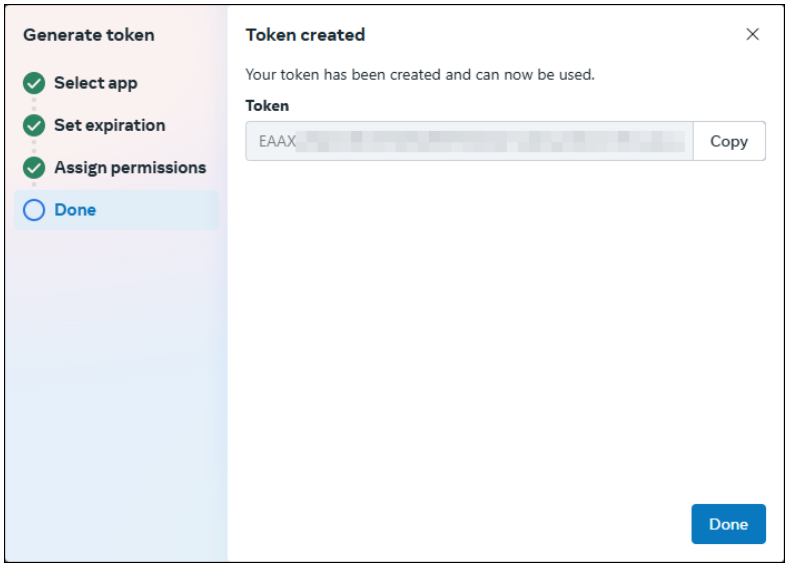
f. Generate token according to the settings as shown in the following figures.







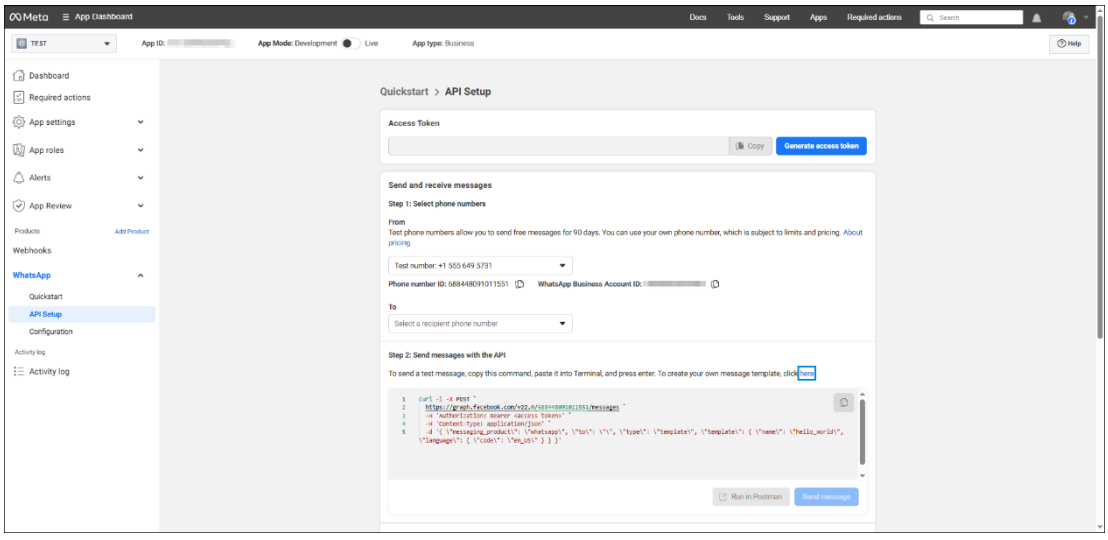
g. Copy token and record it for subsequent use in DIAScreen, and then click **Done**.



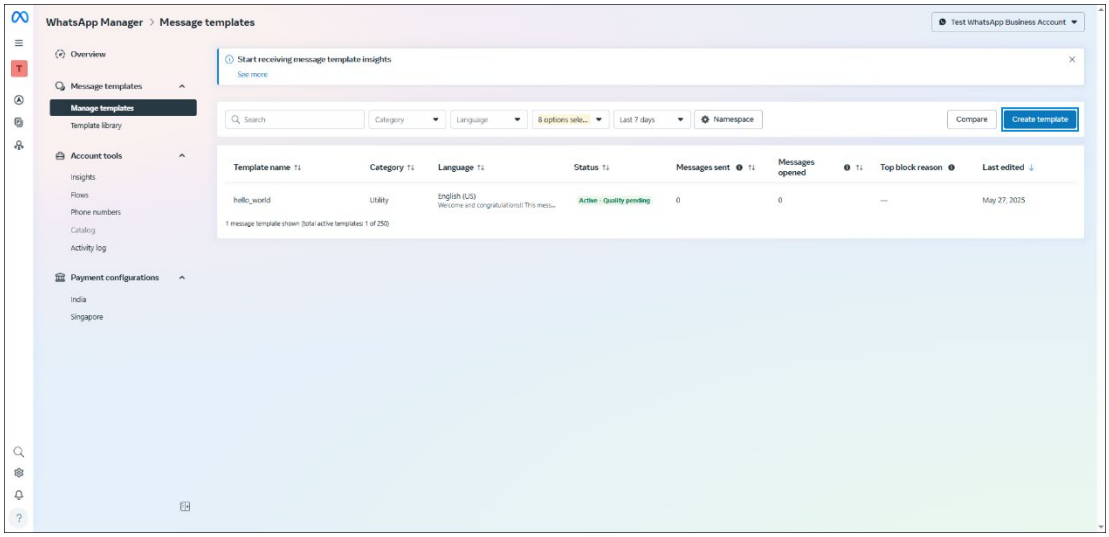
5. Create a Meta template.

a. Go to developers.facebook.com/apps/549547851136042/whatsapp-business/wa-dev-console/?business_id=917902311909443/.

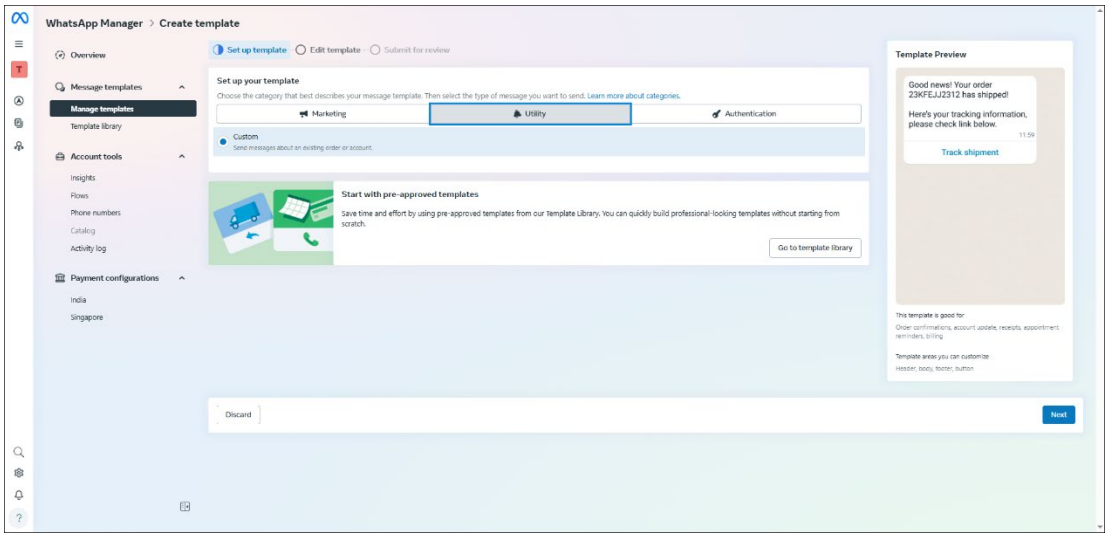
b. Click **here** to create a message template in the **API Setup** tab of the **App Dashboard** dialog.



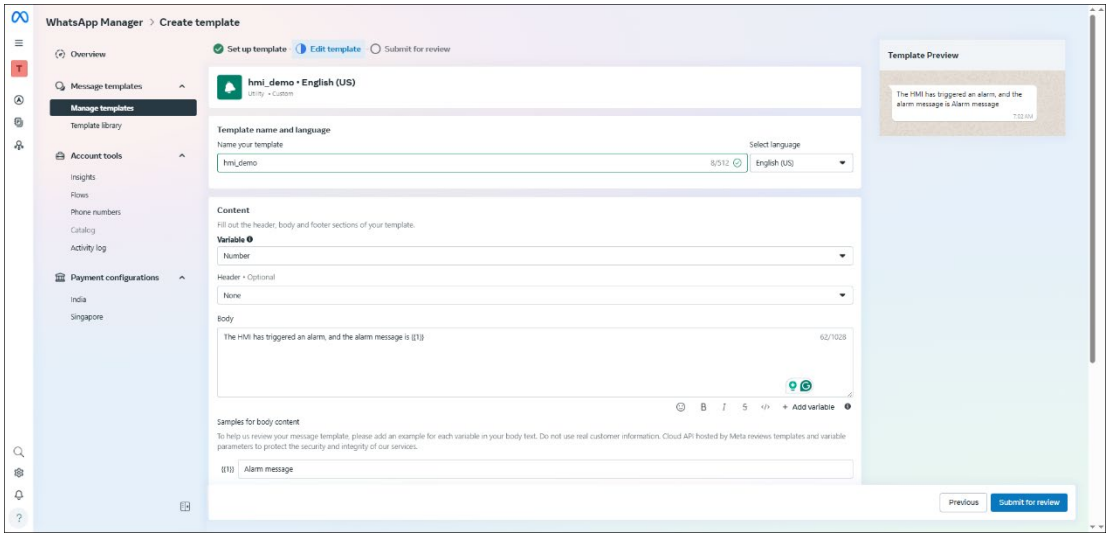
c. In the **Manage templates** page of **WhatsApp Manager**, click **Create template**.



d. Click **Utility**, and then click **Next**.



e. Enter **Template name** and select English (US) for **Select language**. The content example is shown in the following figure.



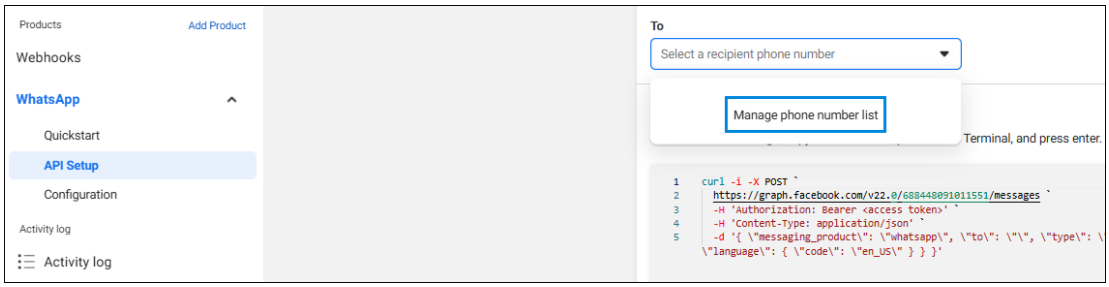
Note:

- The template content must contain {{1}} to correctly transmit the content.
- This template name needs to be recorded for subsequent use in DIAScreen.

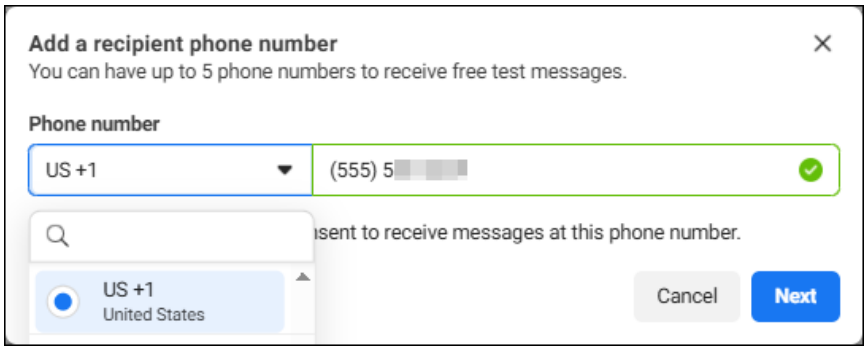
f. Click **Submit for review**.

6. Add the recipient's phone number.

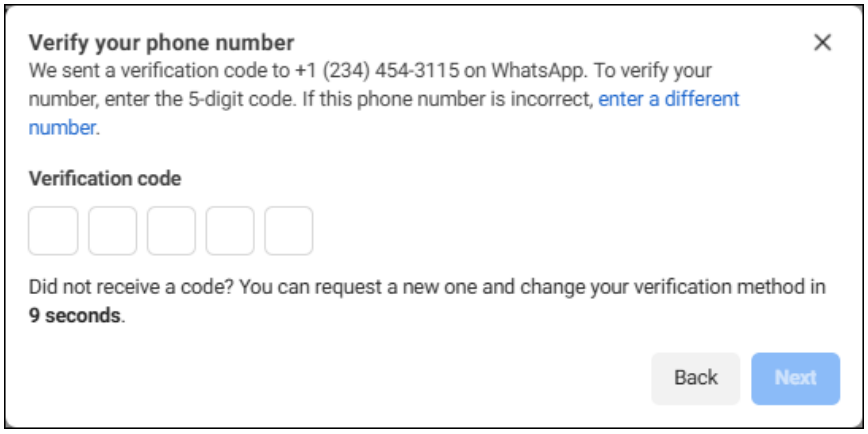
a. In the **To** drop-down menu, click **Manage phone number list**.



b. Select the country code from the **Phone number** drop-down menu, enter phone number, and click **Next**.



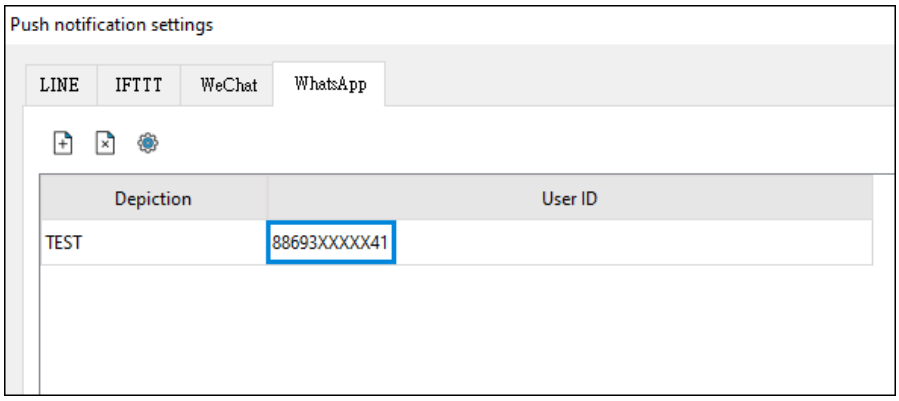
c. After adding, you will be asked to verify the phone number. After verification, record the number for subsequent use in DIAScreen.



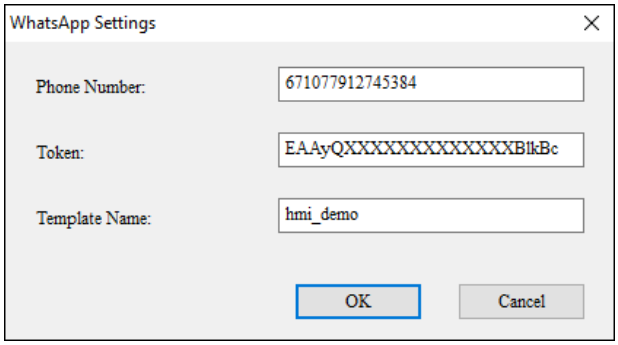
7. Create a project in DIAScreen.

a. After the project is created, go to **IIoT** > **Notification**.

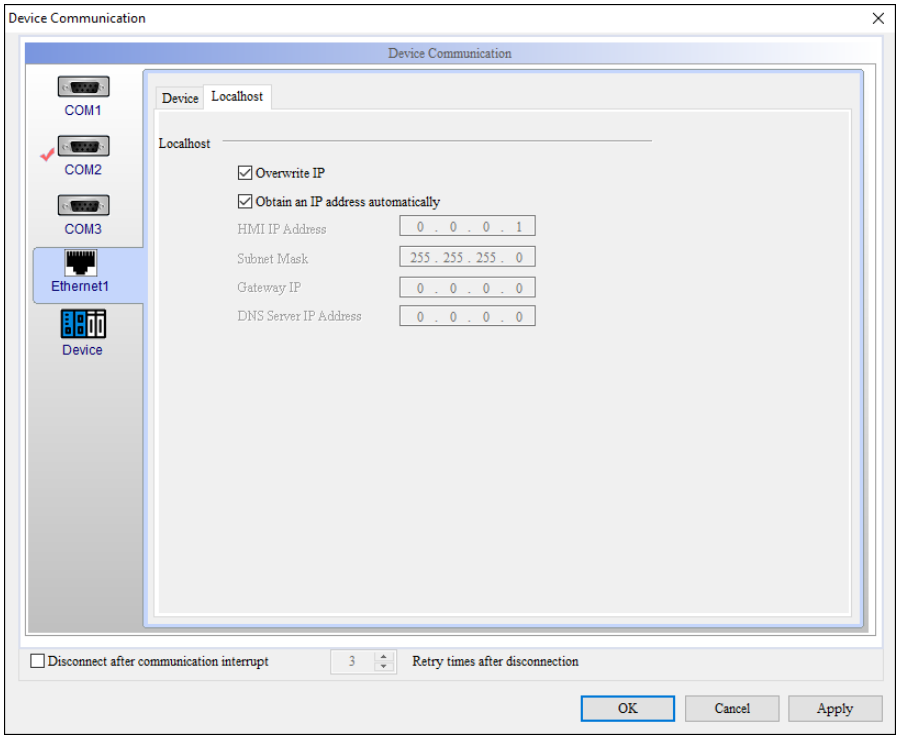
b. In the **WhatsApp** tab of the **Push notification settings** dialog, click , and then enter the phone number you just set in **User ID** field (without +) and enter the **Description** according to your need.



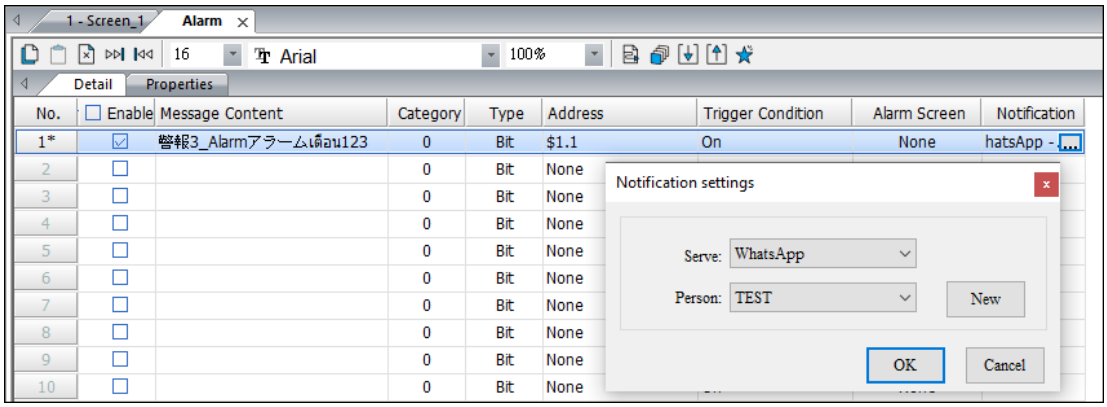
c. Click , and then enter **Phone Number**, **Token** and **Template Name** in the **WhatsApp Settings** dialog.



d. In the **Ethernet 1** page of the **Device Communication** dialog, select **Overwrite IP** and **Obtain IP address automatically** checkboxes in the **Localhost** tab.

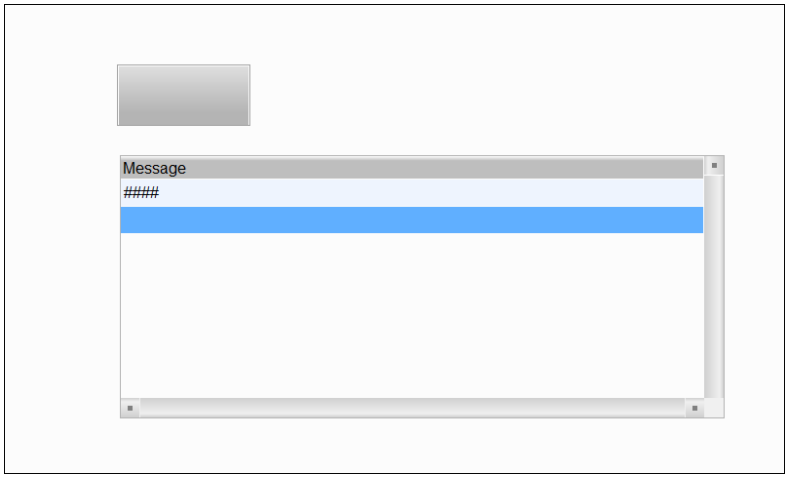


e. Go to **Data Management** > **Alarm Settings**, set up alarm in the **Detail** tab of the **Alarm** page, and then set **Notification**.

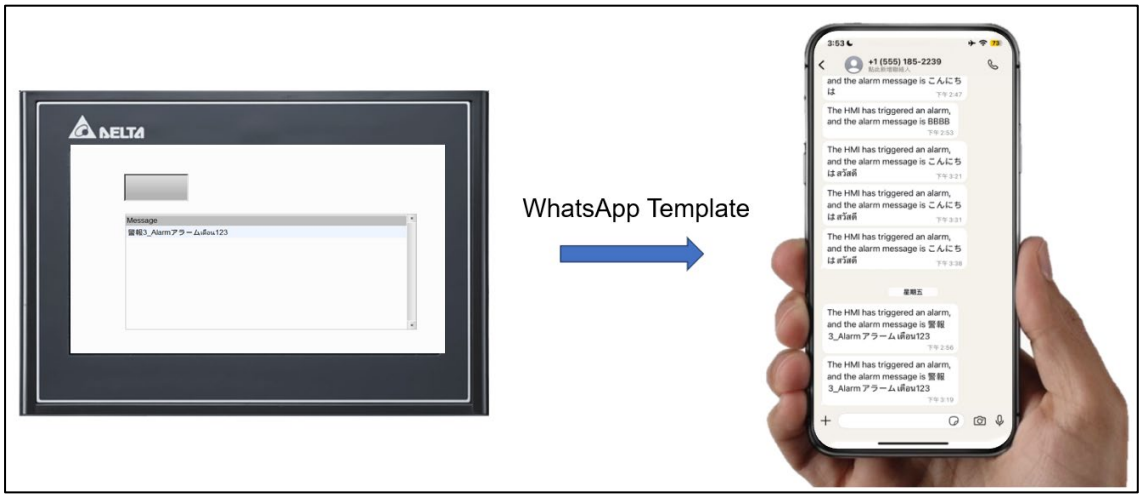


f. Create an **Alarm History Table** element and a **Maintained Button** element on the screen.

g. Set up **Alarm History Table** element and set **Write Address** of the **Maintained Button** element to $1.0 (for triggering the alarm).



8. After completing the above settings, download the screen to the HMI and connect the HMI to the external network. After the alarm is triggered, you can view the corresponding alarm message on WhatsApp.



Note: Alarm messages support multiple languages.

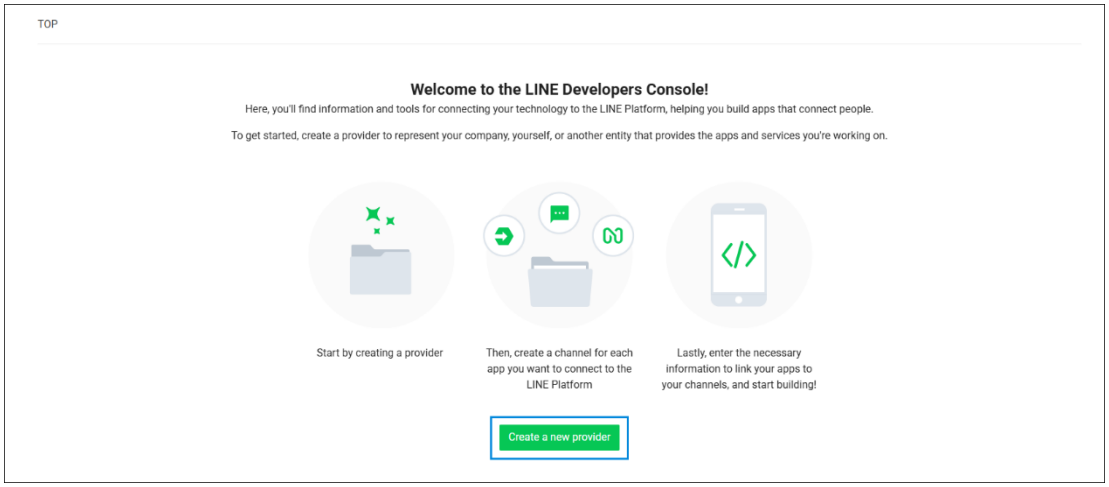
Line Push Notification

This section describes how to send an alarm message to the LINE App through the push notification function when an alarm occurs on the HMI.

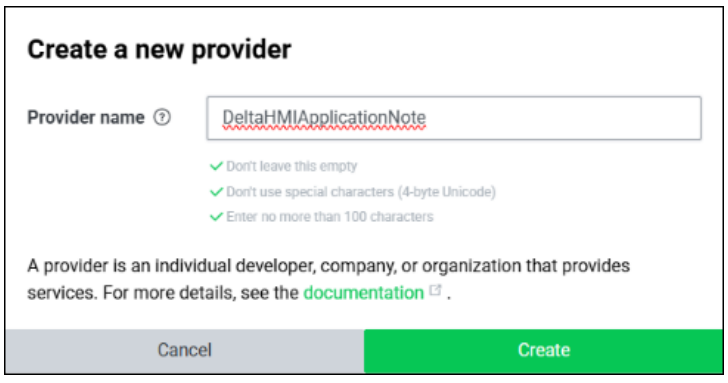
Follow these steps to set up the Line push notification function.

Set up Line push notification function

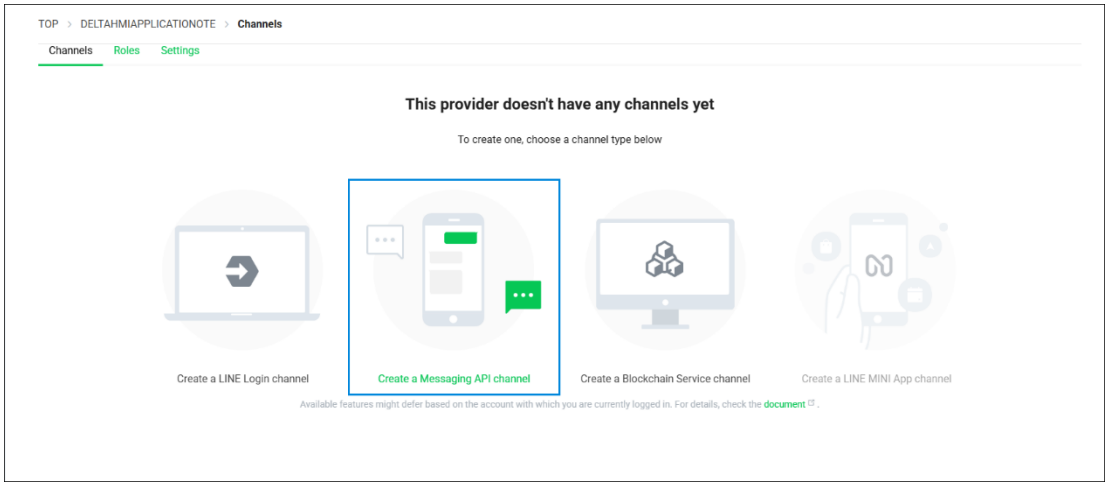
1. Go to developers.line.biz/console/ and log in to the LINE official account with your existing LINE account.
2. Create a provider.
   a. Click **Create a new provider** in **LINE Developer**.



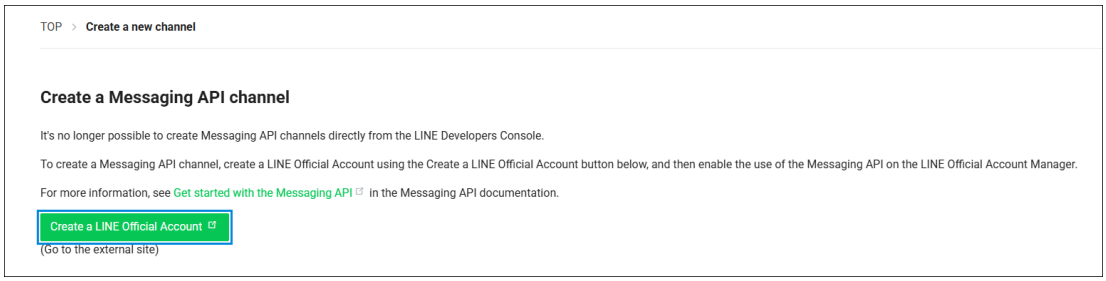
   b. Enter the Provider name in the **Create a new provider** dialog, create **Create**.



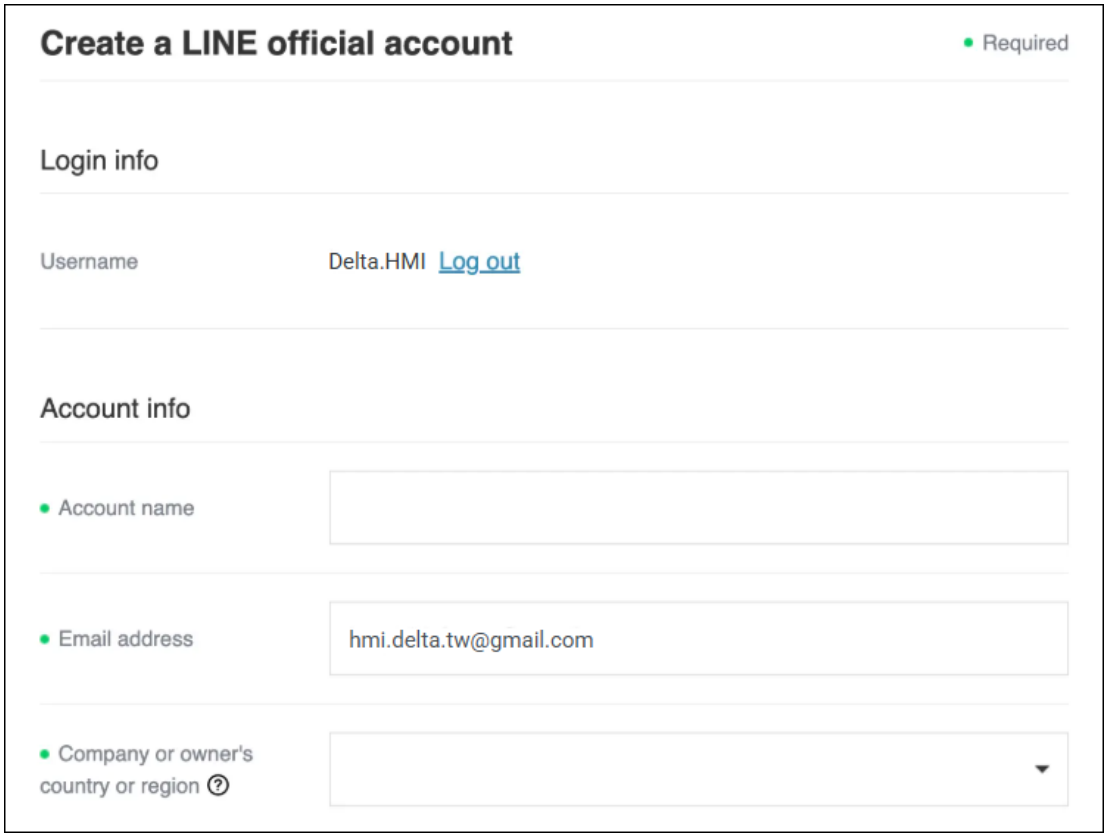
   c. Click **Create a Messaging API channel**.

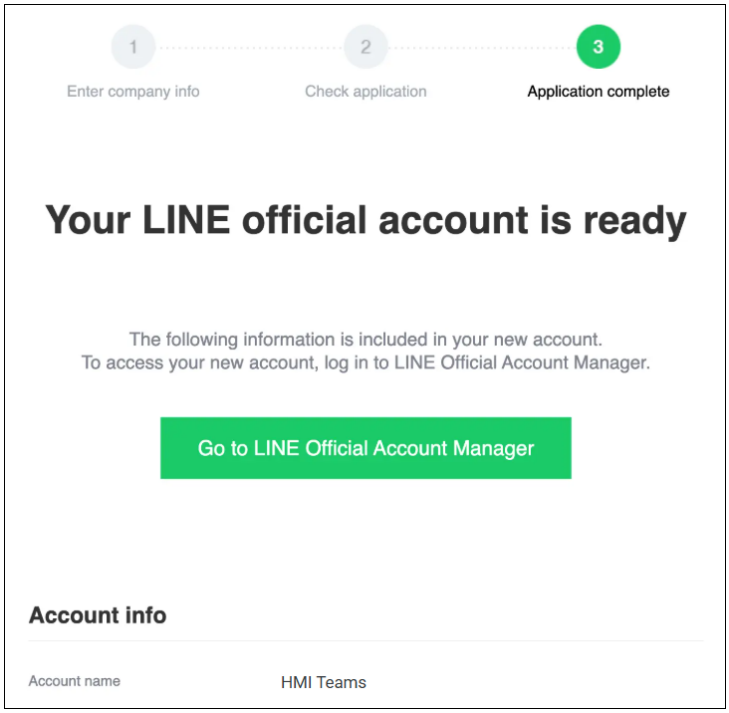


d. Click **Create a LINE Official Account**.



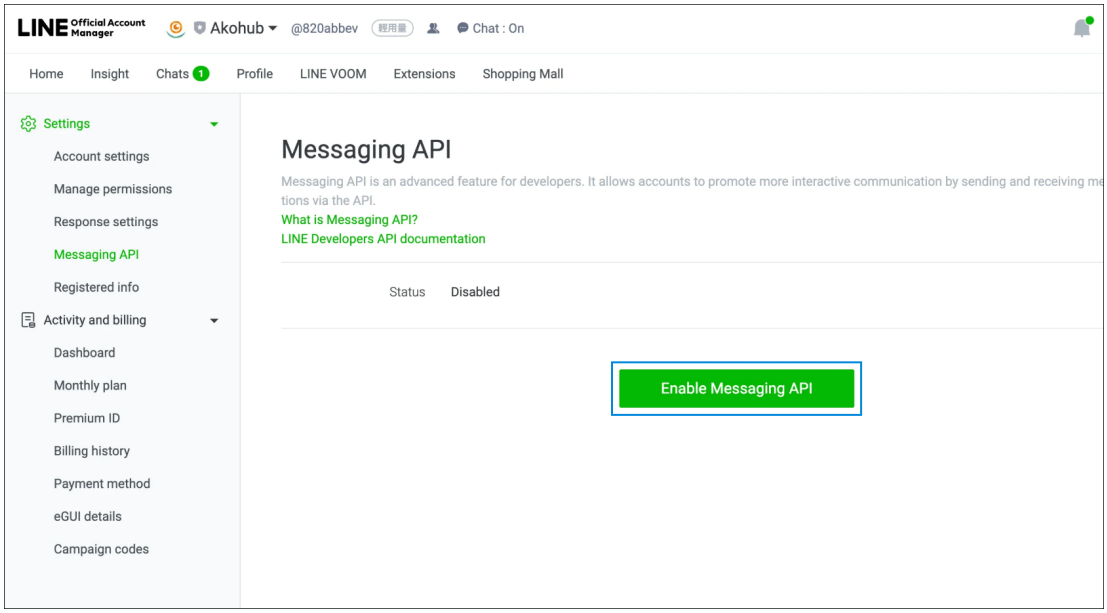
e. Enter relevant information.



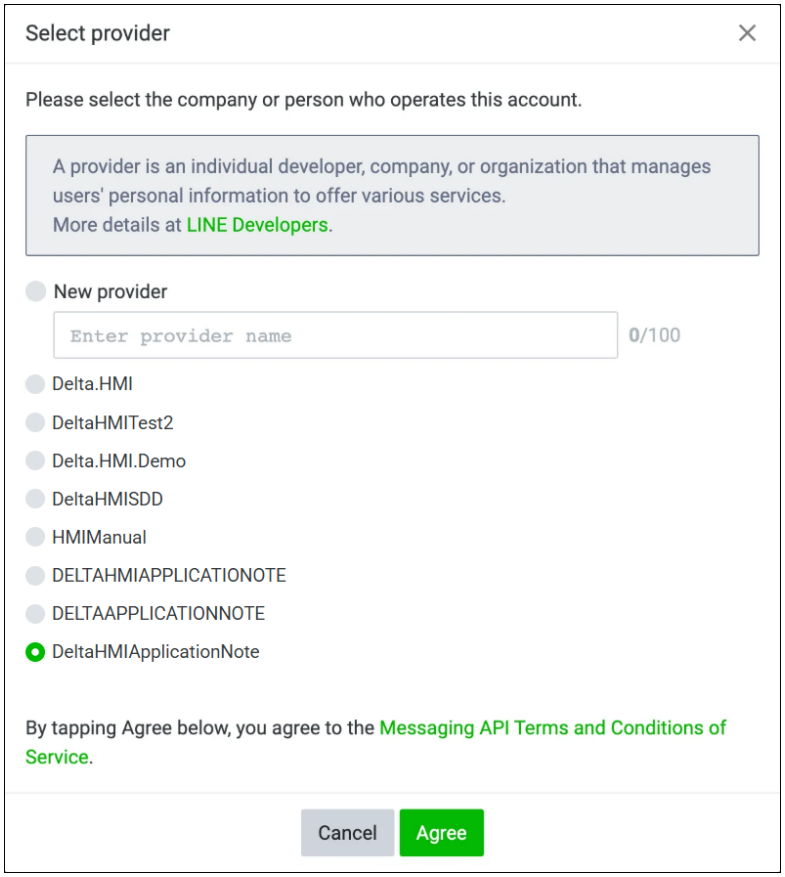


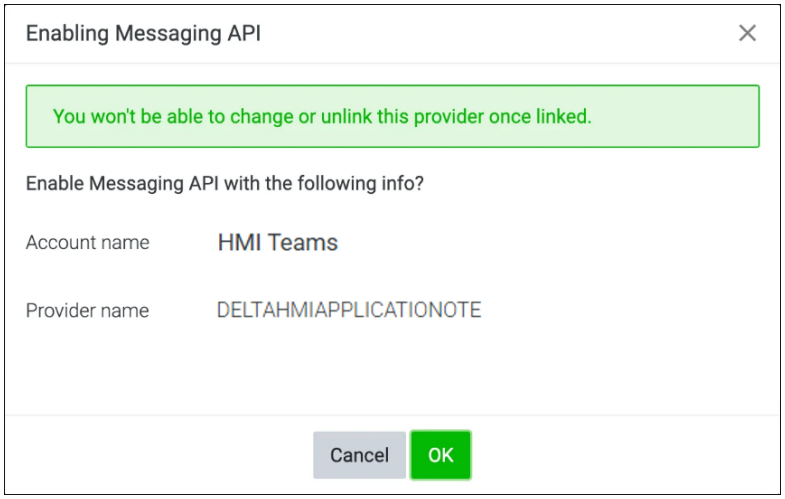
3. Enable Messaging API.

   a. After the account is created, the relevant management page automatically opens, and then click **Enable Messaging API**.

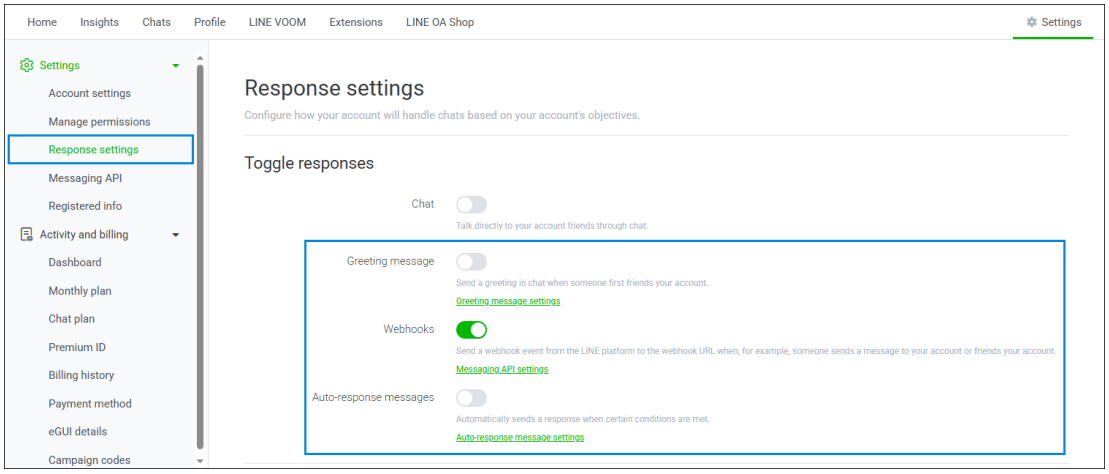


   b. Select the LINE Provider you just created, and then click **Agree** and **OK** in sequence.

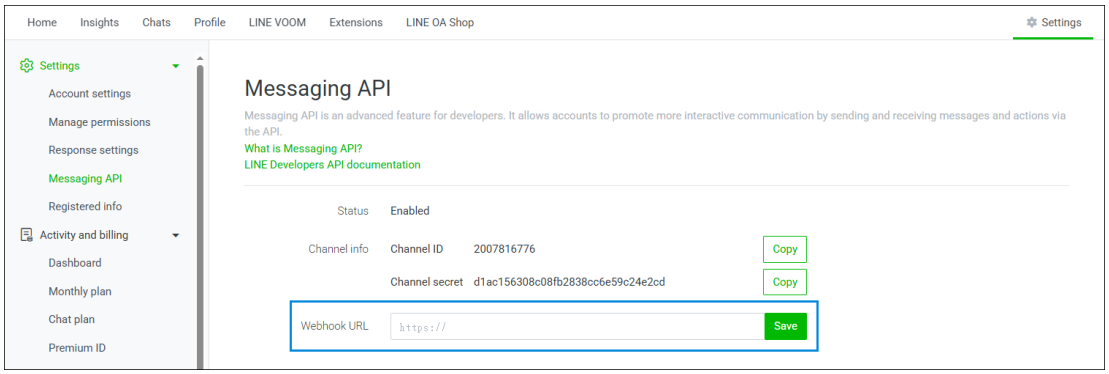




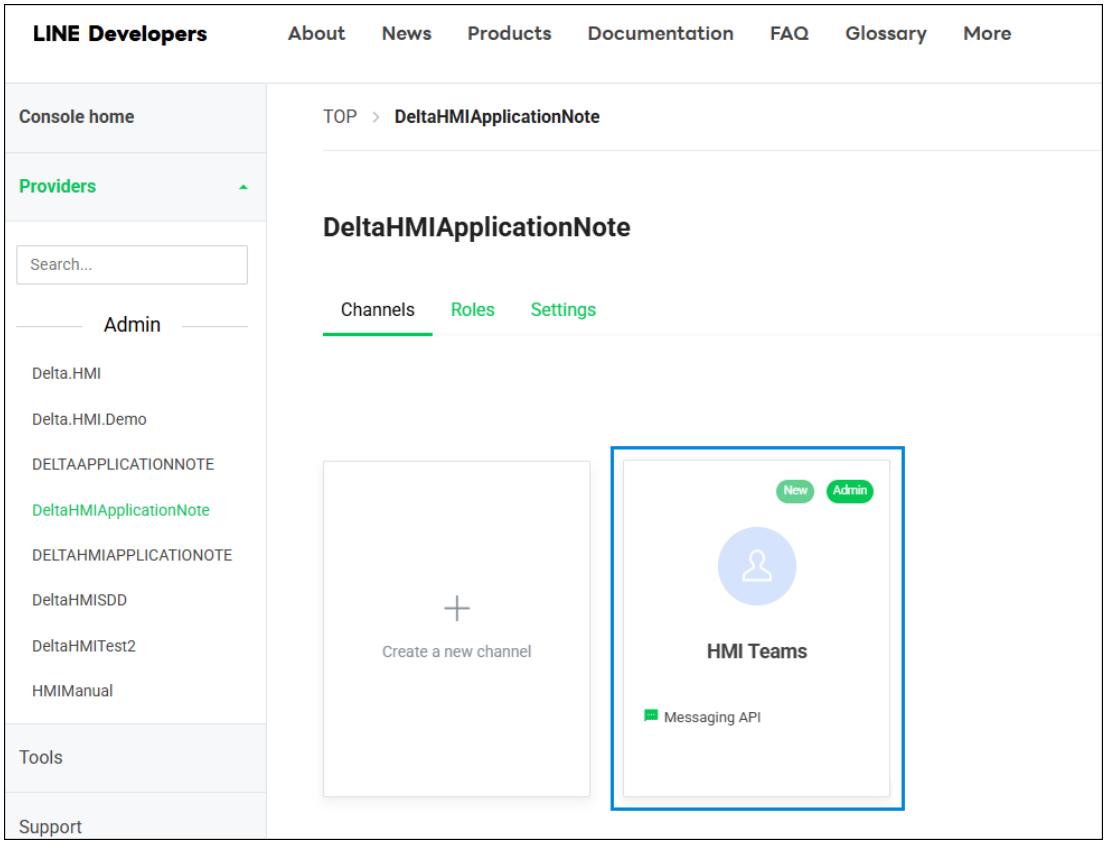
c. Go to **Response settings** tab and disable **Greeting message** and **Auto-response messages**, and then enable **Webhooks**.



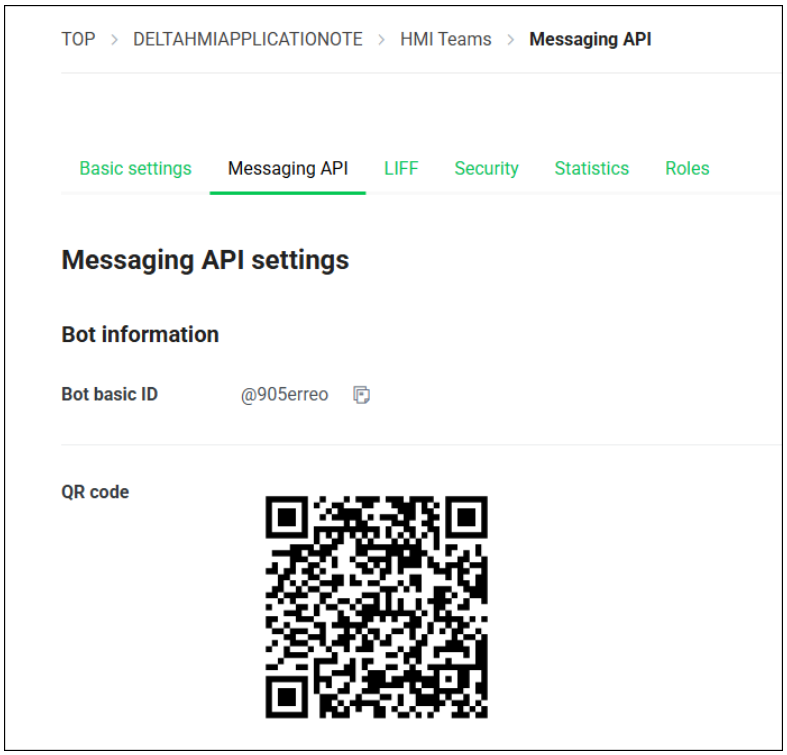
Note: To enable Webhooks, set the **Webhook URL** in the **Messaging API** tab.

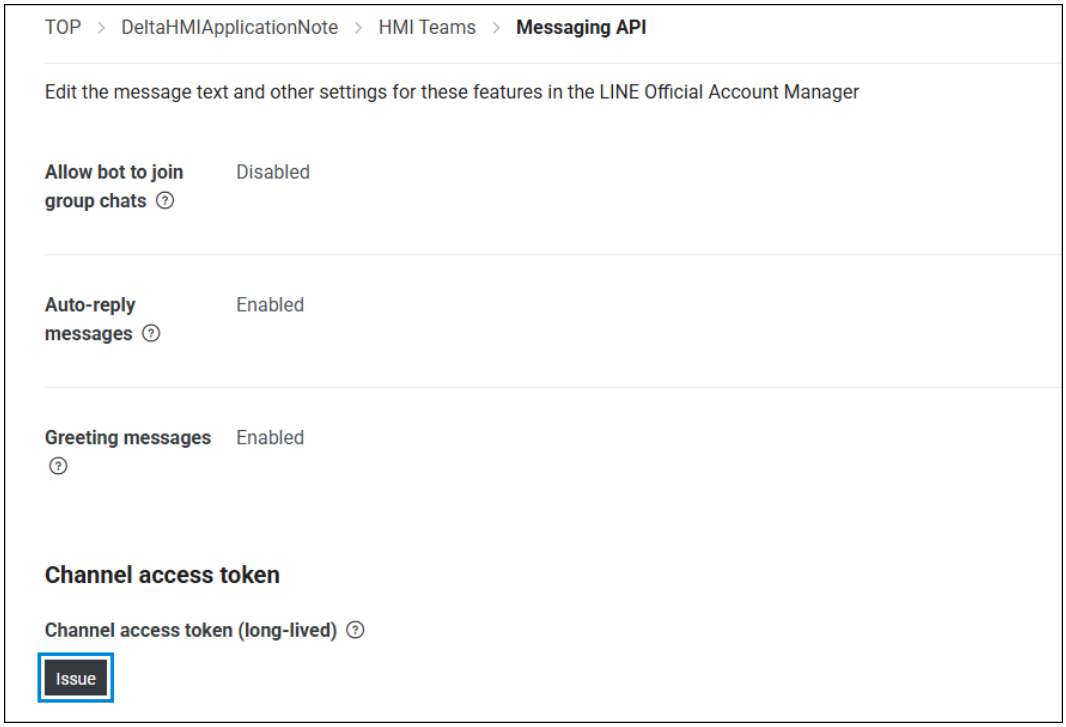


a. Go ahead and return to **LINE Developers** and click the channel you just created.



b. You can find the QR Code (for users to join the official account) and the **Channel access token** in the **Messaging API** tab, and then click **Issue** to issue the token.





c. Copy the **Channel access token** and paste it into the **Webhook URL** field using the following URL format.

URL format:

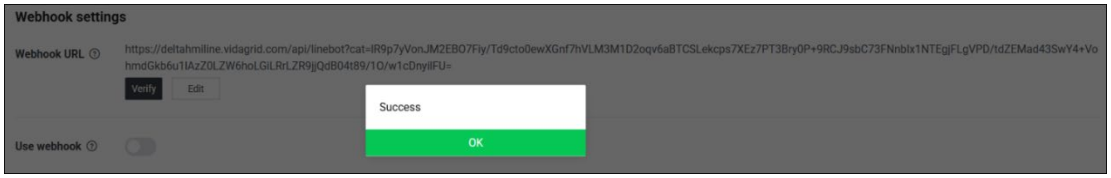
https://deltahmiline.vidagrid.com/api/linebot?cat= + [Channel access token]

Example URL:

https://deltahmiline.vidagrid.com/api/linebot?cat=E1N0ZVy4vLaWZHaly25ABXF7IJLQhVLUkECjaRFkaL8RDjBuObxwWLVhqEVckQJ4wIzyXHIMq1jTMrXDyPSIB0vIPple3ji5cMI54YUmdOCx64icjhEO6SEgiH68s5XMD5Ps4FQNdIP05Mc75tmEYAdB04t89/1O/w1cDnyiIFU=

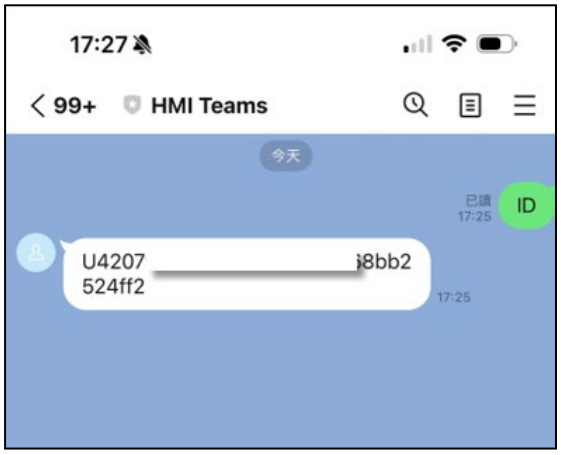
d. Also paste this URL into the **Webhook URL** field in **LINE Official Account Manager** and click **Update**.

e. Click **Verify** to verify the Webhook URL, and a **Success** message displays after successful verification.

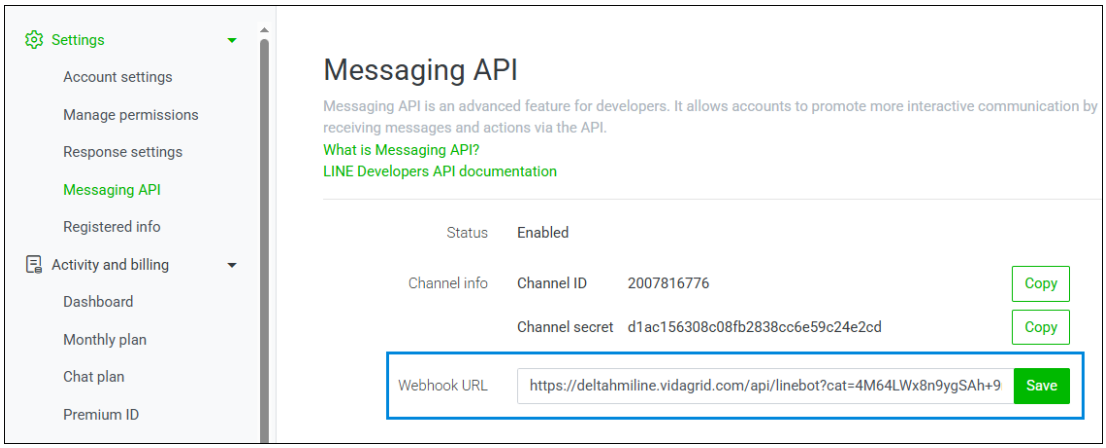


f. Enable **Use webhook**.

g. Use the QR code to add LINE friends and send a message to obtain the ID.



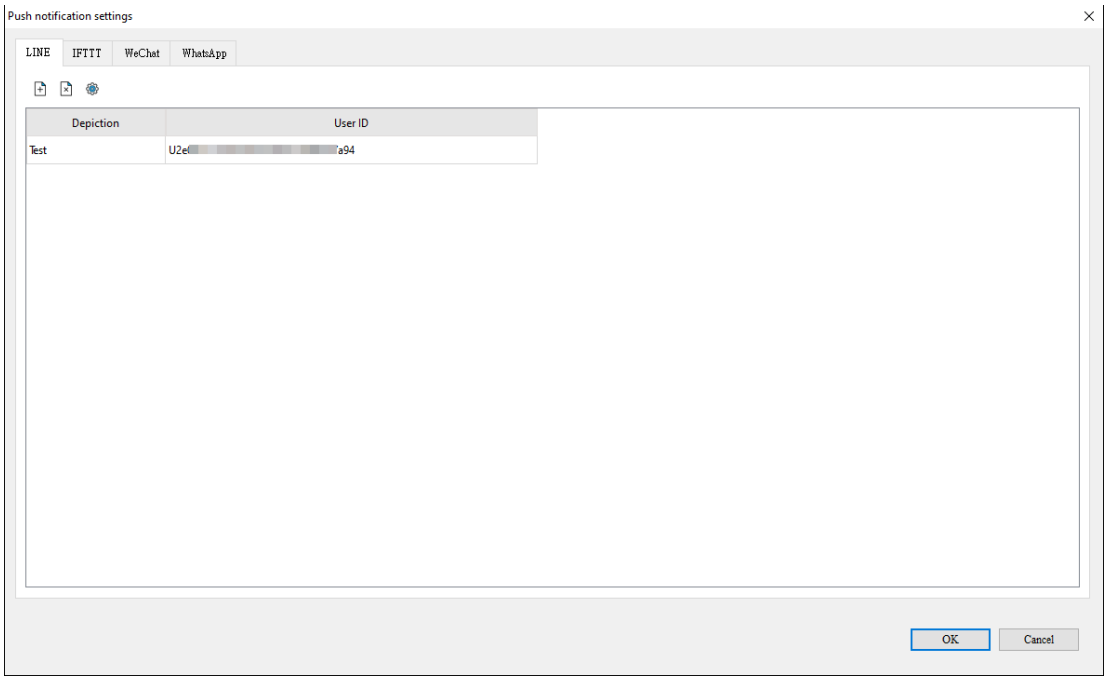
Note: If the ID cannot be obtained, click **Reissue** to reissue the **Channel access token** and reedit the **Webhook URL**.



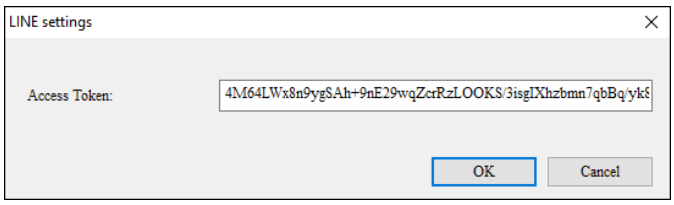
4. Create a project in DIAScreen.

a. After the project is created, go to **IIoT** > **Notification**.

b. Click to add a user in the **LINE** tab of the **Push notification settings** dialog, and then enter the ID obtained from the LINE chatroom in the **User ID** field and enter the **Description** according to your need.



c. Click and enter the **Access Token** (Channel access token) in the **LINE settings** dialog.

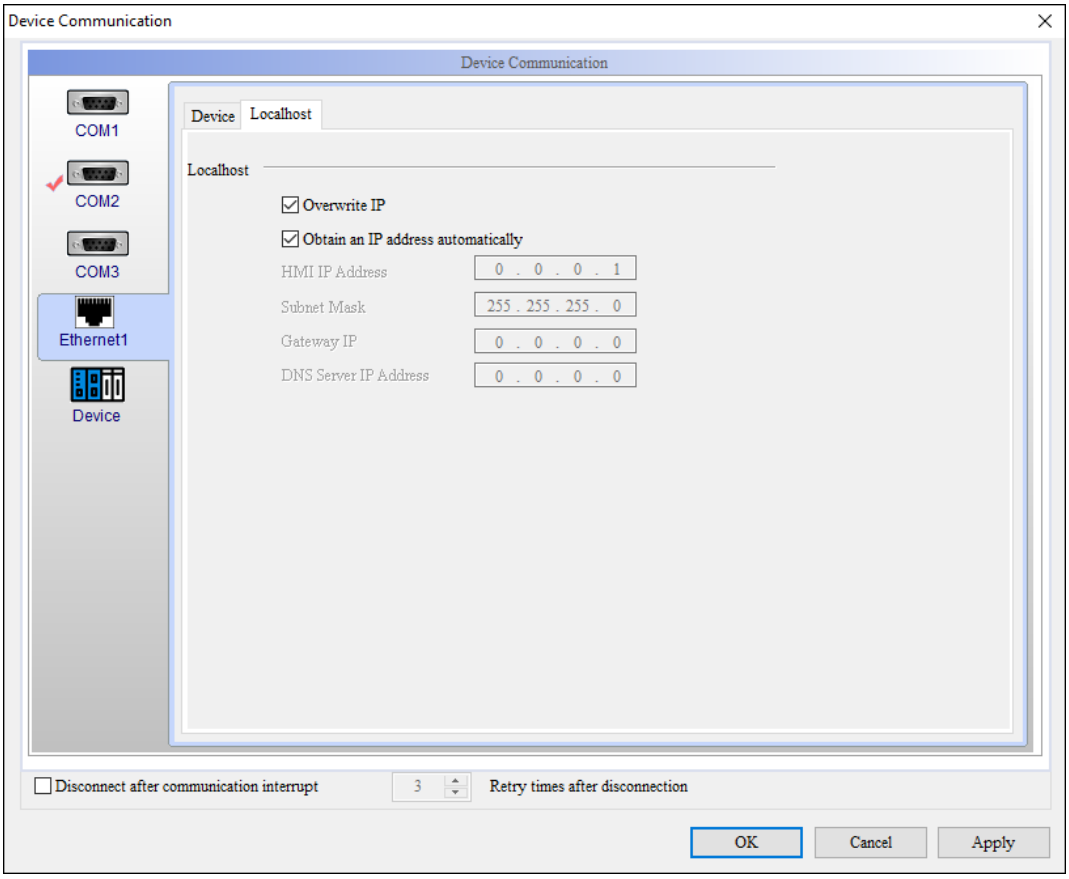


5. Connect to a network.

- **DOP-100 series models (without Wi-Fi Dongle)**

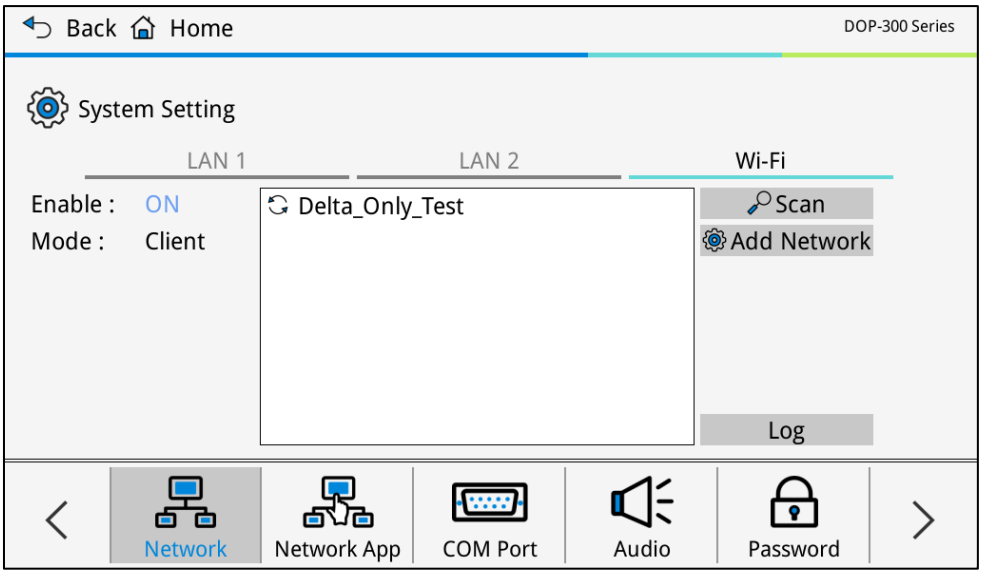
  a. In DIAScreen, go to **General** > **Device Communication**.

  b. In the **Ethernet 1** page of the **Device Communication** dialog, select **Overwrite IP** and **Obtain an IP address automatically** checkboxes in the **Localhost** tab.

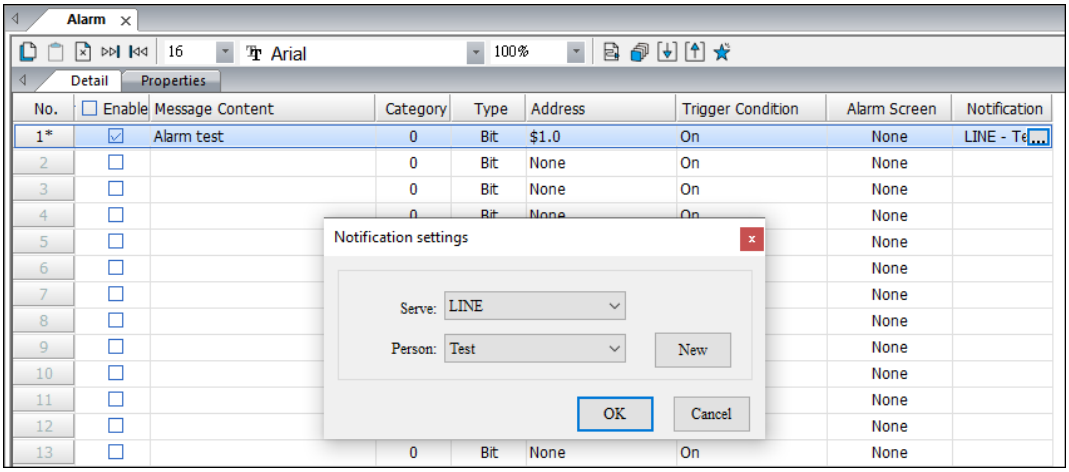


- **DOP-300 series models (Wi-Fi Dongle is connected)**

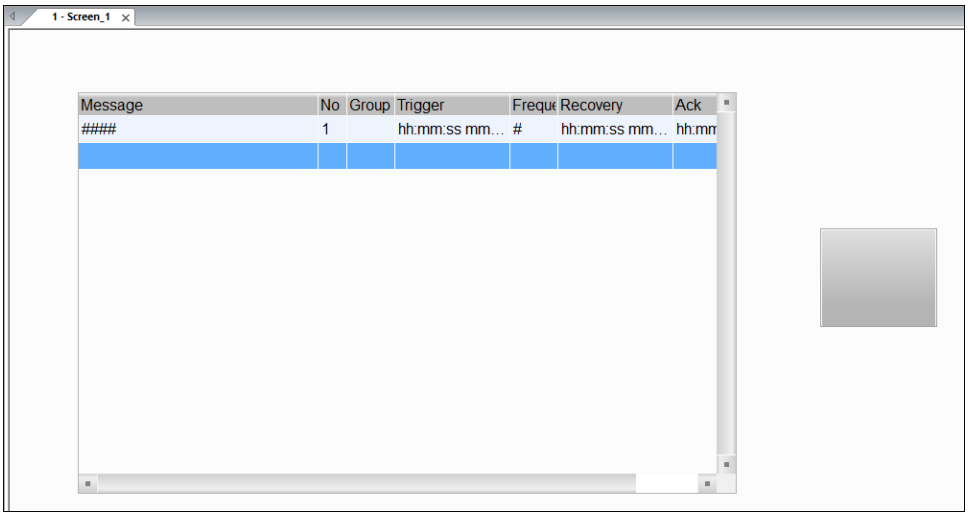
  a. On the HMI screen, go to **System Menu** > **System Setting** > **Network** > **Wi-Fi**, and connect to Wi-Fi.



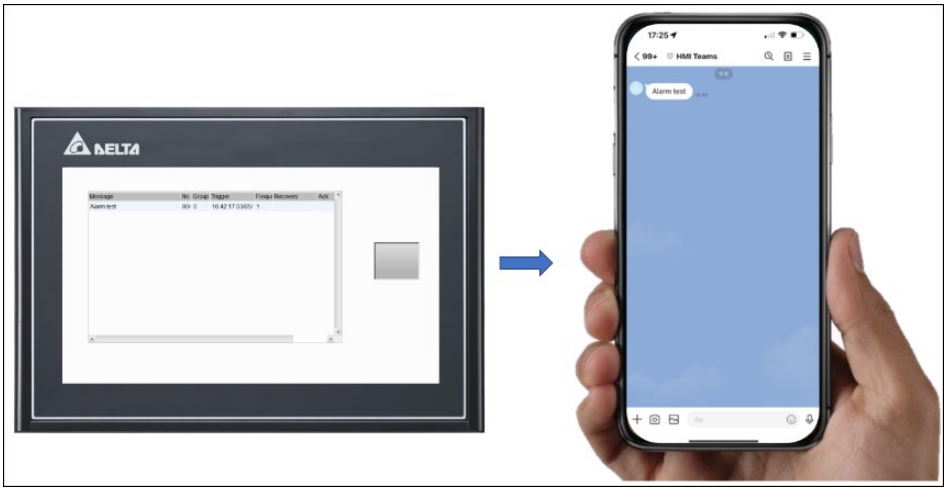
  b. Go to **Data Management** > **Alarm Settings**, set up alarm in the **Detail** tab of the **Alarm** page, and then set **Notification**.



c. Create **Alarm History Table** and **Maintained button** elements on the screen, and then set the **Write Address** to **$1.0**.



6. After completing the above settings, download the screen to the HMI and connect the HMI to the external network. After the alarm is triggered, you can view the corresponding alarm message on LINE App.



History Buffer

The history buffer records specific data in HMI or external storage device, through the buffer attribute to set the sampling data, trigger mode, and file output content, after the buffer data is established. The sampling data can be displayed on the HMI through the sampling element.

History Buffer Settings

How to open the **History Buffer** settings page?

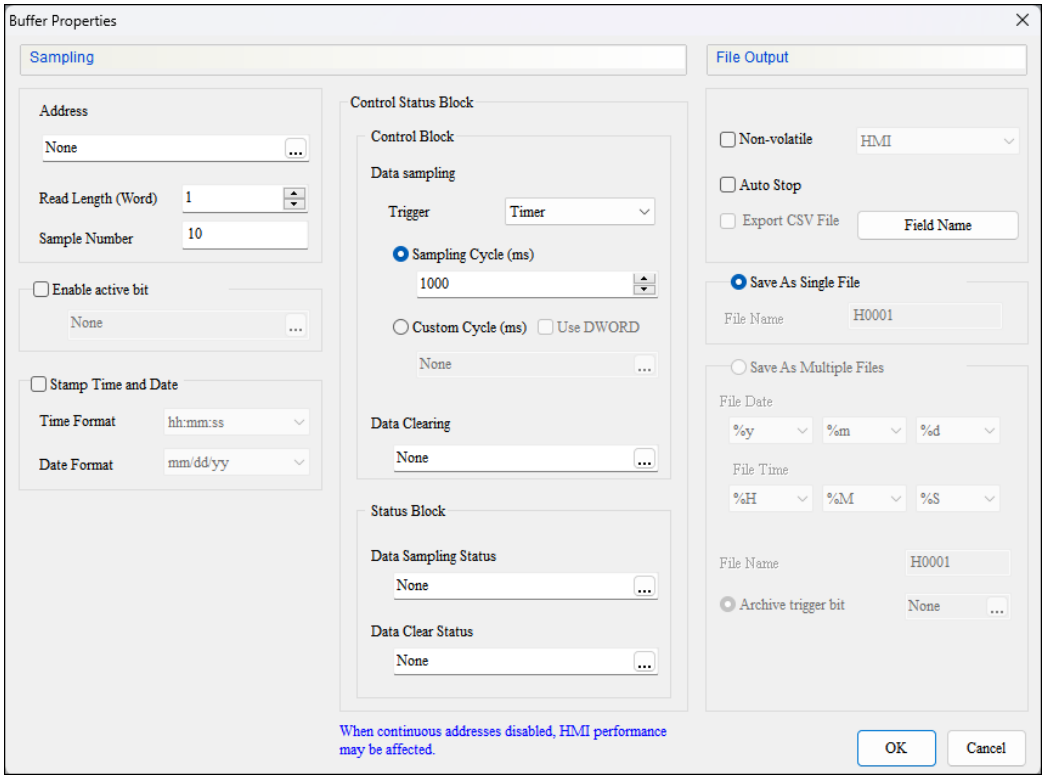
- In **Project** pane, double-click **History Buffer**.

Or

- On the toolbar, click **Data Management** > **History Buffer**.

On the **History Buffer** settings page, click the to open **Buffer Properties** dialog.

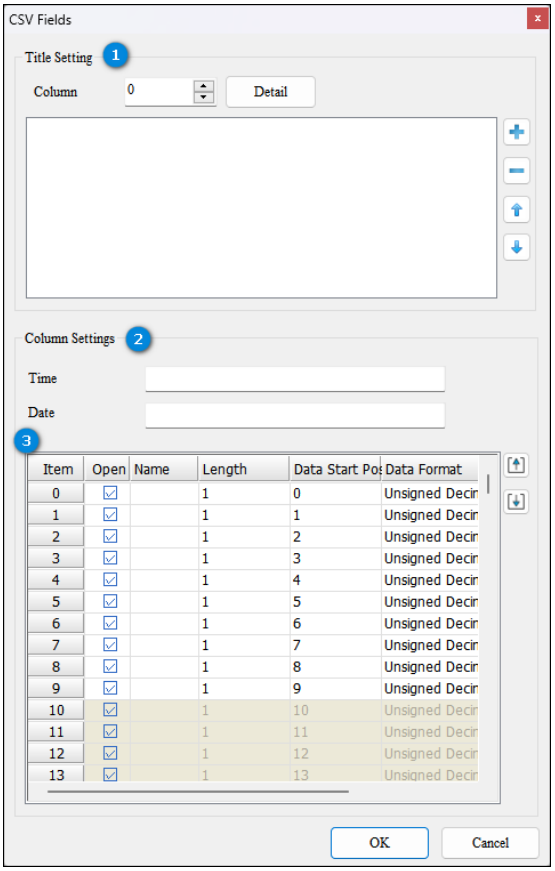
The following table lists the settings in the **Buffer Properties** dialog with their description.



Settings	Description
Address	Select either the internal memory or the controller register address.
Read Length (Word)	Select or enter the number of words that can be sampled. The maximum value is 256.
Sample Number	Enter the number of historical data to be recorded. Up to 9,999,999 sampling points can be This function needs to be used with the **Auto Stop** function. If the **Auto Stop** checkbox is selected, the recording stops automatically when it reaches the set sample number. If the **Auto Stop** checkbox is not selected, the recording does not stop when it reaches the set sample number; instead, the recording restarts from the first record and overwrites the previous data.
Enable active bit	Select to set the active bit. If the **Trigger** is set to **Timer** for **Data sampling**, the HMI immediately records the sampling data as soon as it starts up. The Enable active bit function allows you to determine when to start sampling. When the active bit is On, it starts to sample; when the active bit is Off, it stops sampling.
Stamp Time and Date	Select to set the time format and date format.

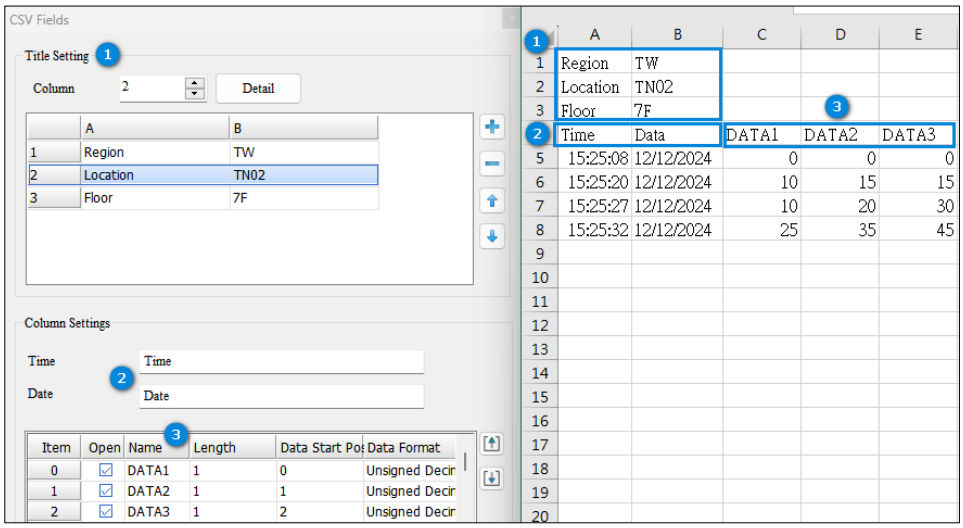
Trigger	Select the trigger source.
	• With **Timer** as the trigger source, you can choose between **Sampling Cycle** and **Custom Cycle**.
	  ○ **Sampling Cycle**: The range value is 100~86400000.
	  ○ **Custom Cycle**: The cycle time of sampling can be dynamically set on HMI through the specified memory address. The maximum value can be set to 32767. If **Use DWORD** is selected, the maximum value can be set to 2147483647.
	• With **Memory Address** as the trigger source, you can configure the bit address to trigger data capture. When the bit is On, it will be sampled.

Data Clearing	Set the address of data clearing as a bit address. When the trigger address is On, the buffer data is cleared.
Status Block	Set the status block address as a bit address to read the sampling / clearing status. When the address is On, data sampling / clearing is performed.
Non-volatile	Check and select the storage location of data. The storage location varies depending on the external device supported by the model, including HMI, USB Disk, and SD. If HMI is selected, the data is recorded in the HMI SRAM when the power is cut off.

<table>
<tr><th>Settings</th><th>Description</th></tr>
<tr><td>Auto Stop</td><td>If selected, the recording stops automatically when the set Sample Number is reached.
Note: Refer to the instructions of Sample Number.</td></tr>
<tr><td>Export CSV File</td><td>Select to export historical data to a CSV file, and you can set the Field Name for a CSV file.
Note: This setting is only available when Non-volatile is selected.
The following table lists the settings in the CSV Fields dialog with their description.
 

<table>
<tr><th>Legend</th><th>Setting</th><th>Description</th></tr>
<tr><td>1</td><td>Title Setting</td><td>Follow these steps to set the title.
1. Title column: Select or enter the Column and click Detail to display the corresponding columns.
Note: Support up to 20 columns.
2. Title row: Click the buttons on the right to add, delete or move the title row.
Note: Support up to 10 rows.
3. Enter the content of the displayed title in the title field.</td></tr>
<tr><td>2</td><td>Time and</td><td>Enter the time and date.</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>Settings</th><th colspan="3">Description</th></tr>
<tr><td></td><td></td><td>Date</td><td></td></tr>
<tr><td></td><td>❸</td><td>Column Settings</td><td>Set for the column item.
The column items are displayed according to the Read Length (Word) setting. You can choose whether to Open the item or not, and set the data style to be presented on the CSV file, including the title name, data format, and integer digits/fractional digits.
You can also set the Data Start Position and Length according to the read data. Different Length supports different data formats. The following table lists the valid range value corresponding to the data format.
<table>
<tr><th colspan="2">Length is 1</th></tr>
<tr><th>Data Format</th><th>Valid Range Value</th></tr>
<tr><td>BCD</td><td>0~9999</td></tr>
<tr><td>Signed BCD</td><td>–999~9999</td></tr>
<tr><td>Signed Decimal</td><td>–32768~32767</td></tr>
<tr><td>Unsigned Decimal</td><td>0~65535</td></tr>
<tr><td>Hex</td><td>0~0xFFFF</td></tr>
<tr><td>Char</td><td>2 words</td></tr>
<tr><th colspan="2">Length is 2</th></tr>
<tr><th>Data format</th><th>Valid Range Value</th></tr>
<tr><td>BCD</td><td>0~99999999</td></tr>
<tr><td>Signed BCD</td><td>–9999999~99999999</td></tr>
<tr><td>Signed Decimal</td><td>–2147483648~2147483647</td></tr>
<tr><td>Unsigned Decimal</td><td>0~4294697295</td></tr>
<tr><td>Hex</td><td>0~0xFFFFFFFF</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>Settings</th><th>Description</th></tr>
<tr><td></td><td>
<table>
<tr><td></td><td></td><td>
<table>
<tr><td>Char</td><td>4 words</td></tr>
<tr><td>Floating</td><td>0~9999999</td></tr>
</table>
Use Export / Import buttons on the right to export the data as an XLSX file. After editing, you can import it into the field settings.
</td></tr>
</table>
The field name of the CSV file and the corresponding output result, as shown in the following figure.


Note: If **Non-volatile** is not enabled, it can be used with the LUA command *history.ExportCSV* to export a CSV file.</td></tr>
<tr><td>File storage</td><td>Select to save the history butter data as single file or multiple file.

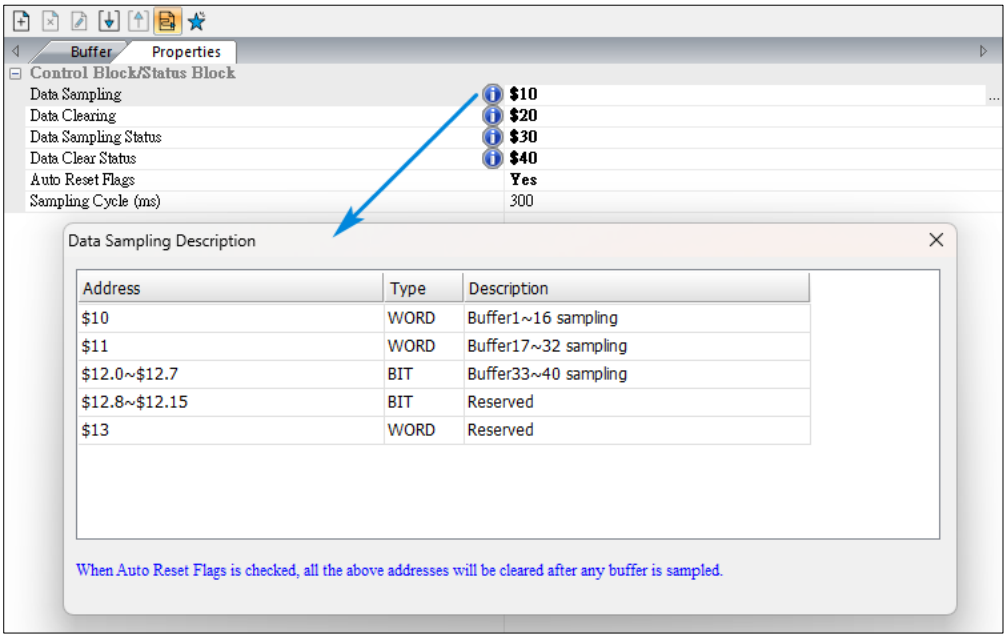
Save As Single File: All data is saved in a single file. You can define the file name (including English and numeric), and the maximum length is 8 characters.
Save As Multiple Files: When the **Sample Number** record of the historical buffer data is full, or the **Archive trigger bit** is triggered, the current data will be saved and the buffer data will be cleared.
The file name format of the multiple files is **File Name_Date_Time**.

Note: **Save As Multiple Files** and **Auto Stop** cannot be used at the same time. To enable **Auto Stop**, select **Save As Single File** option.</td></tr>
</table>

The following table lists the functions on the toolbar in the History Buffer settings page.

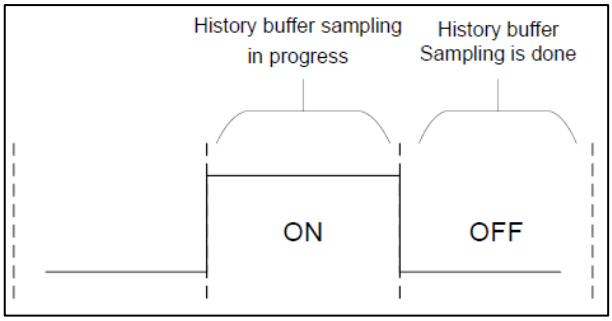
Function	Description
	Click to adds buffers. Up to 40 sets of buffers can be added.
	Click to delete the selected buffer data.
	Click to open the selected buffer properties dialog to modify the buffer properties.
	Click to export buffer data as an XLSX file.
	Click to import buffer data after modification.
	Click to enable / disable the **Continuous historical buffer control command address** function. • **Enabled**: Use continuous address to control the sampling or clearing of data in each buffer. It is set on the **Properties** tab. Refer to the next paragraph. • **Disabled**: Set the respective control command addresses in the **Buffer Properties** dialog. **Note**: If continuous address function is not enabled, it may affect the performance of the HMI.
	Enabling optimization is recommended when the set addresses of multiple buffers are not continuous. If multiple buffer addresses are already continuous, enabling this function may result in the slow read speed.

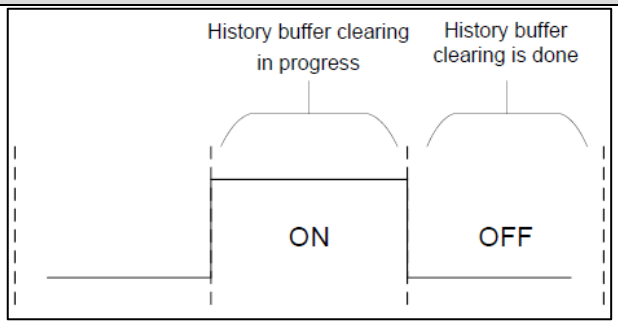
In the **Properties** tab, you can set the control bit of the data sampling or clearing of the history buffer. It controls the actions of each buffer with the set starting address. Take the **Data Sampling** as an example, set the starting address to $10. You can click to query the buffer number of each sampling bit control or view the actual control bits of that buffer in the properties dialog of each buffer.



Note: The **continuous historical buffer control command address** function needs to be enabled to set the Control Block / Status Block address.

The following table lists the settings in the **Properties** tab with their description.

Setting	Description
Data Sampling	Controls the buffer sampling flag. When the control bit is ON, a sampling is performed. **Note**: The **Trigger** of the buffer needs to be **Memory Address**.
Data Clearing	Clears the buffer data flags, including data for sampling elements and non-volatile records.
Data Sampling Status	When the HMI samples the history buffer, it turns the corresponding history buffer sampling status flag ON. After the sampling is complete, the history buffer sampling status flag is turned OFF in real time. 
Data Clear status	When the HMI clears the history buffer, it turns the corresponding history buffer clear status flag ON. After the clearing is complete, the history buffer clear status flag is turned OFF in real time.

Setting	Description
	
Auto Reset Flags	Set whether to automatically clear flags. When **Auto Cleared Flag** is set to **Yes**, all data sampling or data clearing addresses will be cleared to 0 after sampling or clearing any buffer.
Sampling Cycle (ms)	Set the interval at which the sampling is performed. The default value is 300, and the range value is 100~1000.

Sampling Function

Displays the sampling elements of historical buffer data. The following are sampling elements.

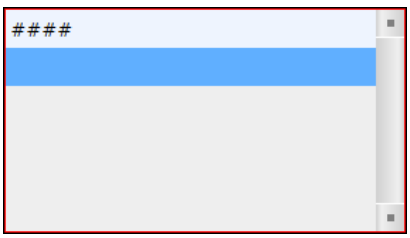
- **Historical Trend Graph**: Displays the sampled data over a period on the element in the form of a trend chart. A maximum of 60 curves can be displayed.
- **Historical Data Table**: Lists the values sampled from the history buffer on the element. A maximum of 60 fields can be displayed, corresponding to 60 curves in the historical trend chart.
- **Historical Event Table**: Converts the sampled values into information and presents them on the element in Word or LSB format.
- **Historical Overview Table**: Draws the historical data stored on an external storage device (USB disk or SD card) on the element in the form of a trend chart.
- **Circular Trend**: Draws the sampled data over a period as a circular trend chart using the polar coordinate system. A maximum of 60 curves can be displayed.

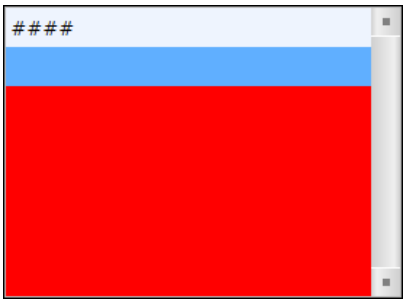
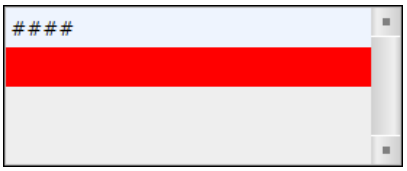
  Note: There can be only one **Circular Trend** element per screen.

Common Properties

- **Style**

  The following table lists the common style properties of **Historical Data Table** and **Historical Event Table**.

Setting	Description
Border Color	Set the border color of the element. The default color is gray. When set to red, it displays as shown in the following figure. 

Setting	Description
Background Color	Set the background color of the element. The default color is white. When set to red, it displays as shown in the following figure. 
Row Color	Set the color of the odd number of rows. The default color is light blue. When set to red, it displays as shown in the following figure.
Alternating Row Color	Set the color of the even number rows. The default color is blue. When set to red, it displays as shown in the following figure. 
Select Row Color	Set the color of the select row to be displayed after clicking the element when executing on the HMI.
Transparency	Select or enter the transparency value of the element. The range value is 50~255. The smaller the value, the higher the transparency of the element.

- **Function Buttons**

The function buttons control the display of charts on the HMI.

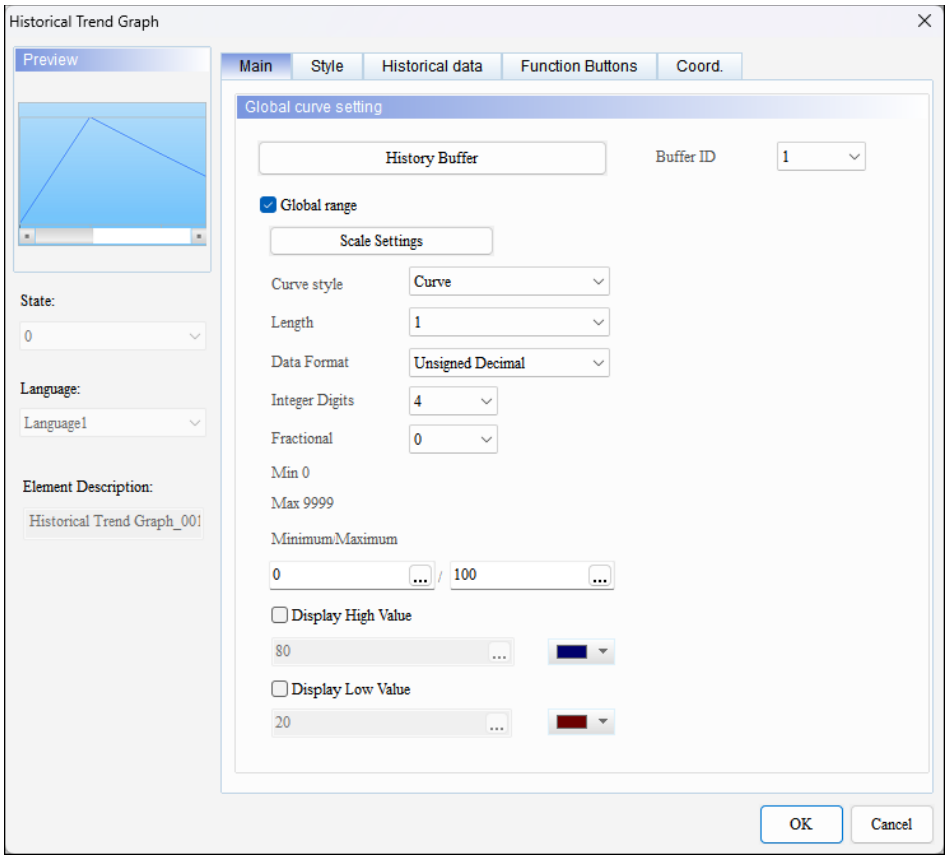
Setting	Description
Zoom In	Select to zoom in or out historical data on the X-axis on the trend chart. **Note**: This function is used for **Historical Trend Graph**.
Zoom Out	
Zoom Reset	Select to reset the X-axis to the default style. **Note**: This function is used for **Historical Trend Graph**.

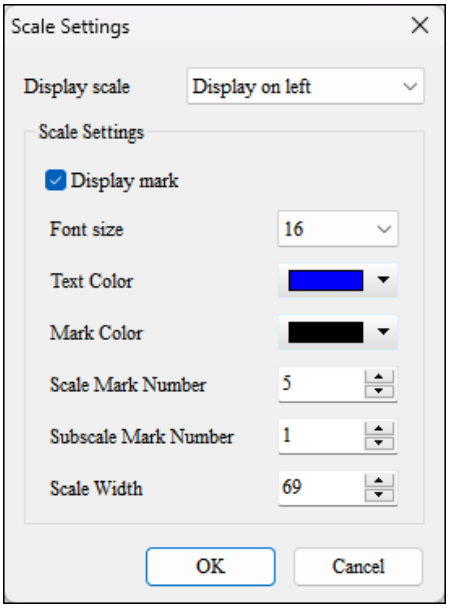
Setting	Description
Scroll	Configure the scroll direction and interval according to the characteristics of the element. The trend chart can be scrolled left and right, and the data table can be scrolled up and down.
Set the time display interval	Displays the corresponding data according to the set time interval.
Set as the default description	Click to fill in the function description as default strings.
Default button width / height	Select or enter the height and width of the function buttons. After changing the height or width, click **Rearrange Function Buttons** in the element context menu to update the button size.

Historical Trend Graph Properties

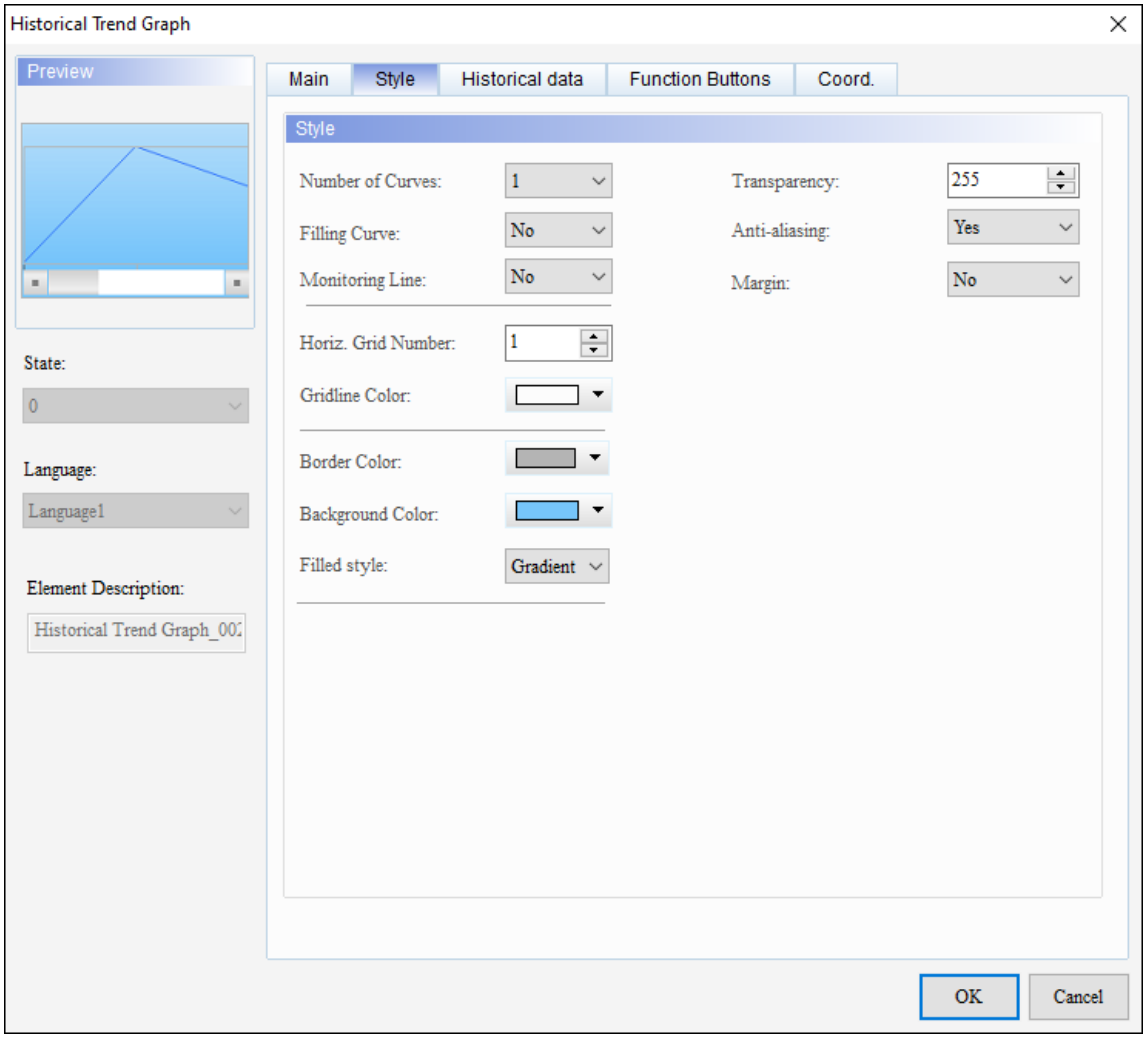
- **Main — Global curve setting**

  The **Global range** checkbox needs to be selected to set the format of all curves on the trend chart uniformly.

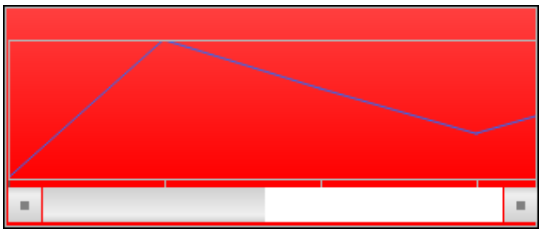
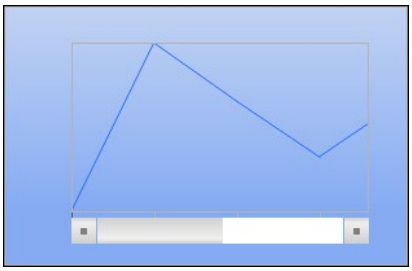


Setting	Description
Scale Settings	Set the position of the scale, the size and color of the markers, and the scale mark number. 
Curve style	Select the curve style.
Length	Select the length. If the read length is 1, the length of the read data is 1 Word. If the read length is 2, the length of the read data is 2 Word.
Data Format	Select the data format. **Note**: **Floating** can only be selected when the read length is 2.
Integer Digits / Fractional Digits	Set integer digits and fractional digits to be displayed.
Minimum / Maximum	Set a valid range value according to the set **Read Length** and **Data Format**. You can enter a constant or a custom memory address to change the minimum / maximum value dynamically.
Display High Value / Display Low Value	Select to display high or low lines. You can enter a constant or custom memory address and set the color of the marking.

- **Style**

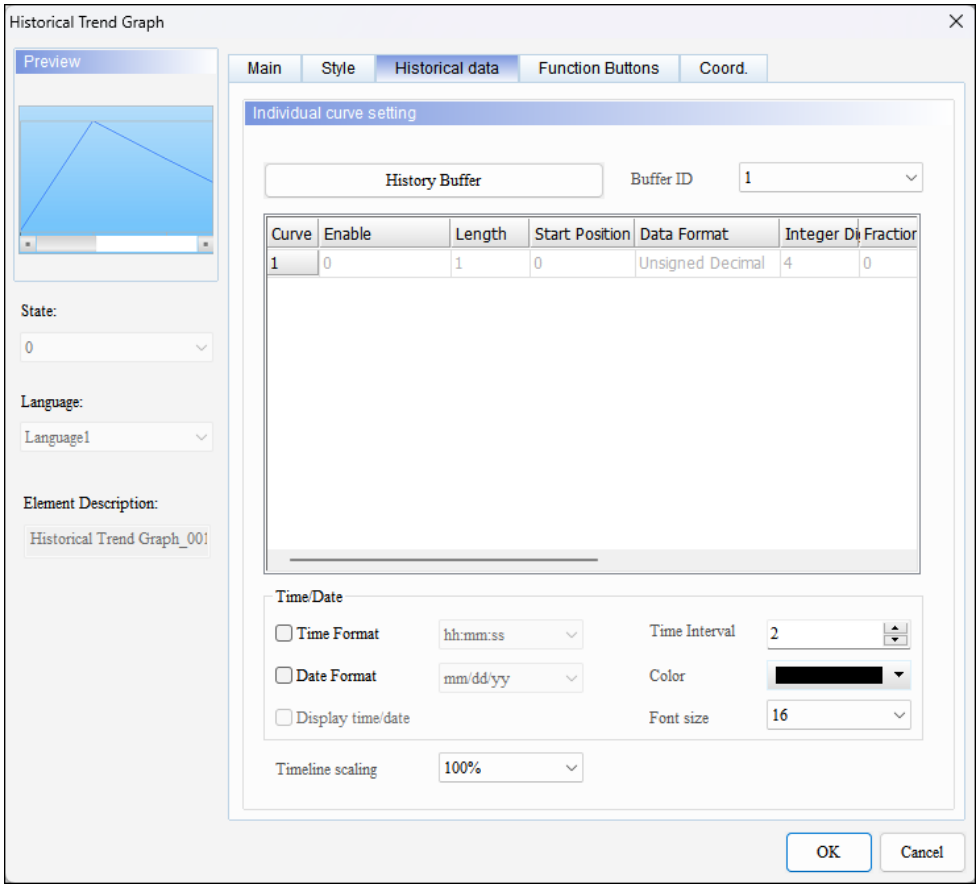


Setting	Description
Number of Curves	Select the number of curves to display. Support up to 60 curves. This setting is synchronized with the number of **Curve** in the **Historical data** tab.
Filling Curve	Select whether to fill the curve. If set to **Yes**, the area below the curve will be filled with the curve color. The filling effect is shown in the following figure.

Setting	Description
Monitoring Line	Select whether to display the monitoring line. If set to **Yes**, the monitoring line will be displayed after clicking the sampling point on the HMI screen.
Horiz. Grid Number / Gridline Color	Set the horizontal grid number and the gridline color. The horizontal grid number divides the number of zones of the Historical Trend Graph element. If set to 2, there is one grid line and 2 zones; if set to 3, there are two grid lines and 3 zones, and so on. Support up to 50 zones.
Background Color	Sets the background color of the element. The default color is blue. When set to red, it displays as shown in the following figure. 
Filled style	Sets the fill effect of **Background Color**.
Transparency	Select or enter the transparency of the element. The range value is 50~255. The lower the number, the higher the transparency.
Anti-aliasing	Select whether to enable the anti-aliasing function. If select **Yes**, the element display becomes more delicate without jagged edges.
Margin	Select whether to enable the margining function. If select **Yes**, it leaves white space around the element, and the trend chart will be indented as shown in the following figure. 

- **Historical data – Individual curve setting**

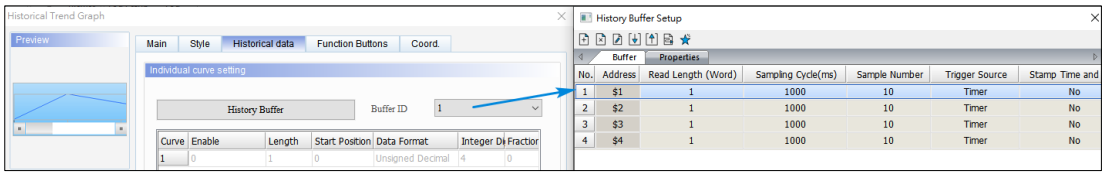
  Make the individual settings for all curves on the trend chart.



Follow these steps to configure the individual curve settings.

To configure the individual curve settings

1. Select the buffer data to be presented by this element, and the **Buffer ID** corresponds to the data number set in the **History Buffer Setup** dialog.



2. In the **Style** tab, set the **Number of Curves**, and then configure the individual curve settings table. The following table lists the settings of the individual curve settings table with their description.

Setting	Description
Enable	Set whether to enable the curve to read the data. • A value of 1 is enabled, and a value of 0 is not enabled. • The bit address can be set to enable curves

Setting	Description
	dynamically on the HMI.
Length	Select the read length according to the data source. **Note**: If **Global range** is selected, it is not possible to set the read length for individual curves.
Start Position	Select the data start position, and it depends on the **Read Length** set in the **History Buffer Setup** dialog.
Data Format	Select the data format. **Note**: • If **Global range** is selected, it is not possible to set the value format for individual curves. • **Floating** can only be selected when the **Read Length** is **2**.
Integer Digits / Fractional Digits	Select the integer digits and fractional digits to be displayed. **Note**: If **Global range** is selected, it is not possible to set the integer digits and fractional digits for individual curves.
Line Color	Set the line color of the curve.
Line Weight	Select the line width of the curve. The range value is 1–8.
Minimum / Maximum	Set the minimum and maximum values according to the **Read Length** and **Data Format**. **Note**: If **Global range** is selected, it is not possible to set the minimum and maximum values for individual curves.
Write Address	Set the write address of the curve value on the monitoring line.
Curve Style	Select the curve style.

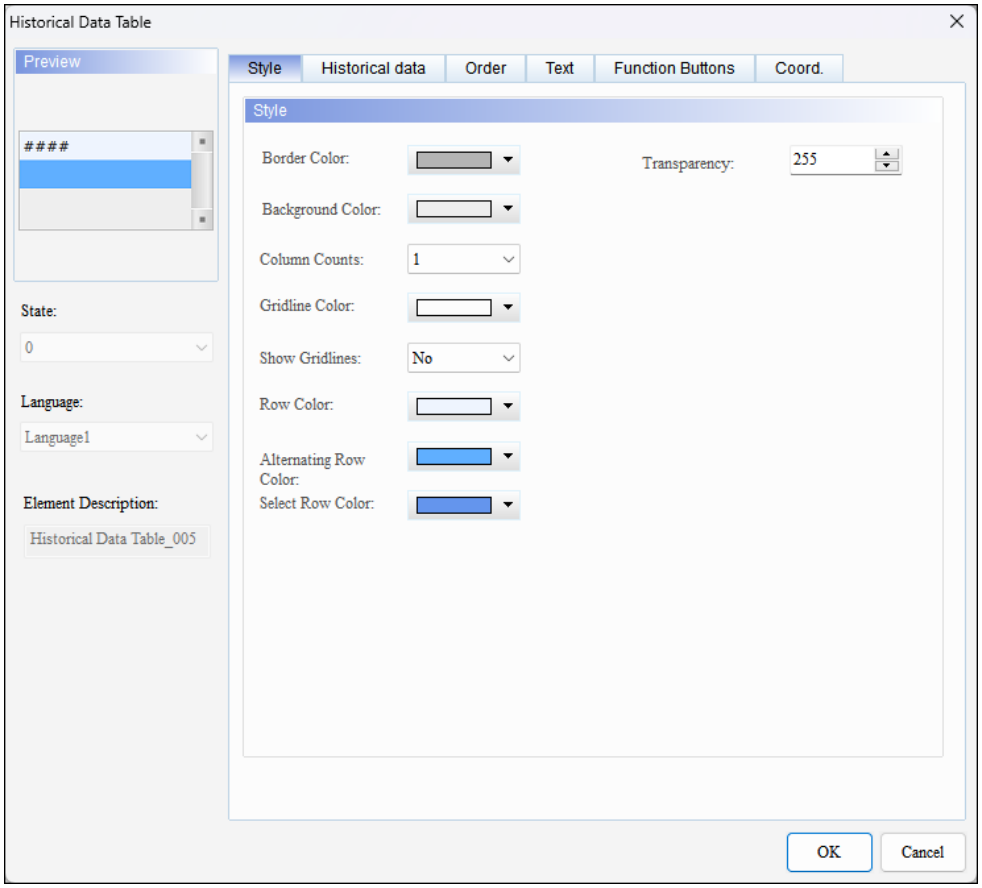
- **Time/Date**
  - **Time / Date Format**: Select to display the sampling time and date, and select the format to be displayed.
  - **Display time/date**: Select to display the timeline and set the time interval.
  - **Time Interval**: Select or enter the interval between the sampling points displayed at the set time. If the interval is 3, the sampling time will be displayed on the timeline at the first and fourth sampling points.

- **Timeline scaling**: Select the zoom ratio of the timeline display. The scale is smaller, the more sample points can be seen.

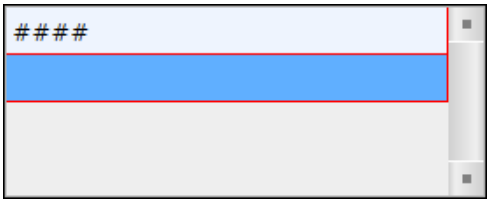
Historical Data Table Properties

- **Style**

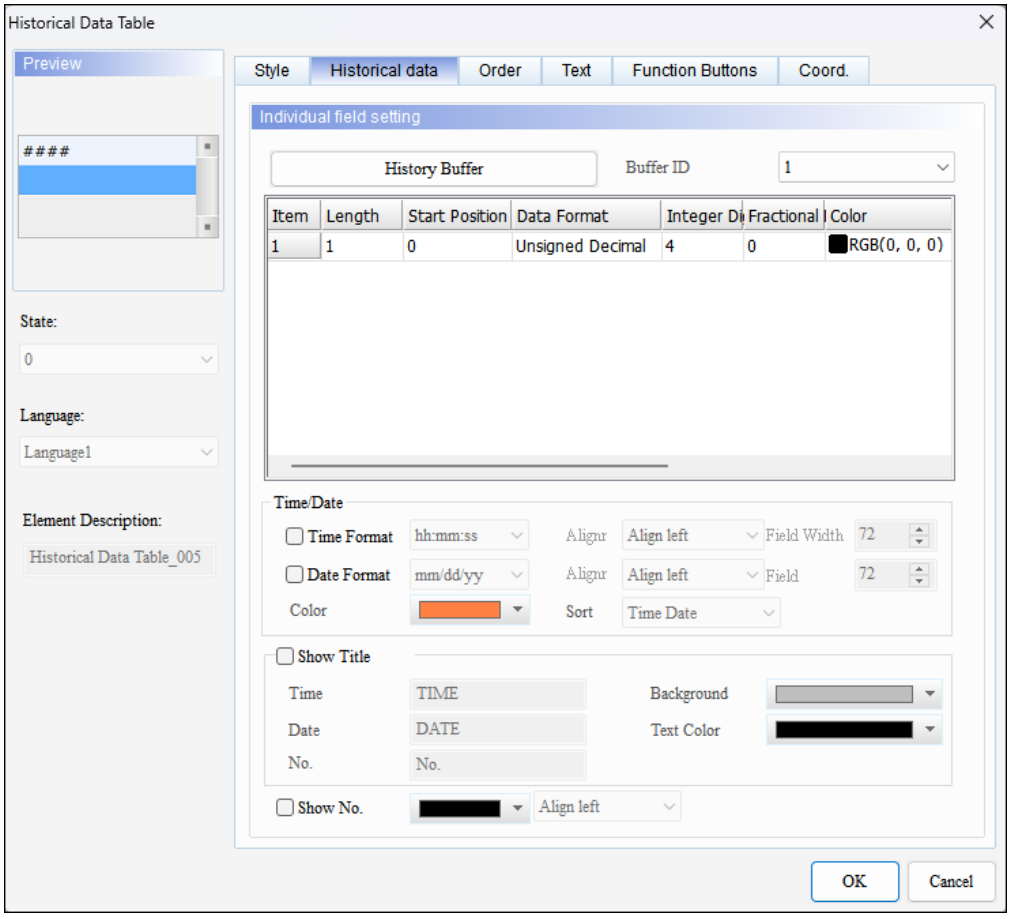
The following table lists the unique properties in the **Historical Data Table**. For other Style properties, refer to Common Properties.



Function	Description
Column Counts	Select the number of curves to be displayed. Support up to 60 columns. This setting is synchronized with the number of curve item in the **Historical data** tab.
Gridline Color / Show Gridlines	Select the gridline color of the element. The default color is white. When set to red, it displays as shown in the following figure. **Note**: **Gridline Color** only works when **Show Gridlines** is

Function	Description
	set to **Yes**. 

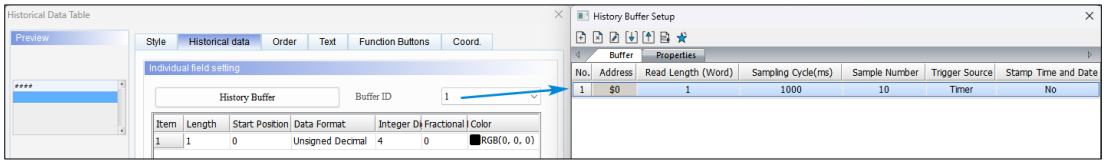
- **Historical data – Individual field setting**



Follow these steps to configure the individual field settings.

To configure the individual field settings

1. Select the buffer data to be presented by this element, and the **Buffer ID** corresponds to the data number set in the **History Buffer Setup** dialog.



2. In the **Style** tab, set the **Column Counts**, and then configure the individual field settings table. The following table lists the settings of the individual field settings table with their description.

Setting	Description
Length	Select the range of length and start position, which depend on the **Read Length** set in the **History Buffer** Setup dialog.
Start Position	
Data Format	Select the data format. **Note**: • **Floating** can only be selected when **Length** is **2**. • If **Length** is greater than 2, you can use only the Char format.
Integer Digits / Fractional Digits	Set the number of integer digits and fractional digits to be displayed. **Note**: There are no integers / fractional digits in the Char format.
Color	Set the text color of this field.
Column Width	Set the display width of this field. The range value is 0~9999.
Prefix Zero	Select to determine the number of digits to prefix according to the set **Integer digits**.
Title bar	Enter the title bar. **Note**: The title bar name is only displayed on the element when the **Show title bar** is selected.
Field alignment	Select to align the text in the field to the left, center, or right.

- **Time/Date**
  - **Time / Date Format**: Select to display sampling time and date, and set the display format, alignment, and field width.
  - **Color**: Select the display color of the text.
- **Show Title**

  Select to display the title bar, and set the time, date, and the title of the index, as well as the background color and text color of the title bar.

Note: The text in the title row of the data field is configured in the **individual field settings**.

- **Show No.**

  Select to display the index, and set the text color and alignment of the index row.

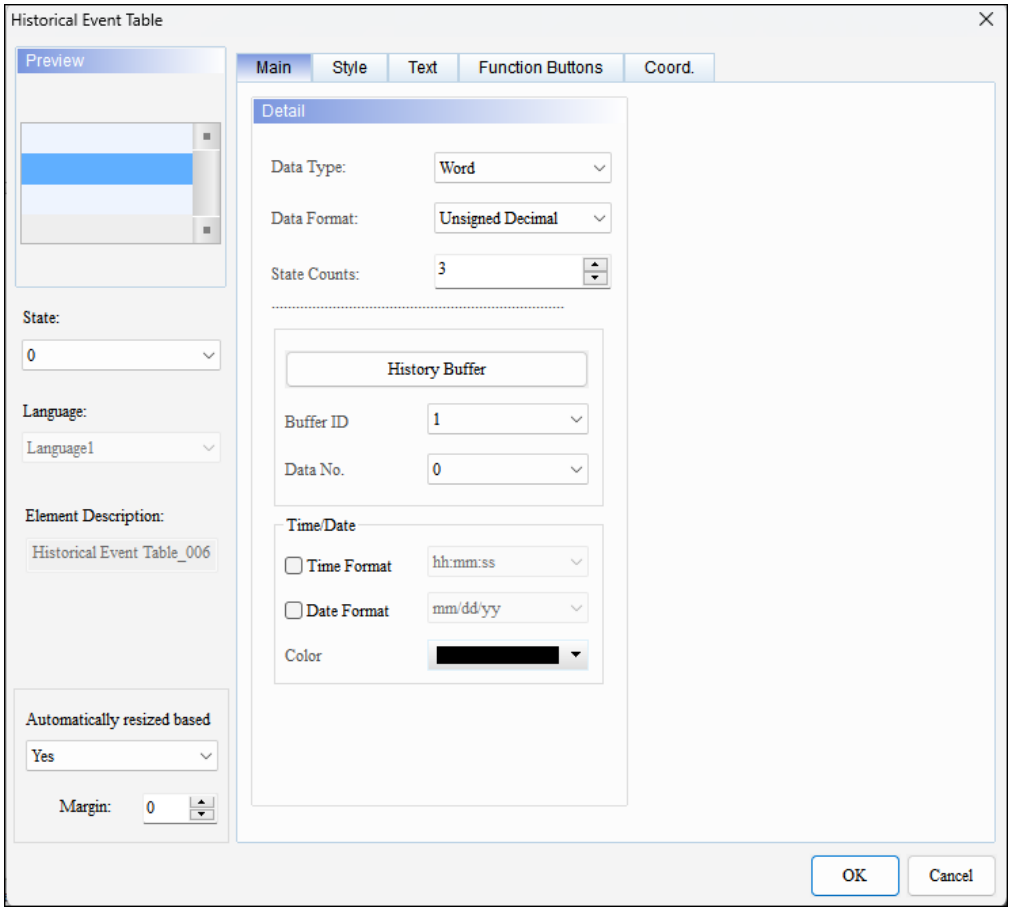
- **Order**

  Select the data sorting method, which is sorted by the sampling time.

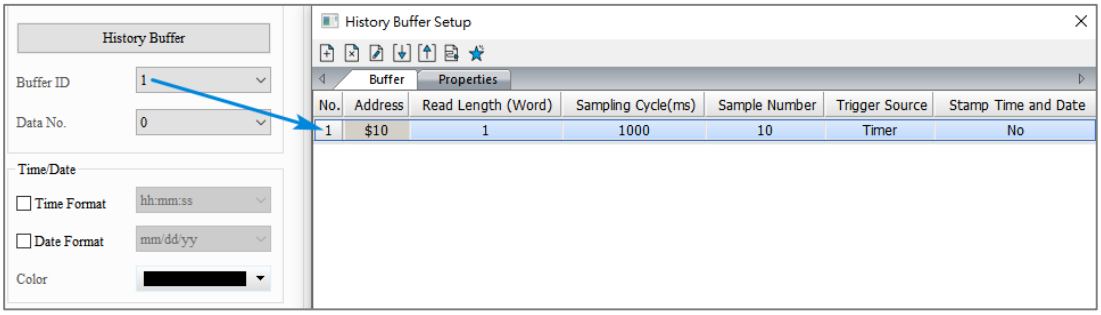
- **Text**

  Select the display font size of the text.

Historical Event Table Properties



- **Main – Detail**
  - **Data Type**: If select **Word**, you can set 1 to 256 states; if select **LSB**, the sampling values of historical data are converted into binary display, and you can set 1 to 16 states.
  - **Note**: To display State 0, select **LSB (Support State 0)**.
- **Main– Buffer setting**
  - **Buffer Number**: Select the buffer data to be presented by this component, which corresponds to the data number set in the **History Buffer Setup** dialog.

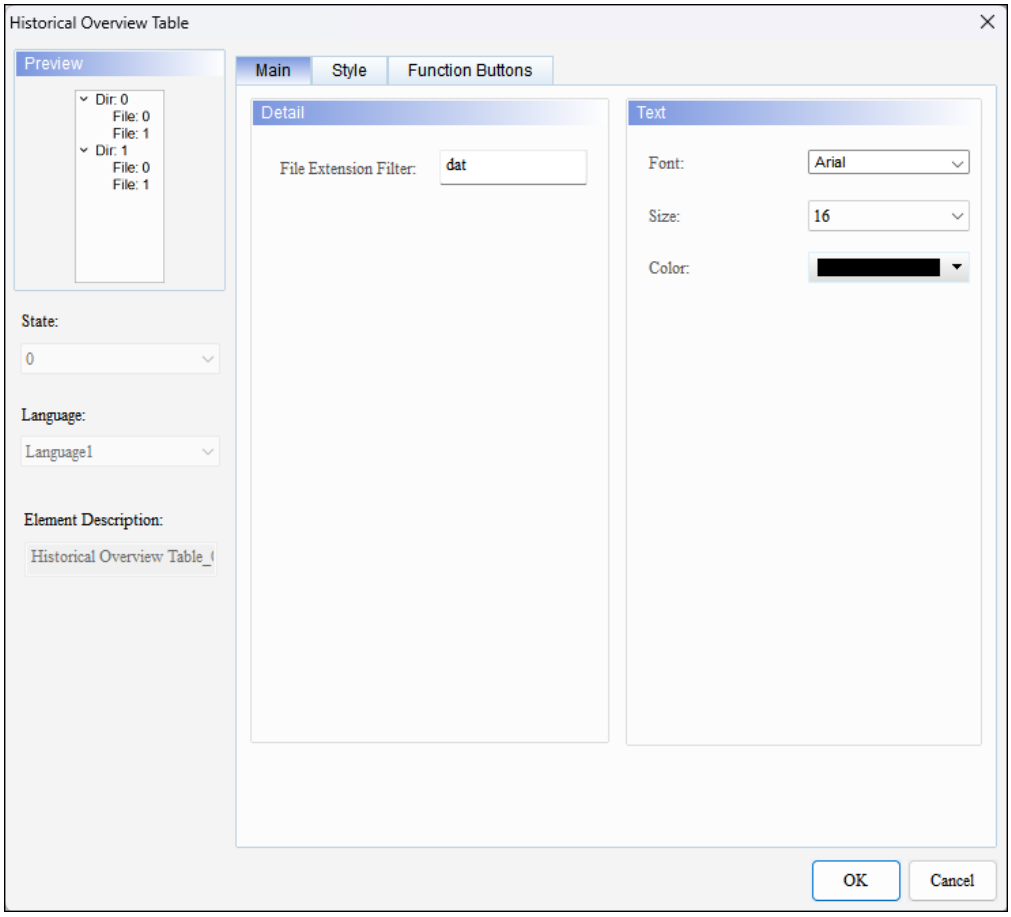


  - **Data No**: Select the data number, which depends on the **Read Length** set in the **History Buffer Setup** dialog. If the read length is 1, the data position is 0.
  - **Time/Date**: **Time / Date Format**: Select to display the sampling time and date, and select the display format and text color.

Historical Overview Table Properties

The **Historical Overview Table** element has a file browsing area on the left and the trend chart on the right, which have different properties.

For the trend chart properties, refer to Historical Trend Graph Properties. The following lists the properties of the **file browsing area** with their description.



- **Main – File Extension Filter**

  Enter the extension of the file you want to browse. The default value is **dat**.

- **Main – Text**

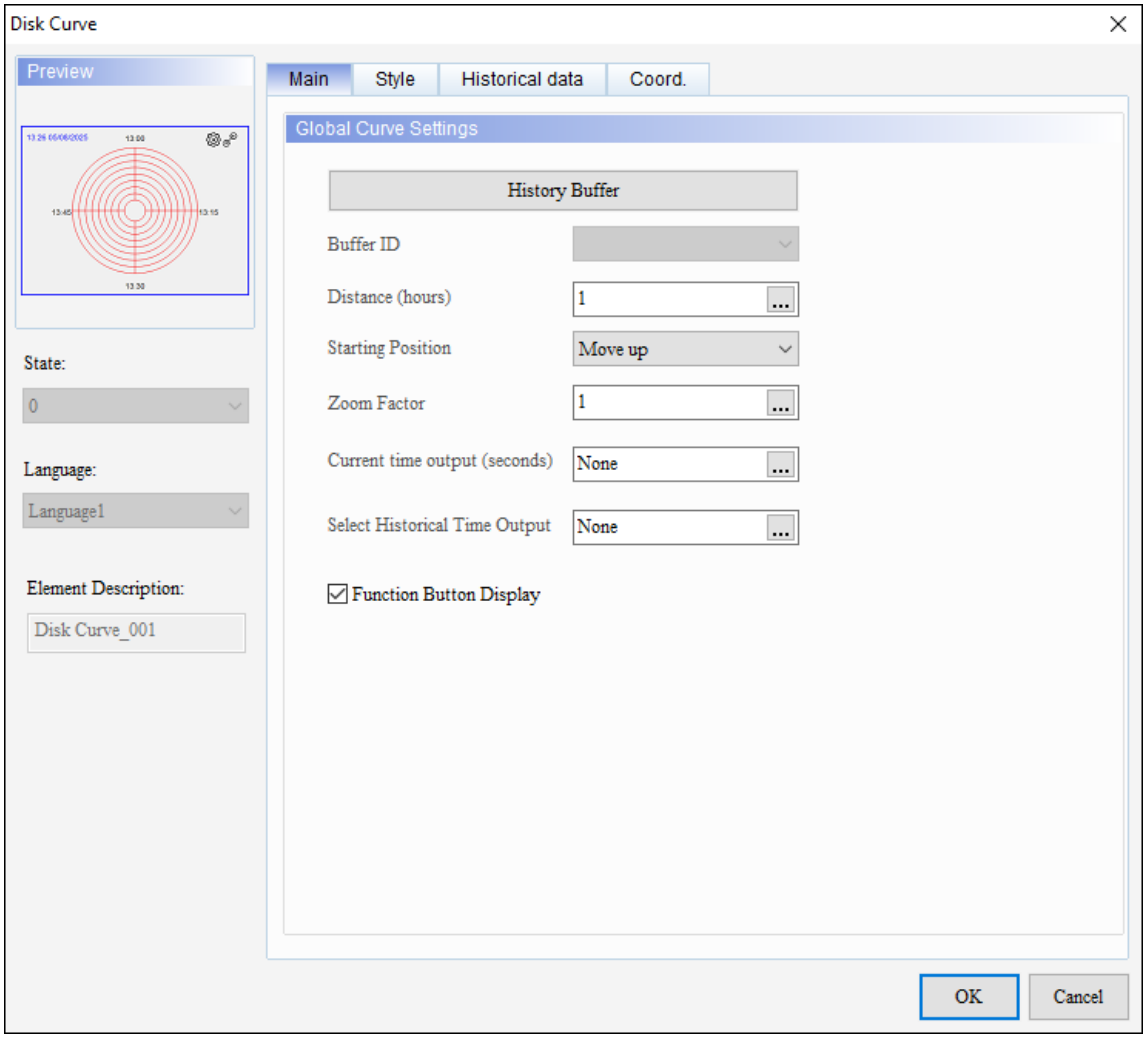
  Set the font, size, and color of the text in the file browsing area.

- **Style**

  Set the border color, background color, and transparency of the file browsing area.

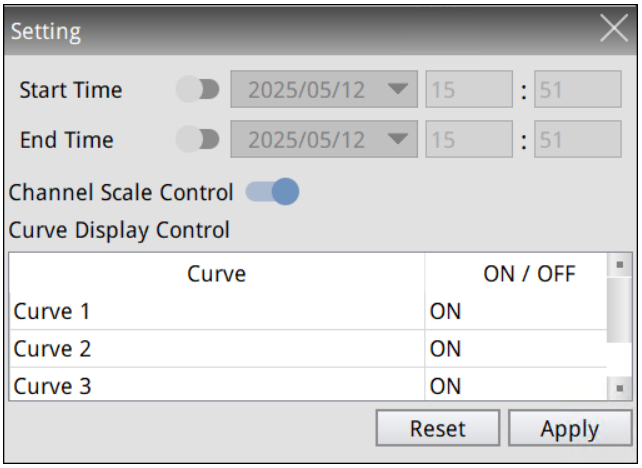
Circular Trend Properties

- **Main — Global curve setting**



Setting	Description
Distance (hours)	Set the length of time for a circle of the circular trend chart. The range value is 1–24. You can also set the memory address to dynamically change the length of time on the HMI.
Starting Position	Select the starting position of the curve.
Zoom Factor	Set the zoom factor. The range value is 1–10. You can also set a memory address to change the displayed zoom factor on the HMI dynamically.
Current time output (seconds)	Set the time gap between the loading time by HMI and the last historical data, which is recorded in seconds.
Select Historical Time Output	Set the time gap between the HMI loading time and monitoring line. Record in seconds.

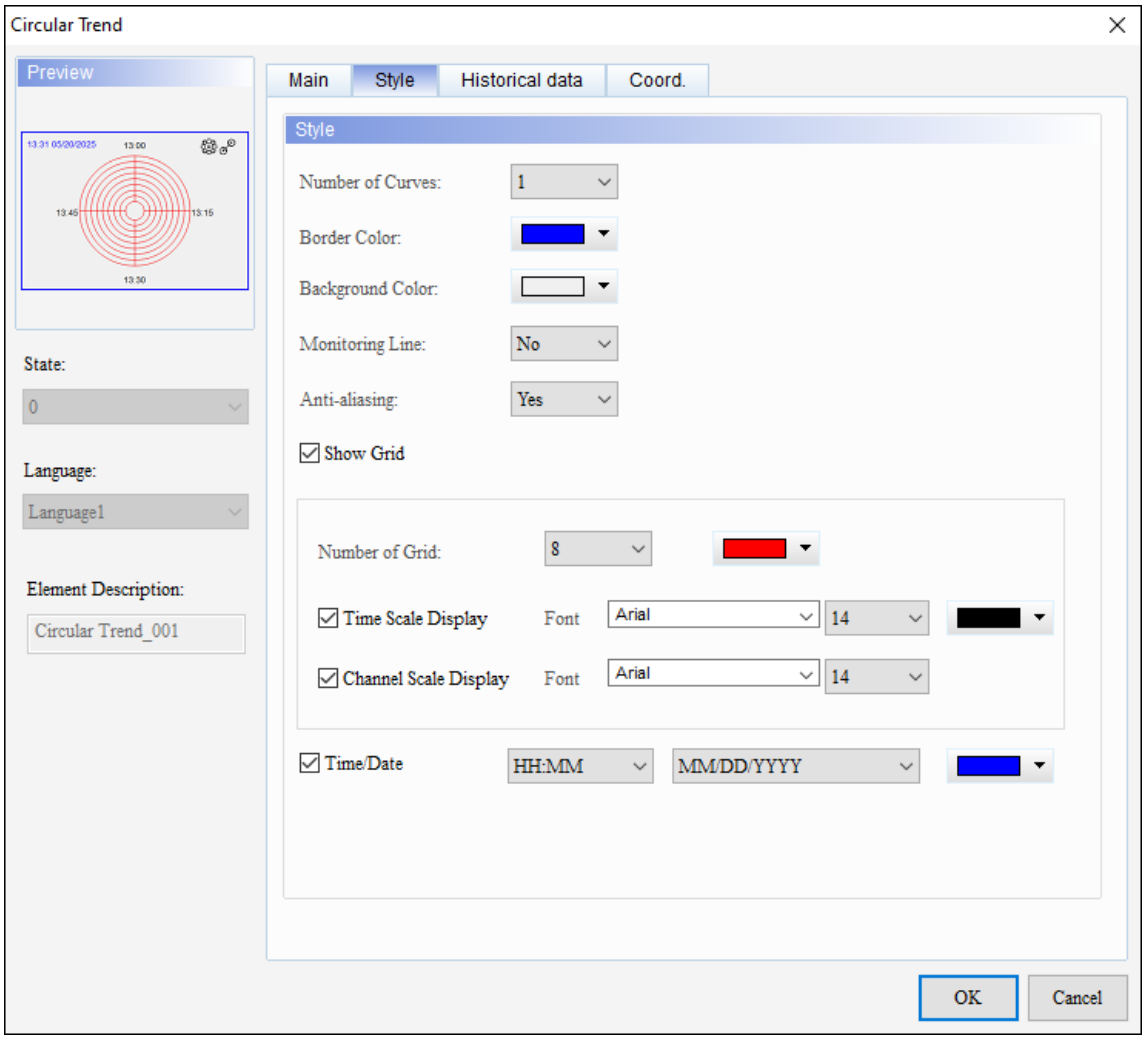
<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Function Button Display</td><td>
If selected, you can open the settings dialog and adjust the circle's display settings on the HMI.

Click to open the setting dialog to set the displayed time range of data and displayed channel.


Click to expand the function menu. The following table lists the functions in the function menu.
<table>
<tr><th>Icon</th><th>Function</th><th>Description</th></tr>
<tr><td> </td><td>Displays the previous data</td><td rowspan="2">Click to toggle the displayed time range. The time range in units of the set time distance.</td></tr>
<tr><td> </td><td>Display the next data</td></tr>
<tr><td> </td><td>Display the last data</td><td>Click to display the last data in units of the time distance.</td></tr>
<tr><td> </td><td>Set the factor for zoom in to 1</td><td>Click to reset the factor for zoom in to 1.</td></tr>
<tr><td> </td><td>Zoom in 1x</td><td>Click once to increase the factor for zoom in by 1 until the factor reaches 10 and no longer zooms in.</td></tr>
<tr><td> </td><td>Zoom out 1x</td><td>Click once to reduce the factor for zoom out by 1 until the factor reaches 1 and no longer zooms out.</td></tr>
</table>

</td></tr>
</table>

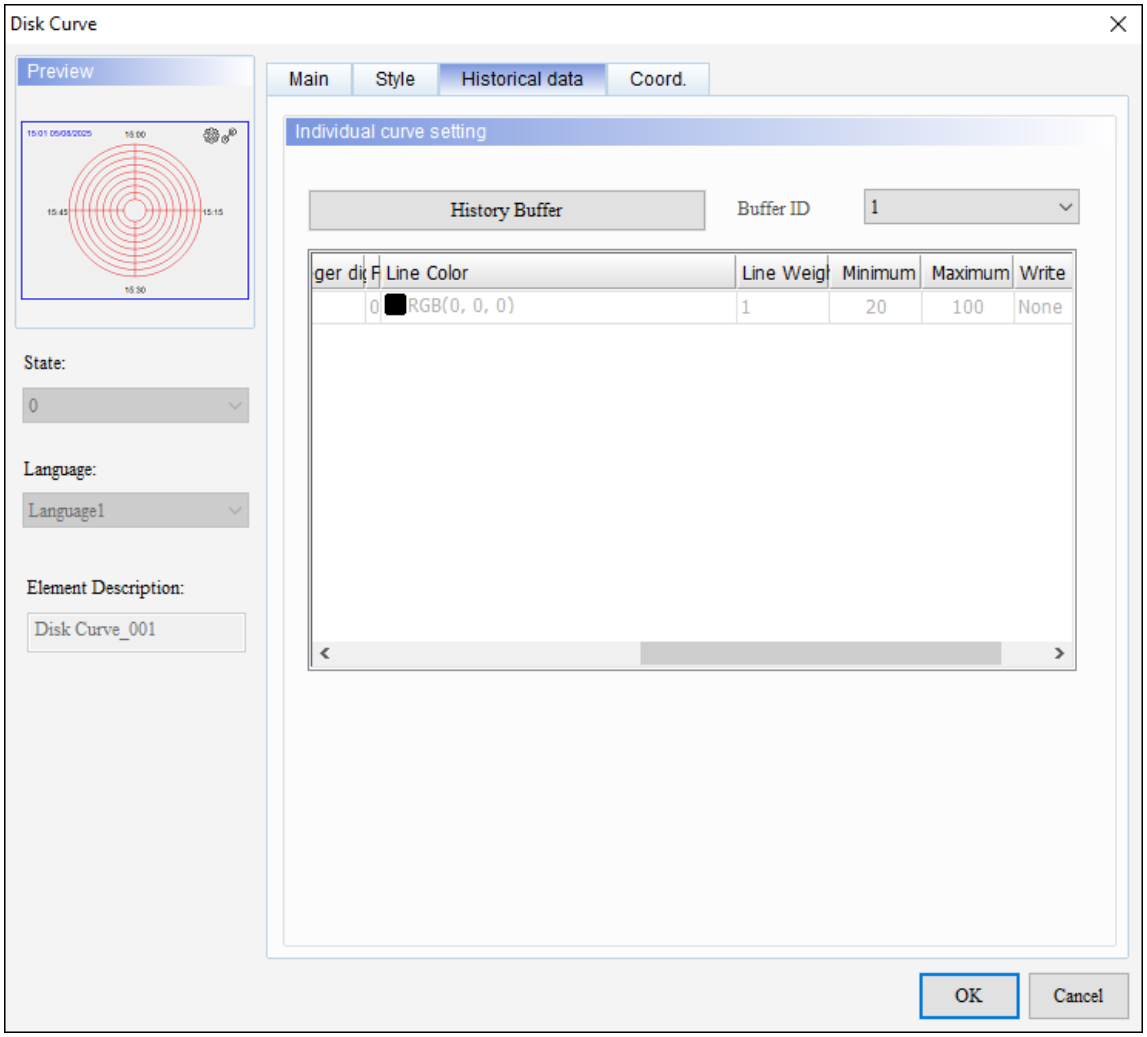
- **Style**



Setting	Description
Number of Curves	Select the number of curves to display. Support up to 8 curves. This setting is synchronized with the number of **Curve** in the **Historical data** tab.
Monitoring Line	Select whether to display the monitoring line. If set to **Yes**, the monitoring line is displayed after clicking the sampling point on the HMI screen, and the monitoring line time is displayed at the upper left.
Anti-aliasing	Select whether to enable the anti-aliasing function. If select **Yes**, the element display becomes more delicate without jagged edges.
Show Grid	Select to display the circular grid, and configure the number of grids and time scale settings.
Time Scale Display	Select to display the time scale on the circular trend chart, and set the scale font and color.

Setting	Description
Channel Scale Display	Select to display the channel scale on the circular trend chart, and set the scale font. **Note**: The scale color is displayed based on **Line Color** in the **Historical data** tab.
Time/Date	Select to display the current time at the upper left of the element, and set the format and color to display.

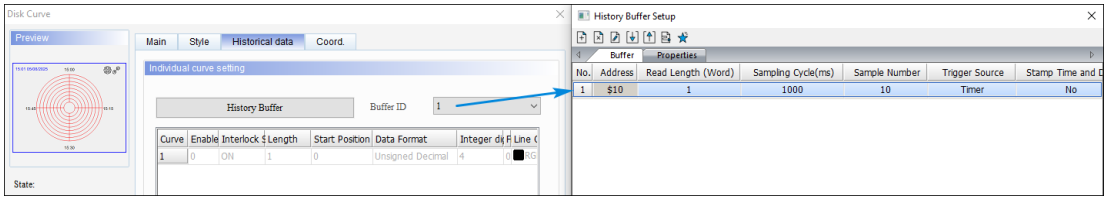
- **Historical data – Individual curve setting**



Follow these steps to configure the individual curve settings.

To configure the individual curve settings

1. Select the buffer data to be presented by this element, and the **Buffer ID** corresponds to the data number set in the **History Buffer Setup** dialog.



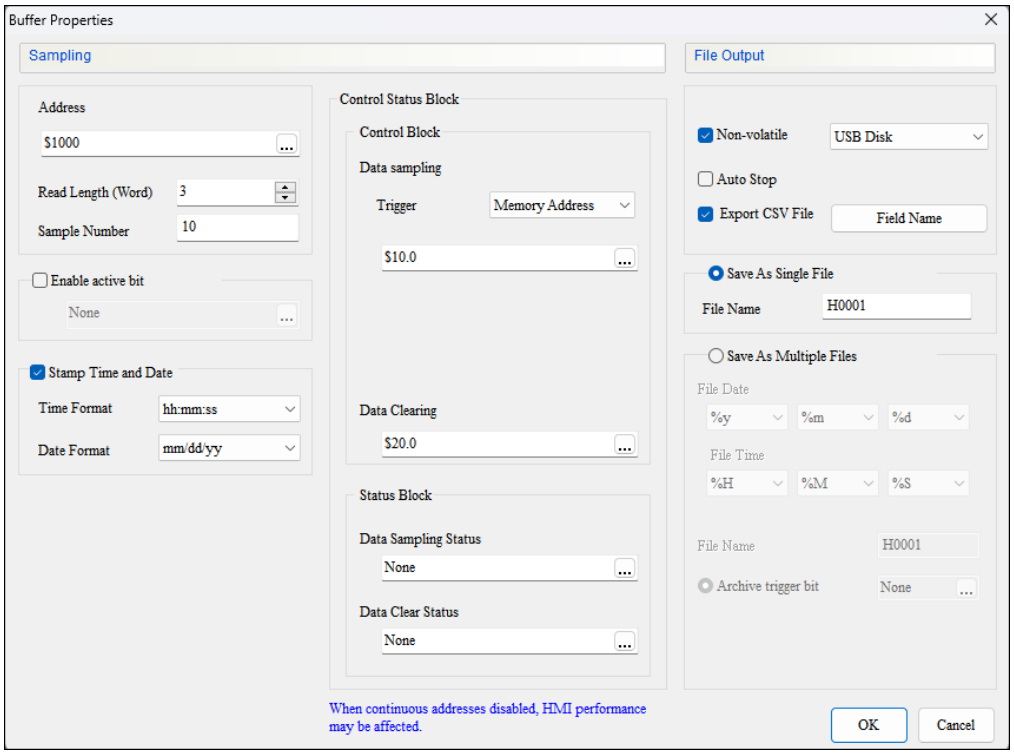
2. In the **Style** tab, set the **Number of Curves**, and then configure the individual curve settings table. The following table lists the settings of the individual curve settings table with their description.

Setting	Description
Enable	Set whether to enable the curve to read the data. • A value of 1 is enabled, and a value of 0 is not enabled. • The bit address can be set to enable curves dynamically on the HMI.
Interlock State	This setting is only available when **Enable** is set as a bit address.
Length	Select the read length according to the data source.
Start Position	Select the data start position, and it depends on the **Read Length** set in the **History Buffer Setup** dialog.
Data Format	Select the data format. **Note**: **Floating** can only be selected when the **Read Length** is **2**.
Integer Digits / Fractional Digits	Select the integer digits and fractional digits to be displayed.
Line Color	Set the line color of the curve.
Line Weight	Select the line width of the curve. The range value is 1–8.
Minimum / Maximum	Set the minimum and maximum values according to the **Read Length** and **Data Format**.
Write Address	Set the write address of the curve value on the monitoring line.

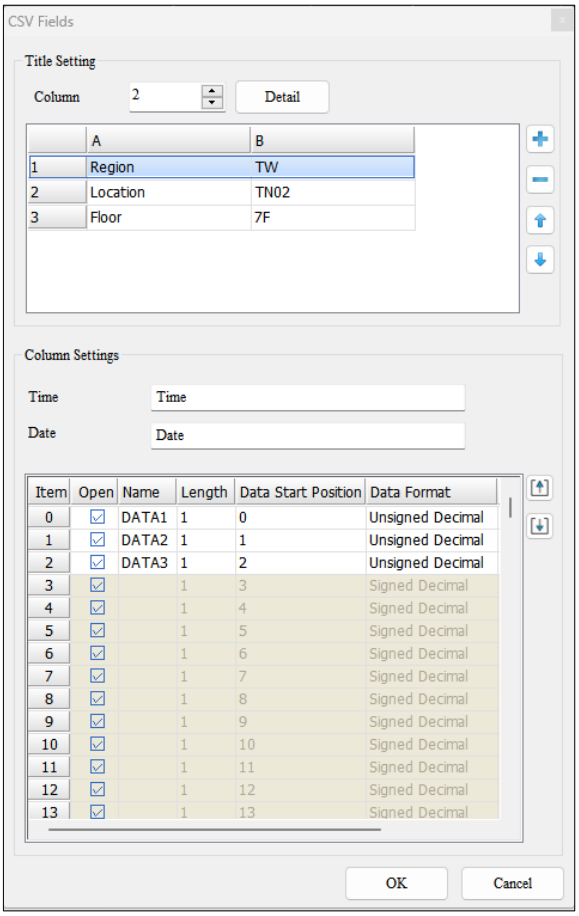
Example of History Buffer Setting

Follow these steps to add a history buffer and view the history data through the sampling element.

1. In the **History Buffer** settings page. Click on the toolbar to add **Buffer Properties** dialog.
2. In the **Buffer Properties** dialog, set the buffer properties as shown in the following figure.



The example of setting the **Field Name** of **Export CSV File** is shown in the following figure.

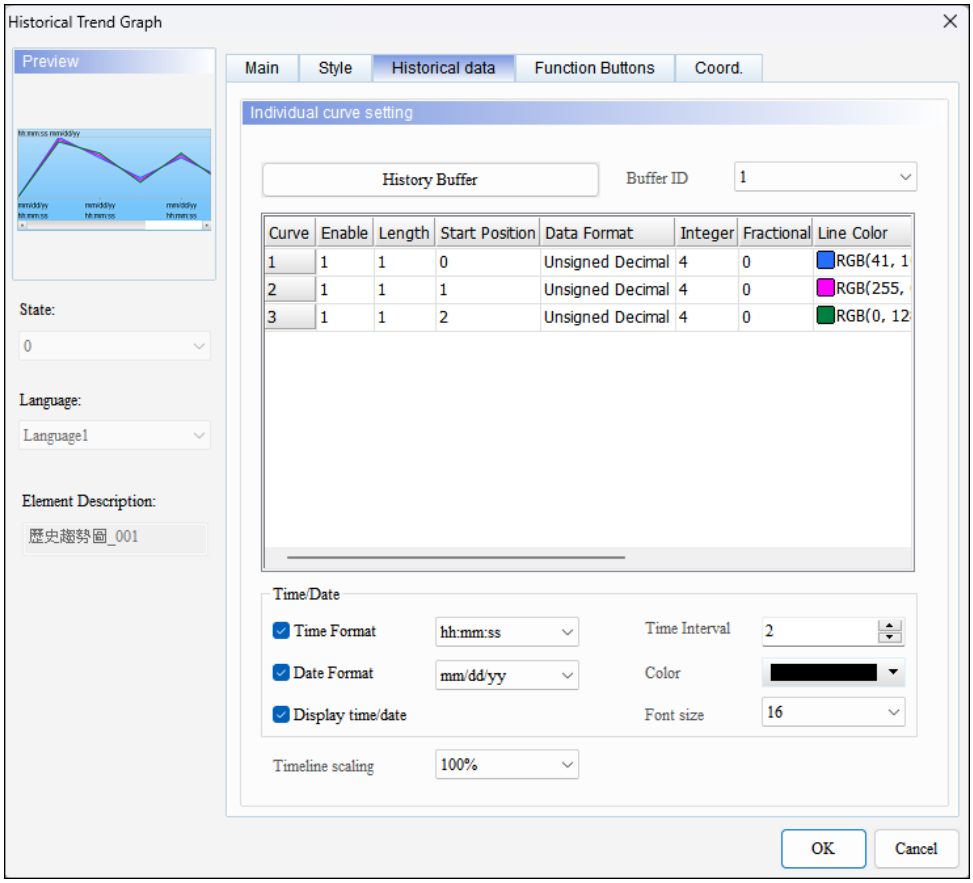


3. Create a **Historical Trend Chart** element on the screen, and then double-click the element to open the properties dialog.

   a. In the **Style** tab, select **3** for **Number of Curves**.

   b. In the **Historical data** tab, set **Enable** in **Individual curve setting** table to **1**, and set the curve properties.

   c. In the **Time/Date** area, set the display method of time and date.

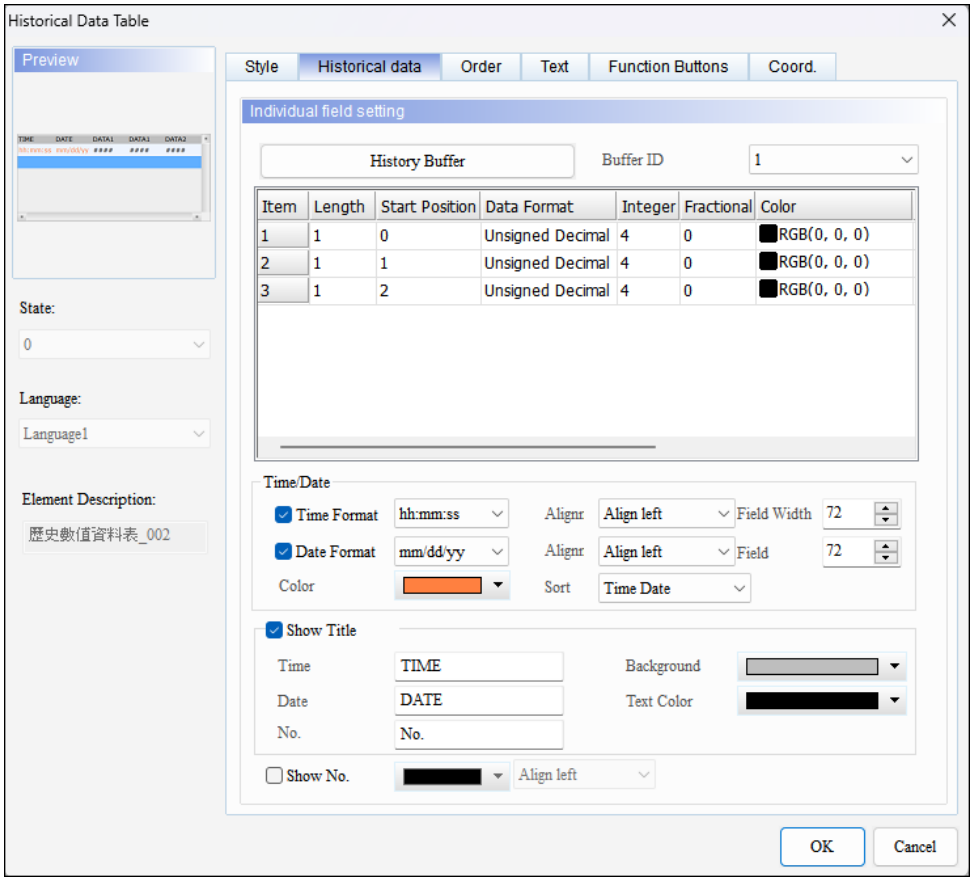


4. Create a **Historical Data Table** element on the screen, and then double-click the element to open the properties dialog.

   a. In the **Style** tab, set 3 for **Column Counts**.

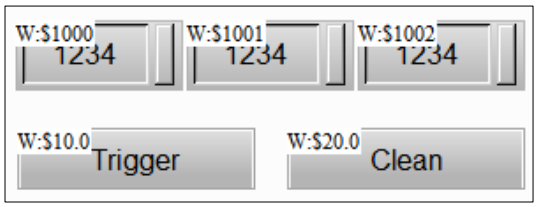
   b. In the **Historical data** tab, set each property of the **Individual curve setting** table.

   c. Select the **Time Format**, **Date Format**, and **Show Title** checkboxes.

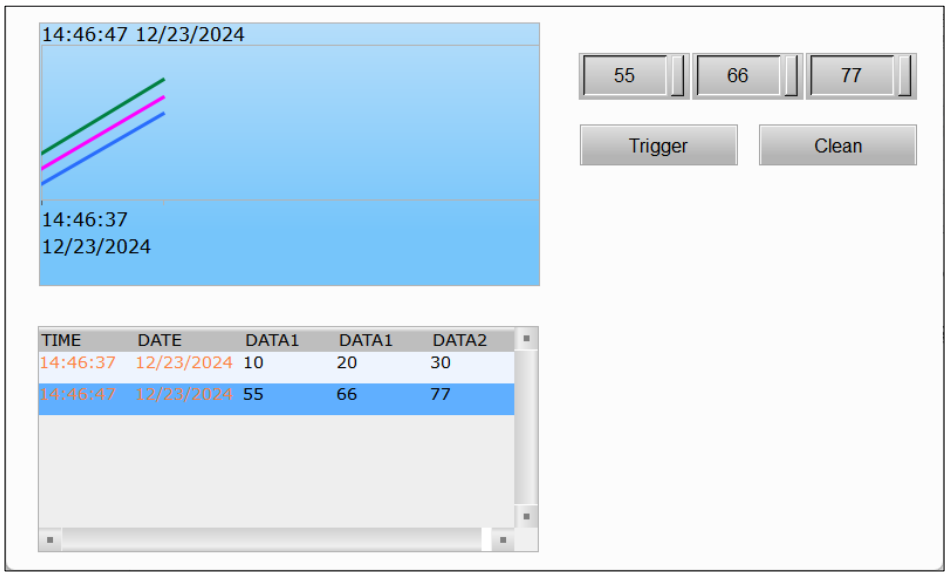


5. Create 3 **Numeric Entry** elements on the screen, and then set the **Write Address** is the read address of the history buffer $1000.

6. Create 2 **Momentary** button elements on the screen, and then set the **Write Address** to $10.0 and $20.0 respectively, corresponding to the trigger bit and data clear bit.



7. Download the screen to the HMI and enter the value within 100 to 3 **Numeric Entry** elements respectively. Click the button with the memory address of $10.0, and the **Historical Data Table** records the current value. Click $10.0 button again, and the historical trend chart draws the data curve. The following figure shows the execution result.



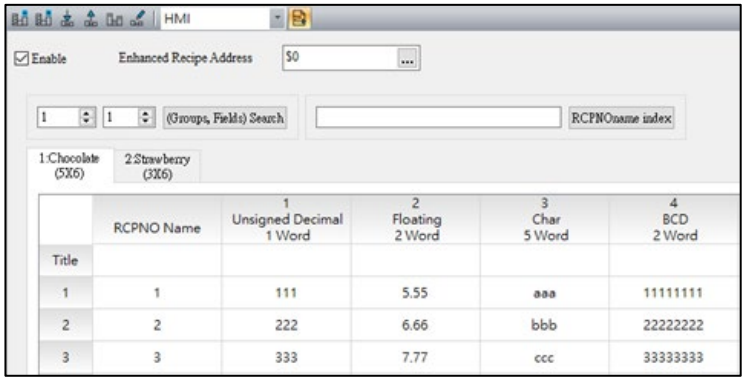
8. The historical data is synchronously written to the USB disk. The file path is `HMI\HMI-000\History\CSV\H0001.csv`, and the content of the CSV file is also written to the data content according to **Field Name**.

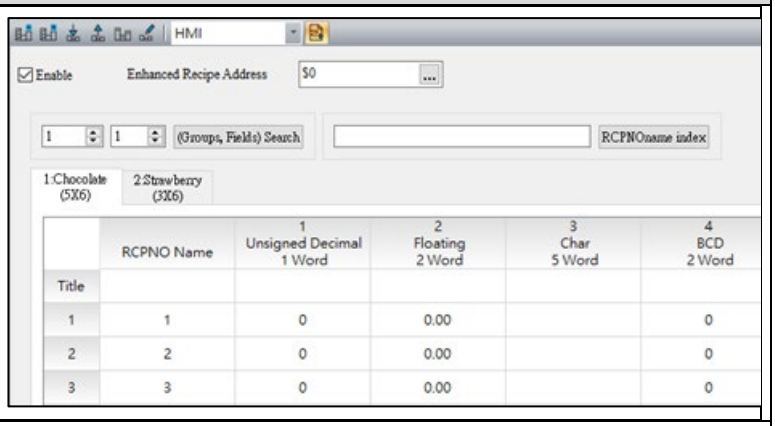
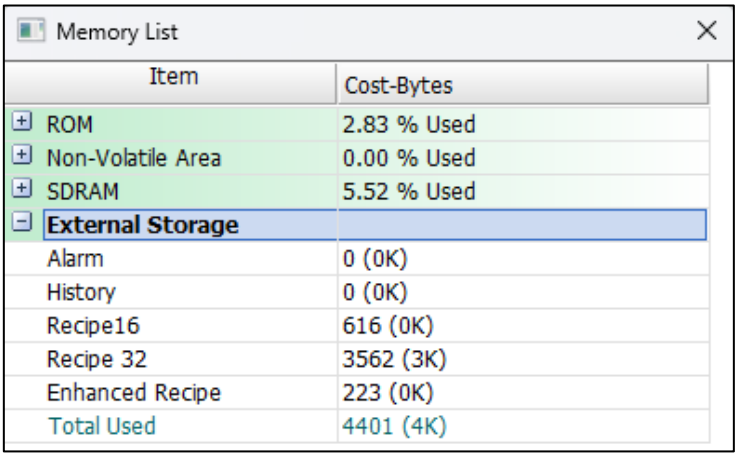
	A	B	C	D	E
1	Region	TW			
2	Location	TN02			
3	Floor	7F			
4	Time	Date	DATA1	DATA2	DATA3
5	14:55:14	12/23/2024	10	20	30
6	14:55:25	12/23/2024	55	66	77

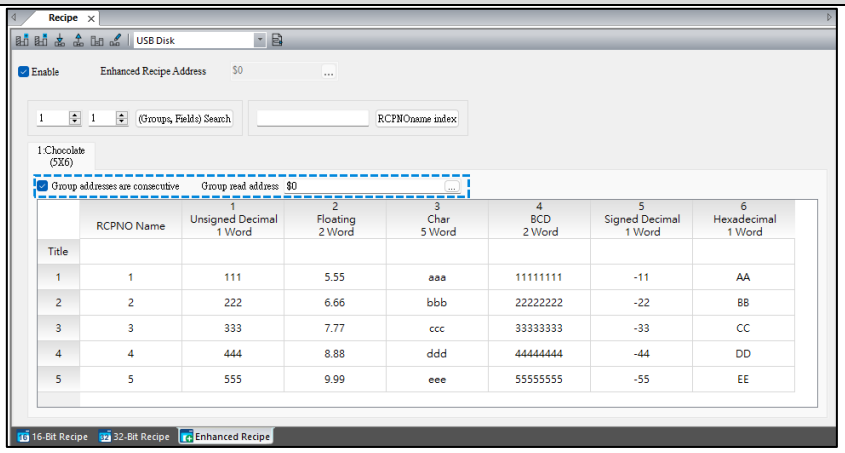
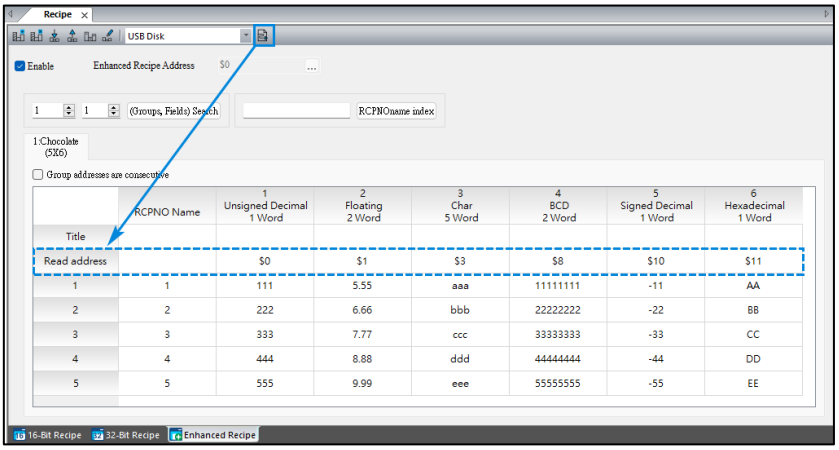
Recipe

Recipe Setting Toolbar

The following table lists the settings in the **Recipe** setting page with their description.

Setting	Description
	Click to add a recipe.
	Click to delete a recipe.
	Click to import the recipe file. Support .rcp and .csv file formats. The **enhanced recipe** support .xlsx file format.
	Click to export the recipe files is .rcp, .csv or.xlsx file formats. • The **.rcp** file is a built-in file format of HMI, which can provide 16-bit recipe download. • The **.csv** file only records the data format of the recipe field and the content of the recipe data. • The **.xlsx** file is for **enhanced recipe** use, that records detailed recipe address information, including data format, recipe title, recipe group address or field read address.
	Click to clear all recipe contents. Before 

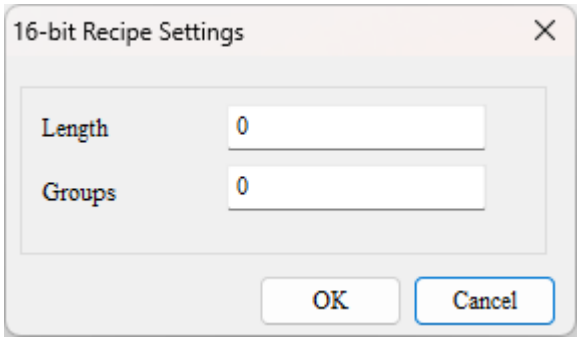
Setting	Description
	After 
	Click to set the **Name**, **Fields**, **Group**, and **Data Format**.
Recipe Storage	Select the recipe storage location. The options vary from the model, including HMI, USB Disk, USB Disk 2,and SD. • If select an external storage device, enable and compile recipe. Go to **View** > Memory List to view the recipe's usage.  • If select **HMI**, the data will be recorded in the ROM of HMI when powering off.
	Click to switch to **Continuous recipe address** or **Non-continuous recipe address**. Unclicking the **Continuous Recipe Address** icon, the **Group addresses are consecutive** checkbox displays on the recipe group page. The read addresses for all recipe groups must be set individually, otherwise the compilation will fail. • If select **Group addresses are consecutive**, you can set the **Group Read Address.** Its address is the starting address of the recipe group.

Setting	Description
	 • If unselect **Group addresses are consecutive**, the **read address** will be displayed in the recipe table, and the **read address** of each recipe field can be set individually. 
Enable	Click to enable the enhanced recipe. If not selected, all recipe settings will be invalid.
Recipe Read Address	Select internal memory or controller register address. No matter how many recipes are created, they all share the same memory address.
(Groups,Fields) Search	After entering the number of groups and fields to be searched, click this button to search.
RCPNO name index	In the **Enhanced Recipe** page, after entering the RCPNO name to be searched, click this button to search.

Recipe Setting

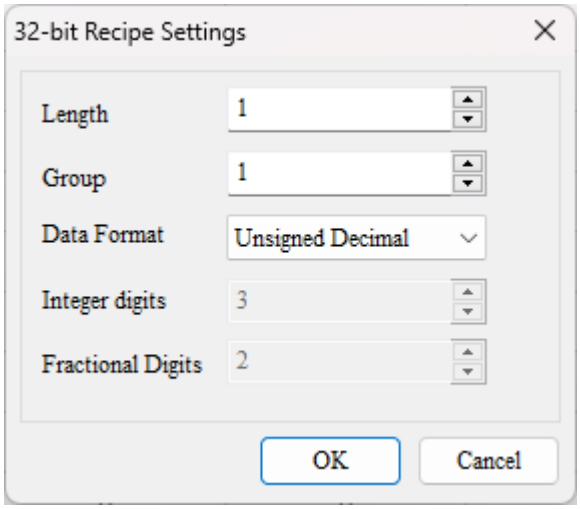
The following describes the setting of the 16-bit recipe, 32-bit recipe and enhanced recipe with their description.

- **16-Bit Recipe**



Setting	Description
Length / Groups	Select or enter the number of length and groups. The value range is 1–65535. **Note**: The recipe length multiplied by groups cannot exceed 50MB.

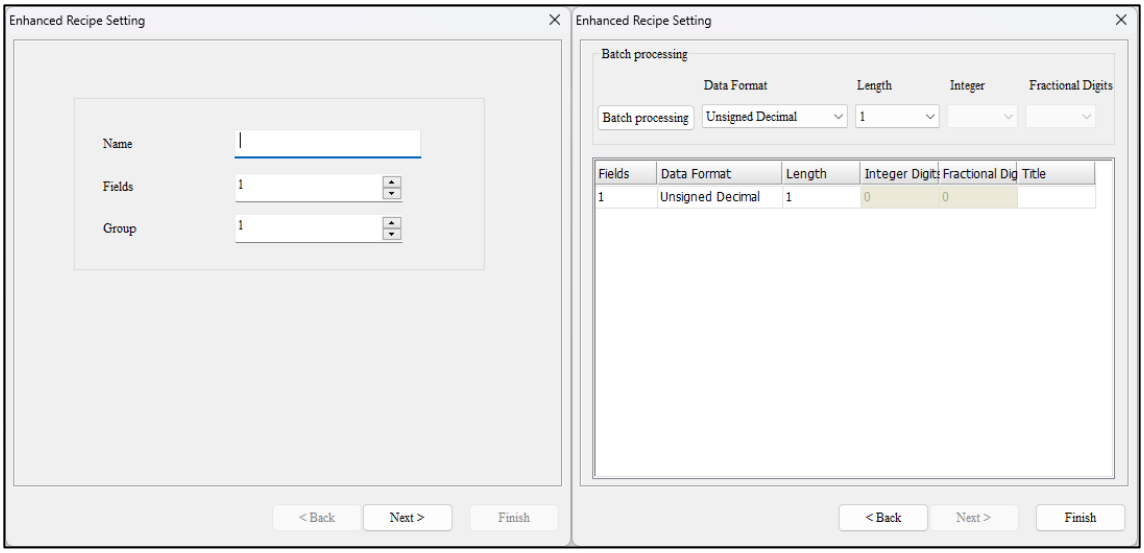
- **32-Bit Recipe**



Setting	Description
Length / Groups	Select or enter the number of length and groups. The value range is 1–65535. **Note**: The recipe of length multiplied by groups cannot exceed 50MB.

Setting	Description
Data Format	Select the data format. The options are Signed Decimal, Unsigned Decimal, and Floating.
Integer Digits	This setting is available only when the Data Format is selected as **Floating**.
Fractional Digits	**Note**: The sum of **Integer Digits** and **Fractional Digits** only supports 7 digits.

- **Enhanced Recipe**



Setting	Description
Name	Enter the group name of the enhanced recipe. Support Unicode input, you can enter the languages of each country.
Field	Select or enter the number of fields. **Fields** indicates the length and of the recipe. The value range is 1–2048.
Group	Select or enter the number of groups. **Group** indicates the length of the enhanced recipe. The value range is 1–65535.
Batch processing	After selecting the data format and length, click **Batch Process** to uniformly change all fields to the set format.
Data Format	Select the data format. The options are BCD, Signed Decimal, Unsigned Decimal, Hex, Floating, and Char. Among them, the read length of the Char format supports up to 32 Word lengths (that is, 64 characters), and supports Unicode input. **Note**: If select **Char**, do not fill in the value with the same character as the delimiter, otherwise, it will cause data errors, and the data will fail to import.

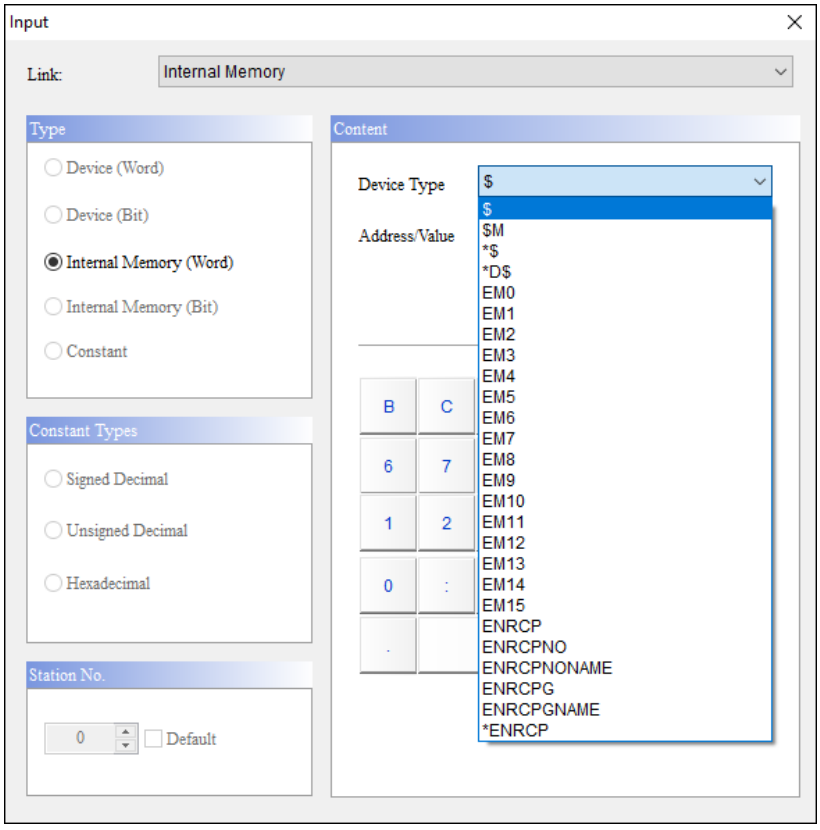
<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Length</td><td>Select the read length. The read length varies from the data format.
If select Char for Data Format and there is still space left after the string, the system will automatically fill in the blank string.
<table>
<tr><th>Data Format</th><th>Length</th></tr>
<tr><td>BCD</td><td rowspan="4">1: Word
2: Double Word
4: Quad Word</td></tr>
<tr><td>Signed Decimal</td></tr>
<tr><td>Unsigned Decimal</td></tr>
<tr><td>Hexadecimal</td></tr>
<tr><td>Floating</td><td>2: Double Word
4: Quad Word</td></tr>
<tr><td>Char</td><td>Supports up to 32 Words (64 characters)</td></tr>
</table></td></tr>
<tr><td>Integer Digits</td><td rowspan="2">This setting is available only when the Data Format is selected as Floating.
Note: The sum of Integer Digits and Fractional Digits only supports 7 digits.</td></tr>
<tr><td>Fractional Digits</td></tr>
</table>

Recipe Register

The following table lists the definition of each recipe special registers.

Register Name	Definition
RCP	Recipe Register
RCPNO	Recipe Number Register
RCPG	Recipe Group Register
*RCP	Indirect recipe index register
ENRCP	Enhanced Recipe Register
ENRCPNO	Enhanced Recipe Number Register
ENRCPG	Enhanced Recipe Group Register
ENRCPGNAME	Enhanced Recipe Group Name Register
*ENRCP	Enhanced indirect recipe index register

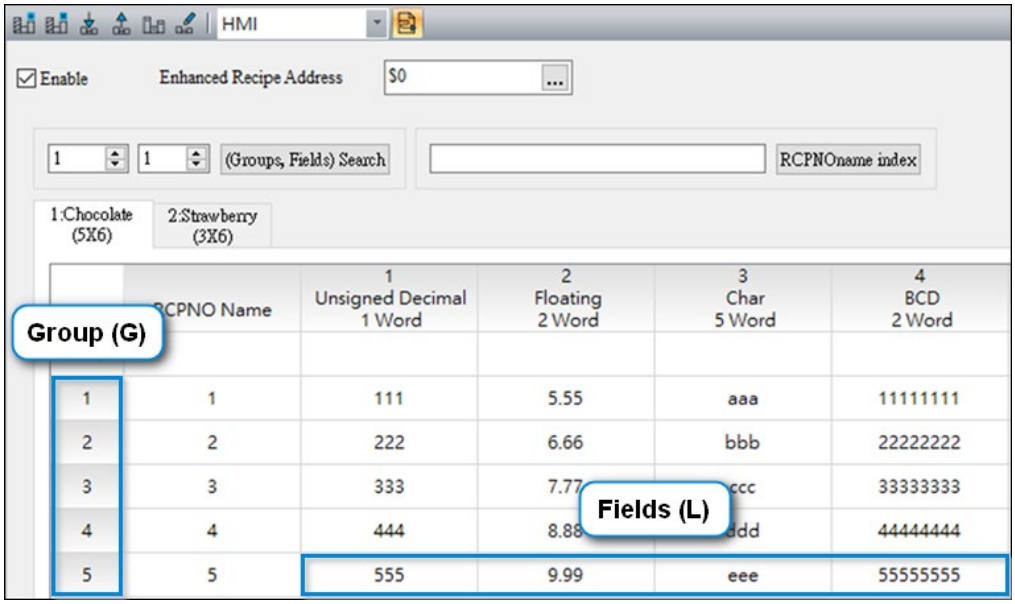
Enable the recipe and set the recipe read address to select the recipe register in the Internal Memory of the element address input dialog.



Recipe Register (RCP)

The 16-Bit / 32-Bit recipe uses **RCP** and enhanced recipe uses **ENRCP**.

The number of recipe buffers occupied by a recipe table is L * (G+1), where G+1 represents recipe buffer. The recipe buffer is arranged in front of the recipe register. This area allows you to place a selected group of recipes. When the number of fields in the recipe buffer is the same as the number of fields in the selected recipe group, it means that the recipe buffer occupies L recipe registers.



	RCPNO Name	1 Unsigned Decimal 1 Word	2 Floating 2 Word	3 Char 5 Word	4 BCD 2 Word
1	1	111	5.55	aaa	11111111
2	2	222	6.66	bbb	22222222
3	3	333	7.77	ccc	33333333
4	4	444	8.88	ddd	44444444
5	5	555	9.99	eee	55555555

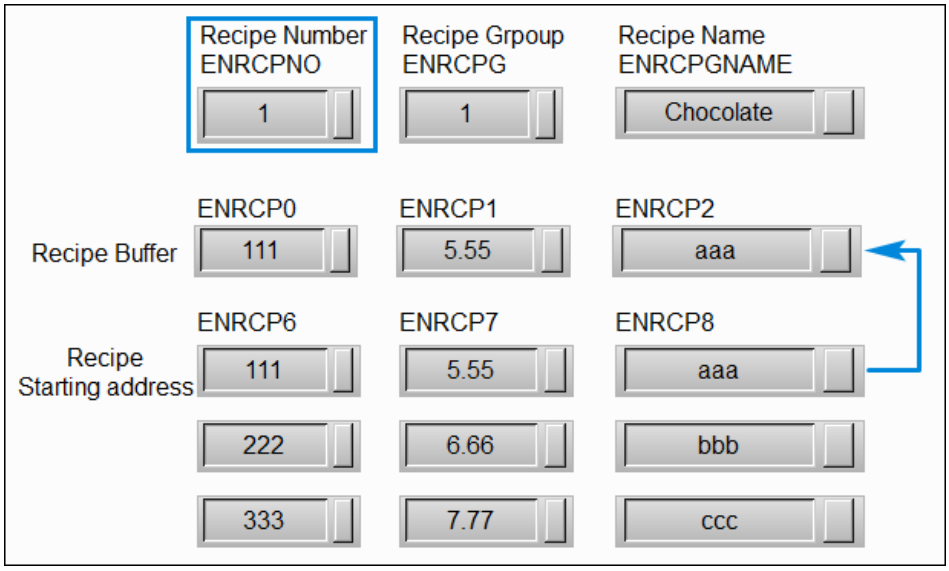
Note: The maximum length of RCP is 65535.

Recipe Number Register (RCPNO)

The 16-Bit / 32-Bit recipe uses **RCPNO** and enhanced recipe uses **ENRCPNO**.

The recipe number register specifies the number of recipes. When reading and writing a recipe, specify the recipe number data through RCPNO and write it to the recipe buffer, and then the recipe buffer reads the data from the specified address or writes the data to the specified address.

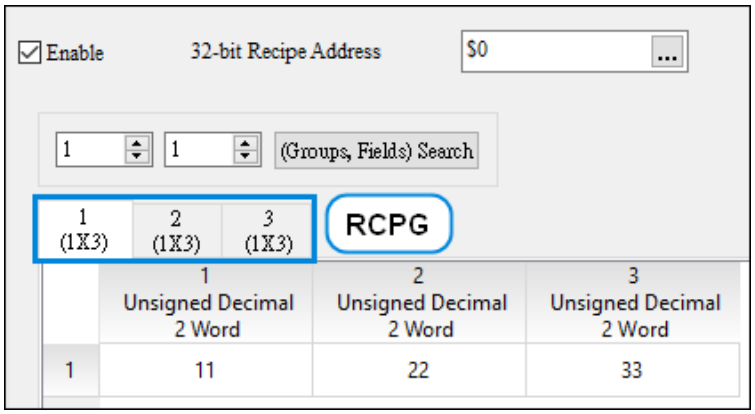
Note: The recipe number register does not have the power-off retention function. When the HMI is powered off, the data in the register cannot be retained.



Recipe Group Register (RCPG)

The 32-Bit recipe uses **RCPG** and the enhanced recipe uses **ENRCPG**.

The recipe group register specifies the recipe group and up to 255 sets of recipe data can be created.

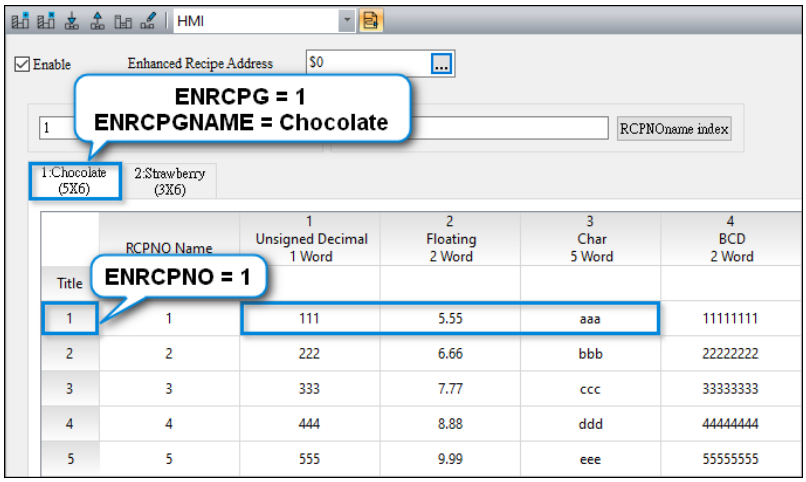


Note: The recipe group register does not have the power-off retention function. When the HMI is powered off, the data in the register cannot be retained.

Enhanced Recipe Group Name Register (ENRCPGNAME)

The enhanced recipe group name register displays the name of the enhanced recipe group register.

ENRCPGNAME names the recipe group name in Unicode, and the name can be entered in multiple languages. With this register, you can enter the recipe name to call the recipe. It must be used with the **Multi-Language Input** element

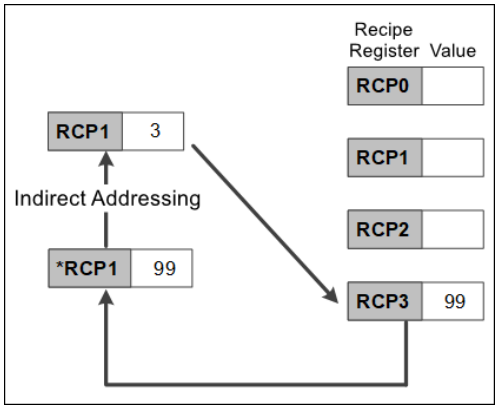


Indirect recipe index register (*RCP)

The 16-Bit / 32-Bit recipe use ***RCP** and enhanced recipe uses ***ENRCP**.

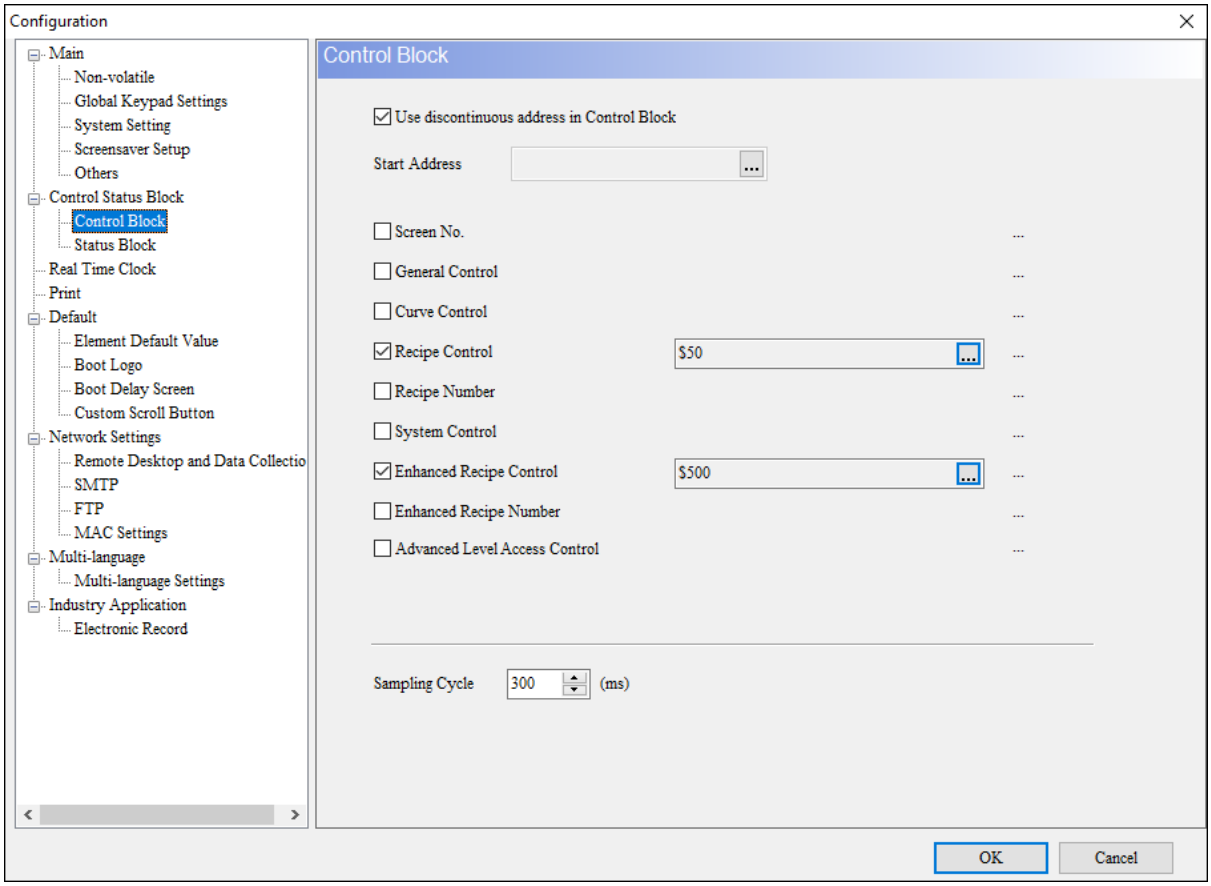
The indirect recipe index register (*RCPn) extracts the value from RCPn and uses this value as a new address, and then accesses the value in this new address. The address access range provided by *RCP is the same as the number of recipe registers, which is L * (G+1).

Note: n is Word (0~65535).



Recipe Operation Example

The reading and writing of recipe are controlled by the control block flag. Go to **General** > **Configuration** > **Control Block** to select the **Recipe Control** or **Enhanced Recipe Control** checkbox to set the control address.



16-Bit Recipe

To set the 16-bit recipe

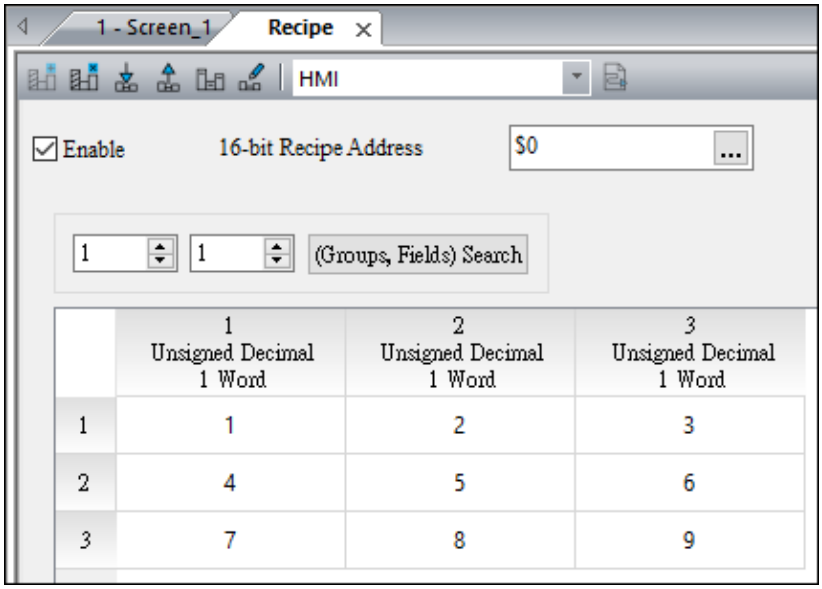
1. Click **Data Management** > **Recipe** > **16-Bit Recipe** on the toolbar.

   Or

   Double-click **Recipe** > **16-Bit Recipe** in the **Project** pane.

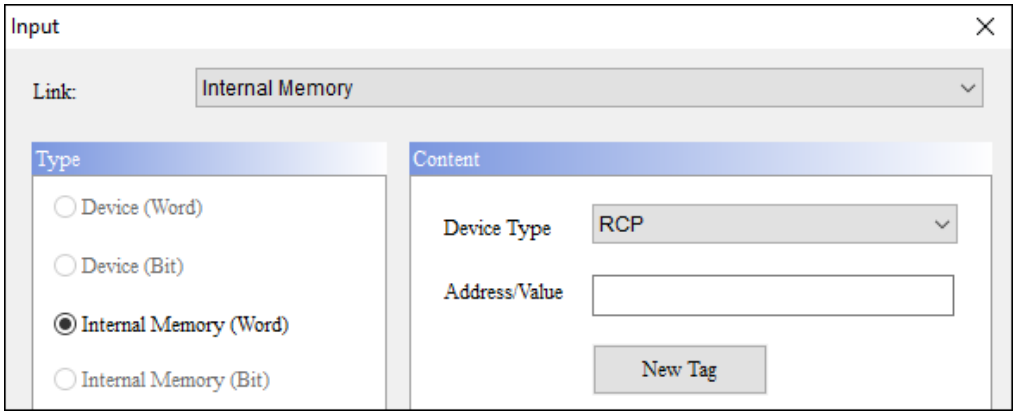
2. Select the **Enable** checkbox in the 16-bit recipe setting page and set **16-bit Recipe Address** to $0.

3. Click to add recipe and set **Length** to **3**, **Group** to **3**, and then enter the recipe data.



To create numeric entry element to read the recipe value

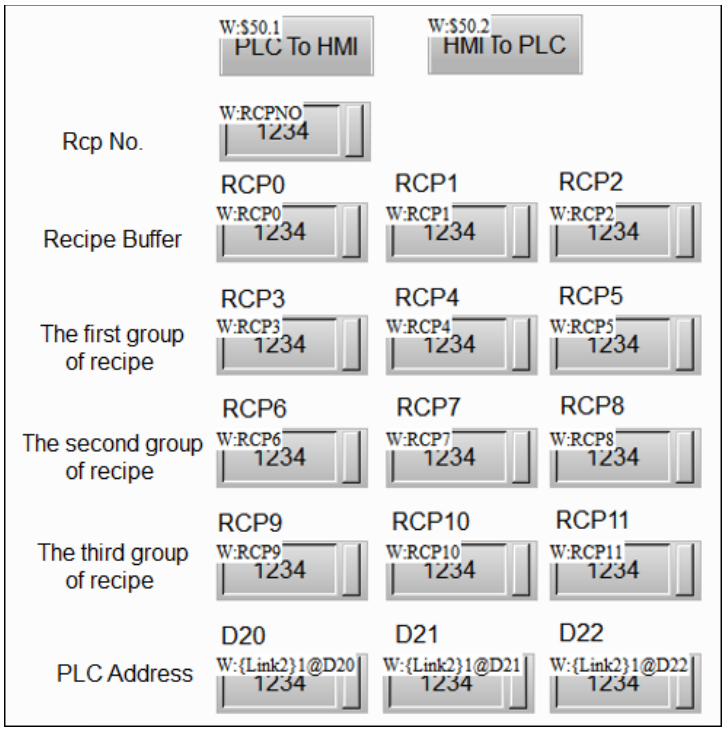
1. Create a **Numeric Entry** element and set the **Write Address** to **RCP** to read the recipe value.



Note: The recipe buffer is RCP0~recipe length (L-1) and the recipe starting address is RCP recipe length (L)。

2. Create a **Numeric Entry** element and set the **Write Address** to **RCPNO** to select the recipe group.
3. Create a **Numeric Entry** element and set the **Write Address** to the external address to set the recipe value.

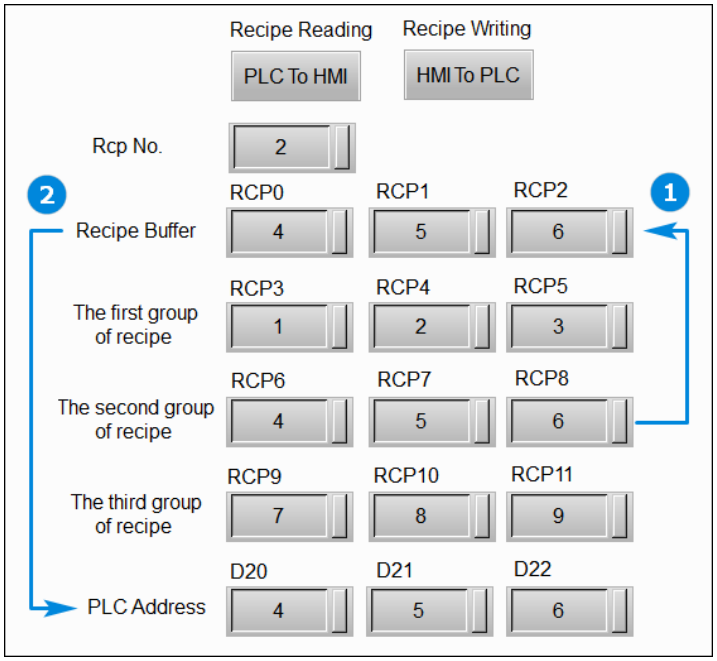
4. Create 2 **Momentary** button element and set the **Write Address** to **$50.1** (recipe reading) and **$50.2** (recipe writing) individually.



Execution Result

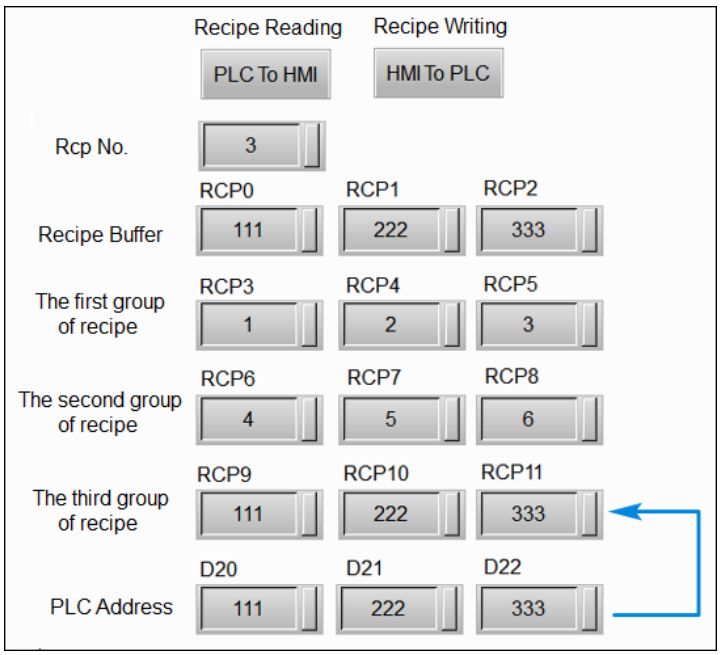
- **Recipe Writing**

1. Set the recipe group to write the recipe value to the recipe buffer.
2. Click **HMI To PLC** to write the recipe buffer data to the PLC address.



- **Recipe Reading**

  1. Enter the value in the PLC address to specify the recipe number.
  2. Click **PLC To HMI** to read back the PLC address data to the specified recipe number.



32-Bit Recipe

To set the 32-bit recipe

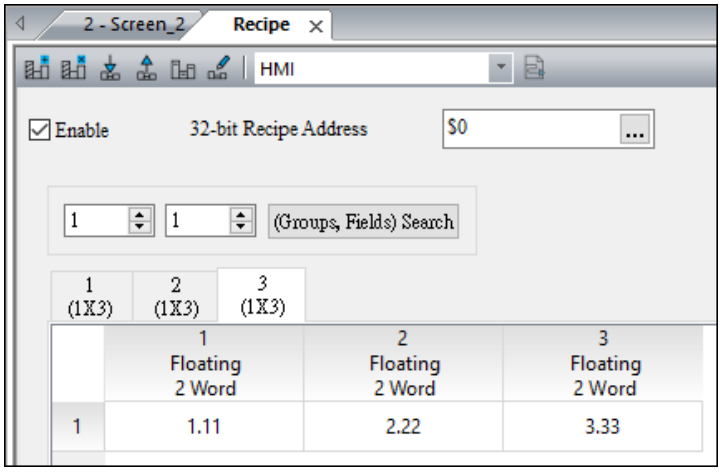
1. Click **Data Management** > **Recipe** > **32-Bit Recipe** on the toolbar.

   Or

   Double-click **Recipe** > **32-Bit Recipe** in the **Project** pane.

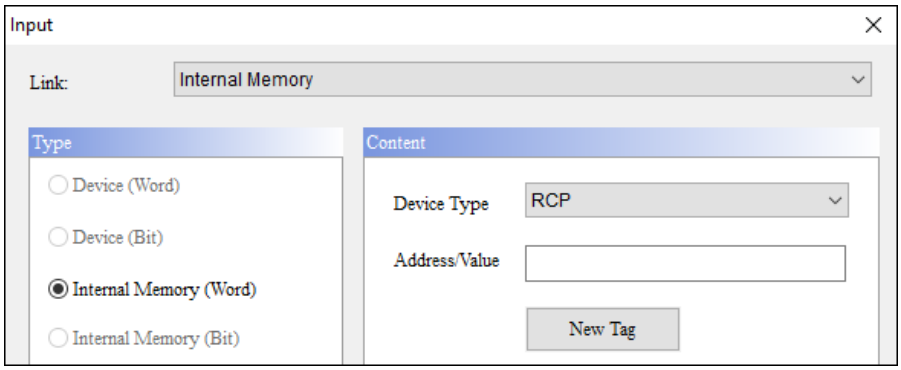
2. Select the **Enable** checkbox in the 32-bit recipe setting page and set **32-bit Recipe Address** to $0.

3. Click to add 3 recipe groups, set **Length** to 3, and select different **Data Format** for these recipe groups, and then enter the recipe data.



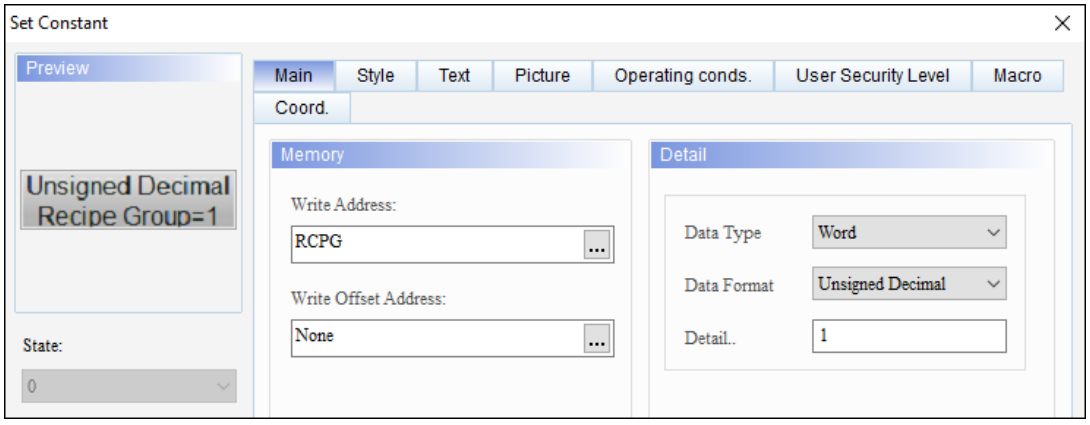
To create numeric entry element to read recipe value

1. Create a **Numeric Entry** element and set the **Write Address** to **RCP**, **Data Type** to **Double Word**, and set the **Data Format** to be the same as in the recipe setting page.

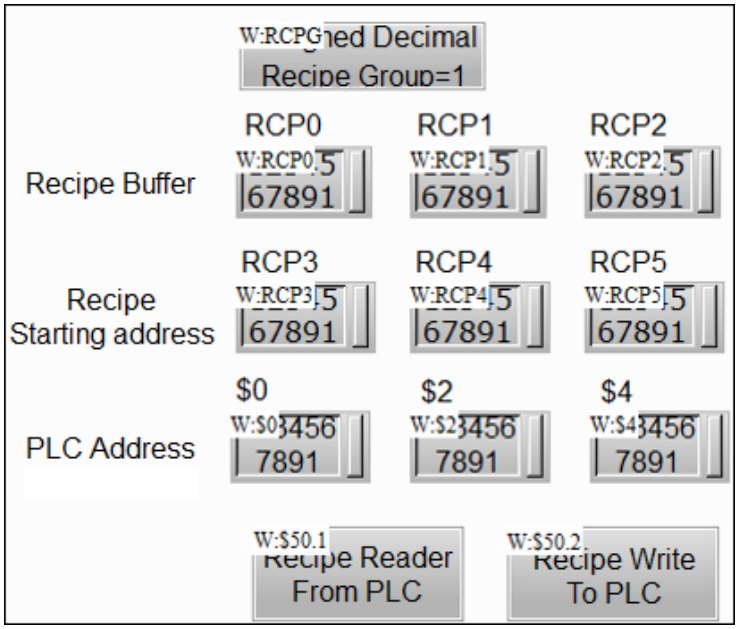


Note: The recipe buffer is RCP0~recipe length (L-1) and the recipe starting address is RCP recipe length (L)。

2. Create a **Set Constant** button element and set the **Write Address** to **RCPG** to specify the recipe group with the set value.



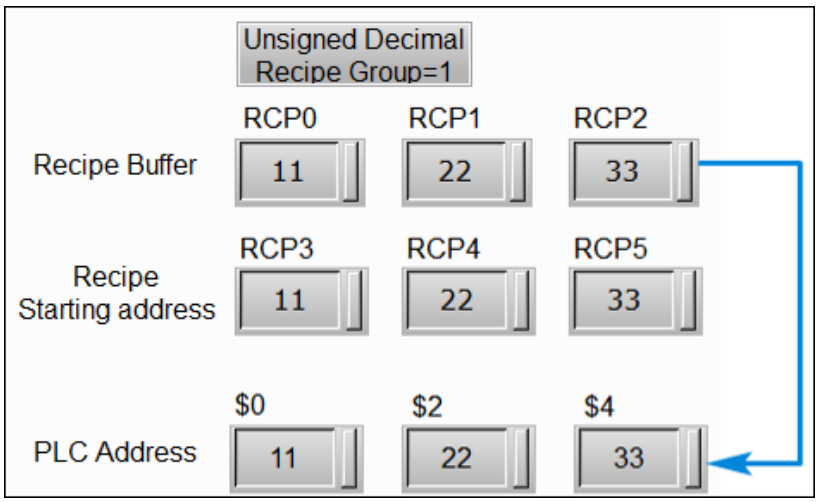
3. Take the recipe reading address as the starting address to set the **Write Address** of these three **Numeric Entry** elements to **$0**, **$2**, and **$4** separately, and then read the recipe value to the memory address.
4. Create 2 **Momentary** button elements and set the **Write Address** to **$50.1** (recipe reading) and **$50.2** (recipe writing).



Execution Result

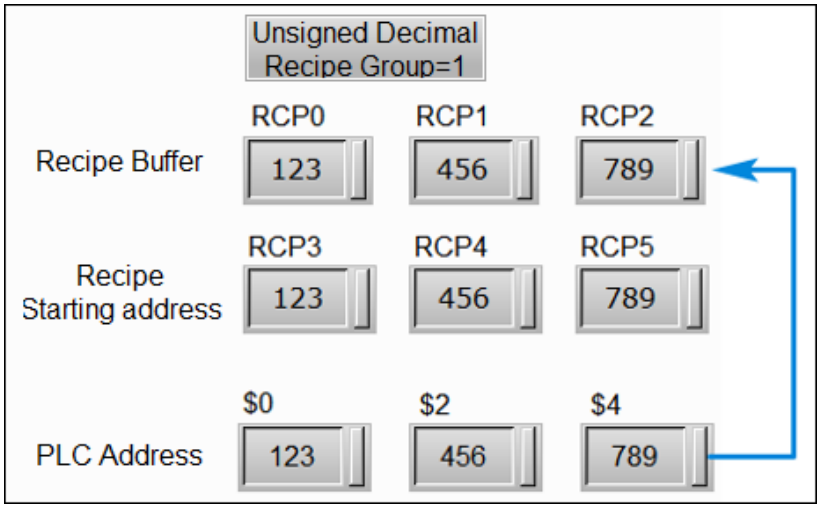
- **Recipe Writing**

1. Click **Recipe Group** to write the recipe value to the recipe buffer.
2. Click **Recipe Write** to write the recipe buffer data to the PLC address.



- **Recipe Reading**

  1. Enter the value in the PLC address to specify the recipe number.
  2. Click **Recipe Read** to read back the PLC address data to the specified recipe number.



Enhanced Recipe

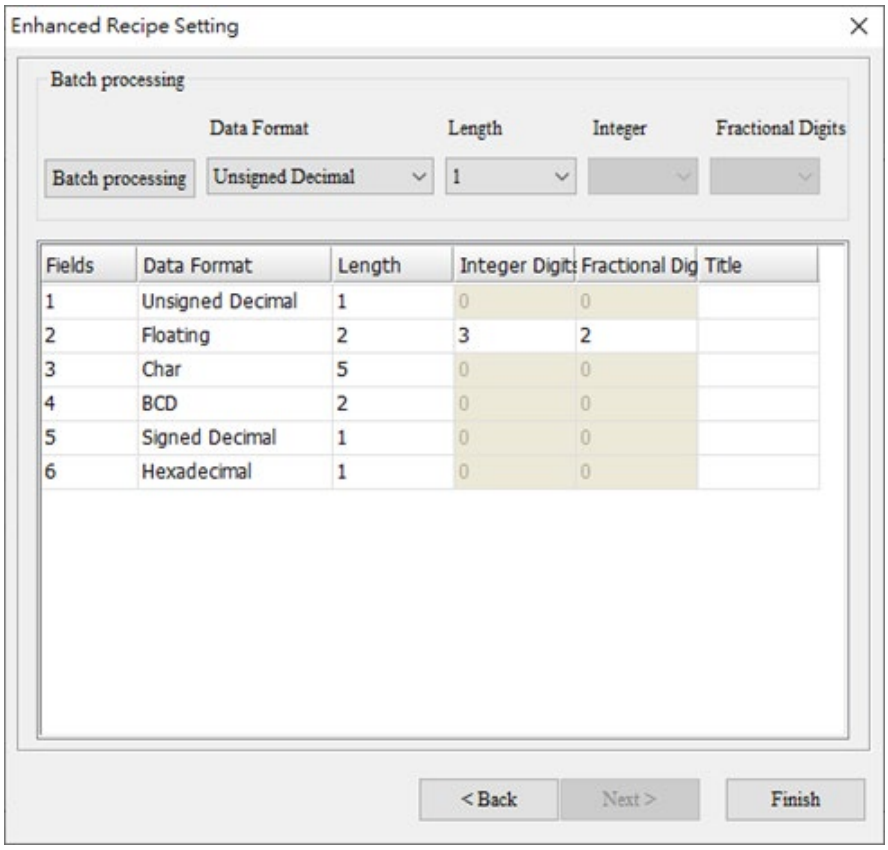
To set the enhanced recipe

1. Click **Data Management** > **Recipe** > **Enhanced Recipe** on the toolbar.

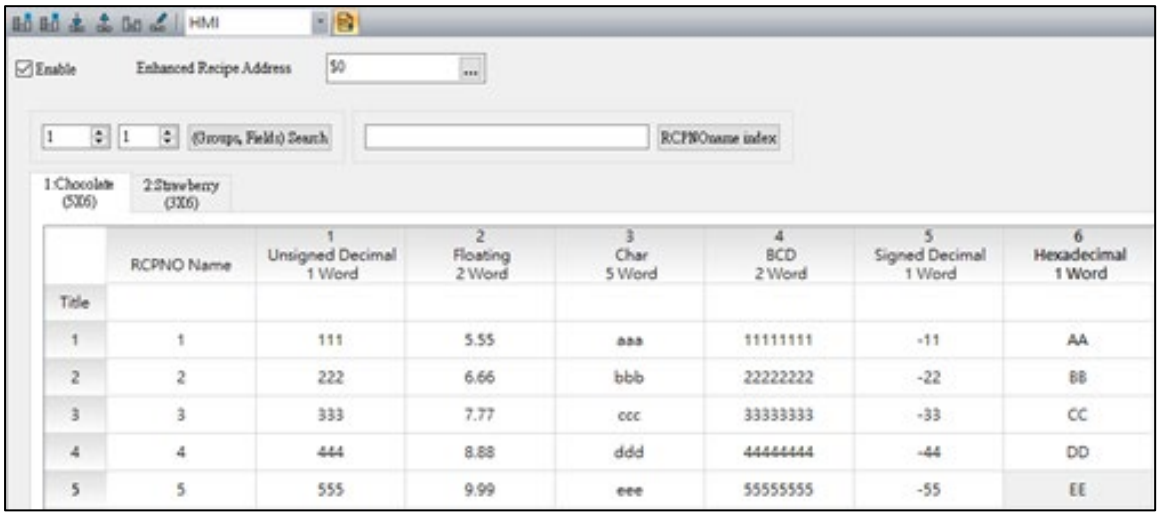
   Or

2. Double-click **Recipe** > **Enhanced Recipe** in the **Project** pane.
3. Select the **Enable** checkbox in the enhanced recipe setting page and set **Enhanced Recipe Address** to $0.
4. Click to add recipe, set **Name** as **Chocolate**, **Field** as **6**, and **Group** to **5**, and then click **Next**.
5. Configure the data format. Do the following:
   - Set the 1st field as **Unsigned Decimal** and **Length** to 1.
   - Set the 2nd field as **Floating**, and **Length** to 2, **Integer Digits** to 3, and **Fractional Digits** to 2.
   - Set the 3rd field as **Char** and **Length** to 5.

- Set the 4th field as **BCD** and **Length** to 2.
- Set the 5th field as **Signed Decimal** and **Length** to 1.
- Set the 6th field as **Hexadecimal** and **Length** to 1.

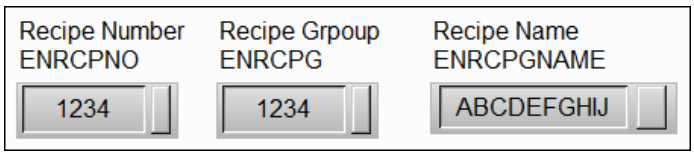


6. Click **Finish** and enter recipe data.



Create element to read the recipe value

1. Create a **Numeric Entry** element and set the **Write Address** to **ENRCPNO** to set the enhanced recipe number.
2. Create a **Numeric Entry** element and set the **Write Address** to **ENRCPG** to set the enhanced recipe group.
3. Create a **Multi-language Input** element and set the **Write Address** to **ENRCPGNAME**, and then set or enter the **String Length** to 10 to enter the recipe name and select the enhanced recipe group name.



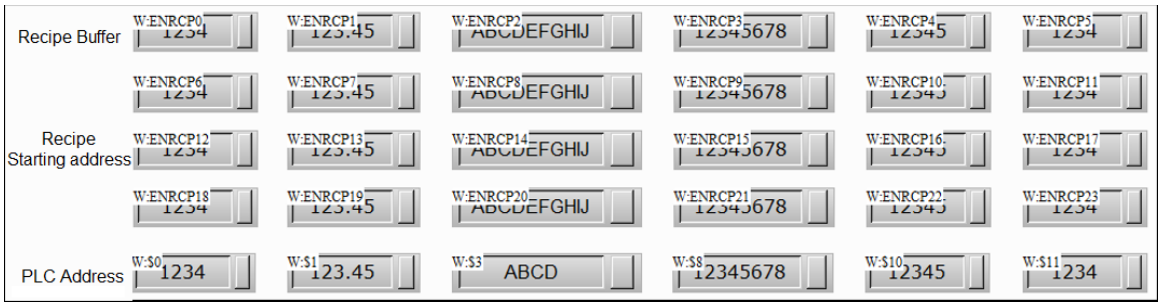
4. Create the reading recipe value of the element. The following table lists the reading element settings corresponding to each field recipe.

Field	Element Setting
1	Use the **Numeric Entry** element, the **Write Address** is ENRCP0, and the **Data Format** is Unsigned Decimal.
2	Use the **Numeric Entry** element, the **Write Address** is ENRCP1, the **Data Format** is Floating, the **Integer Digits** is 3, and the **Fractional** is 2。
3	Use the **Multi-Language Input** element, the **Write Address** is ENRCP2, and the **String Length** is 10. **Note**: One word length can store two characters.
4	Use the **Numeric Entry** element, the **Write Address** is ENRCP3, and the **Data Format** is BCD。
5	Use the **Numeric Entry** element, the **Write Address** is ENRCP4, and the **Data Format** is Signed Decimal。
6	Use the **Numeric Entry** element, the **Write Address** is ENRCP5, and the **Data Format** is Hexadecimal。

Note: The value length of the **Numeric Entry** element refers to the **Length** in the field setting table. Length 1 represents Word and Length 2 represents Double Word.

5. According to the above settings, create elements whose write addresses are ENRCP6~ENRCP23 and PLC address elements starting from the recipe read address, and then read and write the recipe value to the memory address.

Note: The recipe buffer is ENRCP0~ENRCP5, and the actual recipe data is ENRCP6~ENRCP23.

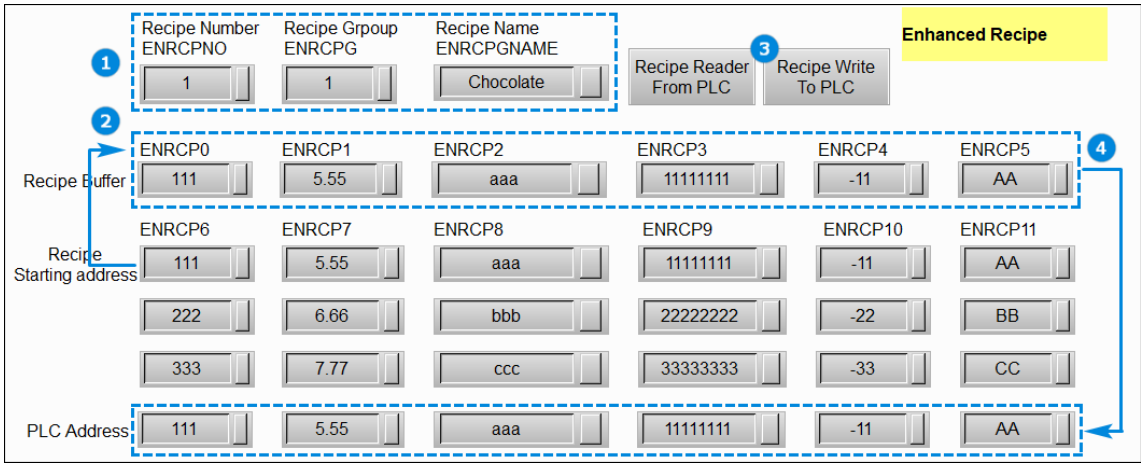


6. Create 2 Momentary button element and set the **Write Address** to **$50.1** (recipe reading) and **$50.2** (recipe writing) individually.

Execution Result

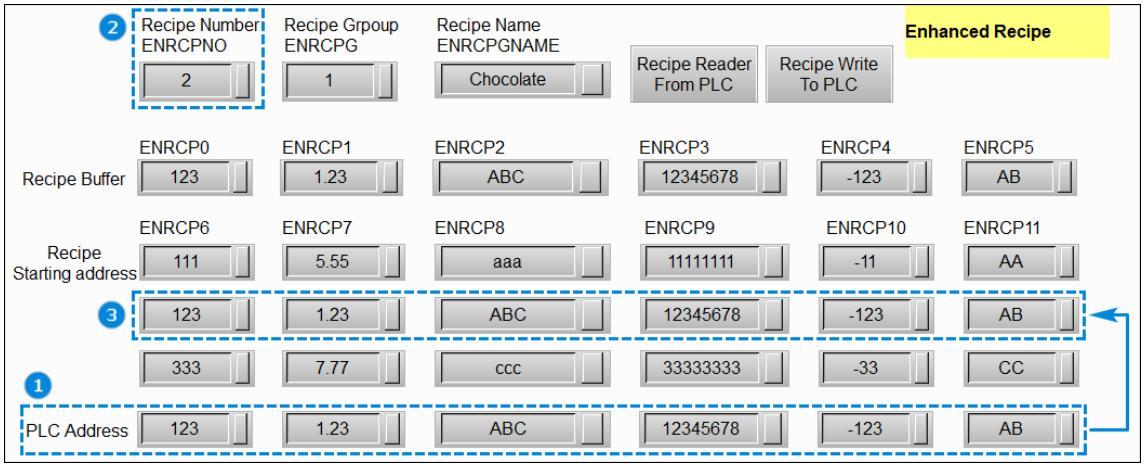
- **Recipe Writing**

  1. Click **Recipe Number** to write the recipe value to the recipe buffer.
  2. Click **Recipe Write** to write the recipe buffer data to the PLC Address.



- **Recipe Reading**

  1. Enter the value in the PLC address to specify the **Recipe Number**.
  2. Click **Recipe Read** to read back the PLC address data to the specified recipe number.

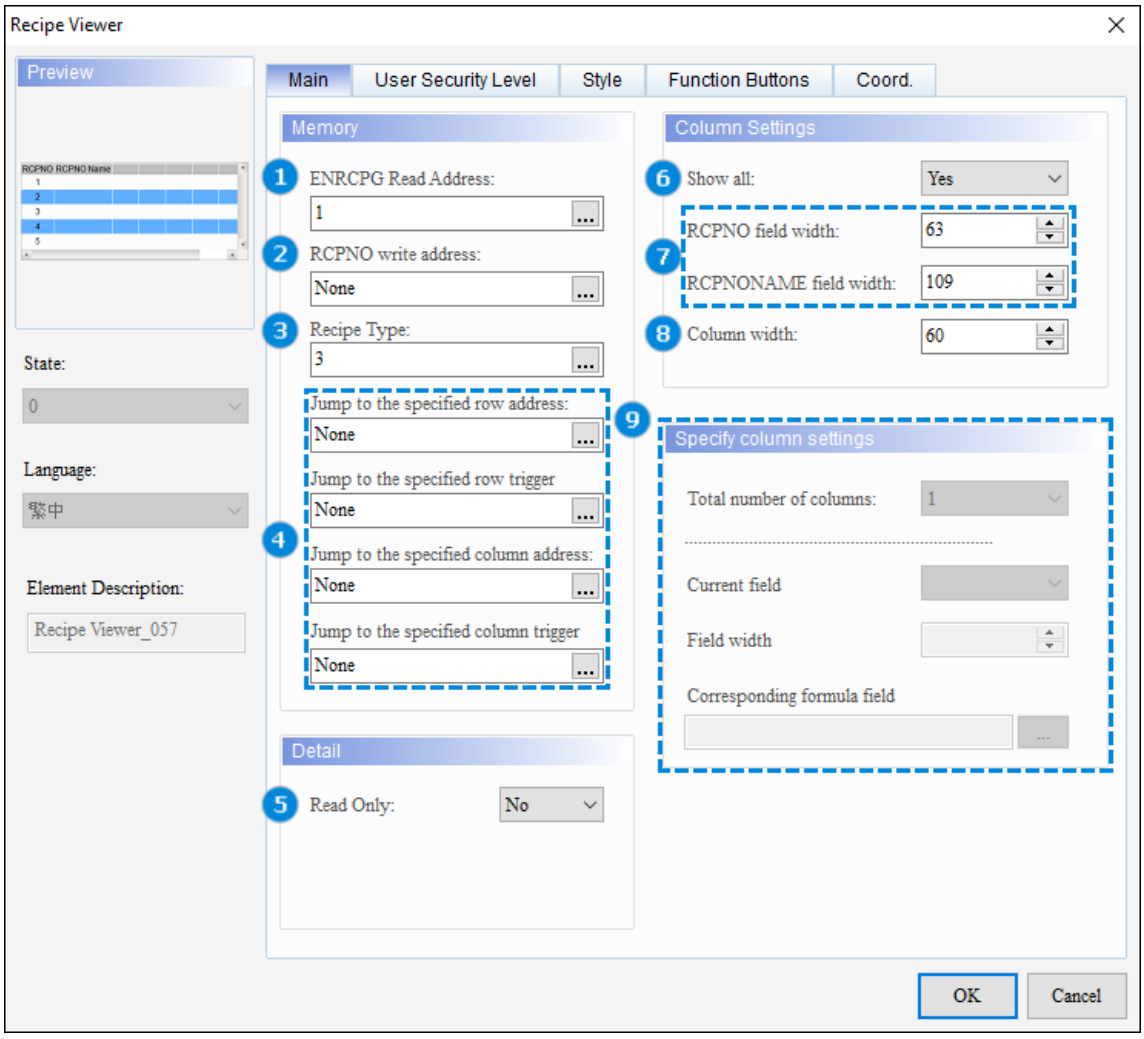

2
Recipe Number
ENRCPNO
2
Recipe Grpoup
ENRCPG
1
Recipe Name
ENRCPGNAME
Chocolate
Recipe Reader From PLC
Recipe Write To PLC
Enhanced Recipe
ENRCP0
ENRCP1
ENRCP2
ENRCP3
ENRCP4
ENRCP5
Recipe Buffer
123
1.23
ABC
12345678
-123
AB
ENRCP6
ENRCP7
ENRCP8
ENRCP9
ENRCP10
ENRCP11
Recipe Starting address
111
5.55
aaa
11111111
-11
AA
3
123
1.23
ABC
12345678
-123
AB
333
7.77
ccc
33333333
-33
CC
1
PLC Address
123
1.23
ABC
12345678
-123
AB

Recipe Viewer

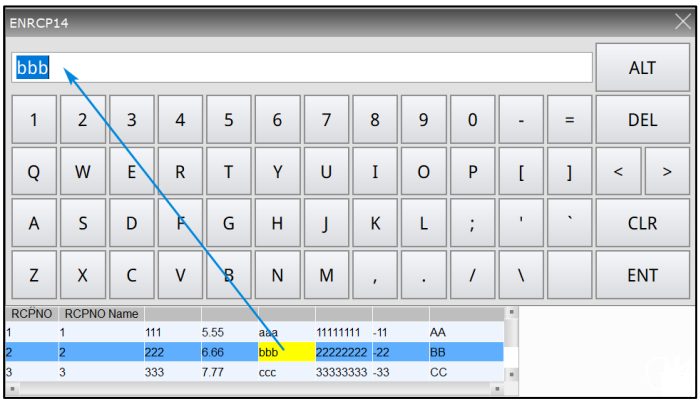
Specify the recipe type to display through the **Recipe Viewer** element and switch the recipe group to display to view the recipe value on the HMI.

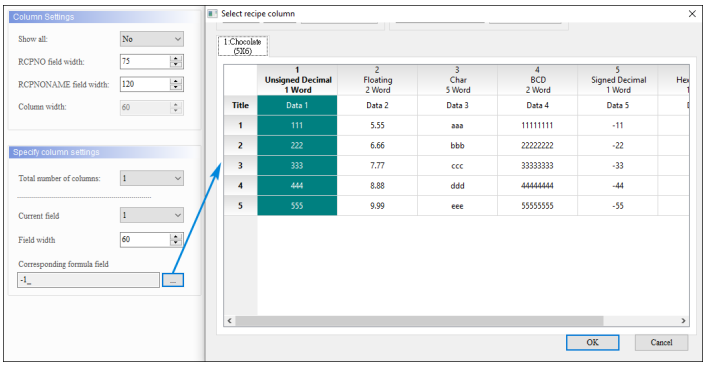
The following table lists the properties of the **Recipe Viewer** element with their description.

- **Main**

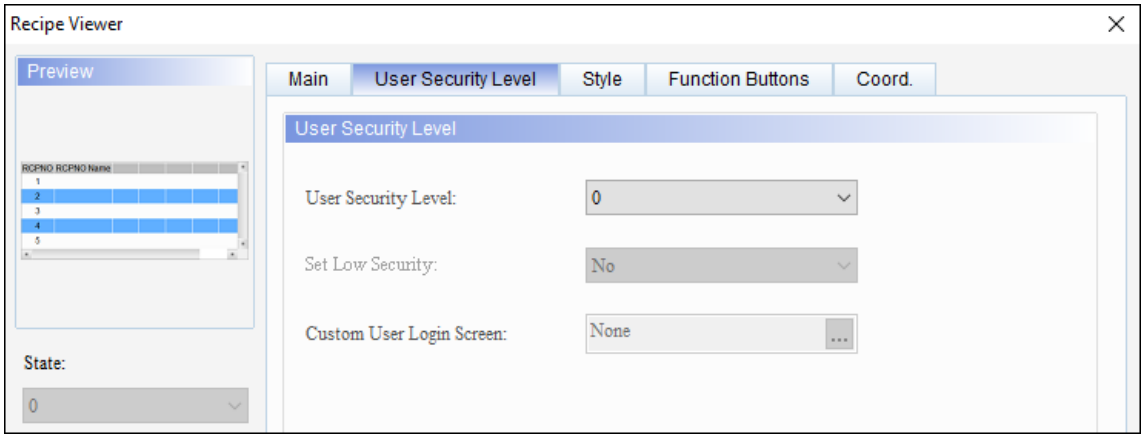


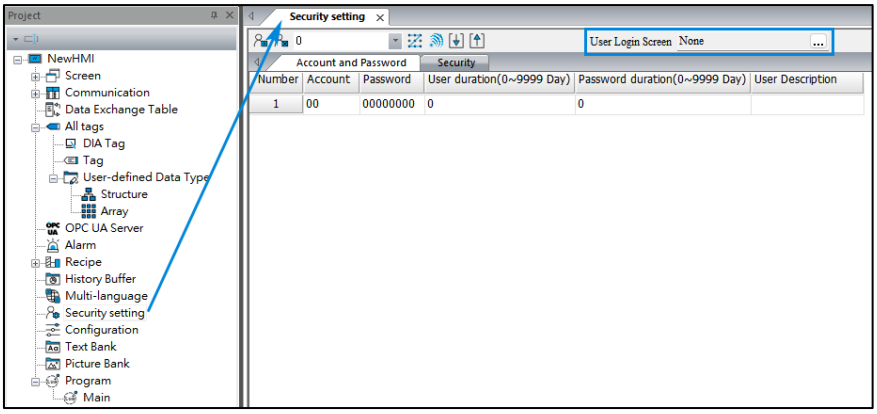
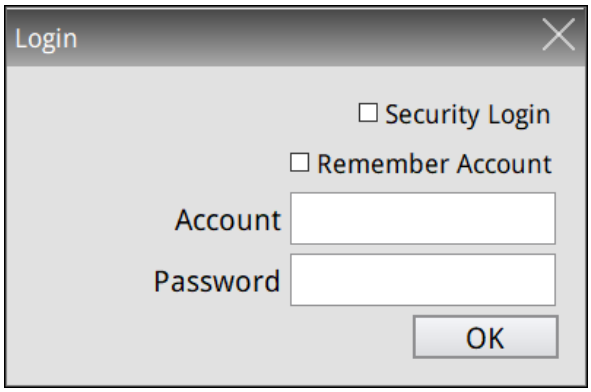
Legend	Setting	Description
1	ENRCPG Read Address	Set a constant to specify the recipe group (RCPG), and this constant is limited by the number of recipe groups, or set memory address to dynamically specify RCPG on the HMI. **Note**: The **16-Bit Recipe** can only be set to 0.

Legend	Setting	Description
②	RCPNO write address	Set the address to read selected RCPNO. Click the RCPNO field of the Recipe Viewer element on the HMI to write the selected RCPNO to the specified address. **Note**: If the element is set to read-only, writing address has no effect.
③	Recipe Type	Set the recipe type to view. The number indicates as follows: • 1: 16-Bit recipe • 2: 32-Bit recipe • 3: Enhanced recipe
④	Jump to the specified row / column address Jump to the specified row / column trigger address	Specify the number of rows and columns through **Jump to the specified row / column address** and trigger the **Jump to the specified row / column trigger address** to On, and the element displays the specify number of rows and columns.
⑤	Read Only	Choose whether to edit the recipe content on the **Recipe Viewer** element. If choose **No**, the corresponding keypad will be displayed according to the data format of the field for you to edit the recipe value after click the recipe field on the HMI. 
⑥	Show all	When the **Recipe Type** is 3, representing specifying as Enhanced Recipe and when the **ENRCPG Read Address** I set to a constant value, you can choose whether to display all recipes. If choose **No**, **Total number of columns** can be selected, and recipe column can be specified to display.

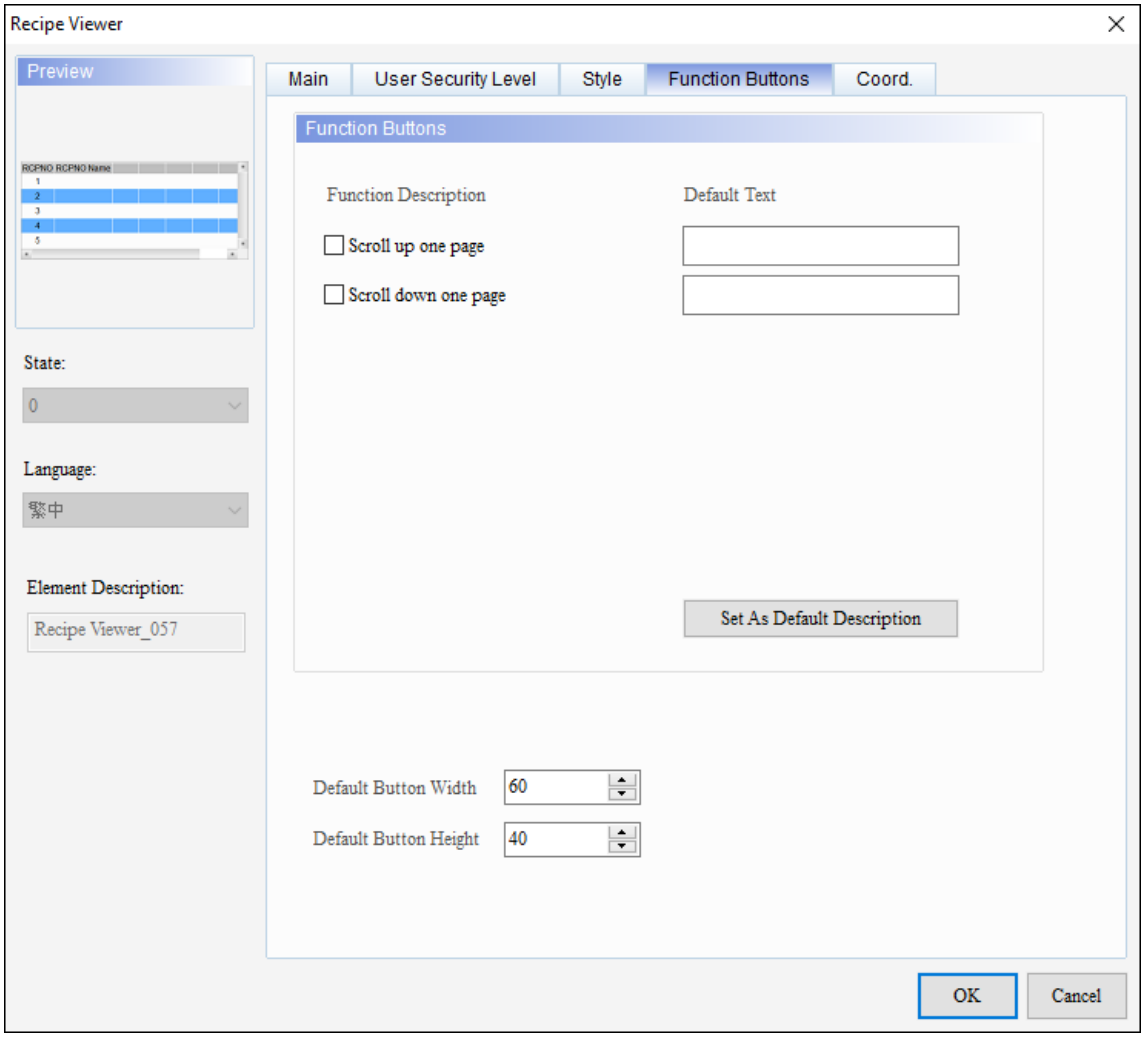
Legend	Setting	Description
❼	RCPNO field width RCPNONAME field width	Select or enter the field width for RCPNO and RCPNONAME.
❽	Column width	When **Show all** is set **Yes**, you can select or enter the column width for the recipe value.
❾	Specify column settings	When **Show all** is set to **No**, you can select **Total number of columns** and switch **Current field** to set the column width and corresponding recipe column. Click **Corresponding formula field** to select the display data for the current field in the **Select recipe column** dialog.  **Note**: This setting is only available for **Enhanced Recipe**.

- **User Security Level**



Setting	Description
User Security Level	Select the user security level. **Note**: When **Read Only** is set to **0**, if the current security level is insufficient after clicking the recipe field on the HMI, a password input dialog will appear. The recipe value can only be written if log in with the higher user security level.
Custom User Login Screen	Set the custom login screen. When **User Security Level** is not set to **0**, the custom login screen can be selected. When triggering the element and the user security level is required the set screen pops up. • Go to **General** > **New Screen** > **User Login Screen** > **Login/Logout Screen** to set the login screen. • If the element does not have a specified login screen, you can go to **Project** > **Security Setting**. Set the **User Login Screen** of each security level in the **Security Settings** page. When the element sets this security level, the login screen configured in this page is displayed.  • If the element or account does not set the login screen, the default login screen is displayed. 

- **Function Buttons**



Setting	Description
Function Description	Select to add function buttons.
Default Description	Enter the description of the function button. Click **Set As Default Description** to insert the default strings to the input fields. You can switch between the different languages to define the default text.
Default Button Width	Select or enter the width of the function buttons. The value range is 25–100, and the default value is **60**. After adjusting the width, right-click the element on the screen and select **Rearrange Function Buttons**, all function buttons display according to the settings.
Default Button Height	Select or enter the width of the function buttons. The value range is 25–100, and the default value is **40**. After adjusting the height, right-click the element on the screen and select **Rearrange Function Buttons**, all function buttons display according to the settings.

Operation Log Setup

The **Operation Log Setup** is used to record the operation of elements on the HMI and the operation date / time of element. The operations include value change, security level change, and bit switching. This function can also be used to find machine malfunctions or analyze production defects.

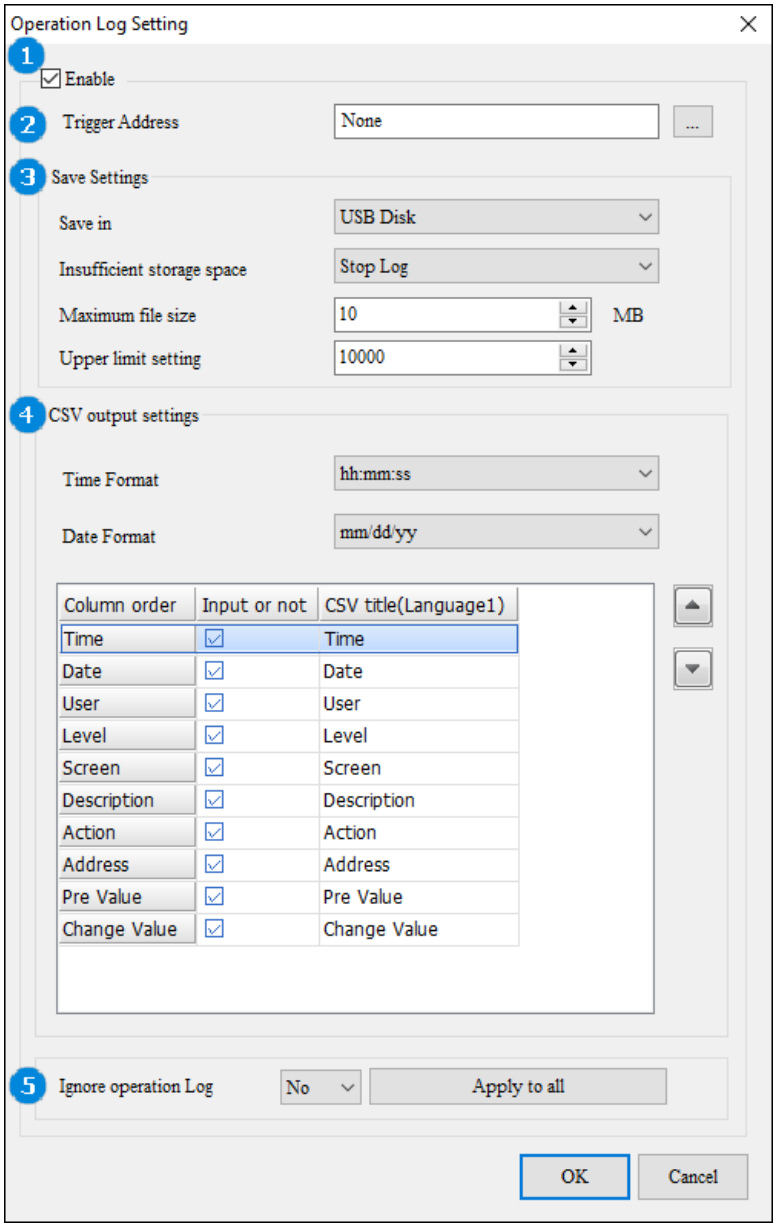
Note:

- The default operation log is a CSV file to record 10,000 pieces of data.
- If data is stored in the HMI, up to 10,000 pieces of data can be stored. If data is stored in an external storage device (USB disk or SD card), when the data reaches the default upper limit, the HMI will overwrite the file or stop recording according to the storage settings.

How to open the **Operation Log Setting** function?

- Click **Data Management** > **Operation Log Setting**.

The following table lists the setting in the **Operation Log Setting** dialog with their description.



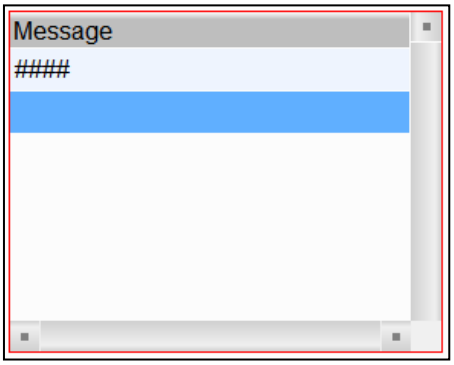
Legend	Setting	Description
1	Enable	If selected, the operation log settings are available. Not enabled by default.
2	Trigger Address	Set the trigger address of the operation log. You can set the **Internal Memory** or external controller address (only Bit triggering is supported). After triggering this address, the operation log starts recording all operations on the HMI.
3	Save Settings	Select the location where to store the CSV format operation log file. When the external storage space is insufficient, you can select Stop Log to stop recording the HMI operations, or **Overwrite Files** to delete the recorded operation data and record the operations

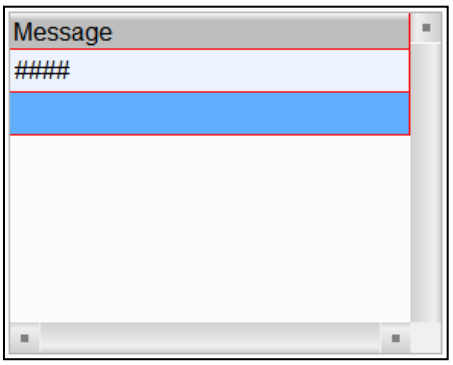
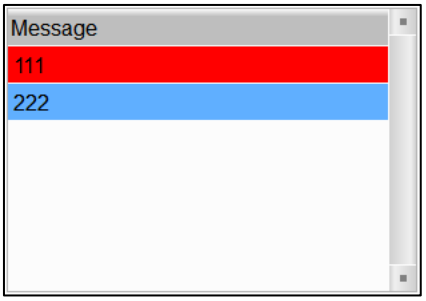
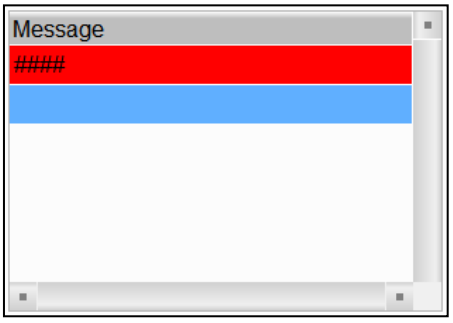
Legend	Setting	Description
		again. You can also set the upper limit of files and the maximum file size. **Note**: The options for **Save in** vary depending on the selected model.
④	CSV output settings	Select the time and date format to be recorded in the output CSV file and the columns to be displayed. You can adjust the order of the columns and set the CSV title to display according to your need.
⑤	Ignore operation Log	Choose whether to ignore the operation log. The default value is **No**.

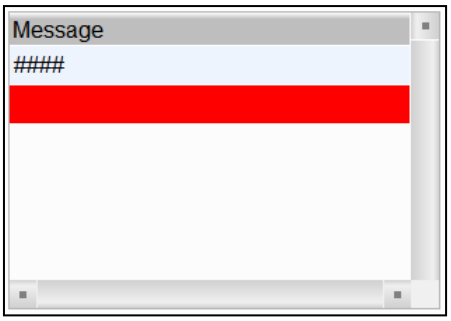
Operation Log

The following table lists the settings in the **Operation Log Setup** dialog with their description.

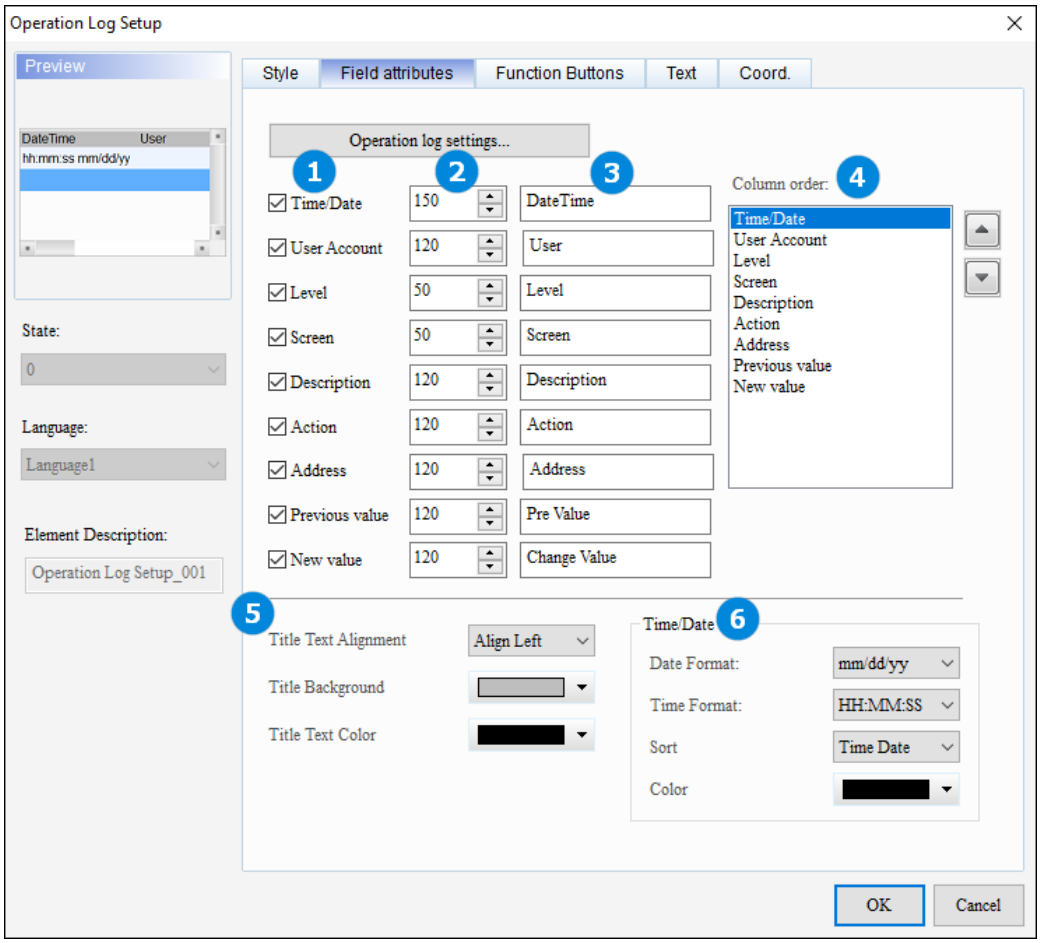
- **Style**

Setting	Description
Background Color	Set the background color of the element. The default is white. When set to red, it displays as follows.
Border Color	Set the border color of the element. The default is gray. When set to red, it displays as follows. 

Setting	Description
Gridline Color / Show Gridlines	Set the gridline color of the element. The default is white. When set to red, it displays as follows. **Note**: The **Gridline Color** setting is valid only when you select **Yes** for **Show Gridlines**. 
Select Row Color	Set the color of the selected row of data when operating the element on the HMI. The default is blue. When set to red, it displays as follows. 
Row Color	Set the color to display for each row of the alarm. The default is light blue. When set to red, it displays as follows 

Setting	Description
Alternating Row Color	Set the color for the alternating row of the alarm. The default is blue. When set to red, it displays as follows. 
Transparency	Set the transparency of the element. The range value is 50–255, and the default value is **255**. The smaller the value, the more transparent the element is.

- **Field attributes**



Legend	Setting	Description	
1	Column display	Select the columns to be displayed in the element.	
2	Column width	Select or enter the width for each column.	
3	Column title	Enter the titles for each column.	
4	Column order	After selecting the columns to be displayed, click ▲ ▼ to adjust the column displaying order.	
5	Title bar	Title Text Alignment	Select the title bar text as left, center or right.
		Title Background	Select the background color of the title bar.
		Title Text Color	Set the text color of the title bar.
6	Date / Time	Date Format	Set the display date format.
		Time Format	Select the display time format.

Legend	Setting	Description	
		Color	Select the display color of date and time. The default is black.

- **Function Buttons**

  You can add function buttons to control the display of the alarm table.

Setting	Description
Function Description	Select to add function buttons.
Default Description	Set the description of the function button. Click **Set As Default Description** to insert the default strings to the input fields. You can switch between the different languages to define the default text.
Default Button Width	Set the width of the function buttons. The value range is 25–100, and the default value is **60**. After adjusting the width, right-click the element on the screen and select **Rearrange Function Buttons**, all function buttons display according to the settings.
Default Button Height	Set the width of the function buttons. The value range is 25–100, and the default value is **40**. After adjusting the height, right-click the element on the screen and select **Rearrange Function Buttons**, all function buttons display according to the settings.

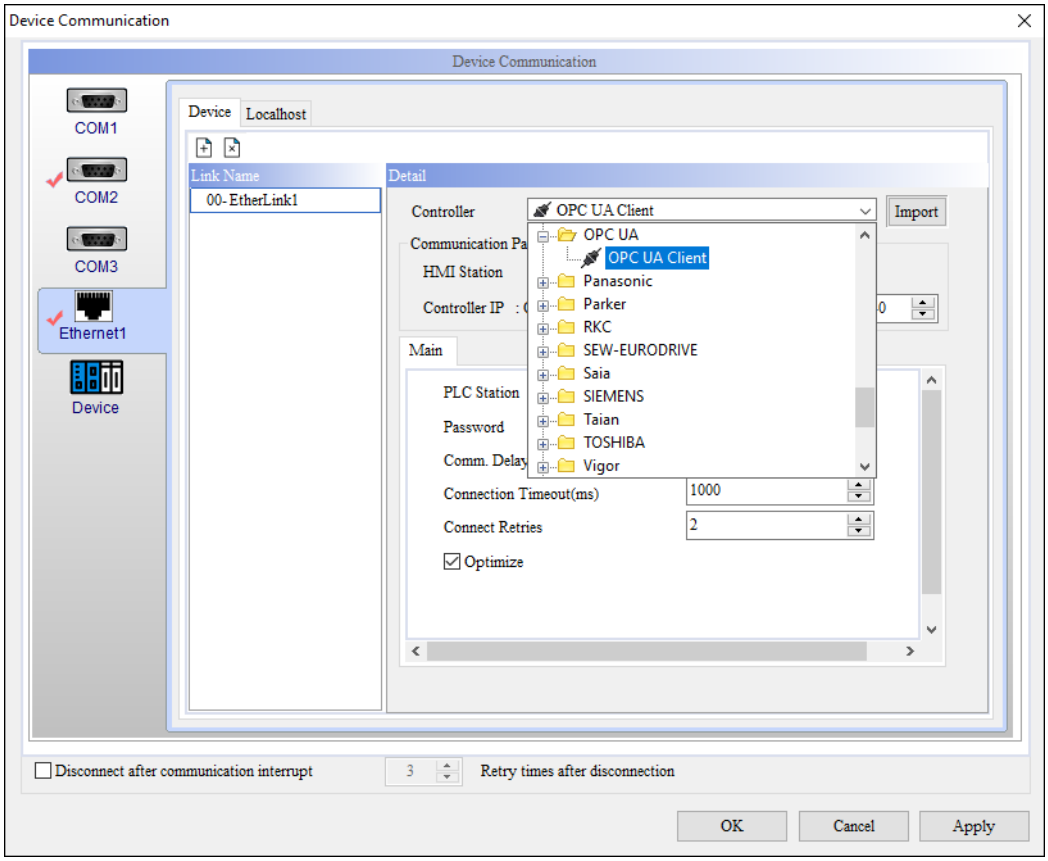
OPC UA

Use the OPC UA Client function to import tags from files or use Ethernet to connect to the OPC UA Server. This section introduces the setting instructions for OPC UA Client and OPC UA Server.

OPC UA Client Settings

To set OPC UA Client

1. Create a project for a model that supports Ethernet.
2. On the menu bar, click **General** > **Device Communication**.
3. In the **Device Communication** dialog, click **Ethernet1**.
4. In the **Device** tab, click to add a new network connection, and then select the controller as **OPC UA Client**.



To import OPC UA Tag

- **Import from file**

1. In the **Device Communication** dialog, set the controller as **OPC UA Client**, and click **Import**.

2. In the **Tag List** dialog, click to import tag (XML or CSV file).

   Note:

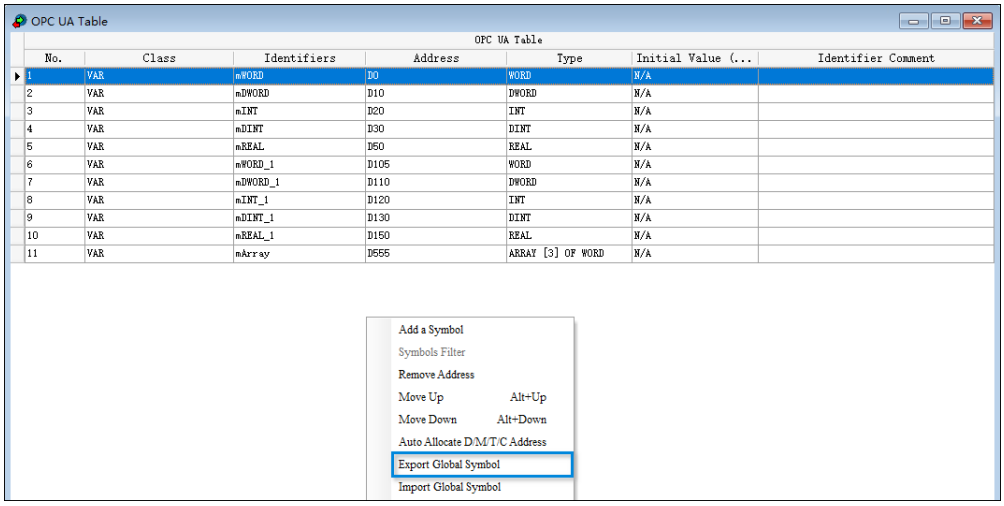
   - The .xml file is a file exported by the OPC UA server.
   - The .csv file is a file exported by ISPSoft software. Follow the steps to export a .csv file.

     a. In ISPSoft software, in **Project** pane, double-click **Global Symbols** > **OPC UA Table**.

     b. In the **OPC UA Table** dialog, right-click an empty space and select **Add a Symbol** to create the symbol.

     c. Export the global symbol table. Do any of the following:

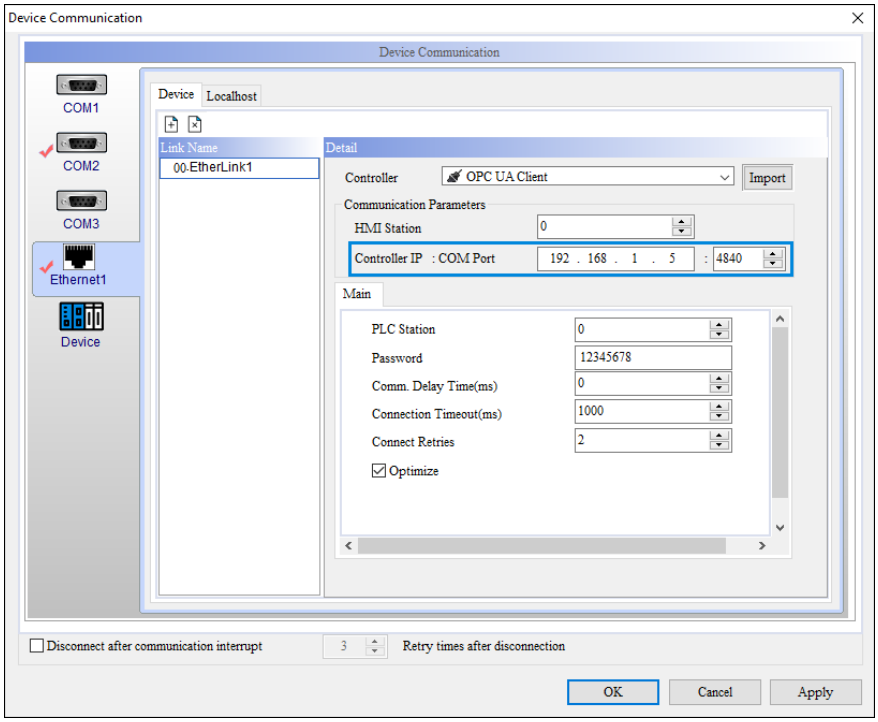
     - Right-click an empty space and select **Export Global Symbol**.
     - In **Project** pane, right-click **OPC UA Table** and select **Export Global Symbol**.



Note: The OPC UA Server function is currently only available for AS-300 series models. It requires an additional AS-FOP02 communication card and ISPSoft version 3.13 and above.

- **Import Tags from OPC UA Server**

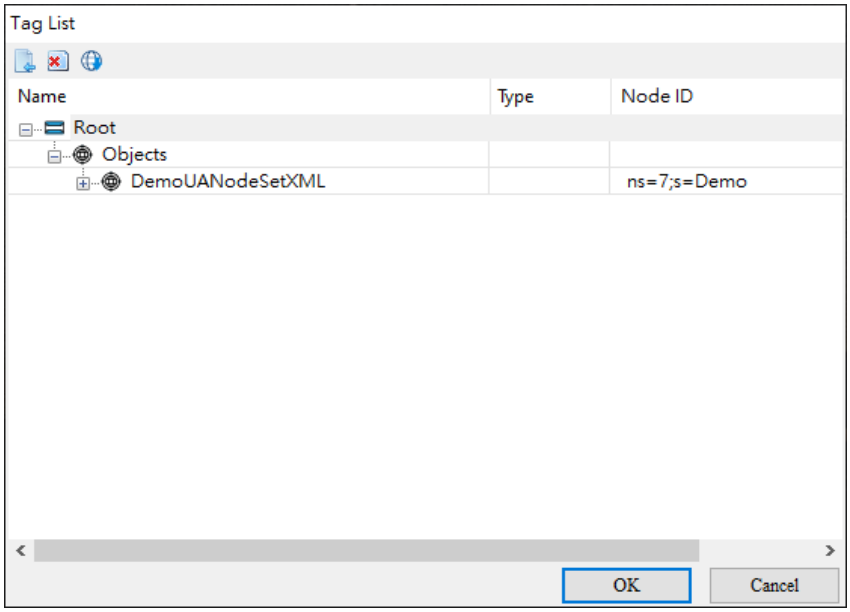
1. In the **Device Communication** dialog, after setting **Controller** as **OPC UA Client**, enter IP address and communication port of the OPC UA server in **Controller IP** field.



2. Click **Import**.

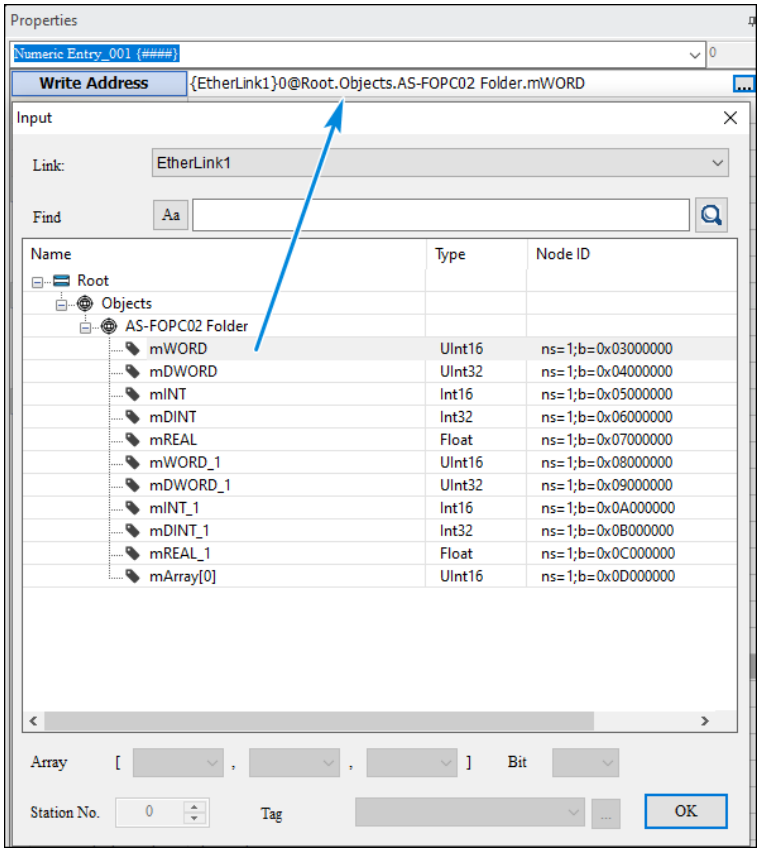
3. In the **Tag List** dialog, click .

   The Tag has been imported to the OPC UA Server through the set IP address. The tag list can be viewed.

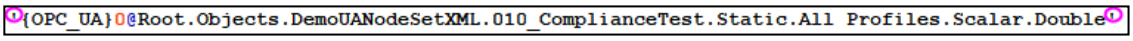


Using OPC UA Tag

- Used for elements: In the **Read Address** / **Write Address** dialog, select **EtherLink1** for **Link**.



- Userd for macro commands: In the macro editing dialog, click and enter macro command. If the variable content selects OPC UA Tag, OPC UA Tag automatically adds single quotes after updating the macro; if you use to input OPC UA Tag, you need to manually add single quotes before and after the OPC UA Tag.



The following table lists the supported OPC UA Tag NodeID types and OPC UA Tag data types.

OPC UA Tag NodeID Type Support List		
NodeID (Identifier Type)	**Scalar**	**Array**
I (Numeric)	V	V
B (Opaque)	V	-
S (String)	V	V
G (GUID)	V	-
Data Type	**Scalar**	**Array**
Boolean	V	V
Byte	V	V
Int32	V	V
Int16	V	V
Int64	V	V
Double	V	V
Float	V	V
SByte	V	V
String	V	V
UInt32	V	V
UInt16	V	V
UInt64	V	V

OPC UA Server Settings

Set the communication address as an OPC UA variable through the OPC UA Server.

Note: This function only supports DOP-107WV, DOP-103WQ, DOP-110WS, DOP-112MX, DOP-115MX, DOP-112WX, DOP-115WX, and DOP-300 series models.

How to open **OPC UA server**?

- In **Project pane**, double-click **OPC UA Server**.

Or

- Click **IIoT** > **OPC UA Server** on the menu bar.

The following table lists the functions in the **OPC UA Server** dialog with their description.

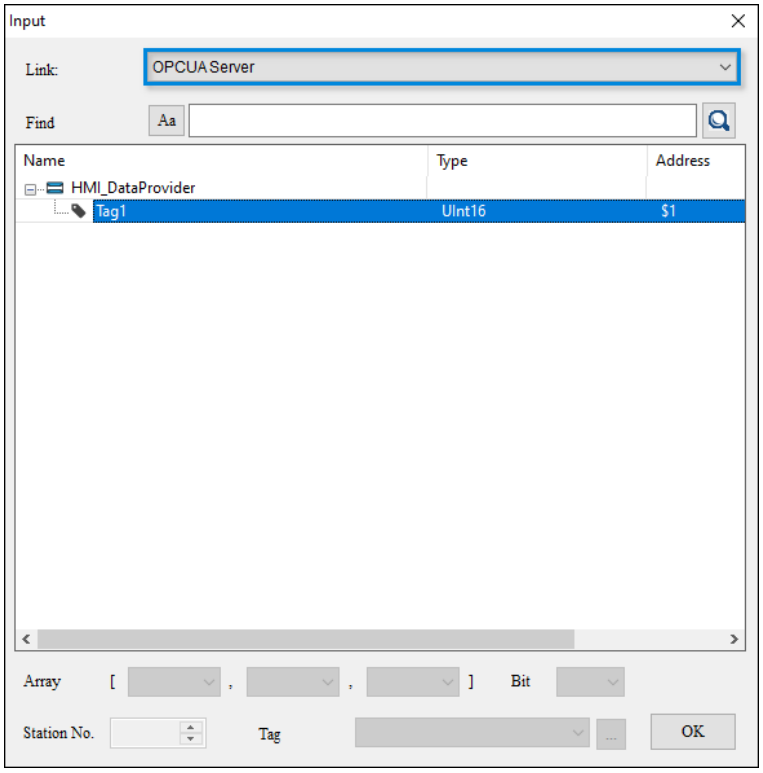
Function	Description
	Click to add a new group.
	Click to add a new tag. **Note**: Address is required. If not set, a message appears to inform you that the address input failed.
	Select the tag or group you want to delete, then click this icon to delete.
	Select the tag or group you want to edit, then click this icon to edit its properties. **Note**: You can double-click a tag or group to edit properties.

Follow these steps to apply the OPC UA server tag to the element.

To apply OPC UA server tag to element

1. Create an element.
2. In the address **Input** dialog of the element, select **OPCUA Server** for **Link**.

   Tags display in the list below.
3. Select the tag you want to apply to the element, and click **OK**.



OPC UA Security Policy and User Authentication

In OPC UA (Open Platform Communications Unified Architecture), security can be ensured through account authentication and encrypted communication. The account authentication is the most basic and simple. Enter the user's account name and password and have the server verified the credentials and allow access if they match.

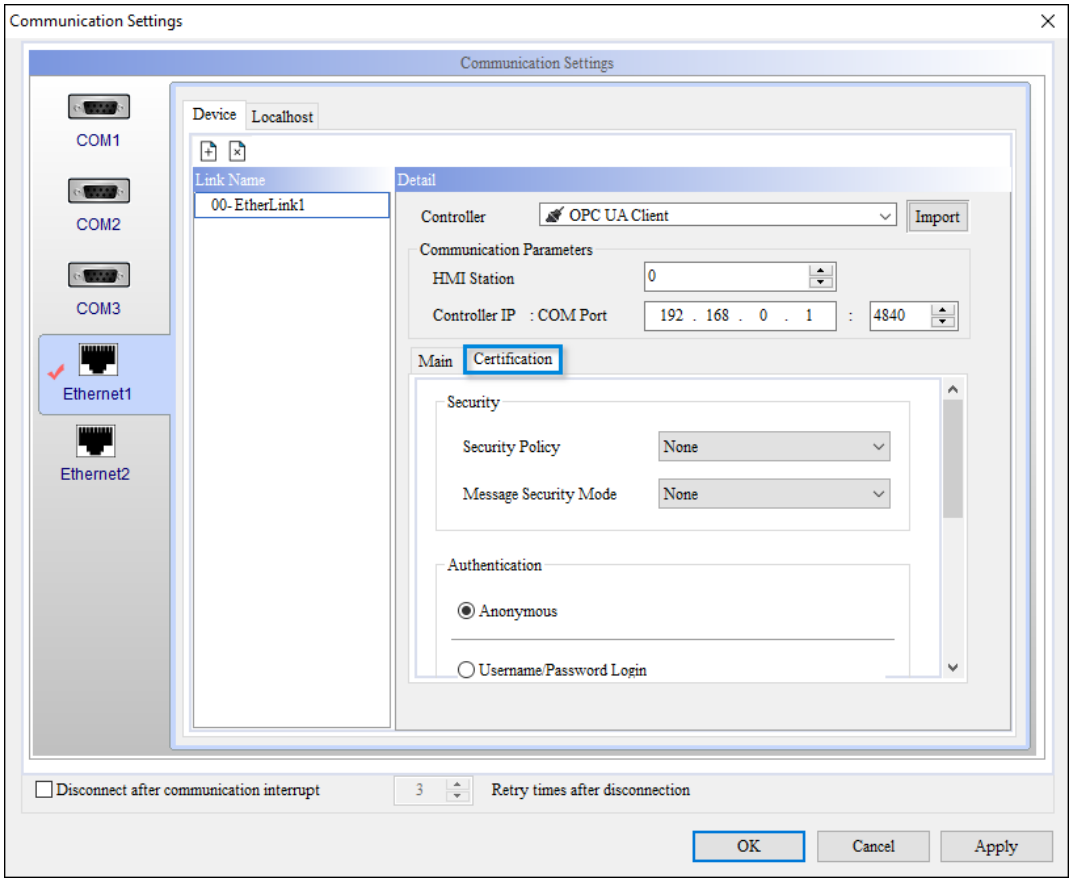
Note: This function is only available for DOP-300 series models.

Follow these steps for OPC UA authentication.

- **OPC UA Client**

  1. In the **Device Communication** dialog, click **Ethernet1**.
  2. In the **Device** tab, select **OPC UA Client** for **Controller**, and the **Certification** tab displays.

The following table lists the functions in the **Certification** tab with their description.



❶ Security

Function	Description
Security Policy	Select a security policy. The available options include None, Basic128Rsa15, Basic256, Basic256Sha256, Aes128Sha256RsaOaep, and Aes256Sha256RsaPss.
Message Security Mode	Select a message security mode. The available options include None, Sign, and Sign & Encrypt.

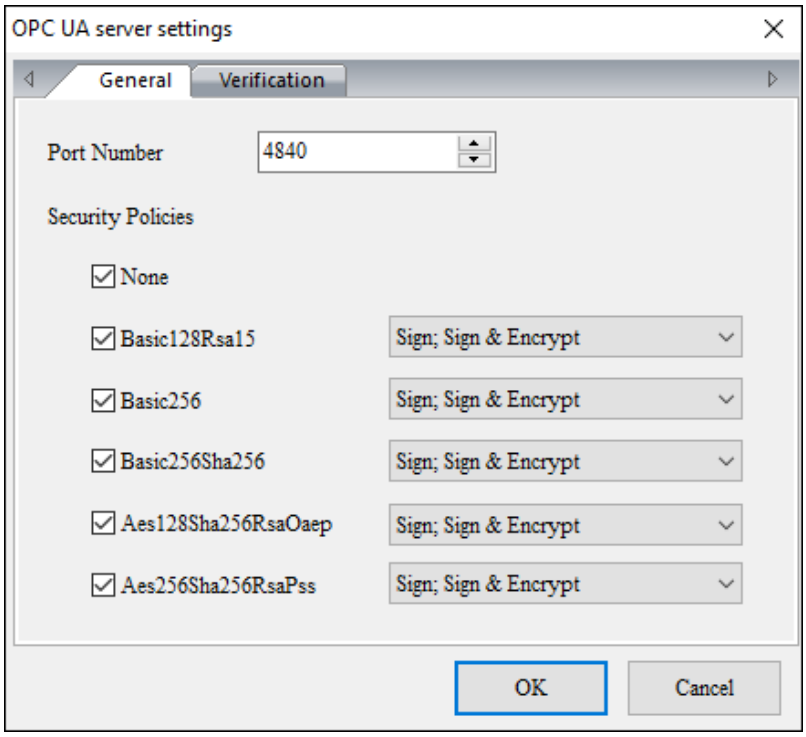
❷ Authentication

Function	Description
Anonymous	Choose anonymous login or user login.
Username/Password Login	

- **OPC UA Server**

In the **OPC UA Server** dialog, click to open **OPC UA server settings** dialog.

The following table lists the functions in the **OPC UA server settings** dialog with their description.



Function		Description
General	Port Number	Select or enter the port number. The range value is 0–65535, and the default value is **4840**.
	Security Policies	Check the security policy and select message security mode. • Security policy options include None, Basic128Rsa15, Basic256, Basic256Sha256, Aes128Sha256RsaOaep, and Aes256Sha256RsaPss. • Message security mode options include None, Sign, and Sign & Encrypt
Verification	Allow anonymity	If selected, check at least one of the options - Browse, Read, or Write.
	Username & Password	If selected, Name and Password should not be vacant.

MQTT Settings

HMI sends messages through an object to the MQTT server or subscribes to a topic from the MQTT server. It can also be an MQTT server. When the HMI is an MQTT server, the messages will not be sent to other servers.

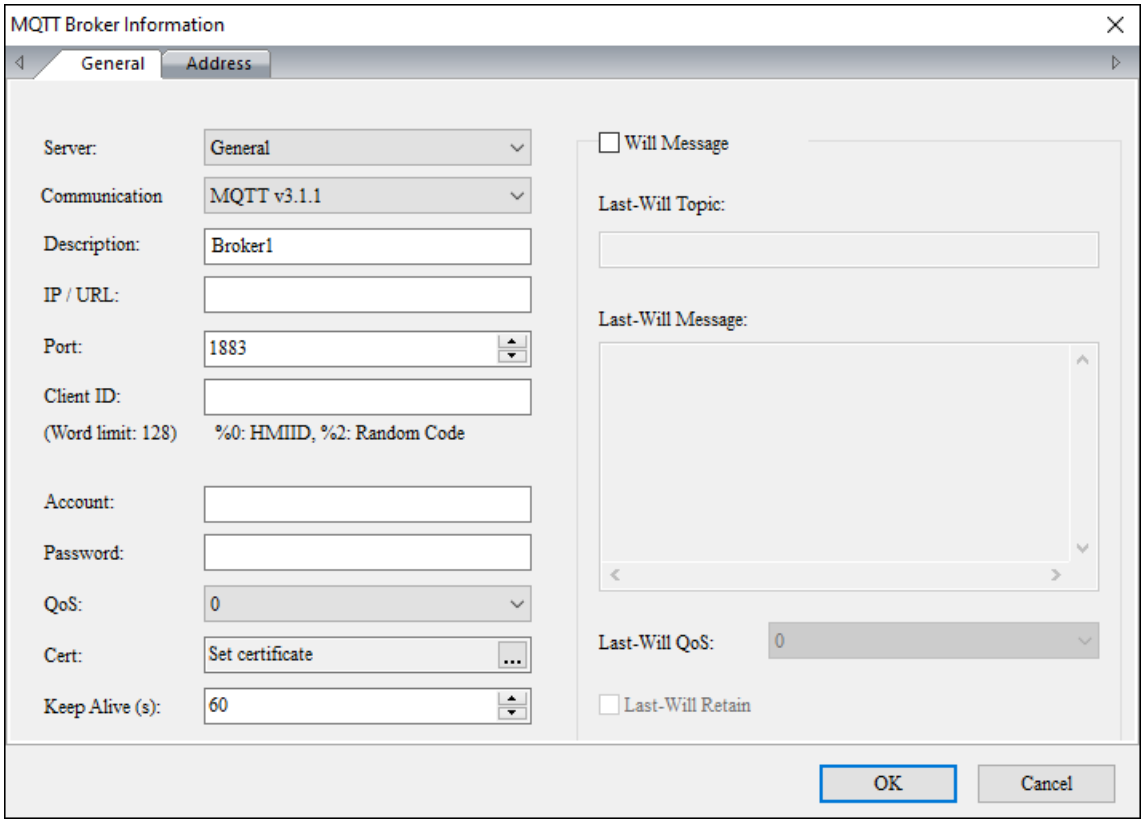
MQTT Broker

You need to add a Broker to enable MQTT functions.

To add a Broker

1. Click **IIoT** > **MQTT Settings** on the toolbar.
2. In the **MQTT Settings** dialog, click to add a Broker on the toolbar.

   Note: Supports up to 5 MQTT Broker for one HMI.
3. In the **MQTT Broker Information** dialog, enter the information for the Broker in the **General** tab.



Note: In the MQTT specification, each client will have a Client ID, allowing the broker to determine where the data is sent, but if the Client ID is duplicated, the client that connects later will fail to connect. In order to avoid duplicate client IDs, HMI provides two variables, %0 (HWID) and %2 (Random Code) settings, where HWID is an internal parameter and is a unique identification code in the HMI, and different machines will have their own HWID.

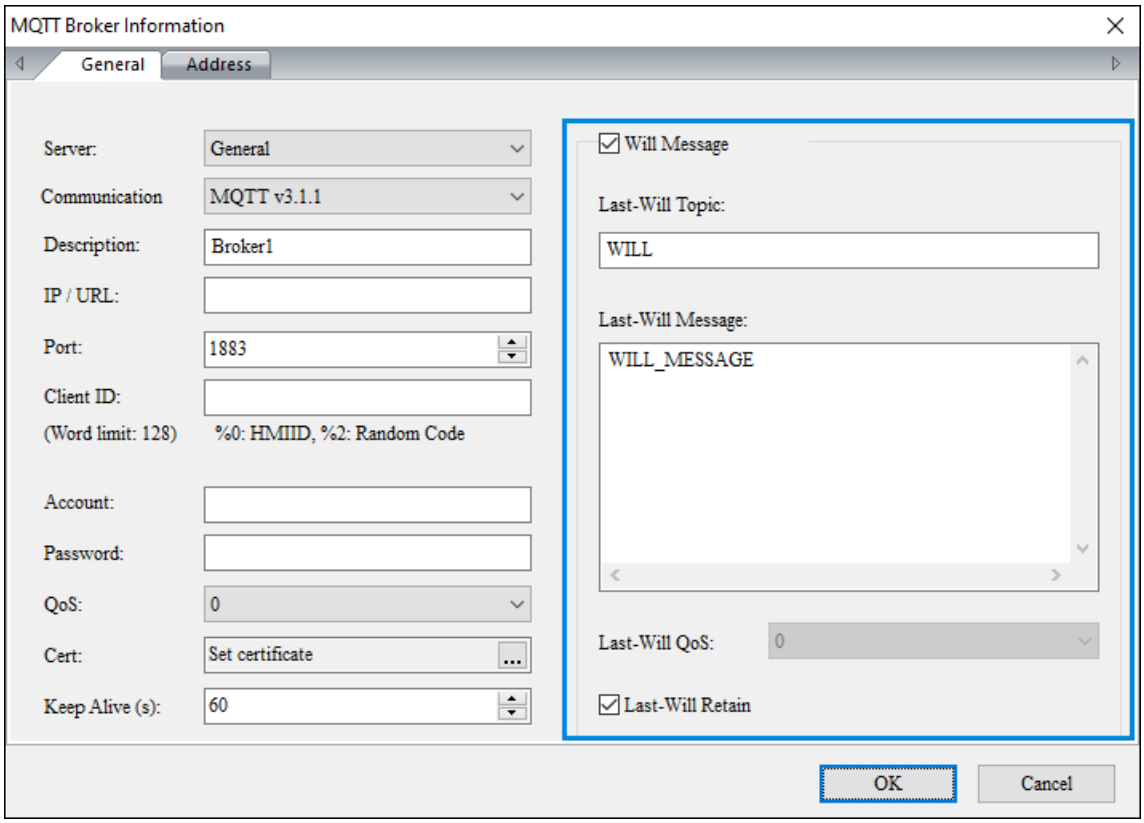
Will Message

The Will Message is a simplified version of the PUBLISH message. It is specified through the CONNECT packet when the device is connected to the server. The server publishes Will Message to the Will Topic specified when the device is unexpectedly disconnected, the server must complete the storage of the Will Message before responding CONNACK to ensure that the message is published.

Follow these steps to set up Will Message.

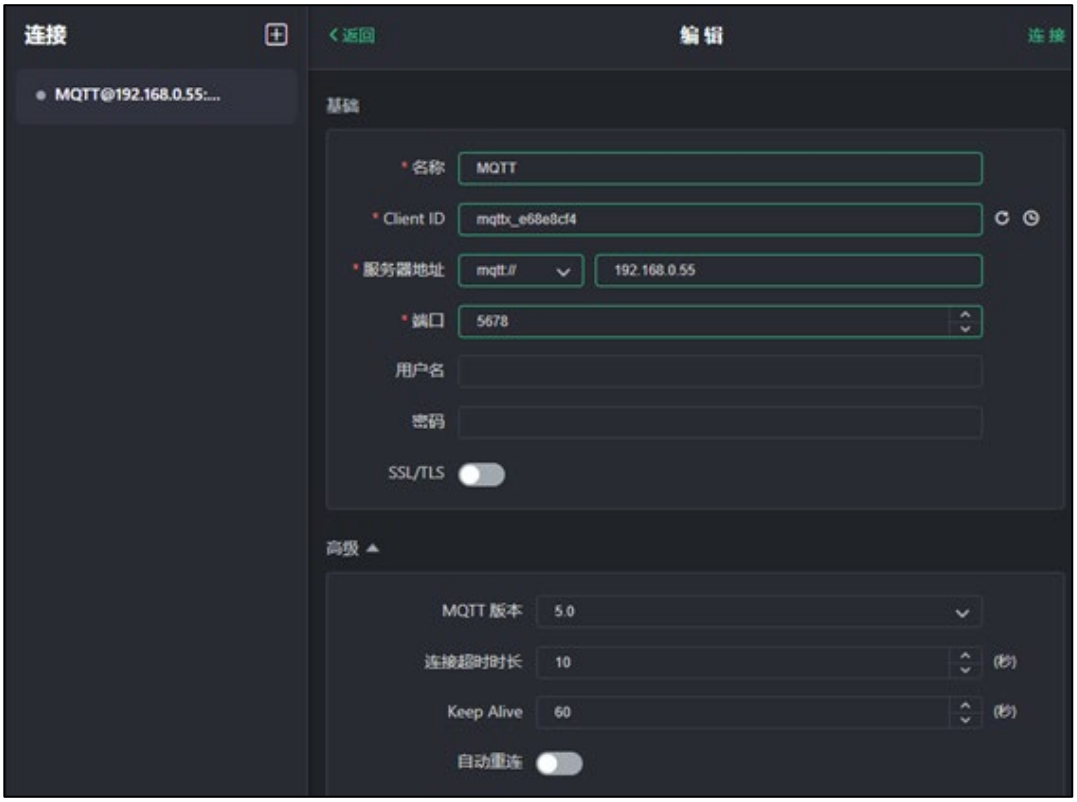
To set up will message

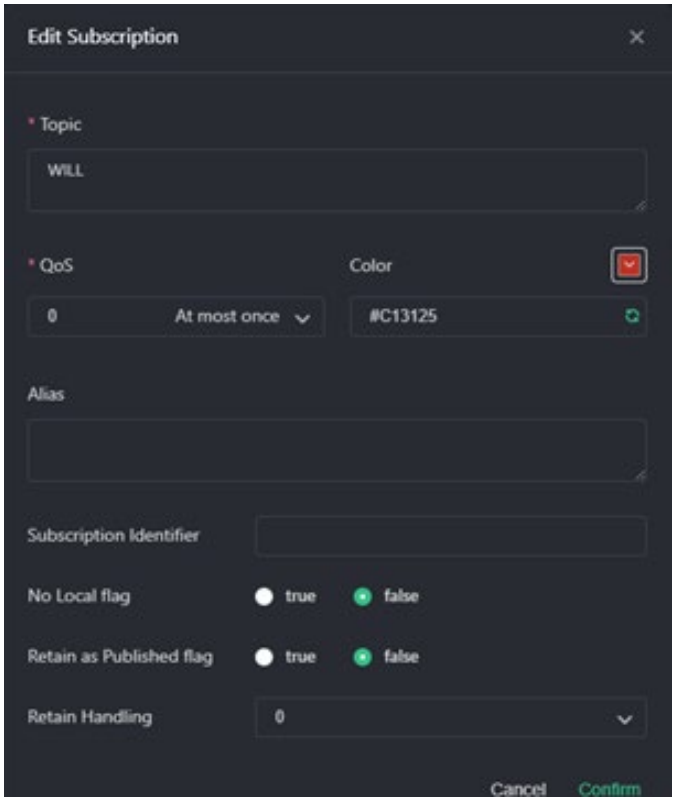
1. In the **MQTT Broker Information** dialog, set up **Will Message**.



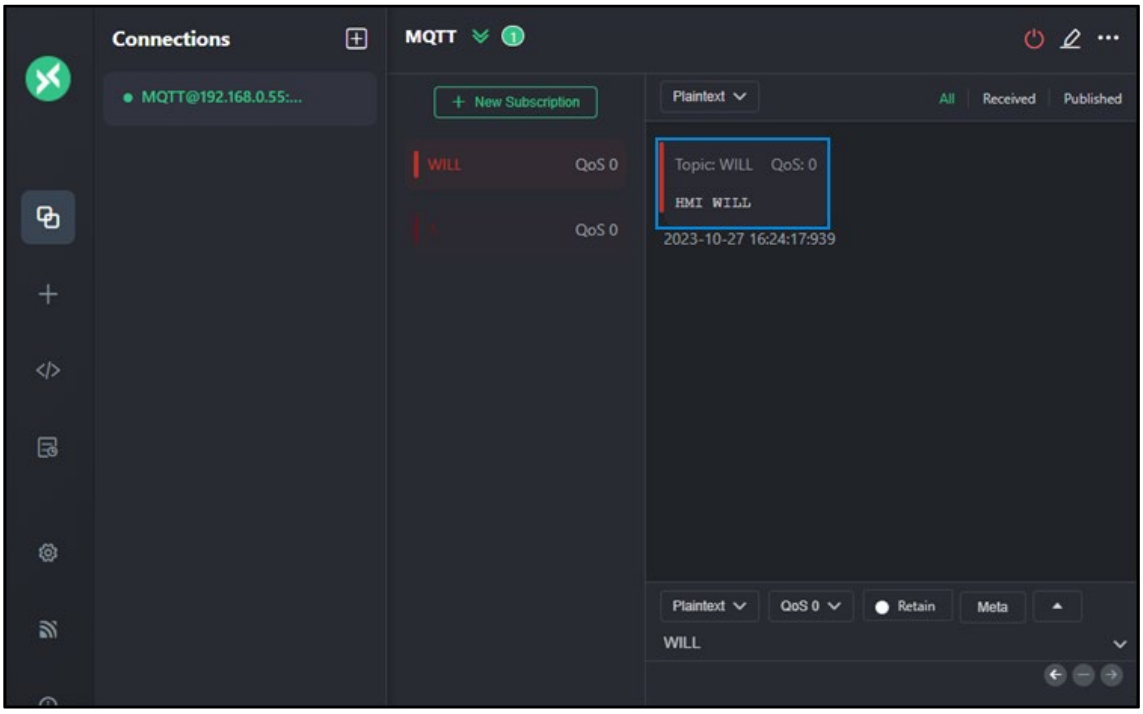
2. Open MQTTX software and set up the Will Message Subscriber of the MQTT connection.

   Note: You can go to mqttx.app/zh to download the MQTTX software.

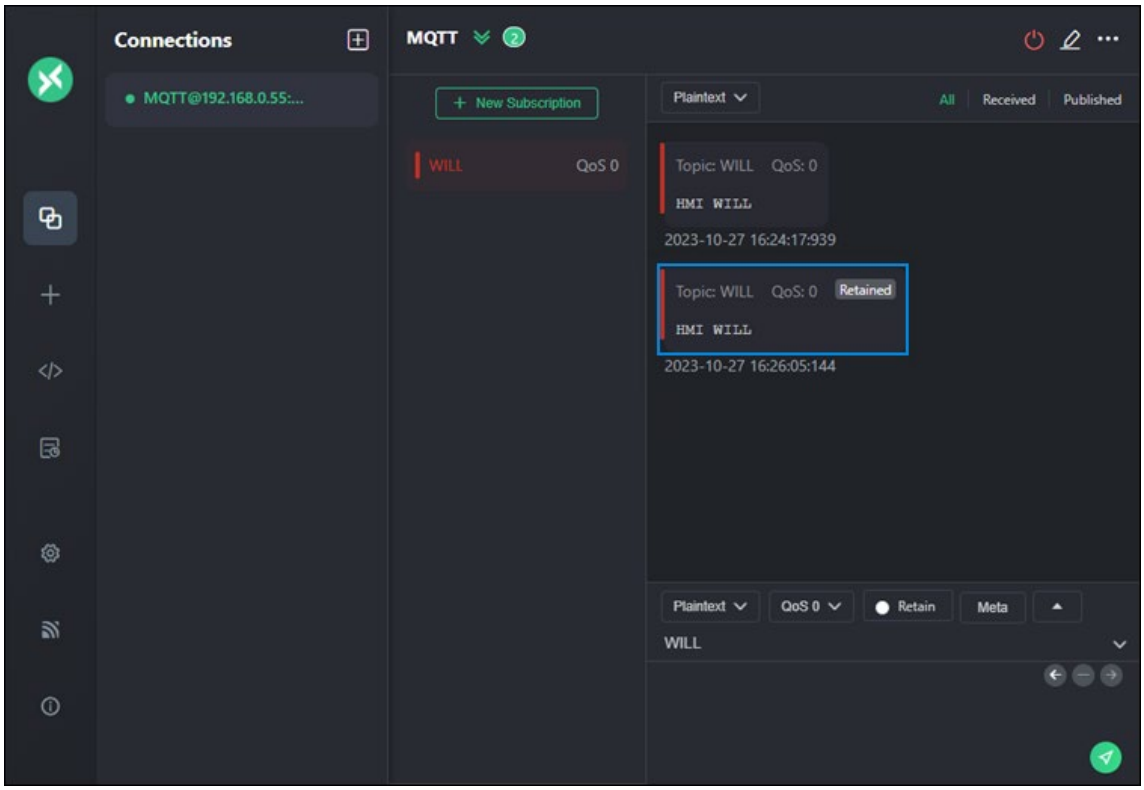




When the HMI is disconnected, MQTTX receives the WILL Message.



3. In the MQTTX tool, click and reconnect to receive the WILL retained message.



Address Setting

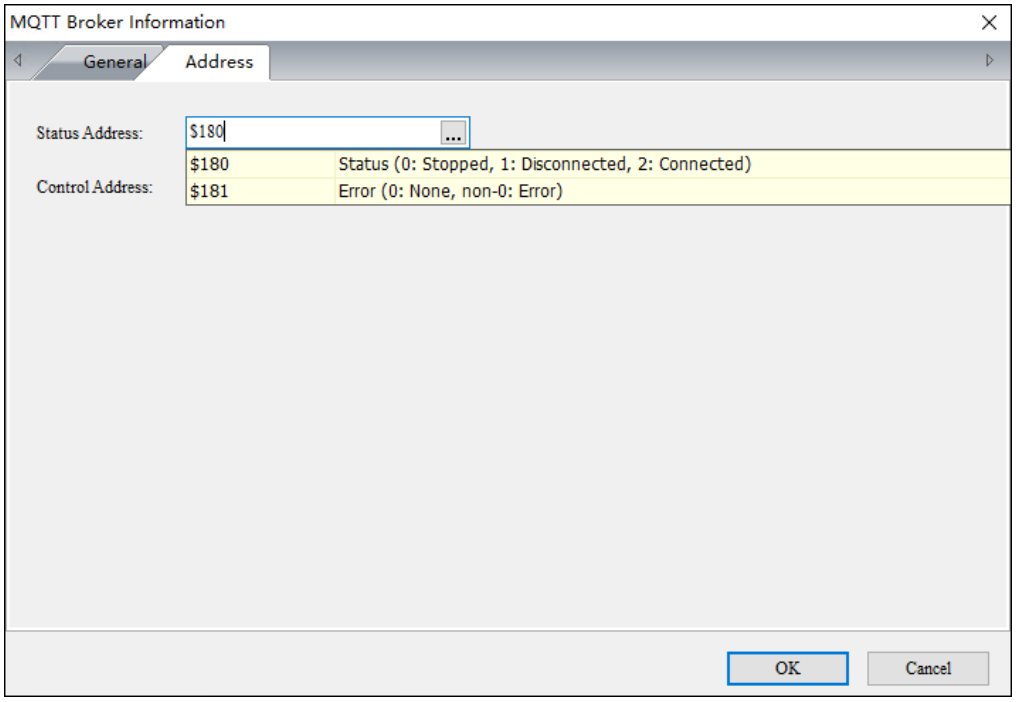
The **States Address** function displays the current connection status and error codes of the Client and Server. The connection state helps you check whether the device is disconnected. If disconnected, you can check the type of the current communication error through the error codes and set the Control Address on the HMI to change the connection message, such as Broker IP, Port, account, password, and so on.

To set states address and control address

- In the **MQTT Broker information** dialog, set the current **States Address** and **Control Address** in the **Address** tab to automatically configure all controls through the continuous address function.

  Note: The **States Address** and **Control Address** limit setting the internal memory address.

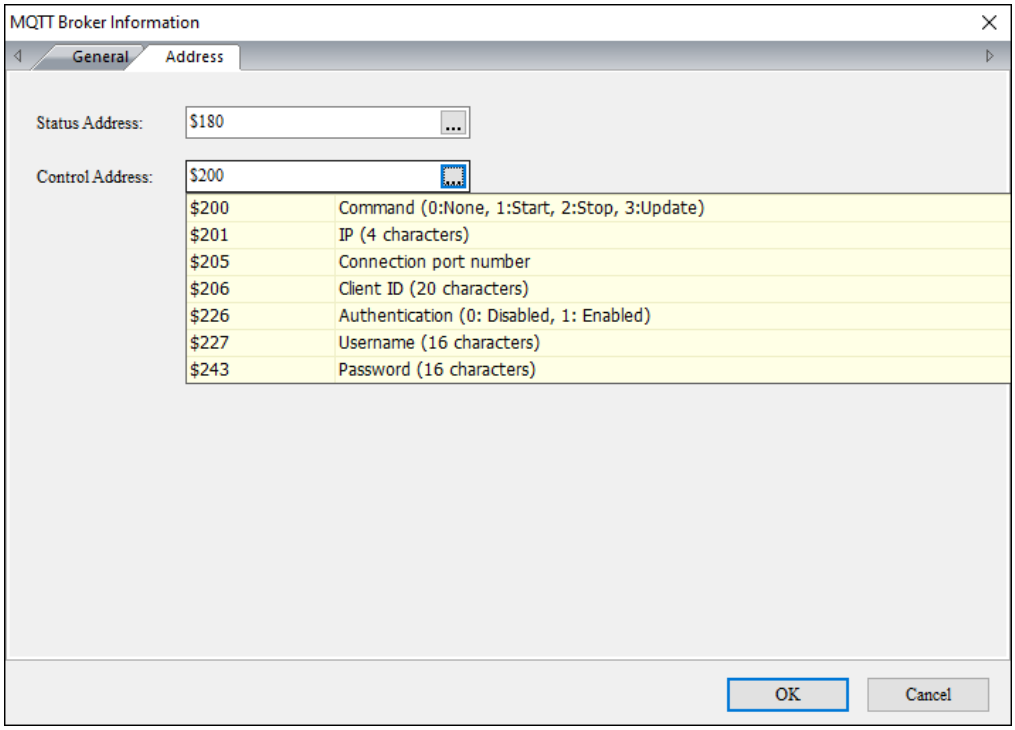
The following table lists the internal addresses limited by **States Address** with their description of error codes. Set the **States Address** to $180 as an example.



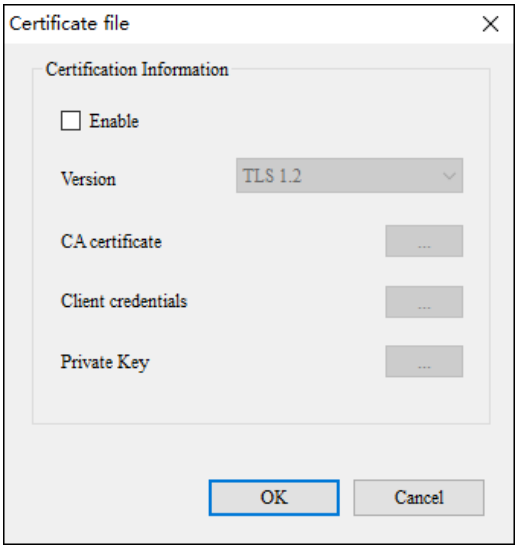
Address	Error Code Description
$180 (Status address)	• 0: Stopped • 1: Disconnected • 2: Connected

Address	Error Code Description
$181 (Error address)	• 0: No errors • 3: Invalid parameter • 4: Client is not connected to Broker • 7: Disconnected • 14: Underlying system calls failed **Note**: Look over and judge from the system log or other relevant messages for detailed error messages. • 15: Network address analysis error or not use the external network to connect • 99: JSON data type is not yet supported

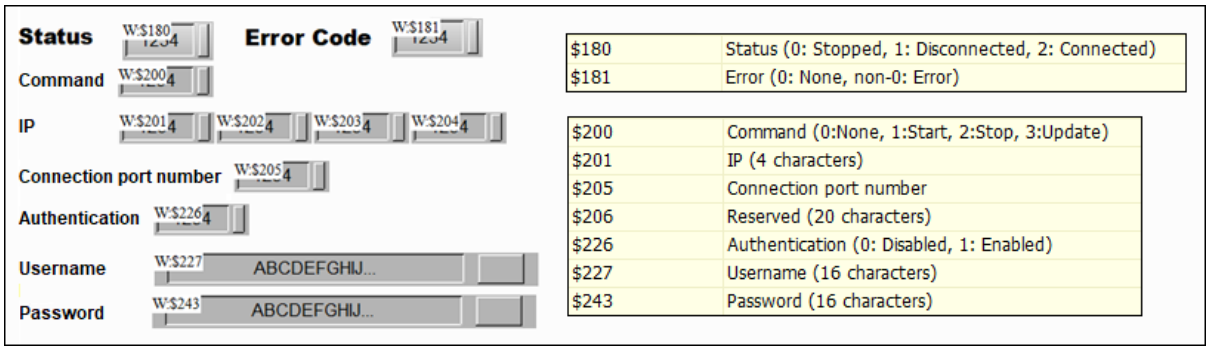
The following table lists the internal address limited by **Control Address** with their description of error codes. Set the **Control Address** to $200 as an example.



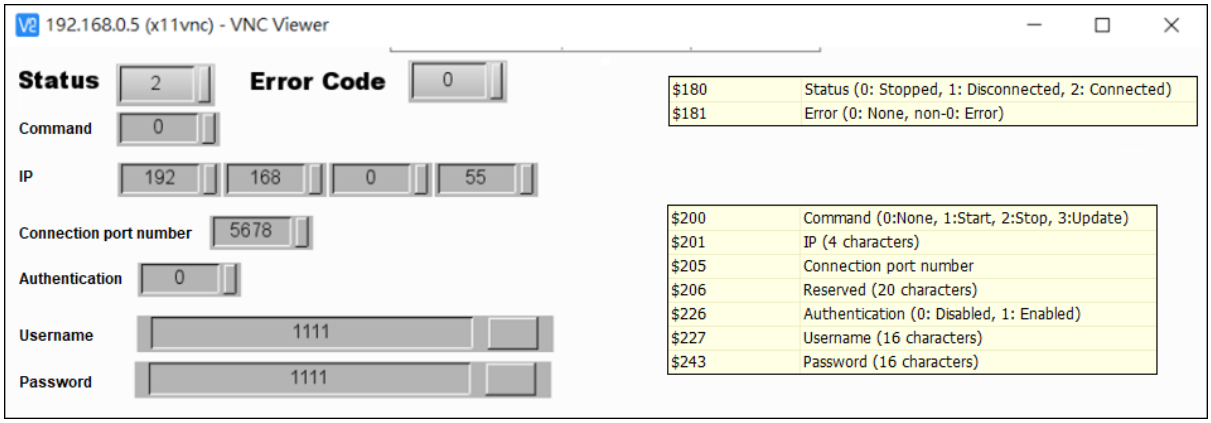
Address	Error Code Description
$200 (Command address)	• 0: None • 1: Start • 2: Stop • 3: Update
$201~204	Used to set the IP address for which the **Data Type** is Word.

Address	Error Code Description
(Broker IP address)	
$206 (Client ID)	A maximum of 20 bits are supported.
$226 (Authentication address)	• 0: Disabled • 1: Enabled **Note**: This address is based on whether the **Certification Information** is enabled. 
$227~242 (Username address)	Used to configure the user account address. Set the **Character Input** element for which the string length is 16.
$243~259 (Password address)	Used to configure the password address. Set the **Character Input** element for which the string length is 16.

The following figure is an example of the screen configuration.



After downloading the screen to the HMI, the addresses are shown in the figure below.

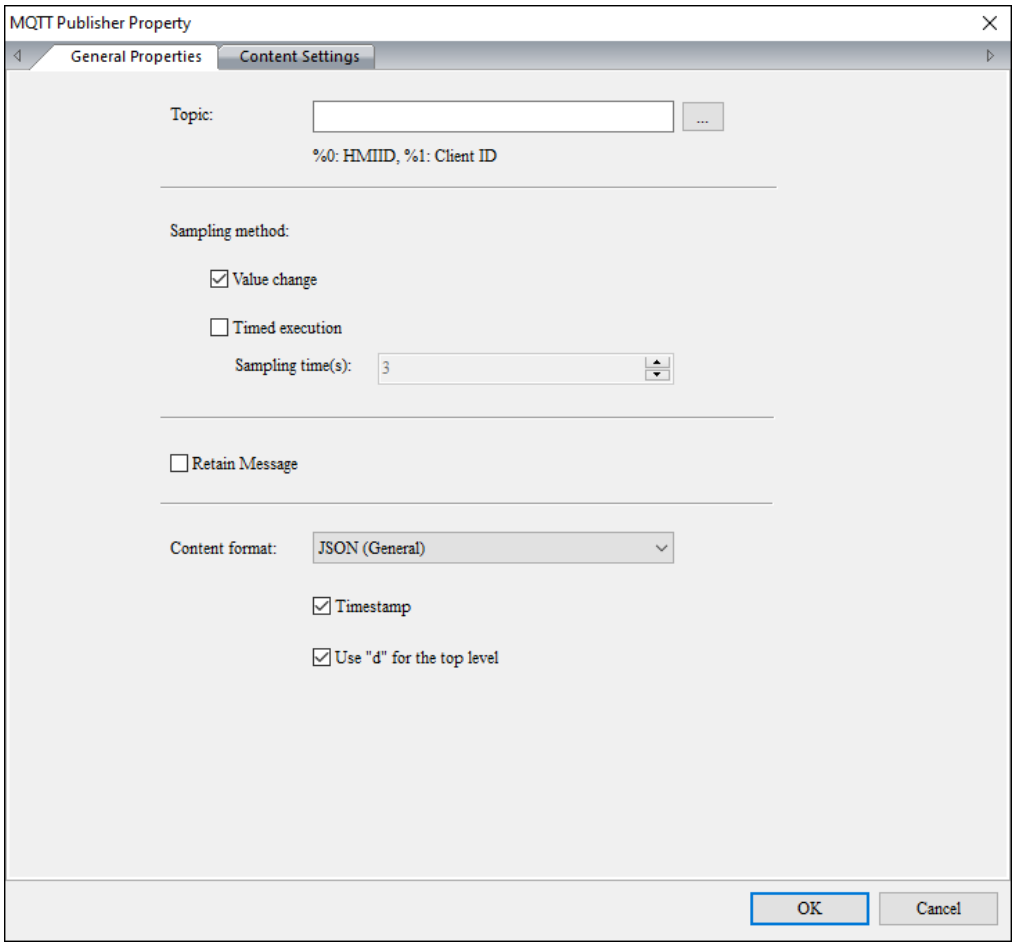


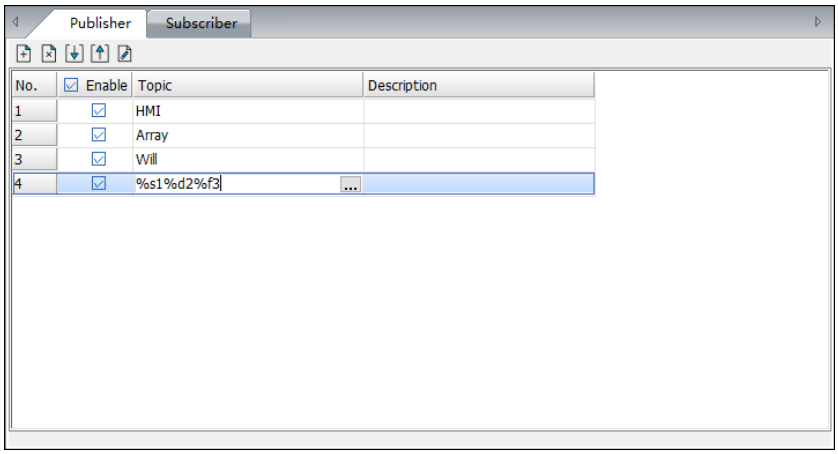
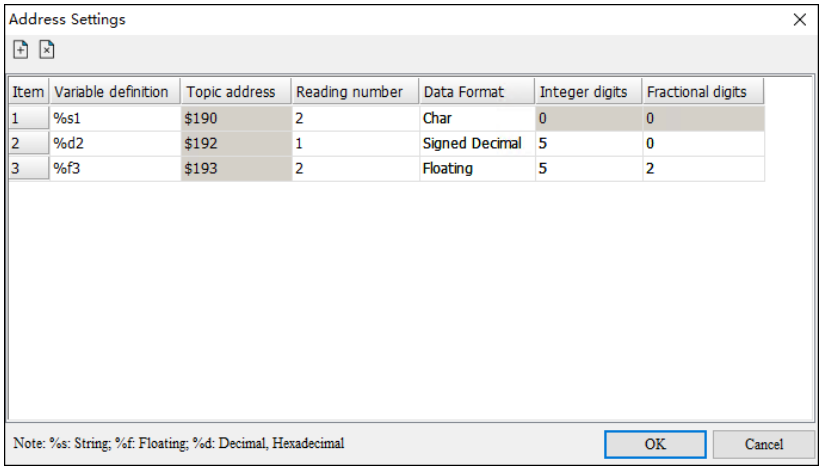
MQTT Publisher & Subscriber

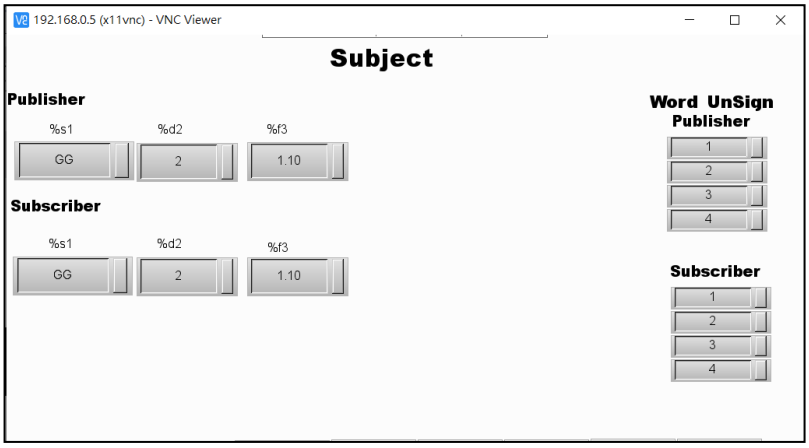
This section describes the properties content of Publisher and Subscriber.

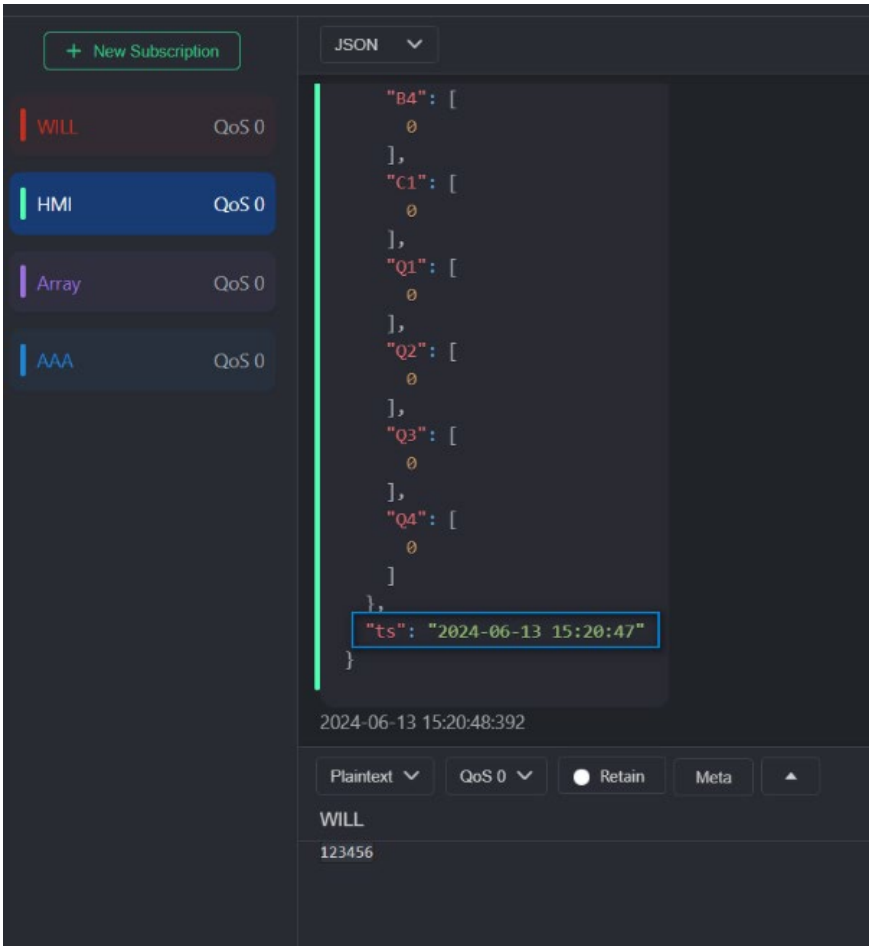
Note: Publisher and Subscriber can add up to 100 topics each.

General Properties



Function	Description
Topic	• In order to reduce the possibility of duplicate topics and make it easier for users to distinguish where data is being sent and received from, you can set the topic as a variable, providing %0 as HWID and %1 as the Client ID. • **DOP-300 series HMI** can dynamically set the topic. After configuring the address variable for the topic content, the topic content can be dynamically changed through the address variable on the HMI to obtain different topic messages. Follow these steps to set the topic. 1. Click [...] in the **Topic** field on the **General Properties** tab or in the **Topic** field on the Publisher or Subscriber list.  2. In the **Address Settings** dialog, set **Variable definition** and **Topic address** and click **OK**. **Note**: ○ The setting formats are %s: String, %f: Floating, %d: Decimal, and Hexadecimal.  ○ The maximum number is 8. 3. Enter the variable definition in the **Topic** field in the **General Properties** tab, for example, %s1%d2%f3.

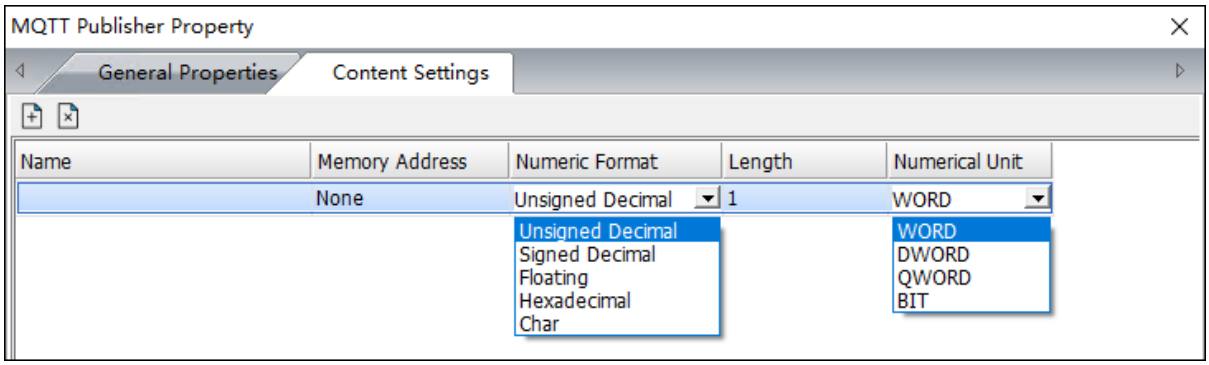
Function	Description
	4. Execute on the HMI. When the topics of Publisher and Subscriber are the same, the HMI sends the corresponding data. 
Sampling method	Select the sampling method during execution. • Value Change: Sampling when the value changes. • Timed Execution: Sampling according to the set sampling time. **Note**: This function is only available in the **Publisher** properties.
Retain Message	When Publisher publishes a message, if the Retained mark in the MQTTX software is set as True, the message is a Retain Message in MQTT. The MQTT server stores the latest retain message for each topic so that clients that come online after the message is published can still receive the message when subscribing to the topic. **Note**: This function is only available in the **Publisher** properties.
Content format	Select the content format. • JSON (General): Basic JSON format. • JSON (Advanced): You can customize the JSON format of the nested structure. **Note**: JSON (Advanced) is only supported in the DOP-300 series HMI.
Timestamp / Use “d” for the top level	• Timestamp and d message format may not exist in different application scenarios. However, the default MQTT Message function of Delta HMI displays these two pieces of information. In case the messages received when receiving data do not contain these two pieces of information, the communication fails. It is suggested to enable **Time stamp** and **Use “d” for the top level** functions. • Confirm whether the timestamp and “d” mark appear in the JSON format of the HMI topic through the MQTTX software. **Note**: Go to mqttx.app/zh to download the MQTTX software.

Function	Description
	Timestamp:  **Use “d” for the top level:**

Content Settings

JSON (General)

DIAScreen provides Unsigned Decimal, Signed Decimal, Floating, Hexadecimal, and Char numeric formats with different numerical units for users to set according to the data size to transfer.

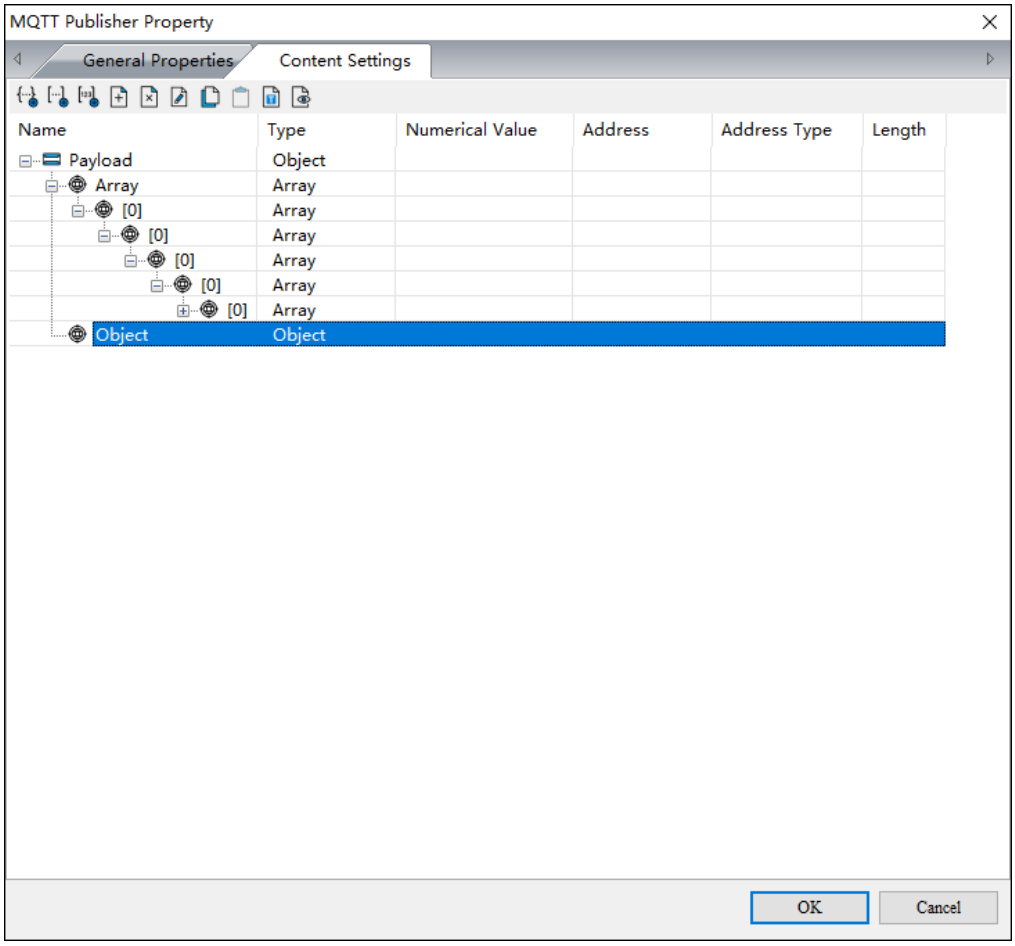


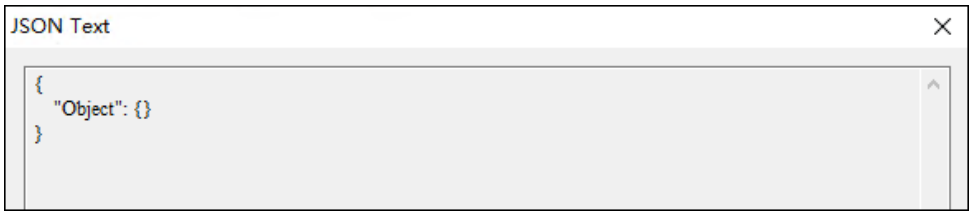
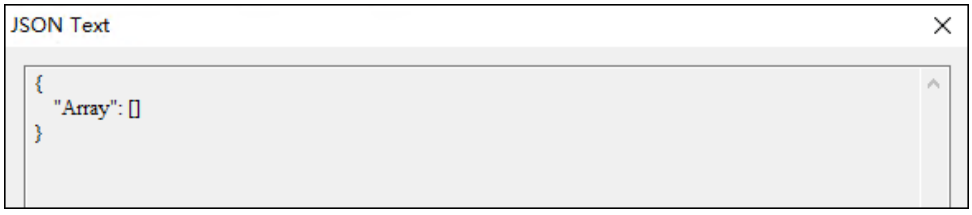
JSON (Advanced)

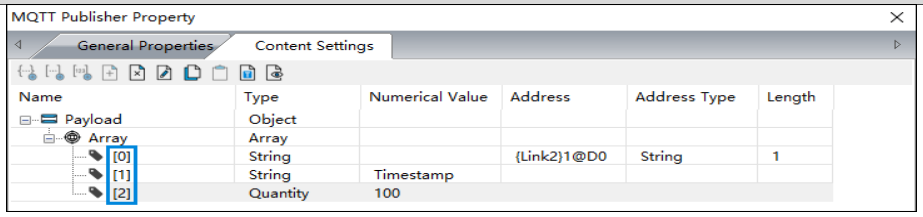
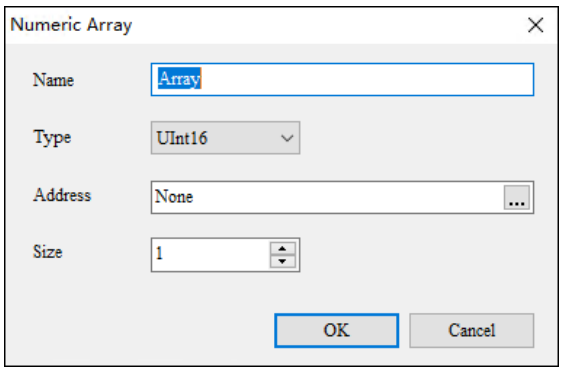
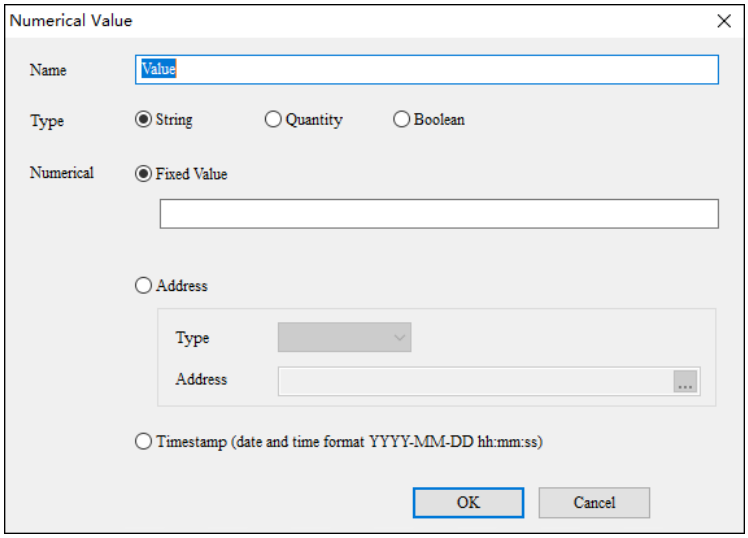
Note:

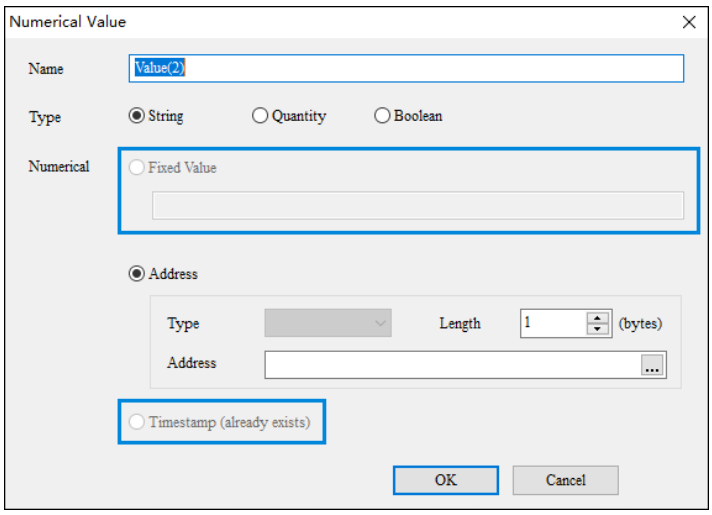
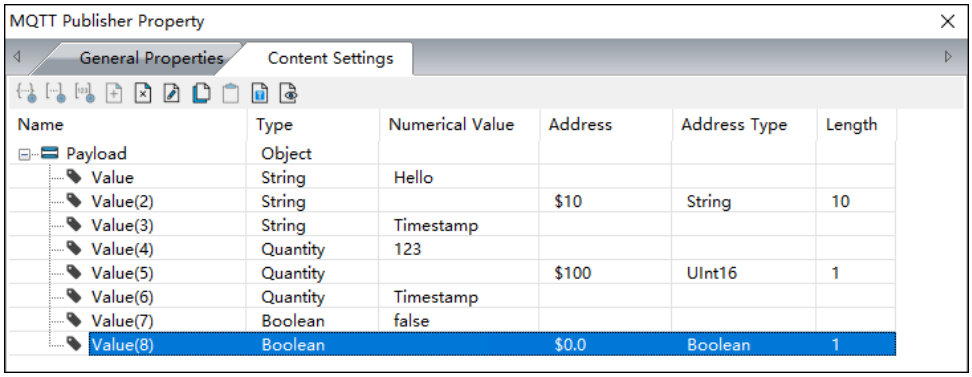
- This function is only supported in the DOP-300 series HMI.
- Advanced JSON format can set up to 8 layers of nodes.
- A maximum of 512 nodes can be set for a topic.

The following lists the toolbar settings in the **Content Settings** tab for JSON (Advance) with their description.

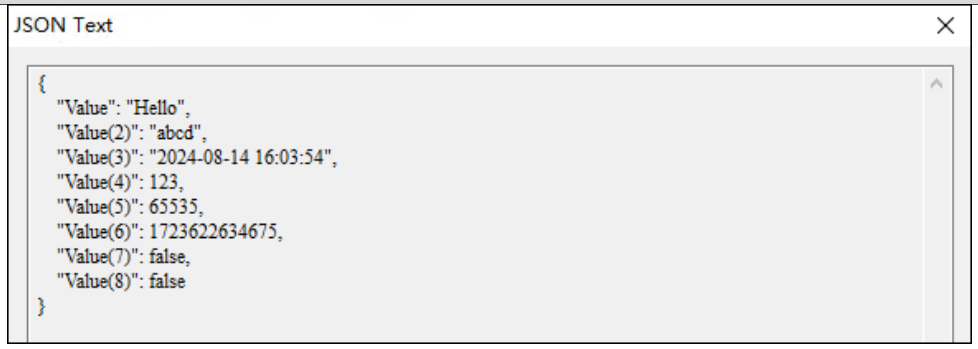
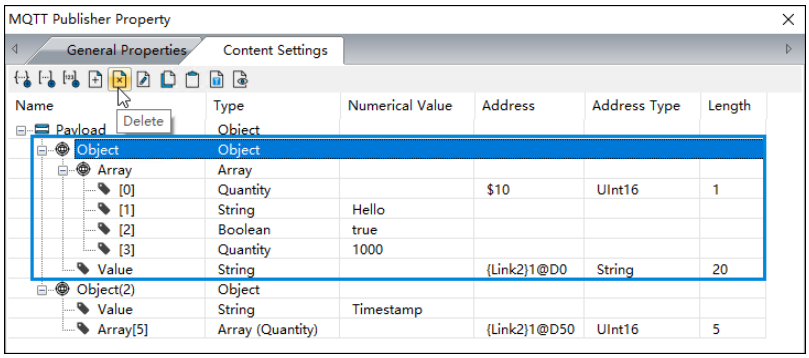
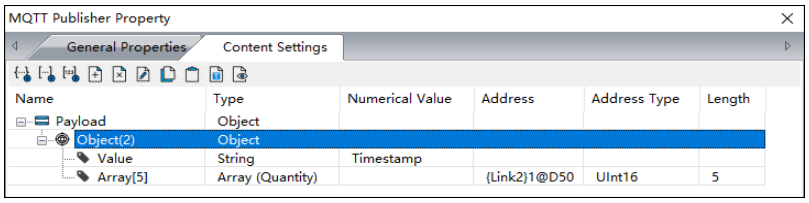


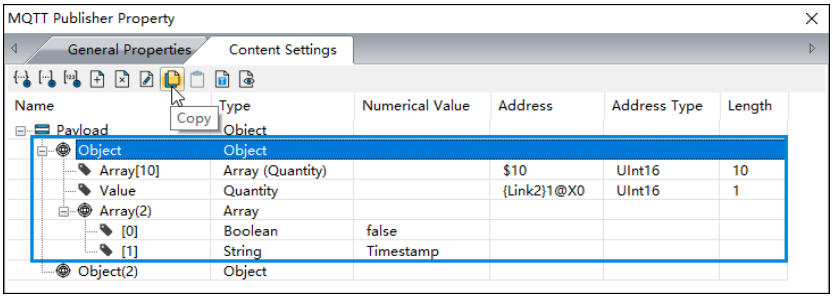
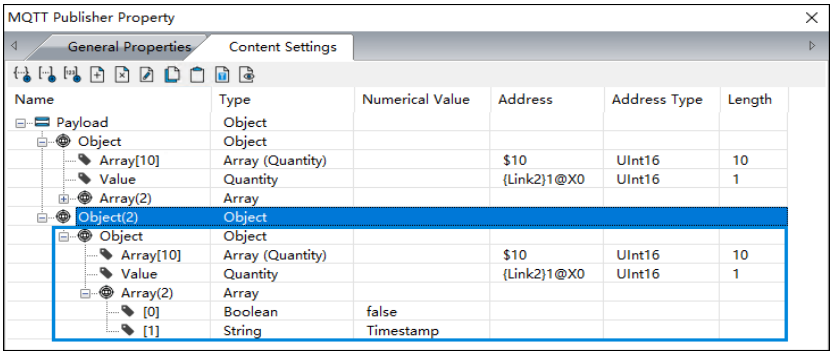
Function	Description
	Click to add an object. The JSON format is represented by **{ }**.  **Note**: The maximum name input length is 256 characters.
	Click to add a general array. The JSON format is represented by **[ ]**.  **Note**: • The maximum name input length is 256 characters • The names of child nodes under general arrays cannot be edited, and the software will automatically name them with [0] ~ [n].

Function	Description
	
	Click to add a Numeric array. In the **Numeric Array** dialog, set **Name**, **Type**, **Address**, and **Size** for a numeric array.  **Note**: The maximum name input length is 256 characters. The following figure shows the JSON format of a numeric array.
	Click to add a value. In the **Numerical Value** dialog, set Name, Type, Timestamp, and so on for a numerical value. 

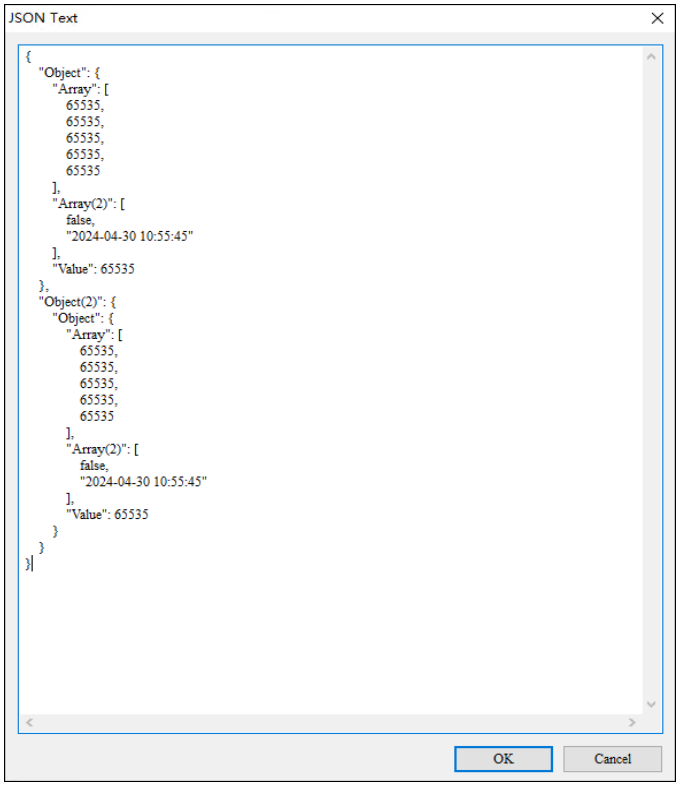
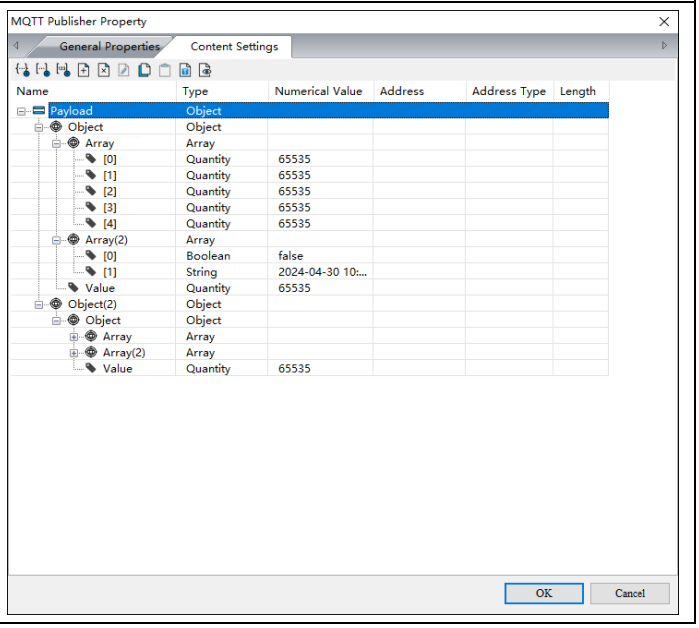
<table>
<tr><th>Function</th><th>Description</th></tr>
<tr><td></td><td>
<table>
<tr><th>Function</th><th>Description</th></tr>
<tr><td>Name</td><td>Enter a value name.
Note: The maximum input length is 256 characters.</td></tr>
<tr><td>Type</td><td>Select the numerical type. The default is String.</td></tr>
<tr><td>Numerical</td><td>Select the numerical format. The default is Address.
Note:
When String is selected as Type, a maximum of 256 characters can be entered for a Fixed Value.
When Quantity is selected as Type, up to 20 characters can be entered for a Fixed Value.
When setting Subscriber topic content, the Fixed Value option is not available and the Timestamp option is limited to one topic selection. 
</td></tr>
</table>
The corresponding JSON format of each numerical type is shown in the following figure.


</td></tr>
</table>

Function	Description
	
	Select the node to be deleted and click this icon to delete. **Note**: Its child nodes will be deleted together. Before delete  After delete 
	Select the node to be edited and click this icon to edit.

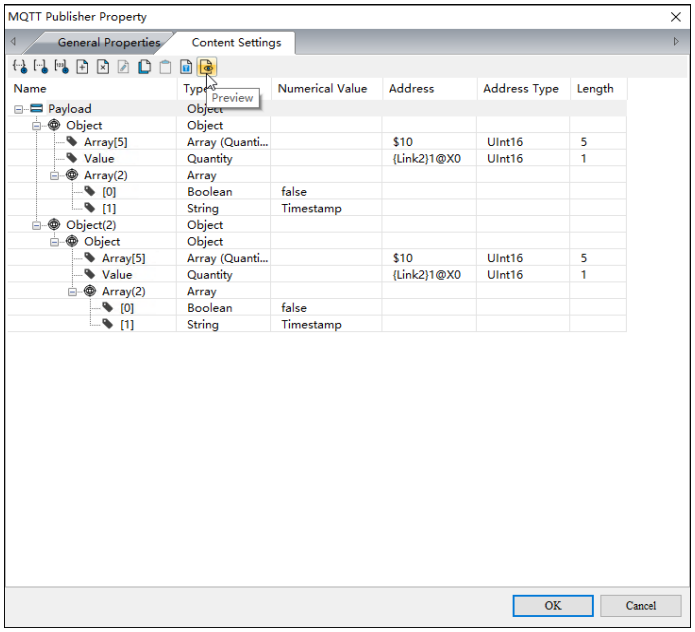
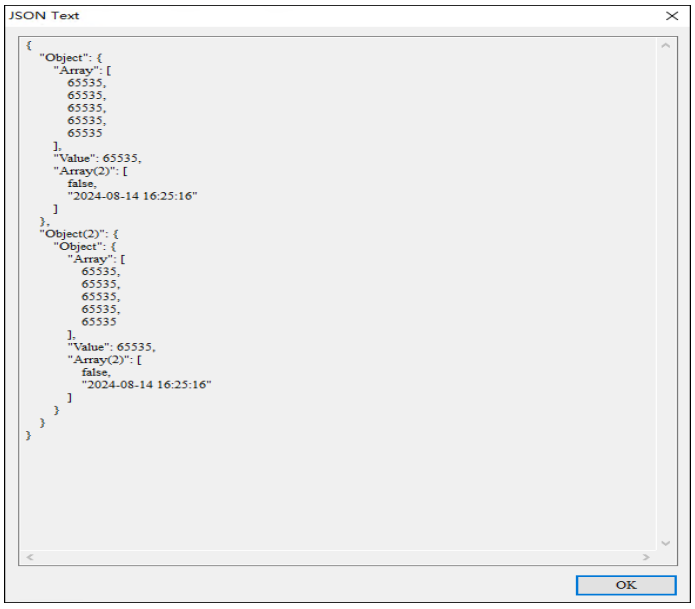
Function	Description
	Select the node to be copied and click this icon to copy and paste. **Note**: Its child nodes will be copied together. Copy  Paste 

Function	Description
	Click to use a template. In the **JSON Text** dialog, fill in the JSON format, the corresponding tree structure will be automatically converted.

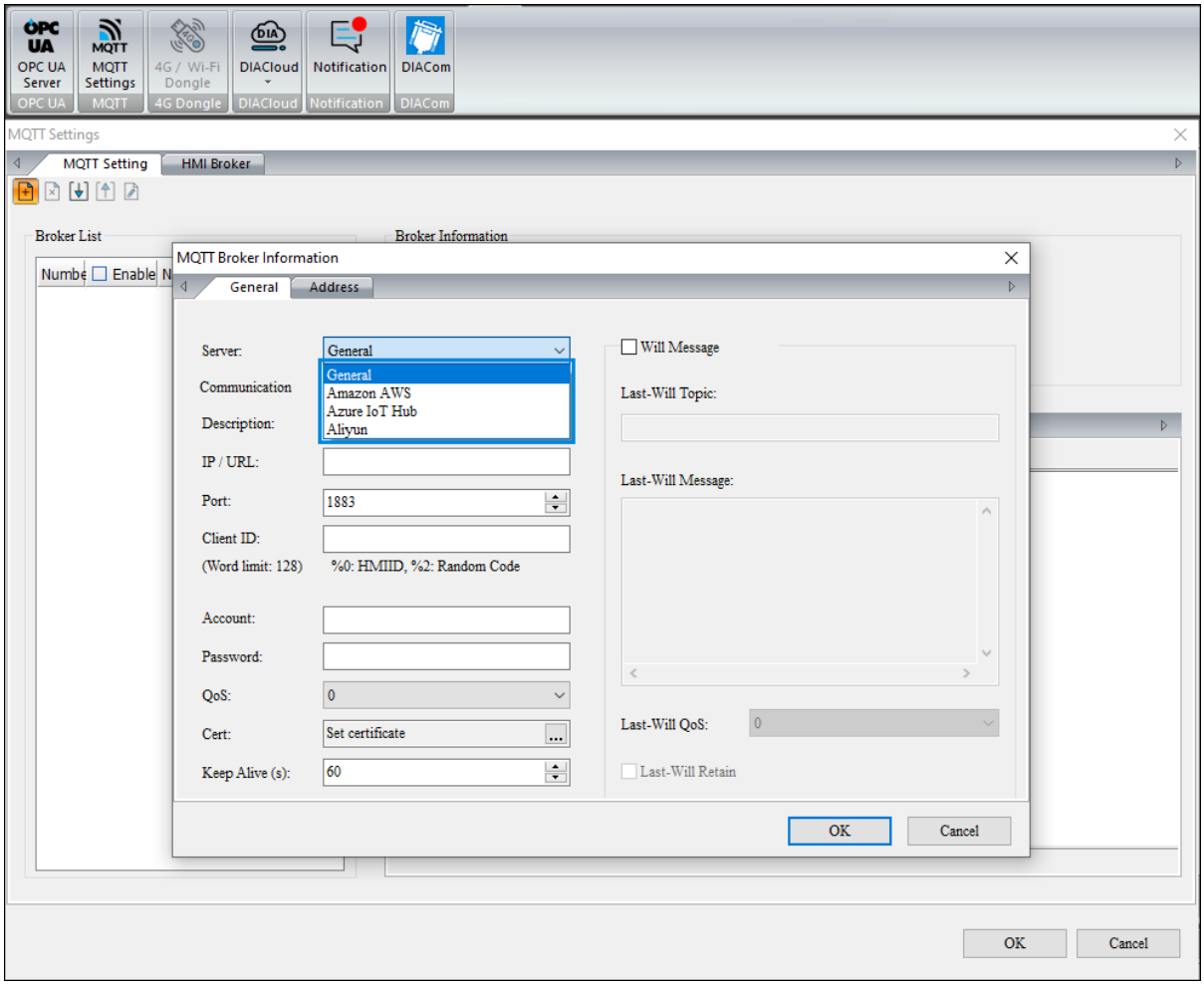
JSON text content	
Tree structure after conversion	

Note:

- The Value node in the Publisher topic settings is converted to a **Fixed Value**.
- The Value node in the Subscriber topic settings is converted to an **Address Value**.

Function	Description
	Click to preview the JSON format. The tree structure content will be converted to the corresponding JSON format. Tree structure content  JSON text content after conversion 

- MQTT supports Azure IoT Hub and Aliyun server. The applicable models are as follows.
  - Azure IoT Hub server: All series models
  - Aliyun server: DOP-300 series models



The comparisons between Azure IoT Hub and Aliyun server are as follows.

	Azure IoT Hub	**Aliyun**
Cert certificate version	TLS 1.2 and above	TLS 1.2 and above
QoS	0, 1	0, 1,2
Communication	MQTT v3.1.1	MQTT v3.1.1
Default port	8883	8883

MQTT Broker Information Settings Limit

- When selecting **Azure IoT Hub** or **Aliyun** for **Server**, **Communication** can only be selected as **MQTT v3.1.1**.
- When selecting **Azure IoT Hub** for **Server**, **QoS** can only be set to **0** or **1**.
- When selecting **Azure IoT Hub** and **Aliyun** for **Server**, **Version** can only be set to **TLS1.2** in the **Certificate** file dialog of the **Cert** field.
- When selecting **Aliyun** for **Server**, **Content format** can only be set to **JSON (Advanced)** in the property dialog of the **Publisher** and **Subscriber** tabs.
- When selecting **Azure IoT Hub** for **Server**, in the **MQTT Subscriber Property** dialog of the **Subscriber** tab, the **Topic** field requires two special open symbols, such as # and #, and # can only be placed in the end of the final level.

  Example:

  - **# symbol**

    Enter devices/Delta_HMI/messages/events/#, other characters must not be entered after the # symbol.

  - **+ symbol**

    Only /single-level message can be entered after the + symbol, for example, devices/+/temperature. If you enter two levels of message, you will not be able to receive the information.

- When selecting **Amazon AWS** for **Server**, you need to verify its credentials. Without a credential, you cannot use the Amazon AWS server.

DIACloud

IIoT integrates the cloud platform for DIAScreen, which can upload DIAScreen projects to the cloud or download projects from the cloud to DIAScreen for editing. At the same time, the HMI can also transmit data to the cloud and monitor the device through the configuration screen of the cloud platform.

Note: This function is only applicable to DOP-300 series models.

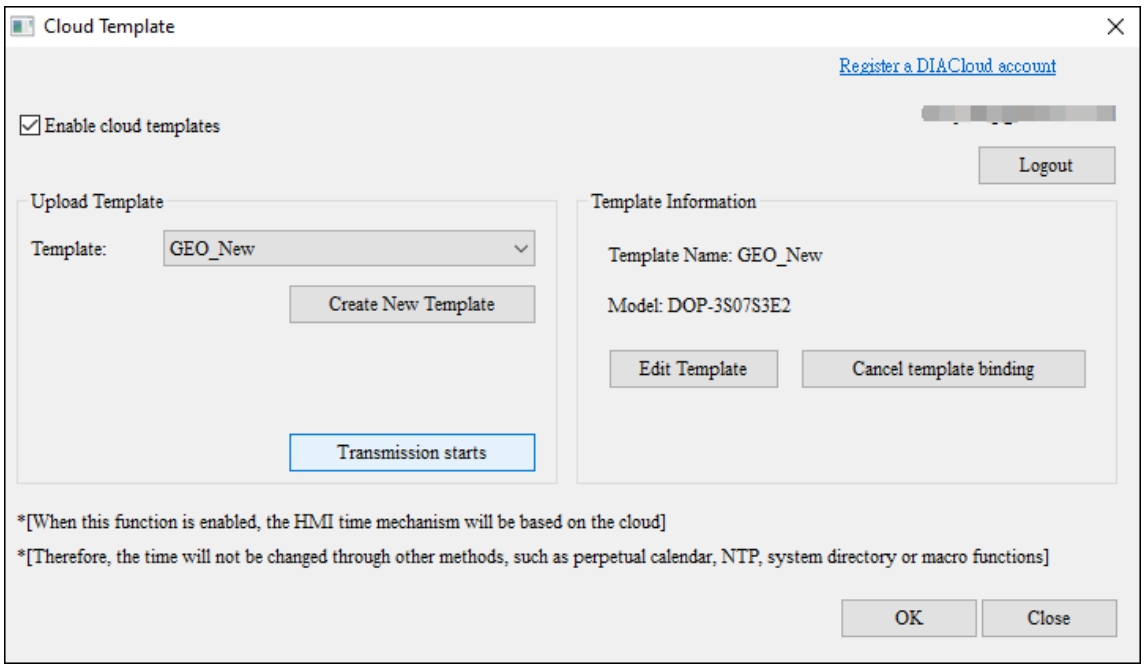
To create a template and upload it to the cloud

1. On the menu bar, click **IIoT** > **DIACloud** > **Cloud Template**.
2. In the **Cloud Template** dialog, check **Enable cloud templates**.

   Note: The VNC function will be enabled at the same time to monitor the HMI screen from the cloud.
3. In the **Upload Template** area, click **Create New Template**.
4. In the **Template Information** area, click **Edit Template** to edit template.
5. In the **Cloud Template Information** dialog, enter the template information and click **OK**.

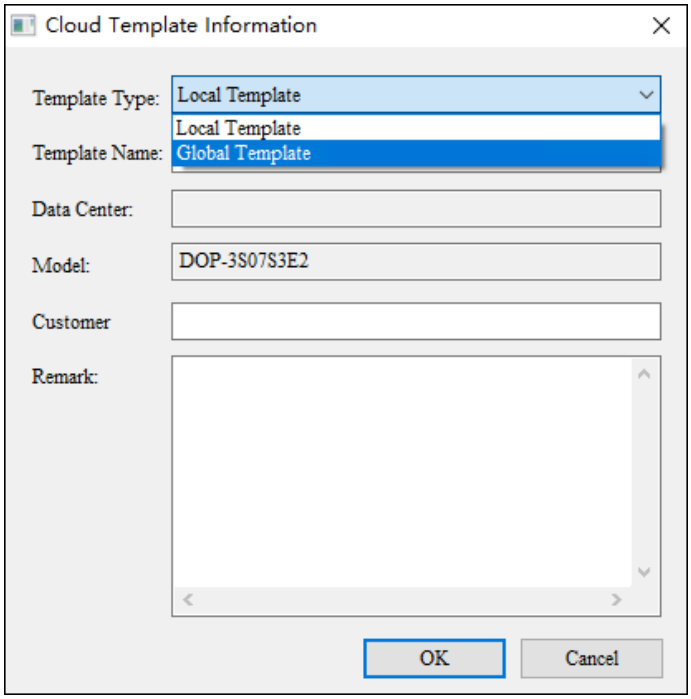
   The template is created.
6. Click **Login** in the upper right corner to open **Login to DIACloud** dialog.
7. In the **Login to DIACloud** dialog, enter your DIACloud Account and Password, and then click **Login**.

   Note: If you have not registered a DIACloud account, go to hms.diacloudsolutions.com to register.
8. Click **Transmission starts** to start to upload the DIAScreen project to the cloud.



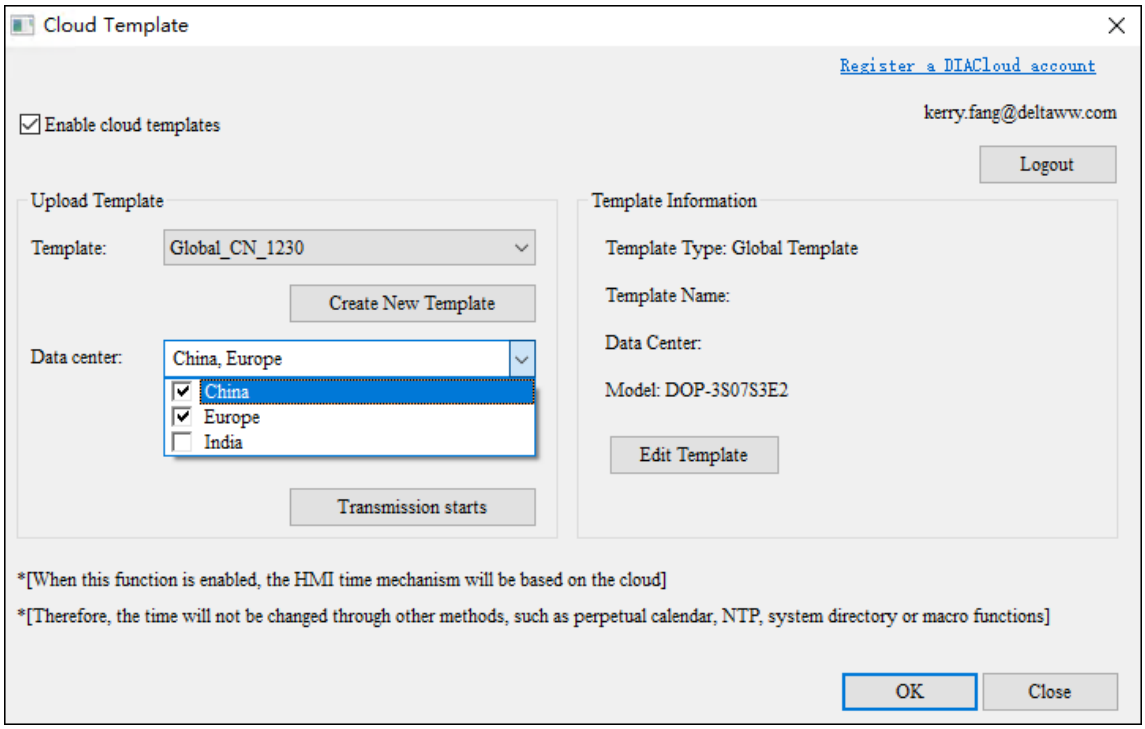
To create data center

1. Click **IIoT** > **DIACloud** > **Cloud Template** on the menu bar.
2. Select **Enable cloud templates** checkbox in the **Cloud Template** dialog.
3. Click **Create New Template** in the **Upload Template** area.
4. Select **Global Template** for the **Template Type** in the **Cloud Template Information** dialog.



5. Select the **Data Center** (multiple selections are available).

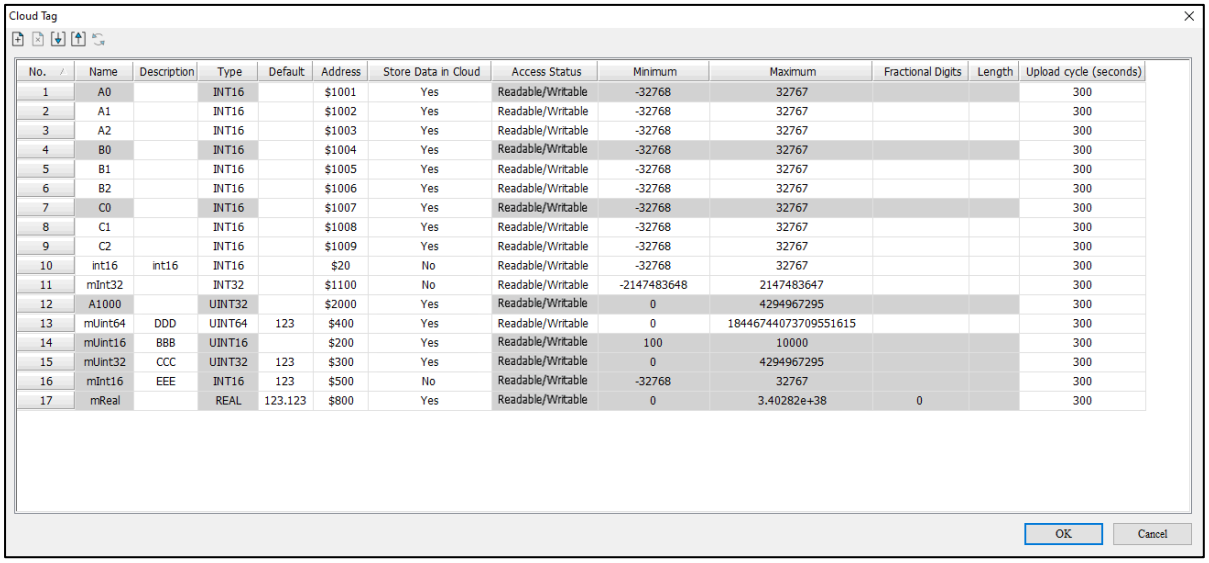
   Note: IIoT data centers include China, Europe, and India regions.



You can also create cloud data in DIAScreen so the configuration screen can reference the corresponding variables to monitor the device.

To create Cloud Tag

1. Click **IIoT** > **DIACloud** > **Cloud Tag** on the toolbar.
2. In the **Cloud Tag** dialog, click to add a new tag.
3. Make relevant settings and click **OK**.



No.	Name	Description	Type	Default	Address	Store Data in Cloud	Access Status	Minimum	Maximum	Fractional Digits	Length	Upload cycle (seconds)
1	A0		INT16		$1001	Yes	Readable/Writable	-32768	32767			300
2	A1		INT16		$1002	Yes	Readable/Writable	-32768	32767			300
3	A2		INT16		$1003	Yes	Readable/Writable	-32768	32767			300
4	B0		INT16		$1004	Yes	Readable/Writable	-32768	32767			300
5	B1		INT16		$1005	Yes	Readable/Writable	-32768	32767			300
6	B2		INT16		$1006	Yes	Readable/Writable	-32768	32767			300
7	C0		INT16		$1007	Yes	Readable/Writable	-32768	32767			300
8	C1		INT16		$1008	Yes	Readable/Writable	-32768	32767			300
9	C2		INT16		$1009	Yes	Readable/Writable	-32768	32767			300
10	int16	int16	INT16		$20	No	Readable/Writable	-32768	32767			300
11	mInt32		INT32		$1100	No	Readable/Writable	-2147483648	2147483647			300
12	A1000		UINT32		$2000	Yes	Readable/Writable	0	4294967295			300
13	mUint64	DDD	UINT64	123	$400	Yes	Readable/Writable	0	18446744073709551615			300
14	mUint16	BBB	UINT16		$200	Yes	Readable/Writable	100	10000			300
15	mUint32	CCC	UINT32	123	$300	Yes	Readable/Writable	0	4294967295			300
16	mInt16	EEE	INT16	123	$500	No	Readable/Writable	-32768	32767			300
17	mReal		REAL	123.123	$800	Yes	Readable/Writable	0	3.40282e+38	0		300

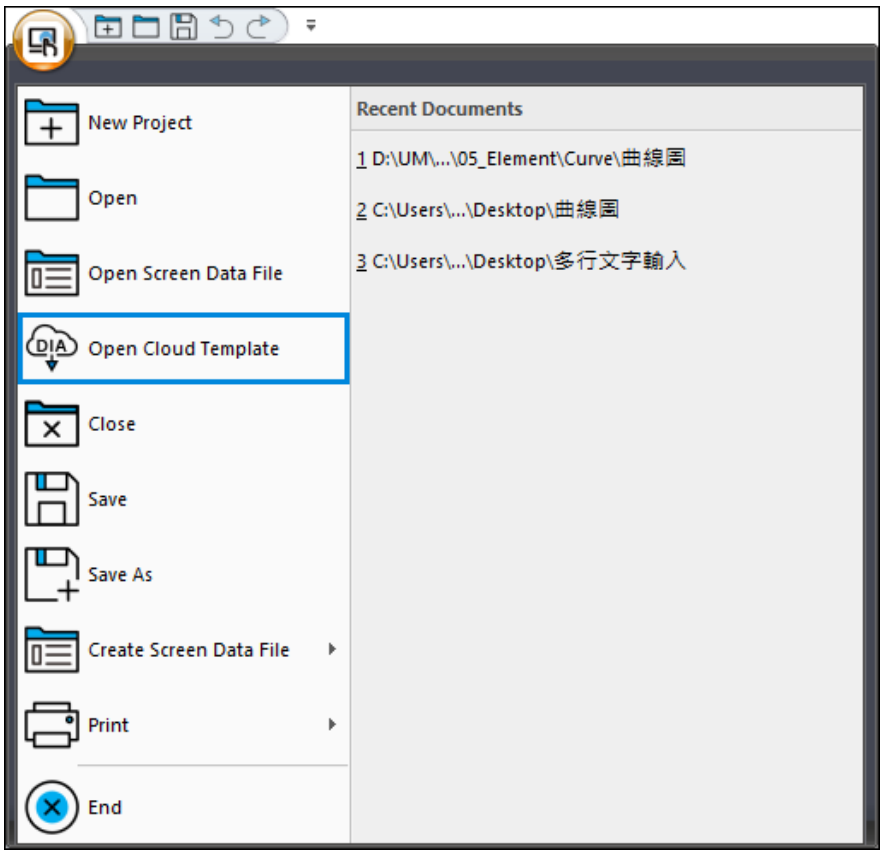
Note:

- Cloud tag supports BOOL, UINT16, UINT32, UINT64, INT16, INT32, INT64, REAL, LREAL and STRING data types. The corresponding data type can be selected based on the actual data value.
- If the currently edited project has been bound to a template, you need to log in to DIACloud before opening the **Cloud Tag** dialog to ensure that the tag content is synchronized with the cloud.
- If the cloud tag is used in the configuration screen, some fields in the **Cloud Tag** will be grayed out and cannot be edited.

Follow these steps to download a project from the cloud to PC.

To download projects to PC from the cloud

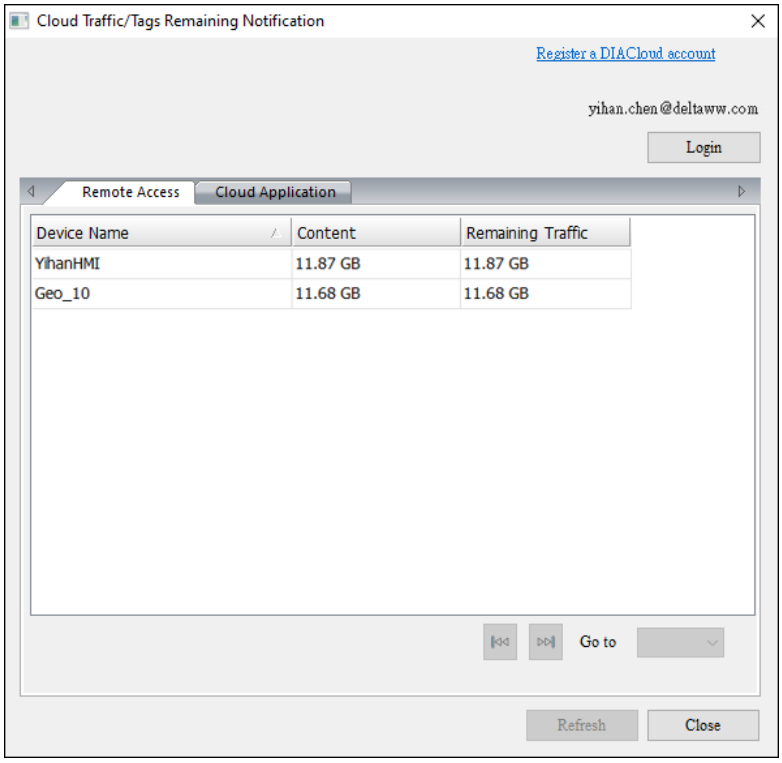
1. In the menu bar, click **Open Cloud Template**.



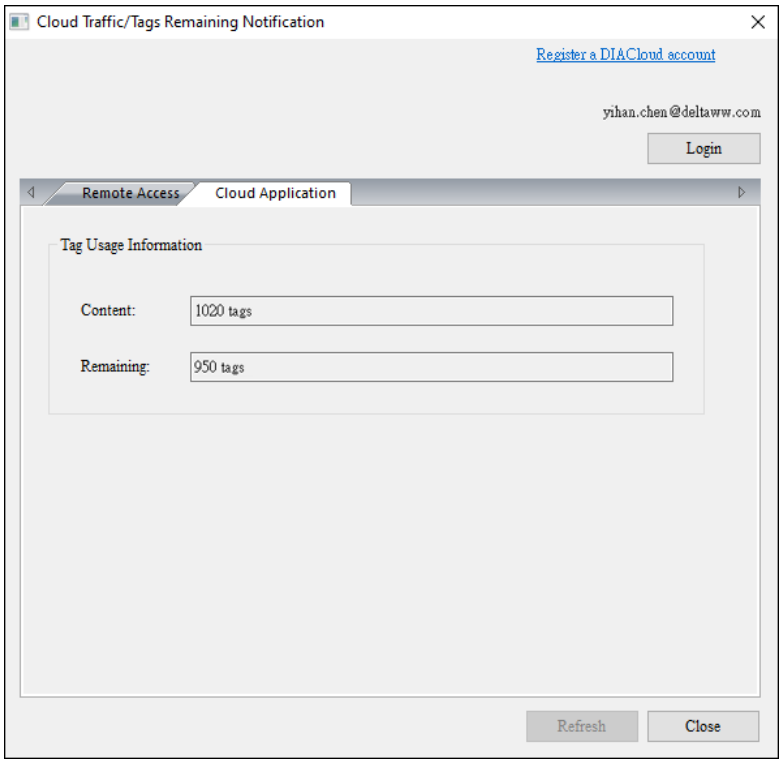
2. Click **Login** to log in to DIACloud.
3. Select the template you want to download.
4. Click **OK** to start the transfer.

To View the cloud traffic / tags renaming notification

1. Click **IIoT** > **DIACloud** > **Cloud Traffic / Tags Renaming Notification** on the toolbar.
2. In the **Cloud Traffic / Tags Renaming Notification** dialog, you can do any of following:
   - Click the **Remote Access** tab and check the remaining of the device bound to this account.



- Click the **Cloud Application** tab and check all tags and remaining tags in this account.



For related devices connected to IoT, refer to *IoT Operation Manual* (go to Delta | Download Center to download.)

Print Settings

Use the **Print Settings** function to perform multi-page printing, planning of print layout, and historical data printing. Refer to Print Settings Example to configure the printing layout.

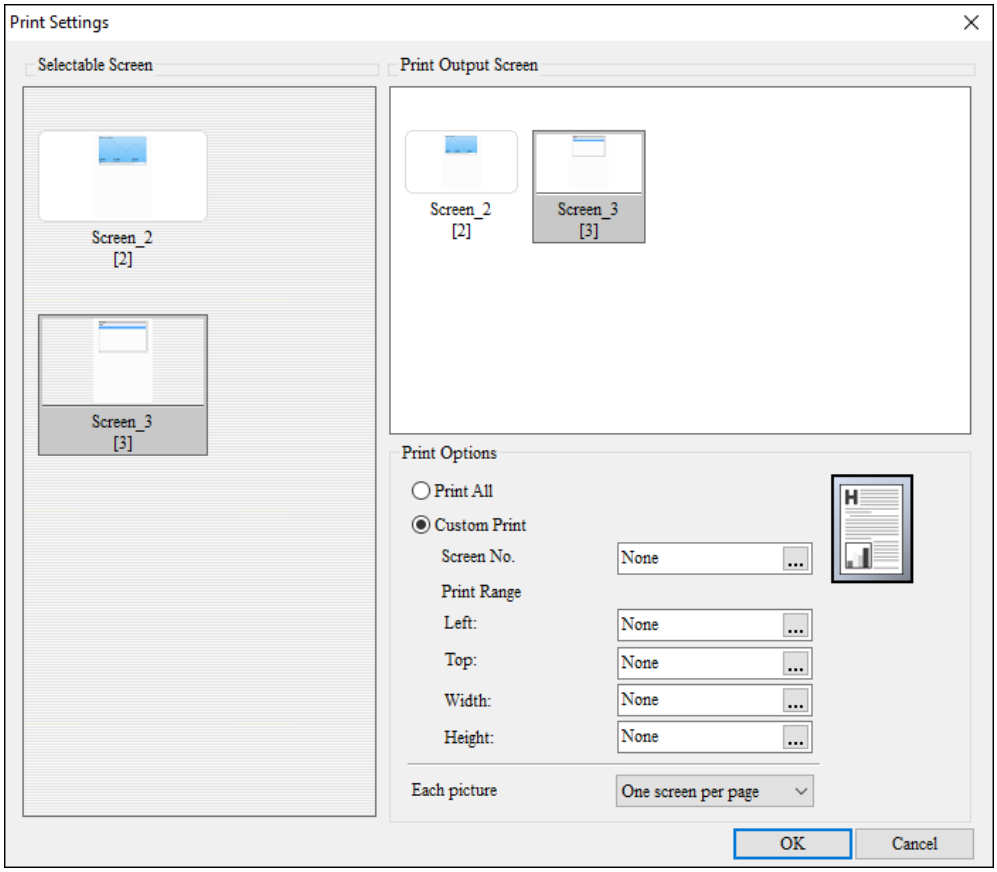
Note:

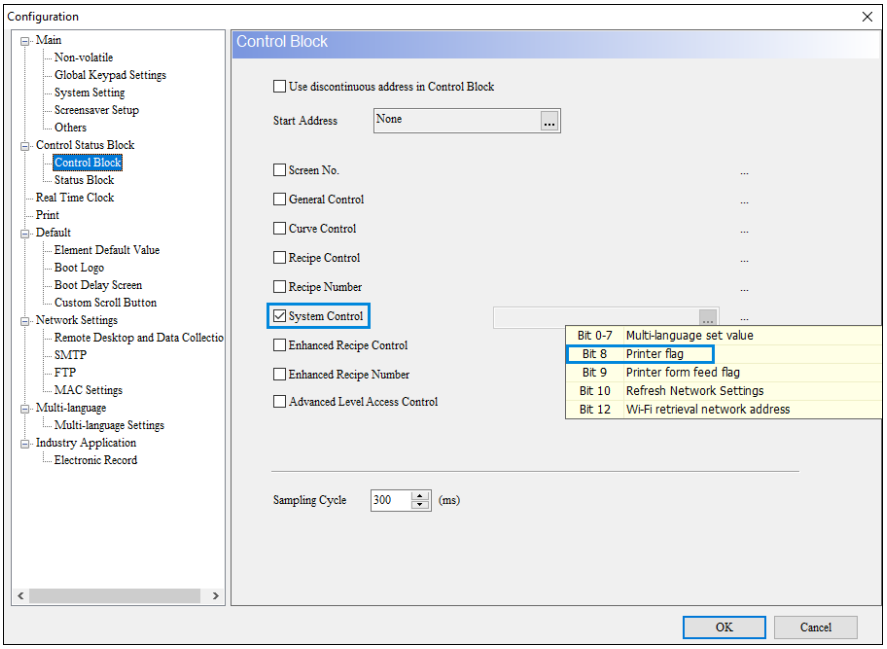
- The screen specified by the **Goto Screen** element cannot be the Print Screen.
- The screen cannot be changed to the Print Screen through the **Goto Screen** element.
- The Print Screen cannot be the default screen, base screen, sub-screen, or screensaver screen.

How to open the **Print Settings** function?

- Click **Tools** > **Print Settings** on the toolbar.

The following table lists the settings in the **Print Settings** dialog with their description.



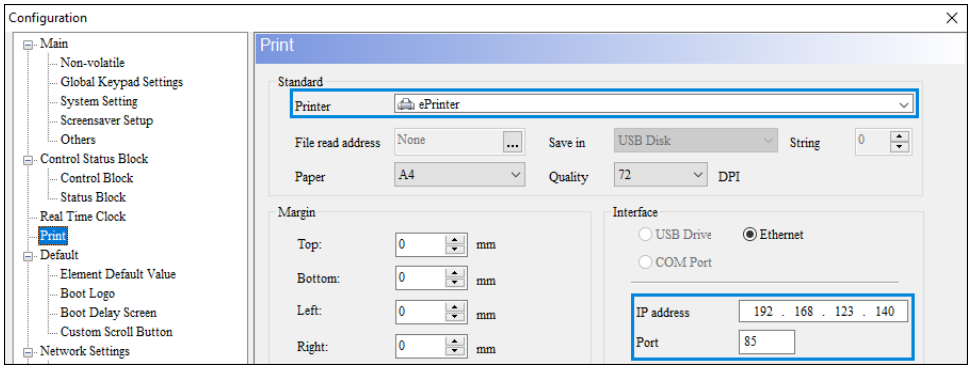
Setting	Description
Print All	Select to print all screens.
Custom Print	Select to print the custom screen. After selecting this item, the settings are available.
Screen No.	Set the number of the custom screen to print. You can individually set the read address of the custom screens. When entering 0, all screens configured in the **Print Settings** dialog will be printed. This setting can also be used with the **Printer flag** in the **Control Block**.  **Note**: This setting is only available when the **Custom Print** is selected.
Print Range	• **Left**: Set the X coordinate of the starting point of the **Print Range**. • **Top**: Set the Y coordinate of the starting point of the **Print Range**. • **Width**: Set the width of the **Print Range**. • **Height**: Set the height of the **Print Range**. **Note**: This setting is only available when the **Custom Print** is selected.

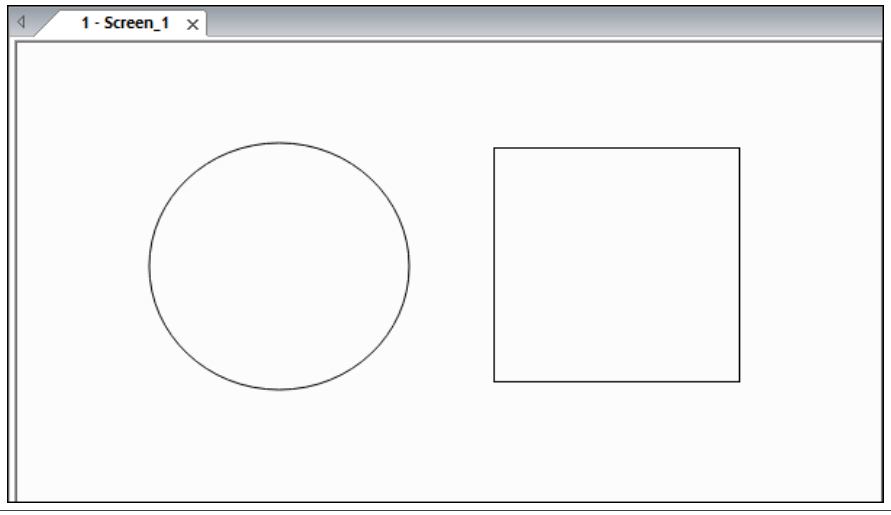
<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Each Picture</td><td>Set the number of screens to be printed per page.
<table>
<tr><th>One screen per page</th><th>Two screens per page</th><th>Four screens per page</th></tr>
<tr><td> </td><td> </td><td> </td></tr>
</table></td></tr>
</table>

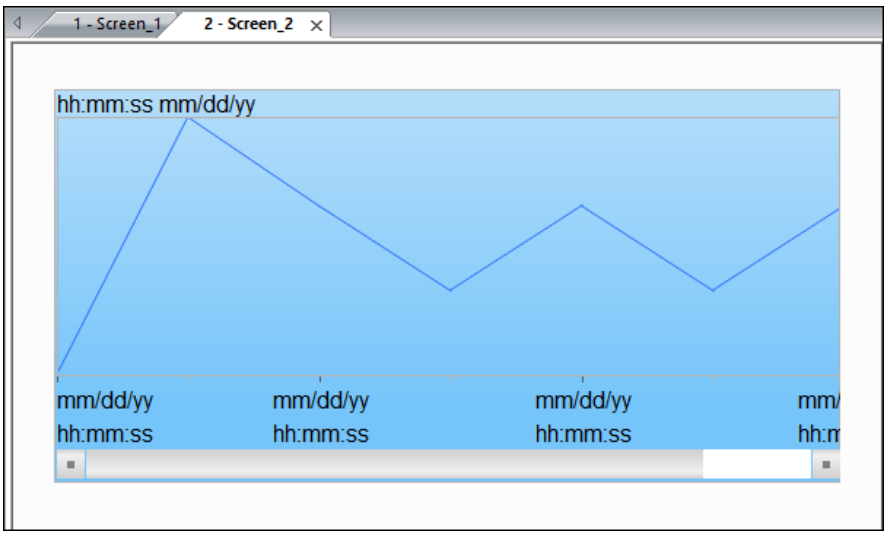
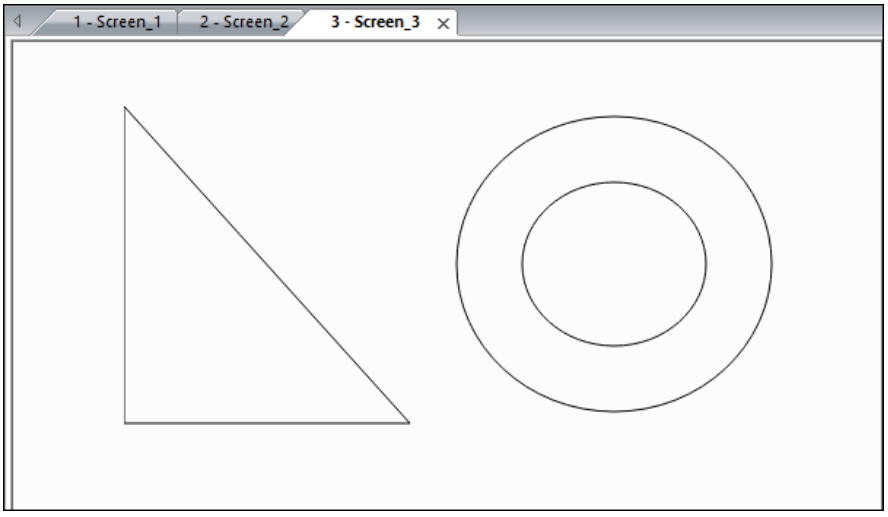
The following table lists an example of the print settings.

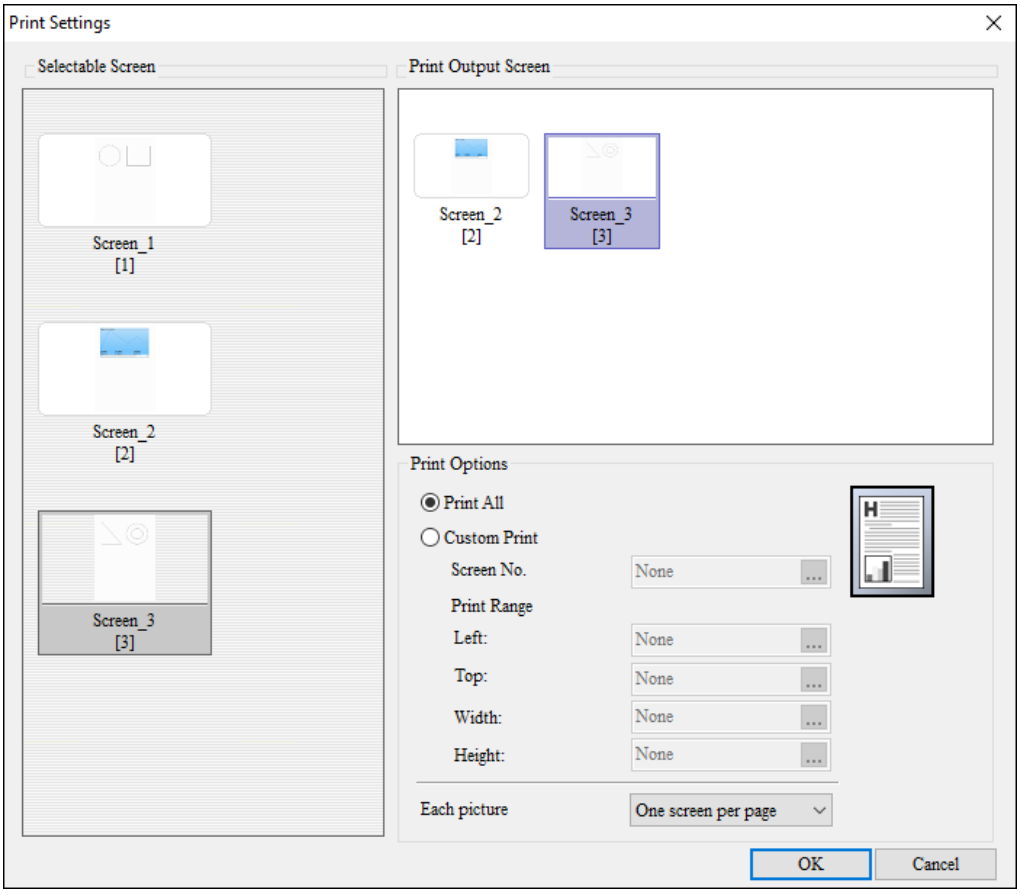
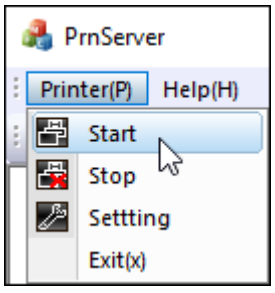
<table>
<tr><th colspan="2">Print Settings Example</th></tr>
<tr><td>Step 1</td><td>1. Create a project and configure the print related settings.

Printer: Select ePrinter.
IP address: Enter the IP address of the computer.
Port: Enter 85.



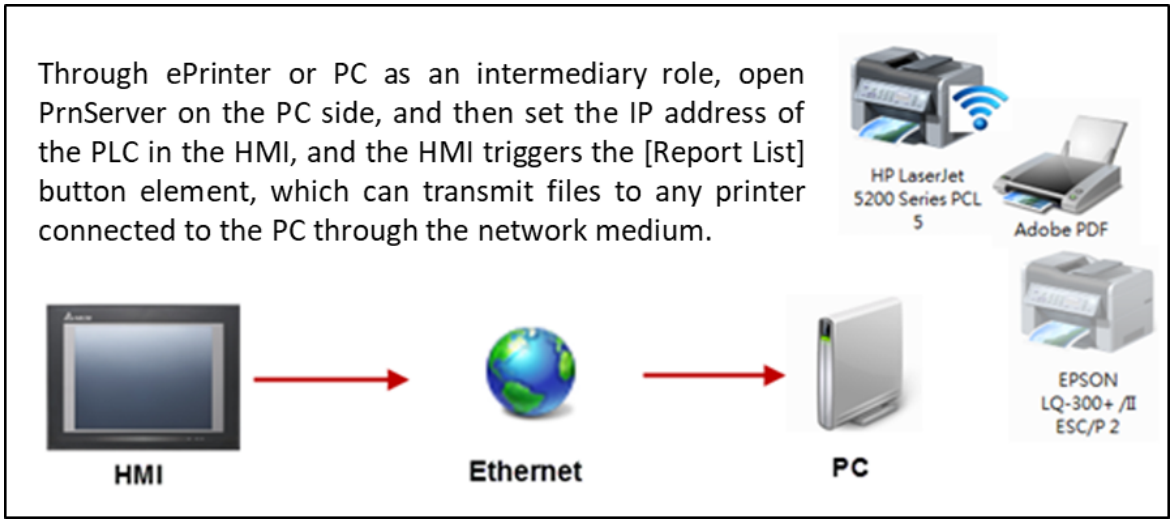
2. Create 3 print screens.
3. On the first screen, create a Circle element and a Rectangle element.

</td></tr>
</table>

Print Settings Example	
	4. On the second screen, create a **Historical Trend Graph** element.  5. On the third screen, create a **Right Triangle** element and a **Hollow Circle** element 

Print Settings Example	
Step 2	In the **Print Settings** dialog, drag Screen_2 and Screen_3 to the **Print Output Screen** area, and then click **OK**. 
Step 3	Create a **Print Output** button element.
Step 4	1. Set the IP address of the HMI and the ePrinter to the same network segment, and then compile and download the screen to the HMI. 2. Open PrnServer, and then click or **Start** in the **Printer** menu. 
Step 5	After PrnServer is connected to port 85, click **Print Output** button element to print the set screens. It takes more time for the HMI to process the print data if there are multiple screens or the history data element to be printed contains a large amount of sampling data.

ePrinter

Use ePrinter to print files online on the HMI without connecting to a physical printer, saving hardware space. The ePrinter architecture is as follows.

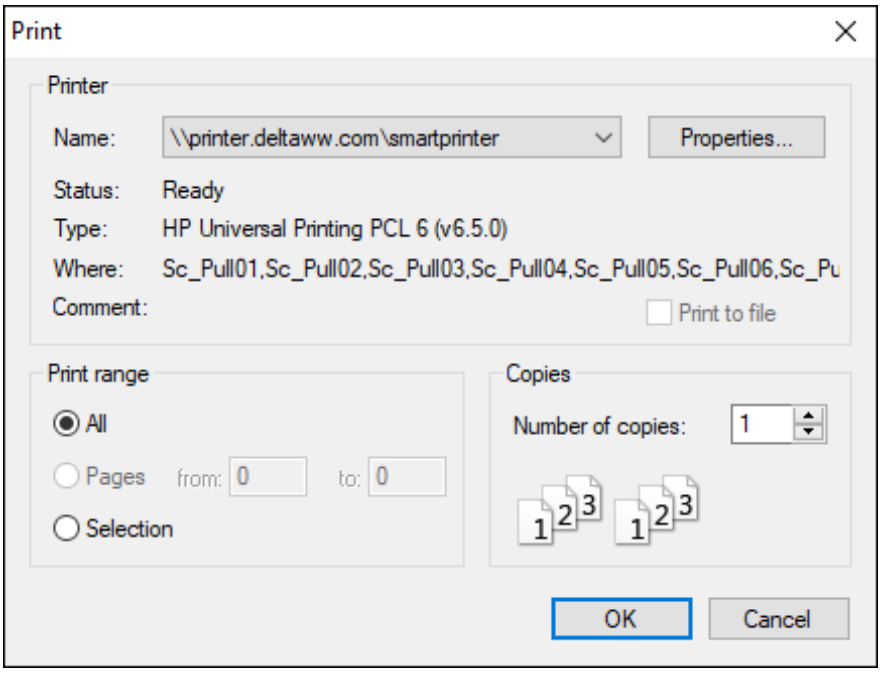
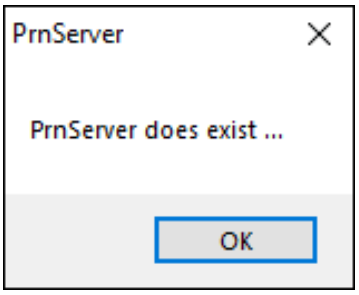
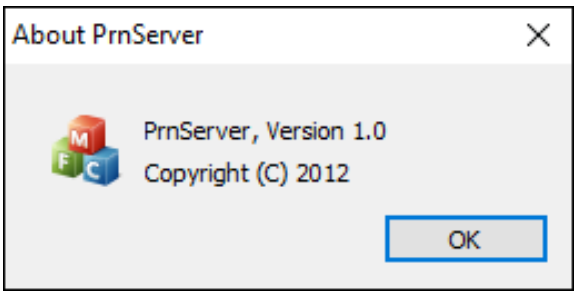


PrnServer

Use PrnServer to print files online. The storage path of the PrnServer execution file is `C:\Program Files (x86)\Delta Industrial Automation\DIAStudio\DIAScreen (version)\PrnServer.exe`.

The following table lists the icons in the **PrnServer** dialog with their description.

Menu	Icon	Description
Printer		Click to connect to the printer. • Before printing, click this icon to enable the connection. • The default connection port is 85, and the HMI will automatically connect to the default printer of the computer.
		After clicking this icon, the following message will appear.

Menu	Icon	Description
		Click to configure the relevant settings for the printer. 
	Exit(x)	In the menu bar, click Exit(x) to exit PrnServer. When you do not use network printing, be sure to exit. Clicking ☒ does not abort the connection and the following message appears when the PrnServer dialog is opened again. 
Help		Click to view the version information of PrnServer. 

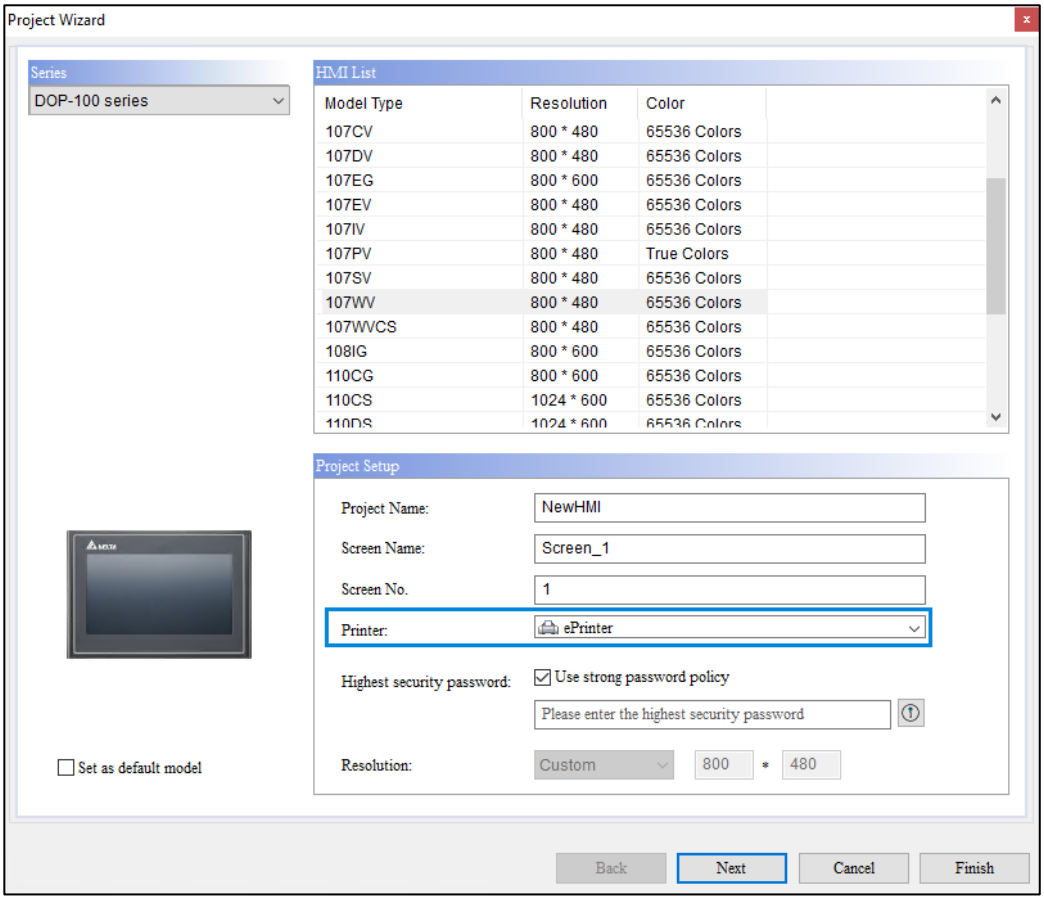
HMI Connection Settings

This section describes how to use the ePrinter function to trigger printing through the HMI.

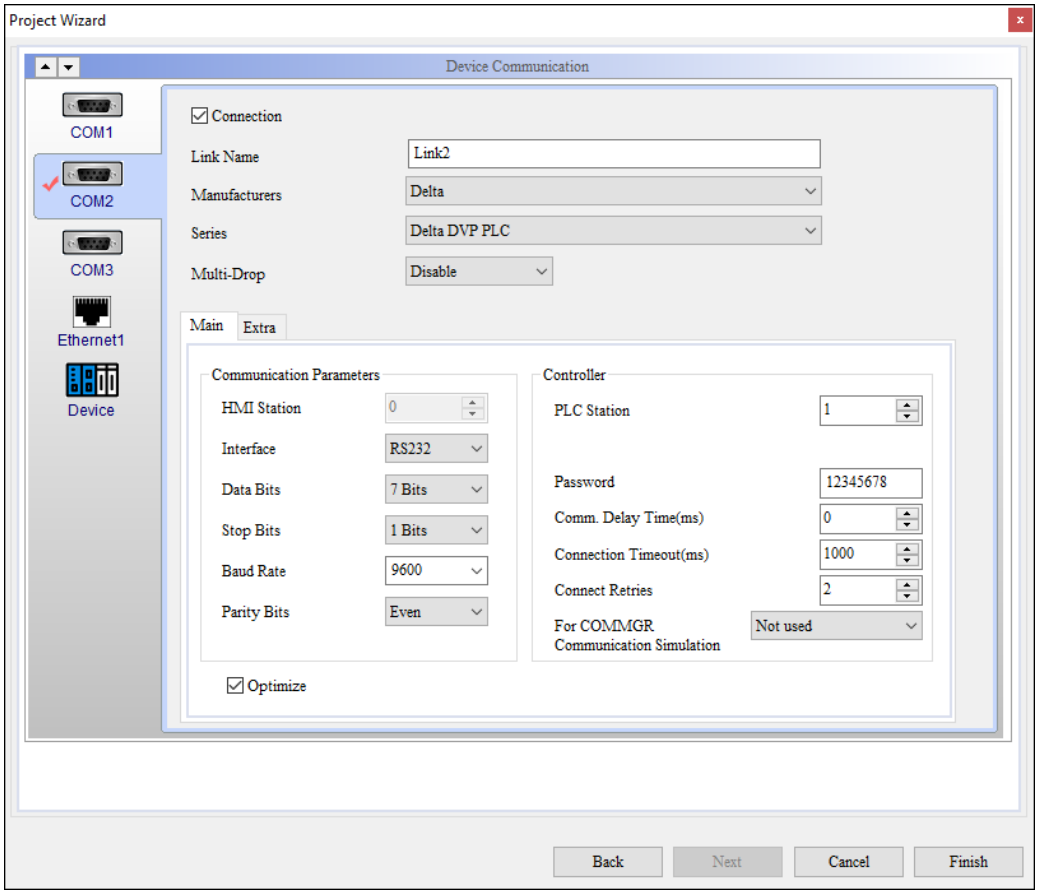
Follow these steps to configure the HMI connection settings.

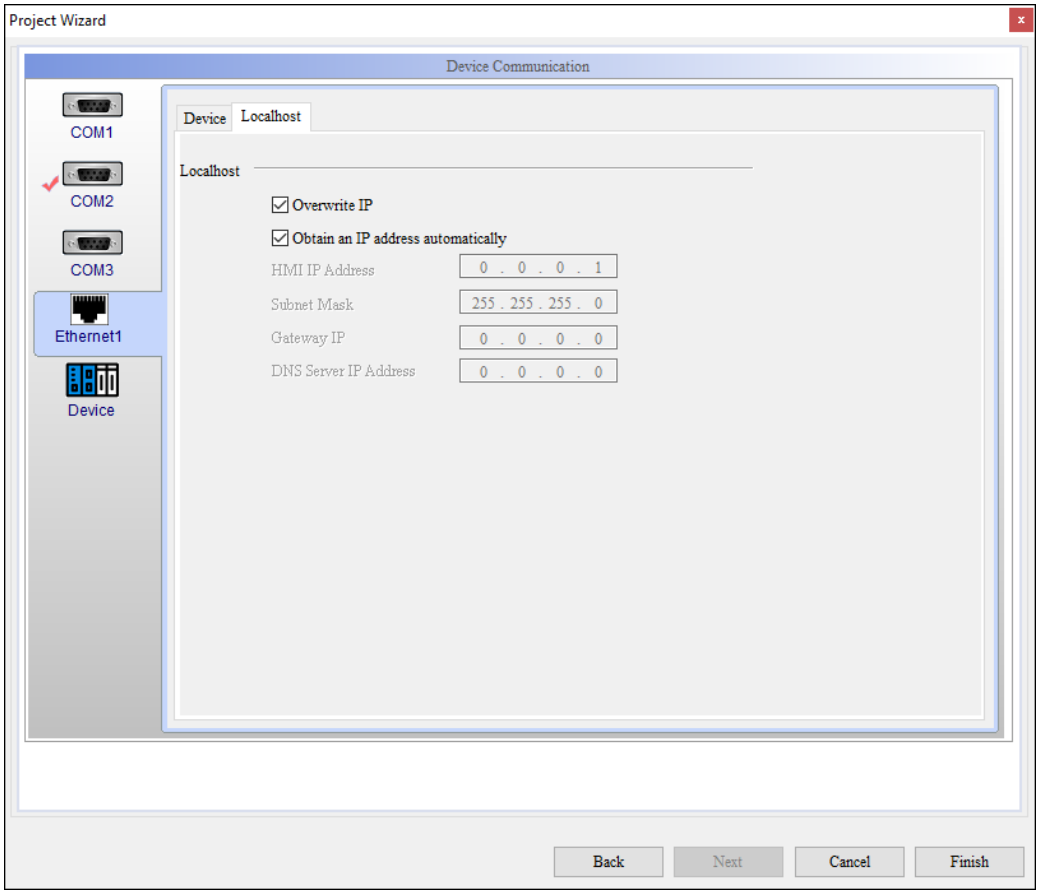
HMI Connection Settings

1. In DIAScreen, create a model project that supported printer function, select **ePrinter** for Printer, and then click **Next**.



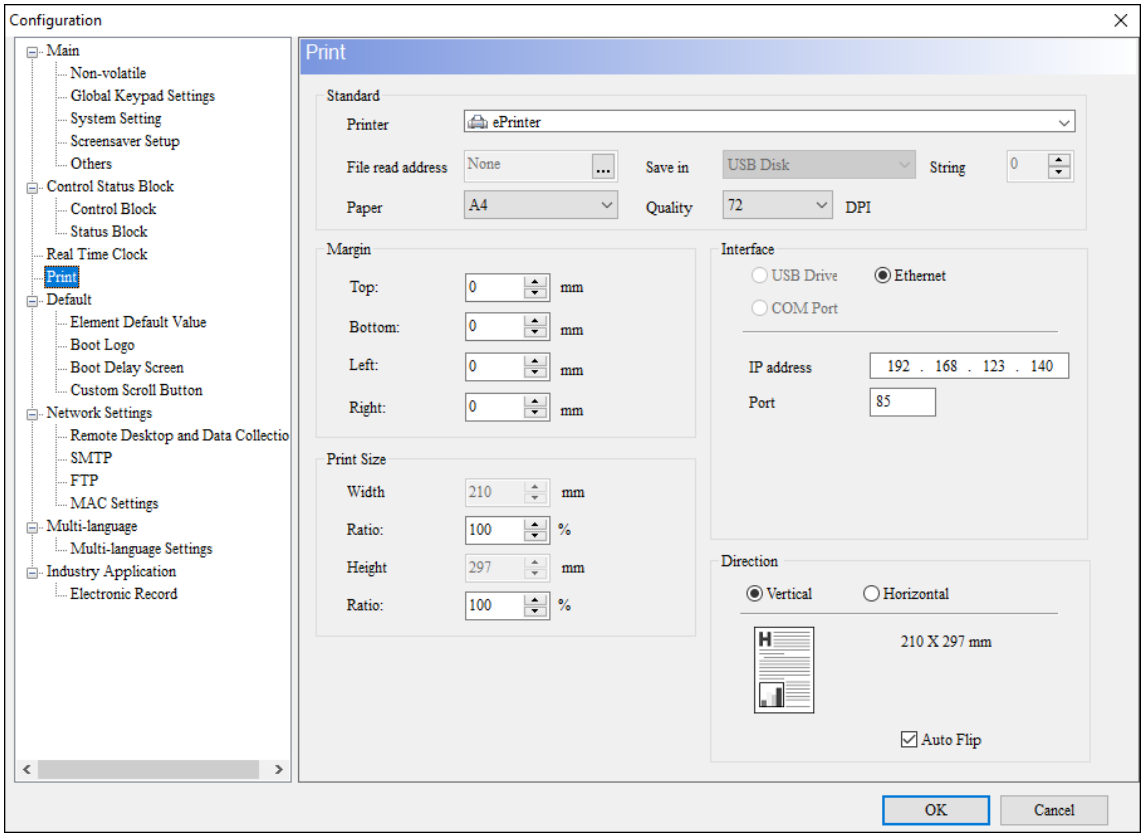
2. Configure communication related settings.


Project Wizard
Device Communication
COM1
COM2
COM3
Ethernet1
Device
Connection
Link Name
Link2
Manufacturers
Delta
Series
Delta DVP PLC
Multi-Drop
Disable
Main
Extra
Communication Parameters
HMI Station
0
Interface
RS232
Data Bits
7 Bits
Stop Bits
1 Bits
Baud Rate
9600
Parity Bits
Even
Controller
PLC Station
1
Password
12345678
Comm. Delay Time(ms)
0
Connection Timeout(ms)
1000
Connect Retries
2
For COMMGR Communication Simulation
Not used
Optimize
Back
Next
Cancel
Finish

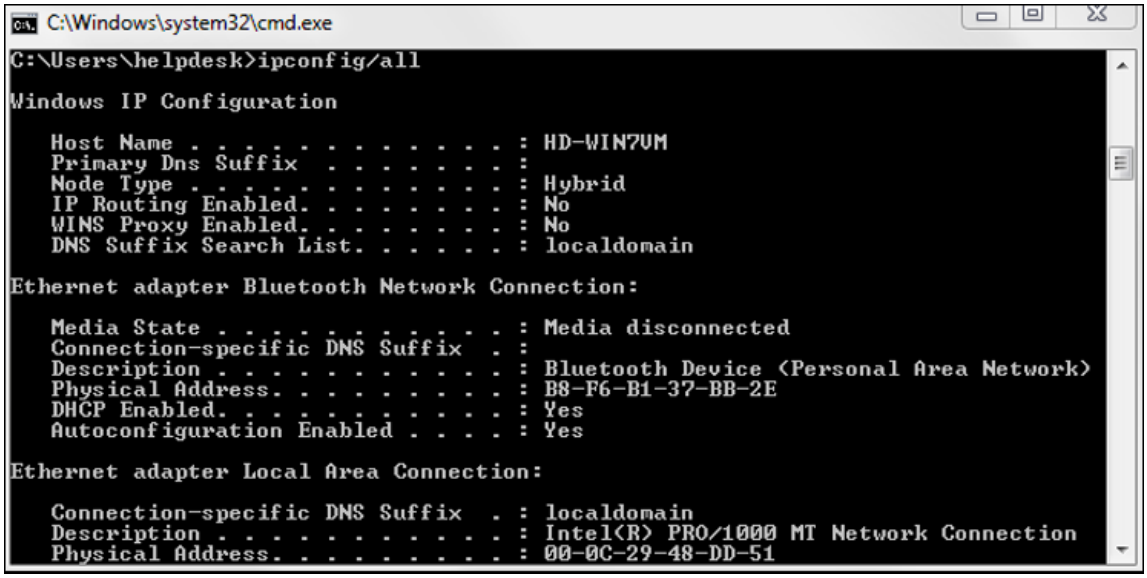

Project Wizard
Device Communication
COM1
COM2
COM3
Ethernet1
Device
Device
Localhost
Localhost
Overwrite IP
Obtain an IP address automatically
HMI IP Address
0 . 0 . 0 . 1
Subnet Mask
255 . 255 . 255 . 0
Gateway IP
0 . 0 . 0 . 0
DNS Server IP Address
0 . 0 . 0 . 0
Back
Next
Cancel
Finish

3. Go to **General** > **Configuration** > **Print**, select **Ethernet** in the **Interface** area, and then enter the computer IP address to enable PrnServer in the **IP address** field. You can enter ipconfig through Command Prompt to query the IP address of the computer.

   The following is an example of entering 192.168.123.140 for the **IP address**.



Note: This IP address and the IP address of HMI screen should be in the same network segment.



4. Enter the default connection port 85 of the printer in the **Port** field. This port is also the port where the printer is connected after the PrnServer connection is enabled.
5. Create **Rectangle**, **Circle**, and **Print Output** button elements on the screen.
6. Compile and download the screen to the HMI.
7. Trigger the **Print Output** button on the HMI to complete printing.

   Note: You must first open PrnServer and enable the connection.

Printing Error Code

If an error occurs when the printing function is performed, refer to the error code displayed by the printer to figure out and troubleshoot the error.

Code	Definition	Cause	Troubleshooting
-2	ERROR_PRINT_PORT Printer port error	USB, parallel port or COM selection error.	Check if the set transmission port is correct.
-3	ERROR_MEMORY_NULL Memory configuration error	HMI memory is insufficient to process print data.	Delete partial screen data to be printed, or select another printer with a larger memory capacity.
-4	ERROR_USB Failed to print due to printer error	The printer fails to print, ink run-out, paper jam.	Print again after troubleshooting.
-5	ERROR_USB_NOT_SELECT Failed to start the printer and unable to connect	The printer has been connected, but HMI failed to recognize it.	The printer driver cannot be connected to the HMI. Contact our Global Customer Service Department.
-6	ERROR_USB_PAPER Failed to print due to paper run-out	The printer displays "out of paper".	Load paper.
-7	ERROR_USB_NOT_CONNECT Failed to connect to the printer	The USB cable is not connected to the printer.	Check if the USB cable is connected to the printer correctly.
-8	ERROR_USB_OPEN Failed to enable USB	Failed to enable USB when starting to print.	Restart the HMI. If printing still fails, contact our Global Customer Service Department.
-9	ERROR_USB_CLOSE Failed to disable USB	Failed to disable USB when printing is finished.	Restart the HMI. If printing still fails, contact our Global Customer Service Department.

Code	Definition	Cause	Troubleshooting
-11	ERROR_NOT_OK The printer has not been initialized	The printer will be initialized when powered on. If you print at this time, the printer will display this error message.	Print after the initialization is completed.
-14	PRINTER ERROR The printer fails to print	CTS pin status error	Check if the CTS communication pin is connected. If printing still fails, contact our Global Customer Service Department.

Template Screen

Use the **Template Screen** function to customize the header and footer of the output PDF file, which contains history data, recipes, alarms, and operation logs.

Follow these steps to set the template screen and print.

To set the template screen and print

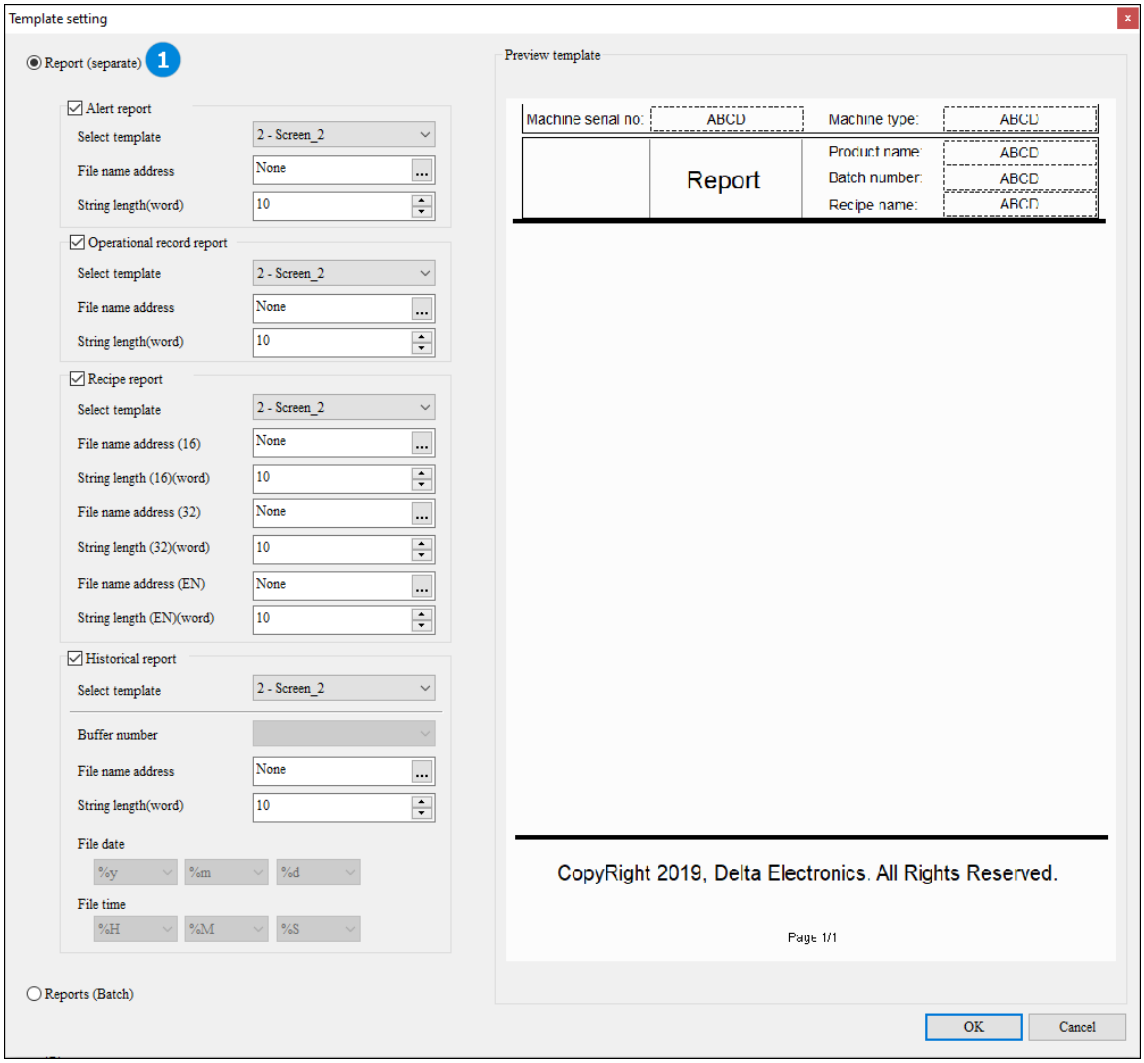
3. Add a template screen. Do any of the following:
   - Click **General** > **New Screen** > **Template Screen** on the toolbar.

   Or

   - In the **Screen Management** pane, right-click the blank space and select **New Screen**, and then select **Template Screen** for the **Screen Type**.
4. On the Template Screen, double-click the blank space.
5. Configure the position, size, and font of the page number in the **Template setting** area of the **Screen property setting** dialog. You can also right-click the **Screen Management** pane and select **Screen Properties** to configure the position of the page number.
6. Create a **Template Output** element, and then double-click the element.
7. Click **Template Setting** in the element properties dialog.
8. Edit the template in the **Template Setting** dialog.

   The following table list the settings in the **Template Setting** dialog with their description.

   ❶ **Report (separate)**



- **Alert report**

Setting	Description
Selection template	Select the corresponding template according to the created template screen.
File name address	Set the file name of the output alarm data file.
String length	Select or enter the string length of the file name.

- **Operational record report**

Setting	Description
Selection template	Select the corresponding template according to the created template screen.
File name address	Set the file name of the output operation log file.
String length	Select or enter the string length of the file name.

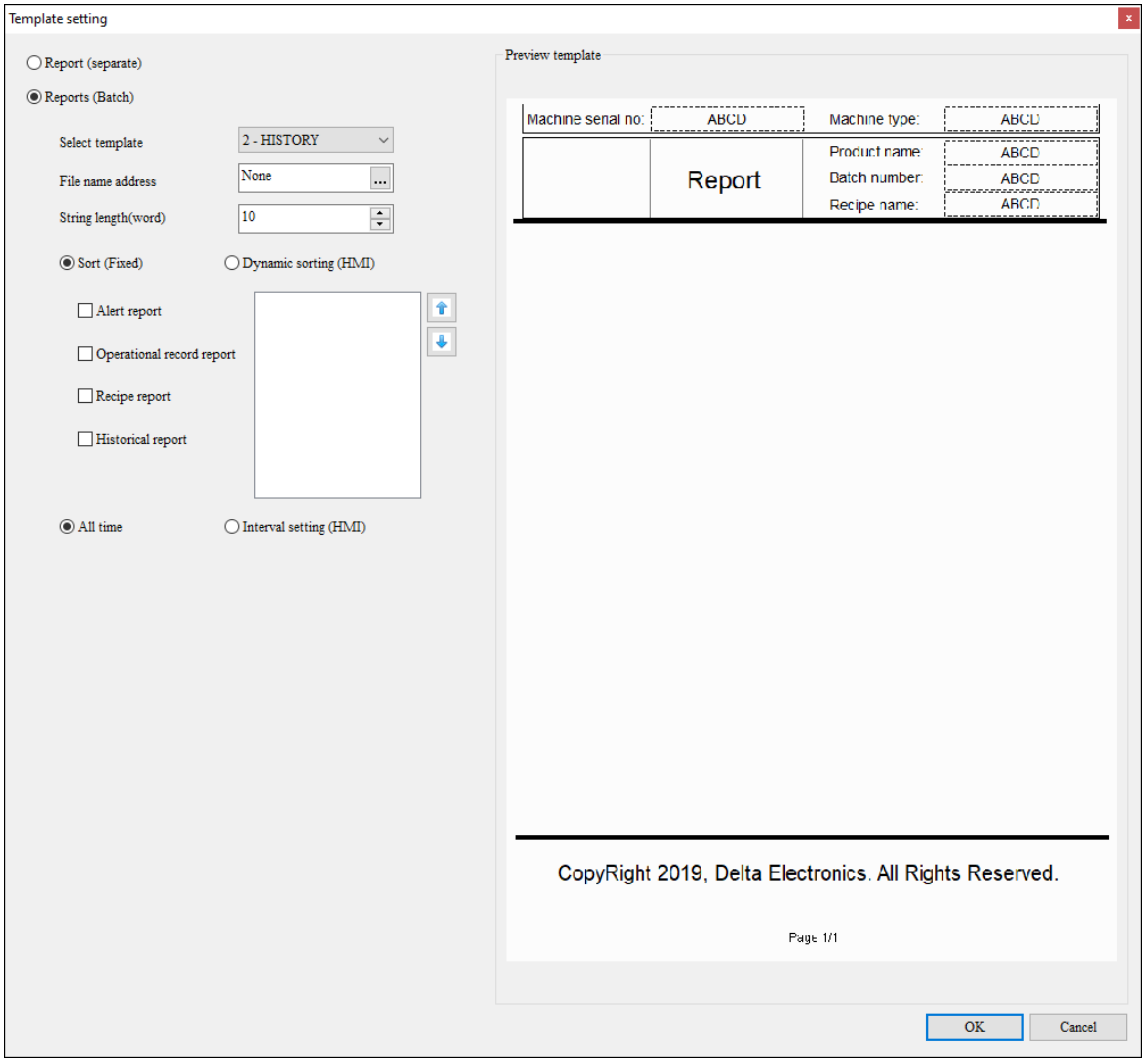
- **Recipe**

Setting	Description
Selection template	Select the corresponding template according to the created template screen.
File name address (16)	Set the file name of the output 16-bit recipe file.
String length (16)	Select or enter the string length of the file name.
File name address (32)	Set the file name of the output 32-bit recipe file.
String length (32)	Select or enter the string length of the file name.
File name address (EN)	Set the file name of the output enhanced recipe file.
String length (EN)	Select or enter the string length of the file name.

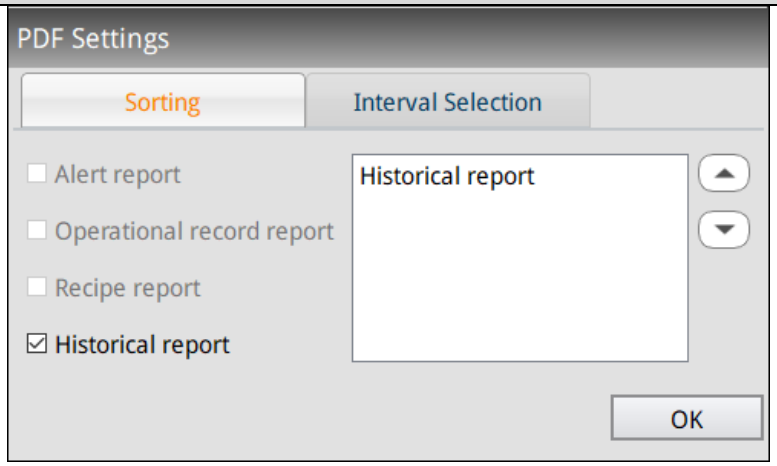
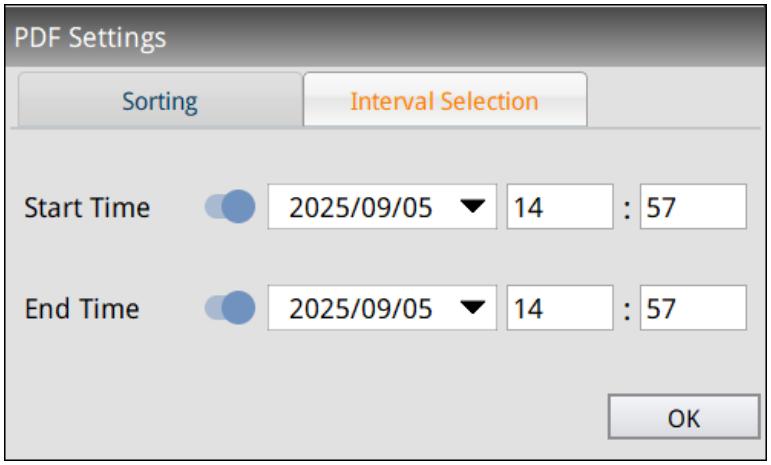
- **Historical report**

Setting	Description
Selection template	Select the corresponding template according to the created template screen.
Buffer number	Select the buffer data to output. • If select 0, all buffer data will be output. • If select 1, the data corresponding to the buffer number 1 will be output, and so forth.
File name address	Set the file name of the output historical data file.
String length	Select or enter the string length of the file name.
File date	Select the year, month, and date of the file. It can also be set to **None** to not output the file date.
File time	Select the hour, minute, and second of the file. It can also be set to **None** to not output the file time.

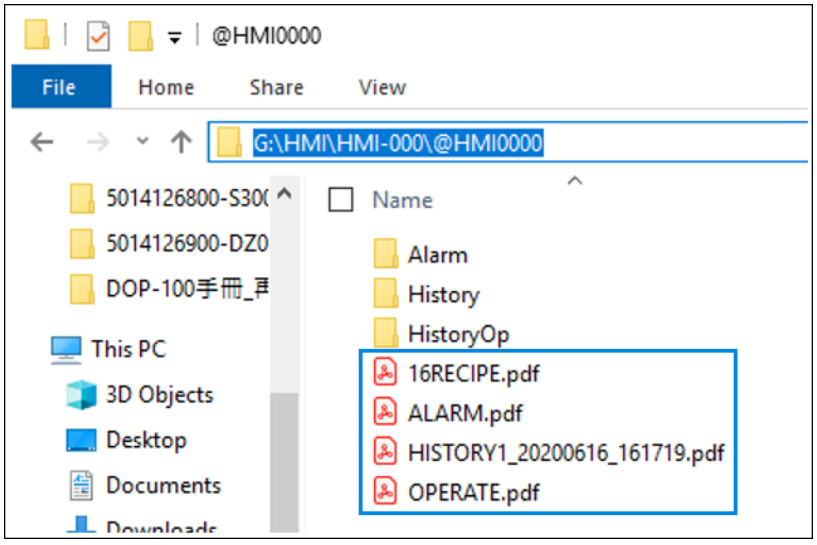
❷ Report (Batch)



Setting	Description
Select template	Select the corresponding template according to the created template screen.
File name address	Set the file name of the output data.
String length	Select or enter the string length of the file name.
Sort (Fixed)	Select to customize the report sorting. Click to change the display order of the report.
Dynamic sorting (HMI)	Select to set the report sorting on the HMI.

Setting	Description
	
All time	Select to record all time data.
Interval setting (HMI)	Select to set the time interval to be recorded on the HMI. 

9. Perform template printing. The PDF files output after printing is stored in the folder `\HMI\HMI-000\@HMI0000`.



Change Model

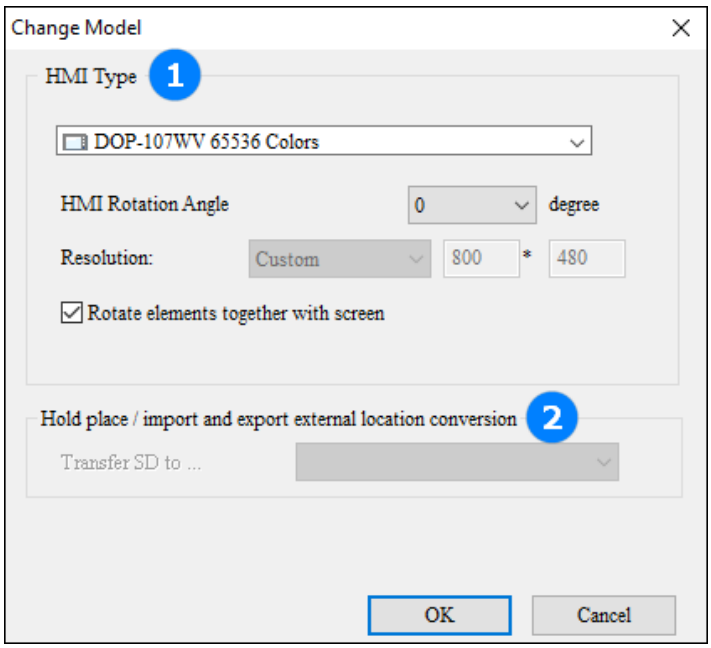
Use the **Change Model** function to change the HMI to another model.

Note: DIAScreen supports opening the HMI screens of DOP-B, DOP-W, and DOP-H models and automatically changes the screens to the DOP-100 model for editing.

How to open the **Change Model** function?

- On the toolbar, click **Tools** > **Change Model**.

The following table lists the settings in the **Change Model** dialog with their description.



1 HMI Type

Setting	Description
Type	Select the HMI type to change.
HMI Rotation Angle	Select the rotation angle of the HMI screen. If the **Rotate elements together with screen** checkbox is not selected, the element will maintain its orientation when the HMI screen rotates. **Note**: This setting only applies to the same type.
Resolution	Select the resolution of the HMI screen. **Note**: This setting is only available for the AX series models.

❷ Hold place / import and export external location conversion

Setting	Description
Transfer SD to	Select the storage location of data after HMI model changes. If the model you are using, such as DOP-110WS, supports SD card, and the SD card is used by an element, this setting will display the USB Disk option when the HMI is changed to another model.

Address Conversion

After selecting an element on the screen, modify all the addresses of the selected element directly in the list through the address conversion function. You can also replace the address for a specific element.

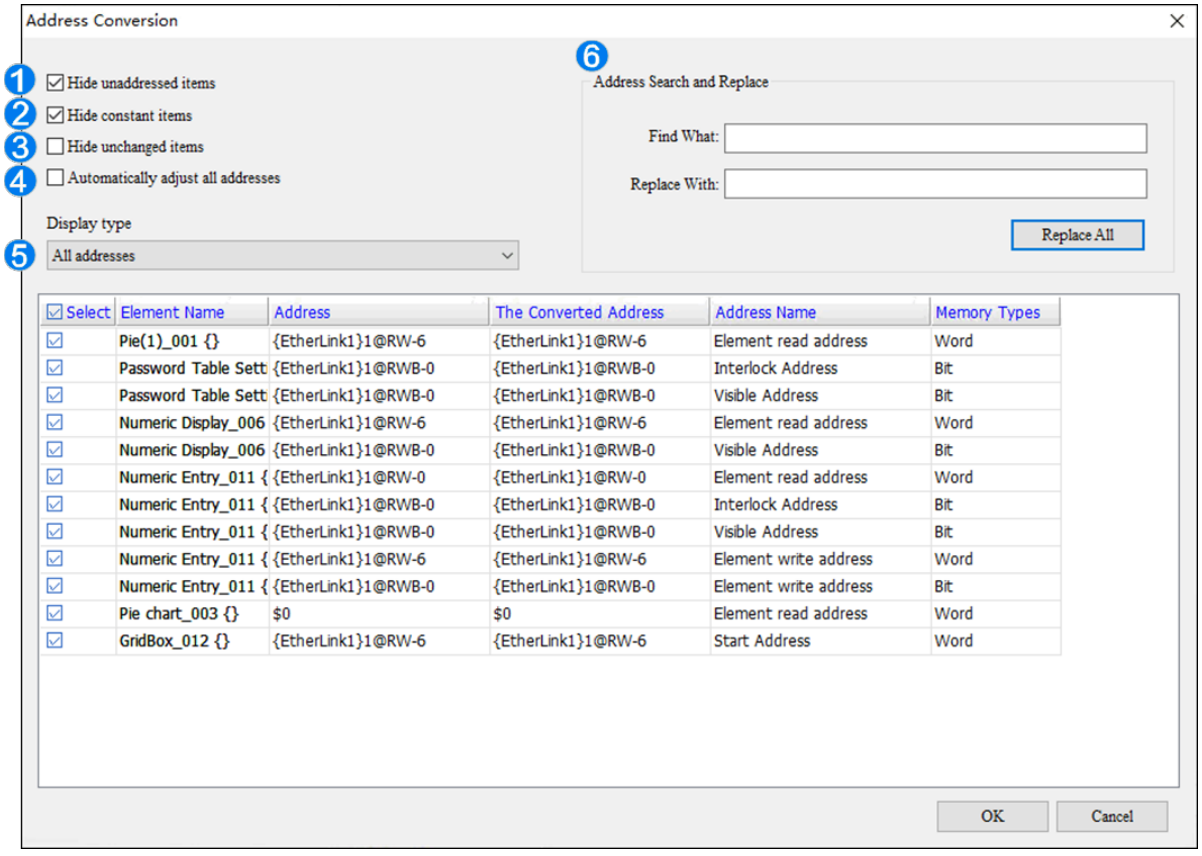
How to open **Address Conversion**?

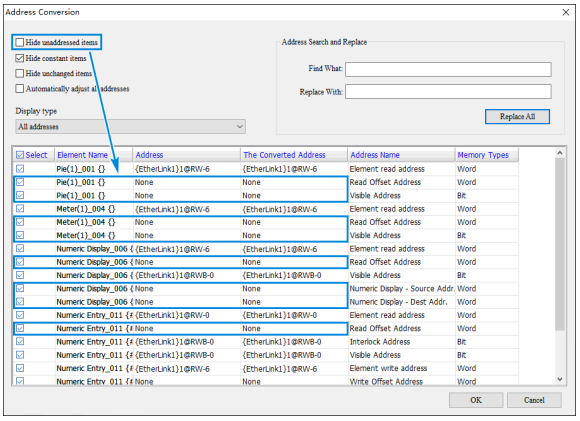
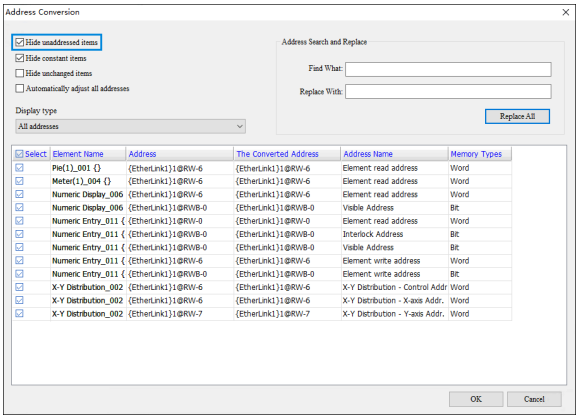
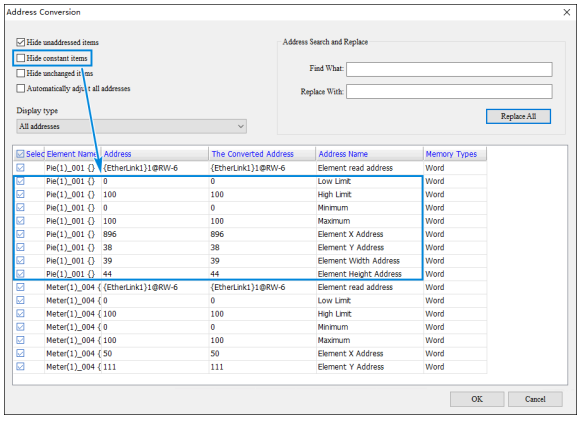
- On the menu bar, click **Tools** > **Address Conversion**.

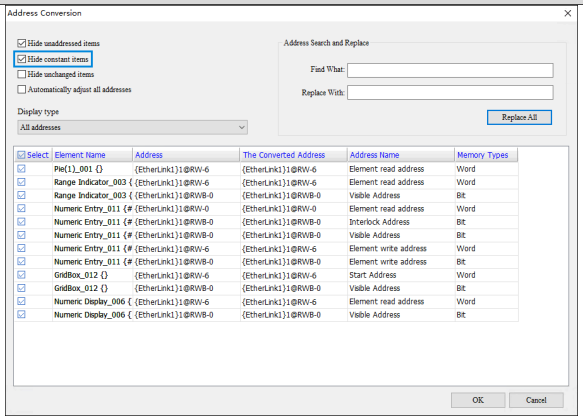
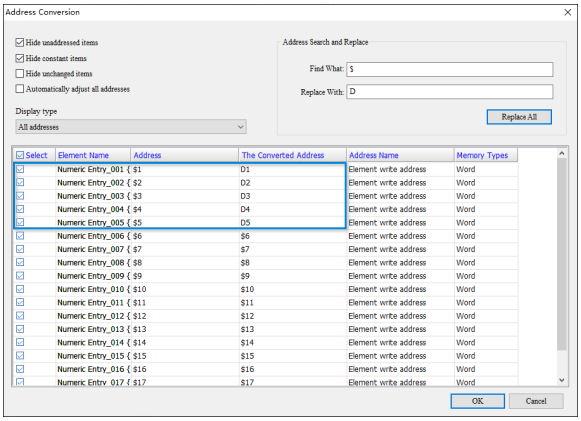
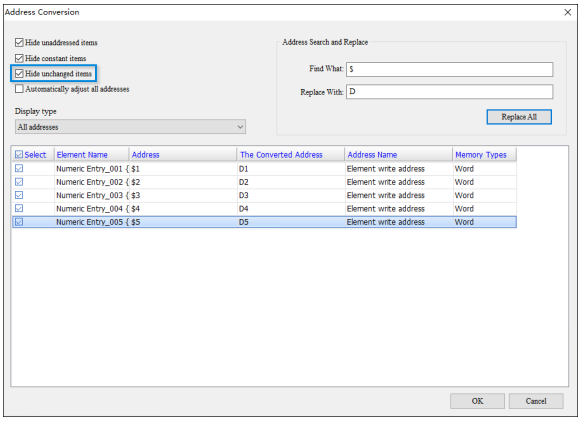
Or

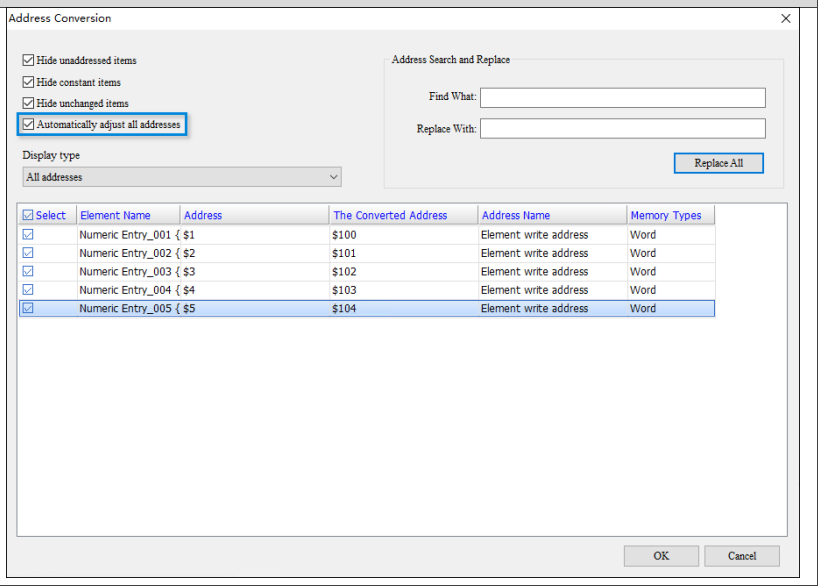
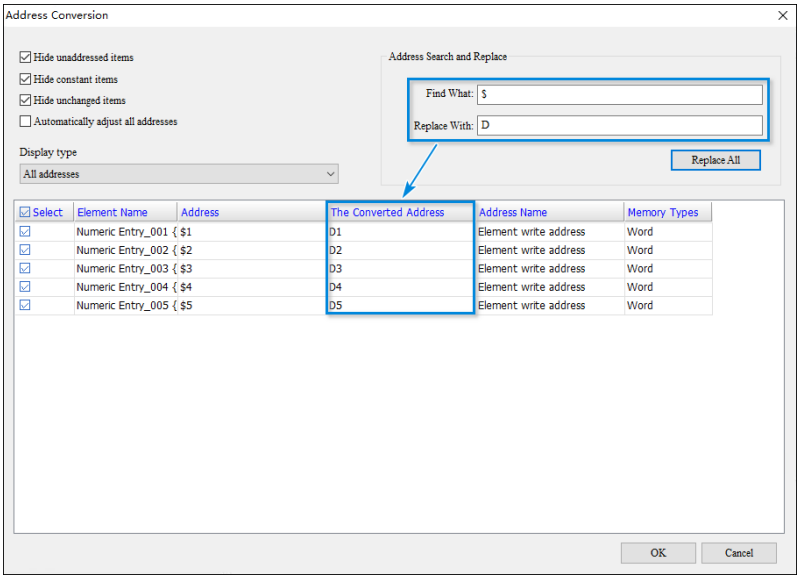
- Right-click the element and select **Address Conversion**.

The following table lists the settings in the **Address Conversion** dialog with their description.



<table>
<tr><th>Legend</th><th>Function</th><th>Description</th></tr>
<tr><td>1</td><td>Hide unaddressed items</td><td>Select to hide items whose addresses are not set.
<table>
<tr><th>Function</th><th>Illustration</th></tr>
<tr><td>Unselected</td><td>Display addresses with None.
  </td></tr>
<tr><td>Selected</td><td>Hide addresses set to None.
  </td></tr>
</table></td></tr>
<tr><td>2</td><td>Hide constant items</td><td>Select to hide addresses set to constant, such as element position and upper/lower limit value.
<table>
<tr><th>Function</th><th>Illustration</th></tr>
<tr><td>Unselected</td><td>Display constant addresses.
  </td></tr>
<tr><td>Selected</td><td>Hide constant addresses.</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>Legend</th><th>Function</th><th>Description</th></tr>
<tr><td></td><td></td><td>  </td></tr>
<tr><td>3</td><td>Hides unchanged items</td><td>If selected, only items with changes to The Converted Address field will be displayed.
<table>
<tr><th>Function</th><th>Illustration</th></tr>
<tr><td>Unselected</td><td>Display all element addresses.
  </td></tr>
<tr><td>Selected</td><td>Hide unchanged items.
  </td></tr>
</table></td></tr>
<tr><td>4</td><td>Automatically adjust all addresses</td><td>Select to adjust all addresses automatically. Changing the address of the first item in The Converted Address column and then selecting the item will automatically offset The converted address according to the original address offset.</td></tr>
</table>

Legend	Function	Description
		
❺	Display type	Select the type of address to display in the address list. The default is **All addresses**.
❻	Address Search and Replace	Enter **Find What** and **Replace With** to find and replace the string with a new string. 

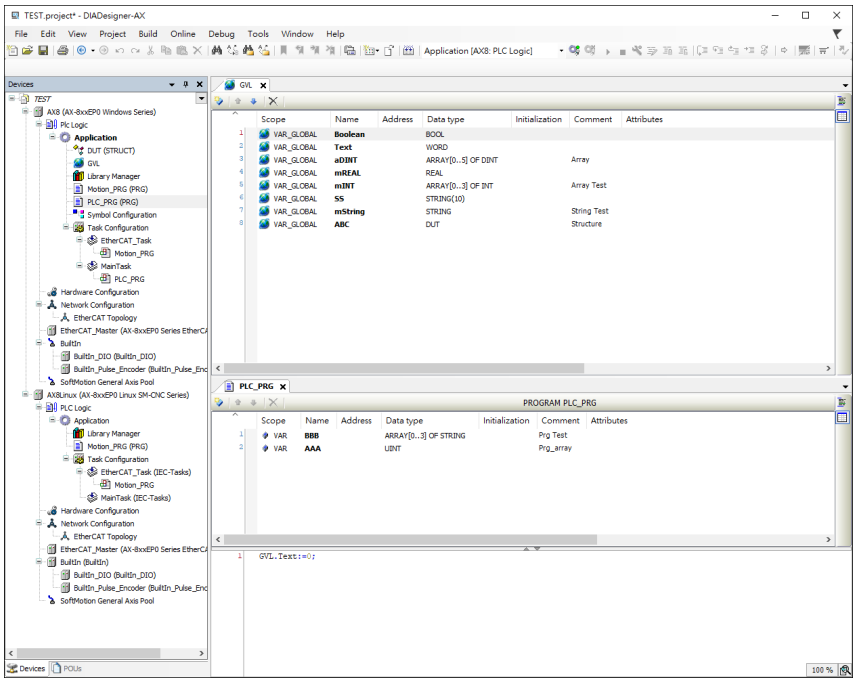
All Tags

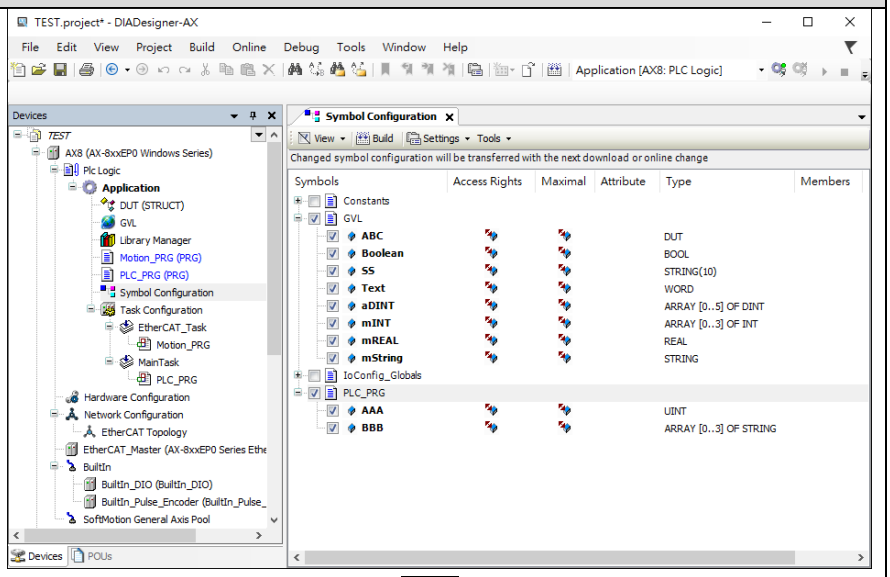
You can convert a memory address to tag through the DIASceen Tag configuring function for use as well as use the import function of different files to import the variables declared in the programming software to DIAScreen.

Built-In PLC Connection Settings

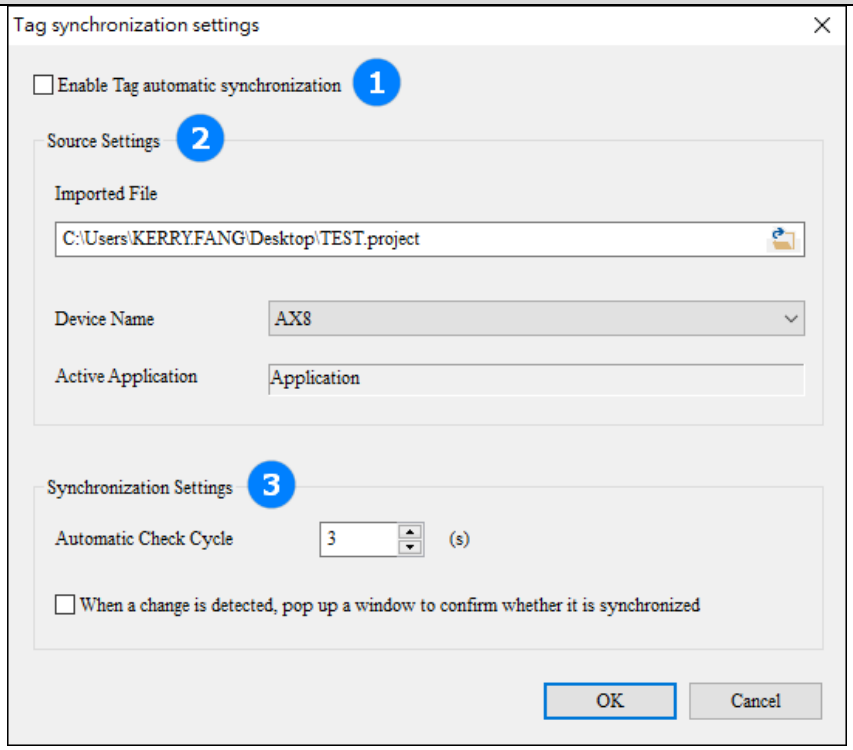
This function is only applied to the AX-8 series HMI. Import CODESYS Tag into DIAScreen, then download the HMI project to the AX-8 Soft-HMI and directly communicate with the internal CODESYS PLC.

The following table lists an example of **Built-in PLC Connection Settings**.

Build-In PLC Connection Settings Example	
Create PLC variables	1. Use DIADesigner-AX or CODESYS software to create a project and create global variables or POU variables.  2. Create a symbol table and select the variables to use.

Build-In PLC Connection Settings Example	
	 3. Select the device and click to compile. If CODESYS software is used, click **Build** > **Generate Code** on the toolbar.
Import Variable to DIAScreen and enable automatic synchronization	1. Create AX-8 series project in DIAScreen. 2. Double-click **Built-In PLC** in the **Project** pane. 3. Click to select project file (.project) or Tag file (.xml)。 4. Click to enable automatic synchronization and set the **Automatic Check Cycle**. **Note**: This function is only used for project file (.project) import.

Build-In PLC Connection Settings Example	

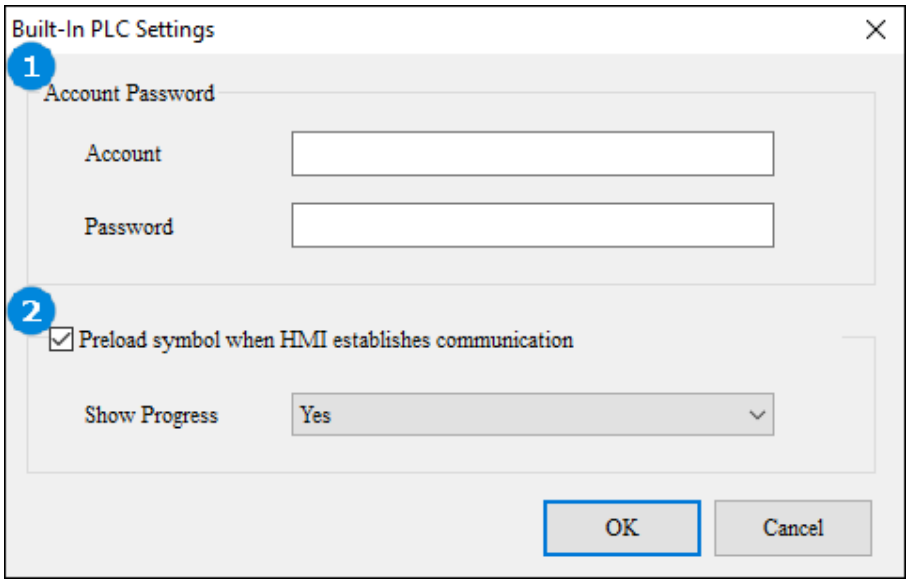


1 **Enable Tag automatic synchronization**: If selected, DIAScreen automatically checks whether the symbol table has been updated according to the set Check Cycle.

2 **Source Settings**: Displays the synchronized project file path, and you can also change the file path here, and then select the Device Name to communicate with the HMI.

3 **Synchronization Settings**: Set the automatic check cycle and the value range is 3–60, and then select the **When a change is detected, pop up a window to confirm whether it is synchronized** checkbox. If not selected, a dialog will not appear when the Tag is updated, and the Tag table will be automatically updated.

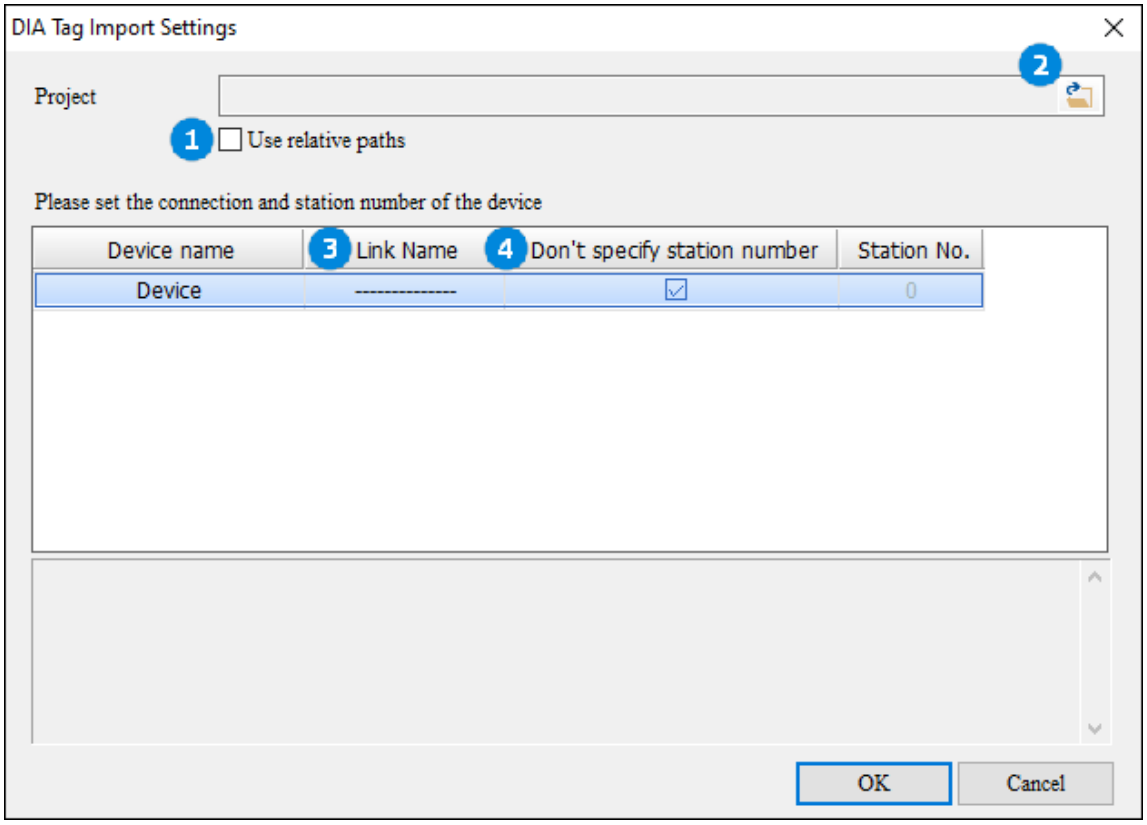
5. Create an element on the screen and select **Built-In PLC** for **Link** in the address Input dialog of the element, and the Tag will be configured in the element address.

Build-In PLC Connection Settings Example	
Built-In PLC Settings	 1 **Account Password**: If PLC has configured security settings, the account and password are required when the HMI communicates with PLC, and then enter the account and password of the PLC in the **Account** and **Password** fields. 2 **Preload symbol when HMI establishes communication**: If selected, symbol information will be loaded when the HMI is started to avoid communication errors caused by executing communication before the symbols are loaded. You can choose whether to display the loading progress bar.

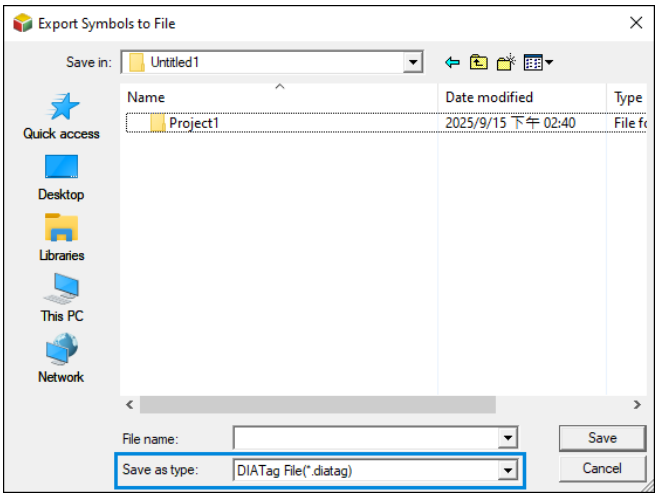
DIA Tag

Synchronize the global variables declared in DIADesigner or DIADesigner-AX through DIA Tag function to DIAScreen and use. The following describes the setting method of DIA Tag.

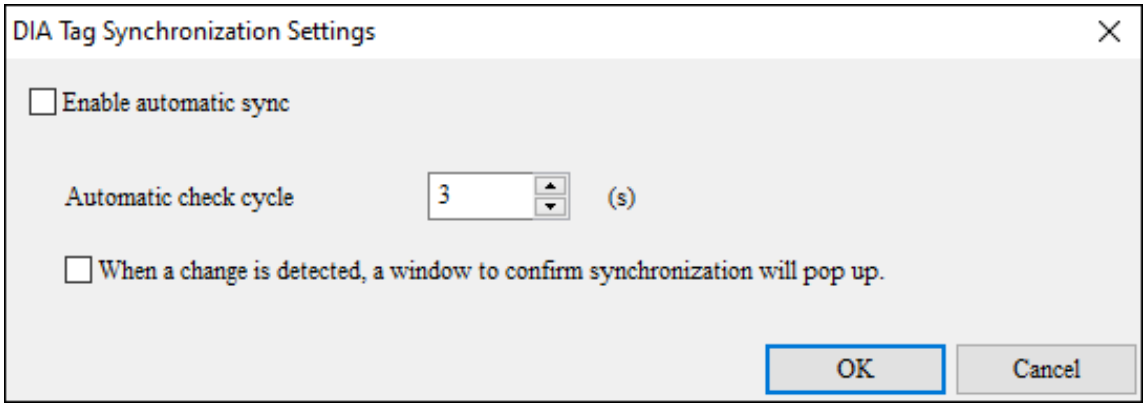
- **Import**



Legend	Function	Description
1	Use relative paths	If selected, the DIAScreen project file (.dpa) must be placed in the same folder as the variable project file (.project or .diade), so that the tags keep synchronized when moving the project folder path.
2	Project Selection	Select the project file of the variable. The supported file types are as follows. • **.project**: DIADesigner-AX project file. After compiling the project, you need to confirm that there is a generated XML file in the project path. • **.diade**: DIADesigner project file. After compiling the project, confirm that there is a DIA Tag folder in the project path. • **.diatag**: The file exported from ISPSoft or DIADesigner.

Legend	Function	Description
		○ When exporting files in ISPSoft, select to save as a diatag file.  ○ In the **Project Explorer** of DIADesigner, right-click **Global Variable** and then select **Export Variable Table** to save all variables in the variable table as a diatag file. • **.xml**: The XML file generated by DIADesigner-AX.
❸	Link Name	After importing the project file, select the link name corresponding to DIAScreen. You need to create a connection in **Device Communication** and select the correct controller. If the controller does not match the imported device, the DIA Tag is not be allowed to be imported.
❹	Don't specify station number	Tags imported by DIADesigner and the station number can be set. • If selected (default), set the station number when the DIA Tag is used on the element. • If not selected, enter the station number in the **Station No.** field.

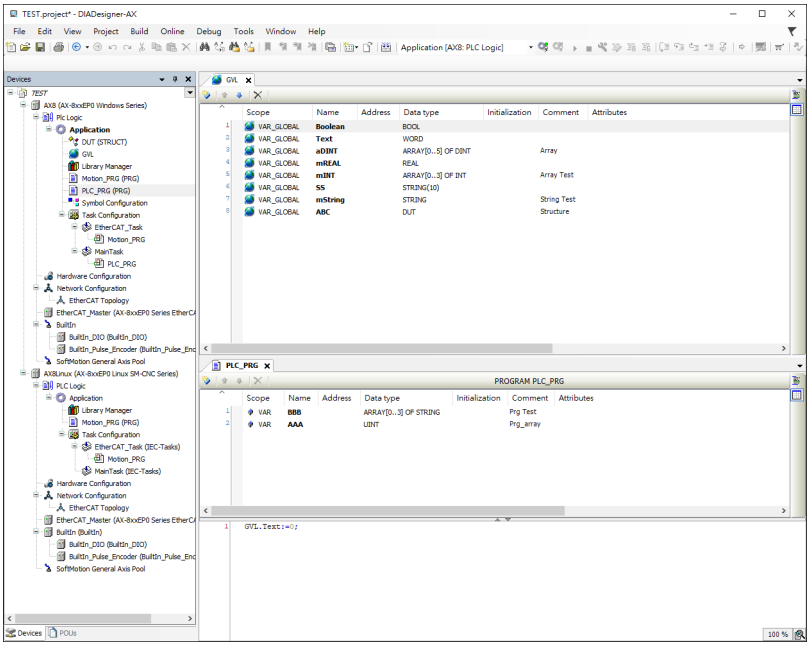
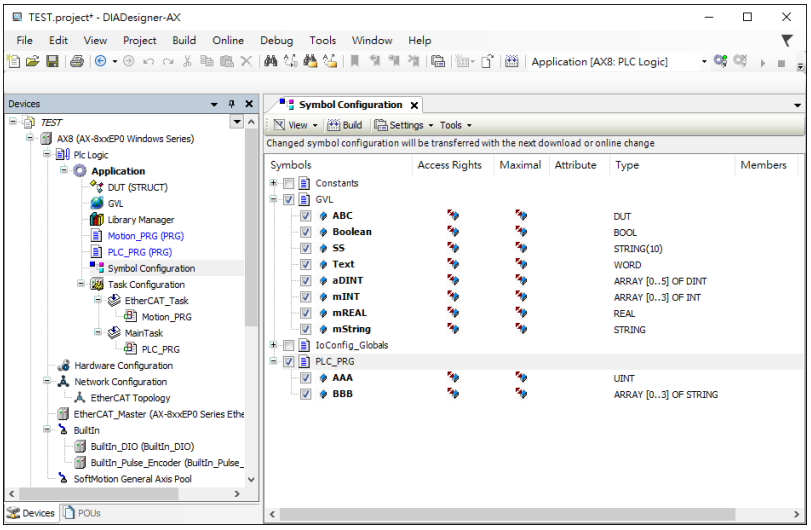
- **Enable DIA Tag automatic synchronization**

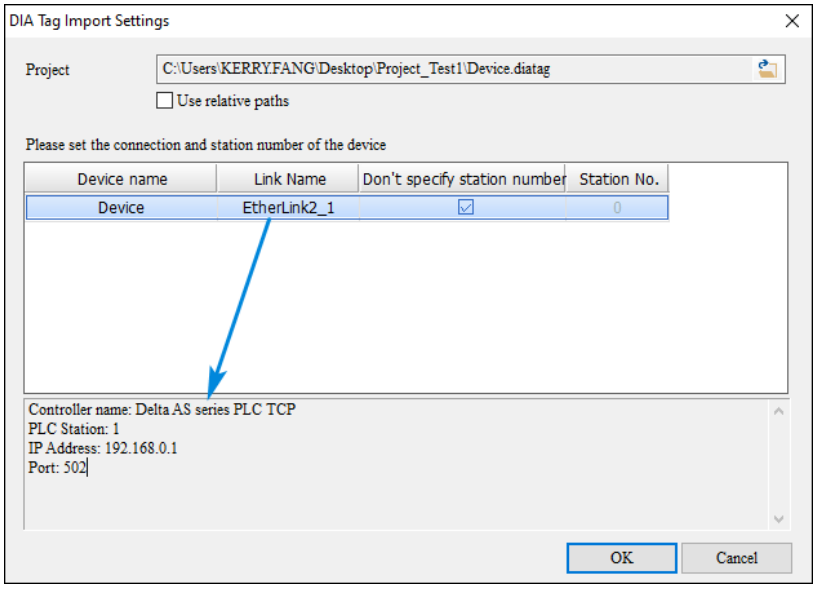


Setting	Description
Enable automatic sync	Select to **Enable Tag automatic synchronization**. Once the code is generated in DIADesigner-AX, DIAScreen automatically checks whether the symbol table has been updated according to the set Check Cycle.
Automatic check cycle	Set the automatic check cycle. The value range is 3–60, and the default value is **3**.
When a change is detected, a window to confirm synchronization will pop up	• If selected, a message dialog appears when a change is detected and asks you whether to synchronize automatically. ○ Select **Yes** to perform automatic synchronization and update the Tag List in DIAScreen. ○ Select **No** to not to update the Tag List in DIAScreen and turn off the automatic synchronization function. • If not selected, a message dialog will not appear when a change is detected, and the Tag List will be automatically updated.

The following describes an example of using the DIA Tag.

DIA Tag Usage Example	
Create variables in DIADesigner	1. Create a DIADesigner project and add a controller device include PLC. 2. Double-click **Programming**, right-click **Global Variable**, and then select **Add Global Variable Table**. 3. In the **Add Global Variable Table** dialog, enter the **Name** and **Comment**, and then click **OK**. 4. In the Global Variable table, add variables. **Note**: The variable address must be configured. If not configured, DIAScreen will not be able to use the Tag.

DIA Tag Usage Example	
	5. Click **Compile** on the toolbar.
Create variables in DIADesigner-AX	1. Create a DIADesigner-AX project and create global variables or POU variables.  2. Create a symbol table and select the variables to use.  3. Select the device and click to compile.
Import variable project file into DIAScreen	1. Create a DIAScreen project. 2. In the Device Communication dialog, select Delta controller. **Note**: • The controller type must be the same as the controller type created in DIADesigner.

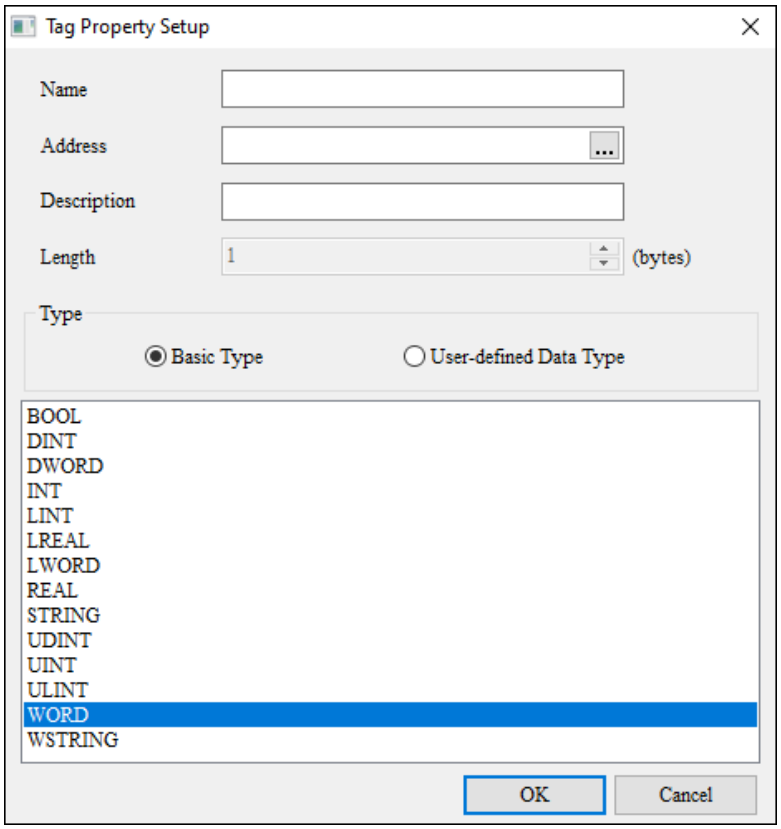
DIA Tag Usage Example	
	• When importing DIADesigner-AX project files, you need to select the Delta AX series Controller TCP (Use Tag) controller. 3. In the **Project** pane, double-click **DIA Tag**. 4. In the **DIA Tag** setting page, click . 5. In the **DIA Tag Import Settings** dialog, select a project file or Tag file. 6. Select a **Link Name**, and you can view the name and default station number of the controller corresponding to the connection.  7. Click **OK** and the imported DIA Tag will be displayed in the **DIA Tag** page. 8. Click to enable DIA Tag automatic synchronization and set the **Automatic check cycle**. 9. Create an element on the screen and select **DIA Tag** for **Link** in the address Input dialog of the element, and the DIA Tag will be configured in the element address.

Tag

You can add tags of different types in the tag page or declare structures and arrays of user-defined data types as tags.

Note: Up to 6,000 tags can be added.

The following table lists the functions of **Tag Property Setup** dialog with their description.



Function	Description
Name	Enter a tag name. The naming convention is as follows: • The name cannot exceed 220 bytes, and one Chinese character occupies 2 bytes. • The name cannot start with a number. • System reserved keywords, such as recipe register address RCP, cannot be used. • Decimal points, square brackets, and @ symbol are not allowed in names. • The name should be in the same language as the operating system (except English and number). Using a language different from the system language will result in garbled characters.

<table>
<tr><th>Function</th><th>Description</th></tr>
<tr><td>Address</td><td>Set the address corresponding to the tag.
Note: Configure an address of sufficient length according to the selected Type.</td></tr>
<tr><td>Description</td><td>Enter a description for the tag.</td></tr>
<tr><td>Length</td><td>Select or enter a length.
Note: This function is only available when Basic Type is STRING or WSTRING.</td></tr>
<tr><td>Type</td><td>

Basic Type
The following table lists the format and description of each Basic Type.
<table>
<tr><th>Type</th><th>Description</th></tr>
<tr><td>BOOL</td><td>1-Bit value format, which can only be used for bit control addresses, such as bit-type button elements.</td></tr>
<tr><td>INT</td><td>Word unit and signed integer format</td></tr>
<tr><td>DINT</td><td>Double Word unit and signed integer format</td></tr>
<tr><td>LINT</td><td>Quad Word unit and signed integer format</td></tr>
<tr><td>WORD</td><td>Word unit value format</td></tr>
<tr><td>DWORD</td><td>Double Word unit value format</td></tr>
<tr><td>LWORD</td><td>Quad Word unit numerical format</td></tr>
<tr><td>REAL</td><td>Floating point value format of Double Word unit</td></tr>
<tr><td>LREAL</td><td>Floating point value format of Quad Word unit</td></tr>
<tr><td>UINT</td><td>Word unit and unsigned integer format</td></tr>
<tr><td>UDINT</td><td>Double Word unit and unsigned integer format</td></tr>
<tr><td>ULINT</td><td>Quad Word unit and unsigned integer format</td></tr>
<tr><td>STRING</td><td>String format, 1 ASCII character is 1 byte, up to 128 bytes.</td></tr>
<tr><td>WSTRING</td><td>String format, 1 ASCII character is 1 byte, up to 256 bytes.</td></tr>
</table>

User-defined Data Type
Create a user-defined type on the structure or array page. You can create a tag for the user-defined type in the tab page.

</td></tr>
</table>

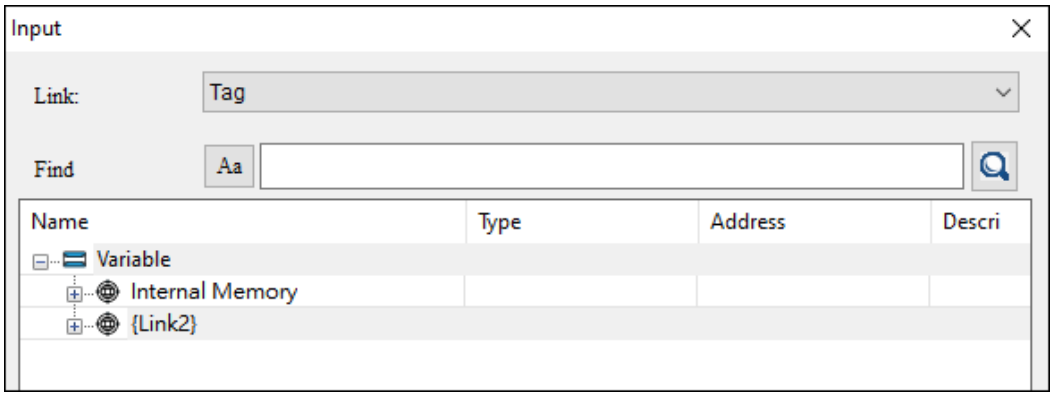
Tag Import

The set tags can be exported as XLSX files and imported into Excel after editing. CSV files of variable tables edited by external software can also be imported into DIAScreen for programming.

Note: You need to clear the current tag content before importing the tag.

Tags usage

In the element address **Input** dialog, select **Tag** for **Link** name, and then select the tag. The element address will be presented as a tag name.



User-defined Data Type

A user-defined data type (also called DUT (Data Type Unit) or UDT (User-defined Data Type)) is a data type that is combined by users using existing basic data types.

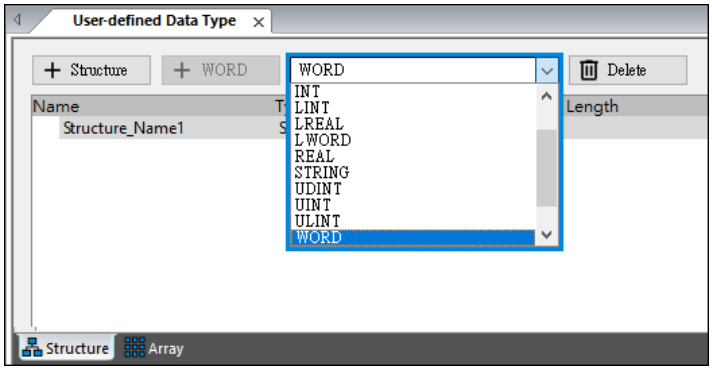
Structure

A structure consists of a fixed number of elements of various data types, and a structure or array element can also serve as an element of another structure. Structures can be used to define grouped data and to transmit parameters as a data unit.

Follow these steps to build a structure.

To build a structure

1. In **User-defined Data Type** page, click **+Structure** to add a structure. You can edit the structure name in the **Name** field.
2. Select the data type.



3. Select **Structure** and click **+WORD**. A new element will be added under the structure, and the element name can be edited in the **Name** field.

The naming rules for structure names and element names are as follows:

- Structure names must not be repeated.
- It is not allowed to add new elements with duplicate names in the same structure.
- The maximum length of the structure name and element name is 32 characters.

- Structure names and element names cannot have spaces, and cannot use special characters such as * # ? \ % @ [] () . , etc.
- Structure names should be in the same language as the operating system (except English and number). Using a language different from the system will result in garbled characters.

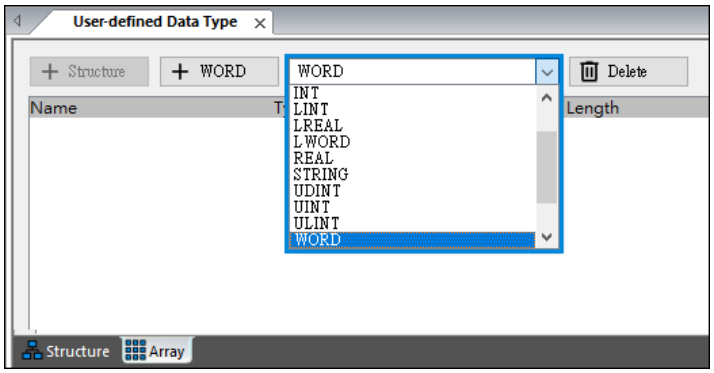
Array

An array consists of fixed length and the same data type.

Follow these steps to create an array.

To create an Array

1. In **User-defined Data Type** page, select a data type.



2. Click **+WORD** to add a new array. In the **Name** field, edit the array name.

The naming rules of array names are as follows:

- Array names must not be repeated.
- The maximum length of the array names is 32 characters.
- Array names cannot have spaces, and cannot use special characters such as * # ? \ % @ [] () . , etc.
- Array names should be in the same language as the operating system (except English and number). Using a language different from the system will result in garbled characters.

Security Setting

Account and Password

Set the security level and the account and password of HMI.

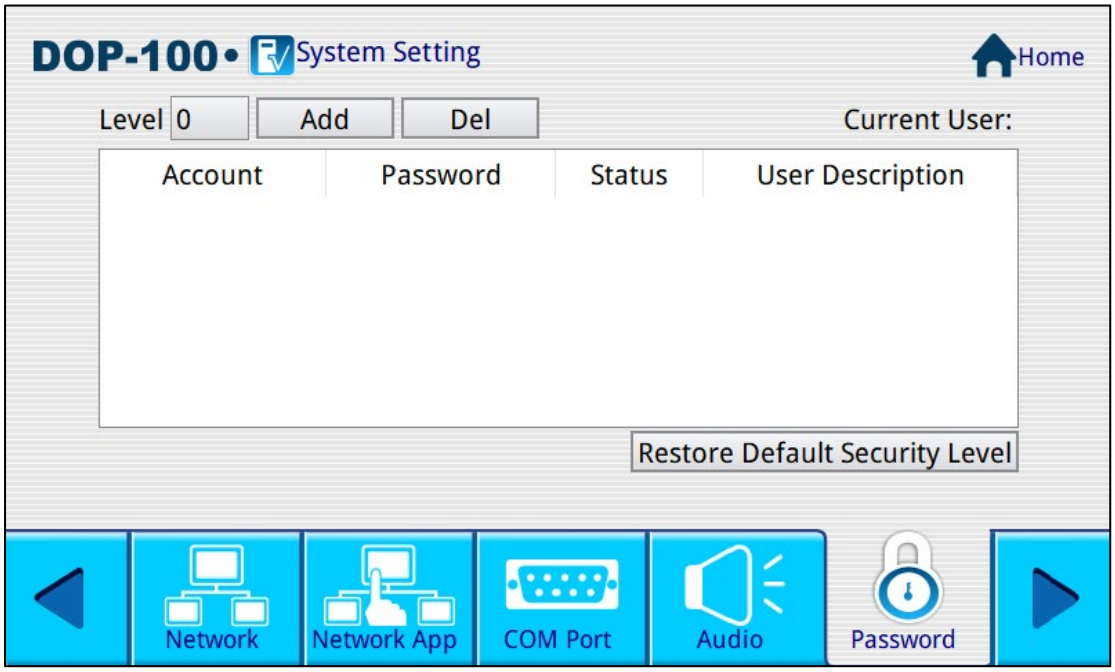
The security level is divided into 0–9, each level can add multiple accounts for users to log in at the same time.

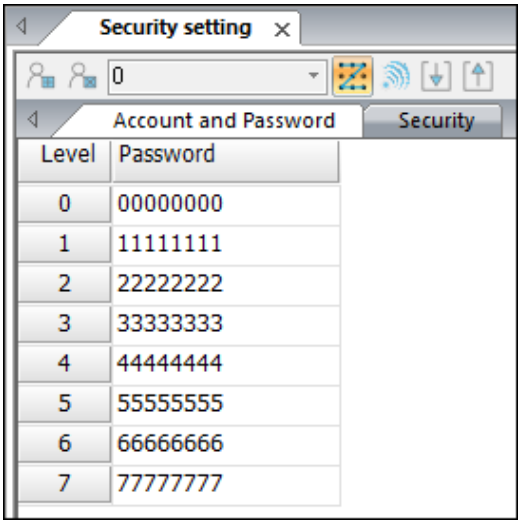
The following describes the setting rules for the account and password:

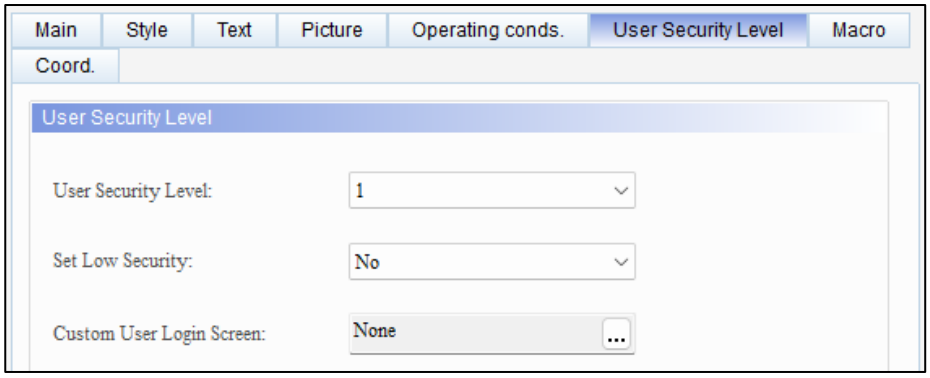
- Password must not be blank.
- The account names cannot be the same, but the passwords can be the same.
- The account length can be set up to 24 characters, and the password length can be set up to 128 characters.
- User description can only be entered in English or numbers.
- Security levels are divided as follows:
  - Level 0: No protection function, anyone can operate.
  - Level 1–9: You need to enter the corresponding password or a higher security level to operate.
  - Level 10: The highest security password and the protection password after project storage. This password is also the password to be entered when executing **Check password when downloading program** and formatting the system file.

Note: The account and password support 1,000 records, regardless of level.

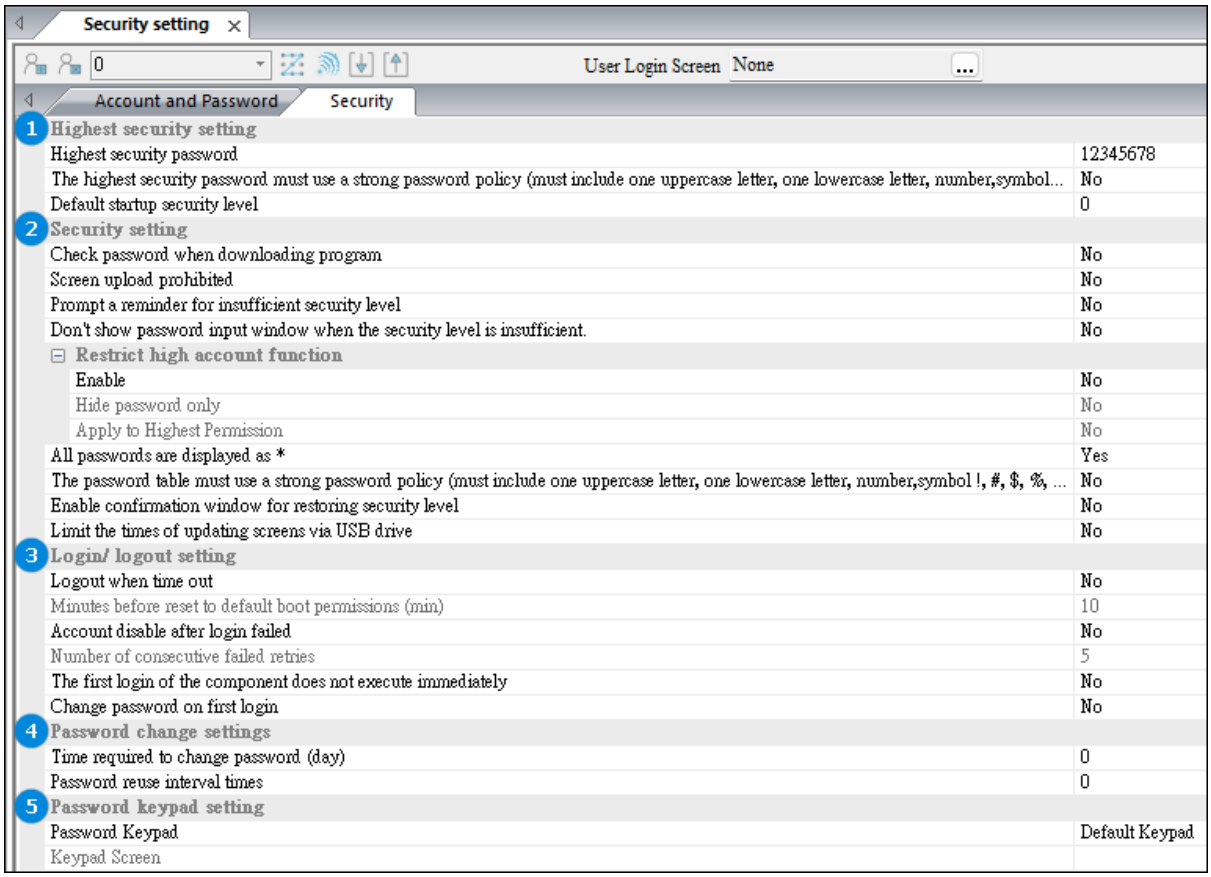
You can use the button element **Password Table** or select **System Menu > System Settings** > **Password** on the HMI to open the account and password table to change the password and account.



Setting	Description
Simple Password	The Simple Password table only has 7 levels of password. The password can be set up to 8 characters and the password text format is Hexadecimal unit from 0 to F. 
Enable RFID login	Click to enable RFID login, you can login your account or add new account and password through external device on the HMI. **Note**: The RFID field in software is read-only. You need to add an RFID code through the HMI and upload all the data before the RFID code can be displayed.
Import/export account and password table	Click to export the account and password table as a CSV file, which can be imported into software after editing. The exported CSV file is named **accounts.csv** and saved in the storage device. The account and password table can be imported into the HMI through the **Import/Export Account** element.

Setting	Description
User Login Screen	Set the login screen for each user level. When the **User Security Level** setting of the element is not 0 and no custom login user screen is specified, the login screen specified here will be displayed. 

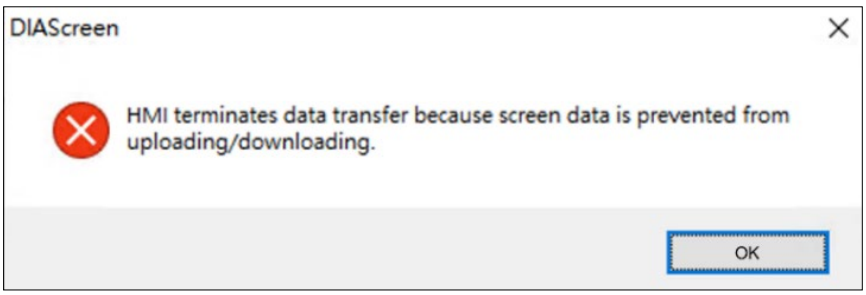
Security

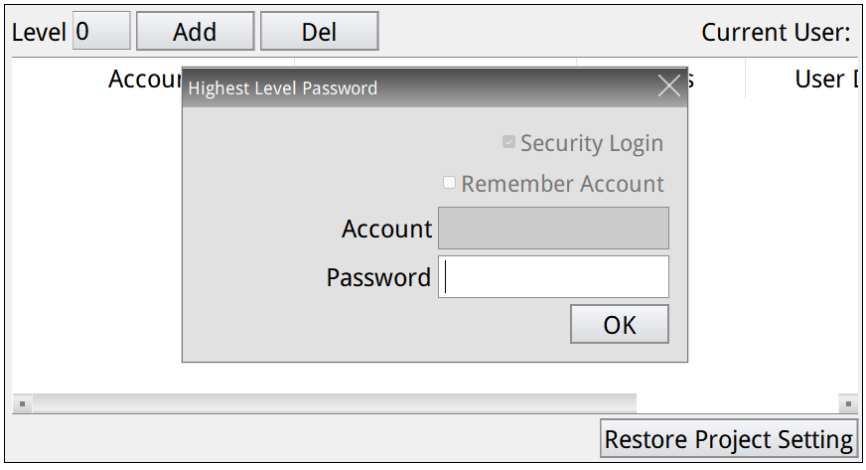


❶ Highest security setting

Setting	Description
Highest security password	Set the password with the highest security level of HMI, which means level 10. This password is used to control upload screen data and recipes, download screen data and recipes (you need to select the **Check password when downloading program** checkbox), password protection, system formatting, system file encryption and copy file functions (you need to select **Limit the times of updating screens via USB drive checkbox**).
The highest security password must use a strong password policy	If set to **Yes**, the highest security password must set more than (inclusive) 8 characters and contain an uppercase letter, a lowercase letter, a number, and a symbol (!, $, #, %).
Default startup security level	Set the default security level when the HMI is turned on. You can set the level from 0 to 10. **Note**: Level 10 is the highest security level.

❷ Security setting

Setting	Description
Check password when downloading program	If set to **Yes**, when executing **Download All Data** and **Download Recipe**, the software asks you to enter the highest security password, and the screen can be downloaded to the HMI normally only after the password is entered successfully **Note**: You need to download the project to the HMI first and execute **Download All Data** after setting to **Yes**.
Screen upload prohibited	If set to **Yes**, when executing **Upload All Data**, the following message will appear.  **Note**: You need to download the project to the HMI, and then execute **Upload All Data for this setting to take effect.**

Setting	Description
Prompt a reminder for insufficient security level	If set to **Yes**, when the user security level set in the element is higher than the default boot security level, the element on the HMI displays that indicates that your current security level is not sufficient.
Don't show password input window when the security level is insufficient	If set to **Yes**, when your current security level is not sufficient, the password input window will not display.
Restrict high account function	• **Enable**: If set to **Yes**, when opening the password table, the account and password of a lower level will not be displayed. • **High password only**: If set to **Yes**, when switching to the low-level accounts, the account is displayed and only password is hidden. • **Apply to highest permission**: If set to **Yes**, when opening the password table and logging in with the highest security level, the password field in the password table will display ********.
All passwords are displayed as *	If set to **Yes**, when opening the password table, the password will be displayed as ******** regardless of its length.
The password table must use a strong password policy	If set to **Yes**, all passwords must be set more than (inclusive) 8 characters and contain an uppercase letter, a lowercase letter, a number, and a symbol (!, $, #, %). **Note**: The strong password policy does not apply to the simple password table.
Enable confirmation window for restoring security level	If set to **Yes**, after pressing **Restore Project Setting** on the HMI, you need to log in with the highest security level to restore the project settings.  **Note**: If using the simple password table, this setting cannot be enabled.

Setting	Description
Limit the times of updating screens via USB drive	If set **Yes**, you can encrypt the screen data files and limit the number of times of copying files. This function is mainly used with the function of file encryption and copying of files.

❸ Login / logout setting

Setting	Description
Logout when time out	If select **Yes**, the system will return to the default boot security level according to the set **Minutes before reset to default boot permissions** when the HMI is in idle status.
Account disable after login failed	If select **Yes**, the account will be disabled according to the set **Number of consecutive failed retries** when login fails consecutively.
The first login of the component does not execute immediately	If select **Yes**, when the current executed security level is not sufficient, click the element to log in and then click it again to execute the element action.
Change password on first login	If select **Yes**, the HMI forces you to change the password after logging in for the first time.

❹ Password change settings

Setting	Description
Time required to change password	Select or enter the required time for changing password. To avoid frequent password changes, you need to wait for the Time required to change password before changing the password again after changing the password on the HMI. For example, if the password is changed on 2025/01/01 12:00 and the **Time required to change password** is set to 3, the password can be changed again on 2025/01/03 12:00. **Note**: If the **Password duration** is not 0, the Password duration must be greater than **Time required to change password**.
Password reuse interval times	Select or enter the number of times the repeated password can be used. You can set the password up to 3 times, that is, the password with 3 days cannot be the same.

⑤ Password keypad setting

<table>
<tr><th>Setting</th><th>Description</th></tr>
<tr><td>Password Keypad</td><td>Select the keypad for entering passwords on the HMI.
<table>
<tr><td>Default Keypad</td><td>
</td></tr>
<tr><td>Simple Keypad</td><td>
</td></tr>
<tr><td>Customized Keypad</td><td>Create Keypad Screen and configure the Keypad element.
</td></tr>
</table>
Note: The Simple Keypad does not apply to a strong password policy.</td></tr>
</table>

Address Settings

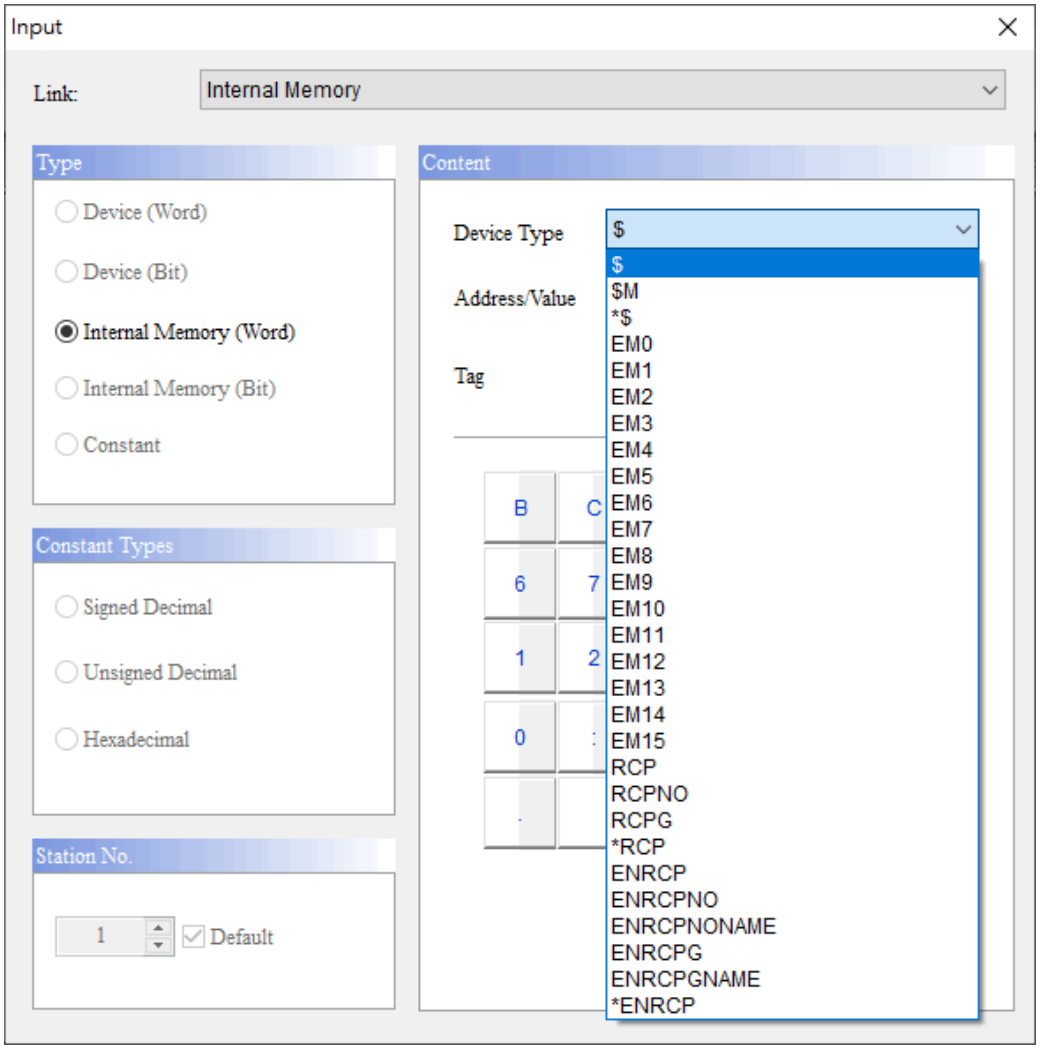
Each HMI function needs to be executed through address concatenation, and you can set the address in the element property dialog. The address settings include Internal Memory Address and External Controller Address. This section describes various address definitions and their setting methods.

Internal Memory

Delta HMI has a total of 13 types of registers with different functions.

1. Internal register $
2. Non-volatile internal register $M
3. Indirect address register *$
4. Extended memory EM0~15
5. Recipe register RCP
6. Recipe number register RCPNO
7. Recipe group register RCPG
8. Recipe indirect addressing register *RCP
9. Enhanced recipe register ENRCP
10. Enhanced recipe number register ENRCPNO
11. Enhanced recipe group register ENRCPG
12. Enhanced recipe group name register ENRCPGNAME
13. Enhanced recipe indirect addressing register *ENRCP

Types 5~13 belong to recipe addresses. For detailed description, see the manual of the recipe.



Internal Register ($)

The Internal Register is the HMI internal memory that freely reads data and supports all kinds of configurations, such as the communication address of the element.

The HMI can access 200,000 sets of 16-bit internal registers.

Note: The internal register does not have a power-off retention function, so the data in the register is not retained when the power is off.

Access Type	Element Type	Access Range
Word	$n	$0~$199999
Bit	$n.b	$0.0~$199999.15

Note: n is Word (0~199999); b is Bit (0~15).

Non-volatile Internal Register ($M)

The Non-volatile Internal Register is data recorded in the non-volatile storage location. The non-volatile storage location records data in a register in the SRAM or NAND Flash IC when the HMI is powered off. You can record important data in this register to avoid data loss. HMI can access 5000 sets of 16-bit non-volatile internal registers.

- $M: The register address where data is stored in SRAM.
- $EML: The register address where data is stored in NAND Flash IC and is applicable to DOP-107SV and DOP-100 series Gen2 models.

Access Type	Element Type	Access Range
Word	$Mn $EMLn	$M0~$M4999 $EML0~$EML4999
Bit	$Mn.b $EMLn.b	$M0.0~$M4999.15 $EML0.0~$EML4999.15

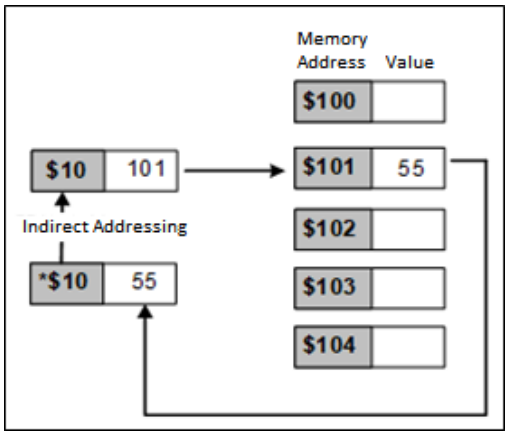
Note: n is Word (0~4999); b is Bit (0~15).

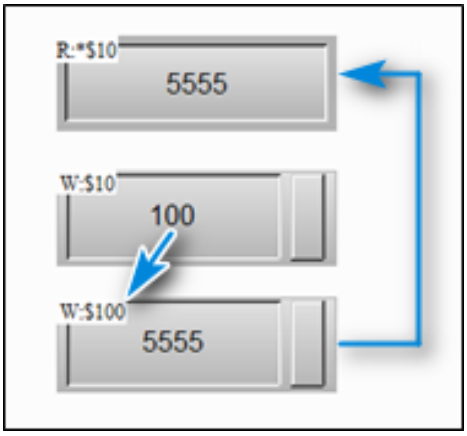
Indirect Address Register (*$ / *D$)

The Indirect Address Register uses the value of the internal memory as the address and assigns the address value to the corresponding **Indirect Address Register** of the internal memory.

Access Type	Element Type	Access Range
Word	*$0~*$199999	$0~$199999
Double Word	*D$0~*D$199998	$0~$199998

Indirect address register *$n or *D$n, obtains the value from the internal memory $n, sets the value as a new address, and then accesses the value from the new address. For example, $10 = 101, $101 = 55, then *$10 = 55.





If the address assigned by the internal memory exceeds the value of one Word (65535), the indirect address register must use ***D$** and obtain the value in Double Word format.

Note: The indirect address register does not have a power-off retention function, so the data in the register is not retained when the power is off.

Extended Memory

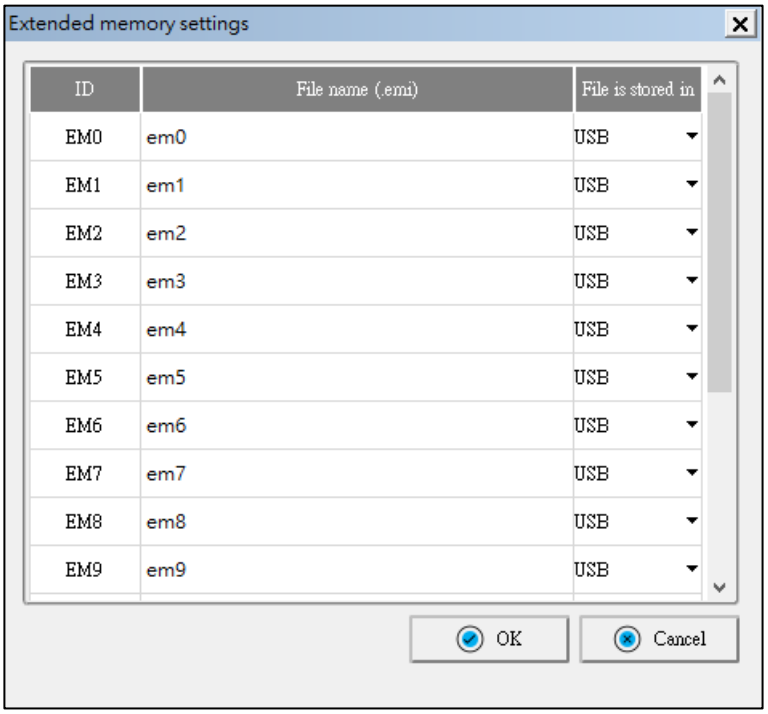
The Extended Memory (EM) can store data on an external storage device so that the data can be retained even after the HMI is powered off. The elements on the HMI screen are also read and written through the extended memory.

- The extended memory has 16 groups (EM0~EM15).
- Supports file storage in external storage devices (USB, USB 2 disk and SD card).
- Each EM is a file, supporting a maximum of 512MB.
- Values can be read and written into the file.
- The maximum Word address is 268435455, and the last Word of the 512MB file can be read.
- The address of the last Bit is EM0_268435455.15.
- EM addresses can be used in elements and macro functions, and can also be used to create data codes.

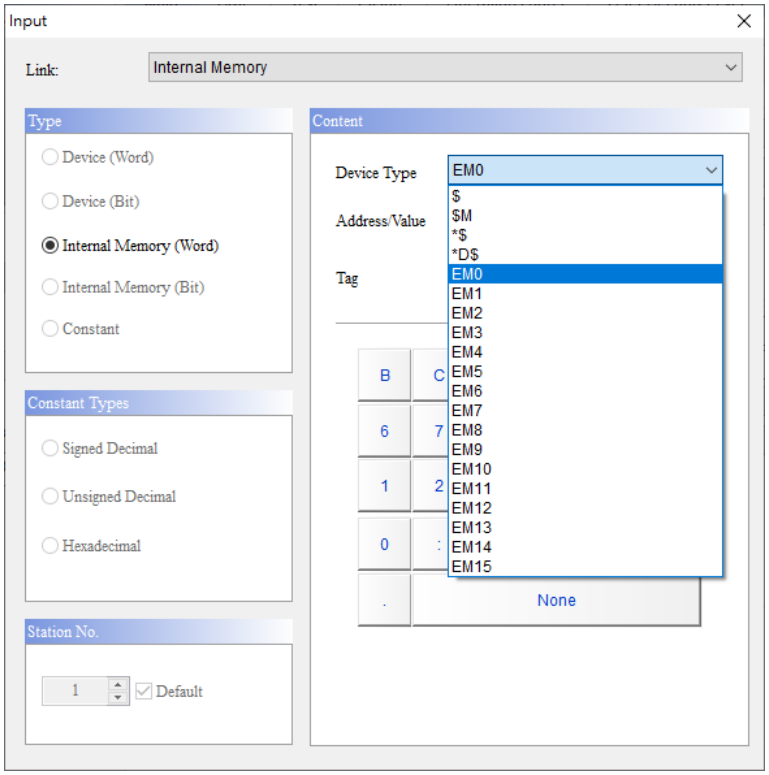
Follow the steps to set the extended memory.

1. On the toolbar, click **Project** > **Other Settings** > **Extended Memory Settings**.
2. In the **Extended memory settings** dialog, set the file name corresponding to each EM area.

   The file name supports multiple languages, and the maximum file name length is (32*4) bytes. The file extension stored on the external storage device is .emi.



3. In the address **Input** dialog of the element, select **Internal Memory** for **Link**, select **EM0~EM15** for **Device Type**, and then enter the read and write address.



The following table lists the API for LUA applications to read and write extended memory.

Command	Description
mem.em.Read (EMID, addr, [FORMAT])	Read 1 WORD from the addr address of EMID
mem.em.Write (EMID, addr, value)	Write 1 WORD from the addr address of EMID
mem.em.ReadDW (EMID, addr, [FORMAT])	Read 1 DWORD from the addr address of EMID
mem.em.WriteDW (EMID, addr, value)	Write 1 DWORD from the addr address of EMID
mem.em.ReadQW (EMID, addr, [FORMAT])	Read 1 QWORD from the addr address of EMID
mem.em.WriteQW (EMID, addr, value)	Write 1 QWORD from the addr address of EMID
mem.em.ReadBit (EMID, addr, bit_index)	Read the bit_index value from the addr address of EMID
mem.em.WriteBit (EMID, addr, bit_index, value)	Write the value from the addr address bit_index of EMID
mem.em.ReadFloat (EMID, addr)	Read 1 Float from the addr address of EMID

Command	Description
mem.em.WriteFloat (EMID, addr, value)	Write 1 Float from the addr address of EMID
mem.em.ReadDouble (EMID, addr)	Read 1 Double from the addr address of EMID
mem.em.WriteDouble (EMID, addr, value)	Write 1 Double from the addr address of EMID
mem.em.ReadASCII (EMID, addr, length)	Read length characters from the addr address of EMID
mem.em.WriteASCII (EMID, addr, string, length)	Write length characters from the addr address of EMID

DXMC Series Internal Memory Address

The following table lists the internal memory address range of Word elements.

Register Type	Symbol Format	Read and Write Address Range	Data Length	Remark
	Word No.(n)			
X_Register	**X**n	**X**0~**X**1FFFFF	Word	Hexadecimal
Y_Register	**Y**n	**Y**0~**Y**1FFFFF	Word	Hexadecimal
DV_Register	**DV**n	**DV**0~**DV**262143	Word	Even addressing
DH_Register	**DH**n	• DXMC-1H08AD-70, DXMC-1H08CD-7S, DXMC-1H08ED-7S: **DH**1000000~**DH**1061439 • DXMC-1P08NE-4S/70/7S, DXMC-1P04NE-7S, DXMC-1FA1RN-70F: **DH**1000000~**DH**1024575 • DXMC-1FA2RN-70F: **DH**1000000~**DH**1049151	Word	Even addressing
SDV_Register	**SDV**n	**SDV**3000000~**SDV**3065535	Word	Even addressing
SDH_Register	**SDH**n	• DXMC-1H08AD-70, DXMC-1H08CD-7S, DXMC-1H08ED-7S:	Word	Even addressing

Register Type	Symbol Format Word No.(n)	Read and Write Address Range	Data Length	Remark
		SDH6000000~S**DH**6065535 • DXMC-1P08NE-4S/70/7S, DXMC-1P04NE-7S, DXMC-1FA1RN-70F: **SDH**6000000~S**DH**6004095 • DXMC-1FA2RN-70F: **SDH**6000000~S**DH**6008191		
RP_Register	**RP**n	**RP**0~**RP**7FFFFF	Word	Hexadecimal
Controller	**CR**n	**CR**0~**CR**14FFFFF	Word	Hexadecimal
MW_Register	**MW**n	**MW**0~**MW**67108863	Word	Even addressing
MD_Register	**MD**n	**MD**0~**MD**67108863	Word	Even addressing
ML_Register	**ML**n	**ML**0~**ML**67108863	Word	Even addressing

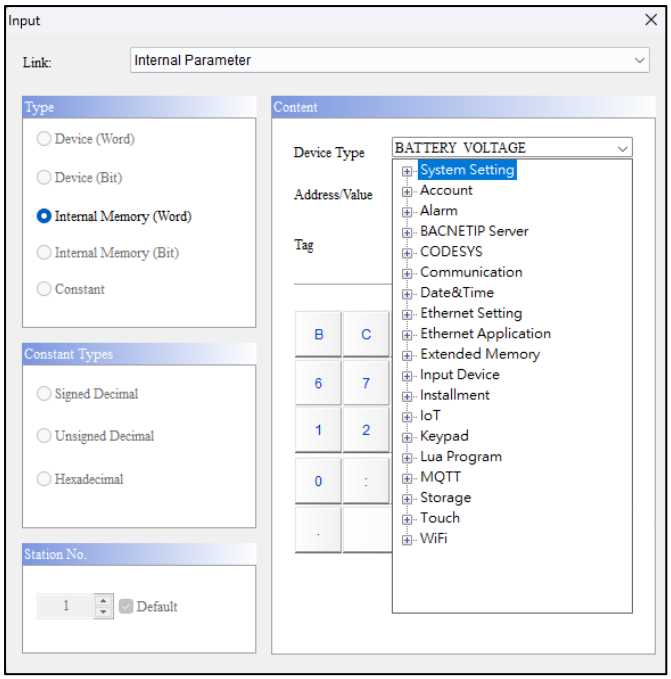
The following table lists the internal memory address range of Bit elements.

Contact Type	Symbol Format Byte No.(n); Bit No.(b)	Read and Write Address Range	Remark
X_Coil	**X**n.b	**X**0.0~**X**1FFFFF.15	Hexadecimal
Y_Coil	**Y**n.b	**Y**0.0~**Y**1FFFFF.15	Hexadecimal
DV_Coil	**DV**n.b	**DV**0.0~**DV**262143.7	-
DH_Coil	**DH**n.b	• DXMC-1H08AD-70, DXMC-1H08CD-7S, DXMC-1H08ED-7S: **DH**1000000.0~**DH**1061439.7 • DXMC-1P08NE-4S/70/7S, DXMC-1P04NE-7S, DXMC-1FA1RN-70F: **DH**1000000.0~**DH**1024575.7 • DXMC-1FA2RN-70F: **DH**1000000.0~**DH**1049151.7	-

Contact Type	Symbol Format Byte No.(n); Bit No.(b)	Read and Write Address Range	Remark
SDV_Coil	**SDV**n.b	**SDV**3000000.0~**SDV**3065535.7	-
SDH_Coil	**SDH**n.b	• DXMC-1H08AD-70, DXMC-1H08CD-7S, DXMC-1H08ED-7S: **SDH**6000000.0~**SDH**6065535.7 • DXMC-1P08NE-4S/70/7S, DXMC-1P04NE-7S, DXMC-1FA1RN-70F: **SDH**6000000.0~**SDH**6004095.7 • DXMC-1FA2RN-70F: **SDH**6000000.0~**SDH**6008191.7	-
SM_Coil	**SM**n.b	**SM**0.0~**SM**8191.15	-
MX_Coil	**MX**n.b	**MX**0.0~**MX**67108863.7	-
MW_Coil	**MW**n.b	**MW**0.0~**MW**67108863.7	-

Internal Parameter

The I**nternal Parameter** can access the HMI internal state values, including system time value, external storage device status, X / Y coordinates when touching, touch status, remaining battery voltage in percentage, network parameters, firmware version, extended memory status, and so on.

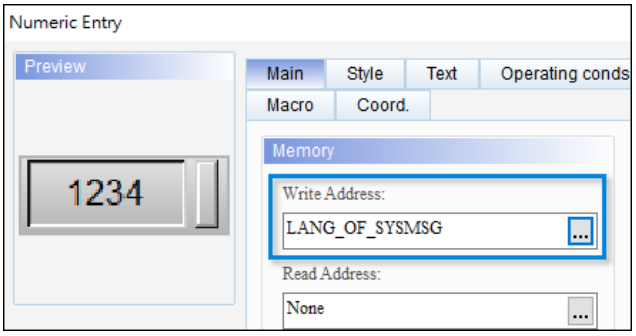
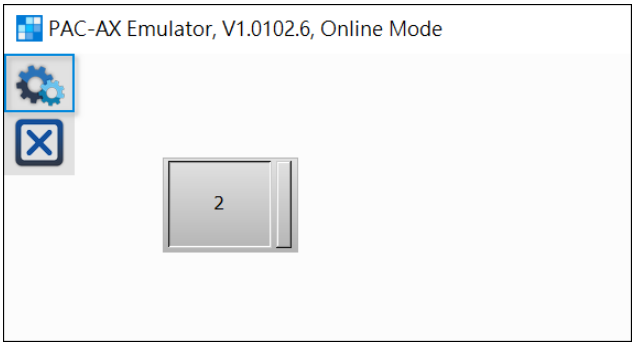
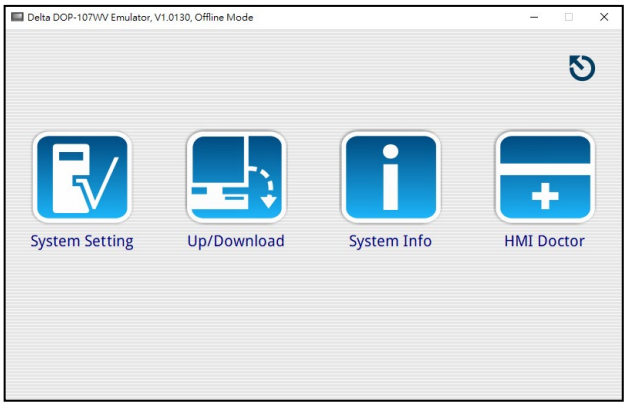


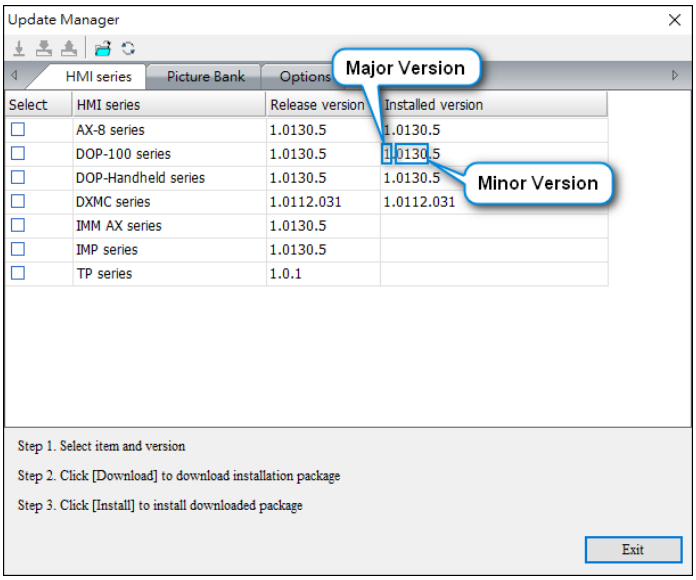
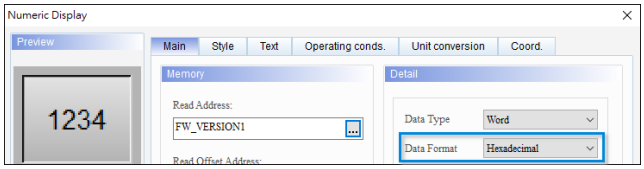
Note: Internal Parameter is available only for Word elements. You are unable to use this function if you create Bit elements.

The following table lists each internal parameter with their descriptions.

System Setting

<table>
<tr><th>Parameter</th><th>Definition</th><th colspan="2">Description</th></tr>
<tr><td>BATTER_
VOLTAGE</td><td>Remaining HMI battery</td><td colspan="2">The display unit is percentage.</td></tr>
<tr><td>LANG_OF_
SYSMSG</td><td>Display language of the HMI directory</td><td colspan="2"><table><tr><th>State Value</th><th>Language</th></tr><tr><td>0</td><td>English</td></tr><tr><td>1</td><td>Tradition Chinese</td></tr><tr><td>2</td><td>Simple Chinese</td></tr><tr><td>3</td><td>Spanish</td></tr><tr><td>4</td><td>French</td></tr><tr><td>5</td><td>Russian</td></tr><tr><td>6</td><td>Turkish</td></tr></table></td></tr>
</table>

Parameter	Definition	Description
		Follow these steps to set the lanaugae of system menu. 1. Create an element such as **Numeric Entry** element. 2. In the address input dialog of the element, select **Link** name as **Internal Parameter**, and the **Device Type** as **LANG_OF_SYSMSG**.  3. Download the screen to the HMI and change the value of the element on the HMI. 4. Press and hold the screen for 2 seconds, and the system menu button appears on the top left of the screen.  After entering the system menu, you can see the display language changes. 

Parameter	Definition	Description
SP_BRIGHT	Backlight brightness of HMI	The display unit is percentage.
SYS_BUZZER_VOLUME	Buzzer volume	The display unit is percentage. The value range is 0–100.
BTN_BUZZER_VOLUME	Touch-tone	The display unit is percentage. The value range is 0–100.
FW_VERSION1	Major version number of HMI firmware version	
FW_VERSION2	Minor version number of HMI firmware version	• When setting this parameter, set the **Data Format** to **Hexadecimal**. 
Loaded CODESYSTags	Total number of loaded CODESYS Tag	Display the number of current loaded tags. HMI displays the progress bar when preloading the tags. After all tags are loaded, the total number of preloaded CODESYS tags can be known.
BOOT_BUZZER_VOLUME	Boot buzzer volume of HMI	The display unit is percentage. The value range is 0–100.
HardwareID	HMI hardware code	Read the HMI hardware code. This parameter is set using the **Character Display** element, and the string length is 32 to ensure that the complete hardware code is read.

4G

Parameter	Definition	Description
RemoveDongleMessage	Display message after removing 4G Dongle	**State Value** / **Result** 0 — The alarm dialog is displayed. 1 — The alarm dialog is not displayed.

Account

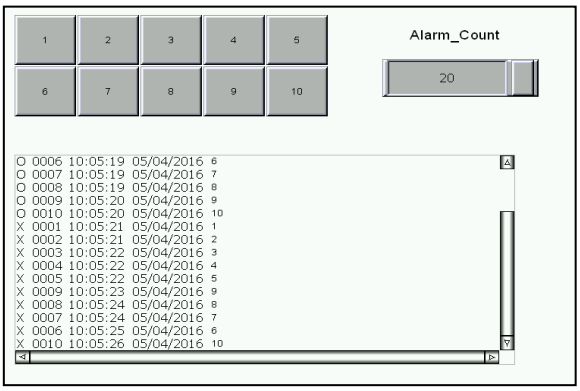
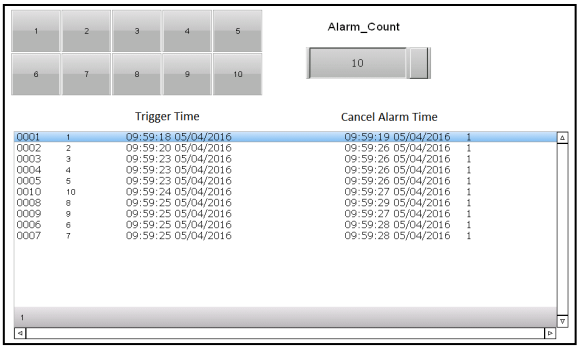
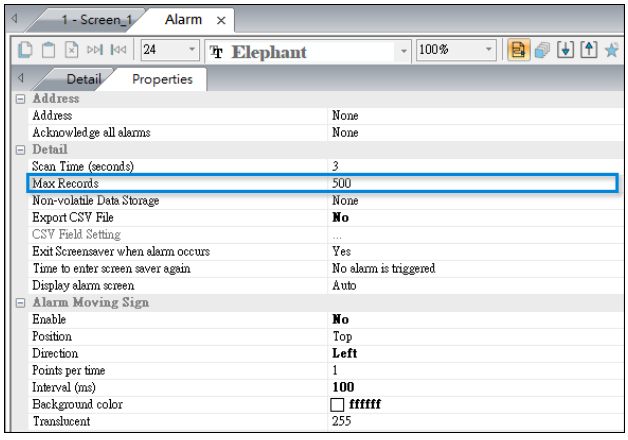
Note: This parameter only applies if the account is set to a **non-simple password**.

Parameter	Definition	Description
ACCOUNT	Currently logged-in account	• Account contains English letters and numbers, use the **Character Display** element. • Account contains non-ASCII characters, use the **Multi-language Input** element.
SetUserAccount	User account	• It is used for user login / logout, adding / deleting an account, and changing a password. • Use the **Character Entry** element to set this parameter.
AdminLogin	Highest authority account number	**State Value** / **Result** 0 — Log in using a non-highest authority account. 1 — Log in with the highest authority account.
SetUser Password	User password	• The password is displayed in ***. • It is used for user login / logout, adding / deleting an account, and changing a password. • Use the **Character Entry** element to set this parameter.
SetUserLevel	User level	• It is used for user login / logout, adding / deleting an account, and changing a password. • The level range is 0–10.

<table>
<tr><th>Parameter</th><th>Definition</th><th>Description</th></tr>
<tr><td>AccountStatus</td><td>Account status</td><td>Read the login status of the current account. The status includes the following:
Unexpired
Unlocked
Expired
Locked
Use the Character Entry element to set this parameter.</td></tr>
<tr><td>CurrentUser Level</td><td>Currently logged in level</td><td>Read the level (0~10) of the current login account.</td></tr>
<tr><td>LoginResult</td><td>Login status</td><td rowspan="2">Read whether the last login / logout of the current account was successful. The status includes the following:
Success
Fail
Use the Character Entry element to set this parameter.</td></tr>
<tr><td>LogoutResult</td><td>Logout status</td></tr>
<tr><td rowspan="2">Login</td><td rowspan="2">Login settings</td><td rowspan="2"><table><tr><th>State Value</th><th>Result</th></tr><tr><td>0</td><td>The action is completed.</td></tr><tr><td>1</td><td>Trigger login / logout</td></tr></table></td></tr>
<tr><td>Logout</td></tr>
<tr><td>DeleteUser Account</td><td>Delete account</td><td><table><tr><th>State Value</th><th>Result</th></tr><tr><td>0</td><td>The action of deleting an account is completed.</td></tr><tr><td>1</td><td>Trigger account deletion</td></tr></table></td></tr>
<tr><td>DeleteUser AccountResult</td><td>Delete account result</td><td>Read whether the deletion of the account is successful. The status includes the following:
Success
Fail
Use the Character Entry element to set this parameter.</td></tr>
<tr><td>AddUser Account</td><td>Add account</td><td><table><tr><th>State Value</th><th>Result</th></tr><tr><td>0</td><td>The action of adding an account is completed.</td></tr><tr><td>1</td><td>Trigger account adding</td></tr></table></td></tr>
</table>

<table>
<tr><th>Parameter</th><th>Definition</th><th>Description</th></tr>
<tr><td>AddUser AccountResult</td><td>Add account result</td><td>Read whether the adding of the account is successfully read. The status includes the following:
Success
Fail
Use the Character Entry element to set this parameter.</td></tr>
<tr><td>ChangeUser Password</td><td>Change password</td><td><table><tr><th>State Value</th><th>Result</th></tr><tr><td>0</td><td>The password change action is completed.</td></tr><tr><td>1</td><td>Trigger password change</td></tr></table></td></tr>
<tr><td>Change PasswordResult</td><td>Change password result</td><td>Read whether the change of the password is successful. The status includes the following:
Success
Fail
Use Character Entry element to read this parameter.</td></tr>
<tr><td>UserDescription</td><td>Get a description of the user who is currently logged in to the account</td><td>The user description only supports English and numbers. Use the Character Display element to read this parameter.</td></tr>
<tr><td>RFIDSerialNumber</td><td>Get the RFID serial number of the current login account</td><td>The RFID serial number uses the Character Display element to read this parameter.</td></tr>
<tr><td>SetRFIDSerialNumber</td><td>Set the RFID serial number</td><td rowspan="2">If is used for Add/Delete User Account Screen, only English and numbers are supported. Use the Character Entry element to set this parameter.</td></tr>
<tr><td>SetUserDescription</td><td>Set a user description</td></tr>
</table>

Alarm

Parameter	Definition	Description
ALARM_COUNT	Alarm triggered The total number of transactions	• Display the total number of alarms that are triggered and de-triggered. • The DOP-B, DOP-H, and HMC series models record the triggered and de-triggered alarms separately. As a result, Alarm_Count displays 20 when there are ten alarms.  • The DOP-W and DOP-100 series models record the triggered and de-triggered alarms within one row of data. As a result, Alarm_Count displays 10 when there are ten alarms.  • This function reminds you to export the alarm data in case the initial alarm data contents are removed when the data reach the set maximum. 

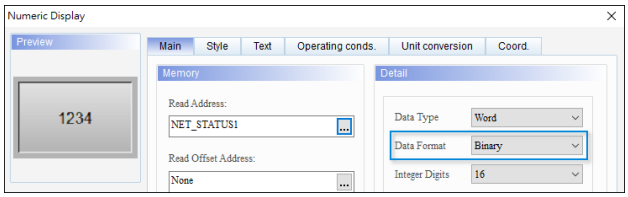
BACNETIP Server

Parameter	Definition	Description
DEVICEID	Device ID of BACnet	The device ID can be changed dynamically on the HMI.

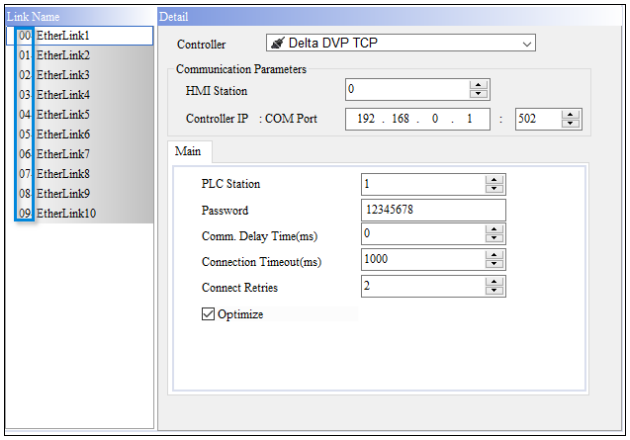
Communication

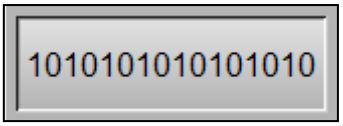
<table>
<tr><th>Parameter</th><th>Definition</th><th>Description</th></tr>
<tr><td>NET_STATUS1</td><td rowspan="4">Network connection status</td><td rowspan="4">

NET_STATUS1 represents the 16 connections of the first controller;
NET_STATUS2 is the 16 connections of the second controller, and so forth for NET_STATUS2 and NET_STATUS4.
Set the Data Format as Binary when setting these internal parameters.



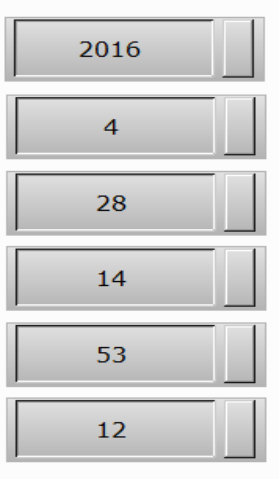
Sort according to the EtherLink device number added by the software.





<table>
<tr><th>State Value</th><th>Result</th></tr>
<tr><td>0</td><td>Not connected</td></tr>
<tr><td>1</td><td>Connected</td></tr>
</table>
</td></tr>
<tr><td>NET_STATUS2</td></tr>
<tr><td>NET_STATUS3</td></tr>
<tr><td>NET_STATUS4</td></tr>
</table>

Date&Time

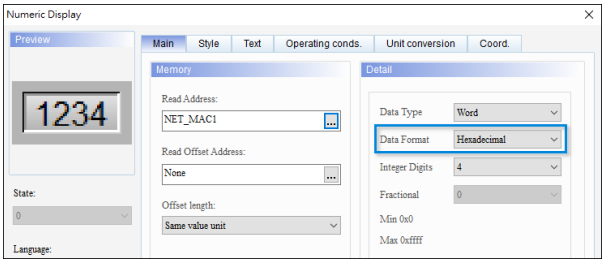
<table>
<tr><th>Parameter</th><th>Definition</th><th>Description</th></tr>
<tr><td>TIME_YEAR</td><td rowspan="6">HMI time value</td><td rowspan="6"><table><tr><th>Time Value</th><th>Result</th></tr><tr><td>TIME_YEAR (yyyy)</td><td rowspan="6"> 
</td></tr><tr><td>TIME_MONTH (mm)</td></tr><tr><td>TIME_DAY (dd)</td></tr><tr><td>TIME_HOUR (hr)</td></tr><tr><td>TIME_MINUTE (min)</td></tr><tr><td>TIME_SECOND (sec)</td></tr></table></td></tr>
<tr><td>TIME_MONTH</td></tr>
<tr><td>TIME_DAY</td></tr>
<tr><td>TIME_HOUR</td></tr>
<tr><td>TIME_MINUTE</td></tr>
<tr><td>TIME_SECOND</td></tr>
</table>

Ethernet Setting

Some DOP-100 series HMI models have two Ethernet ports, so the network parameters are named as: NET1_IP1~NET1_IP4, SUBMASK1_IP1~SUBMASK1_IP4, GWAY1_IP1~GWAY1_IP4 and NET2_IP1~NET2_IP4, SUBMASK2_IP1~SUBMASK2_IP4, GWAY2_IP1~GWAY2_IP4.

For the DOP-100 models to be compatible with the DOP-B projects, the network parameters NET_IP1 to NET_IP4, SUBMASK_IP1 to SUBMASK_IP4, and GWAY_IP1 to GWAY_IP4 are added.

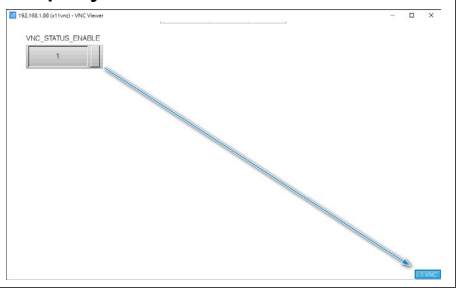
<table>
<tr><th>Parameter</th><th>Definition</th><th>Description</th></tr>
<tr><td>NET1_IP1
NET2_IP1</td><td rowspan="4">IP address of the HMI</td><td rowspan="4">Take 192.168.1.88 as an example.
</td></tr>
<tr><td>NET1_IP2
NET2_IP2</td></tr>
<tr><td>NET1_IP3
NET2_IP3</td></tr>
<tr><td>NET1_IP4
NET2_IP4</td></tr>
</table>

Parameter	Definition	Description
SUBMASK_IP1 SUBMASK1_IP1	Submask IP address of the HMI	Take 255.255.255.0 as an example.
SUBMASK_IP2 SUBMASK1_IP2		
SUBMASK_IP3 SUBMASK1_IP3		
SUBMASK_IP4 SUBMASK1_IP4		
GWAY1_IP1 GWAY2_IP1	Gateway IP address of the HMI	Take 192.168.1.254 as an example.
GWAY1_IP2 GWAY2_IP2		
GWAY1_IP3 GWAY2_IP3		
GWAY1_IP4 GWAY2_IP4		
DNS1_IP1 DNS2_IP1	IP address of the HMI DNS Server	Take 192.168.1.5 as an example.
DNS1_IP2 DNS2_IP2		
DNS1_IP3 DNS2_IP3		
DNS1_IP4 DNS2_IP4		
NET_MAC1 NET2_MAC1	MAC number of the HMI network port	• When setting this parameter, set the **Data Format** to **Hexadecimal**.  • The value is displayed with the high and low bits swapped. If the value is 1800, it is displayed as 00:18.
NET_MAC2 NET2_MAC2		
NET_MAC3 NET2_MAC3		

<table>
<tr><th>Parameter</th><th>Definition</th><th>Description</th></tr>
<tr><td>NET1_IPSETTING
NET2_IPSETTING</td><td>IP Mode Setting</td><td>
<table>
<tr><th>State Value</th><th>Result</th></tr>
<tr><td>0</td><td>DHCP</td></tr>
<tr><td>1</td><td>Static</td></tr>
</table>
Note: After switching modes, the network address can be retrieved again through the control block address. Go to Configuration > Control Block > System Control Register to set b10~b11 Retrieve Ethernet address.
</td></tr>
</table>

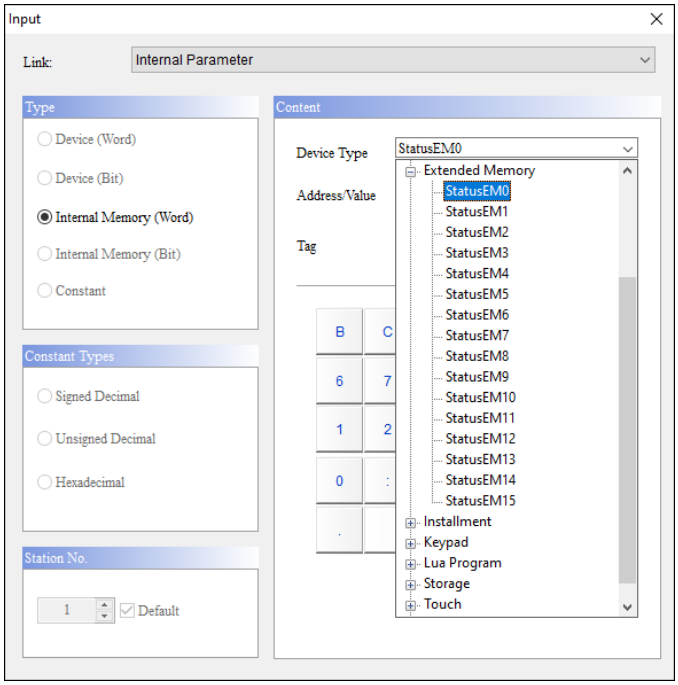
Ethernet Application

<table>
<tr><th>Parameter</th><th>Definition</th><th>Description</th></tr>
<tr><td>SMTP_STATUS</td><td>SMTP connection status</td><td rowspan="2">
<table>
<tr><th>State Value</th><th>State Description</th><th>Corresponding Message in SMTP_INFO</th></tr>
<tr><td>0</td><td>Status unknown</td><td>Unknown</td></tr>
<tr><td>1</td><td>Connection success</td><td>Message Sent</td></tr>
<tr><td>–100</td><td>Connection failed</td><td>Unable to connect to Host</td></tr>
<tr><td>–101</td><td>System disconnected</td><td>The displayed message changes according to the system.</td></tr>
<tr><td>–102</td><td>Authentication required</td><td>530 SMTP Authentication is required.</td></tr>
<tr><td>–103</td><td>Authentication failed</td><td>Fail to send message due to Authentication Fail</td></tr>
<tr><td>–999</td><td>SMTP error</td><td>Fail to send message</td></tr>
</table>
</td></tr>
<tr><td>SMTP_INFO</td><td>SMTP messages</td></tr>
<tr><td>REMO_COUNT</td><td>Number of remote devices connected to the HMI</td><td>Remote devices include eServer, VNC, and LUA online debugging programs.</td></tr>
</table>

<table>
<tr><th>Parameter</th><th>Definition</th><th>Description</th></tr>
<tr><td>VNC_ENABLE</td><td>VNC start</td><td>
<table>
<tr><th>State Value</th><th>Result</th></tr>
<tr><td>0</td><td>Disable VNC</td></tr>
<tr><td>1</td><td>Enable VNC</td></tr>
</table>
</td></tr>
<tr><td>VNC_STATUS_ENABLE</td><td>VNC status display</td><td>
<table>
<tr><th>State Value</th><th>Result</th></tr>
<tr><td>0</td><td>Turn off the VNC status display.</td></tr>
<tr><td>1</td><td>Turn on the VNC status display.
  </td></tr>
</table>
</td></tr>
<tr><td>VNC_ NOPASSWORD</td><td>VNC Password Setting</td><td>
<table>
<tr><th>State Value</th><th>Result</th></tr>
<tr><td>0</td><td>No password is required for VNC connection.</td></tr>
<tr><td>1</td><td>Password is required for VNC connection.</td></tr>
</table>
</td></tr>
<tr><td>VNC_ VIEWONLY</td><td>VNC Read-only Setting</td><td>
<table>
<tr><th>State Value</th><th>Result</th></tr>
<tr><td>0</td><td>After connecting to HMI, you can only view the screen and cannot operate it.</td></tr>
<tr><td>1</td><td>Operation is possible after connecting to HMI.</td></tr>
</table>
</td></tr>
<tr><td>VNC_MULTI_ CONNECT</td><td>VNC Multi-connection Mode</td><td>
<table>
<tr><th>State Value</th><th>Result</th></tr>
<tr><td>0</td><td>Multiple VNC connections are allowed by HMI.</td></tr>
<tr><td>1</td><td>Only one VNC connection is allowed by HMI.</td></tr>
</table>
</td></tr>
<tr><td>VNC_ PASSWORD</td><td>VNC Password Setting</td><td>Set the VNC connection password (up to 8 characters) through the Character Input element.</td></tr>
<tr><td>VNC_PORT</td><td>VNC Connection Port</td><td>Set the VNC connection port.</td></tr>
</table>

Parameter	Definition	Description
InternetConn Status	Internet Connection status	<table><tr><th>State Value</th><th>Result</th></tr><tr><td>0</td><td>No Internet connection</td></tr><tr><td>1</td><td>Internet connection exists</td></tr><tr><td>2</td><td>Detecting</td></tr></table>

Extended Memory

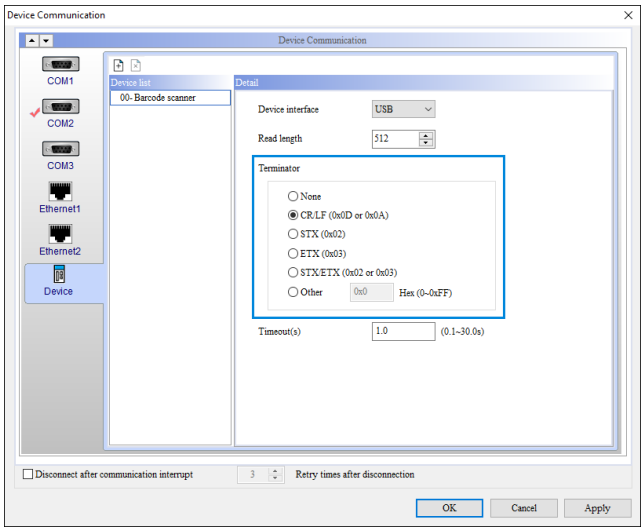
Parameter	Definition	Description
StatusEM0~15	Extended memory status	Displays the current processing status of each extended memory.  <table><tr><th>State Value</th><th>Result</th></tr><tr><td>0</td><td>Writing is complete</td></tr><tr><td>1</td><td>Writing</td></tr><tr><td>2</td><td>The corresponding disk does not exist</td></tr><tr><td>3</td><td>File write error</td></tr><tr><td>4</td><td>File read error</td></tr></table>

Input Device

HMI supports reading data from external input devices. The following parameters control external input devices and read the data.

<table>
<tr><th>Parameter</th><th>Definition</th><th>Description</th></tr>
<tr><td>InputDevID</td><td>The interface of input device</td><td>
<table>
<tr><th>State Value</th><th>Result</th></tr>
<tr><td>1</td><td>COM1</td></tr>
<tr><td>2</td><td>COM2</td></tr>
<tr><td>3</td><td>COM3</td></tr>
<tr><td>4</td><td>COM4</td></tr>
<tr><td>10</td><td>USB (Default)</td></tr>
</table>
</td></tr>
<tr><td>InputDevConnStatus</td><td>The connection status of input device</td><td>
<table>
<tr><th>State Value</th><th>Result</th></tr>
<tr><td>0</td><td>No keypad device detected</td></tr>
<tr><td>1</td><td>Keypad device detected</td></tr>
</table>
</td></tr>
<tr><td>InputDevData</td><td>Read data content</td><td rowspan="2">

Displays the data in InputDevData and reads the data length in InputDevDataLen.
The read data displays the character before the Terminator. If the Terminator is not received, the reading stops when the timeout period expires.

Note: Set the Terminator and Timeout in the Device page on the Device Communication dialog.


</td></tr>
<tr><td>InputDevDataLen</td><td>Read data length</td></tr>
</table>

<table>
<tr><th>Parameter</th><th>Definition</th><th>Description</th></tr>
<tr><td>InputDevClear</td><td>Clear data and data length</td><td>
<table>
<tr><th>State Value</th><th>Result</th></tr>
<tr><td>0</td><td>Idle</td></tr>
<tr><td>1</td><td>Clear the data and data length. After clearing, the device status returns to 0.</td></tr>
</table>
</td></tr>
<tr><td>InputDevInProgress</td><td>Progress of input device</td><td>
<table>
<tr><th>State Value</th><th>Result</th></tr>
<tr><td>0</td><td>Idle</td></tr>
<tr><td>1</td><td>Receiving data</td></tr>
</table>
</td></tr>
<tr><td>InputDevStatus</td><td>Status of input device</td><td>
After clearing the read data, the device status returns to 0.
<table>
<tr><th>State Value</th><th>Result</th></tr>
<tr><td>0</td><td>Idle</td></tr>
<tr><td>1</td><td>Reading successful</td></tr>
<tr><td>2</td><td>Reading failed</td></tr>
<tr><td>3</td><td>The read data size exceeds the specified size.</td></tr>
<tr><td>4</td><td>Open the port failed</td></tr>
</table>
</td></tr>
</table>

Installment

<table>
<tr><th>Parameter</th><th>Definition</th><th>Description</th></tr>
<tr><td>NEXT_INST_ REMAIN_TIME</td><td>Current installment time left</td><td>
Displays the remaining time until the next installment.
It is recommended to use Character Display elements.
Note: The String Length must be set to 23 to display the complete time.
2 days 23 hours 40 min
</td></tr>
<tr><td>INST_LOCK_ STATUS</td><td>Installment locked status</td><td>
<table>
<tr><th>State Value</th><th>Result</th></tr>
<tr><td>0</td><td>Unlocked</td></tr>
<tr><td>1</td><td>To be unlocked</td></tr>
</table>
</td></tr>
</table>

IoT

This parameter is only available for DOP-300 series HMI.

Note: Tunnel related parameters are read-only.

<table>
<tr><th>Parameter</th><th>Definition</th><th>Description</th></tr>
<tr><td>IoTNotify</td><td>Cloud notification</td><td>
<table>
<tr><th>State Value</th><th>Result</th></tr>
<tr><td>0</td><td>Cloud has no new version.</td></tr>
<tr><td>2</td><td>New firmware updates on the cloud.</td></tr>
<tr><td>3</td><td>Cloud has new template updates, but the template ID remains unchanged and the data will not stop updating.</td></tr>
<tr><td>5</td><td>Cloud has new template updates and the template ID changes, and the data will stop updating.</td></tr>
</table>
</td></tr>
<tr><td>IoTDeviceStatus</td><td>Connection state of HMI and Cloud</td><td>
<table>
<tr><th>State Value</th><th>Result</th></tr>
<tr><td>0</td><td>The device is turning on.</td></tr>
<tr><td>1</td><td>The device account is unbinding.</td></tr>
<tr><td>2</td><td>The device account is binding.</td></tr>
<tr><td>3</td><td>Device online</td></tr>
<tr><td>4</td><td>Device offline</td></tr>
</table>
</td></tr>
<tr><td>IoTEnable</td><td>Enable Cloud service</td><td>
<table>
<tr><th>State Value</th><th>Result</th></tr>
<tr><td>0</td><td>Not enabled</td></tr>
<tr><td>1</td><td>Enabled</td></tr>
</table>
</td></tr>
<tr><td>IoTPairingKey</td><td>Cloud template binding verification code</td><td>• Verification code (6 digits) entered when the HMI is bound to the cloud template.
• Use the Numeric Input element to set this parameter.</td></tr>
<tr><td>IoTBindingQRCode</td><td>Bind HMI to DIACloud</td><td>Set to 1, bind the HMI and DIACloud.</td></tr>
<tr><td>IoTUnBinding QRCode</td><td>Unbinding DIACloud</td><td>Set to 1, unbinding the HMI and DIACloud.</td></tr>
</table>

<table>
<tr><th>Parameter</th><th>Definition</th><th>Description</th></tr>
<tr><td>IoTShowQRCode</td><td>Cloud template QR code</td><td>Use the QR Code Display element to set this parameter.</td></tr>
<tr><td>IoTTunnelStatus

IoTTunnel StatusDesc</td><td>State value and description of the HMI connected to FUDA Tunnel</td><td>
<table>
<tr><th>State value</th><th>Description</th></tr>
<tr><td>0</td><td>Tunnel is not connected</td></tr>
<tr><td>1</td><td>Tunnel is connected</td></tr>
<tr><td>2</td><td>The tunnel data has reached the limit.</td></tr>
<tr><td>4</td><td>Fail to load Tunnel configuration file</td></tr>
<tr><td>9</td><td>Fail to open Tunnel interface</td></tr>
<tr><td>–1</td><td>Connection timed out</td></tr>
<tr><td>99</td><td>Establish a connection</td></tr>
</table>
</td></tr>
<tr><td>IoTTunnelID</td><td>The ID of the HMI connected to FUDA Tunnel</td><td>• When not connected, display 0.
• When connected, display the allocated value (1~65535) by the server. The ID is still displayed after disconnection.</td></tr>
<tr><td>IoTTunnelIP</td><td>The Tunnel IP set in the virtual network adapter of HMI</td><td>• IP format is 0.0.0.0~255.255.255.255.
• Use the Character Read element to set this parameter.</td></tr>
<tr><td>IoTTunnelServer</td><td>Tunnel server address</td><td>Displays the server name when connected. The server name is still displayed after disconnection.</td></tr>
<tr><td>IoTTunnelLatency</td><td>Tunnel connection delay time</td><td>• When not connected, display –1.
• When connected, display the delay time.</td></tr>
<tr><td>IOTDataSync Status</td><td>Cloud Template Sync status</td><td>Displays whether the current template is downloaded from DIACloud. The status includes the following:
• Sync.
• Not sync.
Use the Character Entry element to set this parameter.</td></tr>
</table>

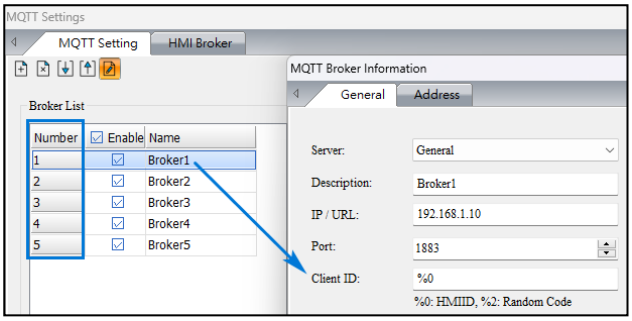
Keypad

Parameter	Definition	Description
KEY_CHAR	Show keyboard text	• When the HMI is connected to an external keypad, the characters can be directly input through the keypad and displayed through this parameter. • It is recommended to use **Character Entry** or **Character Display** elements.

Lua Program

Parameter	Definition	Description
PROGRAM_ STATUS	Current LUA running status	**State Value** / **Description** 0 / Stop 1 / Run 2 / Breakpoint 3 / Pause
PROGRAM_ INFO	LUA execution information	If an error occurs during LUA operation, the current error message can be displayed through this parameter.

MQTT

Parameter	Definition	Description
MQTTClientID	Read Broker ClientID	Set the Broker ID to read the corresponding Client ID.
BrokerNumber	Assign Broker Number	 **Note:** • MQTTClientID is read-only, and this parameter is read by the **Character Display** element. • BrokerNumber corresponds to the Broker List in MQTT Setting.

MQTT Broker

Parameter	Definition	Description
BrokerInfo	Read the execution status of Broker	Displays the current execution status of MQTT Broker. The status includes the following: • Executing • Memory is full and rejects all published messages. • Stopped Use the **Character Entry** element to set this parameter.

Storage

Parameter	Definition	Description
SD_STATUS	State value of external storage device (SD card)	**State Value** / **Result** 0 / SD card is not inserted 1 / SD card is inserted
USB_STATUS	State value of external storage (USB disk)	**State Value** / **Result** 0 / USB disk is not inserted 1 / USB disk is inserted
USB2_STATUS	State value of external storage (USB disk)	**State Value** / **Result** 0 / USB disk is not inserted 1 / USB disk is inserted **Note**: This parameter is only available for DOP-300 series HMI.

Touch

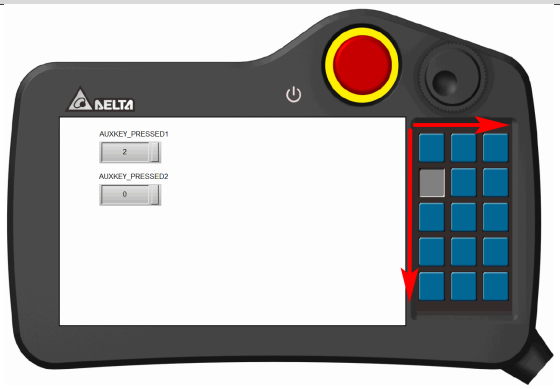
Parameter	Definition	Description
TP_STATUS	State value of HMI panel	**State Value** / **Result** 0 / Panel is not touched 1 / Panel is touched

<table>
<tr><th>Parameter</th><th>Definition</th><th>Description</th></tr>
<tr><td>TP_X</td><td>Touch the X coordinate of HMI panel</td><td rowspan="2"><table><tr><th>Coordinate</th><th>Result</th></tr><tr><td>X</td><td rowspan="2">
</td></tr><tr><td>Y</td></tr></table></td></tr>
<tr><td>TP_Y</td><td>Touch the Y coordinate of HMI panel</td></tr>
<tr><td>TP_FORCE</td><td>Pressing force applied to the HMI</td><td>The display unit is percentage.</td></tr>
<tr><td>TP_DELAY</td><td>Delay time of pressing the HMI screen</td><td>The display unit is percentage.</td></tr>
</table>

Handheld HMI

This parameter is only available for DOP-Handheld series HMI.

<table>
<tr><th>Parameter</th><th>Definition</th><th>Description</th></tr>
<tr><td>EMS_STATUS</td><td>Current status of emergency stop switch</td><td><table><tr><th>State Value</th><th>Result</th></tr><tr><td>0</td><td>Emergency stop switch is not pressed</td></tr><tr><td>1</td><td>Emergency stop switch is pressed</td></tr></table></td></tr>
<tr><td>LSWITCH_ STATUS</td><td>Current status of enabling switch</td><td><table><tr><th>State Value</th><th>Result</th></tr><tr><td>0</td><td>Enabling switch is not pressed</td></tr><tr><td>1</td><td>Enabling switch is pressed</td></tr></table></td></tr>
<tr><td>HANDWHEEL_ COUNTER</td><td>Accumulated number of rotations of handwheel</td><td>Rotate to the right, the value increases by 1; rotate to the left, the value decreases by 1.</td></tr>
<tr><td>AUXKEY_ PRESSED1</td><td rowspan="2">Auxiliary key press status</td><td rowspan="2">Displayed in power of 2, the leftmost button is the first row, and the value is displayed from top to bottom, with the value increasing by 1, 2, 4, and 8, as shown below.
If there are more than 24 auxiliary keys, it will be displayed with parameter AUXKEY_PRESSED2.</td></tr>
<tr><td>AUXKEY_ PRESSED2</td></tr>
</table>

Parameter	Definition	Description
		

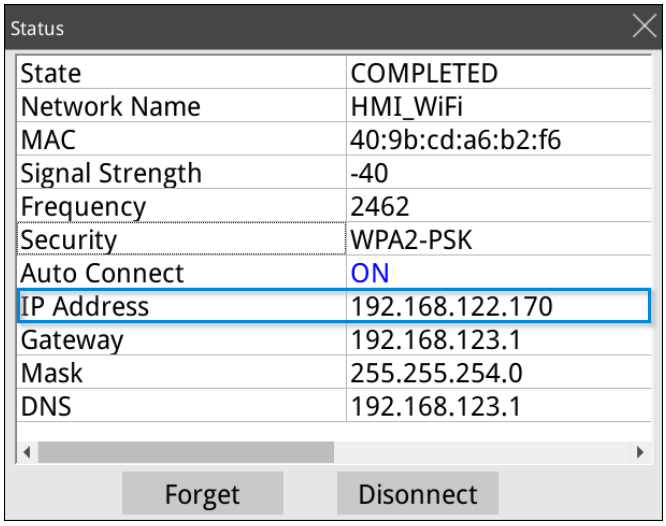
WiFi

Parameter	Definition	Description
WiFi.Country	Wi-Fi country codes	Use the **Character Input** element to set this parameter.
WiFi. DongleStatus	Wi-Fi Dongle configuration status	This parameter is read-only. **State Value** / **Result** 0 – The Dongle is not inserted 1 – The Dongle is inserted
WiFi.Connect Command	Wi-Fi connection configuration	When set to 1 or 2, this parameter will be automatically reset to 0. **State Value** / **Result** 0 – No commands in progress 1 – Disconnected 2 – Connected
WiFi. ErrorMessage	Wi-Fi connection error message	This parameter is read-only. Use the **Character Display** element to read this parameter.
WiFi. ConnectStatus	Wi-Fi connection status	This parameter is read-only. **State Value** / **Result** 0 – Not connected 1 – Connecting 2 – Connected 3 – Fail to connect

<table>
<tr><th>Parameter</th><th>Definition</th><th>Description</th></tr>
<tr><td>WiFi.
SignalLevel</td><td>Wi-Fi
signal level</td><td>This parameter is read-only, and level 4 has the highest signal.<table><tr><th>Level</th><th>Signal Strength</th></tr><tr><td>0</td><td>–100 < signal < –88</td></tr><tr><td>1</td><td>–88 <= signal < –78</td></tr><tr><td>2</td><td>–78 <= signal < –67</td></tr><tr><td>3</td><td>–67 <= signal < –55</td></tr><tr><td>4</td><td>–55 <= signal < 0</td></tr></table></td></tr>
<tr><td>WiFi.
SignalStrength</td><td>Wi-Fi signal strength</td><td>This parameter is read-only, and the unit is dBm.</td></tr>
<tr><td>WiFi.IPSetting</td><td>Wi-Fi
network mode</td><td><table><tr><th>State Value</th><th>Result</th></tr><tr><td>0</td><td>DHCP</td></tr><tr><td>1</td><td>Static</td></tr></table></td></tr>
<tr><td>WiFi.IP1

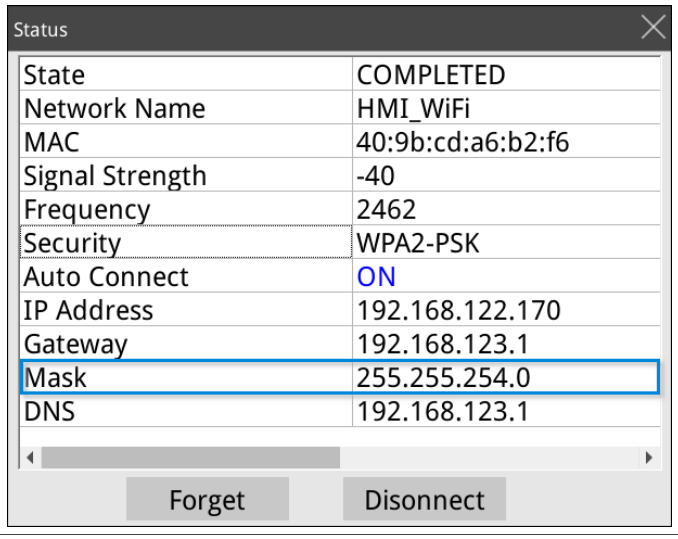
WiFi.IP2

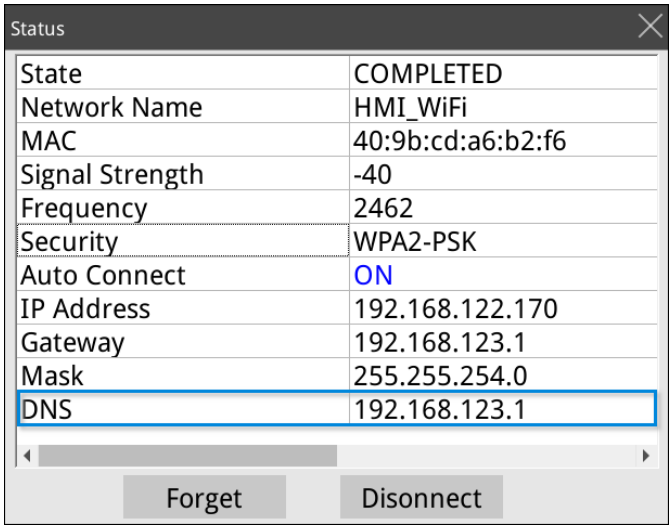
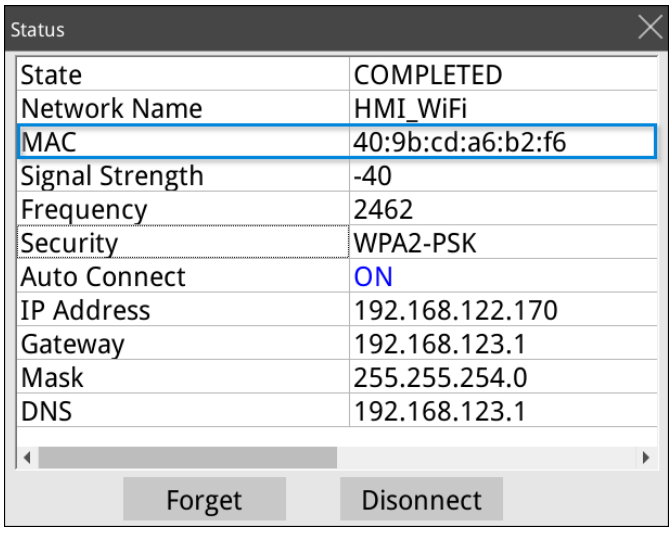
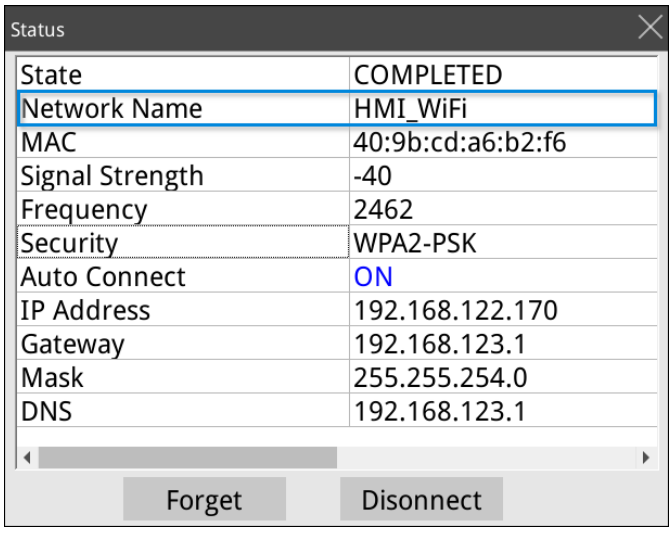
WiFi.IP3

WiFi.IP4</td><td>Wi-Fi
IP address</td><td> 
</td></tr>
<tr><td>WiFi.Mask1

WiFi.Mask2

WiFi.Mask3

WiFi.Mask4</td><td>Wi-Fi
subnet mask</td><td> 
</td></tr>
</table>

Parameter	Definition	Description
WiFi.DNS1 WiFi.DNS2 WiFi.DNS3 WiFi.DNS4	Wi-Fi DNS	
WiFi.MAC1 WiFi.MAC2 WiFi.MAC3 WiFi.MAC4 WiFi.MAC5 WiFi.MAC6	Wi-Fi Dongle MAC	
WiFi. NetworkName WiFi.Password	Name / password of connected AP	

<table>
<tr><th>Parameter</th><th>Definition</th><th>Description</th></tr>
<tr><td>WiFi.Security</td><td>Encryption method</td><td><table><tr><th>State Value</th><th>Result</th></tr><tr><td>0</td><td>WPA2</td></tr><tr><td>1</td><td>WEP</td></tr><tr><td>2</td><td>NONE</td></tr></table></td></tr>
<tr><td>WiFi. AutoConnect</td><td>Whether to automatically connect Wi-Fi</td><td><table><tr><th>State Value</th><th>Result</th></tr><tr><td>0</td><td>Manual</td></tr><tr><td>1</td><td>Auto</td></tr></table></td></tr>
<tr><td>WiFi.Enable</td><td>Whether to enable Wi-Fi Dongle</td><td><table><tr><th>State Value</th><th>Result</th></tr><tr><td>0</td><td>Not enabled</td></tr><tr><td>1</td><td>Enabled</td></tr></table></td></tr>
</table>

External Controller Address

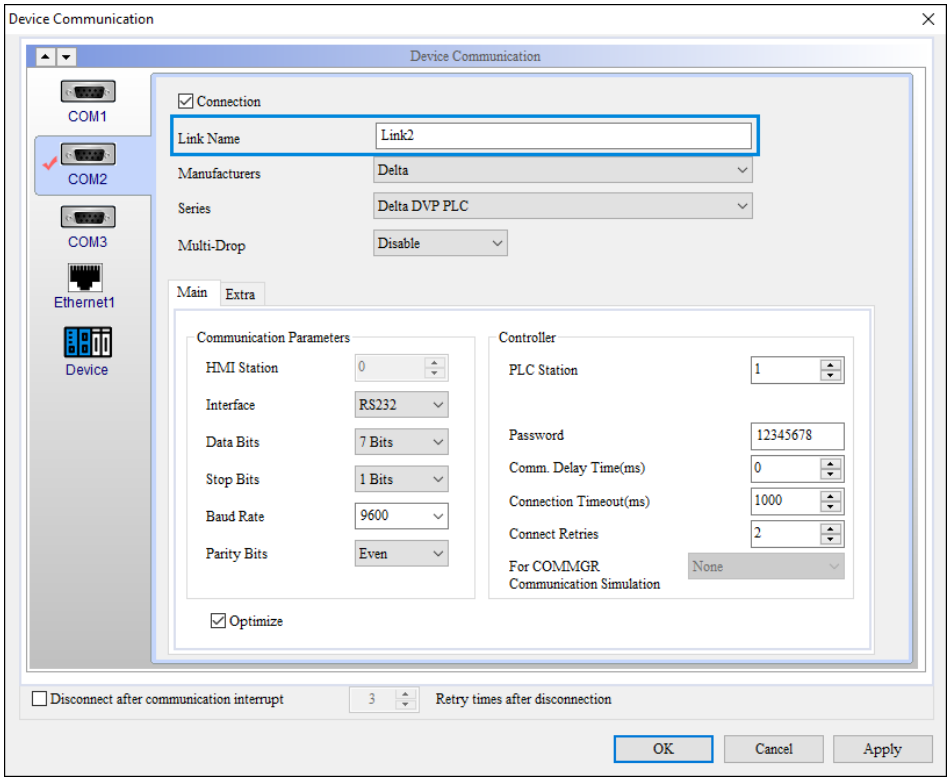
HMI monitors the parameters of the device after it is connected to the controller through COM Port or Ethernet.

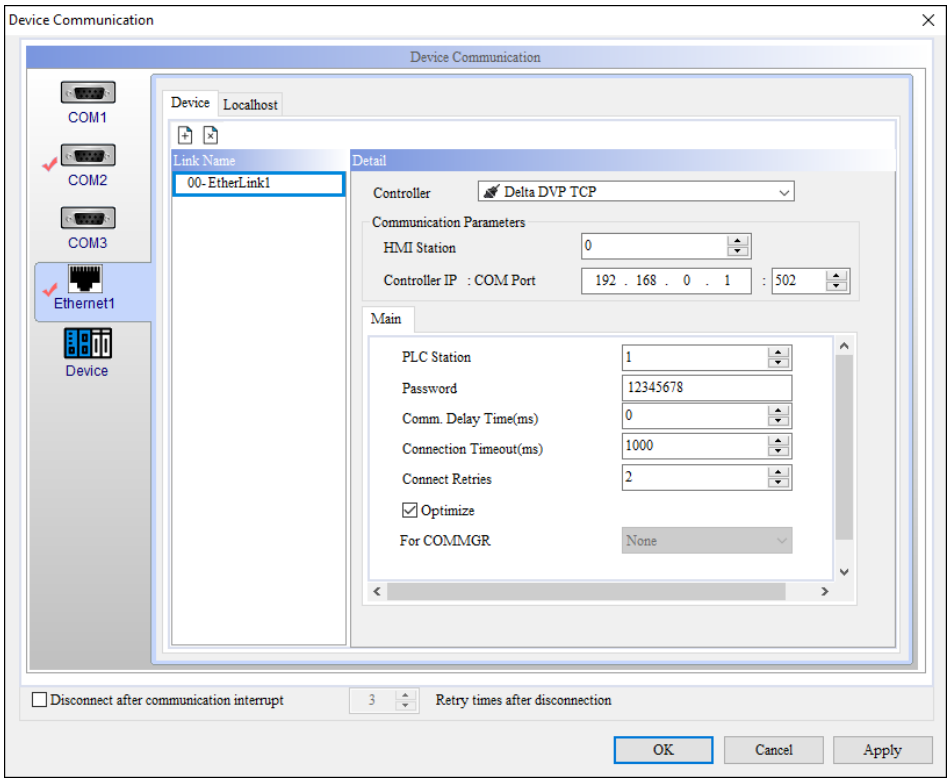
Follow these steps to set the external controller address.

To set the external controller address

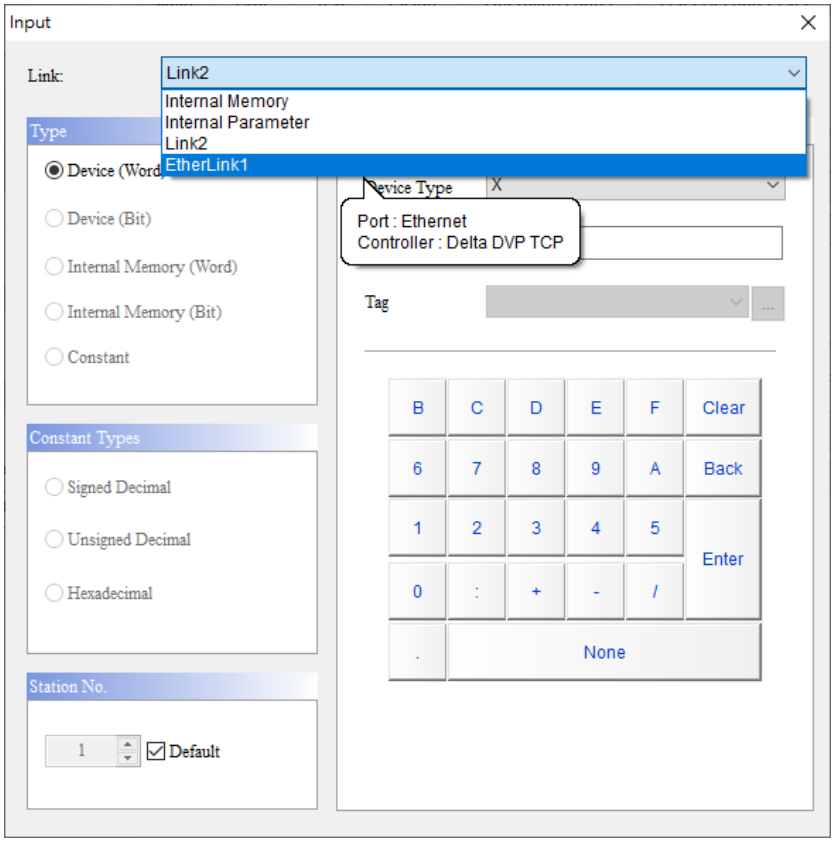
1. Click **General** > **Device Communication** in the menu bar.
2. In the **Device Communication** dialog, set the **Link Name**.

   The following shows an example of setting the COM Port and Ethernet link names.





3. In the address **Input** dialog of the element, select the **Link** name, and then set the controller address.

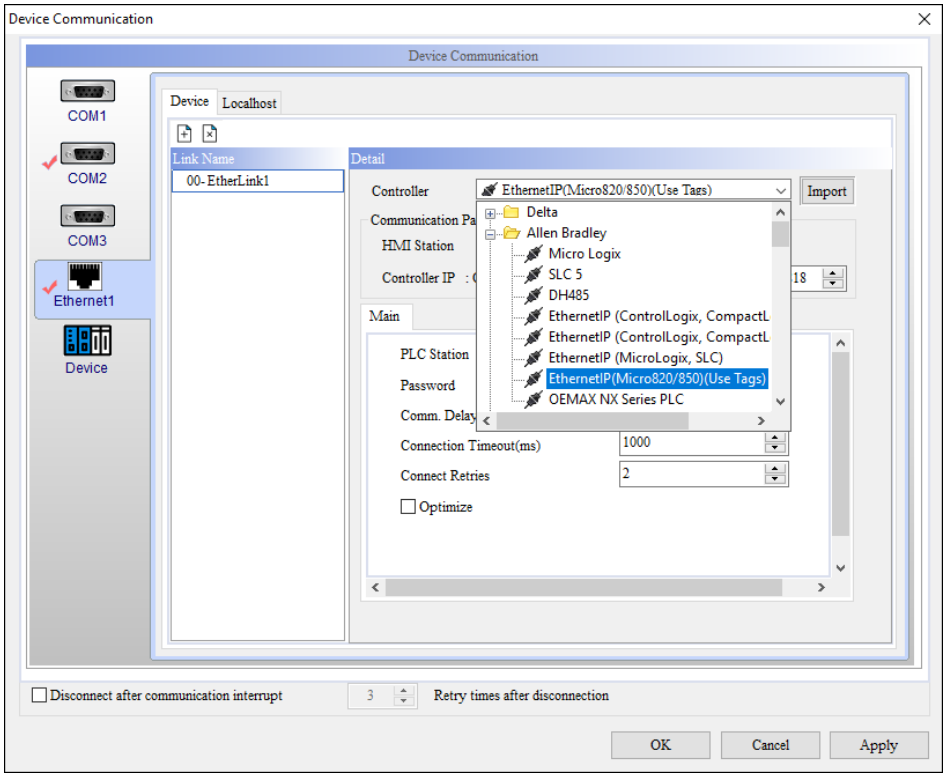


4. Unselect the **Default** checkbox, and then set the **Station No.**

Some Ethernet controllers support variable import function, such as Allen Bradley, Beckoff, Simens, Omron, CODESYS, and so on. After importing the corresponding variable file in the **Device Communication** dialog, the HMI can communicate with the controller through variables (Tag).

How to use Tag?

1. In the **Device Communication** dialog, select **Ethernet1**.
2. In the **Device** tab, click .
3. Select **EthernetIP (Micro820)(Use Tags)** for **Controller**.



4. Click **Import**.
5. In the address **Input** dialog of the element, select the **Link** name with the imported variable file.

   The variables are listed in the dialog.
6. After selecting the variable, click **OK**.

In the variable selection dialog, you can:

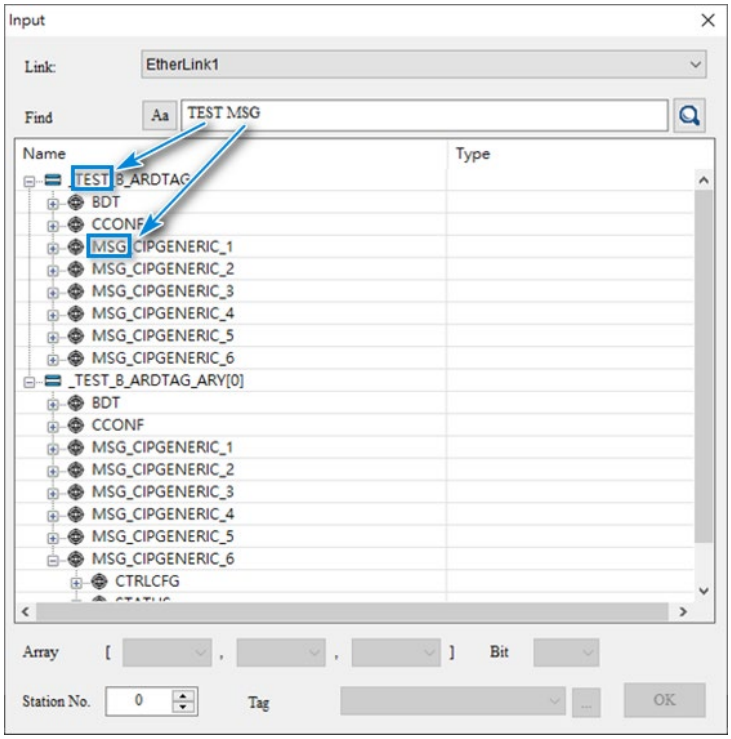
- Use keywords to search for variables.

  In the **Find** field, enter a keyword and click . You can also click to set whether the keyword is case-sensitive.

  The variables containing the keyword are listed in the dialog.
- Use spaces to search for variables under a specific level.

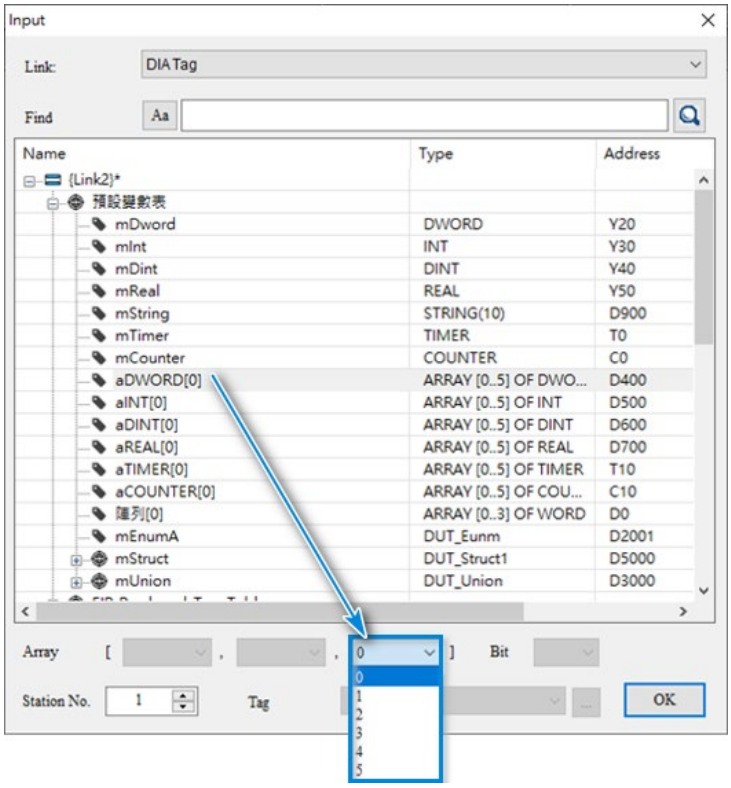
  Enter the upper-level keyword followed by a space, and then enter the lower level keyword.

  The matching variables are listed in the dialog.



7. If array variables are imported, select the number of arrays in the drop-down menu of the **Array** after selecting the array variable.

   Note: If you need to read and write the Bit of a specific variable in the Bit element, select it through the drop-down menu of the **Bit**.



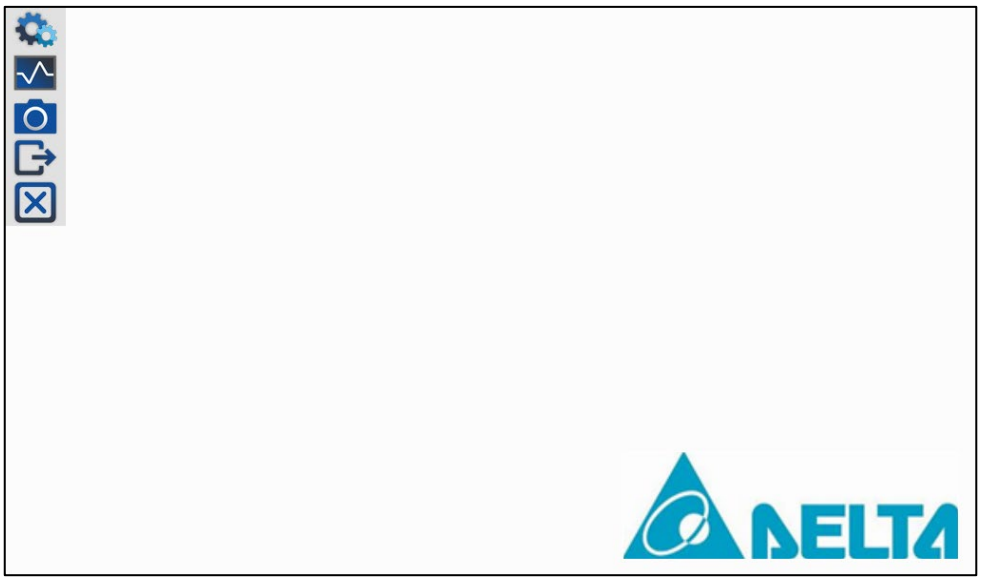
Appendix

System Screen

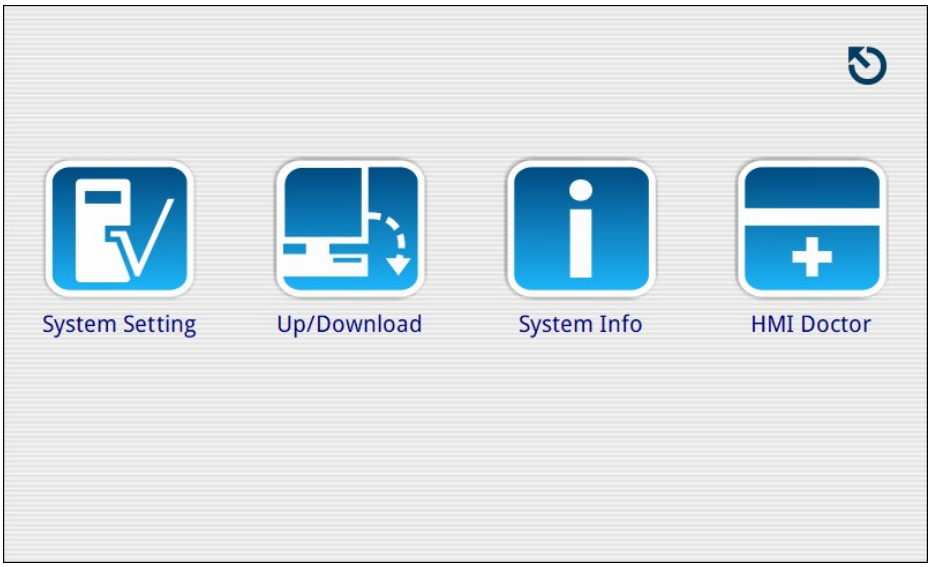
The following describes the system screen functions of the HMI.

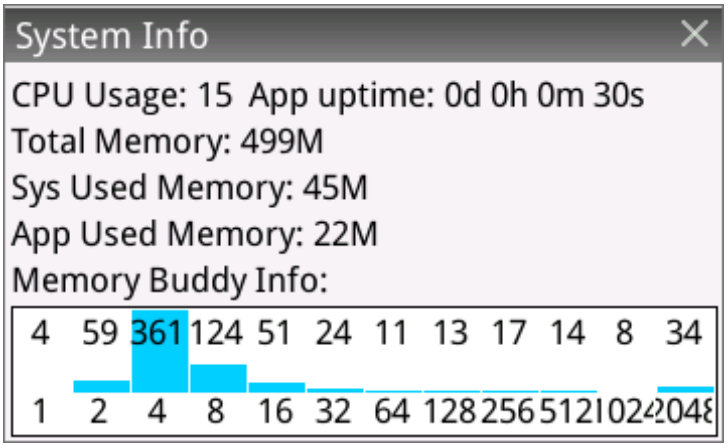
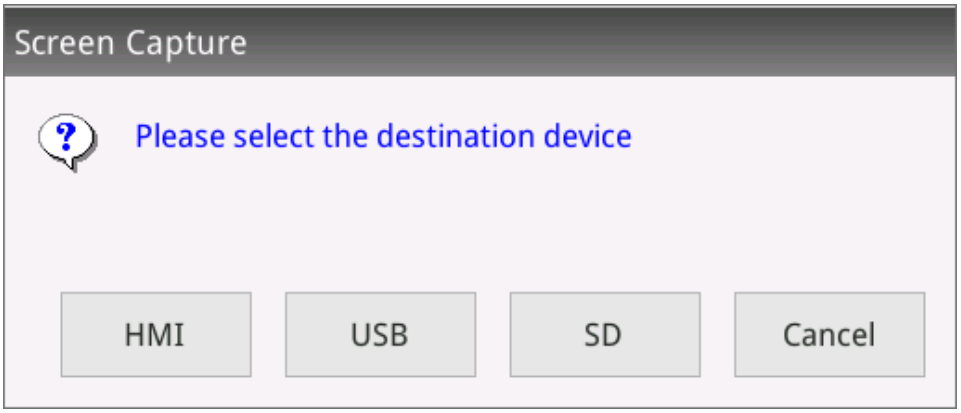
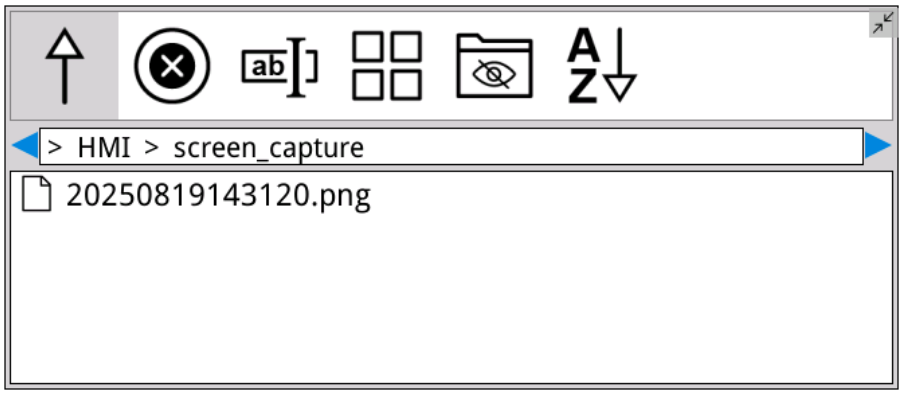
How to enter the system screen?

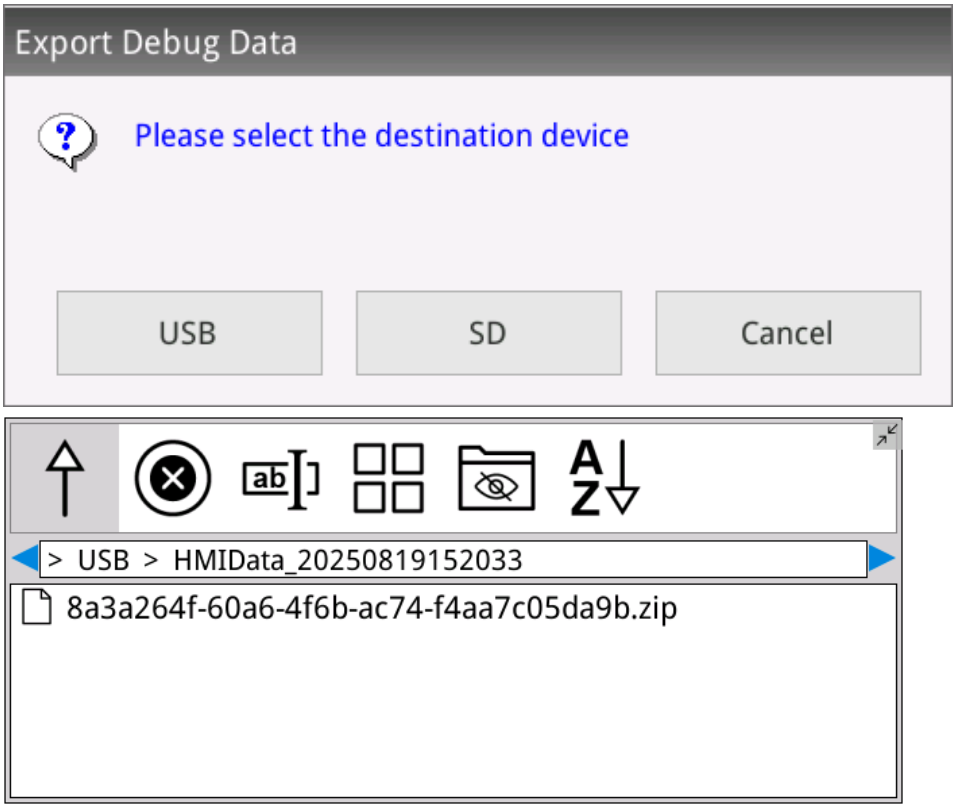
- Press on a blank space for more than 3 seconds, and the system directory appears on the left.



The following table lists the functions on the navigation panel of the **System Menu** with their description.

Setting	Description
Settings	Click to enter the system menu. 

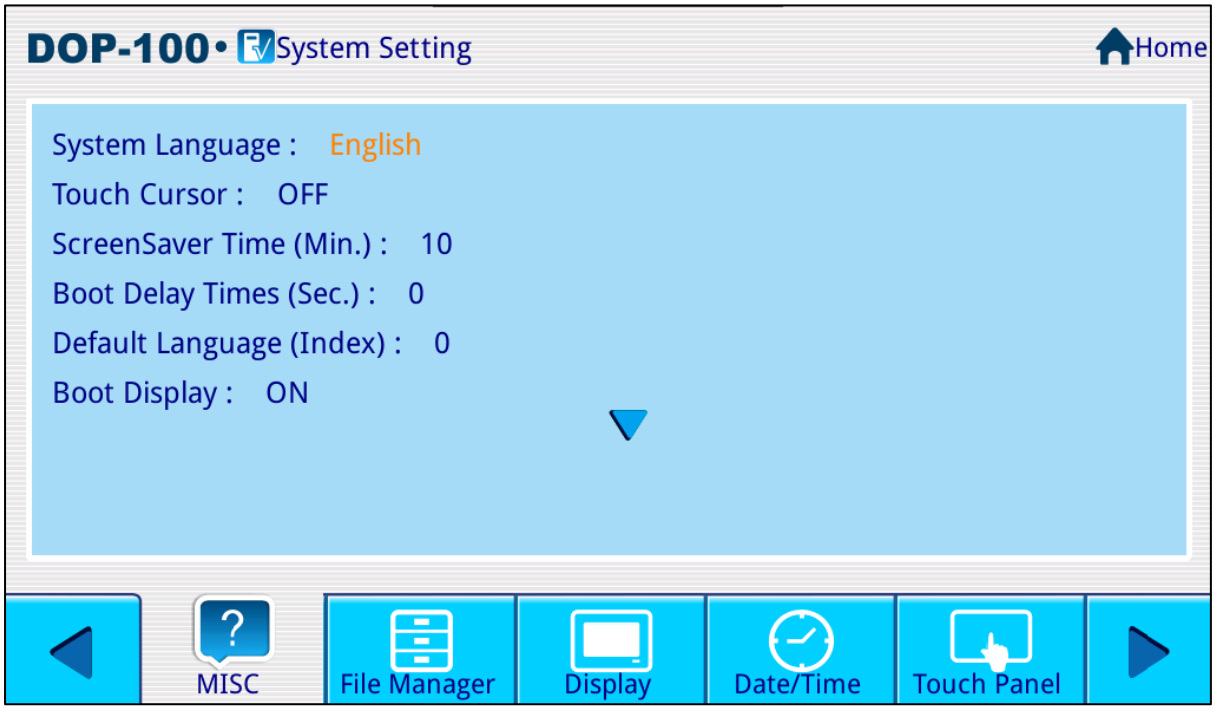
Setting	Description
Monitor	Click to view the current system usage status. 
Capture	Click this icon, select the device to store the screenshot in the **Screen Capture** dialog, and then save the currently captured screen to the screen_capture folder.  The file is named based on the time (hhmmss) on the HMI when the screen is capturing, and the file format is .png. 

Setting	Description
Export	Click this icon, select the device to store the exported file in the **Export Debug Data** dialog, and then export the current system information compressed file to the HMIData folder. 
Close	Click to close the navigation panel of the **System Menu**.

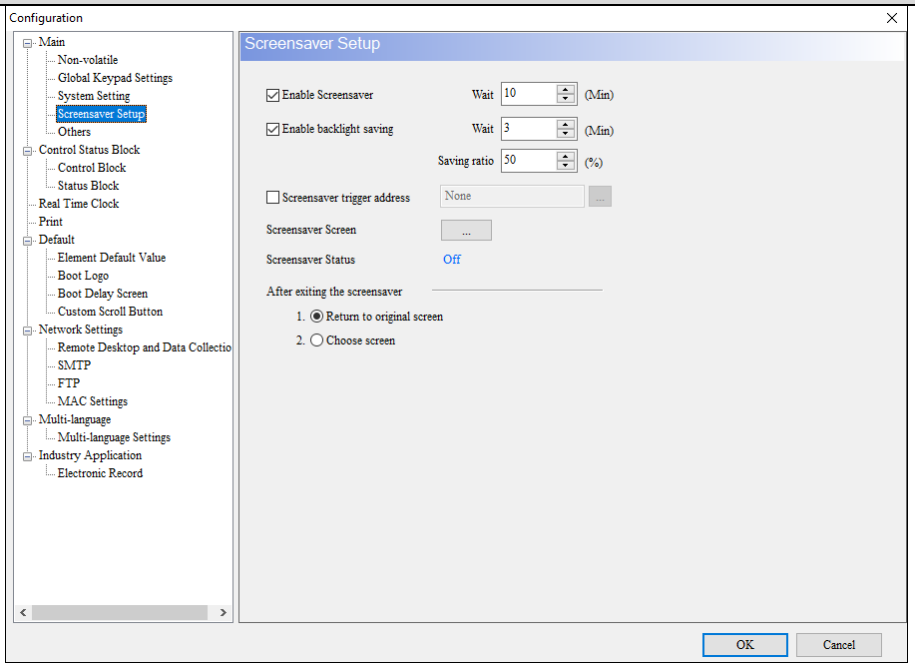
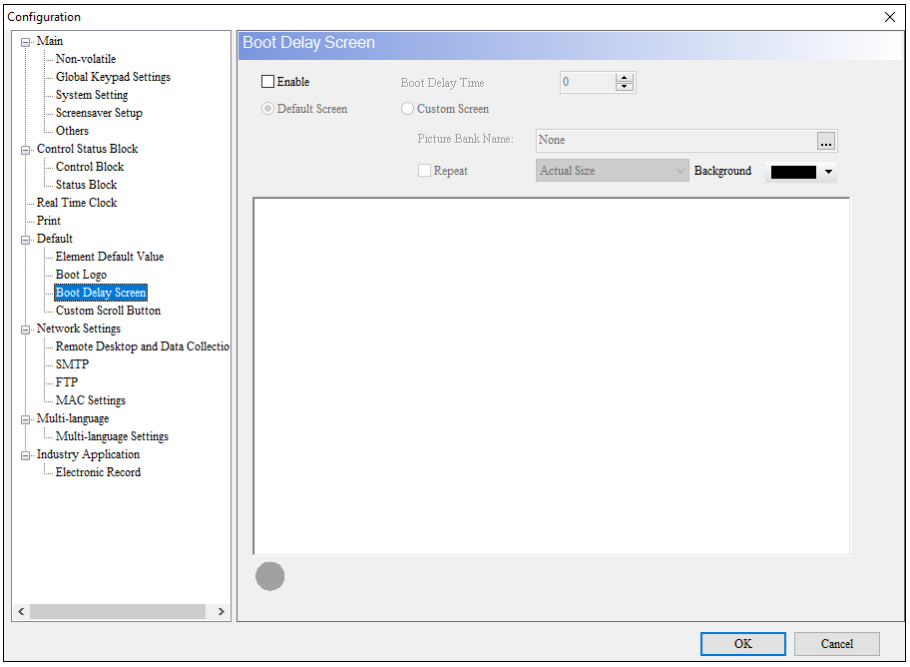
System Setting

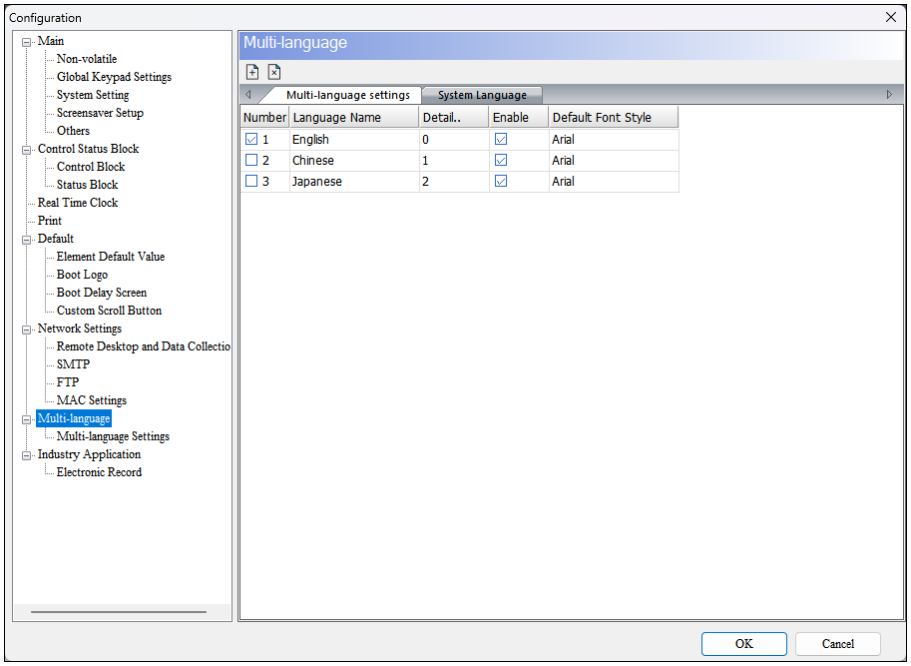
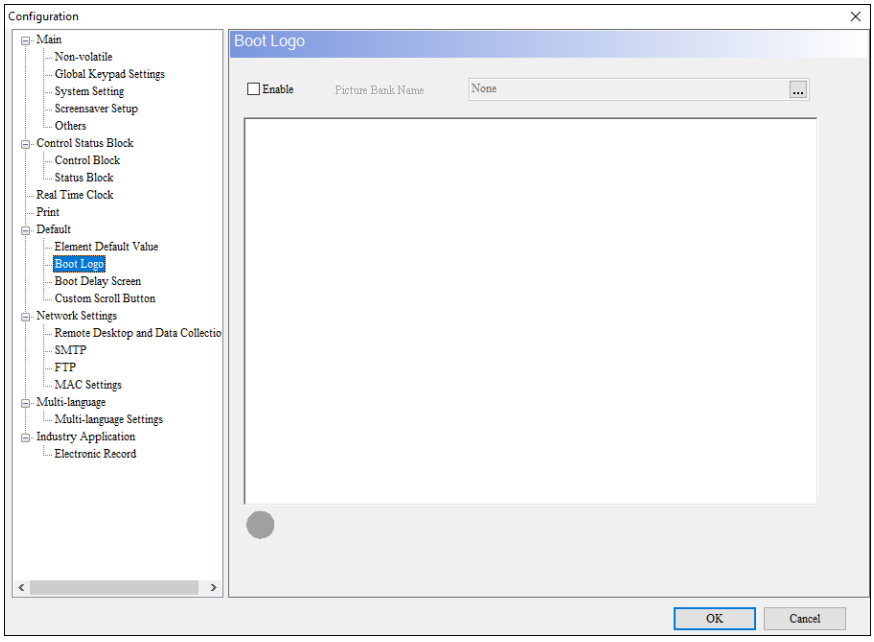
MISC

The following table lists the settings of the **MISC** with their description.



Setting	Description
System Language	Set the display language of the system menu.
Touch Cursor	Set whether to display the touch cursor.
ScreenSaver Time	Set the time to enable the screensaver. If the screensaver is not set, the HMI backlight will be turned off when the enabling time is reached. You can configure the related setting by clicking **Configuration** > **Screensaver Setup** in DIAScreen.

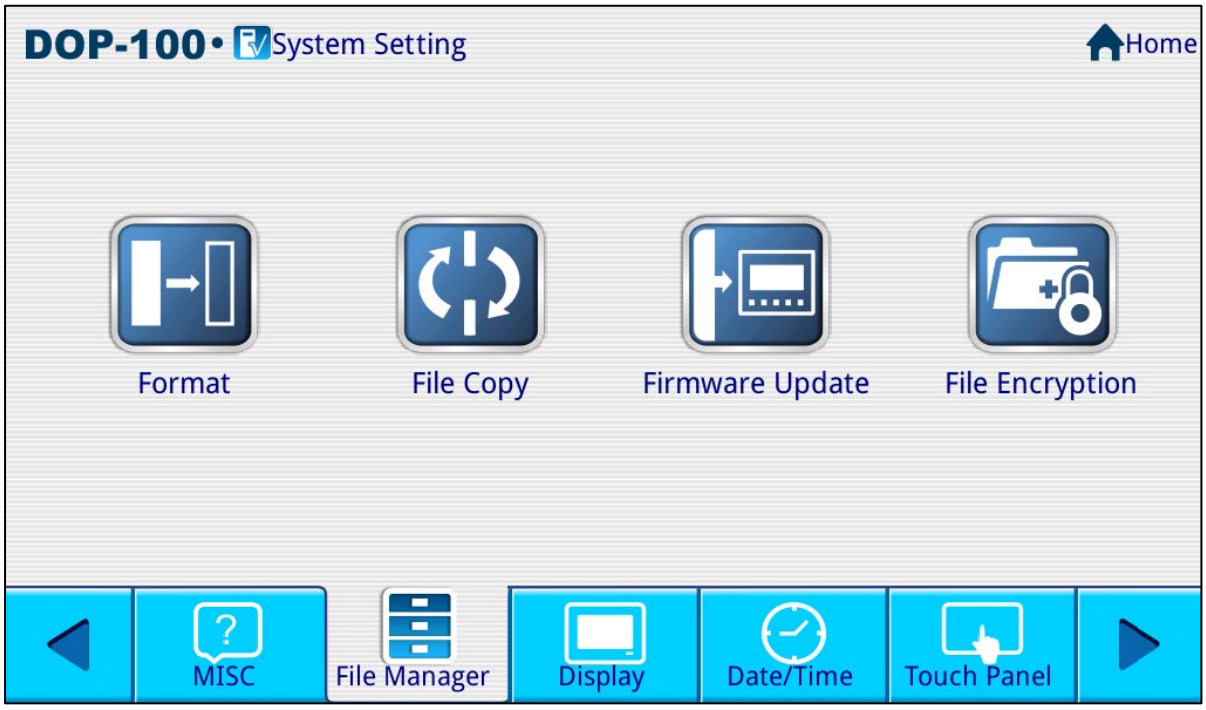
Setting	Description
	
Boot Delay Times	Set the boot delay time, the range value is 0–255. You can configure the related settings by clicking **Configuration** > **Boot Delay Screen** in DIAScreen. 

Setting	Description
Default Language	Select the ID number of the multi-languages. You can add languages by clicking **Configuration** > **Multi-language Settings** in DIAScreen. The selected ID number minus 1 is the default language **Index**. 
Boot Display	Set whether to display the boot screen at startup. Go to DIAScreen > **Configuration** > **Boot Logo** to configure the boot screen. 
USB Comm. Mode	Set the USB upload / download mode. The available options are CDC, DISK, and NONE. **Note**: If select NONE, the USB upload / download mode cannot be used.

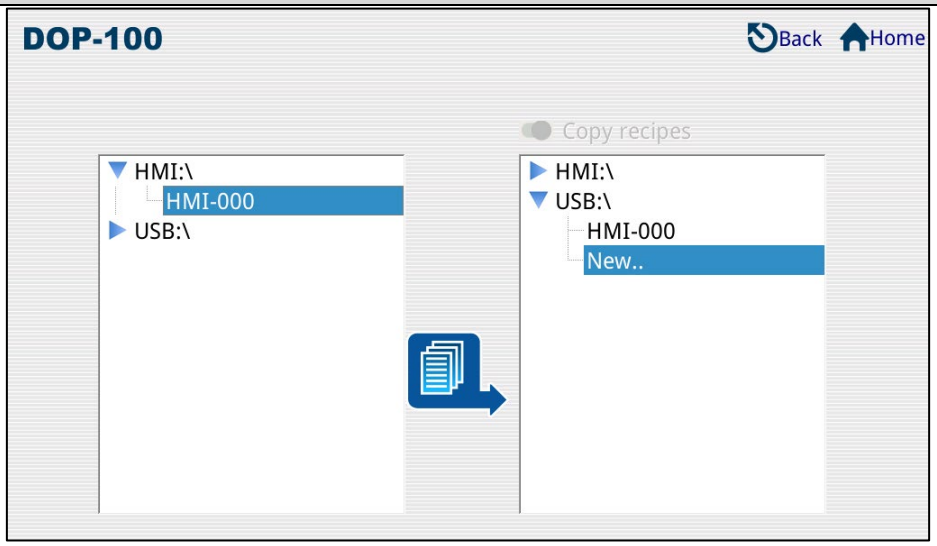
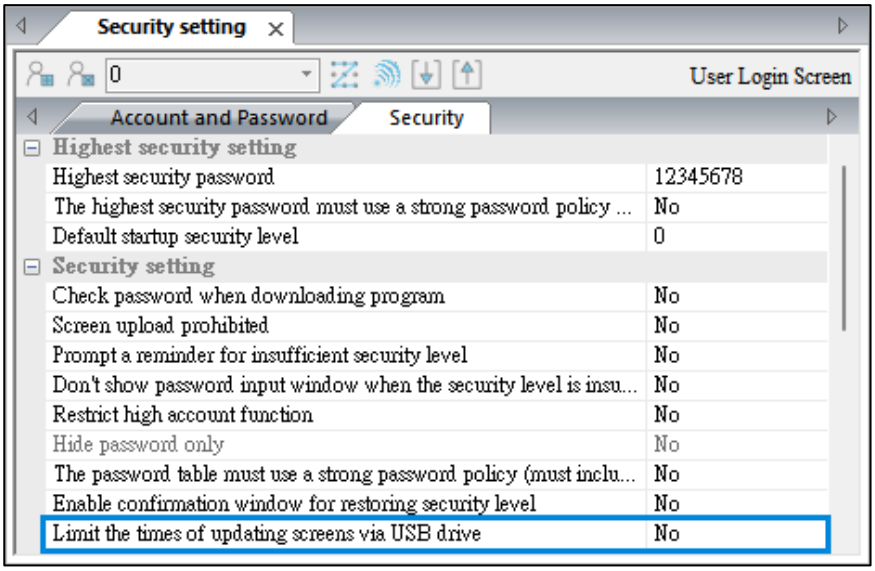
Setting	Description
Network Packet Transmission	Select the network packet transmission.
Default Gateway	Select the gateway. The available options are Auto, LAN, Wi-Fi, and 4G.
USB Storage Device	Set whether to enable USB storage devices. If set to OFF, the USB device will be automatically removed, and the device contents will not be accessed when the USB is plugged in again.

File Manager

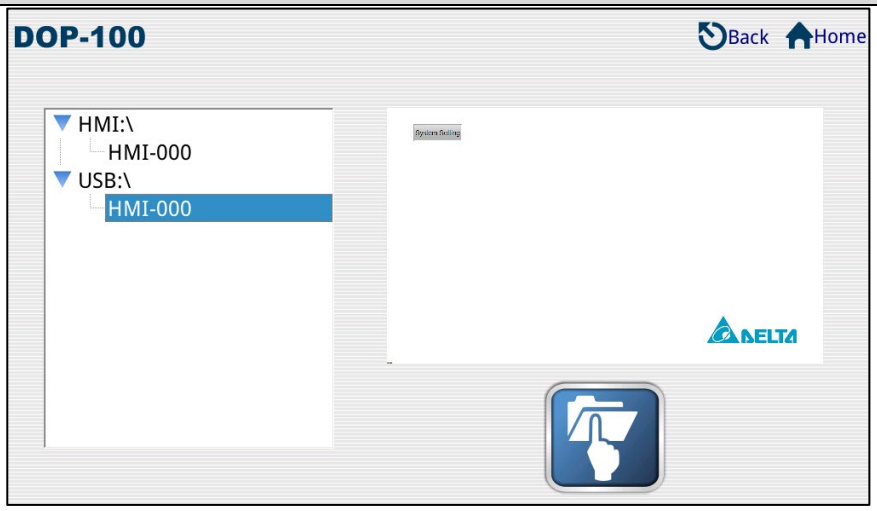
The following table lists the settings of the **File Manager** with their description.



Setting	Description
Format	The file system is cleared after formatting. The available options are HMI, USB Disk, and SD Card. If select HMI, you must enter the highest security password of the current screen file, and the screen file will be cleared after formatting.
File Copy	Copy the screen to external storage devices (USB Disk or SD Card). Select the screen file in the left area and the destination in the right area to copy the screen file.

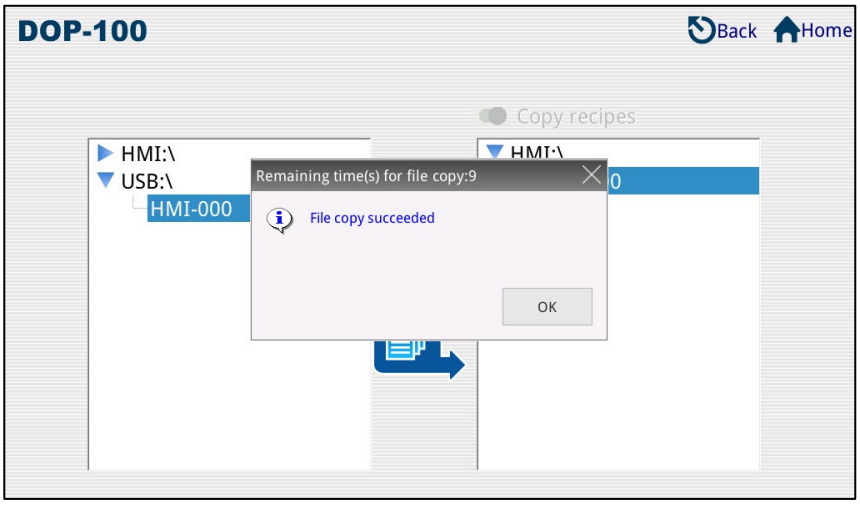
Setting	Description
	
Firmware Update	Update HMI firmware from USB Disk or SD Card.
File Encryption	Follow these steps to encrypt the screen file in the USB Disk or SD Card. 1. Double-click **Security setting** in the DIADScreen **Project** pane and enable **Limit the times of updating screens via USB drive** in the **Security** tab.  2. On the HMI, press **System Setting** > **File Manager** > **File Encryption**. 3. On the **File Encryption** screen, select the screen files folder of the external storage device, and then click .

Setting	Description



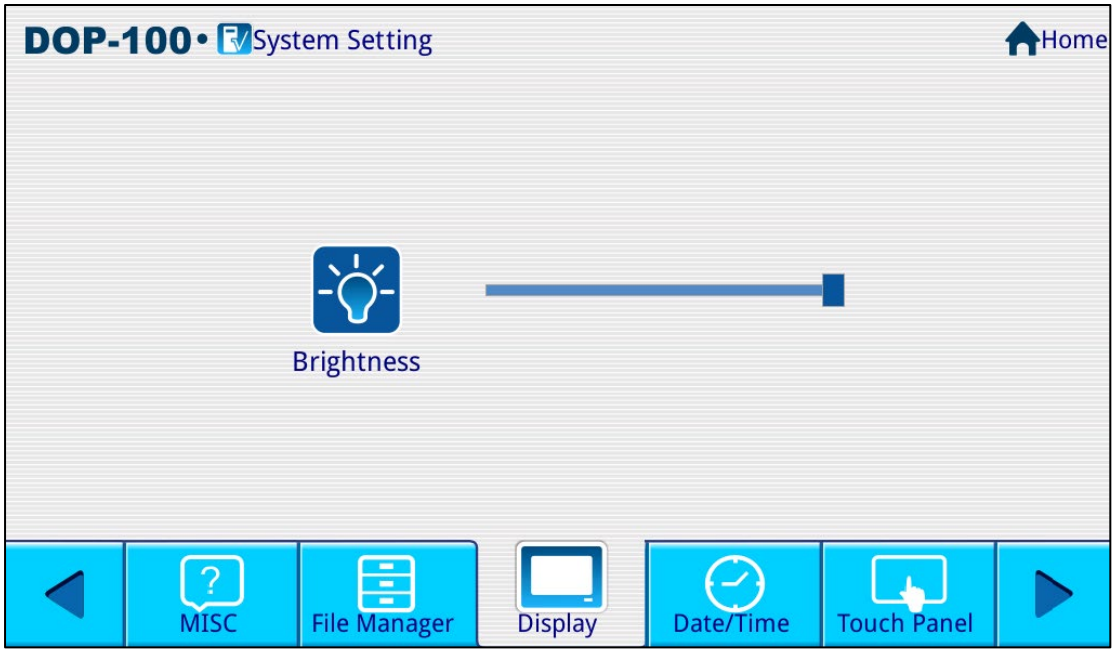
4. After entering the highest security password, enter the maximum number of times the screen file can be copied in the Copt Times dialog.

5. Enter the encryption password, and the file is successfully encrypted.
6. Copy the file (the file password is required), and the Remaining times for file copy will be displayed after the copy is successful.



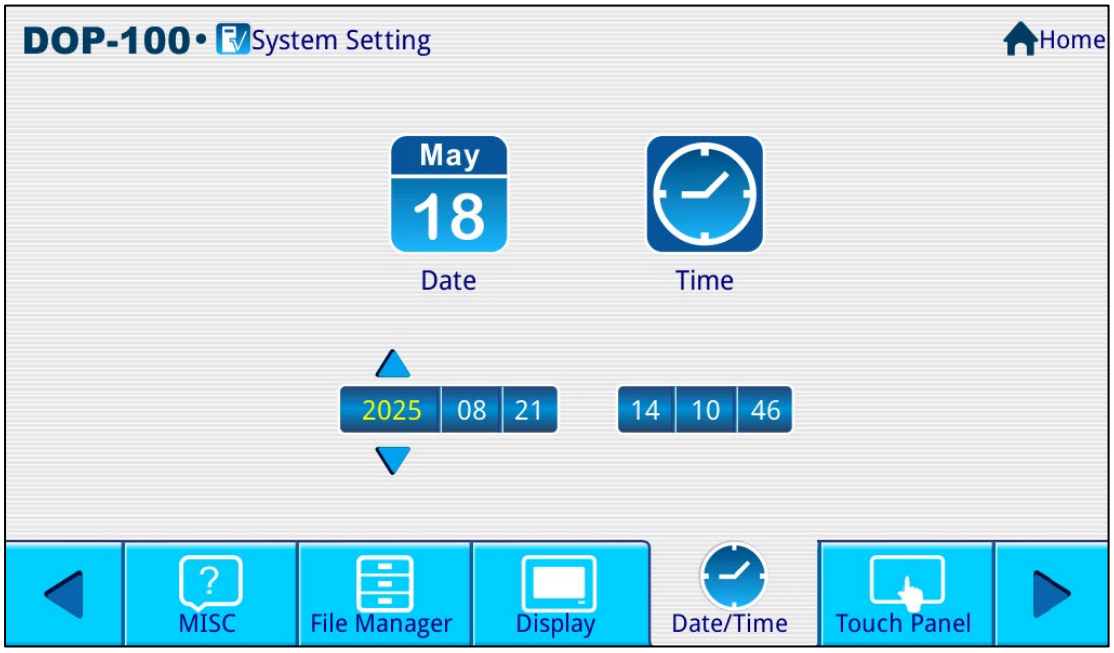
Display

Adjust the brightness of the HMI. You can also configure a **Contrast Brightness** button element on the DIAScreen screen to display the following screen.



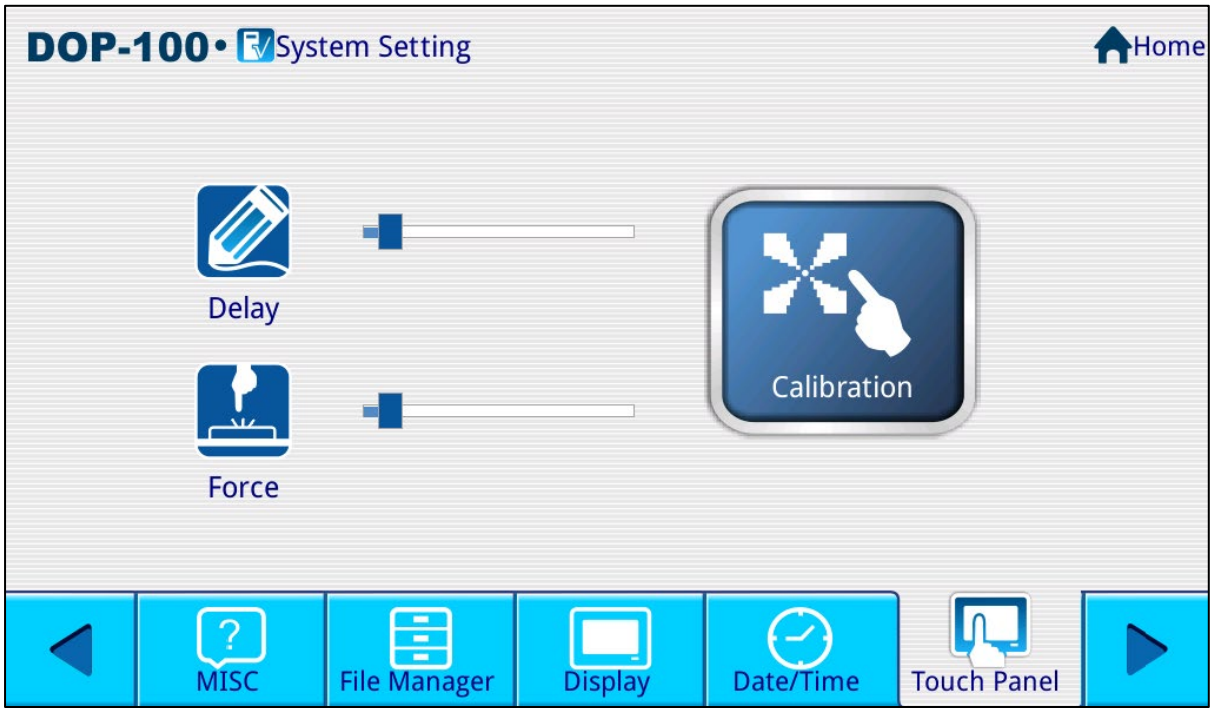
Date / Time

Set the date and time of the HMI system. You can also configure a System Data and Time button element on the DIAScreen screen to display the following screen.



Touch Panel

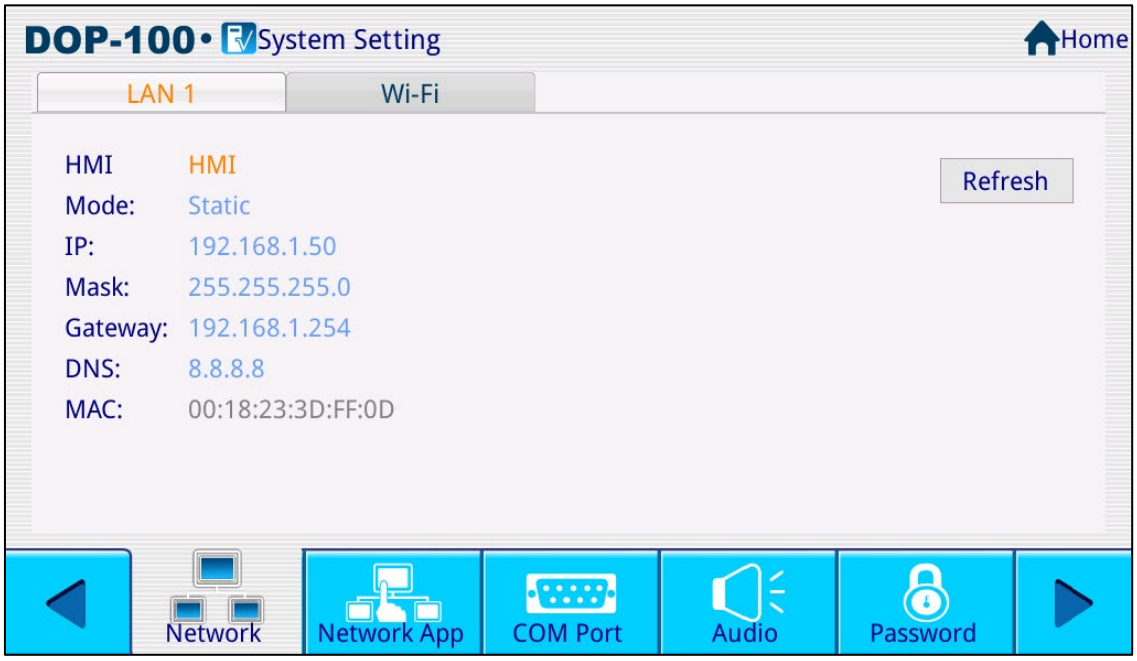
The following table lists the settings of the **Touch Panel** with their description.

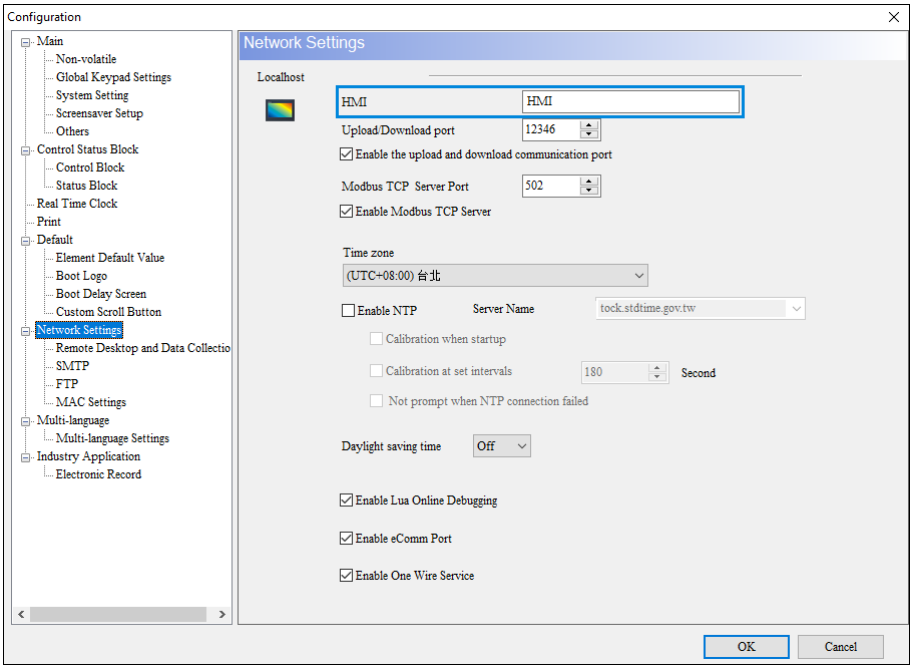


Setting	Description
Delay	Set the delay time for the HMI to process touch movement messages. Dragging to the right reduces the delay and the speed increases; dragging to the left increases the delay and the speed decreases.
Force	Set the pressing force on the touch panel. Dragging to the left makes the force smaller and easier to press; dragging to the right makes the force larger and harder to press.
Calibration	Perform 5-point calibration of the touch panel. Follow the instructions to touch the calibration center point on the screen. If the touch position is away from the center point, there will be a larger deviation. It is suggested to use a stylus to calibrate the screen position.

Network

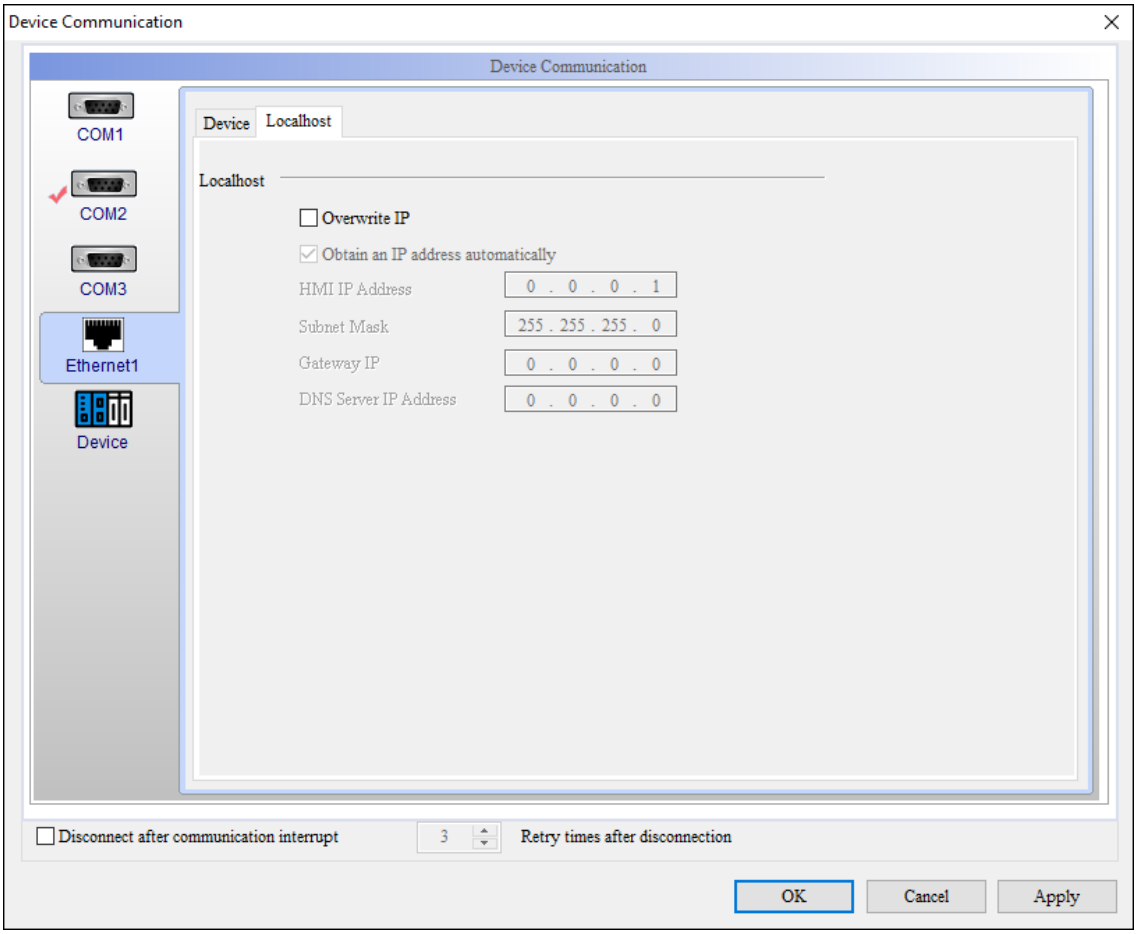
- **LAN**



Setting	Description
HMI	Displays the name set for the HMI. You can change the HMI name by clicking **General** > **Configuration** > **Main** in DIAScreen. 
Mode	Switch the IP address mode. ○ **DHCP**: Obtain an IP address of the HMI automatically. ○ **Static**: Set the IP address of the HMI manually. ○ **BOOTP**: Obtain an IP address of the HMI automatically, but there is no lease period for this IP address.

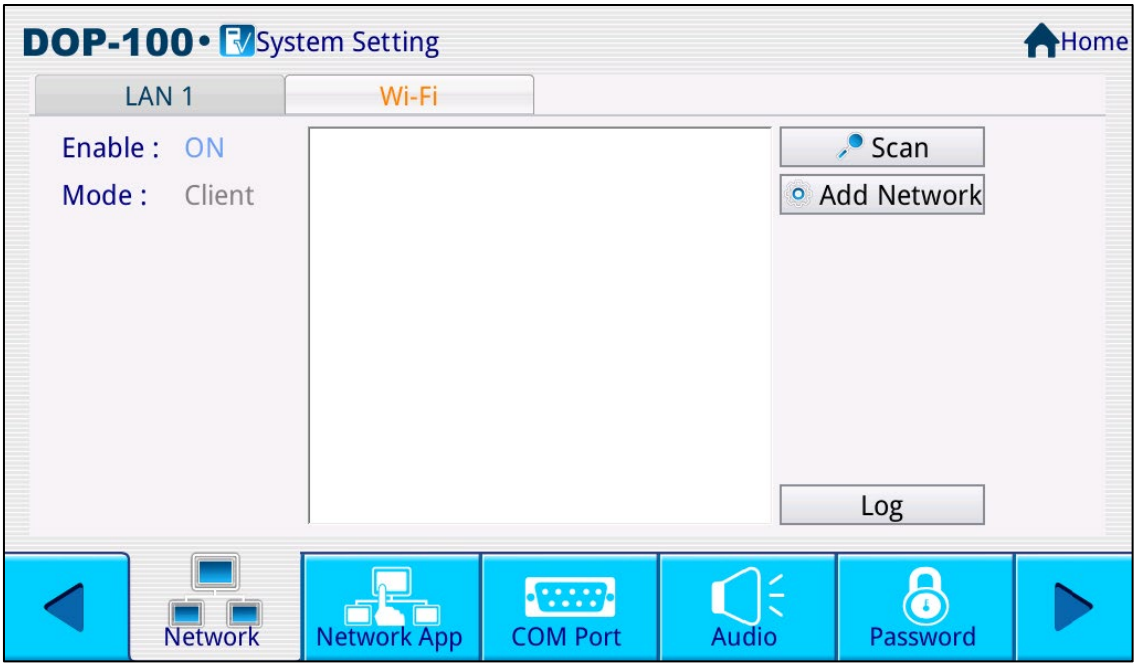
Setting	Description
Mask	Set the network mask. The mask can be set manually when DHCP is OFF.
Gateway	Set the gateway. The gateway can be set manually when DHCP is OFF.
Mask	Set the domain name server. DNS can be set manually when DHCP is OFF.
Gateway	Displays the MAC address of the HMI. This setting cannot be changed.

In DIAScreen, you can set the network of the HMI by clicking **General** > **Device Communication** > **Ethernet** > **Localhost**, and then load the network into the HMI.



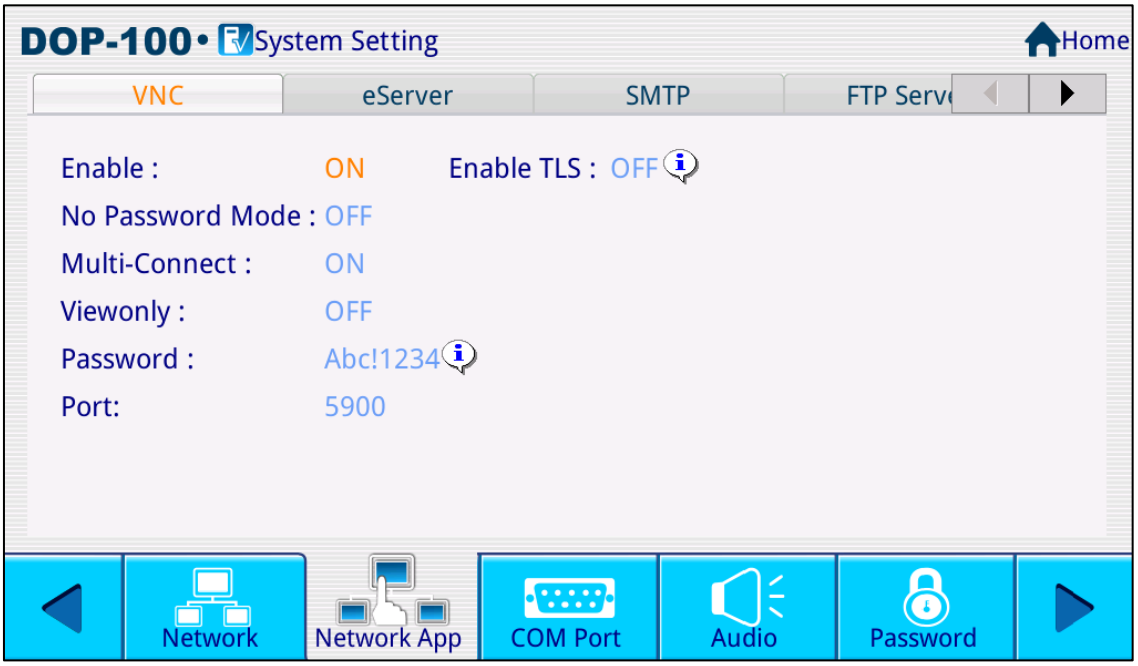
- **Wi-Fi**

  The Wi-Fi tab is only displayed when the HMI is connected to the Wi-Fi Dongle. After enabling the Wi-Fi, you can search for Wi-Fi signal.



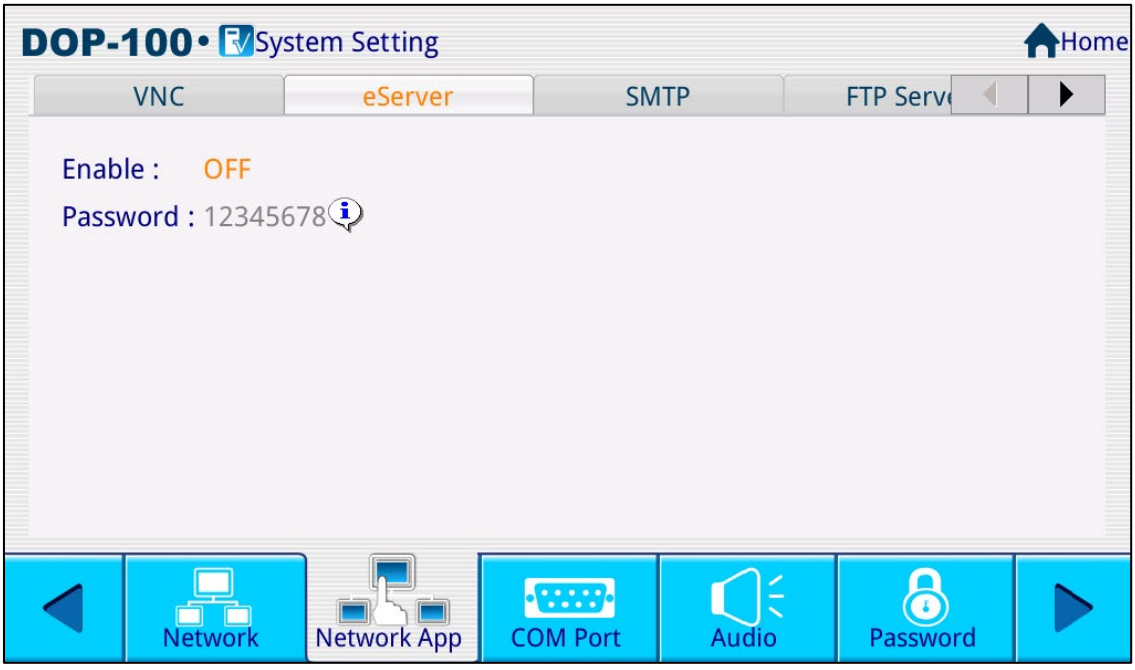
Network App

- **VNC**



Setting	Description
Enable	Choose whether to enable the VNC function.
No Password Mode	Set whether to enable the No Password mode If enabled, you can connect to the VNC remote desktop without entering a password. **Note**: To comply with CyberSecurity standards, it is suggested to enable password checking.
Multi-Connect	Set whether to allow multiple connections. If enabled, up to 64 HMIs can be connected.
Viewonly	Set whether to enable the monitoring mode. If enabled, you can only monitor and cannot operate the HMI when connecting to the VNC remote desktop.
Password	Set the password for connecting to the VNC remote desktop. **Note**: To comply with CyberSecurity standards, passwords must be set to 8 characters and contain at least one uppercase letter, lowercase letter, number, and symbol (!, $, #, %).
Port	Set the connection port.
Enable TLS	Set whether to enable TLS protocol.

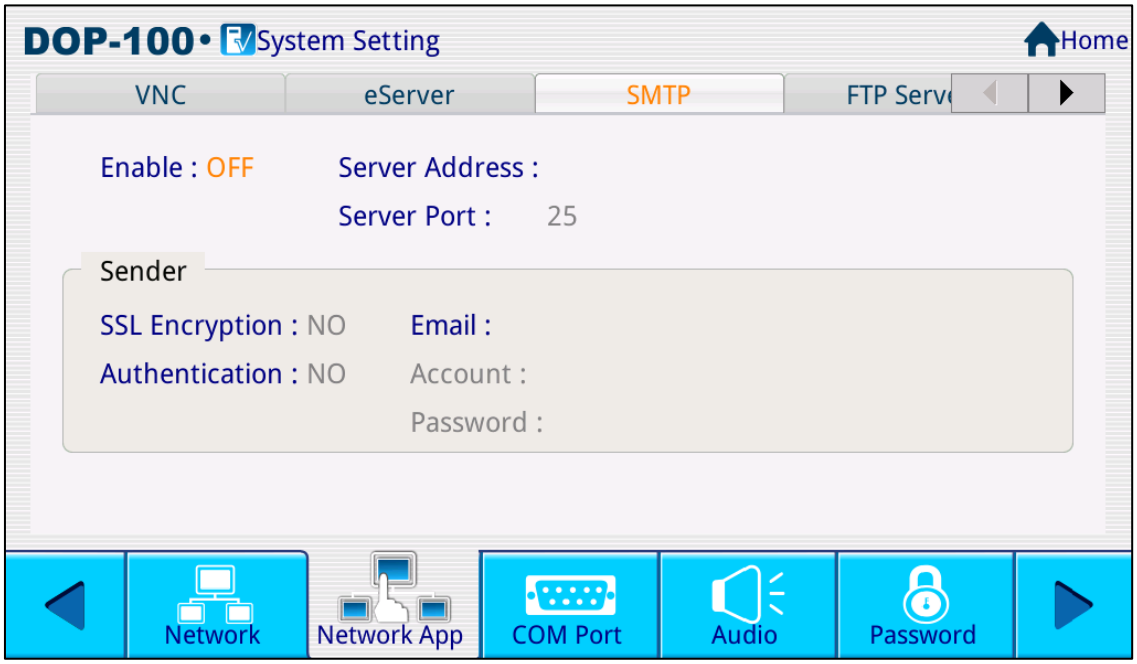
- **eServer**



Setting	Description
Enable	Set whether to enable the eServer function.

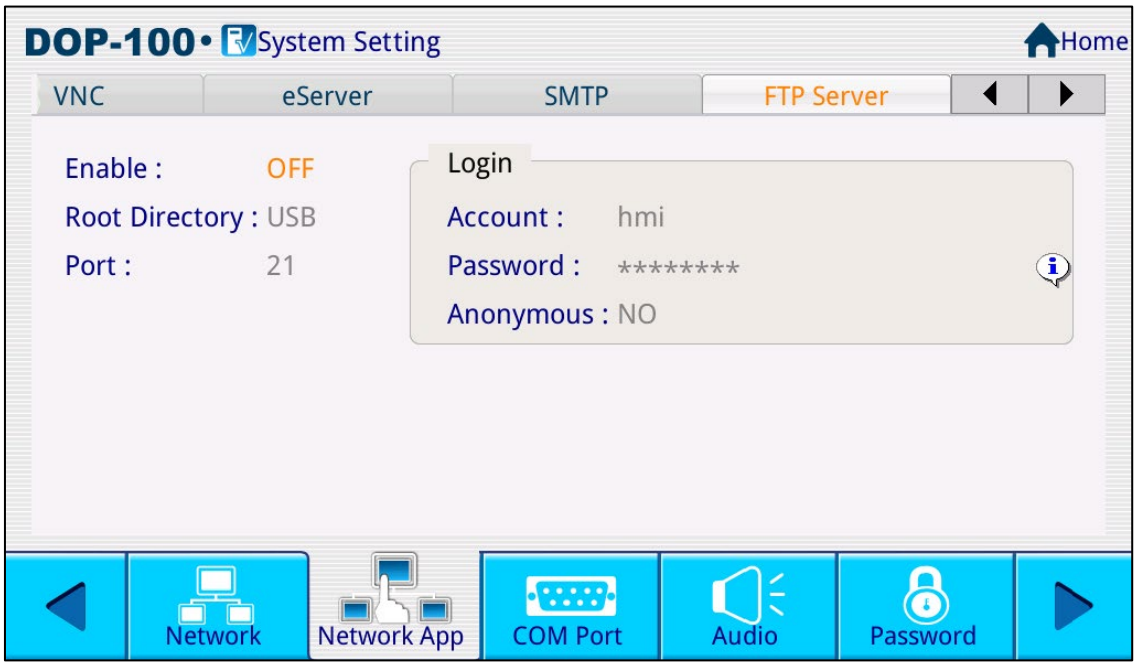
Setting	Description
Password	Set the connection password of Server. **Note**: To comply with CyberSecurity standards, passwords must be set to 8 characters and contain at least one uppercase letter, lowercase letter, number, and symbol (!, $, #, %).

- **SMTP**



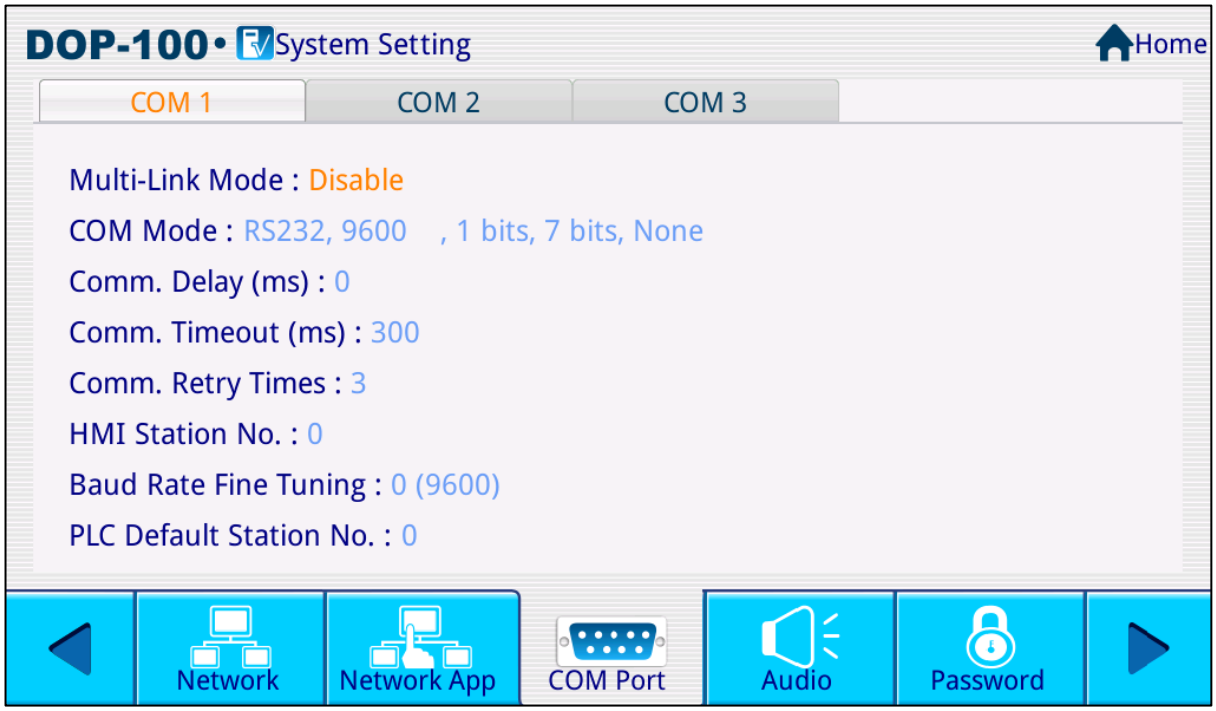
Setting	Description
Enable	Set whether to enable the SMTP function.
Server Address	Set the server address of the mail. This address is the IP address of the mail server created for users.
Server Port	Set the server port of the mail.
SSL Encryption	Set whether to enable the SSL encryption.
Email	Set the mail address of the sender.
Authentication	Set whether to enable the authentication. If enabled, the **Account** and **Password** settings are available.

- **FTP**



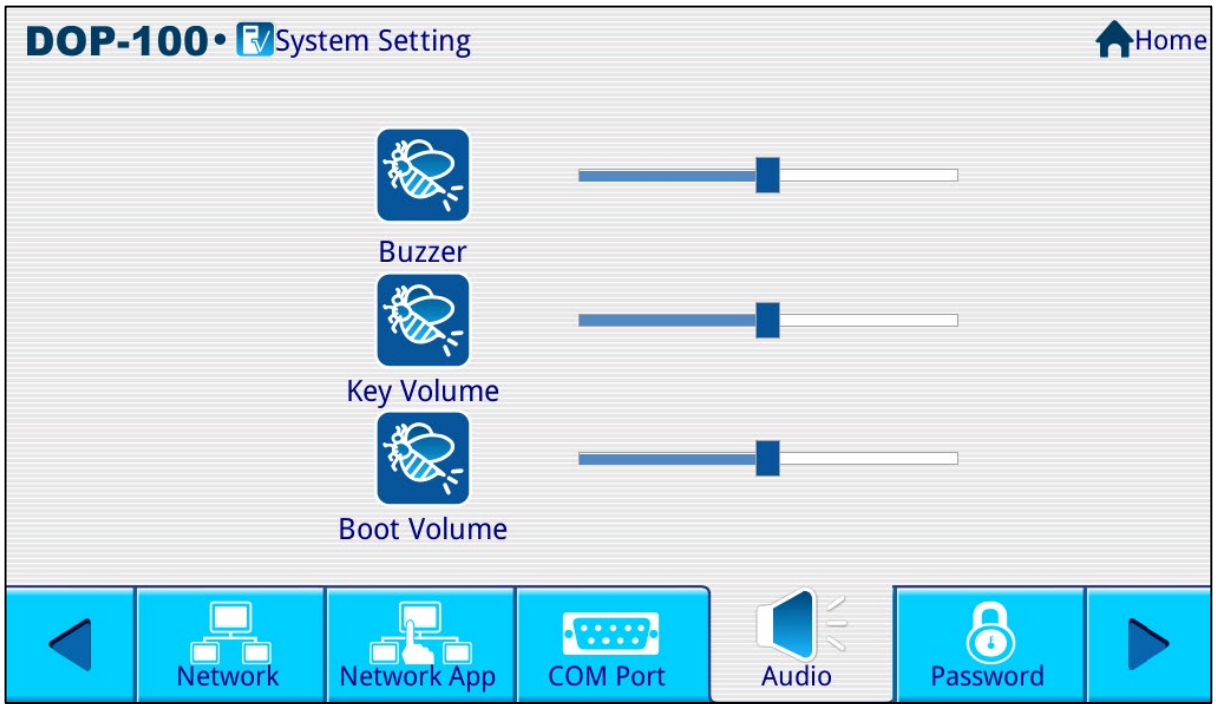
Setting	Description
Enable	Set whether to enable the FTP host function.
Root Directory	Set the storage device.
Port	Set the FTP host port.
Account	Set the login account of FTP.
Password	Set the login password of FTP. **Note**: To comply with CyberSecurity standards, passwords must be set to 8 characters and contain at least one uppercase letter, lowercase letter, number, and symbol (!, $, #, %).
Anonymous	Set whether to log in anonymously. **Note**: To comply with CyberSecurity standards, it is suggested to enable password checking.

COM Port



Setting	Description
Multi-Link Mode	Set whether to use the multi-link mode.
COM Mode	Set the communication interface, baud rate, data bits, stop bits, and parity bits. Press to switch the setting value.
Comm. Delay	Select or enter the interval between each execution of communication. The range value is 0–255, and the default value is **0**.
Comm. Timeout	Select or enter the connection timeout for the communication between the HMI and the device. The range value is 10–2000, and the default value is **1000**.
Comm. Retry Times	Select or enter the retry time. When the device does not respond during the communication process, the HMI will resend the communication command; when the retry time is reached, the HMI will also issue a communication abnormality alarm. The range value is 0–15, and the default value is **2**.

Audio

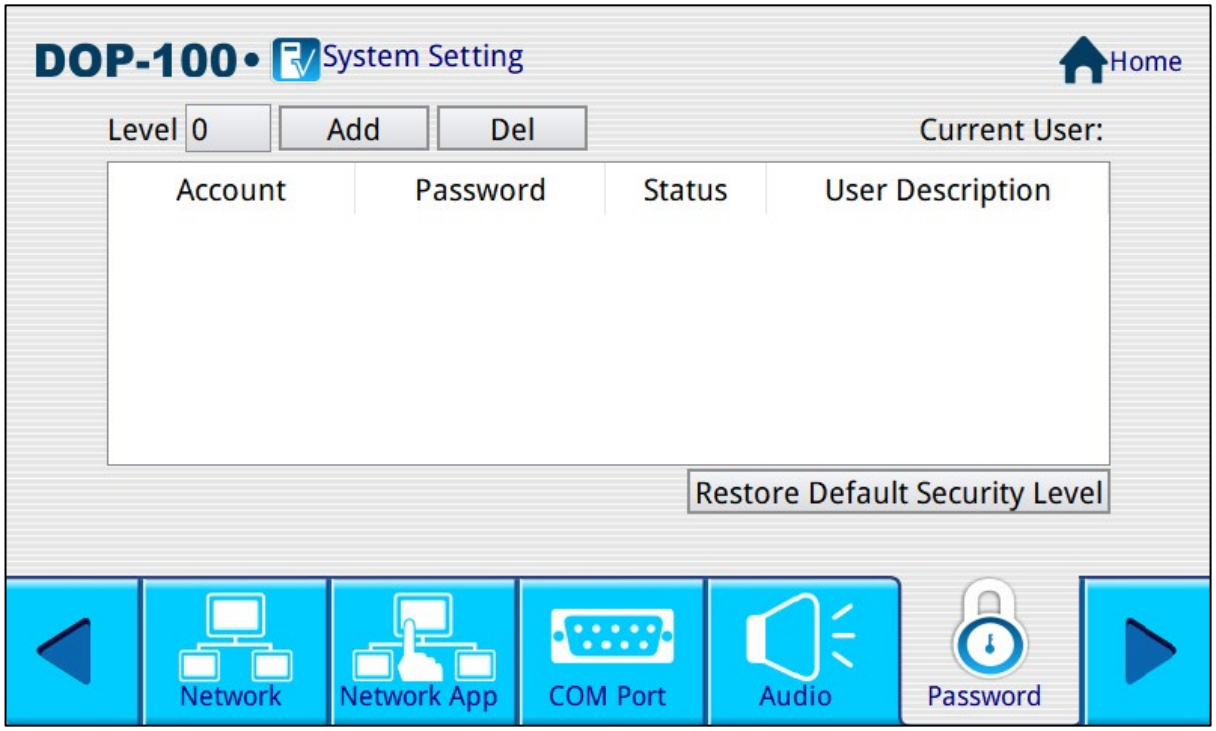


Setting	Description
Buzzer	Set the buzzer volume. Drag to the left to make the volume smaller and drag to the right to make the volume louder.
Key Volume	Set the volume emitted when the HMI button is pressed. Drag to the left to make the volume smaller and drag to the right to make the volume louder.
Boot Volume	Set the volume of the HMI startup sound. Drag to the left to make the volume smaller and drag to the right to make the volume louder.

Password

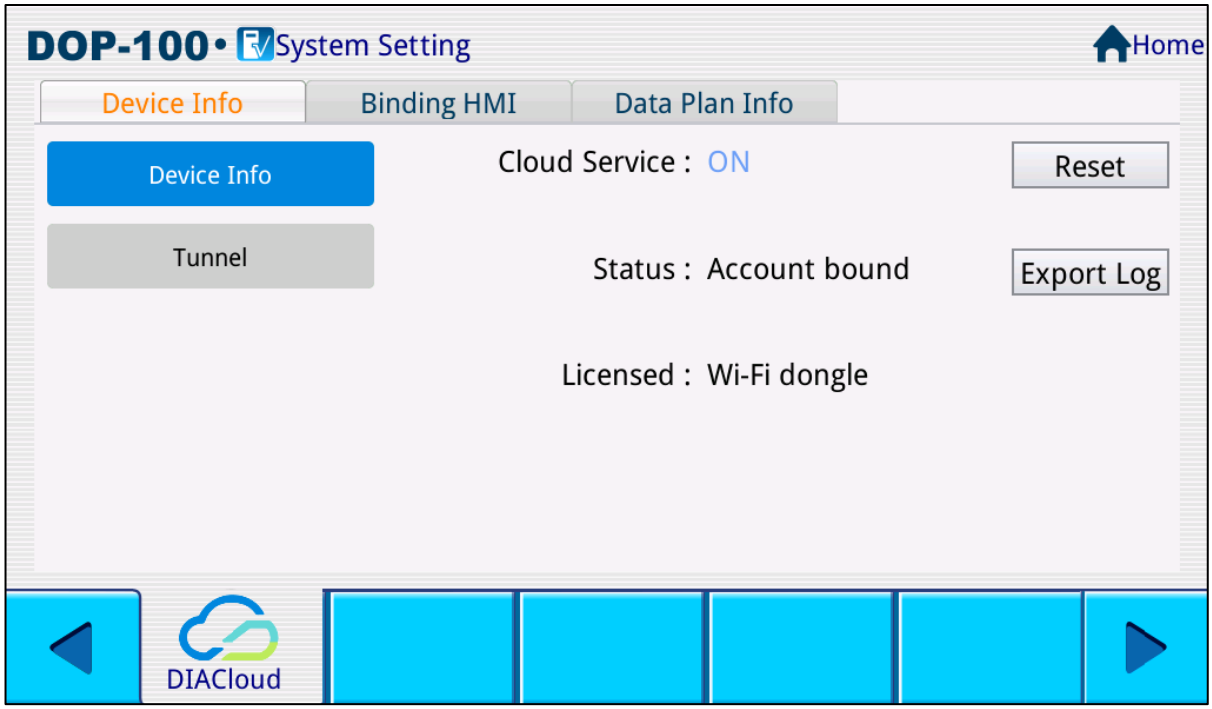
Set the HMI password table. After selecting a **Level** and logging in using the account and password of that level, you can view the password table below the login level (inclusive) and add or delete accounts. Each level can configure up to 1000 accounts.

Click **Restore Default Security Level** to restore the password table contents set the software downloads data.



DIACloud

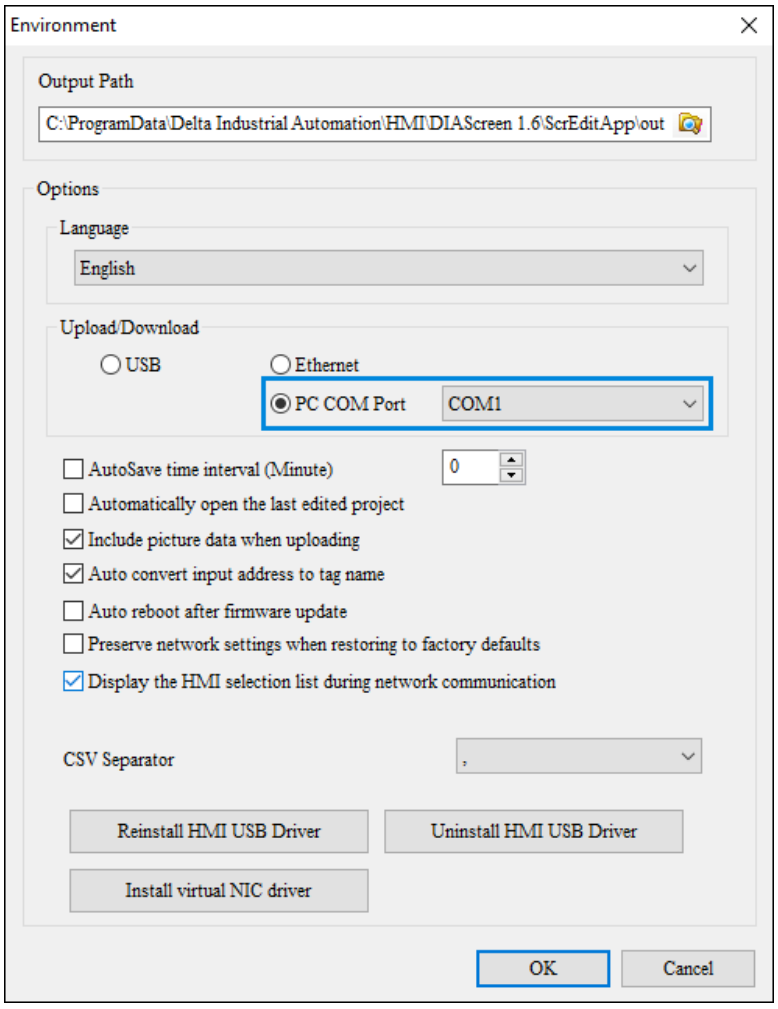
Enable DIACloud to view DIACloud related information.



Up / Download

Standard Mode

Set the COM Port to communicate with DIAScreen and wait for DIAScreen to send the motion command and data packets for Up/Download. In DIAScreen, go to **General** > **Environment**, select the **PC COM Port** in the Upload/Download area of the **Environment** dialog, and then upload / download the screen data and recipe data.



Bypass Mode

With the HMI as an intermediary, it transmits the data received from the source port to the destination port.

- **Mode 1 (COM1 → COM2)**

  With COM1 as the source port and COM2 as the destination port, the data that COM1 received is transmitted using the COM2 protocol.

- **Mode 2 (COM2 → COM1)**

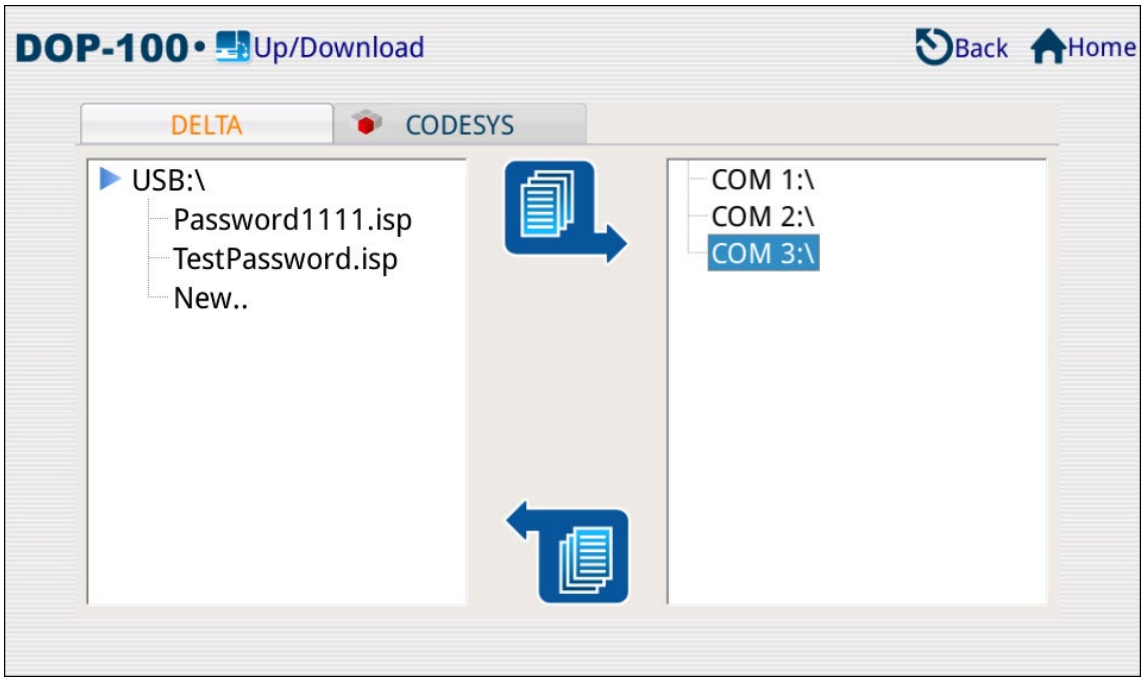
  With COM2 as the source port and COM1 as the destination port, the data that COM2 received is transmitted using the COM1 protocol.

Transfer Mode

Transfer PLC program via HMI.

- **DELTA**

  Support ISPSoft project files. Download the .isp format files stored in the USB to the controller through the COM port, or upload the PLC program to the USB.



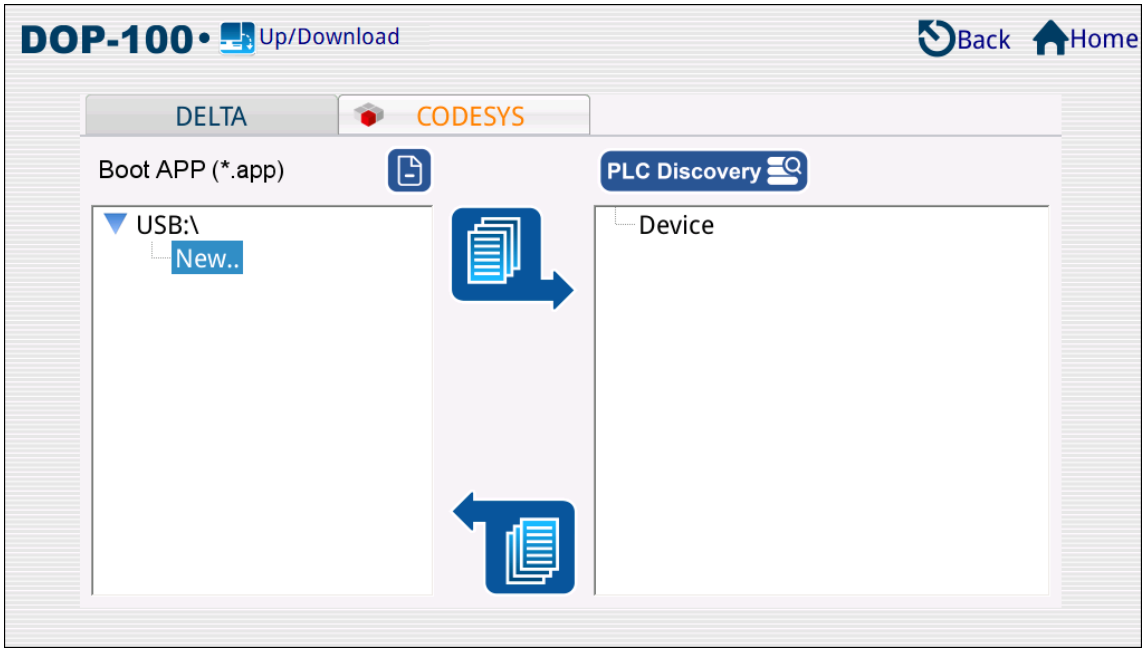
1. When downloading the PLC file, a message dialog will appear, first asking for the project password first and then the PLC password.
2. When uploading the PLC file, a message dialog will appear, first asking for the PLC identification code and then the PLC password.

3. To save the uploaded PLC file as a new file, select **New...** in the left area and click the upload icon.

- **CODESYS**

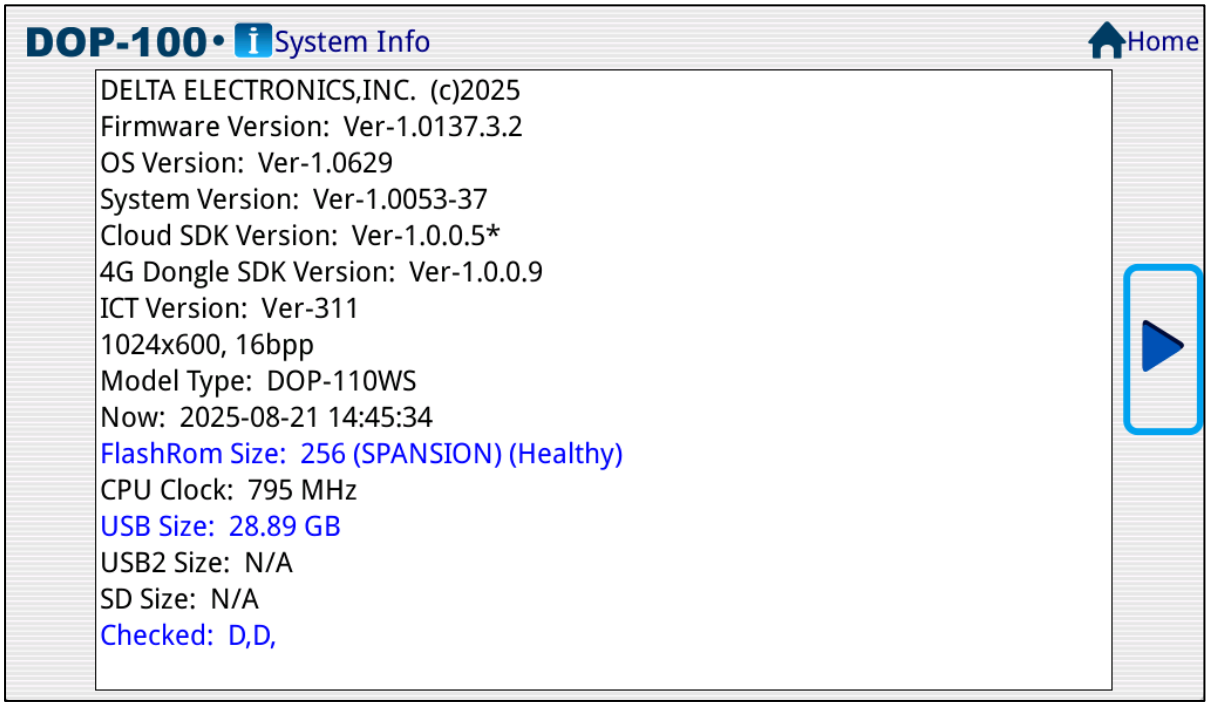
  Store the .app file generated by CODESYS or DIADesigner-AX software on a USB and use the upload and download functions of the HMI to download the program to the CODESYS controller through Ethernet. The HMI can also upload and save the controller program files in .app format to the USB devices.

  Note: The .app format file can be generated through CODESYS software > **Menu** > **Online** > **Create Boot Application**.

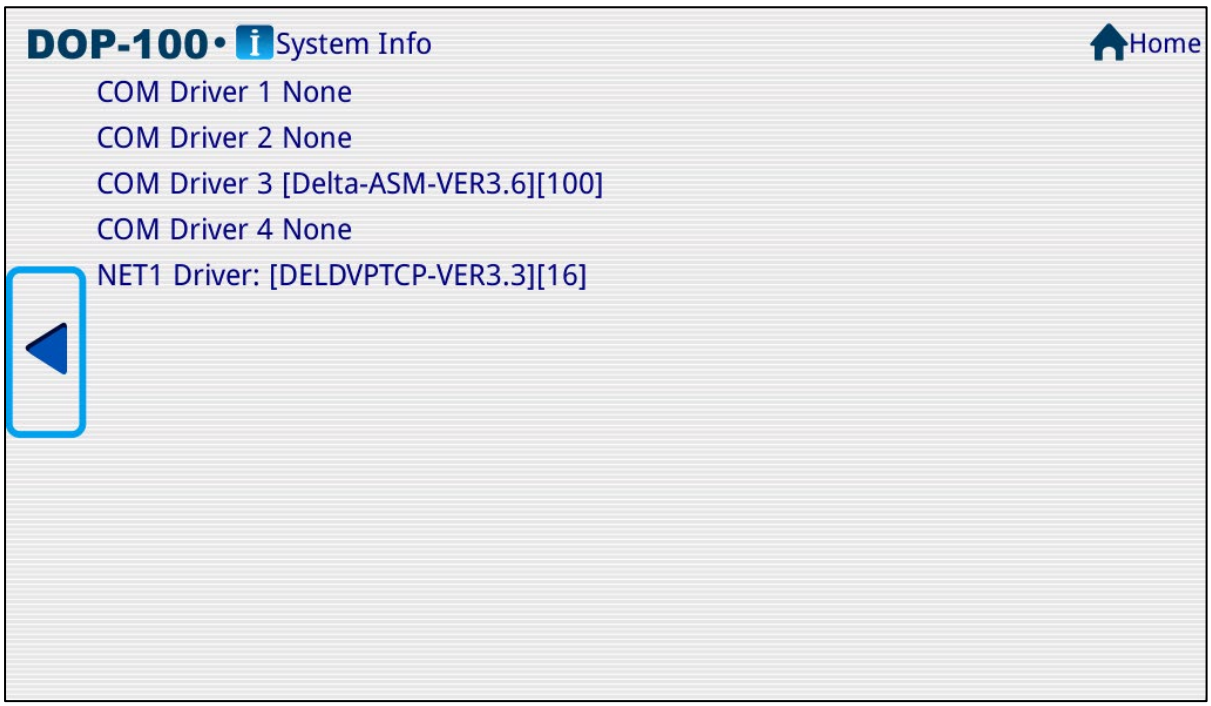


System Info

View the current firmware version, model type, battery capacity, Flash ROM size, CPU clock, system date and time, PLC device, and external storage device used on the HMI.



After switching the screen, you can view the PLC Driver information of the HMI.



HMI Doctor

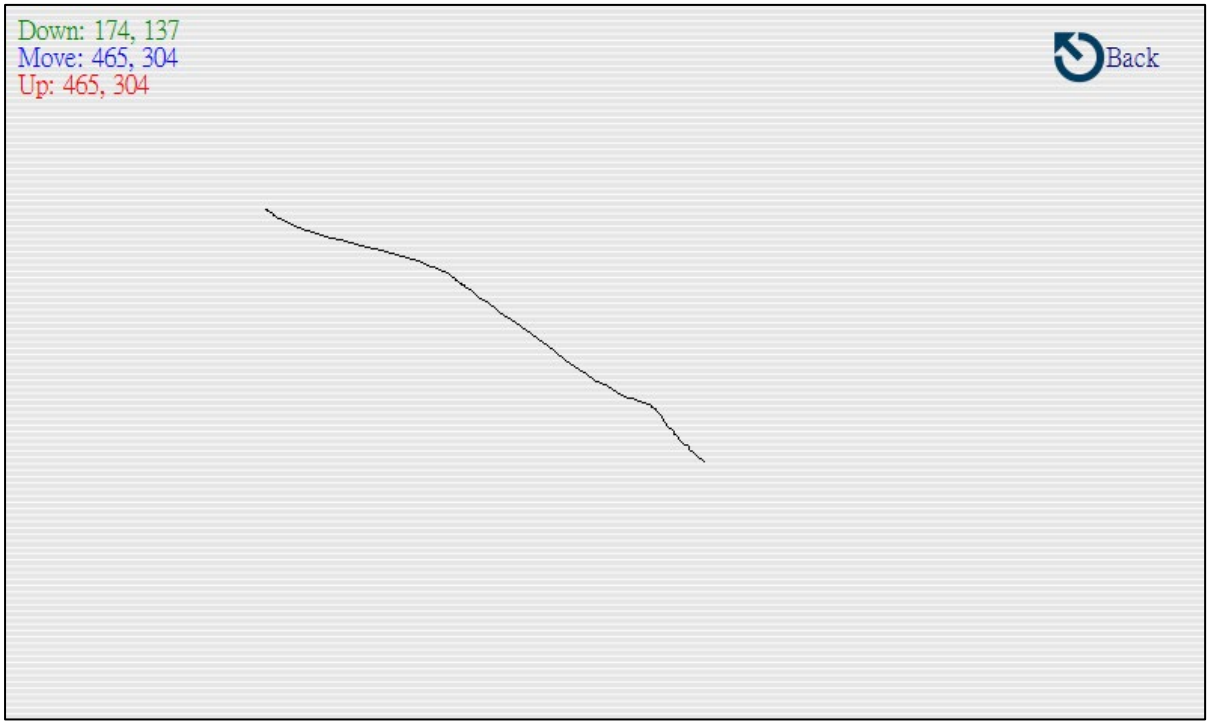
A simple application that enables you to check the hardware interface. Functions include Color (Red, Green, Blue, Black, and White), Draw Line, Buzz, ADC, and Network.

Color

Check the LCD monitor for black spots or stains.

Draw Line

Check whether there is a deviation between the position where the line is drawn and the actual displayed line position on the screen. If the deviation is too large, recalibrate the touch panel.

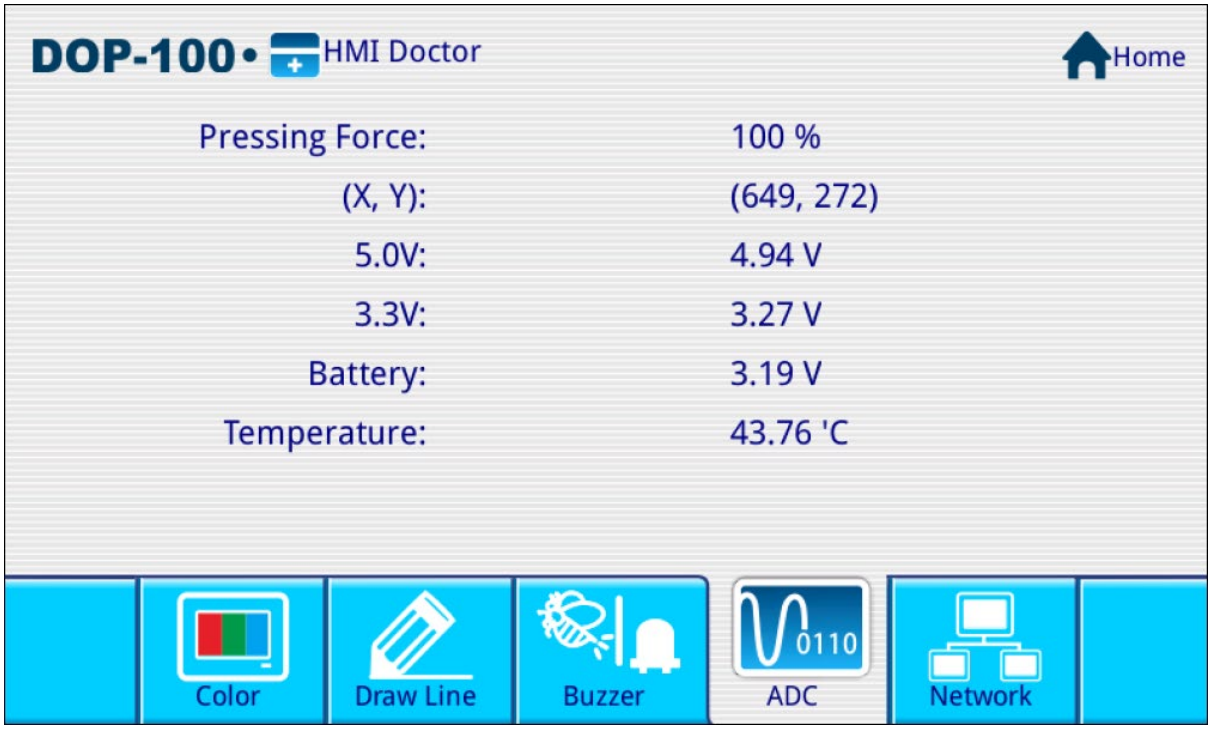


Buzzer

Check whether the buzzer of the HMI rings.

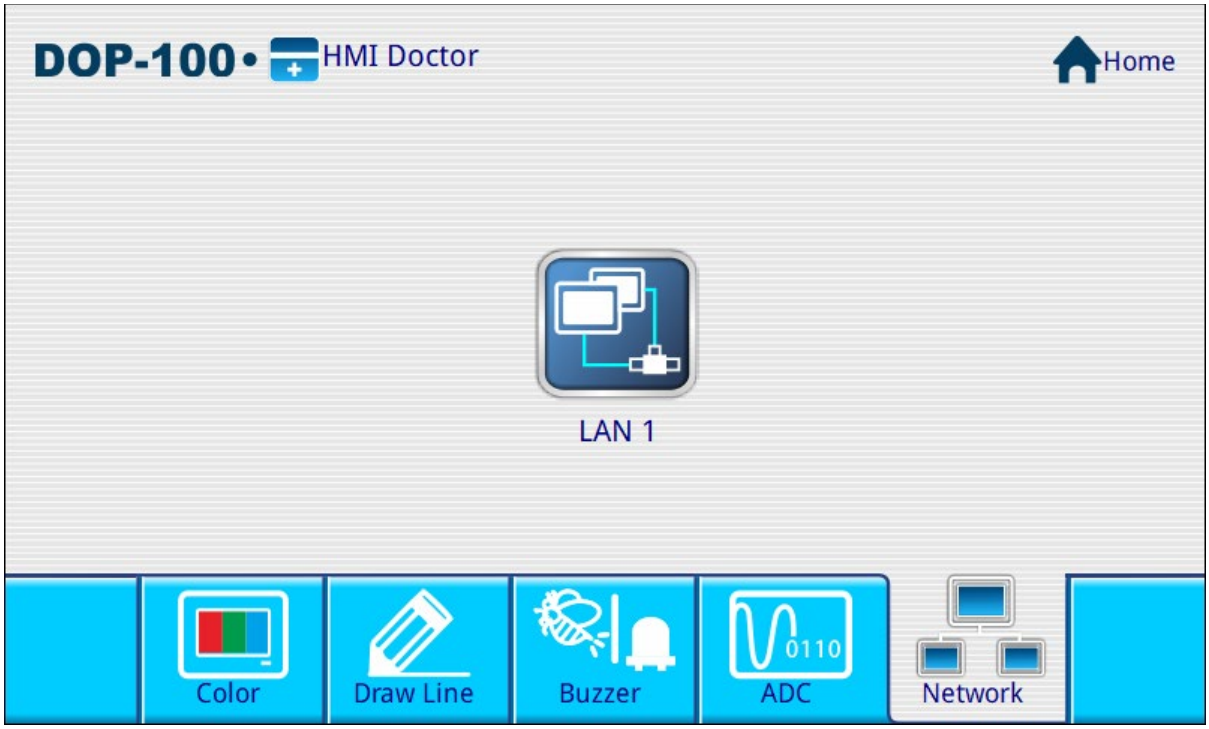
ADC

Check for the Pressing Force, touch position (X and Y coordinates), system voltage, Battery, and Temperature.



Network

Check whether the network is successfully connected. The network icon is initially in light-off status. If the connection is successful, the icon will light up.



Communication Error Code

This appendix describes the communication error messages that display when a communication error occurs for the HMI.

The following figure demonstrates the communication error messages for DOP-100, which register cannot access data normally. COMMUNICATION ERROR 3 displayed is the code for the error.

The following table lists the communication error codes and troubleshooting.

Code	Error Message	Cause	Troubleshooting
0x02	Unknown	Noise interference.	Improve the anti-noise interference equipment and shielding for the transmission cable.
0x03	NoResponse	• Timeout • Incorrect communication cable wiring, PLC station number, and communication parameters including the baud rate, parity, data bits, stop bit, etc.	Check if the setting on the left is incorrect.
0x04	HMICheckSumError	Error identified by PLC from checking the HMI CheckSum.	Improve the anti-noise interference equipment and shielding for the transmission cable.

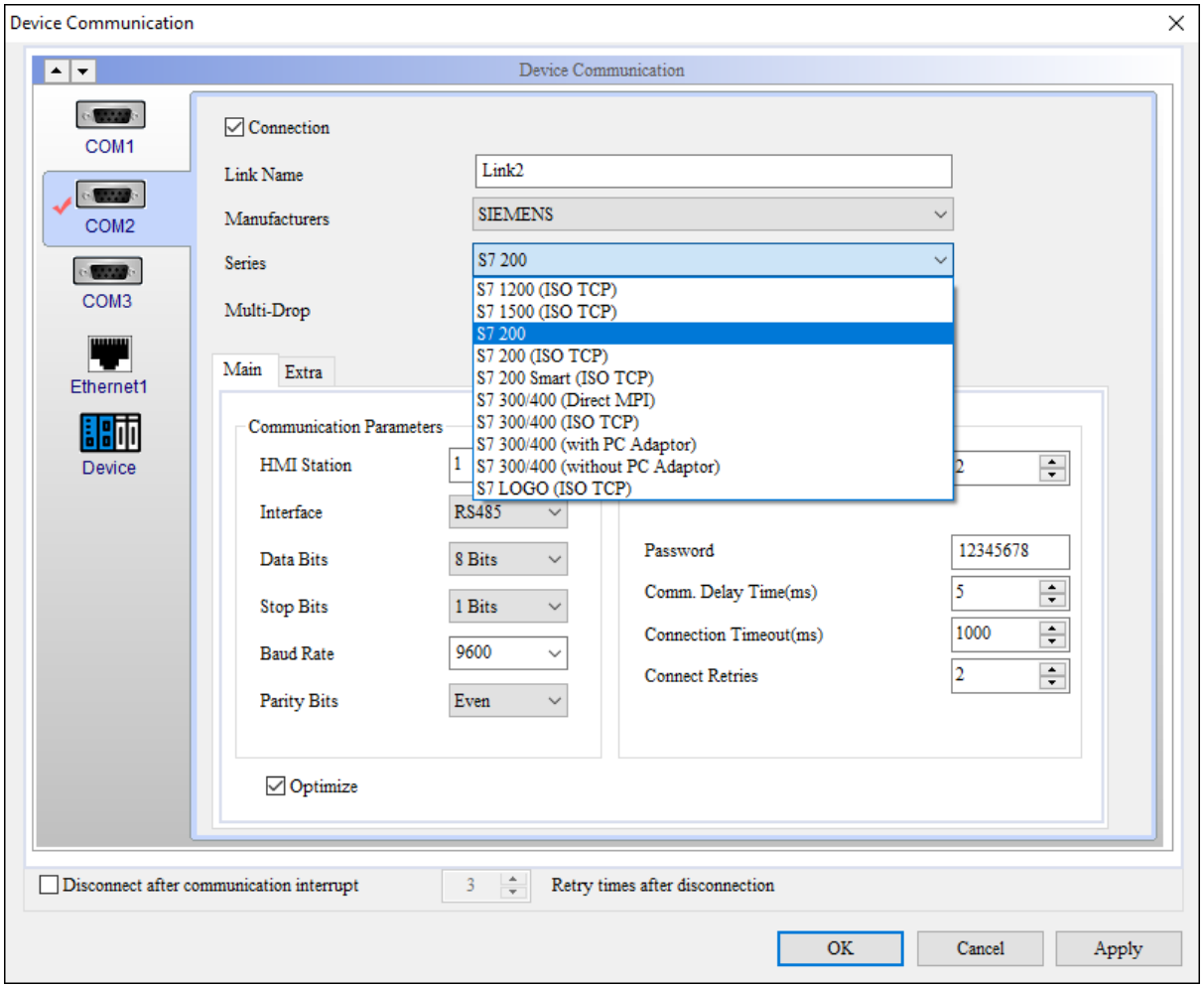
Code	Error Message	Cause	Troubleshooting
0x05	ControllerCheck SumError	Error identified by HMI from checking the PLC CheckSum.	Check if PLC CheckSum has been enabled (PLC software is usually required for the checking).
0x06	CommandError	Read and write PLC command error.	Check if the read and write address for the HMI exceeded the address available for use by the PLC, or if this address cannot be written.
0x07	AddressError	Read and write PLC address error.	Check if the read and write address for the HMI exceeded the address available for use by the PLC, or if this address cannot be written.
0x08	ValueError	Error in data written to the PLC.	Check the range of values accepted by the PLC.
0x09	Controller busy	PLC is busy and unable to process the given command.	PLC is busy, please try again later.
0x0A	NoCTS	HMI CTS pin did not receive PLC RTS signal.	Check if CTS pin on the HMI end and RTS pin on the PLC end are connected, or if the PLC has sent out an RTS signal.
0x0B	NoResource	PLC is overloaded with tasks.	Optimize PLC internal program to reduce workload and prevent such issues from recurring.
0x0C	NoService	PLC is handling an excessive workload	Simplify the internal program of PLC and reduce the workload to eliminate such problems.
0x0D	MustRetry	PLC prompts HMI to resend the command	Wait for HMI to resend the command.

Code	Error Message	Cause	Troubleshooting
0x0E	HMIStationNumberError	HMI station number error.	Check if the HMI station number exceeded the range of valid station numbers, or if it is duplicated with other station numbers.
0x0F	PLCStationNumberError	PLC station number error.	Check if the PLC station number exceeded the range of valid station numbers, or if it is duplicated with other station numbers.
0x10	UARTCommunicateFail	Communication error occurred at the bottom layer of the HMI. COM port was not opened correctly, or task overload on HMI caused abnormal COM port operation.	Make sure the COM port could be used normally, or simplify HMI task load, for example, delete ALARM or MACRO command.
0x1A	RTCSYNCError	PLC does not support this command.	Use the PLC that supports this command.
0x1B	Receive Error	Invalid data format sent by PLC.	Make sure the data format is correct.
0x20	LinkBroken	The network is disconnected (no TCP connection).	Verify that the network cable is connected correctly. If it is correct, check whether the PLC accepts the TCP connection.
0x21	TransIDError	Transaction ID does not match.	The network packet sequence is incorrect. Restart the HMI to restore the correct sequence.
0x22	NoConnection	Network interface mismatch — response does not match the HMI's outgoing interface.	Unplug and reconnect network cable.
0x23	GatewayPathUnavailable	The gateway is unable to route communication between HMI and PLC.	Verify that the network gateway is configured correctly.
0x24	LinksFull	The number of connections between HMI and PLC exceeds 16.	Reduce the number of connections to less than 16.

Code	Error Message	Cause	Troubleshooting
0x2F	ExceptionCode Error	The PLC responded with a code that is not within the Modbus specification.	Contact the PLC manufacturer to confirm the error code.
0x3F	OtherError	The response contains an error code that is not recognized by the protocol.	Verify the valid address range.

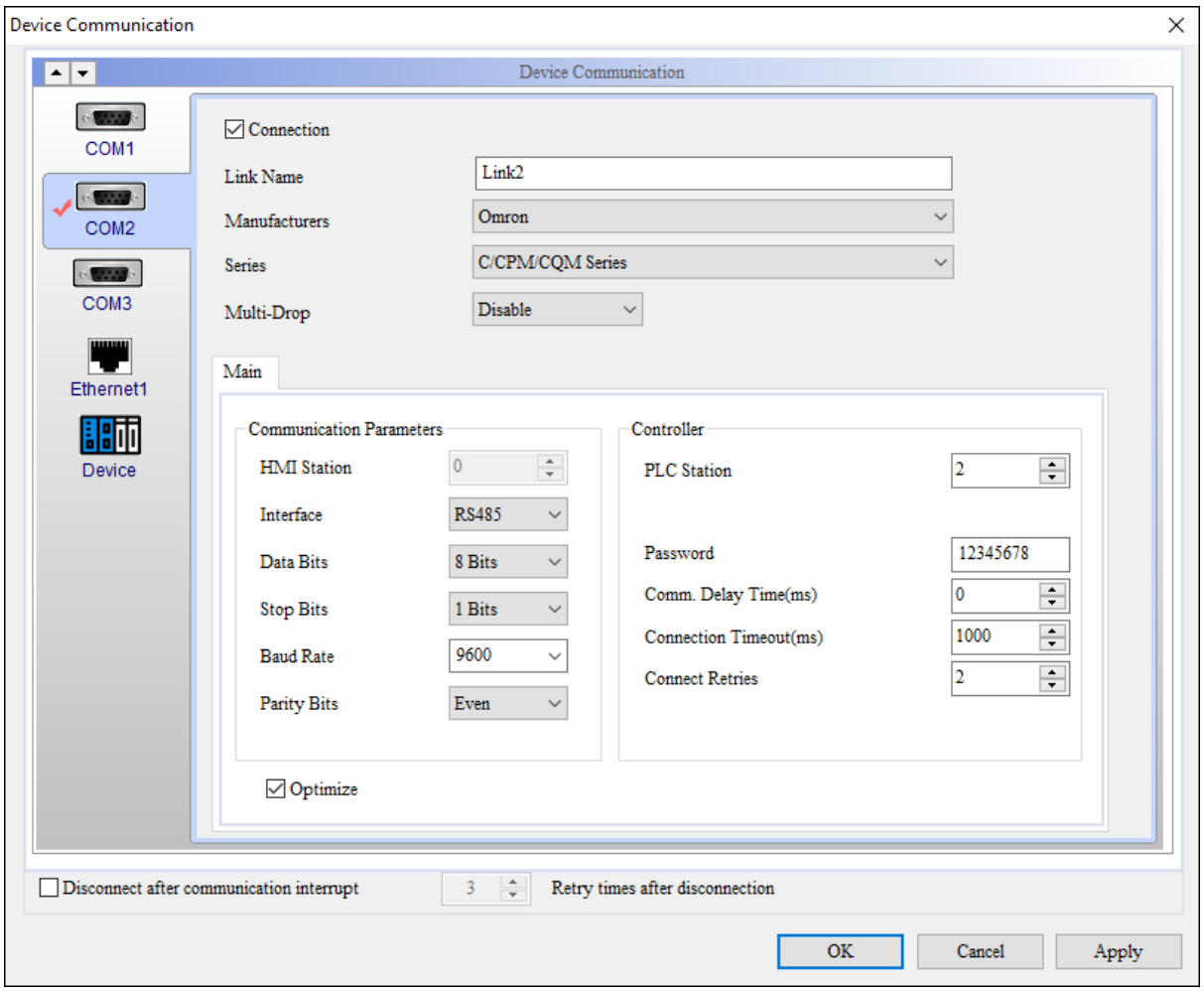
When writing to a register, the error code from the above table is logically OR'ed with 0x40. For example, if a "Value Error" code (0x08) occurs during the write operation, the resulting ValueError will be 0x08 OR 0x40 = 0x48.

The following lists the MPI communication error codes, which are the error messages displayed when the Siemens S7-300 series (Direct MPI), S7-300 series (Without PC Adaptor), and S7-200 series controllers fail to communicate with the HMI.



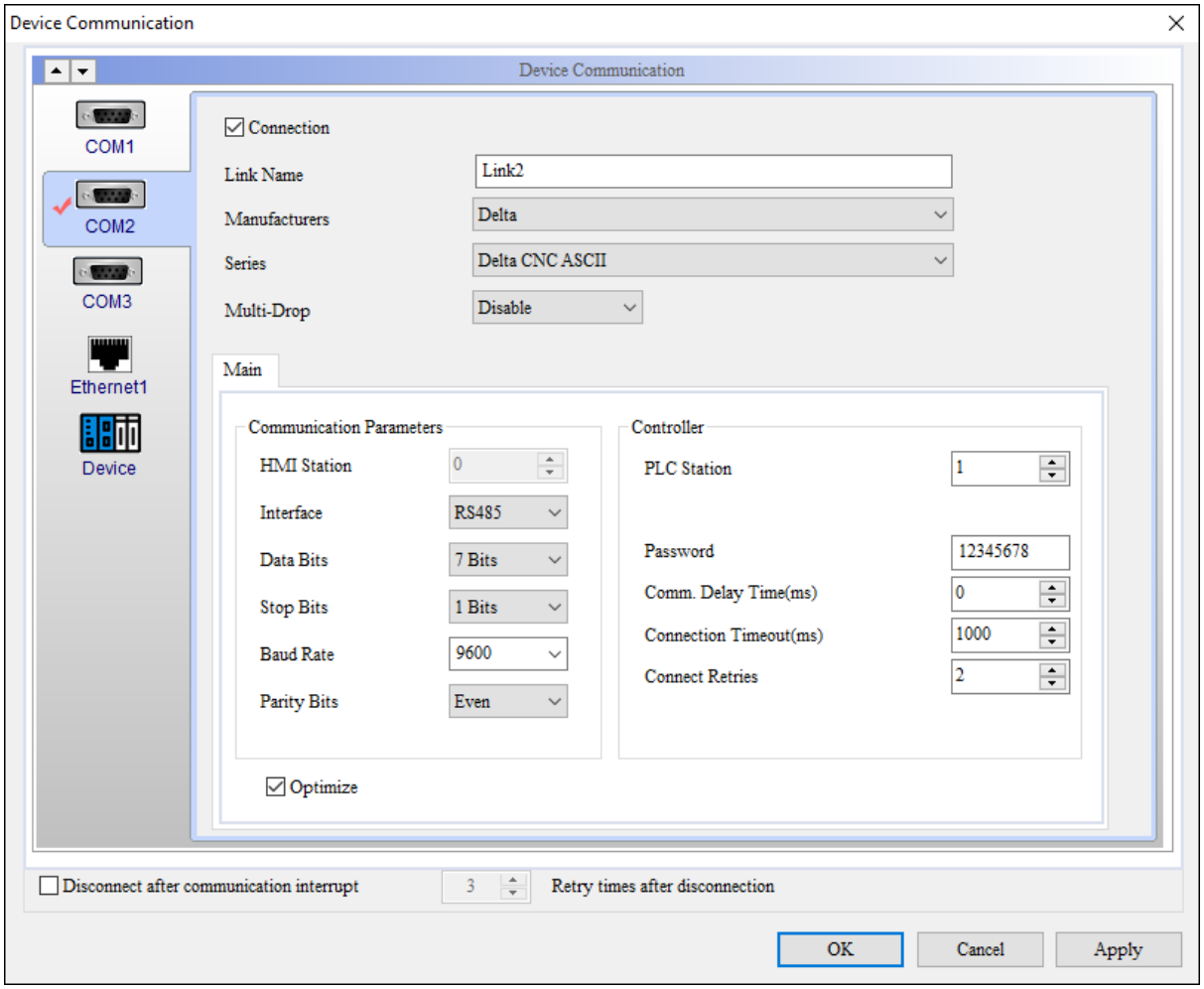
Code	Error Message	Cause	Troubleshooting
0x11	MPI_IDLE (For S7 300 - without PC adaptor, S7 300 - Direct MPI, or S7-200)	This is a special error message for Siemens PLC. HMI cannot acquire packets under idle Internet connection.	Check if there are too many connected modules causing insufficient bandwidth.
0x12	MPI_SN_COLLID (For S7 300 - without PC adaptor, S7 300 - Direct MPI, or S7-200)	This is a special error message for Siemens PLC. Station numbers are duplicated.	Check if there are modules with duplicated stations.
0x14	MPI_NO_SC (For S7 300 - without PC adaptor, S7 300 - Direct MPI, or S7-200)	This is a special error message for Siemens PLC. A maximum number of stations connected, with no more connections available.	Check if the number of the connected modules exceeds the limit.
0x18	MPINoResponse (For S7 300 - without PC adaptor or S7 300 - Direct MPI)	This is a special error message for Siemens PLC. There is no response for the connection requested, as there is no resource allocated.	Check if the network cable is disconnected or if it has a poor connection.
0x0B	NoResource (For S7 300 - without PC adaptor, S7 300 - Direct MPI, or S7-200)	This is a special error message for Siemens PLC. It is caused by task overload on the PLC.	Simplify the programs on the PLC to reduce task load.
0x0C	NoService (For S7 300 - without PC adaptor, S7 300 - Direct MPI, or S7-200)	This is a special error message for Siemens PLC. It is caused by task overload on the PLC.	Simplify the programs on the PLC to reduce task load.
0x3F	Read Error (For S7 300 - without PC adaptor, S7 300 - Direct MPI, or S7-200)	This is a special error message for Siemens PLC. The set PLC address exceeded the accessible range.	Set the PLC address within the accessible range.

The following table lists the error code displayed when using the OMRON C / CPM / CQM series controller to communicate and an error occurs.



Code	Error Message	Cause	Troubleshooting
0x1F	NOTExecutableInRunMode	The HMI is already connected with the PLC, but the PLC is in Run mode and cannot accept the "write data" command.	The PLC must be in Monitor mode to accept "write data" command. This error message will only appear once because when the HMI notices an error with the mode, it will automatically change the PLC mode to Monitor mode.

The following table lists the error code displayed when using the Delta CNC controller to communicate and an error occurs.



Code	Error Message	Cause	Troubleshooting
0x25	ParamPermissionError	This is a Delta CNC controller-specific error message indicating insufficient controller privileges to access the parameters configured for the component.	Access the controller's current permissions to set parameters.

Code	Error Message	Cause	Troubleshooting
0x26	ParamInputError	This is a Delta CNC controller-specific error message indicating that either the value being written to the parameter is invalid, or the parameter is currently unreadable.	• Check whether the parameter is read-only. • Check whether the validity period of the CNC controller has expired. • Check whether the string length of the written value exceeds the allowed range. • Check whether the format of the written value is correct.
0x27	ParamValueRangeExceed	This is a Delta CNC controller-specific error message indicating that the value being written to the parameter exceeds the allowable range for that parameter.	Set the parameter value within the valid range.
0x28	ParamWorkingModeError	This is a Delta CNC controller-specific error message indicating that the current controller operation mode is incorrect, and the parameter cannot be configured in this mode.	Switch the controller to the correct mode.

DOP-300 Series HMI Special Features

The following table lists the special features for DOP-300 series HMI.

Category	Function
Element	VNC Viewer
	Gantt Chart
Protocol	EIP Data Exchange Table
IIoT	OPC UA Server Safety Verification
	OPC UA Client Safety Verification
	MQTT JASON Custom
	Aliyun Server
Internal Parameter	IoT
	Storage - USB2_STATUS

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